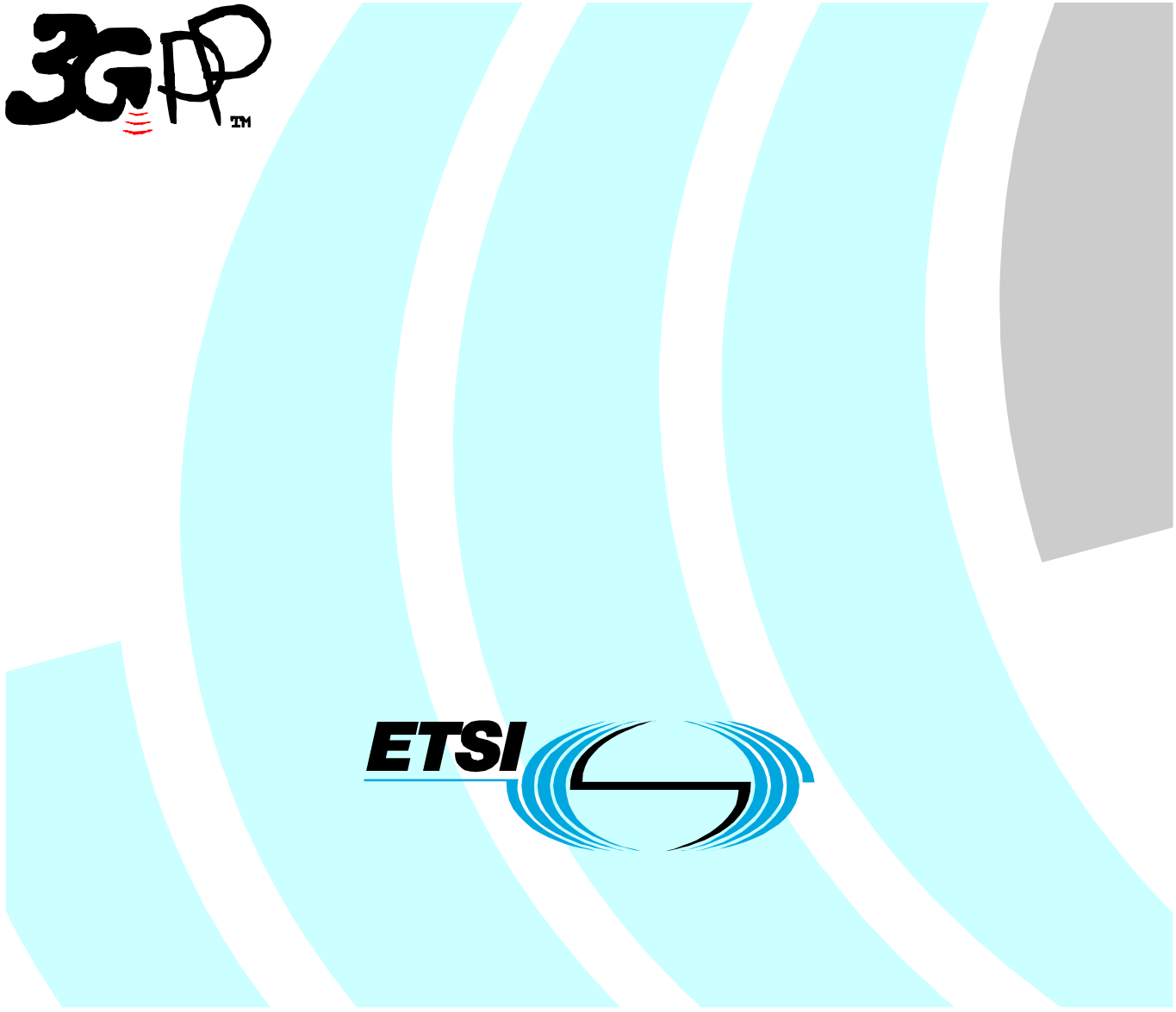
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650 Route des Lucioles
F-06921 Sophia Antipolis Cedex - FRANCE

Tel.: +33 4 92 94 42 00 Fax: +33 4 93 65 47 16

Siret N° 348 623 562 00017 - NAF 742 C
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# Contents

Intellectual Property Rights ..... 2
Foreword ..... 2
Foreword ..... 29
Introduction ..... 29
1 Scope ..... 30
2 References ..... 30
3 Definitions, abbreviations and symbols ..... 31
3.1 Definitions ..... 31
3.2 Abbreviations ..... 32
3.3 Symbols ..... 32
4 Common requirements of test equipment ..... 33
4.1 General Functional Requirements ..... 33
4.2 Minimum performance levels ..... 33
4.2.1 Supported Cell Configuration ..... 33
4.2.1.1 Supported Channels for FDD Mode ..... 34
4.2.1.1.1 Logical channels ..... 34
4.2.1.1.2 Transport channels ..... 34
4.2.1.1.3 Physical channels ..... 34
4.2.1.2 Supported Channels for TDD Mode ..... 35
4.2.1.2.1 Logical channels ..... 35
4.2.1.2.2 Transport channels ..... 35
4.2.1.2.3 Physical channels (3.84 Mcps option) ..... 35
4.2.1.2.4 Physical channels (1.28 Mcps option) ..... 36
4.2.1.3 Support of $T_{cell}$ timing offset ..... 36
4.2.2 RF Performance ..... 36
4.2.2.1 Frequency of Operation ..... 36
4.2.2.2 Power Level Setting Accuracy ..... 37
4.2.2.3 Uplink Power Control ..... 37
4.2.2.4 Uplink Signal Handling ..... 37
4.2.2.5 Uplink Sensitivity ..... 37
4.2.3 Timers Tolerances ..... 37
5 Reference test conditions ..... 37
5.1 Test frequencies ..... 37
5.1.1 FDD Mode Test frequencies ..... 38
5.1.1.1 FDD reference test frequencies for Operating Band I ..... 38
5.1.1.2 FDD reference test frequencies for Operating Band II ..... 38
5.1.1.3 FDD reference test frequencies for Operating Band III ..... 38
5.1.1.4 Void ..... 38
5.1.1.5 FDD reference test frequencies for Operating Band V ..... 38
5.1.1.6 FDD reference test frequencies for Operating Band VI ..... 38
5.1.1.7 FDD reference test frequencies for Operating Band VII ..... 38
5.1.1.8 FDD reference test frequencies for Operating Band VIII ..... 38
5.1.1.9 FDD reference test frequencies for Operating Band IX ..... 39
5.1.2 TDD Mode Test frequencies ..... 39
5.1.2.1 Standard TDD reference test frequencies (3.84 Mcps option) ..... 39
5.1.2.2 Standard TDD reference test frequencies (1.28 Mcps option) ..... 39
5.2 Radio conditions ..... 39
5.2.1 Normal propagation condition ..... 39
5.2.2 Static propagation condition ..... 39
5.2.3 Multi-path fading propagation conditions ..... 40
5.2.4 Moving propagation conditions ..... 40
5.2.5 Birth-Death propagation conditions ..... 40

5.3 Standard test signals ....40
5.4 Signal levels ....40
5.4.1 Downlink signal levels ....40
5.4.2 Uplink signal levels ....40

6 Reference system configurations ....41
6.1 Simulated network environments ....41
6.1.0a Default Master Information Block and Scheduling Block messages ....41
6.1.0a.1 Grouping SIBs for testing ....41
6.1.0a.2 SIB configurations ....41
6.1.0a.3 SIB default schedule ....42
Contents of Master Information Block PLMN type is the case of GSM-MAP ....42
Contents of Scheduling Block 1 (FDD and 1.28 Mcps TDD) ....43
Contents of Scheduling Block 1 (3.84 Mcps TDD) ....44
6.1.0a.4 SIB special schedules ....45
6.1.0a.4.1 SIB schedule for two S-CCPCH or two PRACH (For FDD) ....45
6.1.0a.4.2 SIB schedule for Idle Mode, Measurement and Inter RAT UTRAN to GERAN test cases ....46
6.1.0a.4.3 SIB schedule for Inter RAT handover GERAN to UTRAN test cases ....47
6.1.0b Default System Information Block Messages ....47
Contents of System Information Block type 1 (supported PLMN type is GSM-MAP) ....47
Contents of System Information Block type 2 ....48
Contents of System Information Block type 3 (FDD) ....48
Contents of System Information Block type 3 (3.84 Mcps TDD and 1.28 Mcps TDD) ....49
Contents of System Information Block type 4 in connected mode (FDD) ....50
Contents of System Information Block type 4 in connected mode (similar to SIB type3) (3.84 Mcps TDD and 1.28 Mcps TDD) ....50
Contents of System Information Block type 5 (FDD) ....51
Contents of System Information Block type 5 (3.84 Mcps TDD) ....54
Contents of System Information Block type 5 (1.28 Mcps TDD) ....57
Contents of System Information Block type 6 in connected mode (FDD) ....60
Contents of System Information Block type 6 in connected mode (similar to SIB type 5) (3.84 Mcps TDD) ....60
Contents of System Information Block type6 In connected mode (similar to SIB type5) (1.28 Mcps TDD) ....63
Contents of System Information Block type 7 (FDD) ....66
Contents of System Information Block type 7 (TDD) ....66
Contents of System Information Block type 8, 9 (only for FDD R99 and Rel-4) ....67
Contents of System Information Block type 10 (only for FDD R99 and Rel-4) ....67
Contents of System Information Block type 11 (FDD) ....67
Contents of System Information Block type 11 (3.84 Mcps and 1.28 Mcps TDD) ....70
Contents of System Information Block type 12 in connected mode (FDD) ....73
Contents of System Information Block type 12 in connected mode (3.84 Mcps and 1.28 Mcps TDD) ....73
Contents of System Information Block type 13 (used when supported PLMN type is ANSI-41) ....74
Contents of System Information Block type 14 (3.84 Mcps TDD) ....74
Contents of System Information Block type 16 (FDD) ....75
Contents of System Information Block type17 (3.84 Mcsps TDD and 1.28 Mcps TDD) ....81
Contents of System Information Block type 18 ....81
6.1.1 SCCPCH configuration with Stand-alone SRB for PCCH in the first SCCPCH and Interactive/Background 32 kbps PS RAB + SRBs for CCCH/DCCH/BCCH in the second SCCPCH ....81
Contents of System Information Block type 5 (FDD) ....81
Contents of System Information Block type 5 (3.84 Mcps TDD) ....85
Contents of System Information Block type 5 (1.28 Mcps TDD) ....88
Contents of System Information Block type 6 in connected mode (FDD) ....88
Contents of System Information Block type 6 in connected mode (3.84 Mcps TDD) ....89
Contents of System Information Block type 6 in connected mode (1.28 Mcps TDD) ....89
6.1.2 SCCPCH configuration with Stand-alone SRB for PCCH in the first SCCPCH, RB for CTCH + SRBs for CCCH/BCCH in the second SCCPCH and Interactive/Background 32 kbps PS RAB + SRBs for CCCH/DCCH/BCCH in the third SCCPCH (FDD only) ....89
Contents of System Information Block type 5 (FDD) ....89
Contents of System Information Block type 6 in connected mode (FDD) ....92
6.1.3 SCCPCH configuration with Stand-alone SRB for PCCH in the first SCCPCH and Interactive/Background 32 kbps PS RAB + SRBs for CCCH/DCCH/BCCH in the second and third SCCPCHs ....93
Contents of Scheduling Block 1 (FDD) ....93

Contents of System Information Block type 5 (FDD) ..... 94
Contents of System Information Block type 5 (3.84 Mcps TDD) ..... 98
Contents of System Information Block type 5 (1.28 Mcps TDD) ..... 98
6.1.4 Default parameters for 1 to 8 cell environments ..... 99
Default settings for cell No.1 (FDD) ..... 99
Contents of System Information Block type 11 for cell No.1 (FDD) ..... 99
Contents of System Information Block type 12 in connected mode for cell No.1 (FDD) ..... 99
Default settings for cell No.1 (TDD) ..... 99
Contents of System Information Block type 11 for cell No.1 (TDD) ..... 99
Contents of System Information Block type 12 in connected mode for cell No.1 (TDD) ..... 99
Cell No.2 ..... 99
Default settings for cell No.2 (FDD) ..... 99
Contents of System Information Block type 11 for cell No.2 (FDD) ..... 99
Default settings for cell No.2 (TDD) ..... 101
Contents of System Information Block type 11 for cell No.2 (TDD) ..... 101
Cell No.3 ..... 101
Default settings for cell No.3 (FDD) ..... 102
Contents of System Information Block type 11 for cell No.3 (FDD) ..... 102
Default settings for cell No.3 (TDD) ..... 103
Contents of System Information Block type 11 for cell No.3 (TDD) ..... 103
Cell No.4 ..... 104
Default settings for cell No.4 (FDD) ..... 104
Contents of System Information Block type 11 for cell No.4 (FDD) ..... 104
Default settings for cell No.4 (TDD) ..... 105
Contents of System Information Block type 11 for cell No.4 (TDD) ..... 106
Cell No.5 ..... 107
Default settings for cell No.5 (FDD) ..... 107
Contents of System Information Block type 11 for cell No.5 (FDD) ..... 107
Default settings for cell No.5 (TDD) ..... 108
Contents of System Information Block type 11 for cell No.5 (TDD) ..... 108
Cell No.6 ..... 109
Default settings for cell No.6 (FDD) ..... 110
Contents of System Information Block type 11 for cell No.6 (FDD) ..... 110
Default settings for cell No.6 (TDD) ..... 111
Contents of System Information Block type 11 for cell No.6 (TDD) ..... 111
Cell No.7 ..... 112
Default settings for cell No.7 (FDD) ..... 112
Contents of System Information Block type 11 for cell No.7 (FDD) ..... 113
Default settings for cell No.7 (TDD) ..... 113
Contents of System Information Block type 11 for cell No.7 (TDD) ..... 114
Cell No.8 ..... 114
Default settings for cell No.8 (FDD) ..... 114
Contents of System Information Block type 11 for cell No.8 (FDD) ..... 114
Default settings for cell No.8 (TDD) ..... 115
Contents of System Information Block type 11 for cell No.8 (TDD) ..... 115
Cell No.9 ..... 116
Contents of System Information for cell No.9 (GSM) ..... 116
Default settings for cell No.9 (GSM) ..... 116
Cell No.10 ..... 116
Contents of System Information for cell No.10 (GSM) ..... 116
Default settings for cell No.10 (GSM) ..... 116
Cell No.11 ..... 116
Default settings for cell No.11 (FDD) ..... 117
Contents of System Information Block type 11 for cell No.11 (FDD) ..... 117
6.1.4.1 Default Cell parameters Two PLMN in UTRAN test scenario ..... 117
Contents of System Information Block type 18 for cell No.1, 2, 3, 7, 8 ..... 118
Contents of System Information Block type 18 for cell No.4, 5, 6 ..... 118
6.1.4.2 Default Cell parameters Three PLMN in UTRAN test scenario ..... 118
Contents of System Information Block type 18 for cell No.1, 2, 3 ..... 118
Contents of System Information Block type 18 for cell No.4, 5, 6 ..... 118
Contents of System Information Block type 18 for cell No.7, 8 ..... 119
6.1.5 Reference Radio Conditions (FDD) ..... 119

6.1.6 Reference Radio Conditions (TDD) ..... 120
6.1.7 Reference Radio Conditions (GSM) ..... 121
6.2 Number of neighbour cells ..... 121
6.2.1 Basic Network ..... 121
6.2.2 Soft Handover Network (FDD) ..... 122
6.2.3 Hard Handover Network ..... 122
6.2.4 'Roaming' Network ..... 122
6.3 Cell/BS codes etc ..... 122
6.4 Routing/location area ..... 122
6.5 Network options settings ..... 122
6.6 Power control mode ..... 122
6.6.1 Downlink Power Control ..... 122
6.6.1.1 Outer Loop Power Control ..... 122
6.6.1.2 Inner Loop Power Control ..... 122
6.6.2 Uplink Power Control ..... 122
6.6.2.1 Outer Loop Power Control ..... 122
6.6.2.2 Inner Loop Power Control (FDD) ..... 122
6.7 Tx Diversity modes ..... 123
6.7.1 Non-Diverse Operation ..... 123
6.7.2 Diverse Operation ..... 123
6.7.2.1 Diverse Operation (FDD mode) ..... 123
6.7.2.2 Diverse Operation (TDD mode) ..... 123
6.7.2.2.1 3.84 Mcps option ..... 123
6.7.2.2.2 1.28 Mcps option ..... 123
6.8 Compressed mode parameters ..... 123
6.8.1 Single compressed mode pattern ..... 124
6.8.1.1 Inter Frequency FDD measurement ..... 124
6.8.1.2 Inter Frequency TDD measurement ..... 124
6.8.1.3 Inter RAT measurement (GSM - Carrier RSSI) ..... 125
6.8.1.4 Inter RAT measurement (GSM - Initial BSIC Identification) ..... 125
6.8.1.5 Inter RAT measurement (GSM - BSIC re-confirmation) ..... 126
6.8.2 Multiple compressed mode patterns ..... 126
6.8.2.1 Inter RAT measurement GSM ..... 126
6.8.2.2 Inter Frequency FDD measurement & Inter RAT measurement GSM ..... 127
6.8.2.3 Inter Frequency FDD measurement & Inter Frequency TDD measurement ..... 128
6.8.2.4 Inter Frequency TDD measurement & Inter RAT measurement GSM ..... 128
6.8.2.5 Inter Frequency FDD measurement & Inter Frequency TDD measurement & Inter RAT measurement GSM ..... 128
6.9 BCCH parameters ..... 128
6.10 Reference Radio Bearer configurations used in Radio Bearer interoperability testing ..... 128
6.10.1 QoS Architecture and RAB attributes ..... 129
6.10.2 RAB and signalling RB for FDD ..... 130
6.10.2.1 RABs and signalling RBs ..... 130
6.10.2.2 Combinations of RABs and Signalling RBs ..... 131
Combinations on DPCH ..... 131
Combinations on DSCH and DPCH ..... 135
Combinations on SCCPCH ..... 135
Combinations on PRACH ..... 136
Combinations on DPCH and HS-PDSCH ..... 136
Combinations on HS-PDSCH and E-DPDCH ..... 136
6.10.2.3 Example of linkage between RABs and services ..... 137
6.10.2.4 Typical radio parameter sets ..... 137
6.10.2.4.1 Combinations on DPCH ..... 137
6.10.2.4.1.1 Stand-alone UL:1.7 DL:1.7 kbps SRBs for DCCH ..... 137
6.10.2.4.1.2 Stand-alone UL:3.4 DL:3.4 kbps SRBs for DCCH ..... 139
6.10.2.4.1.3 Stand-alone UL:13.6 DL:13.6 kbps SRBs for DCCH ..... 140
6.10.2.4.1.4 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH ..... 141
6.10.2.4.1.4a Conversational / speech / UL:(12.2 7.95 5.9 4.75) DL:(12.2 7.95 5.9 4.75) kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH ..... 143
6.10.2.4.1.4b Conversational / speech / UL:(12.2 7.4 5.9 4.75) DL:(12.2 7.4 5.9 4.75) kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH + DL:0.15 kbps SRB#5 for DCCH ..... 145

6.10.2.4.1.5 Conversational / speech / UL:10.2 DL:10.2 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....147
6.10.2.4.1.5a Conversational / speech / UL:(10.2, 6.7, 5.9, 4.75) DL:(10.2, 6.7, 5.9, 4.75) kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....149
6.10.2.4.1.6 Conversational / speech / UL:7.95 DL:7.95 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....150
6.10.2.4.1.7 Conversational / speech / UL:7.4 DL:7.4 kbps / CS RAB+ UL:3.4 DL:3.4 kbps SRBs for DCCH....152
6.10.2.4.1.7a Conversational / speech / UL:(7.4, 6.7, 5.9, 4.75) DL:(7.4, 6.7, 5.9, 4.75) kbps / CS RAB+ UL:3.4 DL:3.4 kbps SRBs for DCCH....154
6.10.2.4.1.8 Conversational / speech / UL:6.7 DL:6.7 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....156
6.10.2.4.1.9 Conversational / speech / UL:5.9 DL:5.9 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....157
6.10.2.4.1.10 Conversational / speech / UL:5.15 DL:5.15 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....159
6.10.2.4.1.11 Conversational / speech / UL:4.75 DL:4.75 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....161
6.10.2.4.1.12 Conversational / unknown / UL:28.8/DL:28.8 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....163
6.10.2.4.1.13 Conversational / unknown / UL:64 DL:64 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....164
6.10.2.4.1.14 Conversational / unknown / UL:32 DL:32 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....166
6.10.2.4.1.15 Streaming / unknown / UL:14.4/DL:14.4 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....167
6.10.2.4.1.16 Streaming / unknown / UL:28.8/DL:28.8 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....168
6.10.2.4.1.17 Streaming / unknown / UL:57.6/DL:57.6 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....170
6.10.2.4.1.18 Void....171
6.10.2.4.1.19 Void....171
6.10.2.4.1.20 Void....171
6.10.2.4.1.21 Void....171
6.10.2.4.1.22 Void....171
6.10.2.4.1.23 Interactive or background / UL:32 DL:8 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....171
6.10.2.4.1.23a Interactive or background / UL:8 DL:8 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....173
6.10.2.4.1.23b Interactive or background / UL:16 DL:16 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....174
6.10.2.4.1.23c Interactive or background / UL:32 DL:32 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....176
6.10.2.4.1.23d Interactive or background / UL:32 DL:32 kbps / PS RAB (20 ms TTI)+ UL:3.4 DL:3.4 kbps SRBs for DCCH....177
6.10.2.4.1.24 Void....179
6.10.2.4.1.25 Interactive or background / UL:32 DL: 64 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....179
6.10.2.4.1.26 Interactive or background / UL:64 DL: 64 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....180
6.10.2.4.1.27 Interactive or background / UL:64 DL:128 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....180
6.10.2.4.1.28 Interactive or background / UL:128 DL:128 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....181
6.10.2.4.1.29 Interactive or background / UL:64 DL:144 kbps / PS RAB + UL:3.4 DL: 3.4 kbps SRBs for DCCH....182
6.10.2.4.1.30 Interactive or background / UL:144 DL:144 kbps / PS RAB + UL:3.4 DL: 3.4 kbps SRBs for DCCH....183
6.10.2.4.1.31 Interactive or background / UL:64 DL:256 kbps / PS RAB + UL:3.4 DL: 3.4 kbps SRBs for DCCH....184
6.10.2.4.1.32 Interactive or background / UL:64 DL:384 kbps / PS RAB + UL:3.4 DL: 3.4 kbps SRBs for DCCH....185

6.10.2.4.1.33 Interactive or background / UL:128 DL:384 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....186
6.10.2.4.1.34 Interactive or background / UL:384 DL:384 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....186
6.10.2.4.1.35 Interactive or background / UL:64 DL:2 048 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....187
6.10.2.4.1.36 Void....188
6.10.2.4.1.37 Void....188
6.10.2.4.1.38 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:32 DL:8 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....188
6.10.2.4.1.38a Conversational / speech / 12.2 kbps / CS RAB + Interactive or background / UL:0 DL:0 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....189
6.10.2.4.1.38b Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:8 DL:8 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....191
6.10.2.4.1.38c Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:32 DL:32 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....193
6.10.2.4.1.38d Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:64 DL:64 kbps / PS RAB + Interactive or background / UL:64 DL:64 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....194
6.10.2.4.1.38e Conversational / speech / (12.2 7.95 5.9 4.75) kbps / CS RAB + Interactive or background / UL:0 DL:0 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....196
6.10.2.4.1.38f Conversational / speech / (12.2 7.95 5.9 4.75) kbps / CS RAB + Interactive or background / UL:8 DL:8 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....197
6.10.2.4.1.38g Conversational / speech / (12.2 7.95 5.9 4.75) kbps / CS RAB + Interactive or background / UL:16 DL:16 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....198
6.10.2.4.1.38h Conversational / speech / (12.2 7.95 5.9 4.75) kbps / CS RAB + Interactive or background / UL:32 DL:32 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....199
6.10.2.4.1.38i Conversational / speech / (12.2 7.95 5.9 4.75) kbps / CS RAB + Interactive or background / UL:64 DL:64 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....201
6.10.2.4.1.38j Conversational / speech / (12.2 7.95 5.9 4.75) kbps / CS RAB + Interactive or background / UL:64 DL:128 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....203
6.10.2.4.1.38k Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:32 DL:32 kbps / PS RAB + Interactive or background / UL:32 DL:32 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH (L1 multiplexing)....204
6.10.2.4.1.39 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:32 DL:64 kbps / PS RAB+ UL:3.4 DL: 3.4 kbps SRBs for DCCH....206
6.10.2.4.1.40 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:64 DL:64 kbps / PS RAB+ UL:3.4 DL: 3.4 kbps SRBs for DCCH....207
6.10.2.4.1.41 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:64 DL:128 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....207
6.10.2.4.1.42 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:64 DL:256 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....208
6.10.2.4.1.43 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:64 DL:384 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....209
6.10.2.4.1.44 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:128 DL:2 048 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....210
6.10.2.4.1.45 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Streaming / unknown / UL:57.6 DL:57.6 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....213
6.10.2.4.1.46 Void....214
6.10.2.4.1.47 Void....214
6.10.2.4.1.48 Void....214
6.10.2.4.1.49 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Conversational / unknown / UL:64 DL:64 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....214
6.10.2.4.1.49a Conversational / speech / UL:(12.2 7.95 5.9 4.75) DL:(12.2 7.95 5.9 4.75) kbps / CS RAB + Conversational / unknown / UL:64 DL:64 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....215
6.10.2.4.1.50 Conversational / unknown / UL:64 DL:64 kbps / CS RAB + Conversational / unknown / UL:64 DL:64 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....216
6.10.2.4.1.51 Conversational / unknown / UL:64 DL:64 kbps / CS RAB + Interactive or background / UL:64 DL:64 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....217
6.10.2.4.1.51a Conversational / unknown / UL:64 DL:64 kbps / CS RAB + Interactive or Background / UL:8 DL:8 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....218

6.10.2.4.1.51b Conversational / unknown / UL:64 DL:64 kbps / CS RAB + Interactive or Background / UL:16 DL:64 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH ....219
6.10.2.4.1.52 Conversational / unknown / UL:64 DL:64 kbps / CS RAB + Interactive or background / UL:64 DL:128 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH ....220
6.10.2.4.1.53 Conversational / unknown / UL:64 DL:64 kbps / CS RAB + Interactive or background / UL:128 DL:128 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH ....221
6.10.2.4.1.54 Void ....221
6.10.2.4.1.55 Void ....221
6.10.2.4.1.56 Interactive or background / UL:8 DL:8 kbps / PS RAB + Interactive or background / UL:8 DL:8 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH ....221
6.10.2.4.1.57 Interactive or background / UL:64 DL:64 kbps / PS RAB + Interactive or background / UL:64 DL:64 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH ....223
6.10.2.4.1.58 Streaming / unknown / UL:16 DL:64 kbps / PS RAB + Interactive or background / UL:8 DL:8 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....225
6.10.2.4.1.58a Streaming / unknown / UL:16 DL:128 kbps / PS RAB + Interactive or background / UL:8 DL:8 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....226
6.10.2.4.1.59 Conversational / speech / UL:42.8 DL:42.8 kbps / PS RAB + Interactive / UL:16 DL:16 kbps / PS RAB + Interactive / UL:16 DL:16 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....227
6.10.2.4.1.60 Conversational / speech / UL:42.8 DL:42.8 kbps / PS RAB + Interactive / UL:16 DL:16 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH ....230
6.10.2.4.1.61 Conversational / unknown / UL:8 DL:8 kbps / PS RAB + Interactive or Background / UL:8 DL:8 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH ....231
6.10.2.4.1.62 Conversational / speech / UL:(12.65 8.85 6.6) DL:(12.65 8.85 6.6) kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH + DL:0.15 kbps SRB#5 for DCCH ....234
6.10.2.4.1.63 Interactive or background / UL:64 DL:768 kbps / PS RAB + UL:3.4 DL: 3.4 kbps SRBs for DCCH....236
6.10.2.4.2 Combinations on PDSCH and DPCH....237
6.10.2.4.2.1 Void ....237
6.10.2.4.2.2 Interactive or background / UL:64 DL:384 kbps / PS RAB + UL:3.4 DL: 3.4 kbps SRBs for DCCH....237
6.10.2.4.2.3 Interactive or background / UL:64 DL:2 048 kbps / PS RAB + UL:3.4 DL: 3.4 kbps SRBs for DCCH....238
6.10.2.4.2.4 Void ....240
6.10.2.4.2.5 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:64 DL:384 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH ....240
6.10.2.4.2.6 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:64 DL:2 048 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH ....240
6.10.2.4.3 Combinations on SCCPCH ....241
6.10.2.4.3.1 Stand-alone signalling RB for PCCH ....241
6.10.2.4.3.2 Interactive/Background 32 kbps PS RAB + SRBs for CCCH + SRB for DCCH + SRB for BCCH ....242
6.10.2.4.3.2a Interactive/Background 32 kbps PS RAB + Interactive/Background 32 kbps PS RAB + SRBs for CCCH + SRB for DCCH + SRB for BCCH ....243
6.10.2.4.3.3 Interactive/Background 32 kbps RAB + SRB for PCCH + SRB for CCCH + SRB for DCCH + SRB for BCCH ....244
6.10.2.4.3.4 RB for CTCH + SRB for CCCH + SRB for BCCH ....245
6.10.2.4.4 Combinations on PRACH ....246
6.10.2.4.4.1 Interactive/Background 32 kbps PS RAB + SRB for CCCH + SRB for DCCH ....246
6.10.2.4.4.2 Interactive/Background 32 kbps PS RAB + Interactive/Background 32 kbps PS RAB + SRB for CCCH + SRB for DCCH ....246
6.10.2.4.5 Combinations on DPCH and HS-PDSCH ....247
6.10.2.4.5.1 Interactive or background / UL:64 DL: [max bit rate depending on UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH ....247
6.10.2.4.5.1a Interactive or background / UL:128 DL: [max bit rate depending on UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH ....249
6.10.2.4.5.2 Interactive or background / UL:384 DL: [max bit rate depending on UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH ....250
6.10.2.4.5.3 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:384 DL:[Bit rate depending on the UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....250

6.10.2.4.5.3a Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:64 DL:[Bit rate depending on the UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....252
6.10.2.4.5.4 Conversational / unknown / UL:64 DL:64 kbps / CS RAB + Interactive or background / UL:384 DL:[Bit rate depending on the UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....252
6.10.2.4.5.4a Conversational / unknown / UL:64 DL:64 kbps / CS RAB + Interactive or background / UL:64 DL:[Bit rate depending on the UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....254
6.10.2.4.5.5 Interactive or background / UL:384 DL:[Bit rate depending on the UE category] / PS RAB + Interactive or background / UL:384 DL:[Bit rate depending on the UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH6.10.2.4.5.5.1....Uplink254
6.10.2.4.5.5a Interactive or background / UL:64 DL:[Bit rate depending on the UE category] / PS RAB + Interactive or background / UL:64 DL:[Bit rate depending on the UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....255
6.10.2.4.5.6 Streaming / unknown / UL:128 DL: [guaranteed 128, max bit rate depending on UE category] kbps / PS RAB + Interactive or background / UL:128 DL: [max bit rate depending on UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....257
6.10.2.4.5.7 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Streaming / unknown / UL:128 DL: [guaranteed 128, max bit rate depending on UE category] kbps / PS RAB + Interactive or background / UL:128 DL: [max bit rate depending on UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....258
6.10.2.4.5.8 Conversational / speech / UL:(12.65 8.85 6.6) DL:(12.65 8.85 6.6) kbps / CS RAB + Interactive or Background / UL:384 DL:[Bit rate depending on the UE category] / PS RAB+ UL:3.4 DL:3.4 kbps SRBs for DCCH + DL:0.15 kbps SRB#5 for DCCH....260
6.10.2.4.6 Combinations on HS-PDSCH and E-DPDCH....263
6.10.2.4.6.1 Interactive or background / UL: [max bit rate depending on UE category and TTI] DL: [max bit rate depending on UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH on DCH....263
6.10.2.4.6.2 Interactive or background / UL: [max bit rate depending on UE category and TTI] DL: [max bit rate depending on UE category] / PS RAB + UL:[max bit rate depending on UE category and TTI] DL:3.4 kbps SRBs for DCCH on E-DCH and DL DCH....264
6.10.2.4.6.3 Interactive or background / UL: [max bit rate depending on UE category and TTI] DL: [max bit rate depending on UE category] / PS RAB + UL: [max bit rate depending on UE category and TTI] DL: [max bit rate depending on UE category] SRBs for DCCH on E-DCH and HS-DSCH....266
6.10.2.4.6.4 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL: [max bit rate depending on UE category and TTI] DL: [max bit rate depending on UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....266
6.10.2.4.6.5 Streaming / unknown / UL:[max bit rate depending on UE category and TTI] DL: [max bit rate depending on UE category] kbps / PS RAB + Interactive or background / UL: [max bit rate depending on UE category and TTI] DL: [max bit rate depending on UE category] / PS RAB + UL:[max bit rate depending on UE category and TTI] DL:3.4 kbps SRBs for DCCH on E-DCH and DL DCH....267
6.10.3 RAB and signalling RB for TDD....268
6.10.3.1 RABs and signalling RBs....268
6.10.3.2 Combinations of RABs and Signalling RBs....269
Combinations on DPCH....269
Combinations on PDSCH, SCCPCH, PUSCH and PRACH....273
Combinations on PDSCH, SCCPCH, DPCH, PUSCH and PRACH....273
Combinations on SCCPCH....273
Combinations on PRACH....274
6.10.3.3 Example of linkage between RABs and services....274
6.10.3.4 Typical radio parameter sets....274
6.10.3.4.1 Combinations on DPCH....274
6.10.3.4.1.1 Stand-alone UL:1.7 DL:1.7 kbps SRBs for DCCH....274
6.10.3.4.1.1a Stand-alone UL:1.7 DL:1.7 kbps SRBs for DCCH (multiframe)....276
6.10.3.4.1.2 Stand-alone UL:3.4 DL:3.4 kbps SRBs for DCCH....277
6.10.3.4.1.3 Stand-alone UL:13.6 DL:13.6 kbps SRBs for DCCH....279
6.10.3.4.1.4 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....281

6.10.3.4.1.4a Conversational / speech / UL:(12.2, 7.95, 5.9, 4.75) DL:(12.2, 7.95, 5.9, 4.75) kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH ..... 282
6.10.3.4.1.5 Conversational / speech / UL:10.2 DL:10.2 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH ..... 284
6.10.3.4.1.5a Conversational / speech / UL:(10.2, 6.7, 5.9, 4.75) DL:(10.2, 6.7, 5.9, 4.75) kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH ..... 286
6.10.3.4.1.6 Conversational / speech / UL:7.95 DL:7.95 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH ..... 288
6.10.3.4.1.7 Conversational / speech / UL:7.4 DL:7.4 kbps / CS RAB+ UL:3.4 DL:3.4 kbps SRBs for DCCH ..... 289
6.10.3.4.1.7a Conversational / speech / UL:(7.4, 6.7, 5.9, 4.75) DL:(7.4, 6.7, 5.9, 4.75) kbps / CS RAB+ UL:3.4 DL:3.4 kbps SRBs for DCCH ..... 291
6.10.3.4.1.8 Conversational / speech / UL:6.7 DL:6.7 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH ..... 293
6.10.3.4.1.9 Conversational / speech / UL:5.9 DL:5.9 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH ..... 294
6.10.3.4.1.10 Conversational / speech / UL:5.15 DL:5.15 kbps / CS RAB + UL:1.7 DL:1.7 kbps SRBs for DCCH ..... 296
6.10.3.4.1.11 Conversational / speech / UL:4.75 DL:4.75 kbps / CS RAB + UL:1.7 DL:1.7 kbps SRBs for DCCH ..... 298
6.10.3.4.1.12 Conversational / unknown / UL:28.8/DL:28.8 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH ..... 299
6.10.3.4.1.13 Conversational / unknown / UL:64 DL:64 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH ..... 301
6.10.3.4.1.14 Conversational / unknown / UL:32 DL:32 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH ..... 302
6.10.3.4.1.15 Streaming / unknown / UL:14.4/DL:14.4 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH ..... 304
6.10.3.4.1.16 Streaming / unknown / UL:28.8/DL:28.8 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH ..... 305
6.10.3.4.1.17 Streaming / unknown / UL:57.6/DL:57.6 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH ..... 307
6.10.3.4.1.18 Void ..... 308
6.10.3.4.1.19 Void ..... 308
6.10.3.4.1.20 Void ..... 308
6.10.3.4.1.21 Void ..... 308
6.10.3.4.1.22 Void ..... 308
6.10.3.4.1.23 Interactive or background / UL:32 DL:8 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH ..... 308
6.10.3.4.1.23a Interactive or background / UL:8 DL:8 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH ..... 310
6.10.3.4.1.23b Interactive or background / UL:16 DL:16 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH ..... 311
6.10.3.4.1.23c Interactive or background / UL:32 DL:32 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH ..... 312
6.10.3.4.1.23d Interactive or background / UL:32 DL:32 kbps / PS RAB (20 ms TTI)+ UL:3.4 DL:3.4 kbps SRBs for DCCH ..... 314
6.10.3.4.1.24 Void ..... 315
6.10.3.4.1.25 Interactive or background / UL:32 DL: 64 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH ..... 315
6.10.3.4.1.26 Interactive or background / UL:64 DL: 64 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH ..... 316
6.10.3.4.1.27 Interactive or background / UL:64 DL:128 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH ..... 317
6.10.3.4.1.28 Interactive or background / UL:128 DL:128 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH ..... 318
6.10.3.4.1.29 Interactive or background / UL:64 DL:144 kbps / PS RAB + UL:3.4 DL: 3.4 kbps SRBs for DCCH ..... 319
6.10.3.4.1.30 Interactive or background / UL:144 DL:144 kbps / PS RAB + UL:3.4 DL: 3.4 kbps SRBs for DCCH ..... 320
6.10.3.4.1.31 Interactive or background / UL:64 DL:256 kbps / PS RAB + UL:3.4 DL: 3.4 kbps SRBs for DCCH ..... 321

6.10.3.4.1.32 Interactive or background / UL:64 DL:384 kbps / PS RAB + UL:3.4 DL: 3.4 kbps SRBs for DCCH....322
6.10.3.4.1.33 Interactive or background / UL:128 DL:384 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....323
6.10.3.4.1.34 Interactive or background / UL:384 DL:384 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....323
6.10.3.4.1.35 Interactive or background / UL:64 DL:2 048 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....324
6.10.3.4.1.36 Void....325
6.10.3.4.1.37 Void....325
6.10.3.4.1.38 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:32 DL:8 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....325
6.10.3.4.1.38a Conversational / speech / 12.2 kbps / CS RAB + Interactive or background / UL:0 DL:0 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....327
6.10.3.4.1.38b Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:8 DL:8 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....328
6.10.3.4.1.38c Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:32 DL:32 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....329
6.10.3.4.1.38d Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:64 DL:64 kbps / PS RAB + Interactive or background / UL:64 DL:64 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....331
6.10.3.4.1.38e Conversational / speech / UL:(12.2, 7.95, 5.9, 4.75) DL:(12.2, 7.95, 5.9, 4.75) kbps / CS RAB + Interactive or background / UL:0 DL:0 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....333
6.10.3.4.1.38f Conversational / speech / (12.2, 7.95, 5.9, 4.75) kbps / CS RAB + Interactive or background / UL:8 DL:8 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....334
6.10.3.4.1.38g Conversational / speech / (12.2, 7.95, 5.9, 4.75) kbps / CS RAB + Interactive or background / UL:16 DL:16 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....335
6.10.3.4.1.38h Conversational / speech / (12.2, 7.95, 5.9, 4.75) kbps / CS RAB + Interactive or background / UL:32 DL:32 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....337
6.10.3.4.1.38i Conversational / speech / (12.2, 7.95, 5.9, 4.75) kbps / CS RAB + Interactive or background / UL:64 DL:64 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....338
6.10.3.4.1.38j Conversational / speech / (12.2, 7.95, 5.9, 4.75) kbps / CS RAB + Interactive or background / UL:64 DL:128 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....340
6.10.3.4.1.39 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:32 DL:64 kbps / PS RAB+ UL:3.4 DL: 3.4 kbps SRBs for DCCH....341
6.10.3.4.1.40 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:64 DL:64 kbps / PS RAB+ UL:3.4 DL: 3.4 kbps SRBs for DCCH....342
6.10.3.4.1.41 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:64 DL:128 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....344
6.10.3.4.1.42 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:64 DL:256 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....345
6.10.3.4.1.43 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:64 DL:384 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....347
6.10.3.4.1.44 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:128 DL:2 048 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....348
6.10.3.4.1.45 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Streaming / unknown / UL:57.6 DL:57.6 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....351
6.10.3.4.1.46 Void....352
6.10.3.4.1.47 Void....352
6.10.3.4.1.48 Void....352
6.10.3.4.1.49 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Conversational / unknown / UL:64 DL:64 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....352
6.10.3.4.1.49a Conversational / speech / UL:(12.2 7.95 5.9 4.75) DL:(12.2 7.95 5.9 4.75) kbps / CS RAB + Conversational / unknown / UL:64 DL:64 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....353
6.10.3.4.1.50 Conversational / unknown / UL:64 DL:64 kbps / CS RAB + Conversational / unknown / UL:64 DL:64 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....354
6.10.3.4.1.51 Conversational / unknown / UL:64 DL:64 kbps / CS RAB + Interactive or background / UL:64 DL:64 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....355
6.10.3.4.1.51a Conversational / unknown / UL:64 DL:64 kbps / CS RAB + Interactive or Background / UL:8 DL:8 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....357

6.10.3.4.1.51b Conversational / unknown / UL:64 DL:64 kbps / CS RAB + Interactive or Background / UL:16 DL:64 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH ....358
6.10.3.4.1.52 Conversational / unknown / UL:64 DL:64 kbps / CS RAB + Interactive or background / UL:64 DL:128 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH ....358
6.10.3.4.1.53 Conversational / unknown / UL:64 DL:64 kbps / CS RAB + Interactive or background / UL:128 DL:128 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH ....359
6.10.3.4.1.54 Void ....360
6.10.3.4.1.55 Void ....360
6.10.3.4.1.56 Interactive or background / UL:8 DL:8 kbps / PS RAB + Interactive or background / UL:8 DL:8 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH ....360
6.10.3.4.1.57 Interactive or background / UL:64 DL:64 kbps / PS RAB + Interactive or background / UL:64 DL:64 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH ....361
6.10.3.4.1.58 Streaming / unknown / UL:16 DL:64 kbps / PS RAB + Interactive or background / UL:8 DL:8 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....363
6.10.3.4.1.59 Reserved for future use ....364
6.10.3.4.1.60 Reserved for future use ....364
6.10.3.4.1.61 Conversational / unknown / UL:8 DL:8 kbps / PS RAB + Interactive or Background / UL:8 DL:8 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH ....364
6.10.3.4.2 Combinations on PDSCH, SCCPCH, PUSCH and PRACH ....366
6.10.3.4.2.1 Interactive or background / UL: 64 DL: 256 kbps / PS RAB + UL: 3.4/16.8 DL: 3.4/33.6 kbps SRBs for DCCH, CCCH and BCCH + UL: 16.8 DL: 16 kbps SRBs for SHCCH....366
6.10.3.4.2.2 Interactive or background / UL: 64 DL: 384 kbps / PS RAB + UL: 3.4/16.8 DL: 3.4/33.6 kbps SRBs for DCCH, CCCH and BCCH+ UL: 16.8 DL: 16 kbps SRBs for SHCCH ....372
6.10.3.4.2.3 Interactive or background / UL: 64 DL: 2 048 kbps / PS RAB + UL: 3.4/16.8 DL: 3.4/33.6 kbps SRBs for DCCH, CCCH and BCCH + UL: 16.8 DL: 16 kbps SRBs for SHCCH ....374
6.10.3.4.2.4 Interactive or background / UL: 384 DL: 2 048 kbps / PS RAB + UL: 3.4/16.8 DL: 3.4/33.6 kbps SRBs for DCCH, CCCH and BCCH + UL: 16.8 DL: 16 kbps SRBs for SHCCH ....375
6.10.3.4.3 Combinations on PDSCH, SCCPCH, DPCH, PUSCH and PRACH ....377
6.10.3.4.3.1 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH + Interactive or background / UL: 64 DL: 256 kbps / PS RAB + UL: 16.8 kbps SRBs for CCCH and SHCCH+ DL: 33.6 kbps SRBs for CCCH SHCCH and BCCH....377
6.10.3.4.3.2 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH + Interactive or background / UL: 64 DL: 384 kbps / PS RAB + UL: 16.8 kbps SRBs for CCCH and SHCCH+ DL: 33.6 kbps SRBs for CCCH, SHCCH and BCCH....379
6.10.3.4.3.3 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH + Interactive or background / UL: 64 DL: 2 048 kbps / PS RAB + UL: 16.8 kbps SRBs for CCCH and SHCCH+ DL: 33.6 kbps SRBs for CCCH, SHCCH and BCCH ..380
6.10.3.4.4 Combinations on SCCPCH ....381
6.10.3.4.4.1 Stand-alone signalling RB for PCCH ....381
6.10.3.4.4.2 Interactive/Background 32 kbps PS RAB + SRBs for CCCH + SRB for DCCH + SRB for BCCH ....382
6.10.3.4.4.2a Interactive/Background 32 kbps PS RAB + Interactive/Background 32 kbps PS RAB + SRBs for CCCH + SRB for DCCH + SRB for BCCH ....383
6.10.3.4.4.2b SRBs for CCCH + SRB for DCCH + SRB for BCCH ....384
6.10.3.4.4.3 Interactive/Background 32 kbps RAB + SRB for PCCH + SRB for CCCH + SRB for DCCH + SRB for BCCH ....385
6.10.3.4.4.3a SRB for PCCH + SRB for CCCH + SRB for DCCH + SRB for BCCH ....385
6.10.3.4.4.4 RB for CTCH + SRB for CCCH + SRB for BCCH ....386
6.10.3.4.5 Combinations on PRACH ....387
6.10.3.4.5.1 SRB for CCCH + SRB for DCCH ....387
6.10.3.4.5.2 Interactive/Background 12.8 kbps PS RAB + SRB for CCCH + SRB for DCCH ....387
6.10.3.4.5.3 Interactive/Background 12.8 kbps PS RAB + Interactive/Background 12.8 kbps PS RAB + SRB for CCCH + SRB for DCCH....388
6.11 Common Radio Bearer configurations for other test purposes....388
6.11.1 Unacknowledged Mode Radio Bearer configuration (7 bit Length Indicator) ....389
6.11.1.1 Transport channel parameters for the Uplink RAB....389
6.11.1.2 TFCS ....389
6.11.1.3 Transport channel parameters for the Downlink RAB....389
6.11.1.4 TFCS ....389
6.11.2 Unacknowledged Mode Radio Bearer configuration (15 bit Length Indicator) ....390

6.11.2.1 Transport channel parameters for the Uplink RAB ..... 390
6.11.2.2 Transport channel parameters for the Downlink RAB ..... 390
6.11.3 Acknowledged Mode Radio Bearer configuration (7 bit Length Indicator) ..... 390
6.11.3.1 Transport channel parameters for the Uplink RAB ..... 390
6.11.3.2 TFCS ..... 391
6.11.3.3 Transport channel parameters for the Downlink RAB ..... 391
6.11.3.4 TFCS ..... 391
6.11.4 Acknowledged Mode Radio Bearer configuration (15 bit Length Indicator) ..... 391
6.11.4.1 Transport channel parameters for the Uplink RAB ..... 391
6.11.4.2 Transport channel parameters for the Downlink RAB ..... 392
6.11.4a Reference Radio Bearer configurations used in MAC-hs testing ..... 392
6.11.4a.1 5 x Interactive or background / UL: 8 kbps DL: [max bit rate depending on UE category] / UM PS RAB ..... 392
6.11.4a.1.1 Uplink ..... 392
6.11.4a.1.1.1 Uplink Transport channel parameters for DCH ..... 392
6.11.4a.1.1.2 Uplink physical channel parameters ..... 393
6.11.4a.1.2 Downlink ..... 393
6.11.4a.1.2.1 Transport channel parameters for HS-DSCH ..... 393
6.11.4a.1.2.2 Downlink Transport channel parameters for DCH ..... 394
6.11.4a.1.2.3 Downlink physical channel parameters ..... 394
6.11.4b Interactive or background / UL: 0 kbps DL: 0 kbps PS RAB ..... 395
6.11.4b.1 Uplink ..... 395
6.11.4b.1.1 Uplink Transport channel parameters for DCH ..... 395
6.11.4b.2 Downlink ..... 395
6.11.4b.2.1 Downlink Transport channel parameters for DCH ..... 395
6.11.5 Reference Radio Bearer configurations used in Radio Bearer testing for 1.28 Mcps TDD ..... 396
6.11.5.1 RABs and signalling RBs ..... 396
6.11.5.2 Combinations of RABs and Signalling RBs ..... 396
Combinations on DPCH ..... 396
Combinations on PDSCH, SCCPCH, PUSCH and PRACH ..... 400
Combinations on PDSCH, SCCPCH, DPCH, PUSCH and PRACH ..... 400
Combinations on SCCPCH ..... 400
Combinations on PRACH ..... 401
6.11.5.3 Example of linkage between RABs and services ..... 401
6.11.5.4 Typical radio parameter sets ..... 401
6.11.5.4.1 Combinations on DPCH ..... 401
6.11.5.4.1.1 Stand-alone UL:1.7 DL:1.7 kbps SRBs for DCCH ..... 401
6.11.5.4.1.1a Stand-alone UL:1.7 DL:1.7 kbps SRBs for DCCH (multiframe) ..... 402
6.11.5.4.1.2 Stand-alone UL:3.4 DL:3.4 kbps SRBs for DCCH ..... 403
6.11.5.4.1.3 Stand-alone UL:13.6 DL:13.6 kbps SRBs for DCCH ..... 403
6.11.5.4.1.4 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH ..... 404
6.11.5.4.1.4a Conversational / speech / UL: 12.2 7.95 5.9 4.75 DL:12.2 7.95 5.9 4.75 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH ..... 405
6.11.5.4.1.5 Conversational / speech / UL:10.2 DL:10.2 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH ..... 406
6.11.5.4.1.5a Conversational / speech / UL:10.2 6.7 5.9 4.75 DL:10.2 6.7 5.9 4.75 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH ..... 407
6.11.5.4.1.6 Conversational / speech / UL:7.95 DL:7.95 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH ..... 408
6.11.5.4.1.7 Conversational / speech / UL:7.4 DL:7.4 kbps / CS RAB+ UL:3.4 DL:3.4 kbps SRBs for DCCH ..... 408
6.11.5.4.1.7a Conversational / speech / UL:7.4 6.7 5.9 4.75 DL:7.4 6.7 5.9 4.75 / CS RAB+ UL:3.4 DL:3.4 kbps SRBs for DCCH ..... 409
6.11.5.4.1.8 Conversational / speech / UL:6.7 DL:6.7 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH ..... 410
6.11.5.4.1.9 Conversational / speech / UL:5.9 DL:5.9 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH ..... 411
6.11.5.4.1.10 Conversational / speech / UL:5.15 DL:5.15 kbps / CS RAB + UL:1.7 DL:1.7 kbps SRBs for DCCH ..... 412
6.11.5.4.1.11 Conversational / speech / UL:4.75 DL:4.75 kbps / CS RAB + UL:1.7 DL:1.7 kbps SRBs for DCCH ..... 413

6.11.5.4.1.12 Conversational / unknown / UL:28.8/DL:28.8 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....414
6.11.5.4.1.13 Conversational / unknown / UL:64 DL:64 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....414
6.11.5.4.1.14 Conversational / unknown / UL:32 DL:32 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....415
6.11.5.4.1.15 Streaming / unknown / UL:14.4/DL:14.4 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....416
6.11.5.4.1.16 Streaming / unknown / UL:28.8/DL:28.8 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....417
6.11.5.4.1.17 Streaming / unknown / UL:57.6/DL:57.6 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....418
6.11.5.4.1.18 Void....419
6.11.5.4.1.19 Void....419
6.11.5.4.1.20 Void....419
6.11.5.4.1.21 Void....419
6.11.5.4.1.22 Void....419
6.11.5.4.1.23 Interactive or background / UL:32 DL:8 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....419
6.11.5.4.1.23a Interactive or background / UL:8 DL:8 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....420
6.11.5.4.1.23b Interactive or background / UL:16 DL:16 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....420
6.11.5.4.1.23c Interactive or background / UL:32 DL32 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....421
6.11.5.4.1.23d Interactive or background / UL:32 DL32 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....422
6.11.5.4.1.24 Void....423
6.11.5.4.1.25 Interactive or background / UL:32 DL: 64 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....423
6.11.5.4.1.26 Interactive or background / UL:64 DL: 64 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....423
6.11.5.4.1.27 Interactive or background / UL:64 DL:128 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....424
6.11.5.4.1.28 Interactive or background / UL:128 DL:128 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....424
6.11.5.4.1.29 Interactive or background / UL:64 DL:144 kbps / PS RAB + UL:3.4 DL: 3.4 kbps SRBs for DCCH....425
6.11.5.4.1.30 Interactive or background / UL:144 DL:144 kbps / PS RAB + UL:3.4 DL: 3.4 kbps SRBs for DCCH....425
6.11.5.4.1.31 Interactive or background / UL:64 DL:256 kbps / PS RAB + UL:3.4 DL: 3.4 kbps SRBs for DCCH....426
6.11.5.4.1.32 Interactive or background / UL:64 DL:384 kbps / PS RAB + UL:3.4 DL: 3.4 kbps SRBs for DCCH....426
6.11.5.4.1.33 Interactive or background / UL:128 DL:384 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....427
6.11.5.4.1.34 Interactive or background / UL:384 DL:384 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....427
6.11.5.4.1.35 Interactive or background / UL:64 DL:2 048 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....427
6.11.5.4.1.36 Void....428
6.11.5.4.1.37 Void....428
6.11.5.4.1.38 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:32 DL:8 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....428
6.11.5.4.1.38a Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:0 DL:0 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....429
6.11.2.5.1.38b Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:8 DL:8 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....430
6.11.5.4.1.38c Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:32 DL:32 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....431

6.11.5.4.1.38d Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:64 DL:64 kbps / PS RAB + Interactive or background / UL:64 DL:64 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH ....432
6.11.5.4.1.38e Conversational / speech / UL: 12.2 7.95 5.9 4.75 DL:12.2 7.95 5.9 4.75 kbps / CS RAB + Interactive or background / UL:0 DL:0 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH ....433
6.11.5.4.1.38f Conversational / speech / UL: 12.2 7.95 5.9 4.75 DL:12.2 7.95 5.9 4.75 kbps / CS RAB + Interactive or background / UL:8 DL:8 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....434
6.11.5.4.1.38g Conversational / speech / UL: 12.2 7.95 5.9 4.75 DL:12.2 7.95 5.9 4.75 kbps / CS RAB + Interactive or background / UL:16 DL:16 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....435
6.11.5.4.1.38h Conversational / speech / UL: 12.2 7.95 5.9 4.75 DL:12.2 7.95 5.9 4.75 kbps / CS RAB + Interactive or background / UL:32 DL:32 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....436
6.11.5.4.1.38i Conversational / speech / UL: 12.2 7.95 5.9 4.75 DL:12.2 7.95 5.9 4.75 kbps / CS RAB + Interactive or background / UL:64 DL:64 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....437
6.11.5.4.1.38j Conversational / speech / UL: 12.2 7.95 5.9 4.75 DL:12.2 7.95 5.9 4.75 kbps / CS RAB + Interactive or background / UL:64 DL:128 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....438
6.11.5.4.1.39 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:32 DL:64 kbps / PS RAB+ UL:3.4 DL: 3.4 kbps SRBs for DCCH ....439
6.11.5.4.1.40 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:64 DL:64 kbps / PS RAB+ UL:3.4 DL: 3.4 kbps SRBs for DCCH ....439
6.11.5.4.1.41 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:64 DL:128 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH ....440
6.11.5.4.1.42 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:64 DL:256 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH ....441
6.11.5.4.1.43 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:64 DL:384 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH ....442
6.11.5.4.1.44 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:128 DL:2 048 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH ....442
6.11.5.4.1.45 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Streaming / unknown / UL:57.6 DL:57.6 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....444
6.11.5.4.1.46 Void ....445
6.11.5.4.1.47 Void ....445
6.11.5.4.1.48 Void ....445
6.11.5.4.1.49 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Conversational / unknown / UL:64 DL:64 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....445
6.11.5.4.1.49a Conversational / speech / UL: 12.2 7.95 5.9 4.75 DL: 12.2 7.95 5.9 4.75 kbps / CS RAB + Conversational / unknown / UL:64 DL:64 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....446
6.11.5.4.1.50 Conversational / unknown / UL:64 DL:64 kbps / CS RAB + Conversational / unknown / UL:64 DL:64 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....447
6.11.5.4.1.51 Conversational / unknown / UL:64 DL:64 kbps / CS RAB + Interactive or background / UL:64 DL:64 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH ....448
6.11.5.4.1.51a Conversational / unknown / UL:64 DL:64 kbps / CS RAB + Interactive or background / UL:8 DL:8 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH ....449
6.11.5.4.1.51b Conversational / unknown / UL:64 DL:64 kbps / CS RAB + Interactive or background / UL:16 DL:64 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH ....449
6.11.5.4.1.52 Conversational / unknown / UL:64 DL:64 kbps / CS RAB + Interactive or background / UL:64 DL:128 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH ....450
6.11.5.4.1.53 Conversational / unknown / UL:64 DL:64 kbps / CS RAB + Interactive or background / UL:128 DL:128 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH ....451
6.11.5.4.1.54 Void ....451
6.11.5.4.1.55 Void ....451
6.11.5.4.1.56 Interactive or background / UL:8 DL:8 kbps / PS RAB + Interactive or background / UL:8 DL:8 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH ....451
6.11.5.4.1.57 Interactive or background / UL:64 DL:64 kbps / PS RAB + Interactive or background / UL:64 DL:64 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH ....452

6.11.5.4.1.58 Streaming / unknown / UL:16 DL:64 kbps / PS RAB + Interactive or background / UL:8 DL:8 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....453
6.11.5.4.1.59 Reserved for future use....454
6.11.5.4.1.60 Reserved for future use....454
6.11.5.4.1.61 Conversational / unknown / UL:8 DL:8 kbps / PS RAB + Interactive or Background / UL:8 DL:8 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH....454
6.11.5.4.2 Combinations on PDSCH, SCCPCH, PUSCH and PRACH....455
6.11.5.4.2.1 Interactive or background / UL: 64 DL: 256 kbps / PS RAB + UL: 3.4/16.8 DL: 3.4/33.6 kbps SRBs for DCCH, CCCH and BCCH + UL: 16.8 DL: 16 kbps SRBs for SHCCH....455
6.11.5.4.2.2 Interactive or background / UL: 64 DL: 384 kbps / PS RAB + UL: 3.4/16.8 DL: 3.4/33.6 kbps SRBs for DCCH, CCCH and BCCH+ UL: 16.8 DL: 16 kbps SRBs for SHCCH....458
6.11.5.4.2.3 Interactive or background / UL: 64 DL: 2 048 kbps / PS RAB + UL: 3.4/16.8 DL: 3.4/33.6 kbps SRBs for DCCH, CCCH and BCCH + UL: 16.8 DL: 16 kbps SRBs for SHCCH....459
6.11.5.4.3 Combinations on PDSCH, SCCPCH, DPCH, PUSCH and PRACH....460
6.11.5.4.3.1 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB+ UL:3.4 DL:3.4 kbps SRBs for DCCH + Interactive or background / UL: 64 DL: 256 kbps / PS RAB + UL: 16.8 kbps SRBs for CCCH and SHCCH+ DL: 33.6 kbps SRBs for CCCH, SHCCH and BCCH....460
6.11.5.4.3.2 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB+ UL:3.4 DL:3.4 kbps SRBs for DCCH + Interactive or background / UL: 64 DL: 384 kbps / PS RAB + UL: 16.8 kbps SRBs for CCCH and SHCCH+ DL: 33.6 kbps SRBs for CCCH, SHCCH and BCCH....462
6.11.5.4.3.3 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB+ UL:3.4 DL:3.4 kbps SRBs for DCCH + Interactive or background / UL: 64 DL: 2 048 kbps / PS RAB + UL: 16.8 kbps SRBs for CCCH and SHCCH + DL: 33.6 kbps SRBs for CCCH, SHCCH and BCCH....462
6.11.5.4.4 Combinations on SCCPCH....463
6.11.5.4.4.1 Stand-alone signalling RB for PCCH....463
6.11.5.4.4.2 Interactive/Background 32 kbps PS RAB + SRBs for CCCH + SRB for DCCH + SRB for BCCH....463
6.11.5.4.4.2a Interactive/Background 32 kbps PS RAB + Interactive/Background 32 kbps PS RAB + SRBs for CCCH + SRB for DCCH + SRB for BCCH....464
6.11.5.4.4.2b SRBs for CCCH + SRB for DCCH + SRB for BCCH....465
6.11.5.4.4.3 Interactive/Background 32 kbps RAB + SRB for PCCH + SRB for CCCH + SRB for DCCH + SRB for BCCH....465
6.11.5.4.4.3a SRB for PCCH + SRB for CCCH + SRB for DCCH + SRB for BCCH....466
6.11.5.4.4.4 RB for CTCH + SRB for CCCH + SRB for BCCH....466
6.11.5.4.5 Combinations on PRACH....467
6.11.5.4.5.1 SRB for CCCH + SRBs for DCCH....467
6.11.5.4.5.2 Interactive/Background 12.8 kbps PS RAB + SRB for CCCH + SRBs for DCCH....467
6.11.5.4.5.3 Interactive/Background 12.8 kbps PS RAB + Interactive/Background 12.8 kbps PS RAB + SRB for CCCH + SRB for DCCH....468

7 Generic setup procedures....469
7.1 Basic Generic Procedures....469
7.1.1 UE Test States for Basic Generic Procedures....469
7.1.2 Mobile terminated establishment of Radio Resource Connection....469
7.1.2.1 Initial conditions....470
7.1.2.2 Definition of system information messages....470
7.1.2.3 Procedure....470
7.1.2.4 Specific message contents....470
7.1.2.4.1 PAGING TYPE 1....470
7.1.2.4.2 RRC CONNECTION REQUEST....471
7.1.2.4.3 RRC CONNECTION SETUP....471
7.1.2.4.4 RRC CONNECTION SETUP COMPLETE....471
7.1.3 Radio Bearer Setup Procedure....473
7.1.3.1 Initial conditions....473
7.1.3.2 Definition of system information messages....473
7.1.3.3 Procedure....473
7.1.3.4 Specific message contents....473
7.1.3.4.1 RADIO BEARER SETUP....473
7.1.3.4.2 RADIO BEARER SETUP COMPLETE....474

7.2 Generic setup procedures ....474
7.2.1 UE Test States for Generic setup procedures....474
7.2.2 Registration of UE ....475
7.2.2.1 Registration on CS ....475
7.2.2.1.1 Initial condition ....475
7.2.2.1.2 Definition of system information messages....475
7.2.2.1.3 Procedure....475
7.2.2.1.4 Specific message contents ....476
7.2.2.2 Registration on PS....476
7.2.2.2.1 Initial condition ....476
7.2.2.2.2 Definition of system information messages....476
7.2.2.2.3 Procedure....476
7.2.2.2.4 Specific message contents ....476
7.2.2.3 Registration on CS / PS combined environment ....476
7.2.2.3.1 Initial condition ....476
7.2.2.3.2 Definition of system information messages....477
7.2.2.3.3 Procedure UE establish PS registration immediately after the UE has been switched on ....477
7.2.2.3.3a Procedure UE establish PS registration later the user decides to use the PS services ....477
7.2.2.3.4 Specific message contents ....477
7.2.2.4 Registration on CS / PS non-combined environment....477
7.2.2.4.1 Initial condition ....477
7.2.2.4.2 Definition of system information messages....478
7.2.2.4.3 Procedure....478
7.2.2.4.4 Specific message contents ....478
7.2.3 Call setup ....478
7.2.3.1 Generic call set up procedure for mobile terminating circuit switched calls....478
7.2.3.1.1 Initial conditions....478
7.2.3.1.2 Definition of system information messages....478
7.2.3.1.3 Procedure....478
7.2.3.1.4 Specific message contents ....479
7.2.3.2 Generic call set-up procedure for mobile originating circuit switched calls ....479
7.2.3.2.1 Initial conditions....479
7.2.3.2.2 Definition of system information messages....479
7.2.3.2.3 Procedure....479
7.2.3.2.4 Specific message contents ....479
7.2.4 Session setup....479
7.2.4.1 Generic session set up procedure for mobile terminating packet switched sessions....479
7.2.4.1.1 Initial conditions....479
7.2.4.1.2 Definition of system information messages....480
7.2.4.1.3 Procedure....480
7.2.4.1.4 Specific message contents ....480
7.2.4.2 Generic session set up procedure for mobile originating packet switched sessions....480
7.2.4.2.1 Initial conditions....480
7.2.4.2.2 Definition of system information messages....480
7.2.4.2.3 Procedure....480
7.2.4.2.4 Specific message contents ....481
7.3 Test procedures for RF test....481
7.3.1 UE Test States for RF testing....481
7.3.2 Test procedure for TX, RX and Performance Requirement (without handover) ....481
7.3.2.1 Initial conditions ....481
7.3.2.2 Definition of system information messages ....482
Contents of System information block type 1: RRC ....482
Contents of System Information Block type 5 (FDD)....482
7.3.2.3 Procedure ....482
For UE supporting CS ....482
For UE supporting PS only ....483
7.3.2.4 Specific message contents....483
7.3.2.4.1 ATTACH ACCEPT....483
Contents of Attach Accept message: GMM....483
7.3.2.4.2 Reference measurement channels....483
7.3.2.4.3 Void....483
7.3.2.4.4 Compressed mode ....483

7.3.2.4.5 Transmit diversity mode ..... 483
7.3.3 Test procedure for test cases using Cell_PCH or URA_PCH state ..... 483
7.3.3.1 Initial conditions ..... 483
7.3.3.2 Definition of system information messages ..... 484
Contents of System information block type 1: RRC ..... 484
Contents of System Information Block type 5 (FDD) ..... 484
7.3.3.3 Procedure ..... 484
For UE supporting PS ..... 484
7.3.3.4 Specific message contents ..... 485
Contents of Attach Accept message: GMM ..... 485
7.3.4 Test procedure for Handover ..... 485
7.3.4.1 Initial conditions ..... 485
7.3.4.2 Definition of system information messages ..... 486
Contents of System information block type 1: RRC ..... 486
Contents of System Information Block type 5 (FDD) ..... 486
7.3.4.3 Procedure ..... 486
For UE supporting CS ..... 486
For UE supporting PS only ..... 487
7.3.4.4 Specific message contents ..... 487
Contents of Attach Accept message: GMM ..... 487
7.3.5 Test procedure for test cases using CELL_FACH state ..... 487
7.3.5.1 Initial conditions ..... 487
7.3.5.2 Definition of system information messages ..... 488
Contents of System information block type 1: RRC ..... 488
Contents of System Information Block type 5 (FDD) ..... 488
7.3.5.3 Procedure ..... 488
For UE supporting CS ..... 488
For UE supporting PS only ..... 489
7.3.5.4 Specific message contents ..... 489
Contents of Attach Accept message: GMM ..... 489
7.3.6 Test procedure for HSDPA RF Performance Requirement ..... 489
7.3.6.1 Initial conditions ..... 489
7.3.6.2 Definition of system information messages ..... 489
Contents of System information block type 1: RRC ..... 489
Contents of System Information Block type 5 (FDD) ..... 490
7.3.6.3 Procedure ..... 490
7.3.6.4 Specific message contents ..... 491
7.3.6.4.1 ATTACH ACCEPT ..... 491
Contents of Attach Accept message: GMM ..... 491
7.3.6.4.2 RADIO BEARER SETUP ..... 492
7.3.6.4.3 RRC CONNECTION SETUP ..... 492
Contents of RRC CONNECTION SETUP message: UM ..... 492
7.3.7 Test procedure for inter-RAT handover used in RRM testing ..... 492
7.3.7.1 Initial conditions ..... 492
7.3.7.2 Definition of system information messages ..... 492
Contents of System information block type 1: RRC ..... 492
Contents of System Information Block type 5 (FDD) ..... 493
7.3.7.3 Procedure ..... 493
7.3.7.4 Specific message contents ..... 493
Contents of Attach Accept message: GMM ..... 493
7.3.8 Test procedure for inter-RAT cell FACH reselection used in RRM testing ..... 493
7.3.8.1 Initial conditions ..... 493
7.3.8.2 Definition of system information messages ..... 494
Contents of System information block type 1: RRC ..... 494
Contents of System Information Block type 5 (FDD) ..... 494
7.3.8.3 Procedure ..... 494
7.3.8.4 Specific message contents ..... 495
Contents of Attach Accept message: GMM ..... 495
7.3.9 Test procedure for E-DCH RF test cases ..... 495
7.3.9.1 Initial conditions ..... 495
7.3.9.2 Definition of system information messages ..... 495
7.3.9.3 Procedure ..... 496

7.3.9.4 Specific message contents ..... 497
7.3.9.4.1 ATTACH ACCEPT ..... 497
Contents of Attach Accept message: GMM ..... 497
7.3.9.4.2 RADIO BEARER SETUP ..... 497
7.3.9.4.3 RRC CONNECTION SETUP ..... 497
Contents of RRC CONNECTION SETUP message: UM ..... 497
7.4 Common generic procedures for AS testing ..... 497
7.4.1 UE RRC Test States for common procedures ..... 497
7.4.2 Generic Setup Procedure for RRC test cases ..... 499
7.4.2.1 RRC connection establishment procedure for circuit-switched calls (procedure P3 and P4) ..... 500
7.4.2.1.1 Mobile terminating call ..... 500
7.4.2.1.1.1 Initial conditions ..... 500
7.4.2.1.1.2 Definition of system information messages ..... 500
7.4.2.1.1.3 Procedure ..... 500
7.4.2.1.1.4 Specific message contents ..... 500
7.4.2.1.2 Mobile originating calls ..... 500
7.4.2.1.2.1 Initial conditions ..... 500
7.4.2.1.2.2 Definition of system information messages ..... 500
7.4.2.1.2.3 Procedure ..... 500
7.4.2.1.2.4 Specific message contents ..... 501
7.4.2.2 RRC connection establishment procedure for packet switched sessions (procedure P5 and P6) ..... 501
7.4.2.2.1 Mobile terminating session ..... 501
7.4.2.2.1.1 Initial conditions ..... 501
7.4.2.2.1.2 Definition of system information messages ..... 501
7.4.2.2.1.3 Procedure ..... 501
7.4.2.2.1.4 Specific message contents ..... 501
7.4.2.2.2 Mobile originating sessions ..... 501
7.4.2.2.2.1 Initial conditions ..... 501
7.4.2.2.2.2 Definition of system information messages ..... 501
7.4.2.2.2.3 Procedure ..... 502
7.4.2.2.2.4 Specific message contents ..... 502
7.4.2.3 NAS call set up procedure for circuit switched calls (procedure P7 and P8) ..... 502
7.4.2.3.1 Mobile terminating call ..... 502
7.4.2.3.1.1 Initial conditions ..... 502
7.4.2.3.1.2 Definition of system information messages ..... 502
7.4.2.3.1.3 Procedure ..... 502
7.4.2.3.1.4 Specific message contents ..... 502
7.4.2.3.2 Mobile originating calls ..... 502
7.4.2.3.2.1 Initial conditions ..... 502
7.4.2.3.2.2 Definition of system information messages ..... 503
7.4.2.3.2.3 Procedure ..... 503
7.4.2.3.2.4 Specific message contents ..... 503
7.4.2.4 NAS session activation procedure for packet switched sessions (procedure P9 and P10) ..... 503
7.4.2.4.1 Mobile terminating session ..... 503
7.4.2.4.1.1 Initial conditions ..... 503
7.4.2.4.1.2 Definition of system information messages ..... 503
7.4.2.4.1.3 Procedure ..... 503
7.4.2.4.1.4 Specific message contents ..... 503
7.4.2.4.2 Mobile originating sessions ..... 504
7.4.2.4.2.1 Initial conditions ..... 504
7.4.2.4.2.2 Definition of system information messages ..... 504
7.4.2.4.2.3 Procedure ..... 504
7.4.2.4.2.4 Specific message contents ..... 504
7.4.2.5 Radio access bearer establishment procedure for circuit switched calls (procedure P11 and P12) ..... 504
7.4.2.5.1 Mobile terminating call ..... 504
7.4.2.5.1.1 Initial conditions ..... 504
7.4.2.5.1.2 Definition of system information messages ..... 504
7.4.2.5.1.3 Procedure ..... 504
7.4.2.5.1.4 Specific message contents ..... 505
7.4.2.5.2 Mobile originating calls ..... 505
7.4.2.5.2.1 Initial conditions ..... 505
7.4.2.5.2.2 Definition of system information messages ..... 505

7.4.2.5.2.3 Procedure ..... 505
7.4.2.5.2.4 Specific message contents ..... 505
7.4.2.5a Test loop activation and radio access bearer establishment procedure for circuit switched calls (procedure P7a) ..... 505
7.4.2.5a.1 Initial conditions ..... 505
7.4.2.5a.2 Definition of system information messages ..... 505
7.4.2.5a.3 Procedure ..... 505
7.4.2.5a.4 Specific message contents ..... 506
7.4.2.6 Radio access bearer establishment procedure for packet switched sessions (procedure P13, P14, P25 and P26) ..... 506
7.4.2.6.1 Mobile terminating session ..... 506
7.4.2.6.1.1 Initial conditions ..... 506
7.4.2.6.1.2 Definition of system information messages ..... 506
7.4.2.6.1.3 Procedure ..... 506
7.4.2.6.1.4 Specific message contents ..... 506
7.4.2.6.2 Mobile originating sessions ..... 506
7.4.2.6.2.1 Initial conditions ..... 506
7.4.2.6.2.2 Definition of system information messages ..... 507
7.4.2.6.2.3 Procedure ..... 507
7.4.2.6.2.4 Specific message contents ..... 507
7.4.2.6a Test loop activation and radio access bearer establishment procedure for packet switched sessions (procedure P4a and P9a) ..... 507
7.4.2.6a.1 Initial conditions ..... 507
7.4.2.6a.2 Definition of system information messages ..... 507
7.4.2.6a.3 Procedure ..... 507
7.4.2.6a.4 Specific message contents ..... 508
7.4.2.7 Procedure for transitions to CELL_PCH or URA_PCH state (procedure P15, P16, P17 and P18) ..... 508
7.4.2.7.1 Transition to CELL_PCH (procedure P15 and P16) ..... 508
7.4.2.7.1.1 Initial conditions ..... 508
7.4.2.7.1.2 Definition of system information messages ..... 508
7.4.2.7.1.3 Procedure ..... 508
7.4.2.7.1.4 Specific message contents ..... 508
Contents of PHYSICAL CHANNEL RECONFIGURATION message: DCCH-AM (Step 1) ..... 508
7.4.2.7.2 Transition to URA_PCH (procedure P17 and P18) ..... 508
7.4.2.7.2.1 Initial conditions ..... 508
7.4.2.7.2.2 Definition of system information messages ..... 509
7.4.2.7.2.3 Procedure ..... 509
7.4.2.7.2.4 Specific message contents ..... 509
Contents of PHYSICAL CHANNEL RECONFIGURATION message: DCCH-AM (Step 1) ..... 509
7.4.2.8 Radio access bearer establishment procedure with packet switched sessions for transitions to Multi Call state (procedure P19, 20 and 21) ..... 509
7.4.2.8.1 Transition to PS+CS-DCCH+DTCH DCH (procedure P19) ..... 509
7.4.2.8.1.1 Mobile terminating session ..... 509
7.4.2.8.1.2 Mobile originating sessions ..... 510
7.4.2.8.2 Transition to PS+PS-DCCH+DTCH DCH (procedure P20 and P21) ..... 510
7.4.2.8.2.1 Mobile terminating session ..... 510
7.4.2.8.2.2 Mobile originating sessions ..... 511
7.4.2.9 Radio access bearer establishment procedure with circuit switched calls for transitions to Multi Call state (procedure P22, P23 and P24) ..... 511
7.4.2.9.1 Transition to CS+CS-DCCH+DTCH DCH (procedure P22) ..... 511
7.4.2.9.1.1 Mobile terminating call ..... 511
7.4.2.9.1.2 Mobile originating calls ..... 512
7.4.2.9.2 Transition to PS+CS-DCCH+DTCH DCH (procedure P23 and 24) ..... 513
7.4.2.9.2.1 Mobile terminating call ..... 513
7.4.2.9.2.2 Mobile originating calls ..... 513
7.5 Test procedures for A-GPS Performance requirements testing ..... 514
7.5.1 Normal UE based A-GPS procedure ..... 514
7.5.1.1 Initial conditions ..... 514
7.5.1.2 Procedure ..... 514
7.5.1.3 Specific message contents ..... 515
Contents of RESET UE POSITIONING STORED INFORMATION message: TC ..... 515
Contents of MEASUREMENT CONTROL messages: RRC ..... 515

7.5.2 UE based A-GPS procedure for moving scenario and periodic update test case ..... 517
7.5.2.1 Initial conditions ..... 517
7.5.2.2 Procedure ..... 517
7.5.2.3 Specific message contents ..... 518
Contents of RESET UE POSITIONING STORED INFORMATION message: TC ..... 518
Contents of MEASUREMENT CONTROL message: RRC ..... 518
7.5.3 Void ..... 518
7.5.4 Normal UE assisted GPS procedure ..... 518
7.5.4.1 Initial conditions ..... 518
7.5.4.2 Procedure ..... 518
7.5.4.3 Specific message contents ..... 519
Contents of RESET UE POSITIONING STORED INFORMATION message: TC ..... 519
Contents of MEASUREMENT CONTROL messages: RRC ..... 519
7.5.5 UE assisted A-GPS procedure for moving scenario and periodic update test case ..... 521
7.5.5.1 Initial conditions ..... 521
7.5.5.2 Procedure ..... 521
7.5.5.3 Specific message contents ..... 522
Contents of RESET UE POSITIONING STORED INFORMATION message: TC ..... 522
Contents of MEASUREMENT CONTROL message: RRC ..... 522

8 Test USIM Parameters ..... 522
8.1 Introduction ..... 522
8.1.1 Definitions ..... 522
8.1.2 Definition of the test algorithm for authentication ..... 522
8.1.2.1 Authentication and key derivation in the test USIM and SS ..... 523
8.1.2.2 Generation of re-synchronization parameters in the USIM ..... 524
8.1.2.3 Using the authentication test algorithm for UE conformance testing ..... 524
8.1.2.3.1 Authentication accept case ..... 524
8.1.2.3.2 MAC failure case ..... 525
8.1.2.3.3 SQN failure case ..... 526
8.1.3 Definition of the test algorithm for VGCS/VBS VSTK generation ..... 526
8.1.3.1 VSTK generation in the test USIM and SS ..... 526
8.2 Default Parameters for the test USIM ..... 527
8.3 Default settings for the Elementary Files (EFs) ..... 527
8.3.1 Contents of the EFs at the MF level ..... 527
8.3.1.1 $EF_{DIR}$ ..... 528
8.3.1.2 $EF_{ICCID}$ (ICC Identity) ..... 528
8.3.1.3 $EF_{PL}$ (Preferred Languages) ..... 528
8.3.1.4 $EF_{ARR}$ (Access rule reference) ..... 528
8.3.2 Contents of files at the USIM ADF (Application DF) level ..... 528
8.3.2.1 $EF_{LI}$ (Language Indication) ..... 528
8.3.2.2 $EF_{IMSI}$ (IMSI) ..... 528
8.3.2.3 $EF_{Keys}$ (Ciphering and Integrity Keys) ..... 528
8.3.2.4 $EF_{KeysPS}$ (Ciphering and Integrity Keys for Packet Switched domain) ..... 528
8.3.2.5 $EF_{PLMNwAcT}$ (User controlled PLMN selector with Access Technology) ..... 528
8.3.2.6 $EF_{HPPLMN}$ (Higher Priority PLMN search period) ..... 529
8.3.2.7 $EF_{ACMmax}$ (ACM maximum value) ..... 529
8.3.2.8 $EF_{UST}$ (USIM Service Table) ..... 529
8.3.2.9 $EF_{ACM}$ (Accumulated Call Meter) ..... 530
8.3.2.10 $EF_{GID1}$ (Group Identifier Level 1) ..... 530
8.3.2.11 $EF_{GID2}$ (Group Identifier Level 2) ..... 530
8.3.2.12 $EF_{SPN}$ (Service Provider Name) ..... 530
8.3.2.13 $EF_{PUCT}$ (Price per Unit and Currency Table) ..... 530
8.3.2.14 $EF_{CBMI}$ (Cell Broadcast Message identifier selection) ..... 531
8.3.2.15 $EF_{ACC}$ (Access Control Class) ..... 531
8.3.2.16 $EF_{FPLMN}$ (Forbidden PLMNs) ..... 531
8.3.2.17 $EF_{LOCI}$ (Location Information) ..... 531
8.3.2.18 $EF_{AD}$ (Administrative Data) ..... 531

8.3.2.19 Void....532
8.3.2.20 $EF_{CBMID}$ (Cell Broadcast Message Identifier for Data Download)....532
8.3.2.21 $EF_{ECC}$ (Emergency Call Codes)....532
8.3.2.22 $EF_{CBMIR}$ (Cell Broadcast Message Identifier Range selection)....532
8.3.2.23 $EF_{PSLOCI}$ (Packet Switched location information)....532
8.3.2.24 $EF_{FDN}$ (Fixed Dialling Numbers)....532
8.3.2.25 $EF_{SMS}$ (Short messages)....532
8.3.2.26 $EF_{MSISDN}$ (MSISDN)....532
8.3.2.27 $EF_{SMSP}$ (Short message service parameters)....532
8.3.2.28 $EF_{SMSS}$ (SMS status)....532
8.3.2.29 $EF_{SDN}$ (Service Dialling Numbers)....532
8.3.2.30 $EF_{EXT2}$ (Extension2)....532
8.3.2.31 $EF_{EXT3}$ (Extension3)....532
8.3.2.32 $EF_{SMSR}$ (Short message status reports)....533
8.3.2.33 $EF_{ICI}$ (Incoming Call Information)....533
8.3.2.34 $EF_{OCI}$ (Outgoing Call Information)....533
8.3.2.35 $EF_{ICT}$ (Incoming Call Timer)....533
8.3.2.36 $EF_{OCT}$ (Outgoing Call Timer)....533
8.3.2.37 $EF_{EXT5}$ (Extension5)....533
8.3.2.38 $EF_{CCP2}$ (Capability Configuration Parameters 2)....533
8.3.2.39 $EF_{eMLPP}$ (enhanced Multi Level Precedence and Pre-emption)....533
8.3.2.40 $EF_{AAeM}$ (Automatic Answer for eMLPP Service)....533
8.3.2.41 Void....533
8.3.2.42 $EF_{Hiddenkey}$ (Key for hidden phone book entries)....533
8.3.2.43 Void....533
8.3.2.44 $EF_{BDN}$ (Barred dialling numbers)....533
8.3.2.45 $EF_{EXT4}$ (Extension 4)....533
8.3.2.46 $EF_{CMI}$ (Comparison method information)....533
8.3.2.47 $EF_{EST}$ (Enabled service table)....533
8.3.2.48 $EF_{ACL}$ (Access point name control list)....533
8.3.2.49 $EF_{DCK}$ (Depersonalization control keys)....534
8.3.2.50 $EF_{CNL}$ (Co-operative network list)....534
8.3.2.51 $EF_{START-HFN}$ (Initialisation values for Hyperframe number)....534
8.3.2.52 $EF_{THRESHOLD}$ (Maximum value of START)....534
8.3.2.53 $EF_{OPLMNwACT}$ (Operator controlled PLMN selector with Access Technology)....534
8.3.2.54 $EF_{HPLMNwAcT}$ (HPLMN selector with Access Technology)....534
8.3.2.55 $EF_{ARR}$ (Access rule reference)....534
8.3.2.56 Void....534
8.3.2.57 $EF_{NETPAR}$ (Network Parameters)....534
8.3.2.58 $EF_{PNN}$ (PLMN Network Name)....534
8.3.2.59 $EF_{OPL}$ (Operator PLMN List)....534
8.3.2.60 $EF_{MBDN}$ (Mailbox Dialling Numbers)....534
8.3.2.61 $EF_{EXT6}$ (Extension6)....534
8.3.2.62 $EF_{MBI}$ (Mailbox Identifier)....534
8.3.2.63 $EF_{MWIS}$ (Message Waiting Indication Status)....534
8.3.2.64 $EF_{CFIS}$ (Call Forwarding Indication Status)....534
8.3.2.65 $EF_{EXT7}$ (Extension7)....534
8.3.2.66 $EF_{SPDI}$ (Service Provider Display Information)....535
8.3.2.67 $EF_{MMSN}$ (MMS Notification)....535
8.3.2.68 $EF_{EXT8}$ (Extension 8)....535
8.3.2.69 $EF_{MMSICP}$ (MMS Issuer Connectivity Parameters)....535
8.3.2.70 $EF_{MMSUP}$ (MMS User Preferences)....535
8.3.2.71 $EF_{MMSUCP}$ (MMS User Connectivity Parameters)....535
8.3.2.72 $EF_{NIA}$ (Network's Indication of Alerting)....535
8.3.2.74 $EF_{VGCSS}$ (Voice Group Call Service Status)....537
8.3.2.75 $EF_{VBS}$ (Voice Broadcast Service)....537

8.3.2.76 $EF_{VBSS}$ (Voice Broadcast Service Status) ..... 538
8.3.2.77 $EF_{VGCSCA}$ (Voice Group Call Service Ciphering Algorithm) ..... 539
8.3.2.78 $EF_{VBSCA}$ (Voice Broadcast Service Ciphering Algorithm) ..... 539
8.3.2.79 $EF_{GBABP}$ (GBA Bootstrapping parameters) ..... 539
8.3.2.80 $EF_{MSK}$ (MBMS Service Keys List) ..... 540
8.3.2.81 $EF_{MUK}$ (MBMS User Key) ..... 540
8.3.2.82 Void ..... 540
8.3.2.83 $EF_{GBANL}$ (GBA NAF List) ..... 540
8.3.3 Contents of DFs at the USIM ADF (Application DF) level ..... 540
8.3.3.1 Contents of files at the USIM SoLSA level ..... 540
8.3.3.1.1 $EF_{SAI}$ (SoLSA Access Indicator) ..... 540
8.3.3.1.2 $EF_{SLL}$ (SoLSA LSA List) ..... 540
8.3.3.1.3 LSA Descriptor files ..... 540
8.3.3.1.4 Contents of files at the MExE level ..... 540
8.3.3.1.4.1 $EF_{MExE-ST}$ (MExE Service table) ..... 540
8.3.3.1.4.2 $EF_{ORPK}$ (Operator Root Public Key) ..... 540
8.3.3.1.4.3 $EF_{ARPK}$ (Administrator Root Public Key) ..... 540
8.3.3.1.4.4 $EF_{TPRPK}$ (Third Party Root Public Key) ..... 540
8.3.3.1.4.5 $EF_{TKCDF}$ (Trusted Key/Certificates Data Files) ..... 540
8.3.3.2 Contents of files at the DF PHONEBOOK level ..... 540
8.3.3.2.1 $EF_{PBR}$ (Phone Book Reference file) ..... 540
8.3.3.2.2 $EF_{IAP}$ (Index Administration Phone book) ..... 540
8.3.3.2.3 $EF_{ADN}$ (Abbreviated dialling numbers) ..... 540
8.3.3.2.4 $EF_{EXT1}$ (Extension1) ..... 541
8.3.3.2.5 $EF_{PBC}$ (Phone Book Control) ..... 541
8.3.3.2.6 $EF_{GRP}$ (Grouping file) ..... 541
8.3.3.2.7 $EF_{AAS}$ (Additional number Alpha String) ..... 541
8.3.3.2.8 $EF_{GAS}$ (Grouping information Alpha String) ..... 541
8.3.3.2.9 $EF_{ANR}$ (Additional Number) ..... 541
8.3.3.2.10 $EF_{SNE}$ (Second Name Entry) ..... 541
8.3.3.2.11 $EF_{CCP1}$ (Capability Configuration Parameters 1) ..... 541
8.3.3.2.12 Phone Book Synchronization ..... 541
8.3.3.2.12.1 $EF_{UID}$ (Unique Identifier) ..... 541
8.3.3.2.12.2 $EF_{PSC}$ (Phone book Synchronization Counter) ..... 541
8.3.3.2.12.3 $EF_{CC}$ (Change Counter) ..... 541
8.3.3.2.12.4 $EF_{PUID}$ (Previous Unique Identifier) ..... 541
8.3.3.2.13 $EF_{EMAIL}$ (e-mail address) ..... 541
8.3.3.3 Contents of files at the DF GSM-ACCESS level (Files required for GSM Access) ..... 541
8.3.3.3.1 $EF_{Kc}$ (GSM Ciphering key Kc) ..... 541
8.3.3.3.2 $EF_{KcGPRS}$ (GPRS Ciphering key KcGPRS) ..... 542
8.3.3.3.3 Void ..... 542
8.3.3.3.4 $EF_{CPBCCH}$ (CPBCCH Information) ..... 542
8.3.3.3.5 $EF_{InvScan}$ (Investigation Scan) ..... 542
8.3.4 Contents of EFs at the TELECOM level ..... 542
8.3.4.1 $EF_{ADN}$ (Abbreviated dialling numbers) ..... 542
8.3.4.2 $EF_{EXT1}$ (Extension1) ..... 542
8.3.4.3 $EF_{ECCP}$ (Extended Capability Configuration Parameter) ..... 542
8.3.4.4 $EF_{SUME}$ (SetUpMenu Elements) ..... 542
8.3.4.5 $EF_{ARR}$ (Access rule reference) ..... 542
8.3.5 Contents of DFs at the TELECOM level ..... 542
8.3.5.1 Contents of files at the $DF_{GRAPHICS}$ level ..... 542
8.3.5.1.1 $EF_{IMG}$ (Image) ..... 542
8.3.5.1.2 Image Instance Data Files ..... 542
8.3.5.2 Contents of files at the $DF_{PHONEBOOK}$ under the $DF_{TELECOM}$ ..... 542

9 Default Message Contents ..... 543

9.1 Default Message Contents for Signalling ..... 543
9.1.1 Default RRC Message Contents (FDD) ..... 543
Default SYSTEM INFORMATION: ..... 543
Contents of ACTIVE SET UPDATE message: AM ..... 543
Contents of ACTIVE SET UPDATE COMPLETE message: AM ..... 543
Contents of ACTIVE SET UPDATE FAILURE message: AM ..... 544
Contents of CELL UPDATE message: TM ..... 544
Contents of CELL UPDATE CONFIRM message: UM ..... 544
Contents of DOWNLINK DIRECT TRANSFER message: AM ..... 545
Contents of HANDOVER FROM UTRAN COMMAND-GSM message: AM ..... 545
Contents of HANDOVER FROM UTRAN FAILURE message: AM ..... 546
Contents of INITIAL DIRECT TRANSFER message: AM ..... 546
Contents of MEASUREMENT CONTROL message: AM ..... 547
Contents of MEASUREMENT CONTROL FAILURE message: AM ..... 548
Contents of MEASUREMENT REPORT message: AM ..... 548
Contents of PAGING TYPE 1 message: TM (Speech in CS) ..... 548
Contents of PAGING TYPE 1 message: TM (The others of speech in CS) ..... 549
Contents of PAGING TYPE 1 message: TM (Packet in PS) ..... 549
Contents of PAGING TYPE 1 message: TM (SMS in CS) ..... 549
Contents of PAGING TYPE 1 message: TM (SMS in PS) ..... 549
Contents of PAGING TYPE 2 message: AM (Speech in CS) ..... 549
Contents of PHYSICAL CHANNEL RECONFIGURATION message: AM or UM ..... 550
Contents of PHYSICAL CHANNEL RECONFIGURATION COMPLETE message: AM ..... 553
Contents of PHYSICAL CHANNEL RECONFIGURATION FAILURE message: AM ..... 553
Contents of RADIO BEARER SETUP message: AM or UM ..... 554
Contents of RADIO BEARER SETUP COMPLETE message: AM ..... 587
Contents of RADIO BEARER SETUP FAILURE message: AM ..... 587
Contents of RADIO BEARER RECONFIGURATION message: AM or UM ..... 588
Contents of RADIO BEARER RECONFIGURATION FAILURE message: AM ..... 596
Contents of RADIO BEARER RECONFIGURATION COMPLETE message: AM ..... 596
Contents of RADIO BEARER RELEASE message: AM or UM ..... 597
Contents of RADIO BEARER RELEASE COMPLETE message: AM ..... 602
Contents of RADIO BEARER RELEASE FAILURE message: AM ..... 602
Contents of RRC CONNECTION REQUEST message: TM ..... 602
Contents of RRC CONNECTION REJECT message: UM ..... 603
Contents of RRC CONNECTION RELEASE message: UM ..... 603
Contents of RRC CONNECTION RELEASE COMPLETE message: AM or UM ..... 603
Contents of RRC CONNECTION SETUP message: UM (Transition to CELL_DCH) ..... 604
Contents of RRC CONNECTION SETUP message: UM (Transition to CELL_FACH) ..... 610
Contents of RRC CONNECTION SETUP COMPLETE message: AM ..... 615
Contents of RRC STATUS message: AM ..... 615
Contents of SECURITY MODE COMMAND message: AM ..... 615
Contents of SECURITY MODE COMPLETE message: AM ..... 617
Contents of SECURITY MODE FAILURE message: AM ..... 617
Contents of TRANSPORT CHANNEL RECONFIGURATION message: AM or UM ..... 617
Contents of TRANSPORT CHANNEL RECONFIGURATION COMPLETE message: AM ..... 623
Contents of TRANSPORT CHANNEL RECONFIGURATION FAILURE message: AM ..... 623
Contents of TRANSPORT FORMAT COMBINATION CONTROL message: AM or UM (in CELL_DCH) ..... 624
Contents of TRANSPORT FORMAT COMBINATION CONTROL FAILURE message: AM ..... 624
Contents of UE CAPABILITY ENQUIRY message: AM or UM ..... 624
Contents of UE CAPABILITY INFORMATION message: AM ..... 624
Contents of UE CAPABILITY INFORMATION CONFIRM message: UM ..... 625
Contents of URA UPDATE message: TM ..... 625
Contents of URA UPDATE CONFIRM message: UM ..... 625
Contents of UPLINK DIRECT TRANSFER message: AM ..... 626
Contents of UTRAN MOBILITY INFORMATION message: AM or UM ..... 626
Contents of UTRAN MOBILITY INFORMATION CONFIRM message: AM ..... 627
Contents of UTRAN MOBILITY INFORMATION FAILURE message: AM ..... 627
9.1.2 Default Message Contents for Signalling (TDD) ..... 627
Contents of RRC STATUS message: AM ..... 627
Contents of SECURITY MODE FAILURE message: AM ..... 627
Contents of URA UPDATE message: TM ..... 628

Contents of URA UPDATE CONFIRM message: UM ....628
Contents of UPLINK DIRECT TRANSFER message: AM ....628
Contents of UTRAN MOBILITY INFORMATION message: AM or UM ....629
Contents of UTRAN MOBILITY INFORMATION CONFIRM message: AM ....629
Contents of UE CAPABILITY ENQUIRY message ....630
Contents of UE CAPABILITY INFORMATION message (1.28 Mpcs TDD) ....630
Contents of UE CAPABILITY INFORMATION CONFIRM message ....631
Contents of TRANSPORT CHANNEL RECONFIGURATION message: AM or UM (3.84 Mcps TDD) ....632
Contents of TRANSPORT CHANNEL RECONFIGURATION message: AM or UM (1.28 Mcps TDD) ....641
Contents of TRANSPORT CHANNEL RECONFIGURATION COMPLETE message: AM (3.84 Mcps TDD) ....649
Contents of TRANSPORT CHANNEL RECONFIGURATION COMPLETE message: AM (1.28 Mcps TDD) ....649
Contents of TRANSPORT CHANNEL RECONFIGURATION FAILURE message: AM ....649
Contents of TRANSPORT FORMAT COMBINATION CONTROL message: AM or UM (in CELL_DCH) ....650
Contents of TRANSPORT FORMAT COMBINATION CONTROL FAILURE message: AM ....650
Contents of RRC CONNECTION REJECT message: UM ....650
Contents of CELL UPDATE message: TM ....650
Contents of CELL UPDATE CONFIRM message: UM ....651
Contents of HANDOVER FROM UTRAN COMMAND-GSM message: AM ....651
Contents of HANDOVER FROM UTRAN FAILURE message: AM ....652
Contents of MEASUREMENT CONTROL Message: AM (Intra-frequence measurement) (1.28 Mcps TDD) ....652
Contents of MEASUREMENT CONTROL Message: AM (Inter-frequence measurement) (1.28 Mcps TDD) ....654
Contents of MEASUREMENT CONTROL FAILURE Message: AM ....655
Contents of MEASUREMENT REPORT message: AM（intra-frequency measurement (3.84 Mcps TDD) ....656
Contents of MEASUREMENT REPORT message: AM（intra-frequency measurement）(1.28 Mcps TDD) ....656
Contents of MEASUREMENT REPORT message: AM (inter-frequency measurement) (3.84 Mcps TDD) ....657
Contents of MEASUREMENT REPORT message: AM (inter-frequency measurement) (1.28 Mcps TDD) ....657
Contents of PHYSICAL CHANNEL RECONFIGURATION message: AM or UM (3.84 Mcps TDD) ....658
Contents of PHYSICAL CHANNEL RECONFIGURATION message: AM or UM (1.28 Mcps TDD) ....662
Contents of PHYSICAL CHANNEL RECONFIGURATION COMPLETE message: AM (3.84 Mcps TDD) ....666
Contents of PHYSICAL CHANNEL RECONFIGURATION COMPLETE message: AM (1.28 Mcps TDD) ....666
Contents of PHYSICAL CHANNEL RECONFIGURATION FAILURE message: AM ....666
Contents of RADIO BEARER RECONFIGURATION message: AM or UM (3.84 Mcps TDD) ....666
Contents of RADIO BEARER RECONFIGURATION message: AM or UM (1.28 Mcps TDD) ....675
Contents of RADIO BEARER RECONFIGURATION COMPLETE message: AM (3.84 Mcps TDD) ....684
Contents of RADIO BEARER RECONFIGURATION COMPLETE message: AM (1.28 Mcps TDD) ....685
Contents of RADIO BEARER RECONFIGURATION FAILURE message: AM ....685
Contents of RADIO BEARER RELEASE message: AM or UM (3.84 Mcps TDD) ....685
Contents of RADIO BEARER RELEASE message: AM or UM (1.28 Mcps TDD) ....690
Contents of DOWNLINK DIRECT TRANSFER message: AM ....694
Contents of INITIAL DIRECT TRANSFER message: AM ....695
Contents of PAGING TYPE 1 message: TM (Speech in CS) ....695
Contents of PAGING TYPE 1 message: TM (The others of speech in CS) ....695
Contents of PAGING TYPE 1 message: TM (Packet in PS) ....695
Contents of RADIO BEARER SETUP message: AM or UM (Speech in CS) (3.84 Mcps TDD option) ....696
Contents of RADIO BEARER SETUP message: AM or UM (Packet to CELL_DCH from CELL_DCH in PS) (3.84 Mcps TDD option) ....700
Contents of RADIO BEARER SETUP message: AM or UM (1.28 Mcps TDD) ....705
Contents of RADIO BEARER SETUP COMPLETE message: AM ....716
Contents of RADIO BEARER SETUP FAILURE message: AM ....717
Contents of RADIO BEARER RELEASE COMPLETE message: AM (1.28 Mcps TDD) ....717
Contents of RADIO BEARER RELEASE FAILURE message: AM ....717
Contents of RRC CONNECTION REQUEST message: TM ....718
Contents of RRC CONNECTION REJECT message: UM ....718
Contents of RRC CONNECTION RELEASE message: UM ....718
Contents of RRC CONNECTION RELEASE COMPLETE message: AM or UM ....719
Contents of RRC CONNECTION SETUP message: UM (Transition to CELL_DCH) (3.84 Mcps TDD option) ....719
Contents of RRC CONNECTION SETUP message: UM (Transition to CELL_DCH) (1.28 Mcps TDD option) ....725
Contents of RRC CONNECTION SETUP message: UM (Transition to CELL_FACH) (3.84 Mcps TDD) ....732
Contents of RRC CONNECTION SETUP message: UM (Transition to CELL_FACH) (1.28 Mcps TDD) ....737
Contents of RRC CONNECTION SETUP COMPLETE message: AM ....741
Contents of SECURITY MODE COMMAND message: AM ....742

Contents of SECURITY MODE COMPLETE message: AM ..... 743
Contents of UPLINK DIRECT TRANSFER message: AM ..... 743
9.2 Default Message Contents for RF ..... 743
9.2.1 Default Message Contents for RF (FDD) ..... 743
Contents of Activate RB Test Mode message ..... 743
Contents of Close UE Test Loop message (UE test loop mode 1 without Dummy DCCH transmission) ..... 743
Contents of Close UE Test Loop message (UE test loop mode 2 without Dummy DCCH transmission) ..... 744
Contents of Open UE Test Loop message ..... 744
Contents of PAGING TYPE 1 message: TM (CS) ..... 744
Contents of PAGING TYPE 1 message: TM (PS) ..... 744
Contents of RADIO BEARER SETUP message: AM or UM (Test Loop Mode1) ..... 744
Contents of RADIO BEARER SETUP message: AM or UM (UE supports PS RAB only) ..... 752
Contents of RADIO BEARER SETUP message: AM or UM (UE supports CS RAB for Test Loop Mode 2) ..... 757
Contents of RADIO BEARER SETUP message: AM or UM (HSDPA) ..... 760
Contents of RADIO BEARER SETUP message: AM or UM (HSDPA) ..... 764
Contents of RADIO BEARER SETUP message: BTFD RMC for Test Loop Mode 2 ..... 768
Contents of RADIO BEARER SETUP message: AM or UM (E-DCH and HSDPA) ..... 774
Contents of RRC CONNECTION RELEASE message: UM ..... 777
Contents of RRC CONNECTION SETUP message: UM ..... 778
Contents of SECURITY MODE COMMAND message: AM ..... 783
9.2.2 Default Message Contents for RF (TDD) ..... 784
Contents of Activate RB Test Mode message ..... 784
Contents of Close UE Test Loop message ..... 784
Contents of Open UE Test Loop message ..... 784
Contents of PAGING TYPE 1 message: TM (CS) ..... 785
Contents of PAGING TYPE 1 message: TM (PS) ..... 785
Contents of RADIO BEARER SETUP message: AM or UM (3.84 Mcps TDD) ..... 785
Contents of RADIO BEARER SETUP message: AM or UM (1.28 Mcps TDD) ..... 790
Contents of RRC CONNECTION RELEASE message: UM ..... 795
Contents of RRC CONNECTION SETUP message: UM (3.84 Mcps TDD) ..... 795
Contents of RRC CONNECTION SETUP message: UM (1.28 Mcps TDD) ..... 801
Contents of SECURITY MODE COMMAND message: AM ..... 806

10 A-GPS GPS Scenarios and Assistance Data ..... 808
10.1 General ..... 808
10.1.1 Satellite constellations and assistance data for performance testing ..... 808
10.1.2 GPS Scenarios for performance testing ..... 808
10.1.2.1 GPS Scenario #1 ..... 808
10.1.2.2 GPS Scenario #2 ..... 809
10.1.2.3 GPS Scenario #3 ..... 809
10.1.2.4 UE Location for TTFF test cases ..... 809
10.1.2.4.1 UE Location Offset ..... 809
10.1.2.4.2 UE Altitude ..... 810
10.1.2.5 Satellites to be simulated in each test case ..... 810
10.2 Information elements required for normal UE based testing ..... 810
10.3 Information elements required for UE based Sensitivity Fine Time Assistance test case ..... 811
10.4 Information elements available for normal UE assisted testing ..... 811
10.5 Information elements available for UE assisted Sensitivity Fine Time Assistance test case ..... 812
10.6 Contents of Information elements for performance testing ..... 813
10.6.1 General ..... 813
10.6.2 IE Random Offset Values ..... 813
10.6.2.1 GPS TOW msec ..... 813
10.6.2.1 UTRAN GPS timing of cell frames ..... 814
10.6.3 Assistance Data Reference Time ..... 814
10.6.4 Assistance Data Reference Position ..... 816
10.6.5 Assistance Data Navigation Model ..... 816
10.6.6 Assistance Data Ionospheric Model ..... 817
10.6.7 Assistance Data Almanac ..... 818
10.6.8 Assistance Data Acquisition Assistance ..... 818
10.7 GPS Scenario and values of Information Elements for signalling testing ..... 820
10.7.1 General ..... 820
10.7.2 GPS Scenario ..... 820

10.7.3 Assistance Data Reference Time ....821
10.7.4 Assistance Data Reference Position....821
10.7.5 Assistance Data Navigation Model....821
10.7.6 Assistance Data Ionospheric Model....822
10.7.7 Assistance Data Almanac ....822
10.7.8 Assistance Data Acquisition Assistance ....823

**Annex A (informative): Void ....824**

**Annex B (informative): Void ....825**

**Annex C (normative): GPS Data Files ....826**

C.1 GPS data files for signalling tests....826

C.2 GPS data files for performance tests ....826

**Annex D (informative): Change history ....827**

History ....836

# Foreword

This Technical Specification has been produced by the 3rd Generation Partnership Project (3GPP).

The contents of the present document are subject to continuing work within the TSG and may change following formal TSG approval. Should the TSG modify the contents of the present document, it will be re-released by the TSG with an identifying change of release date and an increase in version number as follows:

Version x.y.z

where:

- x the first digit:
  - 1 presented to TSG for information;
  - 2 presented to TSG for approval;
  - 3 or greater indicates TSG approved document under change control.
- y the second digit is incremented for all changes of substance, i.e. technical enhancements, corrections, updates, etc.
- z the third digit is incremented when editorial only changes have been incorporated in the document.

# Introduction

The definition of the Conformance Tests for UE in 3G will be a complex task as the complete test suite covers RF, EMC and Protocol aspects of the UE.

Each test requires a Test Environment to be defined in which the UE has to operate to defined standards, constraints and performance. The overall task can be simplified if there are a number of well defined and agreed Common Test Environments where every one can be used for a number of tests. Hence the present documents defines testing conditions that are common to several tests avoiding the need to duplicate the same information for every single test.

The present document defines default values for a variety of common areas. Where values are not specified in test cases, the defaults in the present document will apply. If specified, the test case values will take precedence.

The present document addresses the FDD mode as well as the TDD mode.

# 1 Scope

The present document contains definitions of reference conditions and test signals, default parameters, reference radio bearer configurations used in radio bearer interoperability testing, common radio bearer configurations for other test purposes, common requirements for test equipment and generic set-up procedures for use in UE conformance tests.

# 2 References

The following documents contain provisions which, through reference in this text, constitute provisions of the present document.

- References are either specific (identified by date of publication, edition number, version number, etc.) or non-specific.
- For a specific reference, subsequent revisions do not apply.
- For a non-specific reference, the latest version applies. In the case of a reference to a 3GPP document (including a GSM document), a non-specific reference implicitly refers to the latest version of that document *in the same Release as the present document.*

[1] 3GPP TS 34.123-1: "User Equipment (UE) conformance specification; Part 1: Protocol conformance specification".

[2] 3GPP TS 34.121: "Terminal Conformance Specification; Radio Transmission and Reception (FDD)".

[3] 3GPP TS 34.123-2: "User Equipment (UE) conformance specification; Part 2: Implementation Conformance Statement (ICS) proforma specification".

[4] 3GPP TS 34.124: "ElectroMagnetic Compatibility (EMC) requirements for Mobile terminals and ancillary equipment".

[5] 3GPP TS 34.122: "Terminal Conformance Specification; Radio Transmission and Reception (TDD)".

[6] 3GPP TS 34.109: "Terminal logical test interface; Special conformance testing functions".

[7] 3GPP TS 25.301 "Radio interface protocol architecture".

[8] 3GPP TS 25.214: "Physical layer procedures (FDD)".

[9] 3GPP TR 21.905: "Vocabulary for 3GPP Specifications".

[10] 3GPP TR 25.990: "Vocabulary".

[11] 3GPP TS 25.101: "User Equipment (UE) radio transmission and reception (FDD)".

[12] 3GPP TS 25.102: "User Equipment (UE) radio transmission and reception (TDD)".

[13] 3GPP TS 25.211: "Physical channels and mapping of transport channels onto physical channels (FDD)".

[14] 3GPP TS 25.212: "Multiplexing and channel coding (FDD)".

[15] 3GPP TS 23.107: "Quality of Service (QoS) concept and architecture".

[16] 3GPP TS 26.110: "Codec for circuit switched multimedia telephony service; General description".

[17] 3GPP TS 29.007: "General requirements on interworking between the Public Land Mobile Network (PLMN) and the Integrated Services Digital Network (ISDN) or Public Switched Telephone Network (PSTN)".

[18] 3GPP TR 23.910: "Circuit switched data bearer service".

[19] Void.

[20] 3GPP TS 25.104: "Base Station (BS) radio Transmission and Reception (FDD)".

[21] 3GPP TS 25.105: "Base Station (BS) radio Transmission and Reception (TDD)".

[22] 3GPP TS 31.101: "UICC-terminal interface; Physical and logical characteristics".

[23] 3GPP TS 31.102: "Characteristics of the USIM application".

[24] 3GPP TS 33.102: "3G security; Security architecture".

[25] 3GPP TS 33.103: "3G security; Integration guidelines".

[26] 3GPP TS 33.105: "Cryptographic algorithm requirements".

[27] 3GPP TS 25.224: "Physical layer procedures (TDD)".

[28] 3GPP TS 25.221: "Physical channels and mapping of transport channels onto physical channels (TDD)".

[29] 3GPP TS 25.222: "Multiplexing and channel coding (TDD)".

[30] 3GPP TS 25.133: "Requirements for support of radio resource management (FDD)".

[31] 3GPP TS 51.010-1: "Mobile Station (MS) conformance specification; Part 1: Conformance specification".

[32] 3GPP TS 24.008: "Mobile radio interface layer 3 specification; Core network protocols; Stage 3".

[33] 3GPP TS 25.171: "Requirements for support of Assisted Global Positioning System (A-GPS); Frequency Division Duplex (FDD)".

[34] 3GPP TS 25.331: "Radio Resource Control (RRC) protocol specification".

[35] 3GPP TS 25.223: "Spreading and modulation (TDD)".

[36] 3GPP TS 25.304: "User Equipment (UE) procedures in idle mode and procedures for cell reselection in connected mode".

[37] 3GPP TS 25.123: "Requirements for support of radio resource management (TDD)".

[38] 3GPP TS 25.321: "Medium Access Control (MAC) protocol specification".

[39] 3GPP TS 31.120: "UICC-terminal interface; Physical, electrical and logical test specification".

[40] 3GPP TS 31.121: "Base Station System (BSS) equipment specification; Radio aspects".

[41] 3GPP TS 34.171: "Terminal conformance specification; Assisted Global Positioning System (A-GPS); Frequency Division Duplex (FDD)".

[42] 3GPP TS 23.032: "Universal Geographical Area Description (GAD)".

[43] NATO Standard Agreement STANAG 4294 Issue 1

[44] 3GPP TS 43.020: "Security related network functions"

# 3 Definitions, abbreviations and symbols

## 3.1 Definitions

For the purposes of the present document, the terms and definitions given in 3GPP TR 21.905 [9], 3GPP TR 25.990 [10] and the following apply:

**maximum average power:** average transmitter output power obtained over any specified time interval, including periods with no transmission, when the transmit time slots are at the maximum power setting

## 3.2 Abbreviations

For the purposes of the present document, the abbreviations given in 3GPP TR 21.905 [9], 3GPP TR 25.990 [10] and the following apply:

AFC Automatic Frequency Control
A-GPS Assisted - Global Positioning System
AM Acknowledgement Mode
ATT ATTenuator
BCCH Broadcast Control CHannel
CBS Cell Broadcast Service
CC Convolutional Coding
CCCH Common Control CHannel
CCTrCH Coded Composite Transport CHannel
CS Circuit Switching
DCCH Dedicated Control CHannel
DL DownLink
DPCH Dedicated Physical CHannel
DT Direct transfer
DTCH Dedicated Traffic CHannel
FTM File Tunnelling Mode
GPS Global Positioning System
GSS GPS System Simulator
HYB HYBrid
NAS Non-Access Stratum
OBW Occupied BandWidth
OCNS Orthogonal Channel Noise Simulator

NOTE: A mechanism used to simulate the users or control signals on the other orthogonal channels of a downlink.

PRACH Physical Randome Access CHannel
PS Packet Switching
RAB Radio Access Bearer
RB Radio Bearer
RRC Radio Resource Control

NOTE: (for sub-Layer of layer 3) but also Root-Raised Cosine (for Filter shape).

SCCPCH Secondary Common Control Physical CHannel
SMS Short Message Service
SRB Signalling Radio Bearer
SS System Simulator
SSD Source Statistics Descriptor
TC Turbo Coding
TLM TeLeMetry word

NOTE: It contains an 8-bits preamble (10001011).

TM Transparent Mode
TOW Time Of Week
TTFF Time To First Fix
UL UpLink
UM Unacknowledgement Mode

## 3.3 Symbols

For the purposes of the present document, the following symbols apply:

$I_{oc}$ The power spectral density of a band limited white noise source (simulating interference from other cells) as measured at the UE antenna connector.

# 4 Common requirements of test equipment

Mobile conformance testing can be categorized into 3 distinct areas:

- RF Conformance Testing.
- EMC Conformance Testing.
- Signalling Conformance Testing.

The test equipment required for each category of testing may or not be different, depending on the supplier of the test equipment. However, there will be some generic requirements of the test equipment that are essential for all three categories of test, and these are specified in this clause.

In addition, there will be requirements to test operation in multi-system configurations (eg UTRA plus GSM/DCS1800). However, these would not form a common test equipment requirement for the three test areas and are not considered in the present document.

## 4.1 General Functional Requirements

NOTE: This clause has been written such that it does not constrain the implementation of different architectures and designs of test equipment.

All test equipment used to perform conformance testing on a UE shall provide a platform suitable for testing UE's that are either:

a) FDD Mode; or

b) TDD Mode; or

c) both FDD/TDD Modes.

All test equipment shall provide (for the mode(s) supported) the following minimum functionality.

- The capability of emulating a single UTRA cell with the appropriate channels to allow the UE to register on the cell.
- The capability to allow the UE to set up an RRC connection with the System Simulator, and to maintain the connection for the duration of the test.
- The capability (for the specific test):
  - to select and support an appropriate Radio Bearer for the downlink;
  - to set the appropriate downlink power levels;
  - to set up and support the appropriate Radio Bearer for the uplink;
  - to set and control the uplink power levels.

## 4.2 Minimum performance levels

### 4.2.1 Supported Cell Configuration

The System Simulator shall provide the capability to simulate a minimum number of cells (of the appropriate UTRA Mode) whose number and capabilities are governed by the test cases that need to be performed (test cases are defined in 3GPP TS 34.123-1 [1] (Signalling), 3GPP TS 34.121 [2] (RF-FDD) and 3GPP TS 34.122 [5] (RF-TDD)). For this purpose test cases can be split into two different categories: Tests that require only one cell and Tests that require several cells.

To perform test cases requiring one cell, the system simulator must provide a Cell offering the capabilities to perform all the test cases in this category.

To perform test cases requiring several cells, additional cells must be provided by the system simulator. The additional cells, however, need only provide a minimum set of capabilities so as to support the first cell in carrying out the multi-cell test cases.

The type and number of channels (especially physical channels) constitute an important set of capabilities for a cell. The following clauses list possible channels that may be supported by the SS. Each channel type, however, and the minimum number of channels needed are only mandatory if specific test cases require them.

The mapping between Logical and Transport channels is as described in 3GPP TS 25.301 [7]. Similarly the mapping between Transport channels and Physical channels is as described in 3GPP TS 25.211 [13] for the FDD mode, and 3GPP TS 25.221 [28] for the TDD mode. The reference measurement channels (mapping between Transport channels and Physical channels for DTCH/DCCH to be tested) are defined in 3GPP TS 34.121 [2] annex C for FDD and 3GPP TS 34.122 [5] annex C for TDD.

### 4.2.1.1 Supported Channels for FDD Mode

#### 4.2.1.1.1 Logical channels

| Logical channel | Minimum number | Comments |
|---|---|---|
| BCCH | 1 | |
| CCCH | 1 | |
| DCCH | 4 | 2 for RRC testing, 2 for NAS testing |
| PCCH | 1 | |
| DTCH | n <FFS> | Depending on SS's support for RB service testing (See clause 14 of 3GPP TS 34.123-1 [1]) |

#### 4.2.1.1.2 Transport channels

| Transport channel | Minimum number | Comments |
|---|---|---|
| BCH | 1 | |
| FACH | 1 | |
| PCH | 1 | |
| DCH | n <FFS> | |
| DSCH | 1 | Release 99 and Release 4 only. |
| RACH | 2 | |
| CPCH | 1 | Release 99 and Release 4 only. |
| FAUSCH | N/A | Not in Release 1999 |

#### 4.2.1.1.3 Physical channels

| Physical channel | Minimum number | Comments |
|---|---|---|
| P-CCPCH | 1 | Primary Common Control Physical channel. This is used by the Cell to Broadcast System Information messages, it is transmitted using the Primary Scrambling Code for the Cell. |
| P-CPICH | 1 | Primary Common Pilot Channel using the Primary Scrambling Code for the Cell. |
| S-CPICH | 1 (For RF Tests) | Secondary Common Pilot Channel. This signal is used as the phase reference for some RF tests. |
| SCH | 1 | Synchronization Channel (includes P-SCH and S-SCH) |
| S-CCPCH | 2 | Secondary Common Control Physical channel. |
| PICH | 1 | To identify when the UE should access the PCCH for Paging Messages. |
| AICH | 1 | General Acquisition Indicator Channel that can be used for:<br>- Aquisition Indicator Channel, for PRACH<br>- Access Preamble Acquisition Indicator Channel (AP-ICH), for PCPCH. (For release 99 and release 4 only)<br>- Collision-Detection/Channel-Assignment Indicator Channel (CD/CA-ICH), for PCPCH. (For release 99 and release 4 only) |
| DPDCH | 3 | Downlink Physical Data Channel. There will be a single DPCCH associated with all the DPDCHs used for Layer 1 signalling.<br>This number is for the First Cell. Additional Cells may define a lower number which should be at least 1. |
| PDSCH | 1 | Physical Downlink Shared Channel. Release 99 and Release 4 only. |
| DPCH | 1 | Uplink Dedicated Physical channel |

| Physical channel | Minimum number | Comments |
|---|---|---|
| PRACH | 2 | Physical Random Access Channel. |
| PCPCH | 1 | Physical Common Packet Channel. (For release 99 and release 4 only) |
| CSICH | 1 | CPCH Status Indicator Channel. (For release 99 and release 4 only) |

### 4.2.1.2 Supported Channels for TDD Mode

#### 4.2.1.2.1 Logical channels

| Logical channel | Minimum number | Comments |
|---|---|---|
| **Control channels** | | |
| BCCH | 1 | Broadcast Control Channel: DL channel for broadcasting system control information. |
| CCCH | 1 | Common Control Channel: Bi-directional channel for transmitting control information between network and UEs. This channel is commonly used by the UEs having no RRC connection with the network and by the UEs using common transport channels when accessing a new cell after cell reselection. |
| DCCH | 4 | Dedicated Control Channel: A point-to-point bi-directional channel that transmits dedicated control information between a UE and the network. This channel is established through RRC connection setup procedure. 2 channels for RRC testing and 2 channels for NAS testing estimated. |
| PCCH | 1 | Paging Control Channel: DL channel that transfers paging information. This channel is used when the network does not know the location cell of the UE, or, the UE is in the cell connected state |
| SHCCH | 1 | Shared Channel Control Channel: Bi-directional channel that transmits control information for uplink and downlink shared channels between network and UEs. This channel is for TDD only. |
| **Traffic channels** | | |
| DTCH | 1 | Dedicated Traffic Channel is a point-to-point channel, dedicated to one UE, for the transfer of user information. A DTCH can exist in both UL and DL. |
| CTCH | 1 | Common Traffic Channel is a point-to-multipoint unidirectional channel for transfer of dedicated user information for all or a group of specified UEs. |

#### 4.2.1.2.2 Transport channels

| Transport channel | Minimum number | Comments |
|---|---|---|
| BCH | 1 | Broadcast Channel: DL channel used to broadcast system and cell-specific information. |
| FACH | 1 | Forward Access Channel: DL channel used to carry control information to a mobile station when the system knows the location cell of the mobile station (may also carry short user packets). |
| PCH | 1 | Paging Channel: DL channel used to carry control information to a mobile station when the system does not know the location cell of the mobile station. |
| DCH | 2 | Dedicated Channel:UL or DL channel used to carry user or control information between the UTRAN and a UE |
| DSCH | 1 | DL shared channel: DL channel shared by several UEs carrying dedicated control or traffic data. |
| USCH | 1 | UL shared channel: UL channel shared by several UEs carrying dedicated control or traffic data. |
| RACH | 1 | Random Access Channel: UL channel used to carry control information from mobile station. The RACH may also carry short user packets. |

#### 4.2.1.2.3 Physical channels (3.84 Mcps option)

| Physical channel | Minimum number | Comments |
|---|---|---|
| P-CCPCH | 1 | Primary Common Control Physical channel. . The BCH as described in clause 4.2.1 is mapped onto the P-CCPCH. The position (time slot / code) of the P-CCPCH is known from PSCH. |

| | | |
|---|---|---|
| SCH | 1 | Synchronization Channel. Code group of a cell can be derived from the synchronization channel. In order not to limit the uplink/downlink asymmetry the SCH is mapped on one or two downlink slots per frame only. |
| S-CCPCH | 2 | Secondary Common Control Physical channel. PCH and FACH as described in clause 4.2.1 are mapped onto one or more S-CCPCH. |
| PICH | | Paging Indicator Channel is a physical channel used to carry the paging indicators. |
| DPCH (DL) | 3 | Downlink Dedicated Physical channel. DCH channels are mapped onto DPCH |
| PDSCH | 1 | Physical Downlink Shared Channel. DSCH as desribed in clause 4.2.1 is mapped onto one or more PDSCH. |
| DPCH (UL) | 1 | Uplink Dedicated Physical channel. DCH channels are mapped onto DPCH. |
| PUSCH | 1 | Physical Uplink Shared Channel. The USCH as desribed in clause 4.2.1 is mapped onto one or more PUSCH. Timing advance, as described in 3GPP TS 25.224 [27], clause 4.3, is applied to the PUSCH. |
| PRACH | 2 | Physical Random Access Channel. The RACH as described in clause 4.2.1 is mapped onto PRACH |
| PNBSCH | 1 | Physical node B synchronization channel: In case cell sync bursts are used for Node B synchronization the PNBSCH shall be used for the transmission of the cell sync burst 3GPP TS 25.223 [35]. The PNBSCH shall be mapped on the same timeslot as the PRACH. |

#### 4.2.1.2.4 Physical channels (1.28 Mcps option)

| Physical channel | Minimum number | Comments |
|---|---|---|
| P-CCPCH | 2 | Primary Common Control Physical channel.The BCH as described in clause 4.2.1 is mapped onto the P-CCPCH1 and P-CCPCH2. The position (time slot / code) of the P-CCPCHs is fixed in the 1.28 Mcps TDD. The P-CCPCHs are mapped onto the first two code channels of timeslot#0 with spreading factor of 16. |
| DwPCH | 1 | Synchronization Channel for DL. Present in each 5 ms subframe. |
| UpPCH | 1 | Synchronization Channel for UL. Present in each 5 ms subframe. |
| S-CCPCH | 2 | Secondary Common Control Physical channel. PCH and FACH as described in clause 4.2.1 are mapped onto one or more S-CCPCH. |
| PICH | | Paging Indicator Channel is a physical channel used to carry the paging indicators. |
| DPCH (DL) | 3 | Downlink Dedicated Physical channel.DCH channels are mapped onto DPCH |
| PDSCH | 1 | Physical Downlink Shared Channel. PDSCH provides the possibility for transmission of TFCI, SS, and TPC in downlink. |
| DPCH (UL) | 1 | Uplink Dedicated Physical channel. DCH channels are mapped onto DPCH. |
| PUSCH | 1 | Physical Uplink Shared Channel. PUSCH provides the possibility for transmission of TFCI, SS, and TPC in uplink. |
| FPACH | 1 | Fast Physical Access Channel. FPACH is used by the Node B to carry, in a single burst, the acknowledgement of a detected signature with timing and power level adjustment indication to a user equipment. |
| PRACH | 2 | Physical Random Access Channel. The RACH as described in clause 4.2.1 is mapped onto one or more uplink Physical Random Access Channels (PRACH). |

### 4.2.1.3 Support of $T_{cell}$ timing offset

In test case parameter declarations, the parameter $T_{cell}$ may be specified between 0 to 38 399, to allow for extensibility. However, the system simulator is required only to support a maximum $T_{cell}$ value of 2 304, with a step resolution of 256. The SS may limit a $T_{cell}$ value of greater than 2 304, and may round $T_{cell}$ to the nearest multiple of 256.

## 4.2.2 RF Performance

### 4.2.2.1 Frequency of Operation

The System Simulator shall be capable of adjusting the Carrier Frequency of the DL channels to any frequency allowed in the DL frequency band. The DL frequency shall be accurate to the level of accuracy set by the core specications 3GPP TS 25.104 [20] for FDD and 3GPP TS 25.105 [21] for TDD.

For RF tests, the requirement of Test Equipment is described in 3GPP TS 34.121 [2] annex F for FDD and 3GPP TS 34.122 [5] annex F for TDD respectively.

#### 4.2.2.2 Power Level Setting Accuracy

The system simulator shall be able to adjust the average power output of the DL Channels to meet the absolute accuracy of the system simulator DL power levels covered in clause 5.4.1 Downlink Signal Levels.

For RF tests, the requirement of Test Equipment is described in 3GPP TS 34.121 [2] annex F for FDD and 3GPP TS 34.122 [5] annex F for TDD respectively.

The system simulator shall be capable of altering the power of the DL Dedicated channels under control of the UE Layer 1 Signalling information.

#### 4.2.2.3 Uplink Power Control

The system simulator shall be able to command the UE to transmit at the maximum level for its power class or a lower level required for specific tests. The system simulator shall also provide the capability of generating the Layer 1 Signalling information to set the power levels of the Uplink Dedicated Channels from the UE to lower levels if required.

#### 4.2.2.4 Uplink Signal Handling

For FDD mode, the System Simulator shall not be damaged by a Power Class 1 UE transmitting at the maximum power level permitted in 3GPP TS 25.101 [11] and for TDD mode by a Power Class 2 UE transmitting at the maximum power level permitted in 3GPP TS 25.102 [12].

#### 4.2.2.5 Uplink Sensitivity

The simulator shall be able to receive uplink transmissions from the UE when it is transmitting at the minimum power level defined in 3GPP TS 25.101 [11] for FDD mode, and 3GPP TS 25.102 [12] for TDD mode.

Editor's note: this is obviously a useful feature for the system simulator; however it is <ffs> if it should be an essential common requirement for a protocol test system.

### 4.2.3 Timers Tolerances

All the timers used during testing are within a tolerance margin given by the equation below. If for a specific test a different tolerance value is required then this should be specified in the relevant test document (i.e. the document where the test is described).

Timer tolerance = 10%, or $2 \times \text{TTI} + t_{delta}$, whichever value is the greater.

Where $t_{delta}$ is 55 ms.

# 5 Reference test conditions

## 5.1 Test frequencies

The test frequencies are based the UMTS frequency bands defined in the core specifications.

To avoid interference with adjacent frequency bands the lowest test frequency (downlink and uplink) needs to be offset upwardly by at least 2.6 MHz since the channel's width is 5 MHz for FDD and 3.84 Mcps TDD option, and 0.8 MHz for 1.28 Mcps TDD option since the channel's width is 1.6 MHz. The raster spacing is 200KHz. Similarly the highest test frequency (downlink and uplink) needs to be offset downwardly by at least 2.6 MHz for FDD and 3.84 Mcps TDD option, and 0.8 MHz for 1.28 Mcps TDD option.

NOTE1: Additional regulations concerning interferences to frequency bands used by different systems may also exist. Those regulations are specific to the country where the test equipment is used and need to be taken into account if they require a higher offset than 2.6 MHz from the edge frequencies for FDD and 3.84 Mcps TDD option, and 0.8 MHz for 1.28 Mcps TDD option.

NOTE2: In Band VI, to avoid interference with adjacent frequency bands the lowest test frequency (downlink and uplink) needs to be offset upwardly by at least 2.5 MHz, highest test frequency (downlink and uplink) needs to be offset downwardly by at least 2.5 MHz from the edge frequencies since additional center frequencies are specified according to 3GPP TS 25.101 [11].

## 5.1.1 FDD Mode Test frequencies

UTRA/FDD is designed to operate in one of three paired bands 3GPP TS 25.101 [11]. The reference test frequencies for the common test environment for each of the 6 operating bands are defined in the following tables:

### 5.1.1.1 FDD reference test frequencies for Operating Band I

| Test Frequency ID | UARFCN | Frequency of Uplink | UARFCN | Frequency of Downlink |
|---|---|---|---|---|
| Low Range | 9 613 | 1 922.6 MHz | 10 563 | 2 112.6 MHz |
| Mid Range | 9 750 | 1 950.0 MHz | 10 700 | 2 140.0 MHz |
| High Range | 9 887 | 1 977.4 MHz | 10 837 | 2 167.4 MHz |

### 5.1.1.2 FDD reference test frequencies for Operating Band II

| Test Frequency ID | UARFCN | Frequency of Uplink | UARFCN | Frequency of Downlink |
|---|---|---|---|---|
| Low Range | 9 263 | 1 852.6 MHz | 9 663 | 1 932.6 MHz |
| Mid Range | 9 400 | 1 880 MHz | 9 800 | 1 960 MHz |
| High Range | 9 537 | 1 907.4 MHz | 9 937 | 1 987.4 MHz |

### 5.1.1.3 FDD reference test frequencies for Operating Band III

| Test Frequency ID | UARFCN | Frequency of Uplink | UARFCN | Frequency of Downlink |
|---|---|---|---|---|
| Low Range | 938 | 1 712.6 MHz | 1163 | 1 807.6 MHz |
| Mid Range | 1112 | 1 747.4 MHz | 1337 | 1 842.4 MHz |
| High Range | 1287 | 1 782.4 MHz | 1512 | 1 877.4 MHz |

### 5.1.1.4 Void

### 5.1.1.5 FDD reference test frequencies for Operating Band V

| Test Frequency ID | UARFCN | Frequency of Uplink | UARFCN | Frequency of Downlink |
|---|---|---|---|---|
| Low Range | 4 133 | 826.6 MHz | 4 358 | 871.6 MHz |
| Mid Range | 4 175 | 835 MHz | 4 400 | 880 MHz |
| High Range | 4 232 | 846.4 MHz | 4 457 | 891.4 MHz |

### 5.1.1.6 FDD reference test frequencies for Operating Band VI

| Test Frequency ID | UARFCN | Frequency of Uplink | UARFCN | Frequency of Downlink |
|---|---|---|---|---|
| Low Range | 4 163 | 832.6 MHz | 4 388 | 877.6 MHz |
| Mid Range | 4 175 | 835.0MHz | 4 400 | 880.0 MHz |
| High Range | 4 187 | 837.4 MHz | 4 412 | 882.4 MHz |

### 5.1.1.7 FDD reference test frequencies for Operating Band VII

| Test Frequency ID | UARFCN | Frequency of Uplink | UARFCN | Frequency of Downlink |
|---|---|---|---|---|
| Low Range | 2013 | 2502.6 MHz | 2238 | 2622.6 MHz |
| Mid Range | 2175 | 2535.0 MHz | 2400 | 2655.0 MHz |
| High Range | 2337 | 2567.4 MHz | 2562 | 2687.4 MHz |

### 5.1.1.8 FDD reference test frequencies for Operating Band VIII

| Test Frequency ID | UARFCN | Frequency of Uplink | UARFCN | Frequency of Downlink |
|---|---|---|---|---|
| Low Range | 2713 | 882.6 MHz | 2938 | 927.6 MHz |

| | | | | |
|---|---|---|---|---|
| Mid Range | 2788 | 897.6 MHz | 3013 | 942.6 MHz |
| High Range | 2862 | 912.4 MHz | 3087 | 957.4 MHz |

#### 5.1.1.9 FDD reference test frequencies for Operating Band IX

| Test Frequency ID | UARFCN | Frequency of Uplink | UARFCN | Frequency of Downlink |
|---|---|---|---|---|
| Low Range | 8 762 | 1752.4 MHz | 9 237 | 1847.4 MHz |
| Mid Range | 8 837 | 1767.4MHz | 9 312 | 1862.4 MHz |
| High Range | 8 912 | 1782.4 MHz | 9 387 | 1877.4 MHz |

### 5.1.2 TDD Mode Test frequencies

UTRA/TDD is designed to operate in one of three unpaired bands (3GPP TS 25.102 [12]). The reference test frequencies for the common test environment for each of the 3 operating bands are defined in the following tables:

#### 5.1.2.1 Standard TDD reference test frequencies (3.84 Mcps option)

| Test Frequency ID | Band a | | Band b | | Band c | |
|---|---|---|---|---|---|---|
| | UARFCN | Frequency (UL and DL) | UARFCN | Frequency (UL and DL) | UARFCN | Frequency (UL and DL) |
| Low Range | 9 513 | 1 902.6 MHz | 9 263 | 1 852.6 MHz | 9 563 | 1 912.6 MHz |
| Mid Range | 9 550 | 1 910 MHz | 9 400 | 1 880 MHz | 9 600 | 1 920 MHz |
| High Range | 9 587 | 1 917.4 MHz | 9 537 | 1 907.4 MHz | 9 637 | 1 927.4 MHz |
| Low Range | 10 063 | 2 012.6 MHz | 9 663 | 1 932.6 MHz | | |
| Mid Range | 10 087 | 2 017.4 MHz | 9 800 | 1 960 MHz | | |
| High Range | 10 112 | 2 022.4 MHz | 9 937 | 1 987.4 MHz | | |

#### 5.1.2.2 Standard TDD reference test frequencies (1.28 Mcps option)

| Test Frequency ID | Band a | | Band b | | Band c | |
|---|---|---|---|---|---|---|
| | UARFCN | Frequency (UL and DL) | UARFCN | Frequency (UL and DL) | UARFCN | Frequency (UL and DL) |
| Low Range | 9 504 | 1 900.8 MHz | 9 254 | 1 850.8 MHz | 9 554 | 1 910.8 MHz |
| Mid Range | 9 550 | 1 910 MHz | 9 400 | 1 880 MHz | 9 600 | 1 920 MHz |
| High Range | 9 596 | 1 919.2 MHz | 9 546 | 1 909.2 MHz | 9 646 | 1 929.2 MHz |
| Low Range | 10 054 | 2 010.8 MHz | 9 654 | 1 930.8 MHz | | |
| Mid Range | 10 087 | 2 017.4 MHz | 9 800 | 1 960 MHz | | |
| High Range | 10 121 | 2 024.2 MHz | 9 946 | 1 989.2 MHz | | |

## 5.2 Radio conditions

There are a number of radio propagation conditions defined in 3GPP TS 34.121 [2] for FDD mode and 3GPP TS 34.122 [5] for TDD mode, which may be required for a number of tests and hence can be considered as Common Conditions for FDD mode and TDD mode respectively.

NOTE: The System Simulator is required to support at least the normal Propagation Condition; support of the other propagation conditions is optional, depending on the specific test supported by the simulator.

### 5.2.1 Normal propagation condition

This condition provides a connection between the System Simulator that is effectively free from Additive White Gaussian Noise, and where there are no fading or multipath effects. This condition will be used for Signalling tests.

### 5.2.2 Static propagation condition

See 3GPP TS 34.121 [2], annex D for FDD.

For TDD mode, the propagation for the static performance measurement is an Additive White Gaussian Noise (AWGN) environment. No fading and multi-paths exist for this propagation model.

### 5.2.3 Multi-path fading propagation conditions

See 3GPP TS 34.121 [2], annex D for FDD and 3GPP TS 34.122 [5], annex D for TDD.

### 5.2.4 Moving propagation conditions

See 3GPP TS 34.121 [2], annex D for FDD. There are no currently defined Moving propagation conditions for TDD.

### 5.2.5 Birth-Death propagation conditions

See 3GPP TS 34.121 [2], annex D for FDD. There are no currently defined Birth-Death propagation conditions for TDD.

## 5.3 Standard test signals

Reference 3GPP TS 25.101 [11] and 3GPP TS 25102 [12] for definitions of standard test signals.

## 5.4 Signal levels

The power levels given in clauses 5.4.1 and 5.4.2 apply for Signalling tests only. For RF tests power levels are given in 3GPP TS 34.121 [2], annex E for FDD and 3GPP TS 34.122 [5], annex E for TDD.

### 5.4.1 Downlink signal levels

<FFS>

### 5.4.2 Uplink signal levels

<FFS>

# 6 Reference system configurations

This clause defines a number of Reference System Configurations which can be used for different tests.

## 6.1 Simulated network environments

The UE will eventually have to operate in either single mode networks (FDD or TDD), dual mode networks (FDD+TDD), or inter-RAT networks (FDD or TDD + GSM).

The following tables list the default parameters for 1 to 8 cell environments for testing.

To simplify TTCN implementation the total number of simultaneous cells in intra-frequency, inter-frequency and inter-RAT cell information lists (SIB11) have been limited to 8 and a specific cell numbering scheme have been defined to associate cell identifiers with type of cell.

- Cell 1, Cell 2, Cell 3, Cell 7, Cell 8 and Cell 11 are associated with FDD/TDD cells using frequency f1;
- Cell 4, Cell 5 and Cell 6 are associated with FDD/TDD cells using frequency f2; and
- Cell 9 and Cell 10 are associated with GSM cells.

For protocol testing in FDD and TDD intra- and inter-frequency cell environment Cell 1 to Cell 8 are used.

For RF and RRM in FDD and TDD intra- and inter-frequency cell environment Cell 1 to Cell 8 and Cell 11 are used.

For FDD/GSM inter-RAT cell environment Cell 1 to Cell 6, Cell 9 and Cell 10 are used.

In this clause, decimal values are normally used. However, sometimes a hexadecimal value, indicated by an "H", or a binary value, indicated by a "B" is used.

### 6.1.0a Default Master Information Block and Scheduling Block messages

#### 6.1.0a.1 Grouping SIBs for testing

| | | |
|---|---|---|
| **Mandatory in 34.108** | **Used in Idle Mode** | MIB, SB1, (SB2), SIB1, SIB2, SIB3, SIB5, SIB7, SIB11 |
| | **Used in Connected Mode** | SIB4, SIB6, SIB12 |
| **Mandatory for FDD CPCH (R99 and Rel-4 only)** | | SIB8, SIB9 |
| **Mandatory for FDD DRAC** | | SIB10 |
| **Mandatory for TDD** | | SIB14, SIB17 |
| **Mandatory for LCS** | | SIB15, SIB15.1, SIB15.2, SIB15.3 |
| **Mandatory for ANSI-41 system** | | SIB13, SIB13.1, SIB13.2, SIB13.3, SIB13.4 |
| **Mandatory for InterSys HO from GERAN To UTRAN** | | SIB16 |
| **Mandatory for Cell reselection** | | SIB18 |

#### 6.1.0a.2 SIB configurations

Currently three SIB configurations are used.

Configuration 1 is the default. It is used for the following test case scenarios:

- UTRAN/FDD only SYSTEM.
- UTRAN/FDD + GERAN SYSTEM (not involving inter-RAT handover from GERAN to UTRAN).
- UTRAN/TDD only SYSTEM.
- UTRAN/TDD + GERAN SYSTEM (not involving inter-RAT handover from GERAN to UTRAN).
- inter-RAT handover from GERAN to UTRAN test cases.

Configuration 2 is for test cases which need two S_CCPCH or two PRACH.

Configuration 3 is for inter-RAT handover from GERAN to UTRAN test cases.

| | |
|---|---|
| **Configuration 1** | MIB, SB1, SIB1, SIB2, SIB3, SIB4, SIB5, SIB6, SIB7, SIB11, SIB12, SIB18 |
| **Configuration 2** | MIB, SB1, SIB1, SIB2, SIB3, SIB4, SIB5, SIB7, SIB11, SIB12, SIB18 |
| **Configuration 3** | MIB, SB1, SIB1, SIB2, SIB3, SIB4, SIB5, SIB7, SIB11, SIB16, SIB18 |

### 6.1.0a.3 SIB default schedule

| **Block Type** | MIB | SB1 | SIB1 | SIB2 | SIB3 | SIB4 | SIB5 | SIB6 | SIB7 | SIB11 | SIB12 | SIB18 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **SIB_REP** | 8 | 16 | 64 | 64 | 64 | 64 | 64 | 64 | 16 | 64 | 64 | 64 |
| **SEG_ COUNT** | 1 | 1 | 1 | 1 | 1 | 1 | 4(FDD) 5(TDD) | 4 | 1 | 3 | 3 | 1 |

| **Frame No / SIB_POS** | 0 | 2 | 4 | 6 | 8 | 10 | 12 | 14 |
|---|---|---|---|---|---|---|---|---|
| **Block Type** | MIB | SB1 | SIB7 | SIB6 | MIB | SIB6 | SIB6 | SIB6 |

| **Frame No / SIB_POS** | 16 | 18 | 20 | 22 | 24 | 26 | 28 | 30 |
|---|---|---|---|---|---|---|---|---|
| **Block Type** | MIB | SB1 | SIB7/SIB3 | SIB1/SIB2 | MIB | SIB12 | SIB12 | SIB12 |

| **Frame No / SIB_POS** | 32 | 34 | 36 | 38 | 40 | 42 | 44 | 46 |
|---|---|---|---|---|---|---|---|---|
| **Block Type** | MIB | SB1 | SIB7/SIB18 | SIB5 | MIB | SIB5 | SIB5 | SIB5 |

| **Frame No / SIB_POS** | 48 | 50 | 52 | 54 | 56 | 58 | 60 | 62 |
|---|---|---|---|---|---|---|---|---|
| **Block Type** | MIB | SB1 | SIB7/SIB4 | - (FDD) SIB5(LCR TDD) | MIB | SIB11 | SIB11 | SIB11 |

The SEG_COUNT in the table specifies the maximum possible transport BCH blocks scheduled for broadcasting. The more contents a SIB has, the more transport BCH blocks are needed for broadcasting. In order to keep SIB repetition period, SIB_REP, unchanged in different test cases, each specific SIB in the individual test cases after the PER encoding shall not exceed the SEG_COUNT scheduled.

If the transport BCH blocks actually required for a SIB is less than the scheduled SEG_COUNT, the no_segment blocks shall be placed at the rest scheduled transport BCH blocks. In addition, the corresponding SEG_COUNT IE value in MIB or in SB1 shall be set to the number of transport BCH blocks actually required.

Contents of Master Information Block PLMN type is the case of GSM-MAP

| | |
|---|---|
| - MIB value tag | A valid MIB value tag value as defined in TS 25.331 [34] |
| - Supported PLMN types | |
| - PLMN type | GSM-MAP |
| - PLMN identity | |
| - MCC digit | Set to the same Mobile Country Codes stored in the test USIM card (clause 8.3.2.2 EF IMSI(IMSI)). |
| - MNC digit | Set to the same Mobile Network Codesstored in the test USIM card (clause 8.3.2.2 EF IMSI(IMSI)). |
| - ANSI-41 Core Network information | Not Present |
| - References to other system information blocks and scheduling blocks | |
| - References to other system information blocks | |
| - Scheduling information | |
| - CHOICE Value tag | Cell Value Tag |
| - Cell Value tag | A valid Cell value tag value as defined in TS 25.331 [34] |
| - Scheduling | |
| - SEG_COUNT | 1 |
| - SIB_REP | 16 |
| - SIB_POS | 2 |
| - SIB_POS offset info | Not Present - use default |
| - SIB and SB type | Scheduling Block 1 |
| - Scheduling information | |
| - CHOICE Value tag | PLMN Value tag |
| - PLMN Value tag | A valid PLMN value tag value as defined in TS 25.331 [34] |
| - SEG_COUNT | 1 |

| | |
|---|---|
| - SIB_REP | 64 |
| - SIB_POS | 22 |
| - SIB_POS offset info | Not Present - use default |
| - SIB and SB type | System Information Type 1 |
| - Scheduling information | |
| - CHOICE Value tag | Cell Value tag |
| - Cell Value tag | A valid Cell value tag value as defined in TS 25.331 [34] |
| - SEG_COUNT | 1 |
| - SIB_REP | 64 |
| - SIB_POS | 22 |
| - SIB_POS offset info | Not Present - use default |
| - SIB and SB type | System Information Type 2 |
| - Scheduling information | |
| - CHOICE Value tag | Cell Value tag |
| - Cell Value tag | 1 |
| - SEG_COUNT | 1 |
| - SIB_REP | 64 |
| - SIB_POS | 20 |
| - SIB_POS offset info | Not Present - use default |
| - SIB and SB type | System Information Type 3 |
| - Scheduling information | |
| - CHOICE Value tag | Cell Value tag |
| - Cell Value tag | A valid Cell value tag value as defined in TS 25.331 [34] |
| - SEG_COUNT | 1 |
| - SIB_REP | 64 |
| - SIB_POS | 52 |
| - SIB_POS offset info | Not Present - use default |
| - SIB and SB type | System Information Type 4 |
| - Scheduling information | |
| - CHOICE Value tag | Cell Value tag |
| - Cell Value tag | A valid Cell value tag value as defined in TS 25.331 [34] |
| - SEG_COUNT | 4 |
| - SIB_REP | 64 |
| - SIB_POS | 38 |
| - SIB_POS offset info | |
| - SIB_OFF | 4 |
| - SIB_OFF | 2 |
| - SIB_OFF | 2 |
| - SIB and SB type | System Information Type 5 |

Contents of Scheduling Block 1 (FDD and 1.28 Mcps TDD)

| | |
|---|---|
| - References to other system information blocks | |
| - Scheduling information | |
| - CHOICE Value tag | Cell Value tag |
| - Cell Value tag | A valid Cell value tag value as defined in TS 25.331 [34] |
| - SEG_COUNT | 4 |
| - SIB_REP | 64 |
| - SIB_POS | 6 |
| - SIB_POS offset info | |
| - SIB_OFF | 4 |
| - SIB_OFF | 2 |
| - SIB_OFF | 2 |
| - SIB type SIBs only | System Information Type 6 |
| - Scheduling information | |
| - CHOICE Value tag | Not Present |
| - SEG_COUNT | 1 |
| - SIB_REP | 16 |
| - SIB_POS | 4 |
| - SIB_POS offset info | Not Present |
| - SIB type SIBs only | System Information Type 7 |
| - Scheduling information | |
| - CHOICE Value tag | Cell Value tag |
| - Cell Value tag | A valid Cell value tag value as defined in TS 25.331 [34] |
| - SEG_COUNT | 3 |

| | |
|---|---|
| - SIB_REP | 64 |
| - SIB_POS | 58 |
| - SIB_POS offset info | |
| - SIB_OFF | 2 |
| - SIB_OFF | 2 |
| - SIB type SIBs only | System Information Type 11 |
| - Scheduling information | |
| - CHOICE Value tag | Cell Value tag |
| - Cell Value tag | A valid Cell value tag value as defined in TS 25.331 [34] |
| - SEG_COUNT | 3 |
| - SIB_REP | 64 |
| - SIB_POS | 26 |
| - SIB_POS offset info | |
| - SIB_OFF | 2 |
| - SIB_OFF | 2 |
| - SIB type SIBs only | System Information Type 12 |
| - Scheduling information | |
| - CHOICE Value tag | Cell Value tag |
| Cell Value tag | A valid Cell value tag value as defined in TS 25.331 [34] |
| - SEG_COUNT | 1 |
| - SIB_REP | 64 |
| - SIB_POS | 36 |
| - SIB_POS offset info | Not Present |
| - SIB type SIBs only | System Information Type 18 |

Contents of Scheduling Block 1 (3.84 Mcps TDD)

| | |
|---|---|
| - References to other system information blocks | |
| - Scheduling information | |
| - CHOICE Value tag | Cell Value tag |
| - Cell Value tag | A valid Cell value tag value as defined in TS 25.331 [34] |
| - SEG_COUNT | 4 |
| - SIB_REP | 128 |
| - SIB_POS | 3 |
| - SIB_POS offset info | |
| - SIB_OFF | 4 |
| - SIB_OFF | 2 |
| - SIB_OFF | 2 |
| - SIB type SIBs only | System Information Type 6 |
| - Scheduling information | |
| - CHOICE Value tag | Not Present |
| - SEG_COUNT | 1 |
| - SIB_REP | 16 |
| - SIB_POS | 2 |
| - SIB_POS offset info | Not Present |
| - SIB type SIBs only | System Information Type 7 |
| - Scheduling information | |
| - CHOICE Value tag | Cell Value tag |
| - Cell Value tag | A valid Cell value tag value as defined in TS 25.331 [34] |
| - SEG_COUNT | 3 |
| - SIB_REP | 64 |
| - SIB_POS | 29 |
| - SIB_POS offset info | |
| - SIB_OFF | 2 |
| - SIB_OFF | 2 |
| - SIB type SIBs only | System Information Type 11 |
| - Scheduling information | |
| - CHOICE Value tag | Cell Value tag |
| - Cell Value tag | A valid Cell value tag value as defined in TS 25.331 [34] |
| - SEG_COUNT | 3 |
| - SIB_REP | 64 |
| - SIB_POS | 13 |
| - SIB_POS offset info | |
| - SIB_OFF | 2 |
| - SIB_OFF | 2 |

| | |
|---|---|
| - SIB type SIBs only | System Information Type 12 |
| - Scheduling information | |
| - CHOICE Value tag | Cell Value tag |
| - Cell Value tag | A valid Cell value tag value as defined in TS 25.331 [34] |
| - SEG_COUNT | 1 |
| - SIB_REP | 64 |
| - SIB_POS | 54 |
| - SIB_POS offset info | Not Present - use default |
| - SIB type SIBs only | System Information Type 14 |
| - Scheduling information | |
| - CHOICE Value tag | PLMN Value tag |
| - PLMN Value tag | A valid PLMN value tag value as defined in TS 25.331 [34] |
| - SEG_COUNT | 1 |
| - SIB_REP | 64 |
| - SIB_POS | 6 |
| - SIB_POS offset info | Not Present |
| - SIB type SIBs only | System Information Type 18 |

## 6.1.0a.4 SIB special schedules

### 6.1.0a.4.1 SIB schedule for two S-CCPCH or two PRACH (For FDD)

**Table 1**

| **Frame No.** | 0 | 2 | 4 | 6 | 8 | 10 | 12 | 14 |
|---|---|---|---|---|---|---|---|---|
| **REP-POS** | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 |
| **Block Type** | MIB | SB1 | SB1 | | MIB | SIB1 | SIB18 | SIB2 |

| **Frame No.** | 16 | 18 | 20 | 22 | 24 | 26 | 28 | 30 |
|---|---|---|---|---|---|---|---|---|
| **REP-POS** | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 |
| **Block Type** | MIB | SB1 | SB1 | SIB7 | MIB | SIB3 | | SIB4 |

| **Frame No.** | 32 | 34 | 36 | 38 | 40 | 42 | 44 | 46 |
|---|---|---|---|---|---|---|---|---|
| **REP-POS** | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 |
| **Block Type** | MIB | SB1 | SB1 | SIB5 | MIB | SIB5 | SIB5 | SIB5 |

| **Frame No.** | 48 | 50 | 52 | 54 | 56 | 58 | 60 | 62 |
|---|---|---|---|---|---|---|---|---|
| **REP-POS** | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 |
| **Block Type** | MIB | SB1 | SB1 | SIB7 | MIB | SIB11 | SIB11 | SIB11 |

| **Frame No.** | 64 | 66 | 68 | 70 | 72 | 74 | 76 | 78 |
|---|---|---|---|---|---|---|---|---|
| **REP-POS** | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 |
| **Block Type** | MIB | SB1 | SB1 | SIB5 | MIB | SIB5 | SIB5 | SIB5 |

| **Frame No.** | 80 | 82 | 84 | 86 | 88 | 90 | 92 | 94 |
|---|---|---|---|---|---|---|---|---|
| **REP-POS** | 40 | 41 | 42 | 43 | 44 | 45 | 46 | 47 |
| **Block Type** | MIB | SB1 | SB1 | SIB7 | MIB | SIB3 | | SIB4 |

| **Frame No.** | 96 | 98 | 100 | 102 | 104 | 106 | 108 | 110 |
|---|---|---|---|---|---|---|---|---|
| **REP-POS** | 48 | 49 | 50 | 51 | 52 | 53 | 54 | 55 |
| **Block Type** | MIB | SB1 | SB1 | | MIB | | | |

| **Frame No.** | 112 | 114 | 116 | 118 | 120 | 122 | 124 | 126 |
|---|---|---|---|---|---|---|---|---|
| **REP-POS** | 56 | 57 | 58 | 59 | 60 | 61 | 62 | 63 |
| **Block Type** | MIB | SB1 | SB1 | SIB7 | MIB | SIB12 | SIB12 | SIB12 |

SIB-repeat period (in frame)

**Table 2**

| **Block Type** | MIB | SB1 | SIB1 | SIB2 | SIB3 | SIB4 | SIB5 | SIB7 | SIB11 | SIB12 | SIB18 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| **SIB Rep** | 8 | 16 | 128 | 128 | 64 | 64 | 128 | 32 | 128 | 128 | 128 |
| **Max. No of seg.** | 1 | 2 | 1 | 1 | 1 | 1 | 8 | 1 | 3 | 3 | 1 |

### 6.1.0a.4.2 SIB schedule for Idle Mode, Measurement and Inter RAT UTRAN to GERAN test cases

| **Frame No.** | 0 | 2 | 4 | 6 | 8 | 10 | 12 | 14 |
|---|---|---|---|---|---|---|---|---|
| **REP-POS** | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 |
| **Block Type** | MIB | SB1 | SIB6 | SIB6 | MIB | SIB6 | SIB6 | SIB7/SIB 3 |

| **Frame No.** | 16 | 18 | 20 | 22 | 24 | 26 | 28 | 30 |
|---|---|---|---|---|---|---|---|---|
| **REP-POS** | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 |
| **Block Type** | MIB | SB1 | SIB1/SIB2 | SIB12 | MIB | SIB12 | SIB12 | SIB7/SIB 12 |

| **Frame No.** | 32 | 34 | 36 | 38 | 40 | 42 | 44 | 46 |
|---|---|---|---|---|---|---|---|---|
| **REP-POS** | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 |
| **Block Type** | MIB | SB1 | SIB5 | SIB5 | MIB | SIB5 | SIB5 | SIB7/SIB 18 |

| **Frame No.** | 48 | 50 | 52 | 54 | 56 | 58 | 60 | 62 |
|---|---|---|---|---|---|---|---|---|
| **REP-POS** | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 |
| **Block Type** | MIB | SB1 | SIB11 | SIB11 | MIB | SIB11 | SIB11 | SIB7/SIB 4 |

SIB-repeat period (in frame)

| **Block Type** | MIB | SB1 | SIB1 | SIB2 | SIB3 | SIB4 | SIB5 | SIB6 | SIB7 | SIB11 | SIB12 | SIB18 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **SIB Rep** | 8 | 16 | 64 | 64 | 64 | 64 | 64 | 64 | 16 | 64 | 64 | 64 |
| **Max. No of seg.** | 1 | 1 | 1 | 1 | 1 | 1 | 4 | 4 | 1 | 4 | 4 | 1 |

### 6.1.0a.4.3 SIB schedule for Inter RAT handover GERAN to UTRAN test cases

| **Frame No.** | 0 | 2 | 4 | 6 | 8 | 10 | 12 | 14 |
|---|---|---|---|---|---|---|---|---|
| **REP-POS** | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 |
| **Block Type** | MIB | SB1 | SB1 | | MIB | SIB1 | SIB18 | SIB2 |

| **Frame No.** | 16 | 18 | 20 | 22 | 24 | 26 | 28 | 30 |
|---|---|---|---|---|---|---|---|---|
| **REP-POS** | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 |
| **Block Type** | MIB | SB1 | SB1 | SIB7 | MIB | SIB3 | | SIB4 |

| **Frame No.** | 32 | 34 | 36 | 38 | 40 | 42 | 44 | 46 |
|---|---|---|---|---|---|---|---|---|
| **REP-POS** | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 |
| **Block Type** | MIB | SB1 | SB1 | SIB5 | MIB | SIB5 | SIB5 | SIB5 |

| **Frame No.** | 48 | 50 | 52 | 54 | 56 | 58 | 60 | 62 |
|---|---|---|---|---|---|---|---|---|
| **REP-POS** | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 |
| **Block Type** | MIB | SB1 | SB1 | SIB7 | MIB | SIB11 | SIB11 | SIB11 |

| **Frame No.** | 64 | 66 | 68 | 70 | 72 | 74 | 76 | 78 |
|---|---|---|---|---|---|---|---|---|
| **REP-POS** | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 |
| **Block Type** | MIB | SB1 | SB1 | SIB16 | MIB | SIB16 | SIB16 | SIB16 |

| **Frame No.** | 80 | 82 | 84 | 86 | 88 | 90 | 92 | 94 |
|---|---|---|---|---|---|---|---|---|
| **REP-POS** | 40 | 41 | 42 | 43 | 44 | 45 | 46 | 47 |
| **Block Type** | MIB | SB1 | SB1 | SIB7 | MIB | SIB3 | | SIB4 |

| **Frame No.** | 96 | 98 | 100 | 102 | 104 | 106 | 108 | 110 |
|---|---|---|---|---|---|---|---|---|
| **REP-POS** | 48 | 49 | 50 | 51 | 52 | 53 | 54 | 55 |
| **Block Type** | MIB | SB1 | SB1 | SIB16 | MIB | SIB16 | SIB16 | SIB16 |

| **Frame No.** | 112 | 114 | 116 | 118 | 120 | 122 | 124 | 126 |
|---|---|---|---|---|---|---|---|---|
| **REP-POS** | 56 | 57 | 58 | 59 | 60 | 61 | 62 | 63 |
| **Block Type** | MIB | SB1 | SB1 | SIB7 | MIB | | | |

SIB-repeat period (in frame)

| **Block Type** | MIB | SB1 | SIB1 | SIB2 | SIB3 | SIB4 | SIB5 | SIB7 | SIB11 | SIB16 | SIB18 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| **SIB Rep** | 8 | 16 | 128 | 128 | 64 | 64 | 128 | 32 | 128 | 128 | 128 |
| **Max. No of seg.** | 1 | 2 | 1 | 1 | 1 | 1 | 4 | 1 | 3 | 8 | 1 |

## 6.1.0b Default System Information Block Messages

Contents of System Information Block type 1 (supported PLMN type is GSM-MAP)

| | | |
|---|---|---|
| - CN common GSM-MAP NAS system information | A1 | |
| - GSM-MAP NAS system information | | 00 01H |
| - CN domain system information | | |
| - CN domain identity | | PS |
| - CHOICE CN Type | | GSM-MAP |
| - CN domain specific NAS system information | | |
| - GSM-MAP NAS system information | | 05 00H |
| - CN domain specific DRX cycle length coefficient | | 7 |
| - CN domain identity | | CS |
| - CHOICE CN Type | | GSM-MAP |
| - CN domain specific NAS system information | | |
| - GSM-MAP NAS system information | | 1E 01H |
| - CN domain specific DRX cycle length coefficient | | 7 |
| - CN common GSM-MAP NAS system information | A2 | |
| - GSM-MAP NAS system information | | 00 80H (see note) |
| - CN domain system information | | |
| - CN domain identity | | PS |
| - CHOICE CN Type | | GSM-MAP |

| | | |
|---|---|---|
| - CN domain specific NAS system information | | |
| - GSM-MAP NAS system information | | 00 00H (see note) |
| - CN domain specific DRX cycle length coefficient | | 7 |
| - CN domain identity | | CS |
| - CHOICE CN Type | | GSM-MAP |
| - CN domain specific NAS system information | | |
| - GSM-MAP NAS system information | | 1E 01H |
| - CN domain specific DRX cycle length coefficient | | 7 |
| - UE Timers and constants in idle mode | A1, A2 | |
| -T300 | | 4 000 milliseconds |
| -N300 | | 3 |
| -T312 | | 10 seconds |
| - N312 | | 1 |
| - UE Timers and constants in connected mode | | |
| - T301 | | Not Present (2 000 milliseconds: default value) |
| - N301 | | Not Present (2: default value) |
| - T302 | | Not Present (4 000 milliseconds: default value) |
| - N302 | | Not Present (3: default value) |
| - T304 | | Not Present (2 000 milliseconds: default value) |
| - N304 | | Not Present (2: default value) |
| - T305 | | Not Present (30 minutes: default value) |
| - T307 | | Not Present (30 seconds: default value) |
| - T308 | | Not Present (160 milliseconds: default value) |
| - T309 | | Not Present (5 seconds: default value) |
| - T310 | | Not Present (160 milliseconds: default value) |
| - N310 | | Not Present (4: default value) |
| - T311 | | Not Present (2 000 milliseconds: default value) |
| - T312 | | Not Present (1 seconds: default value) |
| - N312 | | Not Present (1: default value) |
| - T313 | | Not Present (3 seconds: default value) |
| - N313 | | Not Present (20: default value) |
| - T314 | | Not Present (12 seconds: default value) |
| - T315 | | Not Present (180 seconds: default value) |
| - N315 | | Not Present (1: default value) |
| - T316 | | Not Present (30 seconds: default value) |
| - T317 | | Not Present (180 seconds: default value) |
| NOTE: For Inter-RAT test cases GERAN and UTRAN cells use different LAC and RAC. | | |

| Condition | Explanation |
|---|---|
| A1 | UTRAN cell environment |
| A2 | UTRAN/GSM inter-RAT cell environment |

Contents of System Information Block type 2

| | |
|---|---|
| - URA identity list | *Only 1 URA identity broadcasted* |
| - URA identity | 0000 0000 0000 0001B |

Contents of System Information Block type 3 (FDD)

| Information Element | Value/remark | Version |
|---|---|---|
| - SIB4 indicator | TRUE | |
| - Cell identity | 0000 0000 0000 0000 0000 0000 0001B | |
| - Cell selection and re-selection info | | |
| - Mapping info | Not Present | |
| - Cell selection and reselection quality measure | CPICH RSCP | |
| - CHOICE mode | FDD | |
| - $S_{intrasearch}$ | 16 dB | |
| - $S_{intersearch}$ | 16 dB | |
| - $S_{searchHCS}$ | Not Present | |
| - RAT List | This parameter is configurable | |
| - RAT identifier | GSM | |
| - $S_{search,RAT}$ | -32 dB | |
| - $S_{HCS,RAT}$ | Not Present | |
| - $S_{limit,SearchRAT}$ | 0 | |
| - Qqualmin | Reference to table 6.1.1 | |

| | | |
|---|---|---|
| - Qrxlevmin | Reference to table 6.1.1 | |
| - $Qhyst1_s$ | 2 dB | |
| - Qhyst2s | Not Present | |
| - Treselections | 0 seconds | |
| - HCS Serving cell information | Not Present | |
| - Maximum allowed UL TX power | Reference to table 6.1.1 | |
| - Cell Access Restriction | | |
| - Cell barred | Not barred | |
| - Intra-frequency cell re-selection indicator | Not present | |
| - $T_{barred}$ | Not present | |
| - Cell Reserved for operator use | Not reserved | |
| - Cell Reservation Extension | Not reserved | |
| - Access Class Barred List | | |
| - Access Class Barred0 | Not barred | |
| - Access Class Barred1 | Not barred | |
| - Access Class Barred2 | Not barred | |
| - Access Class Barred3 | Not barred | |
| - Access Class Barred4 | Not barred | |
| - Access Class Barred5 | Not barred | |
| - Access Class Barred6 | Not barred | |
| - Access Class Barred7 | Not barred | |
| - Access Class Barred8 | Not barred | |
| - Access Class Barred9 | Not barred | |
| - Access Class Barred10 | Not barred | |
| - Access Class Barred11 | Not barred | |
| - Access Class Barred12 | Not barred | |
| - Access Class Barred13 | Not barred | |
| - Access Class Barred14 | Not barred | |
| - Access Class Barred15 | Not barred | |
| - Domain Specific Access Restriction Parameters For PLMN Of MIB | Not present | REL-6 |
| - Domain Specific Access Restriction For Shared Network | Not present | REL-6 |

Contents of System Information Block type 3 (3.84 Mcps TDD and 1.28 Mcps TDD)

| Information Element | Value/remark | Version |
|---|---|---|
| - SIB4 Indicator | TRUE | |
| - Cell identity | 0000 0000 0000 0000 0000 0000 0001B | |
| - Cell selection and re-selection info | | |
| - Mapping info | Not present | |
| - Cell selection and reselection quality measure | (no data) | |
| - CHOICE mode | TDD | |
| - Sintrasearch | 10 dB | |
| - Sintersearch | 10 dB | |
| - SsearchHCS | Not present | |
| - RAT List | This parameter is configurable | |
| - RAT identifier | GSM | |
| - $S_{search,RAT}$ | -32 dB | |
| - $S_{HCS,RAT}$ | Not present | |
| - $S_{limit,SearchRAT}$ | Not Present | |
| - Qrxlevmin | -103 dBm | |
| - $Qhyst1_s$ | 0 dB | |
| - Treselections | 0 seconds | |
| - HCS Serving cell information | Not present | |
| - Maximum allowed UL TX power | 30dBm | |
| - Cell Access Restriction | | |
| - Cell barred | Not barred | |
| - Intra-frequency cell re-selection indicator | Not present | |
| - $T_{barred}$ | Not present | |
| - Cell Reserved for operator use | Not reserved | |
| - Cell Reservation Extension | Not reserved | |
| - Access Class Barred List | | |
| - Access Class Barred0 | Not barred | |
| - Access Class Barred1 | Not barred | |
| - Access Class Barred2 | Not barred | |
| - Access Class Barred3 | Not barred | |

| | | |
|---|---|---|
| - Access Class Barred4 | Not barred | |
| - Access Class Barred5 | Not barred | |
| - Access Class Barred6 | Not barred | |
| - Access Class Barred7 | Not barred | |
| - Access Class Barred8 | Not barred | |
| - Access Class Barred9 | Not barred | |
| - Access Class Barred10 | Not barred | |
| - Access Class Barred11 | Not barred | |
| - Access Class Barred12 | Not barred | |
| - Access Class Barred13 | Not barred | |
| - Access Class Barred14 | Not barred | |
| - Access Class Barred15 | Not barred | |
| - Domain Specific Access Restriction Parameters For PLMN Of MIB | Not present | REL-6 |
| - Domain Specific Access Restriction For Shared Network | Not present | REL-6 |

Contents of System Information Block type 4 in connected mode (FDD)

| | |
|---|---|
| - Cell identity | 0000 0000 0000 0000 0000 0000 0001B |
| - Cell selection and re-selection info | |
| - Mapping Info | Not present |
| - Cell selection and reselection quality measure | CPICH RSCP |
| - CHOICE mode | FDD |
| - Sintrasearch | 16 dB |
| - Sintersearch | 16 dB |
| - SsearchHCS | Not present |
| - RAT List | This parameter is configurable |
| - RAT identifier | GSM |
| - Ssearch,RAT | -32 dB |
| - SHCS,RAT | Not Present |
| - $S_{limit,SearchRAT}$ | 0 |
| - Qqualmin | Reference to table 6.1.1 |
| - Qrxlevmin | Reference to table 6.1.1 |
| - Qhyst1s | 2 dB |
| - Qhyst2s | Not Present |
| - Treselections | 0 seconds |
| - HCS Serving cell information | Not Present |
| - Maximum allowed UL TX power | Reference to table 6.1.1 |
| - Cell Access Restriction | |
| - Cell barred | Not barred |
| - Intra-frequency cell re-selection indicator | Not present |
| - $T_{barred}$ | Not present |
| - Cell Reserved for operator use | Not reserved |
| - Cell Reservation Extension | Not reserved |
| - Access Class Barred List | Not present |

Contents of System Information Block type 4 in connected mode (similar to SIB type3) (3.84 Mcps TDD and 1.28 Mcps TDD)

| | |
|---|---|
| - Cell identity | 0000 0000 0000 0000 0000 0000 0001B |
| - Cell selection and re-selection info | |
| - Mapping info | Not Present |
| - Cell selection and reselection quality measure | CPICH RSCP |
| - CHOICE mode | TDD |
| - Sintrasearch | 11 dB |
| - Sintersearch | 11dB |
| - SsearchHCS | Not present |
| - RAT List | This parameter is configurable |
| - RAT identifier | GSM |
| - Ssearch,RAT | -32 dB |
| - SHCS,RAT | Not present |
| - $S_{limit,ShearchRAT}$ | Not Present |
| - Qrxlevmin | -103 dBm |
| - Qhyst1s | 0 dB |

| | |
|---|---|
| - Treselections | 0 seconds |
| - HCS Serving cell information | Not present |
| - Maximum allowed UL TX power | 30dBm |
| - Cell Access Restriction | |
| - Cell barred | Not barred |
| - Intra-frequency cell re-selection indicator | Not present |
| - $T_{barred}$ | Not present |
| - Cell Reserved for operator use | Not reserved |
| - Cell Reservation Extension | Not reserved |
| - Access Class Barred List | Not present |

Contents of System Information Block type 5 (FDD)

| | |
|---|---|
| - SIB6 indicator | TRUE |
| - PICH Power offset | -5 dB |
| - CHOICE Mode | FDD |
| - AICH Power offset | -5 dB |
| - Primary CCPCH info | Not present |
| - PRACH system information list | |
| - PRACH system information | |
| - PRACH info | |
| - CHOICE mode | FDD |
| - Available Signature | '0000 0000 1111 1111'B |
| - Available SF | 64 |
| - Preamble scrambling code number | 0 |
| - Puncturing Limit | 1.00 |
| - Available Sub Channel number | '1111 1111 1111'B |
| - Transport channel Identity | 15 |
| - RACH TFS | |
| - CHOICE Transport channel type | Common transport channels |
| - Dynamic Transport format information | |
| - RLC size | 168 |
| - Number of TB and TTI List | |
| - Number of Transport blocks | 1 |
| - CHOICE Mode | FDD |
| - CHOICE Logical channel List | Configured |
| - RLC size | 360 |
| - Number of TB and TTI List | |
| - Number of Transport blocks | 1 |
| - CHOICE Mode | FDD |
| - CHOICE Logical channel List | Configured |
| - Semi-static Transport Format information | |
| - Transmission time interval | 20 ms |
| - Type of channel coding | Convolutional |
| - Coding Rate | 1/2 |
| - Rate matching attribute | 150 |
| - CRC size | 16 |
| - RACH TFCS | |
| - CHOICE TFCI signalling | Normal |
| - TFCI Field 1 information | |
| - CHOICE TFCS representation | Complete reconfiguration |
| - TFCS complete reconfiguration information | |
| - CHOICE CTFC Size | 2 bit |
| - CTFC information | 0 |
| - Power offset information | |
| - CHOICE Gain Factors | Computed Gain Factor |
| - Reference TFC ID | 0 |
| - CHOICE Mode | FDD |
| - Power offset Pp-m | 0 dB |
| - CTFC information | 1 |
| - Power offset information | |
| - CHOICE Gain Factors | Signalled Gain Factor |
| - CHOICE mode | FDD |
| - Gain factor ßc | 11 |
| - Gain factor ßd | 15 |
| - Reference TFC ID | 0 |
| - CHOICE Mode | FDD |

| | |
|---|---|
| - Power offset Pp-m | 0 dB |
| - PRACH partitioning | |
| - Access Service Class | |
| - ASC Setting | Not Present |
| - ASC Setting | |
| - CHOICE mode | FDD |
| - Available signature Start Index | 0 (ASC#1) |
| - Available signature End Index | 7 (ASC#1) |
| - Assigned Sub-Channel Number | '1111'B<br>The first/ leftmost bit of the bit string contains the most significant bit of the Assigned Sub-Channel Number. |
| - ASC Setting | Not Present |
| - ASC Setting | |
| - CHOICE mode | FDD |
| - Available signature Start Index | 0 (ASC#3) |
| - Available signature End Index | 7 (ASC#3) |
| - Assigned Sub-Channel Number | '1111'B<br>The first/ leftmost bit of the bit string contains the most significant bit of the Assigned Sub-Channel Number. |
| - ASC Setting | Not Present |
| - ASC Setting | |
| - CHOICE mode | FDD |
| - Available signature Start Index | 0 (ASC#5) |
| - Available signature End Index | 7 (ASC#5) |
| - Assigned Sub-Channel Number | '1111'B<br>The first/ leftmost bit of the bit string contains the most significant bit of the Assigned Sub-Channel Number. |
| - ASC Setting | Not Present |
| - ASC Setting | |
| - CHOICE mode | FDD |
| - Available signature Start Index | 0 (ASC#7) |
| - Available signature End Index | 7 (ASC#7) |
| - Assigned Sub-Channel Number | '1111'B<br>The first/ leftmost bit of the bit string contains the most significant bit of the Assigned Sub-Channel Number. |
| - Persistence scaling factor | |
| - Persistence scaling factor | 0.9 (for ASC#2) |
| - Persistence scaling factor | 0.9 (for ASC#3) |
| - Persistence scaling factor | 0.9 (for ASC#4) |
| - Persistence scaling factor | 0.9 (for ASC#5) |
| - Persistence scaling factor | 0.9 (for ASC#6) |
| - Persistence scaling factor | 0.9 (for ASC#7) |
| - AC-to-ASC mapping table | |
| - AC-to-ASC mapping | 6 (AC0-9) |
| - AC-to-ASC mapping | 5 (AC10) |
| - AC-to-ASC mapping | 4 (AC11) |
| - AC-to-ASC mapping | 3 (AC12) |
| - AC-to-ASC mapping | 2 (AC13) |
| - AC-to-ASC mapping | 1 (AC14) |
| - AC-to-ASC mapping | 0 (AC15) |
| - CHOICE mode | FDD |
| - Primary CPICH TX power | 31 |
| - Constant value | -10 |
| - PRACH power offset | |
| - Power Ramp Step | 3dB |
| - Preamble Retrans Max | 4 |
| - RACH transmission parameters | |
| - Mmax | 2 |
| - NB01min | 3 slot |
| - NB01max | 10 slot |
| - AICH info | |
| - Channelisation code | 3 |
| - STTD indicator | FALSE |
| - AICH transmission timing | 0 |
| - Secondary CCPCH system information | |
| - Secondary CCPCH info | |
| - CHOICE mode | FDD |
| - Secondary scrambling code | Not Present |

| | |
|---|---|
| - STTD indicator | FALSE |
| - Spreading factor | 64 |
| - Code number | 1 |
| - Pilot symbol existence | FALSE |
| - TFCI existence | TRUE (default value) |
| - Fixed or Flexible position | Flexible (default value) |
| - Timing offset | Not Present<br>Absence of this IE is equivalent to default value 0 |
| - TFCS | (This IE is repeated for TFC number for PCH and FACH.) |
| - CHOICE TFCI signalling | Normal |
| - TFCI Field 1 information | |
| - CHOICE TFCS representation | Complete reconfiguration |
| - TFCS complete reconfiguration information | |
| - CHOICE CTFC Size | 4 bit |
| - CTFC information | 0 |
| - Power offset information | Not Present |
| - CTFC information | 1 |
| - Power offset information | Not Present |
| - CTFC information | 2 |
| - Power offset information | Not Present |
| - CTFC information | 3 |
| - Power offset information | Not Present |
| - CTFC information | 4 |
| - Power offset information | Not Present |
| - CTFC information | 5 |
| - Power offset information | Not Present |
| - CTFC information | 6 |
| - Power offset information | Not Present |
| - CTFC information | 8 |
| - Power offset information | Not Present |
| - FACH/PCH information | |
| - TFS | (PCH) |
| - CHOICE Transport channel type | Common transport channels |
| - Dynamic Transport format information | |
| - RLC Size | 240 |
| - Number of TB and TTI List | |
| - Number of Transport blocks | 0 |
| - Number of Transport blocks | 1 |
| - CHOICE Logical channel List | ALL |
| - Semi-static Transport Format information | |
| - Transmission time interval | 10 ms |
| - Type of channel coding | Convolutional |
| - Coding Rate | 1/2 |
| - Rate matching attribute | 230 |
| - CRC size | 16 bit |
| - Transport channel Identity | 12 (for PCH) |
| - CTCH indicator | FALSE |
| - TFS | (FACH) |
| - CHOICE Transport channel type | Common transport channels |
| - Dynamic Transport format information | |
| - RLC Size | 168 |
| - Number of TB and TTI List | |
| - Number of Transport blocks | 0 |
| - Number of Transport blocks | 1 |
| - Number of Transport blocks | 2 |
| - CHOICE Logical channel List | ALL |
| - Semi-static Transport Format information | |
| - Transmission time interval | 10 ms |
| - Type of channel coding | Convolutional |
| - Coding Rate | 1/2 |
| - Rate matching attribute | 220 |
| - CRC size | 16 bit |
| - Transport channel Identity | 13 (for FACH) |
| - CTCH indicator | FALSE |
| - TFS | (FACH) |
| - CHOICE Transport channel type | Common transport channels |
| - Dynamic Transport format information | |
| - RLC Size | 360 |

| | |
|---|---|
| - Number of TB and TTI List | |
| - Number of Transport blocks | 0 |
| - Number of Transport blocks | 1 |
| - CHOICE Logical channel List | ALL |
| - Semi-static Transport Format information | |
| - Transmission time interval | 10 ms |
| - Type of channel coding | Turbo |
| - Rate matching attribute | 130 |
| - CRC size | 16bit |
| - Transport channel Identity | 14 (for FACH) |
| - CTCH indicator | FALSE |
| - PICH info | |
| - CHOICE mode | FDD |
| - Channelisation code | 2 |
| - Number of PI per frame | 18 |
| - STTD indicator | FALSE |
| - CBS DRX Level 1 information | Not Present |

Contents of System Information Block type 5 (3.84 Mcps TDD)

| | |
|---|---|
| - SIB6 indicator | TRUE |
| - PICH Power offset | -5 dB |
| - CHOICE Mode | TDD |
| - PUSCH system information | Not Present |
| - PDSCH system information | Not Present |
| - TDD open loop power control | |
| - Primary CCPCH Tx Power | 30 dbm |
| - CHOICE TDD option | 3.84 Mcps TDD /REL-4/ |
| - Alpha | (1/8) |
| - PRACH Constant Value | -10 |
| - DPCH Constant Value | -10 |
| - PUSCH Constant Value | -10 |
| - UE positioning related parameters | Not Present /REL-4/ |
| - Primary CCPCH info | |
| - CHOICE *mode* | TDD |
| - CHOICE TDD option | 3.84 Mcps TDD /REL-4/ |
| - CHOICE SyncCase | Sync Case 2 |
| - Timeslot | 0 |
| - Cell parameters ID | Not Present |
| - SCTD indicator | FALSE |
| - PRACH system information list | |
| - PRACH system information | |
| - PRACH info | |
| - CHOICE mode | TDD |
| - CHOICE TDD option | 3.84 Mcps TDD /REL-4/ |
| - Timeslot number | 14 |
| - PRACH Channelisation Code List | |
| - CHOICE SF | SF8 |
| - Channelisation Code List | |
| - Channelisation Code | 8/1 |
| - Channelisation Code | 8/2 |
| - Channelisation Code | 8/3 |
| - Channelisation Code | 8/4 |
| - PRACH Midamble | Direct |
| - PNBSCH allocation | Not Present /REL-4/ |
| - Transport channel Identity | 15 |
| - RACH TFS | |
| - CHOICE Transport channel type | Common transport channels |
| - Dynamic Transport format information | |
| - RLC size | Reference clause 6.10 "Parameter Set" |
| - Number of TB and TTI List | Reference clause 6.10 "Parameter Set" |
| - Number of Transport blocks | Reference clause 6.10 "Parameter Set" |
| - CHOICE Mode | TDD |
| - Transmission Time Interval | Not Present |
| - CHOICE Logical channel List | Configured |
| - Semi-static Transport Format information | |
| - Transmission time interval | Reference clause 6.10 "Parameter Set" |

| | |
|---|---|
| - Type of channel coding | Reference clause 6.10 "Parameter Set" |
| - Coding Rate | Reference clause 6.10 "Parameter Set" |
| - Rate matching attribute | Reference clause 6.10 "Parameter Set" |
| - CRC size | Reference clause 6.10 "Parameter Set" |
| - RACH TFCS | Not present |
| - PRACH partitioning | |
| - Access Service Class | |
| - ASC Settings | (ASC#0) |
| - CHOICE mode | TDD |
| - CHOICE TDD option | 3.84 Mcps TDD |
| - Available Channelisation codes indices | Not Present (Default all) |
| - CHOICE subchannel size | Size1 |
| - Available Subchannels | null |
| - ASC Settings | (ASC#1) |
| - CHOICE mode | TDD |
| - CHOICE TDD option | 3.84 Mcps TDD |
| - Available Channelisation codes indices | Not Present (Default all) |
| - CHOICE subchannel size | Size1 |
| - Available Subchannels | null |
| - ASC Settings | (ASC#2) |
| - CHOICE mode | TDD |
| - CHOICE TDD option | 3.84 Mcps TDD |
| - Available Channelisation codes indices | Not Present (Default all) |
| - CHOICE subchannel size | Size1 |
| - Available Subchannels | null |
| - ASC Settings | (ASC#3) |
| - CHOICE mode | TDD |
| - CHOICE TDD option | 3.84 Mcps TDD |
| - Available Channelisation codes indices | Not Present (Default all) |
| - CHOICE subchannel size | Size1 |
| - Available Subchannels | null |
| - ASC Settings | (ASC#4) |
| - CHOICE mode | TDD |
| - CHOICE TDD option | 3.84 Mcps TDD |
| - Available Channelisation codes indices | Not Present (Default all) |
| - CHOICE subchannel size | Size1 |
| - Available Subchannels | null |
| - ASC Settings | (ASC#5) |
| - CHOICE mode | TDD |
| - CHOICE TDD option | 3.84 Mcps TDD |
| - Available Channelisation codes indices | Not Present (Default all) |
| - CHOICE subchannel size | Size1 |
| - Available Subchannels | null |
| - ASC Settings | (ASC#6) |
| - CHOICE mode | TDD |
| - CHOICE TDD option | 3.84 Mcps TDD |
| - Available Channelisation codes indices | Not Present (Default all) |
| - CHOICE subchannel size | Size1 |
| - Available Subchannels | null |
| - Persistence scaling factors | |
| - Access Service Class | |
| - Persistence scaling factor | 0.9 (for ASC#2) |
| - Persistence scaling factor | 0.9 (for ASC#3) |
| - Persistence scaling factor | 0.9 (for ASC#4) |
| - Persistence scaling factor | 0.9 (for ASC#5) |
| - Persistence scaling factor | 0.9 (for ASC#6) |
| - AC-to-ASC mapping | |
| - AC-to-ASC mapping table | |
| - AC-to-ASC mapping | 6 (AC0-9) |
| - AC-to-ASC mapping | 5 (AC10) |
| - AC-to-ASC mapping | 4 (AC11) |
| - AC-to-ASC mapping | 3 (AC12) |
| - AC-to-ASC mapping | 2 (AC13) |
| - AC-to-ASC mapping | 1 (AC14) |
| - AC-to-ASC mapping | 0 (AC15) |
| - CHOICE *mode* | TDD (no data) |
| - Secondary CCPCH system information | |
| - Secondary CCPCH system information | |

| | |
|---|---|
| - Secondary CCPCH info | |
| - CHOICE *mode* | TDD |
| - Offset | 0 |
| - Common timeslot info | |
| - 2nd interleaving mode | Frame |
| - TFCI coding | Reference clause 6.10 "Parameter Set" |
| - Puncturing limit | Reference clause 6.10 "Parameter Set" |
| - Repetition period | Not Present (MD "1") |
| - Repetition length | Not present (empty) |
| - Individual timeslot info | |
| - CHOICE TDD option | 3.84 Mcps TDD |
| - Timeslot number | 1 |
| - TFCI existence | Reference clause 6.10 "Parameter Set" |
| - Midamble Shift and burst type | |
| - CHOICE *TDD option* | 3.84 Mcps TDD |
| - CHOICE Burst Type | Type 1 |
| - Midamble Allocation Mode | Default midamble |
| - Midamble configuration burst type 1 and 3 | 4 |
| - Midamble Shift | Not Present |
| - CHOICE *TDD option* | 3.84 Mcps TDD |
| - no data | |
| - Code List | |
| - Channelisation Code | (This IE is repeated for Code number for PCH and FACH) |
| - TFCS | (This IE is repeated for TFC number for PCH and FACH) |
| -CHOICE *TFCI signalling* | |
| - Normal | |
| - TFCI Field 1 information | |
| - CHOICE TFCS representation | Complete reconfiguration |
| - TFCS complete information | |
| - CHOICE CTFC Size | Number of bits used must be enough to cover all combinations of CTFC from clause 6.10. |
| - CTFC information | Reference clause 6.10 "Parameter Set" |
| - Power offset information | Not Present |
| - FACH/PCH information | |
| - TFS | (PCH) |
| - CHOICE Transport channel type | Common transport channels |
| - Dynamic Transport format information | |
| - RLC Size | Reference clause 6.10 "Parameter Set" |
| - Number of TB and TTI List | Reference clause 6.10 "Parameter Set" |
| - Number of Transport blocks | Reference clause 6.10 "Parameter Set" |
| - CHOICE Mode | TDD |
| - Transmission Time Interval | Reference clause 6.10 "Parameter Set" |
| - CHOICE Logical channel List | ALL |
| - Semi-static Transport Format information | |
| - Transmission time interval | Reference clause 6.10 "Parameter Set" |
| - Type of channel coding | Reference clause 6.10 "Parameter Set" |
| - Coding Rate | Reference clause 6.10 "Parameter Set" |
| - Rate matching attribute | Reference clause 6.10 "Parameter Set" |
| - CRC size | Reference clause 6.10 "Parameter Set" |
| - Transport channel Identity | 12 (for PCH) |
| - CTCH indicator | FALSE |
| - TFS | (FACH) |
| - CHOICE Transport channel type | Common transport channels |
| - Dynamic Transport format information | |
| - RLC Size | Reference clause 6.10 "Parameter Set" |
| - Number of TB and TTI List | Reference clause 6.10 "Parameter Set" |
| - Number of Transport blocks | Reference clause 6.10 "Parameter Set" |
| - CHOICE Mode | TDD |
| - Transmission Time Interval | Reference clause 6.10 "Parameter Set" |
| - CHOICE Logical channel List | ALL |
| - Semi-static Transport Format information | |
| - Transmission time interval | Reference clause 6.10 "Parameter Set" |
| - Type of channel coding | Reference clause 6.10 "Parameter Set" |
| - Coding Rate | Reference clause 6.10 "Parameter Set" |
| - Rate matching attribute | Reference clause 6.10 "Parameter Set" |
| - CRC size | Reference clause 6.10 "Parameter Set" |
| - Transport channel Identity | 13 (for FACH) |
| - CTCH indicator | FALSE |

| | |
|---|---|
| - TFS | (FACH) |
| - CHOICE Transport channel type | Common transport channels |
| - Dynamic Transport format information | |
| - RLC Size | Reference clause 6.10 "Parameter Set" |
| - Number of TB and TTI List | Reference clause 6.10 "Parameter Set" |
| - Number of Transport blocks | Reference clause 6.10 "Parameter Set" |
| - CHOICE Mode | TDD |
| - CHOICE Logical channel List | ALL |
| - Semi-static Transport Format information | |
| - Transmission time interval | Reference clause 6.10 "Parameter Set" |
| - Type of channel coding | Reference clause 6.10 "Parameter Set" |
| - Coding Rate | Reference clause 6.10 "Parameter Set" |
| - Rate matching attribute | Reference clause 6.10 "Parameter Set" |
| - CRC size | Reference clause 6.10 "Parameter Set" |
| - Transport channel Identity | 14 (for FACH) |
| - CTCH indicator | FALSE |
| - PICH info | |
| - CHOICE *mode* | TDD |
| - CHOICE TDD option | 3.84 Mcps TDD |
| - Timeslot number | 0 |
| - Midamble shift and burst type | |
| - CHOICE Burst Type | Type 1 |
| - Midamble Allocation Mode | Default midamble |
| - Midamble configuration burst type 1 and 3 | 8 |
| - Midamble Shift | Not Present |
| - Channelisation code | 16/16 |
| - Repetition period/length | 64/2 |
| - Offset | 0 |
| - Paging indicator length | 4 |
| - $N_{GAP}$ | 4 |
| - $N_{PCH}$ | 2 |
| - CBS DRX Level 1 information | Not Present |

Contents of System Information Block type 5 (1.28 Mcps TDD)

| | |
|---|---|
| - SIB6 indicator | TRUE |
| - PICH Power offset | -5 dB |
| - CHOICE Mode | TDD |
| - PUSCH system information | Not Present |
| - PDSCH system information | Not Present |
| - TDD open loop power control | |
| - Primary CCPCH Tx Power | 30 dbm |
| - CHOICE TDD option | 1.28 Mcps TDD /REL-4/ |
| - no data | |
| - Primary CCPCH info | |
| - CHOICE *mode* | TDD |
| - CHOICE TDD option | 1.28 Mcps TDD /REL-4/ |
| - TSTD indicator | FALSE |
| - Cell parameters ID | Not Present |
| - Block SCTD indicator | FALSE |
| - PRACH system information list | |
| - PRACH system information | |
| - PRACH info | |
| - CHOICE mode | TDD |
| - CHOICE TDD option | 1.28 Mcps TDD /REL-4/ |
| - SYNC_UL info | |
| - SYNC_UL codes bitmap | "11111111" |
| - UL Target SIR | 10 dB |
| - Power Ramping Step | 3 dB |
| - Max SYNC_UL Transmissions | 8 |
| - Mmax | 32 |
| - PRACH definition | |
| - Timeslot number | |
| - CHOICE TDD option | 1.28 Mcps TDD /REL-4/ |
| - Timeslot number | 1 |
| - PRACH Channelisation Code | |
| - Channelisation Code List | |

| | |
|---|---|
| - Channelisation Code | (8/1) |
| - Midamble Shift and burst type | |
| - CHOICE TDD option | 1.28 Mcps TDD /REL-4/ |
| - Midamble Allocation Mode | Default midamble |
| - Midamble configuration | 8 |
| - Midamble Shift | Not present |
| - FPACH info | |
| - Timeslot number | 6 |
| - Channelisation code | (16/16) |
| - Midamble Shift and burst type | |
| - CHOICE TDD option | 1.28 Mcps TDD /REL-4/ |
| - Midamble Allocation Mode | Common Midamble |
| - Midamble configuration | 8 |
| - Midamble Shift | Not present |
| - WT | 4 |
| - Transport channel Identity | 15 |
| - RACH TFS | |
| - CHOICE Transport channel type | Common transport channels |
| - Dynamic Transport format information | |
| - RLC size | Reference clause 6.10 "Parameter Set" |
| - Number of TB and TTI List | Reference clause 6.10 "Parameter Set" |
| - Number of Transport blocks | Reference clause 6.10 "Parameter Set" |
| - CHOICE Mode | TDD |
| - Transmission Time Interval | Not Present |
| - CHOICE Logical channel List | Configured |
| - Semi-static Transport Format information | |
| - Transmission time interval | Reference clause 6.10 "Parameter Set" |
| - Type of channel coding | Reference clause 6.10 "Parameter Set" |
| - Coding Rate | Reference clause 6.10 "Parameter Set" |
| - Rate matching attribute | Reference clause 6.10 "Parameter Set" |
| - CRC size | Reference clause 6.10"Parameter Set" |
| - RACH TFCS | Not present |
| - PRACH partitioning | |
| - Access Service Class | |
| - ASC Settings | (ASC#0) |
| - CHOICE mode | TDD |
| - CHOICE TDD option | 1.28 Mcps TDD |
| - Available SYNC_UL codes indices | "11111111" |
| - CHOICE subchannel size | Size1 |
| - Available Subchannels | Null |
| - ASC Settings | (ASC#1) |
| - CHOICE mode | TDD |
| - CHOICE TDD option | 1.28 Mcps TDD |
| - Available SYNC_UL codes indices | "11111111" |
| - CHOICE subchannel size | Size1 |
| - Available Subchannels | Null |
| - ASC Settings | (ASC#2) |
| - CHOICE mode | TDD |
| - CHOICE TDD option | 1.28 Mcps TDD |
| - Available SYNC_UL codes indices | "11111111" |
| - CHOICE subchannel size | Size1 |
| - Available Subchannels | Null |
| - ASC Settings | (ASC#3) |
| - CHOICE mode | TDD |
| - CHOICE TDD option | 1.28 Mcps TDD |
| - Available SYNC_UL codes indices | "11111111" |
| - CHOICE subchannel size | Size1 |
| - Available Subchannels | Null |
| - ASC Settings | (ASC#4) |
| - CHOICE mode | TDD |
| - CHOICE TDD option | 1.28 Mcps TDD |
| - Available SYNC_UL codes indices | "11111111" |
| - CHOICE subchannel size | Size1 |
| - Available Subchannels | Null |
| - ASC Settings | (ASC#5) |
| - CHOICE mode | TDD |
| - CHOICE TDD option | 1.28 Mcps TDD |
| - Available SYNC_UL codes indices | "11111111" |

| | |
|---|---|
| - CHOICE subchannel size | Size1 |
| - Available Subchannels | Null |
| - ASC Settings | (ASC#6) |
| - CHOICE mode | TDD |
| - CHOICE TDD option | 1.28 Mcps TDD |
| - Available SYNC_UL codes indices | "11111111" |
| - CHOICE subchannel size | Size1 |
| - Available Subchannels | Null |
| - Access Service Class | |
| - Persistence scaling factor | 0.9 (for ASC#2) |
| - Persistence scaling factor | 0.9 (for ASC#3) |
| - Persistence scaling factor | 0.9 (for ASC#4) |
| - Persistence scaling factor | 0.9 (for ASC#5) |
| - Persistence scaling factor | 0.9 (for ASC#6) |
| - AC-to-ASC mapping | |
| - AC-to-ASC mapping table | |
| - AC-to-ASC mapping | 6 (AC0-9) |
| - AC-to-ASC mapping | 5 (AC10) |
| - AC-to-ASC mapping | 4 (AC11) |
| - AC-to-ASC mapping | 3 (AC12) |
| - AC-to-ASC mapping | 2 (AC13) |
| - AC-to-ASC mapping | 1 (AC14) |
| - AC-to-ASC mapping | 0 (AC15) |
| - CHOICE *mode* | TDD (no data) |
| - Secondary CCPCH system information | |
| - Secondary CCPCH info | |
| - CHOICE *mode* | TDD |
| - Offset | 0 |
| - Common timeslot info | |
| - 2nd interleaving mode | Frame |
| - TFCI coding | Reference clause 6.10 "Parameter Set" |
| - Puncturing limit | Reference clause 6.10 "Parameter Set" |
| - Repetition period | 1 |
| - Repetition length | 0 |
| - Individual timeslot info | |
| - CHOICE *TDD option* | 1.28 Mcps TDD |
| - Timeslot number | 0 |
| - TFCI existence | Reference clause 6.10 "Parameter Set" |
| - Midamble Shift and burst type | |
| - CHOICE *TDD option* | 1.28 Mcps TDD |
| - Midamble Allocation Mode | Default midamble |
| - Midamble configuration | 4 |
| - Midamble Shift | Not Present |
| - CHOICE *TDD option* | 1.28 Mcps TDD |
| - Modulation | Reference clause 6.10 "Parameter Set" |
| - SS-TPC Symbols | Reference clause 6.10 "Parameter Set" |
| - Code List | |
| - Channelisation Code | Reference clause 6.10 "Parameter Set" |
| - TFCS | Reference clause 6.10 "Parameter Set" |
| - CHOICE TFCI *signalling* | Normal |
| - TFCI Field 1 information | |
| - CHOICE TFCS representation | Addition |
| - TFCS addition information | |
| - CHOICE CTFC Size | Number of bits used must be enough to cover all combinations of CTFC from clause 6.10. |
| - CTFC information | Reference clause 6.10 "Parameter Set" |
| - Power offset information | Not Present |
| - FACH/PCH information | |
| - Transport channel Identity | 12 (for PCH) |
| - TFS | (PCH) |
| - CHOICE Transport channel type | Common transport channels |
| - Dynamic Transport format information | (This IE is repeated for TFI number.) |
| - RLC Size | Reference clause 6.10 "Parameter Set" |
| - Number of TB and TTI List | Reference clause 6.10 "Parameter Set" |
| - Number of Transport blocks | Reference clause 6.10 "Parameter Set" |
| - CHOICE Mode | TDD |
| - Transmission Time Interval | Not Present |
| - CHOICE Logical channel List | ALL |

| | |
|---|---|
| - Semi-static Transport Format information | |
| - Transmission time interval | Reference clause 6.10 "Parameter Set" |
| - Type of channel coding | Reference clause 6.10 "Parameter Set" |
| - Coding Rate | Reference clause 6.10 "Parameter Set" |
| - Rate matching attribute | Reference clause 6.10 "Parameter Set" |
| - CRC size | Reference clause 6.10 "Parameter Set" |
| - Transport channel Identity | 13 (for FACH) |
| - TFS | (FACH) |
| - CHOICE Transport channel type | Common transport channels |
| - Dynamic Transport format information | (This IE is repeated for TFI number.) |
| - RLC Size | Reference clause 6.10 "Parameter Set" |
| - Number of TB and TTI List | Reference clause 6.10 "Parameter Set" |
| - Number of Transport blocks | Reference clause 6.10 "Parameter Set" |
| - CHOICE Mode | TDD |
| - Transmission Time Interval | Not Present |
| - CHOICE Logical channel List | ALL |
| - Semi-static Transport Format information | |
| - Transmission time interval | Reference clause 6.10 "Parameter Set" |
| - Type of channel coding | Reference clause 6.10 "Parameter Set" |
| - Coding Rate | Reference clause 6.10 "Parameter Set" |
| - Rate matching attribute | Reference clause 6.10 "Parameter Set" |
| - CRC size | Reference clause 6.10 "Parameter Set" |
| - CTCH indicator | FALSE |
| - PICH info | |
| - CHOICE *mode* | TDD |
| - CHOICE *TDD option* | 1.28 Mcps TDD |
| - Timeslot number | 0 |
| - Midamble shift and burst type | |
| - Midamble Allocation Mode | Default midamble |
| - Midamble configuration | 8 |
| - Midamble Shift | Not Present |
| - Channelisation code list | |
| - Channelisation code | (16/1) |
| - Channelisation code | (16/2) |
| - Repetition period/length | 64/2 |
| - Offset | 0 |
| - Paging indicator length | 4 |
| - $N_{GAP}$ | 4 |
| - $N_{PCH}$ | 2 |
| - CBS DRX Level 1 information | Not Present |

Contents of System Information Block type 6 in connected mode (FDD)

| | |
|---|---|
| - PICH power offset | -5 dB |
| - CHOICE Mode | FDD |
| - AICH power offset | -5 dB |
| - Primary CCPCH info | Not Present |
| - PRACH system information list | Not present |
| - Secondary CCPCH system info | Not Present |
| - CBS DRX Level 1 information | Not Present |

Contents of System Information Block type 6 in connected mode (similar to SIB type 5) (3.84 Mcps TDD)

| | |
|---|---|
| - PICH Power offset | -5 dB |
| - CHOICE Mode | TDD |
| - PUSCH system information | Not Present |
| - PDSCH system information | Not Present |
| - TDD open loop power control | |
| - Primary CCPCH Tx Power | 30 dbm |
| - CHOICE TDD option | 3.84 Mcps TDD /REL-4/ |
| - Alpha | (1/8) |
| - PRACH Constant Value | -10 |
| - DPCH Constant Value | -10 |
| - PUSCH Constant Value | -10 |
| - Primary CCPCH info | |
| - CHOICE *mode* | TDD |

| | |
|---|---|
| - CHOICE TDD option | 3.84 Mcps TDD /REL-4/ |
| - CHOICE SyncCase | Sync Case 2 |
| - Timeslot | 0 |
| - Cell parameters ID | Not Present |
| - SCTD indicator | FALSE |
| - PRACH system information list | |
| - PRACH system information | |
| - PRACH info | |
| - CHOICE mode | TDD |
| - CHOICE TDD option | 3.84 Mcps TDD /REL-4/ |
| - Timeslot number | 14 |
| - PRACH Channelisation Code List | |
| - CHOICE SF | SF8 |
| - Channelisation Code List | |
| - Channelisation Code | 8/1 |
| - Channelisation Code | 8/2 |
| - Channelisation Code | 8/3 |
| - Channelisation Code | 8/4 |
| - PRACH Midamble | Direct |
| - Transport channel Identity | 15 |
| - RACH TFS | |
| - CHOICE Transport channel type | Common transport channels |
| - Dynamic Transport format information | |
| - RLC size | Reference clause 6.10 "Parameter Set" |
| - Number of TB and TTI List | Reference clause 6.10 "Parameter Set" |
| - Number of Transport blocks | Reference clause 6.10 "Parameter Set" |
| - CHOICE Mode | TDD |
| - Transmission Time Interval | Not Present |
| - CHOICE Logical channel List | Configured |
| - Semi-static Transport Format information | |
| - Transmission time interval | Reference clause 6.10 "Parameter Set" |
| - Type of channel coding | Reference clause 6.10 "Parameter Set" |
| - Coding Rate | Reference clause 6.10 "Parameter Set" |
| - Rate matching attribute | Reference clause 6.10 "Parameter Set" |
| - CRC size | Reference clause 6.10 "Parameter Set" |
| - RACH TFCS | Not present |
| - PRACH partitioning | |
| - Access Service Class | |
| - ASC Settings | (ASC#0) |
| - CHOICE mode | TDD |
| - CHOICE TDD option | 3.84 Mcps TDD /REL-4/ |
| - Available Channelisation codes indices | Not Present (Default all) |
| - CHOICE subchannel size | Size1 |
| - Available Subchannels | null |
| - ASC Settings | (ASC#1) |
| - CHOICE mode | TDD |
| - CHOICE TDD option | 3.84 Mcps TDD /REL-4/ |
| - Available Channelisation codes indices | Not Present (Default all) |
| - CHOICE subchannel size | Size1 |
| - Available Subchannels | null |
| - ASC Settings | (ASC#2) |
| - CHOICE mode | TDD |
| - CHOICE TDD option | 3.84 Mcps TDD /REL-4/ |
| - Available Channelisation codes indices | Not Present (Default all) |
| - CHOICE subchannel size | Size1 |
| - Available Subchannels | null |
| - ASC Settings | (ASC#3) |
| - CHOICE mode | TDD |
| - CHOICE TDD option | 3.84 Mcps TDD /REL-4/ |
| - Available Channelisation codes indices | Not Present (Default all) |
| - CHOICE subchannel size | Size1 |
| - Available Subchannels | null |
| - ASC Settings | (ASC#4) |
| - CHOICE mode | TDD |
| - CHOICE TDD option | 3.84 Mcps TDD /REL-4/ |
| - Available Channelisation codes indices | Not Present (Default all) |
| - CHOICE subchannel size | Size1 |
| - Available Subchannels | null |

| | |
|---|---|
| - ASC Settings | (ASC#5) |
| - CHOICE mode | TDD |
| - Available Channelisation codes indices | Not Present (Default all) |
| - CHOICE subchannel size | Size1 |
| - Available Subchannels | null |
| - ASC Settings | (ASC#6) |
| - CHOICE mode | TDD |
| - CHOICE TDD option | 3.84 Mcps TDD /REL-4/ |
| - Available Channelisation codes indices | Not Present (Default all) |
| - CHOICE subchannel size | Size1 |
| - Available Subchannels | null |
| - Persistence scaling factors | |
| - Access Service Class | |
| - Persistence scaling factor | 0.9 (for ASC#2) |
| - Persistence scaling factor | 0.9 (for ASC#3) |
| - Persistence scaling factor | 0.9 (for ASC#4) |
| - Persistence scaling factor | 0.9 (for ASC#5) |
| - Persistence scaling factor | 0.9 (for ASC#6) |
| - AC-to-ASC mapping | Not Present |
| - CHOICE *mode* | TDD (no data) |
| - Secondary CCPCH system information | |
| - Secondary CCPCH system information | |
| - Secondary CCPCH info | |
| - CHOICE *mode* | TDD |
| - Offset | 0 |
| - Common timeslot info | |
| - 2nd interleaving mode | Not Present (MD "Frame") |
| - TFCI coding | Reference clause 6.10 "Parameter Set" |
| - Puncturing limit | Reference clause 6.10 "Parameter Set" |
| - Repetition period | Not Present (MD "1") |
| - Repetition length | Not present |
| - Individual timeslot info | |
| - CHOICE TDD option | 3.84 Mcps TDD /REL-4/ |
| - Timeslot number | 1 |
| - TFCI existence | Reference clause 6.10 "Parameter Set" |
| - Midamble Shift and burst type | |
| - CHOICE Burst Type | Type 1 |
| - Midamble Allocation Mode | Default midamble |
| - Midamble configuration burst type 1 and 3 | 4 |
| - Midamble Shift | Not Present |
| - Code List | |
| - Channelisation Code | Reference clause 6.10 "Parameter Set" |
| - TFCS | (This IE is repeated for TFC number for PCH and FACH.) |
| - Normal | |
| - TFCI Field 1 information | |
| - CHOICE TFCS representation | Complete reconfiguration |
| - TFCS complete reconfiguration information | |
| - CHOICE CTFC Size | Number of bits used must be enough to cover all combinations of CTFC from clause 6.10. |
| - CTFC information | Reference clause 6.10 "Parameter Set" |
| - Power offset information | Not Present |
| - FACH/PCH information | |
| - TFS | (PCH) |
| - CHOICE Transport channel type | Common transport channels |
| - Dynamic Transport format information | |
| - RLC Size | Reference clause 6.10 "Parameter Set" |
| - Number of TB and TTI List | Reference clause 6.10 "Parameter Set" |
| - Number of Transport blocks | Reference clause 6.10 "Parameter Set" |
| - CHOICE Mode | TDD |
| - Transmission Time Interval | Reference clause 6.10 "Parameter Set" |
| - CHOICE Logical channel List | ALL |
| - Semi-static Transport Format information | |
| - Transmission time interval | Reference clause 6.10 "Parameter Set" |
| - Type of channel coding | Reference clause 6.10 "Parameter Set" |
| - Coding Rate | Reference clause 6.10 "Parameter Set" |
| - Rate matching attribute | Reference clause 6.10 "Parameter Set" |
| - CRC size | Reference clause 6.10 "Parameter Set" |
| - Transport channel Identity | 12 (for PCH) |

| | |
|---|---|
| - CTCH indicator | FALSE |
| - TFS | (FACH) |
| - CHOICE Transport channel type | Common transport channels |
| - Dynamic Transport format information | |
| - RLC Size | Reference clause 6.10 "Parameter Set" |
| - Number of TB and TTI List | Reference clause 6.10 "Parameter Set" |
| - Number of Transport blocks | Reference clause 6.10 "Parameter Set" |
| - CHOICE Mode | TDD |
| - Transmission Time Interval | Reference clause 6.10 "Parameter Set" |
| - CHOICE Logical channel List | ALL |
| - Semi-static Transport Format information | |
| - Transmission time interval | Reference clause 6.10 "Parameter Set" |
| - Type of channel coding | Reference clause 6.10 "Parameter Set" |
| - Coding Rate | Reference clause 6.10 "Parameter Set" |
| - Rate matching attribute | Reference clause 6.10 "Parameter Set" |
| - CRC size | Reference clause 6.10 "Parameter Set" |
| - Transport channel Identity | 13 (for FACH) |
| - TFS | (FACH) |
| - CHOICE Transport channel type | Common transport channels |
| - Dynamic Transport format information | (This IE is repeated for TFI number.) |
| - RLC Size | Reference clause 6.10 "Parameter Set" |
| - Number of TB and TTI List | Reference clause 6.10 "Parameter Set" |
| - Number of Transport blocks | Reference clause 6.10 "Parameter Set" |
| - CHOICE Mode | TDD |
| - CHOICE Logical channel List | ALL |
| - Semi-static Transport Format information | |
| - Transmission time interval | Reference clause 6.10 "Parameter Set" |
| - Type of channel coding | Reference clause 6.10 "Parameter Set" |
| - Coding Rate | Reference clause 6.10 "Parameter Set" |
| - Rate matching attribute | Reference clause 6.10 "Parameter Set" |
| - CRC size | Reference clause 6.10 "Parameter Set" |
| - Transport channel Identity | 14 (for FACH) |
| - CTCH indicator | FALSE |
| - CTCH indicator | FALSE |
| - PICH info | |
| - CHOICE *mode* | TDD |
| - CHOICE TDD option | 3.84 Mcps TDD |
| - Timeslot number | 0 |
| - Midamble shift and burst type | |
| - CHOICE Burst Type | Type 1 |
| - Midamble Allocation Mode | Default midamble |
| - Midamble configuration burst type 1 and 3 | 8 |
| - Midamble Shift | Not Present |
| - Channelisation code | 16/16 |
| - Repetition period/length | 64/2 |
| - Offset | 0 |
| - Paging indicator length | 4 |
| - $N_{GAP}$ | 4 |
| - $N_{PCH}$ | 2 |
| - CBS DRX Level 1 information | Not Present |

Contents of System Information Block type6 In connected mode (similar to SIB type5) (1.28 Mcps TDD)

| | |
|---|---|
| - SIB6 indicator | TRUE |
| - PICH Power offset | -5 dB |
| - CHOICE Mode | TDD |
| - PUSCH system information | Not Present |
| - PDSCH system information | Not Present |
| - TDD open loop power control | |
| - Primary CCPCH Tx Power | 30 dbm |
| - CHOICE TDD option | 1.28 Mcps TDD /REL-4/ |
| - no data | |
| - Primary CCPCH info | |
| - CHOICE *mode* | TDD |
| - CHOICE TDD option | 1.28 Mcps TDD /REL-4/ |
| - TSTD indicator | FALSE |
| - Cell parameters ID | Not Present |

| | |
|---|---|
| - Block SCTD indicator | FALSE |
| - PRACH system information list | |
| - PRACH system information | |
| - PRACH info | |
| - CHOICE mode | TDD |
| - CHOICE TDD option | 1.28 Mcps TDD /REL-4/ |
| - SYNC_UL info | |
| - SYNC_UL codes bitmap | "11111111" |
| - UL Target SIR | 10 dB |
| - Power Ramping Step | 3 dB |
| - Max SYNC_UL Transmissions | 8 |
| - Mmax | 32 |
| - PRACH definition | |
| - Timeslot number | |
| - CHOICE TDD option | 1.28 Mcps TDD /REL-4/ |
| - Timeslot number | 1 |
| - PRACH Channelisation Code List | |
| - Channelisation Code List | |
| - Channelisation Code | (8/1) |
| - Midamble Shift and burst type | |
| - CHOICE TDD option | 1.28 Mcps TDD /REL-4/ |
| - Midamble Allocation Mode | Default midamble |
| - Midamble configuration | 8 |
| - Midamble Shift | Not present |
| - FPACH info | |
| - Timeslot number | 6 |
| - Channelisation code | (16/16) |
| - Midamble Shift and burst type | |
| - CHOICE TDD option | 1.28 Mcps TDD /REL-4/ |
| - Midamble Allocation Mode | Common Midamble |
| - Midamble configuration | 8 |
| - Midamble Shift | Not present |
| - WT | 4 |
| - PNBSCH allocation | Not Present /REL-4/ |
| - Transport channel Identity | 15 |
| - RACH TFS | |
| - CHOICE Transport channel type | Common transport channels |
| - Dynamic Transport format information | |
| - RLC size | Reference clause 6.10 "Parameter Set" |
| - Number of TB and TTI List | Reference clause 6.10 "Parameter Set" |
| - Number of Transport blocks | Reference clause 6.10 "Parameter Set" |
| - CHOICE Mode | TDD |
| - Transmission Time Interval | Not Present |
| - CHOICE Logical channel List | Configured |
| - Semi-static Transport Format information | |
| - Transmission time interval | Reference clause 6.10 "Parameter Set" |
| - Type of channel coding | Reference clause 6.10 "Parameter Set" |
| - Coding Rate | Reference clause 6.10 "Parameter Set" |
| - Rate matching attribute | Reference clause 6.10 "Parameter Set" |
| - CRC size | Reference clause 6.10 "Parameter Set" |
| - RACH TFCS | Not present |
| - PRACH partitioning | |
| - Access Service Class | |
| - ASC Settings | (ASC#0) |
| - CHOICE mode | TDD |
| - CHOICE TDD option | 1.28 Mcps TDD |
| - Available SYNC_UL codes indices | "111111111" |
| - CHOICE subchannel size | Size1 |
| - Available Subchannels | Null |
| - ASC Settings | (ASC#1) |
| - CHOICE mode | TDD |
| - CHOICE TDD option | 1.28 Mcps TDD |
| - Available SYNC_UL codes indices | "111111111" |
| - CHOICE subchannel size | Size1 |
| - Available Subchannels | Null |
| - ASC Settings | (ASC#2) |
| - CHOICE mode | TDD |
| - CHOICE TDD option | 1.28 Mcps TDD |
| - Available SYNC_UL codes indices | "111111111" |

| | |
|---|---|
| - CHOICE subchannel size | Size1 |
| - Available Subchannels | Null |
| - ASC Settings | (ASC#3) |
| - CHOICE mode | TDD |
| - CHOICE TDD option | 1.28 Mcps TDD |
| - Available SYNC_UL codes indices | "111111111" |
| - CHOICE subchannel size | Size1 |
| - Available Subchannels | Null |
| - ASC Settings | (ASC#4) |
| - CHOICE mode | TDD |
| - CHOICE TDD option | 1.28 Mcps TDD |
| - Available SYNC_UL codes indices | "111111111" |
| - CHOICE subchannel size | Size1 |
| - Available Subchannels | Null |
| - ASC Settings | (ASC#5) |
| - CHOICE mode | TDD |
| - CHOICE TDD option | 1.28 Mcps TDD |
| - Available SYNC_UL codes indices | "111111111" |
| - CHOICE subchannel size | Size1 |
| - Available Subchannels | Null |
| - ASC Settings | (ASC#6) |
| - CHOICE mode | TDD |
| - CHOICE TDD option | 1.28 Mcps TDD |
| - Available SYNC_UL codes indices | "111111111" |
| - CHOICE subchannel size | Size1 |
| - Available Subchannels | Null |
| - Access Service Class | |
| - Persistence scaling factor | 0.9 (for ASC#2) |
| - Persistence scaling factor | 0.9 (for ASC#3) |
| - Persistence scaling factor | 0.9 (for ASC#4) |
| - Persistence scaling factor | 0.9 (for ASC#5) |
| - Persistence scaling factor | 0.9 (for ASC#6) |
| - AC-to-ASC mapping | Not Present |
| - CHOICE *mode* | TDD (no data) |
| - Secondary CCPCH system information | |
| - Secondary CCPCH system information | |
| - Secondary CCPCH info | |
| - CHOICE *mode* | TDD |
| - Offset | 0 |
| - Common timeslot info | |
| - 2nd interleaving mode | Frame |
| - TFCI coding | Reference clause 6.10 "Parameter Set" |
| - Puncturing limit | Reference clause 6.10 "Parameter Set" |
| - Repetition period | 1 |
| - Repetition length | 0 |
| - Individual timeslot info | |
| - CHOICE *TDD option* | 1.28 Mcps TDD |
| - Timeslot number | 0 |
| - TFCI existence | Reference clause 6.10 "Parameter Set" |
| - Midamble Shift and burst type | |
| - CHOICE *TDD option* | 1.28 Mcps TDD |
| - Midamble Allocation Mode | Default midamble |
| - Midamble configuration | 4 |
| - Midamble Shift | Not Present |
| - CHOICE *TDD option* | 1.28 Mcps TDD |
| - Modulation | Reference clause 6.10 "Parameter Set" |
| - SS-TPC Symbols | Reference clause 6.10 "Parameter Set" |
| - Code List | |
| - Channelisation Code | Reference clause 6.10 "Parameter Set" |
| - TFCS | Reference clause 6.10 "Parameter Set" |
| - Normal | |
| - TFCI Field 1 information | |
| - CHOICE TFCS representation | Complete reconfiguration |
| - TFCS complete reconfiguration information | |
| - CHOICE CTFC Size | Number of bits used must be enough to cover all combinations of CTFC from clause 6.10. |
| - CTFC information | Reference clause 6.10 "Parameter Set" |
| - Power offset information | Not Present |

| | |
|---|---|
| - FACH/PCH information | |
| - Transport channel Identity | 12 (for PCH) |
| - TFS | (PCH) |
| - CHOICE Transport channel type | Common transport channels |
| - Dynamic Transport format information | |
| - RLC Size | Reference clause 6.10 "Parameter Set" |
| - Number of TB and TTI List | Reference clause 6.10 "Parameter Set" |
| - Number of Transport blocks | Reference clause 6.10 "Parameter Set" |
| - CHOICE Mode | TDD |
| - Transmission Time Interval | Not Present |
| - CHOICE Logical channel List | ALL |
| - Semi-static Transport Format information | |
| - Transmission time interval | Reference clause 6.10 "Parameter Set" |
| - Type of channel coding | Reference clause 6.10 "Parameter Set" |
| - Coding Rate | Reference clause 6.10 "Parameter Set" |
| - Rate matching attribute | Reference clause 6.10 "Parameter Set" |
| - CRC size | Reference clause 6.10 "Parameter Set" |
| - Transport channel Identity | 13 (for FACH) |
| - TFS | (FACH) |
| - CHOICE Transport channel type | Common transport channels |
| - Dynamic Transport format information | |
| - RLC Size | Reference clause 6.10 "Parameter Set" |
| - Number of TB and TTI List | Reference clause 6.10 "Parameter Set" |
| - Number of Transport blocks | Reference clause 6.10 "Parameter Set" |
| - CHOICE Mode | TDD |
| - Transmission Time Interval | Not Present |
| - CHOICE Logical channel List | ALL |
| - Semi-static Transport Format information | |
| - Transmission time interval | Reference clause 6.10 "Parameter Set" |
| - Type of channel coding | Reference clause 6.10 "Parameter Set" |
| - Coding Rate | Reference clause 6.10 "Parameter Set" |
| - Rate matching attribute | Reference clause 6.10 "Parameter Set" |
| - CRC size | Reference clause 6.10 "Parameter Set" |
| - CTCH indicator | FALSE |
| - PICH info | |
| - CHOICE *mode* | TDD |
| - CHOICE TDD option | 1.28 Mcps TDD |
| - Timeslot number | 0 |
| - Midamble shift and burst type | |
| - Midamble Allocation Mode | Default midamble |
| - Midamble configuration | 8 |
| - Midamble Shift | Not Present |
| - Channelisation code list | |
| - Channelisation code | (16/1) |
| - Channelisation code | (16/2) |
| - Repetition period/length | 64/2 |
| - Offset | 0 |
| - Paging indicator length | 4 |
| - $N_{GAP}$ | 4 |
| - $N_{PCH}$ | 2 |
| - CBS DRX Level 1 information | Not Present |

Contents of System Information Block type 7 (FDD)

| | |
|---|---|
| CHOICE Mode | FDD |
| - UL interference | -100 dBm |
| - PRACHs listed in system information block type5 | |
| - Dynamic persistence level | 2 |
| - PRACHs listed in system information block type6 | |
| - Dynamic persistence level | 2 |
| - Expiration Time Factor | Not Present - use default value of 1 |

Contents of System Information Block type 7 (TDD)

| | |
|---|---|
| CHOICE Mode | TDD |
| PRACHs listed in system information block type5 | |
| - Dynamic persistence level | 2 |
| PRACHs listed in system information block type6 | |
| - Dynamic persistence level | 2 |
| Expiration Time Factor | Not Present - use default value of 1 |

Contents of System Information Block type 8, 9 (only for FDD R99 and Rel-4)

This information is used for static CPCH in the cell, so this is not present.

Contents of System Information Block type 10 (only for FDD R99 and Rel-4)

This information is used for DRAC, so this is not present.

Contents of System Information Block type 11 (FDD)

This is the default message content of SIB 11 for cell 1.

See clause 6.1.4 for the difference in message contents of System Information Block type 11 (FDD) for cell 2 to 8.

| | | |
|---|---|---|
| - SIB12 indicator | A1, A2, A3 | TRUE |
| - FACH measurement occasion info | | Not Present |
| - Measurement control system information | | |
| - Use of HCS | | Not used |
| - Cell selection and reselection quality measure | | CPICH RSCP |
| **- Intra-frequency measurement system information** | A1, A2, A3 | |
| - Intra-frequency measurement identity | | Not Present<br>Absence of this IE is equivalent to default value 1 |
| - Intra-frequency cell info list | | |
| - CHOICE intra-frequency cell removal | | Not present<br>(This IE shall be ignored by the UE for SIB11) |
| - New intra-frequency cells | | |
| - Intra-frequency cell id | | 1 |
| - Cell info | | |
| - Cell individual offset | | Not present<br>Absence of this IE is equivalent to default value 0 dB |
| - Reference time difference to cell | | Not Present |
| - Read SFN indicator | | FALSE |
| - CHOICE mode | | FDD |
| - Primary CPICH info | | |
| - Primary scrambling code | | Refer to clause titled "Default settings for cell No.1 (FDD)" in clause 6.1.4 |
| - Primary CPICH TX power | | Not Present |
| - TX Diversity indicator | | FALSE |
| - Cell Selection and Re-selection info | | Not Present<br>(The IE shall be absent as this is the serving cell) |
| - Intra-frequency cell id | | 2 |
| - Cell info | | |
| - Cell individual offset | | Not present<br>Absence of this IE is equivalent to default value 0dB |
| - Reference time difference to cell | | Not present |
| - Read SFN indicator | | TRUE |
| - CHOICE mode | | FDD |
| - Primary CPICH info | | |
| - Primary scrambling code | | Refer to clause titled "Default settings for cell No.2 (FDD)" in clause 6.1.4 |
| - Primary CPICH TX power | | Not Present |
| - TX Diversity indicator | | FALSE |
| - Cell Selection and Re-selection info | | Not present<br>For neigbouring cell, if HCS is not used and all the parameters in cell selection and re-selection info are Default value, this IE is absent. |
| - Intra-frequency cell id | | 3 |

| | | |
|---|---|---|
| - Cell info | | Same content as specified for Intra-frequency cell id=2 with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.3 (FDD)" in clause 6.1.4 |
| - Intra-frequency cell id | A1, A3 | 7 |
| - Cell info | | Same content as specified for Intra-frequency cell id=2 with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.7 (FDD)" in clause 6.1.4 |
| - Intra-frequency cell id | A1, A3 | 8 |
| - Cell info | | Same content as specified for Intra-frequency cell id=2 with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.8 (FDD)" in clause 6.1.4 |
| - Intra-frequency cell id | A3 | 11 |
| - Cell info | | Same content as specified for Intra-frequency cell id=2 with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.11 (FDD)" in clause 6.1.4 |
| - Cells for measurement | A1, A2, A3 | Not Present |
| - Intra-frequency measurement quantity | A1, A2, A3 | |
| - Filter coefficient | | Not present<br>Absence of this IE is equivalent to the default value 0 |
| - CHOICE mode | | FDD |
| - Measurement quantity | | CPICH RSCP |
| - Intra-frequency reporting quantity for RACH Reporting | | Not Present |
| - Maximum number of reported cells on RACH | | Not Present |
| - Reporting information for state CELL_DCH | | |
| - Intra-frequency reporting quantity | | |
| - Reporting quantities for active set cells | | |
| - Cell synchronization information reporting indicator | | FALSE |
| - Cell identity reporting indicator | | TRUE |
| - CHOICE mode | | FDD |
| - CPICH Ec/N0 reporting indicator | | FALSE |
| - CPICH RSCP reporting indicator | | TRUE |
| - Pathloss reporting indicator | | FALSE |
| - Reporting quantities for monitored set cells | | |
| - Cell synchronization information reporting indicator | | TRUE |
| - Cell identity reporting indicator | | TRUE |
| - CHOICE mode | | FDD |
| - CPICH Ec/N0 reporting indicator | | FALSE |
| - CPICH RSCP reporting indicator | | TRUE |
| - Pathloss reporting indicator | | FALSE |
| - Reporting quantities for detected set cells | | Not Present |
| - Measurement reporting mode | | |
| - Measurement Report Transfer Mode | | Acknowledged mode RLC |
| - Periodic Reporting/Event Trigger Reporting Mode | | Event trigger |
| - CHOICE report criteria | | Intra-frequency measurement reporting criteria |
| - Intra-frequency measurement reporting criteria | | |
| - Parameters required for each event | | 3 kinds |
| - Intra-frequency event identity | | 1a |
| - Triggering condition 1 | | Not Present |
| - Triggering condition 2 | | Monitored set cells |
| - Reporting Range Constant | | 5dB |
| - Cells forbidden to affect Reporting range | | Not Present |
| - W | | 1.0 |
| - Hysteresis | | 0.0 |
| - Threshold Used Frequency | | Not Present |
| - Reporting deactivation threshold | | 2 |
| - Replacement activation threshold | | Not Present |
| - Time to trigger | | 640 |
| - Amount of reporting | | 4 |

| | | |
|---|---|---|
| - Reporting interval | | 4 000 |
| - Reporting cell status | | |
| - CHOICE reported cell | | Report cell within active set and/or monitored set cells on used frequency |
| - Maximum number of reported cells | | 3 |
| - Intra-frequency event identity | | 1b |
| - Triggering condition 1 | | Active set cells |
| - Triggering condition 2 | | Not Present |
| - Reporting Range Constant | | 5dB |
| - Cells forbidden to affect Reporting range | | Not Present |
| - W | | 1.0 |
| - Hysteresis | | 0.0 |
| - Threshold Used Frequency | | Not Present |
| - Reporting deactivation threshold | | Not Present |
| - Replacement activation threshold | | Not Present |
| - Time to trigger | | 640 |
| - Amount of reporting | | Not Present |
| - Reporting interval | | Not Present |
| - Reporting cell status | | |
| - CHOICE reported cell | | Report cell within active set and/or monitored set cells on used frequency |
| - Maximum number of reported cells | | 3 |
| - Intra-frequency event identity | | 1c |
| - Triggering condition 1 | | Not Present |
| - Triggering condition 2 | | Not Present |
| - Reporting Range Constant | | Not Present |
| - Cells forbidden to affect Reporting range | | Not Present |
| - W | | Not Present |
| - Hysteresis | | 0.0 |
| - Threshold Used Frequency | | Not Present |
| - Reporting deactivation threshold | | Not Present |
| - Replacement activation threshold | | 3 |
| - Time to trigger | | 640 |
| - Amount of reporting | | 4 |
| - Reporting interval | | 4 000 |
| - Reporting cell status | | |
| - CHOICE reported cell | | Report cell within active set and/or monitored set cells on used frequency |
| - Maximum number of reported cells | | 3 |
| **- Inter-frequency measurement system information** | A1, A2 | |
| - Inter-frequency cell info list | | |
| - CHOICE Inter-frequency cell removal | | Not present<br>(This IE shall be ignored by the UE for SIB11) |
| - New inter-frequency cells | | |
| - Inter frequency cell id | | 4 |
| - Frequency info | | |
| - CHOICE mode | | FDD |
| - UARFCN uplink(Nu) | | Not present<br>Absence of this IE is equivalent to apply the default duplex distance defined for the operating frequency according to 3GPP TS 25.101 [11] |
| - UARFCN downlink(Nd) | | Reference to table 6.1.2 for Cell 4 |
| - Cell info | | |
| - Cell individual offset | | Not present<br>Absence of this IE is equivalent to default value 0 dB |
| - Reference time difference to cell | | Not present |
| - Read SFN indicator | | FALSE |
| - CHOICE mode | | FDD |
| - Primary CPICH info | | |
| - Primary scrambling code | | Refer to clause titled "Default settings for cell No.4 (FDD)" in clause 6.1.4 |
| - Primary CPICH Tx power | | Not present |
| - TX Diversity Indicator | | FALSE |
| - Cell Selection and Re-selection Info | | Not present (same values as for serving cell applies) |
| - Inter frequency cell id | | 5 |

| | | |
|---|---|---|
| - Frequency info | | Not Present<br>Absence of this IE is equivalent to value of the previous "frequency info" in the list. |
| - Cell info | | Same content as specified for Inter-frequency cell id=4 with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.5 (FDD)" in clause 6.1.4 |
| - Inter frequency cell id | | 6 |
| - Frequency info | | Not Present<br>Absence of this IE is equivalent to value of the previous "frequency info" in the list. |
| - Cell info | | Same content as specified for Inter-frequency cell id=4 with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.6 (FDD)" in clause 6.1.4 |
| - Cell for measurement | | Not present |
| - Inter-RAT measurement system information | A1, A3 | Not Present |
| **- Inter-RAT measurement system information** | A2 | |
| **- Inter-RAT cell info list** | | |
| - CHOICE *Inter-RAT cell removal* | | Not Present<br>(This IE shall be ignored by the UE for SIB11) |
| - New inter-RAT cells | | |
| - Inter-RAT cell id | | 9 |
| - CHOICE *Radio Access Technology* | | GSM |
| - GSM | | |
| - Cell individual offset | | 0 |
| - Cell selection and re-selection info | | Not Present |
| - BSIC | | |
| - Base transceiver Station Identity Code (BSIC) | | Reference to table 6.1.10 for Cell 9 |
| - Band indicator | | According to PICS/PIXIT |
| - BCCH ARFCN | | Reference to table 6.1.10 for Cell 9 |
| - Inter-RAT cell id | | 10 |
| - CHOICE *Radio Access Technology* | | GSM |
| - GSM | | |
| - Cell individual offset | | 0 |
| - Cell selection and re-selection info | | Not Present |
| - BSIC | | |
| - Base transceiver Station Identity Code (BSIC) | | Reference to table 6.1.10 for Cell 10 |
| - Band indicator | | According to PICS/PIXITs |
| - BCCH ARFCN | | Reference to table 6.1.10 for Cell 10 |
| - Cell for measurement | | Not present |
| - Traffic volume measurement system information | A1, A2, A3 | Not Present |

| Condition | Explanation |
|---|---|
| A1 | FDD cell environment |
| A2 | FDD/GSM inter-RAT cell environment |
| A3 | FDD intra-frequency cell environment (6 intra-frequency cells without inter-frequency cells) |

Contents of System Information Block type 11 (3.84 Mcps and 1.28 Mcps TDD)

This is the default message content of SIB 11 for cell 1.

See clause 6.1.4 for the difference in message contents of System Information Block type 11 (TDD) for cell 2 to 8.

| | | |
|---|---|---|
| - SIB 12 Indicator | A1, A2 | TRUE |
| - FACH measurement occasion info | | Not Present |
| - Measurement control system information | | |
| - Use of HCS | | Not used |
| - Cell selection and reselection quality measureCell | | CPICH RSCP |
| - Intra-frequency measurement system information | A1, A2 | |
| - Intra-frequency measurement identity | | Not Present<br>Absence of this IE is equivalent to default value 1 |
| - Intra-frequency cell info list | | |

| | | |
|---|---|---|
| - CHOICE intra-frequency cell removal | | Not present<br>(This IE shall be ignored by the UE for SIB11) |
| - New intra-frequency cells | | |
| - Intra-frequency cell id | | 1 |
| - Cell info | | |
| - Cell individual offset | | Not present<br>Absence of this IE is equivalent to default value 0dB |
| - Reference time difference to cell | | Not Present |
| - Read SFN Indicator | | FALSE |
| - CHOICE mode | | TDD |
| - Primary CCPCH info | | |
| - Cell parameters ID | | Reference clause 6.1.4 Default settings for cell |
| - Primary CCPCH TX power | | Not Present |
| - Timeslot list | | Not Present |
| - CHOICE TDD option | | |
| - 3.84 Mcps TDD | | |
| - Timeslot number | | Not Present |
| - Burst type | | Not Present |
| - 1.28 Mcps TDD | | |
| - Timeslot number | | Not Present |
| - Cell Selection and Re-selection info | | Not Present<br>(The IE shall be absent as this is the serving cell) |
| - Intra-frequency cell id | | 2 |
| - Cell info | | |
| - Cell individual offset | | Not present<br>Absence of this IE is equivalent to default value 0dB |
| - Reference time difference to cell | | Not Present |
| - Read SFN Indicator | | FALSE |
| - CHOICE mode | | TDD |
| - Primary CCPCH info | | |
| - Cell parameters ID | | Refer to clause titled “Default setting for cell No.2 (TDD)” in clause 6.1.4 |
| - Primary CCPCH TX power | | Not Present |
| - Timeslot list | | Not Present |
| - CHOICE TDD option | | |
| - 3.84 Mcps TDD | | |
| - Timeslot number | | Not Present |
| - Burst type | | Not Present |
| - 1.28 Mcps TDD | | |
| - Timeslot number | | Not Present |
| - Cell Selection and Re-selection info | | Not Present |
| - Intra-frequency cell id | | 3 |
| - Cell info | | Same content as specified for intra-frequency cell id=2 with the exception that value for Cell Parameters ID shall be according to clause titled “Default settings for cell No.3(TDD)” in clause 6.1.4 |
| - Intra-frequency cell id | | 7 |
| - Cell info | | Same content as specified for intra-frequency cell id=2 with the exception that value for Cell Parameters ID shall be according to clause titled “Default settings for cell No.7(TDD)” in clause 6.1.4 |
| - Intra-frequency cell id | | 8 |
| - Cell info | | Same content as specified for intra-frequency cell id=2 with the exception that value for Cell Parameters ID shall be according to clause titled “Default settings for cell No.8(TDD)” in clause 6.1.4 |
| - Cell for measurement | A1, A2 | Not Present |
| - Intra-frequency measurement quantity | A1, A2 | |
| - Filter coefficient | | Not present<br>Absence of this IE is equivalent to the default value 0 |
| - CHOICE mode | | TDD |
| - Measurement quantity list | | |
| - Measurement quantity | | P-CCPCH RSCP |
| - Intra-frequency reporting quantity for RACH Reporting | | Not Present |
| - Maximum number of reported cells on RACH | | Not Present |
| - Reporting information for state CELL_DCH | | |
| - Intra-frequency reporting quantity | | |

| | | |
|---|---|---|
| - Reporting quantities for active set cells | | |
| - Cell synchronization information reporting indicator | | TRUE |
| - Cell identity reporting indicator | | TRUE |
| - CHOICE mode | | TDD |
| - Timeslot ISCP reporting indicator | | FALSE |
| - Proposed TSGN reporting required | | FALSE |
| - P-CCPCH RSCP reporting indicator | | TRUE |
| - Pathloss reporting indicator | | FALSE |
| - Reporting quantities for monitored set cells | | |
| - Cell synchronization information reporting indicator | | FALSE |
| - Cell identity reporting indicator | | TRUE |
| - CHOICE mode | | TDD |
| - Timeslot ISCP reporting indicator | | FALSE |
| - Proposed TSGN reporting required | | FALSE |
| - P-CCPCH RSCP reporting indicator | | TRUE |
| - Pathloss reporting indicator | | FALSE |
| - Reporting quantities for detected set cells | | Not Present |
| - Measurement reporting mode | | |
| - Measurement Report Transfer Mode | | Acknowledged mode RLC |
| - Periodical Reporting / Event Trigger Reporting Mode | | Event trigger |
| -CHOICE report criteria | | |
| - Intra-frequency measurement reporting criteria | | |
| - Parameters required for each event | | |
| - Intra-frequency event identity | | 1g |
| - Triggering condition1 | | Not Present |
| - Triggering condition2 | | Not Present |
| - Reporting Range Constant | | Not Present |
| - cells forbidden to affect reporting range | | Not Present |
| - W(optional in case of 1a,1b) | | Not Present |
| - Hysteresis | | 0.0 |
| - Threshold used frequency | | Not Present |
| - Reporting deactivation threshold | | 3 |
| - Replacement activation threshold | | Not Present |
| - Time to trigger | | 640 |
| - Amount of reporting | | 4 |
| - Reporting interval | | 4000 |
| - Reporting cell status | | |
| - CHOICE reported cells | | Report cell within active set and/or monitored cells on used frequency |
| - Maximum number of reported cells | | 3 |
| - Inter-frequency measurement system information | A1, A2 | |
| - Inter-frequency cell info list | | |
| - CHOICE Inter-frequency cell removal | | Not present<br>(This IE shall be ignored by the UE for SIB11) |
| - New inter-frequency cells | | |
| - Inter frequency cell id | | 4 |
| - Frequency info | | |
| - CHOICE mode | | TDD |
| - UARFCN (Nt) | | Reference to table 6.1.7 for Cell 4 |
| - Cell info | | |
| - Cell individual offset | | Not present<br>Absence of this IE is equivalent to default value 0dB |
| - Reference time difference to cell | | Not present |
| - Read SFN indicator | | FALSE |
| - CHOICE mode | | TDD |
| - Primary CCPCH info | | Refer to clause titled "Default settings for cell No.4 (TDD)" in clause 6.1.4 |
| - Primary CCPCH Tx power | | Not present |
| - TX Diversity Indicator | | FALSE |
| - Cell Selection and Re-selection Info | | Not present (same values as for serving cell applies) |
| - Inter frequency cell id | | 5 |
| - Frequency info | | Not Present<br>Absence of this IE is equivalent to value of the previous "frequency info" in the list. |

| | | |
|---|---|---|
| - Cell info | | Same content as specified for Inter-frequency cell id=4 with the exception that value for Cell parameters ID shall be according to clause titled "Default settings for cell No.5 (TDD)" in clause 6.1.4 |
| - Inter frequency cell id | | 6 |
| - Frequency info | | Not Present<br>Absence of this IE is equivalent to value of the previous "frequency info" in the list. |
| - Cell info | | Same content as specified for Inter-frequency cell id=4 with the exception that value for Cell parameters ID shall be according to clause titled "Default settings for cell No.6 (TDD)" in clause 6.1.4 |
| - Cell for measurement | | Not present |
| - Inter-RAT measurement system information | A1 | Not Present |
| - Inter-RAT measurement system information | A2 | |
| - Inter-RAT cell info list | | |
| - CHOICE *Inter-RAT cell removal* | | Not Present<br>(This IE shall be ignored by the UE for SIB11) |
| - New inter-RAT cells | | |
| - Inter-RAT cell id | | 9 |
| - CHOICE *Radio Access Technology* | | GSM |
| - GSM | | |
| - Cell individual offset | | 0 |
| - Cell selection and re-selection info | | Not Present |
| - BSIC | | |
| - Base transceiver Station Identity Code (BSIC) | | Reference to table 6.1.10 for Cell 9 |
| - Band indicator | | According to PICS/PIXIT |
| - BCCH ARFCN | | Reference to table 6.1.10 for Cell 9 |
| - Inter-RAT cell id | | 10 |
| - CHOICE *Radio Access Technology* | | GSM |
| - GSM | | |
| - Cell individual offset | | 0 |
| - Cell selection and re-selection info | | Not Present |
| - BSIC | | |
| - Base transceiver Station Identity Code (BSIC) | | Reference to table 6.1.10 for Cell 10 |
| - Band indicator | | According to PICS/PIXITs |
| - BCCH ARFCN | | Reference to table 6.1.10 for Cell 10 |
| - Cell for measurement | | Not present |
| - Traffic volume measurement system information | A1, A2 | Not Present |

| Condition | Explanation |
|---|---|
| A1 | TDD cell environment |
| A2 | TDD/GSM inter-RAT cell environment |

Contents of System Information Block type 12 in connected mode (FDD)

This is the default message content of SIB 12 for cell 1.

See clause 6.1.4 for the difference in message contents of System Information Block type 12 (FDD) for cell 2 to 8.

| | |
|---|---|
| - FACH measurement occasion info | Not Present |
| - Measurement control system information | |
| - Use of HCS | Not used |
| - Cell selection and reselection quality measure | CPICH RSCP |
| - Intra-frequency measurement system information | Not Present |
| **- Inter-frequency measurement system information** | Not Present |
| - Inter-RAT measurement system information | Not Present |
| - Traffic volume measurement system information | Not Present |

Contents of System Information Block type 12 in connected mode (3.84 Mcps and 1.28 Mcps TDD)

This is the default message content of SIB 12 for cell 1.

See clause 6.1.4 for the difference in message contents of System Information Block type 12 (TDD) for cell 2 to 8.

| | |
|---|---|
| - FACH measurement occasion info | Not Present |
| - Measurement control system information | |
| - Use of HCS | Not used |
| - Cell selection and reselection quality measure | CPICH RSCP |
| - Intra-frequency measurement system information | Not Present |
| - Inter-RAT measurement system information | Not Present |
| - Traffic volume measurement system information | Not Present |

Contents of System Information Block type 13 (used when supported PLMN type is ANSI-41)

| | |
|---|---|
| - CN Domain system information list | |
| - CN Domain system information | *For Packet-Switched domain* |
| - CN domain identity | PS |
| - CHOICE CN Type | ANSI-41 |
| - CN domain specific NAS system information | |
| - NAS (ANSI-41) system information | T.B.D |
| - CN domain specific DRX cycle length coefficient | 7 |
| - CN Domain system information | *For Circuit-Switched domain* |
| - CN domain identity | CS |
| - CHOICE CN Type | ANSI-41 |
| - CN domain specific NAS system information | |
| - NAS (ANSI-41) system information | T.B.D |
| - CN domain specific DRX cycle length coefficient | 7 |
| - UE timers and constants in idle mode | |
| - T300 | 400 milliseconds |
| - N300 | 3 |
| - T312 | 10 seconds |
| - N312 | 200 |
| - Capability update requirement | |
| - UE radio access FDD capability update requirement | TRUE |
| - UE radio access TDD capability update requirement | FALSE |
| - System specific capability update requirement list | Not Present |

Contents of System Information Block type 14 (3.84 Mcps TDD)

| | |
|---|---|
| - Individual Timeslot interference list | |
| - Individual Timeslot interference | |
| - Timeslot number | 2 |
| - UL Timeslot Interference | -90 dbm |
| - Individual Timeslot interference | |
| - Timeslot number | 3 |
| - UL Timeslot Interference | -90 dbm |
| - Individual Timeslot interference | |
| - Timeslot number | 4 |
| - UL Timeslot Interference | -90 dbm |
| - Individual Timeslot interference | |
| - Timeslot number | 5 |
| - UL Timeslot Interference | -90 dbm |
| - Individual Timeslot interference | |
| - Timeslot number | 6 |
| - UL Timeslot Interference | -90 dbm |
| - Individual Timeslot interference | |
| - Timeslot number | 7 |
| - UL Timeslot Interference | -90 dbm |
| - Individual Timeslot interference | |
| - Timeslot number | 9 |
| - UL Timeslot Interference | -90 dbm |
| - Individual Timeslot interference | |
| - Timeslot number | 10 |
| - UL Timeslot Interference | -90 dbm |
| - Individual Timeslot interference | |
| - Timeslot number | 11 |
| - UL Timeslot Interference | -90 dbm |
| - Individual Timeslot interference | |

| | |
|---|---|
| - Timeslot number | 12 |
| - UL Timeslot Interference | -90 dbm |
| - Individual Timeslot interference | |
| - Timeslot number | 13 |
| - UL Timeslot Interference | -90 dbm |
| - Individual Timeslot interference | |
| - Timeslot number | 14 |
| - UL Timeslot Interference | -90 dbm |
| - Expiration Time Factor | Not Present (MD "1") |

Contents of System Information Block type 16 (FDD)

| | |
|---|---|
| - Pre-Defined Radio Configuration | (12.2 KBPS AMR) |
| - Pre-defined RB configuration | |
| - Re-establishment timer | useT315 |
| - SRB InformationList | |
| - Signalling RB information to setup | (UM DCCH for RRC) |
| - RB identity | 1 |
| - CHOICE RLC info type | |
| - RLC info | |
| - CHOICE Uplink RLC mode | UM RLC |
| - Transmission RLC discard | timerBasedNoExplicit : dt100 |
| - CHOICE Downlink RLC mode | UM RLC |
| - RB mapping info | |
| - Information for each multiplexing option | |
| - RLC logical channel mapping indicator | Not Present |
| - Number of RLC logical channels | 1 |
| - Uplink transport channel type | DCH |
| - UL Transport channel identity | 5 |
| - Logical channel identity | 1 |
| - CHOICE RLC size list | Configured |
| - MAC logical channel priority | 1 |
| - Downlink RLC logical channel info | |
| - Number of RLC logical channels | 1 |
| - Downlink transport channel type | DCH |
| - DL DCH Transport channel identity | 10 |
| - DL DSCH Transport channel identity | Not Present |
| - Logical channel identity | 1 |
| - Signalling RB information to setup | (AM DCCH for RRC) |
| - RB identity | 2 |
| - CHOICE RLC info type | |
| - RLC info | |
| - CHOICE Uplink RLC mode | AM RLC |
| - Transmission RLC discard | |
| - SDU discard mode | Max DAT retransmissions |
| - MAX_DAT | 4 |
| - Timer_MRW | 100 |
| - MaxMRW | 4 |
| - Transmission window size | 8 |
| - Timer_RST | 500 |
| - Max_RST | 4 |
| - Polling info | |
| - Timer_poll_prohibit | 200 |
| - Timer_poll | 200 |
| - Poll_PDU | Not Present |
| - Poll_SDU | 1 |
| - Last transmission PDU poll | TRUE |
| - Last retransmission PDU poll | TRUE |
| - Poll_Window | 99 |
| - Timer_poll_periodic | Not Present |
| - CHOICE Downlink RLC mode | AM RLC |
| - In-sequence delivery | TRUE |
| - Receiving window size | 8 |
| - Downlink RLC status info | |
| - Timer_status_prohibit | 200 |
| - Timer_EPC | 200 |
| - Missing PDU indicator | TRUE |
| - Timer_STATUS_periodic | Not Present |
| - RB mapping info | |
| - Information for each multiplexing option | |
| - RLC logical channel mapping indicator | Not Present |
| - Number of RLC logical channels | 1 |
| - Uplink transport channel type | DCH |
| - UL Transport channel identity | 5 |
| - Logical channel identity | 2 |
| - CHOICE RLC size list | Configured |
| - MAC logical channel priority | 2 |
| - Downlink RLC logical channel info | |
| - Number of RLC logical channels | 1 |
| - Downlink transport channel type | DCH |

| | |
|---|---|
| - DL DCH Transport channel identity | 10 |
| - DL DSCH Transport channel identity | Not Present |
| - Logical channel identity | 2 |
| - Signalling RB information to setup | (AM DCCH for RRC) |
| - RB identity | 3 |
| - CHOICE RLC info type | |
| - RLC info | |
| - CHOICE Uplink RLC mode | AM RLC |
| - Transmission RLC discard | |
| - SDU discard mode | Max DAT retransmissions |
| - MAX_DAT | 4 |
| - Timer_MRW | 100 |
| - MaxMRW | 4 |
| - Transmission window size | 8 |
| - Timer_RST | 500 |
| - Max_RST | 4 |
| - Polling info | |
| - Timer_poll_prohibit | 200 |
| - Timer_poll | 200 |
| - Poll_PDU | Not Present |
| - Poll_SDU | 1 |
| - Last transmission PDU poll | TRUE |
| - Last retransmission PDU poll | TRUE |
| - Poll_Window | 99 |
| - Timer_poll_periodic | Not Present |
| - CHOICE Downlink RLC mode | AM RLC |
| - In-sequence delivery | TRUE |
| - Receiving window size | 8 |
| - Downlink RLC status info | |
| - Timer_status_prohibit | 200 |
| - Timer_EPC | 200 |
| - Missing PDU indicator | TRUE |
| - Timer_STATUS_periodic | Not Present |
| - RB mapping info | |
| - Information for each multiplexing option | |
| - RLC logical channel mapping indicator | Not Present |
| - Number of RLC logical channels | 1 |
| - Uplink transport channel type | DCH |
| - UL Transport channel identity | 5 |
| - Logical channel identity | 3 |
| - CHOICE RLC size list | Configured |
| - MAC logical channel priority | 3 |
| - Downlink RLC logical channel info | |
| - Number of RLC logical channels | 1 |
| - Downlink transport channel type | DCH |
| - DL DCH Transport channel identity | 10 |
| - DL DSCH Transport channel identity | Not Present |
| - Logical channel identity | 3 |
| - Signalling RB information to setup | (AM DCCH for RRC) |
| - RB identity | 4 |
| - CHOICE RLC info type | |
| - RLC info | |
| - CHOICE Uplink RLC mode | AM RLC |
| - Transmission RLC discard | |
| - SDU discard mode | Max DAT retransmissions |
| - MAX_DAT | 4 |
| - Timer_MRW | 100 |
| - MaxMRW | 4 |
| - Transmission window size | 8 |
| - Timer_RST | 500 |
| - Max_RST | 4 |
| - Polling info | |
| - Timer_poll_prohibit | 200 |
| - Timer_poll | 200 |
| - Poll_PDU | Not Present |
| - Poll_SDU | 1 |
| - Last transmission PDU poll | TRUE |
| - Last retransmission PDU poll | TRUE |

| | |
|---|---|
| - Poll_Window | 99 |
| - Timer_poll_periodic | Not Present |
| - CHOICE Downlink RLC mode | AM RLC |
| - In-sequence delivery | TRUE |
| - Receiving window size | 8 |
| - Downlink RLC status info | |
| - Timer_status_prohibit | 200 |
| - Timer_EPC | 200 |
| - Missing PDU indicator | TRUE |
| - Timer_STATUS_periodic | Not Present |
| - RB mapping info | |
| - Information for each multiplexing option | |
| - RLC logical channel mapping indicator | Not Present |
| - Number of RLC logical channels | 1 |
| - Uplink transport channel type | DCH |
| - UL Transport channel identity | 5 |
| - Logical channel identity | 4 |
| - CHOICE RLC size list | Configured |
| - MAC logical channel priority | 4 |
| - Downlink RLC logical channel info | |
| - Number of RLC logical channels | 1 |
| - Downlink transport channel type | DCH |
| - DL DCH Transport channel identity | 10 |
| - DL DSCH Transport channel identity | Not Present |
| - Logical channel identity | 4 |
| - RAB information for setup | |
| - RB information to setup | |
| - RB identity | 10 |
| - PDCP info | Not Present |
| - CHOICE RLC info type | RLC info |
| - CHOICE Uplink RLC mode | TM RLC |
| - Transmission RLC discard | Not Present |
| - Segmentation indication | TRUE |
| - CHOICE Downlink RLC mode | TM RLC |
| - Segmentation indication | TRUE |
| - RB mapping info | |
| - Information for each multiplexing option | |
| - RLC logical channel mapping indicator | Not Present |
| - Number of uplink RLC logical channels | 1 |
| - Uplink transport channel type | DCH |
| - UL Transport channel identity | 1 |
| - Logical channel identity | 7 |
| - CHOICE RLC size list | Configured |
| - MAC logical channel priority | 6 |
| - Downlink RLC logical channel info | |
| - Number of downlink RLC logical channels | 1 |
| - Downlink transport channel type | DCH |
| - DL DCH Transport channel identity | 6 |
| - DL DSCH Transport channel identity | Not Present |
| - Logical channel identity | 7 |
| - RB identity | 11 |
| - PDCP info | Not Present |
| - CHOICE RLC info type | RLC info |
| - CHOICE Uplink RLC mode | TM RLC |
| - Transmission RLC discard | Not Present |
| - Segmentation indication | TRUE |
| - CHOICE Downlink RLC mode | TM RLC |
| - Segmentation indication | TRUE |
| - RB mapping info | |
| - Information for each multiplexing option | |
| - RLC logical channel mapping indicator | Not Present |
| - Number of uplink RLC logical channels | 1 |
| - Uplink transport channel type | DCH |
| - UL Transport channel identity | 2 |
| - Logical channel identity | 8 |
| - CHOICE RLC size list | Configured |
| - MAC logical channel priority | 6 |
| - Downlink RLC logical channel info | |

| | |
|---|---|
| - Number of downlink RLC logical channels | 1 |
| - Downlink transport channel type | DCH |
| - DL DCH Transport channel identity | 7 |
| - DL DSCH Transport channel identity | Not Present |
| - Logical channel identity | 8 |
| - RB identity | 12 |
| - PDCP info | Not Present |
| - CHOICE RLC info type | RLC info |
| - CHOICE Uplink RLC mode | TM RLC |
| - Transmission RLC discard | Not Present |
| - Segmentation indication | TRUE |
| - CHOICE Downlink RLC mode | TM RLC |
| - Segmentation indication | TRUE |
| - RB mapping info | |
| - Information for each multiplexing option | |
| - RLC logical channel mapping indicator | Not Present |
| - Number of uplink RLC logical channels | 1 |
| - Uplink transport channel type | DCH |
| - UL Transport channel identity | 3 |
| - Logical channel identity | 9 |
| - CHOICE RLC size list | Configured |
| - MAC logical channel priority | 6 |
| - Downlink RLC logical channel info | |
| - Number of downlink RLC logical channels | 1 |
| - Downlink transport channel type | DCH |
| - DL DCH Transport channel identity | 8 |
| - DL DSCH Transport channel identity | Not Present |
| - Logical channel identity | 9 |
| - Pre-Defined Transport Channel Configuration | |
| - UL CommonTransChInfo | |
| - UL TFCS | |
| - TFC subset | Default value is the complete existing set of transport format combinations |
| - Allowed Transport Format combination | 0,1,2,3,4,5 |
| - PRACH TFCS | Not Present |
| - CHOICE mode | FDD |
| - CHOICE TFCI signalling | Normal |
| - TFCI Field 1 information | |
| - CHOICE TFCS representation | Addition |
| - TFCS addition configure information | |
| - CHOICE TFCS Size | Number of bits used must be enough to cover all combinations of CTFC from clause 6.10.2.4.1.4.1 Parameter Set. |
| - CTFC information | This IE is repeated for TFC numbers and reference to clause 6.10.2.4.1.4.1 Parameter Set |
| - Power offset information | |
| - CHOICE Gain Factors | Signalled Gain Factor |
| - CHOICE mode | FDD |
| - Gain factor ßc | 0 |
| - Gain factor ßd | 0 |
| - Reference TFC ID | 0 |
| - Power offset Pp-m | 0 dB |
| - Reference TFC ID | 0 |
| - Power offset Pp-m | 0 dB |
| - Added or Reconfigured UL TrCH information | 4 TrCHs(DCH for DCCH and 3DCHs for DTCH) |
| - Uplink transport channel type | DCH |
| - UL Transport channel identity | 1 |
| - TFS | |
| - CHOICE Transport channel type | Dedicated transport channels |
| - Dynamic Transport format information | |
| - RLC Size | Reference to clause 6.10.2.4.1.4.1 Parameter Set |
| - Number of TBs and TTI List | (This IE is repeated for TFI number.) |
| - Transmission Time Interval | Not Present |
| - Number of Transport blocks | Reference to clause 6.10.2.4.1.4.1 Parameter Set |
| - CHOICE Logical channel list | All |
| - Semi-static Transport Format information | |
| - Transmission time interval | Reference to clause 6.10.2.4.1.4.1 Parameter Set |
| - Type of channel coding | Reference to clause 6.10.2.4.1.4.1 Parameter Set |

| | |
|---|---|
| - Coding Rate | Reference to clause 6.10.2.4.1.4.1 Parameter Set |
| - Rate matching attribute | Reference to clause 6.10.2.4.1.4.1 Parameter Set |
| - CRC size | Reference to clause 6.10.2.4.1.4.1 Parameter Set |
| - Uplink transport channel type | DCH |
| - UL Transport channel identity | 2 |
| - TFS | |
| - CHOICE Transport channel type | Dedicated transport channels |
| - Dynamic Transport format information | |
| - RLC Size | Reference to clause 6.10.2.4.1.4.1 Parameter Set |
| - Number of TBs and TTI List | (This IE is repeated for TFI number.) |
| - Transmission Time Interval | Not Present |
| - Number of Transport blocks | Reference to clause 6.10.2.4.1.4.1 Parameter Set |
| - CHOICE Logical channel list | All |
| - Semi-static Transport Format information | |
| - Transmission time interval | Reference to clause 6.10.2.4.1.4.1 Parameter Set |
| - Type of channel coding | Reference to clause 6.10.2.4.1.4.1 Parameter Set |
| - Coding Rate | Reference to clause 6.10.2.4.1.4.1 Parameter Set |
| - Rate matching attribute | Reference to clause 6.10.2.4.1.4.1 Parameter Set |
| - CRC size | Reference to clause 6.10.2.4.1.4.1 Parameter Set |
| - Uplink transport channel type | DCH |
| - UL Transport channel identity | 3 |
| - TFS | |
| - CHOICE Transport channel type | Dedicated transport channels |
| - Dynamic Transport format information | |
| - RLC Size | Reference to clause 6.10.2.4.1.4.1 Parameter Set |
| - Number of TBs and TTI List | (This IE is repeated for TFI number.) |
| - Transmission Time Interval | Not Present |
| - Number of Transport blocks | Reference to clause 6.10.2.4.1.4.1 Parameter Set |
| - CHOICE Logical channel list | All |
| - Semi-static Transport Format information | |
| - Transmission time interval | Reference to clause 6.10.2.4.1.4.1 Parameter Set |
| - Type of channel coding | Reference to clause 6.10.2.4.1.4.1 Parameter Set |
| - Coding Rate | Reference to clause 6.10.2.4.1.4.1 Parameter Set |
| - Rate matching attribute | Reference to clause 6.10.2.4.1.4.1 Parameter Set |
| - CRC size | Reference to clause 6.10.2.4.1.4.1 Parameter Set |
| - Uplink transport channel type | DCH |
| - UL Transport channel identity | 5 |
| - TFS | |
| - CHOICE Transport channel type | Dedicated transport channels |
| - Dynamic Transport format information | |
| - RLC Size | Reference to clause 6.10.2.4.1.2.1 Parameter Set |
| - Number of TBs and TTI List | (This IE is repeated for TFI number.) |
| - Transmission Time Interval | Not Present |
| - Number of Transport blocks | Reference to clause 6.10.2.4.1.2.1 Parameter Set |
| - CHOICE Logical channel list | All |
| - Semi-static Transport Format information | |
| - Transmission time interval | Reference to clause 6.10.2.4.1.2.1 Parameter Set |
| - Type of channel coding | Reference to clause 6.10.2.4.1.2.1 Parameter Set |
| - Coding Rate | Reference to clause 6.10.2.4.1.2.1 Parameter Set |
| - Rate matching attribute | Reference to clause 6.10.2.4.1.2.1 Parameter Set |
| - CRC size | Reference to clause 6.10.2.4.1.2.1 Parameter Set |
| - DL CommonTransChInfo | |
| - SCCPCH TFCS | Not Present |
| - CHOICE mode | FDD |
| - CHOICE DL parameters | SameasUL |
| - Added or Reconfigured DL TrCH information | 4 TrCHs(DCH for DCCH and 3DCHs for DTCH) |
| - Downlink transport channel type | DCH |
| - DL Transport channel identity | 6 |
| - CHOICE DL parameters | Same as UL |
| - Uplink transport channel type | DCH |
| - UL TrCH identity | 1 |
| - DCH quality target | |
| - BLER Quality value | 0 |
| - Downlink transport channel type | DCH |
| - DL Transport channel identity | 7 |
| - CHOICE DL parameters | Same as UL |
| - Uplink transport channel type | DCH |
| - UL TrCH identity | 2 |

| | |
|---|---|
| - DCH quality target | |
| - BLER Quality value | 0 |
| - Downlink transport channel type | DCH |
| - DL Transport channel identity | 8 |
| - CHOICE DL parameters | Same as UL |
| - Uplink transport channel type | DCH |
| - UL TrCH identity | 3 |
| - DCH quality target | |
| - BLER Quality value | 0 |
| - Downlink transport channel type | DCH |
| - DL Transport channel identity | 10 |
| - CHOICE DL parameters | Same as UL |
| - Uplink transport channel type | DCH |
| - UL TrCH identity | 5 |
| - DCH quality target | |
| - BLER Quality value | 0 |
| - Pre-Defined Physical Channel Configuration | |
| - Uplink DPCH power control info Predef | |
| - CHOICE mode | FDD |
| - Power Control Algorithm | Algorithm1 |
| - CHOICE mode | FDD |
| - TFCI existence | FALSE |
| - Puncturing Limit | 0.88 |
| - Downlink DPCH power control info Predef | |
| - CHOICE mode | FDD |
| - Spreading factor | 128 |
| - Fixed or Flexible Position | Fixed |
| - TFCI existence | FALSE |

Contents of System Information Block type17 (3.84 Mcsps TDD and 1.28 Mcps TDD)

This system information block contains fast changing parameters for the configuration of the shared physical channels to be used in connected mode, so this is not present.

Contents of System Information Block type 18

| | |
|---|---|
| - Idle mode PLMN identities | |
| - PLMNs of intra-frequency cells list | Not present |
| - PLMNs of inter-frequency cells list | Not present |
| - PLMNs of inter-RAT cells list | Not present |
| - Connected mode PLMN identities | Not present |

## 6.1.1 SCCPCH configuration with Stand-alone SRB for PCCH in the first SCCPCH and Interactive/Background 32 kbps PS RAB + SRBs for CCCH/DCCH/BCCH in the second SCCPCH

Two SCCPCHs are used in this SYSTEM INFORMATION configuration. The first SCCPCH carries the PCH and the second SCCPCH carries the FACH for Interactive/Background 32 kbps PS RAB and the FACH for SRBs on CCCH/ DCCH/ BCCH.

This Reference System Configuration is the same as defined in clause 6.1, except for the following SIBs.

Contents of System Information Block type 5 (FDD)

| | |
|---|---|
| - SIB6 indicator | TRUE |
| - PICH Power offset | -5 dB |
| - CHOICE Mode | FDD |
| - AICH Power offset | -5 dB |
| - Primary CCPCH info | Not Present |
| - PRACH system information list | |
| - PRACH system information | |
| - PRACH info | |
| - CHOICE mode | FDD |
| - Available Signature | '0000 0000 1111 1111'B |
| - Available SF | 64 |

| | |
|---|---|
| - Preamble scrambling code number | 0 |
| - Puncturing Limit | 1.00 |
| - Available Sub Channel number | '1111 1111 1111'B |
| - Transport channel Identity | 15 |
| - RACH TFS | |
| - CHOICE Transport channel type | Common transport channels |
| - Dynamic Transport format information | |
| - RLC size | 168 |
| - Number of TB and TTI List | |
| - Number of Transport blocks | 1 |
| - CHOICE Mode | FDD |
| - CHOICE Logical channel List | Configured |
| - RLC size | 360 |
| - Number of TB and TTI List | |
| - Number of Transport blocks | 1 |
| - CHOICE Mode | FDD |
| - CHOICE Logical channel List | Configured |
| - Semi-static Transport Format information | |
| - Transmission time interval | 20 ms |
| - Type of channel coding | Convolutional |
| - Coding Rate | 1/2 |
| - Rate matching attribute | 150 |
| - CRC size | 16 |
| - RACH TFCS | |
| - CHOICE TFCI signalling | Normal |
| - TFCI Field 1 information | |
| - CHOICE TFCS representation | Complete reconfiguration |
| - TFCS complete reconfiguration information | |
| - CHOICE CTFC Size | 2 bit |
| - CTFC information | 0 |
| - Power offset information | |
| - CHOICE Gain Factors | Computed Gain Factor |
| - Reference TFC ID | 0 |
| - CHOICE Mode | FDD |
| - Power offset Pp-m | 0 dB |
| - CTFC information | 1 |
| - Power offset information | |
| - CHOICE Gain Factors | Signalled Gain Factor |
| - CHOICE mode | FDD |
| - Gain factor ßc | 11 |
| - Gain factor ßd | 15 |
| - Reference TFC ID | 0 |
| - CHOICE Mode | FDD |
| - Power offset Pp-m | 0 dB |
| - PRACH partitioning | |
| - Access Service Class | |
| - ASC Setting | Not Present |
| - ASC Setting | |
| - CHOICE mode | FDD |
| - Available signature Start Index | 0 (ASC#1) |
| - Available signature End Index | 7 (ASC#1) |
| - Assigned Sub-Channel Number | '1111'B<br>The first/ leftmost bit of the bit string contains the most significant bit of the Assigned Sub-Channel Number. |
| - ASC Setting | Not Present |
| - ASC Setting | |
| - CHOICE mode | FDD |
| - Available signature Start Index | 0 (ASC#3) |
| - Available signature End Index | 7 (ASC#3) |
| - Assigned Sub-Channel Number | '1111'B<br>The first/ leftmost bit of the bit string contains the most significant bit of the Assigned Sub-Channel Number. |
| - ASC Setting | Not Present |
| - ASC Setting | |
| - CHOICE mode | FDD |
| - Available signature Start Index | 0 (ASC#5) |
| - Available signature End Index | 7 (ASC#5) |
| - Assigned Sub-Channel Number | '1111'B |

| | |
|---|---|
| | The first/ leftmost bit of the bit string contains the most significant bit of the Assigned Sub-Channel Number. |
| - ASC Setting | Not Present |
| - ASC Setting | |
| - CHOICE mode | FDD |
| - Available signature Start Index | 0 (ASC#7) |
| - Available signature End Index | 7 (ASC#7) |
| - Assigned Sub-Channel Number | '1111'B<br>The first/ leftmost bit of the bit string contains the most significant bit of the Assigned Sub-Channel Number. |
| - Persistence scaling factor | |
| - Persistence scaling factor | 0.9 (for ASC#2) |
| - Persistence scaling factor | 0.9 (for ASC#3) |
| - Persistence scaling factor | 0.9 (for ASC#4) |
| - Persistence scaling factor | 0.9 (for ASC#5) |
| - Persistence scaling factor | 0.9 (for ASC#6) |
| - Persistence scaling factor | 0.9 (for ASC#7) |
| - AC-to-ASC mapping table | |
| - AC-to-ASC mapping | 6 (AC0-9) |
| - AC-to-ASC mapping | 5 (AC10) |
| - AC-to-ASC mapping | 4 (AC11) |
| - AC-to-ASC mapping | 3 (AC12) |
| - AC-to-ASC mapping | 2 (AC13) |
| - AC-to-ASC mapping | 1 (AC14) |
| - AC-to-ASC mapping | 0 (AC15) |
| - CHOICE mode | FDD |
| - Primary CPICH TX power | 31 |
| - Constant value | -10 |
| - PRACH power offset | |
| - Power Ramp Step | 3dB |
| - Preamble Retrans Max | 4 |
| - RACH transmission parameters | |
| - Mmax | 2 |
| - NB01min | 3 slot |
| - NB01max | 10 slot |
| - AICH info | |
| - Channelisation code | 3 |
| - STTD indicator | FALSE |
| - AICH transmission timing | 0 |
| - Secondary CCPCH system information | (For 2 SCCPCHs) |
| - Secondary CCPCH info | (SCCPCH for standalone PCH) |
| - CHOICE mode | FDD |
| - Secondary scrambling code | Not Present |
| - STTD indicator | FALSE |
| - Spreading factor | 128 |
| - Code number | 4 |
| - Pilot symbol existence | FALSE |
| - TFCI existence | FALSE |
| - Fixed or Flexible position | Fixed |
| - Timing offset | 30 |
| - TFCS | |
| - CHOICE TFCI signalling | Normal |
| - TFCI Field 1 information | |
| - CHOICE TFCS representation | Complete reconfiguration |
| - TFCS complete reconfiguration information | |
| - CHOICE CTFC Size | 2 bit |
| - CTFC information | 0 |
| - Power offset information | Not Present |
| - CTFC information | 1 |
| - Power offset information | Not Present |
| - FACH/PCH information | |
| - TFS | (PCH) |
| - CHOICE Transport channel type | Common transport channels |
| - Dynamic Transport format information | |
| - RLC Size | 240 |
| - Number of TB and TTI List | |
| - Number of Transport blocks | 0 |
| - Number of Transport blocks | 1 |

| | |
|---|---|
| - CHOICE Mode | FDD |
| - CHOICE Logical channel List | ALL |
| - Semi-static Transport Format information | |
| - Transmission time interval | 10 ms |
| - Type of channel coding | Convolutional |
| - Coding Rate | 1/2 |
| - Rate matching attribute | 230 |
| - CRC size | 16 bit |
| - Transport channel Identity | 12 (for PCH) |
| - CTCH indicator | FALSE |
| - PICH info | |
| - CHOICE mode | FDD |
| - Channelisation code | 2 |
| - Number of PI per frame | 18 |
| - STTD indicator | FALSE |
| - Secondary CCPCH info | (SCCPCH including two FACHs) |
| - CHOICE mode | FDD |
| - Secondary scrambling code | Not Present |
| - STTD indicator | FALSE |
| - Spreading factor | 64 |
| - Code number | 1 |
| - Pilot symbol existence | FALSE |
| - TFCI existence | TRUE (default value) |
| - Fixed or Flexible position | Flexible (default value) |
| - Timing offset | Not Present<br>Absence of this IE is equivalent to default value 0 |
| - TFCS | |
| - CHOICE TFCI signalling | Normal |
| - TFCI Field 1 information | |
| - CHOICE TFCS representation | Complete reconfiguration |
| - TFCS complete reconfiguration information | |
| - CHOICE CTFC Size | 4 bit |
| - CTFC information | 0 |
| - Power offset information | Not Present |
| - CTFC information | 1 |
| - Power offset information | Not Present |
| - CTFC information | 2 |
| - Power offset information | Not Present |
| - CTFC information | 3 |
| - Power offset information | Not Present |
| - CTFC information | 4 |
| - Power offset information | Not Present |
| - FACH/PCH information | |
| - TFS | (FACH) |
| - CHOICE Transport channel type | Common transport channels |
| - Dynamic Transport format information | |
| - RLC Size | 168 |
| - Number of TB and TTI List | |
| - Number of Transport blocks | 0 |
| - Number of Transport blocks | 1 |
| - Number of Transport blocks | 2 |
| - CHOICE Mode | FDD |
| - CHOICE Logical channel List | ALL |
| - Semi-static Transport Format information | |
| - Transmission time interval | 10 ms |
| - Type of channel coding | Convolutional |
| - Coding Rate | 1/2 |
| - Rate matching attribute | 220 |
| - CRC size | 16 bit |
| - Transport channel Identity | 13 (for FACH) |
| - CTCH indicator | FALSE |
| - TFS | (FACH) |
| - CHOICE Transport channel type | Common transport channels |
| - Dynamic Transport format information | |
| - RLC Size | 360 |
| - Number of TB and TTI List | |
| - Number of Transport blocks | 0 |
| - Number of Transport blocks | 1 |

| | |
|---|---|
| - CHOICE Mode | FDD |
| - CHOICE Logical channel List | ALL |
| - Semi-static Transport Format information | |
| - Transmission time interval | 10 ms |
| - Type of channel coding | Turbo |
| - Rate matching attribute | 130 |
| - CRC size | 16bit |
| - Transport channel Identity | 14 (for FACH) |
| - CTCH indicator | FALSE |
| - CBS DRX Level 1 information | Not Present |

Contents of System Information Block type 5 (3.84 Mcps TDD)

| | |
|---|---|
| - SIB6 indicator | FALSE |
| - CHOICE Mode | TDD |
| - TDD open loop power control | |
| - PUSCH system information | Not Present |
| - PDSCH system information | Not Present |
| - TDD open loop power control | |
| - Primary CCPCH Tx Power | 30 dbm |
| - CHOICE TDD option | 3.84 Mcps TDD /REL-4/ |
| - Alpha | (1/8) |
| - PRACH Constant Value | -10 |
| - DPCH Constant Value | -10 |
| - PUSCH Constant Value | -10 |
| - UE positioning related parameters | Not Present /REL-4/ |
| - Primary CCPCH info | |
| - CHOICE *mode* | TDD |
| - CHOICE TDD option | 3.84 Mcps TDD /REL-4/ |
| - CHOICE SyncCase | Sync Case 2 |
| - Timeslot | 0 |
| - Cell parameters ID | Not Present |
| - SCTD indicator | FALSE |
| - PRACH system information list | |
| - PRACH system information | |
| - PRACH info | |
| - CHOICE mode | TDD |
| - CHOICE TDD option | 3.84 Mcps TDD /REL-4/ |
| - Timeslot number | 14 |
| - PRACH Channelisation Code List | |
| - CHOICE SF | SF8 |
| - Channelisation Code List | |
| - Channelisation Code | 8/1 |
| - Channelisation Code | 8/2 |
| - Channelisation Code | 8/3 |
| - Channelisation Code | 8/4 |
| - PRACH Midamble | Direct |
| - PNBSCH allocation | Not Present /REL-4/ |
| - Transport channel Identity | 15 |
| - RACH TFS | |
| - CHOICE Transport channel type | Common transport channels |
| - Dynamic Transport format information | |
| - RLC size | 168 |
| - Number of TB and TTI List | |
| - Number of Transport blocks | 1 |
| - CHOICE Mode | TDD |
| - CHOICE Logical channel List | Configured |
| - RLC size | 360 |
| - Number of TB and TTI List | |
| - Number of Transport blocks | 1 |
| - CHOICE Mode | TDD |
| - CHOICE Logical channel List | Configured |
| - Semi-static Transport Format information | |
| - Transmission time interval | 20 ms |
| - Type of channel coding | Convolutional |
| - Coding Rate | 1/2 |
| - Rate matching attribute | 150 |

| | |
|---|---|
| - CRC size | 16 |
| - RACH TFCS | |
| - CHOICE TFCI signalling | Normal |
| - TFCI Field 1 information | |
| - CHOICE TFCS representation | Complete reconfiguration |
| - TFCS complete reconfiguration information | |
| - CHOICE CTFC Size | 2 bit |
| - CTFC information | 0 |
| - Power offset information | |
| - CHOICE Gain Factors | Computed Gain Factor |
| - Reference TFC ID | 0 |
| - CHOICE Mode | TDD |
| - Power offset Pp-m | 0 dB |
| - CTFC information | 1 |
| - Power offset information | |
| - CHOICE Gain Factors | Signalled Gain Factor |
| - CHOICE mode | TDD |
| - Gain factor ßc | 11 |
| - Gain factor ßd | 15 |
| - Reference TFC ID | 0 |
| - CHOICE Mode | TDD |
| - Power offset Pp-m | 0 dB |
| - PRACH partitioning | |
| - Access Service Class | |
| - ASC Setting | Not Present |
| - ASC Setting | |
| - CHOICE mode | TDD |
| - CHOICE TDD option | 3.84 Mcps TDD |
| - Available Channelisation codes indices | Not Present (Default all) |
| - CHOICE subchannel size | Size1 |
| - Available Subchannels | null |
| - ASC Setting | Not Present |
| - ASC Setting | |
| - CHOICE mode | TDD |
| - CHOICE TDD option | 3.84 Mcps TDD |
| - Available Channelisation codes indices | Not Present (Default all) |
| - CHOICE subchannel size | Size1 |
| - Available Subchannels | null |
| - ASC Setting | Not Present |
| - ASC Setting | |
| - CHOICE mode | TDD |
| - CHOICE TDD option | 3.84 Mcps TDD |
| - Available Channelisation codes indices | Not Present (Default all) |
| - CHOICE subchannel size | Size1 |
| - Available Subchannels | null |
| - ASC Setting | Not Present |
| - ASC Setting | |
| - CHOICE mode | TDD |
| - CHOICE TDD option | 3.84 Mcps TDD |
| - Available Channelisation codes indices | Not Present (Default all) |
| - CHOICE subchannel size | Size1 |
| - Available Subchannels | null |
| - Persistence scaling factor | |
| - Persistence scaling factor | 0.9 (for ASC#2) |
| - Persistence scaling factor | 0.9 (for ASC#3) |
| - Persistence scaling factor | 0.9 (for ASC#4) |
| - Persistence scaling factor | 0.9 (for ASC#5) |
| - Persistence scaling factor | 0.9 (for ASC#6) |
| - Persistence scaling factor | 0.9 (for ASC#7) |
| - AC-to-ASC mapping table | |
| - AC-to-ASC mapping | 6 (AC0-9) |
| - AC-to-ASC mapping | 5 (AC10) |
| - AC-to-ASC mapping | 4 (AC11) |
| - AC-to-ASC mapping | 3 (AC12) |
| - AC-to-ASC mapping | 2 (AC13) |
| - AC-to-ASC mapping | 1 (AC14) |
| - AC-to-ASC mapping | 0 (AC15) |
| - CHOICE mode | TDD (no data) |

| | |
|---|---|
| - Secondary CCPCH system information | (For 2 SCCPCHs) |
| - Secondary CCPCH info | (SCCPCH for standalone PCH) |
| - CHOICE mode | TDD |
| - Secondary scrambling code | Not Present |
| - STTD indicator | FALSE |
| - Spreading factor | 128 |
| - Code number | 4 |
| - Pilot symbol existence | FALSE |
| - TFCI existence | FALSE |
| - Fixed or Flexible position | Fixed |
| - Timing offset | 30 |
| - TFCS | |
| - CHOICE TFCI signalling | Normal |
| - TFCI Field 1 information | |
| - CHOICE TFCS representation | Complete reconfiguration |
| - TFCS complete reconfiguration information | |
| - CHOICE CTFC Size | 2 bit |
| - CTFC information | 0 |
| - Power offset information | Not Present |
| - CTFC information | 1 |
| - Power offset information | Not Present |
| - FACH/PCH information | |
| - TFS | (PCH) |
| - CHOICE Transport channel type | Common transport channels |
| - Dynamic Transport format information | |
| - RLC Size | 240 |
| - Number of TB and TTI List | |
| - Number of Transport blocks | 0 |
| - Number of Transport blocks | 1 |
| - CHOICE Mode | TDD |
| - CHOICE Logical channel List | ALL |
| - Semi-static Transport Format information | |
| - Transmission time interval | 10 ms |
| - Type of channel coding | Convolutional |
| - Coding Rate | 1/2 |
| - Rate matching attribute | 230 |
| - CRC size | 16 bit |
| - Transport channel Identity | 12 (for PCH) |
| - CTCH indicator | FALSE |
| - PICH info | |
| - CHOICE mode | TDD |
| - CHOICE TDD option | 3.84 Mcps TDD |
| - Timeslot number | 0 |
| - Midamble shift and burst type | 4 |
| - CHOICE Burst Type | Type 1 |
| - Midamble Allocation Mode | Default midamble |
| - Midamble configuration burst type 1 and 3 | 8 |
| - Midamble Shift | Not Present |
| - Channelisation code | 16/16 |
| - Repetition period/length | 64/2 |
| - Offset | 0 |
| - Paging indicator length | 4 |
| - NGAP | 4 |
| - NPCH | 2 |
| - Secondary CCPCH info | (SCCPCH including two FACHs) |
| - CHOICE mode | TDD |
| - Secondary scrambling code | Not Present |
| - STTD indicator | FALSE |
| - Spreading factor | 64 |
| - Code number | 1 |
| - Pilot symbol existence | FALSE |
| - TFCI existence | TRUE (default value) |
| - Fixed or Flexible position | Flexible (default value) |
| - Timing offset | Not Present<br>Absence of this IE is equivalent to default value 0 |

| | |
|---|---|
| - TFCS | |
| - CHOICE TFCI signalling | Normal |
| - TFCI Field 1 information | |
| - CHOICE TFCS representation | Complete reconfiguration |
| - TFCS complete reconfiguration information | |
| - CHOICE CTFC Size | 4 bit |
| - CTFC information | 0 |
| - Power offset information | Not Present |
| - CTFC information | 1 |
| - Power offset information | Not Present |
| - CTFC information | 2 |
| - Power offset information | Not Present |
| - CTFC information | 3 |
| - Power offset information | Not Present |
| - CTFC information | 4 |
| - Power offset information | Not Present |
| - FACH/PCH information | |
| - TFS | (FACH) |
| - CHOICE Transport channel type | Common transport channels |
| - Dynamic Transport format information | |
| - RLC Size | 168 |
| - Number of TB and TTI List | |
| - Number of Transport blocks | 0 |
| - Number of Transport blocks | 1 |
| - Number of Transport blocks | 2 |
| - CHOICE Mode | TDD |
| - CHOICE Logical channel List | ALL |
| - Semi-static Transport Format information | |
| - Transmission time interval | 10 ms |
| - Type of channel coding | Convolutional |
| - Coding Rate | 1/2 |
| - Rate matching attribute | 220 |
| - CRC size | 16 bit |
| - Transport channel Identity | 13 (for FACH) |
| - CTCH indicator | FALSE |
| - TFS | (FACH) |
| - CHOICE Transport channel type | Common transport channels |
| - Dynamic Transport format information | |
| - RLC Size | 360 |
| - Number of TB and TTI List | |
| - Number of Transport blocks | 0 |
| - Number of Transport blocks | 1 |
| - CHOICE Mode | TDD |
| - CHOICE Logical channel List | ALL |
| - Semi-static Transport Format information | |
| - Transmission time interval | 10 ms |
| - Type of channel coding | Turbo |
| - Rate matching attribute | 130 |
| - CRC size | 16bit |
| - Transport channel Identity | 14 (for FACH) |
| - CTCH indicator | FALSE |
| - CBS DRX Level 1 information | Not Present |

Contents of System Information Block type 5 (1.28 Mcps TDD)

<FFS>

Contents of System Information Block type 6 in connected mode (FDD)

| | |
|---|---|
| - PICH Power offset | -5 dB |
| - CHOICE Mode | FDD |
| - AICH Power offset | -5 dB |
| - Primary CCPCH info | Not Present |
| - PRACH system information list | Not Present |
| - Secondary CCPCH system information | Not Present |
| - CBS DRX Level 1 information | Not Present |

Contents of System Information Block type 6 in connected mode (3.84 Mcps TDD)

None

Contents of System Information Block type 6 in connected mode (1.28 Mcps TDD)

<FFS>

## 6.1.2 SCCPCH configuration with Stand-alone SRB for PCCH in the first SCCPCH, RB for CTCH + SRBs for CCCH/BCCH in the second SCCPCH and Interactive/Background 32 kbps PS RAB + SRBs for CCCH/DCCH/BCCH in the third SCCPCH (FDD only)

Three SCCPCHs are used in this SYSTEM INFORMATION configuration. The first SCCPCH carries the PCH. The second SCCPCH carries the FACH for CTCH (Cell Broadcast Service) and the FACH for SRBs on CCCH/ BCCH for idle mode UEs. The third SCCPCH carries the FACH for Interactive/Background 32 kbps PS RAB and the FACH for SRBs on CCCH/ DCCH/ BCCH for connected mode UEs.

This Reference System Configuration is the same as defined in clause 6.1, except for the following SIBs.

Contents of System Information Block type 5 (FDD)

| | |
|---|---|
| - SIB6 indicator | TRUE |
| - PICH Power offset | -5 dB |
| - CHOICE Mode | FDD |
| - AICH Power offset | -5 dB |
| - Primary CCPCH info | Not Present |
| - PRACH system information list | |
| - PRACH system information | |
| - PRACH info | |
| - CHOICE mode | FDD |
| - Available Signature | '0000 0000 1111 1111'B |
| - Available SF | 64 |
| - Preamble scrambling code number | 0 |
| - Puncturing Limit | 1.00 |
| - Available Sub Channel number | '1111 1111 1111'B |
| - Transport channel Identity | 15 |
| - RACH TFS | |
| - CHOICE Transport channel type | Common transport channels |
| - Dynamic Transport format information | |
| - RLC size | 168 |
| - Number of TB and TTI List | |
| - Number of Transport blocks | 1 |
| - CHOICE Mode | FDD |
| - CHOICE Logical channel List | Configured |
| - RLC size | 360 |
| - Number of TB and TTI List | |
| - Number of Transport blocks | 1 |
| - CHOICE Mode | FDD |
| - CHOICE Logical channel List | Configured |
| - Semi-static Transport Format information | |
| - Transmission time interval | 20 ms |
| - Type of channel coding | Convolutional |
| - Coding Rate | 1/2 |
| - Rate matching attribute | 150 |
| - CRC size | 16 |
| - RACH TFCS | |
| - CHOICE TFCI signalling | Normal |
| - TFCI Field 1 information | |
| - CHOICE TFCS representation | Complete reconfiguration |
| - TFCS complete reconfiguration information | |
| - CHOICE CTFC Size | 2 bit |
| - CTFC information | 0 |
| - Power offset information | |
| - CHOICE Gain Factors | Computed Gain Factor |
| - Reference TFC ID | 0 |

| | |
|---|---|
| - CHOICE mode | FDD |
| - Power offset Pp-m | 0 dB |
| - CTFC information | 1 |
| - Power offset information | |
| - CHOICE Gain Factors | Signalled Gain Factor |
| - CHOICE mode | FDD |
| - Gain factor ßc | 11 |
| - Gain factor ßd | 15 |
| - Reference TFC ID | 0 |
| - CHOICE Mode | FDD |
| - Power offset Pp-m | 0 dB |
| - PRACH partitioning | |
| - Access Service Class | |
| - ASC Setting | Not Present |
| - ASC Setting | |
| - CHOICE mode | FDD |
| - Available signature Start Index | 0 (ASC#1) |
| - Available signature End Index | 7 (ASC#1) |
| - Assigned Sub-Channel Number | '1111'B<br>The first/ leftmost bit of the bit string contains the most significant bit of the Assigned Sub-Channel Number. |
| - ASC Setting | Not Present |
| - ASC Setting | |
| - CHOICE mode | FDD |
| - Available signature Start Index | 0 (ASC#3) |
| - Available signature End Index | 7 (ASC#3) |
| - Assigned Sub-Channel Number | '1111'B<br>The first/ leftmost bit of the bit string contains the most significant bit of the Assigned Sub-Channel Number. |
| - ASC Setting | Not Present |
| - ASC Setting | |
| - CHOICE mode | FDD |
| - Available signature Start Index | 0 (ASC#5) |
| - Available signature End Index | 7 (ASC#5) |
| - Assigned Sub-Channel Number | '1111'B<br>The first/ leftmost bit of the bit string contains the most significant bit of the Assigned Sub-Channel Number. |
| - ASC Setting | Not Present |
| - ASC Setting | |
| - CHOICE mode | FDD |
| - Available signature Start Index | 0 (ASC#7) |
| - Available signature End Index | 7 (ASC#7) |
| - Assigned Sub-Channel Number | '1111'B<br>The first/ leftmost bit of the bit string contains the most significant bit of the Assigned Sub-Channel Number. |
| - Persistence scaling factor | |
| - Persistence scaling factor | 0.9 (for ASC#2) |
| - Persistence scaling factor | 0.9 (for ASC#3) |
| - Persistence scaling factor | 0.9 (for ASC#4) |
| - Persistence scaling factor | 0.9 (for ASC#5) |
| - Persistence scaling factor | 0.9 (for ASC#6) |
| - Persistence scaling factor | 0.9 (for ASC#7) |
| - AC-to-ASC mapping table | |
| - AC-to-ASC mapping | 6 (AC0-9) |
| - AC-to-ASC mapping | 5 (AC10) |
| - AC-to-ASC mapping | 4 (AC11) |
| - AC-to-ASC mapping | 3 (AC12) |
| - AC-to-ASC mapping | 2 (AC13) |
| - AC-to-ASC mapping | 1 (AC14) |
| - AC-to-ASC mapping | 0 (AC15) |
| - CHOICE mode | FDD |
| - Primary CPICH TX power | 31 |
| - Constant value | -10 |
| - PRACH power offset | |
| - Power Ramp Step | 3dB |
| - Preamble Retrans Max | 4 |
| - RACH transmission parameters | |
| - Mmax | 2 |

| | |
|---|---|
| - NB01min | 3 slot |
| - NB01max | 10 slot |
| - AICH info | |
| - Channelisation code | 3 |
| - STTD indicator | FALSE |
| - AICH transmission timing | 0 |
| - Secondary CCPCH system information | (For 2 SCCPCHs) |
| - Secondary CCPCH info | (SCCPCH for standalone PCH) |
| - CHOICE mode | FDD |
| - Secondary scrambling code | Not Present |
| - STTD indicator | FALSE |
| - Spreading factor | 128 |
| - Code number | 4 |
| - Pilot symbol existence | FALSE |
| - TFCI existence | FALSE |
| - Fixed or Flexible position | Fixed |
| - Timing offset | 30 |
| - TFCS | |
| - CHOICE TFCI signalling | Normal |
| - TFCI Field 1 information | |
| - CHOICE TFCS representation | Complete reconfiguration |
| - TFCS complete reconfiguration information | |
| - CHOICE CTFC Size | 2 bit |
| - CTFC information | 0 |
| - Power offset information | Not Present |
| - CTFC information | 1 |
| - Power offset information | Not Present |
| - FACH/PCH information | |
| - TFS | (PCH) |
| - CHOICE Transport channel type | Common transport channels |
| - Dynamic Transport format information | |
| - RLC Size | 240 |
| - Number of TB and TTI List | |
| - Number of Transport blocks | 0 |
| - Number of Transport blocks | 1 |
| - CHOICE Mode | FDD |
| - CHOICE Logical channel List | ALL |
| - Semi-static Transport Format information | |
| - Transmission time interval | 10 ms |
| - Type of channel coding | Convolutional |
| - Coding Rate | 1/2 |
| - Rate matching attribute | 230 |
| - CRC size | 16 bit |
| - Transport channel Identity | 12 (for PCH) |
| - CTCH indicator | FALSE |
| - PICH info | |
| - CHOICE mode | FDD |
| - Channelisation code | 2 |
| - Number of PI per frame | 18 |
| - STTD indicator | FALSE |
| - Secondary CCPCH info | (SCCPCH including two FACHs) |
| - CHOICE mode | FDD |
| - Secondary scrambling code | Not Present |
| - STTD indicator | FALSE |
| - Spreading factor | 128 |
| - Code number | 5 |
| - Pilot symbol existence | FALSE |
| - TFCI existence | TRUE (default value) |
| - Fixed or Flexible position | Flexible (default value) |
| - Timing offset | Not Present<br>Absence of this IE is equivalent to default value 0 |
| - TFCS | |
| - CHOICE TFCI signalling | Normal |
| - TFCI Field 1 information | |
| - CHOICE TFCS representation | Complete reconfiguration |
| - TFCS complete reconfiguration information | |

| | |
|---|---|
| - CHOICE CTFC Size | 2 bit |
| - CTFC information | 0 |
| - Power offset information | Not Present |
| - CTFC information | 1 |
| - Power offset information | Not Present |
| - CTFC information | 2 |
| - Power offset information | Not Present |
| - FACH/PCH information | |
| - TFS | (FACH) |
| - CHOICE Transport channel type | Common transport channels |
| - Dynamic Transport format information | |
| - RLC Size | 168 |
| - Number of TB and TTI List | |
| - Number of Transport blocks | 0 |
| - Number of Transport blocks | 1 |
| - CHOICE Mode | FDD |
| - CHOICE Logical channel List | ALL |
| - Semi-static Transport Format information | |
| - Transmission time interval | 10 ms |
| - Type of channel coding | Convolutional |
| - Coding Rate | 1/3 |
| - Rate matching attribute | 220 |
| - CRC size | 16 bit |
| - Transport channel Identity | 13 (for FACH) |
| - CTCH indicator | FALSE |
| - TFS | (FACH) |
| - CHOICE Transport channel type | Common transport channels |
| - Dynamic Transport format information | |
| - RLC Size | 168 |
| - Number of TB and TTI List | |
| - Number of Transport blocks | 0 |
| - Number of Transport blocks | 1 |
| - CHOICE Mode | FDD |
| - CHOICE Logical channel List | ALL |
| - Semi-static Transport Format information | |
| - Transmission time interval | 10 ms |
| - Type of channel coding | Convolutional |
| - Coding Rate | 1/3 |
| - Rate matching attribute | 220 |
| - CRC size | 16bit |
| - Transport channel Identity | 14 (for FACH) |
| - CTCH indicator | TRUE |
| - CBS DRX Level 1 information | |
| - Period of CTCH allocation (N) | 2 |
| - CBS frame offset (K) | 0 |

Contents of System Information Block type 6 in connected mode (FDD)

| | |
|---|---|
| - PICH Power offset | -5 dB |
| - CHOICE Mode | FDD |
| - AICH Power offset | -5 dB |
| - Primary CCPCH info | Not present |
| - PRACH system information list | Not Present |
| - Secondary CCPCH system information | |
| - Secondary CCPCH info | (SCCPCH including two FACHs) |
| - CHOICE mode | FDD |
| - Secondary scrambling code | Not Present |
| - STTD indicator | FALSE |
| - Spreading factor | 64 |
| - Code number | 1 |
| - Pilot symbol existence | FALSE |
| - TFCI existence | TRUE (default value) |
| - Fixed or Flexible position | Flexible (default value) |
| - Timing offset | 90 |
| - TFCS | |
| - CHOICE TFCI signalling | Normal |
| - TFCI Field 1 information | |

| | |
|---|---|
| - CHOICE TFCS representation | Complete reconfiguration |
| - TFCS complete reconfiguration information | |
| - CHOICE CTFC Size | 4 bit |
| - CTFC information | 0 |
| - Power offset information | Not Present |
| - CTFC information | 1 |
| - Power offset information | Not Present |
| - CTFC information | 2 |
| - Power offset information | Not Present |
| - CTFC information | 3 |
| - Power offset information | Not Present |
| - CTFC information | 4 |
| - Power offset information | Not Present |
| - FACH/PCH information | |
| - TFS | (FACH) |
| - CHOICE Transport channel type | Common transport channels |
| - Dynamic Transport format information | |
| - RLC Size | 168 |
| - Number of TB and TTI List | |
| - Number of Transport blocks | 0 |
| - Number of Transport blocks | 1 |
| - Number of Transport blocks | 2 |
| - CHOICE Mode | FDD |
| - CHOICE Logical channel List | ALL |
| - Semi-static Transport Format information | |
| - Transmission time interval | 10 ms |
| - Type of channel coding | Convolutional |
| - Coding Rate | 1/2 |
| - Rate matching attribute | 220 |
| - CRC size | 16 bit |
| - Transport channel Identity | 16 (for FACH) |
| - CTCH indicator | FALSE |
| - TFS | (FACH) |
| - CHOICE Transport channel type | Common transport channels |
| - Dynamic Transport format information | |
| - RLC Size | 360 |
| - Number of TB and TTI List | |
| - Number of Transport blocks | 0 |
| - Number of Transport blocks | 1 |
| - CHOICE Mode | FDD |
| - CHOICE Logical channel List | ALL |
| - Semi-static Transport Format information | |
| - Transmission time interval | 10 ms |
| - Type of channel coding | Turbo |
| - Rate matching attribute | 130 |
| - CRC size | 16bit |
| - Transport channel Identity | 17 (for FACH) |
| - CTCH indicator | FALSE |
| - CBS DRX Level 1 information | Not Present |

## 6.1.3 SCCPCH configuration with Stand-alone SRB for PCCH in the first SCCPCH and Interactive/Background 32 kbps PS RAB + SRBs for CCCH/DCCH/BCCH in the second and third SCCPCHs

Three SCCPCHs are used in this SYSTEM INFORMATION configuration. The first SCCPCH carries the PCH and both the second and third SCCPCHs carry the FACH for Interactive/Background 32 kbps PS RAB and the FACH for SRBs on CCCH/ DCCH/ BCCH.

This Reference System Configuration is the same as defined in clause 6.1, except for the following SIBs. (SIB6 is not used in this configuration.)

Contents of Scheduling Block 1 (FDD)

| | |
|---|---|
| - References to other system information blocks | |
| - Scheduling information | |
| - CHOICE Value tag | Not Present |

| | |
|---|---|
| - SEG_COUNT | 1 |
| - SIB_REP | 16 |
| - SIB_POS | 4 |
| - SIB_POS offset info | Not Present |
| - SIB type SIBs only | System Information Type 7 |
| - Scheduling information | |
| - CHOICE Value tag | Cell Value tag |
| - Cell Value tag | A valid Cell value tag value as defined in TS 25.331 [34] |
| - SEG_COUNT | 3 |
| - SIB_REP | 64 |
| - SIB_POS | 58 |
| - SIB_POS offset info | |
| - SIB_OFF | 2 |
| - SIB_OFF | 2 |
| - SIB type SIBs only | System Information Type 11 |
| - Scheduling information | |
| - CHOICE Value tag | Cell Value tag |
| - Cell Value tag | A valid Cell value tag value as defined in TS 25.331 [34] |
| - SEG_COUNT | 3 |
| - SIB_REP | 64 |
| - SIB_POS | 26 |
| - SIB_POS offset info | |
| - SIB_OFF | 2 |
| - SIB_OFF | 2 |
| - SIB type SIBs only | System Information Type 12 |
| - Scheduling information | |
| - CHOICE Value tag | PLMN Value tag |
| - PLMN Value tag | A valid PLMN value tag value as defined in TS 25.331 [34] |
| - SEG_COUNT | 1 |
| - SIB_REP | 64 |
| - SIB_POS | 36 |
| - SIB_POS offset info | Not Present |
| - SIB type SIBs only | System Information Type 18 |

Contents of System Information Block type 5 (FDD)

| | |
|---|---|
| - SIB6 indicator | FALSE |
| - PICH Power offset | -5 dB |
| - CHOICE Mode | FDD |
| - AICH Power offset | -5 dB |
| - Primary CCPCH info | Not Present |
| - PRACH system information list | |
| - PRACH system information | |
| - PRACH info | |
| - CHOICE mode | FDD |
| - Available Signature | '0000 0000 1111 1111'B |
| - Available SF | 64 |
| - Preamble scrambling code number | 0 |
| - Puncturing Limit | 1.00 |
| - Available Sub Channel number | '1111 1111 1111'B |
| - Transport channel Identity | 15 |
| - RACH TFS | |
| - CHOICE Transport channel type | Common transport channels |
| - Dynamic Transport format information | |
| - RLC size | 168 |
| - Number of TB and TTI List | |
| - Number of Transport blocks | 1 |
| - CHOICE Mode | FDD |
| - CHOICE Logical channel List | Configured |
| - RLC size | 360 |
| - Number of TB and TTI List | |
| - Number of Transport blocks | 1 |
| - CHOICE Mode | FDD |
| - CHOICE Logical channel List | Configured |

| | |
|---|---|
| - Semi-static Transport Format information | |
| - Transmission time interval | 20 ms |
| - Type of channel coding | Convolutional |
| - Coding Rate | 1/2 |
| - Rate matching attribute | 150 |
| - CRC size | 16 |
| - RACH TFCS | |
| - CHOICE TFCI signalling | Normal |
| - TFCI Field 1 information | |
| - CHOICE TFCS representation | Complete reconfiguration |
| - TFCS complete reconfiguration information | |
| - CHOICE CTFC Size | 2 bit |
| - CTFC information | 0 |
| - Power offset information | |
| - CHOICE Gain Factors | Computed Gain Factor |
| - Reference TFC ID | 0 |
| - CHOICE mode | FDD |
| - Power offset Pp-m | 0 dB |
| - CTFC information | 1 |
| - Power offset information | |
| - CHOICE Gain Factors | Signalled Gain Factor |
| - CHOICE mode | FDD |
| - Gain factor ßc | 11 |
| - Gain factor ßd | 15 |
| - Reference TFC ID | 0 |
| - CHOICE Mode | FDD |
| - Power offset Pp-m | 0 dB |
| - PRACH partitioning | |
| - Access Service Class | |
| - ASC Setting | Not Present |
| - ASC Setting | |
| - CHOICE mode | FDD |
| - Available signature Start Index | 0 (ASC#1) |
| - Available signature End Index | 7 (ASC#1) |
| - Assigned Sub-Channel Number | '1111'B<br>The first/ leftmost bit of the bit string contains the most significant bit of the Assigned Sub-Channel Number. |
| - ASC Setting | Not Present |
| - ASC Setting | |
| - CHOICE mode | FDD |
| - Available signature Start Index | 0 (ASC#3) |
| - Available signature End Index | 7 (ASC#3) |
| - Assigned Sub-Channel Number | '1111'B<br>The first/ leftmost bit of the bit string contains the most significant bit of the Assigned Sub-Channel Number. |
| - ASC Setting | Not Present |
| - ASC Setting | |
| - CHOICE mode | FDD |
| - Available signature Start Index | 0 (ASC#5) |
| - Available signature End Index | 7 (ASC#5) |
| - Assigned Sub-Channel Number | '1111'B<br>The first/ leftmost bit of the bit string contains the most significant bit of the Assigned Sub-Channel Number. |
| - ASC Setting | Not Present |
| - ASC Setting | |
| - CHOICE mode | FDD |
| - Available signature Start Index | 0 (ASC#7) |
| - Available signature End Index | 7 (ASC#7) |
| - Assigned Sub-Channel Number | '1111'B<br>The first/ leftmost bit of the bit string contains the most significant bit of the Assigned Sub-Channel Number. |
| - Persistence scaling factor | |
| - Persistence scaling factor | 0.9 (for ASC#2) |
| - Persistence scaling factor | 0.9 (for ASC#3) |

| | |
|---|---|
| - Persistence scaling factor | 0.9 (for ASC#4) |
| - Persistence scaling factor | 0.9 (for ASC#5) |
| - Persistence scaling factor | 0.9 (for ASC#6) |
| - Persistence scaling factor | 0.9 (for ASC#7) |
| - AC-to-ASC mapping table | |
| - AC-to-ASC mapping | 6 (AC0-9) |
| - AC-to-ASC mapping | 5 (AC10) |
| - AC-to-ASC mapping | 4 (AC11) |
| - AC-to-ASC mapping | 3 (AC12) |
| - AC-to-ASC mapping | 2 (AC13) |
| - AC-to-ASC mapping | 1 (AC14) |
| - AC-to-ASC mapping | 0 (AC15) |
| - CHOICE mode | FDD |
| - Primary CPICH TX power | 31 |
| - Constant value | -10 |
| - PRACH power offset | |
| - Power Ramp Step | 3dB |
| - Preamble Retrans Max | 4 |
| - RACH transmission parameters | |
| - Mmax | 2 |
| - NB01min | 3 slot |
| - NB01max | 10 slot |
| - AICH info | |
| - Channelisation code | 3 |
| - STTD indicator | FALSE |
| - AICH transmission timing | 0 |
| - Secondary CCPCH system information | (For 3 SCCPCHs) |
| - Secondary CCPCH info | (SCCPCH for standalone PCH) |
| - CHOICE mode | FDD |
| - Secondary scrambling code | Not Present |
| - STTD indicator | FALSE |
| - Spreading factor | 128 |
| - Code number | 6 |
| - Pilot symbol existence | FALSE |
| - TFCI existence | FALSE |
| - Fixed or Flexible position | Fixed |
| - Timing offset | 30 |
| - TFCS | |
| - CHOICE TFCI signalling | Normal |
| - TFCI Field 1 information | |
| - CHOICE TFCS representation | Complete reconfiguration |
| - TFCS complete reconfiguration information | |
| - CHOICE CTFC Size | 2 bit |
| - CTFC information | 0 |
| - Power offset information | Not Present |
| - CTFC information | 1 |
| - Power offset information | Not Present |
| - FACH/PCH information | |
| - TFS | (PCH) |
| - CHOICE Transport channel type | Common transport channels |
| - Dynamic Transport format information | |
| - RLC Size | 240 |
| - Number of TB and TTI List | |
| - Number of Transport blocks | 0 |
| - Number of Transport blocks | 1 |
| - CHOICE Mode | FDD |
| - CHOICE Logical channel List | ALL |
| - Semi-static Transport Format information | |
| - Transmission time interval | 10 ms |
| - Type of channel coding | Convolutional |
| - Coding Rate | 1/2 |
| - Rate matching attribute | 230 |
| - CRC size | 16 bit |
| - Transport channel Identity | 12 (for PCH) |
| - CTCH indicator | FALSE |
| - PICH info | |
| - CHOICE mode | FDD |
| - Channelisation code | 2 |

| | |
|---|---|
| - Number of PI per frame | 18 |
| - STTD indicator | FALSE |
| - Secondary CCPCH info | (SCCPCH including two FACHs) |
| - CHOICE mode | FDD |
| - Secondary scrambling code | Not Present |
| - STTD indicator | FALSE |
| - Spreading factor | 64 |
| - Code number | 1 |
| - Pilot symbol existence | FALSE |
| - TFCI existence | TRUE (default value) |
| - Fixed or Flexible position | Flexible (default value) |
| - Timing offset | Not Present<br>Absence of this IE is equivalent to default value 0 |
| - TFCS | |
| - CHOICE TFCI signalling | Normal |
| - TFCI Field 1 information | |
| - CHOICE TFCS representation | Complete reconfiguration |
| - TFCS complete reconfiguration information | |
| - CHOICE CTFC Size | 4 bit |
| - CTFC information | 0 |
| - Power offset information | Not Present |
| - CTFC information | 1 |
| - Power offset information | Not Present |
| - CTFC information | 2 |
| - Power offset information | Not Present |
| - CTFC information | 3 |
| - Power offset information | Not Present |
| - CTFC information | 4 |
| - Power offset information | Not Present |
| - FACH/PCH information | |
| - TFS | (FACH) |
| - CHOICE Transport channel type | Common transport channels |
| - Dynamic Transport format information | |
| - RLC Size | 168 |
| - Number of TB and TTI List | |
| - Number of Transport blocks | 0 |
| - Number of Transport blocks | 1 |
| - Number of Transport blocks | 2 |
| - CHOICE Mode | FDD |
| - CHOICE Logical channel List | ALL |
| - Semi-static Transport Format information | |
| - Transmission time interval | 10 ms |
| - Type of channel coding | Convolutional |
| - Coding Rate | 1/2 |
| - Rate matching attribute | 220 |
| - CRC size | 16 bit |
| - Transport channel Identity | 13 (for FACH) |
| - CTCH indicator | FALSE |
| - TFS | (FACH) |
| - CHOICE Transport channel type | Common transport channels |
| - Dynamic Transport format information | |
| - RLC Size | 360 |
| - Number of TB and TTI List | |
| - Number of Transport blocks | 0 |
| - Number of Transport blocks | 1 |
| - CHOICE Mode | FDD |
| - CHOICE Logical channel List | ALL |
| - Semi-static Transport Format information | |
| - Transmission time interval | 10 ms |
| - Type of channel coding | Turbo |
| - Rate matching attribute | 130 |
| - CRC size | 16bit |
| - Transport channel Identity | 14 (for FACH) |
| - CTCH indicator | FALSE |
| - Secondary CCPCH info | (SCCPCH including two FACHs) |
| - CHOICE mode | FDD |
| - Secondary scrambling code | Not Present |
| - STTD indicator | FALSE |

| | |
|---|---|
| - Spreading factor | 64 |
| - Code number | 2 |
| - Pilot symbol existence | FALSE |
| - TFCI existence | TRUE (default value) |
| - Fixed or Flexible position | Flexible (default value) |
| - Timing offset | 90 |
| - TFCS | |
| - CHOICE TFCI signalling | Normal |
| - TFCI Field 1 information | |
| - CHOICE TFCS representation | Complete reconfiguration |
| - TFCS complete reconfiguration information | |
| - CHOICE CTFC Size | 4 bit |
| - CTFC information | 0 |
| - Power offset information | Not Present |
| - CTFC information | 1 |
| - Power offset information | Not Present |
| - CTFC information | 2 |
| - Power offset information | Not Present |
| - CTFC information | 3 |
| - Power offset information | Not Present |
| - CTFC information | 4 |
| - Power offset information | Not Present |
| - FACH/PCH information | |
| - TFS | (FACH) |
| - CHOICE Transport channel type | Common transport channels |
| - Dynamic Transport format information | |
| - RLC Size | 168 |
| - Number of TB and TTI List | |
| - Number of Transport blocks | 0 |
| - Number of Transport blocks | 1 |
| - Number of Transport blocks | 2 |
| - CHOICE Mode | FDD |
| - CHOICE Logical channel List | ALL |
| - Semi-static Transport Format information | |
| - Transmission time interval | 10 ms |
| - Type of channel coding | Convolutional |
| - Coding Rate | 1/2 |
| - Rate matching attribute | 220 |
| - CRC size | 16 bit |
| - Transport channel Identity | 16 (for FACH) |
| - CTCH indicator | FALSE |
| - TFS | (FACH) |
| - CHOICE Transport channel type | Common transport channels |
| - Dynamic Transport format information | |
| - RLC Size | 360 |
| - Number of TB and TTI List | |
| - Number of Transport blocks | 0 |
| - Number of Transport blocks | 1 |
| - CHOICE Mode | FDD |
| - CHOICE Logical channel List | ALL |
| - Semi-static Transport Format information | |
| - Transmission time interval | 10 ms |
| - Type of channel coding | Turbo |
| - Rate matching attribute | 130 |
| - CRC size | 16bit |
| - Transport channel Identity | 17 (for FACH) |
| - CTCH indicator | FALSE |
| - CBS DRX Level 1 information | Not Present |

Contents of System Information Block type 5 (3.84 Mcps TDD)

<FFS>

Contents of System Information Block type 5 (1.28 Mcps TDD)

<FFS>

## 6.1.4 Default parameters for 1 to 8 cell environments

Default settings for cell No.1 (FDD)

| | |
|---|---|
| Downlink input level | Reference clause 6.10 "Parameter Set" |
| Uplink output power | Minimum supported by the UE's power class. |
| PCCPCH/PCPICH carrier number | Reference clause 6.10 "Parameter Set" |
| Cell Channel Description | |
| - Primary CPICH info | |
| - Primary scrambling code | 100 |

Contents of System Information Block type 11 for cell No.1 (FDD)

See clause 6.1.0b for contents of System Information Block type 11 (FDD) for cell 1.

Contents of System Information Block type 12 in connected mode for cell No.1 (FDD)

See clause 6.1.0b for contents of System Information Block type 12 (FDD) for cell 1.

Default settings for cell No.1 (TDD)

| | |
|---|---|
| Downlink input level | Reference clause 6 Parameter Set |
| Uplink output power | Minimum supported by the UE's power class. |
| PCCPCH/PCPICH carrier number | Reference clause 6 Parameter Set |
| Cell Channel Description | |
| - Primary CCPCH info | |
| - Cell parameters ID | 0 |

Contents of System Information Block type 11 for cell No.1 (TDD)

See clause 6.1.0b for contents of System Information Block type 11 (TDD) for cell 1.

Contents of System Information Block type 12 in connected mode for cell No.1 (TDD)

See clause 6.1.0b for contents of System Information Block type 12 (TDD) for cell 1.

Cell No.2

The contents of SYSTEM INFORMATION BLOCK TYPE 1 to 16 messages for cell No.2 are identical to those of cell No.1 with the following exceptions.

| | |
|---|---|
| Cell identity | 0000 0000 0000 0000 0000 0000 0010B |
| URA identity | 0000 0000 0000 0001B |

Default settings for cell No.2 (FDD)

| | |
|---|---|
| Downlink input level | Reference clause 6.10 "Parameter Set" |
| Uplink output power | Minimum supported by the UE's power class. |
| PCCPCH/PCPICH carrier number | Reference clause 6.10 "Parameter Set" |
| Cell Channel Description | |
| - Primary CPICH info | |
| - Primary scrambling code | 150 |

Contents of System Information Block type 11 for cell No.2 (FDD)

| | | |
|---|---|---|
| **- Intra-frequency measurement system information** | A1, A2, A3 | |
| …. | | |
| - New intra-frequency cells | | |
| - Intra-frequency cell id | | 2 |

| | | |
|---|---|---|
| - Cell info | | Same content as specified for Intra-frequency cell id=1 (serving cell) in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.2 (FDD)" in clause 6.1.4 |
| - Intra-frequency cell id | | 1 |
| - Cell info | | Same content as specified for Intra-frequency cell id=2 in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.1 (FDD)" in clause 6.1.4 |
| - Intra-frequency cell id | | 3 |
| - Cell info | | Same content as specified for Intra-frequency cell id=3 in SIB11 for Cell 1 in clause 6.1.0b |
| - Intra-frequency cell id | A1, A3 | 7 |
| - Cell info | | Same content as specified for Intra-frequency cell id=7 in SIB11 for Cell 1 in clause 6.1.0b |
| - Intra-frequency cell id | | 8 |
| - Cell info | | Same content as specified for Intra-frequency cell id=8 in SIB11 for Cell 1 in clause 6.1.0b |
| - Intra-frequency cell id | A3 | 11 |
| - Cell info | | Same content as specified for Intra-frequency cell id=2 with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.11 (FDD)" in clause 6.1.4 |
| ….. | | |
| **- Inter-frequency measurement system information** | A1, A2 | |
| ….. | | |
| - New inter-frequency cells | | |
| - Inter frequency cell id | | 4 |
| - Frequency info | | Same content as specified for Inter-frequency cell id=4 in SIB11 for Cell 1 in clause 6.1.0b |
| - Cell info | | Same content as specified for Inter-frequency cell id=4 in SIB11 for Cell 1 in clause 6.1.0b |
| - Inter frequency cell id | | 5 |
| - Frequency info | | Same content as specified for Inter-frequency cell id=5 in SIB11 for Cell 1 in clause 6.1.0b |
| - Cell info | | Same content as specified for Inter-frequency cell id=5 in SIB11 for Cell 1 in clause 6.1.0b |
| - Inter frequency cell id | | 6 |
| - Frequency info | | Same content as specified for Inter-frequency cell id=6 in SIB11 for Cell 1 in clause 6.1.0b |
| - Cell info | | Same content as specified for Inter-frequency cell id=6 in SIB11 for Cell 1 in clause 6.1.0b |
| ….. | | |
| **- Inter-RAT cell info list** | A2 | |
| …. | | |
| - New inter-RAT cells | | |
| - Inter-RAT cell id | | 9 |
| - CHOICE *Radio Access Technology* | | GSM |
| - GSM | | Same content as specified for inter-RAT cell id=9 in SIB11 for Cell 1 in clause 6.1.0b |
| - Inter-RAT cell id | | 10 |
| - CHOICE *Radio Access Technology* | | GSM |
| - GSM | | Same content as specified for inter-RAT cell id=10 in SIB11 for Cell 1 in clause 6.1.0b |

| | | |
|---|---|---|
| …. | | |

| Condition | Explanation |
|---|---|
| A1 | FDD cell environment |
| A2 | FDD/GSM inter-RAT cell environment |
| A3 | FDD intra-frequency cell environment (6 intra-frequency cells without inter-frequency cells) |

Default settings for cell No.2 (TDD)

| | |
|---|---|
| Downlink input level | Reference clause 6 Parameter Set |
| Uplink output power | Minimum supported by the UE's power class. |
| PCCPCH/PCPICH carrier number | Reference clause 6 Parameter Set |
| Cell Channel Description | |
| - Primary CCPCH info | |
| - Cell parameters ID | 4 |

Contents of System Information Block type 11 for cell No.2 (TDD)

| | |
|---|---|
| **- Intra-frequency measurement system information** | |
| …. | |
| - New intra-frequency cells | |
| - Intra-frequency cell id | 2 |
| - Cell info | Same content as specified for Intra-frequency cell id=1 (serving cell) in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Cell parameters ID shall be according to clause titled "Default settings for cell No.2 (TDD)" in clause 6.1.4 |
| - Intra-frequency cell id | 1 |
| - Cell info | Same content as specified for Intra-frequency cell id=2 in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Cell parameters ID shall be according to clause titled "Default settings for cell No.1 (TDD)" in clause 6.1.4 |
| - Intra-frequency cell id | 3 |
| - Cell info | Same content as specified for Intra-frequency cell id=3 in SIB11 for Cell 1 in clause 6.1.0b |
| - Intra-frequency cell id | 7 |
| - Cell info | Same content as specified for Intra-frequency cell id=7 in SIB11 for Cell 1 in clause 6.1.0b |
| - Intra-frequency cell id | 8 |
| - Cell info | Same content as specified for Intra-frequency cell id=8 in SIB11 for Cell 1 in clause 6.1.0b |
| ….. | |
| **- Inter-frequency measurement system information** | |
| ….. | |
| - New inter-frequency cells | |
| - Inter frequency cell id | 4 |
| - Frequency info | Same content as specified for Inter-frequency cell id=4 in SIB11 for Cell 1 in clause 6.1.0b |
| - Cell info | Same content as specified for Inter-frequency cell id=4 in SIB11 for Cell 1 in clause 6.1.0b |
| - Inter frequency cell id | 5 |
| - Frequency info | Same content as specified for Inter-frequency cell id=5 in SIB11 for Cell 1 in clause 6.1.0b |
| - Cell info | Same content as specified for Inter-frequency cell id=5 in SIB11 for Cell 1 in clause 6.1.0b |
| - Inter frequency cell id | 6 |
| - Frequency info | Same content as specified for Inter-frequency cell id=6 in SIB11 for Cell 1 in clause 6.1.0b |
| - Cell info | Same content as specified for Inter-frequency cell id=6 in SIB11 for Cell 1 in clause 6.1.0b |
| ….. | |

Cell No.3

The contents of SYSTEM INFORMATION BLOCK TYPE 1 to 16 messages for cell No.3 are identical to those of cell No.1 with the following exceptions.

| | |
|---|---|
| Cell identity | 0000 0000 0000 0000 0000 0000 0011B |
| URA identity | 0000 0000 0000 0010B |

Default settings for cell No.3 (FDD)

| | |
|---|---|
| Downlink input level | Reference clause 6.10 "Parameter Set" |
| Uplink output power | Minimum supported by the UE's power class. |
| PCCPCH/PCPICH carrier number | Reference clause 6.10 "Parameter Set" |
| Cell Channel Description | |
| - Primary CPICH info | |
| - Primary scrambling code | 200 |

Contents of System Information Block type 11 for cell No.3 (FDD)

| | | |
|---|---|---|
| **- Intra-frequency measurement system information** | A1, A2, A3 | |
| …. | | |
| - New intra-frequency cells | | |
| - Intra-frequency cell id | | 3 |
| - Cell info | | Same content as specified for Intra-frequency cell id=1 (serving cell) in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.3 (FDD)" in clause 6.1.4 |
| - Intra-frequency cell id | | 1 |
| - Cell info | | Same content as specified for Intra-frequency cell id=2 (neigbour cell) in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.1 (FDD)" in clause 6.1.4 |
| - Intra-frequency cell id | | 2 |
| - Cell info | | Same content as specified for Intra-frequency cell id=2 in SIB11 for Cell 1 in clause 6.1.0b |
| - Intra-frequency cell id | A1, A3 | 7 |
| - Cell info | | Same content as specified for Intra-frequency cell id=7 in SIB11 for Cell 1 in clause 6.1.0b |
| - Intra-frequency cell id | | 8 |
| - Cell info | | Same content as specified for Intra-frequency cell id=8 in SIB11 for Cell 1 in clause 6.1.0b |
| - Intra-frequency cell id | A3 | 11 |
| - Cell info | | Same content as specified for Intra-frequency cell id=11 in SIB11 for Cell 1 in clause 6.1.0b |
| ….. | | |
| **- Inter-frequency measurement system information** | A1, A2 | |
| ….. | | |
| - New inter-frequency cells | | |
| - Inter frequency cell id | | 4 |
| - Frequency info | | Same content as specified for Inter-frequency cell id=4 in SIB11 for Cell 1 in clause 6.1.0b |
| - Cell info | | Same content as specified for Inter-frequency cell id=4 in SIB11 for Cell 1 in clause 6.1.0b |
| - Inter frequency cell id | | 5 |
| - Frequency info | | Not Present<br>Absence of this IE is equivalent to value of the previous "frequency info" in the list. |
| - Cell info | | Same content as specified for Inter-frequency cell id=5 in SIB11 for Cell 1 in clause 6.1.0b |
| - Inter frequency cell id | | 6 |
| - Frequency info | | Not Present<br>Absence of this IE is equivalent to value of the previous "frequency info" in the list. |

| | | |
|---|---|---|
| - Cell info | | Same content as specified for Inter-frequency cell id=6 in SIB11 for Cell 1 in clause 6.1.0b |
| ..... | | |
| **- Inter-RAT cell info list** | A2 | |
| .... | | |
| - New inter-RAT cells | | |
| - Inter-RAT cell id | | 9 |
| - CHOICE *Radio Access Technology* | | GSM |
| - GSM | | Same content as specified for inter-RAT cell id=9 in SIB11 for Cell 1 in clause 6.1.0b |
| - Inter-RAT cell id | | 10 |
| - CHOICE *Radio Access Technology* | | GSM |
| - GSM | | Same content as specified for inter-RAT cell id=10 in SIB11 for Cell 1 in clause 6.1.0b |
| .... | | |

| Condition | Explanation |
|---|---|
| A1 | FDD cell environment |
| A2 | FDD/GSM inter-RAT cell environment |
| A3 | FDD intra-frequency cell environment (6 intra-frequency cells without inter-frequency cells) |

Default settings for cell No.3 (TDD)

| | |
|---|---|
| Downlink input level | Reference clause 6 Parameter Set |
| Uplink output power | Minimum supported by the UE's power class. |
| PCCPCH/PCPICH carrier number | Reference clause 6 Parameter Set |
| Cell Channel Description | |
| - Primary CCPCH info | |
| - Cell parameters ID | 8 |

Contents of System Information Block type 11 for cell No.3 (TDD)

| | |
|---|---|
| **- Intra-frequency measurement system information** | |
| .... | |
| - New intra-frequency cells | |
| - Intra-frequency cell id | 3 |
| - Cell info | Same content as specified for Intra-frequency cell id=1 (serving cell) in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Cell parameters ID shall be according to clause titled "Default settings for cell No.3 (TDD)" in clause 6.1.4 |
| - Intra-frequency cell id | 1 |
| - Cell info | Same content as specified for Intra-frequency cell id=2 (neigbour cell) in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Cell parameters ID shall be according to clause titled "Default settings for cell No.1 (TDD)" in clause 6.1.4 |
| - Intra-frequency cell id | 2 |
| - Cell info | Same content as specified for Intra-frequency cell id=2 in SIB11 for Cell 1 in clause 6.1.0b |
| - Intra-frequency cell id | 7 |
| - Cell info | Same content as specified for Intra-frequency cell id=7 in SIB11 for Cell 1 in clause 6.1.0b |
| - Intra-frequency cell id | 8 |
| - Cell info | Same content as specified for Intra-frequency cell id=8 in SIB11 for Cell 1 in clause 6.1.0b |
| ..... | |
| **- Inter-frequency measurement system information** | |
| ..... | |
| - New inter-frequency cells | |
| - Inter frequency cell id | 4 |
| - Frequency info | Same content as specified for Inter-frequency cell id=4 in SIB11 for Cell 1 in clause 6.1.0b |
| - Cell info | Same content as specified for Inter-frequency cell id=4 in SIB11 for Cell 1 in clause 6.1.0b |

| | |
|---|---|
| - Inter frequency cell id | 5 |
| - Frequency info | Not Present<br>Absence of this IE is equivalent to value of the previous "frequency info" in the list. |
| - Cell info | Same content as specified for Inter-frequency cell id=5 in SIB11 for Cell 1 in clause 6.1.0b |
| - Inter frequency cell id | 6 |
| - Frequency info | Not Present<br>Absence of this IE is equivalent to value of the previous "frequency info" in the list. |
| - Cell info | Same content as specified for Inter-frequency cell id=6 in SIB11 for Cell 1 in clause 6.1.0b |
| ….. | |

Cell No.4

The contents of SYSTEM INFORMATION BLOCK TYPE 1 to 16 messages for cell No.4 are identical to those of cell No.1 with the following exceptions.

| | |
|---|---|
| Cell identity | 0000 0000 0000 0000 0000 0000 0100B |
| URA identity | 0000 0000 0000 0010B |

Default settings for cell No.4 (FDD)

| | |
|---|---|
| Downlink input level | Reference clause 6.10 "Parameter Set" |
| Uplink output power | Minimum supported by the UE's power class. |
| PCCPCH/PCPICH carrier number | Reference clause 6.10 "Parameter Set" |
| Cell Channel Description | |
| - Primary CPICH info | |
| - Primary scrambling code | 250 |

Contents of System Information Block type 11 for cell No.4 (FDD)

| | | |
|---|---|---|
| **- Intra-frequency measurement system information** | A1, A2 | |
| …. | | |
| - New intra-frequency cells | | |
| - Intra-frequency cell id | | 4 |
| - Cell info | | Same content as specified for Intra-frequency cell id=1 (serving cell) in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.4 (FDD)" in clause 6.1.4 |
| - Intra-frequency cell id | | 5 |
| - Cell info | | Same content as specified for Intra-frequency cell id=2 in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.5 (FDD)" in clause 6.1.4 |
| - Intra-frequency cell id | | 6 |
| - Cell info | | Same content as specified for Intra-frequency cell id=2 in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.6 (FDD)" in clause 6.1.4 |
| ….. | | |
| **- Inter-frequency measurement system information** | A1, A2 | |
| ….. | | |
| - New inter-frequency cells | | |
| - Inter-frequency cell id | | 1 |
| - Frequency info | | |

| | | |
|---|---|---|
| - UARFCN uplink(Nu) | | Not present<br>Absence of this IE is equivalent to apply the default duplex distance defined for the operating frequency according to 25.101 |
| - UARFCN downlink(Nd) | | Reference to table 6.1.2 for Cell 1 |
| - Cell info | | Same content as specified for Inter-frequency cell id=4 in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.1 (FDD)" in clause 6.1.4 |
| - Inter-frequency cell id | | 2 |
| - Frequency info | | Not Present<br>Absence of this IE is equivalent to value of the previous "frequency info" in the list. |
| - Cell info | | Same content as specified for Inter-frequency cell id=4 in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.2 (FDD)" in clause 6.1.4 |
| - Inter-frequency cell id | | 3 |
| - Frequency info | | Not Present<br>Absence of this IE is equivalent to value of the previous "frequency info" in the list. |
| - Cell info | | Same content as specified for Inter-frequency cell id=4 in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.3 (FDD)" in clause 6.1.4 |
| - Inter-frequency cell id | A1 | 7 |
| - Frequency info | | Not Present<br>Absence of this IE is equivalent to value of the previous "frequency info" in the list. |
| - Cell info | | Same content as specified for Inter-frequency cell id=4 in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.7 (FDD)" in clause 6.1.4 |
| - Inter-frequency cell id | | 8 |
| - Frequency info | | Not Present<br>Absence of this IE is equivalent to value of the previous "frequency info" in the list. |
| - Cell info | | Same content as specified for Inter-frequency cell id=4 in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.8 (FDD)" in clause 6.1.4 |
| **- Inter-RAT cell info list** | A2 | |
| …. | | |
| - New inter-RAT cells | | |
| - Inter-RAT cell id | | 9 |
| - CHOICE *Radio Access Technology* | | GSM |
| - GSM | | Same content as specified for inter-RAT cell id=9 in SIB11 for Cell 1 in clause 6.1.0b |
| - Inter-RAT cell id | | 10 |
| - CHOICE *Radio Access Technology* | | GSM |
| - GSM | | Same content as specified for inter-RAT cell id=10 in SIB11 for Cell 1 in clause 6.1.0b |
| …. | | |

| Condition | Explanation |
|---|---|
| A1 | FDD cell environment |
| A2 | FDD/GSM inter-RAT cell environment |

Default settings for cell No.4 (TDD)

| Downlink input level | Reference clause 6 Parameter Set |
|---|---|

| | |
|---|---|
| Uplink output power | Minimum supported by the UE's power class. |
| PCCPCH/PCPICH carrier number | Reference clause 6 Parameter Set |
| Cell Channel Description | |
| - Primary CCPCH info | |
| - Cell parameters ID | 12 |

Contents of System Information Block type 11 for cell No.4 (TDD)

| | |
|---|---|
| **- Intra-frequency measurement system information** | |
| …. | |
| - New intra-frequency cells | |
| - Intra-frequency cell id | 4 |
| - Cell info | Same content as specified for Intra-frequency cell id=1 (serving cell) in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Cell parameters ID shall be according to clause titled "Default settings for cell No.4 (TDD)" in clause 6.1.4 |
| - Intra-frequency cell id | 5 |
| - Cell info | Same content as specified for Intra-frequency cell id=2 in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Cell parameters ID shall be according to clause titled "Default settings for cell No.5 (TDD)" in clause 6.1.4 |
| - Intra-frequency cell id | 6 |
| - Cell info | Same content as specified for Intra-frequency cell id=2 in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Cell parameters ID shall be according to clause titled "Default settings for cell No.6 (TDD)" in clause 6.1.4 |
| ….. | |
| **- Inter-frequency measurement system information** | |
| ….. | |
| - New inter-frequency cells | |
| - Inter-frequency cell id | 1 |
| - Frequency info | |
| - UARFCN downlink(Nt) | Reference to table 6.1.7 for Cell 1 |
| - Cell info | Same content as specified for Inter-frequency cell id=4 in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Cell parameters ID shall be according to clause titled "Default settings for cell No.1 (TDD)" in clause 6.1.4 |
| - Inter-frequency cell id | 2 |
| - Frequency info | Not Present<br>Absence of this IE is equivalent to value of the previous "frequency info" in the list. |
| - Cell info | Same content as specified for Inter-frequency cell id=4 in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Cell parameters ID shall be according to clause titled "Default settings for cell No.2 (TDD)" in clause 6.1.4 |
| - Inter-frequency cell id | 3 |
| - Frequency info | Not Present<br>Absence of this IE is equivalent to value of the previous "frequency info" in the list. |
| - Cell info | Same content as specified for Inter-frequency cell id=4 in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Cell parameters ID shall be according to clause titled "Default settings for cell No.3 (TDD)" in clause 6.1.4 |
| - Inter-frequency cell id | 7 |
| - Frequency info | Not Present<br>Absence of this IE is equivalent to value of the previous "frequency info" in the list. |
| - Cell info | Same content as specified for Inter-frequency cell id=4 in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Cell parameters ID shall be according to clause titled "Default settings for cell No.7 (TDD)" in clause 6.1.4 |
| - Inter-frequency cell id | 8 |
| - Frequency info | Not Present<br>Absence of this IE is equivalent to value of the previous "frequency info" in the list. |

| | |
|---|---|
| - Cell info | Same content as specified for Inter-frequency cell id=4 in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Cell parameters ID shall be according to clause titled "Default settings for cell No.8 (TDD)" in clause 6.1.4 |

Cell No.5

The contents of SYSTEM INFORMATION BLOCK TYPE 1 to 16 messages for cell No.5 are identical to those of cell No.4 with the following exceptions.

| | |
|---|---|
| Cell identity | 0000 0000 0000 0000 0000 0000 0101B |
| URA identity | 0000 0000 0000 0011B |

Default settings for cell No.5 (FDD)

| | |
|---|---|
| Downlink input level | Reference clause 6.10 "Parameter Set" |
| Uplink output power | Minimum supported by the UE's power class. |
| PCCPCH/PCPICH carrier number | Reference clause 6.10 "Parameter Set" |
| Cell Channel Description | |
| - Primary CPICH info | |
| - Primary scrambling code | 300 |

Contents of System Information Block type 11 for cell No.5 (FDD)

| | | |
|---|---|---|
| **- Intra-frequency measurement system information** | A1, A2 | |
| …. | | |
| - New intra-frequency cells | | |
| - Intra-frequency cell id | | 5 |
| - Cell info | | Same content as specified for Intra-frequency cell id=1 (serving cell) in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.5 (FDD)" in clause 6.1.4 |
| - Intra-frequency cell id | | 4 |
| - Cell info | | Same content as specified for Intra-frequency cell id=2 in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.4 (FDD)" in clause 6.1.4 |
| - Intra-frequency cell id | | 6 |
| - Cell info | | Same content as specified for Intra-frequency cell id=2 in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.6 (FDD)" in clause 6.1.4 |
| ….. | | |
| **- Inter-frequency measurement system information** | A1, A2 | |
| ….. | | |
| - New inter-frequency cells | | |
| - Inter-frequency cell id | | 1 |
| - Frequency info | | |
| - UARFCN uplink(Nu) | | Not present<br>Absence of this IE is equivalent to apply the default duplex distance defined for the operating frequency according to 25.101 |
| - UARFCN downlink(Nd) | | Reference to table 6.1.2 for Cell 1 |
| - Cell info | | Same content as specified for Inter-frequency cell id=4 in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.1 (FDD)" in clause 6.1.4 |
| - Inter-frequency cell id | | 2 |
| - Frequency info | | Not Present<br>Absence of this IE is equivalent to value of the previous "frequency info" in the list. |

| | | |
|---|---|---|
| - Cell info | | Same content as specified for Inter-frequency cell id=4 in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.2 (FDD)" in clause 6.1.4 |
| - Inter-frequency cell id | | 3 |
| - Frequency info | | Not Present<br>Absence of this IE is equivalent to value of the previous "frequency info" in the list. |
| - Cell info | | Same content as specified for Inter-frequency cell id=4 in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.3 (FDD)" in clause 6.1.4 |
| - Inter-frequency cell id | A1 | 7 |
| - Frequency info | | Not Present<br>Absence of this IE is equivalent to value of the previous "frequency info" in the list. |
| - Cell info | | Same content as specified for Inter-frequency cell id=4 in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.7 (FDD)" in clause 6.1.4 |
| - Inter-frequency cell id | | 8 |
| - Frequency info | | Not Present<br>Absence of this IE is equivalent to value of the previous "frequency info" in the list. |
| - Cell info | | Same content as specified for Inter-frequency cell id=4 in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.8 (FDD)" in clause 6.1.4 |
| ….. | | |
| **- Inter-RAT cell info list** | A2 | |
| …. | | |
| - New inter-RAT cells | | |
| - Inter-RAT cell id | | 9 |
| - CHOICE *Radio Access Technology* | | GSM |
| - GSM | | Same content as specified for inter-RAT cell id=9 in SIB11 for Cell 1 in clause 6.1.0b |
| - Inter-RAT cell id | | 10 |
| - CHOICE *Radio Access Technology* | | GSM |
| - GSM | | Same content as specified for inter-RAT cell id=10 in SIB11 for Cell 1 in clause 6.1.0b |
| …. | | |

| Condition | Explanation |
|---|---|
| A1 | FDD cell environment |
| A2 | FDD/GSM inter-RAT cell environment |

Default settings for cell No.5 (TDD)

| | |
|---|---|
| Downlink input level | Reference clause 6 Parameter Set |
| Uplink output power | Minimum supported by the UE's power class. |
| PCCPCH/PCPICH carrier number | Reference clause 6 Parameter Set |
| Cell Channel Description | |
| - Primary CCPCH info | |
| - Cell parameters ID | 114 |

Contents of System Information Block type 11 for cell No.5 (TDD)

| | |
|---|---|
| **- Intra-frequency measurement system information** | |
| …. | |
| - New intra-frequency cells | |
| - Intra-frequency cell id | 5 |

| | |
|---|---|
| - Cell info | Same content as specified for Intra-frequency cell id=1 (serving cell) in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Cell parameters ID shall be according to clause titled "Default settings for cell No.5 (TDD)" in clause 6.1.4 |
| - Intra-frequency cell id | 4 |
| - Cell info | Same content as specified for Intra-frequency cell id=2 in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Cell parameters ID shall be according to clause titled "Default settings for cell No.4 (TDD)" in clause 6.1.4 |
| - Intra-frequency cell id | 6 |
| - Cell info | Same content as specified for Intra-frequency cell id=2 in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Cell parameters ID shall be according to clause titled "Default settings for cell No.6 (TDD)" in clause 6.1.4 |
| ….. | |
| **- Inter-frequency measurement system information** | |
| ….. | |
| - New inter-frequency cells | |
| - Inter-frequency cell id | 1 |
| - Frequency info | |
| - UARFCN downlink(Nt) | Reference to table 6.1.7 for Cell 1 |
| - Cell info | Same content as specified for Inter-frequency cell id=4 in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Cell parameters ID shall be according to clause titled "Default settings for cell No.1 (TDD)" in clause 6.1.4 |
| - Inter-frequency cell id | 2 |
| - Frequency info | Not Present<br>Absence of this IE is equivalent to value of the previous "frequency info" in the list. |
| - Cell info | Same content as specified for Inter-frequency cell id=4 in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Cell parameters ID shall be according to clause titled "Default settings for cell No.2 (TDD)" in clause 6.1.4 |
| - Inter-frequency cell id | 3 |
| - Frequency info | Not Present<br>Absence of this IE is equivalent to value of the previous "frequency info" in the list. |
| - Cell info | Same content as specified for Inter-frequency cell id=4 in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Cell parameters ID shall be according to clause titled "Default settings for cell No.3 (TDD)" in clause 6.1.4 |
| - Inter-frequency cell id | 7 |
| - Frequency info | Not Present<br>Absence of this IE is equivalent to value of the previous "frequency info" in the list. |
| - Cell info | Same content as specified for Inter-frequency cell id=4 in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Cell parameters ID shall be according to clause titled "Default settings for cell No.7 (TDD)" in clause 6.1.4 |
| - Inter-frequency cell id | 8 |
| - Frequency info | Not Present<br>Absence of this IE is equivalent to value of the previous "frequency info" in the list. |
| - Cell info | Same content as specified for Inter-frequency cell id=4 in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Cell parameters ID shall be according to clause titled "Default settings for cell No.8 (TDD)" in clause 6.1.4 |

Cell No.6

The contents of SYSTEM INFORMATION BLOCK TYPE 1 to 16 messages for cell No.6 are identical to those of cell No.4 with the following exceptions:

| | |
|---|---|
| Cell identity | 0000 0000 0000 0000 0000 0000 0110B |
| URA identity | 0000 0000 0000 0011B |

Default settings for cell No.6 (FDD)

| | |
|---|---|
| Downlink input level | Reference clause 6 Parameter Set |
| Uplink output power | Minimum supported by the UE's power class. |
| PCCPCH/PCPICH carrier number | Reference clause 6 Parameter Set |
| Cell Channel Description | |
| - Primary CPICH info | |
| - Primary scrambling code | 350 |

Contents of System Information Block type 11 for cell No.6 (FDD)

| | | |
|---|---|---|
| **- Intra-frequency measurement system information** | A1, A2 | |
| …. | | |
| - New intra-frequency cells | | |
| - Intra-frequency cell id | | 6 |
| - Cell info | | Same content as specified for Intra-frequency cell id=1 (serving cell) in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.6 (FDD)" in clause 6.1.4 |
| - Intra-frequency cell id | | 4 |
| - Cell info | | Same content as specified for Intra-frequency cell id=2 in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.4 (FDD)" in clause 6.1.4 |
| - Intra-frequency cell id | | 5 |
| - Cell info | | Same content as specified for Intra-frequency cell id=2 in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.5 (FDD)" in clause 6.1.4 |
| ….. | | |
| **- Inter-frequency measurement system information** | A1, A2 | |
| ….. | | |
| - New inter-frequency cells | | |
| - Inter-frequency cell id | | 1 |
| - Frequency info | | |
| - UARFCN uplink(Nu) | | Not present<br>Absence of this IE is equivalent to apply the default duplex distance defined for the operating frequency according to 3GPP TS 25.101 [11] |
| - UARFCN downlink(Nd) | | Reference to table 6.1.2 for Cell 1 |
| - Cell info | | Same content as specified for Inter-frequency cell id=4 in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.1 (FDD)" in clause 6.1.4 |
| - Inter-frequency cell id | | 2 |
| - Frequency info | | Not Present<br>Absence of this IE is equivalent to value of the previous "frequency info" in the list. |
| - Cell info | | Same content as specified for Inter-frequency cell id=4 in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.2 (FDD)" in clause 6.1.4 |
| - Inter-frequency cell id | | 3 |
| - Frequency info | | Not Present<br>Absence of this IE is equivalent to value of the previous "frequency info" in the list. |
| - Cell info | | Same content as specified for Inter-frequency cell id=4 in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.3 (FDD)" in clause 6.1.4 |

| | | |
|---|---|---|
| - Inter-frequency cell id | A1 | 7 |
| - Frequency info | | Not Present<br>Absence of this IE is equivalent to value of the previous "frequency info" in the list. |
| - Cell info | | Same content as specified for Inter-frequency cell id=4 in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.7 (FDD)" in clause 6.1.4 |
| - Inter-frequency cell id | | 8 |
| - Frequency info | | Not Present<br>Absence of this IE is equivalent to value of the previous "frequency info" in the list. |
| - Cell info | | Same content as specified for Inter-frequency cell id=4 in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.8 (FDD)" in clause 6.1.4 |
| ….. | | |
| **- Inter-RAT cell info list** | A2 | |
| …. | | |
| - New inter-RAT cells | | |
| - Inter-RAT cell id | | 9 |
| - CHOICE *Radio Access Technology* | | GSM |
| - GSM | | Same content as specified for inter-RAT cell id=9 in SIB11 for Cell 1 in clause 6.1.0b |
| - Inter-RAT cell id | | 10 |
| - CHOICE *Radio Access Technology* | | GSM |
| - GSM | | Same content as specified for inter-RAT cell id=10 in SIB11 for Cell 1 in clause 6.1.0b |
| …. | | |

| Condition | Explanation |
|---|---|
| A1 | FDD cell environment |
| A2 | FDD/GSM inter-RAT cell environment |

Default settings for cell No.6 (TDD)

| | |
|---|---|
| Downlink input level | Reference clause 6 Parameter Set |
| Uplink output power | Minimum supported by the UE's power class. |
| PCCPCH/PCPICH carrier number | Reference clause 6 Parameter Set |
| Cell Channel Description | |
| - Primary CCPCH info | |
| - Cell parameters ID | 119 |

Contents of System Information Block type 11 for cell No.6 (TDD)

| | |
|---|---|
| **- Intra-frequency measurement system information** | |
| …. | |
| - New intra-frequency cells | |
| - Intra-frequency cell id | 6 |
| - Cell info | Same content as specified for Intra-frequency cell id=1 (serving cell) in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Cell parameters ID shall be according to clause titled "Default settings for cell No.6 (TDD)" in clause 6.1.4 |
| - Intra-frequency cell id | 4 |
| - Cell info | Same content as specified for Intra-frequency cell id=2 in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Cell parameters ID shall be according to clause titled "Default settings for cell No.4 (TDD)" in clause 6.1.4 |
| - Intra-frequency cell id | 5 |

| | |
|---|---|
| - Cell info | Same content as specified for Intra-frequency cell id=2 in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Cell parameters ID shall be according to clause titled "Default settings for cell No.5 (TDD)" in clause 6.1.4 |
| ….. | |
| **- Inter-frequency measurement system information** | |
| ….. | |
| - New inter-frequency cells | |
| - Inter-frequency cell id | 1 |
| - Frequency info | |
| - UARFCN downlink(Nt) | Reference to table 6.1.7 for Cell 1 |
| - Cell info | Same content as specified for Inter-frequency cell id=4 in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Cell parameters ID shall be according to clause titled "Default settings for cell No.1 (TDD)" in clause 6.1.4 |
| - Inter-frequency cell id | 2 |
| - Frequency info | Not Present<br>Absence of this IE is equivalent to value of the previous "frequency info" in the list. |
| - Cell info | Same content as specified for Inter-frequency cell id=4 in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Cell parameters ID shall be according to clause titled "Default settings for cell No.2 (TDD)" in clause 6.1.4 |
| - Inter-frequency cell id | 3 |
| - Frequency info | Not Present<br>Absence of this IE is equivalent to value of the previous "frequency info" in the list. |
| - Cell info | Same content as specified for Inter-frequency cell id=4 in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Cell parameters ID shall be according to clause titled "Default settings for cell No.3 (TDD)" in clause 6.1.4 |
| - Inter-frequency cell id | 7 |
| - Frequency info | Not Present<br>Absence of this IE is equivalent to value of the previous "frequency info" in the list. |
| - Cell info | Same content as specified for Inter-frequency cell id=4 in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Cell parameters ID shall be according to clause titled "Default settings for cell No.7 (TDD)" in clause 6.1.4 |
| - Inter-frequency cell id | 8 |
| - Frequency info | Not Present<br>Absence of this IE is equivalent to value of the previous "frequency info" in the list. |
| - Cell info | Same content as specified for Inter-frequency cell id=4 in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Cell parameters ID shall be according to clause titled "Default settings for cell No.8 (TDD)" in clause 6.1.4 |
| ….. | |

Cell No.7

The contents of SYSTEM INFORMATION BLOCK TYPE 1 to 16 messages for cell No.7 are identical to those of cell No.1 with the following exceptions.

| | |
|---|---|
| Cell identity | 0000 0000 0000 0000 0000 0000 0111B |
| URA identity | 0000 0000 0000 0100B |

Default settings for cell No.7 (FDD)

| | |
|---|---|
| Downlink input level | Reference clause 6.10 "Parameter Set" |
| Uplink output power | Minimum supported by the UE's power class. |
| PCCPCH/PCPICH carrier number | Reference clause 6.10 "Parameter Set" |
| Cell Channel Description | |
| - Primary CPICH info | |
| - Primary scrambling code | 400 |

Contents of System Information Block type 11 for cell No.7 (FDD)

| | | |
|---|---|---|
| **- Intra-frequency measurement system information** | A1, A3 | |
| …. | | |
| - New intra-frequency cells | | |
| - Intra-frequency cell id | | 7 |
| - Cell info | | Same content as specified for Intra-frequency cell id=1 (serving cell) in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.7 (FDD)" in clause 6.1.4 |
| - Intra-frequency cell id | | 1 |
| - Cell info | | Same content as specified for Intra-frequency cell id=2 (neigbour cell) in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.1 (FDD)" in clause 6.1.4 |
| - Intra-frequency cell id | | 2 |
| - Cell info | | Same content as specified for Intra-frequency cell id=2 in SIB11 for Cell 1 in clause 6.1.0b |
| - Intra-frequency cell id | | 3 |
| - Cell info | | Same content as specified for Intra-frequency cell id=3 in SIB11 for Cell 1 in clause 6.1.0b |
| - Intra-frequency cell id | | 8 |
| - Cell info | | Same content as specified for Intra-frequency cell id=8 in SIB11 for Cell 1 in clause 6.1.0b |
| - Intra-frequency cell id | A3 | 11 |
| - Cell info | | Same content as specified for Intra-frequency cell id=11 in SIB11 for Cell 1 in clause 6.1.0b |
| ….. | | |
| **- Inter-frequency measurement system information** | A1 | |
| ….. | | |
| - New inter-frequency cells | | |
| - Inter frequency cell id | | 4 |
| - Frequency info | | Same content as specified for Inter-frequency cell id=4 in SIB11 for Cell 1 in clause 6.1.0b |
| - Cell info | | Same content as specified for Inter-frequency cell id=4 in SIB11 for Cell 1 in clause 6.1.0b |
| - Inter frequency cell id | | 5 |
| - Frequency info | | Same content as specified for Inter-frequency cell id=5 in SIB11 for Cell 1 in clause 6.1.0b |
| - Cell info | | Same content as specified for Inter-frequency cell id=5 in SIB11 for Cell 1 in clause 6.1.0b |
| - Inter frequency cell id | | 6 |
| - Frequency info | | Same content as specified for Inter-frequency cell id=6 in SIB11 for Cell 1 in clause 6.1.0b |
| - Cell info | | Same content as specified for Inter-frequency cell id=6 in SIB11 for Cell 1 in clause 6.1.0b |
| ….. | | |

| Condition | Explanation |
|---|---|
| A1 | FDD cell environment |
| A2 | FDD/GSM inter-RAT cell environment |
| A3 | FDD intra-frequency cell environment (6 intra-frequency cells without inter-frequency cells) |

Default settings for cell No.7 (TDD)

| | |
|---|---|
| Downlink input level | Reference clause 6 Parameter Set |
| Uplink output power | Minimum supported by the UE's power class. |
| PCCPCH/PCPICH carrier number | Reference clause 6 Parameter Set |
| Cell Channel Description | |
| - Primary CCPCH info | |
| - Cell parameters ID | 123 |

Contents of System Information Block type 11 for cell No.7 (TDD)

| | |
|---|---|
| **- Intra-frequency measurement system information** | |
| …. | |
| - New intra-frequency cells | |
| - Intra-frequency cell id | 7 |
| - Cell info | Same content as specified for Intra-frequency cell id=1 (serving cell) in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Cell parameters ID shall be according to clause titled "Default settings for cell No.7 (TDD)" in clause 6.1.4 |
| - Intra-frequency cell id | 1 |
| - Cell info | Same content as specified for Intra-frequency cell id=2 (neigbour cell) in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Cell parameters ID shall be according to clause titled "Default settings for cell No.1 (TDD)" in clause 6.1.4 |
| - Intra-frequency cell id | 2 |
| - Cell info | Same content as specified for Intra-frequency cell id=2 in SIB11 for Cell 1 in clause 6.1.0b |
| - Intra-frequency cell id | 3 |
| - Cell info | Same content as specified for Intra-frequency cell id=3 in SIB11 for Cell 1 in clause 6.1.0b |
| - Intra-frequency cell id | 8 |
| - Cell info | Same content as specified for Intra-frequency cell id=8 in SIB11 for Cell 1 in clause 6.1.0b |
| ….. | |
| **- Inter-frequency measurement system information** | |
| ….. | |
| - New inter-frequency cells | |
| - Inter frequency cell id | 4 |
| - Frequency info | Same content as specified for Inter-frequency cell id=4 in SIB11 for Cell 1 in clause 6.1.0b |
| - Cell info | Same content as specified for Inter-frequency cell id=4 in SIB11 for Cell 1 in clause 6.1.0b |
| - Inter frequency cell id | 5 |
| - Frequency info | Same content as specified for Inter-frequency cell id=5 in SIB11 for Cell 1 in clause 6.1.0b |
| - Cell info | Same content as specified for Inter-frequency cell id=5 in SIB11 for Cell 1 in clause 6.1.0b |
| - Inter frequency cell id | 6 |
| - Frequency info | Same content as specified for Inter-frequency cell id=6 in SIB11 for Cell 1 in clause 6.1.0b |
| - Cell info | Same content as specified for Inter-frequency cell id=6 in SIB11 for Cell 1 in clause 6.1.0b |
| ….. | |

Cell No.8

The contents of SYSTEM INFORMATION BLOCK TYPE 1 to 16 messages for cell No.8 are identical to those of cell No.1 with the following exceptions:

| | |
|---|---|
| Cell identity | 0000 0000 0000 0000 0000 0000 1000B |
| URA identity | 0000 0000 0000 0100B |

Default settings for cell No.8 (FDD)

| | |
|---|---|
| Downlink input level | Reference clause 6.10 "Parameter Set" |
| Uplink output power | Minimum supported by the UE's power class. |
| PCCPCH/PCPICH carrier number | Reference clause 6.10 "Parameter Set" |
| Cell Channel Description | |
| - Primary CPICH info | |
| - Primary scrambling code | 450 |

Contents of System Information Block type 11 for cell No.8 (FDD)

| | | |
|---|---|---|
| **- Intra-frequency measurement system information** | A1, A3 | |
| …. | | |
| - New intra-frequency cells | | |
| - Intra-frequency cell id | | 8 |
| - Cell info | | Same content as specified for Intra-frequency cell id=1 (serving cell) in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.8 (FDD)" in clause 6.1.4 |
| - Intra-frequency cell id | | 1 |
| - Cell info | | Same content as specified for Intra-frequency cell id=2 (neigbour cell) in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.1 (FDD)" in clause 6.1.4 |
| - Intra-frequency cell id | | 2 |
| - Cell info | | Same content as specified for Intra-frequency cell id=2 in SIB11 for Cell 1 in clause 6.1.0b |
| - Intra-frequency cell id | | 3 |
| - Cell info | | Same content as specified for Intra-frequency cell id=3 in SIB11 for Cell 1 in clause 6.1.0b |
| - Intra-frequency cell id | | 7 |
| - Cell info | | Same content as specified for Intra-frequency cell id=7 in SIB11 for Cell 1 in clause 6.1.0b |
| - Intra-frequency cell id | A3 | 11 |
| - Cell info | | Same content as specified for Intra-frequency cell id=11 in SIB11 for Cell 1 in clause 6.1.0b |
| ….. | | |
| **- Inter-frequency measurement system information** | A1 | |
| ….. | | |
| - New inter-frequency cells | | |
| - Inter frequency cell id | | 4 |
| - Frequency info | | Same content as specified for Inter-frequency cell id=4 in SIB11 for Cell 1 in clause 6.1.0b |
| - Cell info | | Same content as specified for Inter-frequency cell id=4 in SIB11 for Cell 1 in clause 6.1.0b |
| - Inter frequency cell id | | 5 |
| - Frequency info | | Same content as specified for Inter-frequency cell id=5 in SIB11 for Cell 1 in clause 6.1.0b |
| - Cell info | | Same content as specified for Inter-frequency cell id=5 in SIB11 for Cell 1 in clause 6.1.0b |
| - Inter frequency cell id | | |
| - Frequency info | | Same content as specified for Inter-frequency cell id=6 in SIB11 for Cell 1 in clause 6.1.0b |
| - Cell info | | Same content as specified for Inter-frequency cell id=6 in SIB11 for Cell 1 in clause 6.1.0b |
| ….. | | |

| Condition | Explanation |
|---|---|
| A1 | FDD cell environment |
| A2 | FDD/GSM inter-RAT cell environment |
| A3 | FDD intra-frequency cell environment (6 intra-frequency cells without inter-frequency cells) |

Default settings for cell No.8 (TDD)

| | |
|---|---|
| Downlink input level | Reference clause 6 Parameter Set |
| Uplink output power | Minimum supported by the UE's power class. |
| PCCPCH/PCPICH carrier number | Reference clause 6 Parameter Set |
| Cell Channel Description | |
| - Primary CCPCH info | |
| - Cell parameters ID | 127 |

Contents of System Information Block type 11 for cell No.8 (TDD)

| | |
|---|---|
| **- Intra-frequency measurement system information** | |
| …. | |
| - New intra-frequency cells | |
| - Intra-frequency cell id | 8 |
| - Cell info | Same content as specified for Intra-frequency cell id=1 (serving cell) in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Cell parameters ID shall be according to clause titled "Default settings for cell No.8 (TDD)" in clause 6.1.4 |
| - Intra-frequency cell id | 1 |
| - Cell info | Same content as specified for Intra-frequency cell id=2 (neigbour cell) in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Cell parameters ID shall be according to clause titled "Default settings for cell No.1 (TDD)" in clause 6.1.4 |
| - Intra-frequency cell id | 2 |
| - Cell info | Same content as specified for Intra-frequency cell id=2 in SIB11 for Cell 1 in clause 6.1.0b |
| - Intra-frequency cell id | 3 |
| - Cell info | Same content as specified for Intra-frequency cell id=3 in SIB11 for Cell 1 in clause 6.1.0b |
| - Intra-frequency cell id | 7 |
| - Cell info | Same content as specified for Intra-frequency cell id=7 in SIB11 for Cell 1 in clause 6.1.0b |
| ….. | |
| **- Inter-frequency measurement system information** | |
| ….. | |
| - New inter-frequency cells | |
| - Inter frequency cell id | 4 |
| - Frequency info | Same content as specified for Inter-frequency cell id=4 in SIB11 for Cell 1 in clause 6.1.0b |
| - Cell info | Same content as specified for Inter-frequency cell id=4 in SIB11 for Cell 1 in clause 6.1.0b |
| - Inter frequency cell id | 5 |
| - Frequency info | Same content as specified for Inter-frequency cell id=5 in SIB11 for Cell 1 in clause 6.1.0b |
| - Cell info | Same content as specified for Inter-frequency cell id=5 in SIB11 for Cell 1 in clause 6.1.0b |
| - Inter frequency cell id | 6 |
| - Frequency info | Same content as specified for Inter-frequency cell id=6 in SIB11 for Cell 1 in clause 6.1.0b |
| - Cell info | Same content as specified for Inter-frequency cell id=6 in SIB11 for Cell 1 in clause 6.1.0b |
| ….. | |

Cell No.9

Contents of System Information for cell No.9 (GSM)

See 3GPP TS 51.010-1 [31], clause 10.1.2.

Default settings for cell No.9 (GSM)

See table 6.1.10.

Cell No.10

Contents of System Information for cell No.10 (GSM)

See 3GPP TS 51.010-1 [31], clause 10.1.2.

Default settings for cell No.10 (GSM)

See table 6.1.10

Cell No.11

The contents of SYSTEM INFORMATION BLOCK TYPE 1 to 16 messages for cell No.11 are identical to those of cell No.1 with the following exceptions.

| | |
|---|---|
| Cell identity | 0000 0000 0000 0000 0000 0000 1011B |
| URA identity | 0000 0000 0000 0010B |

Default settings for cell No.11 (FDD)

| | |
|---|---|
| Downlink input level | Reference clause 6.10 "Parameter Set" |
| Uplink output power | Minimum supported by the UE's power class. |
| PCCPCH/PCPICH carrier number | Reference clause 6.10 "Parameter Set" |
| Cell Channel Description | |
| - Primary CPICH info | |
| - Primary scrambling code | 500 |

Contents of System Information Block type 11 for cell No.11 (FDD)

| | | |
|---|---|---|
| **- Intra-frequency measurement system information** | A3 | |
| …. | | |
| - New intra-frequency cells | | |
| - Intra-frequency cell id | | 11 |
| - Cell info | | Same content as specified for Intra-frequency cell id=1 (serving cell) in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.11 (FDD)" in clause 6.1.4 |
| - Intra-frequency cell id | | 1 |
| - Cell info | | Same content as specified for Intra-frequency cell id=2 (neigbour cell) in SIB11 for Cell 1 in clause 6.1.0b with the exception that value for Primary scrambling code shall be according to clause titled "Default settings for cell No.1 (FDD)" in clause 6.1.4 |
| - Intra-frequency cell id | | 2 |
| - Cell info | | Same content as specified for Intra-frequency cell id=2 in SIB11 for Cell 1 in clause 6.1.0b |
| - Intra-frequency cell id | | 3 |
| - Cell info | | Same content as specified for Intra-frequency cell id=3 in SIB11 for Cell 1 in clause 6.1.0b |
| - Intra-frequency cell id | | 7 |
| - Cell info | | Same content as specified for Intra-frequency cell id=7 in SIB11 for Cell 1 in clause 6.1.0b |
| - Intra-frequency cell id | | 8 |
| - Cell info | | Same content as specified for Intra-frequency cell id=8 in SIB11 for Cell 1 in clause 6.1.0b |
| ….. | | |

| Condition | Explanation |
|---|---|
| A1 | FDD cell environment |
| A2 | FDD/GSM inter-RAT cell environment |
| A3 | FDD intra-frequency cell environment (6 intra-frequency cells without inter-frequency cells) |

### 6.1.4.1 Default Cell parameters Two PLMN in UTRAN test scenario

In this scenario two cell groups belong to two different PLMN, Cell 1, 2, 3, 7, 8 (for PLMN1) and Cell 4,5,6 (for PLMN2) shall be configured on two different frequencies.

The contents of SYSTEM INFORMATION BLOCK TYPE 1 to 16 messages for cell No.1 to 8 are identical to those of cell No.1-8 in clause 6.1.4. Exceptions are found in SYSTEM INFORMATION BLOCK TYPE 11:

- SYSTEM INFORMATION BLOCK TYPE 11 for cell No.1, 2, 3, 7, 8 contains cell No.1, 2, 3, 7, 8 in Intra-frequency measurement system information, and cell No.4, 5, 6 in Inter-frequency measurement system information.

- SYSTEM INFORMATION BLOCK TYPE 11 for cell No.4,5,6 contains cell No.4,5,6 in Intra-frequency measurement system information, and cell No. 1, 2, 3, 7, 8 in Inter-frequency measurement system information.
- All other parameters in SYSTEM INFORMATION BLOCK TYPE 11 are set to identical to clause 6.1.4.

Contents of System Information Block type 18 for cell No.1, 2, 3, 7, 8

| | |
|---|---|
| - Idle mode PLMN identities | |
| - PLMNs of intra-frequency cells list | Not Present |
| - PLMNs of inter-frequency cells list | |
| - PLMN identity | Set to PLMN2 |
| - PLMNs of inter-RAT cells list | Not present |
| - Connected mode PLMN identities | Not present |

Contents of System Information Block type 18 for cell No.4, 5, 6

| | |
|---|---|
| - Idle mode PLMN identities | |
| - PLMNs of intra-frequency cells list | Not Present |
| - PLMNs of inter-frequency cells list | |
| - PLMN identity | Set to PLMN1 |
| - PLMNs of inter-RAT cells list | Not present |
| - Connected mode PLMN identities | Not present |

### 6.1.4.2 Default Cell parameters Three PLMN in UTRAN test scenario

In this scenario three cell groups belong to three different PLMN, Cell 1, 2, 3 (for PLMN1), Cell 4, 5, 6 (for PLMN2) and Cell 7, 8 (for PLMN3) shall be configured on three different frequencies.

The contents of SYSTEM INFORMATION BLOCK TYPE 1 to 16 messages for cell No.1 to 8 are identical to those of cell No.1-8 in clause 6.1.4. Exceptions are found in SYSTEM INFORMATION BLOCK TYPE 11:

- SYSTEM INFORMATION BLOCK TYPE 11 for cell No.1, 2, 3 contains cell No.1, 2, 3 in Intra-frequency measurement system information, and cell No.4, 5, 6, 7, 8 in Inter-frequency measurement system information.
- SYSTEM INFORMATION BLOCK TYPE 11 for cell No.4, 5, 6 contains cell No. 4, 5, 6 in Intra-frequency measurement system information, and cell No. 1, 2, 3, 7, 8 in Inter-frequency measurement system information.
- SYSTEM INFORMATION BLOCK TYPE 11 for cell No. 7, 8 contains cell No. 7, 8 in Intra-frequency measurement system information, and cell No. 1, 2, 3, 4, 5, 6 in Inter-frequency measurement system information.
- All other parameters in SYSTEM INFORMATION BLOCK TYPE 11 are set to identical to clause 6.1.4.

Contents of System Information Block type 18 for cell No.1, 2, 3

| | |
|---|---|
| - Idle mode PLMN identities | |
| - PLMNs of intra-frequency cells list | Not Present |
| - PLMNs of inter-frequency cells list | |
| - PLMN identity | Set to PLMN2 |
| - PLMN identity | Set to PLMN2 |
| - PLMN identity | Set to PLMN2 |
| - PLMN identity | Set to PLMN3 |
| - PLMNs of inter-RAT cells list | Not present |
| - Connected mode PLMN identities | Not present |

Contents of System Information Block type 18 for cell No.4, 5, 6

| | |
|---|---|
| - Idle mode PLMN identities | |
| - PLMNs of intra-frequency cells list | Not Present |
| - PLMNs of inter-frequency cells list | |
| - PLMN identity | Set to PLMN1 |
| - PLMN identity | Set to PLMN1 |
| - PLMN identity | Set to PLMN1 |
| - PLMN identity | Set to PLMN3 |

| | |
|---|---|
| - PLMNs of inter-RAT cells list | Not present |
| - Connected mode PLMN identities | Not present |

Contents of System Information Block type 18 for cell No.7, 8

| | |
|---|---|
| - Idle mode PLMN identities | |
| - PLMNs of intra-frequency cells list | Not Present |
| - PLMNs of inter-frequency cells list | |
| - PLMN identity | Set to PLMN1 |
| - PLMN identity | Set to PLMN1 |
| - PLMN identity | Set to PLMN1 |
| - PLMN identity | Set to PLMN2 |
| - PLMNs of inter-RAT cells list | Not present |
| - Connected mode PLMN identities | Not present |

## 6.1.5 Reference Radio Conditions (FDD)

The following transmission parameters shall be used unless otherwise stated in the description of the individual test case.

Table 6.1.3 are the default settings for a non-suitable cell which is configured and always present whereas table 6.1.4 is for a cell that is switched off. Cells configured according to table 6.1.3 are for test cases in which it is necessary to make a cell unsuitable, and then subsequently make it suitable. This could be achieved by switching the cell off and then reconfiguration as in table 6.1.4, but this takes a lot of time to do.

**Table 6.1.1: Default settings for a serving cell in a single cell environment**

| Parameter | Unit | Cell 1 |
|---|---|---|
| Cell type | | Serving cell |
| UTRA RF Channel Number | | Channel 1 |
| Qqualmin | dB | -24 |
| Qrxlevmin | dBm | -79 |
| UE_TXPWR_MAX_RACH | dBm | 21 |
| CPICH Ec (see notes 1 and 2) | dBm/3.84 MHz | -60 |
| NOTE 1: The power level is specified in terms of CPICH_Ec instead of CPICH_RSCP as RSCP is a receiver measurement and only CPICH_Ec can be directly controlled by the SS.<br>NOTE 2: The cell fulfils 3GPP TS 25.304 [36], clause 5.2.3.1.2 and 3GPP TS 25.133 [30], clause 8.1.2.2.1. | | |

**Table 6.1.2: Default settings for a serving cell and a suitable neighbour cell in a multi-cell environment**

| Parameter | Unit | Cell 1 | Cell 2 | Cell 4 |
|---|---|---|---|---|
| Cell type | | Serving cell | Suitable neighbour intra-frequency cell | Suitable neighbour inter-frequency cell |
| UTRA RF Channel Number | | Channel 1 | Channel 1 | Channel 2 |
| Qqualmin | dB | -24 | -24 | |
| Qrxlevmin | dBm | -79 | -79 | |
| UE_TXPWR_MAX_RACH | dBm | 21 | 21 | |
| CPICH Ec (see notes 1 and 2) | dBm/3.84 MHz | -60 | -70 | |
| NOTE 1: The power level is specified in terms of CPICH_Ec instead of CPICH_RSCP as RSCP is a receiver measurement and only CPICH_Ec can be directly controlled by the SS.<br>NOTE 2: Both cells fulfil 3GPP TS 25.304 [36], clause 5.2.3.1.2 and 3GPP TS 25.133 [30], clause 8.1.2.2.1. | | | | |

**Table 6.1.3: Default settings for a non-suitable cell**

| Parameter | Unit | Level |
|---|---|---|
| Qqualmin | dB | -24 |
| Qrxlevmin | dBm | -79 |
| UE_TXPWR_MAX_RACH | dBm | 21 |
| CPICH_Ec | dBm/3.84 MHz | -90 |

NOTE 1: The power level is specified in terms of CPICH_Ec instead of CPICH_RSCP as RSCP is a receiver measurement and only CPICH_Ec can be directly controlled by the SS.
NOTE 2: The cell is not suitable according to 3GPP TS 25.304 [36], clause 5.2.3.1.2.

**Table 6.1.4: Default settings for a non-suitable "Off" cell**

| Parameter | Unit | Level |
|---|---|---|
| Qqualmin | dB | -24 |
| Qrxlevmin | dBm | -79 |
| UE_TXPWR_MAX_RACH | dBm | 21 |
| CPICH_Ec | dBm/3.84 MHz | ≤ -122 |
| NOTE 1: The power level is specified in terms of CPICH_Ec instead of CPICH_RSCP as RSCP is a receiver measurement and only CPICH_Ec can be directly controlled by the SS.<br>NOTE 2: The cell is not suitable according to 3GPP TS 25.304 [36], clause 5.2.3.1.2. | | |

**Table 6.1.5: Default power levels of physical channels relative to CPICH_Ec**

| Parameter | Unit | Level Idle mode | Level Connected mode |
|---|---|---|---|
| DPCH_Ec | dB | (see note) | See table 6.1.6 |
| PCCPCH_Ec | dB | -2 | |
| SCCPCH_Ec | dB | -2 | |
| AICH_Ec | dB | -5 | |
| SCH_Ec | dB | -5 | |
| PICH_Ec | dB | -5 | |
| NOTE: This shall be less than -122 dBm to ensure the channel is considered as "off". | | | |

**Table 6.1.6: Default power levels of DPCH_Ec relative to CPICH_Ec**

| Data transmission rate | Level |
|---|---|
| 12.2 kbps | -5 |
| 64 kbps | -2 |
| 144 kbps | +1 |
| 384 kbps | +5 |

## 6.1.6 Reference Radio Conditions (TDD)

The following transmission parameters shall be used for unless otherwise stated in the description of the individual test case.

**Table 6.1.6a: Default settings for a serving cell in a single cell environment**

| Parameter | Unit | Cell 1 |
|---|---|---|
| Cell type | | Serving cell |
| UTRA RF Channel Number | | Channel 1 |
| Qrxlevmin | dBm | -81 |
| UE_TXPWR_MAX_RACH | dBm | 21 |
| PCCPCH RSCP | dBm | -60 |
| NOTE: The cell fulfils 3GPP TS 25.304 [36], clause 5.2.3.1.2 and 3GPP TS 25.123 [37]. | | |

**Table 6.1.7: Default settings for a serving cell and a suitable neighbour cell in a multi-cell environment**

| Parameter | Unit | Cell 1 | Cell 2 | Cell 4 |
|---|---|---|---|---|
| Cell type | | Serving cell | Suitable neighbour intra-frequency cell | Suitable neighbour inter-frequency cell |
| UTRA RF Channel Number | | Channel 1 | Channel 1 | Channel 2 |
| Qrxlevmin | dBm | -81 | -81 | |
| UE_TXPWR_MAX_RACH | dBm | 21 | 21 | |
| PCCPCH RSCP | dBm | -60 | -70 | |

NOTE: Both cells fulfil 3GPP TS 25.304 [36], clause 5.2.3.1.2 and 3GPP TS 25.123 [37].

**Table 6.1.8: Default settings for a non-suitable cell**

| Parameter | Unit | Level |
|---|---|---|
| Qrxlevmin | dBm | -81 |
| UE_TXPWR_MAX_RACH | dBm | 21 |
| PCCPCH RSCP | dBm | -91 |
| NOTE: The cell is not suitable according to 3GPP TS 25.304 [36], clause 5.2.3.1.2. | | |

**Table 6.1.9: Default settings for a non-suitable "Off" cell**

| Parameter | Unit | Level |
|---|---|---|
| Qrxlevmin | dBm | -81 |
| UE_TXPWR_MAX_RACH | dBm | 21 |
| PCCPCH RSCP | dBm | ≤ -110 |
| NOTE: The cell is not suitable according to 3GPP TS 25.304 [36], clause 5.2.3.1.2. | | |

## 6.1.7 Reference Radio Conditions (GSM)

The following transmission parameters shall be used unless otherwise stated in the description of the individual test case.

**Table 6.1.10: Default settings for a serving cell and a suitable neighbour cell in a multi-cell environment**

| Parameter | Unit | Cell 9 | Cell 10 |
|---|---|---|---|
| Cell type | | Serving cell | Suitable neighbour cell |
| BCCH ARFCN | | As defined in the initial conditions in clause 26.6.5.1 of TS 51.010-1 [31] for cell A and the GSM band under test. | As defined in the initial conditions in clause 26.6.5.1 of TS 51.010-1 [31] for cell B and the GSM band under test. |
| Base transceiver Station Identity Code (BSIC) | | BSIC1 | BSIC2 |
| Qrxlevmin | dBm | -81 | -81 |
| MS_TXPWR_MAX_CCH | dBm | According to maximum output power for the power class of the MS under test | |
| RF level | dBm | -48 | -54 |
| NOTE: Both cells fulfil 3GPP TS 25.304 [36], clause 5.2.6.1.4 and 3GPP TS 25.133 [37], clause 8.1.2.5. | | | |

**Table 6.1.11: Default settings for a non-suitable cell**

| Parameter | Unit | Level |
|---|---|---|
| Qrxlevmin | dBm | -81 |
| MS_TXPWR_MAX_CCH | dBm | According to maximum output power for the power class of the MS under test |
| RF level | dBm | -90 |
| NOTE: The cell is not suitable according to 3GPP TS 25.304 [36], clause 5.2.6.1.4 | | |

# 6.2 Number of neighbour cells

The options for the number of neighbour cells (ie the total number of active cells in the simulated network) are given below. See clause 6.1 for cell configurations.

## 6.2.1 Basic Network

| Number of Cells | Use of Network Configuration |
|---|---|
| 1 | Basic UE registration; RRC Connection Establishment and Release; operation of dedicated channels in non-handover modes; general RF and EMC testing |

### 6.2.2 Soft Handover Network (FDD)

| Number of Cells | Use of Network Configuration/Constraints |
|---|---|
| 2 | Can be used in place of basic network, plus offering operation of dedicated channels in 2 way soft handover or in 2 way SSDT (R99 and Rel-4 only) handover for RF or signalling tests; simple cell reselection tests |

### 6.2.3 Hard Handover Network

| Number of Cells | Use of Network Configuration |
|---|---|
| 2 | Can be used in place of basic network, plus offering operation in 2 cell hard handover (inter-frequency) |

### 6.2.4 'Roaming' Network

| Number of Cells | Use of Network Configuration |
|---|---|
| 7 | This configuration is intended to provide the capability for extensive cell selection and reselection testing, as defined under Idle Mode Testing.<br>It is <ffs> if 7 is the correct number of cells and also <ffs> is the number of separate RF channels to be supported by the 'Roaming Network' |

## 6.3 Cell/BS codes etc

See clause 6.1.

## 6.4 Routing/location area

See clause 6.1.

## 6.5 Network options settings

See clause 6.1.

## 6.6 Power control mode

### 6.6.1 Downlink Power Control

#### 6.6.1.1 Outer Loop Power Control

This is used to set the SIR requirements from the given BER/BLER requirements for the dedicated channel - the reference configuration is for the BER/BLER and SIR requirements to be fixed, ie Outer Loop Power Control is disabled.

#### 6.6.1.2 Inner Loop Power Control

The inner loop power control adjusts the power of the dedicated channel to meet the SIR requirements. The reference condition is for the Inner Loop Power Control to be disabled.

### 6.6.2 Uplink Power Control

#### 6.6.2.1 Outer Loop Power Control

This is used to set the SIR requirements from the given BER/BLER requirements for the dedicated channel - the reference configuration is for the BER/BLER and SIR requirements to be fixed, ie Outer Loop Power Control is disabled.

#### 6.6.2.2 Inner Loop Power Control (FDD)

The inner loop power control adjusts the power of the dedicated channel to meet the SIR requirements.

# 6.7 Tx Diversity modes

The reference settings for Tx Diversity Mode shall be:

## 6.7.1 Non-Diverse Operation

DL Transmit Diversity shall be disabled on all cells in the simulated network.

## 6.7.2 Diverse Operation

### 6.7.2.1 Diverse Operation (FDD mode)

The diversity options applied to the DL channels shall be as below for all cells in the simulated network.

| Channel | Open loop mode | | Closed loop Mode |
|---|---|---|---|
| | TSTD | STTD | |
| P-CCPCH | - | X | - |
| SCH | X | - | - |
| S-CCPCH | - | X | - |
| DPCH | - | X | - |
| PICH | - | X | - |
| AICH | - | X | - |

### 6.7.2.2 Diverse Operation (TDD mode)

The diversity options applied to the DL channels shall be as below for all cells in the simulated network.

#### 6.7.2.2.1 3.84 Mcps option

**Table 6.7.1: Application of Tx diversity schemes on downlink physical channel types in 3.84 Mcps TDD "X" - can be applied, "-" - must not be applied**

| Physical channel type | Open loop TxDiversity | | Closed loop TxDiversity |
|---|---|---|---|
| | TSTD | SCTD (see note) | |
| P-CCPCH | - | X | - |
| S-CCPCH | -- | X | -- |
| SCH | X | - | - |
| DPCH | - | - | X |
| PDSCH | - | X | X |
| PICH | - | X | - |
| NOTE: SCTD may only be applied to physical channels when they are allocated to beacon locations. | | | |

#### 6.7.2.2.2 1.28 Mcps option

**Table 6.7.2: Application of Tx diversity schemes on downlink physical channel types in 1.28 Mcps TDD "X" - can be applied, "-" - must not be applied**

| Physical channel type | Open loop TxDiversity | | Closed loop TxDiversity |
|---|---|---|---|
| | TSTD | SCTD (see note) | |
| P-CCPCH | X | X | - |
| S-CCPCH | X | X | - |
| DwPCH | X | - | - |
| DPCH | X | - | X |
| PDSCH | X | X | X |
| PICH | X | X | - |
| NOTE: SCTD may only be applied to physical channels when they are allocated to beacon locations. | | | |

# 6.8 Compressed mode parameters

In this clause, Parameters for reference compressed mode patterns are defined which are used in signalling test cases such as inter frequency FDD measurement, inter frequency TDD measurement and inter RAT measurement in 3GPP TS 34.123-1 [1]. These parameters are defined in 3GPP TS 25.133 [30] for measurment performance tests.

Depending on UE capability, there are four methods constructed of three types using of compressed mode such as UL only, DL only and both UL and DL, and using without application of compressed for the above measurement purposes. As test requirement is the same even if the test methods are different, ICS/IXIT statement is applied to the test cases so that the test procedure and specific message contents specified in 3GPP TS 34.123-1 [1] can be distinguished.

## 6.8.1 Single compressed mode pattern

Configuration parameters in single compressed mode pattern for one type of measurement objects are described in the following clauses.

### 6.8.1.1 Inter Frequency FDD measurement

The configuration parameters for an inter frequency FDD measurement is shown in table 6.8.1.

**Table 6.8.1: Compressed mode parameters (Inter Frequency FDD measurement)**

| Parameter | Value | Note |
|---|---|---|
| TGSN (Transmission Gap Starting Slot Number) | 4 | |
| TGL1 (Transmission Gap Length 1) | 7 | |
| TGL2 (Transmission Gap Length 2) | - | Only one gap in use. |
| TGD (Transmission Gap Distance) | undefined | |
| TGPL1 (Transmission Gap Pattern Length) | 3 | |
| TGPL2 (Transmission Gap Pattern Length) | - | R99 and Rel-4:<br>Only one pattern in use.<br>Rel-5 and onwards:<br>Not applicable |
| TGCFN (Transmission Gap Connection Frame Number) | (Current CFN + (256 - TTI/10msec))mod 256 | |
| UL/DL compressed mode selection | DL, UL or DL & UL | 3 configurations possible.<br>DL, UL or both DL and UL |
| UL compressed mode method | SF/2 | |
| DL compressed mode method | SF/2 | |
| Scrambling code change | No | |
| RPP (Recovery period power control mode) | 0 | |
| ITP (Initial transmission power control mode) | 0 | |

### 6.8.1.2 Inter Frequency TDD measurement

The configuration parameters for an inter frequency TDD measurement is shown in table 6.8.2.

**Table 6.8.2: Compressed mode parameters (Inter Frequency TDD measurement)**

| Parameter | Value | Note |
|---|---|---|
| TGSN (Transmission Gap Starting Slot Number) | 10 | |
| TGL1 (Transmission Gap Length 1) | 10 | |
| TGL2 (Transmission Gap Length 2) | - | Only one gap in use. |
| TGD (Transmission Gap Distance) | undefined | |
| TGPL1 (Transmission Gap Pattern Length) | 11 | |
| TGPL2 (Transmission Gap Pattern Length) | - | R99 and Rel-4:<br>Only one pattern in use.<br>Rel-5 and onwards:<br>Not applicable |
| TGCFN (Transmission Gap Connection Frame Number) | (Current CFN + (256 - TTI/10msec))mod 256 | |
| UL/DL compressed mode selection | DL, UL or DL & UL | 3 configurations possible.<br>DL, UL or both DL and UL |
| UL compressed mode method | SF/2 | |
| DL compressed mode method | Puncturing | |

| Parameter | Value | Note |
|---|---|---|
| Scrambling code change | No | |
| RPP (Recovery period power control mode) | 0 | |
| ITP (Initial transmission power control mode) | 0 | |

### 6.8.1.3 Inter RAT measurement (GSM - Carrier RSSI)

The configuration parameters for an Inter RAT measurement (GSM - Carrier RSSI) is shown in table 6.8.3.

**Table 6.8.3: Compressed mode parameters (Inter RAT measurement - GSM Carrier RSSI)**

| Parameter | Value | Note |
|---|---|---|
| TGSN (Transmission Gap Starting Slot Number) | 4 | |
| TGL1 (Transmission Gap Length 1) | 7 | |
| TGL2 (Transmission Gap Length 2) | - | Only one gap in use. |
| TGD (Transmission Gap Distance) | undefined | |
| TGPL1 (Transmission Gap Pattern Length) | 12 | |
| TGPL2 (Transmission Gap Pattern Length) | - | R99 and Rel-4:<br>Only one pattern in use.<br>Rel-5 and onwards:<br>Not applicable |
| TGCFN (Transmission Gap Connection Frame Number) | (Current CFN + (256 - TTI/10msec))mod 256 | |
| UL/DL compressed mode selection | DL, UL or DL & UL | 3 configurations possible.<br>DL, UL or both DL and UL |
| UL compressed mode method | SF/2 | |
| DL compressed mode method | SF/2 | |
| Scrambling code change | No | |
| RPP (Recovery period power control mode) | 0 | |
| ITP (Initial transmission power control mode) | 0 | |

### 6.8.1.4 Inter RAT measurement (GSM - Initial BSIC Identification)

The configuration parameters for an inter frequency RAT measurement ( GSM - Initial BSIC Identification ) is shown in table 6.8.4.

**Table 6.8.4: Compressed mode parameters (Inter RAT measurement - GSM Initial BSIC Identification)**

| Parameter | Value | Note |
|---|---|---|
| TGSN (Transmission Gap Starting Slot Number) | 4 | |
| TGL1 (Transmission Gap Length 1) | 7 | |
| TGL2 (Transmission Gap Length 2) | - | Only one gap in use. |
| TGD (Transmission Gap Distance) | undefined | |
| TGPL1 (Transmission Gap Pattern Length) | 8 | |
| TGPL2 (Transmission Gap Pattern Length) | - | R99 and Rel-4:<br>Only one pattern in use.<br>Rel-5 and onwards:<br>Not applicable |
| TGCFN (Transmission Gap Connection Frame Number) | (Current CFN + (256 - TTI/10msec))mod 256 | |
| UL/DL compressed mode selection | DL, UL or DL & UL | 3 configurations possible.<br>DL, UL or both DL and UL |
| UL compressed mode method | SF/2 | |
| DL compressed mode method | SF/2 | |
| Scrambling code change | No | |
| RPP (Recovery period power control mode) | 0 | |
| ITP (Initial transmission power control mode) | 0 | |

### 6.8.1.5 Inter RAT measurement (GSM - BSIC re-confirmation)

The configuration parameters for an inter RAT measurement ( GSM - BSIC re-confirmation) is shown in table 6.8.5.

**Table 6.8.5: Compressed mode parameters (Inter RAT measurement - GSM BSIC re-confirmation)**

| Parameter | Value | Note |
|---|---|---|
| TGSN (Transmission Gap Starting Slot Number) | 4 | |
| TGL1 (Transmission Gap Length 1) | 7 | |
| TGL2 (Transmission Gap Length 2) | - | Only one gap in use. |
| TGD (Transmission Gap Distance) | undefined | |
| TGPL1 (Transmission Gap Pattern Length) | 8 | |
| TGPL2 (Transmission Gap Pattern Length) | - | R99 and Rel-4:<br>Only one pattern in use.<br>Rel-5 and onwards:<br>Not applicable |
| TGCFN (Transmission Gap Connection Frame Number) | (Current CFN + (256 - TTI/10msec))mod 256 | |
| UL/DL compressed mode selection | DL, UL or DL & UL | 3 configurations possible.<br>DL, UL or both DL and UL |
| UL compressed mode method | SF/2 | |
| DL compressed mode method | SF/2 | |
| Scrambling code change | No | |
| RPP (Recovery period power control mode) | 0 | |
| ITP (Initial transmission power control mode) | 0 | |

## 6.8.2 Multiple compressed mode patterns

Configuration parameters in multiple compressed mode patterns for several types of measurement objects are described in the following clauses.

### 6.8.2.1 Inter RAT measurement GSM

The configuration parameters for an inter RAT measurement (GSM - Carrier RSSI, Initial BSIC Identification and BSIC Re-confirmation) is shown in table 6.8.6.

**Table 6.8.6: Compressed mode parameters (Inter RAT measurement - GSM Carrier RSSI and Initial BSIC identification and BSIC re-confirmation)**

| Parameter | GSM Carrier RSSI | GSM Initial BSIC identification | GSM BSIC re-confirmation | Note |
|---|---|---|---|---|
| TGSN (Transmission Gap Starting Slot Number) | 4 | 4 | 4 | |
| TGL1 (Transmission Gap Length 1) | 7 | 7 | 7 | |
| TGL2 (Transmission Gap Length 2) | - | - | - | Only one gap in use. |
| TGD (Transmission Gap Distance) | undefined | undefined | undefined | |
| TGPL1 (Transmission Gap Pattern Length) | 12 | 8 | 8 | |
| TGPL2 (Transmission Gap Pattern Length) | - | - | - | R99 and Rel-4:<br>Only one pattern in use.<br>Rel-5 and onwards:<br>Not applicable. |
| TGCFN (Transmission Gap Connection Frame Number): | (Current CFN + (252 - TTI/10msec))mod 256 | (Current CFN + (254 - TTI/10msec))mod 256 | (Current CFN + (250 - TTI/10msec))mod 256 | Defined by higher layers |
| UL/DL compressed mode selection | DL, UL or DL & UL | DL, UL or DL & UL | DL, UL or DL & UL | 3 configurations possible.<br>DL, UL or both DL and UL |
| UL compressed mode method | SF/2 | SF/2 | SF/2 | |
| DL compressed mode method | SF/2 | SF/2 | SF/2 | |
| Scrambling code change | No | No | No | |

| Parameter | GSM Carrier RSSI | GSM Initial BSIC identification | GSM BSIC re-confirmation | Note |
|---|---|---|---|---|
| RPP (Recovery period power control mode) | 0 | 0 | 0 | |
| ITP (Initial transmission power control mode) | 0 | 0 | 0 | |

### 6.8.2.2 Inter Frequency FDD measurement & Inter RAT measurement GSM

The configuration parameters for Inter Frequency FDD measurement and Inter RAT measurement (GSM - Carrier RSSI, Initial BSIC Identification and BSIC Re-confirmation) is shown in table 6.8.7.

The pattern is illustrated by Figure 6.8.2.2.

**Table 6.8.7: Compressed mode parameters (Inter Frequency and Inter RAT measurement - GSM Carrier RSSI and Initial BSIC identification and BSIC re-confirmation)**

| Parameter | Inter Frequency FDD | GSM Carrier RSSI | GSM Initial BSIC identification | GSM BSIC re-confirmation | Note |
|---|---|---|---|---|---|
| TGSN (Transmission Gap Starting Slot Number) | 8 | 8 | 8 | 8 | |
| TGL1 (Transmission Gap Length 1) | 14 | 14 | 14 | 14 | |
| TGL2 (Transmission Gap Length 2) | 14 | 14 | 14 | 14 | |
| TGD (Transmission Gap Distance) | 0 | 60 | 45 | 0 | |
| TGPL1 (Transmission Gap Pattern Length) | 12 | 24 | 24 | 24 | |
| TGPL2 (Transmission Gap Pattern Length) | - | - | - | - | R99 and Rel-4: Only one pattern in use. Rel-5 and onwards: Not applicable |
| TGCFN (Transmission Gap Connection Frame Number): | (Current CFN + (238 - TTI/10msec)) mod 256 | (Current CFN + (242 - TTI/10msec)) mod 256 | (Current CFN + (256 - TTI/10msec)) mod 256 | (Current CFN + (253 - TTI/10msec)) mod 256 | Defined by higher layers |
| UL/DL compressed mode selection | DL, UL or DL & UL | DL, UL or DL & UL | DL, UL or DL & UL | DL, UL or DL & UL | 3 configurations possible. DL, UL or both DL and UL |
| UL compressed mode method | SF/2 | SF/2 | SF/2 | SF/2 | |
| DL compressed mode method | SF/2 | SF/2 | SF/2 | SF/2 | |
| Scrambling code change | No | No | No | No | |
| RPP (Recovery period power control mode) | 0 | 0 | 0 | 0 | |
| ITP (Initial transmission power control mode) | 0 | 0 | 0 | 0 | |

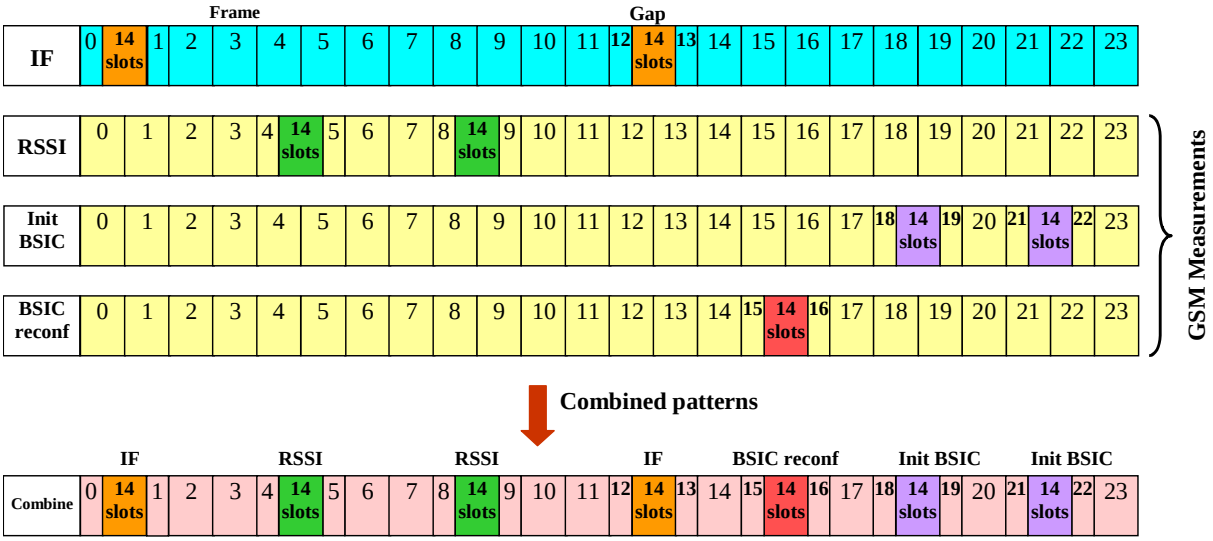



**Figure 6.8.2.2: Inter-frequency (IF) and Inter-RAT (IRAT) measurement gaps during 24 frames cycle for the compressed mode pattern as specified in Table 6.8.7**

### 6.8.2.3 Inter Frequency FDD measurement & Inter Frequency TDD measurement

FFS

### 6.8.2.4 Inter Frequency TDD measurement & Inter RAT measurement GSM

FFS

### 6.8.2.5 Inter Frequency FDD measurement & Inter Frequency TDD measurement & Inter RAT measurement GSM

FFS

## 6.9 BCCH parameters

See clause 6.1.

## 6.10 Reference Radio Bearer configurations used in Radio Bearer interoperability testing

The reference radio bearer configurations are typical configurations of the radio interface. This sub-set of the mandatory set of radio bearer configurations supported by the UE is intended to be used as test configurations for testing of the UE. The purpose of the reference radio bearer configurations is to ensure interoperability of UE's in different regions and networks.

The reference radio bearer configurations are used in the radio bearer interoperability test cases, clause 14 of 3GPP TS 34.123-1 [1]. The reference radio bearer configurations are also intended to be the first choice for other test cases where a radio bearer configuration is needed. For test cases requiring alternative configurations not provided by the reference radio bearer configurations then these specific radio bearer configurations are either specified in the actual test case itself; or in case the configurations are used by more than one test case then these common radio bearer configurations are specified in clause 6.11 of the present document.

NOTE If not specifically specified then the mid-value of the RM attribute value range as specified by the actual reference radio bearer configuration shall be applied for testing. However, in the case of UL and DL:3.4 kbps SRBs for DCCH and where the Choice "Same As UL" is used for the IE "DL Transport channel information common for all transport channel", the RM attribute for the "DL:3.4 kbps SRBs for DCCH" shall be set to the same value as that used in the Uplink.

## 6.10.1 QoS Architecture and RAB attributes

From a user point-of-view services are considered end-to-end, this means from a Terminal Equipment (TE) to another TE. An End-to-End Service may have a certain Quality of Service (QoS) which is provided for the user through the different networks. In UMTS, it is the UMTS Bearer Service that provides the requested QoS through the use of different QoS classes as defined in 3GPP TS 23.107 [15].

The UMTS Bearer Service consists of two parts, the Radio Access Bearer Service, RAB, and the Core Network Bearer Service. The Radio Access Bearer Service is realized by a Radio Bearer Service and an Iu-Bearer Service. The relationship between the services is illustrated in figure 6.10.1.1.

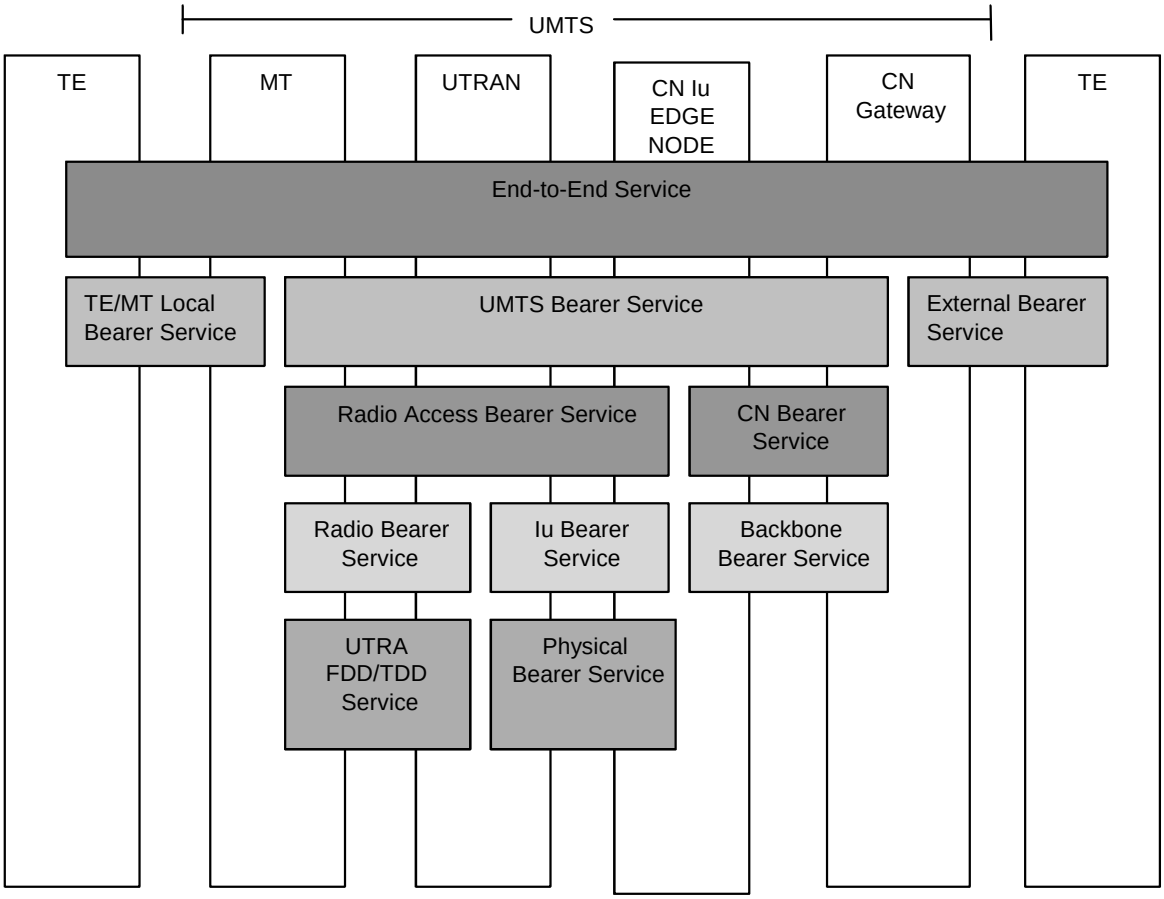



**Figure 6.10.1.1: UMTS QoS Architecture**

The Radio Access Bearer Service is characterized by a number of attributes such as Traffic class, Maximum bit rate, Guaranteed bit rate, SDU error ratio, Residual BER, Transfer Delay etc. As a first approach the four following attributes have been considered to come up with the parameter settings in clause 6.10.2.4 for FDD mode and clause 6.10.3.4 for TDD mode:

- Traffic class;
- SSD;
- Maximum bit rate;
- Residual BER.

The Traffic classes are explained in table 6.10.1.1. The Maximum bit rate has been considered at RLC layer and Physical Layer for the acknowledged and unacknowledged modes respectively. The Residual BER is understood as BER at RLC layer and Transport BLER for the acknowledged and unacknowledged modes respectively.

NOTE: The maximum bit rate in clause 6.10.2.4 for FDD mode and clause 6.10.3.4 for TDD mode is one of the RAB attribute as described above. For Interactive/Background PS RABs, however, the maximum bit rate of Radio Bearer can be lower than the maximum bit rate of RAB attributes due to radio resource

management. Bit rates of Interactive/Background PS RABs described in 6.10.2.4 for FDD mode and clause 6.10.3.4 for TDD mode may represent the maximum bit rate of Radio Bearer taking account into this management.

**Table 6.10.1.1: Traffic classes**

| Traffic class | Conversational class<br>conversational RT | Streaming class<br>streaming RT | Interactive class<br>Interactive best effort | Background<br>Background best effort |
|---|---|---|---|---|
| **Fundamental characteristics** | - Preserve time relation (variation) between information entities of the stream<br>Conversational pattern (stringent and low delay) | - Preserve time relation (variation) between information entities of the stream (i.e. some but constant delay) | Request response pattern<br>Preserve payload content | Destination is not expecting the data within a certain time<br>Preserve payload content |
| **Example of the application** | - speech, video, etc. | - facsimile (NT)<br>- streaming audio and video | - Web browsing | - background download of emails |

## 6.10.2 RAB and signalling RB for FDD

### 6.10.2.1 RABs and signalling RBs

In the following clauses, the typical parameter sets are presented for reference RABs, signalling RBs and important combinations of them. The data rate given for each RAB is the maximum data rate that can be supported by that RAB.

NOTE: The granularity for each RAB needs to be clarified.

**Table 6.10.2.1.1: Prioritized RABs**

| # | Traffic class<br>3GPP TS 23.107 [15] | SSD<br>3GPP TS 23.107 [15] | Max. rate, kbps | CS/PS | Version |
|---|---|---|---|---|---|
| 1 | Conversational | Speech | UL:12.2 DL:12.2 | CS | R99 |
| 1a | Conversational | Speech | UL:(12.2 7.95 5.9 4.75)<br>DL:(12.2 7.95 5.9 4.75) | CS | R99 |
| 1b | Conversational | Speech | UL:(12.2 7.4 5.9 4.75)<br>DL:(12.2 7.4 5.9 4.75) | CS | R99 |
| 2 | Conversational | Speech | UL:10.2 DL:10.2 | CS | R99 |
| 2a | Conversational | Speech | UL:(10.2, 6.7, 5.9, 4.75)<br>DL:(10.2, 6.7, 5.9, 4.75) | CS | R99 |
| 3 | Conversational | Speech | UL:7.95 DL:7.95 | CS | R99 |
| 4 | Conversational | Speech | UL:7.4 DL:7.4 | CS | R99 |
| 4a | Conversational | Speech | UL:(7.4, 6.7, 5.9, 4.75)<br>DL:(7.4, 6.7, 5.9, 4.75) | CS | R99 |
| 5 | Conversational | Speech | UL:6.7 DL:6.7 | CS | R99 |
| 6 | Conversational | Speech | UL:5.9 DL:5.9 | CS | R99 |
| 7 | Conversational | Speech | UL:5.15 DL:5.15 | CS | R99 |
| 8 | Conversational | Speech | UL:4.75 DL:4.75 | CS | R99 |
| 9 | Conversational | Unknown | UL:28.8 DL:28.8 | CS | R99 |
| 10 | Conversational | Unknown | UL:64 DL:64 | CS | R99 |
| 11 | Conversational | Unknown | UL:32 DL:32 | CS | R99 |
| 11a | Conversational | Unknown | UL:8 DL:8 | PS | R99 |
| 12 | Streaming | Unknown | UL:14.4 DL:14.4 | CS | R99 |
| 13 | Streaming | Unknown | UL:28.8 DL:28.8 | CS | R99 |
| 14 | Streaming | Unknown | UL:57.6 DL:57.6 | CS | R99 |
| 15 | Void | | | | |
| 15a | Streaming | Unknown | UL:16 DL:64 | PS | R99 |
| 15b | Streaming | Unknown | UL:16 DL:128 | PS | R99 |
| 16 | Void | | | | |
| 17 | Void | | | | |
| 18 | Void | | | | |
| 19 | Void | | | | |
| 20 | Interactive or Background | N/A | UL:32 DL:8 | PS | R99 |
| 20a | Interactive or Background | N/A | UL:8 DL:8 | PS | R99 |
| 20b | Interactive or Background | N/A | UL:16 DL:16 | PS | R99 |
| 20c | Interactive or Background | N/A | UL:32 DL:32 | PS | R99 |

| # | Traffic class 3GPP TS 23.107 [15] | SSD 3GPP TS 23.107 [15] | Max. rate, kbps | CS/PS | Version |
|---|---|---|---|---|---|
| 21 | Void | | | | |
| 22 | Interactive or Background | N/A | UL:32 DL:64 | PS | R99 |
| 23 | Interactive or Background | N/A | UL:64 DL:64 | PS | R99 |
| 24 | Interactive or Background | N/A | UL:64 DL:128 | PS | R99 |
| 25 | Interactive or Background | N/A | UL:128 DL:128 | PS | R99 |
| 26 | Interactive or Background | N/A | UL:64 DL:384 | PS | R99 |
| 27 | Interactive or Background | N/A | UL:128 DL:384 | PS | R99 |
| 28 | Interactive or Background | N/A | UL:384 DL:384 | PS | R99 |
| 29 | Interactive or Background | N/A | UL:64 DL:2048 | PS | R99 |
| 30 | Interactive or Background | N/A | UL:128 DL:2048 | PS | R99 |
| 31 | Void | | | | |
| 32 | Interactive or Background | N/A | UL:64 DL:256 | PS | R99 |
| 33 | Interactive or Background | N/A | UL:0 DL:32 | PS | R99 |
| 34 | Interactive or Background | N/A | UL:32 DL: 0 | PS | R99 |
| 35 | Interactive or Background | N/A | UL:64 DL:144 | PS | R99 |
| 36 | Interactive or Background | N/A | UL:144 DL:144 | PS | R99 |
| 37 | Conversational | N/A | UL:42.8 DL:42.8 | PS | REL-5 |
| 38 | Conversational | Speech | UL:(12.65 8.85 6.6) DL:(12.65 8.85 6.6) | CS | REL-5 |
| 39 | Interactive or Background | N/A | UL:64 DL:768 | PS | REL-5 |

**Table 6.10.2.1.2: Signalling RBs**

| # | Maximum rate, kbps | Logical channel | PhyCh onto which SRBs are mapped | Version |
|---|---|---|---|---|
| 1 | UL:1.7 DL:1.7 | DCCH | DPCH | R99 and Rel-4 only |
| 2 | UL:3.4 DL:3.4 | DCCH | DPCH | R99 |
| 3 | UL:13.6 DL:13.6 | DCCH | DPCH | R99 |
| 4 | DL:27.2 (alt. 40.8) | DCCH | SCCPCH | R99 |
| 5 | UL:16.6 | CCCH | PRACH | R99 |
| 6 | DL:30.4 (alt. 45.6) | CCCH | SCCPCH | R99 |
| 7 | DL:33.2 (alt. 49.8) | BCCH | SCCPCH | R99 |
| 8 | DL:24 (alt. 6.4) | PCCH | SCCPCH | R99 |
| 9 | DL: 0.15 | DCCH | DPCH | REL-5 |

### 6.10.2.2 Combinations of RABs and Signalling RBs

In the present document, physical channel parameters for following combinations of RABs and signalling RBs on a CCTrCH are described.

NOTE: It is understood that for speech service the AMR mode may be operated asymmetrically for the uplink and downlink.

Combinations on DPCH

1) Stand-alone UL:1.7 DL:1.7 kbps SRBs for DCCH (R99 and Rel-4 only).

2) Stand-alone UL:3.4 DL:3.4 kbps SRBs for DCCH.

3) Stand-alone UL:13.6 DL:13.6 kbps SRBs for DCCH.

4) Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH.

4a) Conversational / speech / UL:(12.2 7.95 5.9 4.75) DL:(12.2 7.95 5.9 4.75) kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH.

4b) Conversational / speech / UL:(12.2 7.4 5.9 4.75) DL:(12.2 7.4 5.9 4.75) kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH + DL:0.15 kbps SRB#5 for DCCH (REL-4).

5) Conversational / speech / UL:10.2 DL:10.2 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH.

5a) Conversational / speech / UL:(10.2, 6.7, 5.9, 4.75) DL:(10.2, 6.7, 5.9, 4.75) kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH.

6) Conversational / speech / UL:7.95 DL:7.95 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH.

7) Conversational / speech / UL:7.4 DL:7.4 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH.

7a) Conversational / speech / UL:(7.4, 6.7, 5.9, 4.75) DL:(7.4, 6.7, 5.9, 4.75) kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH.

8) Conversational / speech / UL:6.7 DL:6.7 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH.

9) Conversational / speech / UL:5.9 DL:5.9 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH.

10) Conversational / speech / UL:5.15 DL:5.15 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH.

11) Conversational / speech / UL:4.75 DL:4.75 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH.

12) Conversational / unknown / UL:28.8 DL:28.8 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH.

13) Conversational / unknown / UL:64 DL:64 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH.

14) Conversational / unknown / UL:32 DL:32 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH.

15) Streaming / unknown / UL:14.4/DL:14.4 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH.

16) Streaming / unknown / UL:28.8/DL:28.8 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH.

17) Streaming / unknown / UL:57.6/DL:57.6 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH.

18) Void

19) Void.

20) Void.

21) Void.

22) Void.

23) Interactive or background / UL:32 DL:8 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH.

23a) Interactive or background / UL:8 DL:8 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH.

23b) Interactive or background / UL:16 DL:16 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH.

23c) Interactive or background / UL:32 DL:32 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH.

23d) Interactive or background / UL:32 DL:32 kbps / PS RAB (20 ms TTI) + UL:3.4 DL:3.4 kbps SRBs for DCCH.

24) Void

25) Interactive or background / UL:32 DL: 64 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH.

26) Interactive or background / UL:64 DL: 64 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH.

27) Interactive or background / UL:64 DL:128 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH.

28) Interactive or background / UL:128 DL:128 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH.

29) Interactive or background / UL:64 DL:144 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH.

30) Interactive or background / UL:144 DL:144 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH.

31) Interactive or background / UL:64 DL:256 kbps / PS RAB + UL:3.4 DL: 3.4 kbps SRBs for DCCH.

32) Interactive or background / UL:64 DL:384 kbps / PS RAB + UL:3.4 DL: 3.4 kbps SRBs for DCCH.

33) Interactive or background / UL:128 DL:384 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH.

34) Interactive or background / UL:384 DL:384 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH.

35) Interactive or background / UL:64 DL:2 048 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH.

36) Void

37) Void

38) Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB
+ Interactive or background / UL:32 DL:8 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

38a) Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB
+ Interactive or background / UL:0 DL:0 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

38b) Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB
+ Interactive or background / UL:8 DL:8 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

38c) Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB
+ Interactive or background / UL:32 DL:32 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

38d) Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB
+ Interactive or background / UL:64 DL:64 kbps / PS RAB
+ Interactive or background / UL:64 DL:64 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

38e) Conversational / speech / UL:(12.2 7.95 5.9 4.75) DL:(12.2 7.95 5.9 4.75) kbps / CS RAB
+ Interactive or background / UL:0 DL:0 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

38f) Conversational / speech / UL:(12.2 7.95 5.9 4.75) DL:(12.2 7.95 5.9 4.75) kbps / CS RAB
+ Interactive or background / UL:8 DL:8 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

38g) Conversational / speech / UL:(12.2 7.95 5.9 4.75) DL:(12.2 7.95 5.9 4.75) kbps / CS RAB
+ Interactive or background / UL:16 DL:16 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

38h) Conversational / speech / UL:(12.2 7.95 5.9 4.75) DL:(12.2 7.95 5.9 4.75) kbps / CS RAB
+ Interactive or background / UL:32 DL:32 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

38i) Conversational / speech / UL:(12.2 7.95 5.9 4.75) DL:(12.2 7.95 5.9 4.75) kbps / CS RAB
+ Interactive or background / UL:64 DL:64 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

38j) Conversational / speech / UL:(12.2 7.95 5.9 4.75) DL:(12.2 7.95 5.9 4.75) kbps / CS RAB
+ Interactive or background / UL:64 DL:128 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

38k) Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB
+ Interactive or background / UL:32 DL:32 kbps / PS RAB
+ Interactive or background / UL:32 DL:32 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH (L1 multiplexing).

39) Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB
+ Interactive or background / UL:32 DL:64 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

40) Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB
+ Interactive or background / UL:64 DL:64 kbps / PS RAB
+ UL:3.4 DL: 3.4 kbps SRBs for DCCH.

41) Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB
+ Interactive or background / UL:64 DL:128 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

42) Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB
+ Interactive or background / UL:64 DL:256 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

43) Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB
+ Interactive or background / UL:64 DL:384 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

44) Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB
+ Interactive or background / UL:128 DL:2 048 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

45) Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB
+ Streaming / unknown / UL:57.6 DL:57.6 kbps / CS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

46) Void

47) Void.

48) Void.

49) Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB
+ Conversational / unknown / UL:64 DL:64 kbps / CS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

49a) Conversational / speech / UL:(12.2 7.95 5.9 4.75) DL:(12.2 7.95 5.9 4.75) kbps / CS RAB
+ Conversational / unknown / UL:64 DL:64 kbps / CS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

50) Conversational / unknown / UL:64 DL:64 kbps / CS RAB
+ Conversational / unknown / UL:64 DL:64 kbps / CS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

51) Conversational / unknown / UL:64 DL:64 kbps / CS RAB
+ Interactive or background / UL:64 DL:64 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

51a) Conversational / unknown / UL:64 DL:64 kbps / CS RAB
+ Interactive or Background / UL:8 DL:8 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

51b) Conversational / unknown / UL:64 DL:64 kbps / CS RAB
+ Interactive or Background / UL:16 DL:64 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

52) Conversational / unknown / UL:64 DL:64 kbps / CS RAB
+ Interactive or background / UL:64 DL:128 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

53) Conversational / unknown / UL:64 DL:64 kbps / CS RAB
+ Interactive or background / UL:128 DL:128 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

54) Void

55) Void.

56) Interactive or background / UL:8 DL:8 kbps / PS RAB
+ Interactive or background / UL:8 DL:8 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

57) Interactive or background / UL:64 DL:64 kbps / PS RAB
+ Interactive or background / UL:64 DL:64 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

58) Streaming / unknown / UL:16 DL:64 kbps / PS RAB
+ Interactive or background / UL:8 DL:8 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

58a) Streaming / unknown / UL:16 DL:128 kbps / PS RAB + Interactive or background / UL:8 DL:8 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

59) Conversational / Speech / UL:42.8 DL:42.8 kbps / PS RAB
+ Interactive or background / UL:16 DL:16 kbps / PS RAB
+ Interactive or background / UL:16 DL:16 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH (REL-5).

60) Conversational / Speech / UL:42.8 DL:42.8 kbps / PS RAB
+ Interactive or background / UL:16 DL:16 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH (REL-5).

61) Conversational / unknown / UL:8 DL:8 kbps / PS RAB
+ Interactive or Background / UL:8 DL:8 kbps / PS RAB +
+ UL:3.4 DL:3.4 kbps SRBs for DCCH

62) Conversational / speech / UL:(12.65 8.85 6.6) DL:(12.65 8.85 6.6) kbps / CS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH + DL:0.15 kbps SRB#5 for DCCH (REL-5).

63) Interactive or background / UL:64 DL:768 kbps / PS RAB
+ UL:3.4 DL: 3.4 kbps SRBs for DCCH (REL-5).

### Combinations on DSCH and DPCH

1) Void

2) Interactive or background / UL:64 DL:384 kbps / PS RAB
+ UL:3.4 DL: 3.4 kbps SRBs for DCCH (R99 and Rel-4 only).

3) Interactive or background / UL:64 DL:2 048 kbps / PS RAB
+ UL:3.4 DL: 3.4 kbps SRBs for DCCH (R99 and Rel-4 only).

4) Void

5) Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB
+ Interactive or background / UL:64 DL:384 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH (R99 and Rel-4 only).

6) Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB
+ Interactive or background / UL:64 DL:2 048 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH (R99 and Rel-4 only).

### Combinations on SCCPCH

1) Stand-alone 24 kbps SRB for PCCH.

2) Interactive or background / DL:32 kbps / PS RAB
+ SRB for CCCH
+ SRBs for DCCH
+ SRB for BCCH.

3) Interactive or background / DL:32 kbps / PS RAB
+ SRB for PCCH
+ SRB for CCCH
+ SRBs for DCCH
+ SRB for BCCH.

4) RB for CTCH
+ SRB for CCCH
+SRB for BCCH

Combinations on PRACH

1) Interactive or background / UL:32 kbps / PS RAB
+ SRB for CCCH
+ SRBs for DCCH.

Combinations on DPCH and HS-PDSCH

1) Interactive or background / UL:64 DL: [max bit rate depending on UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH (REL-5)

1a) Interactive or background / UL:128 DL: [max bit rate depending on UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH (REL-5)

2) Interactive or background / UL:384 DL: [max bit rate depending on UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH (REL-5)

3) Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:384 DL:[Bit rate depending on the UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH (REL-5)

3a) Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:64 DL:[Bit rate depending on the UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH (REL-5)

4) Conversational / unknown / UL:64 DL:64 kbps / CS RAB + Interactive or background / UL:384 DL:[Bit rate depending on the UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH (REL-5)

4a) Conversational / unknown / UL:64 DL:64 kbps / CS RAB + Interactive or background / UL:64 DL:[Bit rate depending on the UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH (REL-5)

5) Interactive or background / UL:384 DL:[Bit rate depending on the UE category] / PS RAB + Interactive or background / UL:384 DL:[Bit rate depending on the UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH (REL-5)

5a) Interactive or background / UL:64 DL:[Bit rate depending on the UE category] / PS RAB + Interactive or background / UL:64 DL:[Bit rate depending on the UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH (REL-5)

6) Streaming / unknown / UL:128 DL: [guaranteed 128, max bit rate depending on UE category] kbps / PS RAB + Interactive or background / UL:128 DL: [max bit rate depending on UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH (REL-5)

7) Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Streaming / unknown / UL:128 DL: [guaranteed 128, max bit rate depending on UE category] kbps / PS RAB + Interactive or background / UL:128 DL: [max bit rate depending on UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH (REL-5)

Combinations on HS-PDSCH and E-DPDCH

1) Interactive or background / UL: [max bit rate depending on UE category and TTI] DL: [max bit rate depending on UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH on DCH (REL-6)

2) Interactive or background / UL: [max bit rate depending on UE category and TTI] DL: [max bit rate depending on UE category] / PS RAB + UL:[max bit rate depending on UE category and TTI] DL:3.4 kbps SRBs for DCCH on E-DCH and DL DCH (REL-6)

3) Interactive or background / UL: [max bit rate depending on UE category and TTI] DL: [max bit rate depending on UE category] / PS RAB + UL: [max bit rate depending on UE category and TTI] DL: [max bit rate depending on UE category] SRBs for DCCH on E-DCH and HS-DSCH (REL-6)

4) Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL: [max bit rate depending on UE category and TTI] DL: [max bit rate depending on UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

5) Streaming / unknown / UL:[max bit rate depending on UE category and TTI] DL: [max bit rate depending on UE category] kbps / PS RAB + Interactive or background / UL: [max bit rate depending on UE category and TTI] DL: [max bit rate depending on UE category] / PS RAB + UL:[max bit rate depending on UE category and TTI] DL:3.4 kbps SRBs for DCCH on E-DCH and DL DCH

### 6.10.2.3 Example of linkage between RABs and services

RABs, which are included in the present document, can provide the services as shown in table 6.10.1.1. Furthermore, the required BER for each RAB, which is assumed in the present document, is shown in table 6.10.2.3.1.

**Table 6.10.2.3.1: Example of linkage between RABs and services**

| RAB | | | | Residual BER [15] | Services |
|---|---|---|---|---|---|
| **Traffic class [15]** | **SSD [15]** | **Max. rate, kbps** | **CS/PS** | | |
| Conversational | Speech | UL:4.75-12.2 DL:4.75-12.2 | CS | $5x10^{-4}$, $1x10^{-3}$, $5x10^{-3}$ | AMR speech |
| Conversational | Unknown | UL:64 DL:64 | CS | $1x10^{-4}$ or $1x10^{-6}$ | UDI 1B, 64k 3G-324M [15] |
| Conversational | Unknown | UL:32 DL:32 | CS | $1x10^{-4}$ or $1x10^{-6}$ | 32k 3G-324M [15] |
| Conversational | Unknown | UL:28.8 DL:28.8 | CS | $1x10^{-3}$ | Transparent modem |
| Streaming | Unknown | UL:14.4 DL:14.4 | CS | $1x10^{-3}$ | FAX[6] |
| Streaming | Unknown | UL:28.8 DL:28.8 | CS | $1x10^{-3}$ | FAX [18] PIAFS 32 kbps |
| Streaming | Unknown | UL:57.6 DL:57.6 | CS | $1x10^{-3}$ | Modem [18], FTM [17] PIAFS 64 kbps |
| Streaming | Unknown | UL:64-128 or DL:64-384 | CS | $1x10^{-3}$ or $1x10^{-4}$ | Streaming video, uni-directional |
| Interactive or Background | N/A | UL:32-384 DL:8-2048 | PS | $1x10^{-3}$ or $1x10^{-4}$ | Packet |

NOTE 1: SMS can be provided via the signalling RB (DCCH) on DPCH or SCCPCH.

NOTE 2: CBS can be provided via the signalling RB (CTCH) on SCCPCH.

NOTE 3: UDI *n*B can be provided via *n* RABs of conversational 64 kbps.

### 6.10.2.4 Typical radio parameter sets

#### 6.10.2.4.1 Combinations on DPCH

##### 6.10.2.4.1.1 Stand-alone UL:1.7 DL:1.7 kbps SRBs for DCCH

###### 6.10.2.4.1.1.1 Uplink

6.10.2.4.1.1.1.1 Transport channel parameters

6.10.2.4.1.1.1.1.1 Transport channel parameters for UL:1.7 kbps SRBs for DCCH

| Higher layer | RAB/signalling RB | | SRB#1 | SRB#2 | SRB#3 | SRB#4 |
|---|---|---|---|---|---|---|
| | User of Radio Bearer | | RRC | RRC | NAS_DT High prio | NAS_DT Low prio |
| RLC | Logical channel type | | DCCH | DCCH | DCCH | DCCH |
| | RLC mode | | UM | AM | AM | AM |
| | Payload sizes, bit | | 136 | 128 | 128 | 128 |
| | Max data rate, bps | | 1 700 | 1 600 | 1 600 | 1 600 |
| | AMD/UMD PDU header, bit | | 8 | 16 | 16 | 16 |
| MAC | MAC header, bit | | 4 | 4 | 4 | 4 |
| | MAC multiplexing | | 4 logical channel multiplexing | | | |
| Layer 1 | TrCH type | | DCH | | | |
| | TB sizes, bit | | 148 (alt 0, 148) | | | |
| | TFS | TF0, bits | 0x148 (alt 1x0) | | | |
| | | TF1, bits | 1x148 | | | |

| | | |
|---|---|---|
| | TTI, ms | 80 |
| | Coding type | CC 1/3 |
| | CRC, bit | 16 |
| | Max number of bits/TTI before rate matching | 516 |
| | Uplink: Max number of bits/radio frame before rate matching | 65 |
| | RM attribute | 155 to 185 |

6.10.2.4.1.1.1.2 TFCS

| | |
|---|---|
| TFCS size | 2 |
| TFCS | SRBs for DCCH = TF0, TF1 |

6.10.2.4.1.1.1.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Uplink | Min spreading factor | 256 |
| | Max number of DPDCH data bits/radio frame | 150 |
| | Puncturing Limit | 1 |

6.10.2.4.1.1.2 Downlink

6.10.2.4.1.1.2.1 Transport channel parameters

6.10.2.4.1.1.2.1.1 Transport channel parameters for DL:1.7 kbps SRBs for DCCH

| | | | | | | |
|---|---|---|---|---|---|---|
| Higher layer | RAB/signalling RB | | SRB#1 | SRB#2 | SRB#3 | SRB#4 |
| | User of Radio Bearer | | RRC | RRC | NAS_DT High prio | NAS_DT Low prio |
| RLC | Logical channel type | | DCCH | DCCH | DCCH | DCCH |
| | RLC mode | | UM | AM | AM | AM |
| | Payload sizes, bit | | 136 | 128 | 128 | 128 |
| | Max data rate, bps | | 1 700 | 1 600 | 1 600 | 1 600 |
| | AMD/UMD PDU header, bit | | 8 | 16 | 16 | 16 |
| MAC | MAC header, bit | | 4 | 4 | 4 | 4 |
| | MAC multiplexing | | 4 logical channel multiplexing | | | |
| Layer 1 | TrCH type | | DCH | | | |
| | TB sizes, bit | | 148 (alt 0, 148) (note) | | | |
| | TFS | TF0, bits | 0 x148 (alt 1x0) (note) | | | |
| | | TF1, bits | 1x148 | | | |
| | TTI, ms | | 80 | | | |
| | Coding type | | CC 1/3 | | | |
| | CRC, bit | | 16 | | | |
| | Max number of bits/TTI before rate matching | | 516 | | | |
| | RM attribute | | 155 to 185 | | | |
| NOTE: Alternative parameters enable the measurement "transport channel BLER" in the UE. | | | | | | |

6.10.2.4.1.1.2.1.2 TFCS

| | |
|---|---|
| TFCS size | 2 |
| TFCS | SRBs for DCCH = TF0, TF1 |

6.10.2.4.1.1.2.2 Physical channel parameters

| | | | |
|---|---|---|---|
| DPCH Downlink | DTX position | | Fixed |
| | Spreading factor | | 512 |
| | DPCCH | Number of TFCI bits/slot | 0 |
| | | Number of TPC bits/slot | 2 |
| | | Number of Pilot bits/slot | 4 |
| | DPDCH | Number of data bits/slot | 4 |
| | | Number of data bits/frame | 60 |

6.10.2.4.1.2 Stand-alone UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.2.4.1.2.1 Uplink

6.10.2.4.1.2.1.1 Transport channel parameters

6.10.2.4.1.2.1.1.1 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

| Higher layer | RAB/signalling RB | | SRB#1 | SRB#2 | SRB#3 | SRB#4 |
|---|---|---|---|---|---|---|
| | User of Radio Bearer | | RRC | RRC | NAS_DT High prio | NAS_DT Low prio |
| RLC | Logical channel type | | DCCH | DCCH | DCCH | DCCH |
| | RLC mode | | UM | AM | AM | AM |
| | Payload sizes, bit | | 136 | 128 | 128 | 128 |
| | Max data rate, bps | | 3 400 | 3 200 | 3 200 | 3 200 |
| | AMD/UMD PDU header, bit | | 8 | 16 | 16 | 16 |
| MAC | MAC header, bit | | 4 | 4 | 4 | 4 |
| | MAC multiplexing | | 4 logical channel multiplexing | | | |
| Layer 1 | TrCH type | | DCH | | | |
| | TB sizes, bit | | 148 (alt 0, 148) | | | |
| | TFS | TF0, bits | 0x148 (alt 1x0) | | | |
| | | TF1, bits | 1x148 | | | |
| | TTI, ms | | 40 | | | |
| | Coding type | | CC 1/3 | | | |
| | CRC, bit | | 16 | | | |
| | Max number of bits/TTI before rate matching | | 516 | | | |
| | Uplink: Max number of bits/radio frame before rate matching | | 129 | | | |
| | RM attribute | | 155 to 185 | | | |

6.10.2.4.1.2.1.1.2 TFCS

| TFCS size | 2 |
|---|---|
| TFCS | SRBs for DCCH = TF0, TF1 |

6.10.2.4.1.2.1.2 Physical channel parameters

| DPCH Uplink | Min spreading factor | 256 |
|---|---|---|
| | Max number of DPDCH data bits/radio frame | 150 |
| | Puncturing Limit | 1 |

6.10.2.4.1.2.2 Downlink

6.10.2.4.1.2.2.1 Transport channel parameters

6.10.2.4.1.2.2.1.1 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

| Higher layer | RAB/signalling RB | | SRB#1 | SRB#2 | SRB#3 | SRB#4 |
|---|---|---|---|---|---|---|
| | User of Radio Bearer | | RRC | RRC | NAS_DT High prio | NAS_DT Low prio |
| RLC | Logical channel type | | DCCH | DCCH | DCCH | DCCH |
| | RLC mode | | UM | AM | AM | AM |
| | Payload sizes, bit | | 136 | 128 | 128 | 128 |
| | Max data rate, bps | | 3 400 | 3 200 | 3 200 | 3 200 |
| | AMD/UMD PDU header, bit | | 8 | 16 | 16 | 16 |
| MAC | MAC header, bit | | 4 | 4 | 4 | 4 |
| | MAC multiplexing | | 4 logical channel multiplexing | | | |
| Layer 1 | TrCH type | | DCH | | | |
| | TB sizes, bit | | 148 (alt 0, 148) (note) | | | |
| | TFS | TF0, bits | 0x148 (alt 1x0) (note) | | | |
| | | TF1, bits | 1x148 | | | |

| | | |
|---|---|---|
| | TTI, ms | 40 |
| | Coding type | CC 1/3 |
| | CRC, bit | 16 |
| | Max number of bits/TTI before rate matching | 516 |
| | RM attribute | 155 to 230 |
| NOTE: Alternative parameters enable the measurement "transport channel BLER" in the UE. | | |

6.10.2.4.1.2.2.1.2 TFCS

| | |
|---|---|
| TFCS size | 2 |
| TFCS | SRBs for DCCH = TF0, TF1 |

6.10.2.4.1.2.2.2 Physical channel parameters

| | | | |
|---|---|---|---|
| DPCH Downlink | DTX position | | Fixed |
| | Spreading factor | | 256 |
| | DPCCH | Number of TFCI bits/slot | 0 |
| | | Number of TPC bits/slot | 2 |
| | | Number of Pilot bits/slot | 4 |
| | DPDCH | Number of data bits/slot | 14 |
| | | Number of data bits/frame | 210 |

6.10.2.4.1.3 Stand-alone UL:13.6 DL:13.6 kbps SRBs for DCCH

6.10.2.4.1.3.1 Uplink

6.10.2.4.1.3.1.1 Transport channel parameters

6.10.2.4.1.3.1.1.1 Transport channel parameters for UL:13.6 kbps SRBs for DCCH

| | | | | | | |
|---|---|---|---|---|---|---|
| Higher layer | RAB/signalling RB | | SRB#1 | SRB#2 | SRB#3 | SRB#4 |
| | User of Radio Bearer | | RRC | RRC | NAS_DT High prio | NAS_DT Low prio |
| RLC | Logical channel type | | DCCH | DCCH | DCCH | DCCH |
| | RLC mode | | UM | AM | AM | AM |
| | Payload sizes, bit | | 136 | 128 | 128 | 128 |
| | Max data rate, bps | | 13 600 | 12 800 | 12 800 | 12 800 |
| | AMD/UMD PDU header, bit | | 8 | 16 | 16 | 16 |
| MAC | MAC header, bit | | 4 | 4 | 4 | 4 |
| | MAC multiplexing | | 4 logical channel multiplexing | | | |
| Layer 1 | TrCH type | | DCH | | | |
| | TB sizes, bit | | 148 (alt 0, 148) | | | |
| | TFS | TF0, bits | 0x148 (alt 1x0) | | | |
| | | TF1, bits | 1x148 | | | |
| | TTI, ms | | 10 | | | |
| | Coding type | | CC 1/3 | | | |
| | CRC, bit | | 16 | | | |
| | Max number of bits/TTI before rate matching | | 516 | | | |
| | Uplink: Max number of bits/radio frame before rate matching | | 516 | | | |

6.10.2.4.1.3.1.1.2 TFCS

| | |
|---|---|
| TFCS size | 2 |
| TFCS | SRBs for DCCH = TF0, TF1 |

6.10.2.4.1.3.1.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Uplink | Min spreading factor | 64 |
| | Max number of DPDCH data bits/radio frame | 600 |

| | | |
|---|---|---|
| | Puncturing Limit | 1 |

6.10.2.4.1.3.2 Downlink

6.10.2.4.1.3.2.1 Transport channel parameters

6.10.2.4.1.3.2.1.1 Transport channel parameters for DL:13.6 kbps SRBs for DCCH

| Higher layer | RAB/signalling RB | | SRB#1 | SRB#2 | SRB#3 | SRB#4 |
|---|---|---|---|---|---|---|
| | User of Radio Bearer | | RRC | RRC | NAS_DT High prio | NAS_DT Low prio |
| RLC | Logical channel type | | DCCH | DCCH | DCCH | DCCH |
| | RLC mode | | UM | AM | AM | AM |
| | Payload sizes, bit | | 136 | 128 | 128 | 128 |
| | Max data rate, bps | | 13 600 | 12 800 | 12 800 | 12 800 |
| | AMD/UMD PDU header, bit | | 8 | 16 | 16 | 16 |
| MAC | MAC header, bit | | 4 | 4 | 4 | 4 |
| | MAC multiplexing | | 4 logical channel multiplexing | | | |
| Layer 1 | TrCH type | | DCH | | | |
| | TB sizes, bit | | 148 (alt 0, 148) (note) | | | |
| | TFS | TF0, bits | 0x148 (alt 1x0) (note) | | | |
| | | TF1, bits | 1x148 | | | |
| | TTI, ms | | 10 | | | |
| | Coding type | | CC 1/3 | | | |
| | CRC, bit | | 16 | | | |
| | Max number of bits/TTI before rate matching | | 516 | | | |
| NOTE: alternative parameters enable the measurement "transport channel BLER" in the UE. | | | | | | |

6.10.2.4.1.3.2.1.2 TFCS

| | |
|---|---|
| TFCS size | 2 |
| TFCS | SRBs for DCCH = TF0, TF1 |

6.10.2.4.1.3.2.2 Physical channel parameters

| DPCH Downlink | DTX position | | Fixed |
|---|---|---|---|
| | Spreading factor | | 128 |
| | DPCCH | Number of TFCI bits/slot | 0 |
| | | Number of TPC bits/slot | 2 |
| | | Number of Pilot bits/slot | 4 |
| | DPDCH | Number of data bits/slot | 34 |
| | | Number of data bits/frame | 510 |

6.10.2.4.1.4 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.2.4.1.4.1 Uplink

6.10.2.4.1.4.1.1 Transport channel parameters

6.10.2.4.1.4.1.1.1 Transport channel parameters for Conversational / speech / UL:12.2 kbps / CS RAB

| Higher layer | RAB/Signalling RB | RAB subflow #1 | RAB subflow #2 | RAB subflow #3 |
|---|---|---|---|---|
| RLC | Logical channel type | DTCH | | |
| | RLC mode | TM | TM | TM |
| | Payload sizes, bit | 39, 81 (alt. 0, 39, 81) | 103 | 60 |
| | Max data rate, bps | 12 200 | | |
| | TrD PDU header, bit | 0 | | |
| MAC | MAC header, bit | 0 | | |
| | MAC multiplexing | N/A | | |
| Layer 1 | TrCH type | DCH | DCH | DCH |

| | | | | | |
|---|---|---|---|---|---|
| | TB sizes, bit | | 39, 81<br>(alt. 0, 39, 81) | 103 | 60 |
| | TFS | TF0, bits | 0x81(alt. 1x0) (note) | 0x103 | 0x60 |
| | | TF1, bits | 1x39 | 1x103 | 1x60 |
| | | TF2, bits | 1x81 | N/A | N/A |
| | TTI, ms | | 20 | 20 | 20 |
| | Coding type | | CC 1/3 | CC 1/3 | CC 1/2 |
| | CRC, bit | | 12 | N/A | N/A |
| | Max number of bits/TTI after channel coding | | 303 | 333 | 136 |
| | Uplink: Max number of bits/radio frame before rate matching | | 152 | 167 | 68 |
| | RM attribute | | 180 to 220 | 170 to 210 | 215 to 256 |
| NOTE: In case of using this alternative, CRC parity bits are to be attached to RAB subflow#1 any time since number of TrBlks are 1 even if there is no data on RAB subflow#1 (see clause 4.2.1.1 in 3GPP TS 25.212 [14]). | | | | | |

6.10.2.4.1.4.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.1.1.

6.10.2.4.1.4.1.1.3 TFCS

| | |
|---|---|
| TFCS size | 6 |
| TFCS | (RAB subflow#1, RAB subflow#2, RAB subflow#3,DCCH)=<br>(TF0, TF0, TF0, TF0), (TF1, TF0, TF0, TF0), (TF2, TF1, TF1, TF0),<br>(TF0, TF0, TF0, TF1), (TF1, TF0, TF0, TF1), (TF2, TF1, TF1, TF1) |

6.10.2.4.1.4.1.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Uplink | Min spreading factor | 64 |
| | Max number of DPDCH data bits/radio frame | 600 |
| | Puncturing Limit | 0.84 |

6.10.2.4.1.4.2 Downlink

6.10.2.4.1.4.2.1 Transport channel parameters

6.10.2.4.1.4.2.1.1 Transport channel parameters for Conversational / speech / DL:12.2 kbps / CS RAB

| | | | | | |
|---|---|---|---|---|---|
| Higher layer | RAB/Signalling RB | | RAB subflow #1 | RAB subflow #2 | RAB subflow #3 |
| RLC | Logical channel type | | DTCH | | |
| | RLC mode | | TM | TM | TM |
| | Payload sizes, bit | | 0<br>39<br>81 | 103 | 60 |
| | Max data rate, bps | | 12 200 | | |
| | TrD PDU header, bit | | 0 | | |
| MAC | MAC header, bit | | 0 | | |
| | MAC multiplexing | | N/A | | |
| Layer 1 | TrCH type | | DCH | DCH | DCH |
| | TB sizes, bit | | 0<br>39<br>81 | 103 | 60 |
| | TFS (note 1) | TF0, bits | 1x0 (note 2) | 0x103 | 0x60 |
| | | TF1, bits | 1x39 | 1x103 | 1x60 |
| | | TF2, bits | 1x81 | N/A | N/A |
| | TTI, ms | | 20 | 20 | 20 |
| | Coding type | | CC 1/3 | CC 1/3 | CC 1/2 |
| | CRC, bit | | 12 | N/A | N/A |
| | Max number of bits/TTI after channel coding | | 303 | 333 | 136 |
| | RM attribute | | 180 to 220 | 170 to 210 | 215 to 256 |

| |
|---|
| NOTE 1: The TrCH corresponding to RAB subflow #1 should be used as the guiding TrCH, (see clause 4.3 in 3GPP TS 25.212 [14]). |
| NOTE 2: CRC parity bits are to be attached to RAB subflow#1 any time since number of TrBlks are 1 even if there is no data on RAB subflow#1 (see clause 4.2.1.1 in 3GPP TS 25.212 [14]). |

#### 6.10.2.4.1.4.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.2.1.1.

#### 6.10.2.4.1.4.2.1.3 TFCS

| | |
|---|---|
| TFCS size | 6 |
| TFCS | (RAB subflow#1, RAB subflow#2, RAB subflow#3,DCCH)=<br>(TF0, TF0, TF0, TF0), (TF1, TF0, TF0, TF0), (TF2, TF1, TF1, TF0),<br>(TF0, TF0, TF0, TF1), (TF1, TF0, TF0, TF1), (TF2, TF1, TF1, TF1) |

#### 6.10.2.4.1.4.2.2 Physical channel parameters

| | | | |
|---|---|---|---|
| DPCH Downlink | DTX position | | Fixed |
| | Spreading factor | | 128 |
| | DPCCH | Number of TFCI bits/slot | 0 |
| | | Number of TPC bits/slot | 2 |
| | | Number of Pilot bits/slot | 4 |
| | DPDCH | Number of data bits/slot | 34 |
| | | Number of data bits/frame | 510 |

### 6.10.2.4.1.4a Conversational / speech / UL:(12.2 7.95 5.9 4.75) DL:(12.2 7.95 5.9 4.75) kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

#### 6.10.2.4.1.4a.1 Uplink

#### 6.10.2.4.1.4a.1.1 Transport channel parameters

#### 6.10.2.4.1.4a.1.1.1 Transport channel parameters for Conversational / speech / UL: (12.2 7.95 5.9 4.75) kbps / CS RAB

| | | | | | |
|---|---|---|---|---|---|
| Higher layer | RAB/Signalling RB | | RAB subflow #1 | RAB subflow #2 | RAB subflow #3 |
| RLC | Logical channel type | | DTCH | | |
| | RLC mode | | TM | TM | TM |
| | Payload sizes, bit | | 39, 42, 55, 75, 81<br>(alt. 0, 39, 81) | 53, 63, 84, 103 | 60 |
| | Max data rate, bps | | 12 200 | | |
| | TrD PDU header, bit | | 0 | | |
| MAC | MAC header, bit | | 0 | | |
| | MAC multiplexing | | N/A | | |
| Layer 1 | TrCH type | | DCH | DCH | DCH |
| | TB sizes, bit | | 39, 42, 55, 75, 81<br>(alt. 0, 39, 42, 55, 75, 81) | 53, 63, 84, 103 | 60 |
| | TFS | TF0, bits | 0x81(alt. 1x0) (note) | 0x103 | 0x60 |
| | | TF1, bits | 1x39 | 1x53 | 1x60 |
| | | TF2 bits | 1x42 | 1x63 | N/A |
| | | TF3, bits | 1x55 | 1x84 | N/A |
| | | TF4, bits | 1x75 | 1x103 | N/A |
| | | TF5, bits | 1x81 | N/A | N/A |
| | TTI, ms | | 20 | 20 | 20 |
| | Coding type | | CC 1/3 | CC 1/3 | CC 1/2 |
| | CRC, bit | | 12 | N/A | N/A |
| | Max number of bits/TTI after channel coding | | 303 | 333 | 136 |
| | Uplink: Max number of bits/radio frame before rate matching | | 152 | 167 | 68 |
| | RM attribute | | 180 to 220 | 170 to 210 | 215 to 256 |

| NOTE: | In case of using this alternative, CRC parity bits are to be attached to RAB subflow#1 any time since number of TrBlks are 1 even if there is no data on RAB subflow#1 (see clause 4.2.1.1 in 3GPP TS 25.212 [14]). |
|---|---|

6.10.2.4.1.4a.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.1.1.1.

6.10.2.4.1.4a.1.1.3 TFCS

| TFCS size | 12 |
|---|---|
| TFCS | (RAB subflow#1, RAB subflow#2, RAB subflow#3, DCCH)=<br>(TF0,TF0,TF0,TF0), (TF1,TF0,TF0,TF0), (TF2,TF1,TF0,TF0), (TF3,TF2,TF0,TF0),<br>(TF4,TF3,TF0,TF0), (TF5,TF4,TF1,TF0), (TF0,TF0,TF0,TF1), (TF1,TF0,TF0,TF1),<br>(TF2,TF1,TF0,TF1), (TF3,TF2,TF0,TF1), (TF4,TF3,TF0,TF1), (TF5,TF4,TF1,TF1) |

6.10.2.4.1.4a.1.2 Physical channel parameters

| DPCH Uplink | Min spreading factor | 64 |
|---|---|---|
| | Max number of DPDCH data bits/radio frame | 600 |
| | Puncturing Limit | 0.84 |

6.10.2.4.1.4a.2 Downlink

6.10.2.4.1.4a.2.1 Transport channel parameters

6.10.2.4.1.4a.2.1.1 Transport channel parameters for Conversational / speech / DL: (12.2 7.95 5.9 4.75) kbps / CS RAB

| Higher layer | RAB/Signalling RB | | RAB subflow #1 | RAB subflow #2 | RAB subflow #3 |
|---|---|---|---|---|---|
| RLC | Logical channel type | | DTCH | | |
| | RLC mode | | TM | TM | TM |
| | Payload sizes, bit | | 0, 39, 42, 55, 75, 81 | 53, 63, 84, 103 | 60 |
| | Max data rate, bps | | 12 200 | | |
| | TrD PDU header, bit | | 0 | | |
| MAC | MAC header, bit | | 0 | | |
| | MAC multiplexing | | N/A | | |
| Layer 1 | TrCH type | | DCH | DCH | DCH |
| | TB sizes, bit | | 0, 39, 42, 55, 75, 81 | 53, 63, 84, 103 | 60 |
| | TFS (note 1) | TF0, bits | 1x0 (note 2) | 0x103 | 0x60 |
| | | TF1, bits | 1x39 | 1x53 | 1x60 |
| | | TF2, bits | 1x42 | 1x63 | N/A |
| | | TF3, bits | 1x55 | 1x84 | N/A |
| | | TF4, bits | 1x75 | 1x103 | N/A |
| | | TF5, bits | 1x81 | N/A | N/A |
| | TTI, ms | | 20 | 20 | 20 |
| | Coding type | | CC 1/3 | CC 1/3 | CC 1/2 |
| | CRC, bit | | 12 | N/A | N/A |
| | Max number of bits/TTI after channel coding | | 303 | 333 | 136 |
| | RM attribute | | 180 to 220 | 170 to 210 | 215 to 256 |

NOTE 1: The TrCH corresponding to RAB subflow #1 should be used as the guiding TrCH, (see clause 4.3 in 3GPP TS 25.212 [14]).
NOTE 2: CRC parity bits are to be attached to RAB subflow#1 any time since number of TrBlks are 1 even if there is no data on RAB subflow#1 (see clause 4.2.1.1 in 3GPP TS 25.212 [14]).

6.10.2.4.1.4a.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.2.1.1

6.10.2.4.1.4a.2.1.3 TFCS

| TFCS size | 12 |
|---|---|

| | |
|---|---|
| TFCS | (RAB subflow#1, RAB subflow#2, RAB subflow#3, DCCH)=<br>(TF0,TF0,TF0,TF0), (TF1,TF0,TF0,TF0), (TF2,TF1,TF0,TF0), (TF3,TF2,TF0,TF0),<br>(TF4,TF3,TF0,TF0), (TF5,TF4,TF1,TF0), (TF0,TF0,TF0,TF1), (TF1,TF0,TF0,TF1),<br>(TF2,TF1,TF0,TF1), (TF3,TF2,TF0,TF1), (TF4,TF3,TF0,TF1), (TF5,TF4,TF1,TF1) |

6.10.2.4.1.4a.2.2 Physical channel parameters

| | | | |
|---|---|---|---|
| DPCH Downlink | DTX position | | Fixed |
| | Spreading factor | | 128 |
| | DPCCH | Number of TFCI bits/slot | 0 |
| | | Number of TPC bits/slot | 2 |
| | | Number of Pilot bits/slot | 4 |
| | DPDCH | Number of data bits/slot | 34 |
| | | Number of data bits/frame | 510 |

6.10.2.4.1.4b Conversational / speech / UL:(12.2 7.4 5.9 4.75) DL:(12.2 7.4 5.9 4.75) kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH + DL:0.15 kbps SRB#5 for DCCH

6.10.2.4.1.4b.1.1 Transport channel parameters

6.10.2.4.1.4b.1.1.1 Transport channel parameters for Conversational / speech / UL: (12.2 7.4 5.9 4.75) kbps / CS RAB

| Higher layer | RAB/Signalling RB | | RAB subflow #1 | RAB subflow #2 | RAB subflow #3 |
|---|---|---|---|---|---|
| RLC | Logical channel type | | DTCH | | |
| | RLC mode | | TM | TM | TM |
| | Payload sizes, bit | | 39, 42, 55, 61, 81<br>(alt. 0, 39, 42, 55, 61, 81) | 53, 63, 87, 103 | 60 |
| | Max data rate, bps | | 12 200 | | |
| | TrD PDU header, bit | | 0 | | |
| MAC | MAC header, bit | | 0 | | |
| | MAC multiplexing | | N/A | | |
| Layer 1 | TrCH type | | DCH | DCH | DCH |
| | TB sizes, bit | | 39, 42, 55, 61, 81<br>(alt. 0, 39, 42, 55, 61, 81) | 53, 63, 87, 103 | 60 |
| | TFS | TF0, bits | 0x81(alt. 1x0) (note) | 0x103 | 0x60 |
| | | TF1, bits | 1x39 | 1x53 | 1x60 |
| | | TF2 bits | 1x42 | 1x63 | N/A |
| | | TF3, bits | 1x55 | 1x87 | N/A |
| | | TF4, bits | 1x61 | 1x103 | N/A |
| | | TF5, bits | 1x81 | N/A | N/A |
| | TTI, ms | | 20 | 20 | 20 |
| | Coding type | | CC 1/3 | CC 1/3 | CC 1/2 |
| | CRC, bit | | 12 | N/A | N/A |
| | Max number of bits/TTI after channel coding | | 303 | 333 | 136 |
| | Uplink: Max number of bits/radio frame before rate matching | | 152 | 167 | 68 |
| | RM attribute | | 180 to 220 | 170 to 210 | 215 to 256 |
| NOTE: In case of using this alternative, CRC parity bits are to be attached to RAB subflow#1 any time since number of TrBlks are 1 even if there is no data on RAB subflow#1 (see clause 4.2.1.1 in 3GPP TS 25.212 [14]). | | | | | |

6.10.2.4.1.4b.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See subclause 6.10.2.4.1.2.1.1.1 of [1].

6.10.2.4.1.4b.1.1.3 TFCS

See subclause 6.10.2.4.1.4a.1.1.3 of [1].

6.10.2.4.1.4b.1.1.4 TFC subset list

| | |
|---|---|
| TFC subset list size | 4 |
| TFC subset list | 0 = {(TF0,TF0,TF0,TF0), (TF1,TF0,TF0,TF0), (TF2,TF1,TF0,TF0), (TF0,TF0,TF0,TF1), (TF1,TF0, TF0,TF1), (TF2,TF1,TF0,TF1)},<br>1 = {(TF0,TF0,TF0,TF0), (TF1,TF0,TF0,TF0), (TF2,TF1,TF0,TF0), (TF3,TF2,TF0,TF0), (TF0,TF0,TF0,TF1), (TF1,TF0,TF0,TF1), (TF2,TF1,TF0,TF1), (TF3,TF2,TF0,TF1)},<br>2 = {(TF0,TF0,TF0,TF0), (TF1,TF0,TF0,TF0), (TF2,TF1,TF0,TF0), (TF3,TF2,TF0,TF0), (TF4,TF3,TF0,TF0), (TF0,TF0,TF0,TF1), (TF1,TF0,TF0,TF1), (TF2,TF1,TF0,TF1), (TF3,TF2,TF0,TF1), (TF4,TF3,TF0,TF1)}<br>3 = {(TF0,TF0,TF0,TF0), (TF1,TF0,TF0,TF0), (TF2,TF1,TF0,TF0), (TF3,TF2,TF0,TF0), (TF4,TF3,TF0,TF0), (TF5,TF4,TF1,TF0), (TF0,TF0,TF0,TF1), (TF1,TF0,TF0,TF1), (TF2,TF1,TF0,TF1), (TF3,TF2,TF0,TF1), (TF4,TF3,TF0,TF1), (TF5,TF4,TF1,TF1)} |

#### 6.10.2.4.1.4b.1.2 Physical channel parameters

See subclause 6.10.2.4.1.4a.1.2 of [1].

| | | |
|---|---|---|
| DPCH Uplink | Min spreading factor | 64 |
| | Max number of DPDCH data bits/radio frame | 600 |
| | Puncturing Limit | 0.84 |

### 6.10.2.4.1.4b.2 Downlink

#### 6.10.2.4.1.4b.2.1 Transport channel parameters

##### 6.10.2.4.1.4b.2.1.1 Transport channel parameters for Conversational / speech / DL: (12.2 7.4 5.9 4.75) kbps / CS RAB

| Higher layer | RAB/Signalling RB | | RAB subflow #1 | RAB subflow #2 | RAB subflow #3 |
|---|---|---|---|---|---|
| RLC | Logical channel type | | DTCH | | |
| | RLC mode | | TM | TM | TM |
| | Payload sizes, bit | | 0, 39, 42, 55, 61, 81 | 53, 63, 87, 103 | 60 |
| | Max data rate, bps | | 12 200 | | |
| | TrD PDU header, bit | | 0 | | |
| MAC | MAC header, bit | | 0 | | |
| | MAC multiplexing | | N/A | | |
| Layer 1 | TrCH type | | DCH | DCH | DCH |
| | TB sizes, bit | | 0, 39, 42, 55, 61, 81 | 53, 63, 87, 103 | 60 |
| | TFS (note 1) | TF0, bits | 1x0 (note 2) | 0x103 | 0x60 |
| | | TF1, bits | 1x39 | 1x53 | 1x60 |
| | | TF2, bits | 1x42 | 1x63 | N/A |
| | | TF3, bits | 1x55 | 1x87 | N/A |
| | | TF4, bits | 1x61 | 1x103 | N/A |
| | | TF5, bits | 1x81 | N/A | N/A |
| | TTI, ms | | 20 | 20 | 20 |
| | Coding type | | CC 1/3 | CC 1/3 | CC 1/2 |
| | CRC, bit | | 12 | N/A | N/A |
| | Max number of bits/TTI after channel coding | | 303 | 333 | 136 |
| | RM attribute | | 180 to 220 | 170 to 210 | 215 to 256 |

NOTE 1: The TrCH corresponding to RAB subflow #1 should be used as the guiding TrCH, (see clause 4.3 in 3GPP TS 25.212 [14]).
NOTE 2: CRC parity bits are to be attached to RAB subflow#1 any time since number of TrBlks are 1 even if there is no data on RAB subflow#1 (see clause 4.2.1.1 in 3GPP TS 25.212 [14]).

##### 6.10.2.4.1.4b.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See subclause 6.10.2.4.1.2.2.1.1 of [1].

##### 6.10.2.4.1.4b.2.1.3 Transport channel parameters for DL:0.15 kbps SRB#5 for DCCH

See subclause 6.10.2.4.1.62.2.1.3 of [1].

6.10.2.4.1.4b.2.1.3 TFCS

| | |
|---|---|
| TFCS size | 24 |
| TFCS | (RAB subflow#1, RAB subflow#2, RAB subflow#3, DCCH, DCCH 0.15)=<br>(TF0,TF0,TF0,TF0,TF0), (TF1,TF0,TF0,TF0,TF0), (TF2,TF1,TF0,TF0,TF0),<br>(TF3,TF2,TF0,TF0,TF0), (TF4,TF3,TF0,TF0,TF0), (TF5,TF4,TF1,TF0,TF0),<br>(TF0,TF0,TF0,TF1,TF0), (TF1,TF0,TF0,TF1,TF0), (TF2,TF1,TF0,TF1,TF0),<br>(TF3,TF2,TF0,TF1,TF0), (TF4,TF3,TF0,TF1,TF0), (TF5,TF4,TF1,TF1,TF0),<br>(TF0,TF0,TF0,TF0,TF1), (TF1,TF0,TF0,TF0,TF1), (TF2,TF1,TF0,TF0,TF1),<br>(TF3,TF2,TF0,TF0,TF1), (TF4,TF3,TF0,TF0,TF1), (TF5,TF4,TF1,TF0,TF1),<br>(TF0,TF0,TF0,TF1,TF1), (TF1,TF0,TF0,TF1,TF1), (TF2,TF1,TF0,TF1,TF1),<br>(TF3,TF2,TF0,TF1,TF1), (TF4,TF3,TF0,TF1,TF1), (TF5,TF4,TF1,TF1,TF1) |

6.10.2.4.1.4b.2.2 Physical channel parameters

See subclause 6.10.2.4.1.4a.2.2 of [1].

6.10.2.4.1.5 Conversational / speech / UL:10.2 DL:10.2 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.2.4.1.5.1 Uplink

6.10.2.4.1.5.1.1 Transport channel parameters

6.10.2.4.1.5.1.1.1 Transport channel parameters for Conversational / speech / UL:10.2 kbps / CS RAB

| Higher layer | RAB/Signalling RB | | RAB subflow #1 | RAB subflow #2 | RAB subflow #3 |
|---|---|---|---|---|---|
| RLC | Logical channel type | | DTCH | | |
| | RLC mode | | TM | TM | TM |
| | Payload sizes, bit | | 39, 65<br>(alt. 0, 39, 65) | 99 | 40 |
| | Max data rate, bps | | 10 200 | | |
| | TrD PDU header, bit | | 0 | | |
| MAC | MAC header, bit | | 0 | | |
| | MAC multiplexing | | N/A | | |
| Layer 1 | TrCH type | | DCH | DCH | DCH |
| | TB sizes, bit | | 39, 65<br>(alt. 0, 39, 65) | 99 | 40 |
| | TFS | TF0, bits | 0x65 (alt. 1x0)<br>(note) | 0x99 | 0x40 |
| | | TF1, bits | 1x39 | 1x99 | 1x40 |
| | | TF2, bits | 1x65 | N/A | N/A |
| | TTI, ms | | 20 | 20 | 20 |
| | Coding type | | CC 1/3 | CC 1/3 | CC 1/2 |
| | CRC, bit | | 12 | N/A | N/A |
| | Max number of bits/TTI after channel coding | | 255 | 321 | 96 |
| | Uplink: Max number of bits/radio frame before rate matching | | 128 | 161 | 48 |
| | RM attribute | | 180 to 220 | 170 to 210 | 215 to 256 |
| NOTE: In case of using this alternative, CRC parity bits are to be attached to RAB subflow#1 any time since number of TrBlks are 1 even if there is no data on RAB subflow#1 (see clause 4.2.1.1 in 3GPP TS 25.212 [14]). | | | | | |

6.10.2.4.1.5.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.1.1.1.

6.10.2.4.1.5.1.1.3 TFCS

| | |
|---|---|
| TFCS size | 6 |
| TFCS | (RAB subflow#1, RAB subflow#2, RAB subflow#3,DCCH)=<br>(TF0, TF0, TF0, TF0), (TF1, TF0, TF0, TF0), (TF2, TF1, TF1, TF0),<br>(TF0, TF0, TF0, TF1), (TF1, TF0, TF0, TF1), (TF2, TF1, TF1, TF1) |

6.10.2.4.1.5.1.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Uplink | Min spreading factor | 64 |
| | Max number of DPDCH data bits/radio frame | 600 |
| | Puncturing Limit | 0.96 |

6.10.2.4.1.5.2 Downlink

6.10.2.4.1.5.2.1 Transport channel parameters

6.10.2.4.1.5.2.1.1 Transport channel parameters for Conversational / speech / DL:10.2 kbps / CS RAB

| Higher layer | RAB/Signalling RB | | RAB subflow #1 | RAB subflow #2 | RAB subflow #3 |
|---|---|---|---|---|---|
| RLC | Logical channel type | | DTCH | | |
| | RLC mode | | TM | TM | TM |
| | Payload sizes, bit | | 0<br>39<br>65 | 99 | 40 |
| | Max data rate, bps | | 10 200 | | |
| | TrD PDU header, bit | | 0 | | |
| MAC | MAC header, bit | | 0 | | |
| | MAC multiplexing | | N/A | | |
| Layer 1 | TrCH type | | DCH | DCH | DCH |
| | TB sizes, bit | | 0<br>39<br>65 | 99 | 40 |
| | TFS (note 1) | TF0, bits | 1x0 (note 2) | 0x99 | 0x40 |
| | | TF1, bits | 1x39 | 1x99 | 1x40 |
| | | TF2, bits | 1x65 | N/A | N/A |
| | TTI, ms | | 20 | 20 | 20 |
| | Coding type | | CC 1/3 | CC 1/3 | CC 1/2 |
| | CRC, bit | | 12 | N/A | N/A |
| | Max number of bits/TTI after channel coding | | 255 | 321 | 96 |
| | RM attribute | | 180 to 220 | 170 to 210 | 215 to 256 |

NOTE 1: The TrCH corresponding to RAB subflow #1 should be used as the guiding TrCH, (see clause 4.3 in 3GPP TS 25.212 [14]).
NOTE 2: CRC parity bits are to be attached to RAB subflow#1 any time since number of TrBlks are 1 even if there is no data on RAB subflow#1 (see clause 4.2.1.1 in 3GPP TS 25.212 [14]).

6.10.2.4.1.5.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.2.1.1.

6.10.2.4.1.5.2.1.3 TFCS

| | |
|---|---|
| TFCS size | 6 |
| TFCS | (RAB subflow#1, RAB subflow#2, RAB subflow#3,DCCH)=<br>(TF0, TF0, TF0, TF0), (TF1, TF0, TF0, TF0), (TF2, TF1, TF1, TF0),<br>(TF0, TF0, TF0, TF1), (TF1, TF0, TF0, TF1), (TF2, TF1, TF1, TF1) |

6.10.2.4.1.5.2.2 Physical channel parameters

| | | | |
|---|---|---|---|
| DPCH Downlink | DTX position | | Fixed |
| | Spreading factor | | 128 |
| | DPCCH | Number of TFCI bits/slot | 0 |
| | | Number of TPC bits/slot | 2 |
| | | Number of Pilot bits/slot | 4 |
| | DPDCH | Number of data bits/slot | 34 |
| | | Number of data bits/frame | 510 |

6.10.2.4.1.5a Conversational / speech / UL:(10.2, 6.7, 5.9, 4.75) DL:(10.2, 6.7, 5.9, 4.75) kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.2.4.1.5a.1 Uplink

6.10.2.4.1.5a.1.1 Transport channel parameters

6.10.2.4.1.5a.1.1.1 Transport channel parameters for Conversational / speech / UL:(10.2, 6.7, 5.9, 4.75) kbps / CS RAB

| Higher layer | RAB/Signalling RB | | RAB subflow #1 | RAB subflow #2 | RAB subflow #3 |
|---|---|---|---|---|---|
| RLC | Logical channel type | | DTCH | | |
| | RLC mode | | TM | TM | TM |
| | Payload sizes, bit | | 39, 42, 55, 58, 65 (alt. 0, 39, 42, 55, 58, 65) | 53, 63, 76, 99 | 40 |
| | Max data rate, bps | | 10 200 | | |
| | TrD PDU header, bit | | 0 | | |
| MAC | MAC header, bit | | 0 | | |
| | MAC multiplexing | | N/A | | |
| Layer 1 | TrCH type | | DCH | DCH | DCH |
| | TB sizes, bit | | 39, 42, 55, 58, 65 (alt. 0, 39, 42, 55, 58, 65) | 53, 63, 76, 99 | 40 |
| | TFS | TF0, bits | 0x65 (alt. 1x0) (note) | 0x99 | 0x40 |
| | | TF1, bits | 1x39 | 1x53 | 1x40 |
| | | TF2, bits | 1x42 | 1x63 | N/A |
| | | TF3, bits | 1x55 | 1x76 | N/A |
| | | TF4, bits | 1x58 | 1x99 | N/A |
| | | TF5, bits | 1x65 | N/A | N/A |
| | TTI, ms | | 20 | 20 | 20 |
| | Coding type | | CC 1/3 | CC 1/3 | CC 1/2 |
| | CRC, bit | | 12 | N/A | N/A |
| | Max number of bits/TTI after channel coding | | 255 | 321 | 96 |
| | Uplink: Max number of bits/radio frame before rate matching | | 128 | 161 | 48 |
| | RM attribute | | 180 to 220 | 170 to 210 | 215 to 256 |
| NOTE: In case of using this alternative, CRC parity bits are to be attached to RAB subflow#1 any time since number of TrBlks are 1 even if there is no data on RAB subflow#1 (see clause 4.2.1.1 in 3GPP TS 25.212 [14]). | | | | | |

6.10.2.4.1.5a.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.1.1.1.

6.10.2.4.1.5a.1.1.3 TFCS

| | |
|---|---|
| TFCS size | 12 |
| TFCS | (RAB subflow#1, RAB subflow#2, RAB subflow#3,DCCH)=<br>(TF0,TF0,TF0,TF0), (TF1,TF0,TF0,TF0), (TF2,TF1,TF0,TF0), (TF3,TF2,TF0,TF0),<br>(TF4,TF3,TF0,TF0), (TF5,TF4,TF1,TF0), (TF0,TF0,TF0,TF1), (TF1,TF0,TF0,TF1),<br>(TF2,TF1,TF0,TF1), (TF3,TF2,TF0,TF1), (TF4,TF3,TF0,TF1), (TF5,TF4,TF1,TF1) |

6.10.2.4.1.5a.1.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Uplink | Min spreading factor | 64 |
| | Max number of DPDCH data bits/radio frame | 600 |
| | Puncturing Limit | 0.96 |

6.10.2.4.1.5a.2 Downlink

6.10.2.4.1.5a.2.1 Transport channel parameters

6.10.2.4.1.5a.2.1.1 Transport channel parameters for Conversational / speech / DL: DL:(10.2, 6.7, 5.9, 4.75) kbps / CS RAB

| Higher layer | RAB/Signalling RB | | RAB subflow #1 | RAB subflow #2 | RAB subflow #3 |
|---|---|---|---|---|---|
| RLC | Logical channel type | | DTCH | | |
| | RLC mode | | TM | TM | TM |
| | Payload sizes, bit | | 0, 39, 42, 55, 58, 65 | 0, 53, 63, 76, 99 | 40 |
| | Max data rate, bps | | 10 200 | | |
| | TrD PDU header, bit | | 0 | | |
| MAC | MAC header, bit | | 0 | | |
| | MAC multiplexing | | N/A | | |
| Layer 1 | TrCH type | | DCH | DCH | DCH |
| | TB sizes, bit | | 0, 39, 42, 55, 58, 65 | 0, 53, 63, 76, 99 | 40 |
| | TFS (note 1) | TF0, bits | 1x0 (note 2) | 0x99 | 0x40 |
| | | TF1, bits | 1x39 | 1x53 | 1x40 |
| | | TF2, bits | 1x42 | 1x63 | N/A |
| | | TF3, bits | 1x55 | 1x76 | N/A |
| | | TF4, bits | 1x58 | 1x99 | N/A |
| | | TF5, bits | 1x65 | N/A | N/A |
| | TTI, ms | | 20 | 20 | 20 |
| | Coding type | | CC 1/3 | CC 1/3 | CC 1/2 |
| | CRC, bit | | 12 | N/A | N/A |
| | Max number of bits/TTI after channel coding | | 255 | 321 | 96 |
| | RM attribute | | 180 to 220 | 170 to 210 | 215 to 256 |

NOTE 1: The TrCH corresponding to RAB subflow #1 should be used as the guiding TrCH, (see clause 4.3 in 3GPP TS 25.212 [14]).
NOTE 2: CRC parity bits are to be attached to RAB subflow#1 any time since number of TrBlks are 1 even if there is no data on RAB subflow#1 (see clause 4.2.1.1 in 3GPP TS 25.212 [14]).

6.10.2.4.1.5a.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.2.1.1.

6.10.2.4.1.5a.2.1.3 TFCS

| TFCS size | 12 |
|---|---|
| TFCS | (RAB subflow#1, RAB subflow#2, RAB subflow#3,DCCH)=<br>(TF0,TF0,TF0,TF0), (TF1,TF0,TF0,TF0), (TF2,TF1,TF0,TF0), (TF3,TF2,TF0,TF0),<br>(TF4,TF3,TF0,TF0), (TF5,TF4,TF1,TF0), (TF0,TF0,TF0,TF1), (TF1,TF0,TF0,TF1),<br>(TF2,TF1,TF0,TF1), (TF3,TF2,TF0,TF1), (TF4,TF3,TF0,TF1), (TF5,TF4,TF1,TF1) |

6.10.2.4.1.5a.2.2 Physical channel parameters

| DPCH Downlink | DTX position | | Fixed |
|---|---|---|---|
| | Spreading factor | | 128 |
| | DPCCH | Number of TFCI bits/slot | 0 |
| | | Number of TPC bits/slot | 2 |
| | | Number of Pilot bits/slot | 4 |
| | DPDCH | Number of data bits/slot | 34 |
| | | Number of data bits/frame | 510 |

6.10.2.4.1.6 Conversational / speech / UL:7.95 DL:7.95 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.2.4.1.6.1 Uplink

6.10.2.4.1.6.1.1 Transport channel parameters

6.10.2.4.1.6.1.1.1 Transport channel parameters for Conversational / speech / UL:7.95 kbps / CS RAB

| Higher layer | RAB/Signalling RB | RAB subflow #1 | RAB subflow #2 | RAB subflow #3 |
|---|---|---|---|---|
| RLC | Logical channel type | DTCH | | |
| | RLC mode | TM | TM | TM |

| | | | | | |
|---|---|---|---|---|---|
| | Payload sizes, bit | | 39,75 (alt. 0, 39, 75) | 84 | 60 |
| | Max data rate, bps | | 7950 | | |
| | TrD PDU header, bit | | 0 | | |
| MAC | MAC header, bit | | 0 | | |
| | MAC multiplexing | | N/A | | |
| Layer 1 | TrCH type | | DCH | DCH | DCH |
| | TB sizes, bit | | 39,75 (alt. 0, 39, 75) | 84 | 60 |
| | TFS (note 1) | TF0, bits | 0x75 (alt. 1x0) (note) | 0x84 | 0x60 |
| | | TF1, bits | 1x39 | 1x84 | N/A |
| | | TF2, bits | 1x75 | N/A | N/A |
| | TTI, ms | | 20 | 20 | 20 |
| | Coding type | | CC 1/3 | CC 1/3 | CC 1/2 |
| | CRC, bit | | 12 | N/A | N/A |
| | Max number of bits/TTI after channel coding | | 285 | 276 | 0 |
| | Uplink: Max number of bits/radio frame before rate matching | | 143 | 138 | 0 |
| | RM attribute | | 180 to 220 | 170 to 210 | 215 to 256 |
| NOTE: In case of using this alternative, CRC parity bits are to be attached to RAB subflow#1 any time since number of TrBlks are 1 even if there is no data on RAB subflow#1 (see clauses 4.2.1.1 in 3GPP TS 25.212 [14]). | | | | | |

6.10.2.4.1.6.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.1.1.1.

6.10.2.4.1.6.1.1.3 TFCS

| | |
|---|---|
| TFCS size | 6 |
| TFCS | (RAB subflow#1, RAB subflow#2, RAB subflow #3, DCCH)=<br>(TF0, TF0, TF0, TF0), (TF1, TF0, TF0, TF0), (TF2, TF1, TF0, TF0),<br>(TF0, TF0, TF0, TF1), (TF1, TF0, TF0, TF1), (TF2, TF1, TF0, TF1) |

6.10.2.4.1.6.1.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Uplink | Min spreading factor | 64 |
| | Max number of DPDCH data bits/radio frame | 600 |
| | Puncturing Limit | 0.96 |

6.10.2.4.1.6.2 Downlink

6.10.2.4.1.6.2.1 Transport channel parameters

6.10.2.4.1.6.2.1.1 Transport channel parameters for Conversational / speech / DL:7.95 kbps / CS RAB

| Higher layer | RAB/Signalling RB | | RAB subflow #1 | RAB subflow #2 | RAB subflow #3 |
|---|---|---|---|---|---|
| RLC | Logical channel type | | DTCH | | |
| | RLC mode | | TM | TM | TM |
| | Payload sizes, bit | | 0, 39, 75 | 84 | 60 |
| | Max data rate, bps | | 7950 | | |
| | TrD PDU header, bit | | 0 | | |
| MAC | MAC header, bit | | 0 | | |
| | MAC multiplexing | | N/A | | |
| Layer 1 | TrCH type | | DCH | DCH | DCH |
| | TB sizes, bit | | 0<br>39<br>75 | 84 | 60 |
| | TFS (note 1) | TF0, bits | 1x0 (note 2) | 0x84 | 0x60 |
| | | TF1, bits | 1x39 | 1x84 | N/A |
| | | TF2, bits | 1x75 | N/A | N/A |
| | TTI, ms | | 20 | 20 | 20 |
| | Coding type | | CC 1/3 | CC 1/3 | CC 1/2 |
| | CRC, bit | | 12 | N/A | N/A |

| | | | | |
|---|---|---|---|---|
| | Max number of bits/TTI after channel coding | 285 | 276 | 0 |
| | RM attribute | 180 to 220 | 170 to 210 | 215 to 256 |
| NOTE 1: The TrCH corresponding to RAB subflow #1 should be used as the guiding TrCH, (see clause 4.3 in 3GPP TS 25.212 [14]).<br>NOTE 2: CRC parity bits are to be attached to RAB subflow#1 any time since number of TrBlks are 1 even if there is no data on RAB subflow#1 (see clause 4.2.1.1 in 3GPP TS 25.212 [14]). | | | | |

6.10.2.4.1.6.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.2.1.1.

6.10.2.4.1.6.2.1.3 TFCS

| | |
|---|---|
| TFCS size | 6 |
| TFCS | (RAB subflow#1, RAB subflow#2, RAB subflow#3, DCCH)=<br>(TF0, TF0, TF0, TF0), (TF1, TF0, TF0, TF0), (TF2, TF1, TF0, TF0),<br>(TF0, TF0, TF0, TF1), (TF1, TF0, TF0, TF1), (TF2, TF1, TF0, TF1) |

6.10.2.4.1.6.2.2 Physical channel parameters

| | | | |
|---|---|---|---|
| DPCH Downlink | DTX position | | Fixed |
| | Spreading factor | | 128 |
| | DPCCH | Number of TFCI bits/slot | 0 |
| | | Number of TPC bits/slot | 2 |
| | | Number of Pilot bits/slot | 4 |
| | DPDCH | Number of data bits/slot | 34 |
| | | Number of data bits/frame | 510 |

6.10.2.4.1.7 Conversational / speech / UL:7.4 DL:7.4 kbps / CS RAB+ UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.2.4.1.7.1 Uplink

6.10.2.4.1.7.1.1 Transport channel parameters

6.10.2.4.1.7.1.1.1 Transport channel parameters for Conversational / speech / UL:7.4 kbps / CS RAB

| Higher layer | RAB/Signalling RB | | RAB subflow #1 | RAB subflow #2 | RAB subflow #3 |
|---|---|---|---|---|---|
| RLC | Logical channel type | | DTCH | | |
| | RLC mode | | TM | TM | TM |
| | Payload sizes, bit | | 39, 61 (alt. 0, 39, 61) | 87 | 60 |
| | Max data rate, bps | | 7400 | | |
| | TrD PDU header, bit | | 0 | | |
| MAC | MAC header, bit | | 0 | | |
| | MAC multiplexing | | N/A | | |
| Layer 1 | TrCH type | | DCH | DCH | DCH |
| | TB sizes, bit | | 39, 61 (alt. 0, 39, 61) | 87 | 60 |
| | TFS (note 1) | TF0, bits | 0x61 (alt. 1x0) (note) | 0x87 | 0x60 |
| | | TF1, bits | 1x39 | 1x87 | N/A |
| | | TF2, bits | 1x61 | N/A | N/A |
| | TTI, ms | | 20 | 20 | 20 |
| | Coding type | | CC 1/3 | CC 1/3 | CC 1/2 |
| | CRC, bit | | 12 | N/A | N/A |
| | Max number of bits/TTI after channel coding | | 243 | 285 | 0 |
| | Uplink: Max number of bits/radio frame before rate matching | | 122 | 143 | 0 |
| | RM attribute | | 180 to 220 | 170 to 210 | 215 to 256 |

| NOTE: | In case of using this alternative, CRC parity bits are to be attached to RAB subflow#1 any time since number of TrBlks are 1 even if there is no data on RAB subflow#1 (see clauses 4.2.1.1 in 3GPP TS 25.212 [14]). |
|---|---|

6.10.2.4.1.7.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.1.1.1.

6.10.2.4.1.7.1.1.3 TFCS

| TFCS size | 6 |
|---|---|
| TFCS | (RAB subflow#1, RAB subflow#2, RAB subflow#3, DCCH)=<br>(TF0, TF0, TF0, TF0), (TF1, TF0, TF0, TF0), (TF2, TF1, TF0, TF0),<br>(TF0, TF0, TF0, TF1), (TF1, TF0, TF0, TF1), (TF2, TF1, TF0, TF1) |

6.10.2.4.1.7.1.2 Physical channel parameters

| DPCH Uplink | Min spreading factor | 64 |
|---|---|---|
| | Max number of DPDCH data bits/radio frame | 600 |
| | Puncturing Limit | 0.96 |

6.10.2.4.1.7.2 Downlink

6.10.2.4.1.7.2.1 Transport channel parameters

6.10.2.4.1.7.2.1.1 Transport channel parameters for Conversational / speech / DL:7.4 kbps / CS RAB

| Higher layer | RAB/Signalling RB | | RAB subflow #1 | RAB subflow #2 | RAB subflow #3 |
|---|---|---|---|---|---|
| RLC | Logical channel type | | DTCH | | |
| | RLC mode | | TM | TM | TM |
| | Payload sizes, bit | | 0<br>39<br>61 | 87 | 60 |
| | Max data rate, bps | | 7400 | | |
| | TrD PDU header, bit | | 0 | | |
| MAC | MAC header, bit | | 0 | | |
| | MAC multiplexing | | N/A | | |
| Layer 1 | TrCH type | | DCH | DCH | DCH |
| | TB sizes, bit | | 0<br>39<br>61 | 87 | 60 |
| | TFS (note 1) | TF0, bits | 1x0 (note 2) | 0x87 | 0x60 |
| | | TF1, bits | 1x39 | 1x87 | N/A |
| | | TF2, bits | 1x61 | N/A | N/A |
| | TTI, ms | | 20 | 20 | 20 |
| | Coding type | | CC 1/3 | CC 1/3 | CC 1/2 |
| | CRC, bit | | 12 | N/A | N/A |
| | Max number of bits/TTI after channel coding | | 243 | 285 | 0 |
| | RM attribute | | 180 to 220 | 170 to 210 | 215 to 256 |

NOTE 1: The TrCH corresponding to RAB subflow #1 should be used as the guiding TrCH, (see clause 4.3 in 3GPP TS 25.212 [14]).
NOTE 2: CRC parity bits are to be attached to RAB subflow#1 any time since number of TrBlks are 1 even if there is no data on RAB subflow#1 (see clause 4.2.1.1 in 3GPP TS 25.212 [14]).

6.10.2.4.1.7.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.2.1.1.

6.10.2.4.1.7.2.1.3 TFCS

| TFCS size | 6 |
|---|---|
| TFCS | (RAB subflow#1, RAB subflow#2, RAB subflow#3, DCCH)=<br>(TF0, TF0, TF0, TF0), (TF1, TF0, TF0, TF0), (TF2, TF1, TF0, TF0),<br>(TF0, TF0, TF0, TF1), (TF1, TF0, TF0, TF1), (TF2, TF1, TF0, TF1) |

6.10.2.4.1.7.2.2 Physical channel parameters

| | | | |
|---|---|---|---|
| DPCH Downlink | DTX position | | Fixed |
| | Spreading factor | | 128 |
| | DPCCH | Number of TFCI bits/slot | 0 |
| | | Number of TPC bits/slot | 2 |
| | | Number of Pilot bits/slot | 4 |
| | DPDCH | Number of data bits/slot | 34 |
| | | Number of data bits/frame | 510 |

6.10.2.4.1.7a Conversational / speech / UL:(7.4, 6.7, 5.9, 4.75) DL:(7.4, 6.7, 5.9, 4.75) kbps / CS RAB+ UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.2.4.1.7a.1 Uplink

6.10.2.4.1.7a.1.1 Transport channel parameters

6.10.2.4.1.7a.1.1.1 Transport channel parameters for Conversational / speech / UL:(7.4, 6.7, 5.9, 4.75) kbps / CS RAB

| | | | | |
|---|---|---|---|---|
| Higher layer | RAB/Signalling RB | | RAB subflow #1 | RAB subflow #2 |
| RLC | Logical channel type | | DTCH | |
| | RLC mode | | TM | TM |
| | Payload sizes, bit | | 39, 42, 55, 58, 61 (alt. 0, 39, 42, 55, 58, 61) | 53, 63, 76, 87 |
| | Max data rate, bps | | 7 400 | |
| | TrD PDU header, bit | | 0 | |
| MAC | MAC header, bit | | 0 | |
| | MAC multiplexing | | N/A | |
| Layer 1 | TrCH type | | DCH | DCH |
| | TB sizes, bit | | 39, 42, 55, 58, 61 (alt. 0, 39, 42, 55, 58, 61) | 53, 63, 76, 87 |
| | TFS | TF0, bits | 0x61 (alt. 1x0) (note) | 0x87 |
| | | TF1, bits | 1x39 | 1x53 |
| | | TF2, bits | 1x42 | 1x63 |
| | | TF3, bits | 1x55 | 1x76 |
| | | TF4, bits | 1x58 | 1x87 |
| | | TF5, bits | 1x61 | N/A |
| | TTI, ms | | 20 | 20 |
| | Coding type | | CC 1/3 | CC 1/3 |
| | CRC, bit | | 12 | N/A |
| | Max number of bits/TTI after channel coding | | 243 | 285 |
| | Uplink: Max number of bits/radio frame before rate matching | | 122 | 143 |
| | RM attribute | | 180 to 220 | 170 to 210 |
| NOTE: In case of using this alternative, CRC parity bits are to be attached to RAB subflow#1 any time since number of TrBlks are 1 even if there is no data on RAB subflow#1 (see clause 4.2.1.1 in 3GPP TS 25.212 [14]). | | | | |

6.10.2.4.1.7a.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.1.1.1.

6.10.2.4.1.7a.1.1.3 TFCS

| | |
|---|---|
| TFCS size | 12 |
| TFCS | (RAB subflow#1, RAB subflow#2, DCCH)=<br>(TF0, TF0, TF0), (TF1, TF0, TF0), (TF2, TF1, TF0), (TF3, TF2, TF0), (TF4, TF3, TF0), (TF5, TF4, TF0),<br>(TF0, TF0, TF1), (TF1, TF0, TF1), (TF2, TF1, TF1), (TF3, TF2, TF1), (TF4, TF3, TF1), (TF5, TF4, TF1) |

6.10.2.4.1.7a.1.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Uplink | Min spreading factor | 64 |
| | Max number of DPDCH data bits/radio frame | 600 |
| | Puncturing Limit | 0.96 |

6.10.2.4.1.7a.2 Downlink

6.10.2.4.1.7a.2.1 Transport channel parameters

6.10.2.4.1.7a.2.1.1 Transport channel parameters for Conversational / speech / DL:(7.4, 6.7, 5.9, 4.75) kbps / CS RAB

| Higher layer | RAB/Signalling RB | | RAB subflow #1 | RAB subflow #2 |
|---|---|---|---|---|
| RLC | Logical channel type | | DTCH | |
| | RLC mode | | TM | TM |
| | Payload sizes, bit | | 0, 39, 42, 55, 58, 61 | 53, 63, 76, 87 |
| | Max data rate, bps | | 7 400 | |
| | TrD PDU header, bit | | 0 | |
| MAC | MAC header, bit | | 0 | |
| | MAC multiplexing | | N/A | |
| Layer 1 | TrCH type | | DCH | DCH |
| | TB sizes, bit | | 0, 39, 42, 55, 58, 61 | 53, 63, 76, 87 |
| | TFS (note 1) | TF0, bits | 1x0 (note 2) | 0x87 |
| | | TF1, bits | 1x39 | 1x53 |
| | | TF2, bits | 1x42 | 1x63 |
| | | TF3, bits | 1x55 | 1x76 |
| | | TF4, bits | 1x58 | 1x87 |
| | | TF5, bits | 1x61 | N/A |
| | TTI, ms | | 20 | 20 |
| | Coding type | | CC 1/3 | CC 1/3 |
| | CRC, bit | | 12 | N/A |
| | Max number of bits/TTI after channel coding | | 243 | 285 |
| | RM attribute | | 180 to 220 | 170 to 210 |

NOTE 1: The TrCH corresponding to RAB subflow #1 should be used as the guiding TrCH, (see clause 4.3 in 3GPP TS 25.212 [14]).
NOTE 2: CRC parity bits are to be attached to RAB subflow#1 any time since number of TrBlks are 1 even if there is no data on RAB #1 (see clause 4.2.1.1 in 3GPP TS 25.212 [14]).

6.10.2.4.1.7a.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.2.1.1.

6.10.2.4.1.7a.2.1.3 TFCS

| | |
|---|---|
| TFCS size | 12 |
| TFCS | (RAB subflow#1, RAB subflow#2, DCCH)=<br>(TF0, TF0, TF0), (TF1, TF0, TF0), (TF2, TF1, TF0), (TF3, TF2, TF0), (TF4, TF3, TF0), (TF5, TF4, TF0),<br>(TF0, TF0, TF1), (TF1, TF0, TF1), (TF2, TF1, TF1), (TF3, TF2, TF1), (TF4, TF3, TF1), (TF5, TF4, TF1) |

6.10.2.4.1.7a.2.2 Physical channel parameters

| | | | |
|---|---|---|---|
| DPCH Downlink | DTX position | | Fixed |
| | Spreading factor | | 128 |
| | DPCCH | Number of TFCI bits/slot | 0 |
| | | Number of TPC bits/slot | 2 |
| | | Number of Pilot bits/slot | 4 |
| | DPDCH | Number of data bits/slot | 34 |
| | | Number of data bits/frame | 510 |

6.10.2.4.1.8 Conversational / speech / UL:6.7 DL:6.7 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.2.4.1.8.1 Uplink

6.10.2.4.1.8.1.1 Transport channel parameters

6.10.2.4.1.8.1.1.1 Transport channel parameters for Conversational / speech / UL:6.7 kbps / CS RAB

| Higher layer | RAB/Signalling RB | | RAB subflow #1 | RAB subflow #2 | RAB subflow #3 |
|---|---|---|---|---|---|
| RLC | Logical channel type | | DTCH | | |
| | RLC mode | | TM | TM | TM |
| | Payload sizes, bit | | 39, 58 (alt. 0, 39, 58) | 76 | 60 |
| | Max data rate, bps | | 6700 | | |
| | TrD PDU header, bit | | 0 | | |
| MAC | MAC header, bit | | 0 | | |
| | MAC multiplexing | | N/A | | |
| Layer 1 | TrCH type | | DCH | DCH | DCH |
| | TB sizes, bit | | 39, 58 (alt. 0, 39, 58) | 76 | 60 |
| | TFS (note 1) | TF0, bits | 0x58 (alt. 1x0) (note) | 0x76 | 0x60 |
| | | TF1, bits | 1x39 | 1x76 | N/A |
| | | TF2, bits | 1x58 | N/A | N/A |
| | TTI, ms | | 20 | 20 | 20 |
| | Coding type | | CC 1/3 | CC 1/3 | CC 1/2 |
| | CRC, bit | | 12 | N/A | N/A |
| | Max number of bits/TTI after channel coding | | 234 | 252 | 0 |
| | Uplink: Max number of bits/radio frame before rate matching | | 117 | 126 | 0 |
| | RM attribute | | 180 to 220 | 170 to 210 | 215 to 256 |
| NOTE: In case of using this alternative, CRC parity bits are to be attached to RAB subflow#1 any time since number of TrBlks are 1 even if there is no data on RAB subflow#1 (see clauses 4.2.1.1 in 3GPP TS 25.212 [14]). | | | | | |

6.10.2.4.1.8.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.1.1.1.

6.10.2.4.8.1.1.3 TFCS

| | |
|---|---|
| TFCS size | 6 |
| TFCS | (RAB subflow#1, RAB subflow#2, RAB subflow#3, DCCH)=<br>(TF0, TF0, TF0, TF0), (TF1, TF0, TF0, TF0), (TF2, TF1, TF0, TF0),<br>(TF0, TF0, TF0, TF1), (TF1, TF0, TF0, TF1), (TF2, TF1, TF0, TF1) |

6.10.2.4.1.8.1.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Uplink | Min spreading factor | 64 |
| | Max number of DPDCH data bits/radio frame | 600 |
| | Puncturing Limit | 0.96 |

6.10.2.4.1.8.2 Downlink

6.10.2.4.1.8.2.1 Transport channel parameters

6.10.2.4.1.8.2.1.1 Transport channel parameters for Conversational / speech / DL:6.7 kbps / CS RAB

| Higher layer | RAB/Signalling RB | RAB subflow #1 | RAB subflow #2 | RAB subflow #3 |
|---|---|---|---|---|
| RLC | Logical channel type | DTCH | | |
| | RLC mode | TM | TM | TM |

| | | | | | |
|---|---|---|---|---|---|
| | Payload sizes, bit | | 0<br>39<br>58 | 76 | 60 |
| | Max data rate, bps | | 6700 | | |
| | TrD PDU header, bit | | 0 | | |
| MAC | MAC header, bit | | 0 | | |
| | MAC multiplexing | | N/A | | |
| Layer 1 | TrCH type | | DCH | DCH | DCH |
| | TB sizes, bit | | 0<br>39<br>58 | 76 | 60 |
| | TFS (note 1) | TF0, bits | 1x0 (note 2) | 0x76 | 0x60 |
| | | TF1, bits | 1x39 | 1x76 | N/A |
| | | TF2, bits | 1x58 | N/A | N/A |
| | TTI, ms | | 20 | 20 | 20 |
| | Coding type | | CC 1/3 | CC 1/3 | CC 1/2 |
| | CRC, bit | | 12 | N/A | N/A |
| | Max number of bits/TTI after channel coding | | 234 | 252 | 0 |
| | RM attribute | | 180 to 220 | 170 to 210 | 215 to 256 |
| NOTE 1: The TrCH corresponding to RAB subflow #1 should be used as the guiding TrCH, (see clause 4.3 in 3GPP TS 25.212 [14]).<br>NOTE 2: CRC parity bits are to be attached to RAB subflow#1 any time since number of TrBlks are 1 even if there is no data on RAB subflow#1 (see clause 4.2.1.1 in 3GPP TS 25.212 [14]). | | | | | |

6.10.2.4.1.8.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.2.1.1

6.10.2.4.1.8.2.1.3 TFCS

| | |
|---|---|
| TFCS size | 6 |
| TFCS | (RAB subflow#1, RAB subflow#2, RAB subflow#3, DCCH)=<br>(TF0, TF0, TF0, TF0), (TF1, TF0, TF0, TF0), (TF2, TF1, TF0, TF0),<br>(TF0, TF0, TF0, TF1), (TF1, TF0, TF0, TF1), (TF2, TF1, TF0, TF1) |

6.10.2.4.1.8.2.2 Physical channel parameters

| | | | |
|---|---|---|---|
| DPCH Downlink | DTX position | | Fixed |
| | Spreading factor | | 128 |
| | DPCCH | Number of TFCI bits/slot | 0 |
| | | Number of TPC bits/slot | 2 |
| | | Number of Pilot bits/slot | 4 |
| | DPDCH | Number of data bits/slot | 34 |
| | | Number of data bits/frame | 510 |

6.10.2.4.1.9 Conversational / speech / UL:5.9 DL:5.9 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.2.4.1.9.1 Uplink

6.10.2.4.1.9.1.1 Transport channel parameters

6.10.2.4.1.9.1.1.1 Transport channel parameters for Conversational / speech / UL:5.9 kbps / CS RAB

| | | | | |
|---|---|---|---|---|
| Higher layer | RAB/Signalling RB | RAB subflow #1 | RAB subflow #2 | RAB subflow #3 |
| RLC | Logical channel type | DTCH | | |
| | RLC mode | TM | TM | TM |
| | Payload sizes, bit | 39, 55 (alt. 0, 39, 55) | 63 | 60 |
| | Max data rate, bps | 5900 | | |
| | TrD PDU header, bit | 0 | | |
| MAC | MAC header, bit | 0 | | |
| | MAC multiplexing | N/A | | |
| Layer 1 | TrCH type | DCH | DCH | DCH |

| | | | | | |
|---|---|---|---|---|---|
| | TB sizes, bit | | 39, 55 (alt. 0, 39, 55) | 63 | 60 |
| | TFS | TF0, bits | 0x55 (alt. 1x0) (note) | 0x63 | 0x60 |
| | | TF1, bits | 1x39 | 1x63 | N/A |
| | | TF2, bits | 1x55 | N/A | N/A |
| | TTI, ms | | 20 | 20 | 20 |
| | Coding type | | CC 1/3 | CC 1/3 | CC 1/2 |
| | CRC, bit | | 12 | N/A | N/A |
| | Max number of bits/TTI after channel coding | | 225 | 213 | 0 |
| | Uplink: Max number of bits/radio frame before rate matching | | 113 | 107 | 0 |
| | RM attribute | | 180 to 220 | 170 to 210 | 215 to 256 |
| NOTE: In case of using this alternative, CRC parity bits are to be attached to RAB subflow#1 any time since number of TrBlks are 1 even if there is no data on RAB subflow#1 (see clause 4.2.1.1 in 3GPP TS 25.212 [14]). | | | | | |

6.10.2.4.1.9.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.1.1.1.

6.10.2.4.1.9.1.1.3 TFCS

| | |
|---|---|
| TFCS size | 6 |
| TFCS | (RAB subflow#1, RAB subflow#2, RAB subflow#3, DCCH)=<br>(TF0, TF0, TF0, TF0), (TF1, TF0, TF0, TF0), (TF2, TF1, TF0, TF0),<br>(TF0, TF0, TF0, TF1), (TF1, TF0, TF0, TF1), (TF2, TF1, TF0, TF1) |

6.10.2.4.1.9.1.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Uplink | Min spreading factor | 64 |
| | Max number of DPDCH data bits/radio frame | 600 |
| | Puncturing Limit | 0.96 |

6.10.2.4.1.9.2 Downlink

6.10.2.4.1.9.2.1 Transport channel parameters

6.10.2.4.1.9.2.1.1 Transport channel parameters for Conversational / speech / DL:5.9 kbps / CS RAB

| | | | | | |
|---|---|---|---|---|---|
| Higher layer | RAB/Signalling RB | | RAB subflow #1 | RAB subflow #2 | RAB subflow #3 |
| RLC | Logical channel type | | DTCH | | |
| | RLC mode | | TM | TM | TM |
| | Payload sizes, bit | | 0<br>39<br>55 | 63 | 60 |
| | Max data rate, bps | | 5900 | | |
| | TrD PDU header, bit | | 0 | | |
| MAC | MAC header, bit | | 0 | | |
| | MAC multiplexing | | N/A | | |
| Layer 1 | TrCH type | | DCH | DCH | DCH |
| | TB sizes, bit | | 0<br>39<br>55 | 63 | 60 |
| | TFS (note 1) | TF0, bits | 1x0 (note 2) | 0x63 | 0x60 |
| | | TF1, bits | 1x39 | 1x63 | N/A |
| | | TF2, bits | 1x55 | N/A | N/A |
| | TTI, ms | | 20 | 20 | 20 |
| | Coding type | | CC 1/3 | CC 1/3 | CC ½ |
| | CRC, bit | | 12 | N/A | N/A |
| | Max number of bits/TTI after channel coding | | 225 | 213 | 0 |
| | RM attribute | | 180 to 220 | 170 to 210 | 215 to 256 |

| | |
|---|---|
| NOTE 1: | The TrCH corresponding to RAB subflow #1 should be used as the guiding TrCH, (see clause 4.3 in 3GPP TS 25.212 [14]). |
| NOTE 2: | CRC parity bits are to be attached to RAB subflow#1 any time since number of TrBlks are 1 even if there is no data on RAB subflow#1 (see clause 4.2.1.1 in 3GPP TS 25.212 [14]). |

6.10.2.4.1.9.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.2.1.1.

6.10.2.4.1.9.2.1.3 TFCS

| | |
|---|---|
| TFCS size | 6 |
| TFCS | (RAB subflow#1, RAB subflow#2, RAB subflow#3, DCCH)=<br>(TF0, TF0, TF0, TF0), (TF1, TF0, TF0, TF0), (TF2, TF1, TF0, TF0),<br>(TF0, TF0, TF0, TF1), (TF1, TF0, TF0, TF1), (TF2, TF1, TF0, TF1) |

6.10.2.4.1.9.2.2 Physical channel parameters

| | | | |
|---|---|---|---|
| DPCH Downlink | DTX position | | Fixed |
| | Spreading factor | | 256 |
| | DPCCH | Number of TFCI bits/slot | 0 |
| | | Number of TPC bits/slot | 2 |
| | | Number of Pilot bits/slot | 2 |
| | DPDCH | Number of data bits/slot | 16 |
| | | Number of data bits/frame | 240 |

6.10.2.4.1.10 Conversational / speech / UL:5.15 DL:5.15 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.2.4.1.10.1 Uplink

6.10.2.4.1.10.1.1 Transport channel parameters

6.10.2.4.1.10.1.1 Transport channel parameters for Conversational / speech / UL:5.15 kbps / CS RAB

| Higher layer | RAB/Signalling RB | | RAB subflow #1 | RAB subflow #2 | RAB subflow #3 |
|---|---|---|---|---|---|
| RLC | Logical channel type | | DTCH | | |
| | RLC mode | | TM | TM | TM |
| | Payload sizes, bit | | 39, 49 (alt. 0, 39, 49) | 54 | 60 |
| | Max data rate, bps | | 4750 | | |
| | TrD PDU header, bit | | 0 | | |
| MAC | MAC header, bit | | 0 | | |
| | MAC multiplexing | | N/A | | |
| Layer 1 | TrCH type | | DCH | DCH | DCH |
| | TB sizes, bit | | 39, 49 (alt. 0, 39, 49) | 54 | 60 |
| | TFS | TF0, bits | 0x49 (alt. 1x0) (note) | 0x54 | 0x60 |
| | | TF1, bits | 1x39 | 1x54 | N/A |
| | | TF2, bits | 1x49 | N/A | N/A |
| | TTI, ms | | 20 | 20 | 20 |
| | Coding type | | CC 1/3 | CC 1/3 | CC 1/2 |
| | CRC, bit | | 12 | 12 | N/A |
| | Max number of bits/TTI after channel coding | | 207 | 186 | 0 |
| | Uplink: Max number of bits/radio frame before rate matching | | 104 | 93 | 0 |
| | RM attribute | | 180 to 220 | 170 to 210 | 215 to 256 |

| | |
|---|---|
| NOTE: | In case of usign this alternative, CRC parity bits are to be attached to RAB subflow#1 any time since number of TrBlks are 1 even if there is no data on RAB subflow#1 (see clause 4.2.1.1 in 3GPP TS 25.212 [14]). |

6.10.2.4.1.10.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.1.1.1.

6.10.2.4.1.10.1.1.3 TFCS

| | |
|---|---|
| TFCS size | 6 |
| TFCS | (RAB subflow#1, RAB subflow#2, RAB subflow#3, DCCH)=<br>(TF0, TF0, TF0, TF0), (TF1, TF0, TF0, TF0), (TF2, TF1, TF0, TF0),<br>(TF0, TF0, TF0, TF1), (TF1, TF0, TF0, TF1), (TF2, TF1, TF0, TF1) |

6.10.2.4.1.10.1.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Uplink | Min spreading factor | 128 |
| | Max number of DPDCH data bits/radio frame | 300 |
| | Puncturing Limit | 0.72 |

6.10.2.4.1.10.2 Downlink

6.10.2.4.1.10.2.1 Transport channel parameters

6.10.2.4.1.10.2.1.1 Transport channel parameters for Conversational / speech / DL:5.15 kbps / CS RAB

| Higher layer | RAB/Signalling RB | | RAB subflow #1 | RAB subflow #2 | RAB subflow #3 |
|---|---|---|---|---|---|
| RLC | Logical channel type | | DTCH | | |
| | RLC mode | | TM | TM | TM |
| | Payload sizes, bit | | 0<br>39<br>49 | 54 | 60 |
| | Max data rate, bps | | 5150 | | |
| | TrD PDU header, bit | | 0 | | |
| MAC | MAC header, bit | | 0 | | |
| | MAC multiplexing | | N/A | | |
| Layer 1 | TrCH type | | DCH | DCH | DCH |
| | TB sizes, bit | | 0<br>39<br>49 | 54 | 60 |
| | TFS (note 1) | TF0, bits | 1x0 (note 2) | 0x54 | 0x60 |
| | | TF1, bits | 1x39 | 1x54 | N/A |
| | | TF2, bits | 1x49 | N/A | N/A |
| | TTI, ms | | 20 | 20 | 20 |
| | Coding type | | CC 1/3 | CC 1/3 | CC 1/2 |
| | CRC, bit | | 12 | N/A | N/A |
| | Max number of bits/TTI after channel coding | | 207 | 186 | 0 |
| | RM attribute | | 180 to 220 | 170 to 210 | 215 to 256 |

NOTE 1: The TrCH corresponding to RAB subflow #1 should be used as the guiding TrCH, (see clause 4.3 in 3GPP TS 25.212 [14]).
NOTE 2: CRC parity bits are to be attached to RAB subflow#1 any time since number of TrBlks are 1 even if there is no data on RAB subflow#1 (see clause 4.2.1.1 in 3GPP TS 25.212 [14]).

6.10.2.4.1.10.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.2.1.1.

6.10.2.4.1.10.2.1.3 TFCS

| | |
|---|---|
| TFCS size | 6 |
| TFCS | (RAB subflow#1, RAB subflow#2, RAB subflow#3, DCCH)=<br>(TF0, TF0, TF0, TF0), (TF1, TF0, TF0, TF0), (TF2, TF1, TF0, TF0),<br>(TF0, TF0, TF0, TF1), (TF1, TF0, TF0, TF1), (TF2, TF1, TF0, TF1) |

6.10.2.4.1.10.2.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Downlink | DTX position | Fixed |

| | | | |
|---|---|---|---|
| | Spreading factor | | 256 |
| | DPCCH | Number of TFCI bits/slot | 0 |
| | | Number of TPC bits/slot | 2 |
| | | Number of Pilot bits/slot | 4 |
| | DPDCH | Number of data bits/slot | 14 |
| | | Number of data bits/frame | 210 |

6.10.2.4.1.11 Conversational / speech / UL:4.75 DL:4.75 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.2.4.1.11.1 Uplink

6.10.2.4.1.11.1.1 Transport channel parameters

6.10.2.4.1.11.1.1.1 Transport channel parameters for Conversational / speech / UL:4.75 kbps / CS RAB

| Higher layer | RAB/Signalling RB | | RAB subflow #1 | RAB subflow #2 | RAB subflow #3 |
|---|---|---|---|---|---|
| RLC | Logical channel type | | DTCH | | |
| | RLC mode | | TM | TM | TM |
| | Payload sizes, bit | | 39, 42 (alt. 0, 39, 42) | 53 | 60 |
| | Max data rate, bps | | 4750 | | |
| | TrD PDU header, bit | | 0 | | |
| MAC | MAC header, bit | | 0 | | |
| | MAC multiplexing | | N/A | | |
| Layer 1 | TrCH type | | DCH | DCH | DCH |
| | TB sizes, bit | | 39, 42 (alt. 0, 39, 42) | 53 | 60 |
| | TFS | TF0, bits | 0x42 (alt. 1x0) (note) | 0x53 | 0x60 |
| | | TF1, bits | 1x39 | 1x53 | N/A |
| | | TF2, bits | 1x42 | N/A | N/A |
| | TTI, ms | | 20 | 20 | 20 |
| | Coding type | | CC 1/3 | CC 1/3 | CC 1/2 |
| | CRC, bit | | 12 | N/A | N/A |
| | Max number of bits/TTI after channel coding | | 186 | 183 | 0 |
| | Uplink: Max number of bits/radio frame before rate matching | | 93 | 92 | 0 |
| | RM attribute | | 180 to 220 | 170 to 210 | 215 to 256 |
| NOTE: In case of usign this alternative, CRC parity bits are to be attached to RAB subflow#1 any time since number of TrBlks are 1 even if there is no data on RAB subflow#1 (see clause 4.2.1.1 in 3GPP TS 25.212 [14]). | | | | | |

6.10.2.4.1.11.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.1.1.1.

6.10.2.4.1.11.1.1.3 TFCS

| | |
|---|---|
| TFCS size | 6 |
| TFCS | (RAB subflow#1, RAB subflow#2, RAB subflow#3, DCCH)=<br>(TF0, TF0, TF0, TF0), (TF1, TF0, TF0, TF0), (TF2, TF1, TF0, TF0),<br>(TF0, TF0, TF0, TF1), (TF1, TF0, TF0, TF1), (TF2, TF1, TF0, TF1) |

6.10.2.4.1.11.1.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Uplink | Min spreading factor | 128 |
| | Max number of DPDCH data bits/radio frame | 300 |
| | Puncturing Limit | 0.76 |

6.10.2.4.1.11.2 Downlink

6.10.2.4.1.11.2.1 Transport channel parameters

6.10.2.4.1.11.2.1.1 Transport channel parameters for Conversational / speech / DL:4.75 kbps / CS RAB

| Higher layer | RAB/Signalling RB | | RAB subflow #1 | RAB subflow #2 | RAB subflow #3 |
|---|---|---|---|---|---|
| RLC | Logical channel type | | DTCH | | |
| | RLC mode | | TM | TM | TM |
| | Payload sizes, bit | | 0<br>39<br>42 | 53 | 60 |
| | Max data rate, bps | | 4750 | | |
| | TrD PDU header, bit | | 0 | | |
| MAC | MAC header, bit | | 0 | | |
| | MAC multiplexing | | N/A | | |
| Layer 1 | TrCH type | | DCH | DCH | DCH |
| | TB sizes, bit | | 0<br>39<br>42 | 53 | 60 |
| | TFS (note 1) | TF0, bits | 1x0 (note 2) | 0x53 | 0x60 |
| | | TF1, bits | 1x39 | 1x53 | N/A |
| | | TF2, bits | 1x42 | N/A | N/A |
| | TTI, ms | | 20 | 20 | 20 |
| | Coding type | | CC 1/3 | CC 1/3 | CC 1/2 |
| | CRC, bit | | 12 | N/A | N/A |
| | Max number of bits/TTI after channel coding | | 186 | 183 | 0 |
| | RM attribute | | 180 to 220 | 170 to 210 | 215 to 256 |

NOTE 1: The TrCH corresponding to RAB subflow #1 should be used as the guiding TrCH, (see clause 4.3 in 3GPP TS 25.212 [14]).
NOTE 2: CRC parity bits are to be attached to RAB subflow#1 any time since number of TrBlks are 1 even if there is no data on RAB subflow#1 (see clause 4.2.1.1 in 3GPP TS 25.212 [14]).

6.10.2.4.1.11.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.2.1.1.

6.10.2.4.1.11.2.1.3 TFCS

| TFCS size | 6 |
|---|---|
| TFCS | (RAB subflow#1, RAB subflow#2, RAB subflow#3, DCCH)=<br>(TF0, TF0, TF0, TF0), (TF1, TF0, TF0, TF0), (TF2, TF1, TF0, TF0),<br>(TF0, TF0, TF0, TF1), (TF1, TF0, TF0, TF1), (TF2, TF1, TF0, TF1) |

6.10.2.4.1.11.2.2 Physical channel parameters

| DPCH Downlink | DTX position | | Fixed |
|---|---|---|---|
| | Spreading factor | | 256 |
| | DPCCH | Number of TFCI bits/slot | 0 |
| | | Number of TPC bits/slot | 2 |
| | | Number of Pilot bits/slot | 4 |
| | DPDCH | Number of data bits/slot | 14 |
| | | Number of data bits/frame | 210 |

6.10.2.4.1.12 Conversational / unknown / UL:28.8/DL:28.8 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.2.4.1.12.1 Uplink

6.10.2.4.1.12.1.1 Transport channel parameters

6.10.2.4.1.12.1.1.1 Transport channel parameters for conversational / unknown / UL:28.8 kbps / CS RAB

| | | | |
|---|---|---|---|
| Higher layer | RAB/Signalling RB | | RAB |
| RLC | Logical channel type | | DTCH |
| | RLC mode | | TM |
| | Payload sizes, bit | | 576 |
| | Max data rate, bps | | 28 800 |
| | TrD PDU header, bit | | 0 |
| MAC | MAC header, bit | | 0 |
| | MAC multiplexing | | N/A |
| Layer 1 | TrCH type | | DCH |
| | TB sizes, bit | | 576 |
| | TFS | TF0, bits | 0x576 |
| | | TF1, bits | 1x576 |
| | | TF2, bits | 2x576 |
| | TTI, ms | | 40 |
| | Coding type | | TC |
| | CRC, bit | | 16 |
| | Max number of bits/TTI after channel coding | | 3 564 |
| | Uplink: Max number of bits/radio frame before rate matching | | 891 |
| | RM attribute | | 160 to 200 |

6.10.2.4.1.12.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.1.1.1.

6.10.2.4.1.12.1.1.3 TFCS

| | |
|---|---|
| TFCS size | 6 |
| TFCS | (28.8 kbps RAB, DCCH)=<br>(TF0, TF0), (TF1, TF0), (TF2, TF0), (TF0, TF1), (TF1, TF1), (TF2, TF1) |

6.10.2.4.1.12.1.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Uplink | Min spreading factor | 32 |
| | Max number of DPDCH data bits/radio frame | 1 200 |
| | Puncturing Limit | 0.92 |

6.10.2.4.1.12.2 Downlink

6.10.2.4.1.12.2.1 Transport channel parameters

6.10.2.4.1.12.2.1.1 Transport channel parameters for conversational / unknown / DL:28.8 kbps / CS RAB

| | | | |
|---|---|---|---|
| Higher layer | RAB/Signalling RB | | RAB |
| RLC | Logical channel type | | DTCH |
| | RLC mode | | TM |
| | Payload sizes, bit | | 576 |
| | Max data rate, bps | | 28 800 |
| | TrD PDU header, bit | | 0 |
| MAC | MAC header, bit | | 0 |
| | MAC multiplexing | | N/A |
| Layer 1 | TrCH type | | DCH |
| | TB sizes, bit | | 576 |
| | TFS | TF0, bits | 0x576 |
| | | TF1, bits | 1x576 |

| | | | |
|---|---|---|---|
| | | TF2, bits | 2x576 |
| | TTI, ms | | 40 |
| | Coding type | | TC |
| | CRC, bit | | 16 |
| | Max number of bits/TTI after channel coding | | 3 564 |
| | RM attribute | | 160 to 200 |

6.10.2.4.1.12.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.2.1.1.

6.10.2.4.1.12.2.1.3 TFCS

| | |
|---|---|
| TFCS size | 6 |
| TFCS | (28.8 kbps RAB, DCCH)=<br>(TF0, TF0), (TF1, TF0), (TF2, TF0), (TF0, TF1), (TF1, TF1), (TF2, TF1) |

6.10.2.4.1.12.2.2 Physical channel parameters

| | | | |
|---|---|---|---|
| DPCH Downlink | DTX position | | Flexible |
| | Spreading factor | | 64 |
| | DPCCH | Number of TFCI bits/slot | 8 |
| | | Number of TPC bits/slot | 4 |
| | | Number of Pilot bits/slot | 8 |
| | DPDCH | Number of data bits/slot | 60 |
| | | Number of data bits/frame | 900 |

6.10.2.4.1.13 Conversational / unknown / UL:64 DL:64 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.2.4.1.13.1 Uplink

6.10.2.4.1.13.1.1 Transport channel parameters

6.10.2.4.1.13.1.1.1 Transport channel parameters for Conversational / unknown / UL:64 kbps / CS RAB

| | | | |
|---|---|---|---|
| Higher layer | RAB/Signalling RB | | RAB |
| RLC | Logical channel type | | DTCH |
| | RLC mode | | TM |
| | Payload sizes, bit | | 640 |
| | Max data rate, bps | | 64 000 |
| | TrD PDU header, bit | | 0 |
| MAC | MAC header, bit | | 0 |
| | MAC multiplexing | | N/A |
| Layer 1 | TrCH type | | DCH |
| | TB sizes, bit | | 640 |
| | TFS | TF0, bits | 0x640 |
| | | TF1, bits | 2x640(alt. 4x640) |
| | TTI, ms | | 20(alt. 40) |
| | Coding type | | TC |
| | CRC, bit | | 16 |
| | Max number of bits/TTI after channel coding | | 3 948(alt. 7 884) |
| | Uplink: Max number of bits/radio frame before rate matching | | 1 974(alt. 1 971) |
| | RM attribute | | 150 to 195 |

6.10.2.4.1.13.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.1.1.1.

6.10.2.4.1.13.1.1.3 TFCS

| | |
|---|---|
| TFCS size | 4 |
| TFCS | (64 kbps RAB, DCCH)=(TF0, TF0), (TF1, TF0), (TF0, TF1), (TF1, TF1) |

6.10.2.4.1.13.1.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Uplink | Min spreading factor | 16 |
| | Max number of DPDCH data bits/radio frame | 2 400 |
| | Puncturing Limit | 0.88 |

6.10.2.4.1.13.2 Downlink

6.10.2.4.1.13.2.1 Transport channel parameters

6.10.2.4.1.13.2.1.1 Transport channel parameters for Conversational / unknown / DL:64 kbps / CS RAB

| | | | |
|---|---|---|---|
| Higher layer | RAB/Signalling RB | | RAB |
| RLC | Logical channel type | | DTCH |
| | RLC mode | | TM |
| | Payload sizes, bit | | 640 |
| | Max data rate, bps | | 64 000 |
| | TrD PDU header, bit | | 0 |
| MAC | MAC header, bit | | 0 |
| | MAC multiplexing | | N/A |
| Layer 1 | TrCH type | | DCH |
| | TB sizes, bit | | 640 |
| | TFS | TF0, bits | 0x640 |
| | | TF1, bits | 2x640(alt. 4x640) |
| | TTI, ms | | 20(alt. 40) |
| | Coding type | | TC |
| | CRC, bit | | 16 |
| | Max number of bits/TTI after channel coding | | 3 948(alt. 7 884) |
| | RM attribute | | 150 to 195 |

6.10.2.4.1.13.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.2.1.1.

6.10.2.4.1.13.2.1.3 TFCS

| | |
|---|---|
| TFCS size | 4 |
| TFCS | (64 kbps RAB, DCCH)=(TF0, TF0), (TF1, TF0), (TF0, TF1), (TF1, TF1) |

6.10.2.4.1.13.2.2 Physical channel parameters

| | | | |
|---|---|---|---|
| DPCH Downlink | DTX position | | Flexible |
| | Spreading factor | | 32 |
| | DPCCH | Number of TFCI bits/slot | 8 |
| | | Number of TPC bits/slot | 4 |
| | | Number of Pilot bits/slot | 8 |
| | DPDCH | Number of data bits/slot | 140 |
| | | Number of data bits/frame | 2 100 |

6.10.2.4.1.14 Conversational / unknown / UL:32 DL:32 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.2.4.1.14.1 Uplink

6.10.2.4.1.14.1.1 Transport channel parameters

6.10.2.4.1.14.1.1.1 Transport channel parameters for Conversational / unknown / UL:32 kbps / CS RAB

| | | | |
|---|---|---|---|
| Higher layer | RAB/Signalling RB | | RAB |
| RLC | Logical channel type | | DTCH |
| | RLC mode | | TM |
| | Payload sizes, bit | | 640 |
| | Max data rate, bps | | 32 000 |
| | TrD PDU header, bit | | 0 |
| MAC | MAC header, bit | | 0 |
| | MAC multiplexing | | N/A |
| Layer 1 | TrCH type | | DCH |
| | TB sizes, bit | | 640 |
| | TFS | TF0, bits | 0x640 |
| | | TF1, bits | 1x640(alt. 2x640) |
| | TTI, ms | | 20(alt. 40) |
| | Coding type | | TC |
| | CRC, bit | | 16 |
| | Max number of bits/TTI after channel coding | | 1 980 (alt. 3 948) |
| | Uplink: Max number of bits/radio frame before rate matching | | 990 (alt. 987) |
| | RM attribute | | 165 to 210 |

6.10.2.4.1.14.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.1.1.1.

6.10.2.4.1.13.1.1.3 TFCS

| | |
|---|---|
| TFCS size | 4 |
| TFCS | (32 kbps RAB, DCCH)=(TF0, TF0), (TF1, TF0), (TF0, TF1), (TF1, TF1) |

6.10.2.4.1.14.1.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Uplink | Min spreading factor | 32 |
| | Max number of DPDCH data bits/radio frame | 1 200 |
| | Puncturing Limit | 0.80 |

6.10.2.4.1.14.2 Downlink

6.10.2.4.1.14.2.1 Transport channel parameters

6.10.2.4.1.14.2.1.1 Transport channel parameters for Conversational / unknown / DL:32 kbps / CS RAB

| | | | |
|---|---|---|---|
| Higher layer | RAB/Signalling RB | | RAB |
| RLC | Logical channel type | | DTCH |
| | RLC mode | | TM |
| | Payload sizes, bit | | 640 |
| | Max data rate, bps | | 32 000 |
| | TrD PDU header, bit | | 0 |
| MAC | MAC header, bit | | 0 |
| | MAC multiplexing | | N/A |
| Layer 1 | TrCH type | | DCH |
| | TB sizes, bit | | 640 |
| | TFS | TF0, bits | 0x640 |
| | | TF1, bits | 1x640(alt. 2x640) |
| | TTI, ms | | 20(alt. 40) |
| | Coding type | | TC |
| | CRC, bit | | 16 |

| | | |
|---|---|---|
| | Max number of bits/TTI after channel coding | 1 980(alt. 3 948) |
| | RM attribute | 165 to 210 |

6.10.2.4.1.14.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.2.1.1.

6.10.2.4.1.14.2.1.3 TFCS

| | |
|---|---|
| TFCS size | 4 |
| TFCS | (32 kbps RAB, DCCH)=(TF0, TF0), (TF1, TF0), (TF0, TF1), (TF1, TF1) |

6.10.2.4.1.14.2.2 Physical channel parameters

| | | | |
|---|---|---|---|
| DPCH Downlink | DTX position | | Flexible |
| | Spreading factor | | 64 |
| | DPCCH | Number of TFCI bits/slot | 8 |
| | | Number of TPC bits/slot | 4 |
| | | Number of Pilot bits/slot | 8 |
| | DPDCH | Number of data bits/slot | 60 |
| | | Number of data bits/frame | 900 |

6.10.2.4.1.15 Streaming / unknown / UL:14.4/DL:14.4 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.2.4.1.15.1 Uplink

6.10.2.4.1.15.1.1 Transport channel parameters

6.10.2.4.1.15.1.1.1 Transport channel parameters for Streaming / unknown / UL: 14.4 kbps / CS RAB

| | | | |
|---|---|---|---|
| Higher layer | RAB/Signalling RB | | RAB |
| RLC | Logical channel type | | DTCH |
| | RLC mode | | TM |
| | Payload sizes, bit | | 576 |
| | Max data rate, bps | | 14 400 |
| | TrD PDU header, bit | | 0 |
| MAC | MAC header, bit | | 0 |
| | MAC multiplexing | | N/A |
| Layer 1 | TrCH type | | DCH |
| | TB sizes, bit | | 576 |
| | TFS | TF0, bits | 0x576 |
| | | TF1, bits | 1x576 |
| | TTI, ms | | 40 |
| | Coding type | | TC |
| | CRC, bit | | 16 |
| | Max number of bits/TTI after channel coding | | 1 788 |
| | Uplink: Max number of bits/radio frame before rate matching | | 447 |
| | RM attribute | | 145 to 185 |

6.10.2.4.1.15.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.1.1.1.

6.10.2.4.1.15.1.1.3 TFCS

| | |
|---|---|
| TFCS size | 4 |
| TFCS | (14.4 kbps RAB, DCCH)=(TF0, TF0), (TF1, TF0), (TF0, TF1), (TF1, TF1) |

6.10.2.4.1.15.1.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Uplink | Min spreading factor | 64 |

| | | |
|---|---|---|
| | Max number of DPDCH data bits/radio frame | 600 |
| | Puncturing Limit | 0.88 |

6.10.2.4.1.15.2 Downlink

6.10.2.4.1.15.2.1 Transport channel parameters

6.10.2.4.1.15.2.1.1 Transport channel parameters for Streaming / unknown / DL:14.4 kbps / CS RAB

| | | | |
|---|---|---|---|
| Higher layer | RAB/Signalling RB | | RAB |
| RLC | Logical channel type | | DTCH |
| | RLC mode | | TM |
| | Payload sizes, bit | | 576 |
| | Max data rate, bps | | 14 400 |
| | TrD PDU header, bit | | 0 |
| MAC | MAC header, bit | | 0 |
| | MAC multiplexing | | N/A |
| Layer 1 | TrCH type | | DCH |
| | TB sizes, bit | | 576 |
| | TFS | TF0, bits | 0x576 |
| | | TF1, bits | 1x576 |
| | TTI, ms | | 40 |
| | Coding type | | TC |
| | CRC, bit | | 16 |
| | Max number of bits/TTI after channel coding | | 1 788 |
| | RM attribute | | 145 to 185 |

6.10.2.4.1.15.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.2.1.1.

6.10.2.4.1.15.2.1.3 TFCS

| | |
|---|---|
| TFCS size | 4 |
| TFCS | (14.4 kbps RAB, DCCH)=(TF0, TF0), (TF1, TF0), (TF0, TF1), (TF1, TF1) |

6.10.2.4.1.15.2.2 Physical channel parameters

| | | | |
|---|---|---|---|
| DPCH Downlink | DTX position | | Flexible |
| | Spreading factor | | 128 |
| | DPCCH | Number of TFCI bits/slot | 2 |
| | | Number of TPC bits/slot | 2 |
| | | Number of Pilot bits/slot | 8 |
| | DPDCH | Number of data bits/slot | 28 |
| | | Number of data bits/frame | 420 |

6.10.2.4.1.16 Streaming / unknown / UL:28.8/DL:28.8 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.2.4.1.16.1 Uplink

6.10.2.4.1.16.1.1 Transport channel parameters

6.10.2.4.1.16.1.1.1 Transport channel parameters for Streaming / unknown / UL:28.8 kbps / CS RAB

| | | |
|---|---|---|
| Higher layer | RAB/Signalling RB | RAB |
| RLC | Logical channel type | DTCH |
| | RLC mode | TM |
| | Payload sizes, bit | 576 |
| | Max data rate, bps | 28 800 |
| | TrD PDU header, bit | 0 |
| MAC | MAC header, bit | 0 |
| | MAC multiplexing | N/A |

| | | | |
|---|---|---|---|
| Layer 1 | TrCH type | | DCH |
| | TB sizes, bit | | 576 |
| | TFS | TF0, bits | 0x576 |
| | | TF1, bits | 1x576 |
| | | TF2, bits | 2x576 |
| | TTI, ms | | 40 |
| | Coding type | | TC |
| | CRC, bit | | 16 |
| | Max number of bits/TTI after channel coding | | 3 564 |
| | Uplink: Max number of bits/radio frame before rate matching | | 891 |
| | RM attribute | | 135 to 175 |

6.10.2.4.1.16.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.1.1.1.

6.10.2.4.1.16.1.1.3 TFCS

| | |
|---|---|
| TFCS size | 6 |
| TFCS | (28.8 kbps RAB, DCCH)=<br>(TF0, TF0), (TF1, TF0), (TF2, TF0), (TF0, TF1), (TF1, TF1), (TF2, TF1) |

6.10.2.4.1.16.1.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Uplink | Min spreading factor | 32 |
| | Max number of DPDCH data bits/radio frame | 1 200 |
| | Puncturing Limit | 0.96 |

6.10.2.4.1.16.2 Downlink

6.10.2.4.1.16.2.1 Transport channel parameters

6.10.2.4.1.16.2.1.1 Transport channel parameters for Streaming / unknown / DL:28.8 kbps / CS RAB

| | | | |
|---|---|---|---|
| Higher layer | RAB/Signalling RB | | RAB |
| RLC | Logical channel type | | DTCH |
| | RLC mode | | TM |
| | Payload sizes, bit | | 576 |
| | Max data rate, bps | | 28 800 |
| | TrD PDU header, bit | | 0 |
| MAC | MAC header, bit | | 0 |
| | MAC multiplexing | | N/A |
| Layer 1 | TrCH type | | DCH |
| | TB sizes, bit | | 576 |
| | TFS | TF0, bits | 0x576 (alt. 1x0) (note) |
| | | TF1, bits | 1x576 |
| | | TF2, bits | 2x576 |
| | TTI, ms | | 40 |
| | Coding type | | TC |
| | CRC, bit | | 16 |
| | Max number of bits/TTI after channel coding | | 3 564 |
| | RM attribute | | 135 to 175 |
| NOTE: Alternative 1x0 is used to have CRC present in all transport formats. | | | |

6.10.2.4.1.16.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.2.1.1.

6.10.2.4.1.16.2.1.3 TFCS

| | |
|---|---|
| TFCS size | 6 |
| TFCS | (28.8 kbps RAB, DCCH)=<br>(TF0, TF0), (TF1, TF0), (TF2, TF0), (TF0, TF1), (TF1, TF1), (TF2, TF1) |

6.10.2.4.1.16.2.2 Physical channel parameters

| | | | |
|---|---|---|---|
| DPCH Downlink | DTX position | | Flexible |
| | Spreading factor | | 64 |
| | DPCCH | Number of TFCI bits/slot | 8 |
| | | Number of TPC bits/slot | 4 |
| | | Number of Pilot bits/slot | 8 |
| | DPDCH | Number of data bits/slot | 60 |
| | | Number of data bits/frame | 900 |

6.10.2.4.1.17 Streaming / unknown / UL:57.6/DL:57.6 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.2.4.1.17.1 Uplink

6.10.2.4.1.17.1.1 Transport channel parameters

6.10.2.4.1.17.1.1.1 Transport channel parameters for Streaming / unknown / UL:57.6 kbps / CS RAB

| | | | |
|---|---|---|---|
| Higher layer | RAB/Signalling RB | | RAB |
| RLC | Logical channel type | | DTCH |
| | RLC mode | | TM |
| | Payload sizes, bit | | 576 |
| | Max data rate, bps | | 57 600 |
| | TrD PDU header, bit | | 0 |
| MAC | MAC header, bit | | 0 |
| | MAC multiplexing | | N/A |
| Layer 1 | TrCH type | | DCH |
| | TB sizes, bit | | 576 |
| | TFS | TF0, bits | 0x576 |
| | | TF1, bits | 1x576 |
| | | TF2, bits | 2x576 |
| | | TF3, bits | 3x576 |
| | | TF4, bits | 4x576 |
| | TTI, ms | | 40 |
| | Coding type | | TC |
| | CRC, bit | | 16 |
| | Max number of bits/TTI after channel coding | | 7 116 |
| | Uplink: Max number of bits/radio frame before rate matching | | 1 779 |

6.10.2.4.1.17.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.1.1.1.

6.10.2.4.1.17.1.1.3 TFCS

| | |
|---|---|
| TFCS size | 10 |
| TFCS | (57.6 kbps RAB, DCCH)=<br>(TF0, TF0), (TF1, TF0), (TF2, TF0), (TF3, TF0), (TF4, TF0),<br>(TF0, TF1), (TF1, TF1), (TF2, TF1), (TF3, TF1), (TF4, TF1) |

6.10.2.4.1.17.1.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Uplink | Min spreading factor | 16 |
| | Max number of DPDCH data bits/radio frame | 2 400 |
| | Puncturing Limit | 0.96 |

6.10.2.4.1.17.2 Downlink

6.10.2.4.1.17.2.1 Transport channel parameters

6.10.2.4.1.17.2.1.1 Transport channel parameters for Streaming / unknown / DL:57.6 kbps / CS RAB

| | | | |
|---|---|---|---|
| Higher layer | RAB/Signalling RB | | RAB |
| RLC | Logical channel type | | DTCH |
| | RLC mode | | TM |
| | Payload sizes, bit | | 576 |
| | Max data rate, bps | | 57 600 |
| | TrD PDU header, bit | | 0 |
| MAC | MAC header, bit | | 0 |
| | MAC multiplexing | | N/A |
| Layer 1 | TrCH type | | DCH |
| | TB sizes, bit | | 576 |
| | TFS | TF0, bits | 0x576 |
| | | TF1, bits | 1x576 |
| | | TF2, bits | 2x576 |
| | | TF3, bits | 3x576 |
| | | TF4, bits | 4x576 |
| | TTI, ms | | 40 |
| | Coding type | | TC |
| | CRC, bit | | 16 |
| | Max number of bits/TTI after channel coding | | 7 116 |
| | RM attribute | | 125 to 165 |

6.10.2.4.1.17.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.2.1.1.

6.10.2.4.1.17.2.1.3 TFCS

| | |
|---|---|
| TFCS size | 10 |
| TFCS | (57.6 kbps RAB, DCCH)=<br>(TF0, TF0), (TF1, TF0), (TF2, TF0), (TF3, TF0), (TF4, TF0),<br>(TF0, TF1), (TF1, TF1), (TF2, TF1), (TF3, TF1), (TF4, TF1) |

6.10.2.4.1.17.2.2 Physical channel parameters

| | | | |
|---|---|---|---|
| DPCH Downlink | DTX position | | Flexible |
| | Spreading factor | | 32 |
| | DPCCH | Number of TFCI bits/slot | 8 |
| | | Number of TPC bits/slot | 4 |
| | | Number of Pilot bits/slot | 8 |
| | DPDCH | Number of data bits/slot | 140 |
| | | Number of data bits/frame | 2 100 |

6.10.2.4.1.18 Void

6.10.2.4.1.19 Void

6.10.2.4.1.20 Void

6.10.2.4.1.21 Void

6.10.2.4.1.22 Void

6.10.2.4.1.23 Interactive or background / UL:32 DL:8 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.2.4.1.23.1 Uplink

6.10.2.4.1.23.1.1 Transport channel parameters

6.10.2.4.1.23.1.1.1 Transport channel parameters for Interactive or background / UL:32 kbps / PS RAB

| | | | |
|---|---|---|---|
| Higher layer | RAB/Signalling RB | | RAB |
| RLC | Logical channel type | | DTCH |
| | RLC mode | | AM |
| | Payload sizes, bit | | 320 |
| | Max data rate, bps | | 32 000 |
| | AMD PDU header, bit | | 16 |
| MAC | MAC header, bit | | 0 |
| | MAC multiplexing | | N/A |
| Layer 1 | TrCH type | | DCH |
| | TB sizes, bit | | 336 |
| | TFS | TF0, bits | 0x336 |
| | | TF1, bits | 1x336 |
| | | TF2, bits | 2x336 (alt. N/A) |
| | TTI, ms | | 20 (alt. 10) |
| | Coding type | | TC (alt. CC 1/3) |
| | CRC, bit | | 16 |
| | Max number of bits/TTI after channel coding | | 2 124 (alt. 1 080) |
| | Uplink: Max number of bits/radio frame before rate matching | | 1 062 (alt. 1 080) |
| | RM attribute | | 135 to 175 |

6.10.2.4.1.23.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.1.1.1.

6.10.2.4.1.23.1.1.3 TFCS

| | |
|---|---|
| TFCS size | 6 (alt. 4) |
| TFCS | (32 kbps RAB, DCCH)=<br>(TF0, TF0), (TF1, TF0), (TF2, TF0), (TF0, TF1), (TF1, TF1), (TF2, TF1)<br>(alt. (TF0, TF0), (TF1, TF0), (TF0, TF1), (TF1, TF1)) |

6.10.2.4.1.23.1.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Uplink | Min spreading factor | 32 |
| | Max number of DPDCH data bits/radio frame | 1 200 |
| | Puncturing Limit | 0.88 |

6.10.2.4.1.23.2 Downlink

6.10.2.4.1.23.2.1 Transport channel parameters

6.10.2.4.1.23.2.1.1 Transport channel parameters for Interactive or background / DL:8 kbps / PS RAB

| | | | |
|---|---|---|---|
| Higher layer | RAB/Signalling RB | | RAB |
| RLC | Logical channel type | | DTCH |
| | RLC mode | | AM |
| | Payload sizes, bit | | 320 |
| | Max data rate, bps | | 8 000 |
| | AMD PDU header, bit | | 16 |
| MAC | MAC header, bit | | 0 |
| | MAC multiplexing | | N/A |
| Layer 1 | TrCH type | | DCH |
| | TB sizes, bit | | 336 |
| | TFS | TF0, bits | 0x336 |
| | | TF1, bits | 1x336 |
| | TTI, ms | | 40 |
| | Coding type | | TC (alt. CC 1/3) |
| | CRC, bit | | 16 |
| | Max number of bits/TTI after channel coding | | 1 068 (alt. 1 080) |
| | RM attribute | | 135 to 175 |

6.10.2.4.1.23.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.2.1.1.

6.10.2.4.1.23.2.1.3 TFCS

| | |
|---|---|
| TFCS size | 4 |
| TFCS | (8 kbps RAB, DCCH)=(TF0, TF0), (TF1, TF0), (TF0, TF1), (TF1, TF1) |

6.10.2.4.1.23.2.2 Physical channel parameters

| | | | |
|---|---|---|---|
| DPCH Downlink | DTX position | | Flexible |
| | Spreading factor | | 128 |
| | DPCCH | Number of TFCI bits/slot | 2 |
| | | Number of TPC bits/slot | 2 |
| | | Number of Pilot bits/slot | 4 |
| | DPDCH | Number of data bits/slot | 32 |
| | | Number of data bits/frame | 480 |

6.10.2.4.1.23a Interactive or background / UL:8 DL:8 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.2.4.1.23a.1 Uplink

6.10.2.4.1.23a.1.1 Transport channel parameters

6.10.2.4.1.23a.1.1.1 Transport channel parameters for Interactive or background / UL:8 kbps / PS RAB

| | | | |
|---|---|---|---|
| Higher layer | RAB/Signalling RB | | RAB |
| RLC | Logical channel type | | DTCH |
| | RLC mode | | AM |
| | Payload sizes, bit | | 320 |
| | Max data rate, bps | | 8 000 |
| | AMD PDU header, bit | | 16 |
| MAC | MAC header, bit | | 0 |
| | MAC multiplexing | | N/A |
| Layer 1 | TrCH type | | DCH |
| | TB sizes, bit | | 336 |
| | TFS | TF0, bits | 0x336 |
| | | TF1, bits | 1x336 |
| | TTI, ms | | 40 |
| | Coding type | | CC 1/3 (alt. TC) |
| | CRC, bit | | 16 |
| | Max number of bits/TTI after channel coding | | 1 080 (alt. 1 068) |
| | Uplink: Max number of bits/radio frame before rate matching | | 270 (alt. 267) |
| | RM attribute | | 135 to 175 |

6.10.2.4.1.23a.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.1.1.1.

6.10.2.4.1.23a.1.1.3 TFCS

| | |
|---|---|
| TFCS size | 4 |
| TFCS | (8 kbps RAB, DCCH)=<br>(TF0, TF0), (TF1, TF0), (TF0, TF1), (TF1, TF1) |

6.10.2.4.1.23a.1.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Uplink | Min spreading factor | 64 |
| | Max number of DPDCH data bits/radio frame | 600 |
| | Puncturing Limit | 1.0 |

6.10.2.4.1.23a.2 Downlink

6.10.2.4.1.23a.2.1 Transport channel parameters

6.10.2.4.1.23a.2.1.1 Transport channel parameters for Interactive or background / DL:8 kbps / PS RAB

| | | | |
|---|---|---|---|
| Higher layer | RAB/Signalling RB | | RAB |
| RLC | Logical channel type | | DTCH |
| | RLC mode | | AM |
| | Payload sizes, bit | | 320 |
| | Max data rate, bps | | 8 000 |
| | AMD PDU header, bit | | 16 |
| MAC | MAC header, bit | | 0 |
| | MAC multiplexing | | N/A |
| Layer 1 | TrCH type | | DCH |
| | TB sizes, bit | | 336 |
| | TFS | TF0, bits | 0x336 |
| | | TF1, bits | 1x336 |
| | TTI, ms | | 40 |
| | Coding type | | CC 1/3 (alt. TC) |
| | CRC, bit | | 16 |
| | Max number of bits/TTI after channel coding | | 1 080 (alt. 1 068) |
| | RM attribute | | 135 to 175 |

6.10.2.4.1.23a.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.2.1.1.

6.10.2.4.1.23a.2.1.3 TFCS

| | |
|---|---|
| TFCS size | 4 |
| TFCS | (8 kbps RAB, DCCH)=(TF0, TF0), (TF1, TF0), (TF0, TF1), (TF1, TF1) |

6.10.2.4.1.23a.2.2 Physical channel parameters

| | | | |
|---|---|---|---|
| DPCH Downlink | DTX position | | Flexible |
| | Spreading factor | | 128 |
| | DPCCH | Number of TFCI bits/slot | 2 |
| | | Number of TPC bits/slot | 2 |
| | | Number of Pilot bits/slot | 4 |
| | DPDCH | Number of data bits/slot | 32 |
| | | Number of data bits/frame | 480 |

6.10.2.4.1.23b Interactive or background / UL:16 DL:16 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.2.4.1.23b.1 Uplink

6.10.2.4.1.23b.1.1 Transport channel parameters

6.10.2.4.1.23b.1.1.1 Transport channel parameters for Interactive or background / UL:16 kbps / PS RAB

| | | | |
|---|---|---|---|
| Higher layer | RAB/Signalling RB | | RAB |
| RLC | Logical channel type | | DTCH |
| | RLC mode | | AM |
| | Payload sizes, bit | | 320 |
| | Max data rate, bps | | 16 000 |
| | AMD PDU header, bit | | 16 |
| MAC | MAC header, bit | | 0 |
| | MAC multiplexing | | N/A |
| Layer 1 | TrCH type | | DCH |
| | TB sizes, bit | | 336 |
| | TFS | TF0, bits | 0x336 |
| | | TF1, bits | 1x336 |

| | | | |
|---|---|---|---|
| | | TF2, bits | 2x336 |
| | TTI, ms | | 40 |
| | Coding type | | TC |
| | CRC, bit | | 16 |
| | Max number of bits/TTI after channel coding | | 2 124 |
| | Uplink: Max number of bits/radio frame before rate matching | | 531 |
| | RM attribute | | 135 to 175 |

6.10.2.4.1.23b.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.1.1.1.

6.10.2.4.1.23b.1.1.3 TFCS

| | |
|---|---|
| TFCS size | 6 |
| TFCS | (16 kbps RAB, DCCH)=<br>(TF0, TF0), (TF1, TF0), (TF2, TF0), (TF0, TF1), (TF1, TF1), (TF2, TF1) |

6.10.2.4.1.23b.1.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Uplink | Min spreading factor | 32 |
| | Max number of DPDCH data bits/radio frame | 1 200 |
| | Puncturing Limit | 1.0 |

6.10.2.4.1.23b.2 Downlink

6.10.2.4.1.23b.2.1 Transport channel parameters

6.10.2.4.1.23b.2.1.1 Transport channel parameters for Interactive or background / DL:16 kbps / PS RAB

| | | | |
|---|---|---|---|
| Higher layer | RAB/Signalling RB | | RAB |
| RLC | Logical channel type | | DTCH |
| | RLC mode | | AM |
| | Payload sizes, bit | | 320 |
| | Max data rate, bps | | 16 000 |
| | AMD PDU header, bit | | 16 |
| MAC | MAC header, bit | | 0 |
| | MAC multiplexing | | N/A |
| Layer 1 | TrCH type | | DCH |
| | TB sizes, bit | | 336 |
| | TFS | TF0, bits | 0x336 |
| | | TF1, bits | 1x336 |
| | | TF2, bits | 2x336 |
| | TTI, ms | | 40 |
| | Coding type | | TC |
| | CRC, bit | | 16 |
| | Max number of bits/TTI after channel coding | | 2 124 |
| | RM attribute | | 135 to 175 |

6.10.2.4.1.23b.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.2.1.1.

6.10.2.4.1.23b.2.1.3 TFCS

| | |
|---|---|
| TFCS size | 6 |
| TFCS | (16 kbps RAB, DCCH)=<br>(TF0, TF0), (TF1, TF0), (TF2, TF0), (TF0, TF1), (TF1, TF1), (TF2, TF1) |

6.10.2.4.1.23b.2.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Downlink | DTX position | Flexible |

| | | | |
|---|---|---|---|
| | Spreading factor | | 128 |
| | DPCCH | Number of TFCI bits/slot | 2 |
| | | Number of TPC bits/slot | 2 |
| | | Number of Pilot bits/slot | 4 |
| | DPDCH | Number of data bits/slot | 32 |
| | | Number of data bits/frame | 480 |

6.10.2.4.1.23c Interactive or background / UL:32 DL:32 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.2.4.1.23c.1 Uplink

6.10.2.4.1.23c.1.1 Transport channel parameters

6.10.2.4.1.23c.1.1.1 Transport channel parameters for Interactive or background / UL:32 kbps / PS RAB

| | | | |
|---|---|---|---|
| Higher layer | RAB/Signalling RB | | RAB |
| RLC | Logical channel type | | DTCH |
| | RLC mode | | AM |
| | Payload sizes, bit | | 320 |
| | Max data rate, bps | | 32 000 |
| | AMD PDU header, bit | | 16 |
| MAC | MAC header, bit | | 0 |
| | MAC multiplexing | | N/A |
| Layer 1 | TrCH type | | DCH |
| | TB sizes, bit | | 336 |
| | TFS | TF0, bits | 0x336 |
| | | TF1, bits | 1x336 |
| | | TF2, bits | 2x336 |
| | | TF3, bits | 3x336 |
| | | TF4, bits | 4x336 |
| | TTI, ms | | 40 |
| | Coding type | | TC |
| | CRC, bit | | 16 |
| | Max number of bits/TTI after channel coding | | 4 236 |
| | Uplink: Max number of bits/radio frame before rate matching | | 1 059 |
| | RM attribute | | 135 to 175 |

6.10.2.4.1.23c.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.1.1.1.

6.10.2.4.1.23c.1.1.3 TFCS

| | |
|---|---|
| TFCS size | 10 |
| TFCS | (32 kbps RAB, DCCH)=<br>(TF0,TF0), (TF1,TF0), (TF2,TF0), (TF3,TF0), (TF4,TF0), (TF0,TF1), (TF1,TF1), (TF2,TF1), (TF3,TF1), (TF4,TF1) |

6.10.2.4.1.23c.1.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Uplink | Min spreading factor | 32 |
| | Max number of DPDCH data bits/radio frame | 1 200 |
| | Puncturing Limit | 0.88 |

6.10.2.4.1.23c.2 Downlink

6.10.2.4.1.23c.2.1 Transport channel parameters

6.10.2.4.1.23c.2.1.1 Transport channel parameters for Interactive or background / DL:32 kbps / PS RAB

| | | |
|---|---|---|
| Higher layer | RAB/Signalling RB | RAB |
| RLC | Logical channel type | DTCH |

| | | | |
|---|---|---|---|
| | RLC mode | | AM |
| | Payload sizes, bit | | 320 |
| | Max data rate, bps | | 32 000 |
| | AMD PDU header, bit | | 16 |
| MAC | MAC header, bit | | 0 |
| | MAC multiplexing | | N/A |
| Layer 1 | TrCH type | | DCH |
| | TB sizes, bit | | 336 |
| | TFS | TF0, bits | 0x336 |
| | | TF1, bits | 1x336 |
| | | TF2, bits | 2x336 |
| | | TF3, bits | 3x336 |
| | | TF4, bits | 4x336 |
| | TTI, ms | | 40 |
| | Coding type | | TC |
| | CRC, bit | | 16 |
| | Max number of bits/TTI after channel coding | | 4 236 |
| | RM attribute | | 135 to 175 |

6.10.2.4.1.23c.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.2.1.1.

6.10.2.4.1.23c.2.1.3 TFCS

| | |
|---|---|
| TFCS size | 10 |
| TFCS | (32 kbps RAB, DCCH)=<br>(TF0,TF0), (TF1,TF0), (TF2,TF0), (TF3,TF0), (TF4,TF0), (TF0,TF1), (TF1,TF1), (TF2,TF1), (TF3,TF1), (TF4,TF1) |

6.10.2.4.1.23c.2.2 Physical channel parameters

| | | | |
|---|---|---|---|
| DPCH Downlink | DTX position | | Flexible |
| | Spreading factor | | 64 |
| | DPCCH | Number of TFCI bits/slot | 8 |
| | | Number of TPC bits/slot | 4 |
| | | Number of Pilot bits/slot | 8 |
| | DPDCH | Number of data bits/slot | 60 |
| | | Number of data bits/frame | 900 |

6.10.2.4.1.23d Interactive or background / UL:32 DL:32 kbps / PS RAB (20 ms TTI)+ UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.2.4.1.23d.1 Uplink

6.10.2.4.1.23d.1.1 Transport channel parameters

6.10.2.4.1.23d.1.1.1 Transport channel parameters for Interactive or background / UL:32 kbps / PS RAB

| | | | |
|---|---|---|---|
| Higher layer | RAB/Signalling RB | | RAB |
| RLC | Logical channel type | | DTCH |
| | RLC mode | | AM |
| | Payload sizes, bit | | 320 |
| | Max data rate, bps | | 32 000 |
| | AMD PDU header, bit | | 16 |
| MAC | MAC header, bit | | 0 |
| | MAC multiplexing | | N/A |
| Layer 1 | TrCH type | | DCH |
| | TB sizes, bit | | 336 |
| | TFS | TF0, bits | 0x336 |
| | | TF1, bits | 1x336 |
| | | TF2, bits | 2x336 |
| | TTI, ms | | 20 |

| | | |
|---|---|---|
| | Coding type | TC |
| | CRC, bit | 16 |
| | Max number of bits/TTI after channel coding | 2 124 |
| | Uplink: Max number of bits/radio frame before rate matching | 1 062 |
| | RM attribute | 135 to 175 |

6.10.2.4.1.23d.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.1.1.1.

6.10.2.4.1.23d.1.1.3 TFCS

| | |
|---|---|
| TFCS size | 6 |
| TFCS | (32 kbps RAB, DCCH)=<br>(TF0,TF0), (TF1,TF0), (TF2,TF0), (TF0,TF1), (TF1,TF1), (TF2,TF1) |

6.10.2.4.1.23d.1.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Uplink | Min spreading factor | 32 |
| | Max number of DPDCH data bits/radio frame | 1 200 |
| | Puncturing Limit | 0.88 |

6.10.2.4.1.23d.2 Downlink

6.10.2.4.1.23d.2.1 Transport channel parameters

6.10.2.4.1.23d.2.1.1 Transport channel parameters for Interactive or background / DL:32 kbps / PS RAB

| | | | |
|---|---|---|---|
| Higher layer | RAB/Signalling RB | | RAB |
| RLC | Logical channel type | | DTCH |
| | RLC mode | | AM |
| | Payload sizes, bit | | 320 |
| | Max data rate, bps | | 32 000 |
| | AMD PDU header, bit | | 16 |
| MAC | MAC header, bit | | 0 |
| | MAC multiplexing | | N/A |
| Layer 1 | TrCH type | | DCH |
| | TB sizes, bit | | 336 |
| | TFS | TF0, bits | 0x336 |
| | | TF1, bits | 1x336 |
| | | TF2, bits | 2x336 |
| | TTI, ms | | 20 |
| | Coding type | | TC |
| | CRC, bit | | 16 |
| | Max number of bits/TTI after channel coding | | 2 124 |
| | RM attribute | | 135 to 175 |

6.10.2.4.1.23d.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.2.1.1.

6.10.2.4.1.23d.2.1.3 TFCS

| | |
|---|---|
| TFCS size | 6 |
| TFCS | (32 kbps RAB, DCCH)=<br>(TF0,TF0), (TF1,TF0), (TF2,TF0), (TF0,TF1), (TF1,TF1), (TF2,TF1) |

6.10.2.4.1.23d.2.2 Physical channel parameters

| | | | |
|---|---|---|---|
| DPCH Downlink | DTX position | | Flexible |
| | Spreading factor | | 64 |
| | DPCCH | Number of TFCI bits/slot | 8 |

| | | | |
|---|---|---|---|
| | | Number of TPC bits/slot | 4 |
| | | Number of Pilot bits/slot | 8 |
| | DPDCH | Number of data bits/slot | 60 |
| | | Number of data bits/frame | 900 |

6.10.2.4.1.24 Void

6.10.2.4.1.25 Interactive or background / UL:32 DL: 64 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.2.4.1.25.1 Uplink

See clause 6.10.2.4.1.23.1.

6.10.2.4.1.25.2 Downlink

6.10.2.4.1.25.2.1 Transport channel parameters

6.10.2.4.1.25.2.1.1 Transport channel parameters for Interactive or background / DL:64 kbps / PS RAB

| | | | |
|---|---|---|---|
| Higher layer | RAB/Signalling RB | | RAB |
| RLC | Logical channel type | | DTCH |
| | RLC mode | | AM |
| | Payload sizes, bit | | 320 |
| | Max data rate, bps | | 64 000 |
| | AMD PDU header, bit | | 16 |
| MAC | MAC header, bit | | 0 |
| | MAC multiplexing | | N/A |
| Layer 1 | TrCH type | | DCH |
| | TB sizes, bit | | 336 |
| | TFS | TF0, bits | 0x336 |
| | | TF1, bits | 1x336 |
| | | TF2, bits | 2x336 |
| | | TF3, bits | 3x336 |
| | | TF4, bits | 4x336 |
| | TTI, ms | | 20 |
| | Coding type | | TC |
| | CRC, bit | | 16 |
| | Max number of bits/TTI after channel coding | | 4 236 |
| | RM attribute | | 130 to 170 |

6.10.2.4.1.25.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.2.1.1.

6.10.2.4.1.25.2.1.3 TFCS

| | |
|---|---|
| TFCS size | 10 |
| TFCS | (64 kbps RAB, DCCH)=<br>(TF0, TF0), (TF1, TF0), (TF2, TF0), (TF3, TF0), (TF4, TF0),<br>(TF0, TF1), (TF1, TF1), (TF2, TF1), (TF3, TF1), (TF4, TF1) |

6.10.2.4.1.25.2.2 Physical channel parameters

| | | | |
|---|---|---|---|
| DPCH Downlink | DTX position | | Flexible |
| | Spreading factor | | 32 |
| | DPCCH | Number of TFCI bits/slot | 8 |
| | | Number of TPC bits/slot | 4 |
| | | Number of Pilot bits/slot | 8 |
| | DPDCH | Number of data bits/slot | 140 |
| | | Number of data bits/frame | 2 100 |

6.10.2.4.1.26 Interactive or background / UL:64 DL: 64 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.2.4.1.26.1 Uplink

6.10.2.4.1.26.1.1 Transport channel parameters

6.10.2.4.1.26.1.1.1 Transport channel parameters for Interactive or background / UL:64 kbps / PS RAB

| | | | |
|---|---|---|---|
| Higher layer | RAB/Signalling RB | | RAB |
| RLC | Logical channel type | | DTCH |
| | RLC mode | | AM |
| | Payload sizes, bit | | 320 |
| | Max data rate, bps | | 64 000 |
| | AMD PDU header, bit | | 16 |
| MAC | MAC header, bit | | 0 |
| | MAC multiplexing | | N/A |
| Layer 1 | TrCH type | | DCH |
| | TB sizes, bit | | 336 |
| | TFS | TF0, bits | 0x336 |
| | | TF1, bits | 1x336 |
| | | TF2, bits | 2x336 |
| | | TF3, bits | 3x336 |
| | | TF4, bits | 4x336 |
| | TTI, ms | | 20 |
| | Coding type | | TC |
| | CRC, bit | | 16 |
| | Max number of bits/TTI after channel coding | | 4 236 |
| | Uplink: Max number of bits/radio frame before rate matching | | 2 118 |
| | RM attribute | | 130 to 170 |

6.10.2.4.1.26.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.1.1.1.

6.10.2.4.1.26.1.1.3 TFCS

| | |
|---|---|
| TFCS size | 10 |
| TFCS | (64 kbps RAB, DCCH)=<br>(TF0, TF0), (TF1, TF0), (TF2, TF0), (TF3, TF0), (TF4, TF0),<br>(TF0, TF1), (TF1, TF1), (TF2, TF1), (TF3, TF1), (TF4, TF1) |

6.10.2.4.1.26.1.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Uplink | Min spreading factor | 16 |
| | Max number of DPDCH data bits/radio frame | 2 400 |
| | Puncturing Limit | 0.96 |

6.10.2.4.1.26.2 Downlink

See clause 6.10.2.4.1.25.2.

6.10.2.4.1.27 Interactive or background / UL:64 DL:128 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.2.4.1.27.1 Uplink

See clause 6.10.2.4.1.26.1.

6.10.2.4.1.27.2 Downlink

6.10.2.4.1.27.2.1 Transport channel parameters

6.10.2.4.1.27.2.1.1 Transport channel parameters for Interactive or background / DL:128 kbps / PS RAB

| | | | |
|---|---|---|---|
| Higher layer | RAB/Signalling RB | | RAB |
| RLC | Logical channel type | | DTCH |
| | RLC mode | | AM |
| | Payload sizes, bit | | 320 |
| | Max data rate, bps | | 128 000 |
| | AMD PDU header, bit | | 16 |
| MAC | MAC header, bit | | 0 |
| | MAC multiplexing | | N/A |
| Layer 1 | TrCH type | | DCH |
| | TB sizes, bit | | 336 |
| | TFS | TF0, bits | 0x336 |
| | | TF1, bits | 1x336 |
| | | TF2, bits | 2x336 |
| | | TF3, bits | 4x336 |
| | | TF4, bits | 8x336 |
| | TTI, ms | | 20 |
| | Coding type | | TC |
| | CRC, bit | | 16 |
| | Max number of bits/TTI after channel coding | | 8 460 |
| | RM attribute | | 120 to 160 |

6.10.2.4.1.27.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.2.1.1.

6.10.2.4.1.27.2.1.3 TFCS

| | |
|---|---|
| TFCS size | 10 |
| TFCS | (128 kbps RAB, DCCH)=<br>(TF0, TF0), (TF1, TF0), (TF2, TF0), (TF3, TF0), (TF4, TF0),<br>(TF0, TF1), (TF1, TF1), (TF2, TF1), (TF3, TF1), (TF4, TF1) |

6.10.2.4.1.27.2.2 Physical channel parameters

| | | | |
|---|---|---|---|
| DPCH Downlink | DTX position | | Flexible |
| | Spreading factor | | 16 |
| | DPCCH | Number of TFCI bits/slot | 8 |
| | | Number of TPC bits/slot | 8 |
| | | Number of Pilot bits/slot | 16 |
| | DPDCH | Number of data bits/slot | 288 |
| | | Number of data bits/frame | 4 320 |

6.10.2.4.1.28 Interactive or background / UL:128 DL:128 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.2.4.1.28.1 Uplink

6.10.2.4.1.28.1.1 Transport channel parameters

6.10.2.4.1.28.1.1.1 Transport channel parameters for Interactive or background / UL:128 kbps / PS RAB

| | | |
|---|---|---|
| Higher layer | RAB/Signalling RB | RAB |
| RLC | Logical channel type | DTCH |
| | RLC mode | AM |
| | Payload sizes, bit | 320 |
| | Max data rate, bps | 128 000 |
| | AMD PDU header, bit | 16 |
| MAC | MAC header, bit | 0 |
| | MAC multiplexing | N/A |

| | | | |
|---|---|---|---|
| Layer 1 | TrCH type | | DCH |
| | TB sizes, bit | | 336 |
| | TFS | TF0, bits | 0x336 |
| | | TF1, bits | 1x336 |
| | | TF2, bits | 2x336 |
| | | TF3, bits | 4x336 |
| | | TF4, bits | 8x336 |
| | TTI, ms | | 20 |
| | Coding type | | TC |
| | CRC, bit | | 16 |
| | Max number of bits/TTI after channel coding | | 8 460 |
| | Uplink: Max number of bits/radio frame before rate matching | | 4 230 |
| | RM attribute | | 120 to 160 |

6.10.2.4.1.28.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.1.1.1.

6.10.2.4.1.28.1.1.3 TFCS

| | |
|---|---|
| TFCS size | 10 |
| TFCS | (128 kbps RAB, DCCH)=<br>(TF0, TF0), (TF1, TF0), (TF2, TF0), (TF3, TF0), (TF4, TF0),<br>(TF0, TF1), (TF1, TF1), (TF2, TF1), (TF3, TF1), (TF4, TF1) |

6.10.2.4.1.28.1.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Uplink | Min spreading factor | 8 |
| | Max number of DPDCH data bits/radio frame | 4 800 |
| | Puncturing Limit | 0.96 |

6.10.2.4.1.28.2 Downlink

See clause 6.10.2.4.1.27.2.

6.10.2.4.1.29 Interactive or background / UL:64 DL:144 kbps / PS RAB + UL:3.4 DL: 3.4 kbps SRBs for DCCH

6.10.2.4.1.29.1 Uplink

See clause 6.10.2.4.1.26.1.

6.10.2.4.1.29.2 Downlink

6.10.2.4.1.29.2.1 Transport channel parameters

6.10.2.4.1.29.2.1.1 Transport channel parameters for Interactive or background / DL:144 kbps / PS RAB

| | | | |
|---|---|---|---|
| Higher layer | RAB/Signalling RB | | RAB |
| RLC | Logical channel type | | DTCH |
| | RLC mode | | AM |
| | Payload sizes, bit | | 320 |
| | Max data rate, bps | | 144 000 |
| | AMD PDU header, bit | | 16 |
| MAC | MAC header, bit | | 0 |
| | MAC multiplexing | | N/A |
| Layer 1 | TrCH type | | DCH |
| | TB sizes, bit | | 336 |
| | TFS | TF0, bits | 0x336 |
| | | TF1, bits | 1x336 |
| | | TF2, bits | 2x336 |
| | | TF3, bits | 4x336 |
| | | TF4, bits | 8x336 |

| | | | |
|---|---|---|---|
| | | TF5, bits | 9x336 |
| | TTI, ms | | 20 |
| | Coding type | | TC |
| | CRC, bit | | 16 |
| | Max number of bits/TTI after channel coding | | 9 516 |
| | RM attribute | | 140 to 180 |

6.10.2.4.1.29.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.2.1.1.

6.10.2.4.1.29.2.1.3 TFCS

| | |
|---|---|
| TFCS size | 12 |
| TFCS | (144 kbps RAB, DCCH)=<br>(TF0, TF0), (TF1, TF0), (TF2, TF0), (TF3, TF0), (TF4, TF0), (TF5, TF0)<br>(TF0, TF1), (TF1, TF1), (TF2, TF1), (TF3, TF1), (TF4, TF1), (TF5, TF1) |

6.10.2.4.1.29.2.2 Physical channel parameters

| | | | |
|---|---|---|---|
| DPCH Downlink | DTX position | | Flexible |
| | Spreading factor | | 16 |
| | DPCCH | Number of TFCI bits/slot | 8 |
| | | Number of TPC bits/slot | 8 |
| | | Number of Pilot bits/slot | 16 |
| | DPDCH | Number of data bits/slot | 288 |
| | | Number of data bits/frame | 4 320 |

6.10.2.4.1.30 Interactive or background / UL:144 DL:144 kbps / PS RAB + UL:3.4 DL: 3.4 kbps SRBs for DCCH

6.10.2.4.1.30.1 Uplink

6.10.2.4.1.30.1.1 Transport channel parameters

6.10.2.4.1.30.1.1.1 Transport channel parameters for Interactive or background / UL:144 kbps / PS RAB

| | | | |
|---|---|---|---|
| Higher layer | RAB/Signalling RB | | RAB |
| RLC | Logical channel type | | DTCH |
| | RLC mode | | AM |
| | Payload sizes, bit | | 320 |
| | Max data rate, bps | | 144 000 |
| | AMD PDU header, bit | | 16 |
| MAC | MAC header, bit | | 0 |
| | MAC multiplexing | | N/A |
| Layer 1 | TrCH type | | DCH |
| | TB sizes, bit | | 336 |
| | TFS | TF0, bits | 0x336 |
| | | TF1, bits | 1x336 |
| | | TF2, bits | 2x336 |
| | | TF3, bits | 4x336 |
| | | TF4, bits | 8x336 |
| | | TF5, bits | 9x336 |
| | TTI, ms | | 20 |
| | Coding type | | TC |
| | CRC, bit | | 16 |
| | Max number of bits/TTI after channel coding | | 9 516 |
| | Uplink: Max number of bits/radio frame before rate matching | | 4 758 |
| | RM attribute | | 140 to 180 |

6.10.2.4.1.30.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.1.1.1.

6.10.2.4.1.30.1.1.3 TFCS

| | |
|---|---|
| TFCS size | 12 |
| TFCS | (144 kbps RAB, DCCH)=<br>(TF0, TF0), (TF1, TF0), (TF2, TF0), (TF3, TF0), (TF4, TF0), (TF5, TF0)<br>(TF0, TF1), (TF1, TF1), (TF2, TF1), (TF3, TF1), (TF4, TF1), (TF5, TF1) |

6.10.2.4.1.30.1.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Uplink | Min spreading factor | 8 |
| | Max number of DPDCH data bits/radio frame | 4 800 |
| | Puncturing Limit | 0.84 |

6.10.2.4.1.30.2 Downlink

See clause 6.10.2.4.1.29.2.

6.10.2.4.1.31 Interactive or background / UL:64 DL:256 kbps / PS RAB + UL:3.4 DL: 3.4 kbps SRBs for DCCH

6.10.2.4.1.31.1 Uplink

See clause 6.10.2.4.1.26.1.

6.10.2.4.1.31.2 Downlink

6.10.2.4.1.31.2.1 Transport channel parameters

6.10.2.4.1.31.2.1.1 Transport channel parameters for Interactive or background / DL:256 kbps / PS RAB

| | | | |
|---|---|---|---|
| Higher layer | RAB/Signalling RB | | RAB |
| RLC | Logical channel type | | DTCH |
| | RLC mode | | AM |
| | Payload sizes, bit | | 320 |
| | Max data rate, bps | | 256 000 |
| | AMD PDU header, bit | | 16 |
| MAC | MAC header, bit | | 0 |
| | MAC multiplexing | | N/A |
| Layer 1 | TrCH type | | DCH |
| | TB sizes, bit | | 336 |
| | TFS | TF0, bits | 0x336 |
| | | TF1, bits | 1x336 |
| | | TF2, bits | 2x336 |
| | | TF3, bits | 4x336 |
| | | TF4, bits | 8x336 |
| | | TF5, bits | N/A (alt. 12x336) |
| | | TF6, bits | N/A (alt. 16x336) |
| | TTI, ms | | 10 (alt. 20) |
| | Coding type | | TC |
| | CRC, bit | | 16 |
| | Max number of bits/TTI after channel coding | | 8 460 (alt. 16 920) |
| | RM attribute | | 135 to 175 |

6.10.2.4.1.31.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.2.1.1.

6.10.2.4.1.31.2.1.3 TFCS

| | |
|---|---|
| TFCS size | 10 (alt.14) |
| TFCS | (256 kbps RAB, DCCH)=<br>(TF0, TF0), (TF1, TF0), (TF2, TF0), (TF3, TF0), (TF4, TF0),<br>(TF0, TF1), (TF1, TF1), (TF2, TF1), (TF3, TF1), (TF4, TF1)<br>(alt. (TF0, TF0), (TF1, TF0), (TF2, TF0), (TF3, TF0), (TF4, TF0), (TF5, TF0), (TF6, TF0) |

| | |
|---|---|
| | (TF0, TF1), (TF1, TF1), (TF2, TF1), (TF3, TF1), (TF4, TF1), (TF5, TF1), (TF6, TF1)) |

6.10.2.4.1.31.2.2 Physical channel parameters

| | | | |
|---|---|---|---|
| DPCH Downlink | DTX position | | Flexible |
| | Spreading factor | | 8 |
| | Number od DPDCH | | 1 |
| | DPCCH | Number of TFCI bits/slot | 8 |
| | | Number of TPC bits/slot | 8 |
| | | Number of Pilot bits/slot | 16 |
| | DPDCH | Number of data bits/slot | 608 |
| | | Number of data bits/frame | 9 120 |

6.10.2.4.1.32 Interactive or background / UL:64 DL:384 kbps / PS RAB + UL:3.4 DL: 3.4 kbps SRBs for DCCH

6.10.2.4.1.32.1 Uplink

See clause 6.10.2.4.1.26.1.

6.10.2.4.1.32.2 Downlink

6.10.2.4.1.32.2.1 Transport channel parameters

6.10.2.4.1.32.2.1.1 Transport channel parameters for Interactive or background / DL:384 kbps / PS RAB

| | | | |
|---|---|---|---|
| Higher layer | RAB/Signalling RB | | RAB |
| RLC | Logical channel type | | DTCH |
| | RLC mode | | AM |
| | Payload sizes, bit | | 320 |
| | Max data rate, bps | | 384 000 |
| | AMD PDU header, bit | | 16 |
| MAC | MAC header, bit | | 0 |
| | MAC multiplexing | | N/A |
| Layer 1 | TrCH type | | DCH |
| | TB sizes, bit | | 336 |
| | TFS | TF0, bits | 0x336 |
| | | TF1, bits | 1x336 |
| | | TF2, bits | 2x336 |
| | | TF3, bits | 4x336 |
| | | TF4, bits | 8x336 |
| | | TF5, bits | 12x336 |
| | | TF6, bits | N/A (alt. 16x336) |
| | | TF7, bits | N/A (alt. 20x336) |
| | | TF8, bits | N/A (alt. 24x336) |
| | TTI, ms | | 10 (alt. 20) |
| | Coding type | | TC |
| | CRC, bit | | 16 |
| | Max number of bits/TTI after channel coding | | 12 684 (alt. 25 368) |
| | RM attribute | | 110 to 150 |

6.10.2.4.1.32.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.2.1.1.

6.10.2.4.1.32.2.1.3 TFCS

| | |
|---|---|
| TFCS size | 12 (alt.18) |
| TFCS | (384 kbps RAB, DCCH)=<br>(TF0, TF0), (TF1, TF0), (TF2, TF0), (TF3, TF0), (TF4, TF0), (TF5, TF0)<br>(TF0, TF1), (TF1, TF1), (TF2, TF1), (TF3, TF1), (TF4, TF1), (TF5, TF1)<br>(alt. (TF0, TF0), (TF1, TF0), (TF2, TF0), (TF3, TF0), (TF4, TF0), (TF5, TF0), (TF6, TF0), (TF7, TF0), (TF8, TF0),<br>(TF0, TF1), (TF1, TF1), (TF2, TF1), (TF3, TF1), (TF4, TF1), (TF5, TF1), (TF6, TF1), (TF7, TF1), |

| | |
|---|---|
| | (TF8, TF1)) |

6.10.2.4.1.32.2.2 Physical channel parameters

| | | | |
|---|---|---|---|
| DPCH Downlink | DTX position | | Flexible |
| | Spreading factor | | 8 |
| | Number of DPDCH | | 1 |
| | DPCCH | Number of TFCI bits/slot | 8 |
| | | Number of TPC bits/slot | 8 |
| | | Number of Pilot bits/slot | 16 |
| | DPDCH | Number of data bits/slot | 608 |
| | | Number of data bits/frame | 9 120 |

6.10.2.4.1.33 Interactive or background / UL:128 DL:384 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.2.4.1.33.1 Uplink

See clause 6.10.2.4.1.28.1.

6.10.2.4.1.33.2 Downlink

See clause 6.10.2.4.1.32.2.

6.10.2.4.1.34 Interactive or background / UL:384 DL:384 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.2.4.1.34.1 Uplink

6.10.2.4.1.34.1.1 Transport channel parameters

6.10.2.4.1.34.1.1.1 Transport channel parameters for Interactive or background / UL:384 kbps / PS RAB

| | | | |
|---|---|---|---|
| Higher layer | RAB/Signalling RB | | RAB |
| RLC | Logical channel type | | DTCH |
| | RLC mode | | AM |
| | Payload sizes, bit | | 320 |
| | Max data rate, bps | | 384 000 |
| | AMD PDU header, bit | | 16 |
| MAC | MAC header, bit | | 0 |
| | MAC multiplexing | | N/A |
| Layer 1 | TrCH type | | DCH |
| | TB sizes, bit | | 336 |
| | TFS | TF0, bits | 0x336 |
| | | TF1, bits | 1x336 |
| | | TF2, bits | 2x336 |
| | | TF3, bits | 4x336 |
| | | TF4, bits | 8x336 |
| | | TF5, bits | 12x336 |
| | | TF6, bits | 16x336(alt. N/A) |
| | | TF7, bits | 20x336(alt. N/A) |
| | | TF8, bits | 24x336 (alt. N/A) |
| | TTI, ms | | 20 (alt. 10) |
| | Coding type | | TC |
| | CRC, bit | | 16 |
| | Max number of bits/TTI after channel coding | | 25 368 |
| | Uplink: Max number of bits/radio frame before rate matching | | 12 684 |
| | RM attribute | | 110-180 |

6.10.2.4.1.34.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.1.1.1.

6.10.2.4.1.34.1.1.3 TFCS

| | |
|---|---|
| TFCS size | 18 (alt.12) |
| TFCS | (384 kbps RAB, DCCH)=<br>(TF0, TF0), (TF1, TF0), (TF2, TF0), (TF3, TF0), (TF4, TF0), (TF5, TF0), (TF6, TF0), (TF7, TF0), (TF8, TF0),<br>(TF0, TF1), (TF1, TF1), (TF2, TF1), (TF3, TF1), (TF4, TF1), (TF5, TF1), (TF6, TF1), (TF7, TF1), (TF8, TF1)<br>(alt. (TF0, TF0), (TF1, TF0), (TF2, TF0), (TF3, TF0), (TF4, TF0), (TF5, TF0)<br>(TF0, TF1), (TF1, TF1), (TF2, TF1), (TF3, TF1), (TF4, TF1), (TF5, TF1)) |

6.10.2.4.1.34.1.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Uplink | Min spreading factor | 4 |
| | Max number of DPDCH data bits/radio frame | 9 600 |
| | Number of DPDCH | 1 |
| | Puncturing Limit | 0.64 |

6.10.2.4.1.34.2 Downlink

See clause 6.10.2.4.1.32.2.

6.10.2.4.1.35 Interactive or background / UL:64 DL:2 048 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.2.4.1.35.1 Uplink

See clause 6.10.2.4.1.26.1.

6.10.2.4.1.35.2 Downlink

6.10.2.4.1.35.2.1 Transport channel parameters

6.10.2.4.1.35.2.1.1 Transport channel parameters for Interactive or background / DL:2 048 kbps / PS RAB

| | | | |
|---|---|---|---|
| Higher layer | RAB/Signalling RB | | RAB |
| RLC | Logical channel type | | DTCH |
| | RLC mode | | AM |
| | Payload sizes, bit | | 640 |
| | Max data rate, bps | | 2 048 000 |
| | AMD PDU header, bit | | 16 |
| MAC | MAC header, bit | | 0 |
| | MAC multiplexing | | N/A |
| Layer 1 | TrCH type | | DCH |
| | TB sizes, bit | | 656 |
| | TFS | TF0, bits | 0x656 |
| | | TF1, bits | 1x656 |
| | | TF2, bits | 2x656 |
| | | TF3, bits | 4x656 |
| | | TF4, bits | 8x656 |
| | | TF5, bits | 12x656 |
| | | TF6, bits | 16x656 |
| | | TF7, bits | 20x656 |
| | | TF8, bits | 24x656 |
| | | TF9, bits | 28x656 |
| | | TF10, bits | 32x656 |
| | | TF11, bits | N/A (alt. 36x656) |
| | | TF12, bits | N/A (alt. 40x656) |
| | | TF13, bits | N/A (alt. 44x656) |
| | | TF14, bits | N/A (alt. 48x656) |
| | | TF15, bits | N/A (alt. 52x656) |
| | | TF16, bits | N/A (alt. 56x656) |
| | | TF17, bits | N/A (alt. 60x656) |
| | | TF18, bits | N/A (alt. 64x656) |
| | TTI, ms | | 10 (alt. 20) |
| | Coding type | | TC |
| | CRC, bit | | 16 |

| | | |
|---|---|---|
| | Max number of bits/TTI after channel coding | 64 575 (alt. 129 141) |
| | RM attribute | 130 to 170 |

6.10.2.4.1.35.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.2.1.1.

6.10.2.4.1.35.2.1.3 TFCS

| | |
|---|---|
| TFCS size | 22 (alt.38) |
| TFCS | (2 048 kbps RAB, DCCH)=<br>(TF0, TF0), (TF1, TF0), (TF2, TF0), (TF3, TF0), (TF4, TF0), (TF5, TF0), (TF6, TF0), (TF7, TF0),<br>(TF8, TF0), (TF9, TF0), (TF10, TF0),<br>(TF0, TF1), (TF1, TF1), (TF2, TF1), (TF3, TF1), (TF4, TF1), (TF5, TF1), (TF6, TF1), (TF7, TF1),<br>(TF8, TF1), (TF9, TF1), (TF10, TF1)<br>(alt. TF0, TF0), (TF1, TF0), (TF2, TF0), (TF3, TF0), (TF4, TF0), (TF5, TF0), (TF6, TF0), (TF7, TF0),<br>(TF8, TF0), (TF9, TF0), (TF10, TF0),(TF11, TF0), (TF12, TF0), (TF13, TF0), (TF14, TF0), (TF15,<br>TF0), (TF16, TF0), (TF17, TF0), (TF18, TF0),<br>(TF0, TF1), (TF1, TF1), (TF2, TF1), (TF3, TF1), (TF4, TF1), (TF5, TF1), (TF6, TF1), (TF7, TF1),<br>(TF8, TF1), (TF9, TF1), (TF10, TF1),(TF11, TF0), (TF12, TF0), (TF13, TF0), (TF14, TF0),<br>(TF15, TF0), (TF16, TF0), (TF17, TF0), (TF18, TF0)) |

6.10.2.4.1.35.2.2 Physical channel parameters

| | | | |
|---|---|---|---|
| DPCH Downlink | DTX position | | Flexible |
| | Spreading factor | | 4 |
| | Number of DPCH | | 3 |
| | DPCCH | Number of TFCI bits/slot | 8 |
| | | Number of TPC bits/slot | 8 |
| | | Number of Pilot bits/slot | 16 |
| | DPDCH | Number of data bits/slot | 1 248 |
| | | Number of data bits/frame | 18 720 |

6.10.2.4.1.36 Void

6.10.2.4.1.37 Void

6.10.2.4.1.38 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:32 DL:8 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.2.4.1.38.1 Uplink

6.10.2.4.1.38.1.1 Transport channel parameters

6.10.2.4.1.38.1.1.1 Transport channel parameters for Conversational / speech / UL:12.2 kbps / CS RAB

See clause 6.10.2.4.1.4.1.1.1.

6.10.2.4.1.38.1.1.2 Transport channel parameters for Interactive or background / UL:32 kbps / PS RAB

See clause 6.10.2.4.1.23.1.1.1.

6.10.2.4.1.38.1.1.3 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.1.1.1.

6.10.2.4.1.38.1.1.4 TFCS

| | |
|---|---|
| TFCS size | 18 (alt. 12) |
| TFCS | (RAB subflow#1, RAB subflow#2, RAB subflow#3, 32kbps RAB , DCCH)=<br>(TF0, TF0, TF0, TF0, TF0), (TF1, TF0, TF0, TF0, TF0), (TF2, TF1, TF1, TF0, TF0),<br>(TF0, TF0, TF0, TF1, TF0), (TF1, TF0, TF0, TF1, TF0), (TF2, TF1, TF1, TF1, TF0),<br>(TF0, TF0, TF0, TF2, TF0), (TF1, TF0, TF0, TF2, TF0), (TF2, TF1, TF1, TF2, TF0),<br>(TF0, TF0, TF0, TF0, TF1), (TF1, TF0, TF0, TF0, TF1), (TF2, TF1, TF1, TF0, TF1),<br>(TF0, TF0, TF0, TF1, TF1), (TF1, TF0, TF0, TF1, TF1), (TF2, TF1, TF1, TF1, TF1),<br>(TF0, TF0, TF0, TF2, TF1), (TF1, TF0, TF0, TF2, TF1), (TF2, TF1, TF1, TF2, TF1)<br>(alt. (TF0, TF0, TF0, TF0, TF0), (TF1, TF0, TF0, TF0, TF0), (TF2, TF1, TF1, TF0, TF0),<br>(TF0, TF0, TF0, TF1, TF0), (TF1, TF0, TF0, TF1, TF0), (TF2, TF1, TF1, TF1, TF0),<br>(TF0, TF0, TF0, TF0, TF1), (TF1, TF0, TF0, TF0, TF1), (TF2, TF1, TF1, TF0, TF1),<br>(TF0, TF0, TF0, TF1, TF1), (TF1, TF0, TF0, TF1, TF1), (TF2, TF1, TF1, TF1, TF1)) |

6.10.2.4.1.38.1.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Uplink | Min spreading factor | 16 |
| | Max number of DPDCH data bits/radio frame | 2 400 |
| | Puncturing Limit | 0.96 |

6.10.2.4.1.38.2 Downlink

6.10.2.4.1.38.2.1 Transport channel parameters

6.10.2.4.1.38.2.1.1 Transport channel parameters for Conversational / speech / DL:12.2 kbps / CS RAB

See clause 6.10.2.4.1.4.2.1.1.

6.10.2.4.1.38.2.1.2 Transport channel parameters for Interactive or background / DL:8 kbps / PS RAB

See clause 6.10.2.4.1.23.2.1.1.

6.10.2.4.1.38.2.1.3 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.2.1.

6.10.2.4.1.38.2.1.4 TFCS

| | |
|---|---|
| TFCS size | 12 |
| TFCS | (RAB subflow#1, RAB subflow#2, RAB subflow#3,8kbps RAB, DCCH)=<br>(TF0, TF0, TF0, TF0, TF0), (TF1, TF0, TF0, TF0, TF0), (TF2, TF1, TF1, TF0, TF0),<br>(TF0, TF0, TF0, TF1, TF0), (TF1, TF0, TF0, TF1, TF0), (TF2, TF1, TF1, TF1, TF0),<br>(TF0, TF0, TF0, TF0, TF1), (TF1, TF0, TF0, TF0, TF1), (TF2, TF1, TF1, TF0, TF1),<br>(TF0, TF0, TF0, TF1, TF1), (TF1, TF0, TF0, TF1, TF1), (TF2, TF1, TF1, TF1, TF1) |

6.10.2.4.1.38.2.2 Physical channel parameters

| | | | |
|---|---|---|---|
| DPCH Downlink | DTX position | | Flexible |
| | Spreading factor | | 64 |
| | DPCCH | Number of TFCI bits/slot | 8 |
| | | Number of TPC bits/slot | 4 |
| | | Number of Pilot bits/slot | 8 |
| | DPDCH | Number of data bits/slot | 60 |
| | | Number of data bits/frame | 900 |

6.10.2.4.1.38a Conversational / speech / 12.2 kbps / CS RAB + Interactive or background / UL:0 DL:0 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.2.4.1.38a.1 Uplink

6.10.2.4.1.38a.1.1 Transport channel parameters

6.10.2.4.1.38a.1.1.1 Transport channel parameters for Conversational / speech / UL:12.2 kbps / CS RAB

See clause 6.10.2.4.1.4.1.1.1.

#### 6.10.2.4.1.38a.1.1.2 Transport channel parameters for Interactive or background / UL:0 kbps / PS RAB

| | | | |
|---|---|---|---|
| Higher layer | RAB/Signalling RB | | RAB |
| RLC | Logical channel type | | DTCH |
| | RLC mode | | AM |
| | Payload sizes, bit | | 320 |
| | Max data rate, bps | | 0 |
| | AMD PDU header, bit | | 16 |
| MAC | MAC header, bit | | 0 |
| | MAC multiplexing | | N/A |
| Layer 1 | TrCH type | | DCH |
| | TB sizes, bit | | 336 |
| | TFS | TF0, bits | 0x336 |
| | TTI, ms | | 20 |
| | Coding type | | CC |
| | CRC, bit | | 16 |
| | Max number of bits/TTI after channel coding | | 0 |
| | Uplink: Max number of bits/radio frame before rate matching | | 0 |
| | RM attribute | | 130 to 170 |

#### 6.10.2.4.1.38a.1.1.3 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.1.1.1.

#### 6.10.2.4.1.38a.1.1.4 TFCS

| | |
|---|---|
| TFCS size | 6 |
| TFCS | (RAB subflow#1, RAB subflow#2, RAB subflow#3, 0kbps RAB, DCCH)=<br>(TF0,TF0,TF0,TF0,TF0), (TF1,TF0,TF0,TF0,TF0), (TF2,TF1,TF1,TF0,TF0),<br>(TF0,TF0,TF0,TF0,TF1), (TF1,TF0,TF0,TF0,TF1), (TF2,TF1,TF1,TF0,TF1) |

### 6.10.2.4.1.38a.1.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Uplink | Min spreading factor | 64 |
| | Max number of DPDCH data bits/radio frame | 600 |
| | Puncturing Limit | 0.64 |

## 6.10.2.4.1.38a.2 Downlink

### 6.10.2.4.1.38a.2.1 Transport channel parameters

#### 6.10.2.4.1.38a.2.1.1 Transport channel parameters for Conversational / speech / DL:12.2 kbps / CS RAB

See clause 6.10.2.4.1.4.2.1.1.

#### 6.10.2.4.1.38a.2.1.2 Transport channel parameters for Interactive or background / DL:0 kbps / PS RAB

| | | | |
|---|---|---|---|
| Higher layer | RAB/Signalling RB | | RAB |
| RLC | Logical channel type | | DTCH |
| | RLC mode | | AM |
| | Payload sizes, bit | | 320 |
| | Max data rate, bps | | 0 |
| | AMD PDU header, bit | | 16 |
| MAC | MAC header, bit | | 0 |
| | MAC multiplexing | | N/A |
| Layer 1 | TrCH type | | DCH |
| | TB sizes, bit | | 336 |
| | TFS | TF0, bits | 0x336 |
| | TTI, ms | | 20 |
| | Coding type | | CC |
| | CRC, bit | | 16 |
| | Max number of bits/TTI after channel coding | | 0 |

| | | |
|---|---|---|
| | RM attribute | 130 to 170 |

6.10.2.4.1.38a.2.1.3 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.2.1.1

6.10.2.4.1.38a.2.1.4 TFCS

| | |
|---|---|
| TFCS size | 6 |
| TFCS | (RAB subflow#1, RAB subflow#2, RAB subflow#3, 0kbps RAB, DCCH)=<br>(TF0,TF0,TF0,TF0,TF0), (TF1,TF0,TF0,TF0,TF0), (TF2,TF1,TF1,TF0,TF0),<br>(TF0,TF0,TF0,TF0,TF1), (TF1,TF0,TF0,TF0,TF1), (TF2,TF1,TF1,TF0,TF1) |

6.10.2.4.1.38a.2.2 Physical channel parameters

| | | | |
|---|---|---|---|
| DPCH Downlink | DTX position | | Fixed |
| | Spreading factor | | 128 |
| | DPCCH | Number of TFCI bits/slot | 0 |
| | | Number of TPC bits/slot | 2 |
| | | Number of Pilot bits/slot | 4 |
| | DPDCH | Number of data bits/slot | 34 |
| | | Number of data bits/frame | 510 |

6.10.2.4.1.38b Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:8 DL:8 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.2.4.1.38b.1 Uplink

6.10.2.4.1.38b.1.1 Transport channel parameters

6.10.2.4.1.38b.1.1.1 Transport channel parameters for Conversational / speech / UL:12.2 kbps / CS RAB

See clause 6.10.2.4.1.4.1.1.1.

6.10.2.4.1.38b.1.1.2 Transport channel parameters for Interactive or background / UL:8 kbps / PS RAB

| | | | |
|---|---|---|---|
| Higher layer | RAB/Signalling RB | | RAB |
| RLC | Logical channel type | | DTCH |
| | RLC mode | | AM |
| | Payload sizes, bit | | 320 |
| | Max data rate, bps | | 8 000 |
| | AMD PDU header, bit | | 16 |
| MAC | MAC header, bit | | 0 |
| | MAC multiplexing | | N/A |
| Layer 1 | TrCH type | | DCH |
| | TB sizes, bit | | 336 |
| | TFS | TF0, bits | 0x336 |
| | | TF1, bits | 1x336 |
| | TTI, ms | | 40 |
| | Coding type | | TC |
| | CRC, bit | | 16 |
| | Max number of bits/TTI after channel coding | | 1 068 |
| | Uplink: Max number of bits/radio frame before rate matching | | 267 |
| | RM attribute | | 135 to 175 |

6.10.2.4.1.38b.1.1.3 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.1.1.1.

6.10.2.4.1.38b.1.1.4 TFCS

| | |
|---|---|
| TFCS size | 12 |

| | |
|---|---|
| TFCS | (RAB subflow#1, RAB subflow#2, RAB subflow#3, 8 kbps RAB, DCCH)=<br>(TF0,TF0,TF0,TF0,TF0), (TF1,TF0,TF0,TF0,TF0), (TF2,TF1,TF1,TF0,TF0),<br>(TF0,TF0,TF0,TF1,TF0), (TF1,TF0,TF0,TF1,TF0), (TF2,TF1,TF1,TF1,TF0),<br>(TF0,TF0,TF0,TF0,TF1), (TF1,TF0,TF0,TF0,TF1), (TF2,TF1,TF1,TF0,TF1),<br>(TF0,TF0,TF0,TF1,TF1), (TF1,TF0,TF0,TF1,TF1), (TF2,TF1,TF1,TF1,TF1) |

6.10.2.4.1.38b.1.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Uplink | Min spreading factor | 32 |
| | Max number of DPDCH data bits/radio frame | 1 200 |
| | Puncturing Limit | 1.0 |

6.10.2.4.1.38b.2 Downlink

6.10.2.4.1.38b.2.1 Transport channel parameters

6.10.2.4.1.38b.2.1.1 Transport channel parameters for Conversational / speech / DL:12.2 kbps / CS RAB

See clause 6.10.2.4.1.4.2.1.1.

6.10.2.4.1.38b.2.1.2 Transport channel parameters for Interactive or background / DL:8 kbps / PS RAB

| | | | |
|---|---|---|---|
| Higher layer | RAB/Signalling RB | | RAB |
| RLC | Logical channel type | | DTCH |
| | RLC mode | | AM |
| | Payload sizes, bit | | 320 |
| | Max data rate, bps | | 8 000 |
| | AMD PDU header, bit | | 16 |
| MAC | MAC header, bit | | 0 |
| | MAC multiplexing | | N/A |
| Layer 1 | TrCH type | | DCH |
| | TB sizes, bit | | 336 |
| | TFS | TF0, bits | 0x336 |
| | | TF1, bits | 1x336 |
| | TTI, ms | | 40 |
| | Coding type | | TC |
| | CRC, bit | | 16 |
| | Max number of bits/TTI after channel coding | | 1 068 |
| | RM attribute | | 135 to 175 |

6.10.2.4.1.38b.2.1.3 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.2.1.1.

6.10.2.4.1.38b.2.1.4 TFCS

| | |
|---|---|
| TFCS size | 12 |
| TFCS | (RAB subflow#1, RAB subflow#2, RAB subflow#3, 8 kbps RAB, DCCH)=<br>(TF0,TF0,TF0,TF0,TF0), (TF1,TF0,TF0,TF0,TF0), (TF2,TF1,TF1,TF0,TF0),<br>(TF0,TF0,TF0,TF1,TF0), (TF1,TF0,TF0,TF1,TF0), (TF2,TF1,TF1,TF1,TF0),<br>(TF0,TF0,TF0,TF0,TF1), (TF1,TF0,TF0,TF0,TF1), (TF2,TF1,TF1,TF0,TF1),<br>(TF0,TF0,TF0,TF1,TF1), (TF1,TF0,TF0,TF1,TF1), (TF2,TF1,TF1,TF1,TF1) |

6.10.2.4.1.38b.2.2 Physical channel parameters

| | | | |
|---|---|---|---|
| DPCH Downlink | DTX position | | Flexible |
| | Spreading factor | | 64 |
| | DPCCH | Number of TFCI bits/slot | 8 |
| | | Number of TPC bits/slot | 4 |
| | | Number of Pilot bits/slot | 8 |
| | DPDCH | Number of data bits/slot | 60 |
| | | Number of data bits/frame | 900 |

6.10.2.4.1.38c Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:32 DL:32 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.2.4.1.38c.1 Uplink

6.10.2.4.1.38c.1.1 Transport channel parameters

6.10.2.4.1.38c.1.1.1 Transport channel parameters for Conversational / speech / UL:12.2 kbps / CS RAB

See clause 6.10.2.4.1.4.1.1.1.

6.10.2.4.1.38c.1.1.2 Transport channel parameters for Interactive or background / UL:32 kbps / PS RAB

See clause 6.2.4.1.23c.1.1.1.

6.10.2.4.1.38c.1.1.3 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.1.1.1.

6.10.2.4.1.38c.1.1.4 TFCS

| | |
|---|---|
| TFCS size | 30 |
| TFCS | (RAB subflow#1, RAB subflow#2, RAB subflow#3, 32 kbps RAB, DCCH)=<br>(TF0,TF0,TF0,TF0,TF0), (TF1,TF0,TF0,TF0,TF0), (TF2,TF1,TF1,TF0,TF0),<br>(TF0,TF0,TF0,TF1,TF0), (TF1,TF0,TF0,TF1,TF0), (TF2,TF1,TF1,TF1,TF0),<br>(TF0,TF0,TF0,TF2,TF0), (TF1,TF0,TF0,TF2,TF0), (TF2,TF1,TF1,TF2,TF0),<br>(TF0,TF0,TF0,TF3,TF0), (TF1,TF0,TF0,TF3,TF0), (TF2,TF1,TF1,TF3,TF0),<br>(TF0,TF0,TF0,TF4,TF0), (TF1,TF0,TF0,TF4,TF0), (TF2,TF1,TF1,TF4,TF0),<br>(TF0,TF0,TF0,TF0,TF1), (TF1,TF0,TF0,TF0,TF1), (TF2,TF1,TF1,TF0,TF1),<br>(TF0,TF0,TF0,TF1,TF1), (TF1,TF0,TF0,TF1,TF1), (TF2,TF1,TF1,TF1,TF1),<br>(TF0,TF0,TF0,TF2,TF1), (TF1,TF0,TF0,TF2,TF1), (TF2,TF1,TF1,TF2,TF1),<br>(TF0,TF0,TF0,TF3,TF1), (TF1,TF0,TF0,TF3,TF1), (TF2,TF1,TF1,TF3,TF1),<br>(TF0,TF0,TF0,TF4,TF1), (TF1,TF0,TF0,TF4,TF1), (TF2,TF1,TF1,TF4,TF1) |

6.10.2.4.1.38c.1.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Uplink | Min spreading factor | 16 |
| | Max number of DPDCH data bits/radio frame | 2 400 |
| | Puncturing Limit | 1.0 |

6.10.2.4.1.38c.2 Downlink

6.10.2.4.1.38c.2.1 Transport channel parameters

6.10.2.4.1.38c.2.1.1 Transport channel parameters for Conversational / speech / DL:12.2 kbps / CS RAB

See clause 6.10.2.4.1.4.2.1.1.

6.10.2.4.1.38c.2.1.2 Transport channel parameters for Interactive or background / DL:32 kbps / PS RAB

See clause 6.2.4.1.23c.2.1.1.

6.10.2.4.1.38c.2.1.3 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.2.1.1.

6.10.2.4.1.38c.2.1.4 TFCS

| | |
|---|---|
| TFCS size | 30 |
| TFCS | (RAB subflow#1, RAB subflow#2, RAB subflow#3, 32 kbps RAB, DCCH)=<br>(TF0,TF0,TF0,TF0,TF0), (TF1,TF0,TF0,TF0,TF0), (TF2,TF1,TF1,TF0,TF0),<br>(TF0,TF0,TF0,TF1,TF0), (TF1,TF0,TF0,TF1,TF0), (TF2,TF1,TF1,TF1,TF0),<br>(TF0,TF0,TF0,TF2,TF0), (TF1,TF0,TF0,TF2,TF0), (TF2,TF1,TF1,TF2,TF0),<br>(TF0,TF0,TF0,TF3,TF0), (TF1,TF0,TF0,TF3,TF0), (TF2,TF1,TF1,TF3,TF0), |

| | |
|---|---|
| | (TF0,TF0,TF0,TF4,TF0), (TF1,TF0,TF0,TF4,TF0), (TF2,TF1,TF1,TF4,TF0),<br>(TF0,TF0,TF0,TF0,TF1), (TF1,TF0,TF0,TF0,TF1), (TF2,TF1,TF1,TF0,TF1),<br>(TF0,TF0,TF0,TF1,TF1), (TF1,TF0,TF0,TF1,TF1), (TF2,TF1,TF1,TF1,TF1),<br>(TF0,TF0,TF0,TF2,TF1), (TF1,TF0,TF0,TF2,TF1), (TF2,TF1,TF1,TF2,TF1),<br>(TF0,TF0,TF0,TF3,TF1), (TF1,TF0,TF0,TF3,TF1), (TF2,TF1,TF1,TF3,TF1),<br>(TF0,TF0,TF0,TF4,TF1), (TF1,TF0,TF0,TF4,TF1), (TF2,TF1,TF1,TF4,TF1) |

6.10.2.4.1.38c.2.2 Physical channel parameters

| | | | |
|---|---|---|---|
| DPCH Downlink | DTX position | | Flexible |
| | Spreading factor | | 32 |
| | DPCCH | Number of TFCI bits/slot | 8 |
| | | Number of TPC bits/slot | 4 |
| | | Number of Pilot bits/slot | 8 |
| | DPDCH | Number of data bits/slot | 140 |
| | | Number of data bits/frame | 2 100 |

6.10.2.4.1.38d Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:64 DL:64 kbps / PS RAB + Interactive or background / UL:64 DL:64 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.2.4.1.38d.1 Uplink

6.10.2.4.1.38d.1.1 Transport channel parameters

6.10.2.4.1.38d.1.1.1 Transport channel parameters for Conversational / speech / UL:12.2 kbps / CS RAB

See clause 6.10.2.4.1.4.1.1.1.

6.10.2.4.1.38d.1.1.2 Transport channel parameters for Interactive or background / UL:64 kbps / PS RAB + UL:64 kbps / PS RAB

| | | | | |
|---|---|---|---|---|
| Higher layer | RAB/Signalling RB | | RAB | RAB |
| RLC | Logical channel type | | DTCH | DTCH |
| | RLC mode | | AM | AM |
| | Payload sizes, bit | | 320 | 320 |
| | Max data rate, bps | | 64 000 | 64 000 |
| | AMD PDU header, bit | | 16 | 16 |
| MAC | MAC header, bit | | 4 | 4 |
| | MAC multiplexing | | 2 logical channel multiplexing | |
| Layer 1 | TrCH type | | DCH | |
| | TB sizes, bit | | 340 | |
| | TFS | TF0, bits | 0x340 | |
| | | TF1, bits | 1x340 | |
| | | TF2, bits | 2x340 | |
| | | TF3, bits | 3x340 | |
| | | TF4, bits | 4x340 | |
| | TTI, ms | | 20 | |
| | Coding type | | TC | |
| | CRC, bit | | 16 | |
| | Max number of bits/TTI after channel coding | | 4 284 | |
| | Uplink: Max number of bits/radio frame before rate matching | | 2 142 | |
| | RM attribute | | 130 to 170 | |

6.10.2.4.1.38d.1.1.3 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.1.1.1.

6.10.2.4.1.38d.1.1.4 TFCS

| | |
|---|---|
| TFCS size | 30 |
| TFCS | (RAB subflow#1, RAB subflow#2, RAB subflow#3, 64 kbps RAB + 64 kbps RAB, DCCH)=<br>(TF0,TF0,TF0,TF0,TF0), (TF1,TF0,TF0,TF0,TF0),(TF2,TF1,TF1,TF0,TF0),<br>(TF0,TF0,TF0,TF1,TF0), (TF1,TF0,TF0,TF1,TF0),(TF2,TF1,TF1,TF1,TF0),<br>(TF0,TF0,TF0,TF2,TF0), (TF1,TF0,TF0,TF2,TF0),(TF2,TF1,TF1,TF2,TF0),<br>(TF0,TF0,TF0,TF3,TF0), (TF1,TF0,TF0,TF3,TF0),(TF2,TF1,TF1,TF3,TF0),<br>(TF0,TF0,TF0,TF4,TF0), (TF1,TF0,TF0,TF4,TF0),(TF2,TF1,TF1,TF4,TF0),<br>(TF0,TF0,TF0,TF0,TF1), (TF1,TF0,TF0,TF0,TF1),(TF2,TF1,TF1,TF0,TF1),<br>(TF0,TF0,TF0,TF1,TF1), (TF1,TF0,TF0,TF1,TF1),(TF2,TF1,TF1,TF1,TF1),<br>(TF0,TF0,TF0,TF2,TF1), (TF1,TF0,TF0,TF2,TF1),(TF2,TF1,TF1,TF2,TF1),<br>(TF0,TF0,TF0,TF3,TF1), (TF1,TF0,TF0,TF3,TF1),(TF2,TF1,TF1,TF3,TF1),<br>(TF0,TF0,TF0,TF4,TF1), (TF1,TF0,TF0,TF4,TF1),(TF2,TF1,TF1,TF4,TF1) |

6.10.2.4.1.38d.1.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Uplink | Min spreading factor | 16 |
| | Max number of DPDCH data bits/radio frame | 2 400 |
| | Puncturing Limit | 0.76 |

6.10.2.4.1.38d.2 Downlink

6.10.2.4.1.38d.2.1 Transport channel parameters

6.10.2.4.1.38d.2.1.1 Transport channel parameters for Conversational / speech / DL:12.2 kbps / CS RAB

See clause 6.10.2.4.1.4.2.1.1.

6.10.2.4.1.38d.2.1.2 Transport channel parameters for Interactive or background / DL:64 kbps / PS RAB + DL:64 kbps / PS RAB

| | | | | |
|---|---|---|---|---|
| Higher layer | RAB/Signalling RB | | RAB | RAB |
| RLC | Logical channel type | | DTCH | DTCH |
| | RLC mode | | AM | AM |
| | Payload sizes, bit | | 320 | 320 |
| | Max data rate, bps | | 64 000 | 64 000 |
| | AMD PDU header, bit | | 16 | 16 |
| MAC | MAC header, bit | | 4 | 4 |
| | MAC multiplexing | | 2 logical channel multiplexing | |
| Layer 1 | TrCH type | | DCH | |
| | TB sizes, bit | | 340 | |
| | TFS | TF0, bits | 0x340 | |
| | | TF1, bits | 1x340 | |
| | | TF2, bits | 2x340 | |
| | | TF3, bits | 3x340 | |
| | | TF4, bits | 4x340 | |
| | TTI, ms | | 20 | |
| | Coding type | | TC | |
| | CRC, bit | | 16 | |
| | Max number of bits/TTI after channel coding | | 4 284 | |
| | RM attribute | | 130 to 170 | |

6.10.2.4.1.38d.2.1.3 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.2.1.1.

6.10.2.4.1.38d.2.1.4 TFCS

| | |
|---|---|
| TFCS size | 30 |
| TFCS | (RAB subflow#1, RAB subflow#2, RAB subflow#3, 64 kbps RAB + 64 kbps RAB, DCCH)=<br>(TF0,TF0,TF0,TF0,TF0), (TF1,TF0,TF0,TF0,TF0),(TF2,TF1,TF1,TF0,TF0),<br>(TF0,TF0,TF0,TF1,TF0), (TF1,TF0,TF0,TF1,TF0),(TF2,TF1,TF1,TF1,TF0),<br>(TF0,TF0,TF0,TF2,TF0), (TF1,TF0,TF0,TF2,TF0),(TF2,TF1,TF1,TF2,TF0), |

| | |
|---|---|
| | (TF0,TF0,TF0,TF3,TF0), (TF1,TF0,TF0,TF3,TF0),(TF2,TF1,TF1,TF3,TF0),<br>(TF0,TF0,TF0,TF4,TF0), (TF1,TF0,TF0,TF4,TF0),(TF2,TF1,TF1,TF4,TF0),<br>(TF0,TF0,TF0,TF0,TF1), (TF1,TF0,TF0,TF0,TF1),(TF2,TF1,TF1,TF0,TF1),<br>(TF0,TF0,TF0,TF1,TF1), (TF1,TF0,TF0,TF1,TF1),(TF2,TF1,TF1,TF1,TF1),<br>(TF0,TF0,TF0,TF2,TF1), (TF1,TF0,TF0,TF2,TF1),(TF2,TF1,TF1,TF2,TF1),<br>(TF0,TF0,TF0,TF3,TF1), (TF1,TF0,TF0,TF3,TF1),(TF2,TF1,TF1,TF3,TF1),<br>(TF0,TF0,TF0,TF4,TF1), (TF1,TF0,TF0,TF4,TF1),(TF2,TF1,TF1,TF4,TF1) |

6.10.2.4.1.38d.2.2 Physical channel parameters

| | | | |
|---|---|---|---|
| DPCH Downlink | DTX position | | Flexible |
| | Spreading factor | | 32 |
| | DPCCH | Number of TFCI bits/slot | 8 |
| | | Number of TPC bits/slot | 4 |
| | | Number of Pilot bits/slot | 8 |
| | DPDCH | Number of data bits/slot | 140 |
| | | Number of data bits/frame | 2 100 |

6.10.2.4.1.38e Conversational / speech / (12.2 7.95 5.9 4.75) kbps / CS RAB + Interactive or background / UL:0 DL:0 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.2.4.1.38e.1 Uplink

6.10.2.4.1.38e.1.1 Transport channel parameters

6.10.2.4.1.38e.1.1.1 Transport channel parameters for Conversational / speech / UL: (12.2 7.95 5.9 4.75) kbps / CS RAB

See clause 6.10.2.4.1.4a.1.1.1.

6.10.2.4.1.38e.1.1.2 Transport channel parameters for Interactive or background / UL:0 kbps / PS RAB

See clause 6.10.2.4.1.38a.1.1.2.

6.10.2.4.1.38e.1.1.3 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.1.1.1.

6.10.2.4.1.38e.1.1.4 TFCS

| | |
|---|---|
| TFCS size | 12 |
| TFCS | (RAB subflow#1, RAB subflow#2, RAB subflow#3, 0 kbps RAB, DCCH)=<br>(TF0,TF0,TF0,TF0,TF0), (TF1,TF0,TF0,TF0,TF0), (TF2,TF1,TF0,TF0,TF0),<br>(TF3,TF2,TF0,TF0,TF0), (TF4,TF3,TF0,TF0,TF0), (TF5,TF4,TF1,TF0,TF0),<br>(TF0,TF0,TF0,TF0,TF1), (TF1,TF0,TF0,TF0,TF1), (TF2,TF1,TF0,TF0,TF1),<br>(TF3,TF2,TF0,TF0,TF1), (TF4,TF3,TF0,TF0,TF1), (TF5,TF4,TF1,TF0,TF1) |

6.10.2.4.1.38e.1.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Uplink | Min spreading factor | 64 |
| | Max number of DPDCH data bits/radio frame | 600 |
| | Puncturing Limit | 0.84 |

6.10.2.4.1.38e.2 Downlink

6.10.2.4.1.38e.2.1 Transport channel parameters

6.10.2.4.1.38e.2.1.1 Transport channel parameters for Conversational / speech / DL: (12.2 7.95 5.9 4.75) kbps / CS RAB

See clause 6.10.2.4.1.4a.2.1.1.

6.10.2.4.1.38e.2.1.2 Transport channel parameters for Interactive or background / DL:0 kbps / PS RAB

See clause 6.10.2.4.1.38a.2.1.2

6.10.2.4.1.38e.2.1.3 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.2.1.1

6.10.2.4.1.38e.2.1.4 TFCS

| | |
|---|---|
| TFCS size | 12 |
| TFCS | (RAB subflow#1, RAB subflow#2, RAB subflow#3, 0 kbps RAB, DCCH)=<br>(TF0,TF0,TF0,TF0,TF0), (TF1,TF0,TF0,TF0,TF0), (TF2,TF1,TF0,TF0,TF0),<br>(TF3,TF2,TF0,TF0,TF0), (TF4,TF3,TF0,TF0,TF0), (TF5,TF4,TF1,TF0,TF0),<br>(TF0,TF0,TF0,TF0,TF1), (TF1,TF0,TF0,TF0,TF1), (TF2,TF1,TF0,TF0,TF1),<br>(TF3,TF2,TF0,TF0,TF1), (TF4,TF3,TF0,TF0,TF1), (TF5,TF4,TF1,TF0,TF1), |

6.10.2.4.1.38e.2.2 Physical channel parameters

| | | | |
|---|---|---|---|
| DPCH Downlink | DTX position | | Fixed |
| | Spreading factor | | 128 |
| | DPCCH | Number of TFCI bits/slot | 0 |
| | | Number of TPC bits/slot | 2 |
| | | Number of Pilot bits/slot | 4 |
| | DPDCH | Number of data bits/slot | 34 |
| | | Number of data bits/frame | 510 |

6.10.2.4.1.38f Conversational / speech / (12.2 7.95 5.9 4.75) kbps / CS RAB + Interactive or background / UL:8 DL:8 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.2.4.1.38f.1 Uplink

6.10.2.4.1.38f.1.1 Transport channel parameters

6.10.2.4.1.38f.1.1.1 Transport channel parameters for Conversational / speech / UL: (12.2 7.95 5.9 4.75) kbps / CS RAB

See clause 6.10.2.4.1.4a.1.1.1.

6.10.2.4.1.38f.1.1.2 Transport channel parameters for Interactive or background / UL:8 kbps / PS RAB

See clause 6.10.2.4.1.38b.1.1.2.

6.10.2.4.1.38f.1.1.3 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.1.1.1.

6.10.2.4.1.38f.1.1.4 TFCS

| | |
|---|---|
| TFCS size | 24 |
| TFCS | (RAB subflow#1, RAB subflow#2, RAB subflow#3, 8 kbps RAB, DCCH)=<br>(TF0,TF0,TF0,TF0,TF0), (TF1,TF0,TF0,TF0,TF0), (TF2,TF1,TF0,TF0,TF0),<br>(TF3,TF2,TF0,TF0,TF0), (TF4,TF3,TF0,TF0,TF0), (TF5,TF4,TF1,TF0,TF0),<br>(TF0,TF0,TF0,TF1,TF0), (TF1,TF0,TF0,TF1,TF0), (TF2,TF1,TF0,TF1,TF0),<br>(TF3,TF2,TF0,TF1,TF0), (TF4,TF3,TF0,TF1,TF0), (TF5,TF4,TF1,TF1,TF0),<br>(TF0,TF0,TF0,TF0,TF1), (TF1,TF0,TF0,TF0,TF1), (TF2,TF1,TF0,TF0,TF1),<br>(TF3,TF2,TF0,TF0,TF1), (TF4,TF3,TF0,TF0,TF1), (TF5,TF4,TF1,TF0,TF1),<br>(TF0,TF0,TF0,TF1,TF1), (TF1,TF0,TF0,TF1,TF1), (TF2,TF1,TF0,TF1,TF1),<br>(TF3,TF2,TF0,TF1,TF1), (TF4,TF3,TF0,TF1,TF1), (TF5,TF4,TF1,TF1,TF1) |

6.10.2.4.1.38f.1.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Uplink | Min spreading factor | 32 |
| | Max number of DPDCH data bits/radio frame | 1 200 |
| | Puncturing Limit | 1.0 |

6.10.2.4.1.38f.2 Downlink

6.10.2.4.1.38f.2.1 Transport channel parameters

6.10.2.4.1.38f.2.1.1 Transport channel parameters for Conversational / speech / DL: (12.2 7.95 5.9 4.75) kbps / CS RAB

See clause 6.10.2.4.1.4a.2.1.1.

6.10.2.4.1.38f.2.1.2 Transport channel parameters for Interactive or background / DL:8 kbps / PS RAB

See clause 6.10.2.4.1.38b.2.1.2

6.10.2.4.1.38f.2.1.3 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.2.1.1

6.10.2.4.1.38f.2.1.4 TFCS

| | |
|---|---|
| TFCS size | 24 |
| TFCS | (RAB subflow#1, RAB subflow#2, RAB subflow#3, 8 kbps RAB, DCCH)=<br>(TF0,TF0,TF0,TF0,TF0), (TF1,TF0,TF0,TF0,TF0), (TF2,TF1,TF0,TF0,TF0),<br>(TF3,TF2,TF0,TF0,TF0), (TF4,TF3,TF0,TF0,TF0), (TF5,TF4,TF1,TF0,TF0),<br>(TF0,TF0,TF0,TF1,TF0), (TF1,TF0,TF0,TF1,TF0), (TF2,TF1,TF0,TF1,TF0),<br>(TF3,TF2,TF0,TF1,TF0), (TF4,TF3,TF0,TF1,TF0), (TF5,TF4,TF1,TF1,TF0),<br>(TF0,TF0,TF0,TF0,TF1), (TF1,TF0,TF0,TF0,TF1), (TF2,TF1,TF0,TF0,TF1),<br>(TF3,TF2,TF0,TF0,TF1), (TF4,TF3,TF0,TF0,TF1), (TF5,TF4,TF1,TF0,TF1),<br>(TF0,TF0,TF0,TF1,TF1), (TF1,TF0,TF0,TF1,TF1), (TF2,TF1,TF0,TF1,TF1),<br>(TF3,TF2,TF0,TF1,TF1), (TF4,TF3,TF0,TF1,TF1), (TF5,TF4,TF1,TF1,TF1) |

6.10.2.4.1.38f.2.2 Physical channel parameters

| | | | |
|---|---|---|---|
| DPCH Downlink | DTX position | | Flexible |
| | Spreading factor | | 64 |
| | DPCCH | Number of TFCI bits/slot | 8 |
| | | Number of TPC bits/slot | 4 |
| | | Number of Pilot bits/slot | 8 |
| | DPDCH | Number of data bits/slot | 60 |
| | | Number of data bits/frame | 900 |

6.10.2.4.1.38g Conversational / speech / (12.2 7.95 5.9 4.75) kbps / CS RAB + Interactive or background / UL:16 DL:16 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.2.4.1.38g.1 Uplink

6.10.2.4.1.38g.1.1 Transport channel parameters

6.10.2.4.1.38g.1.1.1 Transport channel parameters for Conversational / speech / UL: (12.2 7.95 5.9 4.75) kbps / CS RAB

See clause 6.10.2.4.1.4a.1.1.1.

6.10.2.4.1.38g.1.1.2 Transport channel parameters for Interactive or background / UL:16 kbps / PS RAB

See clause 6.10.2.4.1.23b.1.1.1.

6.10.2.4.1.38g.1.1.3 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.1.1.1.

6.10.2.4.1.38g.1.1.4 TFCS

| | |
|---|---|
| TFCS size | 32 |
| TFCS | (RAB subflow#1, RAB subflow#2, RAB subflow#3, 16 kbps RAB, DCCH)=<br>(TF0,TF0,TF0,TF0,TF0), (TF1,TF0,TF0,TF0,TF0), (TF2,TF1,TF0,TF0,TF0),<br>(TF3,TF2,TF0,TF0,TF0), (TF4,TF3,TF0,TF0,TF0), (TF5,TF4,TF1,TF0,TF0), |

| | |
|---|---|
| | (TF0,TF0,TF0,TF1,TF0), (TF1,TF0,TF0,TF1,TF0), (TF3,TF2,TF0,TF1,TF0),<br>(TF4,TF3,TF0,TF1,TF0), (TF5,TF4,TF1,TF1,TF0), (TF0,TF0,TF0,TF2,TF0),<br>(TF1,TF0,TF0,TF2,TF0), (TF3,TF2,TF0,TF2,TF0), (TF4,TF3,TF0,TF2,TF0),<br>(TF5,TF4,TF1,TF2,TF0), (TF0,TF0,TF0,TF0,TF1), (TF1,TF0,TF0,TF0,TF1),<br>(TF2,TF1,TF0,TF0,TF1), (TF3,TF2,TF0,TF0,TF1), (TF4,TF3,TF0,TF0,TF1),<br>(TF5,TF4,TF1,TF0,TF1), (TF0,TF0,TF0,TF1,TF1), (TF1,TF0,TF0,TF1,TF1),<br>(TF3,TF2,TF0,TF1,TF1), (TF4,TF3,TF0,TF1,TF1), (TF5,TF4,TF1,TF1,TF1),<br>(TF0,TF0,TF0,TF2,TF1), (TF1,TF0,TF0,TF2,TF1), (TF3,TF2,TF0,TF2,TF1),<br>(TF4,TF3,TF0,TF2,TF1), (TF5,TF4,TF1,TF2,TF1) |

6.10.2.4.1.38g.1.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Uplink | Min spreading factor | 32 |
| | Max number of DPDCH data bits/radio frame | 1 200 |
| | Puncturing Limit | 0.88 |

6.10.2.4.1.38g.2 Downlink

6.10.2.4.1.38g.2.1 Transport channel parameters

6.10.2.4.1.38g.2.1.1 Transport channel parameters for Conversational / speech / DL: (12.2 7.95 5.9 4.75) kbps / CS RAB

See clause 6.10.2.4.1.4a.2.1.1.

6.10.2.4.1.38g.2.1.2 Transport channel parameters for Interactive or background / DL:16 kbps / PS RAB

See clause 6.10.2.4.1.23b.2.1.1.

6.10.2.4.1.38g.2.1.3 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.2.1.1

6.10.2.4.1.38g.2.1.4 TFCS

| | |
|---|---|
| TFCS size | 36 |
| TFCS | (RAB subflow#1, RAB subflow#2, RAB subflow#3, 16 kbps RAB, DCCH)=<br>(TF0,TF0,TF0,TF0,TF0), (TF1,TF0,TF0,TF0,TF0), (TF2,TF1,TF0,TF0,TF0),<br>(TF3,TF2,TF0,TF0,TF0), (TF4,TF3,TF0,TF0,TF0), (TF5,TF4,TF1,TF0,TF0),<br>(TF0,TF0,TF0,TF1,TF0), (TF1,TF0,TF0,TF1,TF0), (TF2,TF1,TF0,TF1,TF0),<br>(TF3,TF2,TF0,TF1,TF0), (TF4,TF3,TF0,TF1,TF0), (TF5,TF4,TF1,TF1,TF0),<br>(TF0,TF0,TF0,TF2,TF0), (TF1,TF0,TF0,TF2,TF0), (TF2,TF1,TF0,TF2,TF0),<br>(TF3,TF2,TF0,TF2,TF0), (TF4,TF3,TF0,TF2,TF0), (TF5,TF4,TF1,TF2,TF0),<br>(TF0,TF0,TF0,TF0,TF1), (TF1,TF0,TF0,TF0,TF1), (TF2,TF1,TF0,TF0,TF1),<br>(TF3,TF2,TF0,TF0,TF1), (TF4,TF3,TF0,TF0,TF1), (TF5,TF4,TF1,TF0,TF1),<br>(TF0,TF0,TF0,TF1,TF1), (TF1,TF0,TF0,TF1,TF1), (TF2,TF1,TF0,TF1,TF1),<br>(TF3,TF2,TF0,TF1,TF1), (TF4,TF3,TF0,TF1,TF1), (TF5,TF4,TF1,TF1,TF1),<br>(TF0,TF0,TF0,TF2,TF1), (TF1,TF0,TF0,TF2,TF1), (TF2,TF1,TF0,TF2,TF1),<br>(TF3,TF2,TF0,TF2,TF1), (TF4,TF3,TF0,TF2,TF1), (TF5,TF4,TF1,TF2,TF1) |

6.10.2.4.1.38g.2.2 Physical channel parameters

| | | | |
|---|---|---|---|
| DPCH Downlink | DTX position | | Flexible |
| | Spreading factor | | 64 |
| | DPCCH | Number of TFCI bits/slot | 8 |
| | | Number of TPC bits/slot | 4 |
| | | Number of Pilot bits/slot | 8 |
| | DPDCH | Number of data bits/slot | 60 |
| | | Number of data bits/frame | 900 |

6.10.2.4.1.38h Conversational / speech / (12.2 7.95 5.9 4.75) kbps / CS RAB + Interactive or background / UL:32 DL:32 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.2.4.1.38h.1 Uplink

6.10.2.4.1.38h.1.1 Transport channel parameters

6.10.2.4.1.38h.1.1.1 Transport channel parameters for Conversational / speech / UL: (12.2 7.95 5.9 4.75) kbps / CS RAB

See clause 6.10.2.4.1.4a.1.1.1.

6.10.2.4.1.38h.1.1.2 Transport channel parameters for Interactive or background / UL:32 kbps / PS RAB

See clause 6.10.2.4.1.23c.1.1.1.

6.10.2.4.1.38h.1.1.3 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.1.1.1.

6.10.2.4.1.38h.1.1.4 TFCS

| TFCS size | 32 |
|---|---|
| TFCS | (RAB subflow#1, RAB subflow#2, RAB subflow#3, 32 kbps RAB, DCCH)=<br>(TF0,TF0,TF0,TF0,TF0), (TF0,TF0,TF0,TF1,TF0), (TF0,TF0,TF0,TF2,TF0),<br>(TF0,TF0,TF0,TF4,TF0), (TF5,TF4,TF1,TF0,TF0), (TF5,TF4,TF1,TF1,TF0),<br>(TF5,TF4,TF1,TF2,TF0), (TF5,TF4,TF1,TF4,TF0), (TF4,TF3,TF0,TF0,TF0),<br>(TF4,TF3,TF0,TF1,TF0), (TF3,TF2,TF0,TF0,TF0), (TF2,TF1,TF0,TF0,TF0),<br>(TF1,TF0,TF0,TF0,TF0), (TF1,TF0,TF0,TF1,TF0), (TF1,TF0,TF0,TF2,TF0),<br>(TF1,TF0,TF0,TF4,TF0), (TF0,TF0,TF0,TF0,TF1), (TF0,TF0,TF0,TF1,TF1),<br>(TF0,TF0,TF0,TF2,TF1), (TF0,TF0,TF0,TF4,TF1), (TF5,TF4,TF1,TF0,TF1),<br>(TF5,TF4,TF1,TF1,TF1), (TF5,TF4,TF1,TF2,TF1), (TF5,TF4,TF1,TF4,TF1),<br>(TF4,TF3,TF0,TF0,TF1), (TF4,TF3,TF0,TF1,TF1), (TF3,TF2,TF0,TF0,TF1),<br>(TF2,TF1,TF0,TF0,TF1), (TF1,TF0,TF0,TF0,TF1), (TF1,TF0,TF0,TF1,TF1),<br>(TF1,TF0,TF0,TF2,TF1), (TF1,TF0,TF0,TF4,TF1) |

6.10.2.4.1.38h.1.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Uplink | Min spreading factor | 16 |
| | Max number of DPDCH data bits/radio frame | 2 400 |
| | Puncturing Limit | 1.0 |

6.10.2.4.1.38h.2 Downlink

6.10.2.4.1.38h.2.1 Transport channel parameters

6.10.2.4.1.38h.2.1.1 Transport channel parameters for Conversational / speech / DL: (12.2 7.95 5.9 4.75) kbps / CS RAB

See clause 6.10.2.4.1.4a.2.1.1.

6.10.2.4.1.38h.2.1.2 Transport channel parameters for Interactive or background / DL:32 kbps / PS RAB

See clause 6.10.2.4.1.23c.2.1.1.

6.10.2.4.1.38h.2.1.3 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.2.1.1

6.10.2.4.1.38h.2.1.4 TFCS

| | |
|---|---|
| TFCS size | 48 |
| TFCS | (RAB subflow#1, RAB subflow#2, RAB subflow#3, 32 kbps RAB, DCCH)=<br>(TF0,TF0,TF0,TF0,TF0), (TF0,TF0,TF0,TF1,TF0), (TF0,TF0,TF0,TF2,TF0),<br>(TF0,TF0,TF0,TF4,TF0), (TF5,TF4,TF1,TF0,TF0), (TF5,TF4,TF1,TF1,TF0),<br>(TF5,TF4,TF1,TF2,TF0), (TF5,TF4,TF1,TF4,TF0), (TF4,TF3,TF0,TF0,TF0),<br>(TF4,TF3,TF0,TF1,TF0), (TF4,TF3,TF0,TF2,TF0), (TF4,TF3,TF0,TF4,TF0),<br>(TF3,TF2,TF0,TF0,TF0), (TF3,TF2,TF0,TF1,TF0), (TF3,TF2,TF0,TF2,TF0),<br>(TF3,TF2,TF0,TF4,TF0), (TF2,TF1,TF0,TF0,TF0), (TF2,TF1,TF0,TF1,TF0),<br>(TF2,TF1,TF0,TF2,TF0), (TF2,TF1,TF0,TF4,TF0), (TF1,TF0,TF0,TF0,TF0),<br>(TF1,TF0,TF0,TF1,TF0), (TF1,TF0,TF0,TF2,TF0), (TF1,TF0,TF0,TF4,TF0),<br>(TF0,TF0,TF0,TF0,TF1), (TF0,TF0,TF0,TF1,TF1), (TF0,TF0,TF0,TF2,TF1),<br>(TF0,TF0,TF0,TF4,TF1), (TF5,TF4,TF1,TF0,TF1), (TF5,TF4,TF1,TF1,TF1),<br>(TF5,TF4,TF1,TF2,TF1), (TF5,TF4,TF1,TF4,TF1), (TF4,TF3,TF0,TF0,TF1),<br>(TF4,TF3,TF0,TF1,TF1), (TF4,TF3,TF0,TF2,TF1), (TF4,TF3,TF0,TF4,TF1),<br>(TF3,TF2,TF0,TF0,TF1), (TF3,TF2,TF0,TF1,TF1), (TF3,TF2,TF0,TF2,TF1),<br>(TF3,TF2,TF0,TF4,TF1), (TF2,TF1,TF0,TF0,TF1), (TF2,TF1,TF0,TF1,TF1),<br>(TF2,TF1,TF0,TF2,TF1), (TF2,TF1,TF0,TF4,TF1), (TF1,TF0,TF0,TF0,TF1),<br>(TF1,TF0,TF0,TF1,TF1), (TF1,TF0,TF0,TF2,TF1), (TF1,TF0,TF0,TF4,TF1) |

6.10.2.4.1.38h.2.2 Physical channel parameters

| | | | |
|---|---|---|---|
| DPCH Downlink | DTX position | | Flexible |
| | Spreading factor | | 32 |
| | DPCCH | Number of TFCI bits/slot | 8 |
| | | Number of TPC bits/slot | 4 |
| | | Number of Pilot bits/slot | 8 |
| | DPDCH | Number of data bits/slot | 140 |
| | | Number of data bits/frame | 2 100 |

6.10.2.4.1.38i Conversational / speech / (12.2 7.95 5.9 4.75) kbps / CS RAB + Interactive or background / UL:64 DL:64 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.2.4.1.38i.1 Uplink

6.10.2.4.1.38i.1.1 Transport channel parameters

6.10.2.4.1.38i.1.1.1 Transport channel parameters for Conversational / speech / UL: (12.2 7.95 5.9 4.75) kbps / CS RAB

See clause 6.10.2.4.1.4a.1.1.1.

6.10.2.4.1.38i.1.1.2 Transport channel parameters for Interactive or background / UL:64 kbps / PS RAB

See clause 6.10.2.4.1.26.1.1.1.

6.10.2.4.1.38i.1.1.3 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.1.1.1.

6.10.2.4.1.38i.1.1.4 TFCS

| | |
|---|---|
| TFCS size | 48 |
| TFCS | (RAB subflow#1, RAB subflow#2, RAB subflow#3, 64 kbps RAB, DCCH)=<br>(TF0,TF0,TF0,TF0,TF0), (TF1,TF0,TF0,TF0,TF0), (TF2,TF1,TF0,TF0,TF0),<br>(TF3,TF2,TF0,TF0,TF0), (TF4,TF3,TF0,TF0,TF0), (TF5,TF4,TF1,TF0,TF0),<br>(TF0,TF0,TF0,TF1,TF0), (TF1,TF0,TF0,TF1,TF0), (TF2,TF1,TF0,TF1,TF0),<br>(TF3,TF2,TF0,TF1,TF0), (TF4,TF3,TF0,TF1,TF0), (TF5,TF4,TF1,TF1,TF0),<br>(TF0,TF0,TF0,TF2,TF0), (TF1,TF0,TF0,TF2,TF0), (TF2,TF1,TF0,TF2,TF0),<br>(TF3,TF2,TF0,TF2,TF0), (TF4,TF3,TF0,TF2,TF0), (TF5,TF4,TF1,TF2,TF0),<br>(TF0,TF0,TF0,TF4,TF0), (TF1,TF0,TF0,TF4,TF0), (TF2,TF1,TF0,TF4,TF0),<br>(TF3,TF2,TF0,TF4,TF0), (TF4,TF3,TF0,TF4,TF0), (TF5,TF4,TF1,TF4,TF0),<br>(TF0,TF0,TF0,TF0,TF1), (TF1,TF0,TF0,TF0,TF1), (TF2,TF1,TF0,TF0,TF1),<br>(TF3,TF2,TF0,TF0,TF1), (TF4,TF3,TF0,TF0,TF1), (TF5,TF4,TF1,TF0,TF1),<br>(TF0,TF0,TF0,TF1,TF1), (TF1,TF0,TF0,TF1,TF1), (TF2,TF1,TF0,TF1,TF1),<br>(TF3,TF2,TF0,TF1,TF1), (TF4,TF3,TF0,TF1,TF1), (TF5,TF4,TF1,TF1,TF1),<br>(TF0,TF0,TF0,TF2,TF1), (TF1,TF0,TF0,TF2,TF1), (TF2,TF1,TF0,TF2,TF1),<br>(TF3,TF2,TF0,TF2,TF1), (TF4,TF3,TF0,TF2,TF1), (TF5,TF4,TF1,TF2,TF1),<br>(TF0,TF0,TF0,TF4,TF1), (TF1,TF0,TF0,TF4,TF1), (TF2,TF1,TF0,TF4,TF1),<br>(TF3,TF2,TF0,TF4,TF1), (TF4,TF3,TF0,TF4,TF1), (TF5,TF4,TF1,TF4,TF1) |

6.10.2.4.1.38i.1.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Uplink | Min spreading factor | 16 |
| | Max number of DPDCH data bits/radio frame | 2 400 |
| | Puncturing Limit | 0.76 |

6.10.2.4.1.38i.2 Downlink

6.10.2.4.1.38i.2.1 Transport channel parameters

6.10.2.4.1.38i.2.1.1 Transport channel parameters for Conversational / speech / DL: (12.2 7.95 5.9 4.75) kbps / CS RAB

See clause 6.10.2.4.1.4a.2.1.1.

6.10.2.4.1.38i.2.1.2 Transport channel parameters for Interactive or background / DL:64 kbps / PS RAB

See clause 6.10.2.4.1.25.2.1.1.

6.10.2.4.1.38i.2.1.3 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.2.1.1

6.10.2.4.1.38i.2.1.4 TFCS

| | |
|---|---|
| TFCS size | 60 |
| TFCS | (RAB subflow#1, RAB subflow#2, RAB subflow#3, 64 kbps RAB, DCCH)=<br>(TF0,TF0,TF0,TF0,TF0), (TF1,TF0,TF0,TF0,TF0), (TF2,TF1,TF0,TF0,TF0),<br>(TF3,TF2,TF0,TF0,TF0), (TF4,TF3,TF0,TF0,TF0), (TF5,TF4,TF1,TF0,TF0),<br>(TF0,TF0,TF0,TF1,TF0), (TF1,TF0,TF0,TF1,TF0), (TF2,TF1,TF0,TF1,TF0),<br>(TF3,TF2,TF0,TF1,TF0), (TF4,TF3,TF0,TF1,TF0), (TF5,TF4,TF1,TF1,TF0),<br>(TF0,TF0,TF0,TF2,TF0), (TF1,TF0,TF0,TF2,TF0), (TF2,TF1,TF0,TF2,TF0),<br>(TF3,TF2,TF0,TF2,TF0), (TF4,TF3,TF0,TF2,TF0), (TF5,TF4,TF1,TF2,TF0),<br>(TF0,TF0,TF0,TF3,TF0), (TF1,TF0,TF0,TF3,TF0), (TF2,TF1,TF0,TF3,TF0),<br>(TF3,TF2,TF0,TF3,TF0), (TF4,TF3,TF0,TF3,TF0), (TF5,TF4,TF1,TF3,TF0),<br>(TF0,TF0,TF0,TF4,TF0), (TF1,TF0,TF0,TF4,TF0), (TF2,TF1,TF0,TF4,TF0),<br>(TF3,TF2,TF0,TF4,TF0), (TF4,TF3,TF0,TF4,TF0), (TF5,TF4,TF1,TF4,TF0),<br>(TF0,TF0,TF0,TF0,TF1), (TF1,TF0,TF0,TF0,TF1), (TF2,TF1,TF0,TF0,TF1),<br>(TF3,TF2,TF0,TF0,TF1), (TF4,TF3,TF0,TF0,TF1), (TF5,TF4,TF1,TF0,TF1),<br>(TF0,TF0,TF0,TF1,TF1), (TF1,TF0,TF0,TF1,TF1), (TF2,TF1,TF0,TF1,TF1),<br>(TF3,TF2,TF0,TF1,TF1), (TF4,TF3,TF0,TF1,TF1), (TF5,TF4,TF1,TF1,TF1),<br>(TF0,TF0,TF0,TF2,TF1), (TF1,TF0,TF0,TF2,TF1), (TF2,TF1,TF0,TF2,TF1),<br>(TF3,TF2,TF0,TF2,TF1), (TF4,TF3,TF0,TF2,TF1), (TF5,TF4,TF1,TF2,TF1),<br>(TF0,TF0,TF0,TF3,TF1), (TF1,TF0,TF0,TF3,TF1), (TF2,TF1,TF0,TF3,TF1),<br>(TF3,TF2,TF0,TF3,TF1), (TF4,TF3,TF0,TF3,TF1), (TF5,TF4,TF1,TF3,TF1),<br>(TF0,TF0,TF0,TF4,TF1), (TF1,TF0,TF0,TF4,TF1), (TF2,TF1,TF0,TF4,TF1),<br>(TF3,TF2,TF0,TF4,TF1), (TF4,TF3,TF0,TF4,TF1), (TF5,TF4,TF1,TF4,TF1) |

6.10.2.4.1.38i.2.2 Physical channel parameters

| | | | |
|---|---|---|---|
| DPCH Downlink | DTX position | | Flexible |
| | Spreading factor | | 32 |
| | DPCCH | Number of TFCI bits/slot | 8 |
| | | Number of TPC bits/slot | 4 |
| | | Number of Pilot bits/slot | 8 |
| | DPDCH | Number of data bits/slot | 140 |
| | | Number of data bits/frame | 2 100 |

6.10.2.4.1.38j Conversational / speech / (12.2 7.95 5.9 4.75) kbps / CS RAB + Interactive or background / UL:64 DL:128 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.2.4.1.38j.1 Uplink

6.10.2.4.1.38j.1.1 Transport channel parameters

See clause 6.10.2.4.1.38i.1.1

6.10.2.4.1.38j.2 Downlink

6.10.2.4.1.38j.2.1 Transport channel parameters

6.10.2.4.1.38j.2.1.1 Transport channel parameters for Conversational / speech / DL: (12.2 7.95 5.9 4.75) kbps / CS RAB

See clause 6.10.2.4.1.4a.2.1.1.

6.10.2.4.1.38j.2.1.2 Transport channel parameters for Interactive or background / DL:128 kbps / PS RAB

See clause 6.10.2.4.1.27.2.1.1.

6.10.2.4.1.38j.2.1.3 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.2.1.1

6.10.2.4.1.38j.2.1.4 TFCS

| | |
|---|---|
| TFCS size | 60 |
| TFCS | (RAB subflow#1, RAB subflow#2, RAB subflow#3, 128 kbps RAB, DCCH)=<br>(TF0,TF0,TF0,TF0,TF0), (TF1,TF0,TF0,TF0,TF0), (TF2,TF1,TF0,TF0,TF0), |

| | |
|---|---|
| | (TF3,TF2,TF0,TF0,TF0), (TF4,TF3,TF0,TF0,TF0), (TF5,TF4,TF1,TF0,TF0),<br>(TF0,TF0,TF0,TF1,TF0), (TF1,TF0,TF0,TF1,TF0), (TF2,TF1,TF0,TF1,TF0),<br>(TF3,TF2,TF0,TF1,TF0), (TF4,TF3,TF0,TF1,TF0), (TF5,TF4,TF1,TF1,TF0),<br>(TF0,TF0,TF0,TF2,TF0), (TF1,TF0,TF0,TF2,TF0), (TF2,TF1,TF0,TF2,TF0),<br>(TF3,TF2,TF0,TF2,TF0), (TF4,TF3,TF0,TF2,TF0), (TF5,TF4,TF1,TF2,TF0),<br>(TF0,TF0,TF0,TF3,TF0), (TF1,TF0,TF0,TF3,TF0), (TF2,TF1,TF0,TF3,TF0),<br>(TF3,TF2,TF0,TF3,TF0), (TF4,TF3,TF0,TF3,TF0), (TF5,TF4,TF1,TF3,TF0),<br>(TF0,TF0,TF0,TF4,TF0), (TF1,TF0,TF0,TF4,TF0), (TF2,TF1,TF0,TF4,TF0),<br>(TF3,TF2,TF0,TF4,TF0), (TF4,TF3,TF0,TF4,TF0), (TF5,TF4,TF1,TF4,TF0),<br>(TF0,TF0,TF0,TF0,TF1), (TF1,TF0,TF0,TF0,TF1), (TF2,TF1,TF0,TF0,TF1),<br>(TF3,TF2,TF0,TF0,TF1), (TF4,TF3,TF0,TF0,TF1), (TF5,TF4,TF1,TF0,TF1),<br>(TF0,TF0,TF0,TF1,TF1), (TF1,TF0,TF0,TF1,TF1), (TF2,TF1,TF0,TF1,TF1),<br>(TF3,TF2,TF0,TF1,TF1), (TF4,TF3,TF0,TF1,TF1), (TF5,TF4,TF1,TF1,TF1),<br>(TF0,TF0,TF0,TF2,TF1), (TF1,TF0,TF0,TF2,TF1), (TF2,TF1,TF0,TF2,TF1),<br>(TF3,TF2,TF0,TF2,TF1), (TF4,TF3,TF0,TF2,TF1), (TF5,TF4,TF1,TF2,TF1),<br>(TF0,TF0,TF0,TF3,TF1), (TF1,TF0,TF0,TF3,TF1), (TF2,TF1,TF0,TF3,TF1),<br>(TF3,TF2,TF0,TF3,TF1), (TF4,TF3,TF0,TF3,TF1), (TF5,TF4,TF1,TF3,TF1),<br>(TF0,TF0,TF0,TF4,TF1), (TF1,TF0,TF0,TF4,TF1), (TF2,TF1,TF0,TF4,TF1),<br>(TF3,TF2,TF0,TF4,TF1), (TF4,TF3,TF0,TF4,TF1), (TF5,TF4,TF1,TF4,TF1) |

6.10.2.4.1.38j.2.2 Physical channel parameters

| | | | |
|---|---|---|---|
| DPCH Downlink | DTX position | | Flexible |
| | Spreading factor | | 16 |
| | DPCCH | Number of TFCI bits/slot | 8 |
| | | Number of TPC bits/slot | 8 |
| | | Number of Pilot bits/slot | 16 |
| | DPDCH | Number of data bits/slot | 288 |
| | | Number of data bits/frame | 4 320 |

6.10.2.4.1.38k Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:32 DL:32 kbps / PS RAB + Interactive or background / UL:32 DL:32 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH (L1 multiplexing)

6.10.2.4.1.38k.1 Uplink

6.10.2.4.1.38k.1.1 Transport channel parameters

6.10.2.4.1.38k.1.1.1 Transport channel parameters for Conversational / speech / UL:12.2 kbps / CS RAB

See clause 6.10.2.4.1.4.1.1.1.

6.10.2.4.1.38k.1.1.2 Transport channel parameters for Interactive or background / UL:32 kbps / PS RAB

See clause 6.10.2.4.1.23.1.1.1.

6.10.2.4.1.38k.1.1.3 Transport channel parameters for Interactive or background / UL:32 kbps / PS RAB

See clause 6.10.2.4.1.23.1.1.1.

6.10.2.4.1.38k.1.1.4 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.1.1.1.

6.10.2.4.1.38k.1.1.5 TFCS

| | |
|---|---|
| TFCS size | 54 (alt. 24) |
| TFCS | (RAB subflow#1, RAB subflow#2, RAB subflow#3, 32kbps RAB, 32kbps RAB, DCCH)=<br>(TF0, TF0, TF0, TF0, TF0, TF0), (TF1, TF0, TF0, TF0, TF0, TF0),<br>(TF2, TF1, TF1, TF0, TF0, TF0), (TF0, TF0, TF0, TF1, TF0, TF0),<br>(TF1, TF0, TF0, TF1, TF0, TF0), (TF2, TF1, TF1, TF1, TF0, TF0),<br>(TF0, TF0, TF0, TF2, TF0, TF0), (TF1, TF0, TF0, TF2, TF0, TF0),<br>(TF2, TF1, TF1, TF2, TF0, TF0), (TF0, TF0, TF0, TF0, TF1, TF0),<br>(TF1, TF0, TF0, TF0, TF1, TF0), (TF2, TF1, TF1, TF0, TF1, TF0),<br>(TF0, TF0, TF0, TF1, TF1, TF0), (TF1, TF0, TF0, TF1, TF1, TF0),<br>(TF2, TF1, TF1, TF1, TF1, TF0), (TF0, TF0, TF0, TF2, TF1, TF0), |

| | |
|---|---|
| | (TF1, TF0, TF0, TF2, TF1, TF0), (TF2, TF1, TF1, TF2, TF1, TF0),<br>(TF0, TF0, TF0, TF0, TF2, TF0), (TF1, TF0, TF0, TF0, TF2, TF0),<br>(TF2, TF1, TF1, TF0, TF2, TF0), (TF0, TF0, TF0, TF1, TF2, TF0),<br>(TF1, TF0, TF0, TF1, TF2, TF0), (TF2, TF1, TF1, TF1, TF2, TF0),<br>(TF0, TF0, TF0, TF2, TF2, TF0), (TF1, TF0, TF0, TF2, TF2, TF0),<br>(TF2, TF1, TF1, TF2, TF2, TF0), (TF0, TF0, TF0, TF0, TF0, TF1),<br>(TF1, TF0, TF0, TF0, TF0, TF1), (TF2, TF1, TF1, TF0, TF0, TF1),<br>(TF0, TF0, TF0, TF1, TF0, TF1), (TF1, TF0, TF0, TF1, TF0, TF1),<br>(TF2, TF1, TF1, TF1, TF0, TF1), (TF0, TF0, TF0, TF2, TF0, TF1),<br>(TF1, TF0, TF0, TF2, TF0, TF1), (TF2, TF1, TF1, TF2, TF0, TF1),<br>(TF0, TF0, TF0, TF0, TF1, TF1), (TF1, TF0, TF0, TF0, TF1, TF1),<br>(TF2, TF1, TF1, TF0, TF1, TF1), (TF0, TF0, TF0, TF1, TF1, TF1),<br>(TF1, TF0, TF0, TF1, TF1, TF1), (TF2, TF1, TF1, TF1, TF1, TF1),<br>(TF0, TF0, TF0, TF2, TF1, TF1), (TF1, TF0, TF0, TF2, TF1, TF1),<br>(TF2, TF1, TF1, TF2, TF1, TF1), (TF0, TF0, TF0, TF0, TF2, TF1),<br>(TF1, TF0, TF0, TF0, TF2, TF1), (TF2, TF1, TF1, TF0, TF2, TF1),<br>(TF0, TF0, TF0, TF1, TF2, TF1), (TF1, TF0, TF0, TF1, TF2, TF1),<br>(TF2, TF1, TF1, TF1, TF2, TF1), (TF0, TF0, TF0, TF2, TF2, TF1),<br>(TF1, TF0, TF0, TF2, TF2, TF1), (TF2, TF1, TF1, TF2, TF2, TF1)<br>(alt. (TF0, TF0, TF0, TF0, TF0, TF0), (TF1, TF0, TF0, TF0, TF0, TF0),<br>(TF2, TF1, TF1, TF0, TF0, TF0), (TF0, TF0, TF0, TF1, TF0, TF0),<br>(TF1, TF0, TF0, TF1, TF0, TF0), (TF2, TF1, TF1, TF1, TF0, TF0),<br>(TF0, TF0, TF0, TF0, TF1, TF0), (TF1, TF0, TF0, TF0, TF1, TF0),<br>(TF2, TF1, TF1, TF0, TF1, TF0), (TF0, TF0, TF0, TF1, TF1, TF0),<br>(TF1, TF0, TF0, TF1, TF1, TF0), (TF2, TF1, TF1, TF1, TF1, TF0),<br>(TF0, TF0, TF0, TF0, TF0, TF1), (TF1, TF0, TF0, TF0, TF0, TF1),<br>(TF2, TF1, TF1, TF0, TF0, TF1), (TF0, TF0, TF0, TF1, TF0, TF1),<br>(TF1, TF0, TF0, TF1, TF0, TF1), (TF2, TF1, TF1, TF1, TF0, TF1),<br>(TF0, TF0, TF0, TF0, TF1, TF1), (TF1, TF0, TF0, TF0, TF1, TF1),<br>(TF2, TF1, TF1, TF0, TF1, TF1), (TF0, TF0, TF0, TF1, TF1, TF1),<br>(TF1, TF0, TF0, TF1, TF1, TF1), (TF2, TF1, TF1, TF1, TF1, TF1)) |

6.10.2.4.1.38k.1.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Uplink | Min spreading factor | 16 |
| | Max number of DPDCH data bits/radio frame | 2 400 |
| | Puncturing Limit | 0.8 |

6.10.2.4.1.38k.2 Downlink

6.10.2.4.1.38k.2.1 Transport channel parameters

6.10.2.4.1.38k.2.1.1 Transport channel parameters for Conversational / speech / DL:12.2 kbps / CS RAB

See clause 6.10.2.4.1.4.2.1.1.

6.10.2.4.1.38k.2.1.2 Transport channel parameters for Interactive or background / DL:32 kbps / PS RAB

See clause 6.10.2.4.1.23d.2.1.1.

6.10.2.4.1.38k.2.1.3 Transport channel parameters for Interactive or background / DL:32 kbps / PS RAB

See clause 6.10.2.4.1.23d.2.1.1.

6.10.2.4.1.38k.2.1.4 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.2.1.

6.10.2.4.1.38k.2.1.5 TFCS

| | |
|---|---|
| TFCS size | 54 |
| TFCS | (RAB subflow#1, RAB subflow#2, RAB subflow#3, 32kbps RAB, 32kbps RAB, DCCH)=<br>(TF0, TF0, TF0, TF0, TF0, TF0), (TF1, TF0, TF0, TF0, TF0, TF0),<br>(TF2, TF1, TF1, TF0, TF0, TF0), (TF0, TF0, TF0, TF1, TF0, TF0),<br>(TF1, TF0, TF0, TF1, TF0, TF0), (TF2, TF1, TF1, TF1, TF0, TF0),<br>(TF0, TF0, TF0, TF2, TF0, TF0), (TF1, TF0, TF0, TF2, TF0, TF0), |

| | |
|---|---|
| | (TF2, TF1, TF1, TF2, TF0, TF0), (TF0, TF0, TF0, TF0, TF1, TF0),<br>(TF1, TF0, TF0, TF0, TF1, TF0), (TF2, TF1, TF1, TF0, TF1, TF0),<br>(TF0, TF0, TF0, TF1, TF1, TF0), (TF1, TF0, TF0, TF1, TF1, TF0),<br>(TF2, TF1, TF1, TF1, TF1, TF0), (TF0, TF0, TF0, TF2, TF1, TF0),<br>(TF1, TF0, TF0, TF2, TF1, TF0), (TF2, TF1, TF1, TF2, TF1, TF0),<br>(TF0, TF0, TF0, TF0, TF2, TF0), (TF1, TF0, TF0, TF0, TF2, TF0),<br>(TF2, TF1, TF1, TF0, TF2, TF0), (TF0, TF0, TF0, TF1, TF2, TF0),<br>(TF1, TF0, TF0, TF1, TF2, TF0), (TF2, TF1, TF1, TF1, TF2, TF0),<br>(TF0, TF0, TF0, TF2, TF2, TF0), (TF1, TF0, TF0, TF2, TF2, TF0),<br>(TF2, TF1, TF1, TF2, TF2, TF0), (TF0, TF0, TF0, TF0, TF0, TF1),<br>(TF1, TF0, TF0, TF0, TF0, TF1), (TF2, TF1, TF1, TF0, TF0, TF1),<br>(TF0, TF0, TF0, TF1, TF0, TF1), (TF1, TF0, TF0, TF1, TF0, TF1),<br>(TF2, TF1, TF1, TF1, TF0, TF1), (TF0, TF0, TF0, TF2, TF0, TF1),<br>(TF1, TF0, TF0, TF2, TF0, TF1), (TF2, TF1, TF1, TF2, TF0, TF1),<br>(TF0, TF0, TF0, TF0, TF1, TF1), (TF1, TF0, TF0, TF0, TF1, TF1),<br>(TF2, TF1, TF1, TF0, TF1, TF1), (TF0, TF0, TF0, TF1, TF1, TF1),<br>(TF1, TF0, TF0, TF1, TF1, TF1), (TF2, TF1, TF1, TF1, TF1, TF1),<br>(TF0, TF0, TF0, TF2, TF1, TF1), (TF1, TF0, TF0, TF2, TF1, TF1),<br>(TF2, TF1, TF1, TF2, TF1, TF1), (TF0, TF0, TF0, TF0, TF2, TF1),<br>(TF1, TF0, TF0, TF0, TF2, TF1), (TF2, TF1, TF1, TF0, TF2, TF1),<br>(TF0, TF0, TF0, TF1, TF2, TF1), (TF1, TF0, TF0, TF1, TF2, TF1),<br>(TF2, TF1, TF1, TF1, TF2, TF1), (TF0, TF0, TF0, TF2, TF2, TF1),<br>(TF1, TF0, TF0, TF2, TF2, TF1), (TF2, TF1, TF1, TF2, TF2, TF1) |

6.10.2.4.1.38k.2.2 Physical channel parameters

| | | | |
|---|---|---|---|
| DPCH Downlink | DTX position | | Flexible |
| | Spreading factor | | 32 |
| | DPCCH | Number of TFCI bits/slot | 8 |
| | | Number of TPC bits/slot | 4 |
| | | Number of Pilot bits/slot | 8 |
| | DPDCH | Number of data bits/slot | 140 |
| | | Number of data bits/frame | 2 100 |

6.10.2.4.1.39 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:32 DL:64 kbps / PS RAB+ UL:3.4 DL: 3.4 kbps SRBs for DCCH

6.10.2.4.1.39.1 Uplink

See clause 6.10.2.4.1.38.1.

6.10.2.4.1.39.2 Downlink

6.10.2.4.1.39.2.1 Transport channel parameters

6.10.2.4.1.39.2.1.1 Transport channel parameters for Conversational / speech / DL:12.2 kbps / CS RAB

See clause 6.10.2.4.1.4.2.1.1.

6.10.2.4.1.39.2.1.2 Transport channel parameters for Interactive or background / DL:64 kbps / PS RAB

See clause 6.10.2.4.1.25.2.1.1.

6.10.2.4.1.39.2.1.3 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.2.1.1.

6.10.2.4.1.39.2.1.4 TFCS

| | |
|---|---|
| TFCS size | 30 |
| TFCS | (RAB subflow#1, RAB subflow#2, RAB subflow#3, 64 kbps RAB , DCCH)=<br>(TF0, TF0, TF0, TF0, TF0), (TF1, TF0, TF0, TF0, TF0), (TF2, TF1, TF1, TF0, TF0),<br>(TF0, TF0, TF0, TF1, TF0), (TF1, TF0, TF0, TF1, TF0), (TF2, TF1, TF1, TF1, TF0),<br>(TF0, TF0, TF0, TF2, TF0), (TF1, TF0, TF0, TF2, TF0), (TF2, TF1, TF1, TF2, TF0),<br>(TF0, TF0, TF0, TF3, TF0), (TF1, TF0, TF0, TF3, TF0), (TF2, TF1, TF1, TF3, TF0),<br>(TF0, TF0, TF0, TF4, TF0), (TF1, TF0, TF0, TF4, TF0), (TF2, TF1, TF1, TF4, TF0), |

| | |
|---|---|
| | (TF0, TF0, TF0, TF0, TF1), (TF1, TF0, TF0, TF0, TF1), (TF2, TF1, TF1, TF0, TF1),<br>(TF0, TF0, TF0, TF1, TF1), (TF1, TF0, TF0, TF1, TF1), (TF2, TF1, TF1, TF1, TF1),<br>(TF0, TF0, TF0, TF2, TF1), (TF1, TF0, TF0, TF2, TF1), (TF2, TF1, TF1, TF2, TF1),<br>(TF0, TF0, TF0, TF3, TF1), (TF1, TF0, TF0, TF3, TF1), (TF2, TF1, TF1, TF3, TF1),<br>(TF0, TF0, TF0, TF4, TF1), (TF1, TF0, TF0, TF4, TF1), (TF2, TF1, TF1, TF4, TF1) |

6.10.2.4.1.39.2.2 Physical channel parameters

| | | | |
|---|---|---|---|
| DPCH Downlink | DTX position | | Flexible |
| | Spreading factor | | 32 |
| | DPCCH | Number of TFCI bits/slot | 8 |
| | | Number of TPC bits/slot | 4 |
| | | Number of Pilot bits/slot | 8 |
| | DPDCH | Number of data bits/slot | 140 |
| | | Number of data bits/frame | 2 100 |

6.10.2.4.1.40 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:64 DL:64 kbps / PS RAB+ UL:3.4 DL: 3.4 kbps SRBs for DCCH

6.10.2.4.1.40.1 Uplink

6.10.2.4.1.40.1.1 Transport channel parameters

6.10.2.4.1.40.1.1.1 Transport channel parameters for Conversational / speech / UL:12.2 kbps / CS RAB

See clause 6.10.2.4.1.4.1.1.1.

6.10.2.4.1.40.1.1.2 Transport channel parameters for Interactive or background / UL:64 kbps / PS RAB

See clause 6.10.2.4.1.26.1.1.1.

6.10.2.4.1.40.1.1.3 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.1.1.1.

6.10.2.4.1.40.1.1.4 TFCS

| | |
|---|---|
| TFCS size | 30 |
| TFCS | (RAB subflow#1, RAB subflow#2, RAB subflow#3, 64 kbps RAB , DCCH)=<br>(TF0, TF0, TF0, TF0, TF0), (TF1, TF0, TF0, TF0, TF0), (TF2, TF1, TF1, TF0, TF0),<br>(TF0, TF0, TF0, TF1, TF0), (TF1, TF0, TF0, TF1, TF0), (TF2, TF1, TF1, TF1, TF0),<br>(TF0, TF0, TF0, TF2, TF0), (TF1, TF0, TF0, TF2, TF0), (TF2, TF1, TF1, TF2, TF0),<br>(TF0, TF0, TF0, TF3, TF0), (TF1, TF0, TF0, TF3, TF0), (TF2, TF1, TF1, TF3, TF0),<br>(TF0, TF0, TF0, TF4, TF0), (TF1, TF0, TF0, TF4, TF0), (TF2, TF1, TF1, TF4, TF0),<br>(TF0, TF0, TF0, TF0, TF1), (TF1, TF0, TF0, TF0, TF1), (TF2, TF1, TF1, TF0, TF1),<br>(TF0, TF0, TF0, TF1, TF1), (TF1, TF0, TF0, TF1, TF1), (TF2, TF1, TF1, TF1, TF1),<br>(TF0, TF0, TF0, TF2, TF1), (TF1, TF0, TF0, TF2, TF1), (TF2, TF1, TF1, TF2, TF1),<br>(TF0, TF0, TF0, TF3, TF1), (TF1, TF0, TF0, TF3, TF1), (TF2, TF1, TF1, TF3, TF1),<br>(TF0, TF0, TF0, TF4, TF1), (TF1, TF0, TF0, TF4, TF1), (TF2, TF1, TF1, TF4, TF1) |

6.10.2.4.1.40.1.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Uplink | Min spreading factor | 16 |
| | Max number of DPDCH data bits/radio frame | 2 400 |
| | Puncturing Limit | 0.76 |

6.10.2.4.1.40.2 Downlink

See clause 6.10.2.4.1.39.2.

6.10.2.4.1.41 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:64 DL:128 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.2.4.1.41.1 Uplink

See clause 6.10.2.4.1.40.1.

6.10.2.4.1.41.2 Downlink

6.10.2.4.1.41.2.1 Transport channel parameters

6.10.2.4.1.41.2.1.1 Transport channel parameters for Conversational / speech / DL:12.2 kbps / CS RAB

See clause 6.10.2.4.1.4.2.1.1.

6.10.2.4.1.41.2.1.2 Transport channel parameters for Interactive or background / DL:128 kbps / PS RAB

See clause 6.10.2.4.1.27.2.1.1.

6.10.2.4.1.41.2.1.3 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.2.1.1.

6.10.2.4.1.41.2.1.4 TFCS

| | |
|---|---|
| TFCS size | 30 |
| TFCS | (RAB subflow#1, RAB subflow#2, RAB subflow#3, 128 kbps RAB , DCCH)=<br>(TF0, TF0, TF0, TF0, TF0), (TF1, TF0, TF0, TF0, TF0), (TF2, TF1, TF1, TF0, TF0),<br>(TF0, TF0, TF0, TF1, TF0), (TF1, TF0, TF0, TF1, TF0), (TF2, TF1, TF1, TF1, TF0),<br>(TF0, TF0, TF0, TF2, TF0), (TF1, TF0, TF0, TF2, TF0), (TF2, TF1, TF1, TF2, TF0),<br>(TF0, TF0, TF0, TF3, TF0), (TF1, TF0, TF0, TF3, TF0), (TF2, TF1, TF1, TF3, TF0),<br>(TF0, TF0, TF0, TF4, TF0), (TF1, TF0, TF0, TF4, TF0), (TF2, TF1, TF1, TF4, TF0),<br>(TF0, TF0, TF0, TF0, TF1), (TF1, TF0, TF0, TF0, TF1), (TF2, TF1, TF1, TF0, TF1),<br>(TF0, TF0, TF0, TF1, TF1), (TF1, TF0, TF0, TF1, TF1), (TF2, TF1, TF1, TF1, TF1),<br>(TF0, TF0, TF0, TF2, TF1), (TF1, TF0, TF0, TF2, TF1), (TF2, TF1, TF1, TF2, TF1),<br>(TF0, TF0, TF0, TF3, TF1), (TF1, TF0, TF0, TF3, TF1), (TF2, TF1, TF1, TF3, TF1),<br>(TF0, TF0, TF0, TF4, TF1), (TF1, TF0, TF0, TF4, TF1), (TF2, TF1, TF1, TF4, TF1) |

6.10.2.4.1.41.2.2 Physical channel parameters

| | | | |
|---|---|---|---|
| DPCH Downlink | DTX position | | Flexible |
| | Spreading factor | | 16 |
| | DPCCH | Number of TFCI bits/slot | 8 |
| | | Number of TPC bits/slot | 8 |
| | | Number of Pilot bits/slot | 16 |
| | DPDCH | Number of data bits/slot | 288 |
| | | Number of data bits/frame | 4 320 |

6.10.2.4.1.42 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:64 DL:256 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.2.4.1.42.1 Uplink

See clause 6.10.2.4.1.40.1.

6.10.2.4.1.42.2 Downlink

6.10.2.4.1.42.2.1 Transport channel parameters

6.10.2.4.1.42.2.1.1 Transport channel parameters for Conversational / speech / DL:12.2 kbps / CS RAB

See clause 6.10.2.4.1.4.2.1.1.

6.10.2.4.1.42.2.1.2 Transport channel parameters for Interactive or background / DL:256 kbps / PS RAB

See clause 6.10.2.4.1.31.2.1.1.

6.10.2.4.1.42.2.1.3 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.2.1.1.

6.10.2.4.1.42.2.1.4 TFCS

| | |
|---|---|
| TFCS size | 30 (alt. 42) |
| TFCS | (RAB subflow#1, RAB subflow#2, RAB subflow#3, 256 kbps RAB , DCCH)=<br>(TF0, TF0, TF0, TF0, TF0), (TF1, TF0, TF0, TF0, TF0), (TF2, TF1, TF1, TF0, TF0),<br>(TF0, TF0, TF0, TF1, TF0), (TF1, TF0, TF0, TF1, TF0), (TF2, TF1, TF1, TF1, TF0),<br>(TF0, TF0, TF0, TF2, TF0), (TF1, TF0, TF0, TF2, TF0), (TF2, TF1, TF1, TF2, TF0),<br>(TF0, TF0, TF0, TF3, TF0), (TF1, TF0, TF0, TF3, TF0), (TF2, TF1, TF1, TF3, TF0),<br>(TF0, TF0, TF0, TF4, TF0), (TF1, TF0, TF0, TF4, TF0), (TF2, TF1, TF1, TF4, TF0),<br>(TF0, TF0, TF0, TF0, TF1), (TF1, TF0, TF0, TF0, TF1), (TF2, TF1, TF1, TF0, TF1),<br>(TF0, TF0, TF0, TF1, TF1), (TF1, TF0, TF0, TF1, TF1), (TF2, TF1, TF1, TF1, TF1),<br>(TF0, TF0, TF0, TF2, TF1), (TF1, TF0, TF0, TF2, TF1), (TF2, TF1, TF1, TF2, TF1),<br>(TF0, TF0, TF0, TF3, TF1), (TF1, TF0, TF0, TF3, TF1), (TF2, TF1, TF1, TF3, TF1),<br>(TF0, TF0, TF0, TF4, TF1), (TF1, TF0, TF0, TF4, TF1), (TF2, TF1, TF1, TF4, TF1)<br>(alt. (TF0, TF0, TF0, TF0, TF0), (TF1, TF0, TF0, TF0, TF0), (TF2, TF1, TF1, TF0, TF0),<br>(TF0, TF0, TF0, TF1, TF0), (TF1, TF0, TF0, TF1, TF0), (TF2, TF1, TF1, TF1, TF0),<br>(TF0, TF0, TF0, TF2, TF0), (TF1, TF0, TF0, TF2, TF0), (TF2, TF1, TF1, TF2, TF0),<br>(TF0, TF0, TF0, TF3, TF0), (TF1, TF0, TF0, TF3, TF0), (TF2, TF1, TF1, TF3, TF0),<br>(TF0, TF0, TF0, TF4, TF0), (TF1, TF0, TF0, TF4, TF0), (TF2, TF1, TF1, TF4, TF0),<br>(TF0, TF0, TF0, TF5, TF0), (TF1, TF0, TF0, TF5, TF0), (TF2, TF1, TF1, TF5, TF0),<br>(TF0, TF0, TF0, TF6, TF0), (TF1, TF0, TF0, TF6, TF0), (TF2, TF1, TF1, TF6, TF0),<br>(TF0, TF0, TF0, TF0, TF1), (TF1, TF0, TF0, TF0, TF1), (TF2, TF1, TF1, TF0, TF1),<br>(TF0, TF0, TF0, TF1, TF1), (TF1, TF0, TF0, TF1, TF1), (TF2, TF1, TF1, TF1, TF1),<br>(TF0, TF0, TF0, TF2, TF1), (TF1, TF0, TF0, TF2, TF1), (TF2, TF1, TF1, TF2, TF1),<br>(TF0, TF0, TF0, TF3, TF1), (TF1, TF0, TF0, TF3, TF1), (TF2, TF1, TF1, TF3, TF1),<br>(TF0, TF0, TF0, TF4, TF1), (TF1, TF0, TF0, TF4, TF1), (TF2, TF1, TF1, TF4, TF1)<br>(TF0, TF0, TF0, TF5, TF1), (TF1, TF0, TF0, TF5, TF1), (TF2, TF1, TF1, TF5, TF1),<br>(TF0, TF0, TF0, TF6, TF1), (TF1, TF0, TF0, TF6, TF1), (TF2, TF1, TF1, TF6, TF1)) |

6.10.2.4.1.42.2.2 Physical channel parameters

| | | | |
|---|---|---|---|
| DPCH Downlink | DTX position | | Flexible |
| | Spreading factor | | 8 |
| | Number of DPDCH | | 1 |
| | DPCCH | Number of TFCI bits/slot | 8 |
| | | Number of TPC bits/slot | 8 |
| | | Number of Pilot bits/slot | 16 |
| | DPDCH | Number of data bits/slot | 608 |
| | | Number of data bits/frame | 9 120 |

6.10.2.4.1.43 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:64 DL:384 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.2.4.1.43.1 Uplink

See clause 6.10.2.4.1.40.1.

6.10.2.4.1.43.2 Downlink

6.10.2.4.1.43.2.1 Transport channel parameters

6.10.2.4.1.43.2.1.1 Transport channel parameters for Conversational / speech / DL:12.2 kbps / CS RAB

See clause 6.10.2.4.1.4.2.1.1.

6.10.2.4.1.43.2.1.2 Transport channel parameters for Interactive or background / DL:384 kbps / PS RAB

See clause 6.10.2.4.1.32.2.1.1.

6.10.2.4.1.43.2.1.3 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.2.1.1.

6.10.2.4.1.43.2.1.4 TFCS

| | |
|---|---|
| TFCS size | 36 (alt. 54) |

| | |
|---|---|
| TFCS | (RAB subflow#1, RAB subflow#2, RAB subflow#3, 384 kbps RAB , DCCH)=<br>(TF0, TF0, TF0, TF0, TF0), (TF1, TF0, TF0, TF0, TF0), (TF2, TF1, TF1, TF0, TF0),<br>(TF0, TF0, TF0, TF1, TF0), (TF1, TF0, TF0, TF1, TF0), (TF2, TF1, TF1, TF1, TF0),<br>(TF0, TF0, TF0, TF2, TF0), (TF1, TF0, TF0, TF2, TF0), (TF2, TF1, TF1, TF2, TF0),<br>(TF0, TF0, TF0, TF3, TF0), (TF1, TF0, TF0, TF3, TF0), (TF2, TF1, TF1, TF3, TF0),<br>(TF0, TF0, TF0, TF4, TF0), (TF1, TF0, TF0, TF4, TF0), (TF2, TF1, TF1, TF4, TF0),<br>(TF0, TF0, TF0, TF5, TF0), (TF1, TF0, TF0, TF5, TF0), (TF2, TF1, TF1, TF5, TF0),<br>(TF0, TF0, TF0, TF0, TF1), (TF1, TF0, TF0, TF0, TF1), (TF2, TF1, TF1, TF0, TF1),<br>(TF0, TF0, TF0, TF1, TF1), (TF1, TF0, TF0, TF1, TF1), (TF2, TF1, TF1, TF1, TF1),<br>(TF0, TF0, TF0, TF2, TF1), (TF1, TF0, TF0, TF2, TF1), (TF2, TF1, TF1, TF2, TF1),<br>(TF0, TF0, TF0, TF3, TF1), (TF1, TF0, TF0, TF3, TF1), (TF2, TF1, TF1, TF3, TF1),<br>(TF0, TF0, TF0, TF4, TF1), (TF1, TF0, TF0, TF4, TF1), (TF2, TF1, TF1, TF4, TF1)<br>(TF0, TF0, TF0, TF5, TF1), (TF1, TF0, TF0, TF5, TF1), (TF2, TF1, TF1, TF5, TF1),<br>(alt. (TF0, TF0, TF0, TF0, TF0), (TF1, TF0, TF0, TF0, TF0), (TF2, TF1, TF1, TF0, TF0),<br>(TF0, TF0, TF0, TF1, TF0), (TF1, TF0, TF0, TF1, TF0), (TF2, TF1, TF1, TF1, TF0),<br>(TF0, TF0, TF0, TF2, TF0), (TF1, TF0, TF0, TF2, TF0), (TF2, TF1, TF1, TF2, TF0),<br>(TF0, TF0, TF0, TF3, TF0), (TF1, TF0, TF0, TF3, TF0), (TF2, TF1, TF1, TF3, TF0),<br>(TF0, TF0, TF0, TF4, TF0), (TF1, TF0, TF0, TF4, TF0), (TF2, TF1, TF1, TF4, TF0),<br>(TF0, TF0, TF0, TF5, TF0), (TF1, TF0, TF0, TF5, TF0), (TF2, TF1, TF1, TF5, TF0),<br>(TF0, TF0, TF0, TF6, TF0), (TF1, TF0, TF0, TF6, TF0), (TF2, TF1, TF1, TF6, TF0),<br>(TF0, TF0, TF0, TF7, TF0), (TF1, TF0, TF0, TF7, TF0), (TF2, TF1, TF1, TF7, TF0),<br>(TF0, TF0, TF0, TF8, TF0), (TF1, TF0, TF0, TF8, TF0), (TF2, TF1, TF1, TF8, TF0),<br>(TF0, TF0, TF0, TF0, TF1), (TF1, TF0, TF0, TF0, TF1), (TF2, TF1, TF1, TF0, TF1),<br>(TF0, TF0, TF0, TF1, TF1), (TF1, TF0, TF0, TF1, TF1), (TF2, TF1, TF1, TF1, TF1),<br>(TF0, TF0, TF0, TF2, TF1), (TF1, TF0, TF0, TF2, TF1), (TF2, TF1, TF1, TF2, TF1),<br>(TF0, TF0, TF0, TF3, TF1), (TF1, TF0, TF0, TF3, TF1), (TF2, TF1, TF1, TF3, TF1),<br>(TF0, TF0, TF0, TF4, TF1), (TF1, TF0, TF0, TF4, TF1), (TF2, TF1, TF1, TF4, TF1)<br>(TF0, TF0, TF0, TF5, TF1), (TF1, TF0, TF0, TF5, TF1), (TF2, TF1, TF1, TF5, TF1)<br>(TF0, TF0, TF0, TF6, TF1), (TF1, TF0, TF0, TF6, TF1), (TF2, TF1, TF1, TF6, TF1),<br>(TF0, TF0, TF0, TF7, TF1), (TF1, TF0, TF0, TF7, TF1), (TF2, TF1, TF1, TF7, TF1)<br>(TF0, TF0, TF0, TF8, TF1), (TF1, TF0, TF0, TF8, TF1), (TF2, TF1, TF1, TF8, TF1)) |

6.10.2.4.1.43.2.2 Physical channel parameters

| | | | |
|---|---|---|---|
| DPCH Downlink | DTX position | | Flexible |
| | Spreading factor | | 8 |
| | Number of DPDCH | | 1 |
| | DPCCH | Number of TFCI bits/slot | 8 |
| | | Number of TPC bits/slot | 8 |
| | | Number of Pilot bits/slot | 16 |
| | DPDCH | Number of data bits/slot | 608 |
| | | Number of data bits/frame | 9 120 |

6.10.2.4.1.44 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:128 DL:2 048 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.2.4.1.44.1 Uplink

6.10.2.4.1.44.1.1 Transport channel parameters

6.10.2.4.1.44.1.1.1 Transport channel parameters for Conversational / speech / UL:12.2 kbps / CS RAB

See clause 6.10.2.4.1.4.1.1.1.

6.10.2.4.1.44.1.1.2 Transport channel parameters for Interactive or background / UL:128 kbps / PS RAB

See clause 6.10.2.4.1.28.1.1.1.

6.10.2.4.1.44.1.1.3 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.1.1.1.

6.10.2.4.1.44.1.1.4 TFCS

| | |
|---|---|
| TFCS size | 30 |
| TFCS | (RAB subflow#1, RAB subflow#2, RAB subflow#3, 128 kbps RAB , DCCH)= |

| | |
|---|---|
| | (TF0, TF0, TF0, TF0, TF0), (TF1, TF0, TF0, TF0, TF0), (TF2, TF1, TF1, TF0, TF0),<br>(TF0, TF0, TF0, TF1, TF0), (TF1, TF0, TF0, TF1, TF0), (TF2, TF1, TF1, TF1, TF0),<br>(TF0, TF0, TF0, TF2, TF0), (TF1, TF0, TF0, TF2, TF0), (TF2, TF1, TF1, TF2, TF0),<br>(TF0, TF0, TF0, TF3, TF0), (TF1, TF0, TF0, TF3, TF0), (TF2, TF1, TF1, TF3, TF0),<br>(TF0, TF0, TF0, TF4, TF0), (TF1, TF0, TF0, TF4, TF0), (TF2, TF1, TF1, TF4, TF0),<br>(TF0, TF0, TF0, TF0, TF1), (TF1, TF0, TF0, TF0, TF1), (TF2, TF1, TF1, TF0, TF1),<br>(TF0, TF0, TF0, TF1, TF1), (TF1, TF0, TF0, TF1, TF1), (TF2, TF1, TF1, TF1, TF1),<br>(TF0, TF0, TF0, TF2, TF1), (TF1, TF0, TF0, TF2, TF1), (TF2, TF1, TF1, TF2, TF1),<br>(TF0, TF0, TF0, TF3, TF1), (TF1, TF0, TF0, TF3, TF1), (TF2, TF1, TF1, TF3, TF1),<br>(TF0, TF0, TF0, TF4, TF1), (TF1, TF0, TF0, TF4, TF1), (TF2, TF1, TF1, TF4, TF1) |

6.10.2.4.1.44.1.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Uplink | Min spreading factor | 8 |
| | Max number of DPDCH data bits/radio frame | 4 800 |
| | Puncturing Limit | 0.92 |

6.10.2.4.1.44.2 Downlink

6.10.2.4.1.44.2.1 Transport channel parameters

6.10.2.4.1.44.2.1.1 Transport channel parameters for Conversational / speech / DL:12.2 kbps / CS RAB

See clause 6.10.2.4.1.4.2.1.1.

6.10.2.4.1.44.2.1.2 Transport channel parameters for Interactive or background / DL:2 048 kbps / PS RAB

See clause 6.10.2.4.1.35.2.1.1.

6.10.2.4.1.44.2.1.3 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.2.1.1.

6.10.2.4.1.44.2.1.4 TFCS

| TFCS size | 66 (alt. 114) |
|---|---|
| TFCS | (RAB subflow#1, RAB subflow#2, RAB subflow#3, 2 048 kbps RAB , DCCH)=<br>(TF0, TF0, TF0, TF0, TF0), (TF1, TF0, TF0, TF0, TF0), (TF2, TF1, TF1, TF0, TF0),<br>(TF0, TF0, TF0, TF1, TF0), (TF1, TF0, TF0, TF1, TF0), (TF2, TF1, TF1, TF1, TF0),<br>(TF0, TF0, TF0, TF2, TF0), (TF1, TF0, TF0, TF2, TF0), (TF2, TF1, TF1, TF2, TF0),<br>(TF0, TF0, TF0, TF3, TF0), (TF1, TF0, TF0, TF3, TF0), (TF2, TF1, TF1, TF3, TF0),<br>(TF0, TF0, TF0, TF4, TF0), (TF1, TF0, TF0, TF4, TF0), (TF2, TF1, TF1, TF4, TF0),<br>(TF0, TF0, TF0, TF5, TF0), (TF1, TF0, TF0, TF5, TF0), (TF2, TF1, TF1, TF5, TF0),<br>(TF0, TF0, TF0, TF6, TF0), (TF1, TF0, TF0, TF6, TF0), (TF2, TF1, TF1, TF6, TF0),<br>(TF0, TF0, TF0, TF7, TF0), (TF1, TF0, TF0, TF7, TF0), (TF2, TF1, TF1, TF7, TF0),<br>(TF0, TF0, TF0, TF8, TF0), (TF1, TF0, TF0, TF8, TF0), (TF2, TF1, TF1, TF8, TF0),<br>(TF0, TF0, TF0, TF9, TF0), (TF1, TF0, TF0, TF9, TF0), (TF2, TF1, TF1, TF9, TF0),<br>(TF0, TF0, TF0, TF10, TF0), (TF1, TF0, TF0, TF10, TF0), (TF2, TF1, TF1, TF10, TF0),<br>(TF0, TF0, TF0, TF0, TF1), (TF1, TF0, TF0, TF0, TF1), (TF2, TF1, TF1, TF0, TF1),<br>(TF0, TF0, TF0, TF1, TF1), (TF1, TF0, TF0, TF1, TF1), (TF2, TF1, TF1, TF1, TF1),<br>(TF0, TF0, TF0, TF2, TF1), (TF1, TF0, TF0, TF2, TF1), (TF2, TF1, TF1, TF2, TF1),<br>(TF0, TF0, TF0, TF3, TF1), (TF1, TF0, TF0, TF3, TF1), (TF2, TF1, TF1, TF3, TF1),<br>(TF0, TF0, TF0, TF4, TF1), (TF1, TF0, TF0, TF4, TF1), (TF2, TF1, TF1, TF4, TF1)<br>(TF0, TF0, TF0, TF5, TF1), (TF1, TF0, TF0, TF5, TF1), (TF2, TF1, TF1, TF5, TF1),<br>TF0, TF0, TF0, TF6, TF1), (TF1, TF0, TF0, TF6, TF1), (TF2, TF1, TF1, TF6, TF1),<br>(TF0, TF0, TF0, TF7, TF1), (TF1, TF0, TF0, TF7, TF1), (TF2, TF1, TF1, TF7, TF1),<br>(TF0, TF0, TF0, TF8, TF1), (TF1, TF0, TF0, TF8, TF1), (TF2, TF1, TF1, TF8, TF1),<br>(TF0, TF0, TF0, TF9, TF1), (TF1, TF0, TF0, TF9, TF1), (TF2, TF1, TF1, TF9, TF1)<br>(TF0, TF0, TF0, TF10, TF1), (TF1, TF0, TF0, TF10, TF1), (TF2, TF1, TF1, TF10, TF1)<br>(alt. (TF0, TF0, TF0, TF0, TF0), (TF1, TF0, TF0, TF0, TF0), (TF2, TF1, TF1, TF0, TF0),<br>(TF0, TF0, TF0, TF1, TF0), (TF1, TF0, TF0, TF1, TF0), (TF2, TF1, TF1, TF1, TF0),<br>(TF0, TF0, TF0, TF2, TF0), (TF1, TF0, TF0, TF2, TF0), (TF2, TF1, TF1, TF2, TF0),<br>(TF0, TF0, TF0, TF3, TF0), (TF1, TF0, TF0, TF3, TF0), (TF2, TF1, TF1, TF3, TF0),<br>(TF0, TF0, TF0, TF4, TF0), (TF1, TF0, TF0, TF4, TF0), (TF2, TF1, TF1, TF4, TF0),<br>(TF0, TF0, TF0, TF5, TF0), (TF1, TF0, TF0, TF5, TF0), (TF2, TF1, TF1, TF5, TF0),<br>(TF0, TF0, TF0, TF6, TF0), (TF1, TF0, TF0, TF6, TF0), (TF2, TF1, TF1, TF6, TF0),<br>(TF0, TF0, TF0, TF7, TF0), (TF1, TF0, TF0, TF7, TF0), (TF2, TF1, TF1, TF7, TF0),<br>(TF0, TF0, TF0, TF8, TF0), (TF1, TF0, TF0, TF8, TF0), (TF2, TF1, TF1, TF8, TF0),<br>(TF0, TF0, TF0, TF9, TF0), (TF1, TF0, TF0, TF9, TF0), (TF2, TF1, TF1, TF9, TF0),<br>(TF0, TF0, TF0, TF10, TF0), (TF1, TF0, TF0, TF10, TF0), (TF2, TF1, TF1, TF10, TF0),<br>(TF0, TF0, TF0, TF11, TF0), (TF1, TF0, TF0, TF11, TF0), (TF2, TF1, TF1, TF11, TF0),<br>(TF0, TF0, TF0, TF12, TF0), (TF1, TF0, TF0, TF12, TF0), (TF2, TF1, TF1, TF12, TF0),<br>(TF0, TF0, TF0, TF13, TF0), (TF1, TF0, TF0, TF13, TF0), (TF2, TF1, TF1, TF13, TF0),<br>(TF0, TF0, TF0, TF14, TF0), (TF1, TF0, TF0, TF14, TF0), (TF2, TF1, TF1, TF14, TF0),<br>(TF0, TF0, TF0, TF15, TF0), (TF1, TF0, TF0, TF15, TF0), (TF2, TF1, TF1, TF15, TF0),<br>(TF0, TF0, TF0, TF16, TF0), (TF1, TF0, TF0, TF16, TF0), (TF2, TF1, TF1, TF16, TF0),<br>(TF0, TF0, TF0, TF17, TF0), (TF1, TF0, TF0, TF17, TF0), (TF2, TF1, TF1, TF17, TF0),<br>(TF0, TF0, TF0, TF18, TF0), (TF1, TF0, TF0, TF18, TF0), (TF2, TF1, TF1, TF18, TF0),<br>(TF0, TF0, TF0, TF0, TF1), (TF1, TF0, TF0, TF0, TF1), (TF2, TF1, TF1, TF0, TF1),<br>(TF0, TF0, TF0, TF1, TF1), (TF1, TF0, TF0, TF1, TF1), (TF2, TF1, TF1, TF1, TF1),<br>(TF0, TF0, TF0, TF2, TF1), (TF1, TF0, TF0, TF2, TF1), (TF2, TF1, TF1, TF2, TF1),<br>(TF0, TF0, TF0, TF3, TF1), (TF1, TF0, TF0, TF3, TF1), (TF2, TF1, TF1, TF3, TF1),<br>(TF0, TF0, TF0, TF4, TF1), (TF1, TF0, TF0, TF4, TF1), (TF2, TF1, TF1, TF4, TF1),<br>(TF0, TF0, TF0, TF5, TF1), (TF1, TF0, TF0, TF5, TF1), (TF2, TF1, TF1, TF5, TF1),<br>(TF0, TF0, TF0, TF6, TF1), (TF1, TF0, TF0, TF6, TF1), (TF2, TF1, TF1, TF6, TF1),<br>(TF0, TF0, TF0, TF7, TF1), (TF1, TF0, TF0, TF7, TF1), (TF2, TF1, TF1, TF7, TF1),<br>(TF0, TF0, TF0, TF8, TF1), (TF1, TF0, TF0, TF8, TF1), (TF2, TF1, TF1, TF8, TF1),<br>(TF0, TF0, TF0, TF9, TF1), (TF1, TF0, TF0, TF9, TF1), (TF2, TF1, TF1, TF9, TF1),<br>(TF0, TF0, TF0, TF10, TF1), (TF1, TF0, TF0, TF10, TF1), (TF2, TF1, TF1, TF10, TF1),<br>(TF0, TF0, TF0, TF11, TF1), (TF1, TF0, TF0, TF11, TF1), (TF2, TF1, TF1, TF11, TF1),<br>(TF0, TF0, TF0, TF12, TF1), (TF1, TF0, TF0, TF12, TF1), (TF2, TF1, TF1, TF12, TF1),<br>(TF0, TF0, TF0, TF13, TF1), (TF1, TF0, TF0, TF13, TF1), (TF2, TF1, TF1, TF13, TF1),<br>(TF0, TF0, TF0, TF14, TF1), (TF1, TF0, TF0, TF14, TF1), (TF2, TF1, TF1, TF14, TF1),<br>(TF0, TF0, TF0, TF15, TF1), (TF1, TF0, TF0, TF15, TF1), (TF2, TF1, TF1, TF15, TF1),<br>(TF0, TF0, TF0, TF16, TF1), (TF1, TF0, TF0, TF16, TF1), (TF2, TF1, TF1, TF16, TF1),<br>(TF0, TF0, TF0, TF17, TF1), (TF1, TF0, TF0, TF17, TF1), (TF2, TF1, TF1, TF17, TF1),<br>(TF0, TF0, TF0, TF18, TF1), (TF1, TF0, TF0, TF18, TF1), (TF2, TF1, TF1, TF18, TF1)) |

6.10.2.4.1.44.2.2 Physical channel parameters

| | | | |
|---|---|---|---|
| DPCH Downlink | DTX position | | Flexible |
| | Spreading factor | | 4 |
| | Number of DPDCH | | 3 |
| | DPCCH | Number of TFCI bits/slot | 8 |
| | | Number of TPC bits/slot | 8 |
| | | Number of Pilot bits/slot | 16 |
| | DPDCH | Number of data bits/slot | 1 248 |
| | | Number of data bits/frame | 18 720 |

6.10.2.4.1.45 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Streaming / unknown / UL:57.6 DL:57.6 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.2.4.1.45.1 Uplink

6.10.2.4.1.45.1.1 Transport channel parameters

6.10.2.4.1.45.1.1.1 Transport channel parameters for Conversational / speech / UL:12.2 kbps / CS RAB

See clause 6.10.2.4.1.4.1.1.1.

6.10.2.4.1.45.1.1.2 Transport channel parameters for Streaming / unknown / UL:57.6 kbps / CS RAB

See clause 6.10.2.4.1.17.1.1.1.

6.10.2.4.1.45.1.1.3 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.1.1.1.

6.10.2.4.1.45.1.1.4 TFCS

| | |
|---|---|
| TFCS size | 30 |
| TFCS | (RAB subflow#1, RAB subflow#2, RAB subflow#3, 57.6 kbps RAB , DCCH)=<br>(TF0, TF0, TF0, TF0, TF0), (TF1, TF0, TF0, TF0, TF0), (TF2, TF1, TF1, TF0, TF0),<br>(TF0, TF0, TF0, TF1, TF0), (TF1, TF0, TF0, TF1, TF0), (TF2, TF1, TF1, TF1, TF0),<br>(TF0, TF0, TF0, TF2, TF0), (TF1, TF0, TF0, TF2, TF0), (TF2, TF1, TF1, TF2, TF0),<br>(TF0, TF0, TF0, TF3, TF0), (TF1, TF0, TF0, TF3, TF0), (TF2, TF1, TF1, TF3, TF0),<br>(TF0, TF0, TF0, TF4, TF0), (TF1, TF0, TF0, TF4, TF0), (TF2, TF1, TF1, TF4, TF0),<br>(TF0, TF0, TF0, TF0, TF1), (TF1, TF0, TF0, TF0, TF1), (TF2, TF1, TF1, TF0, TF1),<br>(TF0, TF0, TF0, TF1, TF1), (TF1, TF0, TF0, TF1, TF1), (TF2, TF1, TF1, TF1, TF1),<br>(TF0, TF0, TF0, TF2, TF1), (TF1, TF0, TF0, TF2, TF1), (TF2, TF1, TF1, TF2, TF1),<br>(TF0, TF0, TF0, TF3, TF1), (TF1, TF0, TF0, TF3, TF1), (TF2, TF1, TF1, TF3, TF1),<br>(TF0, TF0, TF0, TF4, TF1), (TF1, TF0, TF0, TF4, TF1), (TF2, TF1, TF1, TF4, TF1) |

6.10.2.4.1.45.1.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Uplink | Min spreading factor | 16 |
| | Max number of DPDCH data bits/radio frame | 2 400 |
| | Puncturing Limit | 0.88 |

6.10.2.4.1.45.2 Downlink

6.10.2.4.1.45.2.1 Transport channel parameters

6.10.2.4.1.45.2.1.1 Transport channel parameters for Conversational / speech / DL:12.2 kbps / CS RAB

See clause 6.10.2.4.1.4.2.1.1.

6.10.2.4.1.45.2.1.2 Transport channel parameters for Streaming / unknown / DL:57.6 kbps / CS RAB

See clause 6.10.2.4.1.17.2.1.1.

6.10.2.4.1.45.2.1.3 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.2.1.1.

6.10.2.4.1.45.2.1.4 TFCS

| | |
|---|---|
| TFCS size | 30 |
| TFCS | (RAB subflow#1, RAB subflow#2, RAB subflow#3, 57.6 kbps RAB , DCCH)=<br>(TF0, TF0, TF0, TF0, TF0), (TF1, TF0, TF0, TF0, TF0), (TF2, TF1, TF1, TF0, TF0),<br>(TF0, TF0, TF0, TF1, TF0), (TF1, TF0, TF0, TF1, TF0), (TF2, TF1, TF1, TF1, TF0),<br>(TF0, TF0, TF0, TF2, TF0), (TF1, TF0, TF0, TF2, TF0), (TF2, TF1, TF1, TF2, TF0),<br>(TF0, TF0, TF0, TF3, TF0), (TF1, TF0, TF0, TF3, TF0), (TF2, TF1, TF1, TF3, TF0),<br>(TF0, TF0, TF0, TF4, TF0), (TF1, TF0, TF0, TF4, TF0), (TF2, TF1, TF1, TF4, TF0),<br>(TF0, TF0, TF0, TF0, TF1), (TF1, TF0, TF0, TF0, TF1), (TF2, TF1, TF1, TF0, TF1),<br>(TF0, TF0, TF0, TF1, TF1), (TF1, TF0, TF0, TF1, TF1), (TF2, TF1, TF1, TF1, TF1),<br>(TF0, TF0, TF0, TF2, TF1), (TF1, TF0, TF0, TF2, TF1), (TF2, TF1, TF1, TF2, TF1),<br>(TF0, TF0, TF0, TF3, TF1), (TF1, TF0, TF0, TF3, TF1), (TF2, TF1, TF1, TF3, TF1),<br>(TF0, TF0, TF0, TF4, TF1), (TF1, TF0, TF0, TF4, TF1), (TF2, TF1, TF1, TF4, TF1) |

6.10.2.4.1.45.2.2 Physical channel parameters

| | | | |
|---|---|---|---|
| DPCH Downlink | DTX position | | Flexible |
| | Spreading factor | | 32 |
| | DPCCH | Number of TFCI bits/slot | 8 |
| | | Number of TPC bits/slot | 4 |
| | | Number of Pilot bits/slot | 8 |
| | DPDCH | Number of data bits/slot | 140 |
| | | Number of data bits/frame | 2 100 |

6.10.2.4.1.46 Void

6.10.2.4.1.47 Void

6.10.2.4.1.48 Void

6.10.2.4.1.49 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Conversational / unknown / UL:64 DL:64 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.2.4.1.49.1 Uplink

6.10.2.4.1.49.1.1 Transport channel parameters

6.10.2.4.1.49.1.1.1 Transport channel parameters for Conversational / speech / UL:12.2 kbps / CS RAB

See clause 6.10.2.4.1.4.1.1.1.

6.10.2.4.1.49.1.1.2 Transport channel parameters for Conversational / unknown / UL:64 kbps / CS RAB

See clause 6.10.2.4.1.13.1.1.1.

6.10.2.4.1.49.1.1.3 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.1.1.1.

6.10.2.4.1.49.1.1.4 TFCS

| | |
|---|---|
| TFCS size | 12 |
| TFCS | (RAB subflow#1, RAB subflow#2, RAB subflow#3, 64 kbps RAB , DCCH)=<br>(TF0, TF0, TF0, TF0, TF0), (TF1, TF0, TF0, TF0, TF0), (TF2, TF1, TF1, TF0, TF0),<br>(TF0, TF0, TF0, TF1, TF0), (TF1, TF0, TF0, TF1, TF0), (TF2, TF1, TF1, TF1, TF0),<br>(TF0, TF0, TF0, TF0, TF1), (TF1, TF0, TF0, TF0, TF1), (TF2, TF1, TF1, TF0, TF1),<br>(TF0, TF0, TF0, TF1, TF1), (TF1, TF0, TF0, TF1, TF1), (TF2, TF1, TF1, TF1, TF1) |

6.10.2.4.1.49.1.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Uplink | Min spreading factor | 16 |
| | Max number of DPDCH data bits/radio frame | 2 400 |

| | | |
|---|---|---|
| | Puncturing Limit | 0.72 |

6.10.2.4.1.49.2 Downlink

6.10.2.4.1.49.2.1 Transport channel parameters

6.10.2.4.1.49.2.1.1 Transport channel parameters for Conversational / speech / DL:12.2 kbps / CS RAB

See clause 6.10.2.4.1.4.2.1.1.

6.10.2.4.1.49.2.1.2 Transport channel parameters for Conversational / unknown / DL:64 kbps / CS RAB

See clause 6.10.2.4.1.13.2.1.1.

6.10.2.4.1.49.2.1.3 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.2.11.

6.10.2.4.1.49.2.1.4 TFCS

| | |
|---|---|
| TFCS size | 12 |
| TFCS | (RAB subflow#1, RAB subflow#2, RAB subflow#3, 64 kbps RAB , DCCH)=<br>(TF0, TF0, TF0, TF0, TF0), (TF1, TF0, TF0, TF0, TF0), (TF2, TF1, TF1, TF0, TF0),<br>(TF0, TF0, TF0, TF1, TF0), (TF1, TF0, TF0, TF1, TF0), (TF2, TF1, TF1, TF1, TF0),<br>(TF0, TF0, TF0, TF0, TF1), (TF1, TF0, TF0, TF0, TF1), (TF2, TF1, TF1, TF0, TF1),<br>(TF0, TF0, TF0, TF1, TF1), (TF1, TF0, TF0, TF1, TF1), (TF2, TF1, TF1, TF1, TF1) |

6.10.2.4.1.49.2.2 Physical channel parameters

| | | | |
|---|---|---|---|
| DPCH Downlink | DTX position | | Flexible |
| | Spreading factor | | 32 |
| | DPCCH | Number of TFCI bits/slot | 8 |
| | | Number of TPC bits/slot | 4 |
| | | Number of Pilot bits/slot | 8 |
| | DPDCH | Number of data bits/slot | 140 |
| | | Number of data bits/frame | 2 100 |

6.10.2.4.1.49a Conversational / speech / UL:(12.2 7.95 5.9 4.75) DL:(12.2 7.95 5.9 4.75) kbps / CS RAB + Conversational / unknown / UL:64 DL:64 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.2.4.1.49a.1 Uplink

6.10.2.4.1.49a.1.1 Transport channel parameters

6.10.2.4.1.49a.1.1.1 Transport channel parameters for Conversational / speech / UL: (12.2 7.95 5.9 4.75) kbps / CS RAB

See clause 6.10.2.4.1.4a.1.1.1.

6.10.2.4.1.49a.1.1.2 Transport channel parameters for Conversational / unknown / UL:64 kbps / CS RAB

See clause 6.10.2.4.1.13.1.1.1.

6.10.2.4.1.49a.1.1.3 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.1.1.1.

6.10.2.4.1.49a.1.1.4 TFCS

| | |
|---|---|
| TFCS size | 24 |
| TFCS | (RAB subflow#1, RAB subflow#2, RAB subflow#3, 64 kbps RAB, DCCH)=<br>(TF0,TF0,TF0,TF0,TF0), (TF1,TF0,TF0,TF0,TF0), (TF2,TF1,TF0,TF0,TF0),<br>(TF3,TF2,TF0,TF0,TF0), (TF4,TF3,TF0,TF0,TF0), (TF5,TF4,TF1,TF0,TF0),<br>(TF0,TF0,TF0,TF1,TF0), (TF1,TF0,TF0,TF1,TF0), (TF2,TF1,TF0,TF1,TF0), |

| | |
|---|---|
| | (TF3,TF2,TF0,TF1,TF0), (TF4,TF3,TF0,TF1,TF0), (TF5,TF4,TF1,TF1,TF0),<br>(TF0,TF0,TF0,TF0,TF1), (TF1,TF0,TF0,TF0,TF1), (TF2,TF1,TF0,TF0,TF1),<br>(TF3,TF2,TF0,TF0,TF1), (TF4,TF3,TF0,TF0,TF1), (TF5,TF4,TF1,TF0,TF1),<br>(TF0,TF0,TF0,TF1,TF1), (TF1,TF0,TF0,TF1,TF1), (TF2,TF1,TF0,TF1,TF1),<br>(TF3,TF2,TF0,TF1,TF1), (TF4,TF3,TF0,TF1,TF1), (TF5,TF4,TF1,TF1,TF1) |

6.10.2.4.1.49a.1.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Uplink | Min spreading factor | 16 |
| | Max number of DPDCH data bits/radio frame | 2 400 |
| | Puncturing Limit | 0.72 |

6.10.2.4.1.49a.2 Downlink

6.10.2.4.1.49a.2.1 Transport channel parameters

6.10.2.4.1.49a.2.1.1 Transport channel parameters for Conversational / speech / DL: (12.2 7.95 5.9 4.75) kbps / CS RAB

See clause 6.10.2.4.1.4a.2.1.1.

6.10.2.4.1.49a.2.1.2 Transport channel parameters for Conversational / unknown / DL:64 kbps / CS RAB

See clause 6.10.2.4.1.13.2.1.1.

6.10.2.4.1.49a.2.1.3 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.2.1.1

6.10.2.4.1.49a.2.1.4 TFCS

| | |
|---|---|
| TFCS size | 24 |
| TFCS | (RAB subflow#1, RAB subflow#2, RAB subflow#3, 64 kbps RAB, DCCH)=<br>(TF0,TF0,TF0,TF0,TF0), (TF1,TF0,TF0,TF0,TF0), (TF2,TF1,TF0,TF0,TF0),<br>(TF3,TF2,TF0,TF0,TF0), (TF4,TF3,TF0,TF0,TF0), (TF5,TF4,TF1,TF0,TF0),<br>(TF0,TF0,TF0,TF1,TF0), (TF1,TF0,TF0,TF1,TF0), (TF2,TF1,TF0,TF1,TF0),<br>(TF3,TF2,TF0,TF1,TF0), (TF4,TF3,TF0,TF1,TF0), (TF5,TF4,TF1,TF1,TF0),<br>(TF0,TF0,TF0,TF0,TF1), (TF1,TF0,TF0,TF0,TF1), (TF2,TF1,TF0,TF0,TF1),<br>(TF3,TF2,TF0,TF0,TF1), (TF4,TF3,TF0,TF0,TF1), (TF5,TF4,TF1,TF0,TF1),<br>(TF0,TF0,TF0,TF1,TF1), (TF1,TF0,TF0,TF1,TF1), (TF2,TF1,TF0,TF1,TF1),<br>(TF3,TF2,TF0,TF1,TF1), (TF4,TF3,TF0,TF1,TF1), (TF5,TF4,TF1,TF1,TF1) |

6.10.2.4.1.49a.2.2 Physical channel parameters

| | | | |
|---|---|---|---|
| DPCH Downlink | DTX position | | Flexible |
| | Spreading factor | | 32 |
| | DPCCH | Number of TFCI bits/slot | 8 |
| | | Number of TPC bits/slot | 4 |
| | | Number of Pilot bits/slot | 8 |
| | DPDCH | Number of data bits/slot | 140 |
| | | Number of data bits/frame | 2 100 |

6.10.2.4.1.50 Conversational / unknown / UL:64 DL:64 kbps / CS RAB + Conversational / unknown / UL:64 DL:64 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.2.4.1.50.1 Uplink

6.10.2.4.1.50.1.1 Transport channel parameters

6.10.2.4.1.50.1.1.1 Transport channel parameters for Conversational / unknown / UL:64 kbps / CS RAB

See clause 6.10.2.4.1.13.1.1.1.

6.10.2.4.1.50.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.1.1.1.

6.10.2.4.1.50.1.1.3 TFCS

| | |
|---|---|
| TFCS size | 8 |
| TFCS | (64 kbps RAB, 64 kbps RAB, DCCH)=<br>(TF0, TF0, TF0), (TF1, TF0, TF0), (TF0, TF1, TF0), (TF1, TF1, TF0)<br>(TF0, TF0, TF1), (TF1, TF0, TF1), (TF0, TF1, TF1), (TF1, TF1, TF1) |

6.10.2.4.1.50.1.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Uplink | Min spreading factor | 8 |
| | Max number of DPDCH data bits/radio frame | 4 800 |
| | Puncturing Limit | 0.92 |

6.10.2.4.1.50.2 Downlink

6.10.2.4.1.50.2.1 Transport channel parameters

6.10.2.4.1.50.2.1.1 Transport channel parameters for Conversational / unknown / DL:64 kbps / CS RAB

See clause 6.10.2.4.1.13.2.1.1.

6.10.2.4.1.50.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.2.1.1.

6.10.2.4.1.50.2.1.3 TFCS

| | |
|---|---|
| TFCS size | 8 |
| TFCS | (64 kbps RAB, 64 kbps RAB, DCCH)=<br>(TF0, TF0, TF0), (TF1, TF0, TF0), (TF0, TF1, TF0), (TF1, TF1, TF0)<br>(TF0, TF0, TF1), (TF1, TF0, TF1), (TF0, TF1, TF1), (TF1, TF1, TF1) |

6.10.2.4.1.50.2.2 Physical channel parameters

| | | | |
|---|---|---|---|
| DPCH Downlink | DTX position | | Flexible |
| | Spreading factor | | 16 |
| | DPCCH | Number of TFCI bits/slot | 8 |
| | | Number of TPC bits/slot | 8 |
| | | Number of Pilot bits/slot | 16 |
| | DPDCH | Number of data bits/slot | 288 |
| | | Number of data bits/frame | 4 320 |

6.10.2.4.1.51 Conversational / unknown / UL:64 DL:64 kbps / CS RAB + Interactive or background / UL:64 DL:64 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.2.4.1.51.1 Uplink

6.10.2.4.1.51.1.1 Transport channel parameters

6.10.2.4.1.51.1.1.1 Transport channel parameters for Conversational / unknown / UL:64 kbps / CS RAB

See clause 6.10.2.4.1.13.1.1.1.

6.10.2.4.1.51.1.1.2 Transport channel parameters for Interactive or background / UL:64 kbps / PS RAB

See clause 6.10.2.4.1.26.1.1.1.

6.10.2.4.1.51.1.1.3 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.1.1.1.

6.10.2.4.1.51.1.1.4 TFCS

| | |
|---|---|
| TFCS size | 20 |
| TFCS | (Conv. 64 kbps RAB, I/B 64 kbps RAB, DCCH)=<br>(TF0, TF0, TF0), (TF0, TF1, TF0), (TF0, TF2, TF0), (TF0, TF3, TF0), (TF0, TF4, TF0),<br>(TF1, TF0, TF0), (TF1, TF1, TF0), (TF1, TF2, TF0), (TF1, TF3, TF0), (TF1, TF4, TF0),<br>(TF0, TF0, TF1), (TF0, TF1, TF1), (TF0, TF2, TF1), (TF0, TF3, TF1), (TF0, TF4, TF1),<br>(TF1, TF0, TF1), (TF1, TF1, TF1), (TF1, TF2, TF1), (TF1, TF3, TF1), (TF1, TF4, TF1) |

6.10.2.4.1.51.1.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Uplink | Min spreading factor | 8 |
| | Max number of DPDCH data bits/radio frame | 4 800 |
| | Puncturing Limit | 0.88 |

6.10.2.4.1.51.2 Downlink

6.10.2.4.1.51.2.1 Transport channel parameters

6.10.2.4.1.51.2.1.1 Transport channel parameters for Conversational / unknown / DL:64 kbps / CS RAB

See clause 6.10.2.4.1.13.2.1.1.

6.10.2.4.1.51.2.1.2 Transport channel parameters for Interactive or background / DL:64 kbps / PS RAB

See clause 6.10.2.4.1.25.2.1.1.

6.10.2.4.1.51.2.1.3 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.2.1.1.

6.10.2.4.1.51.2.1.4 TFCS

| | |
|---|---|
| TFCS size | 20 |
| TFCS | (Conv. 64 kbps RAB, I/B 64 kbps RAB, DCCH)=<br>(TF0, TF0, TF0), (TF0, TF1, TF0), (TF0, TF2, TF0), (TF0, TF3, TF0), (TF0, TF4, TF0),<br>(TF1, TF0, TF0), (TF1, TF1, TF0), (TF1, TF2, TF0), (TF1, TF3, TF0), (TF1, TF4, TF0),<br>(TF0, TF0, TF1), (TF0, TF1, TF1), (TF0, TF2, TF1), (TF0, TF3, TF1), (TF0, TF4, TF1),<br>(TF1, TF0, TF1), (TF1, TF1, TF1), (TF1, TF2, TF1), (TF1, TF3, TF1), (TF1, TF4, TF1) |

6.10.2.4.1.51.2.2 Physical channel parameters

| | | | |
|---|---|---|---|
| DPCH Downlink | DTX position | | Flexible |
| | Spreading factor | | 16 |
| | DPCCH | Number of TFCI bits/slot | 8 |
| | | Number of TPC bits/slot | 8 |
| | | Number of Pilot bits/slot | 16 |
| | DPDCH | Number of data bits/slot | 288 |
| | | Number of data bits/frame | 4 320 |

6.10.2.4.1.51a Conversational / unknown / UL:64 DL:64 kbps / CS RAB + Interactive or Background / UL:8 DL:8 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.2.4.1.51a.1 Uplink

6.10.2.4.1.51a.1.1 Transport channel parameters

6.10.2.4.1.51a.1.1.1 Transport channel parameters for Conversational / unknown / UL:64 kbps / CS RAB

See clause 6.10.2.4.1.13.1.1.1.

6.10.2.4.1.51a.1.1.2 Transport channel parameters for Interactive or Background / UL:8 kbps / PS RAB

See clause 6.10.2.4.1.38b.1.1.2.

6.10.2.4.1.51a.1.1.3 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.1.1.1.

6.10.2.4.1.51a.1.1.4 TFCS

| | |
|---|---|
| TFCS size | 8 |
| TFCS | (64 kbps Conversational RAB, 8 kbps I/B RAB, DCCH)=<br>(TF0, TF0, TF0), (TF1, TF0, TF0), (TF0, TF1, TF0), (TF1, TF1, TF0),<br>(TF0, TF0, TF1), (TF1, TF0, TF1), (TF0, TF1, TF1), (TF1, TF1, TF1) |

6.10.2.4.1.51a.1.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Uplink | Min spreading factor | 16 |
| | Max number of DPDCH data bits/radio frame | 2 400 |
| | Puncturing Limit | 0.72 |

6.10.2.4.1.51a.2 Downlink

6.10.2.4.1.51a.2.1 Transport channel parameters

6.10.2.4.1.51a.2.1.1 Transport channel parameters for Conversational / unknown / DL:64 kbps / PS RAB

See clause 6.10.2.4.1.13.2.1.1.

6.10.2.4.1.51a.2.1.2 Transport channel parameters for Interactive or Background / DL:8 kbps / PS RAB

See clause 6.10.2.4.1.38b.2.1.2.

6.10.2.4.1.51a.2.1.3 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.2.1.1.

6.10.2.4.1.51a.2.1.4 TFCS

| | |
|---|---|
| TFCS size | 8 |
| TFCS | (64 kbps Conversational RAB, 8 kbps I/B RAB, DCCH)=<br>(TF0, TF0, TF0), (TF1, TF0, TF0), (TF0, TF1, TF0), (TF1, TF1, TF0),<br>(TF0, TF0, TF1), (TF1, TF0, TF1), (TF0, TF1, TF1), (TF1, TF1, TF1) |

6.10.2.4.1.51a.2.2 Physical channel parameters

| | | | |
|---|---|---|---|
| DPCH Downlink | DTX position | | Flexible |
| | Spreading factor | | 32 |
| | DPCCH | Number of TFCI bits/slot | 8 |
| | | Number of TPC bits/slot | 4 |
| | | Number of Pilot bits/slot | 8 |
| | DPDCH | Number of data bits/slot | 140 |
| | | Number of data bits/frame | 2 100 |

6.10.2.4.1.51b Conversational / unknown / UL:64 DL:64 kbps / CS RAB + Interactive or Background / UL:16 DL:64 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.2.4.1.51b.1 Uplink

6.10.2.4.1.51b.1.1 Transport channel parameters

6.10.2.4.1.51b.1.1.1 Transport channel parameters for Conversational / unknown / UL:64 kbps / CS RAB

See clause 6.10.2.4.1.13.1.1.1.

6.10.2.4.1.51b.1.1.2 Transport channel parameters for Interactive or Background / UL:16 kbps / PS RAB

| | | |
|---|---|---|
| Higher layer | RAB/Signalling RB | RAB |

| | | | |
|---|---|---|---|
| RLC | Logical channel type | | DTCH |
| | RLC mode | | AM |
| | Payload sizes, bit | | 320 |
| | Max data rate, bps | | 16 000 |
| | AMD PDU header, bit | | 16 |
| MAC | MAC header, bit | | 0 |
| | MAC multiplexing | | N/A |
| Layer 1 | TrCH type | | DCH |
| | TB sizes, bit | | 336 |
| | TFS | TF0, bits | 0x336 |
| | | TF1, bits | 1x336 |
| | | TF2, bits | 2x336 |
| | TTI, ms | | 40 |
| | Coding type | | TC |
| | CRC, bit | | 16 |
| | Max number of bits/TTI after channel coding | | 2 124 |
| | Uplink: Max number of bits/radio frame before rate matching | | 531 |
| | RM attribute | | 135 to 175 |

6.10.2.4.1.51b.1.1.3 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.1.1.1.

6.10.2.4.1.51b.1.1.4 TFCS

| | |
|---|---|
| TFCS size | 12 |
| TFCS | (64 kbps Conversational RAB, 16 kbps I/B RAB, DCCH)=<br>(TF0, TF0, TF0), (TF0, TF1, TF0), (TF0, TF2, TF0), (TF1, TF0, TF0), (TF1, TF1, TF0), (TF1, TF2, TF0), (TF0, TF0, TF1), (TF0, TF1, TF1), (TF0, TF2, TF1), (TF1, TF0, TF1), (TF1, TF1, TF1), (TF1, TF2, TF1) |

6.10.2.4.1.51b.1.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Uplink | Min spreading factor | 16 |
| | Max number of DPDCH data bits/radio frame | 2 400 |
| | Puncturing Limit | 0.64 |

6.10.2.4.1.51b.2 Downlink

See clause 6.10.2.4.1.51.2.

6.10.2.4.1.52 Conversational / unknown / UL:64 DL:64 kbps / CS RAB + Interactive or background / UL:64 DL:128 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.2.4.1.52.1 Uplink

See clause 6.10.2.4.1.51.1.

6.10.2.4.1.52.2 Downlink

6.10.2.4.1.52.2.1 Transport channel parameters

6.10.2.4.1.52.2.1.1 Transport channel parameters for Conversational / unknown / DL:64 kbps / CS RAB

See clause 6.10.2.4.1.13.2.1.1.

6.10.2.4.1.52.2.1.2 Transport channel parameters for Interactive or background / DL:128 kbps / PS RAB

See clause 6.10.2.4.1.27.2.1.1.

6.10.2.4.1.52.2.1.3 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.2.1.1.

6.10.2.4.1.52.2.1.4 TFCS

| | |
|---|---|
| TFCS size | 20 |
| TFCS | (Conv. 64 kbps RAB, I/B 128 kbps RAB, DCCH)=<br>(TF0, TF0, TF0), (TF0, TF1, TF0), (TF0, TF2, TF0), (TF0, TF3, TF0), (TF0, TF4, TF0),<br>(TF1, TF0, TF0), (TF1, TF1, TF0), (TF1, TF2, TF0), (TF1, TF3, TF0), (TF1, TF4, TF0),<br>(TF0, TF0, TF1), (TF0, TF1, TF1), (TF0, TF2, TF1), (TF0, TF3, TF1), (TF0, TF4, TF1),<br>(TF1, TF0, TF1), (TF1, TF1, TF1), (TF1, TF2, TF1), (TF1, TF3, TF1), (TF1, TF4, TF1) |

6.10.2.4.1.52.2.2 Physical channel parameters

| | | | |
|---|---|---|---|
| DPCH Downlink | DTX position | | Flexible |
| | Spreading factor | | 8 |
| | DPCCH | Number of TFCI bits/slot | 8 |
| | | Number of TPC bits/slot | 8 |
| | | Number of Pilot bits/slot | 16 |
| | DPDCH | Number of data bits/slot | 608 |
| | | Number of data bits/frame | 9 120 |

6.10.2.4.1.53 Conversational / unknown / UL:64 DL:64 kbps / CS RAB + Interactive or background / UL:128 DL:128 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.2.4.1.53.1 Uplink

6.10.2.4.1.53.1.1 Transport channel parameters

6.10.2.4.1.53.1.1.1 Transport channel parameters for Conversational / unknown / UL:64 kbps / CS RAB

See clause 6.10.2.4.1.13.1.1.1.

6.10.2.4.1.53.1.1.2 Transport channel parameters for Interactive or background / UL:128 kbps / PS RAB

See clause 6.10.2.4.1.28.1.1.1.

6.10.2.4.1.53.1.1.3 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.1.1.1.

6.10.2.4.1.53.1.1.4 TFCS

| | |
|---|---|
| TFCS size | 20 |
| TFCS | (Conv. 64 kbps RAB, I/B 128kbps RAB, DCCH)=<br>(TF0, TF0, TF0), (TF0, TF1, TF0), (TF0, TF2, TF0), (TF0, TF3, TF0), (TF0, TF4, TF0),<br>(TF1, TF0, TF0), (TF1, TF1, TF0), (TF1, TF2, TF0), (TF1, TF3, TF0), (TF1, TF4, TF0),<br>(TF0, TF0, TF1), (TF0, TF1, TF1), (TF0, TF2, TF1), (TF0, TF3, TF1), (TF0, TF4, TF1),<br>(TF1, TF0, TF1), (TF1, TF1, TF1), (TF1, TF2, TF1), (TF1, TF3, TF1), (TF1, TF4, TF1) |

6.10.2.4.1.53.1.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Uplink | Min spreading factor | 4 |
| | Max number of DPDCH data bits/radio frame | 9 600 |
| | Puncturing Limit | 0.96 |

6.10.2.4.1.53.2 Downlink

See clause 6.10.2.4.1.52.2.

6.10.2.4.1.54 Void

6.10.2.4.1.55 Void

6.10.2.4.1.56 Interactive or background / UL:8 DL:8 kbps / PS RAB + Interactive or background / UL:8 DL:8 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.2.4.1.56.1 Uplink

6.10.2.4.1.56.1.1 Transport channel parameters

6.10.2.4.1.56.1.1.1 Transport channel parameters for Interactive or background / UL:8 kbps / PS RAB + UL:8 kbps / PS RAB

| | | | | |
|---|---|---|---|---|
| Higher layer | RAB/Signalling RB | | RAB | RAB |
| RLC | Logical channel type | | DTCH | DTCH |
| | RLC mode | | AM | AM |
| | Payload sizes, bit | | 320 | 320 |
| | Max data rate, bps | | 8 000 | 8 000 |
| | AMD PDU header, bit | | 16 | 16 |
| MAC | MAC header, bit | | 4 | 4 |
| | MAC multiplexing | | 2 logical channel multiplexing | |
| Layer 1 | TrCH type | | DCH | |
| | TB sizes, bit | | 340 | |
| | TFS | TF0, bits | 0x340 | |
| | | TF1, bits | 1x340 | |
| | TTI, ms | | 40 | |
| | Coding type | | TC | |
| | CRC, bit | | 16 | |
| | Max number of bits/TTI after channel coding | | 1 080 | |
| | Uplink: Max number of bits/radio frame before rate matching | | 270 | |
| | RM attribute | | 135 to 175 | |

6.10.2.4.1.56.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.1.1.1.

6.10.2.4.1.56.1.1.3 TFCS

| | |
|---|---|
| TFCS size | 4 |
| TFCS | (8 kbps RAB + 8 kbps RAB, DCCH)=<br>(TF0,TF0), (TF1,TF0), (TF0,TF1), (TF1,TF1) |

6.10.2.4.1.56.1.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Uplink | Min spreading factor | 64 |
| | Max number of DPDCH data bits/radio frame | 600 |
| | Puncturing Limit | 1.0 |

6.10.2.4.1.56.2 Downlink

6.10.2.4.1.56.2.1 Transport channel parameters

6.10.2.4.1.56.2.1.1 Transport channel parameters for Interactive or background / DL:8 kbps / PS RAB + DL:8 kbps / PS RAB

| | | | | |
|---|---|---|---|---|
| Higher layer | RAB/Signalling RB | | RAB | RAB |
| RLC | Logical channel type | | DTCH | DTCH |
| | RLC mode | | AM | AM |
| | Payload sizes, bit | | 320 | 320 |
| | Max data rate, bps | | 8 000 | 8 000 |
| | AMD PDU header, bit | | 16 | 16 |
| MAC | MAC header, bit | | 4 | 4 |
| | MAC multiplexing | | 2 logical channel multiplexing | |
| Layer 1 | TrCH type | | DCH | |
| | TB sizes, bit | | 340 | |
| | TFS | TF0, bits | 0x340 | |
| | | TF1, bits | 1x340 | |
| | TTI, ms | | 40 | |
| | Coding type | | TC | |
| | CRC, bit | | 16 | |

| | | |
|---|---|---|
| | Max number of bits/TTI after channel coding | 1 080 |
| | RM attribute | 135 to 175 |

6.10.2.4.1.56.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.2.1.1.

6.10.2.4.1.56.2.1.3 TFCS

| | |
|---|---|
| TFCS size | 4 |
| TFCS | (8 kbps RAB + 8 kbps RAB, DCCH)=<br>(TF0,TF0), (TF1,TF0), (TF0,TF1), (TF1,TF1) |

6.10.2.4.1.56.2.2 Physical channel parameters

| | | | |
|---|---|---|---|
| DPCH Downlink | DTX position | | Flexible |
| | Spreading factor | | 128 |
| | DPCCH | Number of TFCI bits/slot | 2 |
| | | Number of TPC bits/slot | 2 |
| | | Number of Pilot bits/slot | 4 |
| | DPDCH | Number of data bits/slot | 32 |
| | | Number of data bits/frame | 480 |

6.10.2.4.1.57 Interactive or background / UL:64 DL:64 kbps / PS RAB + Interactive or background / UL:64 DL:64 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.2.4.1.57.1 Uplink

6.10.2.4.1.57.1.1 Transport channel parameters

6.10.2.4.1.57.1.1.1 Transport channel parameters for Interactive or background / UL:64 kbps / PS RAB + UL:64 kbps / PS RAB

| | | | | |
|---|---|---|---|---|
| Higher layer | RAB/Signalling RB | | RAB | RAB |
| RLC | Logical channel type | | DTCH | DTCH |
| | RLC mode | | AM | AM |
| | Payload sizes, bit | | 320 | 320 |
| | Max data rate, bps | | 64 000 | 64 000 |
| | AMD PDU header, bit | | 16 | 16 |
| MAC | MAC header, bit | | 4 | 4 |
| | MAC multiplexing | | 2 logical channel multiplexing | |
| Layer 1 | TrCH type | | DCH | |
| | TB sizes, bit | | 340 | |
| | TFS | TF0, bits | 0x340 | |
| | | TF1, bits | 1x340 | |
| | | TF2, bits | 2x340 | |
| | | TF3, bits | 3x340 | |
| | | TF4, bits | 4x340 | |
| | TTI, ms | | 20 | |
| | Coding type | | TC | |
| | CRC, bit | | 16 | |
| | Max number of bits/TTI after channel coding | | 4 284 | |
| | Uplink: Max number of bits/radio frame before rate matching | | 2 142 | |
| | RM attribute | | 130 to 170 | |

6.10.2.4.1.57.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.1.1.1.

6.10.2.4.1.57.1.1.3 TFCS

| | |
|---|---|
| TFCS size | 10 |
| TFCS | (64 kbps RAB + 64 kbps RAB, DCCH)= |

| | |
|---|---|
| | (TF0,TF0), (TF1,TF0), (TF2,TF0), (TF3,TF0), (TF4,TF0),<br>(TF0,TF1), (TF1,TF1), (TF2,TF1), (TF3,TF1), (TF4,TF1) |

6.10.2.4.1.57.1.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Uplink | Min spreading factor | 16 |
| | Max number of DPDCH data bits/radio frame | 2 400 |
| | Puncturing Limit | 0.92 |

6.10.2.4.1.57.2 Downlink

6.10.2.4.1.57.2.1 Transport channel parameters

6.10.2.4.1.57.2.1.1 Transport channel parameters for Interactive or background / DL:64 kbps / PS RAB + DL:64 kbps / PS RAB

| | | | | |
|---|---|---|---|---|
| Higher layer | RAB/Signalling RB | | RAB | RAB |
| RLC | Logical channel type | | DTCH | DTCH |
| | RLC mode | | AM | AM |
| | Payload sizes, bit | | 320 | 320 |
| | Max data rate, bps | | 64 000 | 64 000 |
| | AMD PDU header, bit | | 16 | 16 |
| MAC | MAC header, bit | | 4 | 4 |
| | MAC multiplexing | | 2 logical channel multiplexing | |
| Layer 1 | TrCH type | | DCH | |
| | TB sizes, bit | | 340 | |
| | TFS | TF0, bits | 0x340 | |
| | | TF1, bits | 1x340 | |
| | | TF2, bits | 2x340 | |
| | | TF3, bits | 3x340 | |
| | | TF4, bits | 4x340 | |
| | TTI, ms | | 20 | |
| | Coding type | | TC | |
| | CRC, bit | | 16 | |
| | Max number of bits/TTI after channel coding | | 4 284 | |
| | RM attribute | | 130 to 170 | |

6.10.2.4.1.57.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.2.1.1.

6.10.2.4.1.57.2.1.3 TFCS

| | |
|---|---|
| TFCS size | 10 |
| TFCS | (64 kbps RAB + 64 kbps RAB, DCCH)=<br>(TF0,TF0), (TF1,TF0), (TF2,TF0), (TF3,TF0), (TF4,TF0),<br>(TF0,TF1), (TF1,TF1), (TF2,TF1), (TF3,TF1), (TF4,TF1) |

6.10.2.4.1.57.2.2 Physical channel parameters

| | | | |
|---|---|---|---|
| DPCH Downlink | DTX position | | Flexible |
| | Spreading factor | | 32 |
| | DPCCH | Number of TFCI bits/slot | 8 |
| | | Number of TPC bits/slot | 4 |
| | | Number of Pilot bits/slot | 8 |
| | DPDCH | Number of data bits/slot | 140 |
| | | Number of data bits/frame | 2 100 |

6.10.2.4.1.58 Streaming / unknown / UL:16 DL:64 kbps / PS RAB + Interactive or background / UL:8 DL:8 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.2.4.1.58.1 Uplink

6.10.2.4.1.58.1.1 Transport channel parameters

6.10.2.4.1.58.1.1.1 Transport channel parameters for Streaming / unknown / UL:16 kbps / PS RAB

| | | | |
|---|---|---|---|
| Higher layer | RAB/Signalling RB | | RAB |
| RLC | Logical channel type | | DTCH |
| | RLC mode | | AM |
| | Payload sizes, bit | | 320 |
| | Max data rate, bps | | 16 000 |
| | AMD PDU header, bit | | 16 |
| MAC | MAC header, bit | | 0 |
| | MAC multiplexing | | N/A |
| Layer 1 | TrCH type | | DCH |
| | TB sizes, bit | | 336 |
| | TFS | TF0, bits | 0x336 |
| | | TF1, bits | 1x336 |
| | TTI, ms | | 20 |
| | Coding type | | TC |
| | CRC, bit | | 16 |
| | Max number of bits/TTI after channel coding | | 1 068 |
| | Uplink: Max number of bits/radio frame before rate matching | | 534 |
| | RM attribute | | 135 to 175 |

6.10.2.4.1.58.1.1.2 Transport channel parameters for Interactive or background / UL:8 kbps / PS RAB

See clause 6.10.2.4.1.38b.1.1.2.

6.10.2.4.1.58.1.1.3 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.1.1.1.

6.10.2.4.1.58.1.1.4 TFCS

| | |
|---|---|
| TFCS size | 8 |
| TFCS | (16 kbps RAB, 8 kbps RAB, DCCH)=<br>(TF0,TF0,TF0), (TF1,TF0,TF0), (TF0,TF1,TF0), (TF1,TF1,TF0),<br>(TF0,TF0,TF1), (TF1,TF0,TF1), (TF0,TF1,TF1), (TF1,TF1,TF1) |

6.10.2.4.1.58.1.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Uplink | Min spreading factor | 32 |
| | Max number of DPDCH data bits/radio frame | 1 200 |
| | Puncturing Limit | 1.0 |

6.10.2.4.1.58.2 Downlink

6.10.2.4.1.58.2.1 Transport channel parameters

6.10.2.4.1.58.2.1.1 Transport channel parameters for Streaming / unknown / DL:64 kbps / PS RAB

| | | |
|---|---|---|
| Higher layer | RAB/Signalling RB | RAB |
| RLC | Logical channel type | DTCH |
| | RLC mode | AM |
| | Payload sizes, bit | 640 |
| | Max data rate, bps | 64 000 |
| | AM PDU header, bit | 16 |
| MAC | MAC header, bit | 0 |
| | MAC multiplexing | N/A |
| Layer 1 | TrCH type | DCH |

| | | | |
|---|---|---|---|
| | TB sizes, bit | | 656 |
| | TFS | TF0, bits | 0x656 |
| | | TF1, bits | 1x656 |
| | | TF2, bits | 2x656 |
| | | TF3, bits | 4x656 |
| | TTI, ms | | 40 |
| | Coding type | | TC |
| | CRC, bit | | 16 |
| | Max number of bits/TTI after channel coding | | 8 076 |
| | RM attribute | | 125 to 165 |

6.10.2.4.1.58.2.1.2 Transport channel parameters for Interactive or background / DL:8 kbps / PS RAB

See clause 6.10.2.4.1.38b.2.1.2.

6.10.2.4.1.58.2.1.3 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.2.1.1.

6.10.2.4.1.58.2.1.4 TFCS

| | |
|---|---|
| TFCS size | 16 |
| TFCS | (64 kbps RAB, 8 kbps RAB, DCCH)=<br>(TF0,TF0,TF0), (TF1,TF0,TF0), (TF2,TF0,TF0), (TF3,TF0,TF0),<br>(TF0,TF1,TF0), (TF1,TF1,TF0), (TF2,TF1,TF0), (TF3,TF1,TF0),<br>(TF0,TF0,TF1), (TF1,TF0,TF1), (TF2,TF0,TF1), (TF3,TF0,TF1),<br>(TF0,TF1,TF1), (TF1,TF1,TF1), (TF2,TF1,TF1), (TF3,TF1,TF1) |

6.10.2.4.1.58.2.2 Physical channel parameters

| | | | |
|---|---|---|---|
| DPCH Downlink | DTX position | | Flexible |
| | Spreading factor | | 32 |
| | DPCCH | Number of TFCI bits/slot | 8 |
| | | Number of TPC bits/slot | 4 |
| | | Number of Pilot bits/slot | 8 |
| | DPDCH | Number of data bits/slot | 140 |
| | | Number of data bits/frame | 2 100 |

6.10.2.4.1.58a Streaming / unknown / UL:16 DL:128 kbps / PS RAB + Interactive or background / UL:8 DL:8 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.2.4.1.58a.1 Uplink

6.10.2.4.1.58a.1.1 Transport channel parameters

6.10.2.4.1.58a.1.1.1 Transport channel parameters for Streaming / unknown / UL:16 kbps / PS RAB

See clause 6.10.2.4.1.58.1.1.1.

6.10.2.4.1.58a.1.1.2 Transport channel parameters for Interactive or background / UL:8 kbps / PS RAB

See clause 6.10.2.4.1.38b.1.1.2.

6.10.2.4.1.58a.1.1.3 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.1.1.1.

6.10.2.4.1.58a.1.1.4 TFCS

See clause 6.10.2.4.1.58.1.1.4.

6.10.2.4.1.58a.1.2 Physical channel parameters

See clause 6.10.2.4.1.58.1.2.

6.10.2.4.1.58a.2 Downlink

6.10.2.4.1.58a.2.1 Transport channel parameters

6.10.2.4.1.58a.2.1.1 Transport channel parameters for Streaming / unknown / DL:128 kbps / PS RAB

| | | | |
|---|---|---|---|
| Higher layer | RAB/Signalling RB | | RAB |
| RLC | Logical channel type | | DTCH |
| | RLC mode | | AM |
| | Payload sizes, bit | | 640 |
| | Max data rate, bps | | 128 000 |
| | AM PDU header, bit | | 16 |
| MAC | MAC header, bit | | 0 |
| | MAC multiplexing | | N/A |
| Layer 1 | TrCH type | | DCH |
| | TB sizes, bit | | 656 |
| | TFS | TF0, bits | 0x656 |
| | | TF1, bits | 1x656 |
| | | TF2, bits | 2x656 |
| | | TF3, bits | 3x656 |
| | | TF4, bits | 4x656 |
| | TTI, ms | | 20 |
| | Coding type | | TC |
| | CRC, bit | | 16 |
| | Max number of bits/TTI after channel coding | | 8 076 |
| | RM attribute | | 125 to 165 |

6.10.2.4.1.58a.2.1.2 Transport channel parameters for Interactive or background / DL:8 kbps / PS RAB

See clause 6.10.2.4.1.38b.2.1.2.

6.10.2.4.1.58a.2.1.3 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.2.1.1.

6.10.2.4.1.58a.2.1.4 TFCS

| | |
|---|---|
| TFCS size | 20 |
| TFCS | (128 kbps RAB, 8 kbps RAB, DCCH)=<br>(TF0,TF0,TF0), (TF1,TF0,TF0), (TF2,TF0,TF0), (TF3,TF0,TF0), (TF4,TF0,TF0),<br>(TF0,TF1,TF0), (TF1,TF1,TF0), (TF2,TF1,TF0), (TF3,TF1,TF0), (TF4,TF1,TF0),<br>(TF0,TF0,TF1), (TF1,TF0,TF1), (TF2,TF0,TF1), (TF3,TF0,TF1), (TF4,TF0,TF1),<br>(TF0,TF1,TF1), (TF1,TF1,TF1), (TF2,TF1,TF1), (TF3,TF1,TF1), (TF4,TF1,TF1), |

6.10.2.4.1.58a.2.2 Physical channel parameters

| | | | |
|---|---|---|---|
| DPCH Downlink | DTX position | | Flexible |
| | Spreading factor | | 16 |
| | DPCCH | Number of TFCI bits/slot | 8 |
| | | Number of TPC bits/slot | 8 |
| | | Number of Pilot bits/slot | 16 |
| | DPDCH | Number of data bits/slot | 288 |
| | | Number of data bits/frame | 4 320 |

6.10.2.4.1.59 Conversational / speech / UL:42.8 DL:42.8 kbps / PS RAB + Interactive / UL:16 DL:16 kbps / PS RAB + Interactive / UL:16 DL:16 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.2.4.1.59.1 Uplink

6.10.2.4.1.59.1.1 Transport channel parameters

6.10.2.4.1.59.1.1.1 Transport channel parameters for Conversational / speech / UL:42.8 kbps / PS RAB

| | | | |
|---|---|---|---|
| Higher layer | RAB/Signalling RB | | RAB |
| PDCP | PDCP header size, bit | | 8 |
| RLC | Logical channel type | | DTCH |
| | RLC mode | | UM |
| | Payload sizes, bit | | 920, 304, 96 |
| | Max data rate, bps | | 46 000 |
| | UMD PDU header, bit | | 8 |
| MAC | MAC header, bit | | 0 |
| | MAC multiplexing | | N/A |
| Layer 1 | TrCH type | | DCH |
| | TB sizes, bit | | 928, 312, 104 |
| | TFS | TF0, bits | 0x928 |
| | | TF1, bits | 1x104 |
| | | TF2, bits | 1x312 |
| | | TF3, bits | 1x928 |
| | TTI, ms | | 20 |
| | Coding type | | TC |
| | CRC, bit | | 16 |
| | Max number of bits/TTI after channel coding | | 2 844 |
| | Uplink: Max number of bits/radio frame before rate matching | | 1 422 |
| | RM attribute | | 180 to 220 |

6.10.2.4.1.59.1.1.2 Transport channel parameters for Interactive / UL:16kbps / PS RAB + UL:16 kbps / PS RAB

| | | | | |
|---|---|---|---|---|
| Higher layer | RAB/Signalling RB | | RAB | RAB |
| RLC | Logical channel type | | DTCH | DTCH |
| | RLC mode | | AM | AM |
| | Payload sizes, bit | | 320 | 320 |
| | Max data rate, bps | | 16 000 | 16 000 |
| | AMD PDU header, bit | | 16 | 16 |
| MAC | MAC header, bit | | 4 | 4 |
| | MAC multiplexing | | 2 logical channel multiplexing | |
| Layer 1 | TrCH type | | DCH | |
| | TB sizes, bit | | 340 | |
| | TFS | TF0, bits | 0x340 | |
| | | TF1, bits | 1x340 | |
| | | TF2, bits | 2X340 | |
| | TTI, ms | | 40 | |
| | Coding type | | TC | |
| | CRC, bit | | 16 | |
| | Max number of bits/TTI after channel coding | | 2 148 | |
| | Uplink: Max number of bits/radio frame before rate matching | | 537 | |
| | RM attribute | | 135 to 175 | |

6.10.2.4.1.59.1.1.3 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.1.1.1

6.10.2.4.1.59.1.1.4 TFCS

| | |
|---|---|
| TFCS size | 24 |
| TFCS | (42.8 kbps Conversational RAB, Interactive 16kbps+16kbps RAB, DCCH)=<br>(TF0, TF0, TF0), (TF0, TF0, TF1), (TF0,TF1, TF0),(TF0, TF1,TF1), (TF0,TF2, TF0), (TF0,TF2, TF1)<br>(TF1, TF0, TF0), (TF1, TF0, TF1), (TF1,TF1, TF0), (TF1, TF1,TF1), (TF1,TF2, TF0), (TF1,TF2, TF1)<br>(TF2, TF0, TF0), (TF2, TF0, TF1), (TF2,TF1, TF0), (TF2, TF1,TF1), (TF2,TF2, TF0), (TF2,TF2, TF1)<br>(TF3, TF0, TF0), (TF3, TF0, TF1), (TF3,TF1, TF0), (TF3, TF1,TF1), (TF3,TF2, TF0), (TF3,TF2, TF1) |

6.10.2.4.1.59.1.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Uplink | Min spreading factor | 16 |
| | Max number of DPDCH data bits/radio frame | 2 400 |
| | Puncturing Limit | 0.76 |

6.10.2.4.1.59.2 Downlink

6.10.2.4.1.59.2.1 Transport channel parameters

6.10.2.4.1.59.2.1.1 Transport channel parameters for Conversational / speech / DL:42.8 kbps / PS RAB

| Higher layer | RAB/Signalling RB | | RAB |
|---|---|---|---|
| PDCP | PDCP header size, bit | | 8 |
| RLC | Logical channel type | | DTCH |
| | RLC mode | | UM |
| | Payload sizes, bit | | 920, 304, 96 |
| | Max data rate, bps | | 46 000 |
| | UMD PDU header, bit | | 8 |
| MAC | MAC header, bit | | 0 |
| | MAC multiplexing | | N/A |
| Layer 1 | TrCH type | | DCH |
| | TB sizes, bit | | 928, 312, 104 |
| | TFS | TF0, bits | 0x928 |
| | | TF1, bits | 1x104 |
| | | TF2, bits | 1x312 |
| | | TF3, bits | 1x928 |
| | TTI, ms | | 20 |
| | Coding type | | TC |
| | CRC, bit | | 16 |
| | Max number of bits/TTI after channel coding | | 2 844 |
| | RM attribute | | 180 to 220 |

6.10.2.4.1.59.2.1.2 Transport channel parameters for Interactive / DL:16kbps / PS RAB + DL:16 kbps / PS RAB

| Higher layer | RAB/Signalling RB | | RAB | RAB |
|---|---|---|---|---|
| RLC | Logical channel type | | DTCH | DTCH |
| | RLC mode | | AM | AM |
| | Payload sizes, bit | | 320 | 320 |
| | Max data rate, bps | | 16 000 | 16 000 |
| | AMD PDU header, bit | | 16 | 16 |
| MAC | MAC header, bit | | 4 | 4 |
| | MAC multiplexing | | 2 logical channel multiplexing | |
| Layer 1 | TrCH type | | DCH | |
| | TB sizes, bit | | 340 | |
| | TFS | TF0, bits | 0x340 | |
| | | TF1, bits | 1x340 | |
| | | TF2, bits | 2x340 | |
| | TTI, ms | | 40 | |
| | Coding type | | TC | |
| | CRC, bit | | 16 | |
| | Max number of bits/TTI after channel coding | | 2 148 | |
| | RM attribute | | 135 to 175 | |

6.10.2.4.1.59.2.1.3 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.2.1.1

6.10.2.4.1.59.2.1.4 TFCS

| TFCS size | 24 |
|---|---|
| TFCS | (42.8 kbps Conversational RAB, Interactive 16kbps+16kbps RAB, DCCH)=<br>(TF0, TF0, TF0), (TF0, TF0, TF1), (TF0,TF1, TF0),(TF0, TF1,TF1), (TF0,TF2, TF0), (TF0,TF2, TF1)<br>(TF1, TF0, TF0), (TF1, TF0, TF1), (TF1,TF1, TF0), (TF1, TF1,TF1), (TF1,TF2, TF0), (TF1,TF2, TF1)<br>(TF2, TF0, TF0), (TF2, TF0, TF1), (TF2,TF1, TF0), (TF2, TF1,TF1), (TF2,TF2, TF0), (TF2,TF2, TF1)<br>(TF3, TF0, TF0), (TF3, TF0, TF1), (TF3,TF1, TF0), (TF3, TF1,TF1), (TF3,TF2, TF0), (TF3,TF2, TF1) |

6.10.2.4.1.59.2.2 Physical channel parameters

| | | | |
|---|---|---|---|
| DPCH Downlink | DTX position | | Flexible |
| | Spreading factor | | 32 |
| | DPCCH | Number of TFCI bits/slot | 8 |
| | | Number of TPC bits/slot | 4 |
| | | Number of Pilot bits/slot | 8 |
| | DPDCH | Number of data bits/slot | 140 |
| | | Number of data bits/frame | 2 100 |

6.10.2.4.1.60 Conversational / speech / UL:42.8 DL:42.8 kbps / PS RAB + Interactive / UL:16 DL:16 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.2.4.1.60.1 Uplink

6.10.2.4.1.60.1.1 Transport channel parameters

6.10.2.4.1.60.1.1.1 Transport channel parameters for Conversational / speech / UL:42.8 kbps / PS RAB

| | | | |
|---|---|---|---|
| Higher layer | RAB/Signalling RB | | RAB |
| PDCP | PDCP header size, bit | | 8 |
| RLC | Logical channel type | | DTCH |
| | RLC mode | | UM |
| | Payload sizes, bit | | 920, 304, 96 |
| | Max data rate, bps | | 46 000 |
| | UMD PDU header, bit | | 8 |
| MAC | MAC header, bit | | 0 |
| | MAC multiplexing | | N/A |
| Layer 1 | TrCH type | | DCH |
| | TB sizes, bit | | 928, 312, 104 |
| | TFS | TF0, bits | 0x928 |
| | | TF1, bits | 1x104 |
| | | TF2, bits | 1x312 |
| | | TF3, bits | 1x928 |
| | TTI, ms | | 20 |
| | Coding type | | TC |
| | CRC, bit | | 16 |
| | Max number of bits/TTI after channel coding | | 2 844 |
| | Uplink: Max number of bits/radio frame before rate matching | | 1 422 |
| | RM attribute | | 180 to 220 |

6.10.2.4.1.60.1.1.2 Transport channel parameters for Interactive / UL:16kbps / PS RAB

See clause 6.10.2.4.1.23b.1.1.1

6.10.2.4.1.60.1.1.3 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.1.1.1

6.10.2.4.1.60.1.1.4 TFCS

| | |
|---|---|
| TFCS size | 24 |
| TFCS | (42.8 kbps Conversational RAB, Interactive 16kbps RAB, DCCH)=<br>(TF0, TF0, TF0), (TF0, TF0, TF1), (TF0,TF1, TF0),(TF0, TF1,TF1), (TF0,TF2, TF0), (TF0,TF2, TF1)<br>(TF1, TF0, TF0), (TF1, TF0, TF1), (TF1,TF1, TF0), (TF1, TF1,TF1), (TF1,TF2, TF0), (TF1,TF2, TF1)<br>(TF2, TF0, TF0), (TF2, TF0, TF1), (TF2,TF1, TF0), (TF2, TF1,TF1), (TF2,TF2, TF0), (TF2,TF2, TF1)<br>(TF3, TF0, TF0), (TF3, TF0, TF1), (TF3,TF1, TF0), (TF3, TF1,TF1), (TF3,TF2, TF0), (TF3,TF2, TF1) |

6.10.2.4.1.60.1.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Uplink | Min spreading factor | 16 |
| | Max number of DPDCH data bits/radio frame | 2 400 |
| | Puncturing Limit | 0.76 |

6.10.2.4.1.60.2 Downlink

6.10.2.4.1.60.2.1 Transport channel parameters

6.10.2.4.1.60.2.1.1 Transport channel parameters for Conversational / speech / DL:42.8 kbps / PS RAB

| Higher layer | RAB/Signalling RB | | RAB |
|---|---|---|---|
| PDCP | PDCP header size, bit | | 8 |
| RLC | Logical channel type | | DTCH |
| | RLC mode | | UM |
| | Payload sizes, bit | | 920, 304, 96 |
| | Max data rate, bps | | 46 000 |
| | UMD PDU header, bit | | 8 |
| MAC | MAC header, bit | | 0 |
| | MAC multiplexing | | N/A |
| Layer 1 | TrCH type | | DCH |
| | TB sizes, bit | | 928, 312, 104 |
| | TFS | TF0, bits | 0x928 |
| | | TF1, bits | 1x104 |
| | | TF2, bits | 1x312 |
| | | TF3, bits | 1x928 |
| | TTI, ms | | 20 |
| | Coding type | | TC |
| | CRC, bit | | 16 |
| | Max number of bits/TTI after channel coding | | 2 844 |
| | RM attribute | | 180 to 220 |

6.10.2.4.1.60.2.1.2 Transport channel parameters for Interactive / DL:16kbps PS RAB

See clause 6.10.2.4.1.23b.2.1.1

6.10.2.4.1.60.2.1.3 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.2.1.1

6.10.2.4.1.60.2.1.4 TFCS

| TFCS size | 24 |
|---|---|
| TFCS | (42.8 kbps Conversational RAB, Interactive 16kbps RAB, DCCH)=<br>(TF0, TF0, TF0), (TF0, TF0, TF1), (TF0,TF1, TF0),(TF0, TF1,TF1), (TF0,TF2, TF0), (TF0,TF2, TF1)<br>(TF1, TF0, TF0), (TF1, TF0, TF1), (TF1,TF1, TF0), (TF1, TF1,TF1), (TF1,TF2, TF0), (TF1,TF2, TF1)<br>(TF2, TF0, TF0), (TF2, TF0, TF1), (TF2,TF1, TF0), (TF2, TF1,TF1), (TF2,TF2, TF0), (TF2,TF2, TF1)<br>(TF3, TF0, TF0), (TF3, TF0, TF1), (TF3,TF1, TF0), (TF3, TF1,TF1), (TF3,TF2, TF0), (TF3,TF2, TF1) |

6.10.2.4.1.60.2.2 Physical channel parameters

| DPCH Downlink | DTX position | | Flexible |
|---|---|---|---|
| | Spreading factor | | 32 |
| | DPCCH | Number of TFCI bits/slot | 8 |
| | | Number of TPC bits/slot | 4 |
| | | Number of Pilot bits/slot | 8 |
| | DPDCH | Number of data bits/slot | 140 |
| | | Number of data bits/frame | 2 100 |

6.10.2.4.1.61 Conversational / unknown / UL:8 DL:8 kbps / PS RAB + Interactive or Background / UL:8 DL:8 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.2.4.1.61.1 Uplink

6.10.2.4.1.61.1.1 Transport channel parameters

6.10.2.4.1.61.1.1.1 Transport channel parameters for Conversational / unknown / UL:8 kbps / PS RAB

| Higher layer | RAB/Signalling RB | RAB |
|---|---|---|

| | | | |
|---|---|---|---|
| RLC | Logical channel type | | DTCH |
| | RLC mode | | UM |
| | Payload sizes, bit | | 320 |
| | Max data rate, bps | | 8 000 |
| | UMD PDU header, bit | | 8 |
| MAC | MAC header, bit | | 0 |
| | MAC multiplexing | | N/A |
| Layer 1 | TrCH type | | DCH |
| | TB sizes, bit | | 328 (alt 0, 328) (note) |
| | TFS | TF0, bits | 0x328 (alt 1x0) (note) |
| | | TF1, bits | 1x328 |
| | TTI, ms | | 40 |
| | Coding type | | TC |
| | CRC, bit | | 16 |
| | Max number of bits/TTI after channel coding | | 1 044 |
| | Uplink: Max number of bits/radio frame before rate matching | | 261 |
| | RM attribute | | 135 to 175 |
| NOTE: In case of using this alternative, CRC parity bits are to be attached any time since number of TrBlks are 1 even if there is no data on the RAB (see clause 4.2.1.1 in 3GPP TS 25.212 [14]). | | | |

6.10.2.4.1.61.1.1.2 Transport channel parameters for Interactive or Background / UL:8 kbps / PS RAB

See clause 6.10.2.4.1.38b.1.1.2

6.10.2.4.1.61.1.1.3 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.1.1.1

6.10.2.4.1.61.1.1.4 TFCS

| | |
|---|---|
| TFCS size | 8 |
| TFCS | (8 kbps Conversational RAB, 8 kbps I/B RAB, DCCH)=<br>(TF0, TF0, TF0), (TF0, TF1, TF0), (TF0, TF0, TF1), (TF0, TF1, TF1),<br>(TF1, TF0, TF0), (TF1, TF1, TF0), (TF1, TF0, TF1), (TF1, TF1, TF1) |

6.10.2.4.1.61.1.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Uplink | Min spreading factor | 32 |
| | Max number of DPDCH data bits/radio frame | 1 200 |
| | Puncturing Limit | 1.0 |

6.10.2.4.1.61.2 Downlink

6.10.2.4.1.61.2.1 Transport channel parameters

6.10.2.4.1.61.2.1.1 Transport channel parameters for Conversational / unknown / DL:8 kbps / PS RAB

| | | | |
|---|---|---|---|
| Higher layer | RAB/Signalling RB | | RAB |
| RLC | Logical channel type | | DTCH |
| | RLC mode | | UM |
| | Payload sizes, bit | | 320 |
| | Max data rate, bps | | 8 000 |
| | AMD PDU header, bit | | 8 |
| MAC | MAC header, bit | | 0 |
| | MAC multiplexing | | N/A |
| Layer 1 | TrCH type | | DCH |
| | TB sizes, bit | | 328 (alt 0, 328) (note) |
| | TFS | TF0, bits | 0x328 (alt 1x0) (note) |
| | | TF1, bits | 1x328 |
| | TTI, ms | | 40 |
| | Coding type | | TC |
| | CRC, bit | | 16 |
| | Max number of bits/TTI after channel coding | | 1 044 |
| | RM attribute | | 135 to 175 |

| NOTE: | In case of using this alternative, CRC parity bits are to be attached any time since number of TrBlks are 1 even if there is no data on the RAB (see clause 4.2.1.1 in 3GPP TS 25.212 [14]). |
|---|---|

#### 6.10.2.4.1.61.2.1.2 Transport channel parameters for Interactive or Background / DL:8 kbps / PS RAB

See clause 6.10.2.4.1.38b.2.1.2.

#### 6.10.2.4.1.61.2.1.3 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.2.1.1.

#### 6.10.2.4.1.61.2.1.4 TFCS

| TFCS size | 8 |
|---|---|
| TFCS | (8 kbps Conversational RAB, 8 kbps I/B RAB, DCCH)=<br>(TF0, TF0, TF0), (TF0, TF1, TF0), (TF0, TF0, TF1), (TF0, TF1, TF1),<br>(TF1, TF0, TF0), (TF1, TF1, TF0), (TF1, TF0, TF1), (TF1, TF1, TF1) |

#### 6.10.2.4.1.61.2.2 Physical channel parameters

| DPCH Downlink | DTX position | | Flexible |
|---|---|---|---|
| | Spreading factor | | 64 |
| | DPCCH | Number of TFCI bits/slot | 8 |
| | | Number of TPC bits/slot | 4 |
| | | Number of Pilot bits/slot | 8 |
| | DPDCH | Number of data bits/slot | 60 |
| | | Number of data bits/frame | 900 |

6.10.2.4.1.62 Conversational / speech / UL:(12.65 8.85 6.6) DL:(12.65 8.85 6.6) kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH + DL:0.15 kbps SRB#5 for DCCH

6.10.2.4.1.62.1 Uplink

6.10.2.4.1.62.1.1 Transport channel parameters

6.10.2.4.1.62.1.1.1 Transport channel parameters for Conversational / speech / UL: (12.65 8.85 6.6) kbps / CS RAB

| Higher layer | RAB/Signalling RB | | RAB subflow #1 | RAB subflow #2 | RAB subflow #3 (note 2) |
|---|---|---|---|---|---|
| RLC | Logical channel type | | DTCH | | |
| | RLC mode | | TM | TM | TM |
| | Payload sizes, bit | | 40, 54, 64, 72 (alt. 0, 40, 54, 64, 72) | 78, 113, 181 | 60 |
| | Max data rate, bps | | 12 650 | | |
| | TrD PDU header, bit | | 0 | | |
| MAC | MAC header, bit | | 0 | | |
| | MAC multiplexing | | N/A | | |
| Layer 1 | TrCH type | | DCH | DCH | DCH |
| | TB sizes, bit | | 40, 54, 64, 72 (alt. 0, 40, 54, 64, 72) | 78, 113, 181 | 60 |
| | TFS | TF0, bits | 0x72(alt. 1x0) (note 1) | 0x181 | 0x60 |
| | | TF1, bits | 1x40 | 1x78 | N/A |
| | | TF2 bits | 1x54 | 1x113 | N/A |
| | | TF3, bits | 1x64 | 1x181 | N/A |
| | | TF4, bits | 1x72 | N/A | N/A |
| | TTI, ms | | 20 | 20 | 20 |
| | Coding type | | CC 1/3 | CC 1/3 | CC 1/3 |
| | CRC, bit | | 12 | N/A | N/A |
| | Max number of bits/TTI after channel coding | | 276 | 567 | 0 |
| | Uplink: Max number of bits/radio frame before rate matching | | 138 | 284 | 0 |
| | RM attribute | | 180 to 220 | 170 to 210 | 256 |

NOTE 1: In case of using this alternative, CRC parity bits are to be attached to RAB subflow#1 any time since number of TrBlks are 1 even if there is no data on RAB subflow#1 (see clause 4.2.1.1 in 3GPP TS 25.212 [14]).
NOTE 2: RAB subflow #3 does not exist in Iu interface. UTRAN establishes this additional "dummy" subflow when the RAB for Wideband AMR is assigned.

6.10.2.4.1.62.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.1.1.1.

6.10.2.4.1.62.1.1.3 TFCS

| | |
|---|---|
| TFCS size | 10 |
| TFCS | (RAB subflow#1, RAB subflow#2, RAB subflow #3, DCCH)=<br>(TF0,TF0,TF0,TF0), (TF1,TF0,TF0,TF0), (TF2,TF1,TF0,TF0), (TF3,TF2,TF0,TF0),<br>(TF4,TF3,TF0,TF0), (TF0,TF0,TF0,TF1), (TF1,TF0,TF0,TF1), (TF2,TF1,TF0,TF1),<br>(TF3,TF2,TF0,TF1), (TF4,TF3,TF0,TF1) |

6.10.2.4.1.62.1.1.4 TFC subset list

| | |
|---|---|
| TFC subset list size | 3 |
| TFC subset list | 0 = {(TF0,TF0,TF0,TF0), (TF1,TF0,TF0,TF0), (TF2,TF1,TF0,TF0), (TF0,TF0,TF0,TF1), (TF1,TF0,TF0,TF1), (TF2,TF1,TF0,TF1)},<br>1 = {(TF0,TF0,TF0,TF0), (TF1,TF0,TF0,TF0), (TF2,TF1,TF0,TF0), (TF3,TF2,TF0,TF0), (TF0,TF0,TF0,TF1), (TF1,TF0,TF0,TF1), (TF2,TF1,TF0,TF1), (TF3,TF2,TF0,TF1)},<br>2 = {(TF0,TF0,TF0,TF0), (TF1,TF0,TF0,TF0), (TF2,TF1,TF0,TF0), (TF3,TF2,TF0,TF0), (TF4,TF3,TF0,TF0), (TF0,TF0,TF0,TF1), (TF1,TF0,TF0,TF1), (TF2,TF1,TF0,TF1), (TF3,TF2,TF0,TF1), (TF4,TF3,TF0,TF1)} |

6.10.2.4.1.62.1.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Uplink | Min spreading factor | 64 |
| | Max number of DPDCH data bits/radio frame | 600 |
| | Puncturing Limit | 0.84 |

6.10.2.4.1.62.2 Downlink

6.10.2.4.1.62.2.1 Transport channel parameters

6.10.2.4.1.62.2.1.1 Transport channel parameters for Conversational / speech / DL: (12.65 8.85 6.6) kbps / CS RAB

| Higher layer | RAB/Signalling RB | | RAB subflow #1 | RAB subflow #2 | RAB subflow #3 (note 3) |
|---|---|---|---|---|---|
| RLC | Logical channel type | | DTCH | | |
| | RLC mode | | TM | TM | TM |
| | Payload sizes, bit | | 0, 40, 54, 64, 72 | 78, 113, 181 | 60 |
| | Max data rate, bps | | 12 650 | | |
| | TrD PDU header, bit | | 0 | | |
| MAC | MAC header, bit | | 0 | | |
| | MAC multiplexing | | N/A | | |
| Layer 1 | TrCH type | | DCH | DCH | DCH |
| | TB sizes, bit | | 0, 40, 54, 64, 72 | 78, 113, 181 | 60 |
| | TFS (note 1) | TF0, bits | 1x0 (note 2) | 0x181 | 0x60 |
| | | TF1, bits | 1x40 | 1x78 | N/A |
| | | TF2, bits | 1x54 | 1x113 | N/A |
| | | TF3, bits | 1x64 | 1x181 | N/A |
| | | TF4, bits | 1x72 | N/A | N/A |
| | TTI, ms | | 20 | 20 | 20 |
| | Coding type | | CC 1/3 | CC 1/3 | CC 1/3 |
| | CRC, bit | | 12 | N/A | N/A |
| | Max number of bits/TTI after channel coding | | 276 | 567 | 0 |
| | RM attribute | | 180 to 220 | 170 to 210 | 256 |

NOTE 1: The TrCH corresponding to RAB subflow #1 should be used as the guiding TrCH, (see clause 4.3 in 3GPP TS 25.212 [14]).
NOTE 2: CRC parity bits are to be attached to RAB subflow#1 any time since number of TrBlks are 1 even if there is no data on RAB subflow#1 (see clause 4.2.1.1 in 3GPP TS 25.212 [14]).
NOTE 2: RAB subflow #3 does not exist in Iu interface. UTRAN establishes this additional "dummy" subflow when the RAB for Wideband AMR is assigned

6.10.2.4.1.62.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.2.1.1

6.10.2.4.1.62.2.1.3 Transport channel parameters for DL:0.15 kbps SRB#5 for DCCH

| | | |
|---|---|---|
| Higher layer | RAB/signalling RB | SRB#5 |
| | User of Radio Bearer | RRC |
| RLC | Logical channel type | DCCH |
| | RLC mode | TM |
| | Payload sizes, bit | 3 |

| | | | |
|---|---|---|---|
| | Max data rate, bps | | 150 |
| | TrD PDU header, bit | | 0 |
| MAC | MAC header, bit | | 0 |
| | MAC multiplexing | | N/A |
| Layer 1 | TrCH type | | DCH |
| | TB sizes, bit | | 3 (alt 0, 3) (note) |
| | TFS | TF0, bits | 0x3 (alt 1x0) (note) |
| | | TF1, bits | 1x3 |
| | TTI, ms | | 20 |
| | Coding type | | CC 1/3 |
| | CRC, bit | | 8 |
| | Max number of bits/TTI before rate matching | | 57 |
| | RM attribute | | 155 to 256 |
| NOTE: Alternative parameters enable the measurement "transport channel BLER" in the UE. | | | |

6.10.2.4.1.62.2.1.4 TFCS

| | |
|---|---|
| TFCS size | 20 |
| TFCS | (RAB subflow#1, RAB subflow#2, RAB subflow#3, DCCH 3.4, DCCH 0.15)=<br>(TF0,TF0,TF0,TF0,TF0), (TF1,TF0,TF0,TF0,TF0), (TF2,TF1,TF0,TF0,TF0),<br>(TF3,TF2,TF0,TF0,TF0), (TF4,TF3,TF0TF0,TF0), (TF0,TF0,TF0,TF1,TF0),<br>(TF1,TF0,TF0,TF1,TF0), (TF2,TF1,TF0,TF1,TF0), (TF3,TF2,TF0,TF1,TF0),<br>(TF4,TF3,TF0,TF1,TF0), (TF0,TF0,TF0,TF0,TF1), (TF1,TF0,TF0,TF0,TF1),<br>(TF2,TF1,TF0,TF0,TF1), (TF3,TF2,TF0,TF0,TF1), (TF4,TF3,TF0,TF0,TF1),<br>(TF0,TF0,TF0,TF1,TF1), (TF1,TF0,TF0,TF1,TF1), (TF2,TF1,TF0,TF1,TF1),<br>(TF3,TF2,TF0,TF1,TF1), (TF4,TF3,TF0,TF1,TF1) |

6.10.2.4.1.62.2.2 Physical channel parameters

| | | | |
|---|---|---|---|
| DPCH Downlink | DTX position | | Fixed |
| | Spreading factor | | 128 |
| | DPCCH | Number of TFCI bits/slot | 0 |
| | | Number of TPC bits/slot | 2 |
| | | Number of Pilot bits/slot | 4 |
| | DPDCH | Number of data bits/slot | 34 |
| | | Number of data bits/frame | 510 |

6.10.2.4.1.63 Interactive or background / UL:64 DL:768 kbps / PS RAB + UL:3.4 DL: 3.4 kbps SRBs for DCCH

6.10.2.4.1.63.1 Uplink

See clause 6.10.2.4.1.26.1.

6.10.2.4.1.63.2 Downlink

6.10.2.4.1.63.2.1 Transport channel parameters

6.10.2.4.1.63.2.1.1 Transport channel parameters for Interactive or background / DL:768 kbps / PS RAB

| | | |
|---|---|---|
| Higher layer | RAB/Signalling RB | RAB |
| RLC | Logical channel type | DTCH |
| | RLC mode | AM |
| | Payload sizes, bit | 320 |
| | Max data rate, bps | 768 000 |
| | AMD PDU header, bit | 16 |
| MAC | MAC header, bit | 0 |
| | MAC multiplexing | N/A |

| | | | |
|---|---|---|---|
| Layer 1 | TrCH type | | DCH |
| | TB sizes, bit | | 336 |
| | TFS | TF0, bits | 0x336 |
| | | TF1, bits | 1x336 |
| | | TF2, bits | 2x336 |
| | | TF3, bits | 4x336 |
| | | TF4, bits | 8x336 |
| | | TF5, bits | 12x336 |
| | | TF6, bits | 16x336 |
| | | TF7, bits | 20x336 |
| | | TF8, bits | 24x336 |
| | | TF9, bits | N/A (alt 28x336) |
| | | TF10, bits | N/A (alt 32x336) |
| | | TF11, bits | N/A (alt 36x336) |
| | | TF12, bits | N/A (alt 40x336) |
| | | TF13, bits | N/A (alt 44x336) |
| | | TF14, bits | N/A (alt 48x336) |
| | TTI, ms | | 10 (alt 20) |
| | Coding type | | TC |
| | CRC, bit | | 16 |
| | Max number of bits/TTI after channel coding | | 25 368 (alt 50 736) |
| | RM attribute | | 110 to 150 |

#### 6.10.2.4.1.63.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.2.1.1.

#### 6.10.2.4.1.63.2.1.3 TFCS

| | |
|---|---|
| TFCS size | 18 (alt. 30) |
| TFCS | (768 kbps RAB, DCCH)=<br>(TF0, TF0), (TF1, TF0), (TF2, TF0), (TF3, TF0), (TF4, TF0), (TF5, TF0), (TF6, TF0), (TF7, TF0), (TF8, TF0),<br>(TF0, TF1), (TF1, TF1), (TF2, TF1), (TF3, TF1), (TF4, TF1), (TF5, TF1), (TF6, TF1), (TF7, TF1), (TF8, TF1)<br>(alt . (TF0, TF0), (TF1, TF0), (TF2, TF0), (TF3, TF0), (TF4, TF0), (TF5, TF0), (TF6, TF0), (TF7, TF0), (TF8, TF0), (TF9, TF0), (TF10, TF0), (TF11, TF0), (TF12, TF0), (TF13, TF0), (TF14, TF0)<br>(TF0, TF1), (TF1, TF1), (TF2, TF1), (TF3, TF1), (TF4, TF1), (TF5, TF1), (TF6, TF1), (TF7, TF1), (TF8, TF1) (TF9, TF1), (TF10, TF1), (TF11, TF1), (TF12, TF1), (TF13, TF1), (TF14, TF1)) |

#### 6.10.2.4.1.63.2.2 Physical channel parameters

| | | | |
|---|---|---|---|
| DPCH Downlink | DTX position | | Flexible |
| | Spreading factor | | 8 |
| | Number of DPCH | | 2 |
| | DPCCH | Number of TFCI bits/slot | 8 |
| | | Number of TPC bits/slot | 8 |
| | | Number of Pilot bits/slot | 16 |
| | DPDCH | Number of data bits/slot | 608 |
| | | Number of data bits/frame | 9 120 |

### 6.10.2.4.2 Combinations on PDSCH and DPCH

#### 6.10.2.4.2.1 Void

#### 6.10.2.4.2.2 Interactive or background / UL:64 DL:384 kbps / PS RAB + UL:3.4 DL: 3.4 kbps SRBs for DCCH

##### 6.10.2.4.2.2.1 Uplink

See clause 6.10.2.4.1.26.1.

##### 6.10.2.4.2.2.2 Downlink

6.10.2.4.2.2.1 Transport channel parameters

6.10.2.4.2.2.1.1 Transport channel parameters for Interactive or background / DL:384 kbps / PS RAB

| | | | |
|---|---|---|---|
| Higher layer | RAB/Signalling RB | | RAB |
| RLC | Logical channel type | | DTCH |
| | RLC mode | | AM |
| | Payload sizes, bit | | 320 |
| | Max data rate, bps | | 384 000 |
| | AMD PDU header, bit | | 16 |
| MAC | MAC header, bit | | 18 |
| | MAC multiplexing | | Logical channel multiplexing on a frame by frame basis |
| Layer 1 | TrCH type | | DSCH |
| | TB sizes, bit | | 354 |
| | TFS | TF0, bits | 0x354 |
| | | TF1, bits | 1x354 |
| | | TF2, bits | 2x354 |
| | | TF3, bits | 4 x354 |
| | | TF4, bits | 8 x354 |
| | | TF5, bits | 12 x354 |
| | | TF6, bits | N/A (alt. 16x354) |
| | | TF7, bits | N/A (alt. 20x354) |
| | | TF8, bits | N/A (alt. 24x354) |
| | TTI, ms | | 10 (alt. 20) |
| | Coding type | | TC |
| | CRC, bit | | 16 |
| | Max number of bits/TTI after channel coding | | 13 332 (alt. 26 664) |
| | RM attribute | | 110 to 150 |

6.10.2.4.2.2.1.2 Transport channel parameters for DL: 3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.2.1.1.

6.10.2.4.2.2.1.3 TFCS

| | | |
|---|---|---|
| PDSCH | TFCS size | 6 (alt.9) |
| | TFCS | 384 kbps RAB = TF0, TF1, TF2, TF3, TF4, TF5<br>(alt. TF0, TF1, TF2, TF3, TF4, TF5, TF6, TF7, TF8) |
| DPCH Downlink associated with PDSCH | TFCS size | 2 |
| | TFCS | SRBs for DCCH = TF0, TF1 |

6.10.2.4.2.2.2 Physical channel parameters

| | | | |
|---|---|---|---|
| PDSCH | RAB or SRB, TrCh | | **Interactive or background / 384 kbps / PS RAB, DSCH** |
| | DTX position | | N/A (SingleTrCH) |
| | Minimum spreading factor | | 8 |
| DPCH Downlink associated with PDSCH | RAB or SRB, TrCh | | **3.4 kbps SRB for DCCH, DCH** |
| | DTX position | | N/A (SingleTrCH) |
| | Spreading factor | | 256 |
| | DPCCH | Number of TFCI bits/slot | 2 |
| | | Number of TPC bits/slot | 2 |
| | | Number of Pilot bits/slot | 4 |
| | DPDCH | Number of data bits/slot | 12 |
| | | Number of data bits/frame | 180 |

6.10.2.4.2.3 Interactive or background / UL:64 DL:2 048 kbps / PS RAB + UL:3.4 DL: 3.4 kbps SRBs for DCCH

6.10.2.4.2.3.1 Uplink

See clause 6.10.2.4.1.26.1.

6.10.2.4.2.3.2 Downlink

6.10.2.4.2.3.2.1 Transport channel parameters

6.10.2.4.2.3.2.1.1 Transport channel parameters for Interactive or background / DL:2 048 kbps / PS RAB

| | | | |
|---|---|---|---|
| Higher layer | RAB/Signalling RB | | RAB |
| RLC | Logical channel type | | DTCH |
| | RLC mode | | AM |
| | Payload sizes, bit | | 640 |
| | Max data rate, bps | | 2 048 000 |
| | AMD PDU header, bit | | 16 |
| MAC | MAC header, bit | | 18 |
| | MAC multiplexing | | Logical channel multiplexing on a frame by frame basis |
| Layer 1 | TrCH type | | DSCH |
| | TB sizes, bit | | 674 |
| | TFS | TF0, bits | 0x674 |
| | | TF1, bits | 1x674 |
| | | TF2, bits | 2x674 |
| | | TF3, bits | 4 x674 |
| | | TF4, bits | 8 x674 |
| | | TF5, bits | 12x674 |
| | | TF6, bits | 16x674 |
| | | TF7, bits | 20x674 |
| | | TF8, bits | 24x674 |
| | | TF9, bits | 28x674 |
| | | TF10, bits | 32x674 |
| | | TF11, bits | N/A (alt. 36x674) |
| | | TF12, bits | N/A (alt. 40x674) |
| | | TF13, bits | N/A (alt. 44x674) |
| | | TF14, bits | N/A (alt. 48x674) |
| | | TF15, bits | N/A (alt. 52x674) |
| | | TF16, bits | N/A (alt. 56x674) |
| | | TF17, bits | N/A (alt. 60x674) |
| | | TF18, bits | N/A (alt. 64x674) |
| | TTI, ms | | 10 (alt. 20) |
| | Coding type | | TC |
| | CRC, bit | | 16 |
| | Max number of bits/TTI after channel coding | | 66 300 (alt. 132 588) |
| | RM attribute | | 130 to 170 |

6.10.2.4.2.3.2.1.2 Transport channel parameters for DL: 3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.2.1.1

6.10.2.4.2.3.2.1.3 TFCS

| | | |
|---|---|---|
| PDSCH | TFCS size | 11 (alt.19) |
| | TFCS | 2 048 kbps RAB =<br>TF0, TF1, TF2, TF3, TF4, TF5, TF6, TF7, TF8, TF9, TF10<br>(alt. TF0, TF1, TF2, TF3, TF4, TF5, TF6, TF7, TF8, TF9, TF10, TF11, TF12, TF13, TF14, TF15, TF16, TF17, TF18) |
| DPCH Downlink associated with PDSCH | TFCS size | 2 |
| | TFCS | SRBs for DCCH = TF0, TF1 |

6.10.2.4.2.3.2.2 Physical channel parameters

| | | | |
|---|---|---|---|
| PDSCH | RAB or SRB, TrCh | | **Interactive or background / 2 048 kbps / PS RAB, DSCH** |
| | DTX position | | N/A (SingleTrCH) |
| | Minimum spreading factor | | 4 |
| DPCH Downlink associated with PDSCH | RAB or SRB, TrCh | | 3.4 kbps SRB for DCCH, DCH |
| | DTX position | | N/A (SingleTrCH) |
| | Spreading factor | | 256 |
| | DPCCH | Number of TFCI bits/slot | 2 |
| | | Number of TPC bits/slot | 2 |

| | | | |
|---|---|---|---|
| | | Number of Pilot bits/slot | 4 |
| | DPDCH | Number of data bits/slot | 12 |
| | | Number of data bits/frame | 180 |

6.10.2.4.2.4 Void

6.10.2.4.2.5 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:64 DL:384 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.2.4.2.5.1 Uplink

See clause 6.10.2.4.1.40.1.

6.10.2.4.2.5.2 Downlink

6.10.2.4.2.5.2.1 Transport channel parameters

6.10.2.4.2.5.2.1.1 Transport channel parameters for Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB

See clause 6.10.2.4.1.4.2.1.1.

6.10.2.4.2.5.2.1.2 Transport channel parameters for Interactive or background / DL:384 kbps / PS RAB

See clause 6.10.2.4.2.2.1.1.

6.10.2.4.2.5.2.1.3 Transport channel parameters for DL: 3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.2.1.1.

6.10.2.4.2.5.2.1.4 TFCS

| | | |
|---|---|---|
| PDSCH | TFCS size | 6 (alt.9) |
| | TFCS | 384 kbps RAB = TF0, TF1, TF2, TF3, TF4, TF5<br>(alt. TF0, TF1, TF2, TF3, TF4, TF5, TF6, TF7, TF8) |
| DPCH Downlink associated with PDSCH | TFCS size | 6 |
| | TFCS | (RAB subflow#1, RAB subflow#2, RAB subflow#3, DCCH) =<br>(TF0, TF0, TF0, TF0), (TF1, TF0, TF0, TF0), (TF2, TF1, TF1, TF0),<br>(TF0, TF0, TF0, TF1), (TF1, TF0, TF0, TF1), (TF2, TF1, TF1, TF1) |

6.10.2.4.2.5.2.2 Physical channel parameters

| | | | |
|---|---|---|---|
| PDSCH | RAB or SRB, TrCh | | **Interactive or background / 384 kbps / PS RAB, DSCH** |
| | DTX position | | N/A (SingleTrCH) |
| | Minimum spreading factor | | 8 |
| DPCH Downlink associated with PDSCH | RAB or SRB, TrCh | | Conversational / speech / 12.2 kbps / CS RAB, DCH<br>**+ 3.4 kbps SRBs for DCCH. DCH** |
| | DTX position | | Fixed |
| | Spreading factor | | 128 |
| | DPCCH | Number of TFCI bits/slot | 2 |
| | | Number of TPC bits/slot | 2 |
| | | Number of Pilot bits/slot | 4 |
| | DPDCH | Number of data bits/slot | 32 |
| | | Number of data bits/frame | 480 |

6.10.2.4.2.6 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:64 DL:2 048 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.2.4.2.6.1 Uplink

See clause 6.10.2.4.1.40.1.

6.10.2.4.2.6.2 Downlink

6.10.2.4.2.6.2.1 Transport channel parameters

6.10.2.4.2.6.2.1.1 Transport channel parameters for Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB

See clause 6.10.2.4.1.4.2.1.1.

6.10.2.4.2.6.2.1.2 Transport channel parameters for Interactive or background / DL:2 048 kbps / PS RAB

See clause 6.10.2.4.2.3.2.1.1.

6.10.2.4.2.6.2.1.3 Transport channel parameters for DL: 3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.2.1.1.

6.10.2.4.2.6.2.1.4 TFCS

| | | |
|---|---|---|
| PDSCH | TFCS size | 11 (alt.19) |
| | TFCS | 2 048 kbps RAB =TF0, TF1, TF2, TF3, TF4, TF5, TF6, TF7, TF8, TF9, TF10<br>(alt. TF0, TF1, TF2, TF3, TF4, TF5, TF6, TF7, TF8, TF9, TF10, TF11, TF12, TF13, TF14, TF15, TF16, TF17, TF18) |
| DPCH Downlink associated with PDSCH | TFCS size | 6 |
| | TFCS | (RAB subflow#1, RAB subflow#2, RAB subflow#3, DCCH) =<br>(TF0, TF0, TF0, TF0), (TF1, TF0, TF0, TF0), (TF2, TF1, TF1, TF0),<br>(TF0, TF0, TF0, TF1), (TF1, TF0, TF0, TF1), (TF2, TF1, TF1, TF1) |

6.10.2.4.2.6.2.2 Physical channel parameters

| | | | |
|---|---|---|---|
| PDSCH | RAB or SRB, TrCh | | **Interactive or background / 2 048 kbps / PS RAB, DSCH** |
| | DTX position | | N/A (SingleTrCH) |
| | Minimum spreading factor | | 4 |
| DPCH Downlink associated with PDSCH | RAB or SRB, TrCh | | Conversational / speech / 12.2 kbps / CS RAB, DCH<br>**+ 3.4 kbps SRBs for DCCH. DCH** |
| | DTX position | | Fixed |
| | Spreading factor | | 128 |
| | DPCCH | Number of TFCI bits/slot | 2 |
| | | Number of TPC bits/slot | 2 |
| | | Number of Pilot bits/slot | 4 |
| | DPDCH | Number of data bits/slot | 32 |
| | | Number of data bits/frame | 480 |

## 6.10.2.4.3 Combinations on SCCPCH

6.10.2.4.3.1 Stand-alone signalling RB for PCCH

6.10.2.4.3.1.1 Transport channel parameters

6.10.2.4.3.1.1.1 Transport channel parameter of SRB for PCCH

| | | | |
|---|---|---|---|
| Higher layer | RAB/signalling RB | | SRB |
| | User of Radio Bearer | | RRC |
| RLC | Logical channel type | | PCCH |
| | RLC mode | | TM |
| | Payload sizes, bit | | 240 (alt. 80) |
| | Max data rate, bps | | 24 000 (alt. 8 000) |
| | TrD PDU header, bit | | 0 |
| MAC | MAC header, bit | | 0 |
| | MAC multiplexing | | N/A |
| Layer 1 | TrCH type | | PCH |
| | TB sizes, bit | | 240 (alt. 80) |
| | TFS | TF0, bts | 0x240 (alt. 0x80) |
| | | TF1, bits | 1x240 (alt. 1x80) |
| | TTI, ms | | 10 |
| | Coding type | | CC 1/2 |
| | CRC, bit | | 16 |

| | | |
|---|---|---|
| | Max number of bits/TTI before rate matching | 528 (alt. 208) |
| | RM attribute | 210 to 250 |

6.10.2.4.3.1.1.2 TFCS

| | |
|---|---|
| TFCS size | 2 |
| TFCS | SRBs for PCCH = TF0, TF1 |

6.10.2.4.3.1.2 Physical channel parameters

| | | |
|---|---|---|
| SCCPCH | TFCS size | 2 |
| | DTX position | Fixed |
| | Spreading factor | 128 (alt. 256) |
| | Number of TFCI bits/slot | 0 |
| | Number of Pilot bits/slot | 0 |
| | Number of data bits/slot | 40 (alt. 20) |
| | Number of data bits/frame | 600 (alt. 300) |

6.10.2.4.3.2 Interactive/Background 32 kbps PS RAB + SRBs for CCCH + SRB for DCCH + SRB for BCCH

6.10.2.4.3.2.1 Transport channel parameters

6.10.2.4.3.2.1.1 Transport channel parameters for Interactive/Background 32 kbps PS RAB

| | | | |
|---|---|---|---|
| Higher layer | RAB/signalling RB | | RAB |
| | User of Radio Bearer | | Interactive/ Background RAB |
| RLC | Logical channel type | | DTCH |
| | RLC mode | | AM |
| | Payload sizes, bit | | 320 |
| | Max data rate, bps | | 32 000 |
| | AMD PDU header, bit | | 16 |
| MAC | MAC header, bit | | 24 |
| | MAC multiplexing | | N/A |
| Layer 1 | TrCH type | | FACH |
| | TB sizes, bit | | 360 |
| | TFS | TF0, bits | 0x360 |
| | | TF1, bits | 1x360 |
| | TTI, ms | | 10 |
| | Coding type | | TC |
| | CRC, bit | | 16 |
| | Max number of bits/TTI before rate matching | | 1 140 |
| | RM attribute | | 110 to 150 |

6.10.2.4.3.2.1.2 Transport channel parameters of SRBs for CCCH, SRB for DCCH, and SRB for BCCH

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| Higher layer | RAB/signalling RB | SRB#0 | SRB#1 | SRB#2 | SRB#3 | SRB#4 | SRB#5 |
| | User of Radio Bearer | RRC | RRC | RRC | NAS_DT High prio | NAS_DT Low prio | RRC |
| RLC | Logical channel type | CCCH | DCCH | DCCH | DCCH | DCCH | BCCH |
| | RLC mode | UM | UM | AM | AM | AM | TM |
| | Payload sizes, bit | 152 | 136 or 120 (note) | 128 | 128 | 128 | 166 |
| | Max data rate, bps | 30 400 (alt. 45 600) | 27 200 or 24 000 (alt. 40 800 or 36 000) | 25 600 (alt. 38 400) | 25 600 (alt. 38 400) | 25 600 (alt. 38 400) | 33 200 (alt. 49 800) |
| | AMD/UMD/TrD PDU header, bit | 8 | 8 | 16 | 16 | 16 | 0 |
| MAC | MAC header, bit | 8 | 24 or 40 | 24 | 24 | 24 | 2 |

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| Higher layer | RAB/signalling RB | | SRB#0 | SRB#1 | SRB#2 | SRB#3 | SRB#4 | SRB#5 |
| | User of Radio Bearer | | RRC | RRC | RRC | NAS_DT High prio | NAS_DT Low prio | RRC |
| | MAC multiplexing | | 6 logical channel multiplexing | | | | | |
| Layer 1 | TrCH type | | FACH | | | | | |
| | TB sizes, bit | | 168 | | | | | |
| | TFS | TF0, bits | 0x168 | | | | | |
| | | TF1, bits | 1x168 | | | | | |
| | | TF2, bits | 2x168 | | | | | |
| | | TF3, bits | N/A (alt. 3x168) | | | | | |
| | TTI, ms | | 10 | | | | | |
| | Coding type | | CC 1/2 | | | | | |
| | CRC, bit | | 16 | | | | | |
| | Max number of bits/TTI before rate matching | | 752 (alt. 1 136) | | | | | |
| | RM attribute | | 200 to 240 | | | | | |
| NOTE: MAC header size and PLC payload size depend on use of U-RNTI or C-RNTI. | | | | | | | | |

6.10.2.4.3.2.1.3 TFCS

| | |
|---|---|
| TFCS size | 4 or 5, (alt. 4, 5 or 6) |
| TFCS | (SRBs for CCCH/DCCH/BCCH, 32kbps RAB) =<br>(TF0, TF0), (TF1, TF0), (TF2, TF0), (TF0, TF1), [TF1, TF1] (note)<br>(alt. (TF0, TF0), (TF1, TF0), (TF2, TF0), [TF3, TF0] (note), (TF0, TF1), [TF1, TF1] (note)) |
| NOTE: These TFCs are available only if SCCPCH can be allocated bigger Tx power than required Tx power for TFC of (TF2, TF0). | |

6.10.2.4.3.2.2 Physical channel parameters

| | | |
|---|---|---|
| SCCPCH | DTX position | Flexible |
| | Spreading factor | 64 |
| | Number of TFCI bits/slot | 8 |
| | Number of Pilot bits/slot | 0 |
| | Number of data bits/slot | 72 |
| | Number of data bits/frame | 1 080 |

6.10.2.4.3.2a Interactive/Background 32 kbps PS RAB + Interactive/Background 32 kbps PS RAB + SRBs for CCCH + SRB for DCCH + SRB for BCCH

6.10.2.4.3.2a.1 Transport channel parameters

6.10.2.4.3.2a.1.1 Transport channel parameters for Interactive or background / 32 kbps / PS RAB + 32 kbps / PS RAB

| | | | | |
|---|---|---|---|---|
| Higher layer | RAB/Signalling RB | | RAB | RAB |
| RLC | Logical channel type | | DTCH | DTCH |
| | RLC mode | | AM | AM |
| | Payload sizes, bit | | 320 | 320 |
| | Max data rate, bps | | 32 000 | 32 000 |
| | AMD PDU header, bit | | 16 | 16 |
| MAC | MAC header, bit | | 24 | 24 |
| | MAC multiplexing | | 2 logical channel multiplexing | |
| Layer 1 | TrCH type | | FACH | |
| | TB sizes, bit | | 360 | |
| | TFS | TF0, bits | 0x360 | |
| | | TF1, bits | 1x360 | |
| | TTI, ms | | 10 | |
| | Coding type | | TC | |
| | CRC, bit | | 16 | |
| | Max number of bits/TTI after channel coding | | 1 140 | |
| | RM attribute | | 110 to 150 | |

6.10.2.4.3.2a.1.2 Transport channel parameters of SRBs for CCCH, SRB for DCCH, and SRB for BCCH

See clause 6.10.2.4.3.2.1.2

6.10.2.4.3.2a.1.3 TFCS

| | |
|---|---|
| TFCS size | 4 or 5 (alt. 4, 5 or 6) |
| TFCS | (SRBs for CCCH/DCCH/BCCH, 32kbps RAB + 32kbps RAB) =<br>(TF0, TF0), (TF1, TF0), (TF2, TF0), (TF0, TF1), [TF1, TF1] (note)<br>(alt. (TF0, TF0), (TF1, TF0), (TF2, TF0), [TF3, TF0] (note), (TF0, TF1), [TF1, TF1] (note)) |
| NOTE: These TFCs are available only if SCCPCH can be allocated bigger Tx power than required Tx power for TFC of (TF2, TF0). | |

6.10.2.4.3.2a.2 Physical channel parameters

| | | |
|---|---|---|
| SCCPCH | DTX position | Flexible |
| | Spreading factor | 64 |
| | Number of TFCI bits/slot | 8 |
| | Number of Pilot bits/slot | 0 |
| | Number of data bits/slot | 72 |
| | Number of data bits/frame | 1 080 |

6.10.2.4.3.3 Interactive/Background 32 kbps RAB + SRB for PCCH + SRB for CCCH + SRB for DCCH + SRB for BCCH

6.10.2.4.3.3.1 Transport channel parameters

6.10.2.4.3.3.1.1 Transport channel parameters of SRB for Interactive/Background 32 kbps RAB

See clause 6.10.2.4.3.2.1.

6.10.2.4.3.3.1.2 Transport channel parameters of SRB for PCCH

See clause 6.10.2.4.3.1.1.

6.10.2.4.3.3.1.3 Transport channel parameters of SRBs for CCCH, SRB for DCCH, and SRB for BCCH

See clause 6.10.2.4.3.2.1.2

6.10.2.4.3.3.1.4 TFCS

| | |
|---|---|
| TFCS size | 6, 7 or 8 for 240 bits PCH TrBlk size and TF3 not used<br>(alt 6, 7, 8 or 9 for 80 bits PCH TrBlk size and TF3 not used)<br>(alt 6, 7, 8 or 9 for 240 bits PCH TrBlk size and TF3 used)<br>(alt. 6, 7, 8, 9, 10, or 11 for 80 bits PCH TrBlk size and TF3 used) |
| TFCS | (SRB for PCCH, SRBs for CCCH/ DCCH/ BCCH, 32 kbps RAB) =<br>(TF0, TF0, TF0), (TF1, TF0, TF0), (TF0, TF1, TF0), (TF1, TF1, TF0), (TF0, TF2, TF0), [TF1, TF2, TF0] (see note), (TF0, TF0, TF1), [TF0, TF1, TF1] (see note) for 240 bits PCH TrBlk size and TF3 not used<br>(alt. (TF0, TF0, TF0), (TF1, TF0, TF0), (TF0, TF1, TF0), (TF1, TF1, TF0), (TF0, TF2, TF0), [TF1, TF2, TF0] (see note), (TF0, TF0, TF1), [TF1, TF0, TF1] (see note), [TF0, TF1, TF1] (see note) for 80 bits PCH TrBlk size and TF3 not used)<br>(alt. (TF0, TF0, TF0), (TF1, TF0, TF0), (TF0, TF1, TF0), (TF1, TF1, TF0), (TF0, TF2, TF0), [TF1, TF2, TF0] (see note), [TF0, TF3, TF0] (see note), (TF0, TF0, TF1), [TF0, TF1, TF1] (see note) for 240 bits PCH TrBlk size and TF3 used)<br>(alt. (TF0, TF0, TF0), (TF1, TF0, TF0), (TF0, TF1, TF0), (TF1, TF1, TF0), (TF0, TF2, TF0), [TF1, TF2, TF0] (see note), [TF0, TF3, TF0] (see note), [TF1, TF3, TF0] (see note), (TF0, TF0, TF1), [TF1, TF0, TF1] (see note), [TF0, TF1, TF1] (see note) for 80 bits PCH TrBlk size and TF3 used) |
| NOTE: These TFCs are available only if SCCPCH can be allocated bigger Tx power than required Tx power for TFC of (TF0, TF2, TF0). | |

6.10.2.4.3.3.2 Physical channel parameters

| | | |
|---|---|---|
| SCCPCH | DTX position | Flexible |
| | Spreading factor | 64 |

| | | |
|---|---|---|
| | Number of TFCI bits/slot | 8 |
| | Number of Pilot bits/slot | 0 |
| | Number of data bits/slot | 72 |
| | Number of data bits/frame | 1 080 |

6.10.2.4.3.4 RB for CTCH + SRB for CCCH + SRB for BCCH

6.10.2.4.3.4.1 Transport channel parameters

6.10.2.4.3.4.1.1 Transport channel parameters of RB for CTCH

| | | | |
|---|---|---|---|
| Higher layer | RAB/signalling RB | | N/A |
| | User of Radio Bearer | | BMC |
| RLC | Logical channel type | | CTCH |
| | RLC mode | | UM |
| | Payload sizes, bit | | 152 |
| | Max data rate, bps | | 15 200 |
| | UMD PDU header, bit | | 8 |
| MAC | MAC header, bit | | 8 |
| | MAC multiplexing | | N/A |
| Layer 1 | TrCH type | | FACH |
| | TB sizes, bit | | 168 |
| | TFS | TF0, bts | 0x168 |
| | | TF1, bits | 1x168 |
| | TTI, ms | | 10 |
| | Coding type | | CC 1/3 |
| | CRC, bit | | 16 |
| | Max number of bits/TTI before rate matching | | 576 |
| | RM attribute | | 200 to 240 |

6.10.2.4.3.4.1.2 Transport channel parameters of SRB for CCCH and SRB for BCCH

| | | | | |
|---|---|---|---|---|
| Higher layer | RAB/signalling RB | | SRB#0 | SRB#5 |
| | User of Radio Bearer | | RRC | RRC |
| RLC | Logical channel type | | CCCH | BCCH |
| | RLC mode | | UM | TM |
| | Payload sizes, bit | | 152 | 166 |
| | Max data rate, bps | | 15 200 | 16 600 |
| | AMD/UMD/TrD PDU header, bit | | 8 | 0 |
| MAC | MAC header, bit | | 8 | 2 |
| | MAC multiplexing | | 2 logical channel multiplexing | |
| Layer 1 | TrCH type | | FACH | |
| | TB sizes, bit | | 168 | |
| | TFS | TF0, bits | 0x168 | |
| | | TF1, bits | 1x168 | |
| | TTI, ms | | 10 | |
| | Coding type | | CC 1/3 | |
| | CRC, bit | | 16 | |
| | Max number of bits/TTI before rate matching | | 576 | |
| | RM attribute | | 200 to 240 | |

6.10.2.4.3.4.1.3 TFCS

| | |
|---|---|
| TFCS size | 3 |
| TFCS | (SRBs for CCCH/ BCCH, RB for CTCH) =<br>(TF0, TF0), (TF1, TF0), (TF0, TF1) |

6.10.2.4.3.4.2 Physical channel parameters

| | | |
|---|---|---|
| SCCPCH | DTX position | Flexible |
| | Spreading factor | 128 |
| | Number of TFCI bits/slot | 2 |
| | Number of Pilot bits/slot | 0 |
| | Number of data bits/slot | 38 |
| | Number of data bits/frame | 570 |

## 6.10.2.4.4 Combinations on PRACH

### 6.10.2.4.4.1 Interactive/Background 32 kbps PS RAB + SRB for CCCH + SRB for DCCH

#### 6.10.2.4.4.1.1 Transport channel parameters

##### 6.10.2.4.4.1.1.1 Transport channel parameter for Interactive/Background 32 kbps PS RAB, SRB for CCCH, SRB for DCCH

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| Higher layer | RAB/signalling RB | RAB | SRB#0 | SRB#1 | SRB#2 | SRB#3 | SRB#4 |
| | User of Radio Bearer | Interactive/ Background RAB | RRC | RRC | RRC | NAS_DT High prio | NAS_DT Low prio |
| RLC | Logical channel type | DTCH | CCCH | DCCH | DCCH | DCCH | DCCH |
| | RLC mode | AM | TM | UM | AM | AM | AM |
| | Payload sizes, bit | 320 | 166 | 136 | 128 | 128 | 128 |
| | Max data rate, bps | 32 000 | 16 600 | 13 600 | 12 800 | 12 800 | 12 800 |
| | AMD/UMD/TrD PDU header, bit | 16 | 0 | 8 | 16 | 16 | 16 |
| MAC | MAC header, bit | 24 | 2 | 24 | 24 | 24 | 24 |
| | MAC multiplexing | 6 logical channel multiplexing | | | | | |
| Layer 1 | TrCH type | RACH | | | | | |
| | TB sizes, bit | 360 | 168 | 168 | 168 | 168 | 168 |
| | TFS TF0, bits | 1x168 | | | | | |
| | TFS TF1, bits | 1x360 | | | | | |
| | TTI, ms | 20 (alt. 10) | | | | | |
| | Coding type | CC 1/2 | | | | | |
| | CRC, bit | 16 | | | | | |
| | Max number of bits/TTI after channel coding | 768 | 384 | 384 | 384 | 384 | 384 |
| | Max number of bits/ Radio frame before rate matching | 384 (alt. 768) | 192 (alt. 384) | 192 (alt. 384) | 192 (alt. 384) | 192 (alt. 384) | 192 (alt. 384) |

##### 6.10.2.4.4.1.1.2 TFCS

| | |
|---|---|
| TFCS size | 2 |
| TFCS | 32 kbps + SRBs for CCCH/ DCCH = TF0, TF1 |

#### 6.10.2.4.4.1.2 Physical channel parameters

| | | |
|---|---|---|
| PRACH | Minimum Spreading factor | 64 (alt. 32) |
| | Max number of data bits/radio frame | 600 (alt. 1 200) |
| | Puncturing Limit | 1 |

### 6.10.2.4.4.2 Interactive/Background 32 kbps PS RAB + Interactive/Background 32 kbps PS RAB + SRB for CCCH + SRB for DCCH

#### 6.10.2.4.4.2.1 Transport channel parameters

##### 6.10.2.4.4.2.1.1 Transport channel parameters for Interactive/Background 32 kbps PS RAB, Interactive/Background 32 kbps PS RAB, SRB for CCCH, SRB for DCCH

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| Higher | RAB/signalling RB | RAB | RAB | SRB#0 | SRB#1 | SRB#2 | SRB#3 | SRB#4 |

| | User of Radio Bearer | Interactive/ Background RAB | Interactive/ Background RAB | RRC | RRC | RRC | NAS_DT High prio | NAS_DT Low prio |
|---|---|---|---|---|---|---|---|---|
| RLC | Logical channel type | DTCH | DTCH | CCCH | DCCH | DCCH | DCCH | DCCH |
| | RLC mode | AM | AM | TM | UM | AM | AM | AM |
| | Payload sizes, bit | 320 | 320 | 166 | 136 | 128 | 128 | 128 |
| | Max data rate, bps | 32 000 | 32 000 | 16 600 | 13 600 | 12 800 | 12 800 | 12 800 |
| | AMD/UMD/TrD PDU header, bit | 16 | 16 | 0 | 8 | 16 | 16 | 16 |
| MAC | MAC header, bit | 24 | 24 | 2 | 24 | 24 | 24 | 24 |
| | MAC multiplexing | 7 logical channel multiplexing | | | | | | |
| Layer 1 | TrCH type | RACH | | | | | | |
| | TB sizes, bit | 360 | 360 | 168 | 168 | 168 | 168 | 168 |
| | TFS TF0, bits | 1x168 | | | | | | |
| | TFS TF1, bits | 1x360 | | | | | | |
| | TTI, ms | 20 (alt. 10) | | | | | | |
| | Coding type | CC 1/2 | | | | | | |
| | CRC, bit | 16 | | | | | | |
| | Max number of bits/TTI after channel coding | 768 | 768 | 384 | 384 | 384 | 384 | 384 |
| | Max number of bits/ Radio frame before rate matching | 384 (alt. 768) | 384 (alt 768) | 192 (alt. 384) | 192 (alt. 384) | 192 (alt. 384) | 192 (alt. 384) | 192 (alt. 384) |

6.10.2.4.4.2.1.2 TFCS

| | |
|---|---|
| TFCS size | 2 |
| TFCS | 32 kbps RAB+ 32 kbps RAB + SRBs for CCCH/ DCCH = TF0, TF1 |

6.10.2.4.4.2.2 Physical channel parameters

| | | |
|---|---|---|
| PRACH | Minimum Spreading factor | 64 (alt. 32) |
| | Max number of data bits/radio frame | 600 (alt. 1 200) |
| | Puncturing Limit | 1 |

## 6.10.2.4.5 Combinations on DPCH and HS-PDSCH

### 6.10.2.4.5.1 Interactive or background / UL:64 DL: [max bit rate depending on UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

#### 6.10.2.4.5.1.1 Uplink

See clause 6.10.2.4.1.26.1.

#### 6.10.2.4.5.1.2 Downlink

##### 6.10.2.4.5.1.2.1 Transport channel parameters

###### 6.10.2.4.5.1.2.1.1 Transport channel parameters for HS-DSCH

6.10.2.4.5.1.2.1.1.1 MAC-d flow parameters for Interactive or background / DL: [max bit rate depending on UE category] / PS RAB

| | | |
|---|---|---|
| Higher layer | RAB/Signalling RB | RAB |
| RLC | Logical channel type | DTCH |
| | RLC mode | AM |
| | Payload sizes, bit | 320 (alt. 640) |
| | Max data rate, bps | depends on UE category NOTE1 |
| | AMD PDU header, bit | 16 |

| | | |
|---|---|---|
| MAC | MAC-d header, bit | 0 |
| | MAC multiplexing | N/A |
| | MAC-d PDU size, bit | 336 (alt. 656) |
| | MAC-hs header fixed part, bit | 21 |
| Layer 1 | TrCH type | HS-DSCH |
| | TTI | 2 ms |
| | Coding type | TC |
| | CRC, bit | 24 |
| NOTE: The peak throughput may be limited by the maximum number of MAC-d PDUs that can be included in a single MAC-hs PDU (see 3GPP TS 25.321 [38]). | | |

6.10.2.4.5.1.2.1.2 Transport channel parameters for DCH

6.10.2.4.5.1.2.1.2.1 Transport channel parameters for UL:3.4 DL: 3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.2.1.1.

6.10.2.4.5.1.2.1.2.2 TFCS

See clause 6.10.2.4.1.2.2.1.2.

6.10.2.4.5.1.2.2 Physical channel parameters

6.10.2.4.5.1.2.2.1 Physical channel parameters on DPCH

See clause 6.10.2.4.1.2.2.2.

6.10.2.4.5.1.2.2.2 Physical channel parameters on HS-PDSCH

Note that each alternative configuration in physical channel parameters is stand-alone and can be associated with any of the RAB alternatives in the transport channel parameters.

UE HS-DSCH Physical Layer category 1:

| | | |
|---|---|---|
| HS-PDSCH | Number of processes | 2, (alt. 8) |
| | Process memory size | Split equally among all processes |
| | Max Data Rate | 1.2 Mbps, (alt. 400 kbps) |

UE HS-DSCH Physical Layer category 2:

| | | |
|---|---|---|
| HS-PDSCH | Number of processes | 2, (alt. 8) |
| | Process memory size | Split equally among all processes |
| | Max Data Rate | 1.2 Mbps, (alt. 600 kbps)] |

UE HS-DSCH Physical Layer category 3:

| | | |
|---|---|---|
| HS-PDSCH | Number of processes | 3, (alt. 8) |
| | Process memory size | Split equally among all processes |
| | Max Data Rate | 1.8 Mbps, (alt. 900 kbps) |

UE HS-DSCH Physical Layer category 4:

| | | |
|---|---|---|
| HS-PDSCH | Number of processes | 3, (alt. 8) |
| | Process memory size | Split equally among all processes |
| | Max Data Rate | 1.8 Mbps, (alt. 1.2 Mbps) |

UE HS-DSCH Physical Layer category 5:

| | | |
|---|---|---|
| HS-PDSCH | Number of processes | 6, (alt. 8) |
| | Process memory size | Split equally among all processes |
| | Max Data Rate | 3.65 Mbps, (alt. 3.6 Mbps) |

UE HS-DSCH Physical Layer category 6:

| | | |
|---|---|---|
| HS-PDSCH | Number of processes | 6, (alt. 8) |
| | Process memory size | Split equally among all processes |
| | Max Data Rate | 3.65 Mbps, (alt. 3.65 Mbps) |

UE HS-DSCH Physical Layer category 7:

| | | |
|---|---|---|
| HS-PDSCH | Number of processes | 6, (alt. 8) |
| | Process memory size | Split equally among all processes |
| | Max Data Rate | 7.2 Mbps, (alt. 7.2 Mbps) |

UE HS-DSCH Physical Layer category 8:

| | | |
|---|---|---|
| HS-PDSCH | Number of processes | 6, (alt. 8) |
| | Process memory size | Split equally among all processes |
| | Max Data Rate | 7.2 Mbps, (alt. 7.2 Mbps) |

UE HS-DSCH Physical Layer category 9:

| | | |
|---|---|---|
| HS-PDSCH | Number of processes | 6, (alt. 8) |
| | Process memory size | Split equally among all processes |
| | Max Data Rate | 10.1 Mbps, (alt. 10.1 Mbps) |

UE HS-DSCH Physical Layer category 10:

| | | |
|---|---|---|
| HS-PDSCH | Number of processes | 6, (alt. 8) |
| | Process memory size | Split equally among all processes |
| | Max Data Rate | 14.0 Mbps, (alt. 10.8 Mbps) |

UE HS-DSCH Physical Layer category 11:

| | | |
|---|---|---|
| HS-PDSCH | Number of processes | 3, (alt. 8) |
| | Process memory size | Split equally among all processes |
| | Max Data Rate | 900 kbps, (alt. 450 kbps) |

UE HS-DSCH Physical Layer category 12:

| | | |
|---|---|---|
| HS-PDSCH | Number of processes | 6, (alt. 8) |
| | Process memory size | Split equally among all processes |
| | Max Data Rate | 1.8 Mbps, (alt. 1.8 Mbps) |

6.10.2.4.5.1a Interactive or background / UL:128 DL: [max bit rate depending on UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.2.4.5.1a.1 Uplink

See clause 6.10.2.4.1.28.1.

6.10.2.4.5.1a.2 Downlink

6.10.2.4.5.1a.2.1 Transport channel parameters

6.10.2.4.5.1a.2.1.1 Transport channel parameters for HS-DSCH

6.10.2.4.5.1a.2.1.1.1 MAC-d flow parameters for Interactive or background / DL: [max bit rate depending on UE category] / PS RAB

See clause 6.10.2.4.5.1.2.1.1.1.

6.10.2.4.5.1a.2.1.2 Transport channel parameters for DCH

6.10.2.4.5.1a.2.1.2.1 Transport channel parameters for DL: 3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.2.1.1.

6.10.2.4.5.1a.2.1.2.2 TFCS

See clause 6.10.2.4.1.2.2.1.2.

6.10.2.4.5.1a.2.2 Physical channel parameters

6.10.2.4.5.1a.2.2.1 Physical channel parameters on DPCH

See clause 6.10.2.4.1.2.2.2.

6.10.2.4.5.1a.2.2.2 Physical channel parameters on HS-PDSCH

See clause 6.10.2.4.5.1.2.2.2.

6.10.2.4.5.2 Interactive or background / UL:384 DL: [max bit rate depending on UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.2.4.5.2.1 Uplink

See clause 6.10.2.4.1.34.1.

6.10.2.4.5.2.2 Downlink

6.10.2.4.5.2.2.1 Transport channel parameters

6.10.2.4.5.2.2.1.1 Transport channel parameters for HS-DSCH

6.10.2.4.5.2.2.1.1.1 MAC-d flow parameters for Interactive or background / DL: [max bit rate depending on UE category] / PS RAB

See clause 6.10.2.4.5.1.2.1.1.1.

6.10.2.4.5.2.2.1.2 Transport channel parameters for DCH

6.10.2.4.5.2.2.1.2.1 Transport channel parameters for DL: 3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.2.1.1.

6.10.2.4.5.2.2.1.2.2 TFCS

See clause 6.10.2.4.1.2.2.1.2.

6.10.2.4.5.2.2.2 Physical channel parameters

6.10.2.4.5.2.2.2.1 Physical channel parameters on DPCH

See clause 6.10.2.4.1.2.2.2.

6.10.2.4.5.2.2.2.2 Physical channel parameters on HS-PDSCH

See clause 6.10.2.4.5.1.2.2.2.

6.10.2.4.5.3 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:384 DL:[Bit rate depending on the UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.2.4.5.3.1 Uplink

6.10.2.4.5.3.1.1 Transport channel parameters

6.10.2.4.5.3.1.1.1 Transport channel parameters for Conversational / speech / UL:12.2 kbps / CS RAB

See clause 6.10.2.4.1.4.1.1.1.

6.10.2.4.5.3.1.1.2 Transport channel parameters for Interactive or background / UL:384 kbps / PS RAB

See clause 6.10.2.4.1.34.1.1.1.

6.10.2.4.5.3.1.1.3 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.1.1.1.

6.10.2.4.5.3.1.1.4 TFCS

| TFCS size | 54 (alt. 36) |
|---|---|
| TFCS | (RAB subflow#1, RAB subflow#2, RAB subflow#3, 384 kbps RAB , DCCH)=<br>(TF0, TF0, TF0, TF0, TF0), (TF1, TF0, TF0, TF0, TF0), (TF2, TF1, TF1, TF0, TF0),<br>(TF0, TF0, TF0, TF1, TF0), (TF1, TF0, TF0, TF1, TF0), (TF2, TF1, TF1, TF1, TF0),<br>(TF0, TF0, TF0, TF2, TF0), (TF1, TF0, TF0, TF2, TF0), (TF2, TF1, TF1, TF2, TF0),<br>(TF0, TF0, TF0, TF3, TF0), (TF1, TF0, TF0, TF3, TF0), (TF2, TF1, TF1, TF3, TF0),<br>(TF0, TF0, TF0, TF4, TF0), (TF1, TF0, TF0, TF4, TF0), (TF2, TF1, TF1, TF4, TF0),<br>(TF0, TF0, TF0, TF5, TF0), (TF1, TF0, TF0, TF5, TF0), (TF2, TF1, TF1, TF5, TF0),<br>(TF0, TF0, TF0, TF6, TF0), (TF1, TF0, TF0, TF6, TF0), (TF2, TF1, TF1, TF6, TF0),<br>(TF0, TF0, TF0, TF7, TF0), (TF1, TF0, TF0, TF7, TF0), (TF2, TF1, TF1, TF7, TF0),<br>(TF0, TF0, TF0, TF8, TF0), (TF1, TF0, TF0, TF8, TF0), (TF2, TF1, TF1, TF8, TF0),<br>(TF0, TF0, TF0, TF0, TF1), (TF1, TF0, TF0, TF0, TF1), (TF2, TF1, TF1, TF0, TF1),<br>(TF0, TF0, TF0, TF1, TF1), (TF1, TF0, TF0, TF1, TF1), (TF2, TF1, TF1, TF1, TF1),<br>(TF0, TF0, TF0, TF2, TF1), (TF1, TF0, TF0, TF2, TF1), (TF2, TF1, TF1, TF2, TF1),<br>(TF0, TF0, TF0, TF3, TF1), (TF1, TF0, TF0, TF3, TF1), (TF2, TF1, TF1, TF3, TF1),<br>(TF0, TF0, TF0, TF4, TF1), (TF1, TF0, TF0, TF4, TF1), (TF2, TF1, TF1, TF4, TF1),<br>(TF0, TF0, TF0, TF5, TF1), (TF1, TF0, TF0, TF5, TF1), (TF2, TF1, TF1, TF5, TF1),<br>(TF0, TF0, TF0, TF6, TF1), (TF1, TF0, TF0, TF6, TF1), (TF2, TF1, TF1, TF6, TF1),<br>(TF0, TF0, TF0, TF7, TF1), (TF1, TF0, TF0, TF7, TF1), (TF2, TF1, TF1, TF7, TF1),<br>(TF0, TF0, TF0, TF8, TF1), (TF1, TF0, TF0, TF8, TF1), (TF2, TF1, TF1, TF8, TF1)<br>(alt. (TF0, TF0, TF0, TF0, TF0), (TF1, TF0, TF0, TF0, TF0), (TF2, TF1, TF1, TF0, TF0),<br>(TF0, TF0, TF0, TF1, TF0), (TF1, TF0, TF0, TF1, TF0), (TF2, TF1, TF1, TF1, TF0),<br>(TF0, TF0, TF0, TF2, TF0), (TF1, TF0, TF0, TF2, TF0), (TF2, TF1, TF1, TF2, TF0),<br>(TF0, TF0, TF0, TF3, TF0), (TF1, TF0, TF0, TF3, TF0), (TF2, TF1, TF1, TF3, TF0),<br>(TF0, TF0, TF0, TF4, TF0), (TF1, TF0, TF0, TF4, TF0), (TF2, TF1, TF1, TF4, TF0),<br>(TF0, TF0, TF0, TF5, TF0), (TF1, TF0, TF0, TF5, TF0), (TF2, TF1, TF1, TF5, TF0),<br>(TF0, TF0, TF0, TF0, TF1), (TF1, TF0, TF0, TF0, TF1), (TF2, TF1, TF1, TF0, TF1),<br>(TF0, TF0, TF0, TF1, TF1), (TF1, TF0, TF0, TF1, TF1), (TF2, TF1, TF1, TF1, TF1),<br>(TF0, TF0, TF0, TF2, TF1), (TF1, TF0, TF0, TF2, TF1), (TF2, TF1, TF1, TF2, TF1),<br>(TF0, TF0, TF0, TF3, TF1), (TF1, TF0, TF0, TF3, TF1), (TF2, TF1, TF1, TF3, TF1),<br>(TF0, TF0, TF0, TF4, TF1), (TF1, TF0, TF0, TF4, TF1), (TF2, TF1, TF1, TF4, TF1),<br>(TF0, TF0, TF0, TF5, TF1), (TF1, TF0, TF0, TF5, TF1), (TF2, TF1, TF1, TF5, TF1)) |

6.10.2.4.5.3.1.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Uplink | Min spreading factor | 4 |
| | Max number of DPDCH data bits/radio frame | 9 600 |
| | Number of DPDCH | 1 |
| | Puncturing Limit | 0.60 |

6.10.2.4.5.3.2 Downlink

6.10.2.4.5.3.2.1 Transport channel parameters

6.10.2.4.5.3.2.1.1 Transport channel parameters for HS-DSCH

See clause 6.10.2.4.5.1.2.1.1.1.

6.10.2.4.5.3.2.1.2 Transport channel parameters for DCH

6.10.2.4.5.3.2.1.2.1 Transport channel parameters for Conversational / speech / DL:12.2 kbps / CS RAB

See clause 6.10.2.4.1.4.2.1.1.

6.10.2.4.5.3.2.1.2.2 Transport channel parameters for UL:3.4 DL: 3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.2.1.1.

6.10.2.4.5.3.2.1.2.3 TFCS

See clause 6.10.2.4.1.4.2.1.3.

6.10.2.4.5.3.2.2 Physical channel parameters

6.10.2.4.5.3.2.2.1 Physical channel parameters on DPCH

See clause 6.10.2.4.1.4.2.2.

6.10.2.4.5.3.2.2.2 Physical channel parameters on HS-PDSCH

See clause 6.10.2.4.5.1.2.2.2.

6.10.2.4.5.3a Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:64 DL:[Bit rate depending on the UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.2.4.5.3a.1 Uplink

See clause 6.10.2.4.1.40.1.

6.10.2.4.5.3a.2 Downlink

6.10.2.4.5.3a.2.1 Transport channel parameters

6.10.2.4.5.3a.2.1.1 Transport channel parameters for HS-DSCH

See clause 6.10.2.4.5.1.2.1.1.

6.10.2.4.5.3a.2.1.2 Transport channel parameters for DCH

6.10.2.4.5.3a.2.1.2.1 Transport channel parameters for Conversational / speech / DL:12.2 kbps / CS RAB

See clause 6.10.2.4.1.4.2.1.1.

6.10.2.4.5.3a.2.1.2.2 Transport channel parameters for DL: 3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.2.1.1.

6.10.2.4.5.3a.2.1.2.3 TFCS

See clause 6.10.2.4.1.4.2.1.3.

6.10.2.4.5.3a.2.2 Physical channel parameters

6.10.2.4.5.3a.2.2.1 Physical channel parameters on DPCH

See clause 6.10.2.4.1.4.2.2.

6.10.2.4.5.3a.2.2.2 Physical channel parameters on HS-PDSCH

See clause 6.10.2.4.5.1.2.2.2.

6.10.2.4.5.4 Conversational / unknown / UL:64 DL:64 kbps / CS RAB + Interactive or background / UL:384 DL:[Bit rate depending on the UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.2.4.5.4.1 Uplink

6.10.2.4.5.4.1.1 Transport channel parameters

6.10.2.4.5.4.1.1.1 Transport channel parameters for Conversational / unknown / UL:64 kbps / CS RAB

See clause 6.10.2.4.1.13.1.1.1.

6.10.2.4.5.4.1.1.2 Transport channel parameters for Interactive or background / UL:384 kbps / PS RAB

See clause 6.10.2.4.1.34.1.1.1.

6.10.2.4.5.4.1.1.3 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.1.1.1.

6.10.2.4.5.4.1.1.4 TFCS

| | |
|---|---|
| TFCS size | 36 (alt. 24) |
| TFCS | (64 kbps RAB, 384 kbps RAB , DCCH)=<br>(TF0, TF0, TF0), (TF1, TF0, TF0), (TF0, TF1, TF0), (TF1, TF1, TF0),<br>(TF0, TF2, TF0), (TF1, TF2, TF0), (TF0, TF3, TF0), (TF1, TF3, TF0),<br>(TF0, TF4, TF0), (TF1, TF4, TF0), (TF0, TF5, TF0), (TF1, TF5, TF0),<br>(TF0, TF6, TF0), (TF1, TF6, TF0), (TF0, TF7, TF0), (TF1, TF7, TF0),<br>(TF0, TF8, TF0), (TF1, TF8, TF0), (TF0, TF0, TF1), (TF1, TF0, TF1),<br>(TF0, TF1, TF1), (TF1, TF1, TF1), (TF0, TF2, TF1), (TF1, TF2, TF1),<br>(TF0, TF3, TF1), (TF1, TF3, TF1), (TF0, TF4, TF1), (TF1, TF4, TF1),<br>(TF0, TF5, TF1), (TF1, TF5, TF1), (TF0, TF6, TF1), (TF1, TF6, TF1),<br>(TF0, TF7, TF1), (TF1, TF7, TF1), (TF0, TF8, TF1), (TF1, TF8, TF1)<br>(alt. (TF0, TF0, TF0), (TF1, TF0, TF0), (TF0, TF1, TF0), (TF1, TF1, TF0),<br>(TF0, TF2, TF0), (TF1, TF2, TF0), (TF0, TF3, TF0), (TF1, TF3, TF0),<br>(TF0, TF4, TF0), (TF1, TF4, TF0), (TF0, TF5, TF0), (TF1, TF5, TF0),<br>(TF0, TF0, TF1), (TF1, TF0, TF1), (TF0, TF1, TF1), (TF1, TF1, TF1),<br>(TF0, TF2, TF1), (TF1, TF2, TF1), (TF0, TF3, TF1), (TF1, TF3, TF1),<br>(TF0, TF4, TF1), (TF1, TF4, TF1), (TF0, TF5, TF1), (TF1, TF5, TF1)) |

6.10.2.4.5.4.1.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Uplink | Min spreading factor | 4 |
| | Max number of DPDCH data bits/radio frame | 9 600 |
| | Number of DPDCH | 1 |
| | Puncturing Limit | 0.52 |

6.10.2.4.5.4.2 Downlink

6.10.2.4.5.4.2.1 Transport channel parameters

6.10.2.4.5.4.2.1.1 Transport channel parameters for HS-DSCH

See clause 6.10.2.4.5.1.2.1.1.1.

6.10.2.4.5.4.2.1.2 Transport channel parameters for DCH

6.10.2.4.5.4.2.1.2.1 Transport channel parameters for Conversational / unknown / DL:64 kbps / CS RAB

See clause 6.10.2.4.1.13.2.1.1.

6.10.2.4.5.4.2.1.2.2 Transport channel parameters for UL:3.4 DL: 3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.2.1.1.

6.10.2.4.5.4.2.1.2.3 TFCS

See clause 6.10.2.4.1.13.2.1.3.

6.10.2.4.5.4.2.2 Physical channel parameters

6.10.2.4.5.4.2.2.1 Physical channel parameters on DPCH

See clause 6.10.2.4.1.13.2.2.

6.10.2.4.5.4.2.2.2 Physical channel parameters on HS-PDSCH

See clause 6.10.2.4.5.1.2.2.2.

6.10.2.4.5.4a Conversational / unknown / UL:64 DL:64 kbps / CS RAB + Interactive or background / UL:64 DL:[Bit rate depending on the UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.2.4.5.4a.1 Uplink

See clause 6.10.2.4.1.51.1.

6.10.2.4.5.4a.2 Downlink

6.10.2.4.5.4a.2.1 Transport channel parameters

6.10.2.4.5.4a.2.1.1 Transport channel parameters for HS-DSCH

See clause 6.10.2.4.5.1.2.1.1.

6.10.2.4.5.4a.2.1.2 Transport channel parameters for DCH

6.10.2.4.5.4a.2.1.2.1 Transport channel parameters for Conversational / unknown / DL:64 kbps / CS RAB

See clause 6.10.2.4.1.13.2.1.1.

6.10.2.4.5.4a.2.1.2.2 Transport channel parameters for DL: 3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.2.1.1.

6.10.2.4.5.4a.2.1.2.3 TFCS

See clause 6.10.2.4.1.13.2.1.3.

6.10.2.4.5.4a.2.2 Physical channel parameters

6.10.2.4.5.4a.2.2.1 Physical channel parameters on DPCH

See clause 6.10.2.4.1.13.2.2.

6.10.2.4.5.4a.2.2.2 Physical channel parameters on HS-PDSCH

See clause 6.10.2.4.5.1.2.2.2.

6.10.2.4.5.5 Interactive or background / UL:384 DL:[Bit rate depending on the UE category] / PS RAB + Interactive or background / UL:384 DL:[Bit rate depending on the UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH6.10.2.4.5.5.1 Uplink

6.10.2.4.5.5.1.1 Transport channel parameters

6.10.2.4.5.5.1.1.1 Transport channel parameters for Interactive or background / UL:384 kbps / PS RAB + UL:384 kbps / PS RAB

| | | | | |
|---|---|---|---|---|
| Higher layer | RAB/Signalling RB | | RAB | RAB |
| RLC | Logical channel type | | DTCH | DTCH |
| | RLC mode | | AM | AM |
| | Payload sizes, bit | | 320 | 320 |
| | Max data rate, bps | | 384 000 | 384 000 |
| | AMD PDU header, bit | | 16 | 16 |
| MAC | MAC header, bit | | 4 | 4 |
| | MAC multiplexing | | 2 logical channel multiplexing | |
| Layer 1 | TrCH type | | DCH | |
| | TB sizes, bit | | 340 | |
| | TFS | TF0, bits | 0x340 | |
| | | TF1, bits | 1x340 | |
| | | TF2, bits | 2x340 | |
| | | TF3, bits | 4x340 | |
| | | TF4, bits | 8x340 | |
| | | TF5, bits | 12x340 | |
| | TTI, ms | | 10 | |
| | Coding type | | TC | |

| | | |
|---|---|---|
| | CRC, bit | 16 |
| | Max number of bits/TTI after channel coding | 12 828 |
| | Uplink: Max number of bits/radio frame before rate matching | 12 828 |
| | RM attribute | 110-180 |

6.10.2.4.5.5.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.1.1.1.

6.10.2.4.5.5.1.1.3 TFCS

| | |
|---|---|
| TFCS size | 12 |
| TFCS | (384 kbps RAB + 384 kbps RAB, DCCH)=<br>(TF0, TF0), (TF1, TF0), (TF2, TF0), (TF3, TF0), (TF4, TF0), (TF5, TF0),<br>(TF0, TF1), (TF1, TF1), (TF2, TF1), (TF3, TF1), (TF4, TF1), (TF5, TF1) |

6.10.2.4.5.5.1.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Uplink | Min spreading factor | 4 |
| | Max number of DPDCH data bits/radio frame | 9 600 |
| | Number of DPDCH | 1 |
| | Puncturing Limit | 0.64 |

6.10.2.4.5.5.2 Downlink

6.10.2.4.5.5.2.1 Transport channel parameters

6.10.2.4.5.5.2.1.1 Transport channel parameters for HS-DSCH

6.10.2.4.5.5.2.1.1.1 MAC-d flow parameters for Interactive or background / DL: [max bit rate depending on UE category] / PS RAB

See clause 6.10.2.4.5.1.2.1.1.1.

6.10.2.4.5.5.2.1.1.2 MAC-d flow parameters for Interactive or background / DL: [max bit rate depending on UE category] / PS RAB

See clause 6.10.2.4.5.1.2.1.1.1.

6.10.2.4.5.5.2.1.2 Transport channel parameters for DCH

6.10.2.4.5.5.2.1.2.1 Transport channel parameters for UL:3.4 DL: 3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.2.1.1.

6.10.2.4.5.5.2.1.2.2 TFCS

See clause 6.10.2.4.1.2.2.1.2.

6.10.2.4.5.5.2.2 Physical channel parameters

6.10.2.4.5.5.2.2.1 Physical channel parameters on DPCH

See clause 6.10.2.4.1.2.2.2.

6.10.2.4.5.5.2.2.2 Physical channel parameters on HS-PDSCH

See clause 6.10.2.4.5.1.2.2.2.

6.10.2.4.5.5a Interactive or background / UL:64 DL:[Bit rate depending on the UE category] / PS RAB + Interactive or background / UL:64 DL:[Bit rate depending on the UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.2.4.5.5a.1 Uplink

See clause 6.10.2.4.1.57.1.

6.10.2.4.5.5a.2 Downlink

6.10.2.4.5.5a.2.1 Transport channel parameters

6.10.2.4.5.5a.2.1.1 Transport channel parameters for HS-DSCH

6.10.2.4.5.5a.2.1.1.1 MAC-d flow parameters for Interactive or background / DL: [max bit rate depending on UE category] / PS RAB

See clause 6.10.2.4.5.1.2.1.1.1.

6.10.2.4.5.5a.2.1.1.2 MAC-d flow parameters for Interactive or background / DL: [max bit rate depending on UE category] / PS RAB

See clause 6.10.2.4.5.1.2.1.1.1.

6.10.2.4.5.5a.2.1.2 Transport channel parameters for DCH

6.10.2.4.5.5a.2.1.2.1 Transport channel parameters for DL: 3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.2.1.1.

6.10.2.4.5.5a.2.1.2.2 TFCS

See clause 6.10.2.4.1.2.2.1.2.

6.10.2.4.5.5a.2.2 Physical channel parameters

6.10.2.4.5.5a.2.2.1 Physical channel parameters on DPCH

See clause 6.10.2.4.1.2.2.2.

6.10.2.4.5.5a.2.2.2 Physical channel parameters on HS-PDSCH

See clause 6.10.2.4.5.1.2.2.2.

6.10.2.4.5.6 Streaming / unknown / UL:128 DL: [guaranteed 128, max bit rate depending on UE category] kbps / PS RAB + Interactive or background / UL:128 DL: [max bit rate depending on UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.2.4.5.6.1 Uplink

6.10.2.4.5.6.1.1 Transport channel parameters

6.10.2.4.5.6.1.1.1 Transport channel parameters for Streaming / unknown / UL:128 kbps / PS RAB

| Higher Layer | RAB/Signalling RB | | **RAB** |
|---|---|---|---|
| RLC | Logical channel type | | DTCH |
| | RLC mode | | AM |
| | Payload sizes, bit | | 640 |
| | Max data rate, bps | | 128000 |
| | AM PDU header, bit | | 16 |
| MAC | MAC header, bit | | 0 |
| | MAC multiplexing | | N/A |
| Layer 1 | TrCH type | | DCH |
| | TB sizes, bit | | 656 |
| | TFS | TF0, bits | 0x656 |
| | | TF1, bits | 1x656 |
| | | TF2, bits | 2x656 |
| | | TF3, bits | 4x656 |
| | TTI, ms | | 20 |
| | Coding type | | TC |
| | CRC, bit | | 16 |
| | Max number of bits/TTI after channel coding | | 8076 |
| | Uplink: Max number of bits/radio frame before rate matching | | 4038 |
| | RM attribute | | 125-165 |

6.10.2.4.1.6.1.1.2 Transport channel parameters for Interactive or background / UL:128 kbps / PS RAB

See clause 6.10.2.4.1.28.1.1.1.

6.10.2.4.1.6.1.1.3 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.1.1.1.

6.10.2.4.1.6.1.1.4 TFCS

| | |
|---|---|
| TFCS size | 40 |
| TFCS | (128 kbps RAB, 128 kbps RAB, DCCH)=<br>(TF0, TF0, TF0), (TF1, TF0, TF0), (TF2, TF0, TF0), (TF3, TF0, TF0),<br>(TF0, TF1, TF0), (TF1, TF1, TF0), (TF2, TF1, TF0), (TF3, TF1, TF0),<br>(TF0, TF2, TF0), (TF1, TF2, TF0), (TF2, TF2, TF0), (TF3, TF2, TF0),<br>(TF0, TF3, TF0), (TF1, TF3, TF0), (TF2, TF3, TF0), (TF3, TF3, TF0),<br>(TF0, TF4, TF0), (TF1, TF4, TF0), (TF2, TF4, TF0), (TF3, TF4, TF0),<br>(TF0, TF0, TF1), (TF1, TF0, TF1), (TF2, TF0, TF1), (TF3, TF0, TF1),<br>(TF0, TF1, TF1), (TF1, TF1, TF1), (TF2, TF1, TF1), (TF3, TF1, TF1),<br>(TF0, TF2, TF1), (TF1, TF2, TF1), (TF2, TF2, TF1), (TF3, TF2, TF1),<br>(TF0, TF3, TF1), (TF1, TF3, TF1), (TF2, TF3, TF1), (TF3, TF3, TF1),<br>(TF0, TF4, TF1), (TF1, TF4, TF1), (TF2, TF4, TF1), (TF3, TF4, TF1) |

6.10.2.4.1.6.1.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Uplink | Min spreading factor | 4 |
| | Max number of DPDCH data bits/radio frame | 9600 |
| | Puncturing Limit | 0.96 |

6.10.2.4.5.6.2 Downlink

6.10.2.4.5.6.2.1 Transport channel parameters

6.10.2.4.5.6.2.1.1 Transport channel parameters for HS-DSCH

6.10.2.4.5.6.2.1.1.1 MAC-d flow parameters for Streaming / unknown / DL: [max bit rate depending on UE category] kbps / PS RAB

| Higher Layer | RAB/Signalling RB | **RAB** |
|---|---|---|
| RLC | Logical channel type | DTCH |
| | RLC mode | AM |
| | Payload sizes, bit | 640 (alt. 320) |
| | Max data rate, bps | depends on UE category NOTE1 |
| | AMD PDU header, bit | 16 |
| MAC | MAC-d header, bit | 0 |
| | MAC multiplexing | N/A |
| | MAC-d PDU size, bit | 656 (alt. 336) |
| | MAC-hs header fixed part, bit | 21 |
| Layer 1 | TrCH type | HS-DSCH |
| | TTI | 2 ms |
| | Coding type | TC |
| | CRC, bit | 24 |

NOTE1: The peak throughput may be limited by the maximum number of MAC-d PDUs that can be included in a single MAC-hs PDU (see [25.321]).

6.10.2.4.5.6.2.1.1.2 MAC-d flow parameters for Interactive or background / DL: [max bit rate depending on UE category] / PS RAB

See clause 6.10.2.4.5.1.2.1.1.1.

6.10.2.4.5.6.2.1.2 Transport channel parameters for DCH

6.10.2.4.5.6.2.1.2.1 Transport channel parameters for DL: 3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.2.1.1.

6.10.2.4.5.6.2.1.2.2 TFCS

See clause 6.10.2.4.1.2.2.1.2.

6.10.2.4.5.6.2.2 Physical channel parameters

6.10.2.4.5.6.2.2.1 Physical channel parameters on DPCH

See clause 6.10.2.4.1.2.2.2.

6.10.2.4.5.6.2.2.2 Physical channel parameters on HS-PDSCH

See clause 6.10.2.4.5.1.2.2.2.

6.10.2.4.5.7 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Streaming / unknown / UL:128 DL: [guaranteed 128, max bit rate depending on UE category] kbps / PS RAB + Interactive or background / UL:128 DL: [max bit rate depending on UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.2.4.5.7.1 Uplink

6.10.2.4.5.7.1.1 Transport channel parameters

6.10.2.4.5.7.1.1.1 Transport channel parameters for Conversational / speech / UL:12.2 kbps / CS RAB

See clause 6.10.2.4.1.4.1.1.1.

6.10.2.4.5.7.1.1.2 Transport channel parameters for Streaming / unknown / UL:128 kbps / PS RAB

See clause 6.10.2.4.5.6.1.1.1.

6.10.2.4.1.7.1.1.3 Transport channel parameters for Interactive or background / UL:128 kbps / PS RAB

See clause 6.10.2.4.1.28.1.1.1.

6.10.2.4.1.7.1.1.4 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.1.1.1.

6.10.2.4.1.7.1.1.5 TFCS

| TFCS size | 62 |
|---|---|
| TFCS | (RAB subflow#1, RAB subflow#2, RAB subflow#3, 128 kbps RAB, 128 kbps RAB, DCCH)=<br>(TF0,TF0,TF0,TF0,TF0,TF0), (TF1,TF0,TF0,TF0,TF0,TF0), (TF2,TF1,TF1,TF0,TF0,TF0),<br>(TF0,TF0,TF0,TF1,TF0,TF0), (TF0,TF0,TF0,TF0,TF1,TF0),<br>(TF0,TF0,TF0,TF2,TF1,TF0), (TF1,TF0,TF0,TF2,TF1,TF0), (TF2,TF1,TF1,TF2,TF1,TF0),<br>(TF0,TF0,TF0,TF3,TF1,TF0), (TF1,TF0,TF0,TF3,TF1,TF0), (TF2,TF1,TF1,TF3,TF1,TF0),<br>(TF0,TF0,TF0,TF2,TF2,TF0), (TF1,TF0,TF0,TF2,TF2,TF0), (TF2,TF1,TF1,TF2,TF2,TF0),<br>(TF0,TF0,TF0,TF3,TF2,TF0), (TF1,TF0,TF0,TF3,TF2,TF0), (TF2,TF1,TF1,TF3,TF2,TF0),<br>(TF0,TF0,TF0,TF1,TF3,TF0), (TF1,TF0,TF0,TF1,TF3,TF0), (TF2,TF1,TF1,TF1,TF3,TF0),<br>(TF0,TF0,TF0,TF2,TF3,TF0), (TF1,TF0,TF0,TF2,TF3,TF0), (TF2,TF1,TF1,TF2,TF3,TF0),<br>(TF0,TF0,TF0,TF3,TF3,TF0), (TF1,TF0,TF0,TF3,TF3,TF0), (TF2,TF1,TF1,TF3,TF3,TF0),<br>(TF0,TF0,TF0,TF2,TF4,TF0), (TF1,TF0,TF0,TF2,TF4,TF0), (TF2,TF1,TF1,TF2,TF4,TF0),<br>(TF0,TF0,TF0,TF3,TF4,TF0), (TF1,TF0,TF0,TF3,TF4,TF0), (TF2,TF1,TF1,TF3,TF4,TF0),<br>(TF0,TF0,TF0,TF0,TF0,TF1), (TF1,TF0,TF0,TF0,TF0,TF1), (TF2,TF1,TF1,TF0,TF0,TF1),<br>(TF0,TF0,TF0,TF2,TF1,TF1), (TF1,TF0,TF0,TF2,TF1,TF1), (TF2,TF1,TF1,TF2,TF1,TF1),<br>(TF0,TF0,TF0,TF3,TF1,TF1), (TF1,TF0,TF0,TF3,TF1,TF1), (TF2,TF1,TF1,TF3,TF1,TF1),<br>(TF0,TF0,TF0,TF2,TF2,TF1), (TF1,TF0,TF0,TF2,TF2,TF1), (TF2,TF1,TF1,TF2,TF2,TF1),<br>(TF0,TF0,TF0,TF3,TF2,TF1), (TF1,TF0,TF0,TF3,TF2,TF1), (TF2,TF1,TF1,TF3,TF2,TF1),<br>(TF0,TF0,TF0,TF1,TF3,TF1), (TF1,TF0,TF0,TF1,TF3,TF1), (TF2,TF1,TF1,TF1,TF3,TF1),<br>(TF0,TF0,TF0,TF2,TF3,TF1), (TF1,TF0,TF0,TF2,TF3,TF1), (TF2,TF1,TF1,TF2,TF3,TF1),<br>(TF0,TF0,TF0,TF3,TF3,TF1), (TF1,TF0,TF0,TF3,TF3,TF1), (TF2,TF1,TF1,TF3,TF3,TF1),<br>(TF0,TF0,TF0,TF2,TF4,TF1), (TF1,TF0,TF0,TF2,TF4,TF1), (TF2,TF1,TF1,TF2,TF4,TF1),<br>(TF0,TF0,TF0,TF3,TF4,TF1), (TF1,TF0,TF0,TF3,TF4,TF1), (TF2,TF1,TF1,TF3,TF4,TF1) |

6.10.2.4.1.7.1.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Uplink | Min spreading factor | 4 |
| | Max number of DPDCH data bits/radio frame | 9600 |
| | Puncturing Limit | 0.88 |

6.10.2.4.5.7.2 Downlink

6.10.2.4.5.7.2.1 Transport channel parameters

6.10.2.4.5.7.2.1.1 Transport channel parameters for HS-DSCH

6.10.2.4.5.7.2.1.1.1 MAC-d flow parameters for Streaming / unknown / DL: [max bit rate depending on UE category] kbps / PS RAB

See clause 6.10.2.4.5.6.2.1.1.1.

6.10.2.4.5.7.2.1.1.2 MAC-d flow parameters for Interactive or background / DL: [max bit rate depending on UE category] / PS RAB

See clause 6.10.2.4.5.1.2.1.1.1.

6.10.2.4.5.7.2.1.2 Transport channel parameters for DCH

6.10.2.4.5.7.2.1.2.1 Transport channel parameters for Conversational / speech / DL:12.2 kbps / CS RAB

See clause 6.10.2.4.1.4.2.1.1.

6.10.2.4.5.7.2.1.2.2 Transport channel parameters for DL: 3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.2.1.1.

6.10.2.4.5.7.2.1.2.3 TFCS

See clause 6.10.2.4.1.4.2.1.3.

6.10.2.4.5.7.2.2 Physical channel parameters

6.10.2.4.5.7.2.2.1 Physical channel parameters on DPCH

See clause 6.10.2.4.1.2.2.2.

6.10.2.4.5.7.2.2.2 Physical channel parameters on HS-PDSCH

See clause 6.10.2.4.5.1.2.2.2.

6.10.2.4.5.8 Conversational / speech / UL:(12.65 8.85 6.6) DL:(12.65 8.85 6.6) kbps / CS RAB + Interactive or Background / UL:384 DL:[Bit rate depending on the UE category] / PS RAB+ UL:3.4 DL:3.4 kbps SRBs for DCCH + DL:0.15 kbps SRB#5 for DCCH

6.10.2.4.5.8.1 Uplink

6.10.2.4.5.8.1.1 Transport channel parameters

6.10.2.4.5.8.1.1.1 Transport channel parameters for Conversational / speech / UL: (12.65 8.85 6.6) kbps / CS RAB

See clause 6.10.2.4.1.62.1.1.1

6.10.2.4.5.8.1.1.2 Transport Channel parameters for Interactive or background / UL: 384 kbps / PS RAB

| | | | |
|---|---|---|---|
| Higher layer | RAB/Signalling RB | | RAB |
| RLC | Logical channel type | | DTCH |
| | RLC mode | | AM |
| | Payload sizes, bit | | 320 |
| | Max data rate, bps | | 384 000 |
| | AMD PDU header, bit | | 16 |
| MAC | MAC header, bit | | 0 |
| | MAC multiplexing | | N/A |
| Layer 1 | TrCH type | | DCH |
| | TB sizes, bit | | 336 |
| | TFS | TF0, bits | 0x336 |
| | | TF1, bits | 1x336 |
| | | TF2, bits | 2x336 |
| | | TF3, bits | 4x336 |
| | | TF4, bits | 8x336 |
| | | TF5, bits | 12x336 |
| | TTI, ms | | 10 |
| | Coding type | | TC |
| | CRC, bit | | 16 |
| | Max number of bits/TTI after channel coding | | 12 684 |
| | Uplink: Max number of bits/radio frame before rate matching | | 12 684 |
| | RM attribute | | 110 to 180 |

6.10.2.4.5.8.1.1.3 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.1.1.1.

6.10.2.4.5.8.1.1.4 TFCS

| | |
|---|---|
| TFCS size | 60 |
| TFCS | ((RAB subflow#1, RAB subflow#2, RAB subflow#3, 384 kbps RAB, DCCH)=<br>(TF0,TF0,TF0,TF0,TF0), (TF1,TF0,TF0,TF0,TF0), (TF2,TF1,TF0,TF0,TF0),<br>(TF3,TF2,TF0,TF0,TF0), (TF4,TF3,TF0,TF0,TF0), (TF0,TF0,TF0,TF0,TF1),<br>(TF1,TF0,TF0,TF0,TF1), (TF2,TF1,TF0,TF0,TF1), (TF3,TF2,TF0,TF0,TF1),<br>(TF4,TF3,TF0,TF0,TF1),<br><br>(TF0,TF0,TF0,TF1,TF0), (TF1,TF0,TF0,TF1,TF0), (TF2,TF1,TF0,TF1,TF0),<br>(TF3,TF2,TF0,TF1,TF0), (TF4,TF3,TF0,TF1,TF0), (TF0,TF0,TF0,TF1,TF1),<br>(TF1,TF0,TF0,TF1,TF1), (TF2,TF1,TF0,TF1,TF1), (TF3,TF2,TF0,TF1,TF1),<br>(TF4,TF3,TF0,TF1,TF1),<br><br>(TF0,TF0,TF0,TF2,TF0), (TF1,TF0,TF0,TF2,TF0), (TF2,TF1,TF0,TF2,TF0),<br>(TF3,TF2,TF0,TF2,TF0), (TF4,TF3,TF0,TF2,TF0), (TF0,TF0,TF0,TF2,TF1),<br>(TF1,TF0,TF0,TF2,TF1), (TF2,TF1,TF0,TF2,TF1), (TF3,TF2,TF0,TF2,TF1),<br>(TF4,TF3,TF0,TF2,TF1),<br><br>(TF0,TF0,TF0,TF3,TF0), (TF1,TF0,TF0,TF3,TF0), (TF2,TF1,TF0,TF3,TF0),<br>(TF3,TF2,TF0,TF3,TF0), (TF4,TF3,TF0,TF3,TF0), (TF0,TF0,TF0,TF3,TF1),<br>(TF1,TF0,TF0,TF3,TF1), (TF2,TF1,TF0,TF3,TF1), (TF3,TF2,TF0,TF3,TF1),<br>(TF4,TF3,TF0,TF3,TF1),<br><br>(TF0,TF0,TF0,TF4,TF0), (TF1,TF0,TF0,TF4,TF0), (TF2,TF1,TF0,TF4,TF0),<br>(TF3,TF2,TF0,TF4,TF0), (TF4,TF3,TF0,TF4,TF0), (TF0,TF0,TF0,TF4,TF1),<br>(TF1,TF0,TF0,TF4,TF1), (TF2,TF1,TF0,TF4,TF1), (TF3,TF2,TF0,TF4,TF1),<br>(TF4,TF3,TF0,TF4,TF1),<br><br>(TF0,TF0,TF0,TF5,TF0), (TF1,TF0,TF0,TF5,TF0), (TF2,TF1,TF0,TF5,TF0),<br>(TF3,TF2,TF0,TF5,TF0), (TF4,TF3,TF0,TF5,TF0), (TF0,TF0,TF0,TF5,TF1),<br>(TF1,TF0,TF0,TF5,TF1), (TF2,TF1,TF0,TF5,TF1), (TF3,TF2,TF0,TF5,TF1),<br>(TF4,TF3,TF0,TF5,TF1) |

6.10.2.4.5.8.1.1.5 TFC subset list

| | |
|---|---|
| TFC subset list size | 3 |
| TFC subset list | 0 = {(TF0,TF0,TF0,TF0,TF0), (TF1,TF0,TF0,TF0,TF0), (TF2,TF1,TF0,TF0,TF0),<br>(TF0,TF0,TF0,TF0,TF1), (TF1,TF0,TF0,TF0,TF1), (TF2,TF1,TF0,TF0,TF1),<br>(TF0,TF0,TF0,TF1,TF0), (TF1,TF0,TF0,TF1,TF0), (TF2,TF1,TF0,TF1,TF0),<br>(TF0,TF0,TF0,TF1,TF1), (TF1,TF0,TF0,TF1,TF1), (TF2,TF1,TF0,TF1,TF1),<br>(TF0,TF0,TF0,TF2,TF0), (TF1,TF0,TF0,TF2,TF0), (TF2,TF1,TF0,TF2,TF0),<br>(TF0,TF0,TF0,TF2,TF1), (TF1,TF0,TF0,TF2,TF1), (TF2,TF1,TF0,TF2,TF1),<br>(TF0,TF0,TF0,TF3,TF0), (TF1,TF0,TF0,TF3,TF0), (TF2,TF1,TF0,TF3,TF0),<br>(TF0,TF0,TF0,TF3,TF1), (TF1,TF0,TF0,TF3,TF1), (TF2,TF1,TF0,TF3,TF1),<br>(TF0,TF0,TF0,TF4,TF0), (TF1,TF0,TF0,TF4,TF0), (TF2,TF1,TF0,TF4,TF0),<br>(TF0,TF0,TF0,TF4,TF1), (TF1,TF0,TF0,TF4,TF1), (TF2,TF1,TF0,TF4,TF1),<br>(TF0,TF0,TF0,TF5,TF0), (TF1,TF0,TF0,TF5,TF0), (TF2,TF1,TF0,TF5,TF0),<br>(TF0,TF0,TF0,TF5,TF1), (TF1,TF0,TF0,TF5,TF1), (TF2,TF1,TF0,TF5,TF1),},<br><br>1 = {(TF0,TF0,TF0,TF0,TF0), (TF1,TF0,TF0,TF0,TF0), (TF2,TF1,TF0,TF0,TF0),<br>(TF3,TF2,TF0,TF0,TF0), (TF0,TF0,TF0,TF0,TF1), (TF1,TF0,TF0,TF0,TF1),<br>(TF2,TF1,TF0,TF0,TF1), (TF3,TF2,TF0,TF0,TF1), (TF0,TF0,TF0,TF1,TF0),<br>(TF1,TF0,TF0,TF1,TF0), (TF2,TF1,TF0,TF1,TF0), (TF3,TF2,TF0,TF1,TF0),<br>(TF0,TF0,TF0,TF1,TF1), (TF1,TF0,TF0,TF1,TF1), (TF2,TF1,TF0,TF1,TF1),<br>(TF3,TF2,TF0,TF1,TF1), (TF0,TF0,TF0,TF2,TF0), (TF1,TF0,TF0,TF2,TF0),<br>(TF2,TF1,TF0,TF2,TF0), (TF3,TF2,TF0,TF2,TF0), (TF0,TF0,TF0,TF2,TF1),<br>(TF1,TF0,TF0,TF2,TF1), (TF2,TF1,TF0,TF2,TF1), (TF3,TF2,TF0,TF2,TF1),<br>(TF0,TF0,TF0,TF3,TF0), (TF1,TF0,TF0,TF3,TF0), (TF2,TF1,TF0,TF3,TF0),<br>(TF3,TF2,TF0,TF3,TF0), (TF0,TF0,TF0,TF3,TF1), (TF1,TF0,TF0,TF3,TF1),<br>(TF2,TF1,TF0,TF3,TF1), (TF3,TF2,TF0,TF3,TF1), (TF0,TF0,TF0,TF4,TF0),<br>(TF1,TF0,TF0,TF4,TF0), (TF2,TF1,TF0,TF4,TF0), (TF3,TF2,TF0,TF4,TF0),<br>(TF0,TF0,TF0,TF4,TF1), (TF1,TF0,TF0,TF4,TF1), (TF2,TF1,TF0,TF4,TF1),<br>(TF3,TF2,TF0,TF4,TF1), (TF0,TF0,TF0,TF5,TF0), (TF1,TF0,TF0,TF5,TF0),<br>(TF2,TF1,TF0,TF5,TF0), (TF3,TF2,TF0,TF5,TF0), (TF0,TF0,TF0,TF5,TF1),<br>(TF1,TF0,TF0,TF5,TF1), (TF2,TF1,TF0,TF5,TF1), (TF3,TF2,TF0,TF5,TF1)},<br><br>2 = {(TF0,TF0,TF0,TF0,TF0), (TF1,TF0,TF0,TF0,TF0), (TF2,TF1,TF0,TF0,TF0),<br>(TF3,TF2,TF0,TF0,TF0), (TF4,TF3,TF0,TF0,TF0),(TF0,TF0,TF0,TF0,TF1),<br>(TF1,TF0,TF0,TF0,TF1), (TF2,TF1,TF0,TF0,TF1), (TF3,TF2,TF0,TF0,TF1),<br>(TF4,TF3,TF0,TF0,TF1), (TF0,TF0,TF0,TF1,TF0), (TF1,TF0,TF0,TF1,TF0),<br>(TF2,TF1,TF0,TF1,TF0), (TF3,TF2,TF0,TF1,TF0), (TF4,TF3,TF0,TF1,TF0),<br>(TF0,TF0,TF0,TF1,TF1), (TF1,TF0,TF0,TF1,TF1), (TF2,TF1,TF0,TF1,TF1),<br>(TF3,TF2,TF0,TF1,TF1), (TF4,TF3,TF0,TF1,TF1), (TF0,TF0,TF0,TF2,TF0),<br>(TF1,TF0,TF0,TF2,TF0), (TF2,TF1,TF0,TF2,TF0), (TF3,TF2,TF0,TF2,TF0),<br>(TF4,TF3,TF0,TF2,TF0), (TF0,TF0,TF0,TF2,TF1), (TF1,TF0,TF0,TF2,TF1),<br>(TF2,TF1,TF0,TF2,TF1), (TF3,TF2,TF0,TF2,TF1), (TF4,TF3,TF0,TF2,TF1),<br>(TF0,TF0,TF0,TF3,TF0), (TF1,TF0,TF0,TF3,TF0), (TF2,TF1,TF0,TF3,TF0),<br>(TF3,TF2,TF0,TF3,TF0), (TF4,TF3,TF0,TF3,TF0), (TF0,TF0,TF0,TF3,TF1),<br>(TF1,TF0,TF0,TF3,TF1), (TF2,TF1,TF0,TF3,TF1), (TF3,TF2,TF0,TF3,TF1),<br>(TF4,TF3,TF0,TF3,TF1), (TF0,TF0,TF0,TF4,TF0), (TF1,TF0,TF0,TF4,TF0),<br>(TF2,TF1,TF0,TF4,TF0), (TF3,TF2,TF0,TF4,TF0), (TF4,TF3,TF0,TF4,TF0),<br>(TF0,TF0,TF0,TF4,TF1), (TF1,TF0,TF0,TF4,TF1), (TF2,TF1,TF0,TF4,TF1),<br>(TF3,TF2,TF0,TF4,TF1), (TF4,TF3,TF0,TF4,TF1), (TF0,TF0,TF0,TF5,TF0),<br>(TF1,TF0,TF0,TF5,TF0), (TF2,TF1,TF0,TF5,TF0), (TF3,TF2,TF0,TF5,TF0),<br>(TF4,TF3,TF0,TF5,TF0), (TF0,TF0,TF0,TF5,TF1), (TF1,TF0,TF0,TF5,TF1),<br>(TF2,TF1,TF0,TF5,TF1), (TF3,TF2,TF0,TF5,TF1), (TF4,TF3,TF0,TF5,TF1)} |

6.10.2.4.5.8.1.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Uplink | Min spreading factor | 4 |
| | Max number of DPDCH data bits/radio frame | 9 600 |
| | Number of DPDCH | 1 |
| | Puncturing Limit | 0.60 |

6.10.2.4.5.8.2 Downlink

6.10.2.4.5.8.2.1 Transport channel parameters

6.10.2.4.5.8.2.1.1 Transport channel parameters for HS-DSCH

See clause 6.10.2.4.5.1.2.1.1

6.10.2.4.5.8.2.1.2 Transport channel parameters for DCH

6.10.2.4.5.8.2.1.2.1 Transport channel parameters for Conversational / speech / DL: (12.65 8.85 6.6) kbps / CS RAB

See clause 6.10.2.4.1.62.2.1.1

6.10.2.4.5.8.2.1.2.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.2.1.1

6.10.2.4.5.8.2.1.2.3 Transport channel parameters for DL:0.15 kbps SRB#5 for DCCH

See clause 6.10.2.4.1.62.2.1.3

6.10.2.4.5.8.2.1.2.4 TFCS

See clause 6.10.2.4.1.62.2.1.4

6.10.2.4.5.8.2.2 Physical channel parameters

6.10.2.4.5.8.2.2.1 Physical channel parameters on DPCH

See clause 6.10.2.4.1.62.2.2

6.10.2.4.5.8.2.2.2 Physical Channel parameters on HS-PDSCH

See clause 6.10.2.4.5.1.2.2.1

### 6.10.2.4.6 Combinations on HS-PDSCH and E-DPDCH

6.10.2.4.6.1 Interactive or background / UL: [max bit rate depending on UE category and TTI] DL: [max bit rate depending on UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH on DCH

6.10.2.4.6.1.1 Uplink

6.10.2.4.6.1.1.1 Transport channel parameters

6.10.2.4.6.1.1.1.1 Transport channel parameters for E-DCH

6.10.2.4.6.1.1.1.1.1 MAC-d flow parameters for Interactive or background / UL: [max bit rate depending on UE category and TTI] / PS RAB

| | | |
|---|---|---|
| Higher layer | RAB/Signalling RB | RAB |
| RLC | Logical channel type | DTCH |
| | RLC mode | AM |
| | Payload sizes, bit | 320 |
| | Max data rate, bps | Depends on UE category and TTI |
| | AMD PDU header, bit | 16 |
| MAC | MAC multiplexing | N/A |
| | MAC-d PDU size, bit | 336 |
| | MAC-e/es header fixed part, bit | 18 |
| Layer 1 | TrCH type | E-DCH |
| | TTI | 10ms (alt. 2ms) (NOTE) |
| | Coding type | TC |
| | CRC, bit | 24 |
| NOTE: The support of 2ms TTI depends on the UE category | | |

6.10.2.4.6.1.1.2 Transport channel parameters for DCH

6.10.2.4.6.1.1.2.1 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.1.1.1.

6.10.2.4.6.1.1.2 Physical channel parameters

6.10.2.4.6.1.1.2.1 Physical channel parameters on E-DPDCH

Note that each alternative configuration in physical channel parameters is stand-alone and can be associated with any of the RAB alternatives in the transport channel parameters.

UE E-DPDCH Physical Layer category 1:

| | | |
|---|---|---|
| E-DPDCH | Number of processes | 4 |
| | TTI | 10 ms |
| | Max Data Rate | 0.7296 Mbps |

UE E-DPDCH Physical Layer category 2:

| | | | |
|---|---|---|---|
| E-DPDCH | Number of processes | 4 | 8 |
| | TTI | 10 ms | 2 ms |
| | Max Data Rate | 1.4592 Mbps | 1.4595 Mbps |

UE E-DPDCH Physical Layer category 3:

| | | |
|---|---|---|
| E-DPDCH | Number of processes | 4 |
| | TTI | 10 ms |
| | Max Data Rate | 1.4592 Mbps |

UE E-DPDCH Physical Layer category 4:

| | | | |
|---|---|---|---|
| E-DPDCH | Number of processes | 4 | 8 |
| | TTI | 10 ms | 2 ms |
| | Max Data Rate | 2.0 Mbps | 2.918500 Mbps |

UE E-DPDCH Physical Layer category 5:

| | | |
|---|---|---|
| E-DPDCH | Number of processes | 4 |
| | TTI | 10 ms |
| | Max Data Rate | 2.0 Mbps |

UE E-DPDCH Physical Layer category 6:

| | | | |
|---|---|---|---|
| E-DPDCH | Number of processes | 4 | 8 |
| | TTI | 10 ms | 2 ms |
| | Max Data Rate | 2.0 Mbps | 5.76 Mbps |

6.10.2.4.6.1.1.2.2 Physical channel parameters for DPCH

See clause 6.10.2.4.1.2.1.2

6.10.2.4.6.1.2 Downlink

See clause 6.10.2.4.5.1.2.

6.10.2.4.6.2 Interactive or background / UL: [max bit rate depending on UE category and TTI] DL: [max bit rate depending on UE category] / PS RAB + UL:[max bit rate depending on UE category and TTI] DL:3.4 kbps SRBs for DCCH on E-DCH and DL DCH

6.10.2.4.6.2.1 Uplink

6.10.2.4.6.2.1.1 Transport channel parameters

6.10.2.4.6.2.1.1.1 Transport channel parameters for E-DCH

6.10.2.4.6.2.1.1.1.1 MAC-d flow#1 parameters for Interactive or background / UL: [max bit rate depending on UE category and TTI] / PS RAB

See clause 6.10.2.4.6.1.1.1.1.1.

6.10.2.4.6.2.1.1.1.2 MAC-d flow#2 parameters for UL: [max bit rate depending on UE category and TTI] SRBs for E-DCH

| Higher layer | RAB/Signalling RB | SRB#1 | SRB#2 | SRB#3 | SRB#4 |
|---|---|---|---|---|---|
| RLC | Logical channel type | DCCH | DCCH | DCCH | DCCH |
| | RLC mode | UM | AM | AM | AM |
| | Payload sizes, bit | 136 | 128 | 128 | 128 |
| | Max data rate, bps | Depends on UE category and TTI | | | |
| | AMD PDU header, bit | 8 | 16 | 16 | 16 |
| MAC | MAC-es multiplexing | 4 logical channel multiplexing | | | |
| | MAC-d PDU size, bit | 144 | | | |
| | MAC-e/es header fixed part, bit | 18 | | | |
| Layer 1 | TrCH type | E-DCH | | | |
| | TTI | 10ms (alt. 2ms) (NOTE) | | | |
| | Coding type | TC | | | |
| | CRC, bit | 24 | | | |
| NOTE: The support of 2ms TTI depends on the UE category. | | | | | |

6.10.2.4.6.2.1.2 Physical channel parameters

6.10.2.4.6.2.1.2.1 Physical channel parameters on E-DPDCH

See clause 6.10.2.4.6.1.1.2.1.

6.10.2.4.6.2.2 Downlink

See clause 6.10.2.4.5.1.2.

6.10.2.4.6.3 Interactive or background / UL: [max bit rate depending on UE category and TTI] DL: [max bit rate depending on UE category] / PS RAB + UL: [max bit rate depending on UE category and TTI] DL: [max bit rate depending on UE category] SRBs for DCCH on E-DCH and HS-DSCH

6.10.2.4.6.3.1 Uplink

See clause 6.10.2.4.6.1.1.

6.10.2.4.6.3.1.2 Physical channel parameters

6.10.2.4.6.3.1.2.1 Physical channel parameters on E-DPDCH

See clause 6.10.2.4.6.1.1.2.1.

6.10.2.4.6.3.2 Downlink

6.10.2.4.6.3.2.1 Transport channel parameters

6.10.2.4.6.3.2.1.1 Transport channel parameters for HS-DSCH

6.10.2.4.6.3.2.1.1.1 MAC-d flow#1 parameters for Interactive or background / DL: [max bit rate depending on UE category] / PS RAB

See clause 6.10.2.4.5.1.2.1.1.1.

6.10.2.4.6.3.2.1.1.1 MAC-d flow#2 parameters for DL: [max bit rate depending on UE category] SRBs for HS-DSCH

| Higher layer | RAB/Signalling RB | SRB#1 | SRB#2 | SRB#3 | SRB#4 |
|---|---|---|---|---|---|
| RLC | Logical channel type | DCCH | DCCH | DCCH | DCCH |
| | RLC mode | UM | AM | AM | AM |
| | Payload sizes, bit | 136 | 128 | 128 | 128 |
| | Max data rate, bps | Depends on UE category (NOTE) | | | |
| | AMD PDU header, bit | 8 | 16 | 16 | 16 |
| MAC | MAC-d header, bit | 4 | 4 | 4 | 4 |
| | MAC multiplexing | 4 logical channel multiplexing | | | |
| | MAC-d PDU size, bit | 148 | | | |
| | MAC-hs header fixed part, bit | 21 | | | |
| Layer 1 | TrCH type | HS-DSCH | | | |
| | TTI | 2 ms | | | |
| | Coding type | TC | | | |
| | CRC, bit | 24 | | | |
| NOTE: The peak throughput may be limited by the maximum number of MAC-d PDUs that can be included in a single MAC-hs PDU (see 3GPP TS 25.321 [38]). | | | | | |

6.10.2.4.6.3.2.2 Physical channel parameters

The physical channel configuration shall use F-DPCH.

6.10.2.4.6.3.2.2.1 Physical channel parameters on HS-PDSCH.

See clause 6.10.2.4.5.1.2.2.2.

6.10.2.4.6.4 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL: [max bit rate depending on UE category and TTI] DL: [max bit rate depending on UE category] / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.2.4.6.4.1 Uplink

6.10.2.4.6.4.1.1 Transport channel parameters

6.10.2.4.6.4.1.1.1 Transport channel parameters for E-DCH

6.10.2.4.6.4.1.1.1.1 MAC-d flow parameters for Interactive or background / UL: [max bit rate depending on UE category and TTI] / PS RAB

See clause 6.10.2.4.6.1.1.1.1.1.

6.10.2.4.6.4.1.1.2 Transport channel parameters for Conversational / speech / UL:12.2 kbps / CS RAB

See clause 6.10.2.4.1.4.1.1.1.

6.10.2.4.6.4.1.1.3 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.1.1.1.

6.10.2.4.6.4.1.1.4 TFCS

See clause 6.10.2.4.1.4.1.1.3.

6.10.2.4.6.4.1.2 Physical channel parameters

6.10.2.4.6.4.1.2.1 Physical channel parameters on E-DPDCH

See clause 6.10.2.4.6.1.1.2.1.

6.10.2.4.6.4.1.2.2 Physical channel parameters on DCH

See clause 6.10.2.4.1.4.1.2.

6.10.2.4.6.4.2 Downlink

See clause 6.10.2.4.5.3.2.

6.10.2.4.6.5 Streaming / unknown / UL:[max bit rate depending on UE category and TTI] DL: [max bit rate depending on UE category] kbps / PS RAB + Interactive or background / UL: [max bit rate depending on UE category and TTI] DL: [max bit rate depending on UE category] / PS RAB + UL:[max bit rate depending on UE category and TTI] DL:3.4 kbps SRBs for DCCH on E-DCH and DL DCH

6.10.2.4.6.5.1 Uplink

6.10.2.4.6.5.1.1 Transport channel parameters

6.10.2.4.6.5.1.1.1 Transport channel parameters for E-DCH

MAC-d flow multiplexing between all MAC-d flows in the same MAC-e PDU shall be configured.

6.10.2.4.6.5.1.1.1.1 MAC-d flow #1 parameters for Streaming / unknown / UL: [max bit rate depending on UE category and TTI] / PS RAB

See clause 6.10.2.4.6.1.1.1.1.1.

6.10.2.4.6.5.1.1.1.2 MAC-d flow #2 parameters for Interactive or background / UL: [max bit rate depending on UE category and TTI] / PS RAB

See clause 6.10.2.4.6.1.1.1.1.1.

6.10.2.4.6.5.1.1.1.3 MAC-d flow #3 parameters for UL: [max bit rate depending on UE category and TTI] SRBs for E-DCH

See clause 6.10.2.4.6.2.1.1.1.2.

6.10.2.4.6.5.1.2 Physical channel parameters

6.10.2.4.6.5.1.2.1 Physical channel parameters on E-DPDCH

See clause 6.10.2.4.6.1.1.2.1.

6.10.2.4.6.5.2 Downlink

See clause 6.10.2.4.5.6.2.

## 6.10.3 RAB and signalling RB for TDD

### 6.10.3.1 RABs and signalling RBs

In the following clauses, the typical parameter sets are presented for reference RABs, signalling RBs and important combinations of them. The data rate given for each RAB is the maximum data rate that can be supported by that RAB.

NOTE: The granularity for each RAB needs to be clarified.

**Table 6.10.3.1.1: Prioritized RABs**

| # | Traffic class[3] | SSD[3] | Max. rate, kbps | CS/PS |
|---|---|---|---|---|
| 1 | Conversational | Speech | UL:12.2 DL:12.2 | CS |
| 1a | Conversational | Speech | UL: (12.2 7.95 5.9 4.75) DL(12.2 7.95 5.9 4.75) | CS |
| 2 | Conversational | Speech | UL:10.2 DL:10.2 | CS |
| 2a | Conversational | Speech | UL:(10.2 , 6.7, 5.9, 4.75) DL:10.2, 6.7, 5.9, 4.75) | CS |
| 3 | Conversational | Speech | UL:7.95 DL:7.95 | CS |
| 4 | Conversational | Speech | UL:7.4 DL:7.4 | CS |
| 4a | Conversational | Speech | UL:(12.2 7.95 5.9 4.75, DL:(12.2 7.95 5.9 4.75) | CS |
| 5 | Conversational | Speech | UL:6.7 DL:6.7 | CS |
| 6 | Conversational | Speech | UL:5.9 DL:5.9 | CS |
| 7 | Conversational | Speech | UL:5.15 DL:5.15 | CS |
| 8 | Conversational | Speech | UL:4.75 DL:4.75 | CS |
| 9 | Conversational | Unknown | UL:28.8 DL:28.8 | CS |
| 10 | Conversational | Unknown | UL:64 DL:64 | CS |
| 11 | Conversational | Unknown | UL:32 DL:32 | CS |
| 11a | Conversational | Unknown | UL:8 DL:8 | CS |
| 12 | Streaming | Unknown | UL:14.4 DL:14.4 | CS |
| 13 | Streaming | Unknown | UL:28.8 DL:28.8 | CS |
| 14 | Streaming | Unknown | UL:57.6 DL:57.6 | CS |
| 15 | Void | | | |
| 15a | Streaming | Unknown | UL:16 DL:64 | PS |
| 16 | Void | | | |
| 17 | Void | | | |
| 18 | Void | | | |
| 19 | Void | | | |
| 20 | Interactive or Background | N/A | UL:32 DL:8 | PS |
| 20a | Interactive or Background | N/A | UL:8 DL:8 | PS |
| 20b | Interactive or Background | N/A | UL:16 DL:16 | PS |
| 20c | Interactive or Background | N/A | UL:32 DL:32 | PS |
| 21 | Void | | | |
| 22 | Interactive or Background | N/A | UL:32 DL:64 | PS |
| 23 | Interactive or Background | N/A | UL:64 DL:64 | PS |
| 24 | Interactive or Background | N/A | UL:64 DL:128 | PS |
| 25 | Interactive or Background | N/A | UL:128 DL:128 | PS |
| 26 | Interactive or Background | N/A | UL:64 DL:384 | PS |
| 27 | Interactive or Background | N/A | UL:128 DL:384 | PS |
| 28 | Interactive or Background | N/A | UL:384 DL:384 | PS |
| 29 | Interactive or Background | N/A | UL:64 DL:2048 | PS |
| 30 | Interactive or Background | N/A | UL:128 DL:2048 | PS |
| 31 | Void | | | |
| 32 | Interactive or Background | N/A | UL:64 DL:256 | PS |
| 33 | Interactive or Background | N/A | UL:0 DL:32 | PS |
| 34 | Interactive or Background | N/A | UL:32 DL:0 | PS |
| 35 | Interactive or Background | N/A | UL:64 DL:144 | PS |
| 36 | Interactive or Background | N/A | UL:144 DL:144 | PS |

**Table 6.10.3.1.2: Signalling RBs**

| # | Maximum rate, kbps | Logical channel | PhyCh onto which SRBs are mapped |
|---|---|---|---|

| | | | |
|---|---|---|---|
| 1 | UL:1.7 DL:1.7 | DCCH | DPCH |
| 2 | UL:3.4 DL:3.4 | DCCH | DPCH |
| 3 | UL:13.6 DL:13.6 | DCCH | DPCH |
| 4 | DL:27.2 (alt. 13.6) | DCCH | SCCPCH |
| 5 | UL:16.8 | CCCH | PRACH |
| 6 | DL:32 (alt. 16) | CCCH | SCCPCH |
| 7 | DL:33.6 (alt. 16.8) | BCCH | SCCPCH |
| 8 | DL:12 (alt. 8) | PCCH | SCCPCH |
| 9 | UL:16.8 | SHCCH | PRACH |
| 10 | UL:16.8 | SHCCH | PRACH or PUSCH |
| 11 | DL:32 (alt. 16) | SHCCH | SCCPCH |
| 12 | DL:16 | SHCCH | SCCPCH or PDSCH |

### 6.10.3.2 Combinations of RABs and Signalling RBs

In the present document, physical channel parameters for following combinations of RABs and signalling RBs on a CCTrCH are described.

NOTE: It is understood that for speech service the AMR mode may be operated asymmetrically for the uplink and downlink.

#### Combinations on DPCH

1) Stand-alone UL:1.7 DL:1.7 kbps SRBs for DCCH.

1a) Stand-alone UL:1.7 DL:1.7 kbps SRBs for DCCH (multiframe).

2) Stand-alone UL:3.4 DL:3.4 kbps SRBs for DCCH.

3) Stand-alone UL:13.6 DL:13.6 kbps SRBs for DCCH.

4) Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

4a) Conversational / speech / UL:(12.2 7.95 5.9 4.75) DL:(12.2 7.95 5.9 4.75) kbps / CS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH

5) Conversational / speech / UL:10.2 DL:10.2 kbps / CS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

5a) Conversational / speech / UL:(10.2, 6.7, 5.9, 4.75) DL:(10.2, 6.7, 5.9, 4.75) kbps / CS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH

6) Conversational / speech / UL:7.95 DL:7.95 kbps / CS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

7) Conversational / speech / UL:7.4 DL:7.4 kbps / CS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

7a) Conversational / speech / UL:(7.4, 6.7, 5.9, 4.75) DL:(7.4, 6.7, 5.9, 4.75) kbps / CS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH8) Conversational / speech / UL:6.7 DL:6.7 kbps / CS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

9) Conversational / speech / UL:5.9 DL:5.9 kbps / CS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

10) Conversational / speech / UL:5.15 DL:5.15 kbps / CS RAB
+ UL:1.7 DL:1.7 kbps SRBs for DCCH.

11) Conversational / speech / UL:4.75 DL:4.75 kbps / CS RAB
+ UL:1.7 DL:1.7 kbps SRBs for DCCH.

12) Conversational / unknown / UL:28.8 DL:28.8 kbps / CS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

13) Conversational / unknown / UL:64 DL:64 kbps / CS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

14) Conversational / unknown / UL:32 DL:32 kbps / CS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

15) Streaming / unknown / UL:14.4/DL:14.4 kbps / CS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

16) Streaming / unknown / UL:28.8/DL:28.8 kbps / CS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

17) Streaming / unknown / UL:57.6/DL:57.6 kbps / CS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

18) Void.

19) Void.

20) Void.

21) Void.

22) Void..

23) Interactive or background / UL:32 DL:8 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

23a) Interactive or background / UL:8 DL:8 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH

23b) Interactive or background / UL:16 DL:16 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH

23c) Interactive or background / UL:32 DL:32 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH

23d) Interactive or background / UL:32 DL:32 kbps / PS RAB (20 ms TTI)
+ UL:3.4 DL:3.4 kbps SRBs for DCCH

24) Void..

25) Interactive or background / UL:32 DL: 64 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

26) Interactive or background / UL:64 DL: 64 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

27) Interactive or background / UL:64 DL:128 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

28) Interactive or background / UL:128 DL:128 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

29) Interactive or background / UL:64 DL:144 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

30) Interactive or background / UL:144 DL:144 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

31) Interactive or background / UL:64 DL:256 kbps / PS RAB
+ UL:3.4 DL: 3.4 kbps SRBs for DCCH.

32) Interactive or background / UL:64 DL:384 kbps / PS RAB
+ UL:3.4 DL: 3.4 kbps SRBs for DCCH.

33) Interactive or background / UL:128 DL:384 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

34) Interactive or background / UL:384 DL:384 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

35) Interactive or background / UL:64 DL:2 048 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

36) Interactive or background / UL:128 DL:2 048 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

37) Interactive or background / UL:384 DL:2 048 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

38) Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB
+ Interactive or background / UL:32 DL:8 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

38a) Conversational / speech / 12.2 kbps / CS RAB
+ Interactive or background / UL:0 DL:0 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

38b) Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB
+ Interactive or background/ UL:8 DL:8 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

38c) Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB
+ Interactive or background/ UL:32 DL:32 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

38d) Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB
+ Interactive or background/ UL:64 DL:64 kbps / PS RAB
+ Interactive or background / UL:64 DL:64 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

38e) Conversational / speech / UL:(12.2 7.95 5.9 4.75) DL:(12.2 7.95 5.9 4.75) kbps / CS RAB
+ Interactive or background / UL:0 DL:0 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH

38f) Conversational / speech / (12.2 7.95 5.9 4.75) kbps / CS RAB
+ Interactive or background / UL:8 DL:8 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

38g) Conversational / speech / (12.2 7.95 5.9 4.75) kbps / CS RAB
+ Interactive or background / UL:16 DL:16 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

38h) Conversational / speech / (12.2 7.95 5.9 4.75) kbps / CS RAB
+ Interactive or background / UL:32 DL:32 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

38i) Conversational / speech / (12.2 7.95 5.9 4.75) kbps / CS RAB
+ Interactive or background / UL:64 DL:64 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH

38j) Conversational / speech / (12.2 7.95 5.9 4.75) kbps / CS RAB
+ Interactive or background / UL:64 DL:128 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH

39) Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB
+ Interactive or background / UL:32 DL:64 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

40) Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB
+ Interactive or background / UL:64 DL:64 kbps / PS RAB
+ UL:3.4 DL: 3.4 kbps SRBs for DCCH.

41) Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB
+ Interactive or background / UL:64 DL:128 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

42) Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB
+ Interactive or background / UL:64 DL:256 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

43) Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB
+ Interactive or background / UL:64 DL:384 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

44) Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB
+ Interactive or background / UL:128 DL:2 048 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

45) Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB
+ Streaming / unknown / UL:57.6 DL:57.6 kbps / CS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

46) Void

47) Void

48) Void

49) Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB
+ Conversational / unknown / UL:64 DL:64 kbps / CS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

49a) Conversational / speech / UL:(12.2 7.95 5.9 4.75) DL:(12.2 7.95 5.9 4.75) kbps / CS RAB
+ Conversational / unknown / UL:64 DL:64 kbps / CS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH

50) Conversational / unknown / UL:64 DL:64 kbps / CS RAB
+ Conversational / unknown / UL:64 DL:64 kbps / CS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

51) Conversational / unknown / UL:64 DL:64 kbps / CS RAB
+ Interactive or background / UL:64 DL:64 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

51a) Conversational / unknown / UL:64 DL:64 kbps / CS RAB
+ Interactive or Background / UL:8 DL:8 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH

51b) Conversational / unknown / UL:64 DL:64 kbps / CS RAB
+ Interactive or Background / UL:16 DL:64 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH

52) Conversational / unknown / UL:64 DL:64 kbps / CS RAB
+ Interactive or background / UL:64 DL:128 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

53) Conversational / unknown / UL:64 DL:64 kbps / CS RAB
+ Interactive or background / UL:128 DL:128 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

54) Void.

55) Void

56) Interactive or background / UL:8 DL:8 kbps / PS RAB
+ Interactive or background / UL:8 DL:8 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH

57) Interactive or background / UL:64 DL:64 kbps / PS RAB
+ Interactive or background / UL:64 DL:64 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH

58) Streaming / unknown / UL:16 DL:64 kbps / PS RAB
+ Interactive or background / UL:8 DL:8 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH

59) Reserved for future use

60) Reserved for future use

61) Conversational / unknown / UL:8 DL:8 kbps / PS RAB
+ Interactive or Background / UL:8 DL:8 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH

Combinations on PDSCH, SCCPCH, PUSCH and PRACH

1) Interactive or background / UL:64 DL:256 kbps / PS RAB
+ UL: 3.4/16.8 DL:3.4/ 33.6 kbps SRBs for DCCH, CCCH and BCCH
+ UL:16.8 DL: 16 kbps SRBs for SHCCH.

2) Interactive or background / UL:64 DL:384 kbps / PS RAB
+ UL: 3.4/16.8 DL: 3.4/33.6 kbps SRBs for DCCH, CCCH and BCCH
+ UL: 16.8 DL: 16 kbps SRBs for SHCCH.

3) Interactive or background / UL:64 DL:2 048 kbps / PS RAB
+ UL:3.4/16.8 DL: 3.4/33.6 kbps SRBs for DCCH, CCCH and BCCH
+ UL: 16.8 DL: 16 kbps SRBs for SHCCH.

4) Interactive or background / UL:384 DL:2 048 kbps / PS RAB
+ UL:3.4/16.8 DL: 3.4/33.6 kbps SRBs for DCCH, CCCH and BCCH
+ UL: 16.8 DL: 16 kbps SRBs for SHCCH.

Combinations on PDSCH, SCCPCH, DPCH, PUSCH and PRACH

1) Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH
+ Interactive or background / UL:64 DL:256 kbps / PS RAB
+ UL:16.8 kbps SRBs for CCCH and SHCCH
+ DL: 33.6 kbps SRBs for CCCH, SHCCH and BCCH.

2) Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH
+ Interactive or background / UL:64 DL:384 kbps / PS RAB
+ UL:16.8 kbps SRBs for CCCH and SHCCH
+ DL: 33.6 kbps SRBs for CCCH, SHCCH and BCCH.

3) Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH
+ Interactive or background / UL:64 DL:2 048 kbps / PS RAB
+ UL:16.8 kbps SRBs for CCCH and SHCCH
+ DL: 33.6 kbps SRBs for CCCH, SHCCH and BCCH.

Combinations on SCCPCH

1) Stand-alone 12 kbps SRB for PCCH.

2) Interactive or background / DL:32 kbps / PS RAB
+ SRB for CCCH
+ SRBs for DCCH
+ SRB for BCCH.

2a) Interactive/Background 32 kbps PS RAB + Interactive/Background 32 kbps PS RAB
+ SRBs for CCCH
+ SRB for DCCH
+ SRB for BCCH

2b) SRBs for CCCH
+ SRB for DCCH
+ SRB for BCCH

3) Interactive or background / DL:32 kbps / PS RAB
+ SRB for PCCH
+ SRB for CCCH
+ SRBs for DCCH
+ SRB for BCCH.

3a) SRB for PCCH
+ SRB for CCCH
+ SRB for DCCH
+ SRB for BCCH

4) RB for CTCH
+ SRB for CCCH
+ SRB for BCCH

Combinations on PRACH

1) Interactive or background / UL:12.8 kbps / PS RAB
+ SRB for CCCH
+ SRBs for DCCH.

### 6.10.3.3 Example of linkage between RABs and services

RABs, which are included in the present document, can provide the services as shown in table 6.10.1.1: Traffic classes. Furthermore, the required BER for each RAB, which is assumed in the present document, is shown in table 6.10.3.3.1.

**Table 6.10.3.3.1: Example of linkage between RABs and services**

| RAB | | | | Residual BER[3] | Services |
|---|---|---|---|---|---|
| Traffic class[3] | SSD[3] | Max. rate, kbps | CS/PS | | |
| Conversational | Speech | UL:4.75-12.2 DL:4.75-12.2 | CS | $5x10^{-4}$, $1x10^{-3}$, $5x10^{-3}$ | AMR speech |
| Conversational | Unknown | UL:64 DL:64 | CS | $1x10^{-4}$ or $1x10^{-6}$ | UDI 1B, 64k 3G-324M[4] |
| Conversational | Unknown | UL:32 DL:32 | CS | $1x10^{-4}$ or $1x10^{-6}$ | 32k 3G-324M[4] |
| Conversational | Unknown | UL:28.8 DL:28.8 | CS | $1x10^{-3}$ | Transparent modem |
| Streaming | Unknown | UL:14.4 DL:14.4 | CS | $1x10^{-3}$ | FAX[6] |
| Streaming | Unknown | UL:28.8 DL:28.8 | CS | $1x10^{-3}$ | FAX[6] PIAFS 32 kbps |
| Streaming | Unknown | UL:57.6 DL:57.6 | CS | $1x10^{-3}$ | Modem[6], FTM[5], PIAFS 64 kbps |
| Streaming | Unknown | UL:64-128 or DL:64-384 | CS | $1x10^{-3}$ or $1x10^{-4}$ | Streaming video, uni-directional |
| Interactive or Background | N/A | UL:32-384 DL:8-2048 | PS | $1x10^{-3}$ or $1x10^{-4}$ | Packet |

NOTE 1: SMS can be provided via the signalling RB (DCCH) on DPCH or SCCPCH.

NOTE 2: CBS can be provided via the signalling RB (CTCH) on SCCPCH

NOTE 3: UDI *n*B can be provided via *n* RABs of conversational 64 kbps.

### 6.10.3.4 Typical radio parameter sets

#### 6.10.3.4.1 Combinations on DPCH

##### 6.10.3.4.1.1 Stand-alone UL:1.7 DL:1.7 kbps SRBs for DCCH

6.10.3.4.1.1.1 Uplink

6.10.3.4.1.1.1.1 Transport channel parameters

6.10.3.4.1.1.1.1.1 Transport channel parameters for UL:1.7 kbps SRBs for DCCH

| | | | | | | |
|---|---|---|---|---|---|---|
| Higher layer | RAB/signalling RB | | SRB#1 | SRB#2 | SRB#3 | SRB#4 |
| | User of Radio Bearer | | RRC | RRC | NAS_DT High priority | NAS_DT Low priority |
| RLC | Logical channel type | | DCCH | DCCH | DCCH | DCCH |
| | RLC mode | | UM | AM | AM | AM |
| | Payload sizes, bit | | 136 | 128 | 128 | 128 |
| | Max data rate, bps | | 1 700 | 1 600 | 1 600 | 1 600 |
| | AMD/UMD PDU header, bit | | 8 | 16 | 16 | 16 |
| MAC | MAC header, bit | | 4 | 4 | 4 | 4 |
| | MAC multiplexing | | 4 logical channel multiplexing | | | |
| Layer 1 | TrCH type | | DCH | | | |
| | TB sizes, bit | | 148 (alt. 0,148) (note) | | | |
| | TFS | TF0, bits | 0x148 (alt 1x0) (note) | | | |
| | | TF1, bits | 1x148 | | | |
| | TTI, ms | | 80 | | | |
| | Coding type | | CC 1/3 | | | |
| | CRC, bit | | 16 | | | |
| | Max number of bits/TTI before rate matching | | 516 | | | |
| | Max number of bits/radio frame before rate matching | | 65 | | | |
| | RM attribute | | 155 to 185 | | | |
| NOTE: Alternative parameters enable the measurement "transport channel BLER" in the UTRAN. | | | | | | |

6.10.3.4.1.1.1.1.2 TFCS

| | |
|---|---|
| TFCS size | 2 |
| TFCS | SRBs for DCCH = (TF0), (TF1) |
| NOTE: The first TFC is required for the alt. case, optional otherwise. | |

6.10.3.4.1.1.1.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Uplink | Midamble | 512 chips |
| | Codes and time slots | SF16 x 1 code x 1 time slot |
| | Max. Number of data bits/radio frame | 234 |
| | TFCI code word | 8 bits |
| | TPC | 2 bits |
| | Puncturing Limit | 1 |
| NOTE: In case the first TFC in the TFCS is not configured, the TFCI code word will be 4 bits. | | |

6.10.3.4.1.1.2 Downlink

6.10.3.4.1.1.2.1 Transport channel parameters

6.10.3.4.1.1.2.1.1 Transport channel parameters for DL:1.7 kbps SRBs for DCCH

| | | | | | |
|---|---|---|---|---|---|
| Higher layer | RAB/signalling RB | SRB#1 | SRB#2 | SRB#3 | SRB#4 |
| | User of Radio Bearer | RRC | RRC | NAS_DT High priority | NAS_DT Low priority |
| RLC | Logical channel type | DCCH | DCCH | DCCH | DCCH |
| | RLC mode | UM | AM | AM | AM |
| | Payload sizes, bit | 136 | 128 | 128 | 128 |
| | Max data rate, bps | 1 700 | 1 600 | 1 600 | 1 600 |
| | AMD/UMD PDU header, bit | 8 | 16 | 16 | 16 |
| MAC | MAC header, bit | 4 | 4 | 4 | 4 |
| | MAC multiplexing | 4 logical channel multiplexing | | | |
| Layer 1 | TrCH type | DCH | | | |
| | TB sizes, bit | 148 (alt. 0,148) (note) | | | |

| | | | |
|---|---|---|---|
| | TFS | TF0, bits | 0 x148 (alt. 1x0) (note) |
| | | TF1, bits | 1x148 |
| | TTI, ms | | 80 |
| | Coding type | | CC 1/3 |
| | CRC, bit | | 16 |
| | Max number of bits/TTI before rate matching | | 516 |
| | Max number of bits/radio frame before rate matching | | 65 |
| | RM attribute | | 155 to 185 |
| NOTE: Alternative parameters enable the measurement "transport channel BLER" in the UE. | | | |

6.10.3.4.1.1.2.1.2 TFCS

| | |
|---|---|
| TFCS size | 2 |
| TFCS | SRBs for DCCH = (TF0), (TF1) |
| NOTE: The first TFC is required for the alt. case, optional otherwise. | |

6.10.3.4.1.1.2.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Downlink | Midamble | 512 chips |
| | Codes and time slots | SF16 x 1 code x 1 time slot |
| | Max. Number of data bits/radio frame | 236 bits |
| | TFCI code word | 8 bits |
| | Puncturing limit | 1 |
| NOTE: In case the first TFC in the TFCS is not configured, the TFCI code word will be 4 bits. | | |

6.10.3.4.1.1a Stand-alone UL:1.7 DL:1.7 kbps SRBs for DCCH (multiframe)

6.10.3.4.1.1a.1 Uplink

6.10.3.4.1.1a.1.1 Transport channel parameters

6.10.3.4.1.1a.1.1.1 Transport channel parameters for UL:1.7 kbps SRBs for DCCH

| | | | | | | |
|---|---|---|---|---|---|---|
| Higher layer | RAB/signalling RB | | SRB#1 | SRB#2 | SRB#3 | SRB#4 |
| | User of Radio Bearer | | RRC | RRC | NAS_DT High priority | NAS_DT Low priority |
| RLC | Logical channel type | | DCCH | DCCH | DCCH | DCCH |
| | RLC mode | | UM | AM | AM | AM |
| | Payload sizes, bit | | 136 | 128 | 128 | 128 |
| | Max data rate, bps | | 1 700 | 1 600 | 1 600 | 1 600 |
| | AMD/UMD PDU header, bit | | 8 | 16 | 16 | 16 |
| MAC | MAC header, bit | | 4 | 4 | 4 | 4 |
| | MAC multiplexing | | 4 logical channel multiplexing | | | |
| Layer 1 | TrCH type | | DCH | | | |
| | TB sizes, bit | | 148 | | | |
| | TFS | TF0, bits | 0x148 | | | |
| | | TF1, bits | 1x148 | | | |
| | TTI, ms | | 20 | | | |
| | Coding type | | CC 1/3 | | | |
| | CRC, bit | | 16 | | | |
| | Max number of bits/TTI before rate matching | | 516 | | | |
| | Max number of bits/radio frame before rate matching | | 258 | | | |

6.10.3.4.1.1a.1.1.2 TFCS

| | |
|---|---|
| TFCS size | 2 |
| TFCS | SRBs for DCCH = (TF0), (TF1) |

6.10.3.4.1.1a.1.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Uplink | Midamble | 256 chips |
| | Codes and time slots | SF16 x 1 code x 1 time slot |
| | Max. Number of data bits/radio frame | 266 |
| | TFCI code word | 8 bits |
| | TPC | 2 bits |
| | Puncturing Limit | 1 |
| | Repetition period | 8 |
| | Repetition length | 2 |
| NOTE: In case the first TFC in the TFCS is not configured, the TFCI code word will be 4 bits. | | |

6.10.3.4.1.1a.2 Downlink

6.10.3.4.1.1a.2.1 Transport channel parameters

6.10.3.4.1.1a.2.1.1 Transport channel parameters for DL:1.7 kbps SRBs for DCCH

| | | | | | | |
|---|---|---|---|---|---|---|
| Higher layer | RAB/signalling RB | | SRB#1 | SRB#2 | SRB#3 | SRB#4 |
| | User of Radio Bearer | | RRC | RRC | NAS_DT High priority | NAS_DT Low priority |
| RLC | Logical channel type | | DCCH | DCCH | DCCH | DCCH |
| | RLC mode | | UM | AM | AM | AM |
| | Payload sizes, bit | | 136 | 128 | 128 | 128 |
| | Max data rate, bps | | 1 700 | 1 600 | 1 600 | 1 600 |
| | AMD/UMD PDU header, bit | | 8 | 16 | 16 | 16 |
| MAC | MAC header, bit | | 4 | 4 | 4 | 4 |
| | MAC multiplexing | | 4 logical channel multiplexing | | | |
| Layer 1 | TrCH type | | DCH | | | |
| | TB sizes, bit | | 148 | | | |
| | TFS | TF0, bits | 0 x148 | | | |
| | | TF1, bits | 1x148 | | | |
| | TTI, ms | | 20 | | | |
| | Coding type | | CC 1/3 | | | |
| | CRC, bit | | 16 | | | |
| | Max number of bits/TTI before rate matching | | 516 | | | |
| | Max number of bits/radio frame before rate matching | | 258 | | | |

6.10.3.4.1.1a.2.1.2 TFCS

| | |
|---|---|
| TFCS size | 2 |
| TFCS | SRBs for DCCH = (TF0), (TF1) |
| NOTE: The first TFC is optional. | |

6.10.3.4.1.1a.2.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Downlink | Midamble | 256 chips |
| | Codes and time slots | SF16 x 1 code x 1 time slot |
| | Max. Number of data bits/radio frame | 268 bits |
| | TFCI code word | 8 bits |
| | Puncturing limit | 1 |
| | Repetition period | 8 |
| | Repetition length | 2 |
| NOTE: In case the first TFC in the TFCS is not configured, the TFCI code word will be 4 bits. | | |

6.10.3.4.1.2 Stand-alone UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.3.4.1.2.1 Uplink

6.10.3.4.1.2.1.1 Transport channel parameters

6.10.3.4.1.2.1.1.1 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

| Higher layer | RAB/signalling RB | | SRB#1 | SRB#2 | SRB#3 | SRB#4 |
|---|---|---|---|---|---|---|
| | User of Radio Bearer | | RRC | RRC | NAS_DT High priority | NAS_DT Low priority |
| RLC | Logical channel type | | DCCH | DCCH | DCCH | DCCH |
| | RLC mode | | UM | AM | AM | AM |
| | Payload sizes, bit | | 136 | 128 | 128 | 128 |
| | Max data rate, bps | | 3 400 | 3 200 | 3 200 | 3 200 |
| | AMD/UMD PDU header, bit | | 8 | 16 | 16 | 16 |
| MAC | MAC header, bit | | 4 | 4 | 4 | 4 |
| | MAC multiplexing | | 4 logical channel multiplexing | | | |
| Layer 1 | TrCH type | | DCH | | | |
| | TB sizes, bit | | 148 (alt. 0,148) (note) | | | |
| | TFS | TF0, bits | 0x148 (alt. 1x0) (note) | | | |
| | | TF1, bits | 1x148 | | | |
| | TTI, ms | | 40 | | | |
| | Coding type | | CC 1/3 | | | |
| | CRC, bit | | 16 | | | |
| | Max number of bits/TTI before rate matching | | 516 | | | |
| | Max number of bits/radio frame before rate matching | | 129 | | | |
| | RM attribute | | 155 to 165 | | | |
| NOTE: Alternative parameters enable the measurement "transport channel BLER" in the UTRAN. | | | | | | |

6.10.3.4.1.2.1.1.2 TFCS

| TFCS size | 2 |
|---|---|
| TFCS | SRBs for DCCH = (TF0), (TF1) |
| NOTE: The first TFC is required for the alt. case, optional otherwise. | |

6.10.3.4.1.2.1.2 Physical channel parameters

| DPCH Uplink | Midamble | 512 chips |
|---|---|---|
| | Codes and time slots | SF16 x 1 code x 1 time slot |
| | Max. Number of data bits/radio frame | 234 bits |
| | TFCI code word | 8 bits |
| | TPC | 2 bit |
| | Puncturing Limit | 1 |
| NOTE: In case the first TFC in the TFCS is not configured, the TFCI code word will be 4 bits. | | |

6.10.3.4.1.2.2 Downlink

6.10.3.4.1.2.2.1 Transport channel parameters

6.10.3.4.1.2.2.1.1 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

| Higher layer | RAB/signalling RB | | SRB#1 | SRB#2 | SRB#3 | SRB#4 |
|---|---|---|---|---|---|---|
| | User of Radio Bearer | | RRC | RRC | NAS_DT High priority | NAS_DT Low priority |
| RLC | Logical channel type | | DCCH | DCCH | DCCH | DCCH |
| | RLC mode | | UM | AM | AM | AM |
| | Payload sizes, bit | | 136 | 128 | 128 | 128 |
| | Max data rate, bps | | 3 400 | 3 200 | 3 200 | 3 200 |
| | AMD/UMD PDU header, bit | | 8 | 16 | 16 | 16 |
| MAC | MAC header, bit | | 4 | 4 | 4 | 4 |
| | MAC multiplexing | | 4 logical channel multiplexing | | | |
| Layer 1 | TrCH type | | DCH | | | |
| | TB sizes, bit | | 148 (alt. 0, 148) (note) | | | |
| | TFS | TF0, bits | 0x148 (alt. 1x0) (note) | | | |
| | | TF1, bits | 1x148 | | | |
| | TTI, ms | | 40 | | | |

| | | |
|---|---|---|
| | Coding type | CC 1/3 |
| | CRC, bit | 16 |
| | Max number of bits/TTI before rate matching | 516 |
| | Max number of bits/radio frame before rate matching | 129 |
| | RM attribute | 155 to 165 |
| NOTE: Alternative parameters enable the measurement "transport channel BLER" in the UE. | | |

6.10.3.4.1.2.2.1.2 TFCS

| | |
|---|---|
| TFCS size | 2 |
| TFCS | SRBs for DCCH = (TF0), (TF1) |
| NOTE: The first TFC is required for the alt. case, optional otherwise. | |

6.10.3.4.1.2.2.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Downlink | Midamble | 512 chips |
| | Codes and time slots | SF16 x 1 code x 1 time slot |
| | Max. Number of data bits/radio frame | 236 |
| | TFCI code word | 8 bits |
| | Puncturing limit | 1 |
| NOTE: In case the first TFC in the TFCS is not configured, the TFCI code word will be 4 bits. | | |

6.10.3.4.1.3 Stand-alone UL:13.6 DL:13.6 kbps SRBs for DCCH

6.10.3.4.1.3.1 Uplink

6.10.3.4.1.3.1.1 Transport channel parameters

6.10.3.4.1.3.1.1.1 Transport channel parameters for UL:13.6 kbps SRBs for DCCH

| | | | | | | |
|---|---|---|---|---|---|---|
| Higher layer | RAB/signalling RB | | SRB#1 | SRB#2 | SRB#3 | SRB#4 |
| | User of Radio Bearer | | RRC | RRC | NAS_DT High priority | NAS_DT Low priority |
| RLC | Logical channel type | | DCCH | DCCH | DCCH | DCCH |
| | RLC mode | | UM | AM | AM | AM |
| | Payload sizes, bit | | 136 | 128 | 128 | 128 |
| | Max data rate, bps | | 13 600 | 12 800 | 12 800 | 12 800 |
| | AMD/UMD PDU header, bit | | 8 | 16 | 16 | 16 |
| MAC | MAC header, bit | | 4 | 4 | 4 | 4 |
| | MAC multiplexing | | 4 logical channel multiplexing | | | |
| Layer 1 | TrCH type | | DCH | | | |
| | TB sizes, bit | | 148 (alt. 0,148) (note) | | | |
| | TFS | TF0, bits | 0x148 (alt. 1x0) (note) | | | |
| | | TF1, bits | 1x148 | | | |
| | TTI, ms | | 10 | | | |
| | Coding type | | CC 1/3 | | | |
| | CRC, bit | | 16 | | | |
| | Max number of bits/TTI before rate matching | | 516 | | | |
| | Max number of bits/radio frame before rate matching | | 516 | | | |
| NOTE: Alternative parameters enable the measurement "transport channel BLER" in the UTRAN. | | | | | | |

6.10.3.4.1.3.1.1.2 TFCS

| | |
|---|---|
| TFCS size | 2 |
| TFCS | SRBs for DCCH = (TF0), (TF1) |
| NOTE: The first TFC is required for the alt. case, optional otherwise. | |

6.10.3.4.1.3.1.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Uplink | Midamble | 512 chips |
| | Codes and time slots | SF8 x 1 code x 1 time slot |
| | Max. Number of data bits/radio frame | 468 bits |
| | TFCI code word | 8 bits |
| | TPC | 2 bits |
| | Puncturing Limit | 0.88 |
| NOTE: In case the first TFC in the TFCS is not configured, the TFCI code word will be 4 bits. | | |

6.10.3.4.1.3.2 Downlink

6.10.3.4.1.3.2.1 Transport channel parameters

6.10.3.4.1.3.2.1.1 Transport channel parameters for DL:13.6 kbps SRBs for DCCH

| | | | | | | |
|---|---|---|---|---|---|---|
| Higher layer | RAB/signalling RB | | SRB#1 | SRB#2 | SRB#3 | SRB#4 |
| | User of Radio Bearer | | RRC | RRC | NAS_DT High priority | NAS_DT Low priority |
| RLC | Logical channel type | | DCCH | DCCH | DCCH | DCCH |
| | RLC mode | | UM | AM | AM | AM |
| | Payload sizes, bit | | 136 | 128 | 128 | 128 |
| | Max data rate, bps | | 13 600 | 12 800 | 12 800 | 12 800 |
| | AMD/UMD PDU header, bit | | 8 | 16 | 16 | 16 |
| MAC | MAC header, bit | | 4 | 4 | 4 | 4 |
| | MAC multiplexing | | 4 logical channel multiplexing | | | |
| Layer 1 | TrCH type | | DCH | | | |
| | TB sizes, bit | | 148 (alt. 0,148) (note) | | | |
| | TFS | TF0, bits | 0x148 (alt. 1x0) (note) | | | |
| | | TF1, bits | 1x148 | | | |
| | TTI, ms | | 10 | | | |
| | Coding type | | CC 1/3 | | | |
| | CRC, bit | | 16 | | | |
| | Max number of bits/TTI before rate matching | | 516 | | | |
| | Max number of bits/radio frame before rate matching | | 516 | | | |
| NOTE: Alternative parameters enable the measurement "transport channel BLER" in the UE. | | | | | | |

6.10.3.4.1.3.2.1.2 TFCS

| | |
|---|---|
| TFCS size | 2 |
| TFCS | SRBs for DCCH = (TF0), (TF1) |
| NOTE: The first TFC is required for the alt. case, optional otherwise. | |

6.10.3.4.1.3.2.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Downlink | Midamble | 512 chips |
| | Codes and time slots | SF16 x 2 codes x 1 time slot |
| | Max. Number of data bits/radio frame | 480 bits |
| | TFCI code word | 8 bits |
| | Puncturing limit | 0.92 |
| NOTE: In case the first TFC in the TFCS is not configured, the TFCI code word will be 4 bits. | | |

6.10.3.4.1.4 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.3.4.1.4.1 Uplink

6.10.3.4.1.4.1.1 Transport channel parameters

6.10.3.4.1.4.1.1.1 Transport channel parameters for Conversational / speech / UL:12.2 kbps / CS RAB

| Higher layer | RAB/Signalling RB | | RAB subflow #1 | RAB subflow #2 | RAB subflow #3 |
|---|---|---|---|---|---|
| RLC | Logical channel type | | DTCH | | |
| | RLC mode | | TM | TM | TM |
| | Payload sizes, bit | | 39, 81 (alt. 0, 39, 81) | 103 | 60 |
| | Max data rate, bps | | 12 200 | | |
| | TrD PDU header, bit | | 0 | | |
| MAC | MAC header, bit | | 0 | | |
| | MAC multiplexing | | N/A | | |
| Layer 1 | TrCH type | | DCH | DCH | DCH |
| | TB sizes, bit | | 39, 81 (alt. 0, 39, 81) | 103 | 60 |
| | TFS | TF0, bits | 0x81(alt. 1x0) (note) | 0x103 | 0x60 |
| | | TF1, bits | 1x39 | 1x103 | 1x60 |
| | | TF2, bits | 1x81 | N/A | N/A |
| | TTI, ms | | 20 | 20 | 20 |
| | Coding type | | CC 1/3 | CC 1/3 | CC 1/2 |
| | CRC, bit | | 12 | N/A | N/A |
| | Max number of bits/TTI after channel coding | | 303 | 333 | 136 |
| | Max number of bits/radio frame before rate matching | | 152 | 167 | 68 |
| | RM attribute | | 180 to 220 | 170 to 210 | 215 to 256 |
| NOTE: In case of using this alternative, CRC parity bits are to be attached to RAB subflow#1 any time since number of TrBlks are 1 even if there is no data on RAB subflow#1 (see clause 4.2.1.1 in 3GPP TS 25.222 [29]). | | | | | |

6.10.3.4.1.4.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.10.3.4.1.4.1.1.3 TFCS

| TFCS size | 6 |
|---|---|
| TFCS | (RAB subflow#1, RAB subflow#2, RAB subflow#3,DCCH)=<br>(TF0, TF0, TF0, TF0), (TF1, TF0, TF0, TF0), (TF2, TF1, TF1, TF0),<br>(TF0, TF0, TF0, TF1), (TF1, TF0, TF0, TF1), (TF2, TF1, TF1, TF1) |
| NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise. | |

6.10.3.4.1.4.1.2 Physical channel parameters

| DPCH Uplink | Midamble | 512 chips |
|---|---|---|
| | Codes and time slots | SF8 x 1 code x 1 time slot |
| | Max. Number of data bits/radio frame | 452 bits |
| | TFCI code word | 16 bits |
| | TPC | 2 bit |
| | Puncturing Limit | 0.72 |

6.10.3.4.1.4.2 Downlink

6.10.3.4.1.4.2.1 Transport channel parameters

6.10.3.4.1.4.2.1.1 Transport channel parameters for Conversational / speech / DL:12.2 kbps / CS RAB

| Higher layer | RAB/Signalling RB | RAB subflow #1 | RAB subflow #2 | RAB subflow #3 |
|---|---|---|---|---|

| | | | | | |
|---|---|---|---|---|---|
| RLC | Logical channel type | | DTCH | | |
| | RLC mode | | TM | TM | TM |
| | Payload sizes, bit | | 39,81 (alt. 0, 39, 81) | 103 | 60 |
| | Max data rate, bps | | 12 200 | | |
| | TrD PDU header, bit | | 0 | | |
| MAC | MAC header, bit | | 0 | | |
| | MAC multiplexing | | N/A | | |
| Layer 1 | TrCH type | | DCH | DCH | DCH |
| | TB sizes, bit | | 39,81 (alt. 0,39,81) | 103 | 60 |
| | TFS | TF0, bits | 0x81 (alt. 1x0) (note) | 0x103 | 0x60 |
| | | TF1, bits | 1x39 | 1x103 | 1x60 |
| | | TF2, bits | 1x81 | N/A | N/A |
| | TTI, ms | | 20 | 20 | 20 |
| | Coding type | | CC 1/3 | CC 1/3 | CC 1/2 |
| | CRC, bit | | 12 | N/A | N/A |
| | Max number of bits/TTI after channel coding | | 303 | 333 | 136 |
| | Max number of bits/radio frame before rate matching | | 152 | 167 | 68 |
| | RM attribute | | 180 to 220 | 170 to 210 | 215 to 256 |
| NOTE: CRC parity bits are to be attached to RAB subflow#1 any time since number of TrBlks are 1 even if there is no data on RAB subflow#1 (see clause 4.2.1.1 in 3GPP TS 25.222 [29]). | | | | | |

6.10.3.4.1.4.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.10.3.4.1.4.2.1.3 TFCS

| | |
|---|---|
| TFCS size | 6 |
| TFCS | (RAB subflow#1, RAB subflow#2, RAB subflow#3,DCCH)=<br>(TF0, TF0, TF0, TF0), (TF1, TF0, TF0, TF0), (TF2, TF1, TF1, TF0),<br>(TF0, TF0, TF0, TF1), (TF1, TF0, TF0, TF1), (TF2, TF1, TF1, TF1) |
| NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise. | |

6.10.3.4.1.4.2.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Downlink | Midamble | 512 chips |
| | Codes and time slots | SF16 x 2 codes x 1 time slot |
| | Max. Number of data bits/radio frame | 472 bits |
| | TFCI code word | 16 bits |
| | Puncturing limit | 0.76 |

6.10.3.4.1.4a Conversational / speech / UL:(12.2, 7.95, 5.9, 4.75) DL:(12.2, 7.95, 5.9, 4.75) kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.3.4.1.4a.1 Uplink

6.10.3.4.1.4a.1.1 Transport channel parameters

6.10.3.4.1.4a.1.1.1 Transport channel parameters for Conversational / speech / UL: (12.2, 7.95, 5.9, 4.75) kbps / CS RAB

| | | | | |
|---|---|---|---|---|
| Higher layer | RAB/Signalling RB | RAB subflow #1 | RAB subflow #2 | RAB subflow #3 |
| RLC | Logical channel type | DTCH | | |
| | RLC mode | TM | TM | TM |
| | Payload sizes, bit | 39, 42, 55, 75, 81 (alt. 0, 39, 42, 55, 75, 81) | 53, 63, 84, 103 | 60 |
| | Max data rate, bps | 12 200 | | |
| | TrD PDU header, bit | 0 | | |
| MAC | MAC header, bit | 0 | | |
| | MAC multiplexing | N/A | | |
| Layer 1 | TrCH type | DCH | DCH | DCH |

| | | | | | |
|---|---|---|---|---|---|
| | TB sizes, bit | | 39, 42, 55, 75, 81 (alt. 0, 39, 42, 55, 75, 81) | 53, 63, 84, 103 | 60 |
| | TFS | TF0, bits | 0x81(alt. 1x0) (note) | 0x103 | 0x60 |
| | | TF1, bits | 1x39 | 1x53 | 1x60 |
| | | TF2 bits | 1x42 | 1x63 | N/A |
| | | TF3, bits | 1x55 | 1x84 | N/A |
| | | TF4, bits | 1x75 | 1x103 | N/A |
| | | TF5, bits | 1x81 | N/A | N/A |
| | TTI, ms | | 20 | 20 | 20 |
| | Coding type | | CC 1/3 | CC 1/3 | CC 1/2 |
| | CRC, bit | | 12 | N/A | N/A |
| | Max number of bits/TTI after channel coding | | 303 | 333 | 136 |
| | Max number of bits/radio frame before rate matching | | 152 | 167 | 68 |
| | RM attribute | | 180 to 220 | 170 to 210 | 215 to 256 |
| NOTE: In case of using this alternative, CRC parity bits are to be attached to RAB subflow#1 any time since number of TrBlks are 1 even if there is no data on RAB subflow#1 (see clause 4.2.1.1 in 3GPP TS 25.222 [29]). | | | | | |

6.10.3.4.1.4a.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.10.3.4.1.4a.1.1.3 TFCS

| | |
|---|---|
| TFCS size | 12 |
| TFCS | (RAB subflow#1, RAB subflow#2, RAB subflow#3, DCCH)= (TF0,TF0,TF0,TF0), (TF1,TF0,TF0,TF0), (TF2,TF1,TF0,TF0), (TF3,TF2,TF0,TF0), (TF4,TF3,TF0,TF0), (TF5,TF4,TF1,TF0), (TF0,TF0,TF0,TF1), (TF1,TF0,TF0,TF1), (TF2,TF1,TF0,TF1), (TF3,TF2,TF0,TF1), (TF4,TF3,TF0,TF1), (TF5,TF4,TF1,TF1) |
| NOTE: In case TB size zero is configured for any transport channel the first TFC is required; it is optional otherwise. | |

6.10.3.4.1.4a.1.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Uplink | Midamble | 512 chips |
| | Codes and time slots | SF8 x 1 code x 1 time slot |
| | Max. Number of data bits/radio frame | 452 bits |
| | TFCI code word | 16 bits |
| | TPC | 2 bit |
| | Puncturing Limit | 0.72 |

6.10.3.4.1.4a.2 Downlink

6.10.3.4.1.4a.2.1 Transport channel parameters

6.10.3.4.1.4a.2.1.1 Transport channel parameters for Conversational / speech / DL: (12.2, 7.95, 5.9, 4.75) kbps / CS RAB

| | | | | | |
|---|---|---|---|---|---|
| Higher layer | RAB/Signalling RB | | RAB subflow #1 | RAB subflow #2 | RAB subflow #3 |
| RLC | Logical channel type | | DTCH | | |
| | RLC mode | | TM | TM | TM |
| | Payload sizes, bit | | 39, 42, 55, 75, 81 (alt. 0, 39, 42, 55, 75, 81) | 53, 63, 84, 103 | 60 |
| | Max data rate, bps | | 12 200 | | |
| | TrD PDU header, bit | | 0 | | |
| MAC | MAC header, bit | | 0 | | |
| | MAC multiplexing | | N/A | | |
| Layer 1 | TrCH type | | DCH | DCH | DCH |
| | TB sizes, bit | | 39, 42, 55, 75, 81 (alt. 0, 39, 42, 55, 75, 81) | 53, 63, 84, 103 | 60 |
| | TFS | TF0, bits | 0x81(alt. 1x0) (note) | 0x103 | 0x60 |
| | | TF1, bits | 1x39 | 1x53 | 1x60 |
| | | TF2, bits | 1x42 | 1x63 | N/A |

| | | | | | |
|---|---|---|---|---|---|
| | | TF3, bits | 1x55 | 1x84 | N/A |
| | | TF4, bits | 1x75 | 1x103 | N/A |
| | | TF5, bits | 1x81 | N/A | N/A |
| | TTI, ms | | 20 | 20 | 20 |
| | Coding type | | CC 1/3 | CC 1/3 | CC 1/2 |
| | CRC, bit | | 12 | N/A | N/A |
| | Max number of bits/TTI after channel coding | | 303 | 333 | 136 |
| | Max number of bits/radio frame before rate matching | | 152 | 167 | 68 |
| | RM attribute | | 180 to 220 | 170 to 210 | 215 to 256 |
| NOTE: In case of using this alternative, CRC parity bits are to be attached to RAB subflow#1 any time since number of TrBlks are 1 even if there is no data on RAB subflow#1 (see clause 4.2.1.1 in 3GPP TS 25.222 [29]). | | | | | |

6.10.3.4.1.4a.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.10.3.4.1.4a.2.1.3 TFCS

| | |
|---|---|
| TFCS size | 12 |
| TFCS | (RAB subflow#1, RAB subflow#2, RAB subflow#3, DCCH)=<br>(TF0,TF0,TF0,TF0), (TF1,TF0,TF0,TF0), (TF2,TF1,TF0,TF0), (TF3,TF2,TF0,TF0),<br>(TF4,TF3,TF0,TF0), (TF5,TF4,TF1,TF0), (TF0,TF0,TF0,TF1), (TF1,TF0,TF0,TF1),<br>(TF2,TF1,TF0,TF1), (TF3,TF2,TF0,TF1), (TF4,TF3,TF0,TF1), (TF5,TF4,TF1,TF1) |
| NOTE: In case TB size zero is configured for any transport channel the first TFC is required; it is optional otherwise. | |

6.10.3.4.1.4a.2.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Downlink | Midamble | 512 chips |
| | Codes and time slots | SF16 x 2 codes x 1 time slot |
| | Max. Number of data bits/radio frame | 472 bits |
| | TFCI code word | 16 bits |
| | Puncturing limit | 0.76 |

6.10.3.4.1.5 Conversational / speech / UL:10.2 DL:10.2 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.3.4.1.5.1 Uplink

6.10.3.4.1.5.1.1 Transport channel parameters

6.10.3.4.1.5.1.1.1 Transport channel parameters for Conversational / speech / UL:10.2 kbps / CS RAB

| Higher layer | RAB/Signalling RB | | RAB subflow #1 | RAB subflow #2 | RAB subflow #3 |
|---|---|---|---|---|---|
| RLC | Logical channel type | | DTCH | | |
| | RLC mode | | TM | TM | TM |
| | Payload sizes, bit | | 39, 65<br>(alt. 0, 39, 65) | 99 | 40 |
| | Max data rate, bps | | 10 200 | | |
| | TrD PDU header, bit | | 0 | | |
| MAC | MAC header, bit | | 0 | | |
| | MAC multiplexing | | N/A | | |
| Layer 1 | TrCH type | | DCH | DCH | DCH |
| | TB sizes, bit | | 39, 65<br>(alt. 0, 39, 65) | 99 | 40 |
| | TFS | TF0, bits | 0x65 (alt. 1x0)<br>(note) | 0x99 | 0x40 |
| | | TF1, bits | 1x39 | 1x99 | 1x40 |
| | | TF2, bits | 1x65 | N/A | N/A |
| | TTI, ms | | 20 | 20 | 20 |
| | Coding type | | CC 1/3 | CC 1/3 | CC 1/2 |
| | CRC, bit | | 12 | N/A | N/A |

| | | | | |
|---|---|---|---|---|
| | Max number of bits/TTI after channel coding | 255 | 321 | 96 |
| | Max number of bits/radio frame before rate matching | 128 | 161 | 48 |
| | RM attribute | 180 to 220 | 170 to 210 | 215 to 256 |
| NOTE: In case of using this alternative, CRC parity bits are to be attached to RAB subflow#1 any time since number of TrBlks are 1 even if there is no data on RAB subflow#1 (see clause 4.2.1.1 in 3GPP TS 25.222 [29]). | | | | |

6.10.3.4.1.5.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.10.3.4.1.5.1.1.3 TFCS

| | |
|---|---|
| TFCS size | 6 |
| TFCS | (RAB subflow#1, RAB subflow#2, RAB subflow#3,DCCH)=<br>(TF0, TF0, TF0, TF0), (TF1, TF0, TF0, TF0), (TF2, TF1, TF1, TF0),<br>(TF0, TF0, TF0, TF1), (TF1, TF0, TF0, TF1), (TF2, TF1, TF1, TF1) |
| NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise. | |

6.10.3.4.1.5.1.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Uplink | Midamble | 512 chips |
| | Codes and time slots | SF16 x 1 code x 1 time slot |
| | Max. Number of data bits/radio frame | 226 bits |
| | TFCI code word | 16 bits |
| | TPC | 2 bit |
| | Puncturing Limit | 0.40 |

6.10.3.4.1.5.2 Downlink

6.10.3.4.1.5.2.1 Transport channel parameters

6.10.3.4.1.5.2.1.1 Transport channel parameters for Conversational / speech / DL:10.2 kbps / CS RAB

| Higher layer | RAB/Signalling RB | | RAB subflow #1 | RAB subflow #2 | RAB subflow #3 |
|---|---|---|---|---|---|
| RLC | Logical channel type | | DTCH | | |
| | RLC mode | | TM | TM | TM |
| | Payload sizes, bit | | 39,65 (alt. 0, 39, 65) | 99 | 40 |
| | Max data rate, bps | | 10 200 | | |
| | TrD PDU header, bit | | 0 | | |
| MAC | MAC header, bit | | 0 | | |
| | MAC multiplexing | | N/A | | |
| Layer 1 | TrCH type | | DCH | DCH | DCH |
| | TB sizes, bit | | 39, 65 (alt.0,39,65) | 99 | 40 |
| | TFS | TF0, bits | 0x65 (alt,1x0) (note) | 0x99 | 0x40 |
| | | TF1, bits | 1x39 | 1x99 | 1x40 |
| | | TF2, bits | 1x65 | N/A | N/A |
| | TTI, ms | | 20 | 20 | 20 |
| | Coding type | | CC 1/3 | CC 1/3 | CC 1/2 |
| | CRC, bit | | 12 | N/A | N/A |
| | Max number of bits/TTI after channel coding | | 255 | 321 | 96 |
| | Max number of bits/radio frame before rate matching | | 128 | 161 | 48 |
| | RM attribute | | 180 to 220 | 170 to 210 | 215 to 256 |
| NOTE: CRC parity bits are to be attached to RAB subflow#1 any time since number of TrBlks are 1 even if there is no data on RAB subflow#1 (see clause 4.2.1.1 in 3GPP TS 25.222 [29]). | | | | | |

6.10.3.4.1.5.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.10.3.4.1.5.2.1.3 TFCS

| | |
|---|---|
| TFCS size | 6 |
| TFCS | (RAB subflow#1, RAB subflow#2, RAB subflow#3,DCCH)=<br>(TF0, TF0, TF0, TF0), (TF1, TF0, TF0, TF0), (TF2, TF1, TF1, TF0),<br>(TF0, TF0, TF0, TF1), (TF1, TF0, TF0, TF1), (TF2, TF1, TF1, TF1) |
| NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise. | |

6.10.3.4.1.5.2.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Downlink | Midamble | 512 chips |
| | Codes and time slots | SF16 x 1 code x 1 time slot |
| | Max. Number of data bits/radio frame | 228 bits |
| | TFCI code word | 16 bits |
| | Puncturing limit | 0.40 |

6.10.3.4.1.5a Conversational / speech / UL:(10.2, 6.7, 5.9, 4.75) DL:(10.2, 6.7, 5.9, 4.75) kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.3.4.1.5a.1 Uplink

6.10.3.4.1.5a.1.1 Transport channel parameters

6.10.3.4.1.5a.1.1.1 Transport channel parameters for Conversational / speech / UL:(10.2, 6.7, 5.9, 4.75) kbps / CS RAB

| Higher layer | RAB/Signalling RB | | RAB subflow #1 | RAB subflow #2 | RAB subflow #3 |
|---|---|---|---|---|---|
| RLC | Logical channel type | | DTCH | | |
| | RLC mode | | TM | TM | TM |
| | Payload sizes, bit | | 39, 42, 55, 58, 65<br>(alt. 0, 39, 42, 55, 58, 65) | 53, 63, 76, 99 | 40 |
| | Max data rate, bps | | 10 200 | | |
| | TrD PDU header, bit | | 0 | | |
| MAC | MAC header, bit | | 0 | | |
| | MAC multiplexing | | N/A | | |
| Layer 1 | TrCH type | | DCH | DCH | DCH |
| | TB sizes, bit | | 39, 42, 55, 58, 65<br>(alt. 0, 39, 42, 55, 58, 65) | 53, 63, 76, 99 | 40 |
| | TFS | TF0, bits | 0x65 (alt. 1x0) (note) | 0x99 | 0x40 |
| | | TF1, bits | 1x39 | 1x53 | 1x40 |
| | | TF2, bits | 1x42 | 1x63 | N/A |
| | | TF3, bits | 1x55 | 1x76 | N/A |
| | | TF4, bits | 1x58 | 1x99 | N/A |
| | | TF5, bits | 1x65 | N/A | N/A |
| | TTI, ms | | 20 | 20 | 20 |
| | Coding type | | CC 1/3 | CC 1/3 | CC 1/2 |
| | CRC, bit | | 12 | N/A | N/A |
| | Max number of bits/TTI after channel coding | | 255 | 321 | 96 |
| | Max number of bits/radio frame before rate matching | | 128 | 161 | 48 |
| | RM attribute | | 180 to 220 | 170 to 210 | 215 to 256 |
| NOTE: In case of using this alternative, CRC parity bits are to be attached to RAB subflow#1 any time since number of TrBlks are 1 even if there is no data on RAB subflow#1 (see clause 4.2.1.1 in TS 25.222). | | | | | |

6.10.3.4.1.5a.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.10.3.4.1.5a.1.1.3 TFCS

| | |
|---|---|
| TFCS size | 12 |
| TFCS | (RAB subflow#1, RAB subflow#2, RAB subflow#3,DCCH)=<br>(TF0,TF0,TF0,TF0), (TF1,TF0,TF0,TF0), (TF2,TF1,TF0,TF0), (TF3,TF2,TF0,TF0), |

| | |
|---|---|
| | (TF4,TF3,TF0,TF0), (TF5,TF4,TF1,TF0), (TF0,TF0,TF0,TF1), (TF1,TF0,TF0,TF1), (TF2,TF1,TF0,TF1), (TF3,TF2,TF0,TF1), (TF4,TF3,TF0,TF1), (TF5,TF4,TF1,TF1) |
| NOTE: In case TB size zero is configured for any transport channel the first TFC is required; it is optional otherwise. | |

6.10.3.4.1.5a.1.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Uplink | Midamble | 512 chips |
| | Codes and time slots | SF16 x 1 code x 1 time slot |
| | Max. Number of data bits/radio frame | 226 bits |
| | TFCI code word | 16 bits |
| | TPC | 2 bit |
| | Puncturing Limit | 0.40 |

6.10.3.4.1.5a.2 Downlink

6.10.3.4.1.5a.2.1 Transport channel parameters

6.10.3.4.1.5a.2.1.1 Transport channel parameters for Conversational / speech / DL: DL:(10.2, 6.7, 5.9, 4.75) kbps / CS RAB

| Higher layer | RAB/Signalling RB | | RAB subflow #1 | RAB subflow #2 | RAB subflow #3 |
|---|---|---|---|---|---|
| RLC | Logical channel type | | DTCH | | |
| | RLC mode | | TM | TM | TM |
| | Payload sizes, bit | | 39, 42, 55, 58, 65 (alt. 0, 39, 42, 55, 58, 65) | 0, 53, 63, 76, 99 | 40 |
| | Max data rate, bps | | 10 200 | | |
| | TrD PDU header, bit | | 0 | | |
| MAC | MAC header, bit | | 0 | | |
| | MAC multiplexing | | N/A | | |
| Layer 1 | TrCH type | | DCH | DCH | DCH |
| | TB sizes, bit | | 39, 42, 55, 58, 65 (alt. 0, 39, 42, 55, 58, 65) | 0, 53, 63, 76, 99 | 40 |
| | TFS | TF0, bits | 0x65 (alt. 1x0) (note) | 0x99 | 0x40 |
| | | TF1, bits | 1x39 | 1x53 | 1x40 |
| | | TF2, bits | 1x42 | 1x63 | N/A |
| | | TF3, bits | 1x55 | 1x76 | N/A |
| | | TF4, bits | 1x58 | 1x99 | N/A |
| | | TF5, bits | 1x65 | N/A | N/A |
| | TTI, ms | | 20 | 20 | 20 |
| | Coding type | | CC 1/3 | CC 1/3 | CC 1/2 |
| | CRC, bit | | 12 | N/A | N/A |
| | Max number of bits/TTI after channel coding | | 255 | 321 | 96 |
| | Max number of bits/radio frame before rate matching | | 128 | 161 | 48 |
| | RM attribute | | 180 to 220 | 170 to 210 | 215 to 256 |
| NOTE: In case of using this alternative, CRC parity bits are to be attached to RAB subflow#1 any time since number of TrBlks are 1 even if there is no data on RAB subflow#1 (see clause 4.2.1.1 in 3GPP TS 25.222 [29]). | | | | | |

6.10.3.4.1.5a.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.10.3.4.1.5a.2.1.3 TFCS

| | |
|---|---|
| TFCS size | 12 |
| TFCS | (RAB subflow#1, RAB subflow#2, RAB subflow#3,DCCH)= (TF0,TF0,TF0,TF0), (TF1,TF0,TF0,TF0), (TF2,TF1,TF0,TF0), (TF3,TF2,TF0,TF0), (TF4,TF3,TF0,TF0), (TF5,TF4,TF1,TF0), (TF0,TF0,TF0,TF1), (TF1,TF0,TF0,TF1), (TF2,TF1,TF0,TF1), (TF3,TF2,TF0,TF1), (TF4,TF3,TF0,TF1), (TF5,TF4,TF1,TF1) |
| NOTE: In case TB size zero is configured for any transport channel the first TFC is required; it is optional otherwise. | |

6.10.3.4.1.5a.2.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Downlink | Midamble | 512 chips |
| | Codes and time slots | SF16 x 1 codes x 1 time slot |
| | Max. Number of data bits/radio frame | 228 bits |
| | TFCI code word | 16 bits |
| | Puncturing limit | 0.40 |

6.10.3.4.1.6 Conversational / speech / UL:7.95 DL:7.95 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.3.4.1.6.1 Uplink

6.10.3.4.1.6.1.1 Transport channel parameters

6.10.3.4.1.6.1.1.1 Transport channel parameters for Conversational / speech / UL:7.95 kbps / CS RAB

| Higher layer | RAB/Signalling RB | | RAB subflow #1 | RAB subflow #2 |
|---|---|---|---|---|
| RLC | Logical channel type | | DTCH | |
| | RLC mode | | TM | TM |
| | Payload sizes, bit | | 39, 75 (alt. 0, 39, 75) | 84 |
| | Max data rate, bps | | 7 950 | |
| | TrD PDU header, bit | | 0 | |
| MAC | MAC header, bit | | 0 | |
| | MAC multiplexing | | N/A | |
| Layer 1 | TrCH type | | DCH | DCH |
| | TB sizes, bit | | 39, 75 (alt. 0, 39, 75) | 84 |
| | TFS | TF0, bits | 0x75 (alt. 1x0) (note) | 0x84 |
| | | TF1, bits | 1x39 | 1x84 |
| | | TF2, bits | 1x75 | N/A |
| | TTI, ms | | 20 | 20 |
| | Coding type | | CC 1/3 | CC 1/3 |
| | CRC, bit | | 12 | N/A |
| | Max number of bits/TTI after channel coding | | 285 | 276 |
| | Max number of bits/radio frame before rate matching | | 143 | 138 |
| | RM attribute | | 180 to 220 | 170 to 210 |
| NOTE: In case of using this alternative, CRC parity bits are to be attached to RAB subflow#1 any time since number of TrBlks are 1 even if there is no data on RAB subflow#1 (see clauses 4.2.1.1 in 3GPP TS 25.222 [29]). | | | | |

6.10.3.4.1.6.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.10.3.4.1.6.1.1.3 TFCS

| | |
|---|---|
| TFCS size | 6 |
| TFCS | (RAB subflow#1, RAB subflow#2, DCCH)=<br>(TF0, TF0, TF0), (TF1, TF0, TF0), (TF2, TF1, TF0),<br>(TF0, TF0, TF1), (TF1, TF0, TF1), (TF2, TF1, TF1) |
| NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise. | |

6.10.3.4.1.6.1.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Uplink | Midamble | 512 chips |
| | Codes and time slots | SF16 x 1 code x 1 time slot |
| | Max. Number of data bits/radio frame | 226 bits |
| | TFCI code word | 16 bits |
| | TPC | 2 bits |
| | Puncturing Limit | 0.44 |

6.10.3.4.1.6.2 Downlink

6.10.3.4.1.6.2.1 Transport channel parameters

6.10.3.4.1.6.2.1.1 Transport channel parameters for Conversational / speech / DL:7.95 kbps / CS RAB

| Higher layer | RAB/Signalling RB | | RAB subflow #1 | RAB subflow #2 |
|---|---|---|---|---|
| RLC | Logical channel type | | DTCH | |
| | RLC mode | | TM | TM |
| | Payload sizes, bit | | 39, 75 (alt. 0, 39, 75) | 84 |
| | Max data rate, bps | | 7 950 | |
| | TrD PDU header, bit | | 0 | |
| MAC | MAC header, bit | | 0 | |
| | MAC multiplexing | | N/A | |
| Layer 1 | TrCH type | | DCH | DCH |
| | TB sizes, bit | | 39, 75 (alt. 0, 39, 75) | 84 |
| | TFS | TF0, bits | 0x75 (alt. 1x0) (note) | 0x84 |
| | | TF1, bits | 1x39 | 1x84 |
| | | TF2, bits | 1x75 | N/A |
| | TTI, ms | | 20 | 20 |
| | Coding type | | CC 1/3 | CC 1/3 |
| | CRC, bit | | 12 | N/A |
| | Max number of bits/TTI after channel coding | | 285 | 276 |
| | Max number of bits/radio frame before rate matching | | 143 | 138 |
| | RM attribute | | 180 to 220 | 170 to 210 |
| NOTE: CRC parity bits are to be attached to RAB subflow#1 any time since number of TrBlks are 1 even if there is no data on RAB subflow#1 (see clause 4.2.1.1 in 3GPP TS 25.222 [29]). | | | | |

6.10.3.4.1.6.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.10.3.4.1.6.2.1.3 TFCS

| TFCS size | 6 |
|---|---|
| TFCS | (RAB subflow#1, RAB subflow#2, DCCH)=<br>(TF0, TF0, TF0), (TF1, TF0, TF0), (TF2, TF1, TF0),<br>(TF0, TF0, TF1), (TF1, TF0, TF1), (TF2, TF1, TF1) |
| NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise. | |

6.10.3.4.1.6.2.2 Physical channel parameters

| DPCH Downlink | Midamble | 512 chips |
|---|---|---|
| | Codes and time slots | SF16 x 1 code x 1 time slot |
| | Max. Number of data bits/radio frame | 228 bits |
| | TFCI code word | 16 bits |
| | Puncturing limit | 0.48 |

6.10.3.4.1.7 Conversational / speech / UL:7.4 DL:7.4 kbps / CS RAB+ UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.3.4.1.7.1 Uplink

6.10.3.4.1.7.1.1 Transport channel parameters

6.10.3.4.1.7.1.1.1 Transport channel parameters for Conversational / speech / UL:7.4 kbps / CS RAB

| Higher layer | RAB/Signalling RB | RAB subflow #1 | RAB subflow #2 |
|---|---|---|---|
| RLC | Logical channel type | DTCH | |
| | RLC mode | TM | TM |
| | Payload sizes, bit | 39, 61 (alt. 0, 39, 61) | 87 |
| | Max data rate, bps | 7 400 | |
| | TrD PDU header, bit | 0 | |

| | | | | |
|---|---|---|---|---|
| MAC | MAC header, bit | | 0 | |
| | MAC multiplexing | | N/A | |
| Layer 1 | TrCH type | | DCH | DCH |
| | TB sizes, bit | | 39, 61 (alt. 0, 39, 61) | 87 |
| | TFS | TF0, bits | 0x61 (alt. 1x0) (note) | 0x87 |
| | | TF1, bits | 1x39 | 1x87 |
| | | TF2, bits | 1x61 | N/A |
| | TTI, ms | | 20 | 20 |
| | Coding type | | CC 1/3 | CC 1/3 |
| | CRC, bit | | 12 | N/A |
| | Max number of bits/TTI after channel coding | | 243 | 285 |
| | Max number of bits/radio frame before rate matching | | 122 | 143 |
| | RM attribute | | 180 to 220 | 170 to 210 |
| NOTE: CRC parity bits are to be attached to RAB subflow#1 any time since number of TrBlks are 1 even if there is no data on RAB subflow#1 (see clause 4.2.1.1 in 3GPP TS 25.222 [29]). | | | | |

6.10.3.4.1.7.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.10.3.4.1.7.1.1.3 TFCS

| | |
|---|---|
| TFCS size | 6 |
| TFCS | (RAB subflow#1, RAB subflow#2, DCCH)=<br>(TF0, TF0, TF0), (TF1, TF0, TF0), (TF2, TF1, TF0),<br>(TF0, TF0, TF1), (TF1, TF0, TF1), (TF2, TF1, TF1) |
| NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise. | |

6.10.3.4.1.7.1.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Uplink | Midamble | 512 chips |
| | Codes and time slots | SF16 x 1 code x 1 time slot |
| | Max. Number of data bits/radio frame | 226 bits |
| | TFCI code word | 16 bits |
| | TPC | 2 bits |
| | Puncturing Limit | 0.48 |

6.10.3.4.1.7.2 Downlink

6.10.3.4.1.7.2.1 Transport channel parameters

6.10.3.4.1.7.2.1.1 Transport channel parameters for Conversational / speech / DL:7.4 kbps / CS RAB

| | | | | |
|---|---|---|---|---|
| Higher layer | RAB/Signalling RB | | RAB subflow #1 | RAB subflow #2 |
| RLC | Logical channel type | | DTCH | |
| | RLC mode | | TM | TM |
| | Payload sizes, bit | | 39, 61 (alt. 0, 39, 61) | 87 |
| | Max data rate, bps | | 7 400 | |
| | TrD PDU header, bit | | 0 | |
| MAC | MAC header, bit | | 0 | |
| | MAC multiplexing | | N/A | |
| Layer 1 | TrCH type | | DCH | DCH |
| | TB sizes, bit | | 39, 61 (alt. 0, 39, 61) | 87 |
| | TFS | TF0, bits | 0x61(alt. 1x0) (note) | 0x87 |
| | | TF1, bits | 1x39 | 1x87 |
| | | TF2, bits | 1x61 | N/A |
| | TTI, ms | | 20 | 20 |
| | Coding type | | CC 1/3 | CC 1/3 |
| | CRC, bit | | 12 | N/A |
| | Max number of bits/TTI after channel coding | | 243 | 285 |
| | Max number of bits/radio frame before rate matching | | 122 | 143 |
| | RM attribute | | 180 to 220 | 170 to 210 |

| NOTE: | CRC parity bits are to be attached to RAB subflow#1 any time since number of TrBlks are 1 even if there is no data on RAB #1 (see clause 4.2.1.1 in 3GPP TS 25.222 [29]). |
|---|---|

6.10.3.4.1.7.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.10.3.4.1.7.2.1.3 TFCS

| TFCS size | 6 |
|---|---|
| TFCS | (RAB subflow#1, RAB subflow#2, DCCH)=<br>(TF0, TF0, TF0), (TF1, TF0, TF0), (TF2, TF1, TF0),<br>(TF0, TF0, TF1), (TF1, TF0, TF1), (TF2, TF1, TF1) |
| NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; optional otherwise. | |

6.10.3.4.1.7.2.2 Physical channel parameters

| DPCH Downlink | Midamble | 512 chips |
|---|---|---|
| | Codes and time slots | SF16 x 1 code x 1 time slot |
| | Max. Number of data bits/radio frame | 228 bits |
| | TFCI code word | 16 bits |
| | Puncturing limit | 0.48 |

6.10.3.4.1.7a Conversational / speech / UL:(7.4, 6.7, 5.9, 4.75) DL:(7.4, 6.7, 5.9, 4.75) kbps / CS RAB+ UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.3.4.1.7a.1 Uplink

6.10.3.4.1.7a.1.1 Transport channel parameters

6.10.3.4.1.7a.1.1.1 Transport channel parameters for Conversational / speech / UL:(7.4, 6.7, 5.9, 4.75) kbps / CS RAB

| Higher layer | RAB/Signalling RB | | RAB subflow #1 | RAB subflow #2 |
|---|---|---|---|---|
| RLC | Logical channel type | | DTCH | |
| | RLC mode | | TM | TM |
| | Payload sizes, bit | | 39, 42, 55, 58, 61<br>(alt. 0, 39, 42, 55, 58, 61) | 53, 63, 76, 87 |
| | Max data rate, bps | | 7 400 | |
| | TrD PDU header, bit | | 0 | |
| MAC | MAC header, bit | | 0 | |
| | MAC multiplexing | | N/A | |
| Layer 1 | TrCH type | | DCH | DCH |
| | TB sizes, bit | | 39, 42, 55, 58, 61<br>(alt. 0, 39, 42, 55, 58, 61) | 53, 63, 76, 87 |
| | TFS | TF0, bits | 0x61 (alt. 1x0)<br>(note) | 0x87 |
| | | TF1, bits | 1x39 | 1x53 |
| | | TF2, bits | 1x42 | 1x63 |
| | | TF3, bits | 1x55 | 1x76 |
| | | TF4, bits | 1x58 | 1x87 |
| | | TF5, bits | 1x61 | N/A |
| | TTI, ms | | 20 | 20 |
| | Coding type | | CC 1/3 | CC 1/3 |
| | CRC, bit | | 12 | N/A |
| | Max number of bits/TTI after channel coding | | 243 | 285 |
| | Max number of bits/radio frame before rate matching | | 122 | 143 |
| | RM attribute | | 180 to 220 | 170 to 210 |
| NOTE: In case of using this alternative, CRC parity bits are to be attached to RAB subflow#1 any time since number of TrBlks are 1 even if there is no data on RAB subflow#1 (see clause 4.2.1.1 in 3GPP TS 25.222 [29]). | | | | |

6.10.3.4.1.7a.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.10.3.4.1.7a.1.1.3 TFCS

| | |
|---|---|
| TFCS size | 12 |
| TFCS | (RAB subflow#1, RAB subflow#2, DCCH)=<br>(TF0, TF0, TF0), (TF1, TF0, TF0), (TF2, TF1, TF0), (TF3, TF2, TF0), (TF4, TF3, TF0), (TF5, TF4, TF0),<br>(TF0, TF0, TF1), (TF1, TF0, TF1), (TF2, TF1, TF1), (TF3, TF2, TF1), (TF4, TF3, TF1), (TF5, TF4, TF1) |
| NOTE: In case TB size zero is configured for any transport channel the first TFC is required; it is optional otherwise. | |

6.10.3.4.1.7a.1.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Uplink | Midamble | 512 chips |
| | Codes and time slots | SF16 x 1 code x 1 time slot |
| | Max. Number of data bits/radio frame | 226 bits |
| | TFCI code word | 16 bits |
| | TPC | 2 bits |
| | Puncturing Limit | 0.48 |

6.10.3.4.1.7a.2 Downlink

6.10.3.4.1.7a.2.1 Transport channel parameters

6.10.3.4.1.7a.2.1.1 Transport channel parameters for Conversational / speech / DL:(7.4, 6.7, 5.9, 4.75) kbps / CS RAB

| Higher layer | RAB/Signalling RB | | RAB subflow #1 | RAB subflow #2 |
|---|---|---|---|---|
| RLC | Logical channel type | | DTCH | |
| | RLC mode | | TM | TM |
| | Payload sizes, bit | | 39, 42, 55, 58, 61<br>(alt. 0, 39, 42, 55, 58, 61) | 53, 63, 76, 87 |
| | Max data rate, bps | | 7 400 | |
| | TrD PDU header, bit | | 0 | |
| MAC | MAC header, bit | | 0 | |
| | MAC multiplexing | | N/A | |
| Layer 1 | TrCH type | | DCH | DCH |
| | TB sizes, bit | | 39, 42, 55, 58, 61<br>(alt. 0, 39, 42, 55, 58, 61) | 53, 63, 76, 87 |
| | TFS | TF0, bits | 0x61 (alt. 1x0)<br>(note) | 0x87 |
| | | TF1, bits | 1x39 | 1x53 |
| | | TF2, bits | 1x42 | 1x63 |
| | | TF3, bits | 1x55 | 1x76 |
| | | TF4, bits | 1x58 | 1x87 |
| | | TF5, bits | 1x61 | N/A |
| | TTI, ms | | 20 | 20 |
| | Coding type | | CC 1/3 | CC 1/3 |
| | CRC, bit | | 12 | N/A |
| | Max number of bits/TTI after channel coding | | 243 | 285 |
| | Max number of bits/radio frame before rate matching | | 122 | 143 |
| | RM attribute | | 180 to 220 | 170 to 210 |
| NOTE: In case of using this alternative, CRC parity bits are to be attached to RAB subflow#1 any time since number of TrBlks are 1 even if there is no data on RAB subflow#1 (see clause 4.2.1.1 in 3GPP TS 25.222 [29]). | | | | |

6.10.3.4.1.7a.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.10.3.4.1.7a.2.1.3 TFCS

| | |
|---|---|
| TFCS size | 12 |
| TFCS | (RAB subflow#1, RAB subflow#2, DCCH)=<br>(TF0, TF0, TF0), (TF1, TF0, TF0), (TF2, TF1, TF0), (TF3, TF2, TF0), (TF4, TF3, TF0), (TF5, TF4, TF0), |

| | |
|---|---|
| | (TF0, TF0, TF1), (TF1, TF0, TF1), (TF2, TF1, TF1), (TF3, TF2, TF1), (TF4, TF3, TF1), (TF5, TF4, TF1) |
| NOTE: | In case TB size zero is configured for any transport channel the first TFC is required; it is optional otherwise. |

6.10.3.4.1.7a.2.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Downlink | Midamble | 512 chips |
| | Codes and time slots | SF16 x 1 code x 1 time slot |
| | Max. Number of data bits/radio frame | 228 bits |
| | TFCI code word | 16 bits |
| | Puncturing limit | 0.48 |

6.10.3.4.1.8 Conversational / speech / UL:6.7 DL:6.7 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.3.4.1.8.1 Uplink

6.10.3.4.1.8.1.1 Transport channel parameters

6.10.3.4.1.8.1.1.1 Transport channel parameters for Conversational / speech / UL:6.7 kbps / CS RAB

| Higher layer | RAB/Signalling RB | | RAB subflow #1 | RAB subflow #2 |
|---|---|---|---|---|
| RLC | Logical channel type | | DTCH | |
| | RLC mode | | TM | TM |
| | Payload sizes, bit | | 39, 58 (alt. 0, 39, 58) | 76 |
| | Max data rate, bps | | 6 700 | |
| | TrD PDU header, bit | | 0 | |
| MAC | MAC header, bit | | 0 | |
| | MAC multiplexing | | N/A | |
| Layer 1 | TrCH type | | DCH | DCH |
| | TB sizes, bit | | 39, 58 (alt. 0, 39, 58) | 76 |
| | TFS | TF0, bits | 0x58 (alt. 1x0) (note) | 0x76 |
| | | TF1, bits | 1x39 | 1x76 |
| | | TF2, bits | 1x58 | N/A |
| | TTI, ms | | 20 | 20 |
| | Coding type | | CC 1/3 | CC 1/3 |
| | CRC, bit | | 12 | N/A |
| | Max number of bits/TTI after channel coding | | 234 | 252 |
| | Max number of bits/radio frame before rate matching | | 117 | 126 |
| | RM attribute | | 180 to 220 | 170 to 210 |
| NOTE: | In case of using this alternative, CRC parity bits are to be attached to RAB subflow#1 any time since number of TrBlks are 1 even if there is no data on RAB subflow#1 (see clause 4.2.1.1 in 3GPP TS 25.222 [29]). | | | |

6.10.3.4.1.8.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.10.3.4.1.8.1.1.3 TFCS

| | |
|---|---|
| TFCS size | 6 |
| TFCS | (RAB subflow#1, RAB subflow#2, DCCH)=<br>(TF0, TF0, TF0), (TF1, TF0, TF0), (TF2, TF1, TF0),<br>(TF0, TF0, TF1), (TF1, TF0, TF1), (TF2, TF1, TF1) |
| NOTE: | In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise. |

6.10.3.4.1.8.1.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Uplink | Midamble | 512 chips |
| | Codes and time slots | SF16 x 1 code x 1 time slot |
| | Max. Number of data bits/radio frame | 226 bits |
| | TFCI code word | 16 bits |
| | TPC | 2 bits |
| | Puncturing Limit | 0.52 |

6.10.3.4.1.8.2 Downlink

6.10.3.4.1.8.2.1 Transport channel parameters

6.10.3.4.1.8.2.1.1 Transport channel parameters for Conversational / speech / DL:6.7 kbps / CS RAB

| Higher layer | RAB/Signalling RB | | RAB subflow #1 | RAB subflow #2 |
|---|---|---|---|---|
| RLC | Logical channel type | | DTCH | |
| | RLC mode | | TM | TM |
| | Payload sizes, bit | | 39, 58 (alt. 0, 39, 58) | 76 |
| | Max data rate, bps | | 6 700 | |
| | TrD PDU header, bit | | 0 | |
| MAC | MAC header, bit | | 0 | |
| | MAC multiplexing | | N/A | |
| Layer 1 | TrCH type | | DCH | DCH |
| | TB sizes, bit | | 39, 58 (alt. 0,39,58) | 76 |
| | TFS | TF0, bits | 0x58 (alt.1x0) (note) | 0x76 |
| | | TF1, bits | 1x39 | 1x76 |
| | | TF2, bits | 1x58 | N/A |
| | TTI, ms | | 20 | 20 |
| | Coding type | | CC 1/3 | CC 1/3 |
| | CRC, bit | | 12 | N/A |
| | Max number of bits/TTI after channel coding | | 234 | 252 |
| | Max number of bits/radio frame before rate matching | | 117 | 126 |
| | RM attribute | | 180 to 220 | 170 to 210 |
| NOTE : CRC parity bits are to be attached to RAB subflow#1 any time since number of TrBlks are 1 even if there is no data on RAB subflow#1 (see clause 4.2.1.1 in 3GPP TS 25.222 [29]). | | | | |

6.10.3.4.1.8.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.10.3.4.1.8.2.1.3 TFCS

| TFCS size | 6 |
|---|---|
| TFCS | (RAB subflow#1, RAB subflow#2, DCCH)=<br>(TF0, TF0, TF0), (TF1, TF0, TF0), (TF2, TF1, TF0),<br>(TF0, TF0, TF1), (TF1, TF0, TF1), (TF2, TF1, TF1) |
| NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise. | |

6.10.3.4.1.8.2.2 Physical channel parameters

| DPCH Downlink | Midamble | 512 chips |
|---|---|---|
| | Codes and time slots | SF16 x 1 code x 1 time slot |
| | Max. Number of data bits/radio frame | 228 bits |
| | TFCI code word | 16 bits |
| | Puncturing limit | 0.52 |

6.10.3.4.1.9 Conversational / speech / UL:5.9 DL:5.9 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.3.4.1.9.1 Uplink

6.10.3.4.1.9.1.1 Transport channel parameters

6.10.3.4.1.9.1.1.1 Transport channel parameters for Conversational / speech / UL:5.9 kbps / CS RAB

| Higher layer | RAB/Signalling RB | RAB subflow #1 | RAB subflow #2 |
|---|---|---|---|
| RLC | Logical channel type | DTCH | |
| | RLC mode | TM | TM |

| | | | | |
|---|---|---|---|---|
| | Payload sizes, bit | | 39, 55 (alt. 0, 39, 55) | 63 |
| | Max data rate, bps | | 5 900 | |
| | TrD PDU header, bit | | 0 | |
| MAC | MAC header, bit | | 0 | |
| | MAC multiplexing | | N/A | |
| Layer 1 | TrCH type | | DCH | DCH |
| | TB sizes, bit | | 39, 55 (alt. 0, 39, 55) | 63 |
| | TFS | TF0, bits | 0x55 (alt. 1x0) (note) | 0x63 |
| | | TF1, bits | 1x39 | 1x63 |
| | | TF2, bits | 1x55 | N/A |
| | TTI, ms | | 20 | 20 |
| | Coding type | | CC 1/3 | CC 1/3 |
| | CRC, bit | | 12 | N/A |
| | Max number of bits/TTI after channel coding | | 225 | 213 |
| | Max number of bits/radio frame before rate matching | | 113 | 107 |
| | RM attribute | | 180 to 220 | 170 to 210 |
| NOTE: In case of using this alternative, CRC parity bits are to be attached to RAB subflow#1 any time since number of TrBlks are 1 even if there is no data on RAB subflow#1 (see clause 4.2.1.1 in 3GPP TS 25.222 [29]). | | | | |

6.10.3.4.1.9.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.10.3.4.1.9.1.1.3 TFCS

| | |
|---|---|
| TFCS size | 6 |
| TFCS | (RAB subflow#1, RAB subflow#2, DCCH)=<br>(TF0, TF0, TF0), (TF1, TF0, TF0), (TF2, TF1, TF0),<br>(TF0, TF0, TF1), (TF1, TF0, TF1), (TF2, TF1, TF1) |
| NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise. | |

6.10.3.4.1.9.1.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Uplink | Midamble | 512 chips |
| | Codes and time slots | SF16 x 1 code x 1 time slot |
| | Max. Number of data bits/radio frame | 226 bits |
| | TFCI code word | 16 bits |
| | TPC | 2 bits |
| | Puncturing Limit | 0.56 |

6.10.3.4.1.9.2 Downlink

6.10.3.4.1.9.2.1 Transport channel parameters

6.10.3.4.1.9.2.1.1 Transport channel parameters for Conversational / speech / DL:5.9 kbps / CS RAB

| | | | | |
|---|---|---|---|---|
| Higher layer | RAB/Signalling RB | | RAB subflow #1 | RAB subflow #2 |
| RLC | Logical channel type | | DTCH | |
| | RLC mode | | TM | TM |
| | Payload sizes, bit | | 39, 55 (alt. 0, 39, 55) | 63 |
| | Max data rate, bps | | 5 900 | |
| | TrD PDU header, bit | | 0 | |
| MAC | MAC header, bit | | 0 | |
| | MAC multiplexing | | N/A | |
| Layer 1 | TrCH type | | DCH | DCH |
| | TB sizes, bit | | 39, 55 (alt. 0, 39, 55) | 63 |
| | TFS | TF0, bits | 0x55 (alt. 1x0) (note) | 0x63 |
| | | TF1, bits | 1x39 | 1x63 |
| | | TF2, bits | 1x55 | N/A |
| | TTI, ms | | 20 | 20 |
| | Coding type | | CC 1/3 | CC 1/3 |
| | CRC, bit | | 12 | N/A |

| | | | |
|---|---|---|---|
| | Max number of bits/TTI after channel coding | 225 | 213 |
| | Max number of bits/radio frame before rate matching | 113 | 107 |
| | RM attribute | 180 to 220 | 170 to 210 |
| NOTE: CRC parity bits are to be attached to RAB subflow#1 any time since number of TrBlks are 1 even if there is no data on RAB subflow#1 (see clause 4.2.1.1 in 3GPP TS 25.222 [29]). | | | |

6.10.3.4.1.9.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.10.3.4.1.9.2.1.3 TFCS

| | |
|---|---|
| TFCS size | 6 |
| TFCS | (RAB subflow#1, RAB subflow#2, DCCH)=<br>(TF0, TF0, TF0), (TF1, TF0, TF0), (TF2, TF1, TF0),<br>(TF0, TF0, TF1), (TF1, TF0, TF1), (TF2, TF1, TF1) |
| NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise. | |

6.10.3.4.1.9.2.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Downlink | Midamble | 512 chips |
| | Codes and time slots | SF16 x 1 code x 1 time slot |
| | Max. Number of data bits/radio frame | 228 bits |
| | TFCI code word | 16 bits |
| | Puncturing limit | 0.56 |

6.10.3.4.1.10 Conversational / speech / UL:5.15 DL:5.15 kbps / CS RAB + UL:1.7 DL:1.7 kbps SRBs for DCCH

6.10.3.4.1.10.1 Uplink

6.10.3.4.1.10.1.1 Transport channel parameters

6.10.3.4.1.10.1.1.1 Transport channel parameters for Conversational / speech / UL:5.15 kbps / CS RAB

| | | | | |
|---|---|---|---|---|
| Higher layer | RAB/Signalling RB | | RAB subflow #1 | RAB subflow #2 |
| RLC | Logical channel type | | DTCH | |
| | RLC mode | | TM | TM |
| | Payload sizes, bit | | 39, 49 (alt. 0, 39, 49) | 54 |
| | Max data rate, bps | | 5 150 | |
| | TrD PDU header, bit | | 0 | |
| MAC | MAC header, bit | | 0 | |
| | MAC multiplexing | | N/A | |
| Layer 1 | TrCH type | | DCH | DCH |
| | TB sizes, bit | | 39, 49 (alt. 0, 39, 49) | 54 |
| | TFS | TF0, bits | 0x49 (alt. 1x0) (note) | 0x54 |
| | | TF1, bits | 1x39 | 1x54 |
| | | TF2, bits | 1x49 | N/A |
| | TTI, ms | | 20 | 20 |
| | Coding type | | CC 1/3 | CC 1/3 |
| | CRC, bit | | 12 | N/A |
| | Max number of bits/TTI after channel coding | | 207 | 186 |
| | Max number of bits/radio frame before rate matching | | 104 | 93 |
| | RM attribute | | 180 to 220 | 170 to 210 |
| NOTE: In case of using this alternative, CRC parity bits are to be attached to RAB subflow#1 any time since number of TrBlks are 1 even if there is no data on RAB subflow#1 (see clause 4.2.1.1 in 3GPP TS 25.222 [29]). | | | | |

6.10.3.4.1.10.1.1.2 Transport channel parameters for UL:1.7 kbps SRBs for DCCH

See clause 6.10.3.4.1.1.1.1.1.

6.10.3.4.1.10.1.1.3 TFCS

| | |
|---|---|
| TFCS size | 6 |
| TFCS | (RAB subflow#1, RAB subflow#2, DCCH)=<br>(TF0, TF0, TF0), (TF1, TF0, TF0), (TF2, TF1, TF0),<br>(TF0, TF0, TF1), (TF1, TF0, TF1), (TF2, TF1, TF1) |
| NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise. | |

6.10.3.4.1.10.1.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Uplink | Midamble | 512 chips |
| | Codes and time slots | SF16 x 1 code x 1 time slot |
| | Max. Number of data bits/radio frame | 226 bits |
| | TFCI code word | 16 bits |
| | TPC | 2 bits |
| | Puncturing Limit | 0.72 |

6.10.3.4.1.10.2 Downlink

6.10.3.4.1.10.2.1 Transport channel parameters

6.10.3.4.1.10.2.1.1 Transport channel parameters for Conversational / speech / DL:5.15 kbps / CS RAB

| Higher layer | RAB/Signalling RB | | RAB subflow #1 | RAB subflow #2 |
|---|---|---|---|---|
| RLC | Logical channel type | | DTCH | |
| | RLC mode | | TM | TM |
| | Payload sizes, bit | | 39, 49 (alt. 0, 39, 49) | 54 |
| | Max data rate, bps | | 5 150 | |
| | TrD PDU header, bit | | 0 | |
| MAC | MAC header, bit | | 0 | |
| | MAC multiplexing | | N/A | |
| Layer 1 | TrCH type | | DCH | DCH |
| | TB sizes, bit | | 39, 49 (alt. 0, 39, 49) | 54 |
| | TFS | TF0, bits | 0x49 (alt. 1x0) (note) | 0x54 |
| | | TF1, bits | 1x39 | 1x54 |
| | | TF2, bits | 1x49 | N/A |
| | TTI, ms | | 20 | 20 |
| | Coding type | | CC 1/3 | CC 1/3 |
| | CRC, bit | | 12 | N/A |
| | Max number of bits/TTI after channel coding | | 207 | 186 |
| | Max number of bits/radio frame before rate matching | | 104 | 93 |
| | RM attribute | | 180 to 220 | 170 to 210 |
| NOTE: CRC parity bits are to be attached to RAB subflow#1 any time since number of TrBlks are 1 even if there is no data on RAB subflow#1 (see clause 4.2.1.1 in 3GPP TS 25.222 [29]). | | | | |

6.10.3.4.1.10.2.1.2 Transport channel parameters for DL: 1.7 kbps SRBs for DCCH

See clause 6.10.3.4.1.1.2.1.1.

6.10.3.4.1.10.2.1.3 TFCS

| | |
|---|---|
| TFCS size | 6 |
| TFCS | (RAB subflow#1, RAB subflow#2, DCCH)=<br>(TF0, TF0, TF0), (TF1, TF0, TF0), (TF2, TF1, TF0),<br>(TF0, TF0, TF1), (TF1, TF0, TF1), (TF2, TF1, TF1) |
| NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise. | |

6.10.3.4.1.10.2.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Downlink | Midamble | 512 chips |
| | Codes and time slots | SF16 x 1 code x 1 time slot |
| | Max. Number of data bits/radio frame | 228 bits |

| | | |
|---|---|---|
| | TFCI code word | 16 bits |
| | Puncturing limit | 0.72 |

6.10.3.4.1.11 Conversational / speech / UL:4.75 DL:4.75 kbps / CS RAB + UL:1.7 DL:1.7 kbps SRBs for DCCH

6.10.3.4.1.11.1 Uplink

6.10.3.4.1.11.1.1 Transport channel parameters

6.10.3.4.1.11.1.1.1 Transport channel parameters for Conversational / speech / UL:4.75 kbps / CS RAB

| Higher layer | RAB/Signalling RB | | RAB subflow #1 | RAB subflow #2 |
|---|---|---|---|---|
| RLC | Logical channel type | | DTCH | |
| | RLC mode | | TM | TM |
| | Payload sizes, bit | | 39, 42 (alt. 0, 39, 42) | 53 |
| | Max data rate, bps | | 4 750 | |
| | TrD PDU header, bit | | 0 | |
| MAC | MAC header, bit | | 0 | |
| | MAC multiplexing | | N/A | |
| Layer 1 | TrCH type | | DCH | DCH |
| | TB sizes, bit | | 39, 42 (alt. 0, 39, 42) | 53 |
| | TFS | TF0, bits | 0x42 (alt. 1x0) (note) | 0x53 |
| | | TF1, bits | 1x39 | 1x53 |
| | | TF2, bits | 1x42 | N/A |
| | TTI, ms | | 20 | 20 |
| | Coding type | | CC 1/3 | CC 1/3 |
| | CRC, bit | | 12 | N/A |
| | Max number of bits/TTI after channel coding | | 186 | 183 |
| | Max number of bits/radio frame before rate matching | | 93 | 92 |
| | RM attribute | | 180 to 220 | 170 to 210 |
| NOTE: In case of using this alternative, CRC parity bits are to be attached to RAB subflow#1 any time since number of TrBlks are 1 even if there is no data on RAB subflow#1 (see clause 4.2.1.1 in 3GPP TS 25.222 [29]). | | | | |

6.10.3.4.1.11.1.1.2 Transport channel parameters for UL:1.7 kbps SRBs for DCCH

See clause 6.10.3.4.1.1.1.1.

6.10.3.4.1.11.1.1.3 TFCS

| | |
|---|---|
| TFCS size | 6 |
| TFCS | (RAB subflow#1, RAB subflow#2, DCCH)=<br>(TF0, TF0, TF0), (TF1, TF0, TF0), (TF2, TF1, TF0),<br>(TF0, TF0, TF1), (TF1, TF0, TF1), (TF2, TF1, TF1) |
| NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise. | |

6.10.3.4.1.11.1.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Uplink | Midamble | 512 chips |
| | Codes and time slots | SF16 x 1 code x 1 time slot |
| | Max. Number of data bits/radio frame | 226 bits |
| | TFCI code word | 16 bits |
| | TPC | 2 bits |
| | Puncturing Limit | 0.76 |

6.10.3.4.1.11.2 Downlink

6.10.3.4.1.11.2.1 Transport channel parameters

6.10.3.4.1.11.2.1.1 Transport channel parameters for Conversational / speech / DL:4.75 kbps / CS RAB

| Higher layer | RAB/Signalling RB | RAB subflow #1 | RAB subflow #2 |
|---|---|---|---|

| | | | | |
|---|---|---|---|---|
| RLC | Logical channel type | | DTCH | |
| | RLC mode | | TM | TM |
| | Payload sizes, bit | | 39, 42 (alt. 0, 39, 42) | 53 |
| | Max data rate, bps | | 4 750 | |
| | TrD PDU header, bit | | 0 | |
| MAC | MAC header, bit | | 0 | |
| | MAC multiplexing | | N/A | |
| Layer 1 | TrCH type | | DCH | DCH |
| | TB sizes, bit | | 39, 42 (alt. 0, 39, 42) | 53 |
| | TFS | TF0, bits | 0X42 (alt.1x0 )(note) | 0x53 |
| | | TF1, bits | 1x39 | 1x53 |
| | | TF2, bits | 1x42 | N/A |
| | TTI, ms | | 20 | 20 |
| | Coding type | | CC 1/3 | CC 1/3 |
| | CRC, bit | | 12 | N/A |
| | Max number of bits/TTI after channel coding | | 186 | 183 |
| | Max number of bits/radio frame before rate matching | | 93 | 92 |
| | RM attribute | | 180 to 220 | 170 to 210 |
| NOTE: CRC parity bits are to be attached to RAB subflow#1 any time since number of TrBlks are 1 even if there is no data on RAB subflow#1 (see clause 4.2.1.1 in 3GPP TS 25.222 [29]). | | | | |

6.10.3.4.1.11.2.1.2 Transport channel parameters for DL:1.7 kbps SRBs for DCCH

See clause 6.10.3.4.1.1.2.1.1.

6.10.3.4.1.11.2.1.3 TFCS

| | |
|---|---|
| TFCS size | 6 |
| TFCS | (RAB subflow#1, RAB subflow#2, DCCH)=<br>(TF0, TF0, TF0), (TF1, TF0, TF0), (TF2, TF1, TF0),<br>(TF0, TF0, TF1), (TF1, TF0, TF1), (TF2, TF1, TF1) |
| NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise. | |

6.10.3.4.1.11.2.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Downlink | Midamble | 512 chips |
| | Codes and time slots | SF16 x 1 code x 1 time slot |
| | Max. Number of data bits/radio frame | 228 bits |
| | TFCI code word | 16 bits |
| | Puncturing limit | 0.76 |

6.10.3.4.1.12 Conversational / unknown / UL:28.8/DL:28.8 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.3.4.1.12.1 Uplink

6.10.3.4.1.12.1.1 Transport channel parameters

6.10.3.4.1.12.1.1.1 Transport channel parameters for conversational / unknown / UL:28.8 kbps / CS RAB

| | | | |
|---|---|---|---|
| Higher layer | RAB/Signalling RB | | RAB |
| RLC | Logical channel type | | DTCH |
| | RLC mode | | TM |
| | Payload sizes, bit | | 576 |
| | Max data rate, bps | | 28 800 |
| | TrD PDU header, bit | | 0 |
| MAC | MAC header, bit | | 0 |
| | MAC multiplexing | | N/A |
| Layer 1 | TrCH type | | DCH |
| | TB sizes, bit | | 576 |
| | TFS | TF0, bits | 0x576 |
| | | TF1, bits | 1x576 |
| | TTI, ms | | 20 |

| | | |
|---|---|---|
| | Coding type | TC |
| | CRC, bit | 16 |
| | Max number of bits/TTI after channel coding | 3 564 |
| | Max number of bits/radio frame before rate matching | 891 |
| | RM attribute | 160 to 200 |

6.10.3.4.1.12.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.10.3.4.1.12.1.1.3 TFCS

| | |
|---|---|
| TFCS size | 4 |
| TFCS | (28.8 kbps RAB, DCCH)=<br>(TF0, TF0), (TF1, TF0), (TF0, TF1), (TF1, TF1) |
| NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise. | |

6.10.3.4.1.12.1.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Uplink | Midamble | 512 chips |
| | Codes and time slots | SF4 x 1 code x 1 time slot |
| | Max. Number of data bits/radio frame | 904 bits |
| | TFCI code word | 16 bits |
| | TPC | 2 bits |
| | Puncturing Limit | 0.76 |
| NOTE: In case the first TFC in a TFCS is not configured, the TFCI code word will be 8 bits. | | |

6.10.3.4.1.12.2 Downlink

6.10.3.4.1.12.2.1 Transport channel parameters

6.10.3.4.1.12.2.1.1 Transport channel parameters for conversational / unknown / DL:28.8 kbps / CS RAB

| | | | |
|---|---|---|---|
| Higher layer | RAB/Signalling RB | | RAB |
| RLC | Logical channel type | | DTCH |
| | RLC mode | | TM |
| | Payload sizes, bit | | 576 |
| | Max data rate, bps | | 28 800 |
| | TrD PDU header, bit | | 0 |
| MAC | MAC header, bit | | 0 |
| | MAC multiplexing | | N/A |
| Layer 1 | TrCH type | | DCH |
| | TB sizes, bit | | 576 |
| | TFS | TF0, bits | 0x576 |
| | | TF1, bits | 1x576 |
| | TTI, ms | | 20 |
| | Coding type | | TC |
| | CRC, bit | | 16 |
| | Max number of bits/TTI after channel coding | | 3 564 |
| | Max number of bits/radio frame before rate matching | | 891 |
| | RM attribute | | 160 to 200 |

6.10.3.4.1.12.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.10.3.4.1.12.2.1.3 TFCS

| | |
|---|---|
| TFCS size | 4 |
| TFCS | (28.8 kbps RAB, DCCH)=<br>(TF0, TF0), (TF1, TF0), (TF0, TF1), (TF1, TF1) |
| NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise. | |

6.10.3.4.1.12.2.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Downlink | Midamble | 512 chips |
| | Codes and time slots | SF16 x 2 codes x 1 time slot |
| | Max. Number of data bits/radio frame | 472 bits |
| | TFCI code word | 16 bits |
| | Puncturing limit | 0.40 |
| NOTE: In case the first TFC in the TFCS is not configured, the TFCI code word will be 8 bits. | | |

6.10.3.4.1.13 Conversational / unknown / UL:64 DL:64 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.3.4.1.13.1 Uplink

6.10.3.4.1.13.1.1 Transport channel parameters

6.10.3.4.1.13.1.1.1 Transport channel parameters for Conversational / unknown / UL:64 kbps / CS RAB

| | | | |
|---|---|---|---|
| Higher layer | RAB/Signalling RB | | RAB |
| RLC | Logical channel type | | DTCH |
| | RLC mode | | TM |
| | Payload sizes, bit | | 640 |
| | Max data rate, bps | | 64 000 |
| | TrD PDU header, bit | | 0 |
| MAC | MAC header, bit | | 0 |
| | MAC multiplexing | | N/A |
| Layer 1 | TrCH type | | DCH |
| | TB sizes, bit | | 640 |
| | TFS | TF0, bits | 0x640 |
| | | TF1, bits | 2x640 |
| | TTI, ms | | 20 |
| | Coding type | | TC |
| | CRC, bit | | 16 |
| | Max number of bits/TTI after channel coding | | 3 948 |
| | Max number of bits/radio frame before rate matching | | 1 974 |
| | RM attribute | | 150 to 195 |

6.10.3.4.1.13.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.10.3.4.1.13.1.1.3 TFCS

| | |
|---|---|
| TFCS size | 4 |
| TFCS | (64 kbps RAB, DCCH)=(TF0, TF0), (TF1, TF0), (TF0, TF1), (TF1, TF1) |
| NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise. | |

6.10.3.4.1.13.1.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Uplink | Midamble | 512 chips |
| | Codes and time slots | SF16 x 1 code x 1 time slot + SF4 x 1 code x 1 time slot |
| | Max. Number of data bits/radio frame | 1148 bits |
| | TFCI code word | 16 bits |
| | TPC | 2 bits |
| | Puncturing Limit | 0.48 |
| NOTE: In case the first TFC in the TFCS is not configured, the TFCI code word will be 8 bits. | | |

6.10.3.4.1.13.2 Downlink

6.10.3.4.1.13.2.1 Transport channel parameters

6.10.3.4.1.13.2.1.1 Transport channel parameters for Conversational / unknown / DL:64 kbps / CS RAB

| | | | |
|---|---|---|---|
| Higher layer | RAB/Signalling RB | | RAB |
| RLC | Logical channel type | | DTCH |
| | RLC mode | | TM |
| | Payload sizes, bit | | 640 |
| | Max data rate, bps | | 64 000 |
| | TrD PDU header, bit | | 0 |
| MAC | MAC header, bit | | 0 |
| | MAC multiplexing | | N/A |
| Layer 1 | TrCH type | | DCH |
| | TB sizes, bit | | 640 |
| | TFS | TF0, bits | 0x640 |
| | | TF1, bits | 2x640 |
| | TTI, ms | | 20 |
| | Coding type | | TC |
| | CRC, bit | | 16 |
| | Max number of bits/TTI after channel coding | | 3 948 |
| | Max number of bits/radio frame before rate matching | | 1 974 |
| | RM attribute | | 150 to 195 |

6.10.3.4.1.13.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.10.3.4.1.13.2.1.3 TFCS

| | |
|---|---|
| TFCS size | 4 |
| TFCS | (64 kbps RAB, DCCH)=(TF0, TF0), (TF1, TF0), (TF0, TF1), (TF1, TF1) |
| NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise. | |

6.10.3.4.1.13.2.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Downlink | Midamble | 512 chips |
| | Codes and time slots | SF16 x 5 codes x 1 time slot |
| | Max. Number of data bits/radio frame | 1 204 bits |
| | TFCI code word | 16 bits |
| | Puncturing limit | 0.52 |
| NOTE: In case the first TFC in the TFCS is not configured, the TFCI code word will be 8 bits. | | |

6.10.3.4.1.14 Conversational / unknown / UL:32 DL:32 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.3.4.1.14.1 Uplink

6.10.3.4.1.14.1.1 Transport channel parameters

6.10.3.4.1.14.1.1.1 Transport channel parameters for Conversational / unknown / UL:32 kbps / CS RAB

| | | | |
|---|---|---|---|
| Higher layer | RAB/Signalling RB | | RAB |
| RLC | Logical channel type | | DTCH |
| | RLC mode | | TM |
| | Payload sizes, bit | | 640 |
| | Max data rate, bps | | 32 000 |
| | TrD PDU header, bit | | 0 |
| MAC | MAC header, bit | | 0 |
| | MAC multiplexing | | N/A |
| Layer 1 | TrCH type | | DCH |
| | TB sizes, bit | | 640 |
| | TFS | TF0, bits | 0x640 |

| | | | |
|---|---|---|---|
| | | TF1, bits | 1x640 |
| | TTI, ms | | 20 |
| | Coding type | | TC |
| | CRC, bit | | 16 |
| | Max number of bits/TTI after channel coding | | 1 980 |
| | Max number of bits/radio frame before rate matching | | 990 |
| | RM attribute | | 165 to 210 |

6.10.3.4.1.14.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.10.3.4.1.14.1.1.3 TFCS

| | |
|---|---|
| TFCS size | 4 |
| TFCS | (32 kbps RAB, DCCH)=(TF0, TF0), (TF1, TF0), (TF0, TF1), (TF1, TF1) |
| NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise. | |

6.10.3.4.1.14.1.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Uplink | Midamble | 512 chips |
| | Codes and time slots | SF4 x 1 code x 1 time slot |
| | Max. Number of data bits/radio frame | 904 bits |
| | TFCI code word | 16 bits |
| | TPC | 2 bits |
| | Puncturing Limit | 0.68 |
| NOTE: In case the first TFC in the TFCS is not configured, the TFCI code word will be 8 bits. | | |

6.10.3.4.1.14.2 Downlink

6.10.3.4.1.14.2.1 Transport channel parameters

6.10.3.4.1.14.2.1.1 Transport channel parameters for Conversational / unknown / DL:32 kbps / CS RAB

| | | | |
|---|---|---|---|
| Higher layer | RAB/Signalling RB | | RAB |
| RLC | Logical channel type | | DTCH |
| | RLC mode | | TM |
| | Payload sizes, bit | | 640 |
| | Max data rate, bps | | 32 000 |
| | TrD PDU header, bit | | 0 |
| MAC | MAC header, bit | | 0 |
| | MAC multiplexing | | N/A |
| Layer 1 | TrCH type | | DCH |
| | TB sizes, bit | | 640 |
| | TFS | TF0, bits | 0x640 |
| | | TF1, bits | 1x640 |
| | TTI, ms | | 20 |
| | Coding type | | TC |
| | CRC, bit | | 16 |
| | Max number of bits/TTI after channel coding | | 1 980 |
| | Max number of bits/radio frame before rate matching | | 990 |
| | RM attribute | | 165 to 210 |

6.10.3.4.1.14.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.10.3.4.1.14.2.1.3 TFCS

| | |
|---|---|
| TFCS size | 4 |
| TFCS | (32 kbps RAB, DCCH)=(TF0, TF0), (TF1, TF0), (TF0, TF1), (TF1, TF1) |
| NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise. | |

6.10.3.4.1.14.2.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Downlink | Midamble | 512 chips |
| | Codes and time slots | SF16 x 3 codes x 1 time slot |
| | Max. Number of data bits/radio frame | 716 bits |
| | TFCI code word | 16 bits |
| | Puncturing limit | 0.52 |
| NOTE: In case the first TFC in the TFCS is not configured, the TFCI code word will be 8 bits. | | |

6.10.3.4.1.15 Streaming / unknown / UL:14.4/DL:14.4 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.3.4.1.15.1 Uplink

6.10.3.4.1.15.1.1 Transport channel parameters

6.10.3.4.1.15.1.1.1 Transport channel parameters for Streaming / unknown / UL: 14.4 kbps / CS RAB

| | | | |
|---|---|---|---|
| Higher layer | RAB/Signalling RB | | RAB |
| RLC | Logical channel type | | DTCH |
| | RLC mode | | TM |
| | Payload sizes, bit | | 576 |
| | Max data rate, bps | | 14 400 |
| | TrD PDU header, bit | | 0 |
| MAC | MAC header, bit | | 0 |
| | MAC multiplexing | | N/A |
| Layer 1 | TrCH type | | DCH |
| | TB sizes, bit | | 576 |
| | TFS | TF0, bits | 0x576 |
| | | TF1, bits | 1x576 |
| | TTI, ms | | 40 |
| | Coding type | | TC |
| | CRC, bit | | 16 |
| | Max number of bits/TTI after channel coding | | 1 788 |
| | Max number of bits/radio frame before rate matching | | 447 |
| | RM attribute | | 145 to 185 |

6.10.3.4.1.15.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.10.3.4.1.15.1.1.3 TFCS

| | |
|---|---|
| TFCS size | 4 |
| TFCS | (14.4 kbps RAB, DCCH)=(TF0, TF0), (TF1, TF0), (TF0, TF1), (TF1, TF1) |
| NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise. | |

6.10.3.4.1.15.1.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Uplink | Midamble | 512 chips |
| | Codes and time slots | SF8 x 1 code x 1 time slot |
| | Max. Number of data bits/radio frame | 452bits |
| | TFCI code word | 16 bits |
| | TPC | 2 bits |
| | Puncturing Limit | 0.76 |
| NOTE: In case the first TFC in the TFCS is not configured, the TFCI code word will be 8 bits. | | |

6.10.3.4.1.15.2 Downlink

6.10.3.4.1.15.2.1 Transport channel parameters

6.10.3.4.1.15.2.1.1 Transport channel parameters for Streaming / unknown / DL:14.4 kbps / CS RAB

| | | | |
|---|---|---|---|
| Higher layer | RAB/Signalling RB | | RAB |
| RLC | Logical channel type | | DTCH |
| | RLC mode | | TM |
| | Payload sizes, bit | | 576 |
| | Max data rate, bps | | 14 400 |
| | TrD PDU header, bit | | 0 |
| MAC | MAC header, bit | | 0 |
| | MAC multiplexing | | N/A |
| Layer 1 | TrCH type | | DCH |
| | TB sizes, bit | | 576 |
| | TFS | TF0, bits | 0x576 |
| | | TF1, bits | 1x576 |
| | TTI, ms | | 40 |
| | Coding type | | TC |
| | CRC, bit | | 16 |
| | Max number of bits/TTI after channel coding | | 1 788 |
| | Max number of bits/radio frame before rate matching | | 447 |
| | RM attribute | | 145 to 185 |

6.10.3.4.1.15.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.10.3.4.1.15.2.1.3 TFCS

| | |
|---|---|
| TFCS size | 4 |
| TFCS | (14.4 kbps RAB, DCCH)=(TF0, TF0), (TF1, TF0), (TF0, TF1), (TF1, TF1) |
| NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise. | |

6.10.3.4.1.15.2.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Downlink | Midamble | 512 chips |
| | Codes and time slots | SF16 x 2 codes x 1 time slot |
| | Max. Number of data bits/radio frame | 472 bits |
| | TFCI code word | 16 bits |
| | Puncturing limit | 0.80 |
| NOTE: In case the first TFC in the TFCS is not configured, the TFCI code word will be 8 bits. | | |

6.10.3.4.1.16 Streaming / unknown / UL:28.8/DL:28.8 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.3.4.1.16.1 Uplink

6.10.3.4.1.16.1.1 Transport channel parameters

6.10.3.4.1.16.1.1.1 Transport channel parameters for Streaming / unknown / UL:28.8 kbps / CS RAB

| | | | |
|---|---|---|---|
| Higher layer | RAB/Signalling RB | | RAB |
| RLC | Logical channel type | | DTCH |
| | RLC mode | | TM |
| | Payload sizes, bit | | 576 |
| | Max data rate, bps | | 28 800 |
| | TrD PDU header, bit | | 0 |
| MAC | MAC header, bit | | 0 |
| | MAC multiplexing | | N/A |
| Layer 1 | TrCH type | | DCH |
| | TB sizes, bit | | 576 |
| | TFS | TF0, bits | 0x576 |

| | | | |
|---|---|---|---|
| | | TF1, bits | 1x576 |
| | TTI, ms | | 20 |
| | Coding type | | TC |
| | CRC, bit | | 16 |
| | Max number of bits/TTI after channel coding | | 3 564 |
| | Max number of bits/radio frame before rate matching | | 891 |
| | RM attribute | | 135 to 175 |

6.10.3.4.1.16.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.10.3.4.1.16.1.1.3 TFCS

| | |
|---|---|
| TFCS size | 4 |
| TFCS | (28.8 kbps RAB, DCCH)=<br>(TF0, TF0), (TF1, TF0), (TF0, TF1), (TF1, TF1) |
| NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise. | |

6.10.3.4.1.16.1.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Uplink | Midamble | 512 chips |
| | Codes and time slots | SF8 x 1 code x 1 time slot |
| | Max. Number of data bits/radio frame | 452 bits |
| | TFCI code word | 16 bits |
| | TPC | 2 bits |
| | Puncturing Limit | 0.44 |
| NOTE: In case the first TFC in the TFCS is not configured, the TFCI code word will be 8 bits. | | |

6.10.3.4.1.16.2 Downlink

6.10.3.4.1.16.2.1 Transport channel parameters

6.10.3.4.1.16.2.1.1 Transport channel parameters for Streaming / unknown / DL:28.8 kbps / CS RAB

| | | | |
|---|---|---|---|
| Higher layer | RAB/Signalling RB | | RAB |
| RLC | Logical channel type | | DTCH |
| | RLC mode | | TM |
| | Payload sizes, bit | | 576 |
| | Max data rate, bps | | 28 800 |
| | TrD PDU header, bit | | 0 |
| MAC | MAC header, bit | | 0 |
| | MAC multiplexing | | N/A |
| Layer 1 | TrCH type | | DCH |
| | TB sizes, bit | | 576 |
| | TFS | TF0, bits | 0x576 |
| | | TF1, bits | 1x576 |
| | TTI, ms | | 20 |
| | Coding type | | TC |
| | CRC, bit | | 16 |
| | Max number of bits/TTI after channel coding | | 3 564 |
| | Max number of bits/radio frame before rate matching | | 891 |
| | RM attribute | | 135 to 175 |

6.10.3.4.1.16.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.10.3.4.1.16.2.1.3 TFCS

| | |
|---|---|
| TFCS size | 4 |
| TFCS | (28.8 kbps RAB, DCCH)=<br>(TF0, TF0), (TF1, TF0), (TF0, TF1), (TF1, TF1) |

| NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise. |
|---|

6.10.3.4.1.16.2.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Downlink | Midamble | 512 chips |
| | Codes and time slots | SF16 x 2 codes x 1 time slot |
| | Max. Number of data bits/radio frame | 472 bits |
| | TFCI code word | 16 bits |
| | Puncturing limit | 0.44 |
| NOTE: In case the first TFC in the TFCS is not configured, the TFCI code word will be 8 bits. | | |

6.10.3.4.1.17 Streaming / unknown / UL:57.6/DL:57.6 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.3.4.1.17.1 Uplink

6.10.3.4.1.17.1.1 Transport channel parameters

6.10.3.4.1.17.1.1.1 Transport channel parameters for Streaming / unknown / UL:57.6 kbps / CS RAB

| | | | |
|---|---|---|---|
| Higher layer | RAB/Signalling RB | | RAB |
| RLC | Logical channel type | | DTCH |
| | RLC mode | | TM |
| | Payload sizes, bit | | 576 |
| | Max data rate, bps | | 57 600 |
| | TrD PDU header, bit | | 0 |
| MAC | MAC header, bit | | 0 |
| | MAC multiplexing | | N/A |
| Layer 1 | TrCH type | | DCH |
| | TB sizes, bit | | 576 |
| | TFS | TF0, bits | 0x576 |
| | | TF1, bits | 1x576 |
| | | TF2, bits | 2x576 |
| | | TF3, bits | 3x576 |
| | | TF4, bits | 4x576 |
| | TTI, ms | | 40 |
| | Coding type | | TC |
| | CRC, bit | | 16 |
| | Max number of bits/TTI after channel coding | | 7 116 |
| | Max number of bits/radio frame before rate matching | | 1 779 |
| | RM attribute | | 125 to 165 |

6.10.3.4.1.17.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.10.3.4.1.17.1.1.3 TFCS

| | |
|---|---|
| TFCS size | 10 |
| TFCS | (57.6 kbps RAB, DCCH)=<br>(TF0, TF0), (TF1, TF0), (TF2, TF0), (TF3, TF0), (TF4, TF0),<br>(TF0, TF1), (TF1, TF1), (TF2, TF1), (TF3, TF1), (TF4, TF1) |
| NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise. | |

6.10.3.4.1.17.1.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Uplink | Midamble | 512 chips |
| | Codes and time slots | SF4 x 1 code x 1 time slot |
| | Max. Number of data bits/radio frame | 904 bits |
| | TFCI code word | 16 bits |
| | TPC | 2 bits |
| | Puncturing Limit | 0.44 |

6.10.3.4.1.17.2 Downlink

6.10.3.4.1.17.2.1 Transport channel parameters

6.10.3.4.1.17.2.1.1 Transport channel parameters for Streaming / unknown / DL:57.6 kbps / CS RAB

| | | | |
|---|---|---|---|
| Higher layer | RAB/Signalling RB | | RAB |
| RLC | Logical channel type | | DTCH |
| | RLC mode | | TM |
| | Payload sizes, bit | | 576 |
| | Max data rate, bps | | 57 600 |
| | TrD PDU header, bit | | 0 |
| MAC | MAC header, bit | | 0 |
| | MAC multiplexing | | N/A |
| Layer 1 | TrCH type | | DCH |
| | TB sizes, bit | | 576 |
| | TFS | TF0, bits | 0x576 |
| | | TF1, bits | 1x576 |
| | | TF2, bits | 2x576 |
| | | TF3, bits | 3x576 |
| | | TF4, bits | 4x576 |
| | TTI, ms | | 40 |
| | Coding type | | TC |
| | CRC, bit | | 16 |
| | Max number of bits/TTI after channel coding | | 7 116 |
| | Max number of bits/radio frame before rate matching | | 1 779 |
| | RM attribute | | 125 to 165 |

6.10.3.4.1.17.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.10.3.4.1.17.2.1.3 TFCS

| | |
|---|---|
| TFCS size | 10 |
| TFCS | (57.6 kbps RAB, DCCH)=<br>(TF0, TF0), (TF1, TF0), (TF2, TF0), (TF3, TF0), (TF4, TF0),<br>(TF0, TF1), (TF1, TF1), (TF2, TF1), (TF3, TF1), (TF4, TF1) |
| NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise. | |

6.10.3.4.1.17.2.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Downlink | Midamble | 512 chips |
| | Codes and time slots | SF16 x 4 codes x 1 time slot |
| | Max. Number of data bits/radio frame | 960 bits |
| | TFCI code word | 16 bits |
| | Puncturing limit | 0.48 |

6.10.3.4.1.18 Void

6.10.3.4.1.19 Void

6.10.3.4.1.20 Void

6.10.3.4.1.21 Void

6.10.3.4.1.22 Void

6.10.3.4.1.23 Interactive or background / UL:32 DL:8 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.3.4.1.23.1 Uplink

6.10.3.4.1.23.1.1 Transport channel parameters

6.10.3.4.1.23.1.1.1 Transport channel parameters for Interactive or background / UL:32 kbps / PS RAB

| | | | |
|---|---|---|---|
| Higher layer | RAB/Signalling RB | | RAB |
| RLC | Logical channel type | | DTCH |
| | RLC mode | | AM |
| | Payload sizes, bit | | 320 (alt. 128) |
| | Max data rate, bps | | 32 000 |
| | AMD PDU header, bit | | 16 |
| MAC | MAC header, bit | | 0 |
| | MAC multiplexing | | N/A |
| Layer 1 | TrCH type | | DCH |
| | TB sizes, bit | | 336 (alt.144) |
| | TFS | TF0, bits | 0x336 (alt. 0x144) |
| | | TF1, bits | 1x336 (alt. 1x144) |
| | | TF2, bits | 2x336 (alt. 5x144) |
| | TTI, ms | | 20) |
| | Coding type | | TC |
| | CRC, bit | | 16 |
| | Max number of bits/TTI after channel coding | | 2 124 (alt. 2 412) |
| | Max number of bits/radio frame before rate matching | | 1 062 (alt. 1 206) |
| | RM attribute | | 135 to 175 |

6.10.3.4.1.23.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.10.3.4.1.23.1.1.3 TFCS

| | |
|---|---|
| TFCS size | 6 |
| TFCS | (32 kbps RAB, DCCH)=<br>(TF0, TF0), (TF1, TF0), (TF2, TF0), (TF0, TF1), (TF1, TF1), (TF2, TF1) |
| NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise. | |

6.10.3.4.1.23.1.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Uplink | Midamble | 512 chips |
| | Codes and time slots | SF4 x 1 code x 1 time slot |
| | Max. Number of data bits/radio frame | 904 bits |
| | TFCI code word | 16 bits |
| | TPC | 2 bits |
| | Puncturing Limit | 0.72 (alt. 0.64) |

6.10.3.4.1.23.2 Downlink

6.10.3.4.1.23.2.1 Transport channel parameters

6.10.3.4.1.23.2.1.1 Transport channel parameters for Interactive or background / DL:8 kbps / PS RAB

| | | | |
|---|---|---|---|
| Higher layer | RAB/Signalling RB | | RAB |
| RLC | Logical channel type | | DTCH |
| | RLC mode | | AM |
| | Payload sizes, bit | | 320 |
| | Max data rate, bps | | 8 000 |
| | AMD PDU header, bit | | 16 |
| MAC | MAC header, bit | | 0 |
| | MAC multiplexing | | N/A |
| Layer 1 | TrCH type | | DCH |
| | TB sizes, bit | | 336 |
| | TFS | TF0, bits | 0x336 |
| | | TF1, bits | 1x336 |
| | TTI, ms | | 40 |
| | Coding type | | TC |
| | CRC, bit | | 16 |

| | | |
|---|---|---|
| | Max number of bits/TTI after channel coding | 1 068 |
| | Max number of bits/radio frame before rate matching | 267 |
| | RM attribute | 135 to 175 |

6.10.3.4.1.23.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.10.3.4.1.23.2.1.3 TFCS

| | |
|---|---|
| TFCS size | 4 |
| TFCS | (8 kbps RAB, DCCH)=(TF0, TF0), (TF1, TF0), (TF0, TF1), (TF1, TF1) |
| NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise. | |

6.10.3.4.1.23.2.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Downlink | Midamble | 512 chips |
| | Codes and time slots | SF16 x 1 code x 1 time slot |
| | Max. Number of data bits/radio frame | 228 bits |
| | TFCI code word | 16 bits |
| | Puncturing limit | 0.56 |
| NOTE: In case the first TFC in the TFCS is not configured, the TFCI code word will be 8 bits. | | |

6.10.3.4.1.23a Interactive or background / UL:8 DL:8 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.3.4.1.23a.1 Uplink

6.10.3.4.1.23a.1.1 Transport channel parameters

6.10.3.4.1.23a.1.1.1 Transport channel parameters for Interactive or background / UL:8 kbps / PS RAB

| | | | |
|---|---|---|---|
| Higher layer | RAB/Signalling RB | | RAB |
| RLC | Logical channel type | | DTCH |
| | RLC mode | | AM |
| | Payload sizes, bit | | 320 (alt. 128) |
| | Max data rate, bps | | 8 000 |
| | AMD PDU header, bit | | 16 |
| MAC | MAC header, bit | | 0 |
| | MAC multiplexing | | N/A |
| Layer 1 | TrCH type | | DCH |
| | TB sizes, bit | | 336 (alt. 144) |
| | TFS | TF0, bits | 0x336 (alt. 0x144) |
| | | TF1, bits | 1x336 (alt. 1x144) |
| | | TF2, bits | N/A (alt. 5x144) |
| | TTI, ms | | 40 (alt. 80) |
| | Coding type | | TC |
| | CRC, bit | | 16 |
| | Max number of bits/TTI after channel coding | | 1 068 (alt. 2 412) |
| | Max number of bits/radio frame before rate matching | | 267 (alt. 302) |
| | RM attribute | | 135 to 175 |

6.10.3.4.1.23a.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.10.3.4.1.23a.1.1.3 TFCS

| | |
|---|---|
| TFCS size | 4 (alt. 6) |
| TFCS | (8 kbps RAB, DCCH)=<br>(TF0, TF0), (TF1, TF0), (TF0, TF1), (TF1, TF1)<br>(alt. (TF0, TF0), (TF1, TF0), (TF2, TF0), (TF0, TF1), (TF1, TF1), (TF2, TF1)) |
| NOTE: In case TB size zero is configured for any transport channel the first TFC is required; it is optional otherwise. | |

6.10.3.4.1.23a.1.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Uplink | Midamble | 512 chips |
| | Codes and time slots | SF16 x 1 code x 1 time slot |
| | Max. Number of data bits/radio frame | 226 bits |
| | TFCI code word | 16 bits |
| | TPC | 2 bit |
| | Puncturing Limit | 0.56 (alt. 0.48) |

6.10.3.4.1.23a.2 Downlink

See clause 6.10.3.4.1.23.2.

6.10.3.4.1.23b Interactive or background / UL:16 DL:16 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.3.4.1.23b.1 Uplink

6.10.3.4.1.23b.1.1 Transport channel parameters

6.10.3.4.1.23b.1.1.1 Transport channel parameters for Interactive or background / UL:16 kbps / PS RAB

| | | | |
|---|---|---|---|
| Higher layer | RAB/Signalling RB | | RAB |
| RLC | Logical channel type | | DTCH |
| | RLC mode | | AM |
| | Payload sizes, bit | | 320 (alt. 128) |
| | Max data rate, bps | | 16 000 |
| | AMD PDU header, bit | | 16 |
| MAC | MAC header, bit | | 0 |
| | MAC multiplexing | | N/A |
| Layer 1 | TrCH type | | DCH |
| | TB sizes, bit | | 336 (alt. 144) |
| | TFS | TF0, bits | 0x336 (alt. 0x144) |
| | | TF1, bits | 1x336 (alt. 1x144) |
| | | TF2, bits | 2x336 (alt. 5x144) |
| | TTI, ms | | 40 |
| | Coding type | | TC |
| | CRC, bit | | 16 |
| | Max number of bits/TTI after channel coding | | 2 124 (alt. 2 412) |
| | Max number of bits/radio frame before rate matching | | 531 (alt. 603) |
| | RM attribute | | 135 to 175 |

6.10.3.4.1.23b.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.10.3.4.1.23b.1.1.3 TFCS

| | |
|---|---|
| TFCS size | 6 |
| TFCS | (16 kbps RAB, DCCH)=<br>(TF0, TF0), (TF1, TF0), (TF2, TF0), (TF0, TF1), (TF1, TF1), (TF2, TF1) |
| NOTE: In case TB size zero is configured for any transport channel the first TFC is required; it is optional otherwise. | |

6.10.3.4.1.23b.1.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Uplink | Midamble | 512 chips |
| | Codes and time slots | SF8 x 1 code x 1 time slot |
| | Max. Number of data bits/radio frame | 452 bits |
| | TFCI code word | 16 bits |
| | TPC | 2 bit |
| | Puncturing Limit | 0.68 (alt. 0.60) |

6.10.3.4.1.23b.2 Downlink

6.10.3.4.1.23b.2.1 Transport channel parameters

6.10.3.4.1.23b.2.1.1 Transport channel parameters for Interactive or background / DL:16 kbps / PS RAB

| | | | |
|---|---|---|---|
| Higher layer | RAB/Signalling RB | | RAB |
| RLC | Logical channel type | | DTCH |
| | RLC mode | | AM |
| | Payload sizes, bit | | 320 |
| | Max data rate, bps | | 16 000 |
| | AMD PDU header, bit | | 16 |
| MAC | MAC header, bit | | 0 |
| | MAC multiplexing | | N/A |
| Layer 1 | TrCH type | | DCH |
| | TB sizes, bit | | 336 |
| | TFS | TF0, bits | 0x336 |
| | | TF1, bits | 1x336 |
| | | TF2, bits | 2x336 |
| | TTI, ms | | 40 |
| | Coding type | | TC |
| | CRC, bit | | 16 |
| | Max number of bits/TTI after channel coding | | 2 124 |
| | Max number of bits/radio frame before rate matching | | 531 |
| | RM attribute | | 135 to 175 |

6.10.3.4.1.23b.2.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.10.3.4.1.23b.2.1.3 TFCS

| | |
|---|---|
| TFCS size | 6 |
| TFCS | (16 kbps RAB, DCCH)=<br>(TF0, TF0), (TF1, TF0), (TF2, TF0), (TF0, TF1), (TF1, TF1), (TF2, TF1) |
| NOTE: In case TB size zero is configured for any transport channel the first TFC is required; it is optional otherwise. | |

6.10.3.4.1.23b.2.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Downlink | Midamble | 512 chips |
| | Codes and time slots | SF16 x 2 codes x 1 time slot |
| | Max. Number of data bits/radio frame | 472 bits |
| | TFCI code word | 16 bits |
| | Puncturing limit | 0.68 |

6.10.3.4.1.23c Interactive or background / UL:32 DL:32 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.3.4.1.23c.1 Uplink

6.10.3.4.1.23c.1.1 Transport channel parameters

6.10.3.4.1.23c.1.1.1 Transport channel parameters for Interactive or background / UL:32 kbps / PS RAB

| | | |
|---|---|---|
| Higher layer | RAB/Signalling RB | RAB |
| RLC | Logical channel type | DTCH |
| | RLC mode | AM |
| | Payload sizes, bit | 320 (alt. 128) |
| | Max data rate, bps | 32 000 |
| | AMD PDU header, bit | 16 |
| MAC | MAC header, bit | 0 |
| | MAC multiplexing | N/A |
| Layer 1 | TrCH type | DCH |
| | TB sizes, bit | 336 (alt. 144) |

| | | | |
|---|---|---|---|
| | TFS | TF0, bits | 0x336 (alt. 0x144) |
| | | TF1, bits | 1x336 (alt. 1x144) |
| | | TF2, bits | 2x336 (alt. 5x144) |
| | | TF3, bits | 3x336 (alt. 7x144) |
| | | TF4, bits | 4x336 (alt. 10x144) |
| | TTI, ms | | 40 |
| | Coding type | | TC |
| | CRC, bit | | 16 |
| | Max number of bits/TTI after channel coding | | 4 236 (alt. 4 812) |
| | Max number of bits/radio frame before rate matching | | 1 059 (alt. 1 203) |
| | RM attribute | | 135 to 175 |

6.10.3.4.1.23c.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.10.3.4.1.23c.1.1.3 TFCS

| | |
|---|---|
| TFCS size | 10 |
| TFCS | (32 kbps RAB, DCCH)=<br>(TF0,TF0), (TF1,TF0), (TF2,TF0), (TF3,TF0), (TF4,TF0), (TF0,TF1), (TF1,TF1), (TF2,TF1), (TF3,TF1), (TF4,TF1) |
| NOTE: In case TB size zero is configured for any transport channel the first TFC is required; it is optional otherwise. | |

6.10.3.4.1.23c.1.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Uplink | Midamble | 512 chips |
| | Codes and time slots | SF4 x 1 code x 1 time slot |
| | Max. Number of data bits/radio frame | 904 bits |
| | TFCI code word | 16 bits |
| | TPC | 2 bits |
| | Puncturing Limit | 0.72 (alt. 0.64) |

6.10.3.4.1.23c.2 Downlink

6.10.3.4.1.23c.2.1 Transport channel parameters

6.10.3.4.1.23c.2.1.1 Transport channel parameters for Interactive or background / DL:32 kbps / PS RAB

| | | | |
|---|---|---|---|
| Higher layer | RAB/Signalling RB | | RAB |
| RLC | Logical channel type | | DTCH |
| | RLC mode | | AM |
| | Payload sizes, bit | | 320 |
| | Max data rate, bps | | 32 000 |
| | AMD PDU header, bit | | 16 |
| MAC | MAC header, bit | | 0 |
| | MAC multiplexing | | N/A |
| Layer 1 | TrCH type | | DCH |
| | TB sizes, bit | | 336 |
| | TFS | TF0, bits | 0x336 |
| | | TF1, bits | 1x336 |
| | | TF2, bits | 2x336 |
| | | TF3, bits | 3x336 |
| | | TF4, bits | 4x336 |
| | TTI, ms | | 40 |
| | Coding type | | TC |
| | CRC, bit | | 16 |
| | Max number of bits/TTI after channel coding | | 4 236 |
| | Max number of bits/radio frame before rate matching | | 1 059 |
| | RM attribute | | 135 to 175 |

6.10.3.4.1.23c.2.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.10.3.4.1.23c.2.1.3 TFCS

| | |
|---|---|
| TFCS size | 10 |
| TFCS | (32 kbps RAB, DCCH)=<br>(TF0,TF0), (TF1,TF0), (TF2,TF0), (TF3,TF0), (TF4,TF0), (TF0,TF1), (TF1,TF1), (TF2,TF1), (TF3,TF1), (TF4,TF1) |
| NOTE: In case TB size zero is configured for any transport channel the first TFC is required; it is optional otherwise. | |

6.10.3.4.1.23c.2.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Downlink | Midamble | 512 chips |
| | Codes and time slots | SF16 x 3 codes x 1 time slot |
| | Max. Number of data bits/radio frame | 716 |
| | TFCI code word | 16 bits |
| | Puncturing limit | 0.60 |

6.10.3.4.1.23d Interactive or background / UL:32 DL:32 kbps / PS RAB (20 ms TTI)+ UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.3.4.1.23d.1 Uplink

6.10.3.4.1.23d.1.1 Transport channel parameters

6.10.3.4.1.23d.1.1.1 Transport channel parameters for Interactive or background / UL:32 kbps / PS RAB

| | | | |
|---|---|---|---|
| Higher layer | RAB/Signalling RB | | RAB |
| RLC | Logical channel type | | DTCH |
| | RLC mode | | AM |
| | Payload sizes, bit | | 320 (alt. 128) |
| | Max data rate, bps | | 32 000 |
| | AMD PDU header, bit | | 16 |
| MAC | MAC header, bit | | 0 |
| | MAC multiplexing | | N/A |
| Layer 1 | TrCH type | | DCH |
| | TB sizes, bit | | 336 (alt. 144) |
| | TFS | TF0, bits | 0x336 (alt. 0x144) |
| | | TF1, bits | 1x336 (alt 1x144) |
| | | TF2, bits | 2x336 (alt. 5x144) |
| | TTI, ms | | 20 |
| | Coding type | | TC |
| | CRC, bit | | 16 |
| | Max number of bits/TTI after channel coding | | 2 124 (alt. 2 412) |
| | Max number of bits/radio frame before rate matching | | 1 062 (alt. 1 206) |
| | RM attribute | | 135 to 175 |

6.10.3.4.1.23d.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.10.3.4.1.23d.1.1.3 TFCS

| | |
|---|---|
| TFCS size | 6 |
| TFCS | (32 kbps RAB, DCCH)=<br>(TF0,TF0), (TF1,TF0), (TF2,TF0), (TF0,TF1), (TF1,TF1), (TF2,TF1) |
| NOTE: In case TB size zero is configured for any transport channel the first TFC is required; it is optional otherwise. | |

6.10.3.4.1.23d.1.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Uplink | Midamble | 512 chips |
| | Codes and time slots | SF4 x 1 code x 1 time slot |
| | Max. Number of data bits/radio frame | 904 bits |

| | | |
|---|---|---|
| | TFCI code word | 16 bits |
| | TPC | 2 bits |
| | Puncturing Limit | 0.72 (alt. 0.64) |

6.10.3.4.1.23d.2 Downlink

6.10.3.4.1.23d.2.1 Transport channel parameters

6.10.3.4.1.23d.2.1.1 Transport channel parameters for Interactive or background / DL:32 kbps / PS RAB

| | | | |
|---|---|---|---|
| Higher layer | RAB/Signalling RB | | RAB |
| RLC | Logical channel type | | DTCH |
| | RLC mode | | AM |
| | Payload sizes, bit | | 320 |
| | Max data rate, bps | | 32 000 |
| | AMD PDU header, bit | | 16 |
| MAC | MAC header, bit | | 0 |
| | MAC multiplexing | | N/A |
| Layer 1 | TrCH type | | DCH |
| | TB sizes, bit | | 336 |
| | TFS | TF0, bits | 0x336 |
| | | TF1, bits | 1x336 |
| | | TF2, bits | 2x336 |
| | TTI, ms | | 20 |
| | Coding type | | TC |
| | CRC, bit | | 16 |
| | Max number of bits/TTI after channel coding | | 2 124 |
| | Max number of bits/radio frame before rate matching | | 1 062 |
| | RM attribute | | 135 to 175 |

6.10.3.4.1.23d.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.10.3.4.1.23d.2.1.3 TFCS

| | |
|---|---|
| TFCS size | 6 |
| TFCS | (32 kbps RAB, DCCH)=<br>(TF0,TF0), (TF1,TF0), (TF2,TF0), (TF0,TF1), (TF1,TF1), (TF2,TF1) |
| NOTE: In case TB size zero is configured for any transport channel the first TFC is required; it is optional otherwise. | |

6.10.3.4.1.23d.2.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Downlink | Midamble | 512 chips |
| | Codes and time slots | SF16 x 3 codes x 1 time slot |
| | Max. Number of data bits/radio frame | 716 bits |
| | TFCI code word | 16 bits |
| | Puncturing limit | 0.56 |

6.10.3.4.1.24 Void

6.10.3.4.1.25 Interactive or background / UL:32 DL: 64 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.3.4.1.25.1 Uplink

See clause 6.10.3.4.1.23.1.

6.10.3.4.1.25.2 Downlink

6.10.3.4.1.25.2.1 Transport channel parameters

6.10.3.4.1.25.2.1.1 Transport channel parameters for Interactive or background / DL:64 kbps / PS RAB

| | | | |
|---|---|---|---|
| Higher layer | RAB/Signalling RB | | RAB |
| RLC | Logical channel type | | DTCH |
| | RLC mode | | AM |
| | Payload sizes, bit | | 320 |
| | Max data rate, bps | | 64 000 |
| | AMD PDU header, bit | | 16 |
| MAC | MAC header, bit | | 0 |
| | MAC multiplexing | | N/A |
| Layer 1 | TrCH type | | DCH |
| | TB sizes, bit | | 336 |
| | TFS | TF0, bits | 0x336 |
| | | TF1, bits | 1x336 |
| | | TF2, bits | 2x336 |
| | | TF3, bits | 3x336 |
| | | TF4, bits | 4x336 |
| | TTI, ms | | 20 |
| | Coding type | | TC |
| | CRC, bit | | 16 |
| | Max number of bits/TTI after channel coding | | 4 236 |
| | Max number of bits/radio frame before rate matching | | 2 118 |
| | RM attribute | | 130 to 170 |

6.10.3.4.1.25.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.10.3.4.1.25.2.1.3 TFCS

| | |
|---|---|
| TFCS size | 10 |
| TFCS | (64 kbps RAB, DCCH)=<br>(TF0, TF0), (TF1, TF0), (TF2, TF0), (TF3, TF0), (TF4, TF0),<br>(TF0, TF1), (TF1, TF1), (TF2, TF1), (TF3, TF1), (TF4, TF1) |
| NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise. | |

6.10.3.4.1.25.2.2 Physical channel parameters

| | | | |
|---|---|---|---|
| DPCH Downlink | | Physical Configuration 1 | Physical Configuration 2 |
| | Midamble | 512 chips | 512 chips |
| | Codes and time slots | SF16 x 3 codes x 1 time slot + SF16 x 2 codes x 1 time slot | SF16 x 9 codes x 1 time slot |
| | Max. Number of data bits/radio frame | 1 204 bits | 2 180 bits |
| | TFCI code word | 16 bits | 16 bits |
| | Puncturing limit | 0.52 | 0.96 |

6.10.3.4.1.26 Interactive or background / UL:64 DL: 64 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.3.4.1.26.1 Uplink

6.10.3.4.1.26.1.1 Transport channel parameters

6.10.3.4.1.26.1.1.1 Transport channel parameters for Interactive or background / UL:64 kbps / PS RAB

| | | |
|---|---|---|
| Higher layer | RAB/Signalling RB | RAB |
| RLC | Logical channel type | DTCH |
| | RLC mode | AM |
| | Payload sizes, bit | 320 (alt. 128) |
| | Max data rate, bps | 64 000 |
| | AMD PDU header, bit | 16 |
| MAC | MAC header, bit | 0 |
| | MAC multiplexing | N/A |
| Layer 1 | TrCH type | DCH |
| | TB sizes, bit | 336 (alt. 144) |

| | | | |
|---|---|---|---|
| | TFS | TF0, bits | 0x336 (alt. 0x144) |
| | | TF1, bits | 1x336 (alt. 1x144) |
| | | TF2, bits | 2x336 (alt. 3x144) |
| | | TF3, bits | 3x336 (alt. 7x144) |
| | | TF4, bits | 4x336 (alt. 10x144) |
| | TTI, ms | | 20 |
| | Coding type | | TC |
| | CRC, bit | | 16 |
| | Max number of bits/TTI after channel coding | | 4 236 (alt. 4 812) |
| | Max number of bits/radio frame before rate matching | | 2 118 (alt. 2 406) |
| | RM attribute | | 130 to 170 |

6.10.3.4.1.26.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.10.3.4.1.26.1.1.3 TFCS

| | |
|---|---|
| TFCS size | 10 |
| TFCS | (64 kbps RAB, DCCH)=<br>(TF0, TF0), (TF1, TF0), (TF2, TF0), (TF3, TF0), (TF4, TF0),<br>(TF0, TF1), (TF1, TF1), (TF2, TF1), (TF3, TF1), (TF4, TF1) |

6.10.3.4.1.26.1.2 Physical channel parameters

| | | | |
|---|---|---|---|
| DPCH Uplink | | Physical Configuration 1 | Physical Configuration 2 |
| | Midamble | 512 chips | 512 chips |
| | Codes and time slots | SF16 x 1 code x 1 time slot +<br>SF4 x 1 code x 1 time slot | SF2 x 1 code x 1 time slot +<br>SF4 x 1 code x 1 time slot |
| | Max. Number of data bits/radio frame | 1148 bits | 2 784 bits |
| | TFCI code word | 16 bits | 16 bits |
| | TPC | 2 bits | 2 bits |
| | Puncturing Limit | 0.48 (alt. 0.44) | 1 |

6.10.3.4.1.26.2 Downlink

See clause 6.10.3.4.1.25.2.

6.10.3.4.1.27 Interactive or background / UL:64 DL:128 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.3.4.1.27.1 Uplink

See clause 6.10.3.4.1.26.1.

6.10.3.4.1.27.2 Downlink

6.10.3.4.1.27.2.1 Transport channel parameters

6.10.3.4.1.27.2.1.1 Transport channel parameters for Interactive or background / DL:128 kbps / PS RAB

| | | | |
|---|---|---|---|
| Higher layer | RAB/Signalling RB | | RAB |
| RLC | Logical channel type | | DTCH |
| | RLC mode | | AM |
| | Payload sizes, bit | | 320 |
| | Max data rate, bps | | 128 000 |
| | AMD PDU header, bit | | 16 |
| MAC | MAC header, bit | | 0 |
| | MAC multiplexing | | N/A |
| Layer 1 | TrCH type | | DCH |
| | TB sizes, bit | | 336 |
| | TFS | TF0, bits | 0x336 |
| | | TF1, bits | 1x336 |
| | | TF2, bits | 2x336 |

| | | | |
|---|---|---|---|
| | | TF3, bits | 4x336 |
| | | TF4, bits | 8x336 |
| | TTI, ms | | 20 |
| | Coding type | | TC |
| | CRC, bit | | 16 |
| | Max number of bits/TTI after channel coding | | 8 460 |
| | Max number of bits/radio frame before rate matching | | 4 230 |
| | RM attribute | | 120 to 160 |

6.10.3.4.1.27.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.10.3.4.1.27.2.1.3 TFCS

| | |
|---|---|
| TFCS size | 10 |
| TFCS | (128 kbps RAB, DCCH)=<br>(TF0, TF0), (TF1, TF0), (TF2, TF0), (TF3, TF0), (TF4, TF0),<br>(TF0, TF1), (TF1, TF1), (TF2, TF1), (TF3, TF1), (TF4, TF1) |
| NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise. | |

6.10.3.4.1.27.2.2 Physical channel parameters

| | | | |
|---|---|---|---|
| DPCH Downlink | | Physical Configuration 1 | Physical Configuration 2 |
| | Midamble | 256 chips | 256 chips |
| | Codes and time slots | SF16 x 8 codes x 1 time slot | SF16 x 4 codes x 2 time slots + SF16 x 3 codes x 2 time slots |
| | Max. Number of data bits/radio frame | 2 192 bits | 3848 bits |
| | TFCI code word | 16 bits | 16 bits |
| | Puncturing limit | 0.48 | 0.84 |

6.10.3.4.1.28 Interactive or background / UL:128 DL:128 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.3.4.1.28.1 Uplink

6.10.3.4.1.28.1.1 Transport channel parameters

6.10.3.4.1.28.1.1.1 Transport channel parameters for Interactive or background / UL:128 kbps / PS RAB

| | | | |
|---|---|---|---|
| Higher layer | RAB/Signalling RB | | RAB |
| RLC | Logical channel type | | DTCH |
| | RLC mode | | AM |
| | Payload sizes, bit | | 320 (alt. 128) |
| | Max data rate, bps | | 128 000 |
| | AMD PDU header, bit | | 16 |
| MAC | MAC header, bit | | 0 |
| | MAC multiplexing | | N/A |
| Layer 1 | TrCH type | | DCH |
| | TB sizes, bit | | 336 (alt. 144) |
| | TFS | TF0, bits | 0x336 (alt. 0x144) |
| | | TF1, bits | 1x336 (alt. 1x144) |
| | | TF2, bits | 2x336 (alt. 7x144) |
| | | TF3, bits | 4x336 (alt 14x144) |
| | | TF4, bits | 8x336 (alt. 20x144) |
| | TTI, ms | | 20 |
| | Coding type | | TC |
| | CRC, bit | | 16 |
| | Max number of bits/TTI after channel coding | | 8 460 ( alt. 9 612) |
| | Max number of bits/radio frame before rate matching | | 4 230 (alt. 4 806) |
| | RM attribute | | 120 to 160 |

6.10.3.4.1.28.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.10.3.4.1.28.1.1.3 TFCS

| | |
|---|---|
| TFCS size | 9 (alt.10) |
| TFCS | (128 kbps RAB, DCCH)=<br>(TF0, TF0), (TF1, TF0), (TF2, TF0), (TF3, TF0), (TF4, TF0),<br>(TF0, TF1), (TF1, TF1), (TF2, TF1), (TF3, TF1)<br>(alt. (TF0, TF0), (TF1, TF0), (TF2, TF0), (TF3, TF0), (TF4, TF0),<br>(TF0, TF1), (TF1, TF1), (TF2, TF1), (TF3, TF1), (TF4, TF1)) |
| NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise. | |

6.10.3.4.1.28.1.2 Physical channel parameters

| | | | |
|---|---|---|---|
| DPCH Uplink | | Physical Configuration 1 | Physical Configuration 2 |
| | Midamble | 256 chips | 256 chips |
| | Codes and time slots | SF2 x 1 code x 1 timeslot | SF2 x 1 code x 2 timeslots + SF4 x 1 code x 1 time slot |
| | Max. Number of data bits/radio frame | 2 064 bits | 5 376 bits |
| | TFCI code word | 16 bits | 16 bits |
| | TPC | 2 bits | 2 bits |
| | Puncturing Limit | 0.44 (alt. 0.40) | 1 |

6.10.3.4.1.28.2 Downlink

See clause 6.10.3.4.1.27.2.

6.10.3.4.1.29 Interactive or background / UL:64 DL:144 kbps / PS RAB + UL:3.4 DL: 3.4 kbps SRBs for DCCH

6.10.3.4.1.29.1 Uplink

See clause 6.10.3.4.1.26.1.

6.10.3.4.1.29.2 Downlink

6.10.3.4.1.29.2.1 Transport channel parameters

6.10.3.4.1.29.2.1.1 Transport channel parameters for Interactive or background / DL:144 kbps / PS RAB

| | | | |
|---|---|---|---|
| Higher layer | RAB/Signalling RB | | RAB |
| RLC | Logical channel type | | DTCH |
| | RLC mode | | AM |
| | Payload sizes, bit | | 320 |
| | Max data rate, bps | | 144 000 |
| | AMD PDU header, bit | | 16 |
| MAC | MAC header, bit | | 0 |
| | MAC multiplexing | | N/A |
| Layer 1 | TrCH type | | DCH |
| | TB sizes, bit | | 336 |
| | TFS | TF0, bits | 0x336 |
| | | TF1, bits | 1x336 |
| | | TF2, bits | 2x336 |
| | | TF3, bits | 4x336 |
| | | TF4, bits | 8x336 |
| | | TF5, bits | 9x336 |
| | TTI, ms | | 20 |
| | Coding type | | TC |
| | CRC, bit | | 16 |
| | Max number of bits/TTI after channel coding | | 9 516 |
| | Max number of bits/radio frame before rate matching | | 4 758 |
| | RM attribute | | 140 to 180 |

6.10.3.4.1.29.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.10.3.4.1.29.2.1.3 TFCS

| | |
|---|---|
| TFCS size | 12 |
| TFCS | (144 kbps RAB, DCCH)=<br>(TF0, TF0), (TF1, TF0), (TF2, TF0), (TF3, TF0), (TF4, TF0), (TF5, TF0)<br>(TF0, TF1), (TF1, TF1), (TF2, TF1), (TF3, TF1), (TF4, TF1), (TF5, TF1) |
| NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise. | |

6.10.3.4.1.29.2.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Downlink | Midamble | 256 chips |
| | Codes and time slots | SF16 x 9 codes x 1 time slot |
| | Max. Number of data bits/radio frame | 2468 bits |
| | TFCI code word | 16 bits |
| | Puncturing limit | 0.48 |

6.10.3.4.1.30 Interactive or background / UL:144 DL:144 kbps / PS RAB + UL:3.4 DL: 3.4 kbps SRBs for DCCH

6.10.3.4.1.30.1 Uplink

6.10.3.4.1.30.1.1 Transport channel parameters

6.10.3.4.1.30.1.1.1 Transport channel parameters for Interactive or background / UL:144 kbps / PS RAB

| | | | |
|---|---|---|---|
| Higher layer | RAB/Signalling RB | | RAB |
| RLC | Logical channel type | | DTCH |
| | RLC mode | | AM |
| | Payload sizes, bit | | 320 (alt. 128) |
| | Max data rate, bps | | 144 000 |
| | AMD PDU header, bit | | 16 |
| MAC | MAC header, bit | | 0 |
| | MAC multiplexing | | N/A |
| Layer 1 | TrCH type | | DCH |
| | TB sizes, bit | | 336 (alt. 144) |
| | TFS | TF0, bits | 0x336 (alt. 0x144) |
| | | TF1, bits | 1x336 (alt. 1x144) |
| | | TF2, bits | 2x336 (alt. 10x144) |
| | | TF3, bits | 4x336 (alt. 20x144) |
| | | TF4, bits | 8x336 (alt. 30x144) |
| | | TF5, bits | 9x336 (alt. 45x144) |
| | TTI, ms | | 20 (alt. 40) |
| | Coding type | | TC |
| | CRC, bit | | 16 |
| | Max number of bits/TTI after channel coding | | 9 516 (alt. 21 624) |
| | Max number of bits/radio frame before rate matching | | 4 758 (alt. 5 406) |
| | RM attribute | | 140 to 180 |

6.10.3.4.1.30.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.10.3.4.1.30.1.1.3 TFCS

| | |
|---|---|
| TFCS size | 12 |
| TFCS | (144 kbps RAB, DCCH)=<br>(TF0, TF0), (TF1, TF0), (TF2, TF0), (TF3, TF0), (TF4, TF0), (TF5, TF0)<br>(TF0, TF1), (TF1, TF1), (TF2, TF1), (TF3, TF1), (TF4, TF1), (TF5, TF1) |
| NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise. | |

6.10.3.4.1.30.1.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Uplink | Midamble | 256 chips |
| | Codes and time slots | SF16 x 1 code x 1 time slot + SF2 x 1 codex 1 time slot |
| | Max. Number of data bits/radio frame | 2340 bits |
| | TFCI code word | 16 bits |
| | TPC | 2 bits |
| | Puncturing Limit | 0.44 (alt. 0.40) |

6.10.3.4.1.30.2 Downlink

See clause 6.10.3.4.1.29.2.

6.10.3.4.1.31 Interactive or background / UL:64 DL:256 kbps / PS RAB + UL:3.4 DL: 3.4 kbps SRBs for DCCH

6.10.3.4.1.31.1 Uplink

See clause 6.10.3.4.1.26.1.

6.10.3.4.1.31.2 Downlink

6.10.3.4.1.31.2.1 Transport channel parameters

6.10.3.4.1.31.2.1.1 Transport channel parameters for Interactive or background / DL:256 kbps / PS RAB

| | | | |
|---|---|---|---|
| Higher layer | RAB/Signalling RB | | RAB |
| RLC | Logical channel type | | DTCH |
| | RLC mode | | AM |
| | Payload sizes, bit | | 320 |
| | Max data rate, bps | | 384 000 |
| | AMD PDU header, bit | | 16 |
| MAC | MAC header, bit | | 0 |
| | MAC multiplexing | | N/A |
| Layer 1 | TrCH type | | DCH |
| | TB sizes, bit | | 336 |
| | TFS | TF0, bits | 0x336 |
| | | TF1, bits | 1x336 |
| | | TF2, bits | 2x336 |
| | | TF3, bits | 4x336 |
| | | TF4, bits | 8x336 |
| | | TF5, bits | N/A (alt. 12x336) |
| | | TF6, bits | N/A (alt. 16x336) |
| | TTI, ms | | 10 (alt. 20) |
| | Coding type | | TC |
| | CRC, bit | | 16 |
| | Max number of bits/TTI after channel coding | | 8 460 (alt. 16 920) |
| | Max number of bits/radio frame before rate matching | | 8 460 (alt. 8 460) |
| | RM attribute | | 135 to 175 |

6.10.3.4.1.31.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.10.3.4.1.31.2.1.3 TFCS

| | |
|---|---|
| TFCS size | 10 (alt.14) |
| TFCS | (256 kbps RAB, DCCH)=<br>(TF0, TF0), (TF1, TF0), (TF2, TF0), (TF3, TF0), (TF4, TF0),<br>(TF0, TF1), (TF1, TF1), (TF2, TF1), (TF3, TF1), (TF4, TF1)<br>(alt. (TF0, TF0), (TF1, TF0), (TF2, TF0), (TF3, TF0), (TF4, TF0), (TF5, TF0), (TF6, TF0)<br>(TF0, TF1), (TF1, TF1), (TF2, TF1), (TF3, TF1), (TF4, TF1), (TF5, TF1), (TF6, TF1)) |
| NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise. | |

6.10.3.4.1.31.2.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Downlink | Midamble | 256 chips |
| | Codes and time slots | SF16 x 8 codes x 2 time slots |
| | Max. Number of data bits/radio frame | 4 400 bits |
| | TFCI code word | 16 bits |
| | Puncturing limit | 0.48 |

6.10.3.4.1.32 Interactive or background / UL:64 DL:384 kbps / PS RAB + UL:3.4 DL: 3.4 kbps SRBs for DCCH

6.10.3.4.1.32.1 Uplink

See clause 6.10.3.4.1.26.1.

6.10.3.4.1.32.2 Downlink

6.10.3.4.1.32.2.1 Transport channel parameters

6.10.3.4.1.32.2.1.1 Transport channel parameters for Interactive or background / DL:384 kbps / PS RAB

| | | | |
|---|---|---|---|
| Higher layer | RAB/Signalling RB | | RAB |
| RLC | Logical channel type | | DTCH |
| | RLC mode | | AM |
| | Payload sizes, bit | | 320 |
| | Max data rate, bps | | 384 000 |
| | AMD PDU header, bit | | 16 |
| MAC | MAC header, bit | | 0 |
| | MAC multiplexing | | N/A |
| Layer 1 | TrCH type | | DCH |
| | TB sizes, bit | | 336 |
| | TFS | TF0, bits | 0x336 |
| | | TF1, bits | 1x336 |
| | | TF2, bits | 2x336 |
| | | TF3, bits | 4x336 |
| | | TF4, bits | 8x336 |
| | | TF5, bits | 12x336 |
| | | TF6, bits | N/A (alt. 16x336) |
| | | TF7, bits | N/A (alt. 20x336) |
| | | TF8, bits | N/A (alt. 24x336) |
| | TTI, ms | | 10 (alt. 20) |
| | Coding type | | TC |
| | CRC, bit | | 16 |
| | Max number of bits/TTI after channel coding | | 12 684 (alt. 25 368) |
| | Max number of bits/radio frame before rate matching | | 12 684 (alt. 12 684) |
| | RM attribute | | 110 to 150 |

6.10.3.4.1.32.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.10.3.4.1.32.2.1.3 TFCS

| | |
|---|---|
| TFCS size | 12 (alt.18) |
| TFCS | (384 kbps RAB, DCCH)=<br>(TF0, TF0), (TF1, TF0), (TF2, TF0), (TF3, TF0), (TF4, TF0), (TF5, TF0)<br>(TF0, TF1), (TF1, TF1), (TF2, TF1), (TF3, TF1), (TF4, TF1), (TF5, TF1)<br>(alt. (TF0, TF0), (TF1, TF0), (TF2, TF0), (TF3, TF0), (TF4, TF0), (TF5, TF0), (TF6, TF0), (TF7, TF0), (TF8, TF0),<br>(TF0, TF1), (TF1, TF1), (TF2, TF1), (TF3, TF1), (TF4, TF1), (TF5, TF1), (TF6, TF1), (TF7, TF1), (TF8, TF1)) |
| NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise. | |

6.10.3.4.1.32.2.2 Physical channel parameters

| DPCH Downlink | | Physical Configuration 1 | Physical Configuration 2 |
|---|---|---|---|
| | Midamble | 256 chips | 256 chips |
| | Codes and time slots | SF16 x 8 codes x 3 time slots | SF16 x 6 codes x 4 time slots + SF16 x 4 codes x 1 time slot (alt. SF1 x 1 code x 3 time slots) |
| | Max. Number of data bits/radio frame | 6 608 bits | 7 712 bits (alt. 13232 bits) |
| | TFCI code word | 16 bits | 16 bits |
| | Puncturing Limit | 0.48 | 0.60 (alt. 1) |

6.10.3.4.1.33 Interactive or background / UL:128 DL:384 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.3.4.1.33.1 Uplink

See clause 6.10.3.4.1.28.1.

6.10.3.4.1.33.2 Downlink

See clause 6.10.3.4.1.32.2.

6.10.3.4.1.34 Interactive or background / UL:384 DL:384 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.3.4.1.34.1 Uplink

6.10.3.4.1.34.1.1 Transport channel parameters

6.10.3.4.1.34.1.1.1 Transport channel parameters for Interactive or background / UL:384 kbps / PS RAB

| Higher layer | RAB/Signalling RB | | RAB |
|---|---|---|---|
| RLC | Logical channel type | | DTCH |
| | RLC mode | | AM |
| | Payload sizes, bit | | 320 |
| | Max data rate, bps | | 384 000 |
| | AMD PDU header, bit | | 16 |
| MAC | MAC header, bit | | 0 |
| | MAC multiplexing | | N/A |
| Layer 1 | TrCH type | | DCH |
| | TB sizes, bit | | 336 |
| | TFS | TF0, bits | 0x336 |
| | | TF1, bits | 1x336 |
| | | TF2, bits | 2x336 |
| | | TF3, bits | 4x336 |
| | | TF4, bits | 8x336 |
| | | TF5, bits | 12x336 |
| | | TF6, bits | N/A (alt. 16x336) |
| | | TF7, bits | N/A (alt. 20x336) |
| | | TF8, bits | N/A (alt. 24x336) |
| | TTI, ms | | 10 (alt. 20) |
| | Coding type | | TC |
| | CRC, bit | | 16 |
| | Max number of bits/TTI after channel coding | | 12 684 (alt. 25 368) |
| | Max number of bits/radio frame before rate matching | | 12 684 |
| | RM attribute | | 110 to 150 |

6.10.3.4.1.34.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.10.3.4.1.34.1.1.3 TFCS

| TFCS size | 12 (alt.18) |
|---|---|

| | |
|---|---|
| TFCS | (384 kbps RAB, DCCH)=<br>(TF0, TF0), (TF1, TF0), (TF2, TF0), (TF3, TF0), (TF4, TF0), (TF5, TF0),<br>(TF0, TF1), (TF1, TF1), (TF2, TF1), (TF3, TF1), (TF4, TF1), (TF5, TF1)<br>(alt. (TF0, TF0), (TF1, TF0), (TF2, TF0), (TF3, TF0), (TF4, TF0), (TF5, TF0), (TF6, TF0), (TF7, TF0), (TF8, TF0),<br>(TF0, TF1), (TF1, TF1), (TF2, TF1), (TF3, TF1), (TF4, TF1), (TF5, TF1), (TF6, TF1), (TF7, TF1), (TF8, TF1)) |
| NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise. | |

6.10.3.4.1.34.1.2 Physical channel parameters

| | | | |
|---|---|---|---|
| DPCH Uplink | | Physical Configuration 1 | Physical Configuration 2 |
| | Midamble | 256 chips | 256 chips |
| | Codes and time slots | SF2 x 1 code x 3 time slots | SF2 x 1 code x 5 timeslots + SF4 x 1 code x 2 timeslots (alt. {SF2 x 1 code + SF4 x 1 code} x 4 timeslots) |
| | Max. Number of data bits/radio frame | 6 480 bits | 13 104 bits |
| | TFCI code word | 16 bits | 16 bits |
| | TPC | 2 bits | 2 bits |
| | Puncturing Limit | 0.48 | 1 |

6.10.3.4.1.34.2 Downlink

See clause 6.10.3.4.1.32.2.

6.10.3.4.1.35 Interactive or background / UL:64 DL:2 048 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.3.4.1.35.1 Uplink

6.10.3.4.1.35.1.1 Transport channel parameters

See clause 6.10.3.4.1.26.1.1.

6.10.3.4.1.35.1.1.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Uplink | Midamble | 256 chips |
| | Codes and time slots | SF2 x 1 code x 1 time slot |
| | Max. Number of data bits/radio frame | 2 064 bits |
| | TFCI code word | 16 bits |
| | TPC | 2 bits |
| | Puncturing Limit | 0.88 (alt. 0.80) |

6.10.3.4.1.35.2 Downlink

6.10.3.4.1.35.2.1 Transport channel parameters

6.10.3.4.1.35.2.1.1 Transport channel parameters for Interactive or background / DL:2 048 kbps / PS RAB

| | | | |
|---|---|---|---|
| Higher layer | RAB/Signalling RB | | RAB |
| RLC | Logical channel type | | DTCH |
| | RLC mode | | AM |
| | Payload sizes, bit | | 640 |
| | Max data rate, bps | | 2 048 000 |
| | AMD PDU header, bit | | 16 |
| MAC | MAC header, bit | | 0 |
| | MAC multiplexing | | N/A |
| Layer 1 | TrCH type | | DCH |
| | TB sizes, bit | | 656 |
| | TFS | TF0, bits | 0x656 |
| | | TF1, bits | 1x656 |
| | | TF2, bits | 2x656 |

| Higher layer | RAB/Signalling RB | | RAB |
|---|---|---|---|
| | | TF3, bits | 4x656 |
| | | TF4, bits | 8x656 |
| | | TF5, bits | 12x656 |
| | | TF6, bits | 16x656 |
| | | TF7, bits | 20x656 |
| | | TF8, bits | 24x656 |
| | | TF9, bits | 28x656 |
| | | TF10, bits | 31x656 (alt. 32x656) |
| | | TF11, bits | N/A (alt. 36x656) |
| | | TF12, bits | N/A (alt. 40x656) |
| | | TF13, bits | N/A (alt. 44x656) |
| | | TF14, bits | N/A (alt. 48x656) |
| | | TF15, bits | N/A (alt. 52x656) |
| | | TF16, bits | N/A (alt. 56x656) |
| | | TF17, bits | N/A (alt. 60x656) |
| | | TF18, bits | N/A (alt. 64x656) |
| | TTI, ms | | 10 (alt. 20) |
| | Coding type | | TC |
| | CRC, bit | | 16 |
| | Max number of bits/TTI after channel coding | | 62 565 (alt. 129 141) |
| | Max number of bits/radio frame before rate matching | | 62 565 (alt. 64 571) |
| | RM attribute | | 130 to 170 |

#### 6.10.3.4.1.35.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

#### 6.10.3.4.1.35.2.1.3 TFCS

| TFCS size | 21 (alt.38) |
|---|---|
| TFCS | (2 048 kbps RAB, DCCH)=<br>(TF0, TF0), (TF1, TF0), (TF2, TF0), (TF3, TF0), (TF4, TF0), (TF5, TF0), (TF6, TF0), (TF7, TF0),<br>(TF8, TF0), (TF9, TF0), (TF10, TF0),<br>(TF0, TF1), (TF1, TF1), (TF2, TF1), (TF3, TF1), (TF4, TF1), (TF5, TF1), (TF6, TF1), (TF7, TF1),<br>(TF8, TF1), (TF9, TF1)<br>(alt. TF0, TF0), (TF1, TF0), (TF2, TF0), (TF3, TF0), (TF4, TF0), (TF5, TF0), (TF6, TF0), (TF7, TF0),<br>(TF8, TF0), (TF9, TF0), (TF10, TF0),(TF11, TF0), (TF12, TF0), (TF13, TF0), (TF14, TF0), (TF15, TF0),<br>(TF16, TF0), (TF17, TF0), (TF18, TF0),<br>(TF0, TF1), (TF1, TF1), (TF2, TF1), (TF3, TF1), (TF4, TF1), (TF5, TF1), (TF6, TF1), (TF7, TF1),<br>(TF8, TF1), (TF9, TF1), (TF10, TF1),(TF11, TF1), (TF12, TF1), (TF13, TF1), (TF14, TF1), (TF15, TF1),<br>(TF16, TF1), (TF17, TF1(TF18, TF1)) |
| NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise. | |

#### 6.10.3.4.1.35.2.2 Physical channel parameters

| DPCH Downlink | | Physical Configuration 1 | Physical Configuration 2 |
|---|---|---|---|
| | Midamble | 256 chips | 256 chips |
| | Codes and time slots | SF1 x 1 code x 11 time slots | SF16 x 13 codes x 4 time slots + SF16 x 12 codes x 7 time slot |
| | Max. Number of data bits/radio frame | 48 560 bits (alt. 48 544) | 37 520 bits (alt. 37 504) |
| | TFCI code word | 16 bits (alt. 32 bits) | 16 bits (alt. 32 bits) |
| | Puncturing limit | 0.76 (alt.0.72) | 0.56 |

### 6.10.3.4.1.36 Void

### 6.10.3.4.1.37 Void

### 6.10.3.4.1.38 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:32 DL:8 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

#### 6.10.3.4.1.38.1 Uplink

6.10.3.4.1.38.1.1 Transport channel parameters

6.10.3.4.1.38.1.1.1 Transport channel parameters for Conversational / speech / UL:12.2 kbps / CS RAB

See clause 6.10.3.4.1.4.1.1.1.

6.10.3.4.1.38.1.1.2 Transport channel parameters for Interactive or background / UL:32 kbps / PS RAB

See clause 6.10.3.4.1.23.1.1.1.

6.10.3.4.1.38.1.1.3 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.10.3.4.1.38.1.1.4 TFCS

| | |
|---|---|
| TFCS size | 18 |
| TFCS | (RAB subflow#1, RAB subflow#2, RAB subflow#3, 32kbps RAB , DCCH)=<br>(TF0, TF0, TF0, TF0, TF0), (TF1, TF0, TF0, TF0, TF0), (TF2, TF1, TF1, TF0, TF0),<br>(TF0, TF0, TF0, TF1, TF0), (TF1, TF0, TF0, TF1, TF0), (TF2, TF1, TF1, TF1, TF0),<br>(TF0, TF0, TF0, TF2, TF0), (TF1, TF0, TF0, TF2, TF0), (TF2, TF1, TF1, TF2, TF0),<br>(TF0, TF0, TF0, TF0, TF1), (TF1, TF0, TF0, TF0, TF1), (TF2, TF1, TF1, TF0, TF1),<br>(TF0, TF0, TF0, TF1, TF1), (TF1, TF0, TF0, TF1, TF1), (TF2, TF1, TF1, TF1, TF1),<br>(TF0, TF0, TF0, TF2, TF1), (TF1, TF0, TF0, TF2, TF1), (TF2, TF1, TF1, TF2, TF1) |
| NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise. | |

6.10.3.4.1.38.1.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Uplink | Midamble | 512 chips |
| | Codes and time slots | SF4 x 1 code x 1 time slot |
| | Max. Number of data bits/radio frame | 904 bits |
| | TFCI code word | 16 bits |
| | TPC | 2 bits |
| | Puncturing Limit | 0.52 (alt. 0.48) |

6.10.3.4.1.38.2 Downlink

6.10.3.4.1.38.2.1 Transport channel parameters

6.10.3.4.1.38.2.1.1 Transport channel parameters for Conversational / speech / DL:12.2 kbps / CS RAB

See clause 6.10.3.4.1.4.2.1.1.

6.10.3.4.1.38.2.1.2 Transport channel parameters for Interactive or background / DL:8 kbps / PS RAB

See clause 6.10.3.4.1.23.2.1.1.

6.10.3.4.1.38.2.1.3 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.

6.10.3.4.1.38.2.1.4 TFCS

| | |
|---|---|
| TFCS size | 12 |
| TFCS | (RAB subflow#1, RAB subflow#2, RAB subflow#3,8kbps RAB, DCCH)=<br>(TF0, TF0, TF0, TF0, TF0), (TF1, TF0, TF0, TF0, TF0), (TF2, TF1, TF1, TF0, TF0),<br>(TF0, TF0, TF0, TF1, TF0), (TF1, TF0, TF0, TF1, TF0), (TF2, TF1, TF1, TF1, TF0),<br>(TF0, TF0, TF0, TF0, TF1), (TF1, TF0, TF0, TF0, TF1), (TF2, TF1, TF1, TF0, TF1),<br>(TF0, TF0, TF0, TF1, TF1), (TF1, TF0, TF0, TF1, TF1), (TF2, TF1, TF1, TF1, TF1) |
| NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise. | |

6.10.3.4.1.38.2.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Downlink | Midamble | 512 chips |
| | Codes and time slots | SF16 x 2 codes x 1 time slot |

| | | |
|---|---|---|
| | Max. Number of data bits/radio frame | 472 bits |
| | TFCI code word | 16 bits |
| | Puncturing limit | 0.52 |

6.10.3.4.1.38a Conversational / speech / 12.2 kbps / CS RAB + Interactive or background / UL:0 DL:0 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.3.4.1.38a.1 Uplink

6.10.3.4.1.38a.1.1 Transport channel parameters

6.10.3.4.1.38a.1.1.1 Transport channel parameters for Conversational / speech / UL:12.2 kbps / CS RAB

See clause 6.10.3.4.1.4.1.1.1.

6.10.3.4.1.38a.1.1.2 Transport channel parameters for Interactive or background / UL:0 kbps / PS RAB

| | | | |
|---|---|---|---|
| Higher layer | RAB/Signalling RB | | RAB |
| RLC | Logical channel type | | DTCH |
| | RLC mode | | AM |
| | Payload sizes, bit | | 320 (alt. 128) |
| | Max data rate, bps | | 0 |
| | AMD PDU header, bit | | 16 |
| MAC | MAC header, bit | | 0 |
| | MAC multiplexing | | N/A |
| Layer 1 | TrCH type | | DCH |
| | TB sizes, bit | | 336 (alt. 144) |
| | TFS | TF0, bits | 0x336 (alt 0x144) |
| | TTI, ms | | 20 |
| | Coding type | | TC |
| | CRC, bit | | 16 |
| | Max number of bits/TTI after channel coding | | 0 |
| | Max number of bits/radio frame before rate matching | | 0 |
| | RM attribute | | 130 to 170 |

6.10.3.4.1.38a.1.1.3 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.10.3.4.1.38a.1.1.4 TFCS

| | |
|---|---|
| TFCS size | 6 |
| TFCS | (RAB subflow#1, RAB subflow#2, RAB subflow#3, 0kbps RAB, DCCH)=<br>(TF0,TF0,TF0,TF0,TF0), (TF1,TF0,TF0,TF0,TF0), (TF2,TF1,TF1,TF0,TF0),<br>(TF0,TF0,TF0,TF0,TF1), (TF1,TF0,TF0,TF0,TF1), (TF2,TF1,TF1,TF0,TF1) |
| NOTE: In case TB size zero is configured for any transport channel the first TFC is required; it is optional otherwise. | |

6.10.3.4.1.38a.1.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Uplink | Midamble | 512 chips |
| | Codes and time slots | SF8 x 1 code x 1 time slot |
| | Max. Number of data bits/radio frame | 452 bits |
| | TFCI code word | 16 bits |
| | TPC | 2 bit |
| | Puncturing Limit | 0.68 |

6.10.3.4.1.38a.2 Downlink

6.10.3.4.1.38a.2.1 Transport channel parameters

6.10.3.4.1.38a.2.1.1 Transport channel parameters for Conversational / speech / DL:12.2 kbps / CS RAB

See clause 6.10.3.4.1.4.2.1.1.

6.10.3.4.1.38a.2.1.2 Transport channel parameters for Interactive or background / DL:0 kbps / PS RAB

| | | | |
|---|---|---|---|
| Higher layer | RAB/Signalling RB | | RAB |
| RLC | Logical channel type | | DTCH |
| | RLC mode | | AM |
| | Payload sizes, bit | | 320 |
| | Max data rate, bps | | 0 |
| | AMD PDU header, bit | | 16 |
| MAC | MAC header, bit | | 0 |
| | MAC multiplexing | | N/A |
| Layer 1 | TrCH type | | DCH |
| | TB sizes, bit | | 336 |
| | TFS | TF0, bits | 0x336 |
| | TTI, ms | | 20 |
| | Coding type | | TC |
| | CRC, bit | | 16 |
| | Max number of bits/TTI after channel coding | | 0 |
| | Max number of bits/radio frame before rate matching | | 0 |
| | RM attribute | | 130 to 170 |

6.10.3.4.1.38a.2.1.3 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1

6.10.3.4.1.38a.2.1.4 TFCS

| | |
|---|---|
| TFCS size | 6 |
| TFCS | (RAB subflow#1, RAB subflow#2, RAB subflow#3, 0kbps RAB, DCCH)=<br>(TF0,TF0,TF0,TF0,TF0), (TF1,TF0,TF0,TF0,TF0), (TF2,TF1,TF1,TF0,TF0),<br>(TF0,TF0,TF0,TF0,TF1), (TF1,TF0,TF0,TF0,TF1), (TF2,TF1,TF1,TF0,TF1) |
| NOTE: In case TB size zero is configured for any transport channel the first TFC is required; it is optional otherwise. | |

6.10.3.4.1.38a.2.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Downlink | Midamble | 512 chips |
| | Codes and time slots | SF16 x 2 codes x 1 time slot |
| | Max. Number of data bits/radio frame | 472 bits |
| | TFCI code word | 16 bits |
| | Puncturing limit | 0.68 |

6.10.3.4.1.38b Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:8 DL:8 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.3.4.1.38b.1 Uplink

6.10.3.4.1.38b.1.1 Transport channel parameters

6.10.3.4.1.38b.1.1.1 Transport channel parameters for Conversational / speech / UL:12.2 kbps / CS RAB

See clause 6.10.3.4.1.4.1.1.1.

6.10.3.4.1.38b.1.1.2 Transport channel parameters for Interactive or background / UL:8 kbps / PS RAB

See clause 6.10.3.4.1.23a.1.1.1.

6.10.3.4.1.38b.1.1.3 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.10.3.4.1.38b.1.1.4 TFCS

| | |
|---|---|
| TFCS size | 12 (alt. 17) |
| TFCS | (RAB subflow#1, RAB subflow#2, RAB subflow#3, 8 kbps RAB, DCCH)=<br>(TF0,TF0,TF0,TF0,TF0), (TF1,TF0,TF0,TF0,TF0), (TF2,TF1,TF1,TF0,TF0), |

| | |
|---|---|
| | (TF0,TF0,TF0,TF1,TF0), (TF1,TF0,TF0,TF1,TF0), (TF2,TF1,TF1,TF1,TF0),<br>(TF0,TF0,TF0,TF0,TF1), (TF1,TF0,TF0,TF0,TF1), (TF2,TF1,TF1,TF0,TF1),<br>(TF0,TF0,TF0,TF1,TF1), (TF1,TF0,TF0,TF1,TF1), (TF2,TF1,TF1,TF1,TF1)<br>(alt.<br>(TF0,TF0,TF0,TF0,TF0), (TF1,TF0,TF0,TF0,TF0), (TF2,TF1,TF1,TF0,TF0),<br>(TF0,TF0,TF0,TF1,TF0), (TF1,TF0,TF0,TF1,TF0), (TF2,TF1,TF1,TF1,TF0),<br>(TF0,TF0,TF0,TF2,TF0), (TF1,TF0,TF0,TF2,TF0), (TF2,TF1,TF1,TF2,TF0),<br>(TF0,TF0,TF0,TF0,TF1), (TF1,TF0,TF0,TF0,TF1), (TF2,TF1,TF1,TF0,TF1),<br>(TF0,TF0,TF0,TF1,TF1), (TF1,TF0,TF0,TF1,TF1), (TF2,TF1,TF1,TF1,TF1)<br>(TF0,TF0,TF0,TF2,TF1), (TF1,TF0,TF0,TF2,TF1)) |
| NOTE: In case TB size zero is configured for any transport channel the first TFC is required; it is optional otherwise. | |

6.10.3.4.1.38b.1.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Uplink | Midamble | 512 chips |
| | Codes and time slots | SF8 x 1 code x 1 time slot |
| | Max. Number of data bits/radio frame | 452 bits |
| | TFCI code word | 16 bits |
| | TPC | 2 bits |
| | Puncturing Limit | 0.48 (alt. 0.56) |

6.10.3.4.1.38b.2 Downlink

6.10.3.4.1.38b.2.1 Transport channel parameters

6.10.3.4.1.38b.2.1.1 Transport channel parameters for Conversational / speech / DL:12.2 kbps / CS RAB

See clause 6.10.3.4.1.4.2.1.1.

6.10.3.4.1.38b.2.1.2 Transport channel parameters for Interactive or background / DL:8 kbps / PS RAB

See clause 6.10.3.4.1.23.2.1.1.

6.10.3.4.1.38b.2.1.3 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.10.3.4.1.38b.2.1.4 TFCS

| | |
|---|---|
| TFCS size | 12 |
| TFCS | (RAB subflow#1, RAB subflow#2, RAB subflow#3, 8 kbps RAB, DCCH)=<br>(TF0,TF0,TF0,TF0,TF0), (TF1,TF0,TF0,TF0,TF0), (TF2,TF1,TF1,TF0,TF0),<br>(TF0,TF0,TF0,TF1,TF0), (TF1,TF0,TF0,TF1,TF0), (TF2,TF1,TF1,TF1,TF0),<br>(TF0,TF0,TF0,TF0,TF1), (TF1,TF0,TF0,TF0,TF1), (TF2,TF1,TF1,TF0,TF1),<br>(TF0,TF0,TF0,TF1,TF1), (TF1,TF0,TF0,TF1,TF1), (TF2,TF1,TF1,TF1,TF1) |
| NOTE: In case TB size zero is configured for any transport channel the first TFC is required; it is optional otherwise. | |

6.10.3.4.1.38b.2.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Downlink | Midamble | 512 chips |
| | Codes and time slots | SF16 x 2 codes x 1 time slot |
| | Max. Number of data bits/radio frame | 472 bits |
| | TFCI code word | 16 bits |
| | Puncturing limit | 0.52 |

6.10.3.4.1.38c Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:32 DL:32 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.3.4.1.38c.1 Uplink

6.10.3.4.1.38c.1.1 Transport channel parameters

6.10.3.4.1.38c.1.1.1 Transport channel parameters for Conversational / speech / UL:12.2 kbps / CS RAB

See clause 6.10.3.4.1.4.1.1.1.

6.10.3.4.1.38c.1.1.2 Transport channel parameters for Interactive or background / UL:32 kbps / PS RAB

See clause 6.10.3.4.1.23d.1.1.1.

6.10.3.4.1.38c.1.1.3 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.10.3.4.1.38c.1.1.4 TFCS

| | |
|---|---|
| TFCS size | 18 (alt. 17) |
| TFCS | (RAB subflow#1, RAB subflow#2, RAB subflow#3, 32 kbps RAB, DCCH)=<br>(TF0,TF0,TF0,TF0,TF0), (TF1,TF0,TF0,TF0,TF0), (TF2,TF1,TF1,TF0,TF0),<br>(TF0,TF0,TF0,TF1,TF0), (TF1,TF0,TF0,TF1,TF0), (TF2,TF1,TF1,TF1,TF0),<br>(TF0,TF0,TF0,TF2,TF0), (TF1,TF0,TF0,TF2,TF0), (TF2,TF1,TF1,TF2,TF0),<br>(TF0,TF0,TF0,TF0,TF1), (TF1,TF0,TF0,TF0,TF1), (TF2,TF1,TF1,TF0,TF1),<br>(TF0,TF0,TF0,TF1,TF1), (TF1,TF0,TF0,TF1,TF1), (TF2,TF1,TF1,TF1,TF1),<br>(TF0,TF0,TF0,TF2,TF1), (TF1,TF0,TF0,TF2,TF1), (TF2,TF1,TF1,TF2,TF1)<br>(alt.<br>(TF0,TF0,TF0,TF0,TF0), (TF1,TF0,TF0,TF0,TF0), (TF2,TF1,TF1,TF0,TF0),<br>(TF0,TF0,TF0,TF1,TF0), (TF1,TF0,TF0,TF1,TF0), (TF2,TF1,TF1,TF1,TF0),<br>(TF0,TF0,TF0,TF2,TF0), (TF1,TF0,TF0,TF2,TF0), (TF2,TF1,TF1,TF2,TF0),<br>(TF0,TF0,TF0,TF0,TF1), (TF1,TF0,TF0,TF0,TF1), (TF2,TF1,TF1,TF0,TF1),<br>(TF0,TF0,TF0,TF1,TF1), (TF1,TF0,TF0,TF1,TF1), (TF2,TF1,TF1,TF1,TF1),<br>(TF0,TF0,TF0,TF2,TF1), (TF1,TF0,TF0,TF2,TF1)) |
| NOTE: In case TB size zero is configured for any transport channel the first TFC is required; it is optional otherwise. | |

6.10.3.4.1.38c.1.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Uplink | Midamble | 512 chips |
| | Codes and time slots | SF4 x 1 code x 1 time slot |
| | Max. Number of data bits/radio frame | 904 bits |
| | TFCI code word | 16 bits |
| | TPC | 2 bits |
| | Puncturing Limit | 0.52 (alt. 0.52) |

6.10.3.4.1.38c.2 Downlink

6.10.3.4.1.38c.2.1 Transport channel parameters

6.10.3.4.1.38c.2.1.1 Transport channel parameters for Conversational / speech / DL:12.2 kbps / CS RAB

See clause 6.10.3.4.1.4.2.1.1.

6.10.3.4.1.38c.2.1.2 Transport channel parameters for Interactive or background / DL:32 kbps / PS RAB

See clause 6.10.3.4.1.23d.2.1.1.

6.10.3.4.1.38c.2.1.3 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.10.3.4.1.38c.2.1.4 TFCS

| | |
|---|---|
| TFCS size | 18 |
| TFCS | (RAB subflow#1, RAB subflow#2, RAB subflow#3, 32 kbps RAB, DCCH)=<br>(TF0,TF0,TF0,TF0,TF0), (TF1,TF0,TF0,TF0,TF0), (TF2,TF1,TF1,TF0,TF0),<br>(TF0,TF0,TF0,TF1,TF0), (TF1,TF0,TF0,TF1,TF0), (TF2,TF1,TF1,TF1,TF0),<br>(TF0,TF0,TF0,TF2,TF0), (TF1,TF0,TF0,TF2,TF0), (TF2,TF1,TF1,TF2,TF0),<br>(TF0,TF0,TF0,TF0,TF1), (TF1,TF0,TF0,TF0,TF1), (TF2,TF1,TF1,TF0,TF1),<br>(TF0,TF0,TF0,TF1,TF1), (TF1,TF0,TF0,TF1,TF1), (TF2,TF1,TF1,TF1,TF1),<br>(TF0,TF0,TF0,TF2,TF1), (TF1,TF0,TF0,TF2,TF1), (TF2,TF1,TF1,TF2,TF1) |
| NOTE: In case TB size zero is configured for any transport channel the first TFC is required; it is optional otherwise. | |

6.10.3.4.1.38c.2.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Downlink | Midamble | 512 chips |
| | Codes and time slots | SF16 x 4 codes x 1 time slot |
| | Max. Number of data bits/radio frame | 960 |
| | TFCI code word | 16 bits |
| | Puncturing limit | 0.56 |

6.10.3.4.1.38d Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:64 DL:64 kbps / PS RAB + Interactive or background / UL:64 DL:64 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.3.4.1.38d.1 Uplink

6.10.3.4.1.38d.1.1 Transport channel parameters

6.10.3.4.1.38d.1.1.1 Transport channel parameters for Conversational / speech / UL:12.2 kbps / CS RAB

See clause 6.10.3.4.1.4.1.1.1.

6.10.3.4.1.38d.1.1.2 Transport channel parameters for Interactive or background / UL:64 kbps / PS RAB + UL:64 kbps / PS RAB

| | | | | |
|---|---|---|---|---|
| Higher layer | RAB/Signalling RB | | RAB | RAB |
| RLC | Logical channel type | | DTCH | DTCH |
| | RLC mode | | AM | AM |
| | Payload sizes, bit | | 320 (alt. 128) | 320 (alt. 128) |
| | Max data rate, bps | | 64 000 | 64 000 |
| | AMD PDU header, bit | | 16 | 16 |
| MAC | MAC header, bit | | 4 | 4 |
| | MAC multiplexing | | 2 logical channel multiplexing | |
| Layer 1 | TrCH type | | DCH | |
| | TB sizes, bit | | 340 (alt. 148) | |
| | TFS | TF0, bits | 0x340 (alt 0x148) | |
| | | TF1, bits | 1x340 (alt 1x148) | |
| | | TF2, bits | 2x340 (alt 3x148) | |
| | | TF3, bits | 3x340 (alt 7x148) | |
| | | TF4, bits | 4x340 (alt 10x148) | |
| | TTI, ms | | 20 | |
| | Coding type | | TC | |
| | CRC, bit | | 16 | |
| | Max number of bits/TTI after channel coding | | 4 284 (alt. 4 932) | |
| | Max number of bits/radio frame before rate matching | | 2 142 (alt. 2 466) | |
| | RM attribute | | 130 to 170 | |

6.10.3.4.1.38d.1.1.3 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.10.3.4.1.38d.1.1.4 TFCS

| | |
|---|---|
| TFCS size | 30 |
| TFCS | (RAB subflow#1, RAB subflow#2, RAB subflow#3, 64 kbps RAB + 64 kbps RAB, DCCH)=<br>(TF0,TF0,TF0,TF0,TF0), (TF1,TF0,TF0,TF0,TF0),(TF2,TF1,TF1,TF0,TF0),<br>(TF0,TF0,TF0,TF1,TF0), (TF1,TF0,TF0,TF1,TF0),(TF2,TF1,TF1,TF1,TF0),<br>(TF0,TF0,TF0,TF2,TF0), (TF1,TF0,TF0,TF2,TF0),(TF2,TF1,TF1,TF2,TF0),<br>(TF0,TF0,TF0,TF3,TF0), (TF1,TF0,TF0,TF3,TF0),(TF2,TF1,TF1,TF3,TF0),<br>(TF0,TF0,TF0,TF4,TF0), (TF1,TF0,TF0,TF4,TF0),(TF2,TF1,TF1,TF4,TF0),<br>(TF0,TF0,TF0,TF0,TF1), (TF1,TF0,TF0,TF0,TF1),(TF2,TF1,TF1,TF0,TF1),<br>(TF0,TF0,TF0,TF1,TF1), (TF1,TF0,TF0,TF1,TF1),(TF2,TF1,TF1,TF1,TF1),<br>(TF0,TF0,TF0,TF2,TF1), (TF1,TF0,TF0,TF2,TF1),(TF2,TF1,TF1,TF2,TF1),<br>(TF0,TF0,TF0,TF3,TF1), (TF1,TF0,TF0,TF3,TF1),(TF2,TF1,TF1,TF3,TF1),<br>(TF0,TF0,TF0,TF4,TF1), (TF1,TF0,TF0,TF4,TF1),(TF2,TF1,TF1,TF4,TF1) |
| NOTE: In case TB size zero is configured for any transport channel the first TFC is required; it is optional otherwise. | |

6.10.3.4.1.38d.1.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Uplink | Midamble | 256 chips |
| | Codes and time slots | SF2 x 1 code x 1 time slot |
| | Max. Number of data bits/radio frame | 2 064 bits |
| | TFCI code word | 16 bit |
| | TPC | 2 bits |
| | Puncturing Limit | 0.72 (alt. 0.64) |

6.10.3.4.1.38d.2 Downlink

6.10.3.4.1.38d.2.1 Transport channel parameters

6.10.3.4.1.38d.2.1.1 Transport channel parameters for Conversational / speech / DL:12.2 kbps / CS RAB

See clause 6.10.3.4.1.4.2.1.1.

6.10.3.4.1.38d.2.1.2 Transport channel parameters for Interactive or background / DL:64 kbps / PS RAB + DL:64 kbps / PS RAB

| | | | | |
|---|---|---|---|---|
| Higher layer | RAB/Signalling RB | | RAB | RAB |
| RLC | Logical channel type | | DTCH | DTCH |
| | RLC mode | | AM | AM |
| | Payload sizes, bit | | 320 | 320 |
| | Max data rate, bps | | 64 000 | 64 000 |
| | AMD PDU header, bit | | 16 | 16 |
| MAC | MAC header, bit | | 4 | 4 |
| | MAC multiplexing | | 2 logical channel multiplexing | |
| Layer 1 | TrCH type | | DCH | |
| | TB sizes, bit | | 340 | |
| | TFS | TF0, bits | 0x340 | |
| | | TF1, bits | 1x340 | |
| | | TF2, bits | 2x340 | |
| | | TF3, bits | 3x340 | |
| | | TF4, bits | 4x340 | |
| | TTI, ms | | 20 | |
| | Coding type | | TC | |
| | CRC, bit | | 16 | |
| | Max number of bits/TTI after channel coding | | 4 284 | |
| | Max number of bits/radio frame before rate matching | | 2 142 | |
| | RM attribute | | 130 to 170 | |

6.10.3.4.1.38d.2.1.3 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.10.3.4.1.38d.2.1.4 TFCS

| | |
|---|---|
| TFCS size | 30 |
| TFCS | (RAB subflow#1, RAB subflow#2, RAB subflow#3, 64 kbps RAB + 64 kbps RAB, DCCH)=<br>(TF0,TF0,TF0,TF0,TF0), (TF1,TF0,TF0,TF0,TF0),(TF2,TF1,TF1,TF0,TF0),<br>(TF0,TF0,TF0,TF1,TF0), (TF1,TF0,TF0,TF1,TF0),(TF2,TF1,TF1,TF1,TF0),<br>(TF0,TF0,TF0,TF2,TF0), (TF1,TF0,TF0,TF2,TF0),(TF2,TF1,TF1,TF2,TF0),<br>(TF0,TF0,TF0,TF3,TF0), (TF1,TF0,TF0,TF3,TF0),(TF2,TF1,TF1,TF3,TF0),<br>(TF0,TF0,TF0,TF4,TF0), (TF1,TF0,TF0,TF4,TF0),(TF2,TF1,TF1,TF4,TF0),<br>(TF0,TF0,TF0,TF0,TF1), (TF1,TF0,TF0,TF0,TF1),(TF2,TF1,TF1,TF0,TF1),<br>(TF0,TF0,TF0,TF1,TF1), (TF1,TF0,TF0,TF1,TF1),(TF2,TF1,TF1,TF1,TF1),<br>(TF0,TF0,TF0,TF2,TF1), (TF1,TF0,TF0,TF2,TF1),(TF2,TF1,TF1,TF2,TF1),<br>(TF0,TF0,TF0,TF3,TF1), (TF1,TF0,TF0,TF3,TF1),(TF2,TF1,TF1,TF3,TF1),<br>(TF0,TF0,TF0,TF4,TF1), (TF1,TF0,TF0,TF4,TF1),(TF2,TF1,TF1,TF4,TF1) |
| NOTE: In case TB size zero is configured for any transport channel the first TFC is required; it is optional otherwise. | |

6.10.3.4.1.38d.2.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Downlink | Midamble | 256 chips |
| | Codes and time slots | SF16 x 7 codes x 1 time slot |
| | Max. Number of data bits/radio frame | 1 916 bits |
| | TFCI code word | 16 bits |
| | Puncturing limit | 0.68 |

6.10.3.4.1.38e Conversational / speech / UL:(12.2, 7.95, 5.9, 4.75) DL:(12.2, 7.95, 5.9, 4.75) kbps / CS RAB + Interactive or background / UL:0 DL:0 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.3.4.1.38e.1 Uplink

6.10.3.4.1.38e.1.1 Transport channel parameters

6.10.3.4.1.38e.1.1.1 Transport channel parameters for Conversational / speech / UL: (12.2, 7.95, 5.9, 4.75) kbps / CS RAB

See clause 6.10.3.4.1.4a.1.1.1.

6.10.3.4.1.38e.1.1.2 Transport channel parameters for Interactive or background / UL:0 kbps / PS RAB

See clause 6.10.3.4.1.38a.1.1.2.

6.10.3.4.1.38e.1.1.3 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.10.3.4.1.38e.1.1.4 TFCS

| | |
|---|---|
| TFCS size | 12 |
| TFCS | (RAB subflow#1, RAB subflow#2, RAB subflow#3, 0 kbps RAB, DCCH)=<br>(TF0,TF0,TF0,TF0,TF0), (TF1,TF0,TF0,TF0,TF0), (TF2,TF1,TF0,TF0,TF0),<br>(TF3,TF2,TF0,TF0,TF0), (TF4,TF3,TF0,TF0,TF0), (TF5,TF4,TF1,TF0,TF0),<br>(TF0,TF0,TF0,TF0,TF1), (TF1,TF0,TF0,TF0,TF1), (TF2,TF1,TF0,TF0,TF1),<br>(TF3,TF2,TF0,TF0,TF1), (TF4,TF3,TF0,TF0,TF1), (TF5,TF4,TF1,TF0,TF1) |
| NOTE: In case TB size zero is configured for any transport channel the first TFC is required; it is optional otherwise. | |

6.10.3.4.1.38e.1.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Uplink | Midamble | 512 chips |
| | Codes and time slots | SF8 x 1 code x 1 time slot |
| | Max. Number of data bits/radio frame | 452 bits |
| | TFCI code word | 16 bits |
| | TPC | 2 bit |
| | Puncturing Limit | 0.68 |

6.10.3.4.1.38e.2 Downlink

6.10.3.4.1.38e.2.1 Transport channel parameters

6.10.3.4.1.38e.2.1.1 Transport channel parameters for Conversational / speech / DL: (12.2, 7.95, 5.9, 4.75) kbps / CS RAB

See clause 6.10.3.4.1.4a.2.1.1.

6.10.3.4.1.38e.2.1.2 Transport channel parameters for Interactive or background / DL:0 kbps / PS RAB

See clause 6.10.3.4.1.38a.2.1.2.

6.10.3.4.1.38e.2.1.3 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.10.3.4.1.38e.2.1.4 TFCS

| | |
|---|---|
| TFCS size | 12 |
| TFCS | (RAB subflow#1, RAB subflow#2, RAB subflow#3, 0 kbps RAB, DCCH)=<br>(TF0,TF0,TF0,TF0,TF0), (TF1,TF0,TF0,TF0,TF0), (TF2,TF1,TF0,TF0,TF0),<br>(TF3,TF2,TF0,TF0,TF0), (TF4,TF3,TF0,TF0,TF0), (TF5,TF4,TF1,TF0,TF0),<br>(TF0,TF0,TF0,TF0,TF1), (TF1,TF0,TF0,TF0,TF1), (TF2,TF1,TF0,TF0,TF1),<br>(TF3,TF2,TF0,TF0,TF1), (TF4,TF3,TF0,TF0,TF1), (TF5,TF4,TF1,TF0,TF1), |
| NOTE: In case TB size zero is configured for any transport channel the first TFC is required; it is optional otherwise. | |

6.10.3.4.1.38e.2.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Downlink | Midamble | 512 chips |
| | Codes and time slots | SF16 x 2 codes x 1 time slot |
| | Max. Number of data bits/radio frame | 472 bits |
| | TFCI code word | 16 bits |
| | Puncturing limit | 0.68 |

6.10.3.4.1.38f Conversational / speech / (12.2, 7.95, 5.9, 4.75) kbps / CS RAB + Interactive or background / UL:8 DL:8 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.3.4.1.38f.1 Uplink

6.10.3.4.1.38f.1.1 Transport channel parameters

6.10.3.4.1.38f.1.1.1 Transport channel parameters for Conversational / speech / UL: (12.2, 7.95, 5.9, 4.75) kbps / CS RAB

See clause 6.10.3.4.1.4a.1.1.1.

6.10.3.4.1.38f.1.1.2 Transport channel parameters for Interactive or background / UL:8 kbps / PS RAB

See clause 6.10.3.4.1.23a.1.1.1.

6.10.3.4.1.38f.1.1.3 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.10.3.4.1.38f.1.1.4 TFCS

| | |
|---|---|
| TFCS size | 24 (alt. 32) |
| TFCS | (RAB subflow#1, RAB subflow#2, RAB subflow#3, 8 kbps RAB, DCCH)=<br>(TF0,TF0,TF0,TF0,TF0), (TF1,TF0,TF0,TF0,TF0), (TF2,TF1,TF0,TF0,TF0),<br>(TF3,TF2,TF0,TF0,TF0), (TF4,TF3,TF0,TF0,TF0), (TF5,TF4,TF1,TF0,TF0),<br>(TF0,TF0,TF0,TF1,TF0), (TF1,TF0,TF0,TF1,TF0), (TF2,TF1,TF0,TF1,TF0),<br>(TF3,TF2,TF0,TF1,TF0), (TF4,TF3,TF0,TF1,TF0), (TF5,TF4,TF1,TF1,TF0),<br>(TF0,TF0,TF0,TF0,TF1), (TF1,TF0,TF0,TF0,TF1), (TF2,TF1,TF0,TF0,TF1),<br>(TF3,TF2,TF0,TF0,TF1), (TF4,TF3,TF0,TF0,TF1), (TF5,TF4,TF1,TF0,TF1),<br>(TF0,TF0,TF0,TF1,TF1), (TF1,TF0,TF0,TF1,TF1), (TF2,TF1,TF0,TF1,TF1),<br>(TF3,TF2,TF0,TF1,TF1), (TF4,TF3,TF0,TF1,TF1), (TF5,TF4,TF1,TF1,TF1)<br>(alt.<br>(TF0,TF0,TF0,TF0,TF0), (TF1,TF0,TF0,TF0,TF0), (TF2,TF1,TF0,TF0,TF0),<br>(TF3,TF2,TF0,TF0,TF0), (TF4,TF3,TF0,TF0,TF0), (TF5,TF4,TF1,TF0,TF0),<br>(TF0,TF0,TF0,TF1,TF0), (TF1,TF0,TF0,TF1,TF0), (TF2,TF1,TF0,TF1,TF0),<br>(TF3,TF2,TF0,TF1,TF0), (TF4,TF3,TF0,TF1,TF0), (TF5,TF4,TF1,TF1,TF0),<br>(TF0,TF0,TF0,TF2,TF0), (TF1,TF0,TF0,TF2,TF0), (TF2,TF1,TF0,TF2,TF0),<br>(TF3,TF2,TF0,TF2,TF0), (TF4,TF3,TF0,TF2,TF0), (TF5,TF4,TF1,TF2,TF0),<br>(TF0,TF0,TF0,TF0,TF1), (TF1,TF0,TF0,TF0,TF1), (TF2,TF1,TF0,TF0,TF1),<br>(TF3,TF2,TF0,TF0,TF1), (TF4,TF3,TF0,TF0,TF1), (TF5,TF4,TF1,TF0,TF1),<br>(TF0,TF0,TF0,TF1,TF1), (TF1,TF0,TF0,TF1,TF1), (TF2,TF1,TF0,TF1,TF1),<br>(TF3,TF2,TF0,TF1,TF1), (TF4,TF3,TF0,TF1,TF1), (TF5,TF4,TF1,TF1,TF1)<br>(TF0,TF0,TF0,TF2,TF1), (TF1,TF0, TF0, TF2, TF1)) |
| NOTE: In case TB size zero is configured for any transport channel the first TFC is required; it is optional otherwise. | |

6.10.3.4.1.38f.1.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Uplink | Midamble | 512 chips |

| | | |
|---|---|---|
| | Codes and time slots | SF8 x 1 code x 1 time slot |
| | Max. Number of data bits/radio frame | 452 bits |
| | TFCI code word | 16 bits |
| | TPC | 2 bits |
| | Puncturing Limit | 0.48 (alt.0.56) |

6.10.3.4.1.38f.2 Downlink

6.10.3.4.1.38f.2.1 Transport channel parameters

6.10.3.4.1.38f.2.1.1 Transport channel parameters for Conversational / speech / DL: (12.2, 7.95, 5.9, 4.75) kbps / CS RAB

See clause 6.10.3.4.1.4a.2.1.1.

6.10.3.4.1.38f.2.1.2 Transport channel parameters for Interactive or background / DL:8 kbps / PS RAB

See clause 6.10.3.4.1.23.2.1.1.

6.10.3.4.1.38f.2.1.3 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.10.3.4.1.38f.2.1.4 TFCS

| | |
|---|---|
| TFCS size | 24 |
| TFCS | (RAB subflow#1, RAB subflow#2, RAB subflow#3, 8 kbps RAB, DCCH)=<br>(TF0,TF0,TF0,TF0,TF0), (TF1,TF0,TF0,TF0,TF0), (TF2,TF1,TF0,TF0,TF0),<br>(TF3,TF2,TF0,TF0,TF0), (TF4,TF3,TF0,TF0,TF0), (TF5,TF4,TF1,TF0,TF0),<br>(TF0,TF0,TF0,TF1,TF0), (TF1,TF0,TF0,TF1,TF0), (TF2,TF1,TF0,TF1,TF0),<br>(TF3,TF2,TF0,TF1,TF0), (TF4,TF3,TF0,TF1,TF0), (TF5,TF4,TF1,TF1,TF0),<br>(TF0,TF0,TF0,TF0,TF1), (TF1,TF0,TF0,TF0,TF1), (TF2,TF1,TF0,TF0,TF1),<br>(TF3,TF2,TF0,TF0,TF1), (TF4,TF3,TF0,TF0,TF1), (TF5,TF4,TF1,TF0,TF1),<br>(TF0,TF0,TF0,TF1,TF1), (TF1,TF0,TF0,TF1,TF1), (TF2,TF1,TF0,TF1,TF1),<br>(TF3,TF2,TF0,TF1,TF1), (TF4,TF3,TF0,TF1,TF1), (TF5,TF4,TF1,TF1,TF1) |
| NOTE: In case TB size zero is configured for any transport channel the first TFC is required; it is optional otherwise. | |

6.10.3.4.1.38f.2.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Downlink | Midamble | 512 chips |
| | Codes and time slots | SF16 x 2 codes x 1 time slot |
| | Max. Number of data bits/radio frame | 472 bits |
| | TFCI code word | 16 bits |
| | Puncturing limit | 0.52 |

6.10.3.4.1.38g Conversational / speech / (12.2, 7.95, 5.9, 4.75) kbps / CS RAB + Interactive or background / UL:16 DL:16 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.3.4.1.38g.1 Uplink

6.10.3.4.1.38g.1.1 Transport channel parameters

6.10.3.4.1.38g.1.1.1 Transport channel parameters for Conversational / speech / UL: (12.2, 7.95, 5.9, 4.75) kbps / CS RAB

See clause 6.10.3.4.1.4a.1.1.1.

6.10.3.4.1.38g.1.1.2 Transport channel parameters for Interactive or background / UL:16 kbps / PS RAB

See clause 6.10.3.4.1.23b.1.1.1.

6.10.3.4.1.38g.1.1.3 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.10.3.4.1.38g.1.1.4 TFCS

| | |
|---|---|
| TFCS size | 32 (alt. 31) |
| TFCS | (RAB subflow#1, RAB subflow#2, RAB subflow#3, 16 kbps RAB, DCCH)=<br>(TF0,TF0,TF0,TF0,TF0), (TF1,TF0,TF0,TF0,TF0), (TF2,TF1,TF0,TF0,TF0),<br>(TF3,TF2,TF0,TF0,TF0), (TF4,TF3,TF0,TF0,TF0), (TF5,TF4,TF1,TF0,TF0),<br>(TF0,TF0,TF0,TF1,TF0), (TF1,TF0,TF0,TF1,TF0), (TF3,TF2,TF0,TF1,TF0),<br>(TF4,TF3,TF0,TF1,TF0), (TF5,TF4,TF1,TF1,TF0), (TF0,TF0,TF0,TF2,TF0),<br>(TF1,TF0,TF0,TF2,TF0), (TF3,TF2,TF0,TF2,TF0), (TF4,TF3,TF0,TF2,TF0),<br>(TF5,TF4,TF1,TF2,TF0), (TF0,TF0,TF0,TF0,TF1), (TF1,TF0,TF0,TF0,TF1),<br>(TF2,TF1,TF0,TF0,TF1), (TF3,TF2,TF0,TF0,TF1), (TF4,TF3,TF0,TF0,TF1),<br>(TF5,TF4,TF1,TF0,TF1), (TF0,TF0,TF0,TF1,TF1), (TF1,TF0,TF0,TF1,TF1),<br>(TF3,TF2,TF0,TF1,TF1), (TF4,TF3,TF0,TF1,TF1), (TF5,TF4,TF1,TF1,TF1),<br>(TF0,TF0,TF0,TF2,TF1), (TF1,TF0,TF0,TF2,TF1), (TF3,TF2,TF0,TF2,TF1),<br>(TF4,TF3,TF0,TF2,TF1), (TF5,TF4,TF1,TF2,TF1)<br>(alt.<br>(TF0,TF0,TF0,TF0,TF0), (TF1,TF0,TF0,TF0,TF0), (TF2,TF1,TF0,TF0,TF0),<br>(TF3,TF2,TF0,TF0,TF0), (TF4,TF3,TF0,TF0,TF0), (TF5,TF4,TF1,TF0,TF0),<br>(TF0,TF0,TF0,TF1,TF0), (TF1,TF0,TF0,TF1,TF0), (TF3,TF2,TF0,TF1,TF0),<br>(TF4,TF3,TF0,TF1,TF0), (TF5,TF4,TF1,TF1,TF0), (TF0,TF0,TF0,TF2,TF0),<br>(TF1,TF0,TF0,TF2,TF0), (TF3,TF2,TF0,TF2,TF0), (TF4,TF3,TF0,TF2,TF0),<br>(TF5,TF4,TF1,TF2,TF0), (TF0,TF0,TF0,TF0,TF1), (TF1,TF0,TF0,TF0,TF1),<br>(TF2,TF1,TF0,TF0,TF1), (TF3,TF2,TF0,TF0,TF1), (TF4,TF3,TF0,TF0,TF1),<br>(TF5,TF4,TF1,TF0,TF1), (TF0,TF0,TF0,TF1,TF1), (TF1,TF0,TF0,TF1,TF1),<br>(TF3,TF2,TF0,TF1,TF1), (TF4,TF3,TF0,TF1,TF1), (TF5,TF4,TF1,TF1,TF1),<br>(TF0,TF0,TF0,TF2,TF1), (TF1,TF0,TF0,TF2,TF1), (TF3,TF2,TF0,TF2,TF1),<br>(TF4,TF3,TF0,TF2,TF1)) |
| NOTE 1: In case TB size zero is configured for any transport channel the first TFC is required; it is optional otherwise.<br>NOTE 2: The alt. TFCS is used when the 16Kbps RAB alt. is used. | |

6.10.3.4.1.38g.1.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Uplink | Midamble | 512 chips |
| | Codes and time slots | SF8 x 1 code x 1 time slot +<br>SF16 x 1 code x 1 time slot |
| | Max. Number of data bits/radio frame | 664 bits (alt. 696 bits) |
| | TFCI code word | 32 bits (alt. 16 bits) |
| | TPC | 2 bits |
| | Puncturing Limit | 0.56 (alt. 0.60) |

6.10.3.4.1.38g.2 Downlink

6.10.3.4.1.38g.2.1 Transport channel parameters

6.10.3.4.1.38g.2.1.1 Transport channel parameters for Conversational / speech / DL: (12.2, 7.95, 5.9, 4.75) kbps / CS RAB

See clause 6.10.3.4.1.4a.2.1.1.

6.10.3.4.1.38g.2.1.2 Transport channel parameters for Interactive or background / DL:16 kbps / PS RAB

See clause 6.10.3.4.1.23b.2.1.1.

6.10.3.4.1.38g.2.1.3 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1

6.10.3.4.1.38g.2.1.4 TFCS

| TFCS size | 36 |
|---|---|
| TFCS | (RAB subflow#1, RAB subflow#2, RAB subflow#3, 16 kbps RAB, DCCH)=<br>(TF0,TF0,TF0,TF0,TF0), (TF1,TF0,TF0,TF0,TF0), (TF2,TF1,TF0,TF0,TF0),<br>(TF3,TF2,TF0,TF0,TF0), (TF4,TF3,TF0,TF0,TF0), (TF5,TF4,TF1,TF0,TF0),<br>(TF0,TF0,TF0,TF1,TF0), (TF1,TF0,TF0,TF1,TF0), (TF2,TF1,TF0,TF1,TF0),<br>(TF3,TF2,TF0,TF1,TF0), (TF4,TF3,TF0,TF1,TF0), (TF5,TF4,TF1,TF1,TF0),<br>(TF0,TF0,TF0,TF2,TF0), (TF1,TF0,TF0,TF2,TF0), (TF2,TF1,TF0,TF2,TF0),<br>(TF3,TF2,TF0,TF2,TF0), (TF4,TF3,TF0,TF2,TF0), (TF5,TF4,TF1,TF2,TF0),<br>(TF0,TF0,TF0,TF0,TF1), (TF1,TF0,TF0,TF0,TF1), (TF2,TF1,TF0,TF0,TF1),<br>(TF3,TF2,TF0,TF0,TF1), (TF4,TF3,TF0,TF0,TF1), (TF5,TF4,TF1,TF0,TF1),<br>(TF0,TF0,TF0,TF1,TF1), (TF1,TF0,TF0,TF1,TF1), (TF2,TF1,TF0,TF1,TF1),<br>(TF3,TF2,TF0,TF1,TF1), (TF4,TF3,TF0,TF1,TF1), (TF5,TF4,TF1,TF1,TF1),<br>(TF0,TF0,TF0,TF2,TF1), (TF1,TF0,TF0,TF2,TF1), (TF2,TF1,TF0,TF2,TF1),<br>(TF3,TF2,TF0,TF2,TF1), (TF4,TF3,TF0,TF2,TF1), (TF5,TF4,TF1,TF2,TF1) |
| NOTE: In case TB size zero is configured for any transport channel the first TFC is required; it is optional otherwise. | |

6.10.3.4.1.38g.2.2 Physical channel parameters

| DPCH Downlink | Midamble | 512 chips |
|---|---|---|
| | Codes and time slots | SF16 x 3 codes x 1 time slot |
| | Max. Number of data bits/radio frame | 700 bits |
| | TFCI code word | 32 bits |
| | Puncturing limit | 0.56 |

6.10.3.4.1.38h Conversational / speech / (12.2, 7.95, 5.9, 4.75) kbps / CS RAB + Interactive or background / UL:32 DL:32 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.3.4.1.38h.1 Uplink

6.10.3.4.1.38h.1.1 Transport channel parameters

6.10.3.4.1.38h.1.1.1 Transport channel parameters for Conversational / speech / UL: (12.2, 7.95, 5.9, 4.75) kbps / CS RAB

See clause 6.10.3.4.1.4a.1.1.1.

6.10.3.4.1.38h.1.1.2 Transport channel parameters for Interactive or background / UL:32 kbps / PS RAB

See clause 6.10.3.4.1.23d.1.1.1.

6.10.3.4.1.38h.1.1.3 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.10.3.4.1.38h.1.1.4 TFCS

| TFCS size | 32 |
|---|---|
| TFCS | (RAB subflow#1, RAB subflow#2, RAB subflow#3, 32 kbps RAB, DCCH)=<br>(TF0,TF0,TF0,TF0,TF0), (TF1,TF0,TF0,TF0,TF0), (TF2,TF1,TF0,TF0,TF0),<br>(TF3,TF2,TF0,TF0,TF0), (TF4,TF3,TF0,TF0,TF0), (TF5,TF4,TF1,TF0,TF0),<br>(TF0,TF0,TF0,TF1,TF0), (TF1,TF0,TF0,TF1,TF0), (TF2,TF1,TF0,TF1,TF0),<br>(TF3,TF2,TF0,TF1,TF0), (TF4,TF3,TF0,TF1,TF0), (TF5,TF4,TF1,TF1,TF0),<br>(TF0,TF0,TF0,TF0,TF1), (TF1,TF0,TF0,TF0,TF1), (TF2,TF1,TF0,TF0,TF1),<br>(TF3,TF2,TF0,TF0,TF1), (TF4,TF3,TF0,TF0,TF1), (TF5,TF4,TF1,TF0,TF1),<br>(TF0,TF0,TF0,TF1,TF1), (TF1,TF0,TF0,TF1,TF1), (TF2,TF1,TF0,TF1,TF1),<br>(TF3,TF2,TF0,TF1,TF1), (TF4,TF3,TF0,TF1,TF1), (TF5,TF4,TF1,TF1, TF1),<br>(TF0,TF0,TF0,TF2,TF0), (TF1,TF0,TF0,TF2,TF0), (TF2,TF1,TF0,TF2,TF0),<br>(TF3,TF2,TF0,TF2,TF0), (TF4,TF3,TF0,TF2,TF0), (TF5,TF4,TF1,TF2,TF0),<br>(TF0,TF0,TF0,TF2,TF1), (TF1,TF0,TF0,TF2,TF1) |
| NOTE: In case TB size zero is configured for any transport channel the first TFC is required; it is optional otherwise. | |

6.10.3.4.1.38h.1.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Uplink | Midamble | 512 chips |
| | Codes and time slots | SF4 x 1 code x 1 time slot + SF16 x 1 code x 1 time slot |
| | Max. Number of data bits/radio frame | 1 084 bits |
| | TFCI code word | 32 bits |
| | TPC | 2 bits |
| | Puncturing Limit | 0.68 (alt.0.60) |

6.10.3.4.1.38h.2 Downlink

6.10.3.4.1.38h.2.1 Transport channel parameters

6.10.3.4.1.38h.2.1.1 Transport channel parameters for Conversational / speech / DL: (12.2, 7.95, 5.9, 4.75) kbps / CS RAB

See clause 6.10.3.4.1.4a.2.1.1.

6.10.3.4.1.38h.2.1.2 Transport channel parameters for Interactive or background / DL:32 kbps / PS RAB

See clause 6.10.3.4.1.23d.2.1.1.

6.10.3.4.1.38h.2.1.3 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.10.3.4.1.38h.2.1.4 TFCS

| | |
|---|---|
| TFCS size | 32 |
| TFCS | (RAB subflow#1, RAB subflow#2, RAB subflow#3, 32 kbps RAB, DCCH)= (TF0,TF0,TF0,TF0,TF0), (TF1,TF0,TF0,TF0,TF0), (TF2,TF1,TF0,TF0,TF0), (TF3,TF2,TF0,TF0,TF0), (TF4,TF3,TF0,TF0,TF0), (TF5,TF4,TF1,TF0,TF0), (TF0,TF0,TF0,TF1,TF0), (TF1,TF0,TF0,TF1,TF0), (TF2,TF1,TF0,TF1,TF0), (TF3,TF2,TF1,TF0,TF0), (TF4,TF3,TF0,TF1,TF0), (TF5,TF4,TF1,TF1,TF0), (TF0,TF0,TF0,TF0,TF1), (TF1,TF0,TF0,TF0,TF1), (TF2,TF1,TF0,TF0,TF1), (TF3,TF2,TF0,TF0,TF1), (TF4,TF3,TF0,TF0,TF1), (TF5,TF4,TF1,TF0,TF1), (TF0,TF0,TF0,TF1,TF1), (TF1,TF0,TF0,TF1,TF1), (TF2,TF1,TF0,TF1,TF1), (TF3,TF2,TF0,TF1,TF1), (TF4,TF3,TF0,TF1,TF1), (TF5,TF4,TF1,TF1, TF1), (TF0,TF0,TF0,TF2,TF0), (TF1,TF0,TF0,TF2,TF0), (TF2,TF1,TF0,TF2,TF0), (TF3,TF2,TF0,TF2,TF0), (TF4,TF3,TF0,TF2,TF0), (TF5,TF4,TF1,TF2,TF0), (TF0,TF0,TF0,TF2,TF1), (TF1,TF0,TF0,TF2,TF1) |
| NOTE: In case TB size zero is configured for any transport channel the first TFC is required; it is optional otherwise. | |

6.10.3.4.1.38h.2.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Downlink | Midamble | 512 chips |
| | Codes and time slots | SF16 x 4 codes x 1 time slot |
| | Max. Number of data bits/radio frame | 944 |
| | TFCI code word | 32 bits |
| | Puncturing limit | 0.60 |

6.10.3.4.1.38i Conversational / speech / (12.2, 7.95, 5.9, 4.75) kbps / CS RAB + Interactive or background / UL:64 DL:64 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.3.4.1.38i.1 Uplink

6.10.3.4.1.38i.1.1 Transport channel parameters

6.10.3.4.1.38i.1.1.1 Transport channel parameters for Conversational / speech / UL: (12.2, 7.95, 5.9, 4.75) kbps / CS RAB

See clause 6.10.3.4.1.4a.1.1.1.

6.10.3.4.1.38i.1.1.2 Transport channel parameters for Interactive or background / UL:64 kbps / PS RAB

See clause 6.10.3.4.1.26.1.1.1.

6.10.3.4.1.38i.1.1.3 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.10.3.4.1.38i.1.1.4 TFCS

| | |
|---|---|
| TFCS size | 48 |
| TFCS | (RAB subflow#1, RAB subflow#2, RAB subflow#3, 64 kbps RAB, DCCH)=<br>(TF0,TF0,TF0,TF0,TF0), (TF1,TF0,TF0,TF0,TF0), (TF2,TF1,TF0,TF0,TF0),<br>(TF3,TF2,TF0,TF0,TF0), (TF4,TF3,TF0,TF0,TF0), (TF5,TF4,TF1,TF0,TF0),<br>(TF0,TF0,TF0,TF1,TF0), (TF1,TF0,TF0,TF1,TF0), (TF2,TF1,TF0,TF1,TF0),<br>(TF3,TF2,TF0,TF1,TF0), (TF4,TF3,TF0,TF1,TF0), (TF5,TF4,TF1,TF1,TF0),<br>(TF0,TF0,TF0,TF2,TF0), (TF1,TF0,TF0,TF2,TF0), (TF2,TF1,TF0,TF2,TF0),<br>(TF3,TF2,TF0,TF2,TF0), (TF4,TF3,TF0,TF2,TF0), (TF5,TF4,TF1,TF2,TF0),<br>(TF0,TF0,TF0,TF4,TF0), (TF1,TF0,TF0,TF4,TF0), (TF2,TF1,TF0,TF4,TF0),<br>(TF3,TF2,TF0,TF4,TF0), (TF4,TF3,TF0,TF4,TF0), (TF5,TF4,TF1,TF4,TF0),<br>(TF0,TF0,TF0,TF0,TF1), (TF1,TF0,TF0,TF0,TF1), (TF2,TF1,TF0,TF0,TF1),<br>(TF3,TF2,TF0,TF0,TF1), (TF4,TF3,TF0,TF0,TF1), (TF5,TF4,TF1,TF0,TF1),<br>(TF0,TF0,TF0,TF1,TF1), (TF1,TF0,TF0,TF1,TF1), (TF2,TF1,TF0,TF1,TF1),<br>(TF3,TF2,TF0,TF1,TF1), (TF4,TF3,TF0,TF1,TF1), (TF5,TF4,TF1,TF1,TF1),<br>(TF0,TF0,TF0,TF2,TF1), (TF1,TF0,TF0,TF2,TF1), (TF2,TF1,TF0,TF2,TF1),<br>(TF3,TF2,TF0,TF2,TF1), (TF4,TF3,TF0,TF2,TF1), (TF5,TF4,TF1,TF2,TF1),<br>(TF0,TF0,TF0,TF4,TF1), (TF1,TF0,TF0,TF4,TF1), (TF2,TF1,TF0,TF4,TF1),<br>(TF3,TF2,TF0,TF4,TF1), (TF4,TF3,TF0,TF4,TF1), (TF5,TF4,TF1,TF4,TF1) |
| NOTE: In case TB size zero is configured for any transport channel the first TFC is required; it is optional otherwise. | |

6.10.3.4.1.38i.1.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Uplink | Midamble | 256 chips |
| | Codes and time slots | SF2 x 1 code x 1 time slot |
| | Max. Number of data bits/radio frame | 1 936 bits |
| | TFCI code word | 32 bit |
| | TPC | 2 bits |
| | Puncturing Limit | 0.68 (alt.0.60) |

6.10.3.4.1.38i.2 Downlink

6.10.3.4.1.38i.2.1 Transport channel parameters

6.10.3.4.1.38i.2.1.1 Transport channel parameters for Conversational / speech / DL: (12.2, 7.95, 5.9, 4.75) kbps / CS RAB

See clause 6.10.3.4.1.4a.2.1.1.

6.10.3.4.1.38i.2.1.2 Transport channel parameters for Interactive or background / DL:64 kbps / PS RAB

See clause 6.10.3.4.1.25.2.1.1.

6.10.3.4.1.38i.2.1.3 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.10.3.4.1.38i.2.1.4 TFCS

| | |
|---|---|
| TFCS size | 60 |
| TFCS | (RAB subflow#1, RAB subflow#2, RAB subflow#3, 64 kbps RAB, DCCH)=<br>(TF0,TF0,TF0,TF0,TF0), (TF1,TF0,TF0,TF0,TF0), (TF2,TF1,TF0,TF0,TF0),<br>(TF3,TF2,TF0,TF0,TF0), (TF4,TF3,TF0,TF0,TF0), (TF5,TF4,TF1,TF0,TF0),<br>(TF0,TF0,TF0,TF1,TF0), (TF1,TF0,TF0,TF1,TF0), (TF2,TF1,TF0,TF1,TF0),<br>(TF3,TF2,TF0,TF1,TF0), (TF4,TF3,TF0,TF1,TF0), (TF5,TF4,TF1,TF1,TF0),<br>(TF0,TF0,TF0,TF2,TF0), (TF1,TF0,TF0,TF2,TF0), (TF2,TF1,TF0,TF2,TF0), |

| | |
|---|---|
| | (TF3,TF2,TF0,TF2,TF0), (TF4,TF3,TF0,TF2,TF0), (TF5,TF4,TF1,TF2,TF0), (TF0,TF0,TF0,TF3,TF0), (TF1,TF0,TF0,TF3,TF0), (TF2,TF1,TF0,TF3,TF0), (TF3,TF2,TF0,TF3,TF0), (TF4,TF3,TF0,TF3,TF0), (TF5,TF4,TF1,TF3,TF0), (TF0,TF0,TF0,TF4,TF0), (TF1,TF0,TF0,TF4,TF0), (TF2,TF1,TF0,TF4,TF0), (TF3,TF2,TF0,TF4,TF0), (TF4,TF3,TF0,TF4,TF0), (TF5,TF4,TF1,TF4,TF0), (TF0,TF0,TF0,TF0,TF1), (TF1,TF0,TF0,TF0,TF1), (TF2,TF1,TF0,TF0,TF1), (TF3,TF2,TF0,TF0,TF1), (TF4,TF3,TF0,TF0,TF1), (TF5,TF4,TF1,TF0,TF1), (TF0,TF0,TF0,TF1,TF1), (TF1,TF0,TF0,TF1,TF1), (TF2,TF1,TF0,TF1,TF1), (TF3,TF2,TF0,TF1,TF1), (TF4,TF3,TF0,TF1,TF1), (TF5,TF4,TF1,TF1,TF1), (TF0,TF0,TF0,TF2,TF1), (TF1,TF0,TF0,TF2,TF1), (TF2,TF1,TF0,TF2,TF1), (TF3,TF2,TF0,TF2,TF1), (TF4,TF3,TF0,TF2,TF1), (TF5,TF4,TF1,TF2,TF1), (TF0,TF0,TF0,TF3,TF1), (TF1,TF0,TF0,TF3,TF1), (TF2,TF1,TF0,TF3,TF1), (TF3,TF2,TF0,TF3,TF1), (TF4,TF3,TF0,TF3,TF1), (TF5,TF4,TF1,TF3,TF1), (TF0,TF0,TF0,TF4,TF1), (TF1,TF0,TF0,TF4,TF1), (TF2,TF1,TF0,TF4,TF1), (TF3,TF2,TF0,TF4,TF1), (TF4,TF3,TF0,TF4,TF1), (TF5,TF4,TF1,TF4,TF1) |
| NOTE: In case TB size zero is configured for any transport channel the first TFC is required; it is optional otherwise. | |

6.10.3.4.1.38i.2.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Downlink | Midamble | 256 chips |
| | Codes and time slots | SF16 x 7 codes x 1 time slot |
| | Max. Number of data bits/radio frame | 1 900 bits |
| | TFCI code word | 32 bits |
| | Puncturing limit | 0.68 |

6.10.3.4.1.38j Conversational / speech / (12.2, 7.95, 5.9, 4.75) kbps / CS RAB + Interactive or background / UL:64 DL:128 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.3.4.1.38j.1 Uplink

See clause 6.10.3.4.1.38i.1

6.10.3.4.1.38j.2 Downlink

6.10.3.4.1.38j.2.1 Transport channel parameters

6.10.3.4.1.38j.2.1.1 Transport channel parameters for Conversational / speech / DL: (12.2, 7.95, 5.9, 4.75) kbps / CS RAB

See clause 6.10.3.4.1.4a.2.1.1.

6.10.3.4.1.38j.2.1.2 Transport channel parameters for Interactive or background / DL:128 kbps / PS RAB

See clause 6.10.3.4.1.27.2.1.1.

6.10.3.4.1.38j.2.1.3 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.10.3.4.1.38j.2.1.4 TFCS

| | |
|---|---|
| TFCS size | 60 |
| TFCS | (RAB subflow#1, RAB subflow#2, RAB subflow#3, 128 kbps RAB, DCCH)= (TF0,TF0,TF0,TF0,TF0), (TF1,TF0,TF0,TF0,TF0), (TF2,TF1,TF0,TF0,TF0), (TF3,TF2,TF0,TF0,TF0), (TF4,TF3,TF0,TF0,TF0), (TF5,TF4,TF1,TF0,TF0), (TF0,TF0,TF0,TF1,TF0), (TF1,TF0,TF0,TF1,TF0), (TF2,TF1,TF0,TF1,TF0), (TF3,TF2,TF0,TF1,TF0), (TF4,TF3,TF0,TF1,TF0), (TF5,TF4,TF1,TF1,TF0), (TF0,TF0,TF0,TF2,TF0), (TF1,TF0,TF0,TF2,TF0), (TF2,TF1,TF0,TF2,TF0), (TF3,TF2,TF0,TF2,TF0), (TF4,TF3,TF0,TF2,TF0), (TF5,TF4,TF1,TF2,TF0), (TF0,TF0,TF0,TF3,TF0), (TF1,TF0,TF0,TF3,TF0), (TF2,TF1,TF0,TF3,TF0), (TF3,TF2,TF0,TF3,TF0), (TF4,TF3,TF0,TF3,TF0), (TF5,TF4,TF1,TF3,TF0), (TF0,TF0,TF0,TF4,TF0), (TF1,TF0,TF0,TF4,TF0), (TF2,TF1,TF0,TF4,TF0), (TF3,TF2,TF0,TF4,TF0), (TF4,TF3,TF0,TF4,TF0), (TF5,TF4,TF1,TF4,TF0), (TF0,TF0,TF0,TF0,TF1), (TF1,TF0,TF0,TF0,TF1), (TF2,TF1,TF0,TF0,TF1), (TF3,TF2,TF0,TF0,TF1), (TF4,TF3,TF0,TF0,TF1), (TF5,TF4,TF1,TF0,TF1), |

| | |
|---|---|
| | (TF0,TF0,TF0,TF1,TF1), (TF1,TF0,TF0,TF1,TF1), (TF2,TF1,TF0,TF1,TF1), (TF3,TF2,TF0,TF1,TF1), (TF4,TF3,TF0,TF1,TF1), (TF5,TF4,TF1,TF1,TF1), (TF0,TF0,TF0,TF2,TF1), (TF1,TF0,TF0,TF2,TF1), (TF2,TF1,TF0,TF2,TF1), (TF3,TF2,TF0,TF2,TF1), (TF4,TF3,TF0,TF2,TF1), (TF5,TF4,TF1,TF2,TF1), (TF0,TF0,TF0,TF3,TF1), (TF1,TF0,TF0,TF3,TF1), (TF2,TF1,TF0,TF3,TF1), (TF3,TF2,TF0,TF3,TF1), (TF4,TF3,TF0,TF3,TF1), (TF5,TF4,TF1,TF3,TF1), (TF0,TF0,TF0,TF4,TF1), (TF1,TF0,TF0,TF4,TF1), (TF2,TF1,TF0,TF4,TF1), (TF3,TF2,TF0,TF4,TF1), (TF4,TF3,TF0,TF4,TF1), (TF5,TF4,TF1,TF4,TF1) |
| NOTE: In case TB size zero is configured for any transport channel the first TFC is required; it is optional otherwise. | |

6.10.3.4.1.38j.2.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Downlink | Midamble | 256 chips |
| | Codes and time slots | SF16 x 6 codes x 2 time slots |
| | Max. Number of data bits/radio frame | 3 280 bits |
| | TFCI code word | 32 bits |
| | Puncturing limit | 0.64 |

6.10.3.4.1.39 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:32 DL:64 kbps / PS RAB+ UL:3.4 DL: 3.4 kbps SRBs for DCCH

6.10.3.4.1.39.1 Uplink

See clause 6.10.3.4.1.38.1.

6.10.3.4.1.39.2 Downlink

6.10.3.4.1.39.2.1 Transport channel parameters

6.10.3.4.1.39.2.1.1 Transport channel parameters for Conversational / speech / DL:12.2 kbps / CS RAB

See clause 6.10.3.4.1.4.2.1.1.

6.10.3.4.1.39.2.1.2 Transport channel parameters for Interactive or background / DL:64 kbps / PS RAB

See clause 6.10.3.4.1.25.2.1.1.

6.10.3.4.1.39.2.1.3 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.10.3.4.1.39.2.1.4 TFCS

| | |
|---|---|
| TFCS size | 30 |
| TFCS | (RAB subflow#1, RAB subflow#2, RAB subflow#3, 64 kbps RAB , DCCH)= (TF0, TF0, TF0, TF0, TF0), (TF1, TF0, TF0, TF0, TF0), (TF2, TF1, TF1, TF0, TF0), (TF0, TF0, TF0, TF1, TF0), (TF1, TF0, TF0, TF1, TF0), (TF2, TF1, TF1, TF1, TF0), (TF0, TF0, TF0, TF2, TF0), (TF1, TF0, TF0, TF2, TF0), (TF2, TF1, TF1, TF2, TF0), (TF0, TF0, TF0, TF3, TF0), (TF1, TF0, TF0, TF3, TF0), (TF2, TF1, TF1, TF3, TF0), (TF0, TF0, TF0, TF4, TF0), (TF1, TF0, TF0, TF4, TF0), (TF2, TF1, TF1, TF4, TF0), (TF0, TF0, TF0, TF0, TF1), (TF1, TF0, TF0, TF0, TF1), (TF2, TF1, TF1, TF0, TF1), (TF0, TF0, TF0, TF1, TF1), (TF1, TF0, TF0, TF1, TF1), (TF2, TF1, TF1, TF1, TF1), (TF0, TF0, TF0, TF2, TF1), (TF1, TF0, TF0, TF2, TF1), (TF2, TF1, TF1, TF2, TF1), (TF0, TF0, TF0, TF3, TF1), (TF1, TF0, TF0, TF3, TF1), (TF2, TF1, TF1, TF3, TF1), (TF0, TF0, TF0, TF4, TF1), (TF1, TF0, TF0, TF4, TF1), (TF2, TF1, TF1, TF4, TF1) |
| NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise. | |

6.10.3.4.1.39.2.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Downlink | Midamble | 512 chips |
| | Codes and time slots | SF16 x 8 codes x 1 time slot |
| | Max. Number of data bits/radio frame | 1 936 bits |
| | TFCI code word | 16 bits |

| | | |
|---|---|---|
| | Puncturing limit | 0.68 |

6.10.3.4.1.40 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:64 DL:64 kbps / PS RAB+ UL:3.4 DL: 3.4 kbps SRBs for DCCH

6.10.3.4.1.40.1 Uplink

6.10.3.4.1.40.1.1 Transport channel parameters

6.10.3.4.1.40.1.1.1 Transport channel parameters for Conversational / speech / UL:12.2 kbps / CS RAB

See clause 6.10.3.4.1.4.1.1.1.

6.10.3.4.1.40.1.1.2 Transport channel parameters for Interactive or background / UL:64 kbps / PS RAB

See clause 6.10.3.4.1.26.1.1.1.

6.10.3.4.1.40.1.1.3 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.10.3.4.1.40.1.1.4 TFCS

6.10.3.4.1.40.1.1.4.1 TFCS (one CCTrCH case)

| | |
|---|---|
| TFCS size | 30 |
| TFCS | (RAB subflow#1, RAB subflow#2, RAB subflow#3, 64 kbps RAB , DCCH)=<br>(TF0, TF0, TF0, TF0, TF0), (TF1, TF0, TF0, TF0, TF0), (TF2, TF1, TF1, TF0, TF0),<br>(TF0, TF0, TF0, TF1, TF0), (TF1, TF0, TF0, TF1, TF0), (TF2, TF1, TF1, TF1, TF0),<br>(TF0, TF0, TF0, TF2, TF0), (TF1, TF0, TF0, TF2, TF0), (TF2, TF1, TF1, TF2, TF0),<br>(TF0, TF0, TF0, TF3, TF0), (TF1, TF0, TF0, TF3, TF0), (TF2, TF1, TF1, TF3, TF0),<br>(TF0, TF0, TF0, TF4, TF0), (TF1, TF0, TF0, TF4, TF0), (TF2, TF1, TF1, TF4, TF0),<br>(TF0, TF0, TF0, TF0, TF1), (TF1, TF0, TF0, TF0, TF1), (TF2, TF1, TF1, TF0, TF1),<br>(TF0, TF0, TF0, TF1, TF1), (TF1, TF0, TF0, TF1, TF1), (TF2, TF1, TF1, TF1, TF1),<br>(TF0, TF0, TF0, TF2, TF1), (TF1, TF0, TF0, TF2, TF1), (TF2, TF1, TF1, TF2, TF1),<br>(TF0, TF0, TF0, TF3, TF1), (TF1, TF0, TF0, TF3, TF1), (TF2, TF1, TF1, TF3, TF1),<br>(TF0, TF0, TF0, TF4, TF1), (TF1, TF0, TF0, TF4, TF1), (TF2, TF1, TF1, TF4, TF1) |
| NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise. | |

6.10.3.4.1.40.1.1.4.2 TFCS (two CCTrCH case)

6.10.3.4.1.40.1.1.4.2.1 TFCS (conversational + SRB)

| | |
|---|---|
| TFCS size | 6 |
| TFCS | (RAB subflow#1, RAB subflow#2, RAB subflow#3, 64 kbps RAB, DCCH)=<br>(TF0, TF0, TF0, TF0, TF0), (TF1, TF0, TF0, TF0, TF0), (TF2, TF1, TF1, TF0, TF0),<br>(TF0, TF0, TF0, TF0, TF1), (TF1, TF0, TF0, TF0, TF1), (TF2, TF1, TF1, TF0, TF1) |
| NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise. | |

6.10.3.4.1.40.1.1.4.2.2 TFCS (Interactive or background)

| | |
|---|---|
| TFCS size | 5 |
| TFCS | (RAB subflow#1, RAB subflow#2, RAB subflow#3, 64 kbps RAB, DCCH)=<br>(TF0, TF0, TF0, TF0, TF0), (TF0, TF0, TF0, TF1, TF0), (TF0, TF0, TF0, TF2, TF0),<br>(TF0, TF0, TF0, TF3, TF0), (TF0, TF0, TF0, TF4, TF0) |
| NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise. | |

6.10.3.4.1.40.1.2 Physical channel parameters

6.10.3.4.1.40.1.2.1 Physical channel (one CCTrCH case)

| | | |
|---|---|---|
| DPCH Uplink | Midamble | 512 chips |
| | Codes and time slots | SF2 x 1 code x 1 time slot |

| | | |
|---|---|---|
| | Max. Number of data bits/radio frame | 1 808 bits |
| | TFCI code word | 16 bit |
| | TPC | 2 bits |
| | Puncturing Limit | 0.64 (alt. 0.56) |

6.10.3.4.1.40.1.2.2 Physical channel (two CCTrCH case)

6.10.3.4.1.40.1.2.2.1 Physical channel (conversational + SRB)

| | | |
|---|---|---|
| DPCH Uplink | Midamble | 512 chips |
| | Codes and time slots | SF8 x 1 code x 1 time slot |
| | Max. Number of data bits/radio frame | 452 bits |
| | TFCI code word | 16 bits |
| | TPC | 2 bit |
| | Puncturing Limit | 0.68 |

6.10.3.4.1.40.1.2.2.2 Physical channel (Interactive or background)

| | | |
|---|---|---|
| DPCH Uplink | Midamble | 512 chips |
| | Codes and time slots | SF2 x 1 code x 1 time slot |
| | Max. Number of data bits/radio frame | 1 808 bits |
| | TFCI code word | 16 bits |
| | TPC | 2 bits |
| | Puncturing Limit | 0.84 (alt. 0.72) |

6.10.3.4.1.40.2 Downlink

Transport channel parameters

6.10.3.4.1.40.2.1.1 Transport channel parameters for Conversational / speech / DL:12.2 kbps / CS RAB

See clause 6.10.3.4.1.4.2.1.1.

6.10.3.4.1.40.2.1.2 Transport channel parameters for Interactive or background / DL:64 kbps / PS RAB

See clause 6.10.3.4.1.25.2.1.1.

6.10.3.4.1.40.2.1.3 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.10.3.4.1.40.2.1.4 TFCS

6.10.3.4.1.40.2.1.4.1 TFCS (one CCTrCH case)

See Clause 6.10.3.4.1.39.2.1.4.

6.10.3.4.1.40.2.1.4.2 TFCS (two CCTrCH case)

6.10.3.4.1.40.2.1.4.2.1 TFCS (conversational + SRB)

| | |
|---|---|
| TFCS size | 6 |
| TFCS | (RAB subflow#1, RAB subflow#2, RAB subflow#3, 64 kbps RAB, DCCH)=<br>(TF0, TF0, TF0, TF0, TF0), (TF1, TF0, TF0, TF0, TF0), (TF2, TF1, TF1, TF0, TF0),<br>(TF0, TF0, TF0, TF0, TF1), (TF1, TF0, TF0, TF0, TF1), (TF2, TF1, TF1, TF0, TF1) |
| NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise. | |

6.10.3.4.1.40.2.1.4.2.2 TFCS (Interactive or background)

| | |
|---|---|
| TFCS size | 5 |
| TFCS | (RAB subflow#1, RAB subflow#2, RAB subflow#3, 64 kbps RAB, DCCH)=<br>(TF0, TF0, TF0, TF0, TF0), (TF0, TF0, TF0, TF1, TF0), (TF0, TF0, TF0, TF2, TF0),<br>(TF0, TF0, TF0, TF3, TF0), (TF0, TF0, TF0, TF4, TF0) |
| NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise. | |

6.10.3.4.1.40.2.2 Physical channel parameters

6.10.3.4.1.40.2.2.1 Physical channel parameters (one CCTrCH)

See clause 6.10.3.4.1.39.2.2.

6.10.3.4.1.40.2.2.2 Physical channel parameters (two CCTrCHs)

6.10.3.4.1.40.2.2.2.1 Physical channel parameters (conversational + SRB)

| | | |
|---|---|---|
| DPCH Downlink | Midamble | 512 chips |
| | Codes and time slots | SF16 x 2 codes x 1 time slot |
| | Max. Number of data bits/radio frame | 472 bits |
| | TFCI code word | 16 bits |
| | Puncturing limit | 0.68 |

6.10.3.4.1.40.2.2.2.2 Physical channel parameters (Interactive or background)

| | | |
|---|---|---|
| DPCH Downlink | Midamble | 512 chips |
| | Codes and time slots | SF16 x 5 codes x 1 time slot |
| | Max. Number of data bits/radio frame | 1 204 bits |
| | TFCI code word | 16 bits |
| | Puncturing limit | 0.56 |

6.10.3.4.1.41 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:64 DL:128 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.3.4.1.41.1 Uplink

See clause 6.10.3.4.1.40.1.

6.10.3.4.1.41.2 Downlink

6.10.3.4.1.41.2.1 Transport channel parameters

6.10.3.4.1.41.2.1.1 Transport channel parameters for Conversational / speech / DL:12.2 kbps / CS RAB

See clause 6.10.3.4.1.4.2.1.1.

6.10.3.4.1.41.2.1.2 Transport channel parameters for Interactive or background / DL:128 kbps / PS RAB

See clause 6.10.3.4.1.27.2.1.1.

6.10.3.4.1.41.2.1.3 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.10.3.4.1.41.2.1.4 TFCS

6.10.3.4.1.41.2.1.4.1 TFCS (one CCTrCH case)

| | |
|---|---|
| TFCS size | 30 |
| TFCS | (RAB subflow#1, RAB subflow#2, RAB subflow#3, 128 kbps RAB , DCCH)=<br>(TF0, TF0, TF0, TF0, TF0), (TF1, TF0, TF0, TF0, TF0), (TF2, TF1, TF1, TF0, TF0),<br>(TF0, TF0, TF0, TF1, TF0), (TF1, TF0, TF0, TF1, TF0), (TF2, TF1, TF1, TF1, TF0),<br>(TF0, TF0, TF0, TF2, TF0), (TF1, TF0, TF0, TF2, TF0), (TF2, TF1, TF1, TF2, TF0),<br>(TF0, TF0, TF0, TF3, TF0), (TF1, TF0, TF0, TF3, TF0), (TF2, TF1, TF1, TF3, TF0),<br>(TF0, TF0, TF0, TF4, TF0), (TF1, TF0, TF0, TF4, TF0), (TF2, TF1, TF1, TF4, TF0),<br>(TF0, TF0, TF0, TF0, TF1), (TF1, TF0, TF0, TF0, TF1), (TF2, TF1, TF1, TF0, TF1),<br>(TF0, TF0, TF0, TF1, TF1), (TF1, TF0, TF0, TF1, TF1), (TF2, TF1, TF1, TF1, TF1),<br>(TF0, TF0, TF0, TF2, TF1), (TF1, TF0, TF0, TF2, TF1), (TF2, TF1, TF1, TF2, TF1),<br>(TF0, TF0, TF0, TF3, TF1), (TF1, TF0, TF0, TF3, TF1), (TF2, TF1, TF1, TF3, TF1),<br>(TF0, TF0, TF0, TF4, TF1), (TF1, TF0, TF0, TF4, TF1), (TF2, TF1, TF1, TF4, TF1) |

| NOTE: | In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise. |
|---|---|

6.10.3.4.1.41.2.1.4.2 TFCS (two CCTrCH case)

6.10.3.4.1.41.2.1.4.2.1 TFCS (conversational + SRB)

| TFCS size | 6 |
|---|---|
| TFCS | (RAB subflow#1, RAB subflow#2, RAB subflow#3, 128 kbps RAB , DCCH)=<br>(TF0, TF0, TF0, TF0, TF0), (TF1, TF0, TF0, TF0, TF0), (TF2, TF1, TF1, TF0, TF0),<br>(TF0, TF0, TF0, TF0, TF1), (TF1, TF0, TF0, TF0, TF1), (TF2, TF1, TF1, TF0, TF1) |
| NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise. | |

6.10.3.4.1.41.2.1.4.2.2 TFCS (Interactive or background)

| TFCS size | 5 |
|---|---|
| TFCS | (RAB subflow#1, RAB subflow#2, RAB subflow#3, 128 kbps RAB , DCCH)=<br>(TF0, TF0, TF0, TF0, TF0), (TF0, TF0, TF0, TF1, TF0), (TF0, TF0, TF0, TF2, TF0),<br>(TF0, TF0, TF0, TF3, TF0), (TF0, TF0, TF0, TF4, TF0) |
| NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise. | |

6.10.3.4.1.41.2.2 Physical channel parameters

6.10.3.4.1.41.2.2.1 Physical channel parameters (one CCTrCH)

| DPCH Downlink | Midamble | 256 chips |
|---|---|---|
| | Codes and time slots | SF16 x 5codes x 2time slots |
| | Max. Number of data bits/radio frame | 2 744 bits |
| | TFCI code word | 16 bits |
| | Puncturing limit | 0.52 |

6.10.3.4.1.41.2.2.2 Physical channel parameters (two CCTrCHs)

6.10.3.4.1.41.2.2.2.1 Physical channel parameters (conversational + SRB)

| DPCH Downlink | Midamble | 512 chips |
|---|---|---|
| | Codes and time slots | SF16 x 2 codes x 1 time slot |
| | Max. Number of data bits/radio frame | 472 bits |
| | TFCI code word | 16 bits |
| | Puncturing limit | 0.64 |

6.10.3.4.1.41.2.2.2.2 Physical channel parameters (Interactive or background)

| DPCH Downlink | Midamble | 256 chips |
|---|---|---|
| | Codes and time slots | SF16 x 8 codes x 1 time slot |
| | Max. Number of data bits/radio frame | 2 192 bits |
| | TFCI code word | 16 bits |
| | Puncturing limit | 0,48 |

6.10.3.4.1.42 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:64 DL:256 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.3.4.1.42.1 Uplink

6.10.3.4.1.42.1.1 Transport channel parameters

6.10.3.4.1.42.1.1.1 Transport channel parameters for Conversational / speech / UL:12.2 kbps / CS RAB

See clause 6.10.3.4.1.4.1.1.1.

6.10.3.4.1.42.1.1.2 Transport channel parameters for Interactive or background / UL:64 kbps / PS RAB

See clause 6.10.3.4.1.26.1.1.1.

6.10.3.4.1.42.1.1.3 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.10.3.4.1.42.1.1.4 TFCS

See clause 6.10.3.4.1.40.1.1.4.1.

6.10.3.4.1.42.1.2 Physical channel parameters

See clause 6.10.3.4.1.40.1.2.1.

6.10.3.4.1.42.2 Downlink

6.10.3.4.1.42.2.1 Transport channel parameters

6.10.3.4.1.42.2.1.1 Transport channel parameters for Conversational / speech / DL:12.2 kbps / CS RAB

See clause 6.10.3.4.1.4.2.1.1.

6.10.3.4.1.42.2.1.2 Transport channel parameters for Interactive or background / DL:256 kbps / PS RAB

See clause 6.10.3.4.1.31.2.1.1.

6.10.3.4.1.42.2.1.3 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.10.3.4.1.42.2.1.4 TFCS

| | |
|---|---|
| TFCS size | 30 (alt. 42) |
| TFCS | (RAB subflow#1, RAB subflow#2, RAB subflow#3, 256 kbps RAB, DCCH)=<br>(TF0, TF0, TF0, TF0, TF0), (TF1, TF0, TF0, TF0, TF0), (TF2, TF1, TF1, TF0, TF0),<br>(TF0, TF0, TF0, TF1, TF0), (TF1, TF0, TF0, TF1, TF0), (TF2, TF1, TF1, TF1, TF0),<br>(TF0, TF0, TF0, TF2, TF0), (TF1, TF0, TF0, TF2, TF0), (TF2, TF1, TF1, TF2, TF0),<br>(TF0, TF0, TF0, TF3, TF0), (TF1, TF0, TF0, TF3, TF0), (TF2, TF1, TF1, TF3, TF0),<br>(TF0, TF0, TF0, TF4, TF0), (TF1, TF0, TF0, TF4, TF0), (TF2, TF1, TF1, TF4, TF0),<br>(TF0, TF0, TF0, TF0, TF1), (TF1, TF0, TF0, TF0, TF1), (TF2, TF1, TF1, TF0, TF1),<br>(TF0, TF0, TF0, TF1, TF1), (TF1, TF0, TF0, TF1, TF1), (TF2, TF1, TF1, TF1, TF1),<br>(TF0, TF0, TF0, TF2, TF1), (TF1, TF0, TF0, TF2, TF1), (TF2, TF1, TF1, TF2, TF1),<br>(TF0, TF0, TF0, TF3, TF1), (TF1, TF0, TF0, TF3, TF1), (TF2, TF1, TF1, TF3, TF1),<br>(TF0, TF0, TF0, TF4, TF1), (TF1, TF0, TF0, TF4, TF1), (TF2, TF1, TF1, TF4, TF1)<br>(alt. (TF0, TF0, TF0, TF0, TF0), (TF1, TF0, TF0, TF0, TF0), (TF2, TF1, TF1, TF0, TF0),<br>(TF0, TF0, TF0, TF1, TF0), (TF1, TF0, TF0, TF1, TF0), (TF2, TF1, TF1, TF1, TF0),<br>(TF0, TF0, TF0, TF2, TF0), (TF1, TF0, TF0, TF2, TF0), (TF2, TF1, TF1, TF2, TF0),<br>(TF0, TF0, TF0, TF3, TF0), (TF1, TF0, TF0, TF3, TF0), (TF2, TF1, TF1, TF3, TF0),<br>(TF0, TF0, TF0, TF4, TF0), (TF1, TF0, TF0, TF4, TF0), (TF2, TF1, TF1, TF4, TF0),<br>(TF0, TF0, TF0, TF5, TF0), (TF1, TF0, TF0, TF5, TF0), (TF2, TF1, TF1, TF5, TF0),<br>(TF0, TF0, TF0, TF6, TF0), (TF1, TF0, TF0, TF6, TF0), (TF2, TF1, TF1, TF6, TF0),<br>(TF0, TF0, TF0, TF0, TF1), (TF1, TF0, TF0, TF0, TF1), (TF2, TF1, TF1, TF0, TF1),<br>(TF0, TF0, TF0, TF1, TF1), (TF1, TF0, TF0, TF1, TF1), (TF2, TF1, TF1, TF1, TF1),<br>(TF0, TF0, TF0, TF2, TF1), (TF1, TF0, TF0, TF2, TF1), (TF2, TF1, TF1, TF2, TF1),<br>(TF0, TF0, TF0, TF3, TF1), (TF1, TF0, TF0, TF3, TF1), (TF2, TF1, TF1, TF3, TF1),<br>(TF0, TF0, TF0, TF4, TF1), (TF1, TF0, TF0, TF4, TF1), (TF2, TF1, TF1, TF4, TF1)<br>(TF0, TF0, TF0, TF5, TF1), (TF1, TF0, TF0, TF5, TF1), (TF2, TF1, TF1, TF5, TF1),<br>(TF0, TF0, TF0, TF6, TF1), (TF1, TF0, TF0, TF6, TF1), (TF2, TF1, TF1, TF6, TF1)) |
| NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise. | |

6.10.3.4.1.42.2.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Downlink | Midamble | 256 chips |
| | Codes and time slots | SF16 x 8 codes x 2 time slots +<br>SF16 x 4 codes x 1 time slot |
| | Max. Number of data bits/radio frame | 5 504 bits (alt. 5 488) |

| | | |
|---|---|---|
| | TFCI code word | 16 bits (alt. 32) |
| | Puncturing limit | 0.60 |

6.10.3.4.1.43 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:64 DL:384 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.3.4.1.43.1 Uplink

See clause 6.10.3.4.1.40.1.

6.10.3.4.1.43.2 Downlink

6.10.3.4.1.43.2.1 Transport channel parameters

6.10.3.4.1.43.2.1.1 Transport channel parameters for Conversational / speech / DL:12.2 kbps / CS RAB

See clause 6.10.3.4.1.4.2.1.1.

6.10.3.4.1.43.2.1.2 Transport channel parameters for Interactive or background / DL:384 kbps / PS RAB

See clause 6.10.3.4.1.32.2.1.1.

6.10.3.4.1.43.2.1.3 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.10.3.4.1.43.2.1.4 TFCS

6.10.3.4.1.43.2.1.4.1 TFCS (one CCTrCH case)

| | |
|---|---|
| TFCS size | 36 (alt. 54) |
| TFCS | (RAB subflow#1, RAB subflow#2, RAB subflow#3, 384 kbps RAB, DCCH)=<br>(TF0, TF0, TF0, TF0, TF0), (TF1, TF0, TF0, TF0, TF0), (TF2, TF1, TF1, TF0, TF0),<br>(TF0, TF0, TF0, TF1, TF0), (TF1, TF0, TF0, TF1, TF0), (TF2, TF1, TF1, TF1, TF0),<br>(TF0, TF0, TF0, TF2, TF0), (TF1, TF0, TF0, TF2, TF0), (TF2, TF1, TF1, TF2, TF0),<br>(TF0, TF0, TF0, TF3, TF0), (TF1, TF0, TF0, TF3, TF0), (TF2, TF1, TF1, TF3, TF0),<br>(TF0, TF0, TF0, TF4, TF0), (TF1, TF0, TF0, TF4, TF0), (TF2, TF1, TF1, TF4, TF0),<br>(TF0, TF0, TF0, TF5, TF0), (TF1, TF0, TF0, TF5, TF0), (TF2, TF1, TF1, TF5, TF0),<br>(TF0, TF0, TF0, TF0, TF1), (TF1, TF0, TF0, TF0, TF1), (TF2, TF1, TF1, TF0, TF1),<br>(TF0, TF0, TF0, TF1, TF1), (TF1, TF0, TF0, TF1, TF1), (TF2, TF1, TF1, TF1, TF1),<br>(TF0, TF0, TF0, TF2, TF1), (TF1, TF0, TF0, TF2, TF1), (TF2, TF1, TF1, TF2, TF1),<br>(TF0, TF0, TF0, TF3, TF1), (TF1, TF0, TF0, TF3, TF1), (TF2, TF1, TF1, TF3, TF1),<br>(TF0, TF0, TF0, TF4, TF1), (TF1, TF0, TF0, TF4, TF1), (TF2, TF1, TF1, TF4, TF1)<br>(TF0, TF0, TF0, TF5, TF1), (TF1, TF0, TF0, TF5, TF1), (TF2, TF1, TF1, TF5, TF1),<br>(alt. (TF0, TF0, TF0, TF0, TF0), (TF1, TF0, TF0, TF0, TF0), (TF2, TF1, TF1, TF0, TF0),<br>(TF0, TF0, TF0, TF1, TF0), (TF1, TF0, TF0, TF1, TF0), (TF2, TF1, TF1, TF1, TF0),<br>(TF0, TF0, TF0, TF2, TF0), (TF1, TF0, TF0, TF2, TF0), (TF2, TF1, TF1, TF2, TF0),<br>(TF0, TF0, TF0, TF3, TF0), (TF1, TF0, TF0, TF3, TF0), (TF2, TF1, TF1, TF3, TF0),<br>(TF0, TF0, TF0, TF4, TF0), (TF1, TF0, TF0, TF4, TF0), (TF2, TF1, TF1, TF4, TF0),<br>(TF0, TF0, TF0, TF5, TF0), (TF1, TF0, TF0, TF5, TF0), (TF2, TF1, TF1, TF5, TF0),<br>(TF0, TF0, TF0, TF6, TF0), (TF1, TF0, TF0, TF6, TF0), (TF2, TF1, TF1, TF6, TF0),<br>(TF0, TF0, TF0, TF7, TF0), (TF1, TF0, TF0, TF7, TF0), (TF2, TF1, TF1, TF7, TF0),<br>(TF0, TF0, TF0, TF8, TF0), (TF1, TF0, TF0, TF8, TF0), (TF2, TF1, TF1, TF8, TF0),<br>(TF0, TF0, TF0, TF0, TF1), (TF1, TF0, TF0, TF0, TF1), (TF2, TF1, TF1, TF0, TF1),<br>(TF0, TF0, TF0, TF1, TF1), (TF1, TF0, TF0, TF1, TF1), (TF2, TF1, TF1, TF1, TF1),<br>(TF0, TF0, TF0, TF2, TF1), (TF1, TF0, TF0, TF2, TF1), (TF2, TF1, TF1, TF2, TF1),<br>(TF0, TF0, TF0, TF3, TF1), (TF1, TF0, TF0, TF3, TF1), (TF2, TF1, TF1, TF3, TF1),<br>(TF0, TF0, TF0, TF4, TF1), (TF1, TF0, TF0, TF4, TF1), (TF2, TF1, TF1, TF4, TF1)<br>(TF0, TF0, TF0, TF5, TF1), (TF1, TF0, TF0, TF5, TF1), (TF2, TF1, TF1, TF5, TF1)<br>(TF0, TF0, TF0, TF6, TF1), (TF1, TF0, TF0, TF6, TF1), (TF2, TF1, TF1, TF6, TF1),<br>(TF0, TF0, TF0, TF7, TF1), (TF1, TF0, TF0, TF7, TF1), (TF2, TF1, TF1, TF7, TF1)<br>(TF0, TF0, TF0, TF8, TF1), (TF1, TF0, TF0, TF8, TF1), (TF2, TF1, TF1, TF8, TF1)) |
| NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise. | |

6.10.3.4.1.43.2.1.4.2 TFCS (two CCTrCH case)

6.10.3.4.1.43.2.1.4.2.1 TFCS (conversational + SRB)

| | |
|---|---|
| TFCS size | 6 |
| TFCS | (RAB subflow#1, RAB subflow#2, RAB subflow#3, 384 kbps RAB, DCCH)=<br>(TF0, TF0, TF0, TF0, TF0), (TF1, TF0, TF0, TF0, TF0), (TF2, TF1, TF1, TF0, TF0),<br>(TF0, TF0, TF0, TF0, TF1), (TF1, TF0, TF0, TF0, TF1), (TF2, TF1, TF1, TF0, TF1) |
| NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise. | |

6.10.3.4.1.43.2.1.4.2.2 TFCS (Interactive or background)

| | |
|---|---|
| TFCS size | 6 (alt. 9) |
| TFCS | (RAB subflow#1, RAB subflow#2, RAB subflow#3, 384 kbps RAB, DCCH)=<br>(TF0, TF0, TF0, TF0, TF0), (TF0, TF0, TF0, TF1, TF0), (TF0, TF0, TF0, TF2, TF0),<br>(TF0, TF0, TF0, TF3, TF0), (TF0, TF0, TF0, TF4, TF0), (TF0, TF0, TF0, TF5, TF0)<br>(alt. (TF0, TF0, TF0, TF0, TF0), (TF0, TF0, TF0, TF1, TF0), (TF0, TF0, TF0, TF2, TF0),<br>(TF0, TF0, TF0, TF3, TF0), (TF0, TF0, TF0, TF4, TF0), (TF0, TF0, TF0, TF5, TF0),<br>(TF0, TF0, TF0, TF6, TF0), (TF0, TF0, TF0, TF7, TF0), (TF0, TF0, TF0, TF8, TF0)) |
| NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise. | |

6.10.3.4.1.43.2.2 Physical channel parameters

6.10.3.4.1.43.2.2.1 Physical channel parameters (one CCTrCH)

| | | |
|---|---|---|
| DPCH Downlink | Midamble | 256 chips |
| | Codes and time slots | SF16 x 8 codes x 3 time slots |
| | Max. Number of data bits/radio frame | 6 592 bits |
| | TFCI code word | 32 bits |
| | Puncturing limit | 0.48 |

6.10.3.4.1.43.2.2.2 Physical channel parameters (two CCTrCHs)

6.10.3.4.1.43.2.2.2.1 Physical channel parameters (conversational + SRB)

| | | |
|---|---|---|
| DPCH Downlink | Midamble | 512 chips |
| | Codes and time slots | SF16 x 2 codes x 1 time slot |
| | Max. Number of data bits/radio frame | 472 bits |
| | TFCI code word | 16 bits |
| | Puncturing limit | 0.60 |

6.10.3.4.1.43.2.2.2.2 Physical channel parameters (Interactive or background)

| | | |
|---|---|---|
| DPCH Downlink | Midamble | 256 chips |
| | Codes and time slots | SF16 x 8 codes x 3 time slots |
| | Max. Number of data bits/radio frame | 6 608 bits |
| | TFCI code word | 16 bits |
| | Puncturing limit | 0,52 |

6.10.3.4.1.44 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:128 DL:2 048 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.3.4.1.44.1 Uplink

6.10.3.4.1.44.1.1 Transport channel parameters

6.10.3.4.1.44.1.1.1 Transport channel parameters for Conversational / speech / UL:12.2 kbps / CS RAB

See clause 6.10.3.4.1.4.1.1.1.

6.10.3.4.1.44.1.1.2 Transport channel parameters for Interactive or background / UL:128 kbps / PS RAB

See clause 6.10.3.4.1.28.1.1.1.

6.10.3.4.1.44.1.1.3 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.10.3.4.1.44.1.1.4 TFCS

| | |
|---|---|
| TFCS size | 30 |
| TFCS | (RAB subflow#1, RAB subflow#2, RAB subflow#3, 128 kbps RAB , DCCH)=<br>(TF0, TF0, TF0, TF0, TF0), (TF1, TF0, TF0, TF0, TF0), (TF2, TF1, TF1, TF0, TF0),<br>(TF0, TF0, TF0, TF1, TF0), (TF1, TF0, TF0, TF1, TF0), (TF2, TF1, TF1, TF1, TF0),<br>(TF0, TF0, TF0, TF2, TF0), (TF1, TF0, TF0, TF2, TF0), (TF2, TF1, TF1, TF2, TF0),<br>(TF0, TF0, TF0, TF3, TF0), (TF1, TF0, TF0, TF3, TF0), (TF2, TF1, TF1, TF3, TF0),<br>(TF0, TF0, TF0, TF4, TF0), (TF1, TF0, TF0, TF4, TF0), (TF2, TF1, TF1, TF4, TF0),<br>(TF0, TF0, TF0, TF0, TF1), (TF1, TF0, TF0, TF0, TF1), (TF2, TF1, TF1, TF0, TF1),<br>(TF0, TF0, TF0, TF1, TF1), (TF1, TF0, TF0, TF1, TF1), (TF2, TF1, TF1, TF1, TF1),<br>(TF0, TF0, TF0, TF2, TF1), (TF1, TF0, TF0, TF2, TF1), (TF2, TF1, TF1, TF2, TF1),<br>(TF0, TF0, TF0, TF3, TF1), (TF1, TF0, TF0, TF3, TF1), (TF2, TF1, TF1, TF3, TF1),<br>(TF0, TF0, TF0, TF4, TF1), (TF1, TF0, TF0, TF4, TF1), (TF2, TF1, TF1, TF4, TF1) |
| NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise. | |

6.10.3.4.1.44.1.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Uplink | Midamble | 256 chips |
| | Codes and time slots | {SF8 x 1 code + SF2 x 1 code}<br>x 1 time slot |
| | Max. Number of data bits/radio frame | 2 616 bits |
| | TFCI code word | 16 bits |
| | TPC | 2 bits |
| | Puncturing Limit | 0.52 (alt. 0.44) |

6.10.3.4.1.44.2 Downlink

6.10.3.4.1.44.2.1 Transport channel parameters

6.10.3.4.1.44.2.1.1 Transport channel parameters for Conversational / speech / DL:12.2 kbps / CS RAB

See clause 6.10.3.4.1.4.2.1.1.

6.10.3.4.1.44.2.1.2 Transport channel parameters for Interactive or background / DL:2 048 kbps / PS RAB

See clause 6.10.3.4.1.35.2.1.1.

6.10.3.4.1.44.2.1.3 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.10.3.4.1.44.2.1.4 TFCS

| TFCS size | 66 (alt. 114) |
|---|---|
| TFCS | (RAB subflow#1, RAB subflow#2, RAB subflow#3, 2 048 kbps RAB , DCCH)=<br>(TF0, TF0, TF0, TF0, TF0), (TF1, TF0, TF0, TF0, TF0), (TF2, TF1, TF1, TF0, TF0),<br>(TF0, TF0, TF0, TF1, TF0), (TF1, TF0, TF0, TF1, TF0), (TF2, TF1, TF1, TF1, TF0),<br>(TF0, TF0, TF0, TF2, TF0), (TF1, TF0, TF0, TF2, TF0), (TF2, TF1, TF1, TF2, TF0),<br>(TF0, TF0, TF0, TF3, TF0), (TF1, TF0, TF0, TF3, TF0), (TF2, TF1, TF1, TF3, TF0),<br>(TF0, TF0, TF0, TF4, TF0), (TF1, TF0, TF0, TF4, TF0), (TF2, TF1, TF1, TF4, TF0),<br>(TF0, TF0, TF0, TF5, TF0), (TF1, TF0, TF0, TF5, TF0), (TF2, TF1, TF1, TF5, TF0),<br>(TF0, TF0, TF0, TF6, TF0), (TF1, TF0, TF0, TF6, TF0), (TF2, TF1, TF1, TF6, TF0),<br>(TF0, TF0, TF0, TF7, TF0), (TF1, TF0, TF0, TF7, TF0), (TF2, TF1, TF1, TF7, TF0),<br>(TF0, TF0, TF0, TF8, TF0), (TF1, TF0, TF0, TF8, TF0), (TF2, TF1, TF1, TF8, TF0),<br>(TF0, TF0, TF0, TF9, TF0), (TF1, TF0, TF0, TF9, TF0), (TF2, TF1, TF1, TF9, TF0),<br>(TF0, TF0, TF0, TF10, TF0), (TF1, TF0, TF0, TF10, TF0), (TF2, TF1, TF1, TF10, TF0),<br>(TF0, TF0, TF0, TF0, TF1), (TF1, TF0, TF0, TF0, TF1), (TF2, TF1, TF1, TF0, TF1),<br>(TF0, TF0, TF0, TF1, TF1), (TF1, TF0, TF0, TF1, TF1), (TF2, TF1, TF1, TF1, TF1),<br>(TF0, TF0, TF0, TF2, TF1), (TF1, TF0, TF0, TF2, TF1), (TF2, TF1, TF1, TF2, TF1),<br>(TF0, TF0, TF0, TF3, TF1), (TF1, TF0, TF0, TF3, TF1), (TF2, TF1, TF1, TF3, TF1),<br>(TF0, TF0, TF0, TF4, TF1), (TF1, TF0, TF0, TF4, TF1), (TF2, TF1, TF1, TF4, TF1)<br>(TF0, TF0, TF0, TF5, TF1), (TF1, TF0, TF0, TF5, TF1), (TF2, TF1, TF1, TF5, TF1),<br>TF0, TF0, TF0, TF6, TF1), (TF1, TF0, TF0, TF6, TF1), (TF2, TF1, TF1, TF6, TF1),<br>(TF0, TF0, TF0, TF7, TF1), (TF1, TF0, TF0, TF7, TF1), (TF2, TF1, TF1, TF7, TF1),<br>(TF0, TF0, TF0, TF8, TF1), (TF1, TF0, TF0, TF8, TF1), (TF2, TF1, TF1, TF8, TF1),<br>(TF0, TF0, TF0, TF9, TF1), (TF1, TF0, TF0, TF9, TF1), (TF2, TF1, TF1, TF9, TF1)<br>(TF0, TF0, TF0, TF10, TF1), (TF1, TF0, TF0, TF10, TF1), (TF2, TF1, TF1, TF10, TF1)<br>(alt. (TF0, TF0, TF0, TF0, TF0), (TF1, TF0, TF0, TF0, TF0), (TF2, TF1, TF1, TF0, TF0),<br>(TF0, TF0, TF0, TF1, TF0), (TF1, TF0, TF0, TF1, TF0), (TF2, TF1, TF1, TF1, TF0),<br>(TF0, TF0, TF0, TF2, TF0), (TF1, TF0, TF0, TF2, TF0), (TF2, TF1, TF1, TF2, TF0),<br>(TF0, TF0, TF0, TF3, TF0), (TF1, TF0, TF0, TF3, TF0), (TF2, TF1, TF1, TF3, TF0),<br>(TF0, TF0, TF0, TF4, TF0), (TF1, TF0, TF0, TF4, TF0), (TF2, TF1, TF1, TF4, TF0),<br>(TF0, TF0, TF0, TF5, TF0), (TF1, TF0, TF0, TF5, TF0), (TF2, TF1, TF1, TF5, TF0),<br>(TF0, TF0, TF0, TF6, TF0), (TF1, TF0, TF0, TF6, TF0), (TF2, TF1, TF1, TF6, TF0),<br>(TF0, TF0, TF0, TF7, TF0), (TF1, TF0, TF0, TF7, TF0), (TF2, TF1, TF1, TF7, TF0),<br>(TF0, TF0, TF0, TF8, TF0), (TF1, TF0, TF0, TF8, TF0), (TF2, TF1, TF1, TF8, TF0),<br>(TF0, TF0, TF0, TF9, TF0), (TF1, TF0, TF0, TF9, TF0), (TF2, TF1, TF1, TF9, TF0),<br>(TF0, TF0, TF0, TF10, TF0), (TF1, TF0, TF0, TF10, TF0), (TF2, TF1, TF1, TF10, TF0),<br>(TF0, TF0, TF0, TF11, TF0), (TF1, TF0, TF0, TF11, TF0), (TF2, TF1, TF1, TF11, TF0),<br>(TF0, TF0, TF0, TF12, TF0), (TF1, TF0, TF0, TF12, TF0), (TF2, TF1, TF1, TF12, TF0),<br>(TF0, TF0, TF0, TF13, TF0), (TF1, TF0, TF0, TF13, TF0), (TF2, TF1, TF1, TF13, TF0),<br>(TF0, TF0, TF0, TF14, TF0), (TF1, TF0, TF0, TF14, TF0), (TF2, TF1, TF1, TF14, TF0),<br>(TF0, TF0, TF0, TF15, TF0), (TF1, TF0, TF0, TF15, TF0), (TF2, TF1, TF1, TF15, TF0),<br>(TF0, TF0, TF0, TF16, TF0), (TF1, TF0, TF0, TF16, TF0), (TF2, TF1, TF1, TF16, TF0),<br>(TF0, TF0, TF0, TF17, TF0), (TF1, TF0, TF0, TF17, TF0), (TF2, TF1, TF1, TF17, TF0),<br>(TF0, TF0, TF0, TF18, TF0), (TF1, TF0, TF0, TF18, TF0), (TF2, TF1, TF1, TF18, TF0),<br>(TF0, TF0, TF0, TF0, TF1), (TF1, TF0, TF0, TF0, TF1), (TF2, TF1, TF1, TF0, TF1),<br>(TF0, TF0, TF0, TF1, TF1), (TF1, TF0, TF0, TF1, TF1), (TF2, TF1, TF1, TF1, TF1),<br>(TF0, TF0, TF0, TF2, TF1), (TF1, TF0, TF0, TF2, TF1), (TF2, TF1, TF1, TF2, TF1),<br>(TF0, TF0, TF0, TF3, TF1), (TF1, TF0, TF0, TF3, TF1), (TF2, TF1, TF1, TF3, TF1),<br>(TF0, TF0, TF0, TF4, TF1), (TF1, TF0, TF0, TF4, TF1), (TF2, TF1, TF1, TF4, TF1),<br>(TF0, TF0, TF0, TF5, TF1), (TF1, TF0, TF0, TF5, TF1), (TF2, TF1, TF1, TF5, TF1),<br>(TF0, TF0, TF0, TF6, TF1), (TF1, TF0, TF0, TF6, TF1), (TF2, TF1, TF1, TF6, TF1),<br>(TF0, TF0, TF0, TF7, TF1), (TF1, TF0, TF0, TF7, TF1), (TF2, TF1, TF1, TF7, TF1),<br>(TF0, TF0, TF0, TF8, TF1), (TF1, TF0, TF0, TF8, TF1), (TF2, TF1, TF1, TF8, TF1),<br>(TF0, TF0, TF0, TF9, TF1), (TF1, TF0, TF0, TF9, TF1), (TF2, TF1, TF1, TF9, TF1),<br>(TF0, TF0, TF0, TF10, TF1), (TF1, TF0, TF0, TF10, TF1), (TF2, TF1, TF1, TF10, TF1),<br>(TF0, TF0, TF0, TF11, TF1), (TF1, TF0, TF0, TF11, TF1), (TF2, TF1, TF1, TF11, TF1),<br>(TF0, TF0, TF0, TF12, TF1), (TF1, TF0, TF0, TF12, TF1), (TF2, TF1, TF1, TF12, TF1),<br>(TF0, TF0, TF0, TF13, TF1), (TF1, TF0, TF0, TF13, TF1), (TF2, TF1, TF1, TF13, TF1),<br>(TF0, TF0, TF0, TF14, TF1), (TF1, TF0, TF0, TF14, TF1), (TF2, TF1, TF1, TF14, TF1),<br>(TF0, TF0, TF0, TF15, TF1), (TF1, TF0, TF0, TF15, TF1), (TF2, TF1, TF1, TF15, TF1),<br>(TF0, TF0, TF0, TF16, TF1), (TF1, TF0, TF0, TF16, TF1), (TF2, TF1, TF1, TF16, TF1),<br>(TF0, TF0, TF0, TF17, TF1), (TF1, TF0, TF0, TF17, TF1), (TF2, TF1, TF1, TF17, TF1),<br>(TF0, TF0, TF0, TF18, TF1), (TF1, TF0, TF0, TF18, TF1), (TF2, TF1, TF1, TF18, TF1)) |
| NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise. | |

6.10.3.4.1.44.2.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Downlink | Midamble | 256 chips |
| | Codes and time slots | SF16 x 12 codes x 11 time slots |
| | Max. Number of data bits/radio frame | 36 400 bits |
| | TFCI code word | 32 bits |
| | Puncturing limit | 0.52 |

6.10.3.4.1.45 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Streaming / unknown / UL:57.6 DL:57.6 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.3.4.1.45.1 Uplink

6.10.3.4.1.45.1.1 Transport channel parameters

6.10.3.4.1.45.1.1.1 Transport channel parameters for Conversational / speech / UL:12.2 kbps / CS RAB

See clause 6.10.3.4.1.4.1.1.1.

6.10.3.4.1.45.1.1.2 Transport channel parameters for Streaming / unknown / UL:57.6 kbps / CS RAB

See clause 6.10.3.4.1.17.1.1.1.

6.10.3.4.1.45.1.1.3 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.10.3.4.1.45.1.1.4 TFCS

| | |
|---|---|
| TFCS size | 30 |
| TFCS | (RAB subflow#1, RAB subflow#2, RAB subflow#3, 57.6 kbps RAB , DCCH)=<br>(TF0, TF0, TF0, TF0, TF0), (TF1, TF0, TF0, TF0, TF0), (TF2, TF1, TF1, TF0, TF0),<br>(TF0, TF0, TF0, TF1, TF0), (TF1, TF0, TF0, TF1, TF0), (TF2, TF1, TF1, TF1, TF0),<br>(TF0, TF0, TF0, TF2, TF0), (TF1, TF0, TF0, TF2, TF0), (TF2, TF1, TF1, TF2, TF0),<br>(TF0, TF0, TF0, TF3, TF0), (TF1, TF0, TF0, TF3, TF0), (TF2, TF1, TF1, TF3, TF0),<br>(TF0, TF0, TF0, TF4, TF0), (TF1, TF0, TF0, TF4, TF0), (TF2, TF1, TF1, TF4, TF0),<br>(TF0, TF0, TF0, TF0, TF1), (TF1, TF0, TF0, TF0, TF1), (TF2, TF1, TF1, TF0, TF1),<br>(TF0, TF0, TF0, TF1, TF1), (TF1, TF0, TF0, TF1, TF1), (TF2, TF1, TF1, TF1, TF1),<br>(TF0, TF0, TF0, TF2, TF1), (TF1, TF0, TF0, TF2, TF1), (TF2, TF1, TF1, TF2, TF1),<br>(TF0, TF0, TF0, TF3, TF1), (TF1, TF0, TF0, TF3, TF1), (TF2, TF1, TF1, TF3, TF1),<br>(TF0, TF0, TF0, TF4, TF1), (TF1, TF0, TF0, TF4, TF1), (TF2, TF1, TF1, TF4, TF1) |
| NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise. | |

6.10.3.4.1.45.1.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Uplink | Midamble | 512 chips |
| | Codes and time slots | SF8 x 1 code x 1 time slot + SF4 x 1 codex 1 time slot |
| | Max. Number of data bits/radio frame | 1 392 bits |
| | TFCI code word | 16 bits |
| | TPC | 2 bits |
| | Puncturing Limit | 0.56 |

6.10.3.4.1.45.2 Downlink

6.10.3.4.1.45.2.1 Transport channel parameters

6.10.3.4.1.45.2.1.1 Transport channel parameters for Conversational / speech / DL:12.2 kbps / CS RAB

See clause 6.10.3.4.1.4.2.1.1.

6.10.3.4.1.45.2.1.2 Transport channel parameters for Streaming / unknown / DL:57.6 kbps / CS RAB

See clause 6.10.3.4.1.17.2.1.1.

6.10.3.4.1.45.2.1.3 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.10.3.4.1.45.2.1.4 TFCS

| | |
|---|---|
| TFCS size | 30 |
| TFCS | (RAB subflow#1, RAB subflow#2, RAB subflow#3, 57.6 kbps RAB , DCCH)=<br>(TF0, TF0, TF0, TF0, TF0), (TF1, TF0, TF0, TF0, TF0), (TF2, TF1, TF1, TF0, TF0),<br>(TF0, TF0, TF0, TF1, TF0), (TF1, TF0, TF0, TF1, TF0), (TF2, TF1, TF1, TF1, TF0),<br>(TF0, TF0, TF0, TF2, TF0), (TF1, TF0, TF0, TF2, TF0), (TF2, TF1, TF1, TF2, TF0),<br>(TF0, TF0, TF0, TF3, TF0), (TF1, TF0, TF0, TF3, TF0), (TF2, TF1, TF1, TF3, TF0),<br>(TF0, TF0, TF0, TF4, TF0), (TF1, TF0, TF0, TF4, TF0), (TF2, TF1, TF1, TF4, TF0),<br>(TF0, TF0, TF0, TF0, TF1), (TF1, TF0, TF0, TF0, TF1), (TF2, TF1, TF1, TF0, TF1),<br>(TF0, TF0, TF0, TF1, TF1), (TF1, TF0, TF0, TF1, TF1), (TF2, TF1, TF1, TF1, TF1),<br>(TF0, TF0, TF0, TF2, TF1), (TF1, TF0, TF0, TF2, TF1), (TF2, TF1, TF1, TF2, TF1),<br>(TF0, TF0, TF0, TF3, TF1), (TF1, TF0, TF0, TF3, TF1), (TF2, TF1, TF1, TF3, TF1),<br>(TF0, TF0, TF0, TF4, TF1), (TF1, TF0, TF0, TF4, TF1), (TF2, TF1, TF1, TF4, TF1) |
| NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise. | |

6.10.3.4.1.45.2.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Downlink | Midamble | 512 chips |
| | Codes and time slots | SF16 x 6 codes x 1 time slot |
| | Max. Number of data bits/radio frame | 1 448 bits |
| | TFCI code word | 16 bits |
| | Puncturing limit | 0.56 |

6.10.3.4.1.46 Void

6.10.3.4.1.47 Void

6.10.3.4.1.48 Void

6.10.3.4.1.49 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Conversational / unknown / UL:64 DL:64 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.3.4.1.49.1 Uplink

6.10.3.4.1.49.1.1 Transport channel parameters

6.10.3.4.1.49.1.1.1 Transport channel parameters for Conversational / speech / UL:12.2 kbps / CS RAB

See clause 6.10.3.4.1.4.1.1.1.

6.10.3.4.1.49.1.1.2 Transport channel parameters for Conversational / unknown / UL:64 kbps / CS RAB

See clause 6.10.3.4.1.13.1.1.1.

6.10.3.4.1.49.1.1.3 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.10.3.4.1.49.1.1.4 TFCS

| | |
|---|---|
| TFCS size | 12 |
| TFCS | (RAB subflow#1, RAB subflow#2, RAB subflow#3, 64 kbps RAB, DCCH)=<br>(TF0, TF0, TF0, TF0, TF0), (TF1, TF0, TF0, TF0, TF0), (TF2, TF1, TF1, TF0, TF0),<br>(TF0, TF0, TF0, TF1, TF0), (TF1, TF0, TF0, TF1, TF0), (TF2, TF1, TF1, TF1, TF0),<br>(TF0, TF0, TF0, TF0, TF1), (TF1, TF0, TF0, TF0, TF1), (TF2, TF1, TF1, TF0, TF1),<br>(TF0, TF0, TF0, TF1, TF1), (TF1, TF0, TF0, TF1, TF1), (TF2, TF1, TF1, TF1, TF1) |
| NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise. | |

6.10.3.4.1.49.1.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Uplink | Midamble | 256 chips |
| | Codes and time slots | SF2 x 1 code x 1 time slot |
| | Max. Number of data bits/radio frame | 2 064 bits |
| | TFCI code word | 16 bits |
| | TPC | 2 bits |
| | Puncturing Limit | 0.72 |

6.10.3.4.1.49.2 Downlink

6.10.3.4.1.49.2.1 Transport channel parameters

6.10.3.4.1.49.2.1.1 Transport channel parameters for Conversational / speech / DL:12.2 kbps / CS RAB

See clause 6.10.3.4.1.4.2.1.1.

6.10.3.4.1.49.2.1.2 Transport channel parameters for Conversational / unknown / DL:64 kbps / CS RAB

See clause 6.10.3.4.1.13.2.1.1.

6.10.3.4.1.49.2.1.3 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.10.3.4.1.49.2.1.4 TFCS

| | |
|---|---|
| TFCS size | 12 |
| TFCS | (RAB subflow#1, RAB subflow#2, RAB subflow#3, 64 kbps RAB, DCCH)=<br>(TF0, TF0, TF0, TF0, TF0), (TF1, TF0, TF0, TF0, TF0), (TF2, TF1, TF1, TF0, TF0),<br>(TF0, TF0, TF0, TF1, TF0), (TF1, TF0, TF0, TF1, TF0), (TF2, TF1, TF1, TF1, TF0),<br>(TF0, TF0, TF0, TF0, TF1), (TF1, TF0, TF0, TF0, TF1), (TF2, TF1, TF1, TF0, TF1),<br>(TF0, TF0, TF0, TF1, TF1), (TF1, TF0, TF0, TF1, TF1), (TF2, TF1, TF1, TF1, TF1) |
| NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise. | |

6.10.3.4.1.49.2.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Downlink | Midamble | 256 chips |
| | Codes and time slots | SF16 x 8 codes x 1 time slot |
| | Max. Number of data bits/radio frame | 2 192 bits |
| | TFCI code word | 16 bits |
| | Puncturing limit | 0.76 |

6.10.3.4.1.49a Conversational / speech / UL:(12.2 7.95 5.9 4.75) DL:(12.2 7.95 5.9 4.75) kbps / CS RAB + Conversational / unknown / UL:64 DL:64 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.3.4.1.49a.1 Uplink

6.10.3.4.1.49a.1.1 Transport channel parameters

6.10.3.4.1.49a.1.1.1 Transport channel parameters for Conversational / speech / UL: (12.2 7.95 5.9 4.75) kbps / CS RAB

See clause 6.10.3.4.1.4a.1.1.1.

6.10.3.4.1.49a.1.1.2 Transport channel parameters for Conversational / unknown / UL:64 kbps / CS RAB

See clause 6.10.3.4.1.13.1.1.1.

6.10.3.4.1.49a.1.1.3 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.10.3.4.1.49a.1.1.4 TFCS

| | |
|---|---|
| TFCS size | 24 |

| | |
|---|---|
| TFCS | (RAB subflow#1, RAB subflow#2, RAB subflow#3, 64 kbps RAB, DCCH)=<br>(TF0,TF0,TF0,TF0,TF0), (TF1,TF0,TF0,TF0,TF0), (TF2,TF1,TF0,TF0,TF0),<br>(TF3,TF2,TF0,TF0,TF0), (TF4,TF3,TF0,TF0,TF0), (TF5,TF4,TF1,TF0,TF0),<br>(TF0,TF0,TF0,TF1,TF0), (TF1,TF0,TF0,TF1,TF0), (TF2,TF1,TF0,TF1,TF0),<br>(TF3,TF2,TF0,TF1,TF0), (TF4,TF3,TF0,TF1,TF0), (TF5,TF4,TF1,TF1,TF0),<br>(TF0,TF0,TF0,TF0,TF1), (TF1,TF0,TF0,TF0,TF1), (TF2,TF1,TF0,TF0,TF1),<br>(TF3,TF2,TF0,TF0,TF1), (TF4,TF3,TF0,TF0,TF1), (TF5,TF4,TF1,TF0,TF1),<br>(TF0,TF0,TF0,TF1,TF1), (TF1,TF0,TF0,TF1,TF1), (TF2,TF1,TF0,TF1,TF1),<br>(TF3,TF2,TF0,TF1,TF1), (TF4,TF3,TF0,TF1,TF1), (TF5,TF4,TF1,TF1,TF1) |
| NOTE: In case TB size zero is configured for any transport channel the first TFC is required; it is optional otherwise. | |

6.10.3.4.1.49a.1.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Uplink | Midamble | 256 chips |
| | Codes and time slots | SF2 x 1 code x 1 time slot |
| | Max. Number of data bits/radio frame | 2 064 bits |
| | TFCI code word | 16 bit |
| | TPC | 2 bits |
| | Puncturing Limit | 0.72 |

6.10.3.4.1.49a.2 Downlink

6.10.3.4.1.49a.2.1 Transport channel parameters

6.10.3.4.1.49a.2.1.1 Transport channel parameters for Conversational / speech / DL: (12.2 7.95 5.9 4.75) kbps / CS RAB

See clause 6.10.3.4.1.4a.2.1.1.

6.10.3.4.1.49a.2.1.2 Transport channel parameters for Conversational / unknown / DL:64 kbps / CS RAB

See clause 6.10.3.4.1.13.2.1.1.

6.10.3.4.1.49a.2.1.3 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.10.3.4.1.49a.2.1.4 TFCS

| | |
|---|---|
| TFCS size | 24 |
| TFCS | (RAB subflow#1, RAB subflow#2, RAB subflow#3, 64 kbps RAB, DCCH)=<br>(TF0,TF0,TF0,TF0,TF0), (TF1,TF0,TF0,TF0,TF0), (TF2,TF1,TF0,TF0,TF0),<br>(TF3,TF2,TF0,TF0,TF0), (TF4,TF3,TF0,TF0,TF0), (TF5,TF4,TF1,TF0,TF0),<br>(TF0,TF0,TF0,TF1,TF0), (TF1,TF0,TF0,TF1,TF0), (TF2,TF1,TF0,TF1,TF0),<br>(TF3,TF2,TF0,TF1,TF0), (TF4,TF3,TF0,TF1,TF0), (TF5,TF4,TF1,TF1,TF0),<br>(TF0,TF0,TF0,TF0,TF1), (TF1,TF0,TF0,TF0,TF1), (TF2,TF1,TF0,TF0,TF1),<br>(TF3,TF2,TF0,TF0,TF1), (TF4,TF3,TF0,TF0,TF1), (TF5,TF4,TF1,TF0,TF1),<br>(TF0,TF0,TF0,TF1,TF1), (TF1,TF0,TF0,TF1,TF1), (TF2,TF1,TF0,TF1,TF1),<br>(TF3,TF2,TF0,TF1,TF1), (TF4,TF3,TF0,TF1,TF1), (TF5,TF4,TF1,TF1,TF1) |
| NOTE: In case TB size zero is configured for any transport channel the first TFC is required; it is optional otherwise. | |

6.10.3.4.1.49a.2.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Downlink | Midamble | 256 chips |
| | Codes and time slots | SF16 x 7 codes x 1 time slot |
| | Max. Number of data bits/radio frame | 1 916 bits |
| | TFCI code word | 16 bits |
| | Puncturing limit | 0.68 |

6.10.3.4.1.50 Conversational / unknown / UL:64 DL:64 kbps / CS RAB + Conversational / unknown / UL:64 DL:64 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.3.4.1.50.1 Uplink

6.10.3.4.1.50.1.1 Transport channel parameters

6.10.3.4.1.50.1.1.1 Transport channel parameters for Conversational / unknown / UL:64 kbps / CS RAB

See clause 6.10.3.4.1.13.1.1.1.

6.10.3.4.1.50.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.10.3.4.1.50.1.1.3 TFCS

| | |
|---|---|
| TFCS size | 8 |
| TFCS | (64 kbps RAB, 64 kbps RAB, DCCH)=<br>(TF0, TF0, TF0), (TF1, TF0, TF0), (TF0, TF1, TF0), (TF1, TF1, TF0)<br>(TF0, TF0, TF1), (TF1, TF0, TF1), (TF0, TF1, TF1), (TF1, TF1, TF1) |
| NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise. | |

6.10.3.4.1.50.1.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Uplink | Midamble | 512 chips |
| | Codes and time slots | SF2 x 1 code x 1time slot +<br>SF4 x 1 code x 1 time slot |
| | Max. Number of data bits/radio frame | 2 784 bits |
| | TFCI code word | 16 bits |
| | TPC | 2 bits |
| | Puncturing Limit | 0.60 |

6.10.3.4.1.50.2 Downlink

6.10.3.4.1.50.2.1 Transport channel parameters

6.10.3.4.1.50.2.1.1 Transport channel parameters for Conversational / unknown / DL:64 kbps / CS RAB

See clause 6.10.3.4.1.13.2.1.1.

6.10.3.4.1.50.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.10.3.4.1.50.2.1.3 TFCS

| | |
|---|---|
| TFCS size | 8 |
| TFCS | (64 kbps RAB, 64 kbps RAB, DCCH)=<br>(TF0, TF0, TF0), (TF1, TF0, TF0), (TF0, TF1, TF0), (TF1, TF1, TF0)<br>(TF0, TF0, TF1), (TF1, TF0, TF1), (TF0, TF1, TF1), (TF1, TF1, TF1) |
| NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise. | |

6.10.3.4.1.50.2.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Downlink | Midamble | 512 chips |
| | Codes and time slots | SF16 x 6codes x 2 time slots |
| | Max. Number of data bits/radio frame | 2 912 bits |
| | TFCI code word | 16 bits |
| | Puncturing limit | 0.64 |

6.10.3.4.1.51 Conversational / unknown / UL:64 DL:64 kbps / CS RAB + Interactive or background / UL:64 DL:64 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.3.4.1.51.1 Uplink

6.10.3.4.1.51.1.1 Transport channel parameters

6.10.3.4.1.51.1.1.1 Transport channel parameters for Conversational / unknown / UL:64 kbps / CS RAB

See clause 6.10.3.4.1.13.1.1.1.

6.10.3.4.1.51.1.1.2 Transport channel parameters for Interactive or background / UL:64 kbps / PS RAB

See clause 6.10.3.4.1.26.1.1.1.

6.10.3.4.1.51.1.1.3 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.10.3.4.1.51.1.1.4 TFCS

| | |
|---|---|
| TFCS size | 20 |
| TFCS | (Conv. 64 kbps RAB, I/B 64 kbps RAB, DCCH)=<br>(TF0, TF0, TF0), (TF0, TF1, TF0), (TF0, TF2, TF0), (TF0, TF3, TF0), (TF0, TF4, TF0),<br>(TF1, TF0, TF0), (TF1, TF1, TF0), (TF1, TF2, TF0), (TF1, TF3, TF0), (TF1, TF4, TF0),<br>(TF0, TF0, TF1), (TF0, TF1, TF1), (TF0, TF2, TF1), (TF0, TF3, TF1), (TF0, TF4, TF1),<br>(TF1, TF0, TF1), (TF1, TF1, TF1), (TF1, TF2, TF1), (TF1, TF3, TF1), (TF1, TF4, TF1) |
| NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise. | |

6.10.3.4.1.51.1.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Uplink | Midamble | 256 chips |
| | Codes and time slots | SF2 x 1 code x 1 time slot |
| | Max. Number of data bits/radio frame | 2 064 bits |
| | TFCI code word | 16 bits |
| | TPC | 2 bits |
| | Puncturing Limit | 0.44 (alt. 0.40) |

6.10.3.4.1.51.2 Downlink

6.10.3.4.1.51.2.1 Transport channel parameters

6.10.3.4.1.51.2.1.1 Transport channel parameters for Conversational / unknown / DL:64 kbps / CS RAB

See clause 6.10.3.4.1.13.2.1.1.

6.10.3.4.1.51.2.1.2 Transport channel parameters for Interactive or background / DL:64 kbps / PS RAB

See clause 6.10.3.4.1.25.2.1.1.

6.10.3.4.1.51.2.1.3 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.10.3.4.1.51.2.1.4 TFCS

| | |
|---|---|
| TFCS size | 20 |
| TFCS | (Conv. 64 kbps RAB, I/B 64 kbps RAB, DCCH)=<br>(TF0, TF0, TF0), (TF0, TF1, TF0), (TF0, TF2, TF0), (TF0, TF3, TF0), (TF0, TF4, TF0),<br>(TF1, TF0, TF0), (TF1, TF1, TF0), (TF1, TF2, TF0), (TF1, TF3, TF0), (TF1, TF4, TF0),<br>(TF0, TF0, TF1), (TF0, TF1, TF1), (TF0, TF2, TF1), (TF0, TF3, TF1), (TF0, TF4, TF1),<br>(TF1, TF0, TF1), (TF1, TF1, TF1), (TF1, TF2, TF1), (TF1, TF3, TF1), (TF1, TF4, TF1) |
| NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise. | |

6.10.3.4.1.51.2.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Downlink | Midamble | 256 chips |
| | Codes and time slots | SF16 x 8 codes x 1 time slot |
| | Max. Number of data bits/radio frame | 2 192 bits |
| | TFCI code word | 16 bits |
| | Puncturing limit | 0.48 |

6.10.3.4.1.51a Conversational / unknown / UL:64 DL:64 kbps / CS RAB + Interactive or Background / UL:8 DL:8 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.3.4.1.51a.1 Uplink

6.10.3.4.1.51a.1.1 Transport channel parameters

6.10.3.4.1.51a.1.1.1 Transport channel parameters for Conversational / unknown / UL:64 kbps / CS RAB

See clause 6.10.3.4.1.13.1.1.1.

6.10.3.4.1.51a.1.1.2 Transport channel parameters for Interactive or Background / UL:8 kbps / PS RAB

See clause 6.10.3.4.1.23a.1.1.1.

6.10.3.4.1.51a.1.1.3 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.10.3.4.1.51a.1.1.4 TFCS

| | |
|---|---|
| TFCS size | 8 (alt. 12) |
| TFCS | (64 kbps Conversational RAB, 8 kbps I/B RAB, DCCH)=<br>(TF0, TF0, TF0), (TF0, TF1, TF0), (TF0, TF0, TF1), (TF0, TF1, TF1),<br>(TF1, TF0, TF0), (TF1, TF1, TF0), (TF1, TF0, TF1), (TF1, TF1, TF1)<br>(alt.<br>(TF0, TF0, TF0), (TF0, TF1, TF0), (TF0, TF2, TF0),<br>(TF0, TF0, TF1), (TF0, TF1, TF1), (TF0, TF2, TF1),<br>(TF1, TF0, TF0), (TF1, TF1, TF0), (TF1, TF2, TF0),<br>(TF1, TF0, TF1), (TF1, TF1, TF1), (TF1, TF2, TF1)) |
| NOTE: In case TB size zero is configured for any transport channel the first TFC is required; it is optional otherwise. | |

6.10.3.4.1.51a.1.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Uplink | Midamble | 256 chips |
| | Codes and time slots | SF2 x 1 code x 1 time slot |
| | Max. Number of data bits/radio frame | 2 064 bits |
| | TFCI code word | 16 bit |
| | TPC | 2 bits |
| | Puncturing Limit | 0.76 |

6.10.3.4.1.51a.2 Downlink

6.10.3.4.1.51a.2.1 Transport channel parameters

6.10.3.4.1.51a.2.1.1 Transport channel parameters for Conversational / unknown / DL:64 kbps / PS RAB

See clause 6.10.3.4.1.13.2.1.1.

6.10.3.4.1.51a.2.1.2 Transport channel parameters for Interactive or Background / DL:8 kbps / PS RAB

See clause 6.10.3.4.1.23.2.1.1.

6.10.3.4.1.51a.2.1.3 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.10.3.4.1.51a.2.1.4 TFCS

| | |
|---|---|
| TFCS size | 8 |
| TFCS | (64 kbps Conversational RAB, 8 kbps I/B RAB, DCCH)=<br>(TF0, TF0, TF0), (TF0, TF1, TF0), (TF0, TF0, TF1), (TF0, TF1, TF1),<br>(TF1, TF0, TF0), (TF1, TF1, TF0), (TF1, TF0, TF1), (TF1, TF1, TF1) |
| NOTE: In case TB size zero is configured for any transport channel the first TFC is required; it is optional otherwise. | |

6.10.3.4.1.51a.2.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Downlink | Midamble | 256 chips |
| | Codes and time slots | SF16 x 6 codes x 1 time slot |
| | Max. Number of data bits/radio frame | 1 640 bits |
| | TFCI code word | 16 bits |
| | Puncturing limit | 0.60 |

6.10.3.4.1.51b Conversational / unknown / UL:64 DL:64 kbps / CS RAB + Interactive or Background / UL:16 DL:64 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.3.4.1.51b.1 Uplink

6.10.3.4.1.51b.1.1 Transport channel parameters

6.10.3.4.1.51b.1.1.1 Transport channel parameters for Conversational / unknown / UL:64 kbps / CS RAB

See clause 6.10.3.4.1.13.1.1.1.

6.10.3.4.1.51b.1.1.2 Transport channel parameters for Interactive or Background / UL:16 kbps / PS RAB

See clause 6.10.3.4.1.23b.1.1.1.

6.10.3.4.1.51b.1.1.3 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.10.3.4.1.51b.1.1.4 TFCS

| | |
|---|---|
| TFCS size | 12 |
| TFCS | (64 kbps Conversational RAB, 16 kbps I/B RAB, DCCH)=<br>(TF0, TF0, TF0), (TF0, TF1, TF0), (TF0, TF2, TF0), (TF0, TF0, TF1), (TF0, TF1, TF1), (TF0, TF2, TF1),<br>(TF1, TF0, TF0), (TF1, TF1, TF0), (TF1, TF2, TF0), (TF1, TF0, TF1), (TF1, TF1, TF1), (TF1, TF2, TF1) |
| NOTE: In case TB size zero is configured for any transport channel the first TFC is required; it is optional otherwise. | |

6.10.3.4.1.51b.1.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Uplink | Midamble | 256 chips |
| | Codes and time slots | SF2 x 1 code x 1 time slot |
| | Max. Number of data bits/radio frame | 2 064 bits |
| | TFCI code word | 16 bit |
| | TPC | 2 bits |
| | Puncturing Limit | 0.68 |

6.10.3.4.1.51b.2 Downlink

See clause 6.10.3.4.1.51.2.

6.10.3.4.1.52 Conversational / unknown / UL:64 DL:64 kbps / CS RAB + Interactive or background / UL:64 DL:128 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.3.4.1.52.1 Uplink

See clause 6.10.3.4.1.51.1.

6.10.3.4.1.52.2 Downlink

6.10.3.4.1.52.2.1 Transport channel parameters

6.10.3.4.1.52.2.1.1 Transport channel parameters for Conversational / unknown / DL:64 kbps / CS RAB

See clause 6.10.3.4.1.13.2.1.1.

6.10.3.4.1.52.2.1.2 Transport channel parameters for Interactive or background / DL:128 kbps / PS RAB

See clause 6.10.3.4.1.27.2.1.1.

6.10.3.4.1.52.2.1.3 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.10.3.4.1.52.2.1.4 TFCS

| | |
|---|---|
| TFCS size | 20 |
| TFCS | (Conv. 64 kbps RAB, I/B 128 kbps RAB, DCCH)=<br>(TF0, TF0, TF0), (TF0, TF1, TF0), (TF0, TF2, TF0), (TF0, TF3, TF0), (TF0, TF4, TF0),<br>(TF1, TF0, TF0), (TF1, TF1, TF0), (TF1, TF2, TF0), (TF1, TF3, TF0), (TF1, TF4, TF0),<br>(TF0, TF0, TF1), (TF0, TF1, TF1), (TF0, TF2, TF1), (TF0, TF3, TF1), (TF0, TF4, TF1),<br>(TF1, TF0, TF1), (TF1, TF1, TF1), (TF1, TF2, TF1), (TF1, TF3, TF1), (TF1, TF4, TF1) |
| NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise. | |

6.10.3.4.1.52.2.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Downlink | Midamble | 512 chips |
| | Codes and time slots | {SF16 x 8 codes x 1 time slot} +<br>{SF16 x 5 codes x 1 time slot} |
| | Max. Number of data bits/radio frame | 3 156 bits |
| | TFCI code word | 16 bits |
| | Puncturing limit | 0.44 |

6.10.3.4.1.53 Conversational / unknown / UL:64 DL:64 kbps / CS RAB + Interactive or background / UL:128 DL:128 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.3.4.1.53.1 Uplink

6.10.3.4.1.53.1.1 Transport channel parameters

6.10.3.4.1.53.1.1.1 Transport channel parameters for Conversational / unknown / UL:64 kbps / CS RAB

See clause 6.10.3.4.1.13.1.1.1.

6.10.3.4.1.53.1.1.2 Transport channel parameters for Interactive or background / UL:128 kbps / PS RAB

See clause 6.10.3.4.1.28.1.1.1.

6.10.3.4.1.53.1.1.3 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.10.3.4.1.53.1.1.4 TFCS

| | |
|---|---|
| TFCS size | 20 |
| TFCS | (Conv. 64 kbps RAB, I/B 128kbps RAB, DCCH)=<br>(TF0, TF0, TF0), (TF0, TF1, TF0), (TF0, TF2, TF0), (TF0, TF3, TF0), (TF0, TF4, TF0),<br>(TF1, TF0, TF0), (TF1, TF1, TF0), (TF1, TF2, TF0), (TF1, TF3, TF0), (TF1, TF4, TF0),<br>(TF0, TF0, TF1), (TF0, TF1, TF1), (TF0, TF2, TF1), (TF0, TF3, TF1), (TF0, TF4, TF1),<br>(TF1, TF0, TF1), (TF1, TF1, TF1), (TF1, TF2, TF1), (TF1, TF3, TF1), (TF1, TF4, TF1) |
| NOTE: In case TB size zero is configured for any transport channel, the first TFC is required; it is optional otherwise. | |

6.10.3.4.1.53.1.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Uplink | Midamble | 512 chips |
| | Codes and time slots | SF2 x 1 code x 2 timeslots |
| | Max. Number of data bits/radio frame | 3 760 bits |
| | TFCI code word | 16 bits |
| | TPC | 2 bits |
| | Puncturing Limit | 0.52 (alt. 0.48) |

6.10.3.4.1.53.2 Downlink

See clause 6.10.3.4.1.52.2.

6.10.3.4.1.54 Void

6.10.3.4.1.55 Void

6.10.3.4.1.56 Interactive or background / UL:8 DL:8 kbps / PS RAB + Interactive or background / UL:8 DL:8 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.3.4.1.56.1 Uplink

6.10.3.4.1.56.1.1 Transport channel parameters

6.10.3.4.1.56.1.1.1 Transport channel parameters for Interactive or background / UL:8 kbps / PS RAB + UL:8 kbps / PS RAB

| | | | | |
|---|---|---|---|---|
| Higher layer | RAB/Signalling RB | | RAB | RAB |
| RLC | Logical channel type | | DTCH | DTCH |
| | RLC mode | | AM | AM |
| | Payload sizes, bit | | 320 (alt. 128) | 320 (alt.128) |
| | Max data rate, bps | | 8 000 | 8 000 |
| | AMD PDU header, bit | | 16 | 16 |
| MAC | MAC header, bit | | 4 | 4 |
| | MAC multiplexing | | 2 logical channel multiplexing | |
| Layer 1 | TrCH type | | DCH | |
| | TB sizes, bit | | 340 (alt. 148) | |
| | TFS | TF0, bits | 0x340 (alt. 0x148) | |
| | | TF1, bits | 1x340 (alt. 1x148) | |
| | | TF2, bits | N/A (alt. 5x148) | |
| | TTI, ms | | 40 (alt. 80) | |
| | Coding type | | TC | |
| | CRC, bit | | 16 | |
| | Max number of bits/TTI after channel coding | | 1 080 (alt. 2 472) | |
| | Max number of bits/radio frame before rate matching | | 270 (alt. 309) | |
| | RM attribute | | 135 to 175 | |

6.10.3.4.1.56.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.10.3.4.1.56.1.1.3 TFCS

| | |
|---|---|
| TFCS size | 4 (alt. 6) |
| TFCS | (8 kbps RAB + 8 kbps RAB, DCCH)=<br>(TF0,TF0), (TF1,TF0), (TF0,TF1), (TF1,TF1)<br>(alt. (TF0,TF0), (TF1,TF0), (TF2,TF0), (TF0,TF1), (TF1,TF1), (TF2,TF1)) |
| NOTE: In case TB size zero is configured for any transport channel the first TFC is required; it is optional otherwise. | |

6.10.3.4.1.56.1.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Uplink | Midamble | 512 chips |
| | Codes and time slots | SF16 x 1 code x 1 time slot |
| | Max. Number of data bits/radio frame | 226 bits |
| | TFCI code word | 16 bits |
| | TPC | 2 bits |
| | Puncturing Limit | 0.52 (alt. 0.48) |
| NOTE: In case the first TFC in the TFCS is not configured, the TFCI code word will be 8 bits (alt. 16 bits). | | |

6.10.3.4.1.56.2 Downlink

6.10.3.4.1.56.2.1 Transport channel parameters

6.10.3.4.1.56.2.1.1 Transport channel parameters for Interactive or background / DL:8 kbps / PS RAB + DL:8 kbps / PS RAB

| | | | | |
|---|---|---|---|---|
| Higher layer | RAB/Signalling RB | | RAB | RAB |
| RLC | Logical channel type | | DTCH | DTCH |
| | RLC mode | | AM | AM |
| | Payload sizes, bit | | 320 | 320 |
| | Max data rate, bps | | 8 000 | 8 000 |
| | AMD PDU header, bit | | 16 | 16 |
| MAC | MAC header, bit | | 4 | 4 |
| | MAC multiplexing | | 2 logical channel multiplexing | |
| Layer 1 | TrCH type | | DCH | |
| | TB sizes, bit | | 340 | |
| | TFS | TF0, bits | 0x340 | |
| | | TF1, bits | 1x340 | |
| | TTI, ms | | 40 | |
| | Coding type | | TC | |
| | CRC, bit | | 16 | |
| | Max number of bits/TTI after channel coding | | 1 080 | |
| | Max number of bits/radio frame before rate matching | | 270 | |
| | RM attribute | | 135 to 175 | |

6.10.3.4.1.56.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.10.3.4.1.56.2.1.3 TFCS

| | |
|---|---|
| TFCS size | 4 |
| TFCS | (8 kbps RAB + 8 kbps RAB, DCCH)=<br>(TF0,TF0), (TF1,TF0), (TF0,TF1), (TF1,TF1) |
| NOTE: In case TB size zero is configured for any transport channel the first TFC is required; it is optional otherwise. | |

6.10.3.4.1.56.2.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Downlink | Midamble | 512 chips |
| | Codes and time slots | SF16 x 1 codes x 1 time slot |
| | Max. Number of data bits/radio frame | 228 bits |
| | TFCI code word | 16 bits |
| | Puncturing limit | 0.56 |
| NOTE: In case the first TFC in the TFCS is not configured, the TFCI code word will be 8 bits. | | |

6.10.3.4.1.57 Interactive or background / UL:64 DL:64 kbps / PS RAB + Interactive or background / UL:64 DL:64 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.3.4.1.57.1 Uplink

6.10.3.4.1.57.1.1 Transport channel parameters

6.10.3.4.1.57.1.1.1 Transport channel parameters for Interactive or background / UL:64 kbps / PS RAB + UL:64 kbps / PS RAB

See clause 6.10.3.4.1.38d.1.1.2.

6.10.3.4.1.57.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.10.3.4.1.57.1.1.3 TFCS

| | |
|---|---|
| TFCS size | 10 |
| TFCS | (64 kbps RAB + 64 kbps RAB, DCCH)=<br>(TF0,TF0), (TF1,TF0), (TF2,TF0), (TF3,TF0), (TF4,TF0),<br>(TF0,TF1), (TF1,TF1), (TF2,TF1), (TF3,TF1), (TF4,TF1) |
| NOTE: In case TB size zero is configured for any transport channel the first TFC is required; it is optional otherwise. | |

6.10.3.4.1.57.1.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Uplink | Midamble | 256 chips |
| | Codes and time slots | SF2 x 1 code x 1 time slot |
| | Max. Number of data bits/radio frame | 2 064 bits |
| | TFCI code word | 16 bits |
| | TPC | 2 bits |
| | Puncturing Limit | 0.88 (alt. 0.76) |

6.10.3.4.1.57.2 Downlink

6.10.3.4.1.57.2.1 Transport channel parameters

6.10.3.4.1.57.2.1.1 Transport channel parameters for Interactive or background / DL:64 kbps / PS RAB + DL:64 kbps / PS RAB

| | | | | |
|---|---|---|---|---|
| Higher layer | RAB/Signalling RB | | RAB | RAB |
| RLC | Logical channel type | | DTCH | DTCH |
| | RLC mode | | AM | AM |
| | Payload sizes, bit | | 320 | 320 |
| | Max data rate, bps | | 64 000 | 64 000 |
| | AMD PDU header, bit | | 16 | 16 |
| MAC | MAC header, bit | | 4 | 4 |
| | MAC multiplexing | | 2 logical channel multiplexing | |
| Layer 1 | TrCH type | | DCH | |
| | TB sizes, bit | | 340 | |
| | TFS | TF0, bits | 0x340 | |
| | | TF1, bits | 1x340 | |
| | | TF2, bits | 2x340 | |
| | | TF3, bits | 3x340 | |
| | | TF4, bits | 4x340 | |
| | TTI, ms | | 20 | |
| | Coding type | | TC | |
| | CRC, bit | | 16 | |
| | Max number of bits/TTI after channel coding | | 4 284 | |
| | Max number of bits/radio frame before rate matching | | 2 142 | |
| | RM attribute | | 130 to 170 | |

6.10.3.4.1.57.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.10.3.4.1.57.2.1.3 TFCS

| | |
|---|---|
| TFCS size | 10 |
| TFCS | (64 kbps RAB + 64 kbps RAB, DCCH)=<br>(TF0,TF0), (TF1,TF0), (TF2,TF0), (TF3,TF0), (TF4,TF0),<br>(TF0,TF1), (TF1,TF1), (TF2,TF1), (TF3,TF1), (TF4,TF1) |
| NOTE: In case TB size zero is configured for any transport channel the first TFC is required; it is optional otherwise. | |

6.10.3.4.1.57.2.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Downlink | Midamble | 256 chips |
| | Codes and time slots | SF16 x 5 codes x 1 time slot |
| | Max. Number of data bits/radio frame | 1 364 bits |
| | TFCI code word | 16 bits |
| | Puncturing limit | 0.56 |

6.10.3.4.1.58 Streaming / unknown / UL:16 DL:64 kbps / PS RAB + Interactive or background / UL:8 DL:8 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.3.4.1.58.1 Uplink

6.10.3.4.1.58.1.1 Transport channel parameters

6.10.3.4.1.58.1.1.1 Transport channel parameters for Streaming / unknown / UL:16 kbps / PS RAB

| | | | |
|---|---|---|---|
| Higher layer | RAB/Signalling RB | | RAB |
| RLC | Logical channel type | | DTCH |
| | RLC mode | | AM |
| | Payload sizes, bit | | 320 |
| | Max data rate, bps | | 16 000 |
| | AMD PDU header, bit | | 16 |
| MAC | MAC header, bit | | 0 |
| | MAC multiplexing | | N/A |
| Layer 1 | TrCH type | | DCH |
| | TB sizes, bit | | 336 |
| | TFS | TF0, bits | 0x336 |
| | | TF1, bits | 1x336 |
| | TTI, ms | | 20 |
| | Coding type | | TC |
| | CRC, bit | | 16 |
| | Max number of bits/TTI after channel coding | | 1 068 |
| | Max number of bits/radio frame before rate matching | | 534 |
| | RM attribute | | 135 to 175 |

6.10.3.4.1.58.1.1.2 Transport channel parameters for Interactive or background / UL:8 kbps / PS RAB

See clause 6.10.3.4.1.23a.1.1.1.

6.10.3.4.1.58.1.1.3 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.10.3.4.1.58.1.1.4 TFCS

| | |
|---|---|
| TFCS size | 8 (alt. 12) |
| TFCS | (16 kbps RAB, 8 kbps RAB, DCCH)=<br>(TF0,TF0,TF0), (TF1,TF0,TF0), (TF0,TF1,TF0), (TF1,TF1,TF0),<br>(TF0,TF0,TF1), (TF1,TF0,TF1), (TF0,TF1,TF1), (TF1,TF1,TF1)<br>(alt.<br>(TF0,TF0,TF0), (TF1,TF0,TF0), (TF0,TF1,TF0), (TF1,TF1,TF0), (TF0,TF2,TF0), (TF1,TF2,TF0),<br>(TF0,TF0,TF1), (TF1,TF0,TF1), (TF0,TF1,TF1), (TF1,TF1,TF1), (TF0,TF2,TF1), (TF1,TF2,TF1)) |
| NOTE: In case TB size zero is configured for any transport channel the first TFC is required; it is optional otherwise. | |

6.10.3.4.1.58.1.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Uplink | Midamble | 512 chips |
| | Codes and time slots | SF8 x 1 code x 1 time slot +<br>SF16 x 1code x 1 time slot |
| | Max. Number of data bits/radio frame | 696 bits |
| | TFCI code word | 16 bits |
| | TPC | 2 bits |
| | Puncturing Limit | 0.72 (alt. 0.68) |

6.10.3.4.1.58.2 Downlink

6.10.3.4.1.58.2.1 Transport channel parameters

6.10.3.4.1.58.2.1.1 Transport channel parameters for Streaming / unknown / DL:64 kbps / PS RAB

| | | | |
|---|---|---|---|
| Higher layer | RAB/Signalling RB | | RAB |
| RLC | Logical channel type | | DTCH |
| | RLC mode | | AM |
| | Payload sizes, bit | | 640 |
| | Max data rate, bps | | 64 000 |
| | AM PDU header, bit | | 16 |
| MAC | MAC header, bit | | 0 |
| | MAC multiplexing | | N/A |
| Layer 1 | TrCH type | | DCH |
| | TB sizes, bit | | 656 |
| | TFS | TF0, bits | 0x656 |
| | | TF1, bits | 1x656 |
| | | TF2, bits | 2x656 |
| | | TF3, bits | 4x656 |
| | TTI, ms | | 40 |
| | Coding type | | TC |
| | CRC, bit | | 16 |
| | Max number of bits/TTI after channel coding | | 8 076 |
| | Max number of bits/radio frame before rate matching | | 2 019 |
| | RM attribute | | 125 to 165 |

6.10.3.4.1.58.2.1.2 Transport channel parameters for Interactive or background / DL:8 kbps / PS RAB

See clause 6.10.3.4.1.23.2.1.1.

6.10.3.4.1.58.2.1.3 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.10.3.4.1.58.2.1.4 TFCS

| | |
|---|---|
| TFCS size | 16 |
| TFCS | (64 kbps RAB, 8 kbps RAB, DCCH)=<br>(TF0,TF0,TF0), (TF1,TF0,TF0), (TF2,TF0,TF0), (TF3,TF0,TF0),<br>(TF0,TF1,TF0), (TF1,TF1,TF0), (TF2,TF1,TF0), (TF3,TF1,TF0),<br>(TF0,TF0,TF1), (TF1,TF0,TF1), (TF2,TF0,TF1), (TF3,TF0,TF1),<br>(TF0,TF1,TF1), (TF1,TF1,TF1), (TF2,TF1,TF1), (TF3,TF1,TF1) |
| NOTE: In case TB size zero is configured for any transport channel the first TFC is required; it is optional otherwise. | |

6.10.3.4.1.58.2.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Downlink | Midamble | 256 chips |
| | Codes and time slots | SF16 x 6 codes x 1 time slot |
| | Max. Number of data bits/radio frame | 1 640 bits |
| | TFCI code word | 16 bits |
| | Puncturing limit | 0.64 |

6.10.3.4.1.59 Reserved for future use

6.10.3.4.1.60 Reserved for future use

6.10.3.4.1.61 Conversational / unknown / UL:8 DL:8 kbps / PS RAB + Interactive or Background / UL:8 DL:8 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.10.3.4.1.61.1 Uplink

6.10.3.4.1.61.1.1 Transport channel parameters

6.10.3.4.1.61.1.1.1 Transport channel parameters for Conversational / unknown / UL:8 kbps / PS RAB

| | | |
|---|---|---|
| Higher layer | RAB/Signalling RB | RAB |
| RLC | Logical channel type | DTCH |
| | RLC mode | UM |
| | Payload sizes, bit | 320 |

| | | | |
|---|---|---|---|
| | Max data rate, bps | | 8 000 |
| | UMD PDU header, bit | | 8 |
| MAC | MAC header, bit | | 0 |
| | MAC multiplexing | | N/A |
| Layer 1 | TrCH type | | DCH |
| | TB sizes, bit | | 328 (alt 0, 328) |
| | TFS | TF0, bits | 0x328 (alt 1x0) (note) |
| | | TF1, bits | 1x328 |
| | TTI, ms | | 40 |
| | Coding type | | TC |
| | CRC, bit | | 16 |
| | Max number of bits/TTI after channel coding | | 1 044 |
| | Max number of bits/radio frame before rate matching | | 261 |
| | RM attribute | | 135 to 175 |
| NOTE: In case of using this alternative, CRC parity bits are to be attached any time since number of TrBlks are 1 even if there is no data on the RAB (see clause 4.2.1.1 in 3GPP TS 25.222 [29]). | | | |

6.10.3.4.1.61.1.1.2 Transport channel parameters for Interactive or Background / UL:8 kbps / PS RAB

See clause 6.10.3.4.1.23a.1.1.1.

6.10.3.4.1.61.1.1.3 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.10.3.4.1.61.1.1.4 TFCS

| | |
|---|---|
| TFCS size | 8 (alt. 12) |
| TFCS | (8 kbps Conversational RAB, 8 kbps I/B RAB, DCCH)=<br>(TF0, TF0, TF0), (TF0, TF1, TF0), (TF0, TF0, TF1), (TF0, TF1, TF1),<br>(TF1, TF0, TF0), (TF1, TF1, TF0), (TF1, TF0, TF1), (TF1, TF1, TF1)<br>(alt.<br>(TF0, TF0, TF0), (TF0, TF1, TF0), (TF0, TF2, TF0),<br>(TF0, TF0, TF1), (TF0, TF1, TF1), (TF0, TF2, TF1),<br>(TF1, TF0, TF0), (TF1, TF1, TF0), (TF1, TF2, TF0),<br>(TF1, TF0, TF1), (TF1, TF1, TF1), (TF1, TF2, TF1)) |
| NOTE: In case TB size zero is configured for any transport channel the first TFC is required; it is optional otherwise. | |

6.10.3.4.1.61.1.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Uplink | Midamble | 512 chips |
| | Codes and time slots | SF8 x 1 code x 1 time slot |
| | Max. Number of data bits/radio frame | 452 bits |
| | TFCI code word | 16 bits |
| | TPC | 2 bits |
| | Puncturing Limit | 0.68 (alt. 0.64) |

6.10.3.4.1.61.2 Downlink

6.10.3.4.1.61.2.1 Transport channel parameters

6.10.3.4.1.61.2.1.1 Transport channel parameters for Conversational / unknown / DL:8 kbps / PS RAB

| | | | |
|---|---|---|---|
| Higher layer | RAB/Signalling RB | | RAB |
| RLC | Logical channel type | | DTCH |
| | RLC mode | | UM |
| | Payload sizes, bit | | 320 |
| | Max data rate, bps | | 8 000 |
| | AMD PDU header, bit | | 8 |
| MAC | MAC header, bit | | 0 |
| | MAC multiplexing | | N/A |
| Layer 1 | TrCH type | | DCH |
| | TB sizes, bit | | 328 (alt 0, 328) |
| | TFS | TF0, bits | 0x328 (alt 1x0) (note) |

| | | | |
|---|---|---|---|
| | | TF1, bits | 1x328 |
| | TTI, ms | | 40 |
| | Coding type | | TC |
| | CRC, bit | | 16 |
| | Max number of bits/TTI after channel coding | | 1 044 |
| | Max number of bits/radio frame before rate matching | | 261 |
| | RM attribute | | 135 to 175 |
| NOTE: In case of using this alternative, CRC parity bits are to be attached any time since number of TrBlks are 1 even if there is no data on the RAB (see clause 4.2.1.1 in 3GPP TS 25.222 [29]). | | | |

6.10.3.4.1.61.2.1.2 Transport channel parameters for Interactive or Background / DL:8 kbps / PS RAB

See clause 6.10.3.4.1.23.2.1.1.

6.10.3.4.1.61.2.1.3 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.10.3.4.1.61.2.1.4 TFCS

| | |
|---|---|
| TFCS size | 8 |
| TFCS | (8 kbps Conversational RAB, 8 kbps I/B RAB, DCCH)=<br>(TF0, TF0, TF0), (TF0, TF1, TF0), (TF0, TF0, TF1), (TF0, TF1, TF1),<br>(TF1, TF0, TF0), (TF1, TF1, TF0), (TF1, TF0, TF1), (TF1, TF1, TF1) |
| NOTE: In case TB size zero is configured for any transport channel the first TFC is required; it is optional otherwise. | |

6.10.3.4.1.61.2.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Downlink | Midamble | 512 chips |
| | Codes and time slots | SF16 x 2 codes x 1 time slot |
| | Max. Number of data bits/radio frame | 472 bits |
| | TFCI code word | 16 bits |
| | Puncturing limit | 0.68 |

## 6.10.3.4.2 Combinations on PDSCH, SCCPCH, PUSCH and PRACH

### 6.10.3.4.2.1 Interactive or background / UL: 64 DL: 256 kbps / PS RAB + UL: 3.4/16.8 DL: 3.4/33.6 kbps SRBs for DCCH, CCCH and BCCH + UL: 16.8 DL: 16 kbps SRBs for SHCCH

6.10.3.4.2.1.1 Uplink

6.10.3.4.2.1.1.1 Transport channel parameters

6.10.3.4.2.1.1.1.1 Transport channel parameters for Interactive or background / UL: 64 kbps / PS RAB and UL SRB for SHCCH mapped on USCH

| | | | | |
|---|---|---|---|---|
| Higher layer | RAB/Signalling RB | | RAB | SRB#5 |
| RLC | Logical channel type | | DTCH | SHCCH |
| | RLC mode | | AM | TM |
| | Payload sizes, bit | | 320 (alt. 128) | 168 |
| | Max data rate, bps | | 64 000 | 16 800 |
| | AMD/TrD PDU header, bit | | 16 | 0 |
| MAC | MAC header, bit | | 1 | 1 |
| | MAC multiplexing | | N/A | N/A |
| Layer 1 | TrCH type | | USCH | USCH |
| | TB sizes, bit | | 337 (alt. 145) | 169 |
| | TFS | TF0, bits | 0x337 (alt. 0x145) | 0x169 |
| | | TF1, bits | 1x337 (alt. 1x145) | 1x169 |
| | | TF2, bits | 2x337 (alt. 3x145) | N/A |
| | | TF3, bits | 3x337 (alt. 7x145) | N/A |
| | | TF4, bits | 4x337 (alt. 10x145) | N/A |
| | TTI, ms | | 20 | 10 |
| | Coding type | | TC | CC 1/2 |
| | CRC, bit | | 16 | 16 |

| | | | |
|---|---|---|---|
| | Max number of bits/TTI after channel coding | 4 248 (alt. 4 842) | 386 |
| | Max number of bits/radio frame before rate matching | 2 124 (alt. 2 421) | 386 |
| | RM attribute | 135 to 175 | 230 to 250 |

6.10.3.4.2.1.1.1.2 Transport channel parameters for UL: 3.4 Kbps SRBs for DCCH mapped on USCH

| | | | | | | |
|---|---|---|---|---|---|---|
| Higher layer | RAB/signalling RB | | SRB#1 | SRB#2 | SRB#3 | SRB#4 |
| | User of Radio Bearer | | RRC | RRC | NAS_DT High prio | NAS_DT Low prio |
| RLC | Logical channel type | | DCCH | DCCH | DCCH | DCCH |
| | RLC mode | | UM | AM | AM | AM |
| | Payload sizes, bit | | 136 | 128 | 128 | 128 |
| | Max data rate, bps | | 3 400 | 3 200 | 3 200 | 3 200 |
| | AMD/UMD PDU header, bit | | 8 | 16 | 16 | 16 |
| MAC | MAC header, bit | | 5 | 5 | 5 | 5 |
| | MAC multiplexing | | 4 logical channel multiplexing | | | |
| Layer 1 | TrCH type | | USCH | | | |
| | TB sizes, bit | | 149 | | | |
| | TFS | TF0, bits | 0x149 | | | |
| | | TF1, bits | 1x149 | | | |
| | TTI, ms | | 40 | | | |
| | Coding type | | CC 1/3 | | | |
| | CRC, bit | | 16 | | | |
| | Max number of bits/TTI before rate matching | | 519 | | | |
| | Max number of bits/radio frame before rate matching | | 130 | | | |
| | RM attribute | | 190 to 210 | | | |

6.10.3.4.2.1.1.1.3 TFCS for USCH

| | |
|---|---|
| TFCS size | 20 |
| TFCS | (64 kbps RAB, SHCCH, SRBs for DCCH)=<br>(TF0, TF0, TF0), (TF1, TF0, TF0), (TF2, TF0, TF0), (TF3, TF0, TF0), (TF4, TF0, TF0), (TF0, TF1, TF0), (TF1, TF1, TF0), (TF2, TF1, TF0), (TF3, TF1, TF0), (TF4, TF1, TF0), (TF0, TF0, TF1), (TF1, TF0, TF1), (TF2, TF0, TF1), (TF3, TF0, TF1), (TF4, TF0, TF1),<br>(TF0, TF1, TF1), (TF1, TF1, TF1), (TF2, TF1, TF1), (TF3, TF1, TF1), (TF4, TF1, TF1) |

6.10.3.4.2.1.1.1.4 Transport channel parameters for SRB for CCCH and UL SRBs for DCCH and UL SRB for SHCCH mapped on RACH

6.10.3.4.2.1.1.1.4.1 RACH transport channel configuration without DTCH

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| Higher layer | RAB/signalling RB | SRB#0 | SRB#1 | SRB#2 | SRB#3 | SRB#4 | SRB#5 |
| | User of Radio Bearer | RRC | RRC | RRC | NAS_DT High prio | NAS_DT Low prio | RRC |
| RLC | Logical channel type | CCCH | DCCH | DCCH | DCCH | DCCH | SHCCH |
| | RLC mode | TM | UM | AM | AM | AM | TM |
| | Payload sizes, bit | 168 | 136 | 128 | 128 | 128 | 168 |
| | Max data rate, bps | 16 800 | 13 600 | 12 800 | 12 800 | 12 800 | 16 800 |
| | AMD/UMD/TrD PDU header, bit | 0 | 8 | 16 | 16 | 16 | 0 |
| MAC | MAC header, bit | 2 | 26 | 26 | 26 | 26 | 2 |
| | MAC multiplexing | 6 logical channel multiplexing | | | | | |
| Layer 1 | TrCH type | RACH | | | | | |
| | TB sizes, bit | 170 | | | | | |
| | TFS \| TF0, bits | 1x170 | | | | | |
| | TTI, ms | 10 | | | | | |
| | Coding type | CC 1/2 | | | | | |
| | CRC, bit | 16 | | | | | |
| | Max number of bits/TTI after channel coding | 388 | | | | | |
| | Max number of bits/radio frame before rate matching | 388 | | | | | |

6.10.3.4.2.1.1.1.4.2 RACH transport channel configuration with DTCH

| Higher layer | RAB/signalling RB | RAB | SRB#0 | SRB#1 | SRB#2 | SRB#3 | SRB#4 | SRB#5 |
|---|---|---|---|---|---|---|---|---|
| | User of Radio Bearer | Interactive/ Background RAB | RRC | RRC | RRC | NAS_DT High prio | NAS_DT Low prio | RRC |
| RLC | Logical channel type | DTCH | CCCH | DCCH | DCCH | DCCH | DCCH | SHCCH |
| | RLC mode | AM | TM | UM | AM | AM | AM | TM |
| | Payload sizes, bit | 128 | 168 | 136 | 128 | 128 | 128 | 168 |
| | Max data rate, bps | 12 800 | 16 800 | 13 600 | 12 800 | 12 800 | 12 800 | 16 800 |
| | AMD/UMD/TrD PDU header, bit | 16 | 0 | 8 | 16 | 16 | 16 | 0 |
| MAC | MAC header, bit | 26 | 2 | 26 | 26 | 26 | 26 | 2 |
| | MAC multiplexing | 7 logical channel multiplexing | | | | | | |
| Layer 1 | TrCH type | RACH | | | | | | |
| | TB sizes, bit | 170 | | | | | | |
| | TFS TF0, bits | 1x170 | | | | | | |
| | TTI, ms | 10 | | | | | | |
| | Coding type | CC 1/2 | | | | | | |
| | CRC, bit | 16 | | | | | | |
| | Max number of bits/TTI after channel coding | 388 | | | | | | |
| | Max number of bits/radio frame before rate matching | 388 | | | | | | |

6.10.3.4.2.1.1.2 Physical channel parameters

6.10.3.4.2.1.1.2.1 Physical channel parameters for PUSCH

| PUSCH | Midamble | 512 chips |
|---|---|---|
| | Codes and time slots | SF2 x 1 code x 1 time slot |
| | Max. Number of data bits/radio frame | 1 808 bits |
| | TFCI code word | 16 bits |
| | TPC | 2 bits |
| | Puncturing Limit | 0.60 (alt. 0.56) |

6.10.3.4.2.1.1.2.2 Physical channel parameters for PRACH

| PRACH | Midamble | 512 chips |
|---|---|---|
| | Codes and time slots | SF8 (alt. SF16) x 1 code x 1 time slot |
| | Max. Number of data bits/radio frame | 464 (alt. 232) |
| | Puncturing Limit | 1 (alt. 0.56) |

6.10.3.4.2.1.2 Downlink

6.10.3.4.2.1.2.1 Transport channel parameters

6.10.3.4.2.1.2.1.1 Transport channel parameters for Interactive or background / DL: 256 kbps / PS RAB and DL SRB for SHCCH mapped on DSCH

| Higher layer | RAB/Signalling RB | RAB | SRB#5 |
|---|---|---|---|
| RLC | Logical channel type | DTCH | SHCCH |
| | RLC mode | AM | UM |
| | Payload sizes, bit | 320 | 160 |
| | Max data rate, bps | 256 000 | 16 000 |
| | AMD/UMD PDU header, bit | 16 | 8 |
| MAC | MAC header, bit | 1 | 1 |
| | MAC multiplexing | N/A | N/A |
| Layer 1 | TrCH type | DSCH | DSCH |
| | TB sizes, bit | 337 | 169 |

| | | | | |
|---|---|---|---|---|
| | TFS | TF0, bits | 0x337 | 0x169 |
| | | TF1, bits | 1x337 | 1x169 |
| | | TF2, bits | 2x337 | N/A |
| | | TF3, bits | 4x337 | N/A |
| | | TF4, bits | 8x337 | N/A |
| | | TF5, bits | N/A (alt. 12x337) | N/A |
| | | TF6, bits | N/A (alt. 16x337) | N/A |
| | TTI, ms | | 10 (alt. 20) | 10 |
| | Coding type | | TC | CC 1/2 |
| | CRC, bit | | 16 | 16 |
| | Max number of bits/TTI after channel coding | | 8 484 (alt. 16 968) | 386 |
| | Downlink: Max number of bits/radio frame before rate matching | | 8 484 (alt. 8 484) | 386 |
| | RM attribute | | 135 to 175 | 230 to 250 |

6.10.3.4.2.1.2.1.2 Transport channel parameters for DL: 3.4 Kbps SRBs for DCCH mapped on DSCH

| | | | | | | |
|---|---|---|---|---|---|---|
| Higher layer | RAB/signalling RB | | SRB#1 | SRB#2 | SRB#3 | SRB#4 |
| | User of Radio Bearer | | RRC | RRC | NAS_DT High prio | NAS_DT Low prio |
| RLC | Logical channel type | | DCCH | DCCH | DCCH | DCCH |
| | RLC mode | | UM | AM | AM | AM |
| | Payload sizes, bit | | 136 | 128 | 128 | 128 |
| | Max data rate, bps | | 3 400 | 3 200 | 3 200 | 3 200 |
| | AMD/UMD PDU header, bit | | 8 | 16 | 16 | 16 |
| MAC | MAC header, bit | | 5 | 5 | 5 | 5 |
| | MAC multiplexing | | 4 logical channel multiplexing | | | |
| Layer 1 | TrCH type | | DSCH | | | |
| | TB sizes, bit | | 149 | | | |
| | TFS | TF0, bits | 0x149 | | | |
| | | TF1, bits | 1x149 | | | |
| | TTI, ms | | 40 | | | |
| | Coding type | | CC 1/3 | | | |
| | CRC, bit | | 16 | | | |
| | Max number of bits/TTI before rate matching | | 519 | | | |
| | Max number of bits/radio frame before rate matching | | 130 | | | |
| | RM attribute | | 155 to 165 | | | |

6.10.3.4.2.1.2.1.3 TFCS for DSCH

| | |
|---|---|
| TFCS size | 20 (alt. 28) |
| TFCS | (256 kbps RAB, SHCCH, SRB for DCCH)=<br>(TF0, TF0, TF0), (TF1, TF0, TF0), (TF2, TF0, TF0), (TF3, TF0, TF0), (TF4, TF0, TF0),<br>(TF0, TF1, TF0), (TF1, TF1, TF0), (TF2, TF1, TF0), (TF3, TF1, TF0), (TF4, TF1, TF0),<br>(TF0, TF0, TF1), (TF1, TF0, TF1), (TF2, TF0, TF1), (TF3, TF0, TF1), (TF4, TF0, TF1),<br>(TF0, TF1, TF1), (TF1, TF1, TF1), (TF2, TF1, TF1), (TF3, TF1, TF1), (TF4, TF1, TF1)<br>(alt. (TF0, TF0, TF0), (TF1, TF0, TF0), (TF2, TF0, TF0), (TF3, TF0, TF0), (TF4, TF0, TF0),<br>(TF5, TF0, TF0), (TF6, TF0, TF0), (TF0, TF1, TF0), (TF1, TF1, TF0), (TF2, TF1, TF0),<br>(TF3, TF1, TF0), (TF4, TF1, TF0), (TF5, TF1, TF0), (TF6, TF1, TF0), (TF0, TF0, TF1),<br>(TF1, TF0, TF1), (TF2, TF0, TF1), (TF3, TF0, TF1), (TF4, TF0, TF1), (TF5, TF0, TF1),<br>(TF6, TF0, TF1), (TF0, TF1, TF1), (TF1, TF1, TF1), (TF2, TF1, TF1), (TF3, TF1, TF1),<br>(TF4, TF1, TF1), (TF5, TF1, TF1), (TF6, TF1, TF1)) |

6.10.3.4.2.1.2.1.4 Transport channel parameters for DL SRBs for DCCH and SRB for CCCH and SRB for BCCH and DL SRB for SHCCH mapped on FACH

6.10.3.4.2.1.2.1.4.1 FACH transport channel configuration without DTCH

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| Higher layer | RAB/signalling RB | | SRB#0 | SRB#1 | SRB#2 | SRB#3 | SRB#4 | SRB#5 | SRB#6 |
| | User of Radio Bearer | | RRC | RRC | RRC | NAS_DT High prio | NAS_DT Low prio | RRC | RRC |
| RLC | Logical channel type | | CCCH | DCCH | DCCH | DCCH | DCCH | SHCCH | BCCH |
| | RLC mode | | UM | UM | AM | AM | AM | UM | TM |
| | Payload sizes, bit | | 160 | 136 or 120 (note) | 128 | 128 | 128 | 160 | 168 |
| | Max data rate, bps | | 32 000 (alt. 16 000) | 27 200 or 24 000 (alt. 13 600 or 12 000) | 25 600 (alt. 12 800) | 25 600 (alt. 12 800) | 25 600 (alt. 12 800) | 32 000 (alt. 16 000) | 33 600 (alt. 16 800) |
| | AMD/UMD/TrD PDU header, bit | | 8 | 8 | 16 | 16 | 16 | 8 | 0 |
| MAC | MAC header, bit | | 3 | 27 or 43 | 27 | 27 | 27 | 3 | 3 |
| | MAC multiplexing | | 7 logical channel multiplexing | | | | | | |
| Layer 1 | TrCH type | | FACH | | | | | | |
| | TB sizes, bit | | 171 | | | | | | |
| | TFS | TF0, bits | 0x171 | | | | | | |
| | | TF1, bits | 1x171 | | | | | | |
| | | TF2, bits | 2x171 | | | | | | |
| | | TF3, bits | 3x171(alt. N/A) | | | | | | |
| | | TF4, bits | 4x171(alt. N/A) | | | | | | |
| | TTI, ms | | 20 | | | | | | |
| | Coding type | | TC | | | | | | |
| | CRC, bit | | 16 | | | | | | |
| | Max number of bits/TTI after channel coding | | 2 256 (alt. 1 134) | | | | | | |
| | Max number of bits/radio frame before rate matching | | 1 128 (alt. 567) | | | | | | |
| NOTE: MAC header size and RLC payload size depend on use of U-RNTI or C-RNTI. | | | | | | | | | |

6.10.3.4.2.1.2.1.4.2 FACH transport channel configuration with DTCH

| | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| Higher layer | RAB/signalling RB | | RAB | SRB#0 | SRB#1 | SRB#2 | SRB#3 | SRB#4 | SRB#5 | SRB#6 |
| | User of Radio Bearer | | Interactive/ Background RAB | RRC | RRC | RRC | NAS_DT High prio | NAS_DT Low prio | RRC | RRC |
| RLC | Logical channel type | | DTCH | CCCH | DCCH | DCCH | DCCH | DCCH | SHCCH | BCCH |
| | RLC mode | | AM | UM | UM | AM | AM | AM | UM | TM |
| | Payload sizes, bit | | 320 | 160 | 136 or 120 (note) | 128 | 128 | 128 | 160 | 168 |
| | Max data rate, bps | | 32 000 (alt. 16 000) | 32 000 (alt. 16 000) | 27 200 or 24 000 (alt. 13 600 or 12 000) | 25 600 (alt. 12 800) | 25 600 (alt. 12 800) | 25 600 (alt. 12 800) | 32 000 (alt. 16 000) | 33 600 (alt. 16 800) |
| | AMD/UMD/TrD PDU header, bit | | 16 | 8 | 8 | 16 | 16 | 16 | 8 | 0 |
| MAC | MAC header, bit | | 27 | 3 | 27 or 43 | 27 | 27 | 27 | 3 | 3 |
| | MAC multiplexing | | 8 logical channel multiplexing | | | | | | | |
| Layer 1 | TrCH type | | FACH | | | | | | | |
| | TB sizes, bit | | 171, 363 | | | | | | | |
| | TFS | TF0, bits | 0x171 | | | | | | | |
| | | TF1, bits | 1x171 | | | | | | | |
| | | TF2, bits | 2x171 | | | | | | | |
| | | TF3, bits | 1x363 | | | | | | | |
| | | TF4, bits | 3x171 (alt N/A) | | | | | | | |
| | | TF5, bits | 4x171 (alt. N/A) | | | | | | | |
| | | TF6, bits | 2x363 (alt. N/A) | | | | | | | |
| | TTI, ms | | 20 | | | | | | | |
| | Coding type | | TC | | | | | | | |
| | CRC, bit | | 16 | | | | | | | |
| | Max number of bits/TTI after channel coding | | 2 286 (alt. 1 149) | | | | | | | |
| | Max number of bits/radio frame before rate matching | | 1 143 (alt. 575) | | | | | | | |
| NOTE: MAC header size and RLC payload size depend on use of U-RNTI or C-RNTI. | | | | | | | | | | |

6.10.3.4.2.1.2.1.5 TFCS for FACH

6.10.3.4.2.1.2.1.5.1 TFCS for FACH transport channel configuration without DTCH

| | |
|---|---|
| TFCS size | 5 (alt. 3) |
| TFCS | FACH = (TF0), (TF1), (TF2), (TF3), (TF4) (alt. FACH = (TF0), (TF1), (TF2)) |

6.10.3.4.2.1.2.1.5.2 TFCS for FACH transport channel configuration with DTCH

| | |
|---|---|
| TFCS size | 7 (alt. 4) |
| TFCS | FACH = (TF0), (TF1), (TF2), (TF3), (TF4), (TF5), (TF6) (alt. FACH = (TF0), (TF1), (TF2), (TF3)) |

6.10.3.4.2.1.2.2 Physical channel parameters

6.10.3.4.2.1.2.2.1 Physical channel parameters for PDSCH

| | | |
|---|---|---|
| PDSCH | Midamble | 256 chips |
| | Codes and time slots | SF16 x 8 codes x 2 time slots |
| | Max. Number of data bits/radio frame | 4 400 bits |
| | TFCI code word | 16 bits |
| | Puncturing Limit | 0.44 |

6.10.3.4.2.1.2.2.2 Physical channel parameters for SCCPCH

6.10.3.4.2.1.2.2.2.1 Physical channel parameters for SCCPCH without DTCH

| | | |
|---|---|---|
| SCCPCH | Midamble | 512 chips |
| | Codes and time slots | SF16 x 5 codes x 1 time slot<br>(alt. SF16 x 2 codes x 1 time slot ) |
| | Max. Number of data bits/radio frame | 1 204 bits (alt. 480 bits) |
| | TFCI code word | 16 bits (alt. 8 bits) |
| | Puncturing Limit | 1 (alt. 0.84) |

6.10.3.4.2.1.2.2.2.2 Physical channel parameters for SCCPCH with DTCH

| | | |
|---|---|---|
| SCCPCH | Midamble | 512 chips |
| | Codes and time slots | SF16 x 5 codes x 1 time slot<br>(alt. SF16 x 2 codes x 1 time slot ) |
| | Max. Number of data bits/radio frame | 1 204 bits (alt. 472 bits) |
| | TFCI code word | 16 bits |
| | Puncturing Limit | 1 (alt. 0.80) |

6.10.3.4.2.2 Interactive or background / UL: 64 DL: 384 kbps / PS RAB + UL: 3.4/16.8 DL: 3.4/33.6 kbps SRBs for DCCH, CCCH and BCCH+ UL: 16.8 DL: 16 kbps SRBs for SHCCH

6.10.3.4.2.2.1 Uplink

See clause 6.10.3.4.2.1.1.

6.10.3.4.2.2.2 Downlink

6.10.3.4.2.2.2.1 Transport channel parameters

6.10.3.4.2.2.2.1.1 Transport channel parameters for Interactive or background / DL: 384 kbps / PS RAB and DL SRB for SHCCH mapped on DSCH

| | | | |
|---|---|---|---|
| Higher layer | RAB/Signalling RB | RAB | SRB#5 |
| RLC | Logical channel type | DTCH | SHCCH |
| | RLC mode | AM | UM |
| | Payload sizes, bit | 320 | 160 |
| | Max data rate, bps | 384 000 | 16 000 |
| | AMD/UMD PDU header, bit | 16 | 8 |

| | | | | |
|---|---|---|---|---|
| MAC | MAC header, bit | | 1 | 1 |
| | MAC multiplexing | | N/A | N/A |
| Layer 1 | TrCH type | | DSCH | DSCH |
| | TB sizes, bit | | 337 | 169 |
| | TFS | TF0, bits | 0x337 | 0x169 |
| | | TF1, bits | 1x337 | 1x169 |
| | | TF2, bits | 2x337 | N/A |
| | | TF3, bits | 4x337 | N/A |
| | | TF4, bits | 8x337 | N/A |
| | | TF5, bits | 12x337 | N/A |
| | | TF6, bits | N/A (alt. 16x337) | N/A |
| | | TF7, bits | N/A (alt. 20x337) | N/A |
| | | TF8, bits | N/A (alt. 24x337) | N/A |
| | TTI, ms | | 10 (alt. 20) | 10 |
| | Coding type | | TC | CC 1/2 |
| | CRC, bit | | 16 | 16 |
| | Max number of bits/TTI after channel coding | | 12 720 (alt. 25 440) | 386 |
| | Downlink: Max number of bits/radio frame before rate matching | | 12 720 (alt. 12 720) | 386 |
| | RM attribute | | 135 to 175 | 230 to 250 |

6.10.3.4.2.2.2.1.2 Transport channel parameters for DL: 3.4 Kbps SRBs for DCCH mapped on DSCH

See clause 6.10.3.4.2.1.2.1.2.

6.10.3.4.2.2.2.1.3 TFCS for DSCH

| | |
|---|---|
| TFCS size | 24 (alt. 36) |
| TFCS | (384 kbps RAB, SHCCH, SRBs for DCCH)=<br>(TF0, TF0, TF0), (TF1, TF0, TF0), (TF2, TF0, TF0), (TF3, TF0, TF0), (TF4, TF0, TF0), (TF5, TF0, TF0), (TF0, TF1, TF0), (TF1, TF1, TF0), (TF2, TF1, TF0), (TF3, TF1, TF0), (TF4, TF1, TF0), (TF5, TF1, TF0), (TF0, TF0, TF1), (TF1, TF0, TF1), (TF2, TF0, TF1), (TF3, TF0, TF1), (TF4, TF0, TF1), (TF5, TF0, TF1), (TF0, TF1, TF1), (TF1, TF1, TF1), (TF2, TF1, TF1), (TF3, TF1, TF1), (TF4, TF1, TF1), (TF5, TF1, TF1)<br>(alt. (TF0, TF0, TF0), (TF1, TF0, TF0), (TF2, TF0, TF0), (TF3, TF0, TF0), (TF4, TF0, TF0), (TF5, TF0, TF0), (TF6, TF0, TF0), (TF7, TF0, TF0), (TF8, TF0, TF0), (TF0, TF1, TF0), (TF1, TF1, TF0),<br>(TF2, TF1, TF0), (TF3, TF1, TF0), (TF4, TF1, TF0), (TF5, TF1, TF0), (TF6, TF1, TF0), (TF7, TF1, TF0), (TF8, TF1, TF0), (TF0, TF0, TF1), (TF1, TF0, TF1), (TF2, TF0, TF1), (TF3, TF0, TF1), (TF4, TF0, TF1), (TF5, TF0, TF1), (TF6, TF0, TF1), (TF7, TF0, TF1), (TF8, TF0, TF1), (TF0, TF1, TF1), (TF1, TF1, TF1), (TF2, TF1, TF1), (TF3, TF1, TF1), (TF4, TF1, TF1), (TF5, TF1, TF1), (TF6, TF1, TF1), (TF7, TF1, TF1), (TF8, TF1, TF1)) |

6.10.3.4.2.2.2.1.4 Transport channel parameters for DL SRBs for DCCH and SRB for CCCH and SRB for BCCH and DL SRB for SHCCH mapped on FACH (with & without DTCH)

See clause 6.10.3.4.2.1.2.1.4.

6.10.3.4.2.2.2.1.5 TFCS for FACH

See clause 6.10.3.4.2.1.2.1.5.

6.10.3.4.2.2.2.2 Physical channel parameters

6.10.3.4.2.2.2.2.1 Physical channel parameters for PDSCH

| | | |
|---|---|---|
| PDSCH | Midamble | 256 chips |
| | Codes and time slots | SF16 x 8 codes x 3 time slots |
| | Max. Number of data bits/radio frame | 6 608 bits (alt. 6 592 bits) |
| | TFCI code word | 16 bits (alt. 32 bits) |
| | Puncturing Limit | 0.48 |

6.10.3.4.2.2.2.2.2 Physical channel parameters for SCCPCH

See clause 6.10.3.4.2.1.2.2.2.

6.10.3.4.2.3 Interactive or background / UL: 64 DL: 2 048 kbps / PS RAB + UL: 3.4/16.8 DL: 3.4/33.6 kbps SRBs for DCCH, CCCH and BCCH + UL: 16.8 DL: 16 kbps SRBs for SHCCH

6.10.3.4.2.3.1 Uplink

See clause 6.10.3.4.2.1.1.

6.10.3.4.2.3.2 Downlink

6.10.3.4.2.3.2.1 Transport channel parameters

6.10.3.4.2.3.2.1.1 Transport channel parameters for Interactive or background / DL: 2 048 kbps / PS RAB and DL SRB for SHCCH mapped on DSCH

| Higher layer | RAB/Signalling RB | | RAB | SRB#5 |
|---|---|---|---|---|
| RLC | Logical channel type | | DTCH | SHCCH |
| | RLC mode | | AM | UM |
| | Payload sizes, bit | | 640 | 160 |
| | Max data rate, bps | | 2 048 000 | 16 000 |
| | AMD/UMD PDU header, bit | | 16 | 8 |
| MAC | MAC header, bit | | 1 | 1 |
| | MAC multiplexing | | N/A | N/A |
| Layer 1 | TrCH type | | DSCH | DSCH |
| | TB sizes, bit | | 657 | 169 |
| | TFS | TF0, bits | 0x657 | 0x169 |
| | | TF1, bits | 1x657 | 1x169 |
| | | TF2, bits | 2x657 | N/A |
| | | TF3, bits | 4x657 | N/A |
| | | TF4, bits | 8x657 | N/A |
| | | TF5, bits | 12x657 | N/A |
| | | TF6, bits | 16x657 | N/A |
| | | TF7, bits | 20x657 | N/A |
| | | TF8, bits | 24x657 | N/A |
| | | TF9, bits | 28x657 | N/A |
| | | TF10, bits | 30x657 (alt. 32x657) | N/A |
| | | TF11, bits | N/A (alt. 36x657) | N/A |
| | | TF12, bits | N/A (alt. 40x657) | N/A |
| | | TF13, bits | N/A (alt. 44x657) | N/A |
| | | TF14, bits | N/A (alt. 48x657) | N/A |
| | | TF15, bits | N/A (alt. 52x657) | N/A |
| | | TF16, bits | N/A (alt. 56x657) | N/A |
| | | TF17, bits | N/A (alt. 60x657) | N/A |
| | | TF18, bits | N/A (alt. 64x657) | N/A |
| | TTI, ms | | 10 (alt. 20) | 10 |
| | Coding type | | TC | CC 1/2 |
| | CRC, bit | | 16 | 16 |
| | Max number of bits/TTI after channel coding | | 60 624 (alt. 129 330) | 386 |
| | Downlink: Max number of bits/radio frame before rate matching | | 60 624 (alt. 64 665) | 386 |
| | RM attribute | | 135 to 175 | 230 to 250 |

6.10.3.4.2.3.2.1.2 Transport channel parameters for DL: 3.4 Kbps SRBs for DCCH mapped on DSCH

See clause 6.10.3.4.2.1.2.1.2.

6.10.3.4.2.3.2.1.3 TFCS for DSCH

| TFCS size | 41 (alt.76) |
|---|---|
| TFCS | (2 048 kbps RAB, SHCCH, SRBs for DCCH)=<br>(TF0, TF0, TF0), (TF1, TF0, TF0), (TF2, TF0, TF0), (TF3, TF0, TF0), (TF4, TF0, TF0),<br>(TF5, TF0, TF0), (TF6, TF0, TF0), (TF7, TF0, TF0), (TF8, TF0, TF0), (TF9, TF0, TF0), (TF10, TF0, TF0),<br>(TF0, TF1, TF0), (TF1, TF1, TF0), (TF2, TF1, TF0), (TF3, TF1, TF0), (TF4, TF1, TF0),<br>(TF5, TF1, TF0), (TF6, TF1, TF0), (TF7, TF1, TF0), (TF8, TF1, TF0), (TF9, TF1, TF0),<br>(TF0, TF0, TF1), (TF1, TF0, TF1), (TF2, TF0, TF1), (TF3, TF0, TF1), (TF4, TF0, TF1), |

| | |
|---|---|
| | (TF5, TF0, TF1), (TF6, TF0, TF1), (TF7, TF0, TF1), (TF8, TF0, TF1), (TF9, TF0, TF1),<br>(TF0, TF1, TF1), (TF1, TF1, TF1), (TF2, TF1, TF1), (TF3, TF1, TF1), (TF4, TF1, TF1),<br>(TF5, TF1, TF1), (TF6, TF1, TF1), (TF7, TF1, TF1), (TF8, TF1, TF1), (TF9, TF1, TF1)<br>(alt. (TF0, TF0, TF0), (TF1, TF0, TF0), (TF2, TF0, TF0), (TF3, TF0, TF0), (TF4, TF0, TF0),<br>(TF5, TF0, TF0), (TF6, TF0, TF0), (TF7, TF0, TF0), (TF8, TF0, TF0), (TF9, TF0, TF0),<br>(TF10, TF0, TF0),(TF11, TF0, TF0), (TF12, TF0, TF0), (TF13, TF0, TF0), (TF14, TF0, TF0),<br>(TF15, TF0, TF0), (TF16, TF0, TF0), (TF17, TF0, TF0), (TF18, TF0, TF0),<br>(TF0, TF1, TF0), (TF1, TF1, TF0), (TF2, TF1, TF0), (TF3, TF1, TF0), (TF4, TF1, TF0),<br>(TF5, TF1, TF0), (TF6, TF1, TF0), (TF7, TF1, TF0), (TF8, TF1, TF0), (TF9, TF1, TF0),<br>(TF10, TF1, TF0),(TF11, TF1, TF0), (TF12, TF1, TF0), (TF13, TF1, TF0), (TF14, TF1, TF0),<br>(TF15, TF1, TF0), (TF16, TF1, TF0), (TF17, TF1, TF0), (TF18, TF1, TF0), (TF0, TF0, TF1),<br>(TF1, TF0, TF1), (TF2, TF0, TF1), (TF3, TF0, TF1), (TF4, TF0, TF1), (TF5, TF0, TF1),<br>(TF6, TF0, TF1), (TF7, TF0, TF1), (TF8, TF0, TF1), (TF9, TF0, TF1), (TF10, TF0, TF1),<br>(TF11, TF0, TF1), (TF12, TF0, TF1), (TF13, TF0, TF1), (TF14, TF0, TF1), (TF15, TF0, TF1),<br>(TF16, TF0, TF1), (TF17, TF0, TF1), (TF18, TF0, TF1),<br>(TF0, TF1, TF1), (TF1, TF1, TF1), (TF2, TF1, TF1), (TF3, TF1, TF1), (TF4, TF1, TF1),<br>(TF5, TF1, TF1), (TF6, TF1, TF1), (TF7, TF1, TF1), (TF8, TF1, TF1), (TF9, TF1, TF1),<br>(TF10, TF1, TF1),(TF11, TF1, TF1), (TF12, TF1, TF1), (TF13, TF1, TF1), (TF14, TF1, TF1),<br>(TF15, TF1, TF1), (TF16, TF1, TF1), (TF17, TF1, TF1), (TF18, TF1, TF1)) |

6.10.3.4.2.3.2.1.4 Transport channel parameters for DL SRBs for DCCH and SRB for CCCH and SRB for BCCH and DL SRB for SHCCH mapped on FACH

See clause 6.10.3.4.2.1.2.1.4.1.

6.10.3.4.2.3.2.1.5 TFCS for FACH

See clause 6.10.3.4.2.1.2.1.45.1.

6.10.3.4.2.3.2.2 Physical channel parameters

6.10.3.4.2.3.2.2.1 Physical channel parameters for PDSCH

| | | |
|---|---|---|
| PDSCH | Midamble | 256 chips |
| | Codes and time slots | SF16 x 12 codes x 11 time slots |
| | Max. Number of data bits/radio frame | 36 400 bits |
| | TFCI code word | 32 bits |
| | Puncturing Limit | 0.56 (alt. 0.52) |

6.10.3.4.2.3.2.2.2 Physical channel parameters for SCCPCH

See clause 6.10.3.4.2.1.2.2.2.1.

6.10.3.4.2.4 Interactive or background / UL: 384 DL: 2 048 kbps / PS RAB + UL: 3.4/16.8 DL: 3.4/33.6 kbps SRBs for DCCH, CCCH and BCCH + UL: 16.8 DL: 16 kbps SRBs for SHCCH

6.10.3.4.2.4.1 Uplink

6.10.3.4.2.4.1.1 Transport channel parameters

6.10.3.4.2.4.1.1.1 Transport channel parameters for Interactive or background / UL:384 kbps / PS RAB

| | | | | |
|---|---|---|---|---|
| Higher layer | RAB/Signalling RB | | RAB | SRB#5 |
| RLC | Logical channel type | | DTCH | SHCCH |
| | RLC mode | | AM | TM |
| | Payload sizes, bit | | 320 (alt. 128) | 168 |
| | Max data rate, bps | | 384 000 | 16 800 |
| | AMD/TrD PDU header, bit | | 16 | 0 |
| MAC | MAC header, bit | | 1 | 1 |
| | MAC multiplexing | | N/A | N/A |
| Layer 1 | TrCH type | | USCH | USCH |
| | TB sizes, bit | | 337 (alt. 145) | 169 |
| | TFS | TF0, bits | 0x337 (alt. 0x145) | 0x169 |
| | | TF1, bits | 1x337 (alt. 1x145) | 1x169 |

| | | | | |
|---|---|---|---|---|
| | | TF2, bits | 2x337 (alt. 5x145) | N/A |
| | | TF3, bits | 4x337 (alt. 10x145) | N/A |
| | | TF4, bits | 8x337 (alt. 20x145) | N/A |
| | | TF5, bits | 12x337 (alt. 30x145) | N/A |
| | | TF6, bits | 16x337 (alt. 40x145) | N/A |
| | | TF7, bits | 20x337 (alt. 50x145) | N/A |
| | | TF8, bits | 24x337 (alt. 60x145) | N/A |
| | TTI, ms | | 20 | 10 |
| | Coding type | | TC | CC 1/2 |
| | CRC, bit | | 16 | 16 |
| | Max number of bits/TTI after channel coding | | 25 440 (alt. 29 004) | 386 |
| | Max number of bits/radio frame before rate matching | | 12 720 (alt. 14 502) | 386 |
| | RM attribute | | 135 to 175 | 230 to 250 |

6.10.3.4.2.4.1.1.2 Transport channel parameters for UL: 3.4 Kbps SRBs for DCCH mapped on USCH

See clause 6.10.3.4.2.1.1.1.2.

6.10.3.4.2.4.1.1.3 TFCS for USCH

| | |
|---|---|
| TFCS size | 36 |
| TFCS | (384 kbps RAB, SHCCH, SRBs for DCCH)=<br>(TF0, TF0, TF0), (TF1, TF0, TF0), (TF2, TF0, TF0), (TF3, TF0, TF0), (TF4, TF0, TF0), (TF5, TF0, TF0), (TF6, TF0, TF0), (TF7, TF0, TF0), (TF8, TF0, TF0), (TF0, TF1, TF0), (TF1, TF1, TF0), (TF2, TF1, TF0), (TF3, TF1, TF0), (TF4, TF1, TF0), (TF5, TF1, TF0), (TF6, TF1, TF0), (TF7, TF1, TF0), (TF8, TF1, TF0), (TF0, TF0, TF1), (TF1, TF0, TF1), (TF2, TF0, TF1), (TF3, TF0, TF1), (TF4, TF0, TF1), (TF5, TF0, TF1), (TF6, TF0, TF1), (TF7, TF0, TF1), (TF8, TF0, TF1)<br>(TF0, TF1, TF1), (TF1, TF1, TF1), (TF2, TF1, TF1), (TF3, TF1, TF1), (TF4, TF1, TF1), (TF5, TF1, TF1), (TF6, TF1, TF1), (TF7, TF1, TF1), (TF8, TF1, TF1) |

6.10.3.4.2.4.1.1.4 Transport channel parameters for SRB for CCCH and UL SRBs for DCCH and UL SRB for SHCCH mapped on RACH

See clause 6.10.3.4.2.1.1.1.4.

6.10.3.4.2.4.1.2 Physical channel parameters

6.10.3.4.2.4.1.2.1 Physical channel parameters for PUSCH

| | | |
|---|---|---|
| PUSCH | Midamble | 512 chips |
| | Codes and time slots | SF1 x 1 code x 2 time slots |
| | Max. Number of data bits/radio frame | 7 264 bits |
| | TFCI code word | 32 bits |
| | TPC | 2 bits |
| | Puncturing Limit | 0.52 (alt. 0.44) |

6.10.3.4.2.4.1.2.2 Physical channel parameters for PRACH

See clause 6.10.3.4.2.1.1.2.2.

6.10.3.4.2.4.2 Downlink

6.10.3.4.2.4.2.1 Transport channel parameters

See clause 6.10.3.4.2.3.2.1.

6.10.3.4.2.4.2.2 Physical channel parameters

6.10.3.4.2.4.2.2.1 Physical channel parameters for PDSCH

| | | |
|---|---|---|
| PDSCH | Midamble | 256 chips |

| | | |
|---|---|---|
| | Codes and time slots | SF1 x 1 codes x 9 time slots |
| | Max. Number of data bits/radio frame | 39 712 bits |
| | TFCI code word | 32 bits |
| | Puncturing Limit | 0.64 (alt. 0.60) |

6.10.3.4.2.4.2.2.2 Physical channel parameters for SCCPCH

See clause 6.10.3.4.2.1.2.2.2.1.

### 6.10.3.4.3 Combinations on PDSCH, SCCPCH, DPCH, PUSCH and PRACH

6.10.3.4.3.1 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH + Interactive or background / UL: 64 DL: 256 kbps / PS RAB + UL: 16.8 kbps SRBs for CCCH and SHCCH+ DL: 33.6 kbps SRBs for CCCH SHCCH and BCCH

6.10.3.4.3.1.1 Uplink

6.10.3.4.3.1.1.1 Transport channel parameters

6.10.3.4.3.1.1.1.1 Transport channel parameters for Conversational / speech / UL:12.2 / CS RAB

See clause 6.10.3.4.1.4.1.1.1.

6.10.3.4.3.1.1.1.2 Transport channel parameters for UL SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.10.3.4.3.1.1.1.3 TFCS for DCH

See clause 6.10.3.4.1.4.1.1.3.

6.10.3.4.3.1.1.1.4 Transport channel parameters for Interactive or background / UL: 64 kbps / PS RAB and UL SRB for SHCCH mapped on USCH

See clause 6.10.3.4.2.1.1.1.1.

6.10.3.4.3.1.1.1.5 TFCS for USCH

| | |
|---|---|
| TFCS size | 10 |
| TFCS | (64 kbps RAB, SHCCH)=<br>(TF0, TF0), (TF1, TF0), (TF2, TF0), (TF3, TF0), (TF4, TF0), (TF0, TF1), (TF1, TF1), (TF2, TF1), (TF3, TF1), (TF4, TF1) |

6.10.3.4.3.1.1.1.6 Transport channel parameters for SRB for CCCH and UL SRB for SHCCH mapped on RACH

| | | | | |
|---|---|---|---|---|
| Higher layer | RAB/signalling RB | | SRB#0 | SRB#5 |
| | User of Radio Bearer | | RRC | RRC |
| RLC | Logical channel type | | CCCH | SHCCH |
| | RLC mode | | TM | TM |
| | Payload sizes, bit | | 168 | 168 |
| | Max data rate, bps | | 16 800 | 16 800 |
| | TrD PDU header, bit | | 0 | 0 |
| MAC | MAC header, bit | | 2 | 2 |
| | MAC multiplexing | | 2 logical channel multiplexing | |
| Layer 1 | TrCH type | | RACH | |
| | TB sizes, bit | | 170 | |
| | TFS | TF0, bits | 1x170 | |
| | TTI, ms | | 10 | |
| | Coding type | | CC 1/2 | |
| | CRC, bit | | 16 | |
| | Max number of bits/TTI after channel coding | | 388 | |
| | Max number of bits/radio frame before rate matching | | 388 | |

6.10.3.4.3.1.1.2 Physical channel parameters

6.10.3.4.3.1.1.2.1 Physical channel parameters for DPCH

See clause 6.10.3.4.1.4.1.2.

6.10.3.4.3.1.1.2.2 Physical channel parameters for PUSCH

| | | |
|---|---|---|
| PUSCH | Midamble | 512 chips |
| | Codes and time slots | SF2 x 1 code x 1 time slot |
| | Max. Number of data bits/radio frame | 1 808 bits |
| | TFCI code word | 16 bits |
| | TPC | 2 bits |
| | Puncturing Limit | 0.76 (alt. 0.68) |

6.10.3.4.3.1.1.2.3 Physical channel parameters for PRACH

See clause 6.10.3.4.2.1.1.2.2.

6.10.3.4.3.1.2 Downlink

6.10.3.4.3.1.2.1 Transport channel parameters

6.10.3.4.3.1.2.1.1 Transport channel parameters for Conversational / speech / DL:12.2 kbps / CS RAB

See clause 6.10.3.4.1.4.2.1.1.

6.10.3.4.3.1.2.1.2 Transport channel parameters for DL SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.10.3.4.3.1.2.1.3 TFCS for DCH

See clause 6.10.3.4.1.4.2.1.3.

6.10.3.4.3.1.2.1.4 Transport channel parameters for Interactive or background / DL: 256 kbps / PS RAB and DL SRB for SHCCH mapped on DSCH

See clause 6.10.3.4.2.1.2.1.1.

6.10.3.4.3.1.2.1.5 TFCS for DSCH

| | |
|---|---|
| TFCS size | 10 (alt. 14) |
| TFCS | (256 kbps RAB, SHCCH)=<br>(TF0, TF0), (TF1, TF0), (TF2, TF0), (TF3, TF0), (TF4, TF0), (TF0, TF1), (TF1, TF1), (TF2, TF1), (TF3, TF1), (TF4, TF1)<br>(alt. (TF0, TF0), (TF1, TF0), (TF2, TF0), (TF3, TF0), (TF4, TF0), (TF5, TF0), (TF6, TF0), (TF0, TF1), (TF1, TF1), (TF2, TF1), (TF3, TF1), (TF4, TF1), (TF5, TF1), (TF6, TF1)) |

6.10.3.4.3.1.2.1.6 Transport channel parameters for SRB for CCCH and SRB for BCCH and DL SRB for SHCCH mapped on FACH

| | | | | | |
|---|---|---|---|---|---|
| Higher layer | RAB/Signalling RB | | SRB#0 | SRB#5 | SRB#6 |
| | User of Radio Bearer | | RRC | RRC | RRC |
| RLC | Logical channel type | | CCCH | SHCCH | BCCH |
| | RLC mode | | UM | UM | TM |
| | Payload sizes, bit | | 160 | 160 | 168 |
| | Max data rate, bps | | 32 000 (alt. 16 000) | 32 000 (alt. 16 000) | 33 600 (alt. 16 800) |
| | UMD/TrD PDU header, bit | | 8 | 8 | 0 |
| MAC | MAC header, bit | | 3 | | |
| | MAC multiplexing | | 3 logical channel multiplexing | | |
| Layer 1 | TrCH type | | FACH | | |
| | TB sizes, bit | | 171 | | |
| | TFS | TF0, bits | 0x171 | | |
| | | TF1, bits | 1x171 | | |

| | | | |
|---|---|---|---|
| | | TF2, bits | 2x171 |
| | | TF3, bits | 3x171 (alt. N/A) |
| | | TF4, bits | 4x171 (alt. N/A) |
| | TTI, ms | | 20 |
| | Coding type | | TC |
| | CRC, bit | | 16 |
| | Max number of bits/TTI after channel coding | | 2 256 (alt. 1 134) |
| | Max number of bits/radio frame before rate matching | | 1 128 (alt 567) |

6.10.3.4.3.1.2.1.7 TFCS for FACH

| | |
|---|---|
| TFCS size | 5 (alt. 3) |
| TFCS | FACH = (TF0), (TF1), (TF2), (TF3), (TF4) (alt. FACH = (TF0), (TF1), (TF2)) |

6.10.3.4.3.1.2.2 Physical channel parameters

6.10.3.4.3.1.2.2.1 Physical channel parameters for DPCH

Seeclause 6.10.3.4.1.4.2.2.

6.10.3.4.3.1.2.2.2 Physical channel parameters for PDSCH

| | | |
|---|---|---|
| PDSCH | Midamble | 256 chips |
| | Codes and time slots | SF16 x 8 codes x 2 time slots |
| | Max. Number of data bits/radio frame | 4 400 bits |
| | TFCI code word | 16 bits |
| | Puncturing Limit | 0.48 |

6.10.3.4.3.1.2.2.3 Physical channel parameters for SCCPCH

See clause 6.10.3.4.2.1.2.2.2.1.

6.10.3.4.3.2 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH + Interactive or background / UL: 64 DL: 384 kbps / PS RAB + UL: 16.8 kbps SRBs for CCCH and SHCCH+ DL: 33.6 kbps SRBs for CCCH, SHCCH and BCCH

6.10.3.4.3.2.1 Uplink

See clause 6.10.3.4.3.1.1.

6.10.3.4.3.2.2 Downlink

6.10.3.4.3.2.2.1 Transport channel parameters

6.10.3.4.3.2.2.1.1 Transport channel parameters for Conversational / speech / DL:12.2 kbps / CS RAB

See clause 6.10.3.4.1.4.2.1.1.

6.10.3.4.3.2.2.1.2 Transport channel parameters for DL SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.10.3.4.3.2.2.1.3 TFCS for DCH

See clause 6.10.3.4.1.4.2.1.3.

6.10.3.4.3.2.2.1.4 Transport channel parameters for Interactive or background / DL: 384 kbps / PS RAB and DL SRB for SHCCH mapped on DSCH

See clause 6.10.3.4.2.2.2.1.1.

6.10.3.4.3.2.2.1.5 TFCS for DSCH

| | |
|---|---|
| TFCS size | 12 (alt. 18) |
| TFCS | (384 kbps RAB, SHCCH)=<br>(TF0, TF0), (TF1, TF0), (TF2, TF0), (TF3, TF0), (TF4, TF0), (TF5, TF0),<br>(TF0, TF1), (TF1, TF1), (TF2, TF1), (TF3, TF1), (TF4, TF1), (TF5, TF1)<br>(alt. (TF0, TF0), (TF1, TF0), (TF2, TF0), (TF3, TF0), (TF4, TF0), (TF5, TF0), (TF6, TF0), (TF7, TF0), (TF8, TF0),<br>(TF0, TF1), (TF1, TF1), (TF2, TF1), (TF3, TF1), (TF4, TF1), (TF5, TF1), (TF6, TF1), (TF7, TF1), (TF8, TF1)) |

6.10.3.4.3.2.2.1.6 Transport channel parameters for SRB for CCCH and SRB for BCCH and DL SRB for SHCCH mapped on FACH

See clause 6.10.3.4.3.1.2.1.6.

6.10.3.4.3.2.2.1.7 TFCS for FACH

See clause 6.10.3.4.3.1.2.1.7.

6.10.3.4.3.2.2.2 Physical channel parameters

6.10.3.4.3.2.2.2.1 Physical channel parameters for downlink DPCH

See clause 6.10.3.4.1.4.2.2.

6.10.3.4.3.2.2.2.2 Physical channel parameters for PDSCH

| | | |
|---|---|---|
| PDSCH | Midamble | 256 chips |
| | Codes and time slots | SF16 x 8 codes x 3 time slots |
| | Max. Number of data bits/radio frame | 6 608 bits |
| | TFCI code word | 16 bits |
| | Puncturing Limit | 0.48 |

6.10.3.4.3.2.2.2.3 Physical channel parameters for SCCPCH

See clause 6.10.3.4.2.1.2.2.2.1.

6.10.3.4.3.3 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH + Interactive or background / UL: 64 DL: 2 048 kbps / PS RAB + UL: 16.8 kbps SRBs for CCCH and SHCCH+ DL: 33.6 kbps SRBs for CCCH, SHCCH and BCCH

6.10.3.4.3.3.1 Uplink

See clause 6.10.3.4.3.1.1.

6.10.3.4.3.3.2 Downlink

6.10.3.4.3.3.2.1 Transport channel parameters

6.10.3.4.3.3.2.1.1 Transport channel parameters for Conversational / speech / DL:12.2 kbps / CS RAB

See clause 6.10.3.4.1.4.2.1.1.

6.10.3.4.3.3.2.1.2 Transport channel parameters for DL SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.10.3.4.3.3.2.1.3 TFCS for DCH

See clause 6.10.3.4.1.4.2.1.3.

6.10.3.4.3.3.2.1.4 Transport channel parameters for Interactive or background / DL: 2 048 kbps / PS RAB and DL SRB for SHCCH mapped on DSCH

See clause 6.10.3.4.2.3.2.1.1.

6.10.3.4.3.3.2.1.5 TFCS for DSCH

| | |
|---|---|
| TFCS size | 22 (alt. 38) |
| TFCS | (2 048 kbps RAB, SHCCH)=<br>(TF0, TF0), (TF1, TF0), (TF2, TF0), (TF3, TF0), (TF4, TF0), (TF5, TF0), (TF6, TF0), (TF7, TF0),<br>(TF8, TF0), (TF9, TF0), (TF10, TF0),<br>(TF0, TF1), (TF1, TF1), (TF2, TF1), (TF3, TF1), (TF4, TF1), (TF5, TF1), (TF6, TF1), (TF7, TF1),<br>(TF8, TF1), (TF9, TF1)<br>(alt. (TF0, TF0), (TF1, TF0), (TF2, TF0), (TF3, TF0), (TF4, TF0), (TF5, TF0), (TF6, TF0), (TF7, TF0),<br>(TF8, TF0), (TF9, TF0), (TF10, TF0),(TF11, TF0), (TF12, TF0), (TF13, TF0), (TF14, TF0), (TF15, TF0),<br>(TF16, TF0), (TF17, TF0), (TF18, TF0),<br>(TF0, TF1), (TF1, TF1), (TF2, TF1), (TF3, TF1), (TF4, TF1), (TF5, TF1), (TF6, TF1), (TF7, TF1),<br>(TF8, TF1), (TF9, TF1), (TF10, TF1),(TF11, TF1), (TF12, TF1), (TF13, TF1), (TF14, TF1), (TF15, TF1),<br>(TF16, TF1), (TF17, TF1), (TF18, TF1)) |

6.10.3.4.3.3.2.1.6 Transport channel parameters for SRB for CCCH and SRB for BCCH and DL SRB for SHCCH mapped on FACH

See clause 6.10.3.4.3.1.2.1.6.

6.10.3.4.3.3.2.1.7 TFCS for FACH

See clause 6.10.3.4.3.1.2.1.7.

6.10.3.4.3.3.2.2 Physical channel parameters

6.10.3.4.3.3.2.2.1 Physical channel parameters for downlink DPCH

See clause 6.10.3.4.1.4.2.2.

6.10.3.4.3.3.2.2.2 Physical channel parameters for PDSCH

| | | |
|---|---|---|
| DPCH Downlink | Midamble | 256 chips |
| | Codes and time slots | SF1 x 1 code x 7 time slot |
| | Max. Number of data bits/radio frame | 30 896 bits (alt. 30 880) |
| | TFCI code word | 16 bits (alt. 32 bits) |
| | Puncturing limit | 0.48 (alt. 0.44) |

6.10.3.4.3.3.2.2.3 Physical channel parameters for SCCPCH

See clause 6.10.3.4.2.1.2.2.2.1.

6.10.3.4.4 Combinations on SCCPCH

6.10.3.4.4.1 Stand-alone signalling RB for PCCH

6.10.3.4.4.1.1 Transport channel parameters

6.10.3.4.4.1.1.1 Transport channel parameter of SRB for PCCH

| | | | |
|---|---|---|---|
| Higher layer | RAB/signalling RB | | SRB |
| | User of Radio Bearer | | RRC |
| RLC | Logical channel type | | PCCH |
| | RLC mode | | TM |
| | Payload sizes, bit | | 240 (alt. 80) |
| | Max data rate, bps | | 12 000 (alt. 8 000) |
| | TrD PDU header, bit | | 0 |
| MAC | MAC header, bit | | 0 |
| | MAC multiplexing | | N/A |
| Layer 1 | TrCH type | | PCH |
| | TB sizes, bit | | 240 (alt. 80) |
| | TFS | TF0, bts | 0x240 (alt. 0x80) |
| | | TF1, bits | 1x240 (alt. 1x80) |
| | | TF2, bits | N/A (alt.2x80) |
| | TTI, ms | | 20 |

| | | |
|---|---|---|
| | Coding type | CC 1/2 |
| | CRC, bit | 16 |
| | Max number of bits/TTI before rate matching | 528 (alt. 400) |
| | Max number of bits/radio frame before rate matching | 264 (alt. 200) |
| | RM attribute | 210 to 250 |

6.10.3.4.4.1.1.2 TFCS

| | |
|---|---|
| TFCS size | 2 (alt. 3) |
| TFCS | SRBs for PCCH = (TF0), (TF1) (alt. (TF0), (TF1), (TF2)) |

6.10.3.4.4.1.2 Physical channel parameters

| | | |
|---|---|---|
| S-CCPCH | Midamble | 512 chips |
| | Codes and time slots | SF16 x 2 codes x 1 time slot (alt. SF16 x 1 code x 1 time slot) |
| | Max. Number of data bits/radio frame | 480 bits (alt. 236 bits) |
| | TFCI code word | 8 bits |
| | Puncturing limit | 1 |

6.10.3.4.4.2 Interactive/Background 32 kbps PS RAB + SRBs for CCCH + SRB for DCCH + SRB for BCCH

6.10.3.4.4.2.1 Transport channel parameters

6.10.3.4.4.2.1.1 Transport channel parameters for Interactive/Background 32 kbps PS RAB

| | | | |
|---|---|---|---|
| Higher layer | RAB/signalling RB | | RAB |
| | User of Radio Bearer | | Interactive/ Background RAB |
| RLC | Logical channel type | | DTCH |
| | RLC mode | | AM |
| | Payload sizes, bit | | 320 |
| | Max data rate, bps | | 32 000 (alt. 16 000) |
| | AMD PDU header, bit | | 16 |
| MAC | MAC header, bit | | 27 |
| | MAC multiplexing | | N/A |
| Layer 1 | TrCH type | | FACH |
| | TB sizes, bit | | 363 |
| | TFS | TF0, bits | 0 x363 |
| | | TF1, bits | 1x363 |
| | | TF2, bits | 2x363 (alt. N/A) |
| | TTI, ms | | 20 |
| | Coding type | | TC |
| | CRC, bit | | 16 |
| | Max number of bits/TTI before rate matching | | 2 286 (alt. 1 149) |
| | Max number of bits/radio frame before rate matching | | 1 143 (alt. 575) |
| | RM attribute | | 110 to 150 |

6.10.3.4.4.2.1.2 Transport channel parameters of SRBs for CCCH, SRB for DCCH, and SRB for BCCH

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| Higher layer | RAB/signalling RB | SRB#0 | SRB#1 | SRB#2 | SRB#3 | SRB#4 | SRB#5 |
| | User of Radio Bearer | RRC | RRC | RRC | NAS_DT High prio | NAS_DT Low prio | RRC |
| RLC | Logical channel type | CCCH | DCCH | DCCH | DCCH | DCCH | BCCH |
| | RLC mode | UM | UM | AM | AM | AM | TM |
| | Payload sizes, bit | 160 | 136 or 120 (note) | 128 | 128 | 128 | 168 |

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | Max data rate, bps | | 32 000 (alt. 16 000) | 27 200 or 24 000 (alt. 24 000 or 12 000) | 25 600 (alt. 12 800) | 25 600 (alt. 12 800) | 25 600 (alt. 12 800) | 33 600 (alt. 16 800) |
| | AMD/UMD/TrD PDU header, bit | | 8 | 8 | 16 | 16 | 16 | 0 |
| MAC | MAC header, bit | | 3 | 27 or 43 | 27 | 27 | 27 | 3 |
| | MAC multiplexing | | 6 logical channel multiplexing | | | | | |
| Layer 1 | TrCH type | | FACH | | | | | |
| | TB sizes, bit | | 171 | | | | | |
| | TFS | TF0, bits | 0x171 | | | | | |
| | | TF1, bits | 1x171 | | | | | |
| | | TF2, bits | 2x171 | | | | | |
| | | TF3, bits | 3x171 (alt. N/A) | | | | | |
| | | TF4, bits | 4x171 (alt. N/A) | | | | | |
| | TTI, ms | | 20 | | | | | |
| | Coding type | | TC | | | | | |
| | CRC, bit | | 16 | | | | | |
| | Max number of bits/TTI before rate matching | | 2 256 (alt. 1 134) | | | | | |
| | Max number of bits/radio frame before rate matching | | 1 128 (alt. 567) | | | | | |
| | RM attribute | | 200 to 240 | | | | | |
| NOTE: MAC header size and RLC payload size depend on use of U-RNTI or C-RNTI. | | | | | | | | |

6.10.3.4.4.2.1.3 TFCS

| | |
|---|---|
| TFCS size | 9 (alt. 4) |
| TFCS | (32kbps RAB, SRBs for CCCH/DCCH/BCCH) =<br>(TF0, TF0), (TF0, TF1), (TF0, TF2), (TF0, TF3), (TF0, TF4),(TF1, TF0), (TF1, TF1), (TF1, TF2), (TF2, TF0)<br>(alt. (TF0, TF0), (TF0, TF1), (TF0, TF2),<br>(TF1, TF0)) |
| NOTE: First TFCS applies when the alternative for the 3 2kbps RAB and the alternative for the SRBs for CCCH/DCCH/BCCH are both not configured. The alt. TFCS applies when both the alt. for the 32 kbps RAB and the alt. for the SRBs for CCCH/DCCH/BCCH are configured. All other combinations of these alternatives are not valid. | |

6.10.3.4.4.2.2 Physical channel parameters

| | | |
|---|---|---|
| S-CCPCH | Midamble | 512 chips |
| | Codes and time slots | SF16 x 5 codes x 1 time slot<br>(alt. SF16 x 2 codes x 1 time slot) |
| | Max. Number of data bits/radio frame | 1 204 bits (alt. 472) |
| | TFCI code word | 16 bits |
| | Puncturing limit | 0.60 (alt. 0.48) |

6.10.3.4.4.2a Interactive/Background 32 kbps PS RAB + Interactive/Background 32 kbps PS RAB + SRBs for CCCH + SRB for DCCH + SRB for BCCH

6.10.3.4.4.2a.1 Transport channel parameters

6.10.3.4.4.2a.1.1 Transport channel parameters for Interactive/Background 32 kbps PS RAB + Interactive/Background 32 kbps PS RAB

| | | | |
|---|---|---|---|
| Higher layer | RAB/Signalling RB | RAB | RAB |
| | User of Radio Bearer | Interactive/Background RAB | Interactive/Background RAB |
| RLC | Logical channel type | DTCH | DTCH |
| | RLC mode | AM | AM |
| | Payload sizes, bit | 320 | 320 |
| | Max data rate, bps | 32 000 (alt. 16 000) | 32 000 (alt. 16 000) |
| | AMD PDU header, bit | 16 | 16 |
| MAC | MAC header, bit | 27 | 27 |

| | MAC multiplexing | | 2 logical channel multiplexing |
|---|---|---|---|
| Layer 1 | TrCH type | | FACH |
| | TB sizes, bit | | 363 |
| | TFS | TF0, bits | 0x363 |
| | | TF1, bits | 1x363 |
| | | TF2, bits | 2x363 (alt. N/A) |
| | TTI, ms | | 20 |
| | Coding type | | TC |
| | CRC, bit | | 16 |
| | Max number of bits/TTI before rate matching | | 2 286 (alt. 1 149) |
| | Max number of bits/radio frame before rate matching | | 1 143 (alt. 575) |
| | RM attribute | | 110 to 150 |

6.10.3.4.4.2a.1.2 Transport channel parameters of SRBs for CCCH, SRB for DCCH, and SRB for BCCH

See clause 6.10.3.4.4.2.1.2.

6.10.3.4.4.2a.1.3 TFCS

| TFCS size | 9 (alt. 4) |
|---|---|
| TFCS | (32kbps RAB + 32kbps RAB, SRBs for CCCH/DCCH/BCCH) =<br>(TF0, TF0), (TF0, TF1), (TF0, TF2), (TF0, TF3), (TF0, TF4), (TF1, TF0), (TF1, TF1), (TF1, TF2), (TF2, TF0)<br>(alt. (TF0, TF0), (TF0, TF1), (TF0, TF2), (TF1, TF0)) |
| NOTE: First TFCS applies when the alternative for the 32 kbps RABs and the alternative for the SRBs for CCCH/DCCH/BCCH are both not configured. The alt. TFCS applies when both the alt. for the 32 kbps RABs and the alt. for the SRBs for CCCH/DCCH/BCCH are configured. All other combinations of these alternatives are not valid. | |

6.10.3.4.4.2a.2 Physical channel parameters

| S-CCPCH | Midamble | 512 chips |
|---|---|---|
| | Codes and time slots | SF16 x 5 codes x 1 time slot<br>(alt. SF16 x 2 codes x 1 time slot) |
| | Max. Number of data bits/radio frame | 1 204 bits (alt. 472) |
| | TFCI code word | 16 bits |
| | Puncturing limit | 0.60 (alt. 0.48) |

6.10.3.4.4.2b SRBs for CCCH + SRB for DCCH + SRB for BCCH

6.10.3.4.4.2b.1 Transport channel parameters

6.10.3.4.4.2b.1.1 Transport channel parameters of SRBs for CCCH, SRB for DCCH, and SRB for BCCH

See clause 6.10.3.4.4.2.1.2.

6.10.3.4.4.2b.1.2 TFCS

| TFCS size | 5 (alt. 3) |
|---|---|
| TFCS | (SRBs for CCCH/DCCH/BCCH) =<br>(TF0), (TF1), (TF2), (TF3), (TF4) (alt. (TF0), (TF1), (TF2)) |

6.10.3.4.4.2b.2 Physical channel parameters

| S-CCPCH | Midamble | 512 chips |
|---|---|---|
| | Codes and time slots | SF16 x 5 codes x 1 time slot<br>(alt. SF16 x 2 codes x 1 time slot) |
| | Max. Number of data bits/radio frame | 1 204 bits (alt. 480 bits) |
| | TFCI code word | 16 bits (alt. 8 bits) |

| | | |
|---|---|---|
| | Puncturing limit | 1 (alt. 0.84) |

6.10.3.4.4.3 Interactive/Background 32 kbps RAB + SRB for PCCH + SRB for CCCH + SRB for DCCH + SRB for BCCH

6.10.3.4.4.3.1 Transport channel parameters

6.10.3.4.4.3.1.1 Transport channel parameters for Interactive/Background 32 kbps RAB

See clause 6.10.3.4.4.2.1.1.

6.10.3.4.4.3.1.2 Transport channel parameters of SRB for PCCH

See clause 6.10.3.4.4.1.1.1.

6.10.3.4.4.3.1.3 Transport channel parameters of SRBs for CCCH, SRB for DCCH, and SRB for BCCH

See clause 6.10.3.4.4.2.1.2.

6.10.3.4.4.3.1.4 TFCS

| | |
|---|---|
| TFCS size | 30 (alt. 8) |
| TFCS | (32 kbps RAB, SRB for PCCH, SRBs for CCCH/ DCCH/ BCCH) =<br>(TF0, TF0, TF0), (TF0, TF0, TF1), (TF0, TF0, TF2), (TF0, TF0, TF3), (TF0, TF0, TF4), (TF0, TF1, TF0), (TF0, TF1, TF1), (TF0, TF1, TF2), (TF0, TF1, TF3), (TF0, TF1, TF4), (TF1, TF0, TF0), (TF1, TF0, TF1), (TF1, TF0, TF2), (TF1, TF0, TF3), (TF1, TF0, TF4),(TF1, TF1, TF0), (TF1, TF1, TF1), (TF1, TF1, TF2), (TF1, TF1, TF3), (TF1, TF1, TF4), (TF2, TF0, TF0), (TF2, TF0, TF1), (TF2, TF0, TF2), (TF2, TF0, TF3), (TF2, TF0, TF4),(TF2, TF1, TF0), (TF2, TF1, TF1), (TF2, TF1, TF2), (TF2, TF1, TF3), (TF2, TF1, TF4)<br>(alt. (TF0, TF0, TF0), (TF0, TF0, TF1), (TF0, TF0, TF2), (TF0, TF1, TF0), (TF0, TF1, TF1), (TF0, TF2, TF0), (TF0, TF2, TF1), (TF1, TF0, TF0)) |
| NOTE: Alt. TFCS applies when alts for 32 kbps RAB, SRB for PCCH, and SRBs for CCCH/ DCCH/ BCCH are all configured. | |

6.10.3.4.4.3.2 Physical channel parameters

| | | |
|---|---|---|
| S-CCPCH | Midamble | 512 chips |
| | Codes and time slots | SF16 x 8 codes x 1 time slot<br>(alt. SF16 x 2 codes x 1 time slot) |
| | Max. Number of data bits/radio frame | 1 936 bits (alt. 472 bits) |
| | TFCI code word | 16 bits |
| | Puncturing limit | 0.52 (alt. 0.56) |
| NOTE: Alt. applies when alts for 32 kbps RAB and SRBs for CCCH/ DCCH/ BCCH are both configured. | | |

6.10.3.4.4.3a SRB for PCCH + SRB for CCCH + SRB for DCCH + SRB for BCCH

6.10.3.4.4.3a.1 Transport channel parameters

6.10.3.4.4.3a.1.1 Transport channel parameters of SRB for PCCH

See clause 6.10.3.4.4.1.1.1.

6.10.3.4.4.3a.1.2 Transport channel parameters of SRBs for CCCH, SRB for DCCH, and SRB for BCCH

See clause 6.10.3.4.4.2.1.2.

6.10.3.4.4.3a.1.3 TFCS

| | |
|---|---|
| TFCS size | 10 (alt.7) |
| TFCS | (SRB for PCCH, SRBs for CCCH/ DCCH/ BCCH) =<br>(TF0, TF0), (TF0, TF1), (TF0, TF2), (TF0, TF3), (TF0, TF4), (TF1, TF0), (TF1, TF1), (TF1, TF2), (TF1, TF3), (TF1, TF4)<br>(alt. (TF0, TF0), (TF0, TF1), (TF0, TF2), (TF1, TF0), (TF1, TF1), (TF2, TF0), (TF2, TF1)) |
| NOTE: Alt. TFCS applies when alts for SRB for PCCH and SRBs for CCCH/ DCCH/ BCCH are both configured. | |

6.10.3.4.4.3a.2 Physical channel parameters

| | | |
|---|---|---|
| S-CCPCH | Midamble | 512 chips |
| | Codes and time slots | SF16 x 5 codes x 1 time slot<br>(alt. SF16 x 2 codes x 1 time slot) |
| | Max. Number of data bits/radio frame | 1 204 bits (alt. 480 bits) |
| | TFCI code word | 16 bits (alt. 8 bits) |
| | Puncturing limit | 0.84 (alt. 0.84) |
| NOTE: Alt. applies when alt for SRBs for CCCH/ DCCH/ BCCH is configured. | | |

6.10.3.4.4.4 RB for CTCH + SRB for CCCH + SRB for BCCH

6.10.3.4.4.4.1 Transport channel parameters

6.10.3.4.4.4.1.1 Transport channel parameters of RB for CTCH

| | | | |
|---|---|---|---|
| Higher layer | RAB/signalling RB | | N/A |
| | User of Radio Bearer | | BMC |
| RLC | Logical channel type | | CTCH |
| | RLC mode | | UM |
| | Payload sizes, bit | | 152 |
| | Max data rate, bps | | 15 200 |
| | UMD PDU header, bit | | 8 |
| MAC | MAC header, bit | | 3 |
| | MAC multiplexing | | N/A |
| Layer 1 | TrCH type | | FACH |
| | TB sizes, bit | | 163 |
| | TFS | TF0, bits | 0x163 |
| | | TF1, bits | 1x163 |
| | | TF2, bits | 2x163 |
| | TTI, ms | | 20 |
| | Coding type | | CC 1/3 |
| | CRC, bit | | 16 |
| | Max number of bits/TTI before rate matching | | 1 098 |
| | Max number of bits/radio frame before rate matching | | 549 |
| | RM attribute | | 200 to 240 |

6.10.3.4.4.4.1.2 Transport channel parameters of SRB for CCCH and SRB for BCCH

| | | | | |
|---|---|---|---|---|
| Higher layer | RAB/signalling RB | | SRB#0 | SRB#5 |
| | User of Radio Bearer | | RRC | RRC |
| RLC | Logical channel type | | CCCH | BCCH |
| | RLC mode | | UM | TM |
| | Payload sizes, bit | | 160 | 168 |
| | Max data rate, bps | | 16 000 | 16 800 |
| | AMD/UMD/TrD PDU header, bit | | 8 | 0 |
| MAC | MAC header, bit | | 3 | 3 |
| | MAC multiplexing | | 2 logical channel multiplexing | |
| Layer 1 | TrCH type | | FACH | |
| | TB sizes, bit | | 171 | |
| | TFS | TF0, bits | 0x171 | |
| | | TF1, bits | 1x171 | |
| | | TF2, bits | 2x171 | |
| | TTI, ms | | 20 | |
| | Coding type | | TC | |
| | CRC, bit | | 16 | |
| | Max number of bits/TTI before rate matching | | 1 134 | |
| | Max number of bits/radio frame before rate matching | | 567 | |
| | RM attribute | | 200 to 240 | |

6.10.3.4.4.4.1.3 TFCS

| | |
|---|---|
| TFCS size | 4 |
| TFCS | (RB for CTCH, SRBs for CCCH/BCCH) =<br>(TF0, TF0), (TF0, TF1), (TF0, TF2), (TF1, TF0) , (TF1, TF1), (TF2, TF0) |

6.10.3.4.4.4.2 Physical channel parameters

| | | |
|---|---|---|
| S-CCPCH | Midamble | 512 chips |
| | Codes and time slots | SF16 x 2 codes x 1 time slot |
| | Max. Number of data bits/radio frame | 472 bits |
| | TFCI code word | 16 bits |
| | Puncturing limit | 0.80 |

6.10.3.4.5 Combinations on PRACH

6.10.3.4.5.1 SRB for CCCH + SRB for DCCH

6.10.3.4.5.1.1 Transport channel parameters

6.10.3.4.5.1.1.1 Transport channel parameter for SRB for CCCH, SRB for DCCH

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| Higher layer | RAB/signalling RB | | SRB#0 | SRB#1 | SRB#2 | SRB#3 | SRB#4 |
| | User of Radio Bearer | | RRC | RRC | RRC | NAS_DT High priority | NAS_DT Low priority |
| RLC | Logical channel type | | CCCH | DCCH | DCCH | DCCH | DCCH |
| | RLC mode | | TM | UM | AM | AM | AM |
| | Payload sizes, bit | | 168 | 136 | 128 | 128 | 128 |
| | Max data rate, bps | | 16 800 | 13 600 | 12 800 | 12 800 | 12 800 |
| | AMD/UMD/TrD PDU header, bit | | 0 | 8 | 16 | 16 | 16 |
| MAC | MAC header, bit | | 2 | 26 | 26 | 26 | 26 |
| | MAC multiplexing | | 5 logical channel multiplexing | | | | |
| Layer 1 | TrCH type | | RACH | | | | |
| | TB sizes, bit | | 170 | | | | |
| | TFS | TF0, bits | 1x170 | | | | |
| | TTI, ms | | 10 | | | | |
| | Coding type | | CC 1/2 | | | | |
| | CRC, bit | | 16 | | | | |
| | Max number of bits/TTI after channel coding | | 388 | | | | |
| | Max number of bits/Radio frame before rate matching | | 388 | | | | |

6.10.3.4.5.1.1.2 TFCS

| | |
|---|---|
| TFCS size | 1 |
| TFCS | SRBs for CCCH/ DCCH = (TF0) |

6.10.3.4.5.1.2 Physical channel parameters

| | | |
|---|---|---|
| PRACH | Midamble | 512 chips |
| | Codes and time slots | SF8 (alt. SF16) x 1 code x 1 time slot |
| | Max. Number of data bits/radio frame | 488 bits (alt. 244 bits) |
| | Puncturing Limit | 1.0 (alt. 0.60) |

6.10.3.4.5.2 Interactive/Background 12.8 kbps PS RAB + SRB for CCCH + SRB for DCCH

6.10.3.4.5.2.1 Transport channel parameters

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| Higher | RAB/signalling RB | RAB | SRB#0 | SRB#1 | SRB#2 | SRB#3 | SRB#4 |

| | User of Radio Bearer | Interactive/ Background RAB | RRC | RRC | RRC | NAS_DT High priority | NAS_DT Low priority |
|---|---|---|---|---|---|---|---|
| RLC | Logical channel type | DTCH | CCCH | DCCH | DCCH | DCCH | DCCH |
| | RLC mode | AM | TM | UM | AM | AM | AM |
| | Payload sizes, bit | 128 | 168 | 136 | 128 | 128 | 128 |
| | Max data rate, bps | 12 800 | 16 800 | 13 600 | 12 800 | 12 800 | 12 800 |
| | AMD/UMD/TrD PDU header, bit | 16 | 0 | 8 | 16 | 16 | 16 |
| MAC | MAC header, bit | 26 | 2 | 26 | 26 | 26 | 26 |
| | MAC multiplexing | 6 logical channel multiplexing | | | | | |
| Layer 1 | TrCH type | RACH | | | | | |
| | TB sizes, bit | 170 | | | | | |
| | TFS TF0, bits | 1x170 | | | | | |
| | TTI, ms | 10 | | | | | |
| | Coding type | CC 1/2 | | | | | |
| | CRC, bit | 16 | | | | | |
| | Max number of bits/TTI after channel coding | 388 | | | | | |
| | Max number of bits/ Radio frame before rate matching | 388 | | | | | |

6.10.3.4.5.2.2 Physical channel parameters

See clause 6.10.3.4.5.1.2.

6.10.3.4.5.3 Interactive/Background 12.8 kbps PS RAB + Interactive/Background 12.8 kbps PS RAB + SRB for CCCH + SRB for DCCH

6.10.3.4.5.3.1 Transport channel parameters

| Higher layer | RAB/signalling RB | RAB | RAB | SRB#0 | SRB#1 | SRB#2 | SRB#3 | SRB#4 |
|---|---|---|---|---|---|---|---|---|
| | User of Radio Bearer | Interactive/ Background RAB | Interactive/ Background RAB | RRC | RRC | RRC | NAS_DT High prio | NAS_DT Low prio |
| RLC | Logical channel type | DTCH | DTCH | CCCH | DCCH | DCCH | DCCH | DCCH |
| | RLC mode | AM | AM | TM | UM | AM | AM | AM |
| | Payload sizes, bit | 128 | 128 | 168 | 136 | 128 | 128 | 128 |
| | Max data rate, bps | 12 800 | 12 800 | 16 800 | 13 600 | 12 800 | 12 800 | 12 800 |
| | AMD/UMD/TrD PDU header, bit | 16 | 16 | 0 | 8 | 16 | 16 | 16 |
| MAC | MAC header, bit | 26 | 26 | 2 | 26 | 26 | 26 | 26 |
| | MAC multiplexing | 7 logical channel multiplexing | | | | | | |
| Layer 1 | TrCH type | RACH | | | | | | |
| | TB sizes, bit | 170 | | | | | | |
| | TFS TF0, bits | 1x170 | | | | | | |
| | TTI, ms | 10 | | | | | | |
| | Coding type | CC 1/2 | | | | | | |
| | CRC, bit | 16 | | | | | | |
| | Max number of bits/TTI after channel coding | 388 | | | | | | |
| | Max number of bits/ Radio frame before rate matching | 388 | | | | | | |

6.10.3.4.5.3.2 Physical channel parameters

See clause 6.10.3.4.5.1.2.

# 6.11 Common Radio Bearer configurations for other test purposes

The common radio bearer configurations are used for functional testing of various UE functions. Only common configurations that are used by multiple test cases and are not covered by the reference radio bearer configurations in

clause 6.10 are specified in the present clause. Radio bearer configurations only used by a single test case are specified in the actual test case itself.

NOTE If not specifically specified then the mid-value of the RM attribute value range as specified by the actual reference radio bearer configuration shall be applied for testing.

## 6.11.1 Unacknowledged Mode Radio Bearer configuration (7 bit Length Indicator)

This configuration is based on the Interactive or background / UL:8 DL 8 kbps / PS RAB + UL:3.4 DL 3.4 kbps SRBs for DCCH (see clause 6.10.2.4.1.23a) with the transport channels parameters of the RAB and TFCS defined as follows.

### 6.11.1.1 Transport channel parameters for the Uplink RAB

| Higher layer | RAB/Signalling RB | | RAB |
|---|---|---|---|
| RLC | Logical channel type | | DTCH |
| | RLC mode | | UM |
| | Payload sizes, bit | | 328 |
| | Max data rate, bps | | 8 200 |
| | UMD PDU header, bit | | 8 |
| MAC | MAC header, bit | | 0 |
| | MAC multiplexing | | N/A |
| Layer 1 | TrCH type | | DCH |
| | TB sizes, bit | | 336 |
| | TFS | TF0, bits | 0x336 |
| | | TF1, bits | 1x336 |
| | TTI, ms | | 40 |
| | Coding type | | CC 1/3 |
| | CRC, bit | | 16 |
| | Max number of bits/TTI after channel coding | | 1 080 |
| | Uplink: Max number of bits/radio frame before rate matching | | 270 |
| | RM attribute | | 135 to 175 |

### 6.11.1.2 TFCS

| | |
|---|---|
| TFCS size | 4 |
| TFCS | (8 kbps RAB, DCCH)=<br>(TF0, TF0), (TF1, TF0), (TF0, TF1), (TF1, TF1) |

### 6.11.1.3 Transport channel parameters for the Downlink RAB

| Higher layer | RAB/Signalling RB | | RAB |
|---|---|---|---|
| RLC | Logical channel type | | DTCH |
| | RLC mode | | UM |
| | Payload sizes, bit | | 328 |
| | Max data rate, bps | | 8 200 |
| | UMD PDU header, bit | | 8 |
| MAC | MAC header, bit | | 0 |
| | MAC multiplexing | | N/A |
| Layer 1 | TrCH type | | DCH |
| | TB sizes, bit | | 336 |
| | TFS | TF0, bits | 0x336 |
| | | TF1, bits | 1x336 |
| | TTI, ms | | 40 |
| | Coding type | | CC 1/3 |
| | CRC, bit | | 16 |
| | Max number of bits/TTI after channel coding | | 1 080 |
| | RM attribute | | 135 to 175 |

### 6.11.1.4 TFCS

| | |
|---|---|
| TFCS size | 4 |

| | |
|---|---|
| TFCS | (8 kbps RAB, DCCH)=<br>(TF0, TF0), (TF1, TF0), (TF0, TF1), (TF1, TF1) |

## 6.11.2 Unacknowledged Mode Radio Bearer configuration (15 bit Length Indicator)

This configuration is based on the Interactive or background / UL:64 DL 64 kbps / PS RAB + UL:3.4 DL 3.4 kbps SRBs for DCCH (see clause 6.10.2.4.1.26) with the transport channels parameters of the RAB defined as followed.

### 6.11.2.1 Transport channel parameters for the Uplink RAB

| | | | |
|---|---|---|---|
| Higher layer | RAB/Signalling RB | | RAB |
| RLC | Logical channel type | | DTCH |
| | RLC mode | | UM |
| | Payload sizes, bit | | 1 336 |
| | Max data rate, bps | | 66 800 |
| | UMD PDU header, bit | | 8 |
| MAC | MAC header, bit | | 0 |
| | MAC multiplexing | | N/A |
| Layer 1 | TrCH type | | DCH |
| | TB sizes, bit | | 1 344 |
| | TFS | TF0, bits | 0x1344 |
| | | TF1, bits | 1x1344 |
| | TTI, ms | | 20 |
| | Coding type | | TC |
| | CRC, bit | | 16 |
| | Max number of bits/TTI after channel coding | | 4 092 |
| | Uplink: Max number of bits/radio frame before rate matching | | 2 046 |
| | RM attribute | | 130 to 170 |

### 6.11.2.2 Transport channel parameters for the Downlink RAB

| | | | |
|---|---|---|---|
| Higher layer | RAB/Signalling RB | | RAB |
| RLC | Logical channel type | | DTCH |
| | RLC mode | | UM |
| | Payload sizes, bit | | 1 336 |
| | Max data rate, bps | | 66 800 |
| | UMD PDU header, bit | | 8 |
| MAC | MAC header, bit | | 0 |
| | MAC multiplexing | | N/A |
| Layer 1 | TrCH type | | DCH |
| | TB sizes, bit | | 1 344 |
| | TFS | TF0, bits | 0x1344 |
| | | TF1, bits | 1x1344 |
| | TTI, ms | | 20 |
| | Coding type | | TC |
| | CRC, bit | | 16 |
| | Max number of bits/TTI after channel coding | | 4 092 |
| | RM attribute | | 130 to 170 |

## 6.11.3 Acknowledged Mode Radio Bearer configuration (7 bit Length Indicator)

This configuration is based on the Interactive or background / UL:8 DL 8 kbps / PS RAB + UL:3.4 DL 3.4 kbps SRBs for DCCH (see clause 6.10.2.4.1.23a) with the transport channels parameters of the RAB and TFCS defined as follows.

### 6.11.3.1 Transport channel parameters for the Uplink RAB

| | | |
|---|---|---|
| Higher layer | RAB/Signalling RB | RAB |
| RLC | Logical channel type | DTCH |
| | RLC mode | AM |

| | | | |
|---|---|---|---|
| | Payload sizes, bit | | 128 |
| | Max data rate, bps | | 6 400 |
| | UMD PDU header, bit | | 16 |
| MAC | MAC header, bit | | 0 |
| | MAC multiplexing | | N/A |
| Layer 1 | TrCH type | | DCH |
| | TB sizes, bit | | 144 |
| | TFS | 0x144 | 0x144 |
| | | 1x144 | 1x144 |
| | TTI, ms | | 20 |
| | Coding type | | CC 1/3 |
| | CRC, bit | | 16 |
| | Max number of bits/TTI after channel coding | | 504 |
| | Uplink: Max number of bits/radio frame before rate matching | | 252 |
| | RM attribute | | 135 to 175 |

### 6.11.3.2 TFCS

| | |
|---|---|
| TFCS size | 4 |
| TFCS | (RAB, DCCH)=<br>(TF0, TF0), (TF1, TF0), (TF0, TF1), (TF1, TF1) |

### 6.11.3.3 Transport channel parameters for the Downlink RAB

| | | | |
|---|---|---|---|
| Higher layer | RAB/Signalling RB | | RAB |
| RLC | Logical channel type | | DTCH |
| | RLC mode | | AM |
| | Payload sizes, bit | | 128 |
| | Max data rate, bps | | 6 400 |
| | UMD PDU header, bit | | 16 |
| MAC | MAC header, bit | | 0 |
| | MAC multiplexing | | N/A |
| Layer 1 | TrCH type | | DCH |
| | TB sizes, bit | | 144 |
| | TFS | 0x144 | 0x144 |
| | | 1x144 | 1x144 |
| | TTI, ms | | 20 |
| | Coding type | | CC 1/3 |
| | CRC, bit | | 16 |
| | Max number of bits/TTI after channel coding | | 504 |
| | RM attribute | | 135 to 175 |

### 6.11.3.4 TFCS

| | |
|---|---|
| TFCS size | 4 |
| TFCS | (RAB, DCCH)=<br>(TF0, TF0), (TF1, TF0), (TF0, TF1), (TF1, TF1) |

## 6.11.4 Acknowledged Mode Radio Bearer configuration (15 bit Length Indicator)

This configuration is based on the Interactive or background / UL:64 DL 64 kbps / PS RAB + UL:3.4 DL 3.4 kbps SRBs for DCCH (see clause 6.10.2.4.1.26) with the transport channels parameters of the RAB defined as followed.

### 6.11.4.1 Transport channel parameters for the Uplink RAB

| | | |
|---|---|---|
| Higher layer | RAB/Signalling RB | RAB |
| RLC | Logical channel type | DTCH |
| | RLC mode | AM |
| | Payload sizes, bit | 1 328 |
| | Max data rate, bps | 66 400 |

| | | | |
|---|---|---|---|
| | AMD PDU header, bit | | 16 |
| MAC | MAC header, bit | | 0 |
| | MAC multiplexing | | N/A |
| Layer 1 | TrCH type | | DCH |
| | TB sizes, bit | | 1 344 |
| | TFS | TF0, bits | 0x1344 |
| | | TF1, bits | 1x1344 |
| | TTI, ms | | 20 |
| | Coding type | | TC |
| | CRC, bit | | 16 |
| | Max number of bits/TTI after channel coding | | 4 092 |
| | Uplink: Max number of bits/radio frame before rate matching | | 2 046 |
| | RM attribute | | 130 to 170 |

### 6.11.4.2 Transport channel parameters for the Downlink RAB

| | | | |
|---|---|---|---|
| Higher layer | RAB/Signalling RB | | **RAB** |
| RLC | Logical channel type | | DTCH |
| | RLC mode | | AM |
| | Payload sizes, bit | | 1 328 |
| | Max data rate, bps | | 66 400 |
| | AMD PDU header, bit | | 16 |
| MAC | MAC header, bit | | 0 |
| | MAC multiplexing | | N/A |
| Layer 1 | TrCH type | | DCH |
| | TB sizes, bit | | 1 344 |
| | TFS | TF0, bits | 0x1344 |
| | | TF1, bits | 1x1344 |
| | TTI, ms | | 20 |
| | Coding type | | TC |
| | CRC, bit | | 16 |
| | Max number of bits/TTI after channel coding | | 4 092 |
| | RM attribute | | 130 to 170 |

## 6.11.4a Reference Radio Bearer configurations used in MAC-hs testing

### 6.11.4a.1 5 x Interactive or background / UL: 8 kbps DL: [max bit rate depending on UE category] / UM PS RAB

This reference radio bearer configuration is used by the MAC-hs test case 7.1.5.2 in 3GPP TS 34.123-1 [1].

#### 6.11.4a.1.1 Uplink

##### 6.11.4a.1.1.1 Uplink Transport channel parameters for DCH

6.11.4a.1.1.1.1 Transport channel parameters for 5 x Interactive or background / UL:8 kbps / PS RAB

| Higher layer | RAB/Signalling RB | | **RB5** | **RB6** | **RB7** | **RB8** | **RB9** |
|---|---|---|---|---|---|---|---|
| RLC | Logical channel type | | DTCH | DTCH | DTCH | DTCH | DTCH |
| | RLC mode | | UM | UM | AM | AM | AM |
| | Payload sizes, bit | | 328 | 328 | 320 | 320 | 320 |
| | Max data rate, bps | | 8 200 | 8 200 | 8 000 | 8 000 | 8 000 |
| | UMD/AMD PDU header, bit | | 8 | 8 | 16 | 16 | 16 |
| MAC | MAC header, bit | | 4 | 4 | 4 | 4 | 4 |
| | MAC multiplexing | | 5 logical channel multiplexing | | | | |
| Layer 1 | TrCH type | | DCH | | | | |
| | TB sizes, bit | | 340 | | | | |
| | TFS | TF0, bits | 0x340 | | | | |
| | | TF1, bits | 1x340 | | | | |
| | TTI, ms | | 40 | | | | |
| | Coding type | | TC | | | | |
| | CRC, bit | | 16 | | | | |
| | Max number of bits/TTI after channel coding | | 1 080 | | | | |
| | Uplink: Max number of bits/radio frame before rate matching | | 270 | | | | |
| | RM attribute | | 135 to 175 | | | | |

6.11.4a.1.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.1.1.1.

6.11.4a.1.1.1.3 Uplink TFCS

| | |
|---|---|
| TFCS size | 4 |
| TFCS | (5x8 kbps PS RAB, DCCH)=<br>(TF0, TF0), (TF1, TF0), (TF0, TF1), (TF1, TF1) |

6.11.4a.1.1.2 Uplink physical channel parameters

| | | |
|---|---|---|
| DPCH Uplink | Min spreading factor | 64 |
| | Max number of DPDCH data bits/radio frame | 600 |
| | Puncturing Limit | 1.0 |

## 6.11.4a.1.2 Downlink

6.11.4a.1.2.1 Transport channel parameters for HS-DSCH

6.11.4a.1.2.1.1 MAC-d flow #1 parameters for 2 x Interactive or background / DL: [max bit rate depending on UE category] / PS RAB

| Higher layer | RAB/Signalling RB | **RB5** | **RB6** |
|---|---|---|---|
| RLC | Logical channel type | DTCH | DTCH |
| | RLC mode | UM | UM |
| | Payload sizes, bit | 328 | 328 |
| | Max data rate, bps | depends on UE category | depends on UE category |
| | UMD PDU header, bit | 8 | 8 |
| MAC-d | MAC-d header, bit | 4 | 4 |
| | MAC multiplexing | 2 logical channel multiplexing | |
| | MAC-d PDU size, bit | 340 | |

6.11.4a.1.2.1.2 MAC-d flow #2 parameters for 2 x Interactive or background / DL: [max bit rate depending on UE category] / PS RAB

| Higher layer | RAB/Signalling RB | **RB7** | **RB8** |
|---|---|---|---|
| RLC | Logical channel type | DTCH | DTCH |
| | RLC mode | AM | AM |
| | Payload sizes, bit | 320 | 320 |
| | Max data rate, bps | depends on UE category | depends on UE category |
| | AMD PDU header, bit | 16 | 16 |
| MAC-d | MAC-d header, bit | 4 | 4 |
| | MAC multiplexing | 2 logical channel multiplexing | |
| | MAC-d PDU size, bit | 340 | |

6.11.4a.1.2.1.3 MAC-d flow#3 parameters for Interactive or background / DL: [max bit rate depending on UE category] / PS RAB

| Higher layer | RAB/Signalling RB | **RB9** |
|---|---|---|
| RLC | Logical channel type | DTCH |
| | RLC mode | AM |
| | Payload sizes, bit | 320 |
| | Max data rate, bps | depends on UE category |
| | AMD PDU header, bit | 16 |
| MAC-d | MAC-d header, bit | 0 |
| | MAC multiplexing | N/A |
| | MAC-d PDU size, bit | 336 |

6.11.4a.1.2.1.4 MAC-hs and Layer 1 parameters for Interactive or background / DL: [max bit rate depending on UE category] / PS RAB

| MAC-hs | MAC-hs header fixed part, bit | 21 |
|---|---|---|
| Layer 1 | TrCH type | HS-DSCH |
| | TTI | 2 ms |
| | Coding type | TC |
| | CRC, bit | 24 |

6.11.4a.1.2.2 Downlink Transport channel parameters for DCH

6.11.4a.1.2.2.1 Transport channel parameters for DL: 3.4 kbps SRBs for DCCH

See clause 6.10.2.4.1.2.2.1.1.

6.11.4a.1.2.2.2 Downlink TFCS

| TFCS size | 2 |
|---|---|
| TFCS | SRBs for DCCH = TF0, TF1 |

6.11.4a.1.2.3 Downlink physical channel parameters

6.11.4a.1.2.3.1 Downlink physical channel parameters on DPCH

See clause 6.10.2.4.1.2.2.2.

6.11.4a.1.2.3.2 Physical channel parameters on HS-PDSCH

UE HS-DSCH Physical Layer:

| HS-PDSCH | Number of processes | 2 |
|---|---|---|
| | Process memory size | Split equally among all processes |
| | Max Data Rate | Depending on UE category |

# 6.11.4b Interactive or background / UL: 0 kbps DL: 0 kbps PS RAB

This reference radio bearer configuration is used by the RRC test case 8.4.1.43 in 3GPP TS 34.123-1 [1].

## 6.11.4b.1 Uplink

### 6.11.4b.1.1 Uplink Transport channel parameters for DCH

#### 6.11.4b.1.1.1 Transport channel parameters for Interactive or background / UL:0 kbps / PS RAB

| | | | |
|---|---|---|---|
| Higher layer | RAB/Signalling RB | | RAB |
| RLC | Logical channel type | | DTCH |
| | RLC mode | | AM |
| | Payload sizes, bit | | 320 |
| | Max data rate, bps | | 0 |
| | AMD PDU header, bit | | 16 |
| MAC | MAC header, bit | | 0 |
| | MAC multiplexing | | N/A |
| Layer 1 | TrCH type | | DCH |
| | TB sizes, bit | | 336 |
| | TFS | TF0, bits | 0x336 |
| | TTI, ms | | 20 |
| | Coding type | | CC |
| | CRC, bit | | 16 |
| | Max number of bits/TTI after channel coding | | 0 |
| | Uplink: Max number of bits/radio frame before rate matching | | 0 |
| | RM attribute | | 130 to 170 |

#### 6.11.4b.1.1.2 Uplink TFCS

| | |
|---|---|
| TFCS size | 2 |
| TFCS | (0 kbps RAB, DCCH)=<br>(TF0, TF0), (TF0, TF1) |

### 6.11.4b.1.2 Uplink physical channel parameters

| | | |
|---|---|---|
| DPCH Uplink | Min spreading factor | 64 |
| | Max number of DPDCH data bits/radio frame | 600 |
| | Puncturing Limit | 1.0 |

## 6.11.4b.2 Downlink

### 6.11.4b.2.1 Downlink Transport channel parameters for DCH

#### 6.11.4b.2.1.1 Transport channel parameters for Interactive or background / DL:0 kbps / PS RAB

| | | | |
|---|---|---|---|
| Higher layer | RAB/Signalling RB | | RAB |
| RLC | Logical channel type | | DTCH |
| | RLC mode | | AM |
| | Payload sizes, bit | | 320 |
| | Max data rate, bps | | 0 |
| | AMD PDU header, bit | | 16 |
| MAC | MAC header, bit | | 0 |
| | MAC multiplexing | | N/A |
| Layer 1 | TrCH type | | DCH |
| | TB sizes, bit | | 336 |
| | TFS | TF0, bits | 0x336 |
| | TTI, ms | | 20 |
| | Coding type | | CC |
| | CRC, bit | | 16 |
| | Max number of bits/TTI after channel coding | | 0 |
| | RM attribute | | 130 to 170 |

6.11.4b.2.1.1.2 Downlink TFCS

| | |
|---|---|
| TFCS size | 2 |
| TFCS | (0 kbps RAB, DCCH)=<br>(TF0, TF0), (TF0, TF1) |

6.11.4b.2.2 Downlink physical channel parameters

| | | | |
|---|---|---|---|
| DPCH Downlink | DTX position | | Flexible |
| | Spreading factor | | 128 |
| | DPCCH | Number of TFCI bits/slot | 2 |
| | | Number of TPC bits/slot | 2 |
| | | Number of Pilot bits/slot | 4 |
| | DPDCH | Number of data bits/slot | 34 |
| | | Number of data bits/frame | 510 |

## 6.11.5 Reference Radio Bearer configurations used in Radio Bearer testing for 1.28 Mcps TDD

### 6.11.5.1 RABs and signalling RBs

See clause 6.10.3.1.

### 6.11.5.2 Combinations of RABs and Signalling RBs

In the present document, physical channel parameters for following combinations of RABs and signalling RBs on a CCTrCH are described.

NOTE: It is understood that for speech service the AMR mode may be operated asymmetrically for the uplink and downlink.

Combinations on DPCH

1) Stand-alone UL:1.7 DL:1.7 kbps SRBs for DCCH.

1a) Stand-alone UL:1.7 DL:1.7 kbps SRBs for DCCH (Multiframe).

2) Stand-alone UL:3.4 DL:3.4 kbps SRBs for DCCH.

3) Stand-alone UL:13.6 DL:13.6 kbps SRBs for DCCH.

4) Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

4a) Conversational / speech / UL:(12.2, 7.95, 5.9, 4.75) DL:(12.2, 7.95, 5.9, 4.75) kbps / CS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH

5) Conversational / speech / UL:10.2 DL:10.2 kbps / CS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

5a) Conversational / speech / UL:(10.2, 6.7, 5.9, 4.75) DL:(10.2, 6.7, 5.9, 4.75) kbps / CS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH

6) Conversational / speech / UL:7.95 DL:7.95 kbps / CS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

7a) Conversational / speech / UL:(7.4, 6.7, 5.9, 4.75) DL:(7.4, 6.7, 5.9, 4.75)kbps / CS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH

7) Conversational / speech / UL:7.4 DL:7.4 kbps / CS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

8) Conversational / speech / UL:6.7 DL:6.7 kbps / CS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

9) Conversational / speech / UL:5.9 DL:5.9 kbps / CS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

10) Conversational / speech / UL:5.15 DL:5.15 kbps / CS RAB
+ UL:1.7 DL:1.7 kbps SRBs for DCCH.

11) Conversational / speech / UL:4.75 DL:4.75 kbps / CS RAB
+ UL:1.7 DL:1.7 kbps SRBs for DCCH.

12) Conversational / unknown / UL:28.8 DL:28.8 kbps / CS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

13) Conversational / unknown / UL:64 DL:64 kbps / CS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

14) Conversational / unknown / UL:32 DL:32 kbps / CS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

15) Streaming / unknown / UL:14.4/DL:14.4 kbps / CS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

16) Streaming / unknown / UL:28.8/DL:28.8 kbps / CS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

17) Streaming / unknown / UL:57.6/DL:57.6 kbps / CS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

18) Void.

19) Void.

20) Void.

21) Void.

22) Void.

23) Interactive or background / UL:32 DL:8 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

23a) Interactive or background / UL:8 DL:8 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

23b) Interactive or background / UL:16 DL:16 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

23c) Interactive or background / UL:32 DL:32 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

23d) Interactive or background / UL:32 DL:32 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.(20 msTTI)

24) Void.

25) Interactive or background / UL:32 DL: 64 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

26) Interactive or background / UL:64 DL: 64 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

27) Interactive or background / UL:64 DL:128 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

28) Interactive or background / UL:128 DL:128 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

29) Interactive or background / UL:64 DL:144 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

30) Interactive or background / UL:144 DL:144 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

31) Interactive or background / UL:64 DL:256 kbps / PS RAB
+ UL:3.4 DL: 3.4 kbps SRBs for DCCH.

32) Interactive or background / UL:64 DL:384 kbps / PS RAB
+ UL:3.4 DL: 3.4 kbps SRBs for DCCH.

33) Interactive or background / UL:128 DL:384 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

34) Interactive or background / UL:384 DL:384 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

35) Interactive or background / UL:64 DL:2 048 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

36) Void.

37) Void.

38) Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB
+ Interactive or background / UL:32 DL:8 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

38a) Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB
+ Interactive or background / UL:0 DL:0 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

38b) Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB
+ Interactive or background / UL:8 DL:8 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

38c) Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB
+ Interactive or background / UL:32 DL:32 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

38d) Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB
+ Interactive or background / UL:64 DL:64 kbps / PS RAB
+ Interactive or background / UL:64 DL:64 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

38e) Conversational / speech / UL: (12.2, 7.95, 5.9, 4.75) DL: (12.2, 7.95, 5.9, 4.75) kbps / CS RAB
+ Interactive or background / UL:0 DL:0 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

38f) Conversational / speech / UL: (12.2, 7.95, 5.9, 4.75) DL: (12.2, 7.95, 5.9, 4.75) kbps / CS RAB
+ Interactive or background / UL:8 DL:8 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

38g) Conversational / speech / UL: (12.2, 7.95, 5.9, 4.75) DL: (12.2, 7.95, 5.9, 4.75) kbps / CS RAB
+ Interactive or background / UL:16 DL:16 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

38h) Conversational / speech / UL: (12.2, 7.95, 5.9, 4.75) DL: (12.2, 7.95, 5.9, 4.75) kbps / CS RAB
+ Interactive or background / UL:32 DL:32 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

38i) Conversational / speech / UL: (12.2, 7.95, 5.9, 4.75) DL: (12.2, 7.95, 5.9, 4.75) kbps / CS RAB
+ Interactive or background / UL:64 DL:64 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

38j) Conversational / speech / UL: (12.2, 7.95, 5.9, 4.75) DL: (12.2, 7.95, 5.9, 4.75) kbps / CS RAB
+ Interactive or background / UL:64 DL:128 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

39) Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB
+ Interactive or background / UL:32 DL:64 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

40) Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB
+ Interactive or background / UL:64 DL:64 kbps / PS RAB
+ UL:3.4 DL: 3.4 kbps SRBs for DCCH.

41) Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB
+ Interactive or background / UL:64 DL:128 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

42) Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB
+ Interactive or background / UL:64 DL:256 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

43) Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB
+ Interactive or background / UL:64 DL:384 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

44) Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB
+ Interactive or background / UL:128 DL:2 048 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

45) Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB
+ Streaming / unknown / UL:57.6 DL:57.6 kbps / CS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

46) Void.

47) Void.

48) Void.

49) Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB
+ Conversational / unknown / UL:64 DL:64 kbps / CS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

49a) Conversational / speech / UL:(12.2, 7.95, 5.9, 4.75) DL: (12.2, 7.95, 5.9, 4.75) kbps / CS RAB
+ Conversational / unknown / UL:64 DL:64 kbps / CS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

50) Conversational / unknown / UL:64 DL:64 kbps / CS RAB
+ Conversational / unknown / UL:64 DL:64 kbps / CS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

51) Conversational / unknown / UL:64 DL:64 kbps / CS RAB
+ Interactive or background / UL:64 DL:64 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

51a) Conversational / unknown / UL:64 DL:64 kbps / CS RAB
+ Interactive or background / UL:8 DL:8 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

51b) Conversational / unknown / UL:64 DL:64 kbps / CS RAB
+ Interactive or background / UL:16 DL:64 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

52) Conversational / unknown / UL:64 DL:64 kbps / CS RAB
+ Interactive or background / UL:64 DL:128 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

53) Conversational / unknown / UL:64 DL:64 kbps / CS RAB
+ Interactive or background / UL:128 DL:128 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

54) Void.

55) Void.

56) Interactive or background / UL:8 DL:8 kbps / PS RAB
+ Interactive or background / UL:8 DL:8 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

57) Interactive or Background / UL:64 DL:64 kbps / CS RAB
+ Interactive or background / UL:64 DL:64 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

58) Streaming / Unknown / UL:16 DL:64 kbps / CS RAB
+ Interactive or background / UL:8 DL:8 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

59) Reserved for future use

60) Reserved for future use

61) Conversational / Unknown / UL:8 DL:8 kbps / CS RAB
+ Interactive or background / UL:8 DL:8 kbps / PS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH.

Combinations on PDSCH, SCCPCH, PUSCH and PRACH

1) Interactive or background / UL:64 DL:256 kbps / PS RAB
+ UL: 3.4/16.8 DL: 3.4/33.6 kbps SRBs for DCCH, CCCH and BCCH
+ UL:16.8 DL: 16 kbps SRBs for SHCCH.

2) Interactive or background / UL:64 DL:384 kbps / PS RAB
+ UL: 3.4/16.8 DL: 3.4/33.6 kbps SRBs for DCCH, CCCH and BCCH
+ UL: 16.8 DL: 16 kbps SRBs for SHCCH.

3) Interactive or background / UL:64 DL:2 048 kbps / PS RAB
+ UL:3.4/16.8 DL: 3.4/33.6 kbps SRBs for DCCH, CCCH and BCCH
+ UL: 16.8 DL: 16 kbps SRBs for SHCCH.

Combinations on PDSCH, SCCPCH, DPCH, PUSCH and PRACH

1) Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH
+ Interactive or background / UL:64 DL:256 kbps / PS RAB
+ UL:16.8 kbps SRBs for CCCH and SHCCH
+ DL: 33.6 kbps SRBs for CCCH, SHCCH and BCCH.

2) Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH
+ Interactive or background / UL:64 DL:384 kbps / PS RAB
+ UL:16.8 kbps SRBs for CCCH and SHCCH
+ DL: 33.6 kbps SRBs for CCCH, SHCCH and BCCH.

3) Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB
+ UL:3.4 DL:3.4 kbps SRBs for DCCH
+ Interactive or background / UL:64 DL:2 048 kbps / PS RAB
+ UL:16.8 kbps SRBs for CCCH and SHCCH
+ DL: 33.6 kbps SRBs for CCCH, SHCCH and BCCH.

Combinations on SCCPCH

1) Stand-alone SRB for PCCH.

2) Interactive or background / DL:32 kbps / PS RAB
+ SRB for CCCH
+ SRBs for DCCH
+ SRB for BCCH.

2a) Interactive/Background 32 kbps PS RAB
+ Interactive/Background 32 kbps PS RAB
+ SRB for CCCH
+ SRBs for DCCH
+ SRB for BCCH.

2b) SRBs for CCCH
+ SRB for DCCH
+ SRB for BCCH.

3) Interactive or background / DL:32 kbps / PS RAB
+ SRB for PCCH
+ SRB for CCCH
+ SRBs for DCCH
+ SRB for BCCH.

3a) SRB for PCCH
+ SRB for CCCH
+ SRB for DCCH
+ SRB for BCCH.

4) RB for CTCH
+ SRB for CCCH
+ SRB for BCCH.

Combinations on PRACH

1) SRB for CCCH
+ SRBs for DCCH.

2) Interactive/Background 12.8 kbps PS RAB
+ SRB for CCCH
+ SRBs for DCCH.

3) Interactive/Background 12.8 kbps PS RAB
+ Interactive/Background 12.8 kbps PS RAB
+ SRB for CCCH
+ SRBs for DCCH.

### 6.11.5.3 Example of linkage between RABs and services

See clause 6.10.3.3.

### 6.11.5.4 Typical radio parameter sets

#### 6.11.5.4.1 Combinations on DPCH

##### 6.11.5.4.1.1 Stand-alone UL:1.7 DL:1.7 kbps SRBs for DCCH

###### 6.11.5.4.1.1.1 Uplink

6.11.5.4.1.1.1.1 Transport channel parameters

6.11.5.4.1.1.1.1.1 Transport channel parameters for UL:1.7 kbps SRBs for DCCH

See clause 6.10.3.4.1.1.1.1.1.

6.11.5.4.1.1.1.1.2 TFCS

See clause 6.10.3.4.1.1.1.1.2.

6.11.5.4.1.1.1.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Uplink | Modulation | QPSK |
| | Codes and time slots / radio frame | SF16 x 1 code x 2 time slots |
| | Max. Number of data bits / radio frame | 160 bits |
| | TFCI code word / radio frame | 8 bits |
| | TPC / radio frame | 2x2 bits |
| | SS / radio frame | 2x2 bits |
| | Puncturing Limit | 1 |

6.11.5.4.1.1.2 Downlink

6.11.5.4.1.1.2.1 Transport channel parameters

6.11.5.4.1.1.2.1.1 Transport channel parameters for DL:1.7 kbps SRBs for DCCH

See clause 6.10.3.4.1.1.2.1.1.

6.11.5.4.1.1.2.1.2 TFCS

See clause 6.10.3.4.1.1.2.1.2.

6.11.5.4.1.1.2.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Downlink | Modulation | QPSK |
| | Codes and time slots / radio frame | SF16 x 1 code x 2 time slots |
| | Max. Number of data bits / radio frame | 160 bits |
| | TFCI code word / radio frame | 8 bits |
| | TPC / radio frame | 2x2 bits |
| | SS / radio frame | 2x2 bits |
| | Puncturing Limit | 1 |

6.11.5.4.1.1a Stand-alone UL:1.7 DL:1.7 kbps SRBs for DCCH (multiframe)

6.11.5.4.1.1a.1 Uplink

6.11.5.4.1.1a.1.1 Transport channel parameters

6.11.5.4.1.1a.1.1.1 Transport channel parameters for UL:1.7 kbps SRBs for DCCH

See clause 6.10.3.4.1.1a.1.1.1.

6.11.5.4.1.1a.1.1.2 TFCS

See clause 6.10.3.4.1.1a.1.1.2.

6.11.5.4.1.1a.1.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Uplink | Modulation | QPSK |
| | Codes and time slots / radio frame | SF16 x 1 code x 2 time slots |
| | Max. Number of data bits / radio frame | 160 bits |
| | TFCI code word / radio frame | 8 bits |
| | TPC / radio frame | 2x2 bits |
| | SS / radio frame | 2x2 bits |
| | Puncturing Limit | 0.60 |
| NOTE: In case the first TFC in the TFCS is not configured, the TFCI code word will be 4 bit. | | |

6.11.5.4.1.1a.2 Downlink

6.11.5.4.1.1a.2.1 Transport channel parameters

6.11.5.4.1.1a.2.1.1 Transport channel parameters for DL:1.7 kbps SRBs for DCCH

See clause 6.10.3.4.1.1a.2.1.1.

6.11.5.4.1.1a.2.1.2 TFCS

See clause 6.10.3.4.1.1a.2.1.2.

6.11.5.4.1.1a.2.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Uplink | Modulation | QPSK |
| | Codes and time slots / radio frame | SF16 x 1 code x 2 time slots |
| | Max. Number of data bits / radio frame | 160 bits |
| | TFCI code word / radio frame | 8 bits |
| | TPC / radio frame | 2x2 bits |
| | SS / radio frame | 2x2 bits |
| | Puncturing Limit | 0.60 |
| NOTE: In case the first TFC in the TFCS is not configured, the TFCI code word will be 4 bit. | | |

6.11.5.4.1.2 Stand-alone UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.5.4.1.2.1 Uplink

6.11.5.4.1.2.1.1 Transport channel parameters

6.11.5.4.1.2.1.1.1 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.11.5.4.1.2.1.1.2 TFCS

See clause 6.10.3.4.1.2.1.1.2.

6.11.5.4.1.2.1.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Uplink | Modulation | QPSK |
| | Codes and time slots / radio frame | SF16 x 1 code x 2 time slots |
| | Max. Number of data bits / radio frame | 160 bits |
| | TFCI code word / radio frame | 8 bits |
| | TPC / radio frame | 2x2 bits |
| | SS / radio frame | 2x2 bits |
| | Puncturing Limit | 1 |
| NOTE: In case the first TFCS is not configured, the TFCI code word will be 4 bit. | | |

6.11.5.4.1.2.2 Downlink

6.11.5.4.1.2.2.1 Transport channel parameters

6.11.5.4.1.2.2.1.1 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.11.5.4.1.2.2.1.2 TFCS

See clause 6.10.3.4.1.2.2.1.2.

6.11.5.4.1.2.2.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Downlink | Modulation | QPSK |
| | Codes and time slots / radio frame | SF16 x 1 code x 2 time slots |
| | Max. Number of data bits / radio frame | 160 bits |
| | TFCI code word / radio frame | 8 bits |
| | TPC / radio frame | 2x2 bits |
| | SS / radio frame | 2x2 bits |
| | Puncturing Limit | 1 |
| NOTE: In case the first TFCS is not configured, the TFCI code word will be 4 bit. | | |

6.11.5.4.1.3 Stand-alone UL:13.6 DL:13.6 kbps SRBs for DCCH

6.11.5.4.1.3.1 Uplink

6.11.5.4.1.3.1.1 Transport channel parameters

6.11.5.4.1.3.1.1.1 Transport channel parameters for UL:13.6 kbps SRBs for DCCH

See clause 6.10.3.4.1.3.1.1.1.

6.11.5.4.1.3.1.1.2 TFCS

See clause 6.10.3.4.1.3.1.1.2.

6.11.5.4.1.3.1.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Uplink | Modulation | QPSK |
| | Codes and time slots / radio frame | SF8 x 1 code x 2 time slots |
| | Max. Number of data bits / radio frame | 336 bits |
| | TFCI code word / radio frame | 8 bits |
| | TPC / radio frame | 2x2 bit |
| | SS / radio frame | 2x2 bit |
| | Puncturing Limit | 0.64 |
| NOTE: In case the first TFCS is not configured, the TFCI code word will be 4 bit. | | |

6.11.5.4.1.3.2 Downlink

6.11.5.4.1.3.2.1 Transport channel parameters

6.11.5.4.1.3.2.1.1 Transport channel parameters for DL:13.6 kbps SRBs for DCCH

See clause 6.10.3.4.1.3.2.1.1.

6.11.5.4.1.3.2.1.2 TFCS

See clause 6.10.3.4.1.3.2.1.2.

6.11.5.4.1.3.2.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Downlink | Modulation | QPSK |
| | Codes and time slots / radio frame | SF 16 x 2 code x 2 time slots |
| | Max. Number of data bits / radio frame | 336 bits |
| | TFCI code word / radio frame | 8 bits |
| | TPC / radio frame | 2x2 bits |
| | SS / radio frame | 2x2 bits |
| | Puncturing Limit | 0.64 |
| NOTE: In case the first TFCS is not configured, the TFCI code word will be 4 bit. | | |

6.11.5.4.1.4 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.5.4.1.4.1 Uplink

6.11.5.4.1.4.1.1 Transport channel parameters

6.11.5.4.1.4.1.1.1 Transport channel parameters for Conversational / speech / UL:12.2 kbps / CS RAB

See clause 6.10.3.4.1.4.1.1.1.

6.11.5.4.1.4.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.11.5.4.1.4.1.1.3 TFCS

See clause 6.10.3.4.1.4.1.1.3.

6.11.5.4.1.4.1.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Uplink | Modulation | QPSK |
| | Codes and time slots / radio frame | SF8 x 1 code x 2 time slots |
| | Max. Number of data bits / radio frame | 328 bits |
| | TFCI code word / radio frame | 16 bits |
| | TPC / radio frame | 2x2 bits |
| | SS / radio frame | 2x2 bits |
| | Puncturing Limit | 0.48 |

6.11.5.4.1.4.2 Downlink

6.11.5.4.1.4.2.1 Transport channel parameters

6.11.5.4.1.4.2.1.1 Transport channel parameters for Conversational / speech / DL:12.2 kbps / CS RAB

See clause 6.10.3.4.1.4.2.1.1.

6.11.5.4.1.4.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.11.5.4.1.4.2.1.3 TFCS

See clause 6.10.3.4.1.4.2.1.3.

6.11.5.4.1.4.2.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Downlink | Modulation | QPSK |
| | Codes and time slots / radio frame | SF 16 x 2 code x 2 time slots |
| | Max. Number of data bits / radio frame | 328 bits |
| | TFCI code word / radio frame | 16 bits |
| | TPC / radio frame | 2x2 bits |
| | SS / radio frame | 2x2 bits |
| | Puncturing Limit | 0.48 |

6.11.5.4.1.4a Conversational / speech / UL: 12.2 7.95 5.9 4.75 DL:12.2 7.95 5.9 4.75 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.5.4.1.4a.1 Uplink

6.11.5.4.1.4a.1.1 Transport channel parameters

6.11.5.4.1.4a.1.1.1 Transport channel parameters for Conversational / speech / UL: 12.2 7.95 5.9 4.75 kbps / CS RAB

See clause 6.10.3.4.1.4a.1.1.1.

6.11.5.4.1.4a.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.11.5.4.1.4a.1.1.3 TFCS

See clause 6.10.3.4.1.4a.1.1.3.

6.11.5.4.1.4a.1.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Uplink | Modulation | QPSK |
| | Codes and time slots / radio frame | SF8 x 1 code x 2 time slots |
| | Max. Number of data bits / radio frame | 328 bits |
| | TFCI code word / radio frame | 16 bits |
| | TPC / radio frame | 2x2 bits |
| | SS / radio frame | 2x2 bits |
| | Puncturing Limit | 0.48 |

6.11.5.4.1.4a.2 Downlink

6.11.5.4.1.4a.2.1 Transport channel parameters

6.11.5.4.1.4a.2.1.1 Transport channel parameters for Conversational / speech / DL: (12.2, 7.95, 5.9, 4.75) kbps / CS RAB

See clause 6.10.3.4.1.4a.2.1.1.

6.11.5.4.1.4a.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.11.5.4.1.4a.2.1.3 TFCS

See clause 6.10.3.4.1.4a.1.2.1.3.

6.11.5.4.1.4a.2.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Downlink | Modulation | QPSK |
| | Codes and time slots / radio frame | SF 16 x 2 code x 2 time slots |
| | Max. Number of data bits / radio frame | 328 bits |
| | TFCI code word / radio frame | 16 bits |
| | TPC / radio frame | 2x2 bits |
| | SS / radio frame | 2x2 bits |
| | Puncturing Limit | 0.48 |

6.11.5.4.1.5 Conversational / speech / UL:10.2 DL:10.2 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.5.4.1.5.1 Uplink

6.11.5.4.1.5.1.1 Transport channel parameters

6.11.5.4.1.5.1.1.1 Transport channel parameters for Conversational / speech / UL:10.2 kbps / CS RAB

See clause 6.10.3.4.1.5.1.1.1.

6.11.5.4.1.5.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.11.5.4.1.5.1.1.3 TFCS

See clause 6.10.3.4.1.5.1.1.3.

6.11.5.4.1.5.1.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Uplink | Modulation | QPSK |
| | Codes and time slots / radio frame | SF 8 x 1 code x 2 time slots |
| | Max. Number of data bits / radio frame | 328 bits |
| | TFCI code word / radio frame | 16 bits |
| | TPC / radio frame | 2x2 bits |
| | SS / radio frame | 2x2 bits |
| | Puncturing Limit | 0.52 |

6.11.5.4.1.5.2 Downlink

6.11.5.4.1.5.2.1 Transport channel parameters

6.11.5.4.1.5.2.1.1 Transport channel parameters for Conversational / speech / DL:10.2 kbps / CS RAB

See clause 6.10.3.4.1.5.2.1.1.

6.11.5.4.1.5.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.11.5.4.1.5.2.1.3 TFCS

See clause 6.10.3.4.1.5.2.1.3.

6.11.5.4.1.5.2.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Downlink | Modulation | QPSK |
| | Codes and time slots / radio frame | SF 16 x 2 code x 2 time slots |
| | Max. Number of data bits / radio frame | 328 bits |
| | TFCI code word / radio frame | 16 bits |
| | TPC / radio frame | 2x2 bits |
| | SS / radio frame | 2x2 bits |
| | Puncturing Limit | 0.52 |

6.11.5.4.1.5a Conversational / speech / UL:10.2 6.7 5.9 4.75 DL:10.2 6.7 5.9 4.75 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.5.4.1.5a.1 Uplink

6.11.5.4.1.5a.1.1 Transport channel parameters

6.11.5.4.1.5a.1.1.1 Transport channel parameters for Conversational / speech / UL:10.2 6.7 5.9 4.75 kbps / CS RAB

See clause 6.10.3.4.1.5a.1.1.1.

6.11.5.4.1.5a.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.11.5.4.1.5a.1.1.3 TFCS

See clause 6.10.3.4.1.5a.1.1.3.

6.11.5.4.1.5a.1.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Uplink | Modulation | QPSK |
| | Codes and time slots / radio frame | SF 8 x 1 code x 2 time slots |
| | Max. Number of data bits / radio frame | 328 bits |
| | TFCI code word / radio frame | 16 bits |
| | TPC / radio frame | 2x2 bits |
| | SS / radio frame | 2x2 bits |
| | Puncturing Limit | 0.52 |

6.11.5.4.1.5a.2 Downlink

6.11.5.4.1.5a.2.1 Transport channel parameters

6.11.5.4.1.5a.2.1.1 Transport channel parameters for Conversational / speech / DL: 10.2 6.7 5.9 4.75 kbps / CS RAB

See clause 6.10.3.4.1.5a.2.1.1.

6.11.5.4.1.5a.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.11.5.4.1.5a.2.1.3 TFCS

See clause 6.10.3.4.1.5a.2.1.3.

6.11.5.4.1.5a.2.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Downlink | Modulation | QPSK |

| | | |
|---|---|---|
| | Codes and time slots / radio frame | SF 16 x 2 code x 2 time slots |
| | Max. Number of data bits / radio frame | 328 bits |
| | TFCI code word / radio frame | 16 bits |
| | TPC / radio frame | 2x2 bits |
| | SS / radio frame | 2x2 bits |
| | Puncturing Limit | 0.52 |

6.11.5.4.1.6 Conversational / speech / UL:7.95 DL:7.95 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.5.4.1.6.1 Uplink

6.11.5.4.1.6.1.1 Transport channel parameters

6.11.5.4.1.6.1.1.1 Transport channel parameters for Conversational / speech / UL:7.95 kbps / CS RAB

See clause 6.10.3.4.1.6.1.1.1.

6.11.5.4.1.6.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.11.5.4.1.6.1.1.3 TFCS

See clause 6.10.3.4.1.6.1.1.3.

6.11.5.4.1.6.1.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Uplink | Modulation | QPSK |
| | Codes and time slots / radio frame | SF 8 x 1 code x 2 time slots |
| | Max. Number of data bits / radio frame | 328 bits |
| | TFCI code word / radio frame | 16 bits |
| | TPC / radio frame | 2x2 bits |
| | SS / radio frame | 2x2 bits |
| | Puncturing Limit | 0.60 |

6.11.5.4.1.6.2 Downlink

6.11.5.4.1.6.2.1 Transport channel parameters

6.11.5.4.1.6.2.1.1 Transport channel parameters for Conversational / speech / DL:7.95 kbps / CS RAB

See clause 6.10.3.4.1.6.2.1.1.

6.11.5.4.1.6.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.11.5.4.1.6.2.1.3 TFCS

See clause 6.10.3.4.1.6.2.1.3.

6.11.5.4.1.6.2.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Downlink | Modulation | QPSK |
| | Codes and time slots / radio frame | SF 16 x 2 code x 2 time slots |
| | Max. Number of data bits / radio frame | 328 bits |
| | TFCI code word / radio frame | 16 bits |
| | TPC / radio frame | 2x2 bits |
| | SS / radio frame | 2x2 bits |
| | Puncturing Limit | 0.60 |

6.11.5.4.1.7 Conversational / speech / UL:7.4 DL:7.4 kbps / CS RAB+ UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.5.4.1.7.1 Uplink

6.11.5.4.1.7.1.1 Transport channel parameters

6.11.5.4.1.7.1.1.1 Transport channel parameters for Conversational / speech / UL:7.4 kbps / CS RAB

See clause 6.10.3.4.1.7.1.1.1.

6.11.5.4.1.7.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.11.5.4.1.7.1.1.3 TFCS

See clause 6.10.3.4.1.7.1.1.3.

6.11.5.4.1.7.1.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Uplink | Modulation | QPSK |
| | Codes and time slots / radio frame | SF 8 x 1 code x 2 time slots |
| | Max. Number of data bits / radio frame | 328 bits |
| | TFCI code word / radio frame | 16 bits |
| | TPC / radio frame | 2x2 bits |
| | SS / radio frame | 2x2 bits |
| | Puncturing Limit | 0.64 |

6.11.5.4.1.7.2 Downlink

6.11.5.4.1.7.2.1 Transport channel parameters

6.11.5.4.1.7.2.1.1 Transport channel parameters for Conversational / speech / DL:7.4 kbps / CS RAB

See clause 6.10.3.4.1.7.2.1.1.

6.11.5.4.1.7.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.11.5.4.1.7.2.1.3 TFCS

See clause 6.10.3.4.1.7.2.1.3.

6.11.5.4.1.7.2.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Downlink | Modulation | QPSK |
| | Codes and time slots / radio frame | SF 16 x 2 code x 2 time slots |
| | Max. Number of data bits / radio frame | 328 bits |
| | TFCI code word / radio frame | 16 bits |
| | TPC / radio frame | 2x2 bits |
| | SS / radio frame | 2x2 bits |
| | Puncturing Limit | 0.64 |

6.11.5.4.1.7a Conversational / speech / UL:7.4 6.7 5.9 4.75 DL:7.4 6.7 5.9 4.75 / CS RAB+ UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.5.4.1.7a.1 Uplink

6.11.5.4.1.7a.1.1 Transport channel parameters

6.11.5.4.1.7a.1.1.1 Transport channel parameters for Conversational / speech / UL:7.4 6.7 5.9 4.75 / CS RAB

See clause 6.10.3.4.1.7a.1.1.1.

6.11.5.4.1.7a.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.11.5.4.1.7a.1.1.3 TFCS

See clause 6.10.3.4.1.7a.1.1.3.

6.11.5.4.1.7a.1.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Uplink | Modulation | QPSK |
| | Codes and time slots / radio frame | SF 8 x 1 code x 2 time slots |
| | Max. Number of data bits / radio frame | 328 bits |
| | TFCI code word / radio frame | 16 bits |
| | TPC / radio frame | 2x2 bits |
| | SS / radio frame | 2x2 bits |
| | Puncturing Limit | 0.64 |

6.11.5.4.1.7a.2 Downlink

6.11.5.4.1.7a.2.1 Transport channel parameters

6.11.5.4.1.7a.2.1.1 Transport channel parameters for Conversational / speech / DL:7.4 6.7 5.9 4.75 / CS RAB

See clause 6.10.3.4.1.7a.2.1.1.

6.11.5.4.1.7a.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.11.5.4.1.7a.2.1.3 TFCS

See clause 6.10.3.4.1.7a.2.1.3.

6.11.5.4.1.7a.2.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Downlink | Modulation | QPSK |
| | Codes and time slots / radio frame | SF 16 x 2 code x 2 time slots |
| | Max. Number of data bits / radio frame | 328 bits |
| | TFCI code word / radio frame | 16 bits |
| | TPC / radio frame | 2x2 bits |
| | SS / radio frame | 2x2 bits |
| | Puncturing Limit | 0.64 |

6.11.5.4.1.8 Conversational / speech / UL:6.7 DL:6.7 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.5.4.1.8.1 Uplink

6.11.5.4.1.8.1.1 Transport channel parameters

6.11.5.4.1.8.1.1.1 Transport channel parameters for Conversational / speech / UL:6.7 kbps / CS RAB

See clause 6.10.3.4.1.8.1.1.1.

6.11.5.4.1.8.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.11.5.4.1.8.1.1.3 TFCS

See clause 6.10.3.4.1.8.1.1.3.

6.11.5.4.1.8.1.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Uplink | Modulation | QPSK |
| | Codes and time slots / radio frame | SF 8 x 1 code x 2 time slots |

| | | |
|---|---|---|
| | Max. Number of data bits / radio frame | 328 bits |
| | TFCI code word / radio frame | 16 bits |
| | TPC / radio frame | 2x2 bits |
| | SS / radio frame | 2x2 bits |
| | Puncturing Limit | 0.68 |

6.11.5.4.1.8.2 Downlink

6.11.5.4.1.8.2.1 Transport channel parameters

6.11.5.4.1.8.2.1.1 Transport channel parameters for Conversational / speech / DL:6.7 kbps / CS RAB

See clause 6.10.3.4.1.8.2.1.1.

6.11.5.4.1.8.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.11.5.4.1.8.2.1.3 TFCS

See clause 6.10.3.4.1.8.2.1.3.

6.11.5.4.1.8.2.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Downlink | Modulation | QPSK |
| | Codes and time slots / radio frame | SF 16 x 2 code x 2 time slots |
| | Max. Number of data bits / radio frame | 328 bits |
| | TFCI code word / radio frame | 16 bits |
| | TPC / radio frame | 2x2 bits |
| | SS / radio frame | 2x2 bits |
| | Puncturing Limit | 0.68 |

6.11.5.4.1.9 Conversational / speech / UL:5.9 DL:5.9 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.5.4.1.9.1 Uplink

6.11.5.4.1.9.1.1 Transport channel parameters

6.11.5.4.1.9.1.1.1 Transport channel parameters for Conversational / speech / UL:5.9 kbps / CS RAB

See clause 6.10.3.4.1.9.1.1.1.

6.11.5.4.1.9.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.11.5.4.1.9.1.1.3 TFCS

See clause 6.10.3.4.1.9.1.1.3.

6.11.5.4.1.9.1.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Uplink | Modulation | QPSK |
| | Codes and time slots / radio frame | SF 8 x 1 code x 2 time slots |
| | Max. Number of data bits / radio frame | 328 bits |
| | TFCI code word / radio frame | 16 bits |
| | TPC / radio frame | 2x2 bits |
| | SS / radio frame | 2x2 bits |
| | Puncturing Limit | 0.72 |

6.11.5.4.1.9.2 Downlink

6.11.5.4.1.9.2.1 Transport channel parameters

6.11.5.4.1.9.2.1.1 Transport channel parameters for Conversational / speech / DL:5.9 kbps / CS RAB

See clause 6.10.3.4.1.9.2.1.1.

6.11.5.4.1.9.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.11.5.4.1.9.2.1.3 TFCS

See clause 6.10.3.4.1.9.2.1.3.

6.11.5.4.1.9.2.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Downlink | Modulation | QPSK |
| | Codes and time slots / radio frame | SF 16 x 2 code x 2 time slots |
| | Max. Number of data bits / radio frame | 328 bits |
| | TFCI code word / radio frame | 16 bits |
| | TPC / radio frame | 2x2 bits |
| | SS / radio frame | 2x2 bits |
| | Puncturing Limit | 0.72 |

6.11.5.4.1.10 Conversational / speech / UL:5.15 DL:5.15 kbps / CS RAB + UL:1.7 DL:1.7 kbps SRBs for DCCH

6.11.5.4.1.10.1 Uplink

6.11.5.4.1.10.1.1 Transport channel parameters

6.11.5.4.1.10.1.1.1 Transport channel parameters for Conversational / speech / UL:5.15 kbps / CS RAB

See clause 6.10.3.4.1.10.1.1.1.

6.11.5.4.1.10.1.1.2 Transport channel parameters for UL:1.7 kbps SRBs for DCCH

See clause 6.10.3.4.1.1.1.1.1.

6.11.5.4.1.10.1.1.3 TFCS

See clause 6.10.3.4.1.10.1.1.3.

6.11.5.4.1.10.1.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Uplink | Modulation | QPSK |
| | Codes and time slots / radio frame | SF 8 x 1 code x 2 time slots |
| | Max. Number of data bits / radio frame | 328 bits |
| | TFCI code word / radio frame | 16 bits |
| | TPC / radio frame | 2x2 bits |
| | SS / radio frame | 2x2 bits |
| | Puncturing Limit | 0.96 |

6.11.5.4.1.10.2 Downlink

6.11.5.4.1.10.2.1 Transport channel parameters

6.11.5.4.1.10.2.1.1 Transport channel parameters for Conversational / speech / DL:5.15 kbps / CS RAB

See clause 6.10.3.4.1.10.2.1.1.

6.11.5.4.1.10.2.1.2 Transport channel parameters for DL:1.7 kbps SRBs for DCCH

See clause 6.10.3.4.1.1.2.1.1.

6.11.5.4.1.10.2.1.3 TFCS

See clause 6.10.3.4.1.10.2.1.3.

6.11.5.4.1.10.2.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Downlink | Modulation | QPSK |
| | Codes and time slots / radio frame | SF 16 x 2 code x 2 time slots |
| | Max. Number of data bits / radio frame | 328 bits |
| | TFCI code word / radio frame | 16 bits |
| | TPC / radio frame | 2x2 bits |
| | SS / radio frame | 2x2 bits |
| | Puncturing Limit | 0.96 |

6.11.5.4.1.11 Conversational / speech / UL:4.75 DL:4.75 kbps / CS RAB + UL:1.7 DL:1.7 kbps SRBs for DCCH

6.11.5.4.1.11.1 Uplink

6.11.5.4.1.11.1.1 Transport channel parameters

6.11.5.4.1.11.1.1.1 Transport channel parameters for Conversational / speech / UL:4.75 kbps / CS RAB

See clause 6.10.3.4.1.11.1.1.1.

6.11.5.4.1.11.1.1.2 Transport channel parameters for UL:1.7 kbps SRBs for DCCH

See clause 6.10.3.4.1.1.1.1.1.

6.11.5.4.1.11.1.1.3 TFCS

See clause 6.10.3.4.1.11.1.1.3.

6.11.5.4.1.11.1.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Uplink | Modulation | QPSK |
| | Codes and time slots / radio frame | SF 8 x 1 code x 2 time slots |
| | Max. Number of data bits / radio frame | 328 bits |
| | TFCI code word / radio frame | 16 bits |
| | TPC / radio frame | 2x2 bits |
| | SS / radio frame | 2x2 bits |
| | Puncturing Limit | 1 |

6.11.5.4.1.11.2 Downlink

6.11.5.4.1.11.2.1 Transport channel parameters

6.11.5.4.1.11.2.1.1 Transport channel parameters for Conversational / speech / DL:4.75 kbps / CS RAB

See clause 6.10.3.4.1.11.2.1.1.

6.11.5.4.1.11.2.1.2 Transport channel parameters for DL:1.7 kbps SRBs for DCCH

See clause 6.10.3.4.1.1.2.1.1.

6.11.5.4.1.11.2.1.3 TFCS

See clause 6.10.3.4.1.11.2.1.3.

6.11.5.4.1.11.2.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Downlink | Modulation | QPSK |
| | Codes and time slots / radio frame | SF 16 x 2 code x 2 time slots |
| | Max. Number of data bits / radio frame | 328 bits |
| | TFCI code word / radio frame | 16 bits |
| | TPC / radio frame | 2x2 bits |
| | SS / radio frame | 2x2 bits |
| | Puncturing Limit | 1 |

6.11.5.4.1.12 Conversational / unknown / UL:28.8/DL:28.8 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.5.4.1.12.1 Uplink

6.11.5.4.1.12.1.1 Transport channel parameters

6.11.5.4.1.12.1.1.1 Transport channel parameters for conversational / unknown / UL:28.8 kbps / CS RAB

See clause 6.10.3.4.1.12.1.1.1.

6.11.5.4.1.12.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.11.5.4.1.12.1.1.3 TFCS

See clause 6.10.3.4.1.12.1.1.3.

6.11.5.4.1.12.1.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Uplink | Modulation | QPSK |
| | Codes and time slots / radio frame | SF 4 x 1 code x 2 time slots |
| | Max. Number of data bits / radio frame | 680 bits |
| | TFCI code word / radio frame | 16 bits |
| | TPC / radio frame | 2x2 bits |
| | SS / radio frame | 2x2 bits |
| | Puncturing Limit | 0.52 |

6.11.5.4.1.12.2 Downlink

6.11.5.4.1.12.2.1 Transport channel parameters

6.11.5.4.1.12.2.1.1 Transport channel parameters for conversational / unknown / DL:28.8 kbps / CS RAB

See clause 6.10.3.4.1.12.2.1.1.

6.11.5.4.1.12.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.11.5.4.1.12.2.1.3 TFCS

See clause 6.10.3.4.1.12.2.1.3.

6.11.5.4.1.12.2.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Downlink | Modulation | QPSK |
| | Codes and time slots / radio frame | SF 16 x 4 code x 2 time slots |
| | Max. Number of data bits / radio frame | 680 bits |
| | TFCI code word / radio frame | 16 bits |
| | TPC / radio frame | 2x2 bits |
| | SS / radio frame | 2x2 bits |
| | Puncturing Limit | 0.52 |

6.11.5.4.1.13 Conversational / unknown / UL:64 DL:64 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.5.4.1.13.1 Uplink

6.11.5.4.1.13.1.1 Transport channel parameters

6.11.5.4.1.13.1.1.1 Transport channel parameters for Conversational / unknown / UL:64 kbps / CS RAB

See clause 6.10.3.4.1.13.1.1.1.

6.11.5.4.1.13.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.11.5.4.1.13.1.1.3 TFCS

See clause 6.10.3.4.1.13.1.1.3.

6.11.5.4.1.13.1.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Uplink | Modulation | QPSK |
| | Codes and time slots / radio frame | SF2 x 1 code x 2 time slots |
| | Max. Number of data bits / radio frame | 1 384 bits |
| | TFCI code word / radio frame | 16 bits |
| | TPC / radio frame | 2x2 bits |
| | SS / radio frame | 2x2 bits |
| | Puncturing Limit | 0.52 |

6.11.5.4.1.13.2 Downlink

6.11.5.4.1.13.2.1 Transport channel parameters

6.11.5.4.1.13.2.1.1 Transport channel parameters for Conversational / unknown / DL:64 kbps / CS RAB

See clause 6.10.3.4.1.13.2.1.1.

6.11.5.4.1.13.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.11.5.4.1.13.2.1.3 TFCS

See clause 6.10.3.4.1.13.2.1.3.

6.11.5.4.1.13.2.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Downlink | Modulation | QPSK |
| | Codes and time slots / radio frame | SF 16 x 8 code x 2 time slots |
| | Max. Number of data bits / radio frame | 1 384 bits |
| | TFCI code word / radio frame | 16 bits |
| | TPC / radio frame | 2x2 bits |
| | SS / radio frame | 2x2 bits |
| | Puncturing Limit | 0.52 |

6.11.5.4.1.14 Conversational / unknown / UL:32 DL:32 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.5.4.1.14.1 Uplink

6.11.5.4.1.14.1.1 Transport channel parameters

6.11.5.4.1.14.1.1.1 Transport channel parameters for Conversational / unknown / UL:32 kbps / CS RAB

See clause 6.10.3.4.1.14.1.1.1.

6.11.5.4.1.14.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.11.5.4.1.14.1.1.3 TFCS

See clause 6.10.3.4.1.14.1.1.3.

6.11.5.4.1.14.1.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Uplink | Modulation | QPSK |
| | Codes and time slots / radio frame | SF4 x 1 code x 2 time slots |
| | Max. Number of data bits / radio frame | 680 bits |
| | TFCI code word / radio frame | 16 bits |
| | TPC / radio frame | 2x2 bits |
| | SS / radio frame | 2x2 bits |
| | Puncturing Limit | 0.44 |

6.11.5.4.1.14.2 Downlink

6.11.5.4.1.14.2.1 Transport channel parameters

6.11.5.4.1.14.2.1.1 Transport channel parameters for Conversational / unknown / DL:32 kbps / CS RAB

See clause 6.10.3.4.1.14.2.1.1.

6.11.5.4.1.14.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.11.5.4.1.14.2.1.3 TFCS

See clause 6.10.3.4.1.14.2.1.3.

6.11.5.4.1.14.2.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Downlink | Modulation | QPSK |
| | Codes and time slots / radio frame | SF16 x 4 code x 2 time slots |
| | Max. Number of data bits / radio frame | 680 bits |
| | TFCI code word / radio frame | 8 bits |
| | TPC / radio frame | 2x2 bits |
| | SS / radio frame | 2x2 bits |
| | Puncturing Limit | 0.44 |

6.11.5.4.1.15 Streaming / unknown / UL:14.4/DL:14.4 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.5.4.1.15.1 Uplink

6.11.5.4.1.15.1.1 Transport channel parameters

6.11.5.4.1.15.1.1.1 Transport channel parameters for Streaming / unknown / UL: 14.4 kbps / CS RAB

See clause 6.10.3.4.1.15.1.1.1.

6.11.5.4.1.15.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.11.5.4.1.15.1.1.3 TFCS

See clause 6.10.3.4.1.15.1.1.3.

6.11.5.4.1.15.1.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Uplink | Modulation | QPSK |
| | Codes and time slots / radio frame | SF4 x 1 code x 2 time slots |
| | Max. Number of data bits / radio frame | 680 bits |
| | TFCI code word / radio frame | 16 bits |
| | TPC / radio frame | 2x2 bits |
| | SS / radio frame | 2x2 bits |
| | Puncturing Limit | 1 |

6.11.5.4.1.15.2 Downlink

6.11.5.4.1.15.2.1 Transport channel parameters

6.11.5.4.1.15.2.1.1 Transport channel parameters for Streaming / unknown / DL:14.4 kbps / CS RAB

See clause 6.10.3.4.1.15.2.1.1.

6.11.5.4.1.15.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.11.5.4.1.15.2.1.3 TFCS

See clause 6.10.3.4.1.15.2.1.3.

6.11.5.4.1.15.2.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Downlink | Modulation | QPSK |
| | Codes and time slots / radio frame | SF16 x 3 code x 2 time slots |
| | Max. Number of data bits / radio rame | 504 bits |
| | TFCI code word / radio frame | 16 bits |
| | TPC / radio frame | 2x2 bits |
| | SS / radio frame | 2x2 bits |
| | Puncturing Limit | 0.76 |

6.11.5.4.1.16 Streaming / unknown / UL:28.8/DL:28.8 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.5.4.1.16.1 Uplink

6.11.5.4.1.16.1.1 Transport channel parameters

6.11.5.4.1.16.1.1.1 Transport channel parameters for Streaming / unknown / UL:28.8 kbps / CS RAB

See clause 6.10.3.4.1.16.1.1.1.

6.11.5.4.1.16.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.11.5.4.1.16.1.1.3 TFCS

See clause 6.10.3.4.1.16.1.1.3.

6.11.5.4.1.16.1.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Uplink | Modulation | QPSK |
| | Codes and time slots / frame | SF4 x 1 code x 2 time slots |
| | Max. Number of data bits / radio frame | 680 bits |
| | TFCI code word / radio frame | 16 bits |
| | TPC / radio frame | 2x2 bits |
| | SS / radio frame | 2x2 bits |
| | Puncturing Limit | 0.56 |

6.11.5.4.1.16.2 Downlink

6.11.5.4.1.16.2.1 Transport channel parameters

6.11.5.4.1.16.2.1.1 Transport channel parameters for Streaming / unknown / DL:28.8 kbps / CS RAB

See clause 6.10.3.4.1.16.2.1.1.

6.11.5.4.1.16.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.11.5.4.1.16.2.1.3 TFCS

See clause 6.10.3.4.1.16.2.1.3.

6.11.5.4.1.16.2.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Downlink | Modulation | QPSK |
| | Codes and time slots / radio frame | SF16 x 4 code x 2 time slots |
| | Max. Number of data bits / radio frame | 680 bits |
| | TFCI code word / radio frame | 16 bits |
| | TPC / radio frame | 2x2 bits |
| | SS / radio frame | 2x2 bits |
| | Puncturing Limit | 0.56 |

6.11.5.4.1.17 Streaming / unknown / UL:57.6/DL:57.6 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.5.4.1.17.1 Uplink

6.11.5.4.1.17.1.1 Transport channel parameters

6.11.5.4.1.17.1.1.1 Transport channel parameters for Streaming / unknown / UL:57.6 kbps / CS RAB

See clause 6.10.3.4.1.17.1.1.1.

6.11.5.4.1.17.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.11.5.4.1.17.1.1.3 TFCS

See clause 6.10.3.4.1.17.1.1.3.

6.11.5.4.1.17.1.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Uplink | Modulation | QPSK |
| | Codes and time slots / radio frame | SF2 x 1 code x 2 time slots |
| | Max. Number of data bits / radio frame | 1 384 bits |
| | TFCI code word / radio frame | 16 bits |
| | TPC / radio frame | 2x2 bits |
| | SS / radio frame | 2x2 bits |
| | Puncturing Limit | 0.68 |

6.11.5.4.1.17.2 Downlink

6.11.5.4.1.17.2.1 Transport channel parameters

6.11.5.4.1.17.2.1.1 Transport channel parameters for Streaming / unknown / DL:57.6 kbps / CS RAB

See clause 6.10.3.4.1.17.2.1.1.

6.11.5.4.1.17.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.11.5.4.1.17.2.1.3 TFCS

See clause 6.10.3.4.1.17.2.1.3.

6.11.5.4.1.17.2.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Downlink | Modulation | QPSK |
| | Codes and time slots / radio frame | SF16 x 8 code x 2 time slots |
| | Max. Number of data bits / radio frame | 1 384 bits |
| | TFCI code word / radio frame | 16 bits |
| | TPC / radio frame | 2x2 bits |

| | | |
|---|---|---|
| | SS / radio frame | 2x2 bits |
| | Puncturing Limit | 0.68 |

6.11.5.4.1.18 Void

6.11.5.4.1.19 Void

6.11.5.4.1.20 Void

6.11.5.4.1.21 Void

6.11.5.4.1.22 Void

6.11.5.4.1.23 Interactive or background / UL:32 DL:8 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.5.4.1.23.1 Uplink

6.11.5.4.1.23.1.1 Transport channel parameters

6.11.5.4.1.23.1.1.1 Transport channel parameters for Interactive or background / UL:32 kbps / PS RAB

See clause 6.10.3.4.1.23.1.1.1.

6.11.5.4.1.23.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.11.5.4.1.23.1.1.3 TFCS

See clause 6.10.3.4.1.23.1.1.3.

6.11.5.4.1.23.1.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Uplink | Modulation | QPSK |
| | Codes and time slots / radio frame | SF 4 x 1 code x 2 time slots |
| | Max. Number of data bits/radio frame | 680bits |
| | TFCI code word / radio frame | 16 bits |
| | TPC / radio frame | 2x2 bits |
| | SS / radio frame | 2x2 bits |
| | Puncturing Limit | 0.48 (alt. 0.44) |

6.11.5.4.1.23.2 Downlink

6.11.5.4.1.23.2.1 Transport channel parameters

6.11.5.4.1.23.2.1.1 Transport channel parameters for Interactive or background / DL:8 kbps / PS RAB

See clause 6.10.3.4.1.23.2.1.1.

6.11.5.4.1.23.2.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.11.5.4.1.23.2.1.3 TFCS

See clause 6.10.3.4.1.23.2.1.3.

6.11.5.4.1.23.2.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Downlink | Modulation | QPSK |
| | Codes and time slots / radio frame | SF 16 x 2 codes x 2 time slots |
| | Max. Number of data bits/radio frame | 336 bits |
| | TFCI code word / radio frame | 8 bits |
| | TPC / radio frame | 2x2 bits |
| | SS / radio frame | 2x2 bits |

| | | |
|---|---|---|
| | Puncturing Limit | 0.76 |

6.11.5.4.1.23a Interactive or background / UL:8 DL:8 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.5.4.1.23a.1 Uplink

6.11.5.4.1.23a.1.1 Transport channel parameters

6.11.5.4.1.23a.1.1.1 Transport channel parameters for Interactive or background / UL:8kbps / PS RAB

See clause 6.10.3.4.1.23a.1.1.1.

6.11.5.4.1.23a.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.11.5.4.1.23a.1.1.3 TFCS

See clause 6.10.3.4.1.23a.1.1.3.

6.11.5.4.1.23a.1.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Uplink | Modulation | QPSK |
| | Codes and time slots / radio frame | SF 8 x 1 code x 2 time slots |
| | Max. Number of data bits / radio frame | 328 bits |
| | TFCI code word / radio frame | 16 bits |
| | TPC / radio frame | 2x2 bits |
| | SS / radio frame | 2x2 bits |
| | Puncturing Limit | 0.72 (alt 0.68) |

6.11.5.4.1.23a.2 Downlink

See clause 6.11.5.4.1.23.2.

6.11.5.4.1.23b Interactive or background / UL:16 DL:16 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.5.4.1.23b.1 Uplink

6.11.5.4.1.23b.1.1 Transport channel parameters

6.11.5.4.1.23b.1.1.1 Transport channel parameters for Interactive or background / UL:16 kbps / PS RAB

See clause 6.10.3.4.1.23b.1.1.1.

6.11.5.4.1.23b.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.11.5.4.1.23b.1.1.3 TFCS

See clause 6.10.3.4.1.23b.1.1.3.

6.11.5.4.1.23b.1.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Uplink | Modulation | QPSK |
| | Codes and time slots / radio frame | SF4 x 1 code x 2 time slots |
| | Max. Number of data bits / radio frame | 688 bits |
| | TFCI code word / radio frame | 16bits |
| | TPC / radio frame | 2x2 bits |
| | SS / radio frame | 2x2 bits |
| | Puncturing Limit | 0.92 alt (0.84) |

6.11.5.4.1.23b.2 Downlink

6.11.5.4.1.23b.2.1 Transport channel parameters

6.11.5.4.1.23b.2.1.1 Transport channel parameters for Interactive or background / DL:16 kbps / PS RAB

See clause 6.10.3.4.1.23b.2.1.1.

6.11.5.4.1.23b.2.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.11.5.4.1.23b.2.1.3 TFCS

See clause 6.10.3.4.1.23b.2.1.3.

6.11.5.4.1.23b.2.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Downlink | Modulation | QPSK |
| | Codes and time slots / radio frame | SF16 x 3 code x 2 time slots |
| | Max. Number of data bits / radio rame | 512 bits |
| | TFCI code word / radio frame | 16 bits |
| | TPC / radio frame | 2x2 bits |
| | SS / radio frame | 2x2 bits |
| | Puncturing Limit | 0.68 |

6.11.5.4.1.23c Interactive or background / UL:32 DL32 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.5.4.1.23c.1 Uplink

6.11.5.4.1.23c.1.1 Transport channel parameters

6.11.5.4.1.23c.1.1.1 Transport channel parameters for Interactive or background / UL:32 kbps / PS RAB

See clause 6.10.3.4.1.23c.1.1.1.

6.11.5.4.1.23c.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.11.5.4.1.23c.1.1.3 TFCS

See clause 6.10.3.4.1.23c.1.1.3.

6.11.5.4.1.23c.1.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Uplink | Modulation | QPSK |
| | Codes and time slots / radio frame | SF 4 x 1 code x 2 time slots |
| | Max. Number of data bits/radio frame | 680bits |
| | TFCI code word / radio frame | 16 bits |
| | TPC / radio frame | 2x2 bits |
| | SS / radio frame | 2x2 bits |
| | Puncturing Limit | 0.48 (alt 0.44) |

6.11.5.4.1.23c.2 Downlink

6.11.5.4.1.23c.2.1 Transport channel parameters

6.11.5.4.1.23c.2.1.1 Transport channel parameters for Interactive or background / DL:32 kbps / PS RAB

See clause 6.10.3.4.1.23c.2.1.1.

6.11.5.4.1.23c.2.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.11.5.4.1.23c.2.1.3 TFCS

See clause 6.10.3.4.1.23c.2.1.3.

6.11.5.4.1.23c.2.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Downlink | Modulation | QPSK |
| | Codes and time slots / radio frame | SF16 x 4 code x 2 time slots |
| | Max. Number of data bits/radio frame | 680 bits |
| | TFCI code word / radio frame | 16 bits |
| | TPC / radio frame | 2x2 bits |
| | SS / radio frame | 2x2 bits |
| | Puncturing Limit | 0.48 |

6.11.5.4.1.23d Interactive or background / UL:32 DL32 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.5.4.1.23d.1 Uplink

6.11.5.4.1.23d.1.1 Transport channel parameters

6.11.5.4.1.23d.1.1.1 Transport channel parameters for Interactive or background / UL:32 kbps / PS RAB

See clause 6.10.3.4.1.23d.1.1.1.

6.11.5.4.1.23d.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.11.5.4.1.23d.1.1.3 TFCS

See clause 6.10.3.4.1.23d.1.1.3.

6.11.5.4.1.23d.1.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Uplink | Modulation | QPSK |
| | Codes and time slots / radio frame | SF 4 x 1 code x 2 time slots |
| | Max. Number of data bits/radio frame | 680 bits |
| | TFCI code word / radio frame | 16 bits |
| | TPC / radio frame | 2x2 bits |
| | SS / radio frame | 2x2 bits |
| | Puncturing Limit | 0.48 (alt. 0.44) |

6.11.5.4.1.23d.2 Downlink

6.11.5.4.1.23d.2.1 Transport channel parameters

6.11.5.4.1.23d.2.1.1 Transport channel parameters for Interactive or background / DL:32 kbps / PS RAB

See clause 6.10.3.4.1.23d.2.1.1.

6.11.5.4.1.23d.2.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.11.5.4.1.23d.2.1.3 TFCS

See clause 6.10.3.4.1.23d.2.1.3.

6.11.5.4.1.23d.2.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Downlink | Modulation | QPSK |
| | Codes and time slots / radio frame | SF16 x 4 code x 2 time slots |
| | Max. Number of data bits/radio frame | 680 bits |
| | TFCI code word / radio frame | 16 bits |
| | TPC / radio frame | 2x2 bits |
| | SS / radio frame | 2x2 bits |
| | Puncturing Limit | 0.48 |

6.11.5.4.1.24 Void.

6.11.5.4.1.25 Interactive or background / UL:32 DL: 64 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.5.4.1.25.1 Uplink

See clause 6.11.5.4.1.23.1.

6.11.5.4.1.25.2 Downlink

6.11.5.4.1.25.2.1 Transport channel parameters

6.11.5.4.1.25.2.1.1 Transport channel parameters for Interactive or background / DL:64 kbps / PS RAB

See clause 6.10.3.4.1.25.2.1.1.

6.11.5.4.1.25.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.11.5.4.1.25.2.1.3 TFCS

See clause 6.10.3.4.1.25.2.1.3.

6.11.5.4.1.25.2.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Downlink | Modulation | QPSK |
| | Codes and time slots / radio frame | SF16 x 8 codes x 2 time slots |
| | Max. Number of data bits/radio frame | 1 384 bits |
| | TFCI code word / radio frame | 16 bits |
| | TPC / radio frame | 2x2 bits |
| | SS / radio frame | 2x2 bits |
| | Puncturing Limit/ radio frame | 0.56 |

6.11.5.4.1.26 Interactive or background / UL:64 DL: 64 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.5.4.1.26.1 Uplink

6.11.5.4.1.26.1.1 Transport channel parameters

6.11.5.4.1.26.1.1.1 Transport channel parameters for Interactive or background / UL:64 kbps / PS RAB

See clause 6.10.3.4.1.26.1.1.1.

6.11.5.4.1.26.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.11.5.4.1.26.1.1.3 TFCS

See clause 6.10.3.4.1.26.1.1.3.

6.11.5.4.1.26.1.2 Physical channel parameters

| | | | |
|---|---|---|---|
| DPCH Uplink | | Physical 1 | Physical 2 |
| | Modulation | QPSK | QPSK |
| | Codes and time slots / radio frame | SF2 x 1 code x 2 time slots | SF1 x 1 code x 2 time slots |
| | Max. Number of data bits/radio frame | 1 384 bits | 2 792 bits |
| | TFCI code word / radio frame | 16 bits | 16 bits |
| | TPC / radio frame | 2x2 bits | 2x2 bits |
| | SS / radio frame | 2x2 bits | 2x2 bits |
| | Puncturing Limit | 0.56 (alt 0.48) | 1 |

6.11.5.4.1.26.2 Downlink

See clause 6.11.5.4.1.25.2.

6.11.5.4.1.27 Interactive or background / UL:64 DL:128 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.5.4.1.27.1 Uplink

See clause 6.11.5.4.1.26.1.

6.11.5.4.1.27.2 Downlink

6.11.5.4.1.27.2.1 Transport channel parameters

6.11.5.4.1.27.2.1.1 Transport channel parameters for Interactive or background / DL:128 kbps / PS RAB

See clause 6.10.3.4.1.27.2.1.1.

6.11.5.4.1.27.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.11.5.4.1.27.2.1.3 TFCS

See clause 6.10.3.4.1.27.2.1.3.

6.11.5.4.1.27.2.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Downlink | Modulation | QPSK |
| | Codes and time slots / radio frame | SF 16 x 9 codes x 4 time slots |
| | Max. Number of data bits/radio frame | 3 144 bits |
| | TFCI code word / radio frame | 16 bits |
| | TPC / radio frame | 2x2 bits |
| | SS / radio frame | 2x2 bits |
| | Puncturing Limit | 0.68 |

6.11.5.4.1.28 Interactive or background / UL:128 DL:128 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.5.4.1.28.1 Uplink

6.11.5.4.1.28.1.1 Transport channel parameters

6.11.5.4.1.28.1.1.1 Transport channel parameters for Interactive or background / UL:128 kbps / PS RAB

See clause 6.10.3.4.1.28.1.1.1.

6.11.5.4.1.28.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.11.5.4.1.28.1.1.3 TFCS

See clause 6.10.3.4.1.28.1.1.3.

6.11.5.4.1.28.1.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Uplink | Modulation | QPSK |
| | Codes and time slots / radio frame | SF1 x 1 codes x 2 time slots |
| | Max. Number of data bits/radio frame | 2 792 bits |
| | TFCI code word / radio frame | 16 bits |
| | TPC / radio frame | 2x2 bits |
| | SS / radio frame | 2x2 bits |
| | Puncturing Limit | 0.60 |

6.11.5.4.1.28.2 Downlink

See clause 6.11.5.4.1.27.2.

6.11.5.4.1.29 Interactive or background / UL:64 DL:144 kbps / PS RAB + UL:3.4 DL: 3.4 kbps SRBs for DCCH

6.11.5.4.1.29.1 Uplink

See clause 6.11.5.4.1.26.1.

6.11.5.4.1.29.2 Downlink

6.11.5.4.1.29.2.1 Transport channel parameters

6.11.5.4.1.29.2.1.1 Transport channel parameters for Interactive or background / DL:144 kbps / PS RAB

See clause 6.10.3.4.1.29.2.1.1.

6.11.5.4.1.29.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.11.5.4.1.29.2.1.3 TFCS

See clause 6.10.3.4.1.29.2.1.3.

6.11.5.4.1.29.2.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Downlink | Modulation | QPSK |
| | Codes and time slots / radio frame | SF 16 x 9 codes x 4 time slots |
| | Max. Number of data bits/radio frame | 3 144 bits |
| | TFCI code word / radio frame | 16 bits |
| | TPC / radio frame | 2x2 bits |
| | SS / radio frame | 2x2 bits |
| | Puncturing Limit | 0.52 |

6.11.5.4.1.30 Interactive or background / UL:144 DL:144 kbps / PS RAB + UL:3.4 DL: 3.4 kbps SRBs for DCCH

6.11.5.4.1.30.1 Uplink

6.11.5.4.1.30.1.1 Transport channel parameters

6.11.5.4.1.30.1.1.1 Transport channel parameters for Interactive or background / UL:144 kbps / PS RAB

See clause 6.10.3.4.1.30.1.1.1.

6.11.5.4.1.30.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.11.5.4.1.30.1.1.3 TFCS

See clause 6.10.3.4.1.30.1.1.3.

6.11.5.4.1.30.1.2 Physical channel parameters

| | | | |
|---|---|---|---|
| DPCH Uplink | Modulation | QPSK | 8PSK |
| | Codes and time slots / radio frame | (SF1 x 1 code x 2 time slots) + (SF2 x 1 code x 2 time slots) | SF1 x 1code x 2 time slots |
| | Max. Number of data bits/radio frame | 4 200 bits | 4 188 bits |
| | TFCI code word / radio frame | 16 bits | 24 bits |
| | TPC / radio frame | 2x2 bits | 2x3 bits |
| | SS / radio frame | 2x2 bits | 2x3 bits |
| | Puncturing Limit | 0.72 (alt 0.64) | 0.72 (alt 0.64) |

6.11.5.4.1.30.2 Downlink

See clause 6.11.5.4.1.29.2.

6.11.5.4.1.31 Interactive or background / UL:64 DL:256 kbps / PS RAB + UL:3.4 DL: 3.4 kbps SRBs for DCCH

6.11.5.4.1.31.1 Uplink

See clause 6.11.5.4.1.26.1.

6.11.5.4.1.31.2 Downlink

6.11.5.4.1.31.2.1 Transport channel parameters

6.11.5.4.1.31.2.1.1 Transport channel parameters for Interactive or background / DL:256 kbps / PS RAB

See clause 6.10.3.4.1.31.2.1.1.

6.11.5.4.1.31.2.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.11.5.4.1.31.2.1.3 TFCS

See clause 6.10.3.4.1.31.2.1.3.

6.11.5.4.1.31.2.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Downlink | Modulation | QPSK |
| | Codes and time slots / radio frame | SF 1 x 1 code x 4 time slots |
| | Max. Number of data bits/radio frame | 5 608 bits |
| | TFCI code word / radio frame | 16 bits |
| | TPC / radio frame | 2x2 bits |
| | SS / radio frame | 2x2 bits |
| | Puncturing Limit | 0.56 |

6.11.5.4.1.32 Interactive or background / UL:64 DL:384 kbps / PS RAB + UL:3.4 DL: 3.4 kbps SRBs for DCCH

6.11.5.4.1.32.1 Uplink

See clause 6.11.5.4.1.26.1.

6.11.5.4.1.32.2 Downlink

6.11.5.4.1.32.2.1 Transport channel parameters

6.11.5.4.1.32.2.1.1 Transport channel parameters for Interactive or background / DL:384 kbps / PS RAB

See clause 6.10.3.4.1.32.2.1.1.

6.11.5.4.1.32.2.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.11.5.4.1.32.2.1.3 TFCS

See clause 6.10.3.4.1.32.2.1.3.

6.11.5.4.1.32.2.2 Physical channel parameters

| | | | |
|---|---|---|---|
| DPCH Downlink | Modulation | QPSK | 8PSK |
| | Codes and time slots / radio frame | SF1 x 1 code x 6 time slots | SF1 x 1 code x 4 time slots |
| | Max. Number of data bits/radio frame | 8 424 bits | 8 412 bits |
| | TFCI code word / radio frame | 16 bits | 24 bits |
| | TPC / radio frame | 2x2 bits | 2x3 bits |

| | | | |
|---|---|---|---|
| | SS / radio frame | 2x2 bits | 2x3 bits |
| | Puncturing Limit | 0.64 | 0.64 |

6.11.5.4.1.33 Interactive or background / UL:128 DL:384 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.5.4.1.33.1 Uplink

See clause 6.11.5.4.1.28.1.

6.11.5.4.1.33.2 Downlink

See clause 6.11.5.4.1.32.2.

6.11.5.4.1.34 Interactive or background / UL:384 DL:384 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.5.4.1.34.1 Uplink

6.11.5.4.1.34.1.1 Transport channel parameters

6.11.5.4.1.34.1.1.1 Transport channel parameters for Interactive or background / UL:384 kbps / PS RAB

See clause 6.10.3.4.1.34.1.1.1.

6.11.5.4.1.34.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.11.5.4.1.34.1.1.3 TFCS

See clause 6.10.3.4.1.34.1.1.3.

6.11.5.4.1.34.1.2 Physical channel parameters

| | | | |
|---|---|---|---|
| DPCH Uplink | Modulation | QPSK | 8PSK |
| | Codes and time slots / radio frame | SF 1 x 1 code x 6 time slots | SF 1 x 1 code x 4 time slots |
| | Max. Number of data bits/radio frame | 8 424 bits | 8 412 bits |
| | TFCI code word / radio frame | 16 bits | 24 bits |
| | TPC / radio frame | 2x2 bits | 2x3 bits |
| | SS / radio frame | 2x2 bits | 2x3 bits |
| | Puncturing Limit | 0.64 | 0.64 |

6.11.5.4.1.34.2 Downlink

See clause 6.11.5.4.1.32.2.

6.11.5.4.1.35 Interactive or background / UL:64 DL:2 048 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.5.4.1.35.1 Uplink

See clause 6.11.5.4.1.26.1.

6.11.5.4.1.35.2 Downlink

6.11.5.4.1.35.2.1 Transport channel parameters

6.11.5.4.1.35.2.1.1 Transport channel parameters for Interactive or background / DL:2 048 kbps / PS RAB

| | | |
|---|---|---|
| Higher layer | RAB/Signalling RB | RAB |
| RLC | Logical channel type | DTCH |
| | RLC mode | AM |
| | Payload sizes, bit | 1 704 |
| | Max data rate, bps | 2 048 000 |
| | RLC header, bit | 16 |

| | | | |
|---|---|---|---|
| MAC | MAC header, bit | | 0 |
| | MAC multiplexing | | N/A |
| Layer 1 | TrCH type | | DCH |
| | TB sizes, bit | | 1720 |
| | TFS | TF0, bits | 0x1720 |
| | | TF1, bits | 1x1720 |
| | | TF2, bits | 2x1720 |
| | | TF3, bits | 4x1720 |
| | | TF4, bits | 8x1720 |
| | | TF5, bits | 12x1720 |
| | | TF6, bits | N/A (alt. 16x1720) |
| | | TF7, bits | N/A (alt. 20x1720) |
| | | TF8, bits | N/A (alt. 24x1720) |
| | TTI, ms | | 10 (alt. 20) |
| | Coding type | | No coding |
| | CRC, bit | | 24 |
| | Max number of bits/TTI after channel coding | | 20 928 (alt. 41 856) |
| | Max number of bits/radio frame before rate matching | | 20 928 ( alt. 20 928) |
| | RM attribute | | 130 to 170 |

6.11.5.4.1.35.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.11.5.4.1.35.2.1.3 TFCS

| | |
|---|---|
| TFCS size | 12 (alt.18) |
| TFCS | (2 048 kbps RAB, DCCH)=<br>(TF0, TF0), (TF1, TF0), (TF2, TF0), (TF3, TF0), (TF4, TF0), (TF5, TF0),<br>(TF0, TF1), (TF1, TF1), (TF2, TF1), (TF3, TF1), (TF4, TF1), (TF5, TF1),<br>(alt. (TF0, TF0), (TF1, TF0), (TF2, TF0), (TF3, TF0), (TF4, TF0), (TF5, TF0),<br>(TF6, TF0), (TF7, TF0), (TF8, TF0),<br>(TF0, TF1), (TF1, TF1), (TF2, TF1), (TF3, TF1), (TF4, TF1), (TF5, TF1),<br>(TF6, TF1), (TF7, TF1), (TF8, TF1)) |

6.11.5.4.1.35.2.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Downlink | Modulation | 8PSK |
| | Codes and time slots / radio frame | SF1 x 1 code x 10 time slots |
| | Max. Number of data bits/radio frame | 21 084 bits |
| | TFCI code word / radio frame | 24 bits |
| | TPC / radio frame | 2x3 bits |
| | SS / radio frame | 2x3 bits |
| | Puncturing Limit | 1 |

6.11.5.4.1.36 Void

6.11.5.4.1.37 Void

6.11.5.4.1.38 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:32 DL:8 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.5.4.1.38.1 Uplink

6.11.5.4.1.38.1.1 Transport channel parameters

6.11.5.4.1.38.1.1.1 Transport channel parameters for Conversational / speech / UL:12.2 kbps / CS RAB

See clause 6.10.3.4.1.4.1.1.1.

6.11.5.4.1.38.1.1.2 Transport channel parameters for Interactive or background / UL:32 kbps / PS RAB

See clause 6.10.3.4.1.23.1.1.1.

6.11.5.4.1.38.1.1.3 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.11.5.4.1.38.1.1.4 TFCS

See clause 6.10.3.4.1.38.1.1.4.

6.11.5.4.1.38.1.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Uplink | Modulation | QPSK |
| | Codes and time slots / radio frame | SF 2 x 1 code x 2 time slots |
| | Max. Number of data bits/radio frame | 1 384 bits |
| | TFCI code word / radio frame | 16 bits |
| | TPC / radio frame | 2x2 bits |
| | SS / radio frame | 2x2 bits |
| | Puncturing Limit | 0.72 (alt 0.68) |

6.11.5.4.1.38.2 Downlink

6.11.5.4.1.38.2.1 Transport channel parameters

6.11.5.4.1.38.2.1.1 Transport channel parameters for Conversational / speech / DL:12.2 kbps / CS RAB

See clause 6.10.3.4.1.4.2.1.1.

6.11.5.4.1.38.2.1.2 Transport channel parameters for Interactive or background / DL:8 kbps / PS RAB

See clause 6.10.3.4.1.23.2.1.1.

6.11.5.4.1.38.2.1.3 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.11.5.4.1.38.2.1.4 TFCS

See clause 6.10.3.4.1.38.2.1.4.

6.11.5.4.1.38.2.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Downlink | Modulation | QPSK |
| | Codes and time slots / radio frame | SF16 x 3 codes x 2 time slots |
| | Max. Number of data bits/radio frame | 504 bits |
| | TFCI code word / radio frame | 16 bits |
| | TPC / radio frame | 2x2 bits |
| | SS / radio frame | 2x2 bits |
| | Puncturing Limit | 0.44 |

6.11.5.4.1.38a Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:0 DL:0 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.5.4.1.38a.1 Uplink

6.11.5.4.1.38a.1.1 Transport channel parameters

6.11.5.4.1.38a.1.1.1 Transport channel parameters for Conversational / speech / UL:12.2 kbps / CS RAB

See clause 6.10.3.4.1.4.1.1.1.

6.11.5.4.1.38a.1.1.2 Transport channel parameters for Interactive or background / UL:0 kbps / PS RAB

See clause 6.10.3.4.1.38a.1.1.2.

6.11.5.4.1.38a.1.1.3 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.11.5.4.1.38a.1.1.4 TFCS

See clause 6.10.3.4.1.38a.1.1.4.

6.11.5.4.1.38a.1.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Uplink | Modulation | QPSK |
| | Codes and time slots / radio frame | SF8 x 1 code x 2 time slots |
| | Max. Number of data bits / radio frame | 328 bits |
| | TFCI code word / radio frame | 16 bits |
| | TPC / radio frame | 2x2 bits |
| | SS / radio frame | 2x2 bits |
| | Puncturing Limit | 0.48 |

6.11.5.4.1.38a.2 Downlink

6.11.5.4.1.38a.2.1 Transport channel parameters

6.11.5.4.1.38a.2.1.1 Transport channel parameters for Conversational / speech / DL:12.2 kbps / CS RAB

See clause 6.10.3.4.1.4.2.1.1.

6.11.5.4.1.38a.2.1.2 Transport channel parameters for Interactive or background / DL:0 kbps / PS RAB

See clause 6.10.3.4.1.38a.2.1.2.

6.11.5.4.1.38a.2.1.3 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.11.5.4.1.38a.2.1.4 TFCS

See clause 6.10.3.4.1.38a.2.1.4.

6.11.5.4.1.38a.2.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Downlink | Modulation | QPSK |
| | Codes and time slots / radio frame | SF 16 x 2 code x 2 time slots |
| | Max. Number of data bits / radio frame | 328 bits |
| | TFCI code word / radio frame | 16 bits |
| | TPC / radio frame | 2x2 bits |
| | SS / radio frame | 2x2 bits |
| | Puncturing Limit | 0.48 |

6.11.2.5.1.38b Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:8 DL:8 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.5.4.1.38b.1 Uplink

6.11.5.4.1.38b.1.1 Transport channel parameters

6.11.5.4.1.38b.1.1.1 Transport channel parameters for Conversational / speech / UL:12.2 kbps / CS RAB

See clause 6.10.3.4.1.4.1.1.1.

6.11.5.4.1.38b.1.1.2 Transport channel parameters for Interactive or background / UL:8 kbps / PS RAB

See clause 6.10.3.4.1.23a.1.1.1.

6.11.5.4.1.38b.1.1.3 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.11.5.4.1.38b.1.1.4 TFCS

See clause 6.10.3.4.1.38b.1.1.4.

6.11.5.4.1.38b.1.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Uplink | Modulation | QPSK |
| | Codes and time slots / frame | SF4 x 1 code x 2 time slots |
| | Max. Number of data bits / radio frame | 680 bits |
| | TFCI code word / radio frame | 16 bits |
| | TPC / radio frame | 2x2 bits |
| | SS / radio frame | 2x2 bits |
| | Puncturing Limit | 0.64 (alt. 0.60) |

6.11.5.4.1.38b.2 Downlink

6.11.5.4.1.38b.2.1 Transport channel parameters

6.11.5.4.1.38b.2.1.1 Transport channel parameters for Conversational / speech / DL:12.2 kbps / CS RAB

See clause 6.10.3.4.1.4.2.1.1.

6.11.5.4.1.38b.2.1.2 Transport channel parameters for Interactive or background / DL:8 kbps / PS RAB

See clause 6.10.3.4.1.23.2.1.1.

6.11.5.4.1.38b.2.1.3 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.11.5.4.1.38b.2.1.4 TFCS

See clause 6.10.3.4.1.38b.2.1.4.

6.11.5.4.1.38b.2.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Downlink | Modulation | QPSK |
| | Codes and time slots / radio frame | SF16 x 4 code x 2 time slots |
| | Max. Number of data bits / radio frame | 680 bits |
| | TFCI code word / radio frame | 16 bits |
| | TPC / radio frame | 2x2 bits |
| | SS / radio frame | 2x2 bits |
| | Puncturing Limit | 0.64 |

6.11.5.4.1.38c Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:32 DL:32 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.5.4.1.38c.1 Uplink

6.11.5.4.1.38c.1.1 Transport channel parameters

6.11.5.4.1.38c.1.1.1 Transport channel parameters for Conversational / speech / UL:12.2 kbps / CS RAB

See clause 6.10.3.4.1.4.1.1.1.

6.11.5.4.1.38c.1.1.2 Transport channel parameters for Interactive or background / UL:32 kbps / PS RAB

See clause 6.10.3.4.1.23d.1.1.1.

6.11.5.4.1.38c.1.1.3 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.11.5.4.1.38c.1.1.4 TFCS

See clause 6.10.3.4.1.38c.1.1.4.

6.11.5.4.1.38c.1.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Uplink | Modulation | QPSK |

| | | |
|---|---|---|
| | Codes and time slots / radio frame | SF2 x 1 code x 2 time slots |
| | Max. Number of data bits / radio frame | 1 384 bits |
| | TFCI code word / radio frame | 16 bits |
| | TPC / radio frame | 2x2 bits |
| | SS / radio frame | 2x2 bits |
| | Puncturing Limit | 0.72 (alt 0.64) for TFCS size=18<br>0.80 (alt 0.72) for TFCS size=17 |

6.11.5.4.1.38c.2 Downlink

6.11.5.4.1.38c.2.1 Transport channel parameters

6.11.5.4.1.38c.2.1.1 Transport channel parameters for Conversational / speech / DL:12.2 kbps / CS RAB

See clause 6.10.3.4.1.4.2.1.1.

6.11.5.4.1.38c.2.1.2 Transport channel parameters for Interactive or background / DL:32 kbps / PS RAB

See clause 6.10.3.4.1.23d.2.1.1.

6.11.5.4.1.38c.2.1.3 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.11.5.4.1.38c.2.1.4 TFCS

See clause 6.10.3.4.1.38c.2.1.4.

6.11.5.4.1.38c.2.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Downlink | Modulation | QPSK |
| | Codes and time slots / radio frame | SF16 x 8 code x 2 time slots |
| | Max. Number of data bits / radio frame | 1 384 bits |
| | TFCI code word / radio frame | 16 bits |
| | TPC / radio frame | 2x2 bits |
| | SS / radio frame | 2x2 bits |
| | Puncturing Limit | 0.72 (alt 0.64) |

6.11.5.4.1.38d Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:64 DL:64 kbps / PS RAB + Interactive or background / UL:64 DL:64 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.5.4.1.38d.1 Uplink

6.11.5.4.1.38d.1.1 Transport channel parameters

6.11.5.4.1.38d.1.1.1 Transport channel parameters for Conversational / speech / UL:12.2 kbps / CS RAB

See clause 6.10.3.4.1.4.1.1.1.

6.11.5.4.1.38d.1.1.2 Transport channel parameters for Interactive or background / UL:64 kbps / PS RAB

See clause 6.10.3.4.1.38d.1.1.2.

6.11.5.4.1.38d.1.1.3 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.11.5.4.1.38d.1.1.4 TFCS

See clause 6.10.3.4.1.38d.1.1.4.

6.11.5.4.1.38d.1.2 Physical channel parameters

| | | | |
|---|---|---|---|
| DPCH Uplink | Modulation | QPSK | 8PSK |

| | | | |
|---|---|---|---|
| | Codes and time slots / radio frame | (SF1 x 1 code x 2 time slots) + (SF2 x 1 code x 2 time slots) | SF1 x 1code x 2 time slots |
| | Max. Number of data bits/radio frame | 4 200 bits | 4 188 bits |
| | TFCI code word / radio frame | 16 bits | 24 bits |
| | TPC / radio frame | 2x2 bits | 2x3 bits |
| | SS / radio frame | 2x2 bits | 2x3 bits |
| | Puncturing Limit | 0.72 (alt 0.64) | 0.72 (alt 0.64) |

6.11.5.4.1.38d.2 Downlink

6.11.5.4.1.38d.2.1 Transport channel parameters

6.11.5.4.1.38d.2.1.1 Transport channel parameters for Conversational / speech / DL:12.2 kbps / CS RAB

See clause 6.10.3.4.1.4.2.1.1.

6.11.5.4.1.38d.2.1.2 Transport channel parameters for Interactive or background / DL:64 kbps / PS RAB

See clause 6.10.3.4.1.38d.2.1.2.

6.11.5.4.1.38d.2.1.3 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.11.5.4.1.38d.2.1.4 TFCS

See clause 6.10.3.4.1.38d.2.1.4.

6.11.5.4.1.38d.2.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Downlink | Modulation | QPSK |
| | Codes and time slots / radio frame | SF 16 x 9 codes x 4 time slots |
| | Max. Number of data bits/radio frame | 3 144 bits |
| | TFCI code word / radio frame | 16 bits |
| | TPC / radio frame | 2x2 bits |
| | SS / radio frame | 2x2 bits |
| | Puncturing Limit | 0.52 |

6.11.5.4.1.38e Conversational / speech / UL: 12.2 7.95 5.9 4.75 DL:12.2 7.95 5.9 4.75 kbps / CS RAB + Interactive or background / UL:0 DL:0 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.5.4.1.38e.1 Uplink

6.11.5.4.1.38e.1.1 Transport channel parameters

6.11.5.4.1.38e.1.1.1 Transport channel parameters for Conversational / speech / UL: 12.2 7.95 5.9 4.75 / CS RAB

See clause 6.10.3.4.1.4a.1.1.1.

6.11.5.4.1.38e.1.1.2 Transport channel parameters for Interactive or background / UL:0 kbps / PS RAB

See clause 6.10.3.4.1.38a.1.1.2.

6.11.5.4.1.38e.1.1.3 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.11.5.4.1.38e.1.1.4 TFCS

See clause 6.10.3.4.1.38e.1.1.4.

6.11.5.4.1.38e.1.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Uplink | Modulation | QPSK |
| | Codes and time slots / radio frame | SF8 x 1 code x 2 time slots |
| | Max. Number of data bits / radio frame | 328 bits |
| | TFCI code word / radio frame | 16 bits |
| | TPC / radio frame | 2x2 bits |
| | SS / radio frame | 2x2 bits |
| | Puncturing Limit | 0.48 |

6.11.5.4.1.38e.2 Downlink

6.11.5.4.1.38e.2.1 Transport channel parameters

6.11.5.4.1.38e.2.1.1 Transport channel parameters for Conversational / speech / DL:12.2 7.95 5.9 4.75 / CS RAB

See clause 6.10.3.4.1.4a.2.1.1.

6.11.5.4.1.38e.2.1.2 Transport channel parameters for Interactive or background / DL:0 kbps / PS RAB

See clause 6.10.3.4.1.38a.2.1.2.

6.11.5.4.1.38e.2.1.3 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.11.5.4.1.38e.2.1.4 TFCS

See clause 6.10.3.4.1.38e.2.1.4.

6.11.5.4.1.38e.2.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Downlink | Modulation | QPSK |
| | Codes and time slots / radio frame | SF 16 x 2 code x 2 time slots |
| | Max. Number of data bits / radio frame | 328 bits |
| | TFCI code word / radio frame | 16 bits |
| | TPC / radio frame | 2x2 bits |
| | SS / radio frame | 2x2 bits |
| | Puncturing Limit | 0.48 |

6.11.5.4.1.38f Conversational / speech / UL: 12.2 7.95 5.9 4.75 DL:12.2 7.95 5.9 4.75 kbps / CS RAB + Interactive or background / UL:8 DL:8 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.5.4.1.38f.1 Uplink

6.11.5.4.1.38f.1.1 Transport channel parameters

6.11.5.4.1.38f.1.1.1 Transport channel parameters for Conversational / speech / UL: 12.2 7.95 5.9 4.75 / CS RAB

See clause 6.10.3.4.1.4a.1.1.1.

6.11.5.4.1.38f.1.1.2 Transport channel parameters for Interactive or background / UL:8 kbps / PS RAB

See clause 6.10.3.4.1.23a.1.1.1.

6.11.5.4.1.38f.1.1.3 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.11.5.4.1.38f.1.1.4 TFCS

See clause 6.10.3.4.1.38f.1.1.4.

6.11.5.4.1.38f.1.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Uplink | Modulation | QPSK |
| | Codes and time slots / frame | SF4 x 1 code x 2 time slots |
| | Max. Number of data bits / radio frame | 680 bits |
| | TFCI code word / radio frame | 16 bits |
| | TPC / radio frame | 2x2 bits |
| | SS / radio frame | 2x2 bits |
| | Puncturing Limit | 0.64 (alt 0.60) |

6.11.5.4.1.38f.2 Downlink

6.11.5.4.1.38f.2.1 Transport channel parameters

6.11.5.4.1.38f.2.1.1 Transport channel parameters for Conversational / speech / DL:12.2 7.95 5.9 4.75 / CS RAB

See clause 6.10.3.4.1.4a.2.1.1.

6.11.5.4.1.38f.2.1.2 Transport channel parameters for Interactive or background / DL:8 kbps / PS RAB

See clause 6.10.3.4.1.23.2.1.1.

6.11.5.4.1.38f.2.1.3 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.11.5.4.1.38f.2.1.4 TFCS

See clause 6.10.3.4.1.38f.2.1.4.

6.11.5.4.1.38f.2.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Downlink | Modulation | QPSK |
| | Codes and time slots / radio frame | SF16 x 4 code x 2 time slots |
| | Max. Number of data bits / radio frame | 680 bits |
| | TFCI code word / radio frame | 16 bits |
| | TPC / radio frame | 2x2 bits |
| | SS / radio frame | 2x2 bits |
| | Puncturing Limit | 0.64 |

6.11.5.4.1.38g Conversational / speech / UL: 12.2 7.95 5.9 4.75 DL:12.2 7.95 5.9 4.75 kbps / CS RAB + Interactive or background / UL:16 DL:16 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.5.4.1.38g.1 Uplink

6.11.5.4.1.38g.1.1 Transport channel parameters

6.11.5.4.1.38g.1.1.1 Transport channel parameters for Conversational / speech / UL: 12.2 7.95 5.9 4.75 / CS RAB

See clause 6.10.3.4.1.4a.1.1.1.

6.11.5.4.1.38g.1.1.2 Transport channel parameters for Interactive or background / UL:16 kbps / PS RAB

See clause 6.10.3.4.1.23b.1.1.1.

6.11.5.4.1.38g.1.1.3 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.11.5.4.1.38g.1.1.4 TFCS

See clause 6.10.3.4.1.38g.1.1.4.

6.11.5.4.1.38g.1.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Uplink | Modulation | QPSK |
| | Codes and time slots / radio frame | SF2 x 1 code x 2 time slots |
| | Max. Number of data bits / radio frame | 1 368 bits (alt. 1 384 bits) |
| | TFCI code word / radio frame | 32 bits (alt. 16 bits) |
| | TPC / radio frame | 2x2 bits |
| | SS / radio frame | 2x2 bits |
| | Puncturing Limit | 0.96 (alt 1.0) |
| NOTE: There are 32 bit and 16 bit TFCIs for the two cases. | | |

6.11.5.4.1.38g.2 Downlink

6.11.5.4.1.38g.2.1 Transport channel parameters

6.11.5.4.1.38g.2.1.1 Transport channel parameters for Conversational / speech / DL:12.2 7.95 5.9 4.75 / CS RAB

See clause 6.10.3.4.1.4a.2.1.1.

6.11.5.4.1.38g.2.1.2 Transport channel parameters for Interactive or background / DL:16 kbps / PS RAB

See clause 6.10.3.4.1.23b.2.1.1.

6.11.5.4.1.38g.2.1.3 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.11.5.4.1.38g.2.1.4 TFCS

See clause 6.10.3.4.1.38g.2.1.4.

6.11.5.4.1.38g.2.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Downlink | Modulation | QPSK |
| | Codes and time slots / radio frame | SF16 x 8 code x 2 time slots |
| | Max. Number of data bits / radio frame | 1 368 bits |
| | TFCI code word / radio frame | 32 bits |
| | TPC / radio frame | 2x2 bits |
| | SS / radio frame | 2x2 bits |
| | Puncturing Limit | 1.0 |

6.11.5.4.1.38h Conversational / speech / UL: 12.2 7.95 5.9 4.75 DL:12.2 7.95 5.9 4.75 kbps / CS RAB + Interactive or background / UL:32 DL:32 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.5.4.1.38h.1 Uplink

6.11.5.4.1.38h.1.1 Transport channel parameters

6.11.5.4.1.38h.1.1.1 Transport channel parameters for Conversational / speech / UL: 12.2 7.95 5.9 4.75 / CS RAB

See clause 6.10.3.4.1.4a.1.1.1.

6.11.5.4.1.38h.1.1.2 Transport channel parameters for Interactive or background / UL:32 kbps / PS RAB

See clause 6.10.3.4.1.23d.1.1.1.

6.11.5.4.1.38h.1.1.3 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.11.5.4.1.38h.1.1.4 TFCS

See clause 6.10.3.4.1.38h.1.1.4.

6.11.5.4.1.38h.1.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Uplink | Modulation | QPSK |
| | Codes and time slots / radio frame | SF2 x 1 code x 2 time slots |
| | Max. Number of data bits/radio frame | 1 368 bits |
| | TFCI code word / radio frame | 32 bits |
| | TPC / radio frame | 2x2 bits |
| | SS / radio frame | 2x2 bits |
| | Puncturing Limit | 0.72 (alt 0.64) |

6.11.5.4.1.38h.2 Downlink

6.11.5.4.1.38h.2.1 Transport channel parameters

6.11.5.4.1.38h.2.1.1 Transport channel parameters for Conversational / speech / DL:12.2 7.95 5.9 4.75 / CS RAB

See clause 6.10.3.4.1.4a.2.1.1.

6.11.5.4.1.38h.2.1.2 Transport channel parameters for Interactive or background / DL:32 kbps / PS RAB

See clause 6.10.3.4.1.23d.2.1.1.

6.11.5.4.1.38h.2.1.3 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.11.5.4.1.38h.2.1.4 TFCS

See clause 6.10.3.4.1.38h.2.1.4.

6.11.5.4.1.38h.2.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Downlink | Modulation | QPSK |
| | Codes and time slots / radio frame | SF16 x 8 code x 2 time slots |
| | Max. Number of data bits / radio frame | 1 368 bits |
| | TFCI code word / radio frame | 32 bits |
| | TPC / radio frame | 2x2 bits |
| | SS / radio frame | 2x2 bits |
| | Puncturing Limit | 0.72 |

6.11.5.4.1.38i Conversational / speech / UL: 12.2 7.95 5.9 4.75 DL:12.2 7.95 5.9 4.75 kbps / CS RAB + Interactive or background / UL:64 DL:64 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.5.4.1.38i.1 Uplink

6.11.5.4.1.38i.1.1 Transport channel parameters

6.11.5.4.1.38i.1.1.1 Transport channel parameters for Conversational / speech / UL: 12.2 7.95 5.9 4.75 / CS RAB

See clause 6.10.3.4.1.4a.1.1.1.

6.11.5.4.1.38i.1.1.2 Transport channel parameters for Interactive or background / UL:64 kbps / PS RAB

See clause 6.10.3.4.1.26.1.1.1.

6.11.5.4.1.38i.1.1.3 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.11.5.4.1.38i.1.1.4 TFCS

See clause 6.10.3.4.1.38i.1.1.4.

6.11.5.4.1.38i.1.2 Physical channel parameters

| | | | |
|---|---|---|---|
| DPCH Uplink | Modulation | QPSK | 8PSK |
| | Codes and time slots / radio frame | (SF1 x 1 code x 2 time slots) + (SF2 x 1 code x 2 time slots) | SF1 x 1code x 2 time slots |
| | Max. Number of data bits/radio frame | 4 184 bits | 4 164 bits |
| | TFCI code word / radio frame | 32 bits | 48 bits |
| | TPC / radio frame | 2x2 bits | 2x3 bits |
| | SS / radio frame | 2x2 bits | 2x3 bits |
| | Puncturing Limit | 1 | 1 |

6.11.5.4.1.38i.2 Downlink

6.11.5.4.1.38i.2.1 Transport channel parameters

6.11.5.4.1.38i.2.1.1 Transport channel parameters for Conversational / speech / DL:12.2 7.95 5.9 4.75 / CS RAB

See clause 6.10.3.4.1.4a.2.1.1.

6.11.5.4.1.38i.2.1.2 Transport channel parameters for Interactive or background / DL:64 kbps / PS RAB

See clause 6.10.3.4.1.25.2.1.1.

6.11.5.4.1.38i.2.1.3 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.11.5.4.1.38i.2.1.4 TFCS

See clause 6.10.3.4.1.38i.2.1.4.

6.11.5.4.1.38i.2.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Downlink | Modulation | QPSK |
| | Codes and time slots / radio frame | SF 16 x 9 codes x 4 time slots |
| | Max. Number of data bits/radio frame | 3 128 bits |
| | TFCI code word / radio frame | 32 bits |
| | TPC / radio frame | 2x2 bits |
| | SS / radio frame | 2x2 bits |
| | Puncturing Limit | 1 |

6.11.5.4.1.38j Conversational / speech / UL: 12.2 7.95 5.9 4.75 DL:12.2 7.95 5.9 4.75 kbps / CS RAB + Interactive or background / UL:64 DL:128 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.5.4.1.38j.1 Uplink

See clause 6.11.5.4.1.38i.1.

6.11.5.4.1.38j.2 Downlink

6.11.5.4.1.38j.2.1 Transport channel parameters

6.11.5.4.1.38j.2.1.1 Transport channel parameters for Conversational / speech / DL:12.2 7.95 5.9 4.75 / CS RAB

See clause 6.10.3.4.1.4a.2.1.1.

6.11.5.4.1.38j.2.1.2 Transport channel parameters for Interactive or background / DL:128 kbps / PS RAB

See clause 6.10.3.4.1.27.2.1.1.

6.11.5.4.1.38j.2.1.3 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.11.5.4.1.38j.2.1.4 TFCS

See clause 6.10.3.4.1.38j.2.1.4.

6.11.5.4.1.38j.2.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Downlink | Modulation | QPSK |
| | Codes and time slots / radio frame | SF 16 x 9 codes x 4 time slots |
| | Max. Number of data bits/radio frame | 3 128 bits |
| | TFCI code word / radio frame | 32 bits |
| | TPC / radio frame | 2x2 bits |
| | SS / radio frame | 2x2 bits |
| | Puncturing Limit | 0.60 |

6.11.5.4.1.39 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:32 DL:64 kbps / PS RAB+ UL:3.4 DL: 3.4 kbps SRBs for DCCH

6.11.5.4.1.39.1 Uplink

See clause 6.11.5.4.1.38.1.

6.11.5.4.1.39.2 Downlink

6.11.5.4.1.39.2.1 Transport channel parameters

6.11.5.4.1.39.2.1.1 Transport channel parameters for Conversational / speech / DL:12.2 kbps / CS RAB

See clause 6.10.3.4.1.4.2.1.1.

6.11.5.4.1.39.2.1.2 Transport channel parameters for Interactive or background / DL:64 kbps / PS RAB

See clause 6.10.3.4.1.25.2.1.1.

6.11.5.4.1.39.2.1.3 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.11.5.4.1.39.2.1.4 TFCS

See clause 6.10.3.4.1.39.2.1.4.

6.11.5.4.1.39.2.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Downlink | Modulation | QPSK |
| | Codes and time slots / radio frame | SF 16 x 10 codes x 2 time slots |
| | Max. Number of data bits/radio frame | 1 736 bits |
| | TFCI code word / radio frame | 16 bits |
| | TPC / radio frame | 2x2 bits |
| | SS / radio frame | 2x2 bits |
| | Puncturing Limit | 0.56 |

6.11.5.4.1.40 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:64 DL:64 kbps / PS RAB+ UL:3.4 DL: 3.4 kbps SRBs for DCCH

6.11.5.4.1.40.1 Uplink

6.11.5.4.1.40.1.1 Transport channel parameters

See clause 6.10.3.4.1.40.1.1.

6.11.5.4.1.40.1.2 Physical channel parameters

6.11.5.4.1.40.1.2.1 Physical channel parameters (one CCTrCH case)

| | | |
|---|---|---|
| DPCH Uplink | Modulation | QPSK |
| | Codes and time slots / radio frame | SF1 x 1 code x 2 time slots |

| | | |
|---|---|---|
| | Max. Number of data bits/radio frame | 2 792 bits |
| | TFCI code word / radio frame | 16 bits |
| | TPC / radio frame | 2x2 bits |
| | SS / radio frame | 2x2 bits |
| | Puncturing Limit | 0.92 (alt. 0.84) |

6.11.5.4.1.40.1.2.2 Physical channel parameters (two CCTrCH case)

6.11.5.4.1.40.1.2.2.1 Physical channel parameters (conversational + SRB)

See clause 6.11.5.4.1.4.1.2.

6.11.5.4.1.40.1.2.2.2 Physical channel parameters (Interactive or background)

| | | |
|---|---|---|
| DPCH Uplink | Modulation | QPSK |
| | Codes and time slots / radio frame | SF2 x 1 code x 2 time slots |
| | Max. Number of data bits/radio frame | 1 384 bits |
| | TFCI code word / radio frame | 16 bits |
| | TPC / radio frame | 2x2 bits |
| | SS / radio frame | 2x2 bits |
| | Puncturing Limit | 0.64 (alt. 0.56) |

6.11.5.4.1.40.2 Downlink

6.11.5.4.1.40.2.1 Transport channel parameters

See clause 6.10.3.4.1.40.2.1.

6.11.5.4.1.40.2.2 Physical channel parameters

6.11.5.4.1.40.2.2.1 Physical channel parameters (one CCTrCH)

See clause 6.11.5.4.1.39.2.2.

6.11.5.4.1.40.2.2.2 Physical channel parameters (two CCTrCHs)

6.11.5.4.1.40.2.2.2.1 Physical channel parameters (conversational + SRB)

See clause 6.11.5.4.1.4.2.2.

6.11.5.4.1.40.2.2.2.2 Physical channel parameters (Interactive or background)

| | | |
|---|---|---|
| DPCH Downlink | Modulation | QPSK |
| | Codes and time slots / radio frame | SF16 x 8 codes x 2 time slots |
| | Max. Number of data bits/radio frame | 1 384 bits |
| | TFCI code word / radio frame | 16 bits |
| | TPC / radio frame | 2x2 bits |
| | SS / radio frame | 2x2 bits |
| | Puncturing Limit | 0.64 |

6.11.5.4.1.41 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:64 DL:128 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.5.4.1.41.1 Uplink

See clause 6.11.5.4.1.40.1.

6.11.5.4.1.41.2 Downlink

6.11.5.4.1.41.2.1 Transport channel parameters

See clause 6.10.3.4.1.41.2.1.

6.11.5.4.1.41.2.2 Physical channel parameters

6.11.5.4.1.41.2.2.1 Physical channel parameters (one CCTrCH case)

| | | | |
|---|---|---|---|
| DPCH Downlink | Modulation | QPSK | 8PSK |
| | Codes and time slots / radio frame | SF 16 x 9 codes x 4 time slots | SF 16 x 12 codes x 2 time slots |
| | Max. Number of data bits/radio frame | 3 144 bits | 3 132 bits |
| | TFCI code word / radio frame | 16 bits | 24 bits |
| | TPC / radio frame | 2x2 bits | 2x3 bits |
| | SS / radio frame | 2x2 bits | 2x3 bits |
| | Puncturing Limit | 0.60 | 0.60 |

6.11.5.4.1.41.2.2.2 Physical channel parameters (two CCTrCHs)

6.11.5.4.1.41.2.2.2.1 Physical channel parameters (conversational + SRB)

See clause 6.11.5.4.1.4.2.2.

6.11.5.4.1.41.2.2.2.2 Physical channel parameters (Interactive or background)

| | | | |
|---|---|---|---|
| DPCH Downlink | Modulation | QPSK | 8PSK |
| | Codes and time slots / radio frame | SF 1 x 1 code x 2 time slots | SF 16 x 11 codes x 2 time slots |
| | Max. Number of data bits/radio frame | 2 792 bits | 2868 bits |
| | TFCI code word / radio frame | 16 bits | 24 bits |
| | TPC / radio frame | 2x2 bits | 2x3 bits |
| | SS / radio frame | 2x2 bits | 2x3 bits |
| | Puncturing Limit | 0.64 | 0.64 |

6.11.5.4.1.42 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:64 DL:256 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.5.4.1.42.1 Uplink

6.11.5.4.1.42.1.1 Transport channel parameters

See clause 6.10.3.4.1.42.1.1.

6.11.5.4.1.42.1.2 Physical channel parameters

See clause 6.10.3.4.1.40.1.2.1.

6.11.5.4.1.42.2 Downlink

6.11.5.4.1.42.2.1 Transport channel parameters

6.11.5.4.1.42.2.1.1 Transport channel parameters for Conversational / speech / DL:12.2 kbps / CS RAB

See clause 6.10.3.4.1.4.2.1.1.

6.11.5.4.1.42.2.1.2 Transport channel parameters for Interactive or background / DL:256 kbps / PS RAB

See clause 6.10.3.4.1.31.2.1.1.

6.11.5.4.1.42.2.1.3 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.11.5.4.1.42.2.1.4 TFCS

See clause 6.10.3.4.1.42.2.1.4.

6.11.5.4.1.42.2.2 Physical channel parameters

| | | | |
|---|---|---|---|
| DPCH Downlink | Modulation | QPSK | 8PSK |
| | Codes and time slots / radio frame | SF1 x 1code x 6 time slots | SF1 x 1 code x 4 time slots |
| | Max. Number of data bits/radio frame | 8 408 bits | 8 388 bits |
| | TFCI code word / radio frame | 32 bits | 48 bits |
| | TPC / radio frame | 2x2 bits | 2x3 bits |

| | | | |
|---|---|---|---|
| | SS / radio frame | 2x2 bits | 2x3 bits |
| | Puncturing Limit | 0.80 | 0.80 |

6.11.5.4.1.43 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:64 DL:384 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.5.4.1.43.1 Uplink

See clause 6.11.5.4.1.40.1.

6.11.5.4.1.43.2 Downlink

6.11.5.4.1.43.2.1 Transport channel parameters

See clause 6.10.3.4.1.43.2.1.

6.11.5.4.1.43.2.2 Physical channel parameters

6.11.5.4.1.43.2.2.1 Physical channel parameters (one CCTrCH)

| | | | |
|---|---|---|---|
| DPCH Downlink | Modulation | QPSK | 8PSK |
| | Codes and time slots / radio frame | SF 1 x 1 code x 6 time slots | SF 1 x 1 code x 4 time slots |
| | Max. Number of data bits/radio frame | 8 408 bits | 8 388 bits |
| | TFCI code word / radio frame | 32 bits | 48 bits |
| | TPC / radio frame | 2x2 bits | 2x3 bits |
| | SS / radio frame | 2x2 bits | 2x3 bits |
| | Puncturing Limit | 0.60 | 0.60 |

6.11.5.4.1.43.2.2.2 Physical channel parameters (two CCTrCHs)

6.11.5.4.1.43.2.2.2.1 Physical channel parameters (conversational + SRB)

See clause 6.11.5.4.1.4.2.2.

6.11.5.4.1.43.2.2.2.2 Physical channel parameters (Interactive or background)

| | | | |
|---|---|---|---|
| DPCH Downlink | Modulation | QPSK | 8PSK |
| | Codes and time slots / radio frame | (SF 1 x 1 code x 4 time slots) + (SF 16 x 10 codes x 2 time slots) | SF 1 x 1 code x 4 time slots |
| | Max. Number of data bits/radio frame | 7 368 bits | 8 412 bits |
| | TFCI code word / radio frame | 16 bits | 24 bits |
| | TPC / radio frame | 2x2 bits | 2x3 bits |
| | SS / radio frame | 2x2 bits | 2x3 bits |
| | Puncturing Limit | 0.56 | 0.64 |

6.11.5.4.1.44 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Interactive or background / UL:128 DL:2 048 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.5.4.1.44.1 Uplink

6.11.5.4.1.44.1.1 Transport channel parameters

6.11.5.4.1.44.1.1.1 Transport channel parameters for Conversational / speech / UL:12.2 kbps / CS RAB

See clause 6.10.3.4.1.4.1.1.1.

6.11.5.4.1.44.1.1.2 Transport channel parameters for Interactive or background / UL:128 kbps / PS RAB

See clause 6.10.3.4.1.28.1.1.1.

6.11.5.4.1.44.1.1.3 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.11.5.4.1.44.1.1.4 TFCS

See clause 6.10.3.4.1.44.1.1.4.

6.11.5.4.1.44.1.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Uplink | Modulation | 8PSK |
| | Codes and time slots / radio frame | SF1 x 1 code x 2 time slots |
| | Max. Number of data bits/radio frame | 4 188 bits |
| | TFCI code word / radio frame | 24 bits |
| | TPC / radio frame | 2x3 bits |
| | SS / radio frame | 2x3 bits |
| | Puncturing Limit | 0.80 (alt 0.72) |

6.11.5.4.1.44.2 Downlink

6.11.5.4.1.44.2.1 Transport channel parameters

6.11.5.4.1.44.2.1.1 Transport channel parameters for Conversational / speech / DL:12.2 kbps / CS RAB

See clause 6.10.3.4.1.4.2.1.1.

6.11.5.4.1.44.2.1.2 Transport channel parameters for Interactive or background / DL:2 048 kbps / PS RAB

See clause 6.11.5.4.1.35.2.1.1.

6.11.5.4.1.44.2.1.3 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.11.5.4.1.44.2.1.4 TFCS

| | |
|---|---|
| TFCS size | 32 (alt. 50) |
| TFCS | (RAB subflow#1, RAB subflow#2, RAB subflow#3, 2 048 kbps RAB , DCCH)=<br>((TF0, TF0, TF0, TF0, TF0), (TF1, TF0, TF0, TF0, TF0), (TF2, TF1, TF1, TF0, TF0),<br>(TF0, TF0, TF0, TF1, TF0), (TF1, TF0, TF0, TF1, TF0), (TF2, TF1, TF1, TF1, TF0),<br>(TF0, TF0, TF0, TF2, TF0), (TF1, TF0, TF0, TF2, TF0), (TF2, TF1, TF1, TF2, TF0),<br>(TF0, TF0, TF0, TF3, TF0), (TF1, TF0, TF0, TF3, TF0), (TF2, TF1, TF1, TF3, TF0),<br>(TF0, TF0, TF0, TF4, TF0), (TF1, TF0, TF0, TF4, TF0), (TF2, TF1, TF1, TF4, TF0),<br>(TF0, TF0, TF0, TF5, TF0),<br>(TF0, TF0, TF0, TF0, TF1), (TF1, TF0, TF0, TF0, TF1), (TF2, TF1, TF1, TF0, TF1),<br>(TF0, TF0, TF0, TF1, TF1), (TF1, TF0, TF0, TF1, TF1), (TF2, TF1, TF1, TF1, TF1),<br>(TF0, TF0, TF0, TF2, TF1), (TF1, TF0, TF0, TF2, TF1), (TF2, TF1, TF1, TF2, TF1),<br>(TF0, TF0, TF0, TF3, TF1), (TF1, TF0, TF0, TF3, TF1), (TF2, TF1, TF1, TF3, TF1),<br>(TF0, TF0, TF0, TF4, TF1), (TF1, TF0, TF0, TF4, TF1), (TF2, TF1, TF1, TF4, TF1),<br>(TF0, TF0, TF0, TF5, TF1))<br>(alt. (TF0, TF0, TF0, TF0, TF0), (TF1, TF0, TF0, TF0, TF0), (TF2, TF1, TF1, TF0, TF0),<br>(TF0, TF0, TF0, TF1, TF0), (TF1, TF0, TF0, TF1, TF0), (TF2, TF1, TF1, TF1, TF0),<br>(TF0, TF0, TF0, TF2, TF0), (TF1, TF0, TF0, TF2, TF0), (TF2, TF1, TF1, TF2, TF0),<br>(TF0, TF0, TF0, TF3, TF0), (TF1, TF0, TF0, TF3, TF0), (TF2, TF1, TF1, TF3, TF0),<br>(TF0, TF0, TF0, TF4, TF0), (TF1, TF0, TF0, TF4, TF0), (TF2, TF1, TF1, TF4, TF0),<br>(TF0, TF0, TF0, TF5, TF0), (TF1, TF0, TF0, TF5, TF0), (TF2, TF1, TF1, TF5, TF0),<br>(TF0, TF0, TF0, TF6, TF0), (TF1, TF0, TF0, TF6, TF0), (TF2, TF1, TF1, TF6, TF0),<br>(TF0, TF0, TF0, TF7, TF0), (TF1, TF0, TF0, TF7, TF0), (TF2, TF1, TF1, TF7, TF0),<br>(TF0, TF0, TF0, TF8, TF0),<br>(TF0, TF0, TF0, TF0, TF1), (TF1, TF0, TF0, TF0, TF1), (TF2, TF1, TF1, TF0, TF1),<br>(TF0, TF0, TF0, TF1, TF1), (TF1, TF0, TF0, TF1, TF1), (TF2, TF1, TF1, TF1, TF1),<br>(TF0, TF0, TF0, TF2, TF1), (TF1, TF0, TF0, TF2, TF1), (TF2, TF1, TF1, TF2, TF1),<br>(TF0, TF0, TF0, TF3, TF1), (TF1, TF0, TF0, TF3, TF1), (TF2, TF1, TF1, TF3, TF1),<br>(TF0, TF0, TF0, TF4, TF1), (TF1, TF0, TF0, TF4, TF1), (TF2, TF1, TF1, TF4, TF1),<br>(TF0, TF0, TF0, TF5, TF1), (TF1, TF0, TF0, TF5, TF1), (TF2, TF1, TF1, TF5, TF1),<br>(TF0, TF0, TF0, TF6, TF1), (TF1, TF0, TF0, TF6, TF1), (TF2, TF1, TF1, TF6, TF1),<br>(TF0, TF0, TF0, TF7, TF1), (TF1, TF0, TF0, TF7, TF1), (TF2, TF1, TF1, TF7, TF1),<br>(TF0, TF0, TF0, TF8, TF1)) |

For better understanding of the TFCS please note that the following combinations are not included in the table above:

- (TF2, TF1, TF1, TF5, TF0), (TF1, TF0, TF0, TF5, TF0), (TF1, TF0, TF0, TF5, TF1), (TF2, TF1, TF1, TF5, TF1), (TF2, TF1, TF1, TF8, TF0), (TF1, TF0, TF0, TF8, TF0), (TF1, TF0, TF0, TF8, TF1), (TF2, TF1, TF1, TF8, TF1).

6.11.5.4.1.44.2.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Downlink | Modulation | 8PSK |
| | Codes and time slots / radio frame | SF 1 x 1 code x 10 time slots |
| | Max. Number of data bits/radio frame | 21 060 bits |
| | TFCI code word / radio frame | 48 bits |
| | TPC / radio frame | 2x3 bits |
| | SS / radio frame | 2x3 bits |
| | Puncturing Limit | 1 |

6.11.5.4.1.45 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Streaming / unknown / UL:57.6 DL:57.6 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.5.4.1.45.1 Uplink

6.11.5.4.1.45.1.1 Transport channel parameters

6.11.5.4.1.45.1.1.1 Transport channel parameters for Conversational / speech / UL:12.2 kbps / CS RAB

See clause 6.10.3.4.1.4.1.1.1.

6.11.5.4.1.45.1.1.2 Transport channel parameters for Streaming / unknown / UL:57.6 kbps / CS RAB

See clause 6.10.3.4.1.17.1.1.1.

6.11.5.4.1.45.1.1.3 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.11.5.4.1.45.1.1.4 TFCS

See clause 6.10.3.4.1.45.1.1.4.

6.11.5.4.1.45.1.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Uplink | Modulation | QPSK |
| | Codes and time slots / radio frame | SF2 x 1 code x 2 time slots |
| | Max. Number of data bits/radio frame | 1 384 bits |
| | TFCI code word / radio frame | 16 bits |
| | TPC / radio frame | 2x2 bits |
| | SS / radio frame | 2x2 bits |
| | Puncturing Limit | 0.52 |

6.11.5.4.1.45.2 Downlink

6.11.5.4.1.45.2.1 Transport channel parameters

6.11.5.4.1.45.2.1.1 Transport channel parameters for Conversational / speech / DL:12.2 kbps / CS RAB

See clause 6.10.3.4.1.4.2.1.1.

6.11.5.4.1.45.2.1.2 Transport channel parameters for Streaming / unknown / DL:57.6 kbps / CS RAB

See clause 6.10.3.4.1.17.2.1.1.

6.11.5.4.1.45.2.1.3 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.11.5.4.1.45.2.1.4 TFCS

See clause 6.10.3.4.1.45.2.1.4.

6.11.5.4.1.45.2.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Downlink | Modulation | QPSK |
| | Codes and time slots / radio frame | SF 16 x 9 codes x 2 time slots |
| | Max. Number of data bits/radio frame | 1 560 bits |
| | TFCI code word / radio frame | 16 bits |
| | TPC / radio frame | 2x2 bits |
| | SS / radio frame | 2x2 bits |
| | Puncturing Limit | 0.56 |

6.11.5.4.1.46 Void

6.11.5.4.1.47 Void

6.11.5.4.1.48 Void

6.11.5.4.1.49 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB + Conversational / unknown / UL:64 DL:64 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.5.4.1.49.1 Uplink

6.11.5.4.1.49.1.1 Transport channel parameters

6.11.5.4.1.49.1.1.1 Transport channel parameters for Conversational / speech / UL:12.2 kbps / CS RAB

See clause 6.10.3.4.1.4.1.1.1.

6.11.5.4.1.49.1.1.2 Transport channel parameters for Conversational / unknown / UL:64 kbps / CS RAB

See clause 6.10.3.4.1.13.1.1.1.

6.11.5.4.1.49.1.1.3 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.11.5.4.1.49.1.1.4 TFCS

See clause 6.10.3.4.1.49.1.1.4.

6.11.5.4.1.49.1.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Uplink | Modulation | QPSK |
| | Codes and time slots / radio frame | SF 1 x 1 code x 2 time slots |
| | Max. Number of data bits/radio frame | 2 792 bits |
| | TFCI code word / radio frame | 16 bits |
| | TPC / radio frame | 2x2 bits |
| | SS / radio frame | 2x2 bits |
| | Puncturing Limit | 0.88 |

6.11.5.4.1.49.2 Downlink

6.11.5.4.1.49.2.1 Transport channel parameters

6.11.5.4.1.49.2.1.1 Transport channel parameters for Conversational / speech / DL:12.2 kbps / CS RAB

See clause 6.10.3.4.1.4.2.1.1.

6.11.5.4.1.49.2.1.2 Transport channel parameters for Conversational / unknown / DL:64 kbps / CS RAB

See clause 6.10.3.4.1.13.2.1.1.

6.11.5.4.1.49.2.1.3 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.11.5.4.1.49.2.1.4 TFCS

See clause 6.10.3.4.1.49.2.1.4.

6.11.5.4.1.49.2.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Downlink | Modulation | QPSK |
| | Codes and time slots / radio frame | SF16 x 11 codes x 2 time slots |
| | Max. Number of data bits/radio frame | 1 912 bits |
| | TFCI code word / radio frame | 16 bits |
| | TPC / radio frame | 2x2 bits |
| | SS / radio frame | 2x2 bits |
| | Puncturing Limit | 0.60 |

6.11.5.4.1.49a Conversational / speech / UL: 12.2 7.95 5.9 4.75 DL: 12.2 7.95 5.9 4.75 kbps / CS RAB + Conversational / unknown / UL:64 DL:64 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.5.4.1.49a.1 Uplink

6.11.5.4.1.49a.1.1 Transport channel parameters

6.11.5.4.1.49a.1.1.1 Transport channel parameters for Conversational / speech / UL: 12.2 7.95 5.9 4.75 kbps / CS RAB

See clause 6.10.3.4.1.4a.1.1.1.

6.11.5.4.1.49a.1.1.2 Transport channel parameters for Conversational / unknown / UL:64 kbps / CS RAB

See clause 6.10.3.4.1.13.1.1.1.

6.11.5.4.1.49a.1.1.3 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.11.5.4.1.49a.1.1.4 TFCS

See clause 6.10.3.4.1.49a.1.1.4.

6.11.5.4.1.49a.1.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Uplink | Modulation | QPSK |
| | Codes and time slots / radio frame | SF 1 x 1 code x 2 time slots |
| | Max. Number of data bits/radio frame | 2 792 bits |
| | TFCI code word / radio frame | 16 bits |
| | TPC / radio frame | 2x2 bits |
| | SS / radio frame | 2x2 bits |
| | Puncturing Limit | 0.88 |

6.11.5.4.1.49a.2 Downlink

6.11.5.4.1.49a.2.1 Transport channel parameters

6.11.5.4.1.49a.2.1.1 Transport channel parameters for Conversational / speech / DL: 12.2 7.95 5.9 4.75 kbps / CS RAB

See clause 6.10.3.4.1.4a.2.1.1.

6.11.5.4.1.49a.2.1.2 Transport channel parameters for Conversational / unknown / DL:64 kbps / CS RAB

See clause 6.10.3.4.1.13.2.1.1.

6.11.5.4.1.49a.2.1.3 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.11.5.4.1.49.2.1.4 TFCS

See clause 6.10.3.4.1.49a.2.1.4.

6.11.5.4.1.49a.2.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Downlink | Modulation | QPSK |
| | Codes and time slots / radio frame | SF16 x 11 codes x 2 time slots |
| | Max. Number of data bits/radio frame | 1 912 bits |
| | TFCI code word / radio frame | 16 bits |
| | TPC / radio frame | 2x2 bits |
| | SS / radio frame | 2x2 bits |
| | Puncturing Limit | 0.60 |

6.11.5.4.1.50 Conversational / unknown / UL:64 DL:64 kbps / CS RAB + Conversational / unknown / UL:64 DL:64 kbps / CS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.5.4.1.50.1 Uplink

6.11.5.4.1.50.1.1 Transport channel parameters

6.11.5.4.1.50.1.1.1 Transport channel parameters for Conversational / unknown / UL:64 kbps / CS RAB

See clause 6.10.3.4.1.13.1.1.1.

6.11.5.4.1.50.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.11.5.4.1.50.1.1.3 TFCS

See clause 6.10.3.4.1.50.1.1.3.

6.11.5.4.1.50.1.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Uplink | Modulation | QPSK |
| | Codes and time slots / radio frame | SF 1 x 1 code x 2 time slots |
| | Max. Number of data bits/radio frame | 2 792 bits |
| | TFCI code word / radio frame | 16 bits |
| | TPC / radio frame | 2x2 bits |
| | SS / radio frame | 2x2 bits |
| | Puncturing Limit | 0.52 |

6.11.5.4.1.50.2 Downlink

6.11.5.4.1.50.2.1 Transport channel parameters

6.11.5.4.1.50.2.1.1 Transport channel parameters for Conversational / unknown / DL:64 kbps / CS RAB

See clause 6.10.3.4.1.13.2.1.1.

6.11.5.4.1.50.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.11.5.4.1.50.2.1.3 TFCS

See clause 6.10.3.4.1.50.2.1.3.

6.11.5.4.1.50.2.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Downlink | Modulation | QPSK |
| | Codes and time slots / radio frame | SF 16 x 15 codes x 2 time slots |
| | Max. Number of data bits/radio frame | 2 616 bits |
| | TFCI code word / radio frame | 16 bits |
| | TPC / radio frame | 2x2 bits |
| | SS / radio frame | 2x2 bits |
| | Puncturing Limit | 0.48 |

6.11.5.4.1.51 Conversational / unknown / UL:64 DL:64 kbps / CS RAB + Interactive or background / UL:64 DL:64 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.5.4.1.51.1 Uplink

6.11.5.4.1.51.1.1 Transport channel parameters

6.11.5.4.1.51.1.1.1 Transport channel parameters for Conversational / unknown / UL:64 kbps / CS RAB

See clause 6.10.3.4.1.13.1.1.1.

6.11.5.4.1.51.1.1.2 Transport channel parameters for Interactive or background / UL:64 kbps / PS RAB

See clause 6.10.3.4.1.26.1.1.1.

6.11.5.4.1.51.1.1.3 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.11.5.4.1.51.1.1.4 TFCS

See clause 6.10.3.4.1.51.1.1.4.

6.11.5.4.1.51.1.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Uplink | Modulation | QPSK |
| | Codes and time slots / radio frame | SF 1 x 1 code x 2 time slots |
| | Max. Number of data bits/radio frame | 2 792 bits |
| | TFCI code word / radio frame | 16 bits |
| | TPC / radio frame | 2x2 bits |
| | SS / radio frame | 2x2 bits |
| | Puncturing Limit | 0.52 (alt. 0.48) |

6.11.5.4.1.51.2 Downlink

6.11.5.4.1.51.2.1 Transport channel parameters

6.11.5.4.1.51.2.1.1 Transport channel parameters for Conversational / unknown / DL:64 kbps / CS RAB

See clause 6.10.3.4.1.13.2.1.1.

6.11.5.4.1.51.2.1.2 Transport channel parameters for Interactive or background / DL:64 kbps / PS RAB

See clause 6.10.3.4.1.25.2.1.1.

6.11.5.4.1.51.2.1.3 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.11.5.4.1.51.2.1.4 TFCS

See clause 6.10.3.4.1.51.2.1.4.

6.11.5.4.1.51.2.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Downlink | Modulation | QPSK |
| | Codes and time slots / radio frame | SF1 x 1 code x 2 time slots |
| | Max. Number of data bits/radio frame | 2 792 bits |
| | TFCI code word / radio frame | 16 bits |
| | TPC / radio frame | 2x2 bits |
| | SS / radio frame | 2x2 bits |
| | Puncturing Limit | 0.52 |

6.11.5.4.1.51a Conversational / unknown / UL:64 DL:64 kbps / CS RAB + Interactive or background / UL:8 DL:8 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.5.4.1.51a.1 Uplink

6.11.5.4.1.51a.1.1 Transport channel parameters

6.11.5.4.1.51a.1.1.1 Transport channel parameters for Conversational / unknown / UL:64 kbps / CS RAB

See clause 6.10.3.4.1.13.1.1.1.

6.11.5.4.1.51a.1.1.2 Transport channel parameters for Interactive or background / UL:8 kbps / PS RAB

See clause 6.10.3.4.1.23a.1.1.1.

6.11.5.4.1.51a.1.1.3 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.11.5.4.1.51a.1.1.4 TFCS

See clause 6.10.3.4.1.51a.1.1.4.

6.11.5.4.1.51a.1.2 Physical channel parameters

| | | | |
|---|---|---|---|
| DPCH Uplink | | Physical 1 | Physical 2 |
| | Modulation | QPSK | QPSK |
| | Codes and time slots / radio frame | SF2 x 1 code x 2 time slots | SF1 x 1 code x 2 time slots |
| | Max. Number of data bits/radio frame | 1 384 bits | 2 792 bits |
| | TFCI code word / radio frame | 16 bits | 16 bits |
| | TPC / radio frame | 2x2 bits | 2x2 bits |
| | SS / radio frame | 2x2 bits | 2x2 bits |
| | Puncturing Limit | 0.40 | 0.84 |

6.11.5.4.1.51a.2 Downlink

6.11.5.4.1.51a.2.1 Transport channel parameters

6.11.5.4.1.51a.2.1.1 Transport channel parameters for Conversational / unknown / DL:64 kbps / CS RAB

See clause 6.10.3.4.1.13.2.1.1.

6.11.5.4.1.51a.2.1.2 Transport channel parameters for Interactive or background / DL:8 kbps / PS RAB

See clause 6.10.3.4.1.23.2.1.1.

6.11.5.4.1.51a.2.1.3 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.11.5.4.1.51a.2.1.4 TFCS

See clause 6.10.3.4.1.51.2.1.4.

6.11.5.4.1.51a.2.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Downlink | Modulation | QPSK |
| | Codes and time slots / radio frame | SF1 x 1 code x 2 time slots |
| | Max. Number of data bits/radio frame | 2 792 bits |
| | TFCI code word / radio frame | 16 bits |
| | TPC / radio frame | 2x2 bits |
| | SS / radio frame | 2x2 bits |
| | Puncturing Limit | 0.84 |

6.11.5.4.1.51b Conversational / unknown / UL:64 DL:64 kbps / CS RAB + Interactive or background / UL:16 DL:64 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.5.4.1.51b.1 Uplink

6.11.5.4.1.51b.1.1 Transport channel parameters

6.11.5.4.1.51b.1.1.1 Transport channel parameters for Conversational / unknown / UL:64 kbps / CS RAB

See clause 6.10.3.4.1.13.1.1.1.

6.11.5.4.1.51b.1.1.2 Transport channel parameters for Interactive or background / UL:16 kbps / PS RAB

See clause 6.10.3.4.1.23b.1.1.1.

6.11.5.4.1.51b.1.1.3 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.11.5.4.1.51b.1.1.4 TFCS

See clause 6.10.3.4.1.51b.1.1.4.

6.11.5.4.1.51b.1.2 Physical channel parameters

| DPCH Uplink | | Physical 1 | Physical 2 |
|---|---|---|---|
| | Modulation | QPSK | QPSK |
| | Codes and time slots / radio frame | SF2 x 1 code x 2 time slots | SF1 x 1 code x 2 time slots |
| | Max. Number of data bits/radio frame | 1 384 bits | 2 792 bits |
| | TFCI code word / radio frame | 16 bits | 16 bits |
| | TPC / radio frame | 2x2 bits | 2x2 bits |
| | SS / radio frame | 2x2 bits | 2x2 bits |
| | Puncturing Limit | 0.40 | 0.76 |

6.11.5.4.1.51b.2 Downlink

See clause 6.11.5.4.1.51.2.

6.11.5.4.1.52 Conversational / unknown / UL:64 DL:64 kbps / CS RAB + Interactive or background / UL:64 DL:128 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.5.4.1.52.1 Uplink

See clause 6.11.5.4.1.51.1.

6.11.5.4.1.52.2 Downlink

6.11.5.4.1.52.2.1 Transport channel parameters

6.11.5.4.1.52.2.1.1 Transport channel parameters for Conversational / unknown / DL:64 kbps / CS RAB

See clause 6.10.3.4.1.13.2.1.1.

6.11.5.4.1.52.2.1.2 Transport channel parameters for Interactive or background / DL:128 kbps / PS RAB

See clause 6.10.3.4.1.27.2.1.1.

6.11.5.4.1.52.2.1.3 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.11.5.4.1.52.2.1.4 TFCS

See clause 6.10.3.4.1.52.2.1.4.

6.11.5.4.1.52.2.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Downlink | Modulation | QPSK |
| | Codes and time slots / radio frame | SF 16 x 12 codes x 4 time slots |
| | Max. Number of data bits/radio frame | 4 200 bits |
| | TFCI code word / radio frame | 16 bits |
| | TPC / radio frame | 2x2 bits |
| | SS / radio frame | 2x2 bits |
| | Puncturing Limit | 0.52 |

6.11.5.4.1.53 Conversational / unknown / UL:64 DL:64 kbps / CS RAB + Interactive or background / UL:128 DL:128 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.5.4.1.53.1 Uplink

6.11.5.4.1.53.1.1 Transport channel parameters

6.11.5.4.1.53.1.1.1 Transport channel parameters for Conversational / unknown / UL:64 kbps / CS RAB

See clause 6.10.3.4.1.13.1.1.1.

6.11.5.4.1.53.1.1.2 Transport channel parameters for Interactive or background / UL:128 kbps / PS RAB

See clause 6.10.3.4.1.28.1.1.1.

6.11.5.4.1.53.1.1.3 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.11.5.4.1.53.1.1.4 TFCS

See clause 6.10.3.4.1.53.1.1.4.

6.11.5.4.1.53.1.2 Physical channel parameters

| | | | |
|---|---|---|---|
| DPCH Uplink | Modulation | QPSK | 8PSK |
| | Codes and time slots / radio frame | SF1 x 1 code x 4 time slots | SF1 x 1code x 2 time slots |
| | Max. Number of data bits/radio frame | 5 608 bits | 4 188 bits |
| | TFCI code word / radio frame | 16 bits | 24 bits |
| | TPC / radio frame | 2x2 bits | 2x3 bits |
| | SS / radio frame | 2x2 bits | 2x3 bits |
| | Puncturing Limit | 0.72 (alt 0.68) | 0.52 (alt 0.48) |

6.11.5.4.1.53.2 Downlink

See clause 6.11.5.4.1.52.2.

6.11.5.4.1.54 Void

6.11.5.4.1.55 Void

6.11.5.4.1.56 Interactive or background / UL:8 DL:8 kbps / PS RAB + Interactive or background / UL:8 DL:8 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.5.4.1.56.1 Uplink

6.11.5.4.1.56.1.1 Transport channel parameters

6.11.5.4.1.56.1.1.1 Transport channel parameters for Interactive or background / UL:8 kbps / PS RAB + UL:8 kbps / PS RAB

See clause 6.10.3.4.1.56.1.1.1.

6.11.5.4.1.56.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.11.5.4.1.56.1.1.3 TFCS

See clause 6.10.3.4.1.56.1.1.3.

6.11.5.4.1.56.1.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Uplink | Modulation | QPSK |
| | Codes and time slots / radio frame | SF4 x 1 code x 2 time slots |
| | Max. Number of data bits / radio frame | 680 bits |
| | TFCI code word / radio frame | 16 bits |
| | TPC / radio frame | 2x2 bits |
| | SS / radio frame | 2x2 bits |
| | Puncturing Limit | 0.84 (alt 0.76) |

6.11.5.4.1.56.2 Downlink

6.11.5.4.1.56.2.1 Transport channel parameters

6.11.5.4.1.56.2.1.1 Transport channel parameters for Interactive or background / DL:8 kbps / PS RAB + DL:8 kbps / PS RAB

See clause 6.10.3.4.1.56.2.1.1.

6.11.5.4.1.56.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.11.5.4.1.56.2.1.3 TFCS

See clause 6.10.3.4.1.56.2.1.3.

6.11.5.4.1.56.2.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Downlink | Modulation | QPSK |
| | Codes and time slots / radio frame | SF16 x 4 code x 2 time slots |
| | Max. Number of data bits / radio frame | 680 bits |
| | TFCI code word / radio frame | 16 bits |
| | TPC / radio frame | 2x2 bits |
| | SS / radio frame | 2x2 bits |
| | Puncturing Limit | 0.84 |

6.11.5.4.1.57 Interactive or background / UL:64 DL:64 kbps / PS RAB + Interactive or background / UL:64 DL:64 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.5.4.1.57.1 Uplink

6.11.5.4.1.57.1.1 Transport channel parameters

6.11.5.4.1.57.1.1.1 Transport channel parameters for Interactive or background / UL:64 kbps / PS RAB + UL:64 kbps / PS RAB

See clause 6.10.3.4.1.38d.1.1.2.

6.11.5.4.1.57.1.1.2 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.11.5.4.1.57.1.1.3 TFCS

See clause 6.11.5.4.1.57.1.1.3.

6.11.5.4.1.57.1.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Uplink | | Physical 1 |
| | Modulation | QPSK |
| | Codes and time slots / radio frame | SF1 x 1 code x 2 time slots |

| | | |
|---|---|---|
| | Max. Number of data bits/radio frame | 2 792 bits |
| | TFCI code word / radio frame | 16 bits |
| | TPC / radio frame | 2x2 bits |
| | SS / radio frame | 2x2 bits |
| | Puncturing Limit | 0.52 (alt. 0.44) |

6.11.5.4.1.57.2 Downlink

6.11.5.4.1.57.2.1 Transport channel parameters

6.11.5.4.1.57.2.1.1 Transport channel parameters for Interactive or background / DL:64 kbps / PS RAB + DL:64 kbps / PS RAB

See clause 6.10.3.4.1.57.2.1.1.

6.11.5.4.1.57.2.1.2 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.11.5.4.1.57.2.1.3 TFCS

See clause 6.10.3.4.1.57.2.1.3.

6.11.5.4.1.57.2.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Downlink | Modulation | QPSK |
| | Codes and time slots / radio frame | SF1 x 1 code x 2 time slots |
| | Max. Number of data bits/radio frame | 2 792 bits |
| | TFCI code word / radio frame | 16 bits |
| | TPC / radio frame | 2x2 bits |
| | SS / radio frame | 2x2 bits |
| | Puncturing Limit | 0.52 |

6.11.5.4.1.58 Streaming / unknown / UL:16 DL:64 kbps / PS RAB + Interactive or background / UL:8 DL:8 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.5.4.1.58.1 Uplink

6.11.5.4.1.58.1.1 Transport channel parameters

6.11.5.4.1.58.1.1.1 Transport channel parameters for Streaming / unknown / UL:16 kbps / PS RAB

See clause 6.10.3.4.1.58.1.1.1.

6.11.5.4.1.58.1.1.2 Transport channel parameters for Interactive or background / UL:8 kbps / PS RAB

See clause 6.10.3.4.1.23a.1.1.1.

6.11.5.4.1.58.1.1.3 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.11.5.4.1.58.1.1.4 TFCS

See clause 6.10.3.4.1.58.1.1.4.

6.11.5.4.1.58.1.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Uplink | Modulation | QPSK |
| | Codes and time slots / frame | SF4 x 1 code x 2 time slots |
| | Max. Number of data bits / radio frame | 680 bits |
| | TFCI code word / radio frame | 16 bits |
| | TPC / radio frame | 2x2 bits |
| | SS / radio frame | 2x2 bits |
| | Puncturing Limit | 0.60 (alt 0.56) |

6.11.5.4.1.58.2 Downlink

6.11.5.4.1.58.2.1 Transport channel parameters

6.11.5.4.1.58.2.1.1 Transport channel parameters for Streaming / unknown / DL:64 kbps / PS RAB

See clause 6.10.3.4.1.58.2.1.1.

6.10.5.4.1.58.2.1.2 Transport channel parameters for Interactive or background / DL:8 kbps / PS RAB

See clause 6.10.3.4.1.23.2.1.1.

6.11.5.4.1.58.2.1.3 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.11.5.4.1.58.2.1.4 TFCS

See clause 6.10.3.4.1.58.2.1.4.

6.11.5.4.1.58.2.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Downlink | Modulation | QPSK |
| | Codes and time slots / radio frame | SF16 x 8 code x 2 time slots |
| | Max. Number of data bits / radio frame | 1 384 bits |
| | TFCI code word / radio frame | 16 bits |
| | TPC / radio frame | 2x2 bits |
| | SS / radio frame | 2x2 bits |
| | Puncturing Limit | 0.44 |

6.11.5.4.1.59 Reserved for future use

6.11.5.4.1.60 Reserved for future use

6.11.5.4.1.61 Conversational / unknown / UL:8 DL:8 kbps / PS RAB + Interactive or Background / UL:8 DL:8 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH

6.11.5.4.1.61.1 Uplink

6.11.5.4.1.61.1.1 Transport channel parameters

6.11.5.4.1.61.1.1.1 Transport channel parameters for Conversational / unknown / UL:8 kbps / PS RAB

See clause 6.10.3.4.1.61.1.1.1.

6.10.5.4.1.61.1.1.2 Transport channel parameters for Interactive or Background / UL:8 kbps / PS RAB

See clause 6.10.3.4.1.23a.1.1.1.

6.11.5.4.1.61.1.1.3 Transport channel parameters for UL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.11.5.4.1.61.1.1.4 TFCS

See clause 6.10.3.4.1.61.1.1.4.

6.11.5.4.1.61.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Uplink | Modulation | QPSK |
| | Codes and time slots / radio frame | SF 4 x 1 code x 2 time slots |
| | Max. Number of data bits / radio frame | 680 bits |
| | TFCI code word / radio frame | 16 bits |
| | TPC / radio frame | 2x2 bits |
| | SS / radio frame | 2x2 bits |
| | Puncturing Limit | 0.84 (alt 0.80) |

6.11.5.4.1.61.2 Downlink

6.11.5.4.1.61.2.1 Transport channel parameters

6.11.5.4.1.61.2.1.1 Transport channel parameters for Conversational / unknown / DL:8 kbps / PS RAB

See clause 6.10.3.4.1.61.2.1.1.

6.11.5.4.1.61.2.1.2 Transport channel parameters for Interactive or Background / DL:8 kbps / PS RAB

See clause 6.10.3.4.1.23.2.1.1.

6.11.5.4.1.61.2.1.3 Transport channel parameters for DL:3.4 kbps SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.11.5.4.1.61.2.1.4 TFCS

See clause 6.10.3.4.1.61.2.1.4.

6.11.5.4.1.61.2.2 Physical channel parameters

| | | |
|---|---|---|
| DPCH Downlink | Modulation | QPSK |
| | Codes and time slots / radio frame | SF 16 x 4 code x 2 time slots |
| | Max. Number of data bits / radio frame | 680 bits |
| | TFCI code word / radio frame | 16 bits |
| | TPC / radio frame | 2x2 bits |
| | SS / radio frame | 2x2 bits |
| | Puncturing Limit | 0.84 |

6.11.5.4.2 Combinations on PDSCH, SCCPCH, PUSCH and PRACH

6.11.5.4.2.1 Interactive or background / UL: 64 DL: 256 kbps / PS RAB + UL: 3.4/16.8 DL: 3.4/33.6 kbps SRBs for DCCH, CCCH and BCCH + UL: 16.8 DL: 16 kbps SRBs for SHCCH

6.11.5.4.2.1.1 Uplink

6.11.5.4.2.1.1.1 Transport channel parameters

6.11.5.4.2.1.1.1.1 Transport channel parameters for Interactive or background / UL: 64 kbps / PS RAB and UL SRB for SHCCH mapped on USCH

See clause 6.10.3.4.2.1.1.1.1.

6.11.5.4.2.1.1.1.2 Transport channel parameters for UL: 3.4 Kbps SRBs for DCCH mapped on USCH

See clause 6.10.3.4.2.1.1.1.2.

6.11.5.4.2.1.1.1.3 TFCS for USCH

See clause 6.10.3.4.2.1.1.1.3.

6.11.5.4.2.1.1.1.4 Transport channel parameters for SRB for CCCH and UL SRBs for DCCH and UL SRB for SHCCH mapped on RACH

See clause 6.10.3.4.2.1.1.1.4.

6.11.5.4.2.1.1.2 Physical channel parameters

6.11.5.4.2.1.1.2.1 Physical channel parameters for PUSCH

| | | |
|---|---|---|
| PUSCH | Modulation | QPSK |
| | Codes and time slots / radio frame | SF 1 x 1 code x 2 time slots |
| | Max. Number of data bits/radio frame | 2 792 bits |
| | TFCI code word / radio frame | 16 bits |
| | TPC / radio frame | 2x2 bits |

| | | |
|---|---|---|
| | SS / radio frame | 2x2 bits |
| | Puncturing Limit | 0.88 |

6.11.5.4.2.1.1.2.2 Physical channel parameter for PRACH.

See clause 6.11.5.4.5.1.2.

6.11.5.4.2.1.2 Downlink

6.11.5.4.2.1.2.1 Transport channel parameters

6.11.5.4.2.1.2.1.1 Transport channel parameters for Interactive or background / DL: 256 kbps / PS RAB and DL SRB for SHCCH mapped on DSCH

See clause 6.10.3.4.2.1.2.1.1.

6.11.5.4.2.1.2.1.2 Transport channel parameters for DL: 3.4 Kbps SRBs for DCCH mapped on DSCH

See clause 6.10.3.4.2.1.2.1.2.

6.11.5.4.2.1.2.1.3 TFCS for DSCH

See clause 6.10.3.4.2.1.2.1.3.

6.11.5.4.2.1.2.1.4 Transport channel parameters for DL SRBs for DCCH and SRB for CCCH and SRB for BCCH and DL SRB for SHCCH mapped on FACH

6.11.5.4.2.1.2.1.4.1 FACH transport channel configuration without DTCH

| Higher layer | RAB/signalling RB | | SRB#0 | SRB#1 | SRB#2 | SRB#3 | SRB#4 | SRB#5 | SRB#6 |
|---|---|---|---|---|---|---|---|---|---|
| | User of Radio Bearer | | RRC | RRC | RRC | NAS_DT High prio | NAS_DT Low prio | RRC | RRC |
| RLC | Logical channel type | | CCCH | DCCH | DCCH | DCCH | DCCH | SHCCH | BCCH |
| | RLC mode | | UM | UM | AM | AM | AM | UM | TM |
| | Payload sizes, bit | | 160 | 136 or 120* | 128 | 128 | 128 | 160 | 168 |
| | Max data rate, bps | | 32 000 (alt. 16 000) | 27 200 or 24 000 (alt. 13 600 or 12 000) | 25 600 (alt. 12 800) | 25 600 (alt. 12 800) | 25 600 (alt. 12 800) | 32 000 (alt. 16 000) | 33 600 (alt. 16 000) |
| | RLC header, bit | | 8 | 8 | 16 | 16 | 16 | 8 | 0 |
| MAC | MAC header, bit | | 3 | 27 or 43 | 27 | 27 | 27 | 3 | 3 |
| | MAC multiplexing | | 7 logical channel multiplexing | | | | | | |
| Layer 1 | TrCH type | | FACH | | | | | | |
| | TB sizes, bit | | 171 | 171 | 171 | 171 | 171 | 171 | 171 |
| | TFS | TF0, bits | 0x171 | | | | | | |
| | | TF1, bits | 1x171 | | | | | | |
| | | TF2, bits | 2x171 | | | | | | |
| | | TF3, bits | 3x171( alt. N/A) | | | | | | |
| | | TF4, bits | 4x171( alt. N/A) | | | | | | |
| | | TF5, bits | | | | | | | |
| | | TF6, bits | | | | | | | |
| | TTI, ms | | 20 | | | | | | |
| | Coding type | | CC 1/2 | | | | | | |
| | CRC, bit | | 16 | | | | | | |
| | Max number of bits/TTI after channel coding | | 1 528 (alt. 764) | 1 528 (alt. 764) | 1 528 (alt. 764) | 1 528 (alt. 764) | 1 528 (alt. 764) | 1 528 (alt. 764) | 1 528 (alt. 764) |
| NOTE: MAC header size and RLC payload size depend on use of U-RNTI or C-RNTI. | | | | | | | | | |

6.11.5.4.2.1.2.1.4.2 FACH transport channel configuration with DTCH

| Higher layer | RAB/signalling RB | | RAB | SRB#0 | SRB#1 | SRB#2 | SRB#3 | SRB#4 | SRB#5 | SRB#6 |
|---|---|---|---|---|---|---|---|---|---|---|
| | User of Radio Bearer | | Interactive/ Background RAB | RRC | RRC | RRC | NAS_DT High prio | NAS_DT Low prio | RRC | RRC |
| RLC | Logical channel type | | DTCH | CCCH | DCCH | DCCH | DCCH | DCCH | SHCCH | BCCH |
| | RLC mode | | AM | UM | UM | AM | AM | AM | UM | TM |
| | Payload sizes, bit | | 320 | 160 | 136 or 120 (note) | 128 | 128 | 128 | 160 | 168 |
| | Max data rate, bps | | 32 000 (alt. 16 000) | 32 000 (alt. 16 000) | 27 200 or 24 000 (alt. 13 600 or 12 000) | 25 600 (alt. 12 800) | 25 600 (alt. 12 800) | 25 600 (alt. 12 800) | 32 000 (alt. 16 000) | 33 600 (alt. 16 800) |
| | AMD/UMD/TrD PDU header, bit | | 16 | 8 | 8 | 16 | 16 | 16 | 8 | 0 |
| MAC | MAC header, bit | | 27 | 3 | 27 or 43 | 27 | 27 | 27 | 3 | 3 |
| | MAC multiplexing | | 8 logical channel multiplexing | | | | | | | |
| Layer 1 | TrCH type | | FACH | | | | | | | |
| | TB sizes, bit | | 171, 363 | | | | | | | |
| | TFS | TF0, bits | 0x171 | | | | | | | |
| | | TF1, bits | 1x171 | | | | | | | |
| | | TF2, bits | 2x171 | | | | | | | |
| | | TF3, bits | 1x363 | | | | | | | |
| | | TF4, bits | 3x171 (alt N/A) | | | | | | | |
| | | TF5, bits | 4x171 (alt. N/A) | | | | | | | |
| | | TF6, bits | 2x363 (alt. N/A) | | | | | | | |
| | TTI, ms | | 20 | | | | | | | |
| | Coding type | | CC ½ | | | | | | | |
| | CRC, bit | | 16 | | | | | | | |
| | Max number of bits/TTI after channel coding | | 1 532 (alt. 766) | | | | | | | |
| NOTE: MAC header size and RLC payload size depend on use of U-RNTI or C-RNTI. | | | | | | | | | | |

6.11.5.4.2.1.2.1.5 TFCS for FACH

See clause 6.10.3.4.2.1.2.1.5.

6.11.5.4.2.1.2.2 Physical channel parameters

6.11.5.4.2.1.2.2.1 Physical channel parameters for PDSCH

| PDSCH | Modulation | QPSK | 8PSK |
|---|---|---|---|
| | Codes and time slots / radio frame | SF16 x 11 codes x 6 time slots | SF1 x 1 code x 4 time slots |
| | Max. Number of data bits/radio frame | 5 784 bits | 6 511 bits |
| | TFCI code word / radio frame | 16 bits | 24 bits |
| | TPC / radio frame | 2x2 bits | 2x3 bits |
| | SS / radio frame | 2x2 bits | 2x3 bits |
| | Puncturing Limit | 0.60 | 0.68 |

6.11.5.4.2.1.2.2.2 Physical channel parameters for SCCPCH

6.11.5.4.2.1.2.2.2.1 Physical channel parameters for SCCPCH without DTCH

| S-CCPCH | Modulation | QPSK |
|---|---|---|
| | Codes and time slots / radio frame | SF16 x 5 codes x 2 time slots (alt. SF16 x 2 codes x 2 time slot) |
| | Max. Number of data bits/radio frame | 864 bits (alt. 344 bits) |
| | TFCI code word / radio frame | 16 bits |
| | TP(alt. 8 bits)C/ radio frame | 0 bits |
| | SS / radio frame | 0 bits |

| | | |
|---|---|---|
| | Puncturing Limit | 1 (alt. 0.88) |

6.11.5.4.2.1.2.2.2.2 Physical channel parameters for SCCPCH with DTCH

| | | |
|---|---|---|
| S-CCPCH | Modulation | QPSK |
| | Codes and time slots / radio frame | SF16 x 5 codes x 2 time slots (alt. SF16 x 2 codes x 2 time slot) |
| | Max. Number of data bits/radio frame | 864 bits (alt. 336 bits) |
| | TFCI code word / radio frame | 16 bits |
| | TPC / radio frame | 0 bits |
| | SS / radio frame | 0 bits |
| | Puncturing Limit | 1 (alt. 0.84) |

6.11.5.4.2.2 Interactive or background / UL: 64 DL: 384 kbps / PS RAB + UL: 3.4/16.8 DL: 3.4/33.6 kbps SRBs for DCCH, CCCH and BCCH+ UL: 16.8 DL: 16 kbps SRBs for SHCCH

6.11.5.4.2.2.1 Uplink

See clause 6.11.5.4.2.1.1.

6.11.5.4.2.2.2 Downlink

6.11.5.4.2.2.2.1 Transport channel parameters

6.11.5.4.2.2.2.1.1 Transport channel parameters for Interactive or background / DL: 384 kbps / PS RAB and DL SRB for SHCCH mapped on DSCH

See clause 6.10.3.4.2.2.2.1.1.

6.11.5.4.2.2.2.1.2 Transport channel parameters for DL: 3.4 Kbps SRBs for DCCH mapped on DSCH

See clause 6.10.3.4.2.1.2.1.2.

6.11.5.4.2.2.2.1.3 TFCS for DSCH

See clause 6.10.3.4.2.2.2.1.3.

6.11.5.4.2.2.2.1.4 Transport channel parameters for DL SRBs for DCCH and SRB for CCCH and SRB for BCCH and DL SRB for SHCCH mapped on FACH

See clause 6.11.5.4.2.1.2.1.4.

6.11.5.4.2.2.2.1.5 TFCS for FACH

See clause 6.11.5.4.2.1.2.1.5.

6.11.5.4.2.2.2.2 Physical channel parameters

6.11.5.4.2.2.2.2.1 Physical channel parameters for PDSCH

| | | |
|---|---|---|
| PDSCH | Modulation | QPSK |
| | Codes and time slots / radio frame | SF 1 x 1 code x 6 time slots |
| | Max. Number of data bits/radio frame | 8 424 bits |
| | TFCI code word / radio frame | 16 bits |
| | TPC / radio frame | 2x2 bits |
| | SS / radio frame | 2x2 bits |
| | Puncturing Limit | 0.60 |

6.11.5.4.2.2.2.2.2 Physical channel parameters for SCCPCH

See clause 6.11.5.4.2.1.2.2.2.

6.11.5.4.2.3 Interactive or background / UL: 64 DL: 2 048 kbps / PS RAB + UL: 3.4/16.8 DL: 3.4/33.6 kbps SRBs for DCCH, CCCH and BCCH + UL: 16.8 DL: 16 kbps SRBs for SHCCH

6.11.5.4.2.3.1 Uplink

See clause 6.11.5.4.2.1.1.

6.11.5.4.2.3.2 Downlink

6.11.5.4.2.3.2.1 Transport channel parameters

6.11.5.4.2.3.2.1.1 Transport channel parameters for Interactive or background / DL: 2 048 kbps / PS RAB and DL SRB for SHCCH mapped on DSCH

| Higher layer | RAB/Signalling RB | | RAB | SRB#5 |
|---|---|---|---|---|
| RLC | Logical channel type | | DTCH | SHCCH |
| | RLC mode | | AM | UM |
| | Payload sizes, bit | | 1 704 | 160 |
| | Max data rate, bps | | 2 048 000 | 16 000 |
| | RLC header, bit | | 16 | 8 |
| MAC | MAC header, bit | | 0 | 0 |
| | MAC multiplexing | | N/A | N/A |
| Layer 1 | TrCH type | | DSCH | DSCH |
| | TB sizes, bit | | 1720 | 168 |
| | TFS | TF0, bits | 0x1720 | 0x168 |
| | | TF1, bits | 1x1720 | 1x168 |
| | | TF2, bits | 2x1720 | N/A |
| | | TF3, bits | 4x1720 | N/A |
| | | TF4, bits | 8x1720 | N/A |
| | | TF5, bits | 12x1720 | N/A |
| | | TF6, bits | N/A (alt. 16x1720) | N/A |
| | | TF7, bits | N/A (alt. 20x1720) | N/A |
| | | TF8, bits | N/A (alt. 24x1720) | N/A |
| | TTI, ms | | 10 (alt. 20) | 10 |
| | Coding type | | No Coding | CC 1/2 |
| | CRC, bit | | 24 | 16 |
| | Max number of bits/TTI after channel coding | | 20 928 (alt. 41 856) | 384 |
| | Downlink: Max number of bits/radio frame before rate matching | | 20 928 (alt. 20 928) | 384 |
| | RM attribute | | 135 to 175 | 180 to 220 |

6.11.5.4.2.3.2.1.2 Transport channel parameters for DL: 3.4 Kbps SRBs for DCCH mapped on DSCH

See clause 6.10.3.4.2.1.2.1.2.

6.11.5.4.2.3.2.1.3 TFCS for DSCH

| TFCS size | 22 (alt.34) |
|---|---|
| TFCS | (2 048 kbps RAB, SHCCH, SRBs for DCCH)=<br>(TF0, TF0, TF0), (TF1, TF0, TF0), (TF2, TF0, TF0), (TF3, TF0, TF0), (TF4, TF0, TF0), (TF5, TF0, TF0),<br>(TF0, TF1, TF0), (TF1, TF1, TF0), (TF2, TF1, TF0), (TF3, TF1, TF0), (TF4, TF1, TF0),<br>(TF0, TF0, TF1), (TF1, TF0, TF1), (TF2, TF0, TF1), (TF3, TF0, TF1), (TF4, TF0, TF1), (TF5, TF0, TF1),<br>(TF0, TF1, TF1), (TF1, TF1, TF1), (TF2, TF1, TF1), (TF3, TF1, TF1), (TF4, TF1, TF1),<br>(alt. (TF0, TF0, TF0), (TF1, TF0, TF0), (TF2, TF0, TF0), (TF3, TF0, TF0), (TF4, TF0, TF0),<br>(TF5, TF0, TF0), (TF6, TF0, TF0), (TF7, TF0, TF0), (TF8, TF0, TF0),<br>(TF0, TF1, TF0), (TF1, TF1, TF0), (TF2, TF1, TF0), (TF3, TF1, TF0), (TF4, TF1, TF0), (TF5, TF1, TF0),<br>(TF6, TF1, TF0), (TF7, TF1, TF0),<br>(TF0, TF0, TF1), (TF1, TF0, TF1), (TF2, TF0, TF1), (TF3, TF0, TF1), (TF4, TF0, TF1), (TF5, TF0, TF1),<br>(TF6, TF0, TF1), (TF7, TF0, TF1), (TF8, TF0, TF1),<br>(TF0, TF1, TF1), (TF1, TF1, TF1), (TF2, TF1, TF1), (TF3, TF1, TF1), (TF4, TF1, TF1), (TF5, TF1, TF1),<br>(TF6, TF1, TF1), (TF7, TF1, TF1)) |

For better understanding of the TFCS please note that the following combinations are not included in the table above:

- (TF5, TF1, TF0), (TF5, TF1, TF1), (TF8, TF1, TF0), (TF8, TF1, TF1).

6.11.5.4.2.3.2.1.4 Transport channel parameters for DL SRBs for DCCH and SRB for CCCH and SRB for BCCH and DL SRB for SHCCH mapped on FACH

See clause 6.11.5.4.2.1.2.1.4.

6.11.5.4.2.3.2.1.5 TFCS for FACH

See clause 6.11.5.4.2.1.2.1.5.

6.11.5.4.2.3.2.2 Physical channel parameters

6.11.5.4.2.3.2.2.1 Physical channel parameters for PDSCH

| | | |
|---|---|---|
| PDSCH | Modulation | 8PSK |
| | Codes and time slots / radio frame | SF1 x 1 code x 10 time slots |
| | Max. Number of data bits/radio frame | 21 084 bits |
| | TFCI code word / radio frame | 24 bits |
| | TPC / radio frame | 2x3 bits |
| | SS / radio frame | 2x3 bits |
| | Puncturing Limit | 1 |

6.11.5.4.2.3.2.2.2 Physical channel parameters for S-CCPCH

See clause 6.11.5.4.2.1.2.2.2.

### 6.11.5.4.3 Combinations on PDSCH, SCCPCH, DPCH, PUSCH and PRACH

6.11.5.4.3.1 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB+ UL:3.4 DL:3.4 kbps SRBs for DCCH + Interactive or background / UL: 64 DL: 256 kbps / PS RAB + UL: 16.8 kbps SRBs for CCCH and SHCCH+ DL: 33.6 kbps SRBs for CCCH, SHCCH and BCCH

6.11.5.4.3.1.1 Uplink

6.11.5.4.3.1.1.1 Transport channel parameters

6.11.5.4.3.1.1.1.1 Transport channel parameters for Conversational / speech / UL:12.2 / CS RAB

See clause 6.10.3.4.1.4.1.1.1.

6.11.5.4.3.1.1.1.2 Transport channel parameters for UL SRBs for DCCH

See clause 6.10.3.4.1.2.1.1.1.

6.11.5.4.3.1.1.1.3 TFCS for DCH

See clause 6.10.3.4.1.4.1.1.3.

6.11.5.4.3.1.1.1.4 Transport channel parameters for Interactive or background / UL: 64 kbps / PS RAB and UL SRB for SHCCH mapped on USCH

See clause 6.10.3.4.2.1.1.1.1.

6.11.5.4.3.1.1.1.5 TFCS for USCH

See clause 6.10.3.4.3.1.1.1.5.

6.11.5.4.3.1.1.1.6 Transport channel parameters for SRB for CCCH and UL SRB for SHCCH mapped on RACH

See clause 6.10.3.4.3.1.1.1.6.

6.11.5.4.3.1.1.2 Physical channel parameters

Physical channel parameters for uplink DPCH see clause 6.11.5.4.1.4.1.2.

Physical channel parameters for PUSCH see clause 6.11.5.4.2.1.1.2.

Physical channel parameters for PRACH see clause 6.11.5.4.2.1.1.2.

6.11.5.4.3.1.2 Downlink

6.11.5.4.3.1.2.1 Transport channel parameters

6.11.5.4.3.1.2.1.1 Transport channel parameters for Conversational / speech / DL:12.2 kbps / CS RAB

See clause 6.10.3.4.1.4.2.1.1.

6.11.5.4.3.1.2.1.2 Transport channel parameters for DL SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.11.5.4.3.1.2.1.3 TFCS for DCH

See clause 6.10.3.4.1.4.2.1.3.

6.11.5.4.3.1.2.1.4 Transport channel parameters for Interactive or background / DL: 256 kbps / PS RAB and DL SRB for SHCCH mapped on DSCH

See clause 6.10.3.4.2.1.2.1.1.

6.11.5.4.3.1.2.1.5 TFCS for DSCH

See clause 6.10.3.4.3.1.2.1.5.

6.11.5.4.3.1.2.1.6 Transport channel parameters for SRB for CCCH and SRB for BCCH and DL SRB for SHCCH mapped on FACH

| | | | | | |
|---|---|---|---|---|---|
| Higher layer | RAB/Signalling RB | | SRB#0 | SRB#5 | SRB#6 |
| | User of Radio Bearer | | RRC | RRC | RRC |
| RLC | Logical channel type | | CCCH | SHCCH | BCCH |
| | RLC mode | | UM | UM | TM |
| | Payload sizes, bit | | 160 | 160 | 168 |
| | Max data rate, bps | | 32 000 | 32 000 | 33 600 |
| | RLC header, bit | | 8 | 8 | 0 |
| MAC | MAC header, bit | | 3 | | |
| | MAC multiplexing | | 3 logical channel multiplexing | | |
| Layer 1 | TrCH type | | FACH | | |
| | TB sizes, bit | | 171 | | |
| | TFS | TF0, bits | 0x171 | | |
| | | TF1, bits | 1x171 | | |
| | | TF2, bits | 2x171 | | |
| | | TF3, bits | 3x171 | | |
| | | TF4, bits | 4x171 | | |
| | TTI, ms | | 20 | | |
| | Coding type | | CC 1/2 | | |
| | CRC, bit | | 16 | | |
| | Max number of bits/TTI after channel coding | | 1 528 | | |
| | Max number of bits/radio frame before rate matching | | 764 | | |

6.11.5.4.3.1.2.1.7 TFCS for FACH

| | |
|---|---|
| TFCS size | 5 |
| TFCS | FACH = TF0, TF1,TF2,TF3,TF4 |

6.11.5.4.3.1.2.2 Physical channel parameters

Physical channel parameters for downlink for DPCH see clause 6.11.5.4.1.4.2.2.

Physical channel parameters for downlink for PDSCH see clause 6.11.5.4.2.1.2.2.

Physical channel parameters for SCCPCH see clause 6.11.5.4.2.1.2.2.

6.11.5.4.3.2 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB+ UL:3.4 DL:3.4 kbps SRBs for DCCH + Interactive or background / UL: 64 DL: 384 kbps / PS RAB + UL: 16.8 kbps SRBs for CCCH and SHCCH+ DL: 33.6 kbps SRBs for CCCH, SHCCH and BCCH

6.11.5.4.3.2.1 Uplink

See clause 6.11.5.4.3.1.1.

6.11.5.4.3.2.2 Downlink

6.11.5.4.3.2.2.1 Transport channel parameters

6.11.5.4.3.2.2.1.1 Transport channel parameters for Conversational / speech / DL:12.2 kbps / CS RAB

See clause 6.10.3.1.4.1.4.2.1.1.

6.11.5.4.3.2.2.1.2 Transport channel parameters for DL SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.11.5.4.3.2.2.1.3 TFCS for DCH

See clause 6.10.3.4.1.4.2.1.3.

6.11.5.4.3.2.2.1.4 Transport channel parameters for Interactive or background / DL: 384 kbps / PS RAB and DL SRB for SHCCH mapped on DSCH

See clause 6.10.3.4.2.2.2.1.1.

6.11.5.4.3.2.2.1.5 TFCS for DSCH

See clause 6.10.3.4.3.2.2.1.5.

6.11.5.4.3.2.2.1.6 Transport channel parameters for SRB for CCCH and SRB for BCCH and DL SRB for SHCCH mapped on FACH

See clause 6.11.5.4.3.1.2.1.6.

6.11.5.4.3.2.2.1.7 TFCS for FACH

See clause 6.11.5.4.3.1.2.1.7.

6.11.5.4.3.2.2.2 Physical channel parameters

Physical channel parameters for downlink for DPCH see clause 6.11.5.4.1.4.2.2.

Physical channel parameters for downlink for PDSCH see clause 6.11.5.4.2.2.2.2.

Physical channel parameters for downlink for SCCPCH see clause 6.11.5.4.2.1.2.2.

6.11.5.4.3.3 Conversational / speech / UL:12.2 DL:12.2 kbps / CS RAB+ UL:3.4 DL:3.4 kbps SRBs for DCCH + Interactive or background / UL: 64 DL: 2 048 kbps / PS RAB + UL: 16.8 kbps SRBs for CCCH and SHCCH + DL: 33.6 kbps SRBs for CCCH, SHCCH and BCCH

6.11.5.4.3.3.1 Uplink

See clause 6.11.5.4.3.1.1.

6.11.5.4.3.3.2 Downlink

6.11.5.4.3.3.2.1 Transport channel parameters

6.11.5.4.3.3.2.1.1 Transport channel parameters for Conversational / speech / DL:12.2 kbps / CS RAB

See clause 6.10.3.4.1.4.2.1.1.

6.11.5.4.3.3.2.1.2 Transport channel parameters for DL SRBs for DCCH

See clause 6.10.3.4.1.2.2.1.1.

6.11.5.4.3.3.2.1.3 TFCS for DCH

See clause 6.10.3.4.1.4.2.1.3.

6.11.5.4.3.3.2.1.4 Transport channel parameters for Interactive or background / DL: 2 048 kbps / PS RAB and DL SRB for SHCCH mapped on DSCH

See clause 6.11.5.4.2.3.2.1.2.

6.11.5.4.3.3.2.1.5 TFCS for DSCH

See clause 6.11.5.4.2.3.2.1.4.

6.11.5.4.3.3.2.1.6 Transport channel parameters for SRB for CCCH and SRB for BCCH and DL SRB for SHCCH mapped on FACH

See clause 6.11.5.4.3.1.2.1.6.

6.11.5.4.3.3.2.1.7 TFCS for FACH

See clause 6.11.5.4.3.1.2.1.7.

6.11.5.4.3.3.2.2 Physical channel parameters

Physical channel parameters for downlink DPCH see clause 6.11.5.4.1.4.2.2.

Physical channel parameters for PDSCH see clause 6.11.5.4.2.3.2.2.

Physical channel parameters for SCCPCH see clause 6.11.5.4.2.1.2.2.

### 6.11.5.4.4 Combinations on SCCPCH

6.11.5.4.4.1 Stand-alone signalling RB for PCCH

6.11.5.4.4.1.1 Transport channel parameters

6.11.5.4.4.1.1.1 Transport channel parameter of SRB for PCCH

See clause 6.10.3.4.4.1.1.1.

6.11.5.4.4.1.1.2 TFCS

See clause 6.10.3.4.4.1.1.2.

6.11.5.4.4.1.2 Physical channel parameters

| | | |
|---|---|---|
| S-CCPCH | Modulation | QPSK |
| | Codes and time slots / radio frame | SF16 x 2 codes x 2 time slots<br>(alt. SF16 x 1 codes x 2 time slots) |
| | Max. Number of data bits/radio frame | 344 bits (alt. 168 bits) |
| | TFCI code word / radio frame | 8 bits |
| | TPC / radio frame | 0 bits |
| | SS / radio frame | 0 bits |
| | Puncturing Limit | 1 (alt. 0.84) |
| NOTE: Alt. Puncturing Limit applies when alt. payload sizes and alt. codes and time slots / radio frame are both configured. | | |

6.11.5.4.4.2 Interactive/Background 32 kbps PS RAB + SRBs for CCCH + SRB for DCCH + SRB for BCCH

6.11.5.4.4.2.1 Transport channel parameters

6.11.5.4.4.2.1.1 Transport channel parameters for Interactive/Background 32 kbps PS RAB

See clause 6.10.3.4.4.2.1.1.

6.11.5.4.4.2.1.2 Transport channel parameters of SRBs for CCCH, SRB for DCCH, and SRB for BCCH

| Higher layer | RAB/signalling RB | | SRB#0 | SRB#1 | SRB#2 | SRB#3 | SRB#4 | SRB#5 |
|---|---|---|---|---|---|---|---|---|
| | User of Radio Bearer | | RRC | RRC | RRC | NAS_DT High prio | NAS_DT Low prio | RRC |
| RLC | Logical channel type | | CCCH | DCCH | DCCH | DCCH | DCCH | BCCH |
| | RLC mode | | UM | UM | AM | AM | AM | TM |
| | Payload sizes, bit | | 160 | 136 or 120 | 128 | 128 | 128 | 168 |
| | Max data rate, bps | | 32 000 (alt. 16 000) | 27 200 or 2400 (alt. 24 000 or 12 000) | 25 600 (alt. 12 800) | 25 600 (alt. 12 800) | 25 600 (alt. 12 800) | 33 600 (alt. 16 800) |
| | RLC header, bit | | 8 | 8 | 16 | 16 | 16 | 0 |
| MAC | MAC header, bit | | 3 | 27 or 43 | 27 | 27 | 27 | 3 |
| | MAC multiplexing | | 6 logical channel multiplexing | | | | | |
| Layer 1 | TrCH type | | FACH | | | | | |
| | TB sizes, bit | | 171 | | | | | |
| | TFS | TF0, bits | 0x171 | | | | | |
| | | TF1, bits | 1x171 | | | | | |
| | | TF2, bits | 2x171 | | | | | |
| | | TF3, bits | 3x171 (alt. N/A) | | | | | |
| | | TF4, bits | 4x171 (alt. N/A) | | | | | |
| | TTI, ms | | 20 | | | | | |
| | Coding type | | CC 1/2 | | | | | |
| | CRC, bit | | 16 | | | | | |
| | Max number of bits/TTI before rate matching | | 1 528 (alt. 764) | | | | | |
| | Max number of bits/radio frame before rate matching | | 764 (alt. 382) | | | | | |
| | RM attribute | | 200 to 240 | | | | | |
| NOTE: MAC header size and RLC payload size depend on use of U-RNTI or C-RNTI. | | | | | | | | |

6.11.5.4.4.2.1.3 TFCS

See clause 6.10.3.4.4.2.1.3.

6.11.5.4.4.2.2 Physical channel parameters

| SCCPCH | Modulation | QPSK |
|---|---|---|
| | Codes and time slots / radio frame | SF 16 x 9 codes x 2 time slots (alt. SF16 x 4 codes x 2 time slots) |
| | Max. Number of data bits/radio frame | 1 568 bits (alt. 688 bits) |
| | TFCI code word / radio frame | 16 bits |
| | TPC / radio frame | 0 bits |
| | SS / radio frame | 0 bits |
| | Puncturing Limit | 0.52 (alt. 0.48) |
| NOTE: Alt. Puncturing Limit applies when alt. TFCS and alt. codes and time slots / radio frame are both configured. | | |

6.11.5.4.4.2a Interactive/Background 32 kbps PS RAB + Interactive/Background 32 kbps PS RAB + SRBs for CCCH + SRB for DCCH + SRB for BCCH

6.11.5.4.4.2a.1 Transport channel parameters

6.11.5.4.4.2a.1.1 Transport channel parameters for Interactive/Background 32 kbps PS RAB + Interactive/Background 32 kbps PS RAB

See clause 6.10.3.4.2a.1.1.

6.11.5.4.4.2a.1.2 Transport channel parameters of SRBs for CCCH, SRB for DCCH, and SRB for BCCH

See clause 6.11.5.4.4.2.1.2.

6.11.5.4.4.2a.1.3 TFCS

See clause 6.10.3.4.4.2a.1.3.

6.11.5.4.4.2a.2 Physical channel parameters

See clause 6.11.5.4.4.2.2.

6.11.5.4.4.2b SRBs for CCCH + SRB for DCCH + SRB for BCCH

6.11.5.4.4.2b.1 Transport channel parameters

6.11.5.4.4.2b.1.1 Transport channel parameters of SRBs for CCCH, SRB for DCCH, and SRB for BCCH

See clause 6.11.5.4.4.2.1.2.

6.11.5.4.4.2b.1.2 TFCS

See clause 6.10.3.4.4.2b.1.2.

6.11.5.4.4.2b.2 Physical channel parameters

| | | |
|---|---|---|
| SCCPCH | Modulation | QPSK |
| | Codes and time slots / radio frame | SF 16 x 4 codes x 2 time slots<br>(alt. SF16 x 2 codes x 2 time slots) |
| | Max. Number of data bits/radio frame | 688 bits (alt. 344 bits) |
| | TFCI code word / radio frame | 16 bits (alt. 8 bits) |
| | TPC / radio frame | 0 bits |
| | SS / radio frame | 0 bits |
| | Puncturing Limit | 0.88 |
| NOTE: Alt. Puncturing Limit applies when alt. TFCS and alt. codes and time slots / radio frame are both configured. | | |

6.11.5.4.4.3 Interactive/Background 32 kbps RAB + SRB for PCCH + SRB for CCCH + SRB for DCCH + SRB for BCCH

6.11.5.4.4.3.1 Transport channel parameters

6.11.5.4.4.3.1.1 Transport channel parameters of SRB for Interactive/Background 32 kbps RAB

See clause 6.10.3.4.4.2.1.1.

6.11.5.4.4.3.1.2 Transport channel parameters of SRB for PCCH

See clause 6.10.3.4.4.1.1.1.

6.11.5.4.4.3.1.3 Transport channel parameters of SRBs for CCCH, SRB for DCCH, and SRB for BCCH

See clause 6.11.5.4.4.2.1.2.

6.11.5.4.4.3.1.4 TFCS

See clause 6.10.3.4.4.3.1.4.

6.11.5.4.4.3.2 Physical channel parameters

| | | |
|---|---|---|
| S-CCPCH | Modulation | QPSK |
| | Codes and time slots / radio frame | SF16 x 10 codes x 2 time slots<br>(alt. SF16 x 6 codes x 2 time slots) |
| | Max. Number of data bits/radio frame | 1 744 bits (alt. 1 040 bits) |

| | | |
|---|---|---|
| | TFCI code word / radio frame | 16 bits |
| | TPC / radio frame | 0 bits |
| | SS / radio frame | 0 bits |
| | Puncturing Limit | 0.48 (alt. 0.52) |
| NOTE: Alt. Puncturing Limit applies when alt. TFCS and alt. codes and time slots / radio frame are both configured. | | |

6.11.5.4.4.3a SRB for PCCH + SRB for CCCH + SRB for DCCH + SRB for BCCH

6.11.5.4.4.3a.1 Transport channel parameters

6.11.5.4.4.3a.1.1 Transport channel parameters of SRB for PCCH

See clause 6.10.3.4.4.1.1.1.

6.11.5.4.4.3a.1.2 Transport channel parameters of SRBs for CCCH, SRB for DCCH, and SRB for BCCH

See clause 6.11.5.4.4.2.1.2.

6.11.5.4.4.3a.1.3 TFCS

See clause 6.10.3.4.4.3a.1.3.

6.11.5.4.4.3a.2 Physical channel parameters

| | | |
|---|---|---|
| SCCPCH | Modulation | QPSK |
| | Codes and time slots / radio frame | SF 16 x 4 codes x 2 time slots (alt. SF16 x 2 codes x 2 time slots) |
| | Max. Number of data bits/radio frame | 688 bits (alt. 336 bits) |
| | TFCI code word / radio frame | 16 bits |
| | TPC / radio frame | 0 bits |
| | SS / radio frame | 0 bits |
| | Puncturing Limit | 0.60 (alt. 0.52) |
| NOTE: Alt. applies when alts for SRB for PCCH and SRBs for CCCH/ DCCH/ BCCH are both configured. | | |

6.11.5.4.4.4 RB for CTCH + SRB for CCCH + SRB for BCCH

6.11.5.4.4.4.1 Transport channel parameters

6.11.5.4.4.4.1.1 Transport channel parameters of RB for CTCH

See clause 6.10.3.4.4.4.1.1.

6.11.5.4.4.4.1.2 Transport channel parameters of SRB for CCCH and SRB for BCCH

| | | | | |
|---|---|---|---|---|
| Higher layer | RAB/signalling RB | | SRB#0 | SRB#5 |
| | User of Radio Bearer | | RRC | RRC |
| RLC | Logical channel type | | CCCH | BCCH |
| | RLC mode | | UM | TM |
| | Payload sizes, bit | | 160 | 168 |
| | Max data rate, bps | | 16 000 | 16 800 |
| | AMD/UMD/TrD PDU header, bit | | 8 | 0 |
| MAC | MAC header, bit | | 3 | 3 |
| | MAC multiplexing | | 2 logical channel multiplexing | |
| Layer 1 | TrCH type | | FACH | |
| | TB sizes, bit | | 171 | |
| | TFS | TF0, bits | 0x171 | |
| | | TF1, bits | 1x171 | |
| | | TF2, bits | 2x171 | |
| | TTI, ms | | 20 | |
| | Coding type | | CC 1/3 | |
| | CRC, bit | | 16 | |
| | Max number of bits/TTI before rate matching | | 1 146 | |

| | | | |
|---|---|---|---|
| Higher layer | RAB/signalling RB | SRB#0 | SRB#5 |
| | User of Radio Bearer | RRC | RRC |
| | Max number of bits/radio frame before rate matching | 573 | |
| | RM attribute | 200 to 240 | |

6.11.5.4.4.4.1.3 TFCS

See clause 6.10.3.4.4.4.1.3.

6.11.5.4.4.4.2 Physical channel parameters

| | | |
|---|---|---|
| SCCPCH | Modulation | QPSK |
| | Codes and time slots / radio frame | SF 16 x 4 codes x 2 time slots |
| | Max. Number of data bits/radio frame | 688 bits |
| | TFCI code word / radio frame | 16 bits |
| | TPC / radio frame | 0 bits |
| | SS / radio frame | 0 bits |
| | Puncturing Limit | 0.52 |

6.11.5.4.5 Combinations on PRACH

6.11.5.4.5.1 SRB for CCCH + SRBs for DCCH

6.11.5.4.5.1.1 Transport channel parameters

6.11.5.4.5.1.1.1 Transport channel parameter for SRB for CCCH, SRBs for DCCH

See clause 6.10.3.4.5.1.1.1.

6.11.5.4.5.1.1.2 TFCS

See clause 6.10.3.4.5.1.1.2.

6.11.5.4.5.1.2 Physical channel parameters

| | | |
|---|---|---|
| PRACH | Modulation | QPSK |
| | Codes and time slots / radio frame | SF 8 x 1 code x 2 time slots |
| | Max. Number of data bits/radio frame | 352 bits |
| | TPC / radio frame | 0 bits |
| | SS / radio frame | 0 bits |
| | Puncturing Limit | 0.88 |

6.11.5.4.5.2 Interactive/Background 12.8 kbps PS RAB + SRB for CCCH + SRBs for DCCH

6.11.5.4.5.2.1 Transport channel parameters

6.11.5.4.5.2.1.1 Transport channel parameters for Interactive or background / 12.8 kbps / PS RAB + SRB for CCCH + SRBs for DCCH

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| Higher layer | RAB/signalling RB | RAB | SRB#0 | SRB#1 | SRB#2 | SRB#3 | SRB#4 |
| | User of Radio Bearer | Interactive/ Background RAB | RRC | RRC | RRC | NAS_DT High priority | NAS_DT Low priority |
| RLC | Logical channel type | DTCH | CCCH | DCCH | DCCH | DCCH | DCCH |
| | RLC mode | AM | TM | UM | AM | AM | AM |
| | Payload sizes, bit | 128 | 168 | 136 | 128 | 128 | 128 |
| | Max data rate, bps | 12 800 | 16 800 | 13 600 | 12 800 | 12 800 | 12 800 |
| | AMD/UMD/TrD PDU header, bit | 16 | 0 | 8 | 16 | 16 | 16 |
| MAC | MAC header, bit | 26 | 2 | 26 | 26 | 26 | 26 |
| | MAC multiplexing | 6 logical channel multiplexing | | | | | |
| Layer 1 | TrCH type | RACH | | | | | |
| | TB sizes, bit | 170 | | | | | |
| | TFS TF0, bits | 1x170 | | | | | |

| Higher layer | RAB/signalling RB | RAB | SRB#0 | SRB#1 | SRB#2 | SRB#3 | SRB#4 |
|---|---|---|---|---|---|---|---|
| | User of Radio Bearer | Interactive/ Background RAB | RRC | RRC | RRC | NAS_DT High priority | NAS_DT Low priority |
| | TTI, ms | 10 | | | | | |
| | Coding type | CC 1/2 | | | | | |
| | CRC, bit | 16 | | | | | |
| | Max number of bits/TTI after channel coding | 388 | | | | | |
| | Max number of bits/ Radio frame before rate matching | 388 | | | | | |

6.11.5.4.5.2.1.2 TFCS

| TFCS size | 1 |
|---|---|
| TFCS | 12.8 kbps PS RAB + SRB for CCCH + SRBs for DCCH = (TF0) |

6.11.5.4.5.2.2 Physical channel parameters

See clause 6.11.5.4.5.1.2.

6.11.5.4.5.3 Interactive/Background 12.8 kbps PS RAB + Interactive/Background 12.8 kbps PS RAB + SRB for CCCH + SRB for DCCH

6.11.5.4.5.3.1 Transport channel parameters

6.11.5.4.5.3.1.1 Transport channel parameters for Interactive or background / 12.8 kbps / PS RAB + Interactive or background / 12.8 kbps / PS RAB + SRB for CCCH + SRBs for DCCH

| Higher layer | RAB/signalling RB | RAB | RAB | SRB#0 | SRB#1 | SRB#2 | SRB#3 | SRB#4 |
|---|---|---|---|---|---|---|---|---|
| | User of Radio Bearer | Interactive/ Background RAB | Interactive/ Background RAB | RRC | RRC | RRC | NAS_DT High prio | NAS_DT Low prio |
| RLC | Logical channel type | DTCH | DTCH | CCCH | DCCH | DCCH | DCCH | DCCH |
| | RLC mode | AM | AM | TM | UM | AM | AM | AM |
| | Payload sizes, bit | 128 | 128 | 168 | 136 | 128 | 128 | 128 |
| | Max data rate, bps | 12 800 | 12 800 | 16 800 | 13 600 | 12 800 | 12 800 | 12 800 |
| | AMD/UMD/TrD PDU header, bit | 16 | 16 | 0 | 8 | 16 | 16 | 16 |
| MAC | MAC header, bit | 26 | 26 | 2 | 26 | 26 | 26 | 26 |
| | MAC multiplexing | 7 logical channel multiplexing | | | | | | |
| Layer 1 | TrCH type | RACH | | | | | | |
| | TB sizes, bit | 170 | | | | | | |
| | TFS TF0, bits | 1x170 | | | | | | |
| | TTI, ms | 10 | | | | | | |
| | Coding type | CC ½ | | | | | | |
| | CRC, bit | 16 | | | | | | |
| | Max number of bits/TTI after channel coding | 388 | | | | | | |
| | Max number of bits/ Radio frame before rate matching | 388 | | | | | | |

6.11.5.4.5.3.1.2 TFCS

| TFCS size | 1 |
|---|---|
| TFCS | 12.8 kbps PS RAB + 12.8 kbps PS RAB + SRB for CCCH + SRBs for DCCH = (TF0) |

6.11.5.4.5.3.2 Physical channel parameters

See clause 6.11.5.4.5.1.2.

# 7 Generic setup procedures

## 7.1 Basic Generic Procedures

### 7.1.1 UE Test States for Basic Generic Procedures

This clause describes a set of procedures for use by test cases in 3GPP TS 34.123-1 [1]. Describing these procedures in a generic manner allows their use in many test cases. By using these procedures, test case descriptions need not detail signalling that is not relevant to its purpose or understanding.

The procedures are based upon default values that are adapted to the most common usage. Test cases that require values different from the default will, when specifying the Basic Generic Procedure, also specify those parameters that are modified.

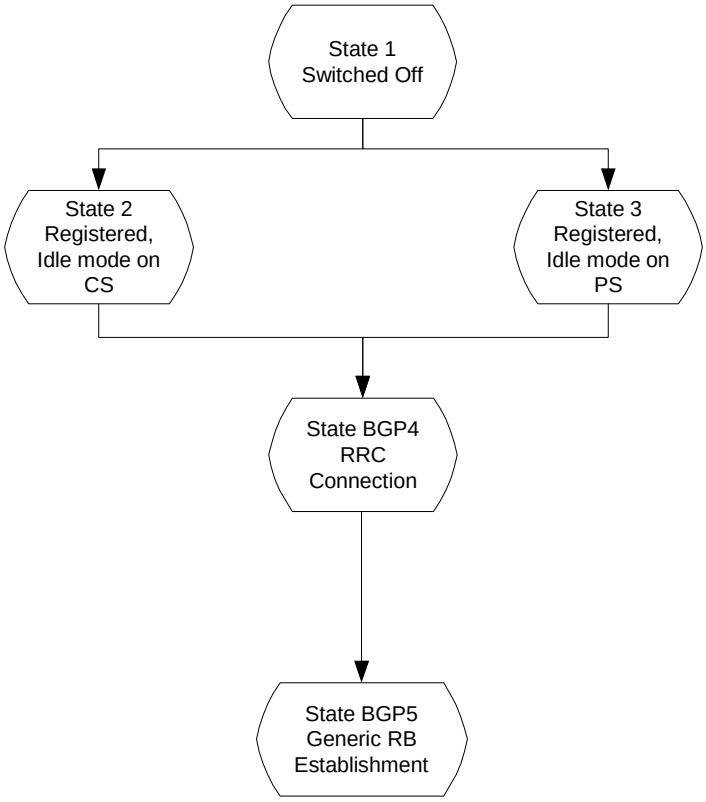



**Figure 7.1.1: UE Test States for Basic Generic Procedures**

In order that the UE can set up a call in UTRAN, there are a number of procedures to be undertaken in a hierachical sequence to move between known states. The sequences are shown in figure 7.1.1 and the status of the relevant protocols in the UE in the different states are given in table 7.1.1.

**Table 7.1.1: The UE states**

| | | RRC | CC | MM | SM | GMM |
|---|---|---|---|---|---|---|
| State 1 | Power OFF | ----- | null | null | pdp-inactive | GMM-null |
| State 2 | CS Registered Idle Mode | idle | null | idle | pdp-inactive | GMM-deregistered |
| State 3 | PS Registered Idle Mode | idle | null | null | pdp-inactive | GMM-registered |
| State BGP4 | RRC Connection | connected | null | as previous | pdp-inactive | as previous |
| State BGP5 | Generic RB Establishment | connected | null | as previous | pdp-inactive | as previous |

### 7.1.2 Mobile terminated establishment of Radio Resource Connection

### 7.1.2.1 Initial conditions

**System Simulator:**

The system simulator will start from the default idle state. Parameters will the default parameters for a single cell, unless otherwise specified in the test case.

**User Equipment:**

Unless otherwise specified in the test case, the UE will be in the following state:

- Default test operating conditions.
- The UE shall have followed the generic registration procedure for CS or PS operations, and will be in Idle Mode, Camped-on (State 2 or State 3).

### 7.1.2.2 Definition of system information messages

The default system information messages are used.

### 7.1.2.3 Procedure

- The SS sends a PAGING TYPE 1 message to the UE on the appropriate paging block, and with the IE "Paging record" containing the TMSI or P-TMSI of the UUT.
- The SS receives an RRC CONNECTION REQUEST message from the UE.
- On receipt of the RRC CONNECTION REQUEST the SS shall transmit a RRC CONNECTION SETUP message to the UE. The SS shall wait for the receipt of an RRC CONNECTION SETUP COMPLETE message from the UE.
- On receipt of an RRC CONNECTION SETUP COMPLETE message, the procedure is complete.

| Step | Direction | | Message | Comments |
|---|---|---|---|---|
| | UE | SS | | |
| 1 | ← | | SYSTEM INFORMATION (BCCH) | Default SI messages |
| 2 | ← | | PAGING TYPE 1 (PCCH) | Sent on appropriate cycle |
| 3 | → | | RRC CONNECTION REQUEST (CCCH) | RRC |
| 4 | ← | | RRC CONNECTION SETUP (CCCH) | RRC |
| 5 | → | | RRC CONNECTION SETUP COMPLETE (DCCH) | RRC |

### 7.1.2.4 Specific message contents

#### 7.1.2.4.1 PAGING TYPE 1

This message is sent from the SS to the UE, using the TM RLC SAP, on the PCCH logical channel.

| Information Element | | | | Value/Remark |
|---|---|---|---|---|
| Message Type | | | | PAGING TYPE 1 |
| **UE Information elements** | | | | |
| Paging record list | Paging record | CN originator | Paging cause | Terminating Speech Call (note) |
| | | | CN domain identity | CS domain (note) |
| | | | UE Identity | TMSI (GSM-MAP)<br>As specified during Registration procedure |
| | | | | |
| **Other information elements** | | | | |
| BCCH modification info | | | | omit |
| NOTE: These defaults are applied if no subsequent procedure is to be run. Otherwise, the Paging cause, CN domain identity and UE Identity are selected in accordance with the requirements of the following procedure. | | | | |

### 7.1.2.4.2 RRC CONNECTION REQUEST

This message is sent by the UE to the SS using the TM-RLC SAP. It is sent on the CCCH Logical channel.

| Information Element | | | Value/Remark |
|---|---|---|---|
| Message Type | | | RRC CONNECTION REQUEST |
| **UE information elements** | | | |
| Initial UE identity | TMSI and LAI | TMSI (GSM-MAP) | As specified during Registration procedure |
| | | LAI (GSM-MAP) | As specified by default 1 cell environment |
| Initial UE capability | Maximum number of AM entities | | As declared in UE ICS |
| Establishment cause | | | As appropriate |
| Protocol error indicator | | | FALSE |
| >UE Specific Behaviour Information 1 idle | | | This IE will not be checked by default behaviour, but in specific test case. |
| | | | |
| **Measurement information elements** | | | |
| Measured results on RACH | | | Not checked |
| NOTE: These defaults are applied if no subsequent procedure is to be run. Otherwise, the UE Identity is selected in accordance with the requirements of the following procedure. | | | |

### 7.1.2.4.3 RRC CONNECTION SETUP

This message is sent from the SS to the UE using the UM-RLC SAP. The message is sent on the CCCH Logical channel.

The default RRC CONNECTION SETUP message for the transition to connected mode CELL_DCH is used except for the IE fields specified below.

| Information Element | | | Value/Remark |
|---|---|---|---|
| Message Type | | | RRC CONNECTION SETUP |
| **UE Information Elements** | | | |
| Initial UE identity | TMSI and LAI | TMSI (GSM-MAP) | As specified during Registration procedure |
| | | LAI (GSM-MAP) | As specified by default 1 cell environment |
| | | | |
| **RB Information Elements** | | | |
| Use default | | | |
| **TrCH Information Elements** | | | |
| Use default | | | |
| **PhyCH Information Elements** | | | |
| Frequency info | | | As specified by default 1 cell environment |
| **Uplink radio resources** | | | |
| Use default | | | |
| **Downlink radio resources** | | | |
| Use default | | | |
| NOTE: These defaults are applied if no subsequent procedure is to be run. Otherwise, the UE Identity is selected in accordance with the requirements of the following procedure. | | | |

### 7.1.2.4.4 RRC CONNECTION SETUP COMPLETE

This message is sent by the UE to the SS using AM-RLC SAP. The message is sent on the DCCH Logical channel.

| Information Element | | Value/Remark |
|---|---|---|
| Message Type | | RRC CONNECTION SETUP COMPLETE |
| **UE Information Elements** | | |
| Hyper frame number | | Not checked |
| UE radio access capability | Conformance test compliance | R99 |

| Information Element | | | Value/Remark |
|---|---|---|---|
| | PDCP capability | Support for lossless SRNS relocation | Not checked |
| | | Supported algorithm types | Not checked |
| | RLC capability | Total RLC AM buffer size | Not checked |
| | | Maximum number of AM entities | Not checked |
| | Transport channel capability | **Downlink** | |
| | | Max no of bits received | Not checked |
| | | Max convolutionally coded bits received | Not checked |
| | | Max turbo coded bits received | Not checked |
| | | Maximum number of simultaneous transport channels | Not checked |
| | | Max no of received transport blocks | Not checked |
| | | Maximum number of TFC in the TFCS | Not checked |
| | | Maximum number of TF | Not checked |
| | | Support for turbo decoding | Not checked |
| | | **Uplink** | |
| | | Max no of bits transmitted | Not checked |
| | | Max convolutionally coded bits received | Not checked |
| | | Max turbo coded bits received | Not checked |
| | | Maximum number of simultaneous transport channels | Not checked |
| | | Max no of transmitted transport blocks | Not checked |
| | | Maximum number of TFC in the TFCS | Not checked |
| | | Maximum number of TF | Not checked |
| | | Support for turbo encoding | Not checked |
| | RF capability | UE power class | As declared for UE |
| | | Tx/Rx frequency separation | Not checked |
| | Physical channel capability | **Downlink** | |
| | | Maximum number of simultaneous CCTrCH | Not checked |
| | | Max no DPCH/PDSCH codes | Not checked (PDSCH: R99 and Rel-4 only) |
| | | Max no physical channel bits received | Not checked |
| | | Support for SF 512 | Not checked |
| | | Support of PDSCH | Not checked (R99 and Rel-4 only) |
| | | Simultaneous reception of SCCPCH and DPCH | Not checked |
| | | Max no of S-CCPCH RL | Not checked |
| | | **Uplink** | |
| | | Maximum number of DPDCH bits transmitted per 10 ms | Not checked |
| | | Support of PCPCH (R99 and Rel-4 only) | Not checked |
| | UE multi-mode/multi-RAT capability | Multi-RAT capability | |
| | | Multi-mode capability | FDD or FDD/TDD |
| | Security capability | Ciphering algorithm capability | Not checked |
| | | Integrity protection algorithm capability | Not checked |

| Information Element | | | Value/Remark |
|---|---|---|---|
| | LCS capability | Standalone location method(s) supported | Not checked |
| | | UE based OTDOA supported | Not checked |
| | | Network Assisted GPS support | Not checked |
| | | GPS reference time capable | Not checked |
| | | Support for IPDL | Not checked |
| | Measurement capability | Need for downlink compressed mode | Not checked |
| | | FDD measurements DL | Not checked |
| | | TDD measurements DL | Not checke |
| | | GSM 900 DL | Not checked |
| | | DCS 1800 DL | Not checked |
| | | GSM 1900 DL | Not checked |
| | | Multi-carrier measurement DL | Not checked |
| | | Need for uplink compressed mode | Not checked |
| | | FDD measurements UL | Not checked |
| | | TDD measurements UL | Not checked |
| | | GSM 900 UL | Not checked |
| | | DCS 1800 UL | Not checked |
| | | GSM 1900 UL | Not checked |
| | | Multi-carrier measurement UL | Not checked |
| UE system specific capability | | | Not checked |

## 7.1.3 Radio Bearer Setup Procedure

### 7.1.3.1 Initial conditions

The procedure specified in clause 7.1.2 will be run. This procedure starts from the successful completion of clause 7.1.2.

### 7.1.3.2 Definition of system information messages

The default system information messages are used.

### 7.1.3.3 Procedure

- The SS sends a RADIO BEARER SETUP message to the UE on the DCCH established by the RRC Connection Establishment procedure.
- The SS receives a RADIO BEARER SETUP COMPLETE message from the UE in RLC Acknowledged mode on the DCCH.

On receiption of the RADIO BEARER SETUP COMPLETE the procedure is complete.

| Step | Direction | | Message | Comments |
|---|---|---|---|---|
| | UE | SS | | |
| 1 | ← | | RADIO BEARER SETUP (DCCH) | RRC |
| 2 | → | | RADIO BEARER SETUP COMPLETE (DCCH) | RRC |

### 7.1.3.4 Specific message contents

#### 7.1.3.4.1 RADIO BEARER SETUP

The RADIO BEARER SETUP message is sent from the System Simulator to the UE, using AM-RLC on the DCCH logical channel.

The default RRC CONNECTION SETUP message for the setup of a speech radio access bearer is used except for the IE fields specified below.

| Information Element | | Value/Remark |
|---|---|---|
| Message Type | | RADIO BEARER SETUP |
| **UE Information Elements** | | |
| **CN Information Elements** | | |
| **RB Information Elements** | | |
| RAB information for setup | Default parameters for 12.2 kbps speech RAB + 3.4 kbps signalling radio bearer according to clause 6.10.2.4.1.4 for FDD, clause 6.10.3.4.1.4 for 3.84 Mcps TDD and 6.11.5.4.1.4 for 1.28 Mcps TDD | |

#### 7.1.3.4.2 RADIO BEARER SETUP COMPLETE

The RADIO BEARER SETUP COMPLETE message is sent from the UE to the System Simulator, using AM-RLC on the DCCH logical channel.

The default RADIO BEARER SETUP COMPLETE message is used.

| Information Element | Value/Remark |
|---|---|
| Message Type | RADIO BEARER SETUP COMPLETE |
| Use default | |

## 7.2 Generic setup procedures

### 7.2.1 UE Test States for Generic setup procedures

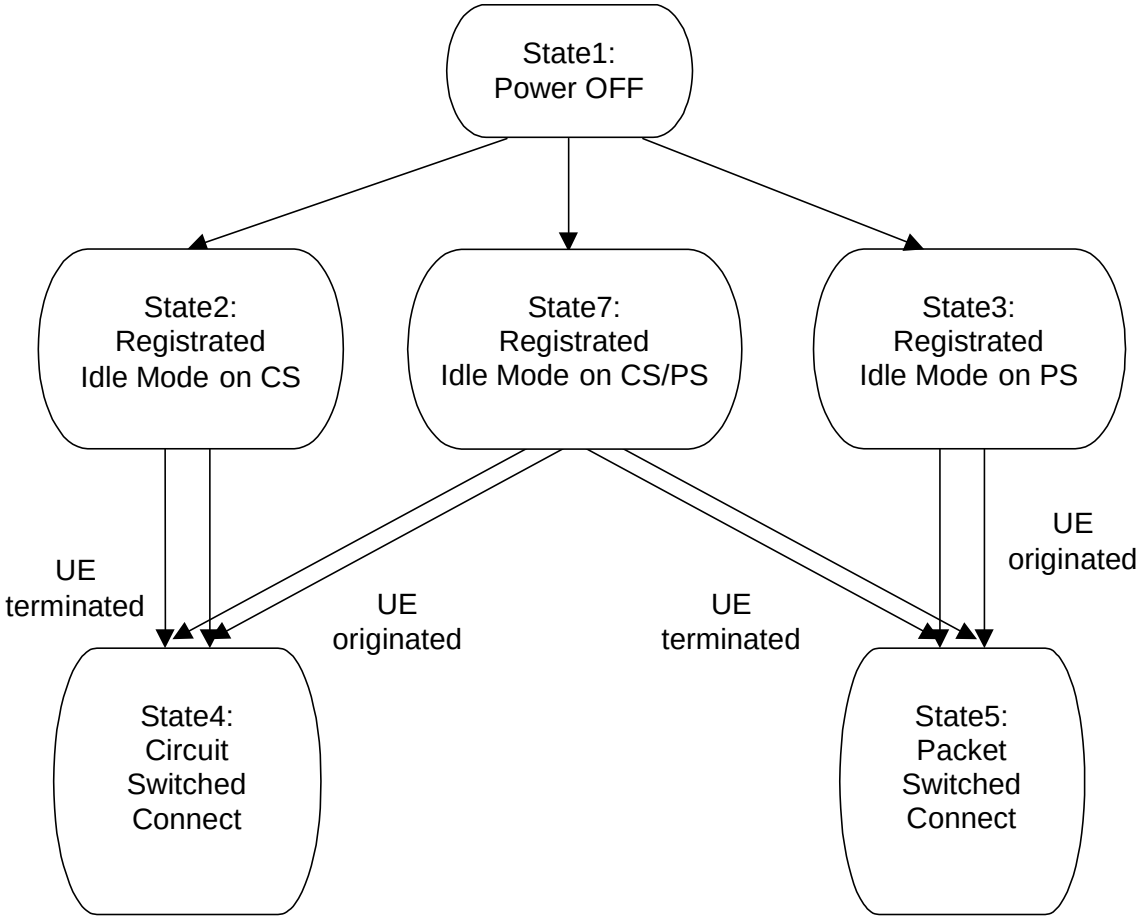



**Figure 7.2.1.1: UE Test States for Generic setup procedures**

In order that the UE can set up a call in UTRAN, there are a number of procedures to be undertaken in a hierarchical sequence to move between known states. The sequences are shown in figure 7.2.1.1 and the status of the relevant protocols in the UE in the different states are given in table 7.2.1.1.

**Table 7.2.1.1: The UE states**

| | | RRC | CC | MM | SM | GMM |
|---|---|---|---|---|---|---|
| State1 | Power OFF | ----- | null | null | pdp-inactive | GMM-null |
| State2 | Registered Idle Mode on CS | idle | null | MM idle | pdp-inactive | GMM-deregistered |
| State3 | Registered Idle Mode on PS | idle | null | null | pdp-inactive | GMM-registered |
| State4 | Circuit Switched Connect | connected | active | MM connection active | pdp-inactive | same as previous state |
| State5 | Packet Switched Connect | connected | null | same as previous state | pdp-active | GMM-registered |
| State7 | Registered Idle Mode on CS/PS | idle | null | MM idle | pdp-inactive | GMM-registered |

## 7.2.2 Registration of UE

The default procedures required to achieve the changes of state between State 1, in clause 7.2.1, and States 2, 3 and 7 are illustrated in the following clauses.

The choice of which procedure to use given a UE supporting packet services is influenced by the Network Mode of Operation being simulated by the SS and by the Operation Mode of the UE, as described in 3GPP TS 24.008 [32] clause 1.7.2.2. Table 7.2.2 shows the appropriate clause number for each combination of these two modes of operation.

**Table 7.2.2: Registration Procedures for UEs Supporting Packet Services**

| Network Mode | | NMO I | NMO II |
|---|---|---|---|
| **UE Mode** | **PS/CS** | 7.2.2.3 | 7.2.2.4 |
| | **PS** | 7.2.2.2 | 7.2.2.2 |

### 7.2.2.1 Registration on CS

#### 7.2.2.1.1 Initial condition

System Simulator:

- 1 cell, default parameters.

User Equipment:

- The UE shall be operated under normal test conditions.
- The Test-USIM shall be inserted.

#### 7.2.2.1.2 Definition of system information messages

The default system information messages are used.

#### 7.2.2.1.3 Procedure

Registration of UE for SS shall be established under ideal radio conditions as defined in clause 5.

| Step | Direction | | Message | Comments |
|---|---|---|---|---|
| | UE | SS | | |
| 1 | | ← | SYSTEM INFORMATION (BCCH) | NW Broadcast |
| 2 | | → | RRC CONNECTION REQUEST (CCCH) | RRC |
| 3 | | ← | RRC CONNECTION SETUP (CCCH) | RRC |
| 4 | | → | RRC CONNECTION SETUP COMPLETE (DCCH) | RRC |
| 5 | | → | LOCATION UPDATING REQUEST | MM |
| 6 | | ← | AUTHENTICATION REQUEST | MM |
| 7 | | → | AUTHENTICATION RESPONSE | MM |
| 8 | | ← | SECURITY MODE COMMAND | RRC |
| 9 | | → | SECURITY MODE COMPLETE | RRC |

| Step | Direction | | Message | Comments |
|---|---|---|---|---|
| | UE | SS | | |
| 10 | ← | | LOCATION UPDATING ACCEPT | MM |
| 11 | → | | TMSI REALLOCATION COMPLETE | MM |
| 12 | ← | | RRC CONNECTION RELEASE | RRC |
| 13 | → | | RRC CONNECTION RELEASE COMPLETE | RRC |

#### 7.2.2.1.4 Specific message contents

All Specific message contents shall be referred to clause 9.

### 7.2.2.2 Registration on PS

#### 7.2.2.2.1 Initial condition

System Simulator:

- 1 cell, default parameters.

User Equipment:

- The UE shall be operated under normal test conditions.
- The Test-USIM shall be inserted.

#### 7.2.2.2.2 Definition of system information messages

The default system information messages are used.

#### 7.2.2.2.3 Procedure

Registration of UE for SS shall be established under ideal radio conditions as defined in clause 5.

| Step | Direction | | Message | Comments |
|---|---|---|---|---|
| | UE | SS | | |
| 1 | ← | | SYSTEM INFORMATION (BCCH) | NW Broadcast |
| 2 | → | | RRC CONNECTION REQUEST (CCCH) | RRC |
| 3 | ← | | RRC CONNECTION SETUP (CCCH) | RRC |
| 4 | → | | RRC CONNECTION SETUP COMPLETE (DCCH) | RRC |
| 5 | → | | ATTACH REQUEST | GMM |
| 6 | ← | | AUTHENTICATION AND CIPHERING REQUEST | GMM |
| 7 | → | | AUTHENTICATION AND CIPHERING RESPONSE | GMM |
| 8 | ← | | SECURITY MODE COMMAND | RRC |
| 9 | → | | SECURITY MODE COMPLETE | RRC |
| 10 | ← | | ATTACH ACCEPT | GMM |
| 11 | → | | ATTACH COMPLETE | GMM |
| 12 | ← | | RRC CONNECTION RELEASE | RRC |
| 13 | → | | RRC CONNECTION RELEASE COMPLETE | RRC |

#### 7.2.2.2.4 Specific message contents

All Specific message contents shall be referred to clause 9.

### 7.2.2.3 Registration on CS / PS combined environment

#### 7.2.2.3.1 Initial condition

System Simulator:

- 1 cell operating in network operation mode I, default parameters.

User Equipment:

- The UE shall be operated under normal test conditions.

- The Test-USIM shall be inserted.

#### 7.2.2.3.2 Definition of system information messages

The default system information messages are used.

#### 7.2.2.3.3 Procedure UE establish PS registration immediately after the UE has been switched on

Registration of UE for SS shall be established under ideal radio conditions as defined in clause 5.

| Step | Direction | | Message | Comments |
|---|---|---|---|---|
| | UE | SS | | |
| 1 | | ← | SYSTEM INFORMATION (BCCH) | NW Broadcast |
| 2 | | → | RRC CONNECTION REQUEST (CCCH) | RRC |
| 3 | | ← | RRC CONNECTION SETUP (CCCH) | RRC |
| 4 | | → | RRC CONNECTION SETUP COMPLETE (DCCH) | RRC |
| 5 | | → | ATTACH REQUEST | GMM |
| 6 | | ← | AUTHENTICATION AND CIPHERING REQUEST | GMM |
| 7 | | → | AUTHENTICATION AND CIPHERING RESPONSE | GMM |
| 8 | | ← | SECURITY MODE COMMAND | RRC |
| 9 | | → | SECURITY MODE COMPLETE | RRC |
| 10 | | ← | ATTACH ACCEPT | GMM |
| 11 | | → | ATTACH COMPLETE | GMM |
| 12 | | ← | RRC CONNECTION RELEASE | RRC |
| 13 | | → | RRC CONNECTION RELEASE COMPLETE | RRC |

#### 7.2.2.3.3a Procedure UE establish PS registration later the user decides to use the PS services

CS registration has been successfully completed and RRC connection is released, cee clause 7.2.2.1. Registration of UE for SS shall be established under ideal radio conditions as defined in clause 5.

| Step | Direction | | Message | Comments |
|---|---|---|---|---|
| | UE | SS | | |
| 1 | | ← | SYSTEM INFORMATION (BCCH) | NW Broadcast |
| 1a | | | | The UE initiates an attach by MMI or by AT command. |
| 2 | | → | RRC CONNECTION REQUEST (CCCH) | RRC |
| 3 | | ← | RRC CONNECTION SETUP (CCCH) | RRC |
| 4 | | → | RRC CONNECTION SETUP COMPLETE (DCCH) | RRC |
| 5 | | → | ATTACH REQUEST | GMM |
| 6 | | ← | AUTHENTICATION AND CIPHERING REQUEST | GMM |
| 7 | | → | AUTHENTICATION AND CIPHERING RESPONSE | GMM |
| 8 | | ← | SECURITY MODE COMMAND | RRC |
| 9 | | → | SECURITY MODE COMPLETE | RRC |
| 10 | | ← | ATTACH ACCEPT | GMM |
| 11 | | → | ATTACH COMPLETE | GMM |
| 12 | | ← | RRC CONNECTION RELEASE | RRC |
| 13 | | → | RRC CONNECTION RELEASE COMPLETE | RRC |

#### 7.2.2.3.4 Specific message contents

All Specific message contents shall be referred to clause 9.

### 7.2.2.4 Registration on CS / PS non-combined environment

#### 7.2.2.4.1 Initial condition

System Simulator:

- 1 cell operating in network operation mode II, default parameters.

User Equipment:

- The UE set to Operation mode A
- The UE shall be operated under normal test conditions.
- The Test-USIM shall be inserted.

#### 7.2.2.4.2 Definition of system information messages

The default system information messages are used.

#### 7.2.2.4.3 Procedure

Registration of UE for SS shall be established under ideal radio conditions as defined in clause 5.

Registrations in the CS domain and in the PS domain shall execute independently. The separate registration procedures may occur sequentially or in parallel. If the procedures occur sequentially PS domain registration can be started immediately after power on or the UE can initiate PS registration by MMI or by AT command. If MMI or AT commands are used, registrations are done with two separate RRC connections. The procedures for CS and PS registration shall be as defined in clauses 7.2.2.1 and 7.2.2.2.

#### 7.2.2.4.4 Specific message contents

All Specific message contents shall be referred to clause 9.

## 7.2.3 Call setup

### 7.2.3.1 Generic call set up procedure for mobile terminating circuit switched calls

#### 7.2.3.1.1 Initial conditions

System Simulator:

- 1 cell, default parameters.

User Equipment:

- The UE shall be operated under normal test conditions.
- The Test-USIM shall be inserted.

#### 7.2.3.1.2 Definition of system information messages

The default system information messages are used.

#### 7.2.3.1.3 Procedure

The Call Set-up procedure shall be performed under Ideal radio conditions as defined in clause 5.

| Step | Direction | | Message | Comments |
|---|---|---|---|---|
| | UE | SS | | |
| 1 | | ← | SYSTEM INFORMATION (BCCH) | Broadcast |
| 2 | | ← | PAGING (PCCH) | Paging |
| 3 | | → | RRC CONNECTION REQUEST (CCCH) | RRC |
| 4 | | ← | RRC CONNECTION SETUP (CCCH) | RRC |
| 5 | | → | RRC CONNECTION SETUP COMPLETE (DCCH) | RRC |
| 6 | | → | PAGING RESPONSE | RR |
| 7 | | ← | AUTHENTICATION REQUEST | MM |
| 8 | | → | AUTHENTICATION RESPONSE | MM |
| 9 | | ← | SECURITY MODE COMMAND | RRC |
| 10 | | → | SECURITY MODE COMPLETE | RRC |
| 11 | | ← | SET UP | CC (see note) |
| 12 | | → | CALL CONFIRMED | CC |
| 13 | | ← | RADIO BEARER SETUP | RRC RAB SETUP |
| 14 | | → | RADIO BEARER SETUP COMPLETE | RRC |

| Step | Direction | | Message | Comments |
|---|---|---|---|---|
| | UE | SS | | |
| 15 | → | | ALERTING | CC (this message is optional) |
| 16 | → | | CONNECT | CC |
| 17 | ← | | CONNECT ACKNOWLEDGE | CC |
| NOTE: The "Signal" information element is not included in the SETUP message. | | | | |

#### 7.2.3.1.4 Specific message contents

All Specific message contents shall be referred to clause 9.

### 7.2.3.2 Generic call set-up procedure for mobile originating circuit switched calls

#### 7.2.3.2.1 Initial conditions

System Simulator:

- 1 cell, default parameters.

User Equipment:

- The UE shall be operated under normal test conditions.
- The Test-USIM shall be inserted.

#### 7.2.3.2.2 Definition of system information messages

The default system information messages are used.

#### 7.2.3.2.3 Procedure

The Call Set-up procedure shall be performed under Ideal radio conditions as defined in clause 5.

| Step | Direction | | Message | Comments |
|---|---|---|---|---|
| | UE | SS | | |
| 1 | ← | | SYSTEM INFORMATION (BCCH) | Broadcast |
| 2 | → | | RRC CONNECTION REQUEST (CCCH) | RRC |
| 3 | ← | | RRC CONNECTION SETUP (CCCH) | RRC |
| 4 | → | | RRC CONNECTION SETUP COMPLETE (DCCH) | RRC |
| 5 | → | | CM SERVICE REQUEST | MM |
| 6 | ← | | AUTHENTICATION REQUEST | MM |
| 7 | → | | AUTHENTICATION RESPONSE | MM |
| 8 | ← | | SECURITY MODE COMMAND | RRC |
| 9 | → | | SECURITY MODE COMPLETE | RRC |
| 10 | → | | SET UP | CC |
| 11 | ← | | CALL PROCEEDING | CC |
| 12 | ← | | RADIO BEARER SETUP | RRC RAB SETUP |
| 13 | → | | RADIO BEARER SETUP COMPLETE | RRC |
| 14 | ← | | ALERTING | CC |
| 15 | ← | | CONNECT | CC |
| 16 | → | | CONNECT ACKOWLEDGE | CC |

#### 7.2.3.2.4 Specific message contents

All Specific message contents shall be referred to clause 9.

## 7.2.4 Session setup

### 7.2.4.1 Generic session set up procedure for mobile terminating packet switched sessions

#### 7.2.4.1.1 Initial conditions

System Simulator:

- 1 cell, default parameters.

User Equipment:

- The UE shall be operated under normal test conditions.
- The Test-USIM shall be inserted.

#### 7.2.4.1.2 Definition of system information messages

The default system information messages are used.

#### 7.2.4.1.3 Procedure

The Session Set-up procedure shall be performed under Ideal radio conditions as defined in clause 5.

| Step | Direction | | Message | Comments |
|---|---|---|---|---|
| | UE | SS | | |
| 1 | | ← | SYSTEM INFORMATION (BCCH) | Broadcast |
| 2 | | ← | PAGING TYPE1 (PCCH) | Paging |
| 3 | | → | RRC CONNECTION REQUEST (CCCH) | RRC |
| 4 | | ← | RRC CONNECTION SETUP (CCCH) | RRC |
| 5 | | → | RRC CONNECTION SETUP COMPLETE (DCCH) | RRC |
| 6 | | → | SERVICE REQUEST | GMM |
| 7 | | ← | AUTHENTICATION AND CIPHERING REQUEST | GMM |
| 8 | | → | AUTHENTICATION AND CIPHERING RESPONSE | GMM |
| 9 | | ← | SECURITY MODE COMMAND | RRC |
| 10 | | → | SECURITY MODE COMPLETE | RRC |
| 11 | | ← | REQUEST PDP CONTEXT ACTIVATION | SM |
| 12 | | → | ACTIVATE PDP CONTEXT REQUEST | SM |
| 13 | | ← | RADIO BEARER SETUP | RRC RAB SETUP |
| 14 | | → | RADIO BEARER SETUP COMPLETE | RRC |
| 15 | | ← | ACTIVATE PDP CONTEXT ACCEPT | SM |

#### 7.2.4.1.4 Specific message contents

All Specific message contents shall be referred to clause 9.

### 7.2.4.2 Generic session set up procedure for mobile originating packet switched sessions

#### 7.2.4.2.1 Initial conditions

System Simulator:

- 1 cell, default parameters.

User Equipment:

- The UE shall be operated under normal test conditions.
- The Test-USIM shall be inserted.

#### 7.2.4.2.2 Definition of system information messages

The default system information messages are used.

#### 7.2.4.2.3 Procedure

The Session Set-up procedure shall be performed under Ideal radio conditions as defined in clause 5.

| Step | Direction | | Message | Comments |
|---|---|---|---|---|
| | UE | SS | | |

| | | | |
|---|---|---|---|
| 1 | ← | SYSTEM INFORMATION (BCCH) | Broadcast |
| 2 | → | RRC CONNECTION REQUEST (CCCH) | RRC |
| 3 | ← | RRC CONNECTION SETUP (CCCH) | RRC |
| 4 | → | RRC CONNECTION SETUP COMPLETE (DCCH) | RRC |
| 5 | → | SERVICE REQUEST | GMM |
| 6 | ← | AUTHENTICATION AND CIPHERING REQUEST | GMM |
| 7 | → | AUTHENTICATION AND CIPHERING RESPONSE | GMM |
| 8 | ← | SECURITY MODE COMMAND | RRC |
| 9 | → | SECURITY MODE COMPLETE | RRC |
| 10 | → | ACTIVATE PDP CONTEXT REQUEST | SM |
| 11 | ← | RADIO BEARER SETUP | RRC RAB SETUP |
| 12 | → | RADIO BEARER SETUP COMPLETE | RRC |
| 13 | ← | ACTIVATE PDP CONTEXT ACCEPT | SM |

#### 7.2.4.2.4 Specific message contents

All Specific message contents shall be referred to clause 9.

## 7.3 Test procedures for RF test

NOTE: In general parameters defined for specific test cases in 3GPP TS 34.121 [2] take priority over the default parameters defined in the present document.

### 7.3.1 UE Test States for RF testing

In this clause, the states of the UE for the test are defined. For RF testing the same UE test states as specified in section 7.2.1 apply plus an additional RB Test Mode State. The RB Test Mode State can be reached from the UE States 2, 3 and 7 according to section 7.2.1. For this RB Test Mode State the different protocols shall be in the following states:

| | RRC | CC | MM | SM | GMM |
|---|---|---|---|---|---|
| RB Test Mode State | connected | null | see Note | pdp-inactive | same as previous state |
| Note: The MM state is "MM connection active" if an RRC connection exists for the CS domain otherwise it is "same as previous state". | | | | | |

### 7.3.2 Test procedure for TX, RX and Performance Requirement (without handover)

#### 7.3.2.1 Initial conditions

System Simulator

- test cases using 1 cell:
  - 1cell, default parameters.
- other test cases using this test procedure:
  - Number of cells and parameters for specific tests are defined in 3GPP TS 34.121 [2] and take priority over the default parameters.

User Equipment:

- The UE shall initially be operated under normal RF test conditions if not otherwise stated in the initial conditions for the actual test case.
- The Test-USIM shall be inserted.
- The UE has a valid TMSI (CS) after the execution of the procedure described in clause 7.2.2.1.
- The UE has a valid P-TMSI (PS) after the execution of the procedure described in clause 7.2.2.2.

### 7.3.2.2 Definition of system information messages

The default system information messages specified in clause 6.1.0b are used with the following exceptions.

Contents of System information block type 1: RRC

| Information Element | Value/remark |
|---|---|
| - CN domain system information | |
| - CN domain identity | PS |
| - CHOICE CN Type | GSM-MAP |
| - CN domain specific NAS system information | |
| - GSM-MAP NAS system information | 00 00 |
| - CN domain specific DRX cycle length coefficient | 7 |
| - CN domain identity | CS |
| - CHOICE CN Type | GSM-MAP |
| - CN domain specific NAS system information | |
| - GSM-MAP NAS system information | 00(T3212 is set to infinity) 01 |
| - CN domain specific DRX cycle length coefficient | 7 |
| - UE Timers and constants in connected mode | |
| - T305 | Infinity |

Contents of System Information Block type 5 (FDD)

| Information Element | Value/remark |
|---|---|
| - Secondary CCPCH system information | |
| - Secondary CCPCH info | |
| - CHOICE mode | FDD |
| - Secondary scrambling code | Not Present |
| - STTD indicator | FALSE |
| - Spreading factor | 64 |
| - Code number | 2 |
| - Pilot symbol existence | FALSE |
| - TFCI existence | TRUE (default value) |
| - Fixed or Flexible position | Flexible (default value) |
| - Timing offset | Not Present<br>Absence of this IE is equivalent to default value 0 |

### 7.3.2.3 Procedure

For UE supporting CS

| Step | Direction | | Message | Comments |
|---|---|---|---|---|
| | UE | SS | | |
| 1 | ← | | SYSTEM INFORMATION (BCCH) | Broadcast |
| 2 | ← | | PAGING TYPE1 (PCCH) | Paging (CS domain, TMSI) |
| 3 | → | | RRC CONNECTION REQUEST (CCCH) | RRC |
| 4 | ← | | RRC CONNECTION SETUP (CCCH) | RRC |
| 5 | → | | RRC CONNECTION SETUP COMPLETE (DCCH) | RRC |
| 6 | → | | PAGING RESPONSE | RR |
| 7 | ← | | AUTHENTICATION REQUEST | MM |
| 8 | → | | AUTHENTICATION RESPONSE | MM |
| 9 | ← | | SECURITY MODE COMMAND | RRC |
| 10 | → | | SECURITY MODE COMPLETE | RRC |
| 11 | ← | | ACTIVATE RB TEST MODE | TC |
| 12 | → | | ACTIVATE RB TEST MODE COMPLETE | TC |
| 13 | ← | | RADIO BEARER SETUP | RRC (RAB SETUP) |
| 14 | → | | RADIO BEARER SETUP COMPLETE | RRC |
| 15 | ← | | CLOSE UE TEST LOOP (DCCH) | TC (UE test loop mode set up) |
| 16 | → | | CLOSE UE TEST LOOP COMPLETE | TC (confirms that loopback entities for the radio bearer(s) have been created and loop back is activated) |
| 17 | ← | | OPEN UE TEST LOOP | TC |
| 18 | → | | OPEN UE TEST LOOP COMPLETE | TC |
| 19 | ← | | RRC CONNECTION RELEASE | RRC |

| | | | |
|---|---|---|---|
| 20 | → | RRC CONNECTION RELEASE COMPLETE | RRC |

For UE supporting PS only

| Step | Direction | | Message | Comments |
|---|---|---|---|---|
| | UE | SS | | |
| 1 | | ← | SYSTEM INFORMATION (BCCH) | Broadcast |
| 2 | | ← | PAGING TYPE1 (PCCH) | Paging (PS domain, P-TMSI) |
| 3 | | → | RRC CONNECTION REQUEST (CCCH) | RRC |
| 4 | | ← | RRC CONNECTION SETUP (CCCH) | RRC |
| 5 | | → | RRC CONNECTION SETUP COMPLETE (DCCH) | RRC |
| 6 | | → | SERVICE REQUEST | GMM |
| 7 | | ← | AUTHENTICATION AND CIPHERING REQUEST | GMM |
| 8 | | → | AUTHENTICATION AND CIPHERING RESPONSE | GMM |
| 9 | | ← | SECURITY MODE COMMAND | RRC |
| 10 | | → | SECURITY MODE COMPLETE | RRC |
| 11 | | ← | ACTIVATE RB TEST MODE | TC |
| 12 | | → | ACTIVATE RB TEST MODE COMPLETE | TC |
| 13 | | ← | RADIO BEARER SETUP | RRC (RAB SETUP) |
| 14 | | → | RADIO BEARER SETUP COMPLETE | RRC |
| 15 | | ← | CLOSE UE TEST LOOP (DCCH) | TC (UE test loop mode set up) |
| 16 | | → | CLOSE UE TEST LOOP COMPLETE | TC (confirms that loopback entities for the radio bearer(s) have been created and loop back is activated) |
| 17 | | ← | OPEN UE TEST LOOP | TC |
| 18 | | → | OPEN UE TEST LOOP COMPLETE | TC |
| 19 | | ← | RRC CONNECTION RELEASE | RRC |
| 20 | | → | RRC CONNECTION RELEASE COMPLETE | RRC |

### 7.3.2.4 Specific message contents

The default message contents specified in clause 9.2 are used with the following exceptions.

#### 7.3.2.4.1 ATTACH ACCEPT

This message is sent from the SS to the UE, used for the UE supporting PS only.

Contents of Attach Accept message: GMM

| Information Element | Value/remark |
|---|---|
| Periodic RA update timer | E0 (timer is deactivated) |

#### 7.3.2.4.2 Reference measurement channels

The configurations of the reference measurement channels for RF tests are described in 3GPP TS 34.121 [2], annex C for FDD and 3GPP TS 34.122 [5], annex C for TDD.

#### 7.3.2.4.3 Void

#### 7.3.2.4.4 Compressed mode

[T.B.D]

#### 7.3.2.4.5 Transmit diversity mode

[T.B.D]

## 7.3.3 Test procedure for test cases using Cell_PCH or URA_PCH state

### 7.3.3.1 Initial conditions

System Simulator:

- Number of cells and parameters for specific tests are defined in 3GPP TS 34.121 [2] and take priority over the default parameters.

User Equipment:

- The UE shall be operated under RF test conditions.
- The Test-USIM shall be inserted.
- The UE has a valid TMSI (CS) after the execution of the procedure described in clause 7.2.2.1.
- The UE has a valid P-TMSI (PS) after the execution of the procedure described in clause 7.2.2.2.

### 7.3.3.2 Definition of system information messages

The default system information messages specified in clause 6.1.0b are used with the following exceptions.

Contents of System information block type 1: RRC

| Information Element | Value/remark |
|---|---|
| - CN domain system information | |
| - CN domain identity | PS |
| - CHOICE CN Type | GSM-MAP |
| - CN domain specific NAS system information | |
| - GSM-MAP NAS system information | 00 00 |
| - CN domain specific DRX cycle length coefficient | 7 |
| - CN domain identity | CS |
| - CHOICE CN Type | GSM-MAP |
| - CN domain specific NAS system information | |
| - GSM-MAP NAS system information | 00(T3212 is set to infinity) 01 |
| - CN domain specific DRX cycle length coefficient | 7 |
| - UE Timers and constants in connected mode | |
| - T305 | Infinity |

Contents of System Information Block type 5 (FDD)

| Information Element | Value/remark |
|---|---|
| - Secondary CCPCH system information | |
| - Secondary CCPCH info | |
| - CHOICE mode | FDD |
| - Secondary scrambling code | Not Present |
| - STTD indicator | FALSE |
| - Spreading factor | 64 |
| - Code number | 2 |
| - Pilot symbol existence | FALSE |
| - TFCI existence | TRUE (default value) |
| - Fixed or Flexible position | Flexible (default value) |
| - Timing offset | Not Present<br>Absence of this IE is equivalent to default value 0 |

### 7.3.3.3 Procedure

For UE supporting PS

| Step | Direction | | Message | Comments |
|---|---|---|---|---|
| | UE | SS | | |
| 1 | ← | | SYSTEM INFORMATION (BCCH) | Broadcast |
| 2 | ← | | PAGING TYPE1 (PCCH) | Paging (PS domain, P-TMSI) |
| 3 | → | | RRC CONNECTION REQUEST (CCCH) | RRC |
| 4 | ← | | RRC CONNECTION SETUP (CCCH) | RRC |
| 5 | → | | RRC CONNECTION SETUP COMPLETE (DCCH) | RRC |
| 6 | → | | SERVICE REQUEST | GMM |
| 7 | ← | | AUTHENTICATION AND CIPHERING REQUEST | GMM |
| 8 | → | | AUTHENTICATION AND CIPHERING RESPONSE | GMM |
| 9 | ← | | SECURITY MODE COMMAND | RRC |

| | | | |
|---|---|---|---|
| 10 | → | SECURITY MODE COMPLETE | RRC |
| 11 | ← | ACTIVATE RB TEST MODE | TC |
| 12 | → | ACTIVATE RB TEST MODE COMPLETE | TC |
| 13 | ← | RADIO BEARER SETUP | RRC<br>- RAB SETUP using Reference Radio Bearer Configuration |
| 14 | → | RADIO BEARER SETUP COMPLETE | RRC |
| 15 | ← | PHYSICAL CHANNEL RECONFIGURATION | RRC<br>- RRC state indicator is set to "Cell_PCH" or "URA_PCH" depending on the test case |
| 16 | → | PHYSICAL CHANNEL RECONFIGURATION COMPLETE | RRC<br>The UE sends this message before it completes state transition. |
| 17 | | Void | SS sends the L2 ack on the PHYSICAL CHANNEL RECONFIGURATION COMPLETE message.<br>NOTE: The SS should continue to keep the dedicated channel configuration during the time when the L2 ack is sent to the UE. |

### 7.3.3.4 Specific message contents

The default message contents specified in clause 9.2 are used with the following exceptions.

The RADIO BEARER SETUP message is defined in clause 9.2.1, "Contents of RADIO BEARER SETUP message: AM or UM (UE supports PS RAB only)".

The PHYSICAL CHANNEL RECONFIGURATION message is defined in clause 9.1.1, "Contents of PHYSICAL CHANNEL RECONFIGURATION message: AM or UM" using condition A8 for URA_PCH and condition A10 for Cell_PCH.

Contents of Attach Accept message: GMM

| Information Element | Value/remark |
|---|---|
| Periodic RA update timer | E0 (timer is deactivated) |

## 7.3.4 Test procedure for Handover

NOTE: This test procedure is also used for some other test cases involving more than 1 cell.

### 7.3.4.1 Initial conditions

System Simulator:

- Intra-frequency hard handover and soft handover (for FDD) case:
  - 2 cells, default parameters according to Cell 1 and Cell 2 in clause 6.1.4.
- Inter-frequency hard handover case:
  - 2 cells, default parameters according to Cell 1 and Cell 4 in clause 6.1.4.
- Inter-system handover UTRAN to GSM case:
  - 2 cells, default parameters according to Cell 1 and Cell 9 in clause 6.1.4.
- other test cases using this test procedure:
  - Number of cells and parameters for specific tests are defined in 3GPP TS 34.121 [2] for FDD and TS 34.122 [5] for TDD and take priority over the default parameters.

UserEquipment:

- The UE shall be initially operated under the normal RF test conditions if not otherwise stated in the initial conditions for the actual test case.
- The Test-USIM shall be inserted.
- The UE has a valid TMSI (CS) after the execution of the procedure described in clause 7.2.2.1.
- The UE has a valid P-TMSI (PS) after the execution of the procedure described in clause 7.2.2.2.

### 7.3.4.2 Definition of system information messages

The default system information messages specified in clause 6.1.0b are used with the following exceptions.

Contents of System information block type 1: RRC

| Information Element | Value/remark |
|---|---|
| - CN domain system information | |
| - CN domain identity | PS |
| - CHOICE CN Type | GSM-MAP |
| - CN domain specific NAS system information | |
| - GSM-MAP NAS system information | 00 00 |
| - CN domain specific DRX cycle length coefficient | 7 |
| - CN domain identity | CS |
| - CHOICE CN Type | GSM-MAP |
| - CN domain specific NAS system information | |
| - GSM-MAP NAS system information | 00(T3212 is set to infinity) 01 |
| - CN domain specific DRX cycle length coefficient | 7 |
| - UE Timers and constants in connected mode | |
| - T305 | Infinity |

Contents of System Information Block type 5 (FDD)

| Information Element | Value/remark |
|---|---|
| - Secondary CCPCH system information | |
| - Secondary CCPCH info | |
| - CHOICE mode | FDD |
| - Secondary scrambling code | Not Present |
| - STTD indicator | FALSE |
| - Spreading factor | 64 |
| - Code number | 2 |
| - Pilot symbol existence | FALSE |
| - TFCI existence | TRUE (default value) |
| - Fixed or Flexible position | Flexible (default value) |
| - Timing offset | Not Present<br>Absence of this IE is equivalent to default value 0 |

For the intra-frequency hard handover and soft handover (for FDD) case the default messages for SIB11 and SIB12 as specified for Cell 1 and Cell 2 in clause 6.1.4 are used.

For the inter-frequency hard handover case the default messages for SIB11 and SIB12 as specified for Cell 1 and Cell 4 in clause 6.1.4 are used.

For the inter-system handover from UTRAN to GSM case the default messages for SIB11 and SIB12 as specified for Cell 1 and Cell 9 in clause 6.1.4 are used.

### 7.3.4.3 Procedure

For UE supporting CS

| Step | Direction | | Message | Comments |
|---|---|---|---|---|
| | UE | SS | | |
| 1 | | ← | SYSTEM INFORMATION (BCCH) | Broadcast |
| 2 | | ← | PAGING TYPE1 (PCCH) | Paging (CS domain, TMSI) |
| 3 | | → | RRC CONNECTION REQUEST (CCCH) | RRC |
| 4 | | ← | RRC CONNECTION SETUP (CCCH) | RRC |

| | | | |
|---|---|---|---|
| 5 | → | RRC CONNECTION SETUP COMPLETE (DCCH) | RRC |
| 6 | → | PAGING RESPONSE | RR |
| 7 | ← | AUTHENTICATION REQUEST | MM |
| 8 | → | AUTHENTICATION RESPONSE | MM |
| 9 | ← | SECURITY MODE COMMAND | RRC |
| 10 | → | SECURITY MODE COMPLETE | RRC |
| 11 | ← | ACTIVATE RB TEST MODE | TC |
| 12 | → | ACTIVATE RB TEST MODE COMPLETE | TC |
| 13 | ← | RADIO BEARER SETUP | RRC<br>- RAB SETUP using Reference Radio Bearer Configuration<br>- RRC state indicator is set to "CELL_DCH" |
| 14 | → | RADIO BEARER SETUP COMPLETE | RRC |
| 15 | ← | RRC CONNECTION RELEASE | RRC |
| 16 | → | RRC CONNECTION RELEASE COMPLETE | RRC |

For UE supporting PS only

| Step | Direction | | Message | Comments |
|---|---|---|---|---|
| | UE | SS | | |
| 1 | ← | | SYSTEM INFORMATION (BCCH) | Broadcast |
| 2 | ← | | PAGING TYPE1 (PCCH) | Paging (PS domain, P-TMSI) |
| 3 | → | | RRC CONNECTION REQUEST (CCCH) | RRC |
| 4 | ← | | RRC CONNECTION SETUP (CCCH) | RRC |
| 5 | → | | RRC CONNECTION SETUP COMPLETE (DCCH) | RRC |
| 6 | → | | SERVICE REQUEST | GMM |
| 7 | ← | | AUTHENTICATION AND CIPHERING REQUEST | GMM |
| 8 | → | | AUTHENTICATION AND CIPHERING RESPONSE | GMM |
| 9 | ← | | SECURITY MODE COMMAND | RRC |
| 10 | → | | SECURITY MODE COMPLETE | RRC |
| 11 | ← | | ACTIVATE RB TEST MODE | TC |
| 12 | → | | ACTIVATE RB TEST MODE COMPLETE | TC |
| 13 | ← | | RADIO BEARER SETUP | RRC<br>- RAB SETUP using Reference Radio Bearer Configuration<br>- RRC state indicator is set to "CELL_DCH" |
| 14 | → | | RADIO BEARER SETUP COMPLETE | RRC |
| 15 | ← | | RRC CONNECTION RELEASE | RRC |
| 16 | → | | RRC CONNECTION RELEASE COMPLETE | RRC |

### 7.3.4.4 Specific message contents

The default message contents specified in clause 9.2 are used with the following exceptions.

Contents of Attach Accept message: GMM

| Information Element | Value/remark |
|---|---|
| Periodic RA update timer | E0 (timer is deactivated) |

## 7.3.5 Test procedure for test cases using CELL_FACH state

### 7.3.5.1 Initial conditions

System Simulator:

- Number of cells and parameters for specific tests are defined in TS 34.121 [2] and take priority over the default parameters.

User Equipment:

- The UE shall be operated under RF test conditions.
- The Test-USIM shall be inserted.
- The UE has a valid TMSI (CS) after the execution of the procedure described in clause 7.2.2.1.
- The UE has a valid P-TMSI (PS) after the execution of the procedure described in clause 7.2.2.2.

### 7.3.5.2 Definition of system information messages

The default system information messages specified in clause 6.1.0b are used with the following exceptions.

Contents of System information block type 1: RRC

| Information Element | Value/remark |
|---|---|
| - CN domain system information | |
| - CN domain identity | PS |
| - CHOICE CN Type | GSM-MAP |
| - CN domain specific NAS system information | |
| - GSM-MAP NAS system information | 00 00 |
| - CN domain specific DRX cycle length coefficient | 7 |
| - CN domain identity | CS |
| - CHOICE CN Type | GSM-MAP |
| - CN domain specific NAS system information | |
| - GSM-MAP NAS system information | 00(T3212 is set to infinity) 01 |
| - CN domain specific DRX cycle length coefficient | 7 |
| - UE Timers and constants in connected mode | |
| - T305 | Infinity |

Contents of System Information Block type 5 (FDD)

| Information Element | Value/remark |
|---|---|
| - Secondary CCPCH system information | |
| - Secondary CCPCH info | |
| - CHOICE mode | FDD |
| - Secondary scrambling code | Not Present |
| - STTD indicator | FALSE |
| - Spreading factor | 64 |
| - Code number | 2 |
| - Pilot symbol existence | FALSE |
| - TFCI existence | TRUE (default value) |
| - Fixed or Flexible position | Flexible (default value) |
| - Timing offset | Not Present<br>Absence of this IE is equivalent to default value 0 |

### 7.3.5.3 Procedure

For UE supporting CS

| Step | Direction | | Message | Comments |
|---|---|---|---|---|
| | UE | SS | | |
| 1 | ← | | SYSTEM INFORMATION (BCCH) | Broadcast |
| 2 | ← | | PAGING TYPE1 (PCCH) | Paging (CS domain, TMSI) |
| 3 | → | | RRC CONNECTION REQUEST (CCCH) | RRC |
| 4 | ← | | RRC CONNECTION SETUP (CCCH) | RRC |
| 5 | → | | RRC CONNECTION SETUP COMPLETE (DCCH) | RRC |
| 6 | → | | PAGING RESPONSE | RR |
| 7 | ← | | AUTHENTICATION REQUEST | MM |
| 8 | → | | AUTHENTICATION RESPONSE | MM |
| 9 | ← | | SECURITY MODE COMMAND | RRC |
| 10 | → | | SECURITY MODE COMPLETE | RRC |
| 11 | ← | | ACTIVATE RB TEST MODE | TC |
| 12 | → | | ACTIVATE RB TEST MODE COMPLETE | TC |
| 13 | ← | | DEACTIVATE RB TEST MODE | TC |
| 14 | → | | DEACTIVATE RB TEST MODE COMPLETE | TC |

| | | | |
|---|---|---|---|
| 15 | ← | RRC CONNECTION RELEASE | RRC |
| 16 | → | RRC CONNECTION RELEASE COMPLETE | RRC |

For UE supporting PS only

| Step | Direction | | Message | Comments |
|---|---|---|---|---|
| | UE | SS | | |
| 1 | ← | | SYSTEM INFORMATION (BCCH) | Broadcast |
| 2 | ← | | PAGING TYPE1 (PCCH) | Paging (PS domain, P-TMSI) |
| 3 | → | | RRC CONNECTION REQUEST (CCCH) | RRC |
| 4 | ← | | RRC CONNECTION SETUP (CCCH) | RRC |
| 5 | → | | RRC CONNECTION SETUP COMPLETE (DCCH) | RRC |
| 6 | → | | SERVICE REQUEST | GMM |
| 7 | ← | | AUTHENTICATION AND CIPHERING REQUEST | GMM |
| 8 | → | | AUTHENTICATION AND CIPHERING RESPONSE | GMM |
| 9 | ← | | SECURITY MODE COMMAND | RRC |
| 10 | → | | SECURITY MODE COMPLETE | RRC |
| 11 | ← | | ACTIVATE RB TEST MODE | TC |
| 12 | → | | ACTIVATE RB TEST MODE COMPLETE | TC |
| 13 | ← | | DEACTIVATE RB TEST MODE | TC |
| 14 | → | | DEACTIVATE RB TEST MODE COMPLETE | TC |
| 15 | ← | | RRC CONNECTION RELEASE | RRC |
| 16 | → | | RRC CONNECTION RELEASE COMPLETE | RRC |

### 7.3.5.4 Specific message contents

The default message contents specified in clause 9.2 are used with the following exceptions.

Contents of Attach Accept message: GMM

| Information Element | Value/remark |
|---|---|
| Periodic RA update timer | E0 (timer is deactivated) |

The RRC connection setup is defined in clause 9.1.1, "Contents of RRC CONNECTION SETUP message: UM (Transition to CELL_FACH)".

## 7.3.6 Test procedure for HSDPA RF Performance Requirement

### 7.3.6.1 Initial conditions

System Simulator:

- 1 HS-DSCH cell, default parameters.

User Equipment:

- The UE shall initially be operated under normal RF test conditions if not otherwise stated in the initial conditions for the actual test case.
- The Test-USIM shall be inserted.
- The UE has a valid TMSI (CS) after the execution of the procedure described in clause 7.2.2.1.
- The UE has a valid P-TMSI (PS) after the execution of the procedure described in clause 7.2.2.2.

### 7.3.6.2 Definition of system information messages

The default system information messages specified in clause 6.1.0b are used with the following exceptions.

Contents of System information block type 1: RRC

| Information Element | Value/remark |
|---|---|

| Information Element | Value/remark |
|---|---|
| - CN domain system information | |
| - CN domain identity | PS |
| - CHOICE CN Type | GSM-MAP |
| - CN domain specific NAS system information | |
| - GSM-MAP NAS system information | 00 00 |
| - CN domain specific DRX cycle length coefficient | 7 |
| - CN domain identity | CS |
| - CHOICE CN Type | GSM-MAP |
| - CN domain specific NAS system information | |
| - GSM-MAP NAS system information | 00(T3212 is set to infinity) 01 |
| - CN domain specific DRX cycle length coefficient | 7 |
| - UE Timers and constants in connected mode | |
| - T305 | Infinity |

Contents of System Information Block type 5 (FDD)

| Information Element | Value/remark |
|---|---|
| - Secondary CCPCH system information | |
| - Secondary CCPCH info | |
| - CHOICE mode | FDD |
| - Secondary scrambling code | Not Present |
| - STTD indicator | FALSE |
| - Spreading factor | 64 |
| - Code number | 2 |
| - Pilot symbol existence | FALSE |
| - TFCI existence | TRUE (default value) |
| - Fixed or Flexible position | Flexible (default value) |
| - Timing offset | Not Present<br>Absence of this IE is equivalent to default value 0 |

### 7.3.6.3 Procedure

| Step | Direction | | Message | Comments |
|---|---|---|---|---|
| | UE | SS | | |
| 1 | ← | | SYSTEM INFORMATION (BCCH) | Broadcast |
| 2 | ← | | PAGING TYPE1 (PCCH) | Paging (CS domain, TMSI) |
| 3 | → | | RRC CONNECTION REQUEST (CCCH) | RRC |
| 4 | ← | | RRC CONNECTION SETUP (CCCH) | RRC |
| 5 | → | | RRC CONNECTION SETUP COMPLETE (DCCH) | RRC |
| 6 | → | | PAGING RESPONSE | RR |
| 7 | ← | | AUTHENTICATION REQUEST | MM |
| 8 | → | | AUTHENTICATION RESPONSE | MM |
| 9 | ← | | SECURITY MODE COMMAND | RRC (CS domain) |
| 10 | → | | SECURITY MODE COMPLETE | RRC |
| 11 | ← | | PAGING TYPE2 (DCCH) | Paging (TMSI (GSM-MAP)/ P-TMSI) |
| 12 | → | | SERVICE REQUEST | GMM |
| 13 | ← | | AUTHENTICATION AND CIPHERING REQUEST | GMM |
| 14 | → | | AUTHENTICATION AND CIPHERING RESPONSE | GMM |
| 15 | ← | | SECURITY MODE COMMAND | RRC (PS domain, IE Integrity protection mode command set to "modify") |
| 16 | → | | SECURITY MODE COMPLETE | RRC |
| 17 | ← | | ACTIVATE RB TEST MODE | TC |
| 18 | → | | ACTIVATE RB TEST MODE COMPLETE | TC |
| 19 | ← | | RADIO BEARER SETUP | RRC (RAB SETUP RMC 12.2 CS) |
| 20 | → | | RADIO BEARER SETUP COMPLETE | RRC |
| 21 | ← | | RADIO BEARER SETUP | RRC (RAB SETUP HSDPA PS) |
| 22 | → | | RADIO BEARER SETUP COMPLETE | RRC |

| | | | |
|---|---|---|---|
| A23 | ← | CLOSE UE TEST LOOP (DCCH) | TC (UE test loop mode set up) . Test steps A23, A24, A26 and A27 are only executed when the test method in TS 34.121 [2] specifies that loopback test shall be used. |
| A24 | → | CLOSE UE TEST LOOP COMPLETE | TC (confirms that loopback entities for the radio bearer(s) have been created and loop back is activated) |
| 25 | <--> | | Perform test |
| A26 | ← | OPEN UE TEST LOOP | TC |
| A27 | → | OPEN UE TEST LOOP COMPLETE | TC |
| 28 | ← | RRC CONNECTION RELEASE | RRC |
| 29 | → | RRC CONNECTION RELEASE COMPLETE | RRC |

For a transition period until RAN5#31 the procedure below can be used. This alternative procedure will be removed at RAN5#31.

| Step | Direction | | Message | Comments |
|---|---|---|---|---|
| | UE | SS | | |
| 1 | ← | | SYSTEM INFORMATION (BCCH) | Broadcast |
| 2 | ← | | PAGING TYPE1 (PCCH) | Paging (PS domain, P-TMSI) |
| 3 | → | | RRC CONNECTION REQUEST (CCCH) | RRC |
| 4 | ← | | RRC CONNECTION SETUP (CCCH) | RRC |
| 5 | → | | RRC CONNECTION SETUP COMPLETE (DCCH) | RRC |
| 6 | → | | SERVICE REQUEST | GMM |
| 7 | ← | | AUTHENTICATION AND CIPHERING REQUEST | GMM |
| 8 | → | | AUTHENTICATION AND CIPHERING RESPONSE | GMM |
| 9 | ← | | SECURITY MODE COMMAND | RRC |
| 10 | → | | SECURITY MODE COMPLETE | RRC |
| 11 | ← | | ACTIVATE RB TEST MODE | TC |
| 12 | → | | ACTIVATE RB TEST MODE COMPLETE | TC |
| 13 | ← | | RADIO BEARER SETUP | RRC (RAB SETUP) |
| 14 | → | | RADIO BEARER SETUP COMPLETE | RRC |
| A15 | ← | | CLOSE UE TEST LOOP (DCCH) | TC (UE test loop mode set up) . Test steps A15, A16, A18 and A19 are only executed when the test method in TS 34.121 [2] specifies that loopback test shall be used. |
| A16 | → | | CLOSE UE TEST LOOP COMPLETE | TC (confirms that loopback entities for the radio bearer(s) have been created and loop back is activated) |
| 17 | <--> | | | Perform test |
| A18 | ← | | OPEN UE TEST LOOP | TC |
| A19 | → | | OPEN UE TEST LOOP COMPLETE | TC |
| 20 | ← | | RRC CONNECTION RELEASE | RRC |
| 21 | → | | RRC CONNECTION RELEASE COMPLETE | RRC |

### 7.3.6.4 Specific message contents

The default message contents specified in clause 9.2 are used with the following exceptions.

#### 7.3.6.4.1 ATTACH ACCEPT

This message is sent from the SS to the UE.

Contents of Attach Accept message: GMM

| Information Element | Value/remark |
|---|---|
| Periodic RA update timer | E0 (timer is deactivated) |

#### 7.3.6.4.2 RADIO BEARER SETUP

For step 19, the message in clause 9.2, "Contents of RADIO BEARER SETUP message: AM or UM (Test Loop Mode1)" is used with condition A1. For step 21, the message in clause 9.2, "Contents of RADIO BEARER SETUP message: AM or UM (HSDPA)" is used.

The configurations of the fixed reference channels for HSDPA RF tests are described in 3GPP TS 34.121[2], annex C for FDD and 3GPP TS 34.122 [5], annex C for TDD.

For a transition period until RAN5#31 the message below can be used. This alternative message will be removed at RAN5#31.

For step 13 in the alternative procedure, the messages in clause 9.2, "Contents of RADIO BEARER SETUP message: AM or UM (HSDPA)" is used.

#### 7.3.6.4.3 RRC CONNECTION SETUP

For step 4 , the message in clause 9.2, "Contents of RRC CONNECTION SETUP message: UM" is used with the following exceptions:

Contents of RRC CONNECTION SETUP message: UM

| Information Element | Value/remark |
|---|---|
| - Default DPCH Offset Value | Arbitrary set to value 1536..306176 by step of 2560 (this corresponds to a 0.5 slot timing offset between the DPCCH and the HS-DPCCH) |

## 7.3.7 Test procedure for inter-RAT handover used in RRM testing

### 7.3.7.1 Initial conditions

System Simulator:

- 2 cells, default parameters according to Cell 1 and Cell 9 in clause 6.1.4.

UserEquipment:

- The UE shall be initially operated under the normal RF test conditions if not otherwise stated in the initial conditions for the actual test case.
- The Test-USIM shall be inserted.
- The UE has a valid TMSI (CS) after the execution of the procedure described in clause 7.2.2.1.
- The UE has a valid P-TMSI (PS) after the execution of the procedure described in clause 7.2.2.2.

### 7.3.7.2 Definition of system information messages

The default system information messages specified in clause 6.1.0b are used with the following exceptions.

Contents of System information block type 1: RRC

| Information Element | Value/remark |
|---|---|
| - CN domain system information | |
| - CN domain identity | PS |
| - CHOICE CN Type | GSM-MAP |
| - CN domain specific NAS system information | |
| - GSM-MAP NAS system information | 00 00 |
| - CN domain specific DRX cycle length coefficient | 7 |
| - CN domain identity | CS |
| - CHOICE CN Type | GSM-MAP |
| - CN domain specific NAS system information | |
| - GSM-MAP NAS system information | 00(T3212 is set to infinity) 01 |
| - CN domain specific DRX cycle length coefficient | 7 |
| - UE Timers and constants in connected mode | |

| Information Element | Value/remark |
|---|---|
| - T305 | Infinity |

Contents of System Information Block type 5 (FDD)

| Information Element | Value/remark |
|---|---|
| - Secondary CCPCH system information | |
| - Secondary CCPCH info | |
| - CHOICE mode | FDD |
| - Secondary scrambling code | Not Present |
| - STTD indicator | FALSE |
| - Spreading factor | 64 |
| - Code number | 2 |
| - Pilot symbol existence | FALSE |
| - TFCI existence | TRUE (default value) |
| - Fixed or Flexible position | Flexible (default value) |
| - Timing offset | Not Present<br>Absence of this IE is equivalent to default value 0 |

For the inter-system handover from UTRAN FDD to GSM case the default messages for SIB11 and SIB12 as specified for Cell 1 and Cell 9 in clause 6.1.4 are used.

### 7.3.7.3 Procedure

| Step | Direction | | Message | Comments |
|---|---|---|---|---|
| | UE | SS | | |
| 1 | ← | | SYSTEM INFORMATION (BCCH) | Broadcast |
| 2 | ← | | PAGING (PCCH) | Paging |
| 3 | → | | RRC CONNECTION REQUEST (CCCH) | RRC |
| 4 | ← | | RRC CONNECTION SETUP (CCCH) | RRC (Transition to cell DCH) |
| 5 | → | | RRC CONNECTION SETUP COMPLETE (DCCH) | RRC |
| 6 | → | | PAGING RESPONSE | RR |
| 7 | ← | | AUTHENTICATION REQUEST | MM |
| 8 | → | | AUTHENTICATION RESPONSE | MM |
| 9 | ← | | SECURITY MODE COMMAND | RRC |
| 10 | → | | SECURITY MODE COMPLETE | RRC |
| 11 | ← | | SET UP | CC (see note) |
| 12 | → | | CALL CONFIRMED | CC |
| 13 | ← | | RADIO BEARER SETUP | RRC RAB SETUP |
| 14 | → | | RADIO BEARER SETUP COMPLETE | RRC |
| 15 | → | | ALERTING | CC (this message is optional) |
| 16 | → | | CONNECT | CC |
| 17 | ← | | CONNECT ACKNOWLEDGE | CC |
| NOTE: The "Signal" information element is not included in the SETUP message. | | | | |

### 7.3.7.4 Specific message contents

The default message contents specified in clause 9.1 are used with the following exceptions.

Contents of Attach Accept message: GMM

| Information Element | Value/remark |
|---|---|
| Periodic RA update timer | E0 (timer is deactivated) |

## 7.3.8 Test procedure for inter-RAT cell FACH reselection used in RRM testing

### 7.3.8.1 Initial conditions

System Simulator:

- Number of cells and parameters for specific tests are defined in TS 34.121 [2] and take priority over the default parameters.

User Equipment:

- The UE shall be operated under RF test conditions.
- The Test-USIM shall be inserted.
- The UE has a valid TMSI (CS) after the execution of the procedure described in clause 7.2.2.1.
- The UE has a valid P-TMSI (PS) after the execution of the procedure described in clause 7.2.2.2.

### 7.3.8.2 Definition of system information messages

The default system information messages specified in clause 6.1.0b are used with the following exceptions.

Contents of System information block type 1: RRC

| Information Element | Value/remark |
|---|---|
| - CN domain system information | |
| - CN domain identity | PS |
| - CHOICE CN Type | GSM-MAP |
| - CN domain specific NAS system information | |
| - GSM-MAP NAS system information | 00 00 |
| - CN domain specific DRX cycle length coefficient | 7 |
| - CN domain identity | CS |
| - CHOICE CN Type | GSM-MAP |
| - CN domain specific NAS system information | |
| - GSM-MAP NAS system information | 00(T3212 is set to infinity) 01 |
| - CN domain specific DRX cycle length coefficient | 7 |
| - UE Timers and constants in connected mode | |
| - T305 | Infinity |

Contents of System Information Block type 5 (FDD)

| Information Element | Value/remark |
|---|---|
| - Secondary CCPCH system information | |
| - Secondary CCPCH info | |
| - CHOICE mode | FDD |
| - Secondary scrambling code | Not Present |
| - STTD indicator | FALSE |
| - Spreading factor | 64 |
| - Code number | 2 |
| - Pilot symbol existence | FALSE |
| - TFCI existence | TRUE (default value) |
| - Fixed or Flexible position | Flexible (default value) |
| - Timing offset | Not Present<br>Absence of this IE is equivalent to default value 0 |

### 7.3.8.3 Procedure

| Step | Direction | | Message | Comments |
|---|---|---|---|---|
| | UE | SS | | |
| 1 | ← | | SYSTEM INFORMATION (BCCH) | Broadcast |
| 2 | → | | RRC CONNECTION REQUEST (CCCH) | RRC |
| 3 | ← | | RRC CONNECTION SETUP (CCCH) | RRC |
| 4 | → | | RRC CONNECTION SETUP COMPLETE (DCCH) | RRC |
| 5 | → | | SERVICE REQUEST | GMM |
| 6 | ← | | AUTHENTICATION AND CIPHERING REQUEST | GMM |
| 7 | → | | AUTHENTICATION AND CIPHERING RESPONSE | GMM |
| 8 | ← | | SECURITY MODE COMMAND | RRC |
| 9 | → | | SECURITY MODE COMPLETE | RRC |
| 10 | → | | ACTIVATE PDP CONTEXT REQUEST | SM |
| 11 | ← | | RADIO BEARER SETUP | RRC RAB SETUP |

| | | | |
|---|---|---|---|
| 12 | → | RADIO BEARER SETUP COMPLETE | RRC |
| 13 | ← | ACTIVATE PDP CONTEXT ACCEPT | SM |

### 7.3.8.4 Specific message contents

The default message contents specified in clause 9.1 are used with the following exceptions.

Contents of Attach Accept message: GMM

| Information Element | Value/remark |
|---|---|
| Periodic RA update timer | E0 (timer is deactivated) |

The RRC connection setup is defined in clause 9.1.1, "Contents of RRC CONNECTION SETUP message: UM (Transition to CELL_FACH)".

## 7.3.9 Test procedure for E-DCH RF test cases

### 7.3.9.1 Initial conditions

System Simulator:

- 1 HS-DSCH plus E-DCH cell, default parameters.

User Equipment:

- The UE shall initially be operated under normal RF test conditions if not otherwise stated in the initial conditions for the actual test case.
- The Test-USIM shall be inserted.
- The UE has a valid TMSI (CS) after the execution of the procedure described in clause 7.2.2.1.
- The UE has a valid P-TMSI (PS) after the execution of the procedure described in clause 7.2.2.2.

### 7.3.9.2 Definition of system information messages

The default system information messages specified in clause 6.1.0b are used with the following exceptions.

Contents of System information block type 1: RRC

| Information Element | Value/remark |
|---|---|
| - CN domain system information | |
| - CN domain identity | PS |
| - CHOICE CN Type | GSM-MAP |
| - CN domain specific NAS system information | |
| - GSM-MAP NAS system information | 00 00 |
| - CN domain specific DRX cycle length coefficient | 7 |
| - CN domain identity | CS |
| - CHOICE CN Type | GSM-MAP |
| - CN domain specific NAS system information | |
| - GSM-MAP NAS system information | 00(T3212 is set to infinity) 01 |
| - CN domain specific DRX cycle length coefficient | 7 |
| - UE Timers and constants in connected mode | |
| - T305 | Infinity |

Contents of System Information Block type 5 (FDD)

| Information Element | Value/remark |
|---|---|
| - Secondary CCPCH system information | |
| - Secondary CCPCH info | |
| - CHOICE mode | FDD |
| - Secondary scrambling code | Not Present |
| - STTD indicator | FALSE |

| | |
|---|---|
| - Spreading factor | 64 |
| - Code number | 2 |
| - Pilot symbol existence | FALSE |
| - TFCI existence | TRUE (default value) |
| - Fixed or Flexible position | Flexible (default value) |
| - Timing offset | Not Present<br>Absence of this IE is equivalent to default value 0 |

### 7.3.9.3 Procedure

| Step | Direction | | Message | Comments |
|---|---|---|---|---|
| | UE | SS | | |
| 1 | | ← | SYSTEM INFORMATION (BCCH) | Broadcast |
| 2 | | ← | PAGING TYPE1 (PCCH) | Paging (CS domain, TMSI) |
| 3 | | → | RRC CONNECTION REQUEST (CCCH) | RRC |
| 4 | | ← | RRC CONNECTION SETUP (CCCH) | RRC |
| 5 | | → | RRC CONNECTION SETUP COMPLETE (DCCH) | RRC |
| 6 | | → | PAGING RESPONSE | RR |
| 7 | | ← | AUTHENTICATION REQUEST | MM |
| 8 | | → | AUTHENTICATION RESPONSE | MM |
| 9 | | ← | SECURITY MODE COMMAND | RRC (CS domain) |
| 10 | | → | SECURITY MODE COMPLETE | RRC |
| 11 | | ← | PAGING TYPE2 (DCCH) | Paging (TMSI (GSM-MAP)/ P-TMSI) |
| 12 | | → | SERVICE REQUEST | GMM |
| 13 | | ← | AUTHENTICATION AND CIPHERING REQUEST | GMM |
| 14 | | → | AUTHENTICATION AND CIPHERING RESPONSE | GMM |
| 15 | | ← | SECURITY MODE COMMAND | RRC (PS domain, IE Integrity protection mode command set to "modify") |
| 16 | | → | SECURITY MODE COMPLETE | RRC |
| 17 | | ← | ACTIVATE RB TEST MODE | TC |
| 18 | | → | ACTIVATE RB TEST MODE COMPLETE | TC |
| 19 | | ← | RADIO BEARER SETUP | RRC (RAB SETUP RMC 12.2 CS) |
| 20 | | → | RADIO BEARER SETUP COMPLETE | RRC |
| 21 | | ← | RADIO BEARER SETUP | RRC (RAB SETUP HSDPA and E-DCH PS) |
| 22 | | → | RADIO BEARER SETUP COMPLETE | RRC |
| A23 | | ← | CLOSE UE TEST LOOP (DCCH) | TC (UE test loop mode set up) . Test steps A23, A24, A26 and A27 are only executed when the test method in TS 34.121 [2] specifies that loopback test shall be used. |
| A24 | | → | CLOSE UE TEST LOOP COMPLETE | TC (confirms that loopback entities for the radio bearer(s) have been created and loop back is activated) |
| 25 | | <--> | | Perform test |
| A26 | | ← | OPEN UE TEST LOOP | TC |
| A27 | | → | OPEN UE TEST LOOP COMPLETE | TC |
| 28 | | ← | RRC CONNECTION RELEASE | RRC |
| 29 | | → | RRC CONNECTION RELEASE COMPLETE | RRC |

#### 7.3.9.4 Specific message contents

The default message contents specified in clause 9.2 are used with the following exceptions.

##### 7.3.9.4.1 ATTACH ACCEPT

This message is sent from the SS to the UE.

Contents of Attach Accept message: GMM

| Information Element | Value/remark |
|---|---|
| Periodic RA update timer | E0 (timer is deactivated) |

##### 7.3.9.4.2 RADIO BEARER SETUP

For step 19, the message in clause 9.2, "Contents of RADIO BEARER SETUP message: AM or UM (Test Loop Mode1)" is used with condition A1. For step 21, the message in clause 9.2, "Contents of RADIO BEARER SETUP message: AM or UM (E-DCH and HSDPA)" is used.

The configurations of the fixed reference channels for HSDPA RF tests are described in 3GPP TS 34.121[2], annex C for FDD and 3GPP TS 34.122 [5], annex C for TDD. The configurations of the reference channels for E-DCH RF tests are described in 3GPP TS 34.121[2].

##### 7.3.9.4.3 RRC CONNECTION SETUP

For step 4 , the messages in clause 9.2, "Contents of RRC CONNECTION SETUP message: UM" is used with the following exceptions:

Contents of RRC CONNECTION SETUP message: UM

| Information Element | Value/remark |
|---|---|
| - Default DPCH Offset Value | Arbitrary set to value 1536..306176 by step of 2560 (this corresponds to a 0.5 slot timing offset between the DPCCH and the HS-DPCCH) |

## 7.4 Common generic procedures for AS testing

### 7.4.1 UE RRC Test States for common procedures

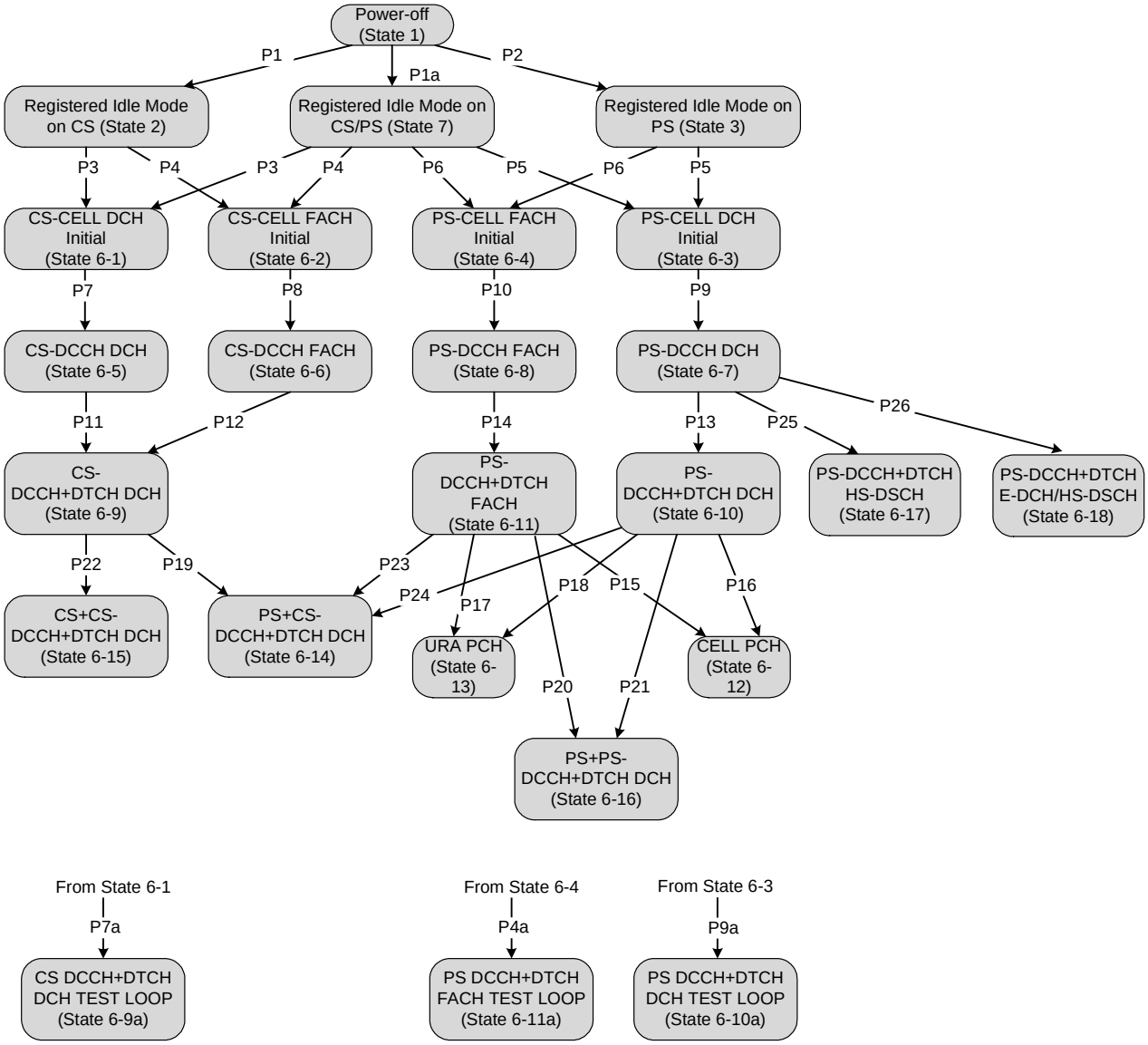



**Figure 7.4.1.1: UE RRC test initial states and common procedures**

For UE to set up a call in UTRAN, there are a number of procedures to be undertaken in a hierarchical sequence to move between known states. The sequences are shown in figure 7.4.1.1, the operating states for various protocols in the UE are given in table 7.4.1.1.

It is noted that figure 7.4.1.1 should not be construed as a formal state transition diagram, in any manner. The intention here is to define the starting state of UE following the execution of the procedures indicated above.

**Table 7.4.1.1: The UE states**

| | | RRC | CC | MM | SM | GMM |
|---|---|---|---|---|---|---|
| State 1 | Power OFF | ----- | Null | Null | Pdp-Inactive | GMM-null |
| State 2 | Registered Idle Mode on CS | Idle | Null | MM Idle | Pdp-Inactive | GMM-deregistered |
| State 3 | Registered Idle Mode on PS | Idle | Null | Null | Pdp-Inactive | GMM-registered |
| State 7 | Registered Idle Mode on CS/PS | Idle | Null | MM Idle | Pdp-Inactive | GMM-registered |
| State BGP6-1 | CS-CELL_DCH_Initial | Connected | Null | MM Idle | Pdp-Inactive | As previous |
| State BGP6-2 | CS-CELL_FACH_Initial | Connected | Null | MM Idle | Pdp-nactive | As previous |
| State BGP6-3 | PS-CELL_DCH_Initial | Connected | Null | As previous | Pdp-Inactive | GMM registered |
| State BGP6-4 | PS-CELL_FACH_Initial | Connected | Null | As previous | Pdp-Inactive | GMM registered |
| State BGP6-5 | CS-DCCH_DCH | Connected (CELL_DCH) | Null | MM Idle | Pdp-Inactive | As previous |
| State BGP6-6 | CS-DCCH_FACH | Connected (CELL_FACH) | Null | MM Idle | Pdp-Inactive | As previous |
| State BGP6-7 | PS-DCCH_DCH | Connected (CELL_DCH) | Null | As previous | Pdp-Active pending | GMM registered |
| State BGP6-8 | PS-DCCH_FACH | Connected (CELL_FACH) | Null | As previous | Pdp-Active pending | GMM registered |
| State BGP6-9 | CS-DCCH+DTCH_DCH | Connected (CELL_DCH) | Active | MM connection active | Pdp-Inactive | As previous |
| State BGP6-9a | CS-DCCH+DTCH_DCH_TEST_LOOP | Connected (CELL_DCH) | Null | MM Idle | Pdp-Inactive | As previous |
| State BGP6-10 | PS-DCCH+DTCH_DCH | Connected (CELL_DCH) | Null | As previous | Pdp-Active | GMM registered |
| State BGP6-10a | PS-DCCH+DTCH_DCH_TEST_LOOP | Connected (CELL_DCH) | Null | As previous | Pdp-Inactive | GMM registered |
| State BGP6-11 | PS-DCCH+DTCH_FACH | Connected (CELL_FACH) | Null | As previous | Pdp-Active | GMM registered |
| State BGP6-11a | PS-DCCH+DTCH_FACH_TEST_LOOP | Connected (CELL_FACH) | Null | As previous | Pdp-Inactive | GMM registered |
| State BGP6-12 | CELL_PCH | Connected (CELL_PCH) | Null | As previous | Pdp-Inactive | GMM registered |
| State BGP6-13 | URA_PCH | Connected (URA_PCH) | Null | As previous | Pdp-Inactive | GMM registered |
| State BGP6-14 | PS+CS-DCCH+DTCH_DCH | Connected (CELL_DCH) | Active | MM connection active | Pdp-Active | GMM registered |
| State BGP6-15 | CS+CS-DCCH+DTCH_DCH | Connected (CELL_DCH) | Active | MM connection active | Pdp-Inactive | As previous |
| State BGP6-16 | PS+PS-DCCH+DTCH_DCH | Connected (CELL_DCH) | Null | As previous | Pdp-Active | GMM registered |
| State BGP6-17 | PS-DCCH+DTCH_HS-DSCH | Connected (CELL_DCH) | Null | As previous | Pdp-Active | GMM registered |
| State BGP6-18 | PS-DCCH+DTCH_E-DCH/HS-DSCH | Connected (CELL_DCH) | Null | As previous | Pdp-Active | GMM registered |

State 1, state 2, state 3, P1, P2 and P1a are described in clause 7.2. States 6-X (for X=1 to 17) are described below.

## 7.4.2 Generic Setup Procedure for RRC test cases

### 7.4.2.1 RRC connection establishment procedure for circuit-switched calls (procedure P3 and P4)

#### 7.4.2.1.1 Mobile terminating call

##### 7.4.2.1.1.1 Initial conditions

System Simulator:

- 1 cell, default parameters.

User Equipment:

- The UE shall be operated under normal test conditions as specified in the present document.
- The Test USIM shall be inserted.

##### 7.4.2.1.1.2 Definition of system information messages

The default system information messages are used as specified in clause 6.1.

##### 7.4.2.1.1.3 Procedure

The Call Set-up procedure shall be performed under ideal radio conditions as defined in clauses 5.2 and 6.1.

| Step | Direction | | Message | Comments |
|---|---|---|---|---|
| | UE | SS | | |
| 1 | | ← | PAGING TYPE 1 (PCCH) | RRC |
| 2 | | → | RRC CONNECTION REQUEST (CCCH) | RRC |
| 3 | | ← | RRC CONNECTION SETUP (CCCH) | RRC |
| 4 | | → | RRC CONNECTION SETUP COMPLETE (DCCH) | RRC |
| 5 | | → | PAGING RESPONSE | RR |

##### 7.4.2.1.1.4 Specific message contents

To execute procedure P3, all specific message contents shall be referred to clause 9.

To execute procedure P4, all specific message contents with the exception of step 3 shall be referred to clause 9. For step 3, the message of the same type titled "Transition to CELL_FACH" in clause 9 is used.

#### 7.4.2.1.2 Mobile originating calls

##### 7.4.2.1.2.1 Initial conditions

System Simulator:

- 1 cell, default parameters.

User Equipment:

- The UE shall be operated under normal test conditions as specified in the present document.
- The Test USIM shall be inserted.

##### 7.4.2.1.2.2 Definition of system information messages

The default system information messages specified in clause 6.1 are used.

##### 7.4.2.1.2.3 Procedure

The Call Set-up procedure shall be performed under ideal radio conditions as defined in clauses 5.2 and 6.1.

| Step | Direction | | Message | Comments |
|---|---|---|---|---|
| | UE | SS | | |
| 1 | | → | RRC CONNECTION REQUEST (CCCH) | RRC |
| 2 | | ← | RRC CONNECTION SETUP (CCCH) | RRC |

| | | | |
|---|---|---|---|
| 3 | → | RRC CONNECTION SETUP COMPLETE (DCCH) | RRC |
| 4 | → | CM SERVICE REQUEST | MM |

#### 7.4.2.1.2.4 Specific message contents

To execute procedure P3, all specific message contents shall be referred to clause 9.

To execute procedure P4, all specific message contents with the exception of step 2 shall be referred to clause 9. For step 2, the message of the same type titled "Transition to CELL_FACH" in clause 9 is used.

### 7.4.2.2 RRC connection establishment procedure for packet switched sessions (procedure P5 and P6)

#### 7.4.2.2.1 Mobile terminating session

##### 7.4.2.2.1.1 Initial conditions

System Simulator:

- 1 cell, default parameters.

User Equipment:

- The UE shall be operated under normal test conditions as specified in the present document.
- The Test USIM shall be inserted.

##### 7.4.2.2.1.2 Definition of system information messages

The default system information messages are used as specified in clause 6.1.

##### 7.4.2.2.1.3 Procedure

The Session Set-up procedure shall be performed under Ideal radio conditions as defined in clauses 5.2 and 6.1.

| Step | Direction | | Message | Comments |
|---|---|---|---|---|
| | UE | SS | | |
| 1 | | ← | PAGING TYPE1 (PCCH) | Paging |
| 2 | | → | RRC CONNECTION REQUEST (CCCH) | RRC |
| 3 | | ← | RRC CONNECTION SETUP (CCCH) | RRC |
| 4 | | → | RRC CONNECTION SETUP COMPLETE (DCCH) | RRC |
| 5 | | → | SERVICE REQUEST | GMM |

##### 7.4.2.2.1.4 Specific message contents

To execute procedure P5, all specific message contents shall be referred to clause 9.

To execute procedure P6, all specific message contents with the exception of step 3 shall be referred to clause 9. For step 3, the message of the same type titled "Transition to CELL_FACH" in clause 9 is used.

#### 7.4.2.2.2 Mobile originating sessions

##### 7.4.2.2.2.1 Initial conditions

System Simulator:

- 1 cell, default parameters.

User Equipment:

- The UE shall be operated under normal test conditions as specified in the present document.
- The Test USIM shall be inserted.

##### 7.4.2.2.2.2 Definition of system information messages

The default system information messages are used as specified in clause 6.1.

#### 7.4.2.2.2.3 Procedure

The Session Set-up procedure shall be performed under Ideal radio conditions as defined in clauses 5.2 and 6.1.

| Step | Direction | | Message | Comments |
|---|---|---|---|---|
| | UE | SS | | |
| 1 | → | | RRC CONNECTION REQUEST (CCCH) | RRC |
| 2 | ← | | RRC CONNECTION SETUP (CCCH) | RRC |
| 3 | → | | RRC CONNECTION SETUP COMPLETE (DCCH) | RRC |
| 4 | → | | SERVICE REQUEST | GMM |

#### 7.4.2.2.2.4 Specific message contents

To execute procedure P5, all specific message contents shall be referred to clause 9.

To execute procedure P6, all specific message contents with the exception of step 2 shall be referred to clause 9. For step 2, the message of the same type titled "Transition to CELL_FACH" in clause 9 is used.

### 7.4.2.3 NAS call set up procedure for circuit switched calls (procedure P7 and P8)

#### 7.4.2.3.1 Mobile terminating call

##### 7.4.2.3.1.1 Initial conditions

System Simulator:

- 1 cell, default parameters.

User Equipment:

- The UE shall be in state 6-1 or state 6-2.
- The Test USIM shall be inserted.

##### 7.4.2.3.1.2 Definition of system information messages

The default system information messages are used as specified in clause 6.1.

##### 7.4.2.3.1.3 Procedure

The Call Set-up procedure shall be performed under Ideal radio conditions as defined in clauses 5.2 and 6.1.

| Step | Direction | | Message | Comments |
|---|---|---|---|---|
| | UE | SS | | |
| 1 | ← | | AUTHENTICATION REQUEST | MM |
| 2 | → | | AUTHENTICATION RESPONSE | MM |
| 3 | ← | | SECURITY MODE COMMAND | RRC |
| 4 | → | | SECURITY MODE COMPLETE | RRC |
| 5 | ← | | SET UP | CC |
| 6 | → | | CALL CONFIRMED | CC |

##### 7.4.2.3.1.4 Specific message contents

All RRC specific message contents shall be referred to clause 9.

#### 7.4.2.3.2 Mobile originating calls

##### 7.4.2.3.2.1 Initial conditions

System Simulator:

- 1 cell, default parameters.

User Equipment:

- The UE shall be in state 6-1or state 6-2.
- The Test USIM shall be inserted.

#### 7.4.2.3.2.2 Definition of system information messages

The default system information messages are used as specified in clause 6.1.

#### 7.4.2.3.2.3 Procedure

The Call Set-up procedure shall be performed under Ideal radio conditions as defined in clauses 5.2 and 6.1.

| Step | Direction | | Message | Comments |
|---|---|---|---|---|
| | UE | SS | | |
| 1 | ← | | AUTHENTICATION REQUEST | MM |
| 2 | → | | AUTHENTICATION RESPONSE | MM |
| 3 | ← | | SECURITY MODE COMMAND | RRC |
| 4 | → | | SECURITY MODE COMPLETE | RRC |
| 5 | → | | SET UP | CC |
| 6 | ← | | CALL PROCEEDING | CC |

#### 7.4.2.3.2.4 Specific message contents

All RRC specific message contents shall be referred to clause 9.

### 7.4.2.4 NAS session activation procedure for packet switched sessions (procedure P9 and P10)

#### 7.4.2.4.1 Mobile terminating session

#### 7.4.2.4.1.1 Initial conditions

System Simulator:

- 1 cell, default parameters.

User Equipment:

- The UE shall be in state 6-3 or state 6-4.
- The Test USIM shall be inserted.

#### 7.4.2.4.1.2 Definition of system information messages

The default system information messages are used as specified in clause 6.1.

#### 7.4.2.4.1.3 Procedure

The Session Set-up procedure shall be performed under Ideal radio conditions as defined in clauses 5.2 and 6.1.

| Step | Direction | | Message | Comments |
|---|---|---|---|---|
| | UE | SS | | |
| 1 | ← | | AUTHENTICATION AND CIPHERING REQUEST | GMM |
| 2 | → | | AUTHENTICATION AND CIPHERING RESPONSE | GMM |
| 3 | ← | | SECURITY MODE COMMAND | RRC |
| 4 | → | | SECURITY MODE COMPLETE | RRC |
| 5 | ← | | REQUEST PDP CONTEXT ACTIVATION | SM |
| 6 | → | | ACTIVATE PDP CONTEXT REQUEST | SM |

#### 7.4.2.4.1.4 Specific message contents

All RRC specific message contents shall be referred to clause 9.

#### 7.4.2.4.2 Mobile originating sessions

##### 7.4.2.4.2.1 Initial conditions

System Simulator:

- 1 cell, default parameters.

User Equipment:

- The UE shall be in state 6-3 or state 6-4.
- The Test USIM shall be inserted.

##### 7.4.2.4.2.2 Definition of system information messages

The default system information messages are used as specified in clause 6.1.

##### 7.4.2.4.2.3 Procedure

The Session Set-up procedure shall be performed under Ideal radio conditions as defined in clauses 5.2 and 6.1.

| Step | Direction | | Message | Comments |
|---|---|---|---|---|
| | UE | SS | | |
| 1 | | ← | AUTHENTICATION AND CIPHERING REQUEST | GMM |
| 2 | | → | AUTHENTICATION AND CIPHERING RESPONSE | GMM |
| 3 | | ← | SECURITY MODE COMMAND | RRC |
| 4 | | → | SECURITY MODE COMPLETE | RRC |
| 5 | | → | ACTIVATE PDP CONTEXT REQUEST | SM |

##### 7.4.2.4.2.4 Specific message contents

All RRC specific message contents shall be referred to clause 9.

### 7.4.2.5 Radio access bearer establishment procedure for circuit switched calls (procedure P11 and P12)

#### 7.4.2.5.1 Mobile terminating call

##### 7.4.2.5.1.1 Initial conditions

System Simulator:

- 1 cell, default parameters.

User Equipment:

- The UE shall be in state 6-5 or state 6-6.
- The Test USIM shall be inserted.

##### 7.4.2.5.1.2 Definition of system information messages

The default system information messages are used as specified in clause 6.1.

##### 7.4.2.5.1.3 Procedure

The Call Set-up procedure shall be performed under Ideal radio conditions as defined in clauses 5.2 and 6.1.

| Step | Direction | | Message | Comments |
|---|---|---|---|---|
| | UE | SS | | |
| 1 | | ← | RADIO BEARER SETUP | RRC RAB SETUP |
| 2 | | → | RADIO BEARER SETUP COMPLETE | RRC |
| 3 | | → | ALERTING | CC (This message is optional) |
| 4 | | → | CONNECT | CC |
| 5 | | ← | CONNECT ACKNOWLEDGE | CC |

#### 7.4.2.5.1.4 Specific message contents

To execute procedure P11, use the message titled "CS speech" (defined in clause 9) for the message in step 1. To execute procedure 12, use the message "The others of speech in CS" (defined in clause 9) for the message in step 1.

### 7.4.2.5.2 Mobile originating calls

#### 7.4.2.5.2.1 Initial conditions

System Simulator:

- 1 cell, default parameters.

User Equipment:

- The UE shall be in state 6-5 or state 6-6.
- The Test USIM shall be inserted.

#### 7.4.2.5.2.2 Definition of system information messages

The default system information messages are used as specified in clause 6.1.

#### 7.4.2.5.2.3 Procedure

The Call Set-up procedure shall be performed under Ideal radio conditions as defined in clauses 5.2 and 6.1.

| Step | Direction | | Message | Comments |
|---|---|---|---|---|
| | UE | SS | | |
| 1 | | ← | RADIO BEARER SETUP | RRC RAB SETUP |
| 2 | | → | RADIO BEARER SETUP COMPLETE | RRC |
| 3 | | ← | ALERTING | CC |
| 4 | | ← | CONNECT | CC |
| 5 | | → | CONNECT ACKOWLEDGE | CC |

#### 7.4.2.5.2.4 Specific message contents

To execute procedure P11, use the message titled "CS speech" (defined in clause 9) for the message in step 1. To execute procedure 12, use the message "The others of speech in CS" (defined in clause 9) for the message in step 1.

## 7.4.2.5a Test loop activation and radio access bearer establishment procedure for circuit switched calls (procedure P7a)

### 7.4.2.5a.1 Initial conditions

System Simulator:

- 1 cell, default parameters.

User Equipment:

- The UE shall be in state 6-1.
- The Test USIM shall be inserted.

### 7.4.2.5a.2 Definition of system information messages

The default system information messages are used as specified in clause 6.1.

### 7.4.2.5a.3 Procedure

The Call Set-up procedure shall be performed under Ideal radio conditions as defined in clauses 5.2 and 6.1.

| Step | Direction | | Message | Comments |
|---|---|---|---|---|
| | UE | SS | | |
| 1 | | ← | AUTHENTICATION REQUEST | MM |
| 2 | | → | AUTHENTICATION RESPONSE | MM |
| 3 | | ← | SECURITY MODE COMMAND | RRC |
| 4 | | → | SECURITY MODE COMPLETE | RRC |
| 5 | | ← | ACTIVATE RB TEST MODE (DCCH) | TC |
| 6 | | → | ACTIVATE RB TEST MODE COMPLETE (DCCH) | TC |
| 1 | | ← | RADIO BEARER SETUP | RRC RAB SETUP |
| 2 | | → | RADIO BEARER SETUP COMPLETE | RRC |
| 14 | | ← | CLOSE UE TEST LOOP (DCCH) | TC<br>UE test mode 1<br>RLC SDU size set as specified for the actual test case. |
| 15 | | → | CLOSE UE TEST LOOP COMPLETE (DCCH) | TC |

#### 7.4.2.5a.4 Specific message contents

To execute procedure P7a, use the message titled "CS speech" (defined in clause 9) for the message in step 1.

### 7.4.2.6 Radio access bearer establishment procedure for packet switched sessions (procedure P13, P14, P25 and P26)

#### 7.4.2.6.1 Mobile terminating session

##### 7.4.2.6.1.1 Initial conditions

System Simulator:

- 1 cell, default parameters.

User Equipment:

- The UE shall be in state 6-7 or state 6-8.
- The Test USIM shall be inserted.

##### 7.4.2.6.1.2 Definition of system information messages

The default system information messages are used as specified in clause 6.1.

##### 7.4.2.6.1.3 Procedure

The Session Set-up procedure shall be performed under Ideal radio conditions as defined in clauses 5.2 and 6.1.

| Step | Direction | | Message | Comments |
|---|---|---|---|---|
| | UE | SS | | |
| 1 | | ← | RADIO BEARER SETUP | RRC RAB SETUP |
| 2 | | → | RADIO BEARER SETUP COMPLETE | RRC |
| 3 | | ← | ACTIVATE PDP CONTEXT ACCEPT | SM |

##### 7.4.2.6.1.4 Specific message contents

For step 1, the messages in clause 9 are used. To execute procedure P13, use the message titled "Packet to CELL_DCH from CELL_DCH in PS". To execute procedure P14, use the message titled "Packet to CELL_FACH from CELL_FACH in PS". To execute procedure P25, use the message titled "Packet to CELL_DCH / HS-DSCH from CELL_DCH in PS". To execute procedure P26, use the RADIO BEARER SETUP message with one of the conditions A12, A13, A14 or A15.

#### 7.4.2.6.2 Mobile originating sessions

##### 7.4.2.6.2.1 Initial conditions

System Simulator:

- 1 cell, default parameters.

User Equipment:

- The UE shall be in state 6-7 or state 6-8.
- The Test USIM shall be inserted.

#### 7.4.2.6.2.2 Definition of system information messages

The default system information messages are used as specified in clause 6.1.

#### 7.4.2.6.2.3 Procedure

The Session Set-up procedure shall be performed under Ideal radio conditions as defined in clauses 5.2 and 6.1.

| Step | Direction | | Message | Comments |
|---|---|---|---|---|
| | UE | SS | | |
| 1 | ← | | RADIO BEARER SETUP | RRC RAB SETUP |
| 2 | → | | RADIO BEARER SETUP COMPLETE | RRC |
| 3 | ← | | ACTIVATE PDP CONTEXT ACCEPT | SM |

#### 7.4.2.6.2.4 Specific message contents

For step 1, the messages in clause 9 are used. To execute procedure P13, use the message titled "Packet to CELL_DCH from CELL_DCH in PS". To execute procedure P14, use the message titled "Packet to CELL_FACH from CELL_FACH in PS". To execute procedure P25, use the message titled "Packet to CELL_DCH / HS-DSCH from CELL_DCH in PS". To execute procedure P26, use the RADIO BEARER SETUP message with one of the conditions A12, A13 or A14.

### 7.4.2.6a Test loop activation and radio access bearer establishment procedure for packet switched sessions (procedure P4a and P9a)

#### 7.4.2.6a.1 Initial conditions

System Simulator:

- 1 cell, default parameters.

User Equipment:

- The UE shall be in state 6-3 or state 6-4.
- The Test USIM shall be inserted.

#### 7.4.2.6a.2 Definition of system information messages

The default system information messages are used as specified in clause 6.1.

#### 7.4.2.6a.3 Procedure

The Session Set-up procedure shall be performed under Ideal radio conditions as defined in clauses 5.2 and 6.1.

| Step | Direction | | Message | Comments |
|---|---|---|---|---|
| | UE | SS | | |
| 1 | ← | | AUTHENTICATION AND CIPHERING REQUEST | GMM |
| 2 | → | | AUTHENTICATION AND CIPHERING RESPONSE | GMM |
| 3 | ← | | SECURITY MODE COMMAND | RRC |
| 4 | → | | SECURITY MODE COMPLETE | RRC |
| 5 | ← | | ACTIVATE RB TEST MODE (DCCH) | TC |
| 6 | → | | ACTIVATE RB TEST MODE COMPLETE (DCCH) | TC |
| 7 | ← | | RADIO BEARER SETUP | RRC RAB SETUP. The 'pdcp info' IE shall be omitted. |
| 8 | → | | RADIO BEARER SETUP COMPLETE | RRC |

| | | | |
|---|---|---|---|
| 14 | ← | CLOSE UE TEST LOOP (DCCH) | TC<br>UE test mode 1<br>RLC SDU size set as specified for the actual test case. |
| 15 | → | CLOSE UE TEST LOOP COMPLETE (DCCH) | TC |

#### 7.4.2.6a.4 Specific message contents

For step 1, the messages in clause 9 are used. To execute procedure P9a, use the message titled "Packet to CELL_DCH from CELL_DCH in PS". To execute procedure 4a, use the message titled "Packet to CELL_FACH from CELL_FACH in PS" with the exception that the 'pdcp info' IE shall be omitted.

### 7.4.2.7 Procedure for transitions to CELL_PCH or URA_PCH state (procedure P15, P16, P17 and P18)

#### 7.4.2.7.1 Transition to CELL_PCH (procedure P15 and P16)

##### 7.4.2.7.1.1 Initial conditions

System Simulator:

- 1 cell, default parameters.

User Equipment:

- The UE shall be in state 6-10 or state 6-11.
- The Test USIM shall be inserted.

##### 7.4.2.7.1.2 Definition of system information messages

The default system information messages are used as specified in clause 6.1.

##### 7.4.2.7.1.3 Procedure

The Call Set-up procedure shall be performed under ideal radio conditions as defined in clauses 5.2 and 6.1.

| Step | Direction | | Message | Comments |
|---|---|---|---|---|
| | UE | SS | | |
| 1 | ← | | PHYSICAL CHANNEL RECONFIGURATION | RRC |
| 2 | → | | PHYSICAL CHANNEL RECONFIGURATION COMPLETE | RRC |

##### 7.4.2.7.1.4 Specific message contents

Contents of PHYSICAL CHANNEL RECONFIGURATION message: DCCH-AM (Step 1)

| Information Element | Value/remark |
|---|---|
| Message Type | |
| RRC State Indicator | CELL_PCH |

#### 7.4.2.7.2 Transition to URA_PCH (procedure P17 and P18)

##### 7.4.2.7.2.1 Initial conditions

System Simulator:

- 1 cell, default parameters.

User Equipment:

- The UE shall be in state 6-10 or state 6-11.
- The Test USIM shall be inserted.

#### 7.4.2.7.2.2 Definition of system information messages

The default system information messages are used as specified in clause 6.1.

#### 7.4.2.7.2.3 Procedure

The Call Set-up procedure shall be performed under ideal radio conditions as defined in clauses 5.2 and 6.1.

| Step | Direction | | Message | Comments |
|---|---|---|---|---|
| | UE | SS | | |
| 1 | | ← | PHYSICAL CHANNEL RECONFIGURATION | RRC |
| 2 | | → | PHYSICAL CHANNEL RECONFIGURATION COMPLETE | RRC |

#### 7.4.2.7.2.4 Specific message contents

Contents of PHYSICAL CHANNEL RECONFIGURATION message: DCCH-AM (Step 1)

| Information Element | Value/remark |
|---|---|
| Message Type | |
| RRC State Indicator | URA_PCH |

### 7.4.2.8 Radio access bearer establishment procedure with packet switched sessions for transitions to Multi Call state (procedure P19, 20 and 21)

#### 7.4.2.8.1 Transition to PS+CS-DCCH+DTCH DCH (procedure P19)

#### 7.4.2.8.1.1 Mobile terminating session

#### 7.4.2.8.1.1.1 Initial conditions

System Simulator:

- 1 cell, default parameters.

User Equipment:

- The UE shall have registered in CS/PS.
- The UE shall be in state 6-9.
- The Test USIM shall be inserted.

#### 7.4.2.8.1.1.2 Definition of system information messages

The default system information messages are used as specified in clause 6.1.

#### 7.4.2.8.1.1.3 Procedure

The Session Set-up procedure shall be performed under Ideal radio conditions as defined in clauses 5.2 and 6.1.

| Step | Direction | | Message | Comments |
|---|---|---|---|---|
| | UE | SS | | |
| 1 | | ← | PAGING TYPE2 (DCCH) | Paging |
| 2 | | → | SERVICE REQUEST | GMM |
| 3 | | ← | AUTHENTICATION AND CIPHERING REQUEST | GMM |
| 4 | | → | AUTHENTICATION AND CIPHERING RESPONSE | GMM |
| 5 | | ← | SECURITY MODE COMMAND | RRC |
| 6 | | → | SECURITY MODE COMPLETE | RRC |
| 7 | | ← | REQUEST PDP CONTEXT ACTIVATION | SM |
| 8 | | → | ACTIVATE PDP CONTEXT REQUEST | SM |
| 9 | | ← | RADIO BEARER SETUP | RRC RAB SETUP |
| 10 | | → | RADIO BEARER SETUP COMPLETE | RRC |
| 11 | | ← | ACTIVATE PDP CONTEXT ACCEPT | SM |

7.4.2.8.1.1.4 Specific message contents

FFS

7.4.2.8.1.2 Mobile originating sessions

7.4.2.8.1.2.1 Initial conditions

System Simulator:

- 1 cell, default parameters.

User Equipment:

- The UE shall be in state 6-9.
- The Test USIM shall be inserted.

7.4.2.8.1.2.2 Definition of system information messages

The default system information messages are used as specified in clause 6.1.

7.4.2.8.1.2.3 Procedure

The Session Set-up procedure shall be performed under Ideal radio conditions as defined in clauses 5.2 and 6.1.

| Step | Direction | | Message | Comments |
|---|---|---|---|---|
| | UE | SS | | |
| 1 | → | | SERVICE REQUEST | GMM |
| 2 | ← | | AUTHENTICATION AND CIPHERING REQUEST | GMM |
| 3 | → | | AUTHENTICATION AND CIPHERING RESPONSE | GMM |
| 4 | ← | | SECURITY MODE COMMAND | RRC |
| 5 | → | | SECURITY MODE COMPLETE | RRC |
| 6 | → | | ACTIVATE PDP CONTEXT REQUEST | SM |
| 7 | ← | | RADIO BEARER SETUP | RRC RAB SETUP |
| 8 | → | | RADIO BEARER SETUP COMPLETE | RRC |
| 9 | ← | | ACTIVATE PDP CONTEXT ACCEPT | SM |

7.4.2.8.1.2.4 Specific message contents

FFS

7.4.2.8.2 Transition to PS+PS-DCCH+DTCH DCH (procedure P20 and P21)

7.4.2.8.2.1 Mobile terminating session

7.4.2.8.2.1.1 Initial conditions

System Simulator:

- 1 cell, default parameters.

User Equipment:

- The UE shall be in state 6-10 or state 6-11.
- The Test USIM shall be inserted.

7.4.2.8.2.1.2 Definition of system information messages

The default system information messages are used as specified in clause 6.1.

7.4.2.8.2.1.3 Procedure

The Session Set-up procedure shall be performed under Ideal radio conditions as defined in clauses 5.2 and 6.1.

| Step | Direction | Message | Comments |
|---|---|---|---|

| | UE | SS | | |
|---|---|---|---|---|
| 1 | | ← | PAGING TYPE2 (DCCH) | Paging |
| 2 | | → | SERVICE REQUEST | GMM |
| 3 | | ← | SERVICE ACCEPT | GMM |
| 4 | | ← | REQUEST PDP CONTEXT ACTIVATION | SM |
| 5 | | → | ACTIVATE PDP CONTEXT REQUEST | SM |
| 6 | | ← | RADIO BEARER SETUP | RRC RAB SETUP |
| 7 | | → | RADIO BEARER SETUP COMPLETE | RRC |
| 8 | | ← | ACTIVATE PDP CONTEXT ACCEPT | SM |

7.4.2.8.2.1.4 Specific message contents

FFS

7.4.2.8.2.2 Mobile originating sessions

7.4.2.8.2.2.1 Initial conditions

System Simulator:

- 1 cell, default parameters.

User Equipment:

- The UE shall be in state 6-10 or state 6-11.
- The Test USIM shall be inserted.

7.4.2.8.2.2.2 Definition of system information messages

The default system information messages are used as specified in clause 6.1.

7.4.2.8.2.2.3 Procedure

The Session Set-up procedure shall be performed under Ideal radio conditions as defined in clauses 5.2 and 6.1.

| Step | Direction | | Message | Comments |
|---|---|---|---|---|
| | UE | SS | | |
| 1 | | → | SERVICE REQUEST | GMM |
| 2 | | ← | SERVICE ACCEPT | GMM |
| 3 | | → | ACTIVATE PDP CONTEXT REQUEST | SM |
| 4 | | ← | RADIO BEARER SETUP | RRC RAB SETUP |
| 5 | | → | RADIO BEARER SETUP COMPLETE | RRC |
| 6 | | ← | ACTIVATE PDP CONTEXT ACCEPT | SM |

7.4.2.8.2.2.4 Specific message contents

FFS

## 7.4.2.9 Radio access bearer establishment procedure with circuit switched calls for transitions to Multi Call state (procedure P22, P23 and P24)

### 7.4.2.9.1 Transition to CS+CS-DCCH+DTCH DCH (procedure P22)

#### 7.4.2.9.1.1 Mobile terminating call

##### 7.4.2.9.1.1.1 Initial conditions

System Simulator:

- 1 cell, default parameters.

User Equipment:

- The UE shall be in state 6-9.
- The Test USIM shall be inserted.

#### 7.4.2.9.1.1.2 Definition of system information messages

The default system information messages are used as specified in clause 6.1.

#### 7.4.2.9.1.1.3 Procedure

The Call Set-up procedure shall be performed under Ideal radio conditions as defined in clauses 5.2 and 6.1.

| Step | Direction | | Message | Comments |
|---|---|---|---|---|
| | UE | SS | | |
| 1 | ← | | PAGING TYPE2 (DCCH) | Paging |
| 2 | → | | PAGING RESPONSE | RR |
| 3 | ← | | SET UP | CC |
| 4 | → | | CALL CONFIRMED | CC |
| 5 | ← | | RADIO BEARER SETUP | RRC RAB SETUP |
| 6 | → | | RADIO BEARER SETUP COMPLETE | RRC |
| 7 | → | | ALERTING | CC (this message is optional) |
| 8 | → | | CONNECT | CC |
| 9 | ← | | CONNECT ACKNOWLEDGE | CC |

#### 7.4.2.9.1.1.4 Specific message contents

FFS

### 7.4.2.9.1.2 Mobile originating calls

#### 7.4.2.9.1.2.1 Initial conditions

System Simulator:

- 1 cell, default parameters.

User Equipment:

- The UE shall be in state 6-9.
- The Test USIM shall be inserted.

#### 7.4.2.9.1.2.2 Definition of system information messages

The default system information messages are used as specified in clause 6.1.

#### 7.4.2.9.1.2.3 Procedure

The Call Set-up procedure shall be performed under Ideal radio conditions as defined in clauses 5.2 and 6.1.

| Step | Direction | | Message | Comments |
|---|---|---|---|---|
| | UE | SS | | |
| 1 | → | | CM SERVICE REQUEST | MM |
| 2 | ← | | CM SERVICE ACCEPT | MM |
| 3 | → | | SET UP | CC |
| 4 | ← | | CALL PROCEEDING | CC |
| 5 | ← | | RADIO BEARER SETUP | RRC RAB SETUP |
| 6 | → | | RADIO BEARER SETUP COMPLETE | RRC |
| 7 | ← | | ALERTING | CC |
| 8 | ← | | CONNECT | CC |
| 9 | → | | CONNECT ACKNOWLEDGE | CC |

#### 7.4.2.9.1.2.4 Specific message contents

FFS

#### 7.4.2.9.2 Transition to PS+CS-DCCH+DTCH DCH (procedure P23 and 24)

##### 7.4.2.9.2.1 Mobile terminating call

###### 7.4.2.9.2.1.1 Initial conditions

System Simulator:

- 1 cell, default parameters.

User Equipment:

- The UE shall have registered in CS/PS.
- The UE shall be in state 6-10 or state 6-11.
- The Test USIM shall be inserted.

###### 7.4.2.9.2.1.2 Definition of system information messages

The default system information messages are used as specified in clause 6.1.

###### 7.4.2.9.2.1.3 Procedure

The Call Set-up procedure shall be performed under Ideal radio conditions as defined in clauses 5.2 and 6.1.

| Step | Direction | | Message | Comments |
|---|---|---|---|---|
| | UE | SS | | |
| 1 | | ← | PAGING TYPE2 (DCCH) | Paging |
| 2 | | → | PAGING RESPONSE | RR |
| 3 | | ← | AUTHENTICATION REQUEST | MM |
| 4 | | → | AUTHENTICATION RESPONSE | MM |
| 5 | | ← | SECURITY MODE COMMAND | RRC |
| 6 | | → | SECURITY MODE COMPLETE | RRC |
| 7 | | ← | SET UP | CC |
| 8 | | → | CALL CONFIRMED | CC |
| 9 | | ← | RADIO BEARER SETUP | RRC RAB SETUP |
| 10 | | → | RADIO BEARER SETUP COMPLETE | RRC |
| 11 | | → | ALERTING | CC (this message is optional) |
| 12 | | → | CONNECT | CC |
| 13 | | ← | CONNECT ACKNOWLEDGE | CC |

###### 7.4.2.9.2.1.4 Specific message contents

FFS

##### 7.4.2.9.2.2 Mobile originating calls

###### 7.4.2.9.2.2.1 Initial conditions

System Simulator:

- 1 cell, default parameters.

User Equipment:

- The UE shall be in state 6-10 or state 6-11.
- The Test USIM shall be inserted.

###### 7.4.2.9.2.2.2 Definition of system information messages

The default system information messages are used as specified in clause 6.1.

###### 7.4.2.9.2.2.3 Procedure

The Call Set-up procedure shall be performed under Ideal radio conditions as defined in clauses 5.2 and 6.1.

| Step | Direction | | Message | Comments |
|---|---|---|---|---|
| | UE | SS | | |
| 1 | | → | CM SERVICE REQUEST | MM |
| 2 | | ← | AUTHENTICATION REQUEST | MM |
| 3 | | → | AUTHENTICATION RESPONSE | MM |
| 4 | | ← | SECURITY MODE COMMAND | RRC |
| 5 | | → | SECURITY MODE COMPLETE | RRC |
| 6 | | → | SET UP | CC |
| 7 | | ← | CALL PROCEEDING | CC |
| 8 | | ← | RADIO BEARER SETUP | RRC RAB SETUP |
| 9 | | → | RADIO BEARER SETUP COMPLETE | RRC |
| 10 | | ← | ALERTING | CC |
| 11 | | ← | CONNECT | CC |
| 12 | | → | CONNECT ACKOWLEDGE | CC |

7.4.2.9.2.2.4 Specific message contents

FFS

# 7.5 Test procedures for A-GPS Performance requirements testing

This clause specifies the procedures that shall be used for testing of A-GPS Performance requirements in TS 34.171 [41].

## 7.5.1 Normal UE based A-GPS procedure

The procedure in this clause shall be used for all UE-based A-GPS TTFF test cases in CELL_DCH and CELL_FACH state as specified in TS 34.171 [41].

### 7.5.1.1 Initial conditions

User Equipment:

The UE is in CELL_DCH or CELL_FACH state after executing the procedure defined in clause D.2 of TS 34.171 [41].

### 7.5.1.2 Procedure

| Step | Direction | | Message | Comments |
|---|---|---|---|---|
| | UE | SS | | |
| 1 | | ← | RESET UE POSITIONING STORED INFORMATION | TC |
| 2 | | ← | RRC MEASUREMENT CONTROL | RRC (Setup, No Reporting, Nav model Satellites 1, 2, 3, 4, 5 (1)) |
| 3 | | ← | RRC MEASUREMENT CONTROL | RRC (Modify, No Reporting, Nav model Satellites 6, 7, 8, 9 (1), Iono Model) |
| 4 | | ← | RRC MEASUREMENT CONTROL | RRC (Modify, Periodical Reporting Criterion, GPS Ref time (1), ReferencePosition (1)) |
| 5 | | → | RRC MEASUREMENT REPORT | RRC (Position Estimate), 1st test instance |
| 6 | | ← | RESET UE POSITIONING STORED INFORMATION | TC |
| 7 | | ← | RRC MEASUREMENT CONTROL | RRC (Setup, No Reporting, Nav model Satellites 1, 2, 3, 4,5 (2)) |
| 8 | | ← | RRC MEASUREMENT CONTROL | RRC (Modify, No Reporting, Nav model Satellites 6, 7, 8, 9 (2), Iono Model) |
| 9 | | ← | RRC MEASUREMENT CONTROL | RRC (Modify, Periodical Reporting Criterion, GPS Ref time (2), ReferencePosition (2)) |
| 10 | | → | RRC MEASUREMENT REPORT | RRC (Position Estimate), 2nd test instance |

| Step | Direction | | Message | Comments |
|---|---|---|---|---|
| | UE | SS | | |
| 11 | ← | | RESET UE POSITIONING STORED INFORMATION | TC |
| …. | …… | | ……. | |
| n | → | | RRC MEASUREMENT REPORT | RRC (Position Estimate), $n^{th}$ test instance |

### 7.5.1.3 Specific message contents

Contents of RESET UE POSITIONING STORED INFORMATION message: TC

| Information Element | Value/remark |
|---|---|
| UE Positioning Technology | AGPS |

Contents of MEASUREMENT CONTROL messages: RRC

MEASUREMENT CONTROL (Steps 2 + (n-1)*5):

| Information element | Value/remark |
|---|---|
| **Measurement Information Elements** | |
| Measurement Identity | 10 |
| Measurement Command | Setup |
| Measurement Reporting Mode | |
| - Measurement report transfer mode | Acknowledged mode RLC |
| - Periodical reporting / Event trigger reporting mode | Periodical reporting |
| Additional Measurements List | Not present |
| CHOICE *Measurement type* | UE positioning measurement |
| - UE positioning measurement | |
| - UE positioning reporting quantity | |
| - Method type | UE based |
| - Positioning methods | GPS |
| - Response time | 128 |
| - Horizontal accuracy | 19 (51 m) |
| - Vertical accuracy | 48 (102 m) |
| - GPS timing of cell wanted | FALSE |
| - Multiple sets | FALSE |
| - Additional assistance data request | FALSE |
| - Environmental characterization | Not present |
| - Measurement validity | |
| - UE state | All states |
| - CHOICE *Reporting criteria* | |
| - No reporting | |
| - UE pos OTDOA assistance data for UE-assisted | Not present |
| - UE pos OTDOA assistance data for UE-based | Not present |
| - UE positioning GPS assistance data | |
| - UE positioning GPS navigation model | Satellites 1-5 as specified in clause 10 |
| **Physical Channel Information Elements** | |
| DPCH compressed mode status info | Not present |

MEASUREMENT CONTROL (Steps 3 + (n-1)*5):

| Information element | Value/remark |
|---|---|
| **Measurement Information Elements** | |
| Measurement Identity | 10 |
| Measurement Command | Modify |
| Measurement Reporting Mode | |
| - Measurement report transfer mode | Acknowledged mode RLC |
| - Periodical reporting / Event trigger reporting mode | Periodical reporting |
| Additional Measurements List | Not present |
| CHOICE *Measurement type* | UE positioning measurement |
| - UE positioning measurement | |
| - UE positioning reporting quantity | |
| - Method type | UE based |
| - Positioning methods | GPS |
| - Response time | 128 |
| - Horizontal accuracy | 19 (51 m) |
| - Vertical accuracy | 48 (102 m) |
| - GPS timing of cell wanted | FALSE |
| - Multiple sets | FALSE |
| - Additional assistance data request | FALSE |
| - Environmental characterization | Not present |
| - Measurement validity | |
| - UE state | All states |
| - CHOICE *Reporting criteria* | |
| - No reporting | |
| - UE pos OTDOA assistance data for UE-assisted | Not present |
| - UE pos OTDOA assistance data for UE-based | Not present |
| - UE positioning GPS assistance data | |
| - UE positioning GPS navigation model | Satellites 6-9 as specified in clause 10 |
| - UE positioning GPS ionospheric model | As specified in clause 10 |
| **Physical Channel Information Elements** | |
| DPCH compressed mode status info | Not present |

MEASUREMENT CONTROL (Steps 4 + (n-1)*5):

| Information element | Value/remark |
|---|---|
| **Measurement Information Elements** | |
| Measurement Identity | 10 |
| Measurement Command | Modify |
| Measurement Reporting Mode | |
| - Measurement report transfer mode | Acknowledged mode RLC |
| - Periodical reporting / Event trigger reporting mode | Periodical reporting |
| Additional Measurements List | Not present |
| CHOICE *Measurement type* | UE positioning measurement |
| - UE positioning measurement | |
| - UE positioning reporting quantity | |
| - Method type | UE based |
| - Positioning methods | GPS |
| - Response time | 128 |
| - Horizontal accuracy | 19 (51 m) |
| - Vertical accuracy | 48 (102 m) |
| - GPS timing of cell wanted | FALSE |
| - Multiple sets | FALSE |
| - Additional assistance data request | FALSE |
| - Environmental characterization | Not present |
| - Measurement validity | |
| - UE state | All states |
| - CHOICE *Reporting criteria* | Periodical Reporting Criteria |
| - Amount of reporting | 1 |
| - Reporting Interval | 20000 |
| - UE pos OTDOA assistance data for UE-assisted | Not present |
| - UE pos OTDOA assistance data for UE-based | Not present |
| - UE positioning GPS assistance data | |
| - UE positioning GPS reference time | As specified in clause 10 |
| - UE positioning GPS reference UE position | As specified in clause 10 |
| **Physical Channel Information Elements** | |
| DPCH compressed mode status info | Not present |

## 7.5.2 UE based A-GPS procedure for moving scenario and periodic update test case

The procedure in this clause shall be used for the UE-based A-GPS moving scenario and periodic update test case in CELL_DCH and CELL_FACH state as specified in TS 34.171 [41].

### 7.5.2.1 Initial conditions

User Equipment:

The UE is in CELL_DCH or CELL_FACH state after executing the procedure defined in clause D.2 of TS 34.171 [41].

### 7.5.2.2 Procedure

| Step | Direction | | Message | Comments |
|---|---|---|---|---|
| | UE | SS | | |
| 1 | ← | | RESET UE POSITIONING STORED INFORMATION | TC |
| 2 | ← | | RRC MEASUREMENT CONTROL | RRC (Setup, No Reporting, Nav model Satellites 1, 2, 3, 4, 5) |
| 3 | ← | | RRC MEASUREMENT CONTROL | RRC (Modify, No Reporting, Nav model Satellites 6, 7, 8, 9, Iono Model) |
| 4 | ← | | RRC MEASUREMENT CONTROL | RRC (Modify, Periodical Reporting Criterion, GPS Ref time, ReferencePosition) |
| 5 | → | | RRC MEASUREMENT REPORT | RRC (Position Estimate) |
| 6 | → | | RRC MEASUREMENT REPORT | RRC (Position Estimate) |
| ….. | → | | …… | |
| n | → | | RRC MEASUREMENT REPORT | RRC (Position Estimate) |

NOTE: In the actual testing the UE may report error messages at step 5 until it has been able to acquire a position estimate.

### 7.5.2.3 Specific message contents

Contents of RESET UE POSITIONING STORED INFORMATION message: TC

The contents of the Reset UE Positioning Stored Information message in Step 1 are the same as specified for Normal UE based A-GPS testing in clause 7.5.1.

Contents of MEASUREMENT CONTROL message: RRC

The contents of the Measurement Control message in steps 2 and 3 are the same as specified for Normal UE based A-GPS testing in clause 7.5.1.

The contents of the Measurement Control message in step 4 are the same as specified for Normal UE based A-GPS testing in clause 7.5.1 with the following exceptions:

| Information Element | Value/remark |
|---|---|
| Amount of reporting<br>Reporting interval | Infinite (see note)<br>2 000 ms |
| NOTE: Infinite means during the complete test time. | |

## 7.5.3 Void

## 7.5.4 Normal UE assisted GPS procedure

The procedure in this clause shall be used for all UE-assisted A-GPS TTFF test cases in CELL_DCH and CELL_FACH state as specified in TS 34.171 [41].

### 7.5.4.1 Initial conditions

User Equipment:

The UE is in CELL_DCH or CELL_FACH state after executing the procedure defined in clause D.2 of TS 34.171 [41].

### 7.5.4.2 Procedure

| Step | Direction | | Message | Comments |
|---|---|---|---|---|
| | UE | SS | | |
| 1 | | ← | RESET UE POSITIONING STORED INFORMATION | TC |
| 2 | | ← | RRC MEASUREMENT CONTROL | RRC (Setup, Periodical Reporting Criteria, GPS Ref time) |
| 3 | | → | RRC MEASUREMENT REPORT | RRC (Additional Assistance Data Request) |
| 4 | | ← | RRC MEASUREMENT CONTROL | RRC (Modify, No Reporting, Assistance Data Satellites 1, 2, 3, 4, 5, 6, 7, 8, 9) |
| 5 | | ← | RRC MEASUREMENT CONTROL | RRC (Modify, Periodical Reporting Criteria) |
| 6 | | → | RRC MEASUREMENT REPORT | RRC (GPS Measured Results), 1st test instance |
| 7 | | ← | RESET UE POSITIONING STORED INFORMATION | TC |
| 8 | | ← | RRC MEASUREMENT CONTROL | RRC (Setup, Periodical Reporting Criteria, GPS Ref time) |
| 9 | | → | RRC MEASUREMENT REPORT | RRC (Additional Assistance Data Request) |
| 10 | | ← | RRC MEASUREMENT CONTROL | RRC (Modify, No Reporting, Assistance Data Satellites 1, 2, 3, 4, 5, 6, 7, 8, 9) |
| 11 | | ← | RRC MEASUREMENT CONTROL | RRC (Modify, Periodical Reporting Criteria) |
| 12 | | → | RRC MEASUREMENT REPORT | RRC (GPS Measured Results), 2nd test instance |

| Step | Direction | | Message | Comments |
|---|---|---|---|---|
| | UE | SS | | |
| 13 | ← | | RESET UE POSITIONING STORED INFORMATION | TC |
| …. | …… | | ……. | |
| n | → | | RRC MEASUREMENT REPORT | RRC (GPS Measured Results), $n^{th}$ test instance |

### 7.5.4.3 Specific message contents

Contents of RESET UE POSITIONING STORED INFORMATION message: TC

| Information Element | Value/remark |
|---|---|
| UE Positioning Technology | AGPS |

Contents of MEASUREMENT CONTROL messages: RRC

MEASUREMENT CONTROL (Steps 2 + (n-1)*6):

| Information element | Value/remark |
|---|---|
| **Measurement Information Elements** | |
| Measurement Identity | 10 |
| Measurement Command | Setup |
| Measurement Reporting Mode | |
| - Measurement report transfer mode | Acknowledged mode RLC |
| - Periodical reporting / Event trigger reporting mode | Periodical reporting |
| Additional Measurements List | Not present |
| CHOICE *Measurement type* | UE positioning measurement |
| - UE positioning measurement | |
| - UE positioning reporting quantity | |
| - Method type | UE assisted |
| - Positioning methods | GPS |
| - Response time | 128 |
| - Horizontal accuracy | 19 (51 m) |
| - Vertical accuracy | 48 (102 m) |
| - GPS timing of cell wanted | FALSE |
| - Multiple sets | FALSE |
| - Additional assistance data request | TRUE |
| - Environmental characterization | Not present |
| - Measurement validity | |
| - UE state | All states |
| - CHOICE *Reporting criteria* | Periodical Reporting Criteria |
| - Amount of reporting | 1 |
| - Reporting Interval | 20000 |
| - UE pos OTDOA assistance data for UE-assisted | Not present |
| - UE pos OTDOA assistance data for UE-based | Not present |
| - UE positioning GPS assistance data | |
| - UE positioning GPS reference time | As specified in clause 10 |
| **Physical Channel Information Elements** | |
| DPCH compressed mode status info | Not present |

MEASUREMENT REPORT (Steps 3 + (n-1)*6):

| Information element | Value/remark |
|---|---|
| **Measurement Information Elements** | |
| Measurement Identity | 10 |
| Measured Results | |
| - CHOICE *Measurement* | |
| - UE positioning measured results | |
| - UE positioning OTDOA measured results | Not present |
| - UE positioning position estimate info | Not present |
| - UE positioning GPS measured results | Not present |
| - UE positioning error | |
| - Error reason | Assistance Data Missing |
| - GPS additional assistance data request | Defines assistance data requested by the UE |
| Measured Results on RACH | Not present |
| Additional Measured Results | Not present |
| Event Results | Not present |

MEASUREMENT CONTROL (Steps 4 + (n-1)*6):

| Information element | Value/remark |
|---|---|
| **Measurement Information Elements** | |
| Measurement Identity | 10 |
| Measurement Command | Modify |
| Measurement Reporting Mode | |
| - Measurement report transfer mode | Acknowledged mode RLC |
| - Periodical reporting / Event trigger reporting mode | Periodical reporting |
| Additional Measurements List | Not present |
| CHOICE *Measurement type* | UE positioning measurement |
| - UE positioning measurement | |
| - UE positioning reporting quantity | |
| - Method type | UE assisted |
| - Positioning methods | GPS |
| - Response time | 128 |
| - Horizontal accuracy | 19 (51 m) |
| - Vertical accuracy | 48 (102 m) |
| - GPS timing of cell wanted | FALSE |
| - Multiple sets | FALSE |
| - Additional assistance data request | FALSE |
| - Environmental characterization | Not present |
| - Measurement validity | |
| - UE state | All states |
| - CHOICE *Reporting criteria* | |
| - No reporting | |
| - UE pos OTDOA assistance data for UE-assisted | Not present |
| - UE pos OTDOA assistance data for UE-based | Not present |
| - UE positioning GPS assistance data | As specified in clause 10 and requested by the UE in Step 3+(n-1)*6 |
| **Physical Channel Information Elements** | |
| DPCH compressed mode status info | Not present |

MEASUREMENT CONTROL (Steps 5 + (n-1)*6):

| Information element | Value/remark |
|---|---|
| **Measurement Information Elements** | |
| Measurement Identity | 10 |
| Measurement Command | Modify |
| Measurement Reporting Mode | |
| - Measurement report transfer mode | Acknowledged mode RLC |
| - Periodical reporting / Event trigger reporting mode | Periodical reporting |
| Additional Measurements List | Not present |
| CHOICE *Measurement type* | UE positioning measurement |
| - UE positioning measurement | |
| - UE positioning reporting quantity | |
| - Method type | UE assisted |
| - Positioning methods | GPS |
| - Response time | 128 |
| - Horizontal accuracy | 19 (51 m) |
| - Vertical accuracy | 48 (102 m) |
| - GPS timing of cell wanted | FALSE |
| - Multiple sets | FALSE |
| - Additional assistance data request | FALSE |
| - Environmental characterization | Not present |
| - Measurement validity | |
| - UE state | All states |
| - CHOICE *Reporting criteria* | Periodical Reporting Criteria |
| - Amount of reporting | 1 |
| - Reporting Interval | 20000 |
| - UE pos OTDOA assistance data for UE-assisted | Not present |
| - UE pos OTDOA assistance data for UE-based | Not present |
| - UE positioning GPS assistance data | Not present |
| **Physical Channel Information Elements** | |
| DPCH compressed mode status info | Not present |

## 7.5.5 UE assisted A-GPS procedure for moving scenario and periodic update test case

The procedure in this clause shall be used for the UE-assisted A-GPS moving scenario and periodic update test case in CELL_DCH and CELL_FACH state as specified in TS 34.171 [41].

### 7.5.5.1 Initial conditions

User Equipment:

The UE is in CELL_DCH or CELL_FACH state after executing the procedure defined in clause D.2 of TS 34.171 [41].

### 7.5.5.2 Procedure

| Step | Direction | | Message | Comments |
|---|---|---|---|---|
| | UE | SS | | |
| 1 | | ← | RESET UE POSITIONING STORED INFORMATION | TC |
| 2 | | ← | RRC MEASUREMENT CONTROL | RRC (Setup, Periodical Reporting Criteria, GPS Ref time) |
| 3 | | → | RRC MEASUREMENT REPORT | RRC (Additional Assistance Data Request) |
| 4 | | ← | RRC MEASUREMENT CONTROL | RRC (Modify, No Reporting, Assistance Data Satellites 1, 2, 3, 4, 5, 6, 7, 8, 9) |
| 5 | | ← | RRC MEASUREMENT CONTROL | RRC (Modify, Periodical Reporting Criteria) |
| 6 | | → | RRC MEASUREMENT REPORT | RRC (GPS Measured Results), 1st test instance |
| 7 | | → | RRC MEASUREMENT REPORT | RRC (GPS Measured Results), 2nd test instance |
| ….. | | → | …… | |

| | | | |
|---|---|---|---|
| n | → | RRC MEASUREMENT REPORT | RRC (GPS Measured Results), $n^{th}$ test instance |
| NOTE: | In the actual testing the UE may report error messages at step 6 until it has been able to acquire GPS measured results. | | |

### 7.5.5.3 Specific message contents

Contents of RESET UE POSITIONING STORED INFORMATION message: TC

The contents of the Reset UE Positioning Stored Information message in Step 1 are the same as specified for Normal UE assisted A-GPS testing in clause 7.5.4.

Contents of MEASUREMENT CONTROL message: RRC

The contents of the Measurement Control message in steps 2 and 4 are the same as specified for Normal UE assisted A-GPS testing in clause 7.5.4.

The contents of the Measurement Control message in step 5 are the same as specified for Normal UE assisted A-GPS testing in clause 7.5.4 with the following exceptions:

| Information Element | Value/remark |
|---|---|
| Amount of reporting | Infinite (see note) |
| Reporting interval | 2 000 ms |
| NOTE: Infinite means during the complete test time. | |

# 8 Test USIM Parameters

## 8.1 Introduction

This clause defines default parameters for programming the elementary files of the test USIM. The requirements of this clause do not apply to the USIM/ME tests of 3GPP TS 31.120 [39] and 3GPP TS 31.121 [40].

### 8.1.1 Definitions

"Test USIM card":

A USIM card supporting the test algorithm for authentication, programmed with the parameters defined in this clause. The electrical, mechanical and environmental requirements of the test USIM card are specified in 3GPP TS 31.101 [22] and 3GPP TS 31.102 [23].

"Test USIM":

Either a test USIM card or the USIM simulator programmed with the parameters defined in this clause.

### 8.1.2 Definition of the test algorithm for authentication

In order to be able to easily test the UMTS authentication and key agreement procedure as specified in 3GPP TS 33.102 [24] and 3GPP TS 33.105 [26] along the whole system, the availability of a test algorithm for generation of authentication vector based on quintets is needed (in GSM triplets was used). Additionally, calculation of the parameters for re-synchronization requests is needed. The definition of the test algorithm are the functions f1, f2, f3, f4, f5 and the corresponding functions for re-synchronization are f1* and f5*.

For test USIM intended to be used for inter-RAT test cases then the test USIM shall support the conversion function c3 according to 3GPP TS 33.102 [24], clause 6.8.1.2 to derive the GSM ciphering key Kc from the UMTS cipher/integrity keys CK and IK.

The test algorithm defined in the present clause shall be implemented in test USIM cards as well in test USIM simulators and SS. The test algorithm may also, for test purposes, be implemented in AUC.

The following procedure employs bit wise modulo 2 addition ("XOR").

The following convention applies:

All data variables in the specification of this test algorithm are presented with the most significant substring on the left hand side and the least significant substring on the right hand side. A substring may be a bit, byte or other arbitrary length bitstring. Where a variable is broken down into a number of substrings, the leftmost (most significant) substring is numbered 0, the next most significant is numbered 1, and so on through to the least significant.

### 8.1.2.1 Authentication and key derivation in the test USIM and SS

The following steps describe sequence of operations for the functions f1, f2, f3, f4 and f5 to perform in the test USIM and SS, in order to obtain the XMAC/MAC, RES/XRES, CK, IK, Kc and AK respectively, to be used in the authentication and key agreement procedure.

Step 1:

XOR to the challenge **RAND**, a predefined number **K** (in which at least one bit is not zero, see clause 8.2), having the same bit length (128 bits) as **RAND**.

The result **XDOUT** of this is:

**XDOUT**[bits 0,1, . . .126,127] = **K** [bits 0,1, . . .126,127] XOR **RAND**[bits 0,1, . . .126,127]

Step 2:

**RES** (test USIM)**, XRES** (SS), **CK, IK** and **AK** are extracted from **XDOUT** this way:

**RES**[bits 0,1, . . .n-1,n] = **f2**(**XDOUT**,n) = **XDOUT**[bits 0,1, . . .n-1,n] (with 30 < n < 128)

NOTE: Suggested length for RES is 128 bits (i.e. n = 127).
In SS and AUC, the XRES calculation is identical to RES.

**CK**[bits 0,1, . . .126,127] = **f3**(**XDOUT**) = **XDOUT**[bits 8,9, . . .126,127,0,1, . . .6,7]

**IK**[bits 0,1, . . . 126,127] = **f4**(**XDOUT**) = **XDOUT**[bits 16,17, . . .126,127,0,1, . . .14,15]

**AK**[bits 0,1, . . . 46,47] = **f4**(**XDOUT**) = **XDOUT**[bits 24,25, . . .70,71]

For test USIM intended for inter-RAT testing the GSM ciphering key Kc shall be derived from the UMTS cipher/integrity keys:

**Kc**[bits 0,1, . . . 62,63] **=** **c3(CK,IK)**, see 3GPP TS 33.102 [24], clause 6.8.1.2.

Step 3:

Concatenate **SQN** with **AMF** to obtain **CDOUT** like this:

**CDOUT**[bits 0,1,. . .62,63] = **SQN**[bits 0,1,. . .46,47] || **AMF**[bits 0,1,. . .14,15]

NOTE: For test USIM the **SQN** = $\mathbf{SQN_{MS}}$ = $\mathbf{SQN_{SS}}$[bits 0,1,. . .46,47] = **AUTN**[bits 0,1,. . .46,47] XOR **AK**[bits 0,1, . . . 46,47] where AUTN is the received authentication token.

Step 4:

**XMAC** (test USIM) and **MAC** (SS) are calculated from **XDOUT** and **CDOUT** this way:

**XMAC**[bits 0,1, . . .62, 63] = **f1**(**XDOUT, CDOUT**) = **XDOUT**[bits 0,1. . .62,63] XOR **CDOUT**[bits 0,1,. . .62,63]

NOTE: In SS and AUC, the MAC calculation is identical to XMAC.

Step 5:

The SS calculates the authentication token **AUTN**:

**AUTN**[bits 0,1,..126,127] = **SQN** ⊕ **AK**[bits 0,1,. . .46,47] || **AMF**[bits 0,1,. . .14,15] || **MAC**[bits 0,1, . . .62, 63]

Where **SQN** ⊕ **AK**[bits 0,1,. . .46,47] = **SQN**[bits 0,1,. . .46,47] XOR **AK**[bits 0,1, . . . 46,47]

### 8.1.2.2 Generation of re-synchronization parameters in the USIM

For SS to be able to initiate an authentication re-synchronization procedure a specific AMF value has been defined.

$AMF_{RESYNCH}$ = AMF[bits 0,1,..14,15] = "1111 1111 1111 1111"

When the test USIM receives an authentication token (AUTN) having the value of AMF field equal to the $AMF_{RESYNCH}$ value then the test USIM shall initiate the re-synchronization procedure.

When the test USIM starts the re-synchronization procedure, the MAC-S and AK have to be calculated using the functions f1* and f5*, which in the test algorithm are identical to f1 and f5, respectively.

Step 1:

XOR to the challenge **RAND**, a predefined number **K** (in which at least one bit is not zero, see 8.2), having the same bit length (128 bits) as **RAND**.

The result **XDOUT** of this is:

**XDOUT**[bits 0,1, . . .126,127] = **K**[bits 0,1, . . .126,127] XOR **RAND**[bits 0,1, . . .126,127]

Step 2:

**AK** is extracted from **XDOUT** this way:

**AK**[bits 0,1, . . . 46,47] = **f5***(**XDOUT**) = **XDOUT**[bits 24,25, . . .70,71]

Step 3:

Concatenate $\mathbf{SQN_{MS}}$ with **AMF*** to obtain **CDOUT** like this:

**CDOUT**[bits 0,1,. . .62,63] = $\mathbf{SQN_{MS}}$[bits 0,1,. . .46,47] || **AMF***[bits 0,1,. . .14,15]

Where **AMF*** assumes a dummy value of all zeros.

NOTE 1: For test USIM the $\mathbf{SQN_{MS}} = \mathbf{SQN_{SS}}$[bits 0,1,. . .46,47] = **AUTN**[bits 0,1,. . .46,47] XOR **AK**[bits 0,1, . . . 46,47] where AUTN is the received authentication token.

NOTE 2: For SS and AUC the $\mathbf{SQN_{MS}}$ = **AUTS**[bits 0,1,. . .46,47] XOR **AK**[bits 0,1, . . . 46,47] where AUTS is the received re-synchronization parameter.

Step 4:

**MAC-S** is calculated from **XDOUT** and **CDOUT** this way:

**MAC-S**[bits 0,1, . . .62, 63] = **f1***(**XDOUT, CDOUT**) = **XDOUT**[bits 0,1. . .62,63] XOR **CDOUT**[bits 0,1,. . .62,63]

NOTE: In SS and AUC, the XMAC-S calculation is identical to MAC-S.

Step 5:

The test USIM calculates the re-synchronization parameter **AUTS**:

**AUTS**[bits 0,1,..110,111] = $\mathbf{SQN_{MS}} \oplus$ **AK**[bits 0,1,. . .46,47] || **MAC-S**[bits 0,1, . . .62, 63]

Where $\mathbf{SQN_{MS}} \oplus$ **AK**[bits 0,1,. . .46,47] = $\mathbf{SQN_{MS}}$ [bits 0,1,. . .46,47] XOR **AK**[bits 0,1, . . . 46,47]

### 8.1.2.3 Using the authentication test algorithm for UE conformance testing

#### 8.1.2.3.1 Authentication accept case

The authentication accept case is illustrated in figures 8.1.2.3.1 and 8.1.2.3.2.

The SS calculates the authentication token AUTN according to the test algorithm as specified in clause 8.1.2.1 (step 1 to step 5) using an AMF value different from the $AMF_{RESYNCH}$ value.

The SS sends an authentication request, including RAND and AUTN parameters, to the ME/USIM.

Based on the received RAND parameter the test USIM calculates the RES, CK, IK, Kc and XMAC parameters according to clause 8.1.2.1 (step 1 to step 4). The test USIM extracts the $SQN_{MS} = SQN_{SS}$, AMF and MAC parameters from the received authentication token AUTN.

The test USIM checks that XMAC = MAC and then return the RES, CK and IK parameters to the ME.

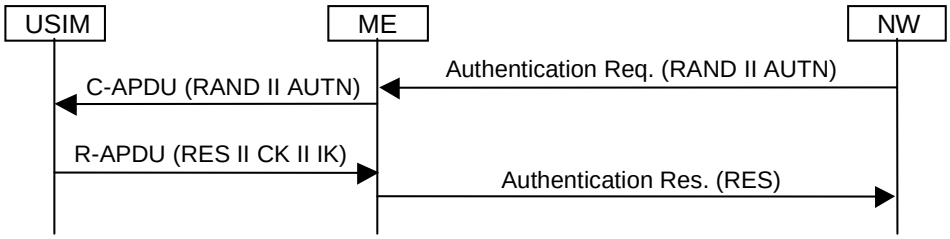



**Figure 8.1.2.3.1: Network accepted by UE (USIM not supporting derivation of GSM cipher key Kc)**

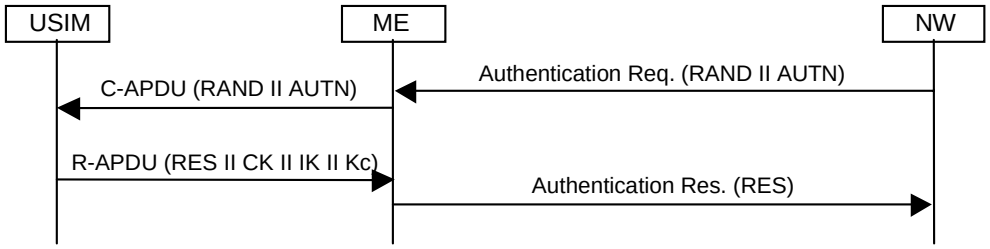



**Figure 8.1.2.3.2: Network accepted by UE (USIM supporting derivation of GSM cipher key Kc)**

### 8.1.2.3.2 MAC failure case

The MAC failure case is illustrated in figure 8.1.2.3.2.

The SS calculates the authentication token AUTN according to the test algorithm as specified in clause 8.1.2.1 (step 1 to step 5) using an AMF value different from the $AMF_{RESYNCH}$ value and a MAC value different from what is calculated in clause 8.1.2.1 step 4.

The SS sends an authentication request, including RAND and AUTN parameters, to the ME/USIM.

Based on the received RAND parameter The test USIM calculates the RES, CK, IK, Kc and XMAC parameters according to clause 8.1.2.1 (step 1 to step 4).

The test USIM extracts the $SQN_{MS} = SQN_{SS}$, AMF and MAC parameters from the received authentication token AUTN.

When the test USIM identifies that the calculated XMAC value is different from the MAC value received in AUTN then the USIM notifies the ME of the MAC failure and the ME sends an AUTENTICATION FAILURE message to the SS (cause "MAC failure").

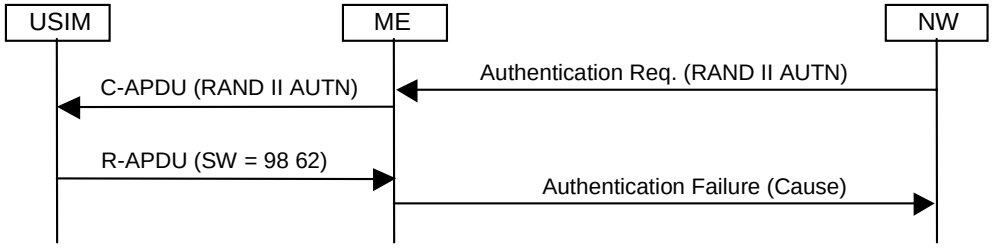


**Figure 8.1.2.3.2: MAC failure cases**

### 8.1.2.3.3 SQN failure case

The SQN failure case is illustrated in figure 8.1.2.3.3.

The SS calculates the authentication token AUTN according to the test algorithm as specified in clause 8.1.2.1 (step 1 to step 5) using an AMF value equal to $AMF_{RESYNCH}$.

The SS sends an authentication request, including RAND and AUTN parameters, to the UE/USIM.

The test USIM extracts the $SQN_{MS} = SQN_{SS}$, AMF and MAC parameters from the received authentication token AUTN.

When the test USIM identifies that the AMF field is equal to the $AMF_{RESYNCH}$ value it calculates the re-synchronization parameter AUTS as specified in clause 8.1.2.2 (step 1 to step 5) and forward it to the ME.

The ME sends an AUTHENTICATION FAILURE message to the SS including the AUTS parameter.

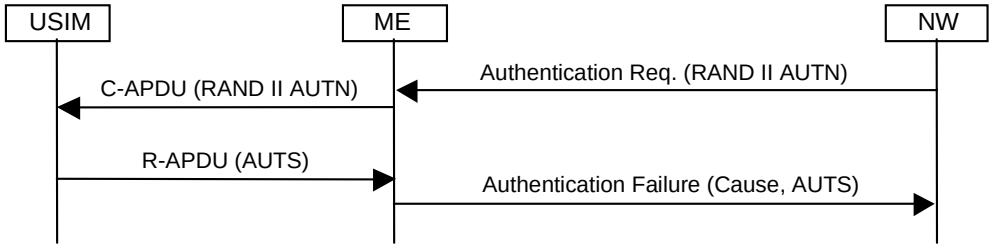



**Figure 8.1.2.3.3: SQN failure case**

## 8.1.3 Definition of the test algorithm for VGCS/VBS VSTK generation

In order to be able to easily test the VGCS/VBS key generation and encryption as specified in TS 43.020 [44] and TS 31.102 [23] along the whole system, the availability of a test algorithm for generation of the VSTK-key is needed.

The test algorithm defined in the present clause shall be implemented in test USIM cards as well in test USIM simulators and SS.

The following procedure employs bit wise modulo 2 addition ("XOR").

The following convention applies:

All data variables in the specification of this test algorithm are presented with the most significant substring on the left hand side and the least significant substring on the right hand side. A substring may be a bit, byte or other arbitrary length bitstring. Where a variable is broken down into a number of substrings, the leftmost (most significant) substring is numbered 0, the next most significant is numbered 1, and so on through to the least significant.

### 8.1.3.1 VSTK generation in the test USIM and SS

The following steps describe the sequence of operations for the function A8_V (TS 43.020 [44]) to be performed in the test USIM and SS, in order to obtain the VSTK, to be used in the subsequent ME/BSS key derivation steps for VGCS/VBS ciphering.

Step 1:

Expand the 36-bit value **VSTK_RAND** to an intermediate 40-bit value **EXPAND**:

**FILLER**[bits 0,..7] = “11111111”

**EXPAND** [bits 0,1, . . .39] **= FILLER**[bits 0,..3] **|| VSTK_RAND**[bits 0,1, . . .35]

Expand the 40-bit value **EXPAND** to a 128-bit value **EXP_RAND**:

**EXP_RAND**[bits 0,1, . . .126,127] = **EXPAND**[bits 0,1, . . .39] || **EXPAND**[bits 0,1, . . .39] || **EXPAND**[bits 0,1, . . .39] || **FILLER**[bits 0,..7]

Step 2:

XOR the expanded 128 bit **EXP_RAND** with a stored **V_Ki** i.e. a 128 bit Voice Group or Broadcast Group Key (128 bit) number taken by the USIM from an internal table indexed by VK_Id and Group_Id

The result **VSTK** of this is:

**VSTK**[bits 0,1, . . .126,127]= **V_Ki** [bits 0,1, . . .126,127] XOR **EXP_RAND**[bits 0,1, . . .126,127]

## 8.2 Default Parameters for the test USIM

K:

Size: 16 Bytes

Default values: Bytes 1 (HEX): 00

Bytes 2 (HEX): 01

Bytes 3 (HEX): 02

Bytes 4 (HEX): 03

Bytes 5 (HEX): 04

Bytes 6 (HEX): 05

Bytes 7 (HEX): 06

Bytes 8 (HEX): 07

Bytes 9 (HEX): 08

Bytes 10 (HEX): 09

Bytes 11 (HEX): 0A

Bytes 12 (HEX): 0B

Bytes 13 (HEX): 0C

Bytes 14 (HEX): 0D

Bytes 15 (HEX): 0E

Bytes 16 (HEX): 0F

PIN Disabling:

The PIN enabled / disabled flag will be set to "PIN Disabled". This ensures that when the Test USIM is inserted into a UE the user will not be prompted for PIN entry.

## 8.3 Default settings for the Elementary Files (EFs)

The format and coding of elementary files of the USIM are defined in 3GPP TS 31.101 [22] and 3GPP TS 31.102 [23]. The following clauses define the default parameters to be programmed into each elementary file. Some files may be updated by the UE based on information received from the SS. These are identified in the following clauses.

If EFs have an unassigned value, it may not be clear from the main text what this value should be. This clause suggests values in these cases.

### 8.3.1 Contents of the EFs at the MF level

#### 8.3.1.1 $EF_{DIR}$

#### 8.3.1.2 $EF_{ICCID}$ (ICC Identity)

The programming of this EF is a test house option.

#### 8.3.1.3 $EF_{PL}$ (Preferred Languages)

The programming of this EF follows default parameter written in 3GPP TS 31.102 [23], annex E.

#### 8.3.1.4 $EF_{ARR}$ (Access rule reference)

The programming of this EF is a test house option.

### 8.3.2 Contents of files at the USIM ADF (Application DF) level

#### 8.3.2.1 $EF_{LI}$ (Language Indication)

The programming of this EF follows default parameter written in 3GPP TS 31.102 [23], annex E.

#### 8.3.2.2 $EF_{IMSI}$ (IMSI)

The IMSI value will be chosen by the test house. The IMSI used by the SS will align this value.

File size: 9 bytes

Default values: Byte 1 (DEC): 8

Bytes 2 to 9 (HEX): 09 10 10 ** ** ** ** **

"*" indicates any number between 0 and 9 subject to the restriction that IMSI mod 1000 (i.e. bytes 7, 8 and 9) lies in one of the following ranges:

- 063 to 125, 189 to 251, 315 to 377, 441 to 503, 567 to 629, 693 to 755, 819 to 881 or 945 to 999.

NOTE: This ensures that the UE can listen to the second CCCH when more than one basic physical channel is configured for the CCCH. This is necessary for the test of "paging re-organization".

#### 8.3.2.3 $EF_{Keys}$ (Ciphering and Integrity Keys)

The programming of this EF follows default parameter written in 3GPP TS 31.102 [23], annex E.

#### 8.3.2.4 $EF_{KeysPS}$ (Ciphering and Integrity Keys for Packet Switched domain)

The programming of this EF follows default parameter written in 3GPP TS 31.102 [23], annex E.

#### 8.3.2.5 $EF_{PLMNwAcT}$ (User controlled PLMN selector with Access Technology)

| | | | |
|---|---|---|---|
| File size: | 5n bytes | | |
| Default values (HEX): | Bytes 1 to 3: | 32 F4 10 | (MCC, MNC) - Translates to 234, 01 |
| | Bytes 4 to 5: | 80 00 | (Access Technology) - Translates to UTRAN |
| | Bytes 6 to 8: | 32 F4 20 | (MCC, MNC) |
| | Bytes 9 to 10: | 80 00 | (Access Technology) |
| | Bytes 11 to 13: | 32 F4 30 | (MCC, MNC) |
| | .... | | |
| | .... | | |
| | .... | | |

Bytes(5n-4) to (5n-2): 32 F4 43 (MCC, MNC)

Bytes (5n-1) to 5n: 80 00 (Access Technology)

PLMNs are shown coded above since this is the largest number required for a test. It is necessary to take this into account since the USIM cards must be dimensioned to cope with this number of records.

### 8.3.2.6 $EF_{HPPLMN}$ (Higher Priority PLMN search period)

File size: 1 byte

Default value (HEX): 00 (No higher priority PLMN search attempts)

### 8.3.2.7 $EF_{ACMmax}$ (ACM maximum value)

File size: 3 bytes

Default: Byte 1: 00

Byte 2: 00

Byte 3: 00

The above translates to: "Not valid".

### 8.3.2.8 $EF_{UST}$ (USIM Service Table)

Services will be allocated and activated as follows.

| Services | | Activated | Version |
|---|---|---|---|
| Service n°1 : | Local Phone Book | Option | |
| Service n°2 : | Fixed Dialling Numbers (FDN) | Option | |
| Service n°3 : | Extension 2 | Option | |
| Service n°4 : | Service Dialling Numbers (SDN) | Option | |
| Service n°5 : | Extension3 | Option | |
| Service n°6 : | Barred Dialling Numbers (BDN) | Option | |
| Service n°7 : | Extension4 | Option | |
| Service n°8 : | Outgoing Call Information (OCI and OCT) | Option | |
| Service n°9 : | Incoming Call Information (ICI and ICT) | Option | |
| Service n°10: | Short Message Storage (SMS) | Yes | |
| Service n°11: | Short Message Status Reports (SMSR) | Option | |
| Service n°12: | Short Message Service Parameters (SMSP) | Yes | |
| Service n°13: | Advice of Charge (AoC) | Yes | |
| Service n°14: | Capability Configuration Parameters (CCP) | Yes | |
| Service n°15: | Cell Broadcast Message Identifier | Yes | |
| Service n°16: | Cell Broadcast Message Identifier Ranges | Yes | |
| Service n°17: | Group Identifier Level 1 | Option | |
| Service n°18: | Group Identifier Level 2 | Option | |
| Service n°19: | Service Provider Name | Option | |
| Service n°20: | User controlled PLMN selector with Access Technology | Yes | |
| Service n°21: | MSISDN | Option | |
| Service n°22: | Image (IMG) | Option | |
| Service n°23: | Not used (reserved for SoLSA) | No | |
| Service n°24: | Enhanced Multi-Level Precedence and Pre-emption Service | Option | |
| Service n°25: | Automatic Answer for Emlpp | Option | |
| Service n°26: | RFU | No | |
| Service n°27: | GSM Access | Yes | |
| Service n°28: | Data download via SMS-PP | Option | |
| Service n°29: | Data download via SMS-CB | Option | |
| Service n°30: | Call Control by USIM | Option | |
| Service n°31: | MO-SMS Control by USIM | Option | |
| Service n°32: | RUN AT COMMAND command | Option | |
| Service n°33: | Packet Switched Domain | Yes | |
| Service n°34: | Enabled Services Table | Yes | |
| Service n°35: | APN Control List (ACL) | Option | |

| Services | | Activated | Version |
|---|---|---|---|
| Service n°36: | Depersonalization Control Keys | Option | |
| Service n°37: | Co-operative Network List | Option | |
| Service n°38: | GSM security context | Yes | |
| Service n°39: | CPBCCH Information | Yes | |
| Service n°40: | Investigation Scan | Yes | |
| Service n°41: | MExE | Option | |
| Service n°42 | Operator controlled PLMN selector with Access Technology | Yes | |
| Service n°43 | HPLMN selector with Access Technology | Yes | |
| Service n°44 | Extension 5 | Option | |
| Service n°45 | PLMN Network Name | Option | |
| Service n°46 | Operator PLMN List | Option | |
| Service n°47 | Mailbox Dialling Numbers | Option | |
| Service n°48 | Message Waiting Indication Status | Option | |
| Service n°49 | Call Forwarding Indication Status | Option | |
| Service n°50 | Reserved and shall be ignored | Option | |
| Service n°51 | Service Provider Display Information | Option | |
| Service n°52 | Multimedia Messaging Service (MMS) | Option | |
| Service n°53 | Extension 8 | Option | |
| Service n°54 | Call control on GPRS by USIM | Option | |
| Service n°55 | MMS User Connectivity Parameters | Option | |
| Service n°56 | Network's indication of alerting in the MS (NIA) | Option | |
| Service n°57 | VGCS Group Identifier List ($EF_{VGCS}$ and $EF_{VGCSS}$) | YES | |
| Service n°58 | VBS Group Identifier List ($EF_{VBS}$ and $EF_{VBSS}$) | YES | |
| Service n°59 | Pseudonym | Option | REL-6 and later |
| Service n°60 | User Controlled PLMN selector for WLAN access | Option | REL-6 and later |
| Service n°61 | Operator Controlled PLMN selector for WLAN access | Option | REL-6 and later |
| Service n°62 | User controlled WSID list | Option | REL-6 and later |
| Service n°63 | Operator controlled WSID list | Option | REL-6 and later |
| Service n°64 | VGCS security | YES | REL-6 and later |
| Service n°65 | VBS security | YES | REL-6 and later |
| Service n°66 | WLAN Reauthentication Identity | Option | REL-6 and later |
| Service n°67 | Multimedia Messages Storage | Option | REL-6 and later |
| Service n°68 | Generic Bootstrapping Architecture (GBA) | Option | REL-6 and later |
| Service n°69 | MBMS security | Option | REL-6 and later |
| Service n°70 | Data download via USSD and USSD application mode | Option | REL-6 and later |
| Service n°72 | Additional TERMINAL PROFILE after UICC activation | Option | REL-6 and later |

### 8.3.2.9 $EF_{ACM}$ (Accumulated Call Meter)

File size: 3 bytes

Default: Byte 1: 00

Byte 2: 00

Byte 3: 00

The above translates to: "Not yet implemented".

### 8.3.2.10 $EF_{GID1}$ (Group Identifier Level 1)

The programming of this EF is a test house option.

### 8.3.2.11 $EF_{GID2}$ (Group Identifier Level 2)

The programming of this EF is a test house option.

### 8.3.2.12 $EF_{SPN}$ (Service Provider Name)

The programming of this EF is a test house option.

### 8.3.2.13 $EF_{PUCT}$ (Price per Unit and Currency Table)

The programming of this EF follows default parameter written in 3GPP TS 31.102 [23], annex E.

### 8.3.2.14 $EF_{CBMI}$ (Cell Broadcast Message identifier selection)

The programming of this EF is a test house option.

The file size is 2n bytes, where n is the number of Cell broadcast message identifier records - each record defining a type of Cell Broadcast message which may be accessed by the UE. Care should be taken when dimensioning the USIM to take into account the number of Cell Broadcast message identifier records required.

### 8.3.2.15 $EF_{ACC}$ (Access Control Class)

The EFACC can be selected by a test house in two types.

Type A;

| | | |
|---|---|---|
| File size: | 2 Bytes | |
| Default values (BIN): | Byte 1: | 000000** |
| | Byte 2: | ******** |

The test house may set any single bit shown by "*" to "1". All remaining bits of byte 2 will be set to "0". This determines the access control class of the USIM.

Type B;

| | | |
|---|---|---|
| Default values (BIN): | Byte 1: | 111110** |
| | Byte 2: | ******** |

The test house may set any single bit shown by "*" to "1". All remaining bits of byte 2 will be set to "0". This determines the access control class of the USIM.

### 8.3.2.16 $EF_{FPLMN}$ (Forbidden PLMNs)

The programming of this EF follows default parameter written in 3GPP TS 31.102 [23], annex E.

### 8.3.2.17 $EF_{LOCI}$ (Location Information)

| | | |
|---|---|---|
| File size: | 11 Bytes | |
| Default values: | Bytes 1 to 4 (HEX): | FF FF FF FF (TMSI) |
| | Bytes 5 to 9 (HEX): | 42 F6 18 FF FE (LAI) |
| | Byte 10 (HEX): | FF (RFU) |
| | Byte 11 (BIN): | 00000001 (Location Update Status = "not updated") |

Bytes 5 to 9: LAI-MCC = 246 (bytes 5 to 6) and LAI-MNC = 81 (byte 7) are frequently used. The LAC (bytes 8 to 9) is set to "FF FE" since this, in conjunction with byte 11 setting of "01", is used to ensure that the UE performs a location update at the beginning of a test.

Bytes in this file (e.g. TMSI in bytes 1 to 4) may be updated as a result of a location update attempt by the UE.

### 8.3.2.18 $EF_{AD}$ (Administrative Data)

| | | |
|---|---|---|
| File size: | 4 bytes | |
| Default values | Byte 1: | 10000000 - (type approval operations) |
| | Byte 2: | 00000000 |
| | Byte 3: | 00000000 |
| | Byte 4: | 00000010 |

### 8.3.2.19 Void

### 8.3.2.20 $EF_{CBMID}$ (Cell Broadcast Message Identifier for Data Download)

The programming of this EF follows default parameter written in 3GPP TS 31.102 [23], annex E.

### 8.3.2.21 $EF_{ECC}$ (Emergency Call Codes)

The programming of this EF is a test house option.

### 8.3.2.22 $EF_{CBMIR}$ (Cell Broadcast Message Identifier Range selection)

The programming of this EF follows default parameter written in 3GPP TS 31.102 [23], annex E.

### 8.3.2.23 $EF_{PSLOCI}$ (Packet Switched location information)

File size: 14 Bytes

| | | |
|---|---|---|
| Default values: | Bytes 1 to 4 (HEX): | FF FF FF FF (P-TMSI) |
| | Bytes 5 to 7 (HEX): | FF FF FF (P-TMSI signature value) |
| | Bytes 8 to 13 (HEX): | 42 F6 18 FF FE FF (RAI) |
| | Byte 14 (BIN): | 00000001 (Routing Area update status = "not updated") |

Bytes 8 to 13: RAI-MCC = 246 (bytes 8 to 9) and RAI-MNC = 81 (byte 10) are frequently used. The LAC (bytes 11 to 12) is set to "FF FE" since this, in conjunction with byte 14 setting of "01", is used to ensure that the UE performs a location update at the beginning of a test.

Bytes in this file (e.g. P-TMSI in bytes 1 to 4) may be updated as a result of a location update attempt by the UE.

### 8.3.2.24 $EF_{FDN}$ (Fixed Dialling Numbers)

The programming of this EF follows default parameter written in 3GPP TS 31.102 [23], annex E.

### 8.3.2.25 $EF_{SMS}$ (Short messages)

The programming of this EF follows default parameter written in 3GPP TS 31.102 [23], annex E.

### 8.3.2.26 $EF_{MSISDN}$ (MSISDN)

The programming of this EF follows default parameter written in 3GPP TS 31.102 [23], annex E.

### 8.3.2.27 $EF_{SMSP}$ (Short message service parameters)

The programming of this EF follows default parameter written in 3GPP TS 31.102 [23], annex E.

### 8.3.2.28 $EF_{SMSS}$ (SMS status)

The programming of this EF follows default parameter written in 3GPP TS 31.102 [23], annex E.

### 8.3.2.29 $EF_{SDN}$ (Service Dialling Numbers)

The programming of this EF follows default parameter written in 3GPP TS 31.102 [23], annex E.

### 8.3.2.30 $EF_{EXT2}$ (Extension2)

The programming of this EF follows default parameter written in 3GPP TS 31.102 [23], annex E.

### 8.3.2.31 $EF_{EXT3}$ (Extension3)

The programming of this EF follows default parameter written in 3GPP TS 31.102 [23], annex E.

#### 8.3.2.32 $EF_{SMSR}$ (Short message status reports)

The programming of this EF follows default parameter written in 3GPP TS 31.102 [23], annex E.

#### 8.3.2.33 $EF_{ICI}$ (Incoming Call Information)

The programming of this EF follows default parameter written in 3GPP TS 31.102 [23], annex E.

#### 8.3.2.34 $EF_{OCI}$ (Outgoing Call Information)

The programming of this EF follows default parameter written in 3GPP TS 31.102 [23], annex E.

#### 8.3.2.35 $EF_{ICT}$ (Incoming Call Timer)

The programming of this EF follows default parameter written in 3GPP TS 31.102 [23], annex E.

#### 8.3.2.36 $EF_{OCT}$ (Outgoing Call Timer)

The programming of this EF follows default parameter written in 3GPP TS 31.102 [23], annex E.

#### 8.3.2.37 $EF_{EXT5}$ (Extension5)

The programming of this EF follows default parameter written in 3GPP TS 31.102 [23], annex E.

#### 8.3.2.38 $EF_{CCP2}$ (Capability Configuration Parameters 2)

The programming of this EF follows default parameter written in 3GPP TS 31.102 [23], annex E.

#### 8.3.2.39 $EF_{eMLPP}$ (enhanced Multi Level Precedence and Pre-emption)

The programming of this EF is a test house option.

#### 8.3.2.40 $EF_{AAeM}$ (Automatic Answer for eMLPP Service)

The programming of this EF follows default parameter written in 3GPP TS 31.102 [23], annex E.

#### 8.3.2.41 Void

#### 8.3.2.42 $EF_{Hiddenkey}$ (Key for hidden phone book entries)

The programming of this EF follows default parameter written in 3GPP TS 31.102 [23], annex E.

#### 8.3.2.43 Void

#### 8.3.2.44 $EF_{BDN}$ (Barred dialling numbers)

The programming of this EF follows default parameter written in 3GPP TS 31.102 [23], annex E.

#### 8.3.2.45 $EF_{EXT4}$ (Extension 4)

The programming of this EF follows default parameter written in 3GPP TS 31.102 [23], annex E.

#### 8.3.2.46 $EF_{CMI}$ (Comparison method information)

The programming of this EF follows default parameter written in 3GPP TS 31.102 [23], annex E.

#### 8.3.2.47 $EF_{EST}$ (Enabled service table)

The programming of this EF is a test house option.

#### 8.3.2.48 $EF_{ACL}$ (Access point name control list)

The programming of this EF follows default parameter written in 3GPP TS 31.102 [23], annex E.

#### 8.3.2.49 $EF_{DCK}$ (Depersonalization control keys)

The programming of this EF follows default parameter written in 3GPP TS 31.102 [23], annex E.

#### 8.3.2.50 $EF_{CNL}$ (Co-operative network list)

The programming of this EF follows default parameter written in 3GPP TS 31.102 [23], annex E.

#### 8.3.2.51 $EF_{START-HFN}$ (Initialisation values for Hyperframe number)

The programming of this EF follows default parameter written in 3GPP TS 31.102 [23], annex E.

#### 8.3.2.52 $EF_{THRESHOLD}$ (Maximum value of START)

The programming of this EF is a test house option.

#### 8.3.2.53 $EF_{OPLMNwACT}$ (Operator controlled PLMN selector with Access Technology)

The programming of this EF follows default parameter written in 3GPP TS 31.102 [23], annex E.

#### 8.3.2.54 $EF_{HPLMNwAcT}$ (HPLMN selector with Access Technology)

The programming of this EF follows default parameter written in 3GPP TS 31.102 [23], annex E.

#### 8.3.2.55 $EF_{ARR}$ (Access rule reference)

The programming of this EF is a test house option.

#### 8.3.2.56 Void

#### 8.3.2.57 $EF_{NETPAR}$ (Network Parameters)

The programming of this EF follows default parameter written in 3GPP TS 31.102 [23], annex E.

#### 8.3.2.58 $EF_{PNN}$ (PLMN Network Name)

The programming of this EF follows default parameter written in 3GPP TS 31.102 [23], annex E.

#### 8.3.2.59 $EF_{OPL}$ (Operator PLMN List)

The programming of this EF follows default parameter written in 3GPP TS 31.102 [23], annex E.

#### 8.3.2.60 $EF_{MBDN}$ (Mailbox Dialling Numbers)

The programming of this EF follows default parameter written in 3GPP TS 31.102 [23], annex E.

#### 8.3.2.61 $EF_{EXT6}$ (Extension6)

The programming of this EF follows default parameter written in 3GPP TS 31.102 [23], annex E.

#### 8.3.2.62 $EF_{MBI}$ (Mailbox Identifier)

The programming of this EF follows default parameter written in 3GPP TS 31.102 [23], annex E.

#### 8.3.2.63 $EF_{MWIS}$ (Message Waiting Indication Status)

The programming of this EF follows default parameter written in 3GPP TS 31.102 [23], annex E.

#### 8.3.2.64 $EF_{CFIS}$ (Call Forwarding Indication Status)

The programming of this EF follows default parameter written in 3GPP TS 31.102 [23], annex E.

#### 8.3.2.65 $EF_{EXT7}$ (Extension7)

The programming of this EF follows default parameter written in 3GPP TS 31.102 [23], annex E.

### 8.3.2.66 $EF_{SPDI}$ (Service Provider Display Information)

The programming of this EF follows default parameter written in 3GPP TS 31.102 [23], annex E.

### 8.3.2.67 $EF_{MMSN}$ (MMS Notification)

The programming of this EF follows default parameter written in 3GPP TS 31.102 [23], annex E.

### 8.3.2.68 $EF_{EXT8}$ (Extension 8)

The programming of this EF follows default parameter written in 3GPP TS 31.102 [23], annex E.

### 8.3.2.69 $EF_{MMSICP}$ (MMS Issuer Connectivity Parameters)

The programming of this EF follows default parameter written in 3GPP TS 31.102 [23], annex E.

### 8.3.2.70 $EF_{MMSUP}$ (MMS User Preferences)

The programming of this EF follows default parameter written in 3GPP TS 31.102 [23], annex E.

### 8.3.2.71 $EF_{MMSUCP}$ (MMS User Connectivity Parameters)

The programming of this EF follows default parameter written in 3GPP TS 31.102 [23], annex E.

### 8.3.2.72 $EF_{NIA}$ (Network's Indication of Alerting)

The programming of this EF follows default parameter written in 3GPP TS 31.102 [23], annex E.

If service n°57 is "available", this file shall be present.

This EF contains a list of those VGCS group identifiers the user has subscribed to. The elementary file is used by the ME for group call establishment and group call reception.

File size: Bytes 200

Default values:

| Bytes | Group ID | Value | BCD encoding in the USIM |
|---|---|---|---|
| 1-4 | 1 | 12 | 21 FF FF FF |
| 5-8 | 2 | 123 | 21 F3 FF FF |
| 9-12 | 3 | 1234 | 21 43 FF FF |
| 13-16 | 4 | 12348 | 21 43 F8 FF |
| 17-20 | 5 | 123491 | 21 43 19 FF |
| 21-24 | 6 | 1235029 | 21 53 20 F9 |
| 25-28 | 7 | 12351 | 21 53 F1 FF |
| 29-32 | 8 | 12352 | 21 53 F2 FF |
| 33-36 | 9 | 12353 | 21 53 F3 FF |
| 37-40 | 10 | 12354 | 21 53 F4 FF |
| 41-44 | 11 | 12355 | 21 53 F5 FF |
| 45-48 | 12 | 12356 | 21 53 F6 FF |
| 49-52 | 13 | 12357 | 21 53 F7 FF |
| 53-56 | 14 | 12358 | 21 53 F8 FF |
| 57-60 | 15 | 12359 | 21 53 F9 FF |
| 61-64 | 16 | 20000 | 02 00 F0 FF |
| 65-68 | 17 | 20001 | 02 00 F1 FF |
| 69-72 | 18 | 20002 | 02 00 F2 FF |
| 73-76 | 19 | 20003 | 02 00 F3 FF |
| 77-80 | 20 | 20004 | 02 00 F4 FF |
| 81-84 | 21 | 20005 | 02 00 F5 FF |
| 85-88 | 22 | 20006 | 02 00 F6 FF |
| 89-92 | 23 | 20007 | 02 00 F7 FF |
| 93-96 | 24 | 20008 | 02 00 F8 FF |
| 97-100 | 25 | 20009 | 02 00 F9 FF |
| 101-104 | 26 | 20010 | 02 10 F0 FF |
| 105-108 | 27 | 66660 | 66 66 F0 FF |

| Bytes | Group ID | Value | BCD encoding in the USIM |
|---|---|---|---|
| 109-112 | 28 | 66661 | 66 66 F1 FF |
| 113-116 | 29 | 66662 | 66 66 F2 FF |
| 117-120 | 30 | 666638 | 66 66 83 FF |
| 121-124 | 31 | 66664 | 66 66 F4 FF |
| 125-128 | 32 | 66665 | 66 66 F5 FF |
| 129-132 | 33 | 66666 | 66 66 F6 FF |
| 133-136 | 34 | 66667 | 66 66 F7 FF |
| 137-140 | 35 | 66668 | 66 66 F8 FF |
| 141-144 | 36 | 66669 | 66 66 F9 FF |
| 145-148 | 37 | 66670 | 66 76 F0 FF |
| 149-152 | 38 | 80120 | 08 21 F0 FF |
| 153-156 | 39 | 80121 | 08 21 F1 FF |
| 157-160 | 40 | 80122 | 08 21 F2 FF |
| 161-164 | 41 | 80123 | 08 21 F3 FF |
| 165-168 | 42 | 80124 | 08 21 F4 FF |
| 169-172 | 43 | 80125 | 08 21 F5 FF |
| 173-176 | 44 | 80126 | 08 21 F6 FF |
| 177-180 | 45 | 80127 | 08 21 F7 FF |
| 181-184 | 46 | 80128 | 08 21 F8 FF |
| 185-188 | 47 | 80129 | 08 21 F9 FF |
| 189-192 | 48 | 80130 | 08 31 F0 FF |
| 193-196 | 49 | 99999 | 99 99 F9 FF |
| 197-200 | 50 | 1111119 | 11 11 11 F9 |

For Group Id = 1 V_Ki with VK_Id = 0:

Size: 16 Bytes

Default values: Bytes 1 (HEX): 00

Bytes 2 (HEX): 01

Bytes 3 (HEX): 02

Bytes 4 (HEX): 03

Bytes 5 (HEX): 04

Bytes 6 (HEX): 05

Bytes 7 (HEX): 06

Bytes 8 (HEX): 07

Bytes 9 (HEX): 08

Bytes 10 (HEX): 09

Bytes 11 (HEX): 0A

Bytes 12 (HEX): 0B

Bytes 13 (HEX): 0C

Bytes 14 (HEX): 0D

Bytes 15 (HEX): 0E

Bytes 16 (HEX): 0F

Group Id= 1: V_Ki with VK_Id = 1:

Size: 16 Bytes

Default values: Bytes 1 (HEX): 01

Bytes 2 (HEX): 02

Bytes 3 (HEX): 03

Bytes 4 (HEX): 04

Bytes 5 (HEX): 05

Bytes 6 (HEX): 06

Bytes 7 (HEX): 07

Bytes 8 (HEX): 08

Bytes 9 (HEX): 09

Bytes 10 (HEX): 0A

Bytes 11 (HEX): 0B

Bytes 12 (HEX): 0C

Bytes 13 (HEX): 0D

Bytes 14 (HEX): 0E

Bytes 15 (HEX): 0F

Bytes 16 (HEX): 00

### 8.3.2.74 $EF_{VGCSS}$ (Voice Group Call Service Status)

.If service n°57 is "available", this file shall be present.

This EF contains the status of activation for the VGCS group identifiers. The elementary file is directly related to the $EF_{VGCS}$. This EF shall always be allocated if $EF_{VGCS}$ is allocated. The following list of group ID are activated: 1, 4, 20, 30, 50.

File size: 7 Bytes

Default value(HEX) : Bytes 1-7: '09 00 08 20 00 00 FE'

### 8.3.2.75 $EF_{VBS}$ (Voice Broadcast Service)

If service n°58 is "available", this file shall be present.

This EF contains a list of those VBS group identifiers the user has subscribed to. The elementary file is used by the ME for broadcast call establishment and broadcast call reception.

File size: Bytes 200

Default values:

| Bytes | Group ID | Value | BCD encoding in the USIM |
|---|---|---|---|
| 1-4 | 1 | 12 | 21 FF FF FF |
| 5-8 | 2 | 123 | 21 F3 FF FF |
| 9-12 | 3 | 1234 | 21 43 FF FF |
| 13-16 | 4 | 12348 | 21 43 F8 FF |
| 17-20 | 5 | 123491 | 21 43 19 FF |
| 21-24 | 6 | 1235029 | 21 53 20 F9 |
| 25-28 | 7 | 12351 | 21 53 F1 FF |
| 29-32 | 8 | 12352 | 21 53 F2 FF |
| 33-36 | 9 | 12353 | 21 53 F3 FF |
| 37-40 | 10 | 12354 | 21 53 F4 FF |
| 41-44 | 11 | 12355 | 21 53 F5 FF |
| 45-48 | 12 | 12356 | 21 53 F6 FF |
| 49-52 | 13 | 12357 | 21 53 F7 FF |
| 53-56 | 14 | 12358 | 21 53 F8 FF |
| 57-60 | 15 | 12359 | 21 53 F9 FF |
| 61-64 | 16 | 20000 | 02 00 F0 FF |
| 65-68 | 17 | 20001 | 02 00 F1 FF |
| 69-72 | 18 | 20002 | 02 00 F2 FF |
| 73-76 | 19 | 20003 | 02 00 F3 FF |
| 77-80 | 20 | 20004 | 02 00 F4 FF |
| 81-84 | 21 | 20005 | 02 00 F5 FF |
| 85-88 | 22 | 20006 | 02 00 F6 FF |

| Bytes | Group ID | Value | BCD encoding in the USIM |
|---|---|---|---|
| 89-92 | 23 | 20007 | 02 00 F7 FF |
| 93-96 | 24 | 20008 | 02 00 F8 FF |
| 97-100 | 25 | 20009 | 02 00 F9 FF |
| 101-104 | 26 | 20010 | 02 10 F0 FF |
| 105-108 | 27 | 66660 | 66 66 F0 FF |
| 109-112 | 28 | 66661 | 66 66 F1 FF |
| 113-116 | 29 | 66662 | 66 66 F2 FF |
| 117-120 | 30 | 666638 | 66 66 83 FF |
| 121-124 | 31 | 66664 | 66 66 F4 FF |
| 125-128 | 32 | 66665 | 66 66 F5 FF |
| 129-132 | 33 | 66666 | 66 66 F6 FF |
| 133-136 | 34 | 66667 | 66 66 F7 FF |
| 137-140 | 35 | 66668 | 66 66 F8 FF |
| 141-144 | 36 | 66669 | 66 66 F9 FF |
| 145-148 | 37 | 66670 | 66 76 F0 FF |
| 149-152 | 38 | 80120 | 08 21 F0 FF |
| 153-156 | 39 | 80121 | 08 21 F1 FF |
| 157-160 | 40 | 80122 | 08 21 F2 FF |
| 161-164 | 41 | 80123 | 08 21 F3 FF |
| 165-168 | 42 | 80124 | 08 21 F4 FF |
| 169-172 | 43 | 80125 | 08 21 F5 FF |
| 173-176 | 44 | 80126 | 08 21 F6 FF |
| 177-180 | 45 | 80127 | 08 21 F7 FF |
| 181-184 | 46 | 80128 | 08 21 F8 FF |
| 185-188 | 47 | 80129 | 08 21 F9 FF |
| 189-192 | 48 | 80130 | 08 31 F0 FF |
| 193-196 | 49 | 99999 | 99 99 F9 FF |
| 197-200 | 50 | 1111119 | 11 11 11 F9 |

### 8.3.2.76 $EF_{VBSS}$ (Voice Broadcast Service Status)

If service n°58 is "available", this file shall be present.

This EF contains the status of activation for the VBS group identifiers. The elementary file is directly related to the $EF_{VBS}$. This EF shall always be allocated if $EF_{VBS}$ is allocated.

The following list of group ID are activated: 1, 4, 20, 30, 50.

File size: 7 Bytes

Default values (HEX): Bytes 1-7: '09 00 08 20 00 00 FE'

For Group ID= 1 V_Ki with VK_Id = 0:

Size: 16 Bytes

Default values: Bytes 1 (HEX): 0F

Bytes 2 (HEX): 0E

Bytes 3 (HEX): 0D

Bytes 4 (HEX): 0C

Bytes 5 (HEX): 0B

Bytes 6 (HEX): 0A

Bytes 7 (HEX): 09

Bytes 8 (HEX): 08

Bytes 9 (HEX): 07

Bytes 10 (HEX): 06

Bytes 11 (HEX): 05

Bytes 12 (HEX): 04

Bytes 13 (HEX): 03

Bytes 14 (HEX): 02

Bytes 15 (HEX): 01

Bytes 16 (HEX): 00

For Group Id=1 V_Ki with VK_Id = 1:

Size: 16 Bytes

Default values: Bytes 1 (HEX): 00

Bytes 2 (HEX): 0F

Bytes 3 (HEX): 0E

Bytes 4 (HEX): 0D

Bytes 5 (HEX): 0C

Bytes 6 (HEX): 0B

Bytes 7 (HEX): 0A

Bytes 8 (HEX): 09

Bytes 9 (HEX): 08

Bytes 10 (HEX): 07

Bytes 11 (HEX): 06

Bytes 12 (HEX): 05

Bytes 13 (HEX): 04

Bytes 14 (HEX): 03

Bytes 15 (HEX): 02

Bytes 16 (HEX): 01

### 8.3.2.77 $EF_{VGCSCA}$ (Voice Group Call Service Ciphering Algorithm)

If service n°64 is "available", this file shall be present.

This EF contains the ciphering algorithm identifiers for each of the Master Group Key (V_Ki) of each VGCS group that the user has subscribed to (defined in $EF_{VGCS}$).

File size: 2 Bytes

Default value: Byte 1 = ‘01’ (i.e. A5/1) and Byte 2 = ‘03’ (i.e. A5/3)

### 8.3.2.78 $EF_{VBSCA}$ (Voice Broadcast Service Ciphering Algorithm)

If service n°65 is "available", this file shall be present.

This EF contains the ciphering algorithm identifiers for each of the Master Group Key (V_Ki) of each VBS group that the user has subscribed to (defined in $EF_{VBS}$).

File size: 2 Bytes

Default value: Byte 1 = ‘01’ (i.e. A5/1) and Byte 2 = ‘03’ (i.e. A5/3)

### 8.3.2.79 $EF_{GBABP}$ (GBA Bootstrapping parameters)

The programming of this EF follows default parameter written in 3GPP TS 31.102 [23], annex E.

#### 8.3.2.80 $EF_{MSK}$ (MBMS Service Keys List)

The programming of this EF follows default parameter written in 3GPP TS 31.102 [23], annex E.

#### 8.3.2.81 $EF_{MUK}$ (MBMS User Key)

The programming of this EF follows default parameter written in 3GPP TS 31.102 [23], annex E.

#### 8.3.2.82 Void

#### 8.3.2.83 $EF_{GBANL}$ (GBA NAF List)

The programming of this EF follows default parameter written in 3GPP TS 31.102 [23], annex E.

### 8.3.3 Contents of DFs at the USIM ADF (Application DF) level

#### 8.3.3.1 Contents of files at the USIM SoLSA level

##### 8.3.3.1.1 $EF_{SAI}$ (SoLSA Access Indicator)

The programming of this EF follows default parameter written in 3GPP TS 31.102 [23], annex E.

##### 8.3.3.1.2 $EF_{SLL}$ (SoLSA LSA List)

The programming of this EF follows default parameter written in 3GPP TS 31.102 [23], annex E.

##### 8.3.3.1.3 LSA Descriptor files

The programming of this EF follows default parameter written in 3GPP TS 31.102 [23], annex E.

##### 8.3.3.1.4 Contents of files at the MExE level

###### 8.3.3.1.4.1 $EF_{MExE-ST}$ (MExE Service table)

The programming of this EF follows default parameter written in 3GPP TS 31.102 [23], annex E.

###### 8.3.3.1.4.2 $EF_{ORPK}$ (Operator Root Public Key)

The programming of this EF follows default parameter written in 3GPP TS 31.102 [23], annex E.

###### 8.3.3.1.4.3 $EF_{ARPK}$ (Administrator Root Public Key)

The programming of this EF follows default parameter written in 3GPP TS 31.102 [23], annex E.

###### 8.3.3.1.4.4 $EF_{TPRPK}$ (Third Party Root Public Key)

The programming of this EF follows default parameter written in 3GPP TS 31.102 [23], annex E.

###### 8.3.3.1.4.5 $EF_{TKCDF}$ (Trusted Key/Certificates Data Files)

The programming of this EF follows default parameter written in 3GPP TS 31.102 [23], annex E.

#### 8.3.3.2 Contents of files at the DF PHONEBOOK level

##### 8.3.3.2.1 $EF_{PBR}$ (Phone Book Reference file)

The programming of this EF is a test house option.

##### 8.3.3.2.2 $EF_{IAP}$ (Index Administration Phone book)

The programming of this EF follows default parameter written in 3GPP TS 31.102 [23], annex E.

##### 8.3.3.2.3 $EF_{ADN}$ (Abbreviated dialling numbers)

The programming of this EF follows default parameter written in 3GPP TS 31.102 [23], annex E.

#### 8.3.3.2.4 $EF_{EXT1}$ (Extension1)

The programming of this EF follows default parameter written in 3GPP TS 31.102 [23], annex E.

#### 8.3.3.2.5 $EF_{PBC}$ (Phone Book Control)

The programming of this EF follows default parameter written in 3GPP TS 31.102 [23], annex E.

#### 8.3.3.2.6 $EF_{GRP}$ (Grouping file)

The programming of this EF follows default parameter written in 3GPP TS 31.102 [23], annex E.

#### 8.3.3.2.7 $EF_{AAS}$ (Additional number Alpha String)

The programming of this EF follows default parameter written in 3GPP TS 31.102 [23], annex E.

#### 8.3.3.2.8 $EF_{GAS}$ (Grouping information Alpha String)

The programming of this EF follows default parameter written in 3GPP TS 31.102 [23], annex E.

#### 8.3.3.2.9 $EF_{ANR}$ (Additional Number)

The programming of this EF follows default parameter written in 3GPP TS 31.102 [23], annex E.

#### 8.3.3.2.10 $EF_{SNE}$ (Second Name Entry)

The programming of this EF follows default parameter written in 3GPP TS 31.102 [23], annex E.

#### 8.3.3.2.11 $EF_{CCP1}$ (Capability Configuration Parameters 1)

The programming of this EF follows default parameter written in 3GPP TS 31.102 [23], annex E.

#### 8.3.3.2.12 Phone Book Synchronization

##### 8.3.3.2.12.1 $EF_{UID}$ (Unique Identifier)

The programming of this EF follows default parameter written in 3GPP TS 31.102 [23], annex E.

##### 8.3.3.2.12.2 $EF_{PSC}$ (Phone book Synchronization Counter)

The programming of this EF follows default parameter written in 3GPP TS 31.102 [23], annex E.

##### 8.3.3.2.12.3 $EF_{CC}$ (Change Counter)

The programming of this EF follows default parameter written in 3GPP TS 31.102 [23], annex E.

##### 8.3.3.2.12.4 $EF_{PUID}$ (Previous Unique Identifier)

The programming of this EF follows default parameter written in 3GPP TS 31.102 [23], annex E.

#### 8.3.3.2.13 $EF_{EMAIL}$ (e-mail address)

The programming of this EF follows default parameter written in 3GPP TS 31.102 [23], annex E.

### 8.3.3.3 Contents of files at the DF GSM-ACCESS level (Files required for GSM Access)

#### 8.3.3.3.1 $EF_{Kc}$ (GSM Ciphering key Kc)

File size: 9 Bytes

Default values (HEX): Bytes 1 to 8: Align with Kc used by SS

Byte 9: 07

Byte 9 is set to 07 to indicate that there is no key available at the start of a test.

The bytes within this elementary file may be updated by the UE as a result of a successful authentication attempt.

#### 8.3.3.3.2 $EF_{KcGPRS}$ (GPRS Ciphering key KcGPRS)

The programming of this EF follows default parameter written in 3GPP TS 31.102 [23], annex E.

#### 8.3.3.3.3 Void

#### 8.3.3.3.4 $EF_{CPBCCH}$ (CPBCCH Information)

The programming of this EF follows default parameter written in 3GPP TS 31.102 [23], annex E.

#### 8.3.3.3.5 $EF_{InvScan}$ (Investigation Scan)

The programming of this EF follows default parameter.

### 8.3.4 Contents of EFs at the TELECOM level

#### 8.3.4.1 $EF_{ADN}$ (Abbreviated dialling numbers)

The programming of this EF is a test house option. It should be noted that sufficient space should be provided on the USIM card for 101 records.

#### 8.3.4.2 $EF_{EXT1}$ (Extension1)

The programming of this EF follows default parameter written in 3GPP TS 31.102 [23], annex E.

#### 8.3.4.3 $EF_{ECCP}$ (Extended Capability Configuration Parameter)

The programming of this EF is a test house option.

#### 8.3.4.4 $EF_{SUME}$ (SetUpMenu Elements)

The programming of this EF is a test house option.

#### 8.3.4.5 $EF_{ARR}$ (Access rule reference)

The programming of this EF follows default parameter written in 3GPP TS 31.102 [23], annex E.

### 8.3.5 Contents of DFs at the TELECOM level

#### 8.3.5.1 Contents of files at the $DF_{GRAPHICS}$ level

##### 8.3.5.1.1 $EF_{IMG}$ (Image)

The programming of this EF follows default parameter written in 3GPP TS 31.102 [23], annex E.

##### 8.3.5.1.2 Image Instance Data Files

The programming of this EF follows default parameter written in 3GPP TS 31.102 [23], annex E.

#### 8.3.5.2 Contents of files at the $DF_{PHONEBOOK}$ under the $DF_{TELECOM}$

The programming of this EF is a test house option.

# 9 Default Message Contents

## 9.1 Default Message Contents for Signalling

### 9.1.1 Default RRC Message Contents (FDD)

This clause contains the default values of common messages, which unless indicated otherwise in specific clauses of 3GPP TS 34.123-1 [1], shall be transmitted and checked by the system simulator.

In this clause, decimal values are normally used. However, sometimes a hexadecimal value, indicated by an "H", or a binary value, indicated by a "B" is used.

The necessary L3 messages are listed in alphabetic order, with the exception of the SYSTEM INFORMATION messages, where it is the information elements which are listed in alphabetic order (this is because some information elements occur in several SYSTEM INFORMATION types).

Default SYSTEM INFORMATION:

NOTE: SYSTEM INFORMATION BLOCK TYPE 1 (except for PLMN type "GSM-MAP"), SYSTEM INFORMATION BLOCK TYPE 8, SYSTEM INFORMATION BLOCK TYPE 9, SYSTEM INFORMATION BLOCK TYPE 10, SYSTEM INFORMATION BLOCK TYPE 14, SYSTEM INFORMATION BLOCK TYPE 15 and SYSTEM INFORMATION BLOCK TYPE 16 messages are not used.

Contents of ACTIVE SET UPDATE message: AM

| Information Element | Value/remark | Version |
|---|---|---|
| Message Type | | |
| RRC transaction identifier | Arbitrarily selects one integer between 0 to 3 | |
| Integrity check info | | |
| - message authentication code | SS calculates the value of MAC-I for this message and writes to this IE. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I. | |
| - RRC message sequence number | SS provides the value of this IE, from its internal counter. | |
| Activation time | now | |
| New U-RNTI | Not Present | |
| New H-RNTI | Not Present | Rel-6 |
| New Primary E-RNTI | Not Present | Rel-6 |
| New Secondary E-RNTI | Not Present | Rel-6 |
| CN information info | Not Present | |
| Maximum allowed UL TX power | Not Present - use default value | |
| Radio link addition information | Not Present | |
| Radio link removal information | Not Present | |
| TX Diversity Mode | None | |
| SSDT information | Not Present | R99 and Rel-4 only |
| DPC Mode | [FFS] | Rel-5 |
| Serving HS-DSCH cell information | Not Present | Rel-6 |
| E-DCH reconfiguration information | Not Present | Rel-6 |

Contents of ACTIVE SET UPDATE COMPLETE message: AM

| Information Element | Value/remark |
|---|---|
| Message Type | |
| RRC transaction identifier | Checked to see if it matches the same value used in the corresponding downlink ACTIVE SET UPDATE message |
| Integrity check info | |
| - Message authentication code | This IE is checked to see if it is present. The value is compared against the XMAC-I value computed by SS. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I. |
| - RRC Message sequence number | This IE is checked to see if it is present. The value is used by SS to compute the XMAC-I value. |

Contents of ACTIVE SET UPDATE FAILURE message: AM

| Information Element | Value/remark |
|---|---|
| Message Type | |
| RRC transaction identifier | Checked to see if it matches the same value used in the corresponding downlink ACTIVE SET UPDATE message |
| Integrity check info | |
| - Message authentication code | This IE is checked to see if it is present. The value is compared against the XMAC-I value computed by SS. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I. |
| - RRC Message sequence number | This IE is checked to see if it is present. The value is used by SS to compute the XMAC-I value. |
| Failure cause | Refer to test requirement |

Contents of CELL UPDATE message: TM

| Information Element | Value/remark | Version |
|---|---|---|
| Message Type | | |
| U-RNTI | Checked to see if it is set to the following values | |
| - SRNC identity | 0000 0000 0001B | |
| - S-RNTI | 0000 0000 0000 0000 0001B | |
| RRC transaction identifier | Checked to see if it is absent | |
| Integrity check info | | |
| - Message authentication code | This IE is checked to see if it is present. The value is compared against the XMAC-I value computed by SS. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I. | |
| - RRC Message sequence number | This IE is checked to see if it is present. The value is used by SS to compute the XMAC-I value. | |
| START List | Checked to see if the 'CN domain identity' and 'START' IEs are present for all CN domains supported by the UE . | |
| - CN domain identity | Checked to see if it is one of the supported CN domains | |
| - START | This IE is checked to see if it is present. The first/ leftmost bit of the bit string contains the most significant bit of the START. | |
| AM_RLC error indication (RB2, RB3 or RB4) | Checked to see if it is set to 'FALSE' | |
| AM_RLC error indication (RB>4) | Checked to see if it is set to 'FALSE' | |
| Cell update cause | See the specific test case | |
| Traffic volume indicator | Checked to see if it is absent | Rel-6 |
| Failure cause | Checked to see if it is absent | |
| RB timer indicator | | |
| - T314 expired | Checked to see if it is set to 'FALSE' | |
| - T315 expired | Checked to see if it is set to 'FALSE' | |
| Establishment cause | This IE is checked to see if it is absent | Rel-5 |
| Reconfiguration Status Indicator | Checked to see if it is absent | Rel-6 |
| Measured results on RACH | Not checked | |

Contents of CELL UPDATE CONFIRM message: UM

| Information Element | Value/remark | Version |
|---|---|---|
| Message Type | | |
| U-RNTI | If this message is sent on CCCH, use the following values. Else, this IE is absent. | |
| - SRNC identity | 0000 0000 0001B | |
| - S-RNTI | 0000 0000 0000 0000 0001B | |
| RRC transaction identifier | Selects an arbitrary integer between 0 to 3 | |
| Integrity check info | | |
| - message authentication code | SS calculates the value of MAC-I for this message and writes to this IE. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I. | |
| - RRC message sequence number | SS provides the value of this IE, from its internal counter. | |
| Integrity protection mode info | Not Present | |
| Ciphering mode info | Not Present | |
| Activation time | Not Present - use default value | |

| Information Element | Value/remark | Version |
|---|---|---|
| New U-RNTI | Not Present | |
| New C-RNTI | Not Present | |
| New DSCH-RNTI | Not Present | R99 and Rel-4 only |
| New H-RNTI | Not Present | Rel-5 |
| New Primary E-RNTI | Not present | Rel-6 |
| New Secondary E-RNTI | Not present | Rel-6 |
| RRC State indicator | CELL_FACH | |
| UTRAN DRX cycle length coefficient | Not Present | |
| RLC re-establish indicator (RB2, RB3 and RB4) | FALSE | |
| RLC re-establish indicator (RB5 and upwards) | FALSE | |
| CN information info | Not Present | |
| URA identity | Not Present | |
| RB information to release list | Not Present | |
| RB information to reconfigure list | Not Present | |
| RB information to be affected list | Not Present | |
| Downlink counter synchronization info | Not Present | |
| PDCP ROHC target mode | Not Present | Rel-5 |
| UL Transport channel information common for all transport channels | Not Present | |
| Deleted TrCH information list | Not Present | |
| Added or Reconfigured TrCH information list | Not Present | |
| CHOICE Mode | FDD | |
| - CPCH set ID | Not Present | R99 and Rel-4 only |
| - Added or Reconfigured TrCH information for DRAC list | Not Present | R99 and Rel-4 only |
| DL Transport channel information common for all transport channels | Not Present | |
| Deleted TrCH information list | Not Present | |
| Added or Reconfigured TrCH information list | Not Present | |
| Frequency info | Not Present | |
| Maximum allowed UL TX power | Not Present | |
| CHOICE channel requirement | Not Present | |
| E-DCH Info | Not Present | Rel-6 |
| CHOICE mode | FDD | R99 and Rel-4 only |
| - Downlink PDSCH information | Not Present | R99 and Rel-4 only |
| Downlink HS-PDSCH Information | Not Present | Rel-5 |
| Downlink information common for all radio links | Not Present | |
| Downlink information per radio link list | Not Present | |
| MBMS PL Service Restriction Information | Not Present | Rel-6 |

Contents of DOWNLINK DIRECT TRANSFER message: AM

| Information Element | Value/remark |
|---|---|
| Message Type | |
| RRC transaction identifier | Arbitrarily selects an integer between 0 and 3 |
| Integrity check info | |
| - Message authentication code | SS calculates the value of MAC-I for this message and writes to this IE. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I. |
| - RRC Message sequence number | SS provides the value of this IE, from its internal counter. |
| CN domain identity | CS domain or PS domain |
| NAS message | See Specific Message Content for each test case |

Contents of HANDOVER FROM UTRAN COMMAND-GSM message: AM

| Information Element | Value/remark |
|---|---|
| Message Type | |
| RRC transaction identifier | Arbitrarily selects one integer between 0 to 3 |

| | |
|---|---|
| Integrity check info | |
| - Message authentication code | SS calculates the value of MAC-I for this message and writes to this IE. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I. |
| - RRC Message sequence number | SS provides the value of this IE, from its internal counter. |
| Activation time | now |
| RAB Info | |
| - RAB identity | 0000 0001B<br>The first/ leftmost bit of the bit string contains the most significant bit of the RAB identity. |
| - CN domain identity | CS domain |
| - NAS Synchronization Indicator | Not present |
| - Re-establishment timer | Use T314 |
| Inter-system message | |
| - CHOICE System type | GSM |
| - Frequency Band | Set to "GSM/ PCS 1900" if GSM/ PCS 1900 is used in this test. Otherwise set to "GSM/DCS 1800 Band" |
| - CHOICE GSM message | Single GSM message |
| - Single GSM message | GSM HANDOVER COMMAND formatted and coded according to GSM specifications as BIT STRING (1..512). The first/ l*eftmost/ most significant* bit of the bit string contains bit 8 of the first octet of the GSM message. The contents of the HANDOVER COMMAND is to be defined in the specific test case. |

Contents of HANDOVER FROM UTRAN FAILURE message: AM

| Information Element | Value/remark |
|---|---|
| Message Type | |
| RRC transaction identifier | Checked to see if it matches the same value used in the corresponding downlink HANDOVER FROM UTRAN COMMAND -GSM message |
| Integrity check info | |
| - Message authentication code | This IE is checked to see if it is present. The value is compared against the XMAC-I value computed by SS. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I. |
| - RRC Message sequence number | This IE is checked to see if it is present. The value is used by SS to compute the XMAC-I value. |
| Inter-RAT handover failure | |
| -Inter-RAT handover failure cause | physical channel failure |
| Inter-system message | Not Checked |

Contents of INITIAL DIRECT TRANSFER message: AM

| Information Element | Value/remark | Version |
|---|---|---|
| Message Type | | |
| Integrity check info | | |
| - Message authentication code | This IE is checked to see if it is present. The value is compared against the XMAC-I value computed by SS. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I. | |
| - RRC Message sequence number | This IE is checked to see if it is present. The value is used by SS to compute the XMAC-I value. | |
| PLMN identity | This IE is checked to see if it is absent | Rel-6 |
| CN domain identity | Checked to see if set to supported CN domain as specified in the IXIT statements. | |
| Intra Domain NAS Node Selector | | |
| - CHOICE version | R99 | |
| - CHOICE CN type | GSM-MAP | |
| - CHOICE Routing basis | Local (P)TMSI | |

| | | |
|---|---|---|
| - Routing parameter | If the IE "CN domain identity" is equal to "CS domain", this bit string is set to to bits b14 through b23 of the TMSI.<br>If the IE "CN domain identity" is equal to "PS domain", this bit string is set to to bits b14 through b23 of the P-TMSI.<br>The TMSI/P-TMSI consists of 4 octets (32bits). This can be represented by a string of bits numbered from b0 to b31, with bit b0 being the least significant<br>The "Routing parameter" bit string consists of bits b14 through b23 of the TMSI/ PTMSI.<br>The first/ leftmost/ most significant bit of the bit string contains bit b23 of the TMSI/ PTMSI. | |
| - Entered parameter | Not checked | |
| NAS message | Set according to that indicated in specific message content for each test case | |
| START | This IE is checked to see if it is present. | |
| Establishment cause | This IE is checked to see if it is absent | Rel-5 |
| Measured results on RACH | Not checked | |
| MBMS joined information | This IE is checked to see if it is absent | Rel-6 |

Contents of MEASUREMENT CONTROL message: AM

| Information Element | Value/remark |
|---|---|
| Message Type | |
| RRC transaction identifier | Arbitrarily selects an unused integer between 0 to 3 |
| Integrity check info | |
| - Message authentication code | SS calculates the value of MAC-I for this message and writes to this IE. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I. |
| - RRC message sequence number | SS provides the value of this IE, from its internal counter. |
| Measurement Identity | 1 |
| Measurement Command | Setup |
| Measurement Reporting Mode | |
| - Measurement Report Transfer Mode | Acknowledged mode RLC |
| - Periodical Reporting/Event Trigger Reporting Mode | Periodical reporting |
| Additional measurement list | Not Present |
| CHOICE Measurement type | Intra-frequency measurement |
| - Intra-frequency measurement | |
| - Intra-frequency cell info list | |
| - CHOICE intra-frequency cell removal | Not present |
| - New intra-frequency cell | |
| - Intra-frequency cell-id | 1 |
| - Cell info | |
| - Cell individual offset | 0dB |
| - Reference time difference to cell | Not Present |
| - Read SFN number | FALSE |
| - CHOICE mode | FDD |
| - Primary CPICH info | |
| - Primary scrambling code | Different from the Default setting in clause 6.1 (FDD) |
| - Primary CPICH Tx power | Not Present |
| - TX Diversity indicator | FALSE |
| - Cells for measurement | Not present |
| - Intra-frequency measurement quantity | Not Present |
| - Intra-frequency reporting quantity | |
| - Reporting quantities for active set cells | |
| - Cell synchronization information reporting indicator | FALSE |
| - Cell Identity reporting indicator | TRUE |
| - CPICH Ec/N0 reporting indicator | FALSE |
| - CPICH RSCP reporting indicator | TRUE |
| - Pathloss reporting indicator | FALSE |
| - Reporting quantities for monitored set cells | |
| - Cell synchronization information reporting indicator | FALSE |
| - Cell Identity reporting indicator | TRUE |
| - CPICH Ec/N0 reporting indicator | FALSE |
| - CPICH RSCP reporting indicator | TRUE |
| - Pathloss reporting indicator | FALSE |

| Information Element | Value/remark |
|---|---|
| - Reporting quantities for detected set cells | Not Present |
| - Reporting cell status | |
| - CHOICE reported cell | Report cell within active set and/or monitored cells on used frequency |
| - Maximum number of reported cells | 2 |
| - Measurement validity | Not Present |
| - CHOICE report criteria | Periodic reporting criteria |
| - Amount of reporting | Infinity |
| - Reporting interval | 64 s |
| DPCH Compressed mode status info | Not Present |

Contents of MEASUREMENT CONTROL FAILURE message: AM

| Information Element | Value/remark |
|---|---|
| Message Type | |
| RRC transaction identifier | Checked to see if it's set to the identical value for the same IE in the downlink MEASUREMENT CONTROL message |
| Integrity check info | |
| - Message authentication code | This IE is checked to see if it is present. The value is compared against the XMAC-I value computed by SS. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I. |
| - RRC Message sequence number | This IE is checked to see if it is present. The value is used by SS to compute the XMAC-I value. |
| Failure cause | See the test content |

Contents of MEASUREMENT REPORT message: AM

| Information Element | Value/remark | Version |
|---|---|---|
| Message Type | | |
| Integrity check info | | |
| - Message authentication code | This IE is checked to see if it is present. The value is compared against the XMAC-I value computed by SS. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I. | |
| - RRC Message sequence number | This IE is checked to see if it is present. The value is used by SS to compute the XMAC-I value. | |
| Measurement identity | 1 | |
| Measured Results | | |
| - Intra-frequency measured results | | |
| - Cell measured results | | |
| - Cell Identity | Not present | |
| - Cell synchronization information | Checked that this IE is absent | |
| - Primary CPICH info | | |
| - Primary scrambling code | Different from the Default setting in clause 6.1 (FDD) | |
| - CPICH Ec/N0 | Checked that this IE is absent | |
| - CPICH RSCP | Checked that this IE is present | |
| - Pathloss | Checked that this IE is absent | |
| Measured results on RACH | Checked that this IE is absent | |
| Additional measured results | Checked that this IE is absent | |
| Event results | Checked that this IE is absent | |
| GSM OTD reference cell | Checked that this IE is absent | Rel-4 |
| Inter-RAT cell info indication | Checked that this IE is absent | Rel-5 |

Contents of PAGING TYPE 1 message: TM (Speech in CS)

| Information Element | Value/remark |
|---|---|
| Message Type | |
| Paging record list | |
| - Paging record | |
| - CHOICE Used paging identity | CN identity |
| - Paging cause | Terminating Conversational Call |
| - CN domain identity | CS domain |
| - CHOICE UE identity | |
| - IMSI (GSM-MAP) | Set to the same octet string as in the IMSI stored in the USIM card |

| | |
|---|---|
| BCCH modification info | Not Present |

Contents of PAGING TYPE 1 message: TM (The others of speech in CS)

| Information Element | Value/remark |
|---|---|
| Message Type | |
| Paging record list | |
| - Paging record | |
| - CHOICE Used paging identity | CN identity |
| - Paging cause | Terminating Streaming Call |
| - CN domain identity | CS domain |
| - CHOICE UE identity | |
| - IMSI (GSM-MAP) | Set to the same octet string as in the IMSI stored in the USIM card |
| BCCH modification info | Not Present |

Contents of PAGING TYPE 1 message: TM (Packet in PS)

| Information Element | Value/remark |
|---|---|
| Message Type | |
| Paging record list | |
| - Paging record | |
| - CHOICE Used paging identity | CN identity |
| - Paging cause | Terminating Interactive Call |
| - CN domain identity | PS domain |
| - CHOICE UE identity | |
| - P-TMSI | Use P-TMSI allocated by SS at initial attach. |
| BCCH modification info | Not Present |

Contents of PAGING TYPE 1 message: TM (SMS in CS)

| Information Element | Value/remark |
|---|---|
| Message Type | |
| Paging record list | |
| - Paging record | |
| - CHOICE Used paging identity | CN identity |
| - Paging cause | Terminating Low Priority Signalling |
| - CN domain identity | CS domain |
| - CHOICE UE identity | |
| - IMSI (GSM-MAP) | Set to the same octet string as in the IMSI stored in the TEST USIM card |
| BCCH modification info | Not Present |

Contents of PAGING TYPE 1 message: TM (SMS in PS)

| Information Element | Value/remark |
|---|---|
| Message Type | |
| Paging record list | |
| - Paging record | |
| - CHOICE Used paging identity | CN identity |
| - Paging cause | Terminating Low Priority Signalling |
| - CN domain identity | PS domain |
| - CHOICE UE identity | |
| - IMSI (GSM-MAP) | Set to the same octet string as in the IMSI stored in the TEST USIM card |
| BCCH modification info | Not Present |

Contents of PAGING TYPE 2 message: AM (Speech in CS)

| Information Element | Value/remark |
|---|---|
| Message Type | |
| RRC transaction identifier | Arbitrarily selects an integer between 0 and 3 |
| Integrity check info | |
| - message authentication code | SS calculates the value of MAC-I for this message and writes to this IE. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I. |

| | |
|---|---|
| - RRC message sequence number | SS provides the value of this IE, from its internal counter. |
| Paging cause | Terminating Conversational Call |
| CN domain identity | CS domain |
| Paging record type identifier | Select the same type as in the IE "Initial UE Identity" in RRC CONNECTION REQUEST" message. |

Contents of PHYSICAL CHANNEL RECONFIGURATION message: AM or UM

| Information Element | Condition | Value/remark | Version |
|---|---|---|---|
| Message Type | A1, A2, A3, A4, A5, A6, A7, A8, A9, A10 | | |
| RRC transaction identifier | | Arbitrarily selects an integer between 0 and 3 | |
| Integrity check info | | | |
| - message authentication code | | SS calculates the value of MAC-I for this message and writes to this IE. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I. | |
| - RRC message sequence number | | SS provides the value of this IE, from its internal counter. | |
| Integrity protection mode info | | Not Present | |
| Ciphering mode info | | Not Present | |
| Activation time | A1, A2, A3 | (256+CFN-(CFN MOD 8 + 8))MOD 256 | |
| Activation time | A4, A5, A6, A7, A8, A9, A10 | Not Present | |
| Delay restriction flag | A1, A2, A3, A4, A5, A6, A7, A8, A9, A10 | Not Present | Rel-6 |
| New U-RNTI | | Not Present | |
| New C-RNTI | A1, A2, A3, A4, A7, A8, A9, A10 | Not Present | |
| New C-RNTI | A5, A6 | '1010 1010 1010 1010' | |
| New DSCH-RNTI | A1, A2, A3, A4, A5, A6, A7, A8, A9, A10 | Not Present | R99 and Rel-4 only |
| New H-RNTI | A1, A2, A3, A4, A5, A6, A7, A8, A9, A10 | Not Present | Rel-5 |
| New Primary E-RNTI | | Not Present | Rel-6 |
| New Secondary E-RNTI | | Not Present | Rel-6 |
| RRC State indicator | A1, A2, A3, A4 | CELL_DCH | |
| RRC State indicator | A5, A6 | CELL_FACH | |
| RRC State indicator | A7, A8 | URA_PCH | |
| RRC State indicator | A9, A10 | CELL_PCH | |
| UTRAN DRX cycle length coefficient | A1, A2, A3, A4, A5, A6 | Not Present | |
| UTRAN DRX cycle length coefficient | A7, A8, A9, A10 | 3 | |
| CN information info | | Not Present | |
| URA identity | | Not Present | |
| Downlink counter synchronization info | | Not Present | |
| Frequency info | A1, A2, A3, A4, A5 | | |
| - UARFCN uplink (Nu) | | Not present<br>Absence of this IE is equivalent to applying the default duplex distance defined for the operating frequency according to 3GPP TS 25.101 [11] | |
| - UARFCN downlink (Nd) | | Reference to clause 5.1 Test frequencies | |
| Frequency info | A6, A7, A8, A9, A10 | Not Present | |
| Maximum allowed UL TX power | | 33dBm | |
| CHOICE *channel requirement* | A5, A6, A7, A8, A9, A10 | Not Present | |
| CHOICE *channel requirement* | A1, A2, A3, A4 | Uplink DPCH info | |
| - Uplink DPCH power control info | | | |
| - DPCCH power offset | | -80dB (i.e. ASN.1 IE value of -40) | |
| - PC Preamble | | 1 frame | |
| - SRB delay | | 7 frames | |
| - Power Control Algorithm | | Algorithm1 | |
| - TPC step size | | 1dB | |
| - $\Delta_{ACK}$ | | Not Present | Rel-5 |

| Information Element | Condition | Value/remark | Version |
|---|---|---|---|
| - $\Delta_{NACK}$ | | Not Present | Rel-5 |
| - Ack-Nack repetition factor | | Not Present | Rel-5 |
| - Scrambling code type | | Long | |
| - Scrambling code number | | 0 (0 to 16777215) | |
| - Number of DPDCH | | Not Present(1) | |
| - spreading factor | | Reference to clause 6.10 Parameter Set | |
| - TFCI existence | | Reference to clause 6.10 Parameter Set | |
| - Number of FBI bit | | Reference to clause 6.10 Parameter Set | |
| - Puncturing Limit | | Reference to clause 6.10 Parameter Set | |
| E-DCH Info | | Not Present | Rel-6 |
| CHOICE Mode | A1, A2, A3, A4, A5, A6, A7, A8, A9, A10 | FDD | |
| - Downlink PDSCH information | | Not Present | R99 and Rel-4 only |
| Downlink HS-PDSCH Information | A1, A2, A3, A4, A5, A6, A7, A8, A9, A10 | Not Present | Rel-5 |
| Downlink information common for all radio links | A1, A2, A3 | | |
| - Downlink DPCH info common for all RL | | | |
| - Timing indicator | | Maintain | |
| - CFN-targetSFN frame offset | | Not Present | |
| - Downlink DPCH power control information | | | |
| - DPC mode | | 0 (single) | |
| - CHOICE mode | | FDD | |
| - Power offset $P_{Pilot-DPDCH}$ | | 0 | |
| - DL rate matching restriction information | | Not Present | |
| - Spreading factor | | Reference to clause 6.10 Parameter Set | |
| - Fixed or Flexible Position | | Reference to clause 6.10 Parameter Set | |
| - TFCI existence | | Reference to clause 6.10 Parameter Set | |
| - CHOICE SF | | Reference to clause 6.10 Parameter Set | |
| - DPCH compressed mode info | | Not Present | |
| - TX Diversity mode | | None | |
| - SSDT information | | Not Present | R99 and Rel-4 only |
| - Default DPCH Offset Value | | Not Present | |
| - MAC-hs reset indicator | | Not Present | Rel-5 |
| Downlink information common for all radio links | A4 | | |
| - Downlink DPCH info common for all RL | | | |
| - Timing indicator | | Initialize | |
| - CFN-targetSFN frame offset | | Not Present | |
| - Downlink DPCH power control information | | | |
| - DPC mode | | 0 (single) | |
| - CHOICE mode | | FDD | |
| - Power offset $P_{Pilot-DPDCH}$ | | 0 | |
| - DL rate matching restriction information | | Not Present | |
| - Spreading factor | | Reference to clause 6.10 Parameter Set | |
| - Fixed or Flexible Position | | Reference to clause 6.10 Parameter Set | |
| - TFCI existence | | Reference to clause 6.10 Parameter Set | |
| - CHOICE SF | | Reference to clause 6.10 Parameter Set | |
| - DPCH compressed mode info | | Not Present | |
| - TX Diversity mode | | None | |
| - SSDT information | | Not Present | R99 and Rel-4 only |
| - Default DPCH Offset Value | | Arbitrary set to value 0..306688 by step of 512 | |
| - MAC-hs reset indicator | | Not Present | Rel-5 |
| Downlink information common for all radio links | A5, A6, A7, A8, A9, A10 | Not Present | |
| Downlink information for each radio links | A1, A2,A3 | | |
| - Choice mode | | FDD | |

| Information Element | Condition | Value/remark | Version |
|---|---|---|---|
| - Primary CPICH info | | | |
| - Primary scrambling code | | Ref. to the Default setting in clause 6.1 (FDD) | |
| - PDSCH with SHO DCH info | | Not Present | R99 and Rel-4 only |
| - PDSCH code mapping | | Not Present | R99 and Rel-4 only |
| - Serving HS-DSCH radio link indicator | | FALSE | Rel-5 |
| - Serving E-DCH radio link indicator | | FALSE | Rel-6 |
| - Downlink DPCH info for each RL | | | |
| - CHOICE mode | | FDD | |
| - Primary CPICH usage for channel estimation | | Primary CPICH may be used | |
| - DPCH frame offset | | Set to value : Default DPCH Offset Value (as currently stored in SS) mod 38 400 | |
| - Secondary CPICH info | | Not Present | |
| - DL channelisation code | | | |
| - Secondary scrambling code | | 5 | |
| - Spreading factor | | Reference to clause 6.10 Parameter Set 0 | |
| - Code number | | | |
| - Scrambling code change | | No change | |
| - TPC combination index | | 0 | |
| - SSDT Cell Identity | | Not Present | R99 and Rel-4 only |
| - Closed loop timing adjustment mode | | Not Present | |
| - E-AGCH Info | | Not Present | Rel-6 |
| - E-HICH Information | | Not Present | Rel-6 |
| - E-RGCH Information | | Not Present | Rel-6 |
| - SCCPCH information for FACH | | Not Present | R99 and Rel-4 only |
| Downlink information for each radio links | A4 | | |
| - Choice mode | | FDD | |
| - Primary CPICH info | | | |
| - Primary scrambling code | | Ref. to the Default setting in clause 6.1 (FDD) | |
| - PDSCH with SHO DCH info | | Not Present | R99 and Rel-4 only |
| - PDSCH code mapping | | Not Present | R99 and Rel-4 only |
| - Serving HS-DSCH radio link indicator | | FALSE | Rel-5 |
| - Serving E-DCH radio link indicator | | FALSE | Rel-6 |
| - Downlink DPCH info for each RL | | | |
| - CHOICE mode | | FDD | |
| - Primary CPICH usage for channel estimation | | Primary CPICH may be used | |
| - DPCH frame offset | | Set to value : Default DPCH Offset Value mod 38 400 | |
| - Secondary CPICH info | | Not Present | |
| - DL channelisation code | | | |
| - Secondary scrambling code | | 5 | |
| - Spreading factor | | Reference to clause 6.10 Parameter Set 0 | |
| - Code number | | | |
| - Scrambling code change | | No change | |
| - TPC combination index | | 0 | |
| - SSDT Cell Identity | | Not Present | R99 and Rel-4 only |
| - Closed loop timing adjustment mode | | Not Present | |
| - E-AGCH Info | | Not Present | Rel-6 |
| - E-HICH Information | | Not Present | Rel-6 |
| - E-RGCH Information | | Not Present | Rel-6 |
| - SCCPCH information for FACH | | Not Present | R99 and Rel-4 only |

| Information Element | Condition | Value/remark | Version |
|---|---|---|---|
| - Downlink information for each radio link | A5 | | |
| - Choice mode | | FDD | |
| - Primary CPICH info | | | |
| - Primary scrambling code | | Ref. to the Default setting in clause 6.1 (FDD) | |
| - PDSCH with SHO DCH info | | Not Present | R99 and Rel-4 only |
| - PDSCH code mapping | | Not Present | R99 and Rel-4 only |
| - Serving HS-DSCH radio link indicator | | FALSE | Rel-5 |
| - Serving E-DCH radio link indicator | | FALSE | Rel-6 |
| - Downlink DPCH info for each RL | | Not Present | |
| - E-AGCH Info | | Not Present | Rel-6 |
| - E-HICH Information | | Not Present | Rel-6 |
| - E-RGCH Information | | Not Present | Rel-6 |
| - SCCPCH Information for FACH | | Not Present | R99 and Rel-4 only |
| - Downlink information for each radio link | A6, A7, A8, A9, A10 | Not Present | |
| MBMS PL Service Restriction Information | A1, A2, A3, A4, A5, A6, A7, A8, A9, A10 | Not Present | Rel-6 |

| Condition | Explanation |
|---|---|
| A1 | This IE need for "Non speech in CS" |
| A2 | This IE need for "Speech in CS" |
| A3 | This IE need for "Packet to CELL_DCH from CELL_DCH in PS" |
| A4 | This IE need for "Packet to CELL_DCH from CELL_FACH in PS" |
| A5 | This IE need for "Packet to CELL_FACH from CELL_DCH in PS" |
| A6 | This IE need for "Packet to CELL_FACH from CELL_FACH in PS" |
| A7 | This IE need for "Packet to URA_PCH from CELL_FACH in PS" |
| A8 | This IE need for "Packet to URA_PCH from CELL_DCH in PS" |
| A9 | This IE need for "Packet to CELL_PCH from CELL_FACH in PS" |
| A10 | This IE need for "Packet to CELL_PCH from CELL_DCH in PS" |

Contents of PHYSICAL CHANNEL RECONFIGURATION COMPLETE message: AM

| Information Element | Value/remark |
|---|---|
| Message Type | |
| RRC transaction identifier | Checked to see if it's set to identical value of the same IE in the downlink PHYSICAL CHANNEL RECONFIGURATION message |
| Integrity check info | |
| - Message authentication code | This IE is checked to see if it is present. The value is compared against the XMAC-I value computed by SS. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I. |
| - RRC Message sequence number | This IE is checked to see if it is present. The value is used by SS to compute the XMAC-I value. |
| Uplink integrity protection activation info | Not checked |
| CHOICE mode | FDD |
| COUNT-C activation time | Not checked |
| Radio bearer uplink ciphering activation time info | Not checked |
| Uplink counter synchronization info | Not present |

Contents of PHYSICAL CHANNEL RECONFIGURATION FAILURE message: AM

| Information Element | Value/remark |
|---|---|
| Message Type | |
| RRC transaction identitifer | Checked to see if it is set to identical value of the same IE in the downlink PHYSICAL CHANNEL RECONFIGURATION message. |
| Integrity check info | |
| - Message authentication code | This IE is checked to see if it is present. The value is compared against the XMAC-I value computed by SS. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I. |
| - RRC Message sequence number | This IE is checked to see if it is present. The value is used by SS to compute the XMAC-I value. |

| | |
|---|---|
| Failure cause | Checked to see if it meets test requirement |

Contents of RADIO BEARER SETUP message: AM or UM

| Information Element | Condition | Value/remark | Version |
|---|---|---|---|
| Message Type | A1, A2, A3, A4, A5, A6, A7, A8, A11 | | |
| | , A9, A10 | | Rel-5 |
| | , A12, A13, A14, A15 | | Rel-6 |
| RRC transaction identifier | | Arbitrarily selects an integer between 0 and 3 | |
| Integrity check info | | | |
| - message authentication code | | SS calculates the value of MAC-I for this message and writes to this IE. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I. | |
| - RRC message sequence number | | SS provides the value of this IE, from its internal counter. | |
| Integrity protection mode info | | Not Present | |
| Ciphering mode info | | Not Present | |
| Activation time | A1, A2, A3, A11 | (256+CFN-(CFN MOD 8 + 8))MOD 256 | |
| | , A9 | | Rel-5 |
| | , A12, A13, A14, A15 | | Rel-6 |
| Activation time | A4, A5, A6, A7, A8 A10 | Not Present | Rel-5 |
| New U-RNTI | A1, A2, A3, A4, A5, A6, A7, A8, A11 | Not Present | |
| | , A9, A10 | | Rel-5 |
| | , A12, A13, A14, A15 | | Rel-6 |
| New C-RNTI | A1, A2, A3, A4, A7, A8, A11 | Not Present | |
| | , A9, A10 | | Rel-5 |
| | , A12, A13, A14, A15 | | Rel-6 |
| New C-RNTI | A5, A6 | '1010 1010 1010 1010' | |
| New DSCH-RNTI | A1, A2, A3, A4, A5, A6, A7, A8, A11 | Not Present | R99 and Rel-4 only |
| New H-RNTI | A1, A2, A3, A4, A5, A6, A7, A8, A11 | Not Present | Rel-5 |
| New H-RNTI | A9, A10 A12, A13, A14, A15 | '1010 1010 1010 1010' | Rel-5 Rel-6 |
| New Primary E-RNTI | A1, A2, A3, A4, A5, A6, A7, A8, A9, A10, A11 | Not Present | Rel-6 |
| New Primary E-RNTI | A12, A13, A14, A15 | '1010 1010 1010 1010' | Rel-6 |
| New Secondary E-RNTI | A1, A2, A3, A4, A5, A6, A7, A8, A9, A10, A11, A12, A13, A14, A15 | Not Present | Rel-6 |
| RRC State indicator | A1, A2, A3, A4, A7, A8, A11 | CELL_DCH | |
| | , A9, A10 | | Rel-5 |
| | , A12, A13, A14, A15 | | Rel-6 |
| RRC State indicator | A5, A6 | CELL_FACH | |
| UTRAN DRX cycle length coefficient | A1, A2, A3, A4, A5, A6, A7, A8, A11 | Not Present | |
| | , A9, A10 | | Rel-5 |
| | , A12, A13, A14, A15 | | Rel-6 |
| CN information info | | Not Present | |
| URA identity | | Not Present | |
| CHOICE Specification mode | | Complete specification | Rel-6 |
| - Signalling RB information to setup | | Not Present | |
| - RAB information for setup | A1, A7 | | |
| - RAB info | | | |
| - RAB identity | | 0000 0001B The first/ leftmost bit of the bit string contains the most significant bit of the RAB identity. | |

| Information Element | Condition | Value/remark | Version |
|---|---|---|---|
| - CN domain identity | | CS domain | |
| - NAS Synchronization Indicator | | Not Present | |
| - Re-establishment timer | | useT314 | |
| - RB information to setup | | | |
| - RB identity | | 10 | |
| - PDCP info | | Not Present | |
| - CHOICE RLC info type | | RLC info | |
| - CHOICE Uplink RLC mode | | TM RLC | |
| - Transmission RLC discard | | Not Present | |
| - Segmentation indication | | FALSE | |
| - CHOICE Downlink RLC mode | | TM RLC | |
| - Segmentation indication | | FALSE | |
| - RB mapping info | | | |
| - Information for each multiplexing option | | | |
| - RLC logical channel mapping indicator | | Not Present | |
| - Number of uplink RLC logical channels | | 1 | |
| - Uplink transport channel type | | DCH | |
| - UL Transport channel identity | | 1 | |
| - Logical channel identity | | Not Present | |
| - CHOICE RLC size list | | Configured | |
| - MAC logical channel priority | | 7 | |
| - Downlink RLC logical channel info | | | |
| - Number of downlink RLC logical channels | | 1 | |
| - Downlink transport channel type | | DCH | |
| - DL DCH Transport channel identity | | 6 | |
| - DL DSCH Transport channel identity | | Not Present | |
| - Logical channel identity | | Not Present | |
| - RAB information for setup | A2, A8 | | |
| - RAB info | | | |
| - RAB identity | | 0000 0001B<br>The first/ leftmost bit of the bit string contains the most significant bit of the RAB identity. | |
| - CN domain identity | | CS domain | |
| - NAS Synchronization Indicator | | Not Present | |
| - Re-establishment timer | | useT314 | |
| - RB information to setup | | | |
| - RB identity | | 10 | |
| - PDCP info | | Not Present | |
| - CHOICE RLC info type | | RLC info | |
| - CHOICE Uplink RLC mode | | TM RLC | |
| - Transmission RLC discard | | Not Present | |
| - Segmentation indication | | FALSE | |
| - CHOICE Downlink RLC mode | | TM RLC | |
| - Segmentation indication | | FALSE | |
| - RB mapping info | | | |
| - Information for each multiplexing option | | | |
| - RLC logical channel mapping indicator | | Not Present | |
| - Number of uplink RLC logical channels | | 1 | |
| - Uplink transport channel type | | DCH | |
| - UL Transport channel identity | | 1 | |
| - Logical channel identity | | Not Present | |
| - CHOICE RLC size list | | Configured | |
| - MAC logical channel priority | | 6 | |
| - Downlink RLC logical channel info | | | |

| Information Element | Condition | Value/remark | Version |
|---|---|---|---|
| - Number of downlink RLC logical channels | | 1 | |
| - Downlink transport channel type | | DCH | |
| - DL DCH Transport channel identity | | 6 | |
| - DL DSCH Transport channel identity | | Not Present | |
| - Logical channel identity | | Not Present | |
| - RB identity | | 11 | |
| - PDCP info | | Not Present | |
| - CHOICE RLC info type | | RLC info | |
| - CHOICE Uplink RLC mode | | TM RLC | |
| - Transmission RLC discard | | Not Present | |
| - Segmentation indication | | FALSE | |
| - CHOICE Downlink RLC mode | | TM RLC | |
| - Segmentation indication | | FALSE | |
| - RB mapping info | | | |
| - Information for each multiplexing option | | | |
| - RLC logical channel mapping indicator | | Not Present | |
| - Number of uplink RLC logical channels | | 1 | |
| - Uplink transport channel type | | DCH | |
| - UL Transport channel identity | | 2 | |
| - Logical channel identity | | Not Present | |
| - CHOICE RLC size list | | Configured | |
| - MAC logical channel priority | | 6 | |
| - Downlink RLC logical channel info | | | |
| - Number of downlink RLC logical channels | | 1 | |
| - Downlink transport channel type | | DCH | |
| - DL DCH Transport channel identity | | 7 | |
| - DL DSCH Transport channel identity | | Not Present | |
| - Logical channel identity | | Not Present | |
| - RB identity | | 12 | |
| - PDCP info | | Not Present | |
| - CHOICE RLC info type | | RLC info | |
| - CHOICE Uplink RLC mode | | TM RLC | |
| - Transmission RLC discard | | Not Present | |
| - Segmentation indication | | FALSE | |
| - CHOICE Downlink RLC mode | | TM RLC | |
| - Segmentation indication | | FALSE | |
| - RB mapping info | | | |
| - Information for each multiplexing option | | | |
| - RLC logical channel mapping indicator | | Not Present | |
| - Number of uplink RLC logical channels | | 1 | |
| - Uplink transport channel type | | DCH | |
| - UL Transport channel identity | | 3 | |
| - Logical channel identity | | Not Present | |
| - CHOICE RLC size list | | Configured | |
| - MAC logical channel priority | | 6 | |
| - Downlink RLC logical channel info | | | |
| - Number of downlink RLC logical channels | | 1 | |
| - Downlink transport channel type | | DCH | |
| - DL DCH Transport channel | | 8 | |

| Information Element | Condition | Value/remark | Version |
|---|---|---|---|
| identity | | | |
| - DL DSCH Transport channel identity | | Not Present | |
| - Logical channel identity | | Not Present | |
| - RAB information for setup | A3, A4, A5, A6 | | |
| - RAB info | | (AM DTCH for PS domain) | |
| - RAB identity | | 0000 0101B<br>The first/ leftmost bit of the bit string contains the most significant bit of the RAB identity. | |
| - CN domain identity | | PS domain | |
| - NAS Synchronization Indicator | | Not Present | |
| - Re-establishment timer | | useT315 | |
| - RB information to setup | | | |
| - RB identity | | 20 | |
| - PDCP info | | | |
| - Support for lossless SRNS relocation | | FALSE | |
| - Max PDCP SN window size | | Not present | |
| - PDCP PDU header | | Absent | |
| - Header compression information | | Not present | |
| - CHOICE RLC info type | | RLC info | |
| - CHOICE Uplink RLC mode | | AM RLC | |
| - Transmission RLC discard | | | |
| - CHOICE SDU discard mode | | No Discard | |
| - MAX_DAT | | 15 | |
| - Transmission window size | | 128 | |
| - Timer_RST | | 500 | |
| - Max_RST | | 4 | |
| - Polling info | | | |
| - Timer_poll_prohibit | | 200 | |
| - Timer_poll | | 200 | |
| - Poll_PDU | | Not Present | |
| - Poll_SDU | | 1 | |
| - Last transmission PDU poll | | TRUE | |
| - Last retransmission PDU poll | | TRUE | |
| - Poll_Windows | | 99 | |
| - Timer_poll_periodic | | Not Present | |
| - CHOICE Downlink RLC mode | | AM RLC | |
| - In-sequence delivery | | TRUE | |
| - Receiving window size | | 128 | |
| - Downlink RLC status info | | | |
| - Timer_status_prohibit | | 200 | |
| - Timer_EPC | | Not Present | |
| - Missing PDU indicator | | TRUE | |
| - Timer_STATUS_periodic | | Not Present | |
| - RB mapping info | | | |
| - Information for each multiplexing option | | 2 RBMuxOptions | |
| - RLC logical channel mapping indicator | | Not Present | |
| - Number of uplink RLC logical channels | | 1 | |
| - Uplink transport channel type | | DCH | |
| - UL Transport channel identity | | 1 | |
| - Logical channel identity | | Not Present | |
| - CHOICE RLC size list | | Configured | |
| - MAC logical channel priority | | 8 | |
| - Downlink RLC logical channel info | | | |
| - Number of downlink RLC logical channels | | 1 | |
| - Downlink transport channel type | | DCH | |
| - DL DCH Transport channel identity | | 6 | |

| Information Element | Condition | Value/remark | Version |
|---|---|---|---|
| - DL DSCH Transport channel identity | | Not Present | |
| - Logical channel identity | | Not Present | |
| - RLC logical channel mapping indicator | | Not Present | |
| - Number of uplink RLC logical channels | | 1 | |
| - Uplink transport channel type | | RACH | |
| - UL Transport channel identity | | Not Present | |
| - Logical channel identity | | 7 | |
| - CHOICE RLC size list | | Explicit list | |
| - RLC size index | | Reference to clause 6 Parameter Set | |
| - MAC logical channel priority | | 8 | |
| - Downlink RLC logical channel info | | | |
| - Number of downlink RLC logical channels | | 1 | |
| - Downlink transport channel type | | FACH | |
| - DL DCH Transport channel identity | | Not Present | |
| - DL DSCH Transport channel identity | | Not Present | |
| - Logical channel identity | | 7 | |
| - RAB information for setup | A9 | | Rel-5 |
| - RAB info | | (high-speed AM DTCH for PS domain) | |
| - RAB identity | | 0000 0101B<br>The first/ leftmost bit of the bit string contains the most significant bit of the RAB identity. | |
| - CN domain identity | | PS domain | |
| - NAS Synchronization Indicator | | Not Present | |
| - Re-establishment timer | | useT315 | |
| - RB information to setup | | | |
| - RB identity | | 25 | |
| - PDCP info | | | |
| - Support for lossless SRNS relocation | | FALSE | |
| - Max PDCP SN window size | | Not present | |
| - PDCP PDU header | | Absent | |
| - Header compression information | | Not present | |
| - CHOICE RLC info type | | RLC info | |
| - CHOICE Uplink RLC mode | | AM RLC | |
| - Transmission RLC discard | | | |
| - CHOICE SDU discard mode | | No Discard | |
| - MAX_DAT | | 15 | |
| - Transmission window size | | 128 | |
| - Timer_RST | | 500 | |
| - Max_RST | | 4 | |
| - Polling info | | | |
| - Timer_poll_prohibit | | 100 | |
| - Timer_poll | | 100 | |
| - Poll_PDU | | Not Present | |
| - Poll_SDU | | 1 | |
| - Last transmission PDU poll | | TRUE | |
| - Last retransmission PDU poll | | TRUE | |
| - Poll_Windows | | 99 | |
| - Timer_poll_periodic | | Not Present | |
| - CHOICE Downlink RLC mode | | AM RLC | |
| - CHOICE Downlink RLC PDU Size | | Reference to clause 6 Parameter Set | |
| - In-sequence delivery | | TRUE | |
| - Receiving window size | | 768 | |
| - Downlink RLC status info | | | |
| - Timer_status_prohibit | | 100 | |
| - Timer_EPC | | Not Present | |

| Information Element | Condition | Value/remark | Version |
|---|---|---|---|
| - Missing PDU indicator | | TRUE | |
| - Timer_STATUS_periodic | | Not Present | |
| - One sided RLC re-establishment | | FALSE | |
| - RB mapping info | | | |
| - Information for each multiplexing option | | 3 RBMuxOptions | |
| - RLC logical channel mapping indicator | | Not Present | |
| - Number of uplink RLC logical channels | | 1 | |
| - Uplink transport channel type | | DCH | |
| - UL Transport channel identity | | 1 | |
| - Logical channel identity | | Not Present | |
| - CHOICE RLC size list | | Configured | |
| - MAC logical channel priority | | 8 | |
| - Downlink RLC logical channel info | | | |
| - Number of downlink RLC logical channels | | 1 | |
| - Downlink transport channel type | | DCH | |
| - DL DCH Transport channel identity | | 6 | |
| - DL DSCH Transport channel identity | | Not Present | |
| - DL HS-DSCH MAC-d flow identity | | Not Present | |
| - Logical channel identity | | Not Present | |
| - RLC logical channel mapping indicator | | Not Present | |
| - Number of uplink RLC logical channels | | 1 | |
| - Uplink transport channel type | | DCH | |
| - UL Transport channel identity | | 1 | |
| - Logical channel identity | | Not Present | |
| - CHOICE RLC size list | | Configured | |
| - MAC logical channel priority | | 8 | |
| - Downlink RLC logical channel info | | | |
| - Number of downlink RLC logical channels | | 1 | |
| - Downlink transport channel type | | HS-DSCH | |
| - DL DCH Transport channel identity | | Not Present | |
| - DL DSCH Transport channel identity | | Not Present | |
| - DL HS-DSCH MAC-d flow identity | | 0 | |
| - Logical channel identity | | Not Present | |
| - RLC logical channel mapping indicator | | Not Present | |
| - Number of uplink RLC logical channels | | 1 | |
| - Uplink transport channel type | | RACH | |
| - UL Transport channel identity | | Not Present | |
| - Logical channel identity | | 7 | |
| - CHOICE RLC size list | | Explicit list | |
| - RLC size index | | Reference to clause 6 Parameter Set | |
| - MAC logical channel priority | | 8 | |
| - Downlink RLC logical channel info | | | |
| - Number of downlink RLC logical channels | | 1 | |
| - Downlink transport channel type | | FACH | |

| Information Element | Condition | Value/remark | Version |
|---|---|---|---|
| - DL DCH Transport channel identity | | Not Present | |
| - DL DSCH Transport channel identity | | Not Present | |
| - Logical channel identity | | 7 | |
| - RAB information for setup | A10 | | Rel-5 |
| - RAB info | | (high-speed AM DTCH for PS domain) | |
| - RAB identity | | 0000 0101B<br>The first/ leftmost bit of the bit string contains the most significant bit of the RAB identity. | |
| - CN domain identity | | PS domain | |
| - NAS Synchronization Indicator | | Not Present | |
| - Re-establishment timer | | useT315 | |
| - RB information to setup | | | |
| - RB identity | | 25 | |
| - PDCP info | | | |
| - Support for lossless SRNS relocation | | FALSE | |
| - Max PDCP SN window size | | Not present | |
| - PDCP PDU header | | Absent | |
| - Header compression information | | Not present | |
| - CHOICE RLC info type | | RLC info | |
| - CHOICE Uplink RLC mode | | AM RLC | |
| - Transmission RLC discard | | | |
| - CHOICE SDU discard mode | | No Discard | |
| - MAX_DAT | | 15 | |
| - Transmission window size | | 128 | |
| - Timer_RST | | 500 | |
| - Max_RST | | 4 | |
| - Polling info | | | |
| - Timer_poll_prohibit | | 100 | |
| - Timer_poll | | 100 | |
| - Poll_PDU | | Not Present | |
| - Poll_SDU | | 1 | |
| - Last transmission PDU poll | | TRUE | |
| - Last retransmission PDU poll | | TRUE | |
| - Poll_Windows | | 99 | |
| - Timer_poll_periodic | | Not Present | |
| - CHOICE Downlink RLC mode | | AM RLC | |
| - CHOICE Downlink RLC PDU Size | | Reference to clause 6 Parameter Set | |
| - In-sequence delivery | | TRUE | |
| - Receiving window size | | 768 | |
| - Downlink RLC status info | | | |
| - Timer_status_prohibit | | 100 | |
| - Timer_EPC | | Not Present | |
| - Missing PDU indicator | | TRUE | |
| - Timer_STATUS_periodic | | Not Present | |
| - One sided RLC re-establishment | | FALSE | |
| - RB mapping info | | | |
| - Information for each multiplexing option | | 1 RBMuxOption | |
| - RLC logical channel mapping indicator | | Not present | |
| - Number of uplink RLC logical channels | | 1 | |
| - Uplink transport channel type | | DCH | |
| - UL Transport channel identity | | 1 | |
| - Logical channel identity | | Not Present | |
| - CHOICE RLC size list | | Configured | |
| - MAC logical channel priority | | 8 | |
| - Downlink RLC logical channel info | | | |
| - Number of downlink RLC | | 1 | |

| Information Element | Condition | Value/remark | Version |
|---|---|---|---|
| logical channels | | | |
| - Downlink transport channel type | | HS-DSCH | |
| - DL DCH Transport channel identity | | Not present | |
| - DL DSCH Transport channel identity | | Not present | |
| - DL HS-DSCH MAC-d flow identity | | 0 | |
| - Logical channel identity | | Not Present | |
| - RAB information for setup | A11 | | |
| - RAB info | | (AM DTCH for PS domain) | |
| - RAB identity | | 0000 0101B<br>The first/ leftmost bit of the bit string contains the most significant bit of the RAB identity. | |
| - CN domain identity | | PS domain | |
| - NAS Synchronization Indicator | | Not Present | |
| - Re-establishment timer | | useT315 | |
| - RB information to setup | | | |
| - RB identity | | 20 | |
| - PDCP info | | | |
| - Support for lossless SRNS relocation | | FALSE | |
| - Max PDCP SN window size | | Not present | |
| - PDCP PDU header | | Absent | |
| - Header compression information | | Not present | |
| - CHOICE RLC info type | | RLC info | |
| - CHOICE Uplink RLC mode | | AM RLC | |
| - Transmission RLC discard | | | |
| - CHOICE SDU discard mode | | No Discard | |
| - MAX_DAT | | 15 | |
| - Transmission window size | | 128 | |
| - Timer_RST | | 500 | |
| - Max_RST | | 4 | |
| - Polling info | | | |
| - Timer_poll_prohibit | | 200 | |
| - Timer_poll | | 200 | |
| - Poll_PDU | | Not Present | |
| - Poll_SDU | | 1 | |
| - Last transmission PDU poll | | TRUE | |
| - Last retransmission PDU poll | | TRUE | |
| - Poll_Windows | | 99 | |
| - Timer_poll_periodic | | Not Present | |
| - CHOICE Downlink RLC mode | | AM RLC | |
| - In-sequence delivery | | TRUE | |
| - Receiving window size | | 128 | |
| - Downlink RLC status info | | | |
| - Timer_status_prohibit | | 200 | |
| - Timer_EPC | | Not Present | |
| - Missing PDU indicator | | TRUE | |
| - Timer_STATUS_periodic | | Not Present | |
| - RB mapping info | | | |
| - Information for each multiplexing option | | 2 RBMuxOptions | |
| - RLC logical channel mapping indicator | | Not Present | |
| - Number of uplink RLC logical channels | | 1 | |
| - Uplink transport channel type | | DCH | |
| - UL Transport channel identity | | 4 | |
| - Logical channel identity | | Not Present | |
| - CHOICE RLC size list | | Configured | |
| - MAC logical channel priority | | 8 | |
| - Downlink RLC logical channel info | | | |

| Information Element | Condition | Value/remark | Version |
|---|---|---|---|
| - Number of downlink RLC logical channels | | 1 | |
| - Downlink transport channel type | | DCH | |
| - DL DCH Transport channel identity | | 9 | |
| - DL DSCH Transport channel identity | | Not Present | |
| - Logical channel identity | | Not Present | |
| - RLC logical channel mapping indicator | | Not Present | |
| - Number of uplink RLC logical channels | | 1 | |
| - Uplink transport channel type | | RACH | |
| - UL Transport channel identity | | Not Present | |
| - Logical channel identity | | 7 | |
| - CHOICE RLC size list | | Explicit list | |
| - RLC size index | | Reference to clause 6 Parameter Set | |
| - MAC logical channel priority | | 8 | |
| - Downlink RLC logical channel info | | | |
| - Number of downlink RLC logical channels | | 1 | |
| - Downlink transport channel type | | FACH | |
| - DL DCH Transport channel identity | | Not Present | |
| - DL DSCH Transport channel identity | | Not Present | |
| - Logical channel identity | | 7 | |
| - RAB information for setup | A12 | | Rel-6 |
| - RAB info | | (high-speed AM DTCH for PS domain) | |
| - RAB identity | | 0000 0101B<br>The first/ leftmost bit of the bit string contains the most significant bit of the RAB identity. | |
| - CN domain identity | | PS domain | |
| - NAS Synchronization Indicator | | Not Present | |
| - Re-establishment timer | | useT315 | |
| - RB information to setup | | | |
| - RB identity | | 25 | |
| - PDCP info | | | |
| - Support for lossless SRNS relocation | | FALSE | |
| - Max PDCP SN window size | | Not present | |
| - PDCP PDU header | | Absent | |
| - Header compression information | | Not present | |
| - CHOICE RLC info type | | RLC info | |
| - CHOICE Uplink RLC mode | | AM RLC | |
| - Transmission RLC discard | | | |
| - CHOICE SDU discard mode | | No Discard | |
| - MAX_DAT | | 15 | |
| - Transmission window size | | 256 | |
| - Timer_RST | | 500 | |
| - Max_RST | | 4 | |
| - Polling info | | | |
| - Timer_poll_prohibit | | 100 | |
| - Timer_poll | | 100 | |
| - Poll_PDU | | Not Present | |
| - Poll_SDU | | 1 | |
| - Last transmission PDU poll | | TRUE | |
| - Last retransmission PDU poll | | TRUE | |
| - Poll_Windows | | 99 | |
| - Timer_poll_periodic | | Not Present | |
| - CHOICE Downlink RLC mode | | AM RLC | |
| - CHOICE Downlink RLC PDU | | Reference to clause 6 Parameter Set | |

| Information Element | Condition | Value/remark | Version |
|---|---|---|---|
| Size | | | |
| - In-sequence delivery | | TRUE | |
| - Receiving window size | | 768 | |
| - Downlink RLC status info | | | |
| - Timer_status_prohibit | | 100 | |
| - Timer_EPC | | Not Present | |
| - Missing PDU indicator | | TRUE | |
| - Timer_STATUS_periodic | | Not Present | |
| - One sided RLC re-establishment | | FALSE | |
| - RB mapping info | | | |
| - Information for each multiplexing option | | 3 RBMuxOptions | |
| - RLC logical channel mapping indicator | | Not Present | |
| - Number of uplink RLC logical channels | | 1 | |
| - Uplink transport channel type | | DCH | |
| - UL Transport channel identity | | 1 | |
| - Logical channel identity | | Not Present | |
| - CHOICE RLC size list | | Configured | |
| - MAC logical channel priority | | 8 | |
| - Downlink RLC logical channel info | | | |
| - Number of downlink RLC logical channels | | 1 | |
| - Downlink transport channel type | | DCH | |
| - DL DCH Transport channel identity | | 6 | |
| - DL DSCH Transport channel identity | | Not Present | |
| - DL HS-DSCH MAC-d flow identity | | Not Present | |
| - Logical channel identity | | Not Present | |
| - RLC logical channel mapping indicator | | Not Present | |
| - Number of uplink RLC logical channels | | 1 | |
| - Uplink transport channel type | | E-DCH | |
| - Logical channel identity | | 7 | |
| - E-DCH MAC-d flow identity | | 2 | |
| - DDI | | 5 | |
| - RLC PDU size list | | 1 RLC PDU size | |
| - RLC PDU size | | 336 bits | |
| - Include in scheduling info | | TRUE | |
| - MAC logical channel priority | | 8 | |
| - Downlink RLC logical channel info | | | |
| - Number of downlink RLC logical channels | | 1 | |
| - Downlink transport channel type | | HS-DSCH | |
| - DL DCH Transport channel identity | | Not Present | |
| - DL DSCH Transport channel identity | | Not Present | |
| - DL HS-DSCH MAC-d flow identity | | 0 | |
| - Logical channel identity | | Not Present | |
| - RLC logical channel mapping indicator | | Not Present | |
| - Number of uplink RLC logical channels | | 1 | |
| - Uplink transport channel type | | RACH | |
| - UL Transport channel identity | | Not Present | |
| - Logical channel identity | | 7 | |

| Information Element | Condition | Value/remark | Version |
|---|---|---|---|
| - CHOICE RLC size list | | Explicit list | |
| - RLC size index | | Reference to clause 6 Parameter Set | |
| - MAC logical channel priority | | 8 | |
| - Downlink RLC logical channel info | | | |
| - Number of downlink RLC logical channels | | 1 | |
| - Downlink transport channel type | | FACH | |
| - DL DCH Transport channel identity | | Not Present | |
| - DL DSCH Transport channel identity | | Not Present | |
| - RAB information for setup | A13, A14, A15 | | Rel-6 |
| - RAB info | | (high-speed AM DTCH for PS domain) | |
| - RAB identity | | 0000 0101B<br>The first/ leftmost bit of the bit string contains the most significant bit of the RAB identity. | |
| - CN domain identity | | PS domain | |
| - NAS Synchronization Indicator | | Not Present | |
| - Re-establishment timer | | useT315 | |
| - RB information to setup | | | |
| - RB identity | | 25 | |
| - PDCP info | | | |
| - Support for lossless SRNS relocation | | FALSE | |
| - Max PDCP SN window size | | Not present | |
| - PDCP PDU header | | Absent | |
| - Header compression information | | Not present | |
| - CHOICE RLC info type | | RLC info | |
| - CHOICE Uplink RLC mode | | AM RLC | |
| - Transmission RLC discard | | | |
| - CHOICE SDU discard mode | | No Discard | |
| - MAX_DAT | | 15 | |
| - Transmission window size | | 256 | |
| - Timer_RST | | 500 | |
| - Max_RST | | 4 | |
| - Polling info | | | |
| - Timer_poll_prohibit | | 100 | |
| - Timer_poll | | 100 | |
| - Poll_PDU | | Not Present | |
| - Poll_SDU | | 1 | |
| - Last transmission PDU poll | | TRUE | |
| - Last retransmission PDU poll | | TRUE | |
| - Poll_Windows | | 99 | |
| - Timer_poll_periodic | | Not Present | |
| - CHOICE Downlink RLC mode | | AM RLC | |
| - CHOICE Downlink RLC PDU Size | | Reference to clause 6 Parameter Set | |
| - In-sequence delivery | | TRUE | |
| - Receiving window size | | 768 | |
| - Downlink RLC status info | | | |
| - Timer_status_prohibit | | 100 | |
| - Timer_EPC | | Not Present | |
| - Missing PDU indicator | | TRUE | |
| - Timer_STATUS_periodic | | Not Present | |
| - One sided RLC re-establishment | | FALSE | |
| - RB mapping info | | | |
| - Information for each multiplexing option | | 1 RBMuxOption | |
| - RLC logical channel mapping indicator | | Not Present | |
| - Number of uplink RLC logical channels | | 1 | |

| Information Element | Condition | Value/remark | Version |
|---|---|---|---|
| - Uplink transport channel type | | E-DCH | |
| - Logical channel identity | | 7 | |
| - E-DCH MAC-d flow identity | | 2 | |
| - DDI | | 5 | |
| - RLC PDU size list | | 1 RLC PDU size | |
| - RLC PDU size | | 336 bits | |
| - Include in scheduling info | | TRUE | |
| - MAC logical channel priority | | 8 | |
| - Downlink RLC logical channel info | | | |
| - Number of downlink RLC logical channels | | 1 | |
| - Downlink transport channel type | | HS-DSCH | |
| - DL DCH Transport channel identity | | Not present | |
| - DL DSCH Transport channel identity | | Not present | |
| - DL HS-DSCH MAC-d flow identity | | 0 | |
| - Logical channel identity | | Not Present | |
| - RAB information for setup | A15 | | Rel-6 |
| - RAB info | | (second high-speed AM DTCH for PS domain) | |
| - RAB identity | | 0000 0110B<br>The first/ leftmost bit of the bit string contains the most significant bit of the RAB identity. | |
| - CN domain identity | | PS domain | |
| - NAS Synchronization Indicator | | Not Present | |
| - Re-establishment timer | | useT315 | |
| - RB information to setup | | | |
| - RB identity | | 26 | |
| - PDCP info | | | |
| - Support for lossless SRNS relocation | | FALSE | |
| - Max PDCP SN window size | | Not present | |
| - PDCP PDU header | | Absent | |
| - Header compression information | | Not present | |
| - CHOICE RLC info type | | RLC info | |
| - CHOICE Uplink RLC mode | | AM RLC | |
| - Transmission RLC discard | | | |
| - CHOICE SDU discard mode | | No Discard | |
| - MAX_DAT | | 15 | |
| - Transmission window size | | 256 | |
| - Timer_RST | | 500 | |
| - Max_RST | | 4 | |
| - Polling info | | | |
| - Timer_poll_prohibit | | 100 | |
| - Timer_poll | | 100 | |
| - Poll_PDU | | Not Present | |
| - Poll_SDU | | 1 | |
| - Last transmission PDU poll | | TRUE | |
| - Last retransmission PDU poll | | TRUE | |
| - Poll_Windows | | 99 | |
| - Timer_poll_periodic | | Not Present | |
| - CHOICE Downlink RLC mode | | AM RLC | |
| - CHOICE Downlink RLC PDU Size | | Reference to clause 6 Parameter Set | |
| - In-sequence delivery | | TRUE | |
| - Receiving window size | | 768 | |
| - Downlink RLC status info | | | |
| - Timer_status_prohibit | | 100 | |
| - Timer_EPC | | Not Present | |
| - Missing PDU indicator | | TRUE | |
| - Timer_STATUS_periodic | | Not Present | |

| Information Element | Condition | Value/remark | Version |
|---|---|---|---|
| - One sided RLC re-establishment | | FALSE | |
| - RB mapping info | | | |
| - Information for each multiplexing option | | 1 RBMuxOption | |
| - RLC logical channel mapping indicator | | Not Present | |
| - Number of uplink RLC logical channels | | 1 | |
| - Uplink transport channel type | | E-DCH | |
| - Logical channel identity | | 8 | |
| - E-DCH MAC-d flow identity | | 3 | |
| - DDI | | 6 | |
| - RLC PDU size list | | 1 RLC PDU size | |
| - RLC PDU size | | 336 bits | |
| - Include in scheduling info | | TRUE | |
| - MAC logical channel priority | | 8 | |
| - Downlink RLC logical channel info | | | |
| - Number of downlink RLC logical channels | | 1 | |
| - Downlink transport channel type | | HS-DSCH | |
| - DL DCH Transport channel identity | | Not present | |
| - DL DSCH Transport channel identity | | Not present | |
| - DL HS-DSCH MAC-d flow identity | | 2 | |
| - Logical channel identity | | Not Present | |
| RB information to reconfigure list | A1, A2, A3, A4, A5, A6, A7, A8 | Not Present | |
| | A9, A10 | | Rel-5 |
| | A12, A13, A14, A15 | | Rel-6 |
| RB information to be affected | A1, A2, A3, A4, A5, A6, A7, A8, A11 | Not Present | |
| | , A9, A10 | | Rel-5 |
| | A12 | | Rel-6 |
| RB information to be affected | A13, A15 | | Rel-6 |
| - RB identity | | 1 (UM DCCH for RRC) | |
| - RB mapping info | | | |
| - Information for each multiplexing option | | 1 RBMuxOption | |
| - RLC logical channel mapping indicator | | Not Present | |
| - Number of uplink RLC logical channels | | 1 | |
| - Uplink transport channel type | | E-DCH | |
| - Logical channel identity | | 1 | |
| - E-DCH MAC-d flow identity | | 1 | |
| - DDI | | 1 | |
| - RLC PDU size list | | 1 RLC PDU size | |
| - RLC PDU size | | 144 bits | |
| - Include in scheduling info | | FALSE | |
| - MAC logical channel priority | | 1 | |
| - Downlink RLC logical channel info | | | |
| - Number of RLC logical channels | | 1 | |
| - Downlink transport channel type | | DCH | |
| - DL DCH Transport channel identity | | 10 | |
| - DL DSCH Transport channel identity | | Not Present | |
| - Logical channel identity | | 1 | |
| - RB identity | | 2 (AM DCCH for RRC) | |

| Information Element | Condition | Value/remark | Version |
|---|---|---|---|
| - RB mapping info | | | |
| - Information for each multiplexing option | | 1 RBMuxOption | |
| - RLC logical channel mapping indicator | | Not Present | |
| - Number of uplink RLC logical channels | | 1 | |
| - Uplink transport channel type | | E-DCH | |
| - Logical channel identity | | 2 | |
| - E-DCH MAC-d flow identity | | 1 | |
| - DDI | | 2 | |
| - RLC PDU size list | | 1 RLC PDU size | |
| - RLC PDU size | | 144 bits | |
| - Include in scheduling info | | FALSE | |
| - MAC logical channel priority | | 2 | |
| - Downlink RLC logical channel info | | | |
| - Number of RLC logical channels | | 1 | |
| - Downlink transport channel type | | DCH | |
| - DL DCH Transport channel identity | | 10 | |
| - DL DSCH Transport channel identity | | Not Present | |
| - Logical channel identity | | 2 | |
| - RB identity | | 3 (AM DCCH for NAS High Priority) | |
| - RB mapping info | | | |
| - Information for each multiplexing option | | 1 RBMuxOption | |
| - RLC logical channel mapping indicator | | Not Present | |
| - Number of uplink RLC logical channels | | 1 | |
| - Uplink transport channel type | | E-DCH | |
| - Logical channel identity | | 3 | |
| - E-DCH MAC-d flow identity | | 1 | |
| - DDI | | 3 | |
| - RLC PDU size list | | 1 RLC PDU size | |
| - RLC PDU size | | 144 bits | |
| - Include in scheduling info | | FALSE | |
| - MAC logical channel priority | | 3 | |
| - Downlink RLC logical channel info | | | |
| - Number of RLC logical channels | | 1 | |
| - Downlink transport channel type | | DCH | |
| - DL DCH Transport channel identity | | 10 | |
| - DL DSCH Transport channel identity | | Not Present | |
| - Logical channel identity | | 3 | |
| - RB identity | | 4 (AM DCCH for NAS Low Priority) | |
| - RB mapping info | | | |
| - Information for each multiplexing option | | 1 RBMuxOption | |
| - RLC logical channel mapping indicator | | Not Present | |
| - Number of uplink RLC logical channels | | 1 | |
| - Uplink transport channel type | | E-DCH | |
| - Logical channel identity | | 4 | |
| - E-DCH MAC-d flow identity | | 1 | |
| - DDI | | 4 | |
| - RLC PDU size list | | 1 RLC PDU size | |
| - RLC PDU size | | 144 bits | |

| Information Element | Condition | Value/remark | Version |
|---|---|---|---|
| - Include in scheduling info | | FALSE | |
| - MAC logical channel priority | | 4 | |
| - Downlink RLC logical channel info | | | |
| - Number of RLC logical channels | | 1 | |
| - Downlink transport channel type | | DCH | |
| - DL DCH Transport channel identity | | 10 | |
| - DL DSCH Transport channel identity | | Not Present | |
| - Logical channel identity | | 4 | |
| RB information to be affected | A14 | | Rel-6 |
| - RB identity | | 1 (UM DCCH for RRC) | |
| - RB mapping info | | | |
| - Information for each multiplexing option | | 1 RBMuxOption | |
| - RLC logical channel mapping indicator | | Not Present | |
| - Number of uplink RLC logical channels | | 1 | |
| - Uplink transport channel type | | E-DCH | |
| - Logical channel identity | | 1 | |
| - E-DCH MAC-d flow identity | | 1 | |
| - DDI | | 1 | |
| - RLC PDU size list | | 1 RLC PDU size | |
| - RLC PDU size | | 144 bits | |
| - Include in scheduling info | | FALSE | |
| - MAC logical channel priority | | 1 | |
| - Downlink RLC logical channel info | | | |
| - Number of RLC logical channels | | 1 | |
| - Downlink transport channel type | | HS-DSCH | |
| - DL DCH Transport channel identity | | Not present | |
| - DL DSCH Transport channel identity | | Not present | |
| - DL HS-DSCH MAC-d flow identity | | 1 | |
| - Logical channel identity | | 1 | |
| - RB identity | | 2 (AM DCCH for RRC) | |
| - RB mapping info | | | |
| - Information for each multiplexing option | | 1 RBMuxOption | |
| - RLC logical channel mapping indicator | | Not Present | |
| - Number of uplink RLC logical channels | | 1 | |
| - Uplink transport channel type | | E-DCH | |
| - Logical channel identity | | 2 | |
| - E-DCH MAC-d flow identity | | 1 | |
| - DDI | | 2 | |
| - RLC PDU size list | | 1 RLC PDU size | |
| - RLC PDU size | | 144 bits | |
| - Include in scheduling info | | FALSE | |
| - MAC logical channel priority | | 2 | |
| - Downlink RLC logical channel info | | | |
| - Number of RLC logical channels | | 1 | |
| - Downlink transport channel type | | HS-DSCH | |
| - DL DCH Transport channel identity | | Not Present | |

| Information Element | Condition | Value/remark | Version |
|---|---|---|---|
| - DL DSCH Transport channel identity | | Not Present | |
| - DL HS-DSCH MAC-d flow identity | | 1 | |
| - Logical channel identity | | 2 | |
| - RB identity | | 3 (AM DCCH for NAS High Priority) | |
| - RB mapping info | | | |
| - Information for each multiplexing option | | 1 RBMuxOption | |
| - RLC logical channel mapping indicator | | Not Present | |
| - Number of uplink RLC logical channels | | 1 | |
| - Uplink transport channel type | | E-DCH | |
| - Logical channel identity | | 3 | |
| - E-DCH MAC-d flow identity | | 1 | |
| - DDI | | 3 | |
| - RLC PDU size list | | 1 RLC PDU size | |
| - RLC PDU size | | 144 bits | |
| - Include in scheduling info | | FALSE | |
| - MAC logical channel priority | | 3 | |
| - Downlink RLC logical channel info | | | |
| - Number of RLC logical channels | | 1 | |
| - Downlink transport channel type | | HS-DSCH | |
| - DL DCH Transport channel identity | | Not Present | |
| - DL DSCH Transport channel identity | | Not Present | |
| - DL HS-DSCH MAC-d flow identity | | 1 | |
| - Logical channel identity | | 3 | |
| - RB identity | | 4 (AM DCCH for NAS Low Priority) | |
| - RB mapping info | | | |
| - Information for each multiplexing option | | 1 RBMuxOption | |
| - RLC logical channel mapping indicator | | Not Present | |
| - Number of uplink RLC logical channels | | 1 | |
| - Uplink transport channel type | | E-DCH | |
| - Logical channel identity | | 4 | |
| - E-DCH MAC-d flow identity | | 1 | |
| - DDI | | 4 | |
| - RLC PDU size list | | 1 RLC PDU size | |
| - RLC PDU size | | 144 bits | |
| - Include in scheduling info | | FALSE | |
| - MAC logical channel priority | | 4 | |
| - Downlink RLC logical channel info | | | |
| - Number of RLC logical channels | | 1 | |
| - Downlink transport channel type | | HS-DSCH | |
| - DL DCH Transport channel identity | | Not Present | |
| - DL DSCH Transport channel identity | | Not Present | |
| - DL HS-DSCH MAC-d flow identity | | 1 | |
| - Logical channel identity | | 4 | |
| Downlink counter synchronization info | A1, A2, A3, A4, A5, A6, A7, A8, A11<br>, A9, A10<br>, A12, A13, A14, A15 | Not Present | <br><br>Rel-5<br>Rel-6 |

| Information Element | Condition | Value/remark | Version |
|---|---|---|---|
| PDCP ROHC target mode | A9, A10<br>, A12, A13, A14, A15 | Not Present | Rel-5<br>Rel-6 |
| UL Transport channel information for all transport channels | A1, A2, A3, A4, A5, A6, A7, A8, A11<br>, A9, A10 | | <br><br><br>Rel-5 |
| - PRACH TFCS | | Not Present | |
| - CHOICE mode | | FDD | |
| - TFC subset | | Not Present | |
| - UL DCH TFCS | | | |
| - CHOICE TFCI signalling | | Normal | |
| - TFCI Field 1 information | | | |
| - CHOICE TFCS representation | | Complete reconfiguration | |
| - TFCS complete reconfigure information | | | |
| - CHOICE CTFC Size | | Number of bits used must be enough to cover all combinations of CTFC from clause 6.10.2.4 Parameter Set. | |
| - CTFC information | | This IE is repeated for TFC numbers and reference to clause 6.10.2.4 Parameter Set | |
| - CTFC | | Reference to clause 6.10.2.4 Parameter Set | |
| - Power offset information | | | |
| - CHOICE Gain Factors | | Computed Gain Factors(The last TFC is set to Signalled Gain Factors) | |
| - Gain factor βc | | 11 (below 64 kbps)<br>9 (higher than 64 kbps) (Not Present if the CHOICE Gain Factors is set to Computed Gain Factors) | |
| - Gain factor βd | | 15<br>(Not Present if the CHOICE Gain Factors is set to Computed Gain Factors) | |
| - Reference TFC ID | | 0 | |
| - CHOICE mode | | FDD | |
| - Power offset P p-m | | Not Present | |
| UL Transport channel information for all transport channels | A12, A13, A14, A15 | Not Present | Rel-6 |
| Deleted UL TrCH information | A1, A2, A3, A4, A5, A6, A7, A8, A11<br>, A9, A10<br>, A12 | Not Present | <br><br>Rel-5<br>Rel-6 |
| Deleted UL TrCH information | A13, A14, A15 | | Rel-6 |
| - Uplink transport channel type | DCH | | |
| - UL transport channel identity | 5 | | |
| Added or Reconfigured UL TrCH information | A1, A3 A4, A5, A6, A7<br>, A9, A10 | 1 DCH added, 1 DCH reconfigured (if from cell_DCH) OR 2 DCHs added (if from cell_FACH) | <br><br>Rel-5 |
| - Uplink transport channel type | | DCH | |
| - UL Transport channel identity | | 1 | |
| - TFS | | | |
| - CHOICE Transport channel type | | Dedicated transport channels | |
| - Dynamic Transport format information | | | |
| - RLC Size | | Reference to clause 6.10 Parameter Set | |
| - Number of TBs and TTI List | | (This IE is repeated for TFI number.) | |
| - Transmission Time Interval | | Not Present | |
| - Number of Transport blocks | | Reference to clause 6.10 Parameter Set | |
| - CHOICE Logical channel list | | All | |
| - Semi-static Transport Format information | | | |
| - Transmission time interval | | Reference to clause 6.10 Parameter Set | |
| - Type of channel coding | | Reference to clause 6.10 Parameter Set | |
| - Coding Rate | | Reference to clause 6.10 Parameter Set | |
| - Rate matching attribute | | Reference to clause 6.10 Parameter Set | |
| - CRC size | | Reference to clause 6.10 Parameter Set | |
| - Uplink transport channel type | | DCH | |

| Information Element | Condition | Value/remark | Version |
|---|---|---|---|
| - UL Transport channel identity | | 5 | |
| - TFS | | | |
| - CHOICE Transport channel type | | Dedicated transport channels | |
| - Dynamic Transport format information | | | |
| - RLC Size | | Reference to clause 6.10 Parameter Set | |
| - Number of TBs and TTI List | | (This IE is repeated for TFI number.) | |
| - Transmission Time Interval | | Not Present | |
| - Number of Transport blocks | | Reference to clause 6.10 Parameter Set | |
| - CHOICE Logical channel list | | All | |
| - Semi-static Transport Format information | | | |
| - Transmission time interval | | Reference to clause 6.10 Parameter Set | |
| - Type of channel coding | | Reference to clause 6.10 Parameter Set | |
| - Coding Rate | | Reference to clause 6.10 Parameter Set | |
| - Rate matching attribute | | Reference to clause 6.10 Parameter Set | |
| - CRC size | | Reference to clause 6.10 Parameter Set | |
| Added or Reconfigured UL TrCH information | A11 | 1 DCH added for DTCH | |
| - Uplink transport channel type | | DCH | |
| - UL Transport channel identity | | 4 | |
| - TFS | | | |
| - CHOICE Transport channel type | | Dedicated transport channels | |
| - Dynamic Transport format information | | | |
| - RLC Size | | Reference to clause 6.10 Parameter Set | |
| - Number of TBs and TTI List | | (This IE is repeated for TFI number.) | |
| - Transmission Time Interval | | Not Present | |
| - Number of Transport blocks | | Reference to clause 6.10 Parameter Set | |
| - CHOICE Logical channel list | | All | |
| - Semi-static Transport Format information | | | |
| - Transmission time interval | | Reference to clause 6.10 Parameter Set | |
| - Type of channel coding | | Reference to clause 6.10 Parameter Set | |
| - Coding Rate | | Reference to clause 6.10 Parameter Set | |
| - Rate matching attribute | | Reference to clause 6.10 Parameter Set | |
| - CRC size | | Reference to clause 6.10 Parameter Set | |
| Added or Reconfigured UL TrCH information | A2, A8 | 4 TrCHs(DCH for DCCH and 3DCHs for DTCH) | |
| - Uplink transport channel type | | DCH | |
| - UL Transport channel identity | | 5 | |
| - TFS | | | |
| - CHOICE Transport channel type | | Dedicated transport channels | |
| - Dynamic Transport format information | | | |
| - RLC Size | | Reference to clause 6.10 Parameter Set | |
| - Number of TBs and TTI List | | (This IE is repeated for TFI number.) | |
| - Transmission Time Interval | | Not Present | |
| - Number of Transport blocks | | Reference to clause 6.10 Parameter Set | |
| - CHOICE Logical channel list | | All | |
| - Semi-static Transport Format information | | | |
| - Transmission time interval | | Reference to clause 6.10 Parameter Set | |
| - Type of channel coding | | Reference to clause 6.10 Parameter Set | |
| - Coding Rate | | Reference to clause 6.10 Parameter Set | |
| - Rate matching attribute | | Reference to clause 6.10 Parameter Set | |
| - CRC size | | Reference to clause 6.10 Parameter Set | |
| - Uplink transport channel type | | DCH | |
| - UL Transport channel identity | | 1 | |
| - TFS | | | |
| - CHOICE Transport channel type | | Dedicated transport channels | |
| - Dynamic Transport format information | | | |
| - RLC Size | | Reference to clause 6.10 Parameter Set | |
| - Number of TBs and TTI List | | (This IE is repeated for TFI number.) | |
| - Transmission Time Interval | | Not Present | |
| - Number of Transport blocks | | Reference to clause 6.10 Parameter Set | |

| Information Element | Condition | Value/remark | Version |
|---|---|---|---|
| - CHOICE Logical channel list | | All | |
| - Semi-static Transport Format information | | | |
| - Transmission time interval | | Reference to clause 6.10 Parameter Set | |
| - Type of channel coding | | Reference to clause 6.10 Parameter Set | |
| - Coding Rate | | Reference to clause 6.10 Parameter Set | |
| - Rate matching attribute | | Reference to clause 6.10 Parameter Set | |
| - CRC size | | Reference to clause 6.10 Parameter Set | |
| - Uplink transport channel type | | DCH | |
| - UL Transport channel identity | | 2 | |
| - TFS | | | |
| - CHOICE Transport channel type | | Dedicated transport channels | |
| - Dynamic Transport format information | | | |
| - RLC Size | | Reference to clause 6.10 Parameter Set | |
| - Number of TBs and TTI List | | (This IE is repeated for TFI number.) | |
| - Transmission Time Interval | | Not Present | |
| - Number of Transport blocks | | Reference to clause 6.10 Parameter Set | |
| - CHOICE Logical channel list | | All | |
| - Semi-static Transport Format information | | | |
| - Transmission time interval | | Reference to clause 6.10 Parameter Set | |
| - Type of channel coding | | Reference to clause 6.10 Parameter Set | |
| - Coding Rate | | Reference to clause 6.10 Parameter Set | |
| - Rate matching attribute | | Reference to clause 6.10 Parameter Set | |
| - CRC size | | Reference to clause 6.10 Parameter Set | |
| - Uplink transport channel type | | DCH | |
| - UL Transport channel identity | | 3 | |
| - TFS | | | |
| - CHOICE Transport channel type | | Dedicated transport channels | |
| - Dynamic Transport format information | | | |
| - RLC Size | | Reference to clause 6.10 Parameter Set | |
| - Number of TBs and TTI List | | (This IE is repeated for TFI number.) | |
| - Transmission Time Interval | | Not Present | |
| - Number of Transport blocks | | Reference to clause 6.10 Parameter Set | |
| - CHOICE Logical channel list | | All | |
| - Semi-static Transport Format information | | | |
| - Transmission time interval | | Reference to clause 6.10 Parameter Set | |
| - Type of channel coding | | Reference to clause 6.10 Parameter Set | |
| - Coding Rate | | Reference to clause 6.10 Parameter Set | |
| - Rate matching attribute | | Reference to clause 6.10 Parameter Set | |
| - CRC size | | Reference to clause 6.10 Parameter Set | |
| Added or Reconfigured UL TrCH information | A12 | 1 E-DCH added | Rel-6 |
| - Uplink transport channel type | | E-DCH | |
| - CHOICE UL parameters | | E-DCH | |
| - E-DCH Transmission Time Interval | | 10 ms | |
| - HARQ info for E-DCH | | | |
| - HARQ RV Configuration | | Not Present | |
| - Added or reconfigured E-DCH MAC-d flow | | | |
| - E-DCH MAC-d flow identity | | 2 | |
| - E-DCH MAC-d flow power offset | | 0 | |
| - E-DCH MAC-d flow maximum number of retransmissions | | 7 | |
| - E-DCH MAC-d flow multiplexing list | | Not Present | |
| - CHOICE transmission grant type | | Scheduled grant info | |
| Added or Reconfigured UL TrCH information | A13, A14 | 1 E-DCH added with one DCCH MAC-d flow and one DTCH MAC-d flow | Rel-6 |
| - Uplink transport channel type | | E-DCH | |

| Information Element | Condition | Value/remark | Version |
|---|---|---|---|
| - CHOICE UL parameters | | E-DCH | |
| - E-DCH Transmission Time Interval | | 10 ms | |
| - HARQ info for E-DCH | | | |
| - HARQ RV Configuration | | Not Present | |
| - Added or reconfigured E-DCH MAC-d flow | | (for DCCH) | |
| - E-DCH MAC-d flow identity | | 1 | |
| - E-DCH MAC-d flow power offset | | 0 | |
| - E-DCH MAC-d flow maximum number of retransmissions | | 7 | |
| - E-DCH MAC-d flow multiplexing list | | Not Present | |
| - CHOICE transmission grant type | | Non-scheduled grant info | |
| - Max MAC-e PDU contents size | | 162 bits | |
| - 2 ms non-scheduled transmission grant HARQ process allocation | | Not Present | |
| - Uplink transport channel type | | E-DCH | |
| - CHOICE UL parameters | | E-DCH | |
| - E-DCH Transmission Time Interval | | 10 ms | |
| - HARQ info for E-DCH | | | |
| - HARQ RV Configuration | | Not Present | |
| - Added or reconfigured E-DCH MAC-d flow | | (for DTCH) | |
| - E-DCH MAC-d flow identity | | 2 | |
| - E-DCH MAC-d flow power offset | | 0 | |
| - E-DCH MAC-d flow maximum number of retransmissions | | 7 | |
| - E-DCH MAC-d flow multiplexing list | | Not Present | |
| - CHOICE transmission grant type | | Scheduled grant info | |
| Added or Reconfigured UL TrCH information | A15 | 1 E-DCH added with one DCCH MAC-d flow and two DTCH MAC-d flows | Rel-6 |
| - Uplink transport channel type | | E-DCH | |
| - CHOICE UL parameters | | E-DCH | |
| - E-DCH Transmission Time Interval | | 10 ms | |
| - HARQ info for E-DCH | | | |
| - HARQ RV Configuration | | Not Present | |
| - Added or reconfigured E-DCH MAC-d flow | | (for DCCH) | |
| - E-DCH MAC-d flow identity | | 1 | |
| - E-DCH MAC-d flow power offset | | 0 | |
| - E-DCH MAC-d flow maximum number of retransmissions | | 7 | |
| - E-DCH MAC-d flow multiplexing list | | Not Present | |
| - CHOICE transmission grant type | | Non-scheduled grant info | |
| - Max MAC-e PDU contents size | | 162 bits | |
| - 2 ms non-scheduled transmission grant HARQ process allocation | | Not Present | |
| - Uplink transport channel type | | E-DCH | |
| - CHOICE UL parameters | | E-DCH | |

| Information Element | Condition | Value/remark | Version |
|---|---|---|---|
| - E-DCH Transmission Time Interval | | 10 ms | |
| - HARQ info for E-DCH | | | |
| - HARQ RV Configuration | | Not Present | |
| - Added or reconfigured E-DCH MAC-d flow | | (for first DTCH) | |
| - E-DCH MAC-d flow identity | | 2 | |
| - E-DCH MAC-d flow power offset | | 0 | |
| - E-DCH MAC-d flow maximum number of retransmissions | | 7 | |
| - E-DCH MAC-d flow multiplexing list | | Not Present | |
| - CHOICE transmission grant type | | Scheduled grant info | |
| - Uplink transport channel type | | E-DCH | |
| - CHOICE UL parameters | | E-DCH | |
| - E-DCH Transmission Time Interval | | 10 ms | |
| - HARQ info for E-DCH | | | |
| - HARQ RV Configuration | | Not Present | |
| - Added or reconfigured E-DCH MAC-d flow | | (for second DTCH) | |
| - E-DCH MAC-d flow identity | | 3 | |
| - E-DCH MAC-d flow power offset | | 0 | |
| - E-DCH MAC-d flow maximum number of retransmissions | | 7 | |
| - E-DCH MAC-d flow multiplexing list | | Not Present | |
| - CHOICE transmission grant type | | Scheduled grant info | |
| CHOICE *mode* | A1, A2, A3, A4, A5, A6, A7, A8, A11<br>, A9, A10<br>, A12, A13 | Not Present | <br><br>Rel-5<br>Rel-6 |
| DL Transport channel information common for all transport channel | A1, A2, A7, A8 | | |
| - SCCPCH TFCS | | Not Present | |
| - CHOICE mode | | FDD | |
| - CHOICE DL parameters | | SameasUL | |
| DL Transport channel information common for all transport channel | A3, A4, A5, A6, A11<br>A10<br>A12, A13, A15 | | <br>Rel-5<br>Rel-6 |
| - SCCPCH TFCS | | Not Present | |
| - CHOICE mode | | FDD | |
| - CHOICE DL parameters | | Explicit | |
| - DL DCH TFCS | | | |
| - CHOICE TFCI Signalling | | Normal | |
| - TFCI Field 1 Information | | | |
| - CHOICE TFCS representation | | Complete reconfiguration | |
| - TFCS complete reconfigure | | | |
| - CHOICE CTFC Size | | Number of bits used must be enough to cover all combinations of CTFC from clause 6.10.2.4 Parameter Set. | |
| - CTFC information | | This IE is repeated for TFC numbers and reference to clause 6.10.2.4 | |
| - CTFC | | Reference to clause 6.10.2.4 Parameter Set | |
| - Power offset information | | Not Present | |
| DL Transport channel information common for all transport channel | A9 | | Rel-5 |
| - SCCPCH TFCS | | Not Present | |
| - CHOICE mode | | FDD | |

| Information Element | Condition | Value/remark | Version |
|---|---|---|---|
| - CHOICE DL parameters | | Explicit | |
| - DL DCH TFCS | | | |
| - CHOICE TFCI Signalling | | Normal | |
| - TFCI Field 1 Information | | | |
| - CHOICE TFCS representation | | Complete reconfiguration | |
| - TFCS complete reconfigure | | | |
| - CHOICE CTFC Size | | ctfc2bit | |
| - CTFC information | | | |
| - CTFC | | 0 | |
| | | ((DL DCH RAB, DCCH)=(TF0, TF0)) | |
| - Power offset information | | Not Present | |
| - CTFC | | 1 | |
| | | ((DL DCH RAB, DCCH)=(TF0, TF1)) | |
| - Power offset information | | Not Present | |
| DL Transport channel information common for all transport channel | A14 | Not Present | Rel-6 |
| Deleted DL TrCH information | A1, A2, A3, A4, A5, A6, A7, A8 | Not Present | |
| | , A9, A10 | | Rel-5 |
| | , A12, A13 | | Rel-6 |
| Deleted DL TrCH information | A14 | | Rel-6 |
| - Downlink transport channel type | | DCH | |
| - DL Transport channel identity | | 10 | |
| Added or Reconfigured DL TrCH information | A1 | 1 DCH added, 1 DCH reconfigured | |
| - Downlink transport channel type | | DCH | |
| - DL Transport channel identity | | 6 | |
| - CHOICE DL parameters | | Same as UL | |
| - Uplink transport channel type | | DCH | |
| - UL TrCH identity | | 1 | |
| - DCH quality target | | | |
| - BLER Quality value | | -2.0 | |
| - Downlink transport channel type | | DCH | |
| - DL Transport channel identity | | 10 | |
| - CHOICE DL parameters | | Same as UL | |
| - Uplink transport channel type | | DCH | |
| - UL TrCH identity | | 5 | |
| - DCH quality target | | | |
| - BLER Quality value | | -2.0 | |
| Added or Reconfigured DL TrCH information | A3, A4, A5, A6, A7 | 2 TrCHs(DCH for DCCH and DCH for DTCH) | |
| - Downlink transport channel type | | DCH | |
| - DL Transport channel identity | | 10 | |
| - CHOICE DL parameters | | Same as UL | |
| - Uplink transport channel type | | DCH | |
| - UL TrCH identity | | 5 | |
| - DCH quality target | | | |
| - BLER Quality value | | -2.0 | |
| - Downlink transport channel type | | DCH | |
| - DL Transport channel identity | | 6 | |
| - CHOICE DL parameters | | Explicit<br>Except for RAB with the symmetric DL and UL rate: Same as UL | |
| - TFS | | | |
| - CHOICE Transport channel type | | Dedicated transport channel | |
| - Dynamic transport format information | | | |
| - RLC Size | | Reference to clause 6.10 Parameter Set | |
| - Number of TBs and TTI List | | (This IE is repeated for TFI number.) | |
| - Transmission Time Interval | | Not Present | |
| - Number of Transport blocks | | Reference to clause 6.10 Parameter Set only including TF0 | |
| - CHOICE Logical channel list | | All | |
| - Semi-static Transport Format information | | | |
| - Transmission time interval | | Reference to clause 6.10 Parameter Set | |
| - Type of channel coding | | Reference to clause 6.10 Parameter Set | |

| Information Element | Condition | Value/remark | Version |
|---|---|---|---|
| - Coding Rate | | Reference to clause 6.10 Parameter Set | |
| - Rate matching attribute | | Reference to clause 6.10 Parameter Set | |
| - CRC size | | Reference to clause 6.10 Parameter Set | |
| - DCH quality target | | | |
| - BLER Quality value | | -2.0 | |
| Added or Reconfigured DL TrCH information | A2, A8 | 4 TrCHs(DCH for DCCH and 3DCHs for DTCH) | |
| - Downlink transport channel type | | DCH | |
| - DL Transport channel identity | | 10 | |
| - CHOICE DL parameters | | Same as UL | |
| - Uplink transport channel type | | DCH | |
| - UL TrCH identity | | 5 | |
| - DCH quality target | | | |
| - BLER Quality value | | 2.0 | |
| - Downlink transport channel type | | DCH | |
| - DL Transport channel identity | | 6 | |
| - CHOICE DL parameters | | Explicit | |
| - TFS | | | |
| - CHOICE Transport channel type | | Dedicated transport channel | |
| - Dynamic transport format information | | | |
| - RLC Size | | Reference to clause 6.10 Parameter Set | |
| - Number of TBs and TTI List | | (This IE is repeated for TFI number.) | |
| - Dynamic transport format information | | | |
| - Transmission Time Interval | | Not Present | |
| - Number of Transport blocks | | Reference to clause 6.10 Parameter Set | |
| - CHOICE Logical channel list | | All | |
| - Semi-static Transport Format information | | | |
| - Transmission time interval | | Reference to clause 6.10 Parameter Set | |
| - Type of channel coding | | Reference to clause 6.10 Parameter Set | |
| - Coding Rate | | Reference to clause 6.10 Parameter Set | |
| - Rate matching attribute | | Reference to clause 6.10 Parameter Set | |
| - CRC size | | Reference to clause 6.10 Parameter Set | |
| - DCH quality target | | | |
| - BLER Quality value | | Not Present | |
| - Downlink transport channel type | | DCH | |
| - DL Transport channel identity | | 7 | |
| - CHOICE DL parameters | | Explicit | |
| - TFS | | | |
| - CHOICE Transport channel type | | Dedicated transport channel | |
| - Dynamic transport format information | | | |
| - RLC Size | | Reference to clause 6.10 Parameter Set | |
| - Number of TBs and TTI List | | (This IE is repeated for TFI number.) | |
| - Dynamic transport format information | | | |
| - Transmission Time Interval | | Not Present | |
| - Number of Transport blocks | | Reference to clause 6.10 Parameter Set | |
| - CHOICE Logical channel list | | All | |
| - Semi-static Transport Format information | | | |
| - Transmission time interval | | Reference to clause 6.10 Parameter Set | |
| - Type of channel coding | | Reference to clause 6.10 Parameter Set | |
| - Coding Rate | | Reference to clause 6.10 Parameter Set | |
| - Rate matching attribute | | Reference to clause 6.10 Parameter Set | |
| - CRC size | | Reference to clause 6.10 Parameter Set | |
| - DCH quality target | | | |
| - BLER Quality value | | Not Present | |
| - Downlink transport channel type | | DCH | |
| - DL Transport channel identity | | 8 | |
| - CHOICE DL parameters | | Explicit | |
| - TFS | | | |
| - CHOICE Transport channel type | | Dedicated transport channel | |
| - Dynamic transport format information | | | |

| Information Element | Condition | Value/remark | Version |
|---|---|---|---|
| - RLC Size | | Reference to clause 6.10 Parameter Set | |
| - Number of TBs and TTI List | | (This IE is repeated for TFI number.) | |
| - Dynamic transport format information | | | |
| - Transmission Time Interval | | Not Present | |
| - Number of Transport blocks | | Reference to clause 6.10 Parameter Set | |
| - CHOICE Logical channel list | | All | |
| - Semi-static Transport Format information | | | |
| - Transmission time interval | | Reference to clause 6.10 Parameter Set | |
| - Type of channel coding | | Reference to clause 6.10 Parameter Set | |
| - Coding Rate | | Reference to clause 6.10 Parameter Set | |
| - Rate matching attribute | | Reference to clause 6.10 Parameter Set | |
| - CRC size | | Reference to clause 6.10 Parameter Set | |
| - DCH quality target | | | |
| - BLER Quality value | | Not Present | |
| Added or Reconfigured DL TrCH information | A9<br>A12 | 3 TrCHs (DCH for DCCH and DCH plus HS-DSCH for DTCH) | Rel-5<br>Rel-6 |
| - Downlink transport channel type | | DCH | |
| - DL Transport channel identity | | 10 | |
| - CHOICE DL parameters | | Same as UL | |
| - Uplink transport channel type | | DCH | |
| - UL TrCH identity | | 5 | |
| - DCH quality target | | | |
| - BLER Quality value | | -2.0 | |
| - Downlink transport channel type | | DCH | |
| - DL Transport channel identity | | 6 | |
| - CHOICE DL parameters | | Explicit | |
| - TFS | | | |
| - CHOICE Transport channel type | | Dedicated transport channel | |
| - Dynamic transport format information | | | |
| - RLC Size | | Reference to clause 6.10 Parameter Set | |
| - Number of TBs and TTI List | | (This IE is repeated for TFI number.) | |
| - Dynamic transport format information | | | |
| - Transmission Time Interval | | Not Present | |
| - Number of Transport blocks | | Reference to clause 6.10 Parameter Set | |
| - CHOICE Logical channel list | | All | |
| - Semi-static Transport Format information | | | |
| - Transmission time interval | | Reference to clause 6.10 Parameter Set | |
| - Type of channel coding | | Reference to clause 6.10 Parameter Set | |
| - Coding Rate | | Reference to clause 6.10 Parameter Set | |
| - Rate matching attribute | | Reference to clause 6.10 Parameter Set | |
| - CRC size | | Reference to clause 6.10 Parameter Set | |
| - DCH quality target | | | |
| - BLER Quality value | | -2.0 | |
| - Downlink transport channel type | | HS-DSCH | |
| - DL Transport channel identity | | Not Present | |
| - CHOICE DL parameters | | HS-DSCH | |
| - HARQ Info | | | |
| - Number of Processes | | Reference to clause 6.10.2.4.5 Parameter Set | |
| - CHOICE *Memory Partitioning* | | Implicit | |
| - Added or reconfigured MAC-d flow | | | |
| - MAC-hs queue to add or reconfigure list | | (one queue) | |
| - MAC-hs queue Id | | 0 | |
| - MAC-d Flow Identity | | 0 | |
| - T1 | | 50 | |
| - MAC-hs window size | | 16 | |
| - MAC-d PDU size Info | | | |
| - MAC-d PDU size | | 336 | |
| - MAC-d PDU size index | | 0 | |

| Information Element | Condition | Value/remark | Version |
|---|---|---|---|
| - MAC-hs queue to delete list | | Not present | |
| - DCH quality target | | Not present | |
| Added or Reconfigured DL TrCH information | A10 | 2 TrCHs (DCH for DCCH and HS-DSCH for DTCH) | Rel-5 |
| - Downlink transport channel type | | DCH | |
| - DL Transport channel identity | | 10 | |
| - CHOICE DL parameters | | Same as UL | |
| - Uplink transport channel type | | DCH | |
| - UL TrCH identity | | 5 | |
| - DCH quality target | | | |
| - BLER Quality value | | -2.0 | |
| - Downlink transport channel type | | HS-DSCH | |
| - DL Transport channel identity | | Not Present | |
| - CHOICE DL parameters | | HS-DSCH | |
| - HARQ Info | | | |
| - Number of Processes | | Reference to clause 6.10.2.4.5 Parameter Set | |
| - CHOICE *Memory Partitioning* | | Implicit | |
| - Added or reconfigured MAC-d flow | | | |
| - MAC-hs queue to add or reconfigure list | | (one queue) | |
| - MAC-hs queue Id | | 0 | |
| - MAC-d Flow Identity | | 0 | |
| - T1 | | 50 | |
| - MAC-hs window size | | 16 | |
| - MAC-d PDU size Info | | | |
| - MAC-d PDU size | | 336 | |
| - MAC-d PDU size index | | 0 | |
| - MAC-hs queue to delete list | | Not present | |
| - DCH quality target | | Not present | |
| Added or Reconfigured DL TrCH information | A11 | 1 DCH for DTCH | |
| - Downlink transport channel type | | DCH | |
| - DL Transport channel identity | | 9 | |
| - CHOICE DL parameters | | Explicit | |
| - TFS | | | |
| - CHOICE Transport channel type | | Dedicated transport channel | |
| - Dynamic transport format information | | | |
| - RLC Size | | Reference to clause 6.10 Parameter Set | |
| - Number of TBs and TTI List | | (This IE is repeated for TFI number.) | |
| - Dynamic transport format information | | | |
| - Transmission Time Interval | | Not Present | |
| - Number of Transport blocks | | Reference to clause 6.10 Parameter Set | |
| - CHOICE Logical channel list | | All | |
| - Semi-static Transport Format information | | | |
| - Transmission time interval | | Reference to clause 6.10 Parameter Set | |
| - Type of channel coding | | Reference to clause 6.10 Parameter Set | |
| - Coding Rate | | Reference to clause 6.10 Parameter Set | |
| - Rate matching attribute | | Reference to clause 6.10 Parameter Set | |
| - CRC size | | Reference to clause 6.10 Parameter Set | |
| - DCH quality target | | | |
| - BLER Quality value | | -2.0 | |
| Added or Reconfigured DL TrCH information | A13 | 2 TrCHs (DCH for DCCH and HS-DSCH for DTCH) | Rel-6 |
| - Downlink transport channel type | | DCH | |
| - DL Transport channel identity | | 10 | |
| - CHOICE DL parameters | | Explicit | |
| - TFS | | | |
| - CHOICE Transport channel type | | Dedicated transport channels | |
| - Dynamic Transport format information | | | |
| - RLC Size | | Reference to clause 6.10 Parameter Set | |

| Information Element | Condition | Value/remark | Version |
|---|---|---|---|
| - Number of TBs and TTI List | | (This IE is repeated for TFI number.) | |
| - Transmission Time Interval | | Not Present | |
| - Number of Transport blocks | | Reference to clause 6.10 Parameter Set | |
| - CHOICE Logical channel list | | All | |
| - Semi-static Transport Format information | | | |
| - Transmission time interval | | Reference to clause 6.10 Parameter Set | |
| - Type of channel coding | | Reference to clause 6.10 Parameter Set | |
| - Coding Rate | | Reference to clause 6.10 Parameter Set | |
| - Rate matching attribute | | Reference to clause 6.10 Parameter Set | |
| - CRC size | | Reference to clause 6.10 Parameter Set | |
| - DCH quality target | | | |
| - BLER Quality value | | -2.0 | |
| - Downlink transport channel type | | HS-DSCH | |
| - DL Transport channel identity | | Not Present | |
| - CHOICE DL parameters | | HS-DSCH | |
| - HARQ Info | | | |
| - Number of Processes | | Reference to clause 6.10.2.4.5 Parameter Set | |
| - CHOICE *Memory Partitioning* | | Implicit | |
| - Added or reconfigured MAC-d flow | | | |
| - MAC-hs queue to add or reconfigure list | | (one queue) | |
| - MAC-hs queue Id | | 0 | |
| - MAC-d Flow Identity | | 0 | |
| - T1 | | 50 | |
| - MAC-hs window size | | 16 | |
| - MAC-d PDU size Info | | | |
| - MAC-d PDU size | | 336 | |
| - MAC-d PDU size index | | 0 | |
| - MAC-hs queue to delete list | | Not present | |
| - DCH quality target | | Not present | |
| Added or Reconfigured DL TrCH information | A14 | 1 TrCH (HS-DSCH for DTCH and DCCH) | Rel-6 |
| - Downlink transport channel type | | HS-DSCH | |
| - DL Transport channel identity | | Not Present | |
| - CHOICE DL parameters | | HS-DSCH | |
| - HARQ Info | | | |
| - Number of Processes | | Reference to clause 6.10.2.4.5 Parameter Set | |
| - CHOICE *Memory Partitioning* | | Implicit | |
| - Added or reconfigured MAC-d flow | | | |
| - MAC-hs queue to add or reconfigure list | | (two queues) | |
| - MAC-hs queue Id | | 0 (for DTCH) | |
| - MAC-d Flow Identity | | 0 | |
| - T1 | | 50 | |
| - MAC-hs window size | | 16 | |
| - MAC-d PDU size Info | | | |
| - MAC-d PDU size | | 336 | |
| - MAC-d PDU size index | | 0 | |
| - MAC-hs queue Id | | 1 (for DCCH) | |
| - MAC-d Flow Identity | | 1 | |
| - T1 | | 50 | |
| - MAC-hs window size | | 16 | |
| - MAC-d PDU size Info | | | |
| - MAC-d PDU size | | 148 | |
| - MAC-d PDU size index | | 0 | |
| - MAC-hs queue to delete list | | Not present | |
| - DCH quality target | | Not present | |
| Added or Reconfigured DL TrCH information | A15 | 2 TrCHs (DCH for DCCH and HS-DSCH for DTCH) | Rel-6 |
| - Downlink transport channel type | | DCH | |

| Information Element | Condition | Value/remark | Version |
|---|---|---|---|
| - DL Transport channel identity | | 10 | |
| - CHOICE DL parameters | | Explicit | |
| - TFS | | | |
| - CHOICE Transport channel type | | Dedicated transport channels | |
| - Dynamic Transport format information | | | |
| - RLC Size | | Reference to clause 6.10 Parameter Set | |
| - Number of TBs and TTI List | | (This IE is repeated for TFI number.) | |
| - Transmission Time Interval | | Not Present | |
| - Number of Transport blocks | | Reference to clause 6.10 Parameter Set | |
| - CHOICE Logical channel list | | All | |
| - Semi-static Transport Format information | | | |
| - Transmission time interval | | Reference to clause 6.10 Parameter Set | |
| - Type of channel coding | | Reference to clause 6.10 Parameter Set | |
| - Coding Rate | | Reference to clause 6.10 Parameter Set | |
| - Rate matching attribute | | Reference to clause 6.10 Parameter Set | |
| - CRC size | | Reference to clause 6.10 Parameter Set | |
| - DCH quality target | | | |
| - BLER Quality value | | -2.0 | |
| - Downlink transport channel type | | HS-DSCH | |
| - DL Transport channel identity | | Not Present | |
| - CHOICE DL parameters | | HS-DSCH | |
| - HARQ Info | | | |
| - Number of Processes | | Reference to clause 6.10.2.4.5 Parameter Set | |
| - CHOICE *Memory Partitioning* | | Implicit | |
| - Added or reconfigured MAC-d flow | | | |
| - MAC-hs queue to add or reconfigure list | | (two queues) | |
| - MAC-hs queue Id | | 0 (for first DTCH) | |
| - MAC-d Flow Identity | | 0 | |
| - T1 | | 50 | |
| - MAC-hs window size | | 16 | |
| - MAC-d PDU size Info | | | |
| - MAC-d PDU size | | 336 | |
| - MAC-d PDU size index | | 0 | |
| - MAC-hs queue Id | | 2 (for second DTCH) | |
| - MAC-d Flow Identity | | 0 | |
| - T1 | | 50 | |
| - MAC-hs window size | | 16 | |
| - MAC-d PDU size Info | | | |
| - MAC-d PDU size | | 336 | |
| - MAC-d PDU size index | | 0 | |
| - MAC-hs queue to delete list | | Not present | |
| - DCH quality target | | Not present | |
| Frequency info | A1, A2, A3, A4, A5, A7, A8, 11<br>, A9, A10<br>, A12, A13, A14 | | <br>Rel-5<br>Rel-6 |
| - UARFCN uplink (Nu) | | Reference to clause 5.1 Test frequencies. This IE should be present, if the default duplex distance defined for the operating frequency band is not used and frequency is different from the current frequency, otherwise set to Not Present. | |
| - UARFCN downlink (Nd) | | Reference to clause 5.1 Test frequencies if frequency is different from the current frequency otherwise set to Not Present. | |
| Frequency info | A6 | Not Present | |
| Maximum allowed UL TX power | A1, A2, A3, A4, A7, A8, A11<br>, A9, A10 | 33dBm | <br>Rel-5 |

| Information Element | Condition | Value/remark | Version |
|---|---|---|---|
| | , A12, A13, A14 | | Rel-6 |
| Maximum allowed UL TX power | A5, A6 | Not Present | |
| CHOICE channel requirement | A1, A2, A3, A4, A7, A8, A11 | Uplink DPCH info | Rel-5 and earlier |
| - Uplink DPCH power control info | | | |
| - DPCCH power offset | | -80dB | |
| - PC Preamble | | 1 frame | |
| - SRB delay | | 7 frames | |
| - Power Control Algorithm | | Algorithm1 | |
| - TPC step size | | 1dB | |
| - $\Delta_{NACK}$ | | Not Present | Rel-5 |
| - $\Delta_{NACK}$ | | Not Present | Rel-5 |
| - Ack-Nack repetition factor | | Not Present | Rel-5 |
| - Scrambling code type | | Long | |
| - Scrambling code number | | 0 (0 to 16777215) | |
| - Number of DPDCH | | Not Present(1) | |
| - spreading factor | | Reference to clause 6.10 Parameter Set | |
| - TFCI existence | | Reference to clause 6.10 Parameter Set | |
| - Number of FBI bit | | Reference to clause 6.10 Parameter Set | |
| - Puncturing Limit | | Reference to clause 6.10 Parameter Set | |
| CHOICE channel requirement | A9, A10 | Uplink DPCH info | Rel-5 |
| - Uplink DPCH power control info | | | |
| - DPCCH power offset | | -80dB | |
| - PC Preamble | | 1 frame | |
| - SRB delay | | 7 frames | |
| - Power Control Algorithm | | Algorithm1 | |
| - TPC step size | | 1dB | |
| - $\Delta_{ACK}$ | | 3 | |
| - $\Delta_{NACK}$ | | 3 | |
| - Ack-Nack repetition factor | | 1 | |
| - Scrambling code type | | Long | |
| - Scrambling code number | | 0 (0 to 16777215) | |
| - Number of DPDCH | | Not Present(1) | |
| - spreading factor | | Reference to clause 6.10 Parameter Set | |
| - TFCI existence | | Reference to clause 6.10 Parameter Set | |
| - Number of FBI bit | | Reference to clause 6.10 Parameter Set | |
| - Puncturing Limit | | Reference to clause 6.10 Parameter Set | |
| CHOICE channel requirement | A5,A6 | Not Present | Rel-5 and earlier |
| Uplink DPCH info | A12 | | Rel-6 |
| - Uplink DPCH power control info | | | |
| - DPCCH power offset | | -80dB | |
| - PC Preamble | | 1 frame | |
| - SRB delay | | 7 frames | |
| - Power Control Algorithm | | Algorithm1 | |
| - TPC step size | | 1dB | |
| - $\Delta_{ACK}$ | | 3 | |
| - $\Delta_{NACK}$ | | 3 | |
| - Ack-Nack repetition factor | | 1 | |
| - Scrambling code type | | Long | |
| - Scrambling code number | | 0 (0 to 16777215) | |
| - Number of DPDCH | | Not Present(1) | |
| - spreading factor | | Reference to clause 6.10 Parameter Set | |
| - TFCI existence | | Reference to clause 6.10 Parameter Set | |
| - Number of FBI bit | | Reference to clause 6.10 Parameter Set | |
| - Puncturing Limit | | Reference to clause 6.10 Parameter Set | |
| Uplink DPCH info | A13, A14, A15 | | Rel-6 |
| - Uplink DPCH power control info | | | |
| - DPCCH power offset | | -80dB | |
| - PC Preamble | | 1 frame | |
| - SRB delay | | 7 frames | |
| - Power Control Algorithm | | Algorithm1 | |
| - TPC step size | | 1dB | |
| - $\Delta_{ACK}$ | | 3 | |
| - $\Delta_{NACK}$ | | 3 | |

| Information Element | Condition | Value/remark | Version |
|---|---|---|---|
| - Ack-Nack repetition factor | | 1 | |
| - Scrambling code type | | Long | |
| - Scrambling code number | | 0 (0 to 16777215) | |
| - Number of DPDCH | | 0 | |
| - spreading factor | | Not Present | |
| - TFCI existence | | FALSE | |
| - Number of FBI bit | | Not Present | |
| - Puncturing Limit | | Not Present | |
| E-DCH info | A12, A13, A14, A15 | | Rel-6 |
| - MAC-es/e reset indicator | | TRUE | |
| - E-DPCCH info | | | |
| - E-DPCCH/DPCCH power offset | | 0 | |
| - Happy bit delay condition | | 100 ms | |
| - E-DPDCH info | | | |
| - E-TFCI table index | | 0 | |
| - E-DCH minimum set E-TFCI | | 9 | |
| - Reference E-TFCIs | | 4 E-TFCIs | |
| - Reference E-TFCI | | 11 | |
| - Reference E-TFCI PO | | 4 | |
| - Reference E-TFCI | | 83 | |
| - Reference E-TFCI PO | | 16 | |
| - Reference E-TFCI | | 110 | |
| - Reference E-TFCI PO | | 19 | |
| - Reference E-TFCI | | 115 | |
| - Reference E-TFCI PO | | 21 | |
| - Maximum channelisation codes | | 2sf4 | |
| - PLnon-max | | 0.84 | |
| - Scheduling Information Configuration | | | |
| - Periodicity for Scheduling Info – no grant | | Not present | |
| - Periodicity for Scheduling Info – grant | | Not present | |
| - Power Offset for Scheduling Info | | 0 | |
| - Scheduled Transmission configuration | | | |
| - 2ms scheduled transmission grant HARQ process allocation | | Not present | |
| - Serving Grant | | Not present | |
| - Primary/Secondary Grant Selector | | Primary | |
| - 3-Index-Step Threshold | | Not present | |
| - 2-Index-Step Threshold | | Not present | |
| CHOICE Mode | A1, A2, A3, A4, A5, A6, A7, A8, A11 | FDD | R99 and Rel-4 only |
| - Downlink PDSCH information | | Not Present | |
| Downlink HS-PDSCH Information | A1, A2, A3, A4, A5, A6, A7, A8, A11 | Not Present | Rel-5 |
| Downlink HS-PDSCH Information | A9, A10<br>A12, A13, A14, A15 | | Rel-5<br>Rel-6 |
| - HS-SCCH Info | | | |
| - CHOICE mode | | FDD | |
| - DL Scrambling Code | | Not present | |
| - HS-SCCH Channelisation Code Information | | | |
| - HS-SCCH Channelisation Code | | 7 | |
| - Measurement Feedback Info | | | |
| - CHOICE mode | | FDD | |
| - POhsdsch | | 6 dB | |
| - CQI Feedback cycle, k | | 4 ms | |
| - CQI repetition factor | | 1 | |
| - $\Delta_{CQI}$ | | 5 (corresponds to 0dB in relative power offset) | |
| - CHOICE mode | | FDD (no data) | |
| Downlink information common for all radio links | A1, A2, A3, A11 | | |

| Information Element | Condition | Value/remark | Version |
|---|---|---|---|
| - Downlink DPCH info common for all RL | | | |
| - Timing indicator | | Maintain | |
| - CFN-targetSFN frame offset | | Not Present | |
| - Downlink DPCH power control information | | | |
| - DPC mode | | 0 (single) | |
| - CHOICE mode | | FDD | |
| - Power offset $P_{Pilot-DPDCH}$ | | 0 | |
| - DL rate matching restriction information | | Not Present | |
| - Spreading factor | | Reference to clause 6.10 Parameter Set | |
| - Fixed or Flexible Position | | Reference to clause 6.10 Parameter Set | |
| - TFCI existence | | Reference to clause 6.10 Parameter Set | |
| - CHOICE SF | | Reference to clause 6.10 Parameter Set | |
| - CHOICE mode | | FDD | |
| - DPCH compressed mode info | | Not Present | |
| - TX Diversity mode | | None | |
| - SSDT information | | Not Present | R99 and Rel-4 only |
| - Default DPCH Offset Value | | Not Present | |
| Downlink information common for all radio links | A9<br>A12, A13, A15 | | Rel-5<br>Rel-6 |
| - Downlink DPCH info common for all RL | | | |
| - Timing indicator | | Maintain | |
| - CFN-targetSFN frame offset | | Not Present | |
| - Downlink DPCH power control information | | | |
| - DPC mode | | 0 (single) | |
| - CHOICE mode | | FDD | |
| - Power offset PPilot-DPDCH | | 0 | |
| - DL rate matching restriction information | | Not Present | |
| - Spreading factor | | Reference to clause 6.10 Parameter Set | |
| - Fixed or Flexible Position | | Reference to clause 6.10 Parameter Set | |
| - TFCI existence | | Reference to clause 6.10 Parameter Set | |
| - CHOICE SF | | Reference to clause 6.10 Parameter Set | |
| - CHOICE mode | | FDD | |
| - DPCH compressed mode info | | Not Present | |
| - TX Diversity mode | | None | |
| - Default DPCH Offset Value | | Not Present | |
| - MAC-hs reset indicator | | TRUE | |
| Downlink information common for all radio links | A4,A7,A8 | | |
| - Downlink DPCH info common for all RL | | | |
| - Timing indicator | | Initialize | |
| - CFN-targetSFN frame offset | | Not Present | |
| - Downlink DPCH power control information | | | |
| - DPC mode | | 0 (single) | |
| - CHOICE mode | | FDD | |
| - Power offset $P_{Pilot-DPDCH}$ | | 0 | |
| - DL rate matching restriction information | | Not Present | |
| - Spreading factor | | Reference to clause 6.10 Parameter Set | |
| - Fixed or Flexible Position | | Reference to clause 6.10 Parameter Set | |
| - TFCI existence | | Reference to clause 6.10 Parameter Set | |
| - CHOICE SF | | Reference to clause 6.10 Parameter Set | |
| - CHOICE mode | | FDD | |
| - DPCH compressed mode info | | Not Present | |
| - TX Diversity mode | | None | |
| - SSDT information | | Not Present | R99 and Rel-4 only |
| - Default DPCH Offset Value | | Arbitrary set to value 0..306688 by step of 512 | |

| Information Element | Condition | Value/remark | Version |
|---|---|---|---|
| Downlink information common for all radio links | A10 | | Rel-5 |
| - Downlink DPCH info common for all RL | | | |
| - Timing indicator | | Initialize | |
| - CFN-targetSFN frame offset | | Not Present | |
| - Downlink DPCH power control information | | | |
| - DPC mode | | 0 (single) | |
| - CHOICE mode | | FDD | |
| - Power offset PPilot-DPDCH | | 0 | |
| - DL rate matching restriction information | | Not Present | |
| - Spreading factor | | Reference to clause 6.10 Parameter Set | |
| - Fixed or Flexible Position | | Reference to clause 6.10 Parameter Set | |
| - TFCI existence | | Reference to clause 6.10 Parameter Set | |
| - CHOICE SF | | Reference to clause 6.10 Parameter Set | |
| - CHOICE mode | | FDD | |
| - DPCH compressed mode info | | Not Present | |
| - TX Diversity mode | | None | |
| - Default DPCH Offset Value | | Arbitrary set to value 0..306688 by step of 512 | |
| - MAC-hs reset indicator | | TRUE | |
| Downlink information common for all radio links | A14 | | Rel-6 |
| - Downlink F-DPCH info common for all RL | | | |
| - Timing Indication | | Maintain | |
| - Timing maintained Synchronization indicator | | FALSE | |
| - Downlink F-DPCH power control information | | | |
| - DPC mode | | 0 (single) | |
| - TPC command error rate target | | 0.04 | |
| - CHOICE mode | | FDD | |
| - DPCH compressed mode info | | Not Present | |
| - TX Diversity mode | | None | |
| - Default DPCH Offset Value | | Arbitrary set to value 0..306688 by step of 512 | |
| - MAC-hs reset indicator | | TRUE | |
| Downlink information common for all radio links | A5,A6 | Not Present | |
| Downlink information for each radio link list | A1, A2, A3, A4, A7, A8, A11 | | |
| - Downlink information for each radio link | | | |
| - Choice mode | | FDD | |
| - Primary CPICH info | | | |
| - Primary scrambling code | | Ref. to the Default setting in clause 6.1 (FDD) | |
| - PDSCH with SHO DCH info | | Not Present | R99 and Rel-4 only |
| - PDSCH code mapping | | Not Present | R99 and Rel-4 only |
| - Serving HS-DSCH radio link indicator | | FALSE | Rel-5 |
| - Downlink DPCH info for each RL | | | |
| - Primary CPICH usage for channel estimation | | Primary CPICH may be used | |
| - DPCH frame offset | | Set to value Default DPCH Offset Value (as currently stored in SS) mod 38 400 | |
| - Secondary CPICH info | | Not Present | |
| - DL channelisation code | | | |
| - Secondary scrambling code | | 1 | |
| - Spreading factor | | Reference to clause 6.10 Parameter Set | |
| - Code number | | 0 | |
| - Scrambling code change | | No code change | |

| Information Element | Condition | Value/remark | Version |
|---|---|---|---|
| - TPC combination index | | 0 | |
| - SSDT Cell Identity | | Not Present | R99 and Rel-4 only |
| - Closed loop timing adjustment mode | | Not Present | |
| - SCCPCH information for FACH | | Not Present | R99 and Rel-4 only |
| Downlink information for each radio link list | A5 | | |
| - Downlink information for each radio link | | | |
| - Choice mode | | FDD | |
| - Primary CPICH info | | | |
| - Primary scrambling code | | Ref. to the Default setting in clause 6.1 (FDD) | |
| - PDSCH with SHO DCH info | | Not Present | R99 and Rel-4 only |
| - PDSCH code mapping | | Not Present | R99 and Rel-4 only |
| - Serving HS-DSCH radio link indicator | | FALSE | Rel-5 |
| - Downlink DPCH info for each RL | | Not present | |
| - SCCPCH information for FACH | | Not Present | R99 and Rel-4 only |
| Downlink information for each radio link list | A9, A10 | | Rel-5 |
| - Downlink information for each radio link | | | |
| - Choice mode | | FDD | |
| - Primary CPICH info | | | |
| - Primary scrambling code | | Ref. to the Default setting in clause 6.1 (FDD) | |
| - PDSCH with SHO DCH info | | Not Present | R99 and Rel-4 only |
| - PDSCH code mapping | | Not Present | R99 and Rel-4 only |
| - Serving HS-DSCH radio link indicator | | TRUE | |
| - Serving E-DCH radio link indicator | | FALSE | Rel-6 |
| - Downlink DPCH info for each RL | | | |
| - Primary CPICH usage for channel estimation | | Primary CPICH may be used | |
| - DPCH frame offset | | Set to value Default DPCH Offset Value (as currently stored in SS) mod 38 400 | |
| - Secondary CPICH info | | Not Present | |
| - DL channelisation code | | | |
| - Secondary scrambling code | | 1 | |
| - Spreading factor | | Reference to clause 6.10 Parameter Set | |
| - Code number | | 0 | |
| - Scrambling code change | | No code change | |
| - TPC combination index | | 0 | |
| - SSDT Cell Identity | | Not Present | R99 and Rel-4 only |
| - Closed loop timing adjustment mode | | Not Present | |
| - E-AGCH Info | | Not Present | Rel-6 |
| - CHOICE E-HICH Information | | Not Present | Rel-6 |
| - CHOICE E-RGCH Information | | Not Present | Rel-6 |
| - SCCPCH information for FACH | | Not Present | R99 and Rel-4 only |
| Downlink information for each radio link list | A12, A13, A15 | | Rel-6 |
| - Downlink information for each radio link | | | |

| Information Element | Condition | Value/remark | Version |
|---|---|---|---|
| - Choice mode | | FDD | |
| - Primary CPICH info | | | |
| - Primary scrambling code | | Ref. to the Default setting in clause 6.1 (FDD) | |
| - Serving HS-DSCH radio link indicator | | TRUE | |
| - Serving E-DCH radio link indicator | | TRUE | |
| - Downlink DPCH info for each RL | | | |
| - Primary CPICH usage for channel estimation | | Primary CPICH may be used | |
| - DPCH frame offset | | Set to value Default DPCH Offset Value (as currently stored in SS) mod 38 400 | |
| - Secondary CPICH info | | Not Present | |
| - DL channelisation code | | | |
| - Secondary scrambling code | | 1 | |
| - Spreading factor | | Reference to clause 6.10 Parameter Set | |
| - Code number | | 0 | |
| - Scrambling code change | | No code change | |
| - TPC combination index | | 0 | |
| - Closed loop timing adjustment mode | | Not Present | |
| - E-AGCH Info | | | |
| - E-AGCH Channelisation Code | | 2 | |
| - CHOICE E-HICH Information | | | |
| - E-HICH Information | | | |
| - DL Scrambling code | | 2 | |
| - Channelisation code | | 0 | |
| - Signature sequence | | 1 | |
| - CHOICE E-RGCH Information | | | |
| - E-RGCH Information | | | |
| - Signature Sequence | | 0 | |
| - RG combination index | | 0 | |
| Downlink information for each radio link list | A14 | | Rel-6 |
| - Downlink information for each radio link | | | |
| - Choice mode | | FDD | |
| - Primary CPICH info | | | |
| - Primary scrambling code | | Ref. to the Default setting in clause 6.1 (FDD) | |
| - Serving HS-DSCH radio link indicator | | TRUE | |
| - Serving E-DCH radio link indicator | | TRUE | |
| - Downlink DPCH info for each RL | | Not Present | |
| - Downlink F-DPCH info for each RL | | | |
| - Primary CPICH usage for channel estimation | | Primary CPICH may be used | |
| - F-DPCH frame offset | | Set to value Default DPCH Offset Value (as currently stored in SS) mod 38 400 | |
| - Secondary CPICH info | | Not Present | |
| - Secondary scrambling code | | 1 | |
| - Code number | | 0 | |
| - TPC combination index | | 0 | |
| - E-AGCH Info | | | |
| - E-AGCH Channelisation Code | | 2 | |
| - CHOICE E-HICH Information | | | |
| - E-HICH Information | | | |
| - DL Scrambling code | | 2 | |
| - Channelisation code | | 0 | |
| - Signature sequence | | 1 | |
| - CHOICE E-RGCH Information | | | |
| - E-RGCH Information | | | |
| - Signature Sequence | | 0 | |
| - RG combination index | | 0 | |

| Information Element | Condition | Value/remark | Version |
|---|---|---|---|
| Downlink information for each radio link list | A6 | Not Present | |
| MBMS PL Service Restriction Information | A1, A2, A3, A4, A5, A6, A7, A8<br>A9, A10<br>A12, A13, A14, A15 | Not Present | <br><br>Rel-5<br>Rel-6 |

| Condition | Explanation | Version |
|---|---|---|
| A1 | This IE is needed for "Non speech to CELL_DCH from CELL_DCH in CS" | |
| A2 | This IE is needed for "Speech to CELL_DCH from CELL_DCH in CS" | |
| A3 | This IE is needed for "Packet to CELL_DCH from CELL_DCH in PS" | |
| A4 | This IE is needed for "Packet to CELL_DCH from CELL_FACH in PS" | |
| A5 | This IE is needed for "Packet to CELL_FACH from CELL_DCH in PS" | |
| A6 | This IE is needed for "Packet to CELL_FACH from CELL_FACH in PS" | |
| A7 | This IE is needed for "Non speech to CELL_DCH from CELL_FACH in CS" | |
| A8 | This IE is needed for "Speech to CELL_DCH from CELL_FACH in CS" | |
| A9 | This IE is needed for "Packet to CELL_DCH / HS-DSCH using three multiplexing options", or when not stated otherwise, for "Packet to CELL_DCH / HS-DSCH from CELL_DCH in PS" | Rel-5 |
| A10 | This IE is needed for "Packet to CELL_DCH / HS-DSCH using one multiplexing option", or when not stated otherwise, for "Packet to CELL_DCH / HS-DSCH from CELL_FACH in PS" | Rel-5 |
| A11 | This IE is needed for " Packet RAB Setup after Speech RAB Setup in CELL_DCH" | |
| A12 | This IE is needed for "Packet to CELL_DCH / E-DCH / HS-DSCH using three multiplexing options (3/3) and SRBs mapped on DCH/DCH" | Rel-6 |
| A13 | This IE is needed for "Packet to CELL_DCH / E-DCH / HS-DSCH using one multiplexing option (1/1) and SRBs mapped on E-DCH/DCH" | Rel-6 |
| A14 | This IE is needed for "Packet to CELL_DCH / E-DCH / HS-DSCH using one multiplexing option (1/1) and SRBs mapped on E-DCH/HS-DSCH" | Rel-6 |
| A15 | This IE is needed for "Packet to CELL_DCH / E-DCH / HS-DSCH with multiple RABs using one multiplexing option (1/1) and SRBs mapped on E-DCH/DCH" | Rel-6 |

Contents of RADIO BEARER SETUP COMPLETE message: AM

| Information Element | Value/remark |
|---|---|
| Message Type | |
| RRC transaction identifier | Checked to see if the value is identical to the same IE in the downlink RADIO BEARER SETUP message. |
| Integrity check info | |
| - Message authentication code | This IE is checked to see if it is present. The value is compared against the XMAC-I value computed by SS. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I. |
| - RRC Message sequence number | This IE is checked to see if it is present. The value is used by SS to compute the XMAC-I value. |
| Uplink integrity protection activation info | Not checked. |
| CHOICE mode | FDD |
| START | Not checked (if ciphering is OFF), check the presence if ciphering is ON. |
| COUNT-C activation time | The UE shall include this IE if the following two conditions are fulfilled: (a) The RADIO BEARER SETUP message did not contain the IE "Ciphering activation time for DPCH" and (b) The RADIO BEARER SETUP message established the first RB(s) mapped to RLC-TM for a CN domain. Else, this IE is absent. |
| Radio bearer uplink ciphering activation time info | Not checked |
| Uplink counter synchronization info | Not present |

Contents of RADIO BEARER SETUP FAILURE message: AM

| Information Element | Value/remark |
|---|---|
| Message Type | |
| RRC transaction identitifer | Checked to see if it is set to identical value of the same IE in the downlink RADIO BEARER SETUP message. |
| Integrity check info | |

| | |
|---|---|
| - Message authentication code | This IE is checked to see if it is present. The value is compared against the XMAC-I value computed by SS. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I. |
| - RRC Message sequence number | This IE is checked to see if it is present. The value is used by SS to compute the XMAC-I value. |
| Failure cause | Checked to see if it meets test requirement |
| Radio bearers for which reconfiguration would have succeeded | Not checked |

Contents of RADIO BEARER RECONFIGURATION message: AM or UM

| Information Element | Condition | Value/remark | Version |
|---|---|---|---|
| Message Type | A1,A2,A3,A4,A5,A6 | | |
| RRC transaction identifier | | Arbitrarily selects an integer between 0 and 3 | |
| Integrity check info | | | |
| - message authentication code | | SS calculates the value of MAC-I for this message and writes to this IE. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I. | |
| - RRC message sequence number | | SS provides the value of this IE, from its internal counter. | |
| Integrity protection mode info | | Not Present | |
| Ciphering mode info | | Not Present | |
| Activation time | A1,A2,A3 | (256+CFN-(CFN MOD 8 + 8))MOD 256 | |
| Activation time | A4, A5,A6 | Not Present | |
| Delay restriction flag | A1,A2,A3,A4,A5,A6 | Not Present | Rel-6 |
| New U-RNTI | | Not Present | |
| New C-RNTI | A1, A2, A3, A4, | Not Present | |
| New C-RNTI | A5, A6 | '1010 1010 1010 1010' | |
| New DSCH-RNTI | A1, A2, A3, A4, A5, A6 | Not Present | R99 and Rel-4 only |
| New H-RNTI | A1, A2, A3, A4, A5, A6 | Not Present | Rel-5 |
| New Primary E-RNTI | | Not Present | Rel-6 |
| New Secondary E-RNTI | | Not Present | Rel-6 |
| RRC State indicator | A1, A2, A3, A4 | CELL_DCH | |
| RRC State indicator | A5, A6 | CELL_FACH | |
| UTRAN DRX cycle length coefficient | A1,A2,A3,A4,A5,A6 | Not Present | |
| CN information info | | Not Present | |
| URA identity | | Not Present | |
| CHOICE specification mode | | Complete specification | Rel-5 |
| RAB information to reconfigure list | | Not Present | |
| RB information to reconfigure list | A1 | TS25.331 specifies that "Although this IE is not always required, need is MP to align with ASN.1". | |
| - RB information to reconfigure | | (UM DCCH for RRC) | |
| - RB identity | | 1 | |
| - PDCP info | | Not Present | |
| - PDCP SN info | | Not Present | |
| - RLC info | | Not Present | |
| - RB mapping info | | Not Present | |
| - RB stop/continue | | Not Present | |
| - RB information to reconfigure | | (AM DCCH for RRC) | |
| - RB identity | | 2 | |
| - PDCP info | | Not Present | |
| - PDCP SN info | | Not Present | |
| - RLC info | | Not Present | |
| - RB mapping info | | Not Present | |
| - RB stop/continue | | Not Present | |
| - RB information to reconfigure | | (AM DCCH for NAS_DT High priority) | |
| - RB identity | | 3 | |
| - PDCP info | | Not Present | |
| - PDCP SN info | | Not Present | |
| - RLC info | | Not Present | |
| - RB mapping info | | Not Present | |

| Information Element | Condition | Value/remark | Version |
|---|---|---|---|
| - RB stop/continue | | Not Present | |
| - RB information to reconfigure | | (AM DCCH for NAS_DT Low priority) | |
| - RB identity | | 4 | |
| - PDCP info | | Not Present | |
| - PDCP SN info | | Not Present | |
| - RLC info | | Not Present | |
| - RB mapping info | | Not Present | |
| - RB stop/continue | | Not Present | |
| - RB information to reconfigure | | (TM DTCH) | |
| - RB identity | | 10 | |
| - PDCP info | | Not Present | |
| - PDCP SN info | | Not Present | |
| - RLC info | | Not Present | |
| - RB mapping info | | Not Present | |
| - RB stop/continue | | Not Present | |
| RB information to reconfigure list | A2 | TS25.331 specifies that "Although this IE is not always required, need is MP to align with ASN.1". | |
| - RB information to reconfigure | | (UM DCCH for RRC) | |
| - RB identity | | 1 | |
| - PDCP info | | Not Present | |
| - PDCP SN info | | Not Present | |
| - RLC info | | Not Present | |
| - RB mapping info | | Not Present | |
| - RB stop/continue | | Not Present | |
| - RB information to reconfigure | | (AM DCCH for RRC) | |
| - RB identity | | 2 | |
| - PDCP info | | Not Present | |
| - PDCP SN info | | Not Present | |
| - RLC info | | Not Present | |
| - RB mapping info | | Not Present | |
| - RB stop/continue | | Not Present | |
| - RB information to reconfigure | | (AM DCCH for NAS_DT High priority) | |
| - RB identity | | 3 | |
| - PDCP info | | Not Present | |
| - PDCP SN info | | Not Present | |
| - RLC info | | Not Present | |
| - RB mapping info | | Not Present | |
| - RB stop/continue | | Not Present | |
| - RB information to reconfigure | | (AM DCCH for NAS_DT Low priority) | |
| - RB identity | | 4 | |
| - PDCP info | | Not Present | |
| - PDCP SN info | | Not Present | |
| - RLC info | | Not Present | |
| - RB mapping info | | Not Present | |
| - RB stop/continue | | Not Present | |
| - RB information to reconfigure | | (TM DTCH) | |
| - RB identity | | 10 | |
| - PDCP info | | Not Present | |
| - PDCP SN info | | Not Present | |
| - RLC info | | Not Present | |
| - RB mapping info | | Not Present | |
| - RB stop/continue | | Not Present | |
| - RB information to reconfigure | | (TM DTCH) | |
| - RB identity | | 11 | |
| - PDCP info | | Not Present | |
| - PDCP SN info | | Not Present | |
| - RLC info | | Not Present | |
| - RB mapping info | | Not Present | |
| - RB stop/continue | | Not Present | |
| - RB information to reconfigure | | (TM DTCH)<br>(This IE is needed for 12.2 kbps and 10.2 kbps) | |
| - RB identity | | 12 | |
| - PDCP info | | Not Present | |
| - PDCP SN info | | Not Present | |
| - RLC info | | Not Present | |

| Information Element | Condition | Value/remark | Version |
|---|---|---|---|
| - RB mapping info | | Not Present | |
| - RB stop/continue | | Not Present | |
| RB information to reconfigure list | A3,A4,A5,A6 | TS25.331 specifies that "Although this IE is not always required, need is MP to align with ASN.1". | |
| - RB information to reconfigure | | (UM DCCH for RRC) | |
| - RB identity | | 1 | |
| - PDCP info | | Not Present | |
| - PDCP SN info | | Not Present | |
| - RLC info | | Not Present | |
| - RB mapping info | | Not Present | |
| - RB stop/continue | | Not Present | |
| - RB information to reconfigure | | (AM DCCH for RRC) | |
| - RB identity | | 2 | |
| - PDCP info | | Not Present | |
| - PDCP SN info | | Not Present | |
| - RLC info | | Not Present | |
| - RB mapping info | | Not Present | |
| - RB stop/continue | | Not Present | |
| - RB information to reconfigure | | (AM DCCH for NAS_DT High priority) | |
| - RB identity | | 3 | |
| - PDCP info | | Not Present | |
| - PDCP SN info | | Not Present | |
| - RLC info | | Not Present | |
| - RB mapping info | | Not Present | |
| - RB stop/continue | | Not Present | |
| - RB information to reconfigure | | (AM DCCH for NAS_DT Low priority) | |
| - RB identity | | 4 | |
| - PDCP info | | Not Present | |
| - PDCP SN info | | Not Present | |
| - RLC info | | Not Present | |
| - RB mapping info | | Not Present | |
| - RB stop/continue | | Not Present | |
| - RB information to reconfigure | | (AM DTCH) | |
| - RB identity | | 20 | |
| - PDCP info | | Not Present | |
| - PDCP SN info | | Not Present | |
| - RLC info | | Not Present | |
| - RB mapping info | | Not Present | |
| - RB stop/continue | | Not Present | |
| RB information to be affected | A1, A2, A3,A4,A5,A6 | Not Present | |
| RB with PDCP context relocation info list | | Not Present | Rel-5 |
| PDCP ROHC target mode | | Not Present | Rel-5 |
| UL Transport channel information common for all transport channels | A1, A2, A5,A6 | Not Present | |
| UL Transport channel information common for all transport channels | A3, A4 | | |
| - PRACH TFCS | | Not Present | |
| - CHOICE mode | | FDD | |
| - TFC subset | | Not Present | |
| - UL DCH TFCS | | | |
| - CHOICE TFCI signalling | | Normal | |
| - TFCI Field 1 information | | | |
| - CHOICE TFCS representation | | Complete reconfiguration | |
| - TFCS complete reconfigure information | | | |
| - CHOICE CTFC Size | | Number of bits used must be enough to cover all combinations of CTFC from clause 6.10.2.4 Parameter Set. | |
| - CTFC information | | This IE is repeated for TFC numbers and reference to clause 6.10.2.4 Parameter Set | |
| - CTFC | | Reference to clause 6.10.2.4 Parameter Set | |
| - Power offset information | | | |

| Information Element | Condition | Value/remark | Version |
|---|---|---|---|
| - CHOICE Gain Factors | | Computed Gain Factors(The last TFC is set to Signalled Gain Factors) | |
| - Gain factor βc | | 11 (below 64 kbps)<br>9 (higher than 64 kbps)<br>(Not Present if the CHOICE Gain Factors is set to ComputedGain Factors) | |
| - Gain factor βd | | 15<br>(Not Present if the CHOICE Gain Factors is set to ComputedGain Factors) | |
| - Reference TFC ID | | 0 | |
| - CHOICE mode | | FDD | |
| - Power offset P p-m | | Not Present | |
| Deleted UL TrCH information | A1, A2, A3, A4, A5,A6 | Not Present | |
| Added or Reconfigured UL TrCH information | A1, A2, A5,A6 | Not Present | |
| Added or Reconfigured UL TrCH information | A4 | 2 TrCHs(DCH for DCCH and DCH for DTCH) | |
| - Uplink transport channel type | | DCH | |
| - UL Transport channel identity | | 5 | |
| - TFS | | | |
| - CHOICE Transport channel type | | Dedicated transport channels | |
| - Dynamic Transport format information | | | |
| - RLC Size | | Reference to clause 6.10 Parameter Set | |
| - Number of TBs and TTI List | | (This IE is repeated for TFI number.) | |
| - Transmission Time Interval | | Not Present | |
| - Number of Transport blocks | | Reference to clause 6.10 Parameter Set | |
| - CHOICE Logical channel list | | All | |
| - Semi-static Transport Format information | | | |
| - Transmission time interval | | Reference to clause 6.10 Parameter Set | |
| - Type of channel coding | | Reference to clause 6.10 Parameter Set | |
| - Coding Rate | | Reference to clause 6.10 Parameter Set | |
| - Rate matching attribute | | Reference to clause 6.10 Parameter Set | |
| - CRC size | | Reference to clause 6.10 Parameter Set | |
| - Uplink transport channel type | | DCH | |
| - UL Transport channel identity | | 1 | |
| - TFS | | | |
| - CHOICE Transport channel type | | Dedicated transport channels | |
| - Dynamic Transport format information | | | |
| - RLC Size | | Reference to clause 6.10 Parameter Set | |
| - Number of TBs and TTI List | | (This IE is repeated for TFI number.) | |
| - Transmission Time Interval | | Not Present | |
| - Number of Transport blocks | | Reference to clause 6.10 Parameter Set | |
| - CHOICE Logical channel list | | All | |
| - Semi-static Transport Format information | | | |
| - Transmission time interval | | Reference to clause 6.10 Parameter Set | |
| - Type of channel coding | | Reference to clause 6.10 Parameter Set | |
| - Coding Rate | | Reference to clause 6.10 Parameter Set | |
| - Rate matching attribute | | Reference to clause 6.10 Parameter Set | |
| - CRC size | | Reference to clause 6.10 Parameter Set | |
| Added or Reconfigured UL TrCH information | A3 | (DCH for DTCH) | |
| - Uplink transport channel type | | DCH | |
| - UL Transport channel identity | | 1 | |
| - TFS | | | |
| - CHOICE Transport channel type | | Dedicated transport channels | |
| - Dynamic Transport format information | | | |
| - RLC Size | | Reference to clause 6.10 Parameter Set | |
| - Number of TBs and TTI List | | (This IE is repeated for TFI number.) | |
| - Transmission Time Interval | | Not Present | |
| - Number of Transport blocks | | Reference to clause 6.10 Parameter Set | |
| - CHOICE Logical channel list | | All | |

| Information Element | Condition | Value/remark | Version |
|---|---|---|---|
| - Semi-static Transport Format information | | | |
| - Transmission time interval | | Reference to clause 6.10 Parameter Set | |
| - Type of channel coding | | Reference to clause 6.10 Parameter Set | |
| - Coding Rate | | Reference to clause 6.10 Parameter Set | |
| - Rate matching attribute | | Reference to clause 6.10 Parameter Set | |
| - CRC size | | Reference to clause 6.10 Parameter Set | |
| CHOICE mode | A1,A2,A3,A4,A5,A6 | Not Present | |
| DL Transport channel information common for all transport channel | A1, A2, A5, A6 | Not Present | |
| DL Transport channel information common for all transport channel | A3,A4 | | |
| - SCCPCH TFCS | | Not Present | |
| - CHOICE mode | | FDD | |
| - CHOICE DL parameters | | Explicit | |
| - DL DCH TFCS | | | |
| - CHOICE TFCI Signalling | | Normal | |
| - TFCI Field 1 Information | | | |
| - CHOICE TFCS representation | | Complete reconfiguration | |
| - TFCS complete reconfigure | | | |
| - CHOICE CTFC Size | | Number of bits used must be enough to cover all combinations of CTFC from clause 6.10.2.4 Parameter Set. | |
| - CTFC information | | This IE is repeated for TFC numbers and reference to clause 6.10.2.4 | |
| - CTFC | | Reference to clause 6.10.2.4 Parameter Set | |
| - Power offset information | | Not Present | |
| Deleted DL TrCH information | A1, A2, A3, A4, A5,A6 | Not Present | |
| Added or Reconfigured DL TrCH information | A1, A2, A5, A6 | Not Present | |
| Added or Reconfigured DL TrCH information | A4 | 2 TrCHs(DCH for DCCH and DCH for DTCH) | |
| - Downlink transport channel type | | DCH | |
| - DL Transport channel identity | | 10 | |
| - CHOICE DL parameters | | Same as UL | |
| - Uplink transport channel type | | DCH | |
| - UL TrCH identity | | 5 | |
| - DCH quality target | | | |
| - BLER Quality value | | Not Present | |
| - Downlink transport channel type | | DCH | |
| - DL Transport channel identity | | 6 | |
| - CHOICE DL parameters | | Explicit | |
| - TFS | | | |
| - CHOICE Transport channel type | | Dedicated transport channel | |
| - Dynamic transport format information | | | |
| - RLC Size | | Reference to clause 6.10 Parameter Set | |
| - Number of TBs and TTI List | | (This IE is repeated for TFI number.) | |
| - Dynamic transport format information | | | |
| - Transmission Time Interval | | Not Present | |
| - Number of Transport blocks | | Reference to clause 6.10 Parameter Set | |
| - Semi-static Transport Format information | | | |
| - Transmission time interval | | Reference to clause 6.10 Parameter Set | |
| - Type of channel coding | | Reference to clause 6.10 Parameter Set | |
| - Coding Rate | | Reference to clause 6.10 Parameter Set | |
| - Rate matching attribute | | Reference to clause 6.10 Parameter Set | |
| - CRC size | | Reference to clause 6.10 Parameter Set | |
| - DCH quality target | | | |
| - BLER Quality value | | -2.0 | |
| Added or Reconfigured DL TrCH information | A3 | | |
| - Downlink transport channel type | | DCH | |
| - DL Transport channel identity | | 6 | |

| Information Element | Condition | Value/remark | Version |
|---|---|---|---|
| - CHOICE DL parameters | | Explicit | |
| - TFS | | | |
| - CHOICE Transport channel type | | Dedicated transport channel | |
| - Dynamic transport format information | | | |
| - RLC Size | | Reference to clause 6.10 Parameter Set | |
| - Number of TBs and TTI List | | (This IE is repeated for TFI number.) | |
| - Dynamic transport format information | | | |
| - Transmission Time Interval | | Not Present | |
| - Number of Transport blocks | | Reference to clause 6.10 Parameter Set | |
| - Semi-static Transport Format information | | | |
| - Transmission time interval | | Reference to clause 6.10 Parameter Set | |
| - Type of channel coding | | Reference to clause 6.10 Parameter Set | |
| - Coding Rate | | Reference to clause 6.10 Parameter Set | |
| - Rate matching attribute | | Reference to clause 6.10 Parameter Set | |
| - CRC size | | Reference to clause 6.10 Parameter Set | |
| - DCH quality target | | | |
| - BLER Quality value | | -2.0 | |
| Frequency info | A1,A2,A3,A4,A5 | | |
| - UARFCN uplink (Nu) | | Not present<br>Absence of this IE is equivalent to applying the default duplex distance defined for the operating frequency according to 3GPP TS 25.101 [11] | |
| - UARFCN downlink (Nd) | | Reference to clause 5.1 Test frequencies | |
| Frequency info | A6 | Not Present | |
| Maximum allowed UL TX power | A1,A2,A3,A4,A5,A6 | 33dBm | |
| CHOICE channel requirement | A1, A2, A3, A4 | Uplink DPCH info | |
| -Uplink DPCH power control info | | | |
| - DPCCH power offset | | -80dB (i.e. ASN.1 IE value of -40) | |
| - PC Preamble | | 1 frame | |
| - SRB delay | | 7 frames | |
| - Power Control Algorithm | | Algorithm1 | |
| - TPC step size | | 1dB | |
| - $\Delta_{ACK}$ | | Not Present | Rel-5 |
| - $\Delta_{NACK}$ | | Not Present | Rel-5 |
| - Ack-Nack repetition factor | | Not Present | Rel-5 |
| - Scrambling code type | | Long | |
| - Scrambling code number | | 0 (0 to 16777215) | |
| - Number of DPDCH | | Not Present(1) | |
| - spreading factor | | Reference to clause 6.10 Parameter Set | |
| - TFCI existence | | Reference to clause 6.10 Parameter Set | |
| - Number of FBI bit | | Reference to clause 6.10 Parameter Set | |
| - Puncturing Limit | | Reference to clause 6.10 Parameter Set | |
| CHOICE channel requirement | A5, A6 | Not Present | |
| E-DCH Info | | Not Present | Rel-6 |
| CHOICE Mode | A1,A2,A3,A4,A5,A6 | FDD | |
| - Downlink PDSCH information | | Not Present | R99 and Rel-4 only |
| Downlink HS-PDSCH Information | A1, A2, A3, A4, A5, A6 | Not Present | Rel-5 |
| Downlink information common for all radio links | A5, A6 | Not Present | |
| Downlink information common for all radio links | A1, A2, A3 | | |
| - Downlink DPCH info common for all RL | | | |
| - Timing indicator | | Maintain | |
| - CFN-targetSFN frame offset | | Not Present | |
| - Downlink DPCH power control information | | | |
| - DPC mode | | 0 (single) | |
| - CHOICE mode | | FDD | |

| Information Element | Condition | Value/remark | Version |
|---|---|---|---|
| - Power offset $P_{Pilot-DPDCH}$ | | 0 | |
| - DL rate matching restriction information | | Not Present | |
| - Spreading factor | | Reference to clause 6.10 Parameter Set | |
| - Fixed or Flexible Position | | Reference to clause 6.10 Parameter Set | |
| - TFCI existence | | Reference to clause 6.10 Parameter Set | |
| - CHOICE SF | | Reference to clause 6.10 Parameter Set | |
| - DPCH compressed mode info | | Not Present | |
| - TX Diversity mode | | None | |
| - SSDT information | | Not Present | R99 and Rel-4 only |
| - Default DPCH Offset Value | | Not Present | |
| - MAC-hs reset indicator | | Not Present | Rel-5 |
| Downlink information common for all radio links | A4 | | |
| - Downlink DPCH info common for all RL | | | |
| - Timing indicator | | Initialize | |
| - CFN-targetSFN frame offset | | Not Present | |
| - Downlink DPCH power control information | | | |
| - DPC mode | | 0 (single) | |
| - CHOICE mode | | FDD | |
| - Power offset $P_{Pilot-DPDCH}$ | | 0 | |
| - DL rate matching restriction information | | Not Present | |
| - Spreading factor | | Reference to clause 6.10 Parameter Set | |
| - Fixed or Flexible Position | | Reference to clause 6.10 Parameter Set | |
| - TFCI existence | | Reference to clause 6.10 Parameter Set | |
| - CHOICE SF | | Reference to clause 6.10 Parameter Set | |
| - DPCH compressed mode info | | Not Present | |
| - TX Diversity mode | | None | |
| - SSDT information | | Not Present | R99 and Rel-4 only |
| - Default DPCH Offset Value | | Present Arbitrary set to value 0..306688 by step of 512 | |
| - MAC-hs reset indicator | | Not Present | Rel-5 |
| Downlink information per radio link list -Downlink information for each radio link | A1, A2, A3 | | |
| - Choice mode | | FDD | |
| - Primary CPICH info | | | |
| - Primary scrambling code | | Ref. to the Default setting in clause 6.1 (FDD) | |
| - PDSCH with SHO DCH info | | Not Present | R99 and Rel-4 only |
| - PDSCH code mapping | | Not Present | R99 and Rel-4 only |
| - Serving HS-DSCH radio link indicator | | FALSE | Rel-5 |
| - Serving E-DCH radio link indicator | | FALSE | Rel-6 |
| - Downlink DPCH info for each RL | | | |
| - Primary CPICH usage for channel estimation | | Primary CPICH may be used | |
| - DPCH frame offset | | Set to value Default DPCH Offset Value (as currently stored in SS) mod 38 400 | |
| - Secondary CPICH info | | Not Present | |
| - Secondary scrambling code | | | |
| - channelisation code | | | |
| - DL channelisation code | | | |
| - Secondary scrambling code | | 2 | |
| - Spreading factor | | Reference to clause 6.10 Parameter Set | |
| - Code number | | 0 | |
| - Scrambling code change | | No change | |
| - TPC combination index | | 0 | |
| - SSDT Cell Identity | | Not Present | R99 and |

| Information Element | Condition | Value/remark | Version |
|---|---|---|---|
| | | | Rel-4 only |
| - Closed loop timing adjustment mode | | Not Present | |
| - E-AGCH Info | | Not present | Rel-6 |
| - E-HICH Information | | Not present | Rel-6 |
| - E-RGCH Information | | Not present | Rel-6 |
| - SCCPCH information for FACH | | Not Present | R99 and Rel-4 only |
| Downlink information per radio link list | A4 | | |
| -Downlink information for each radio link | | | |
| - Choice mode | | FDD | |
| - Primary CPICH info | | | |
| - Primary scrambling code | | Ref. to the Default setting in clause 6.1 (FDD) | |
| - PDSCH with SHO DCH info | | Not Present | R99 and Rel-4 only |
| - PDSCH code mapping | | Not Present | R99 and Rel-4 only |
| - Serving HS-DSCH radio link indicator | | FALSE | Rel-5 |
| - Serving E-DCH radio link indicator | | FALSE | Rel-6 |
| - Downlink DPCH info for each RL | | | |
| - Primary CPICH usage for channel estimation | | Primary CPICH may be used | |
| - DPCH frame offset | | Set to value : Default DPCH Offset Value mod 38 400 | |
| - Secondary CPICH info | | Not Present | |
| - Secondary scrambling code | | | |
| - channelisation code | | | |
| - DL channelisation code | | | |
| - Secondary scrambling code | | 2 | |
| - Spreading factor | | Reference to clause 6.10 Parameter Set 0 | |
| - Code number | | | |
| - Scrambling code change | | No change | |
| - TPC combination index | | 0 | |
| - SSDT Cell Identity | | Not Present | R99 and Rel-4 only |
| - Closed loop timing adjustment mode | | Not Present | |
| - E-AGCH Info | | Not present | Rel-6 |
| - E-HICH Information | | Not present | Rel-6 |
| - E-RGCH Information | | Not present | Rel-6 |
| - SCCPCH information for FACH | | Not Present | R99 and Rel-4 only |
| - Downlink information for each radio link | A5 | | |
| - Choice mode | | FDD | |
| - Primary CPICH info | | | |
| - Primary scrambling code | | Ref. to the Default setting in clause 6.1 (FDD) | |
| - PDSCH with SHO DCH info | | Not Present | R99 and Rel-4 only |
| - PDSCH code mapping | | Not Present | R99 and Rel-4 only |
| - Serving HS-DSCH radio link indicator | | FALSE | Rel-5 |
| - Serving E-DCH radio link indicator | | FALSE | Rel-6 |
| - Downlink DPCH info for each RL | | Not present | |
| - E-AGCH Info | | Not present | Rel-6 |
| - E-HICH Information | | Not present | Rel-6 |
| - E-RGCH Information | | Not present | Rel-6 |
| - SCCPCH Information for FACH | | Not Present | R99 and Rel-4 only |
| - Downlink information for each radio link | A6 | | R99 |

| Information Element | Condition | Value/remark | Version |
|---|---|---|---|
| - Choice mode | | FDD | |
| - Primary CPICH info | | | |
| - Primary scrambling code | | Ref. to the Default setting in clause 6.1 (FDD) | |
| - PDSCH with SHO DCH info | | Not Present | R99 and Rel-4 only |
| - PDSCH code mapping | | Not Present | R99 and Rel-4 only |
| - Serving E-DCH radio link indicator | | FALSE | Rel-6 |
| - Downlink DPCH info for each RL | | Not present | |
| - E-AGCH Info | | Not present | Rel-6 |
| - E-HICH Information | | Not present | Rel-6 |
| - E-RGCH Information | | Not present | Rel-6 |
| - SCCPCH Information for FACH | | Not Present | R99 and Rel-4 only |
| - Downlink information for each radio link | A6 | Not Present | Rel-4 on |
| MBMS PL Service Restriction Information | A1,A2,A3,A4,A5,A6 | Not Present | Rel-6 |

| Condition | Explanation |
|---|---|
| A1 | This IE need for "Non speech in CS" |
| A2 | This IE need for "Speech in CS" |
| A3 | This IE need for "Packet to CELL_DCH from CELL_DCH in PS" |
| A4 | This IE need for "Packet to CELL_DCH from CELL_FACH in PS" |
| A5 | This IE need for "Packet to CELL_FACH from CELL_DCH in PS" |
| A6 | This IE need for "Packet to CELL_FACH from CELL_FACH in PS" |

Contents of RADIO BEARER RECONFIGURATION FAILURE message: AM

| Information Element | Value/remark |
|---|---|
| Message Type | |
| RRC transaction identitifer | Checked to see if it is set to identical value of the same IE in the downlink RADIO BEARER RECONFIGURATION message. |
| Integrity check info | |
| - Message authentication code | This IE is checked to see if it is present. The value is compared against the XMAC-I value computed by SS. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I. |
| - RRC Message sequence number | This IE is checked to see if it is present. The value is used by SS to compute the XMAC-I value. |
| Failure cause | Checked to see if it meets test requirement |
| Radio bearers for which reconfiguration would have succeeded List | Not checked |

Contents of RADIO BEARER RECONFIGURATION COMPLETE message: AM

| Information Element | Value/remark |
|---|---|
| Message Type | |
| RRC transaction identifier | Checked to see if the value is identical to the same IE in the downlink RADIO BEARER RECONFIGURATION COMPLETE message |
| Integrity check info | |
| - Message authentication code | This IE is checked to see if it is present. The value is compared against the XMAC-I value computed by SS. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I. |
| - RRC Message sequence number | This IE is checked to see if it is present. The value is used by SS to compute the XMAC-I value. |
| Uplink integrity protection activation info | Not checked |
| CHOICE mode | FDD |
| COUNT-C activation time | Not checked |
| Radio bearer uplink ciphering activation time info | Not checked |
| Uplink counter synchronization info | Not present |

Contents of RADIO BEARER RELEASE message: AM or UM

| Information Element | Condition | Value/remark | Version |
|---|---|---|---|
| Message Type | A1, A2, A3, A4, A5, A6, A7, A8 , A9, A10 | | Rel-5 |
| RRC transaction identifier | | Arbitrarily selects an integer between 0 and 3 | |
| Integrity check info | | | |
| - message authentication code | | SS calculates the value of MAC-I for this message and writes to this IE. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I. | |
| - RRC message sequence number | | SS provides the value of this IE, from its internal counter. | |
| Integrity protection mode info | | Not Present | |
| Ciphering mode info | | Not Present | |
| Activation time | A1, A2, A3, A7, A8 , A9, A10 | (256+CFN-(CFN MOD 8 + 8))MOD 256 | Rel-5 |
| Activation time | A4, A5, A6 | Not Present | |
| New U-RNTI | | Not Present | |
| New C-RNTI | A1,A2,A3,A4 , A9 | Not Present | Rel-5 |
| New C-RNTI | A5, A6, A7, A8 , A10 | '1010 1010 1010 1010' | Rel-5 |
| New DSCH-RNTI | A1, A2, A3, A4, A5, A6, A7, A8 | Not Present | R99 and Rel-4 only |
| New H-RNTI | A1, A2, A3, A4, A5, A6, A7, A8 , A9, A10, | Not Present | Rel-5 |
| New Primary E-RNTI | | Not Present | Rel-6 |
| New Secondary E-RNTI | | Not Present | Rel-6 |
| RRC State indicator | A1,A2, A3, A4 , A9 | CELL_DCH | Rel-5 |
| RRC State indicator | A5, A6, A7, A8 , A10 | CELL_FACH | Rel-5 |
| UTRAN DRX cycle length coefficient | A1,A2,A3,A4,A5,A6, A7, A8 , A9, A10 | Not Present | Rel-5 |
| CN information info | | Not Present | |
| Signalling Connection release indication | | Not Present | |
| URA identity | | Not Present | |
| RAB information to reconfigure list | | Not Present | |
| RB information to release | A1,A2, A7, A8 | | |
| - RB identity | | 10 | |
| RB information to release | A2, A8 | | |
| - RB identity | | 11 | |
| RB information to release | A2, A8 | | |
| - RB identity | | 12 | |
| RB information to release | A3, A4, A5, A6 | | |
| - RB identity | | 20 | |
| RB information to release | A9, A10 | | Rel-5 |
| - RB identity | | 25 | |
| RB information to reconfigure list | A1,A2, A3,A4,A5, A6, A7, A8, A9, A10 | Not Present | Rel-6 |

| Information Element | Condition | Value/remark | Version |
|---|---|---|---|
| RB information to be affected | A1,A2, A3,A4,A5, A6, A7, A8 , A9, A10 | Not Present | Rel-5 |
| Downlink counter synchronization info | A1,A2,A3,A 4,A5,A6, A7, A8 , A9, A10 | Not Present | Rel-5 |
| UL Transport channel information for all transport channels | A1, A2, A3, A4, A5, A6, A7, A8 , A9, A10 | TFCS reconfigured to fit the new transport channel configuration. | Rel-5 |
| Deleted UL TrCH Information | A1,A2, A3, A4, A5, A6, A7, A8 , A9, A10 | | Rel-5 |
| - Uplink transport channel type | | DCH | |
| - Transport channel identity | | 1 | |
| Deleted UL TrCH Information | A2, A8 | | |
| - Uplink transport channel type | | DCH | |
| - Transport channel identity | | 2 | |
| Deleted UL TrCH Information | A2, A8 | | |
| - Uplink transport channel type | | DCH | |
| - Transport channel identity | | 3 | |
| Added or Reconfigured UL TrCH information | A5, A6, A7, A8 , A10 | Not Present | Rel-5 |
| Added or Reconfigured UL TrCH information | A1, A2, A3, A4 , A9 | TrCHs(DCH for DCCH ) | Rel-5 |
| - Uplink transport channel type | | DCH | |
| - UL Transport channel identity | | 5 | |
| - TFS | | | |
| - CHOICE Transport channel type | | Dedicated transport channels | |
| - Dynamic Transport format information | | | |
| - RLC Size | | According to clause 6.10.2.4.1.3 (standalone 13.6 kbps signalling radio bearer) | |
| - Number of TBs and TTI List | | (This IE is repeated for TFI number.) | |
| - Transmission Time Interval | | According to clause 6.10.2.4.1.3 (standalone 13.6 kbps signalling radio bearer) | |
| - Number of Transport blocks | | According to clause 6.10.2.4.1.3 (standalone 13.6 kbps signalling radio bearer) | |
| - CHOICE Logical channel list | | All | |
| - Semi-static Transport Format information | | | |
| - Transmission time interval | | According to clause 6.10.2.4.1.3 (standalone 13.6 kbps signalling radio bearer) | |
| - Type of channel coding | | According to clause 6.10.2.4.1.3 (standalone 13.6 kbps signalling radio bearer) | |
| - Coding Rate | | According to clause 6.10.2.4.1.3 (standalone 13.6 kbps signalling radio bearer) | |
| - Rate matching attribute | | According to clause 6.10.2.4.1.3 (standalone 13.6 kbps signalling radio bearer) | |
| - CRC size | | According to clause 6.10.2.4.1.3 (standalone 13.6 kbps signalling radio bearer) | |
| DL Transport channel information for all transport channels | A1, A2, A3, A4, A5, A6, A7, A8 , A9, A10 | TFCS reconfigured to fit the new transport channel configuration. | Rel-5 |

| Information Element | Condition | Value/remark | Version |
|---|---|---|---|
| Deleted DL TrCH Information | A1, A2, A3, A4, A5, A6, A7, A8 , A9 | | Rel-5 |
| - Downlink transport channel type | | DCH | |
| - Transport channel identity | | 6 | |
| Deleted DL TrCH Information | A2, A8 | | |
| - Downlink transport channel type | | DCH | |
| - Transport channel identity | | 7 | |
| Deleted DL TrCH Information | A2, A8 | | |
| - Downlink transport channel type | | DCH | |
| - Transport channel identity | | 8 | |
| Deleted DL TrCH Information | A9, A10 | | Rel-5 |
| - Downlink transport channel type | | HS-DSCH | |
| - DL HS-DSCH MAC-d flow identity | | 0 | |
| Added or Reconfigured DL TrCH information | A5, A6, A7, A8 , A10 | Not Present | Rel-5 |
| Added or Reconfigured DL TrCH information | A1, A2, A3, A4 , A9 | 1 TrCHs(DCH for DCCH) | Rel-5 |
| - Downlink transport channel type | | DCH | |
| - DL Transport channel identity | | 10 | |
| - CHOICE DL parameters | | Same as UL | |
| - Uplink transport channel type | | DCH | |
| - UL TrCH identity | | 5 | |
| - DCH quality target | | | |
| - BLER Quality value | | Not Present | |
| Frequency info | A1,A2,A3,A4,A5, A7, A8 , A9, A10 | | Rel-5 |
| - UARFCN uplink (Nu) | | Not present<br>Absence of this IE is equivalent to applying the default duplex distance defined for the operating frequency according to 3GPP TS 25.101 [11] | |
| - UARFCN downlink (Nd) | | Reference to clause 5.1 Test frequencies | |
| Maximum allowed UL TX power | | 33dBm | |
| Frequency info | A6 | Not Present | |
| CHOICE *channel requirement* | A5, A6, A7, A8 , A10 | Not Present | Rel-5 |
| CHOICE channel requirement | A1,A2,A3,A4 , A9 | Uplink DPCH info | Rel-5 |
| - Uplink DPCH power control info | | | |
| - DPCCH power offset | | -80dB (i.e. ASN.1 IE value of -40) | |
| - PC Preamble | | 1 frame | |
| - SRB delay | | 7 frames | |
| - Power Control Algorithm | | Algorithm1 | |
| - $\Delta_{ACK}$ | | Not Present | Rel-5 |
| - $\Delta_{NACK}$ | | Not Present | Rel-5 |
| - Ack-Nack repetition factor | | Not Present | Rel-5 |
| - TPC step size | | 1dB | |
| - Scrambling code type | | Long | |
| - Scrambling code number | | 0 (0 to 16777215) | |
| - Number of DPDCH | | Not Present(1) | |
| - spreading factor | | Reference to clause 6.10 Parameter Set | |
| - TFCI existence | | Reference to clause 6.10 Parameter Set | |
| - Number of FBI bit | | Reference to clause 6.10 Parameter Set | |
| - Puncturing Limit | | Reference to clause 6.10 Parameter Set | |
| E-DCH Info | | Not Present | Rel-6 |
| CHOICE Mode | A1,A2,A3,A4,A5,A6, A7, A8 | FDD | |

| Information Element | Condition | Value/remark | Version |
|---|---|---|---|
| | , A9, A10 | | Rel-5 |
| - Downlink PDSCH information | | Not Present | R99 and Rel-4 only |
| Downlink HS-PDSCH Information | A1, A2, A3, A4, A5, A6, A7, A8, A9, A10 | Not Present | Rel-5 |
| Downlink information common for all radio links | A5, A6, A7, A8 | Not Present | |
| | , A10 | | Rel-5 |
| Downlink information common for all radio links | A1,A2, A3 | | |
| | , A9 | | Rel-5 |
| - Downlink DPCH info common for all RL | | | |
| - Timing indicator | | Maintain | |
| - CFN-targetSFN frame offset | | Not Present | |
| - Downlink DPCH power control information | | | |
| - DPC mode | | 0 (single) | |
| - CHOICE mode | | FDD | |
| - Power offset $P_{Pilot-DPDCH}$ | | 0 | |
| - DL rate matching restriction information | | Not Present | |
| - Spreading factor | | Reference to clause 6.10 Parameter Set | |
| - Fixed or Flexible Position | | Reference to clause 6.10 Parameter Set | |
| - TFCI existence | | Reference to clause 6.10 Parameter Set | |
| - CHOICE SF | | Reference to clause 6.10 Parameter Set | |
| - DPCH compressed mode info | | Not Present | |
| - TX Diversity mode | | None | |
| - SSDT information | | Not Present | R99 and Rel-4 only |
| - Default DPCH Offset Value | | Not Present | |
| - MAC-hs reset indicator | | Not Present | Rel-5 |
| Downlink information common for all radio links | A4 | | |
| - Downlink DPCH info common for all RL | | | |
| - Timing indicator | | Initialize | |
| - CFN-targetSFN frame offset | | Not Present | |
| - Downlink DPCH power control information | | | |
| - DPC mode | | 0 (single) | |
| - CHOICE mode | | FDD | |
| - Power offset $P_{Pilot-DPDCH}$ | | 0 | |
| - DL rate matching restriction information | | Not Present | |
| - Spreading factor | | Reference to clause 6.10 Parameter Set | |
| - Fixed or Flexible Position | | Reference to clause 6.10 Parameter Set | |
| - TFCI existence | | Reference to clause 6.10 Parameter Set | |
| - CHOICE SF | | Reference to clause 6.10 Parameter Set | |
| - DPCH compressed mode info | | Not Present | |
| - TX Diversity mode | | None | |
| - SSDT information | | Not Present | R99 and Rel-4 only |
| - Default DPCH Offset Value | | Arbitrary set to value 0..306688 by step of 512 | |
| - MAC-hs reset indicator | | Not Present | Rel-5 |
| Downlink information for each radio link list | A1,A2,A3 | | |
| | , A9 | | Rel-5 |
| -Downlink information for each radio link | | | |
| - Choice mode | | FDD | |
| - Primary CPICH info | | | |
| - Primary scrambling code | | Ref. to the Default setting in clause 6.1 (FDD) | |
| - PDSCH with SHO DCH info | | Not Present | R99 and Rel-4 only |
| - PDSCH code mapping | | Not Present | R99 and Rel-4 only |
| - Serving HS-DSCH radio link indicator | | FALSE | Rel-5 |
| - Serving E-DCH radio link indicator | | FALSE | Rel-6 |
| - Downlink DPCH info for each RL | | | |
| - Primary CPICH usage for channel estimation | | Primary CPICH may be used | |
| - DPCH frame offset | | Set to value Default DPCH Offset Value | |

| Information Element | Condition | Value/remark | Version |
|---|---|---|---|
| | | (as currently stored in SS) mod 38 400 | |
| - Secondary CPICH info | | Not Present | |
| - Secondary scrambling code | | | |
| - channelisation code | | | |
| - DL channelisation code | | | |
| - Secondary scrambling code | | 3 | |
| - Spreading factor | | Reference to clause 6.10 Parameter Set | |
| - Code number | | 0 | |
| - Scrambling code change | | No change | |
| - TPC combination index | | 0 | |
| - SSDT Cell Identity | | Not Present | R99 and Rel-4 only |
| - Closed loop timing adjustment mode | | Not Present | |
| - E-AGCH Info | | Not present | Rel-6 |
| - E-HICH Information | | Not present | Rel-6 |
| - E-RGCH Information | | Not present | Rel-6 |
| - SCCPCH information for FACH | | Not Present | R99 and Rel-4 only |
| Downlink information for each radio link list | A4 | | |
| -Downlink information for each radio link | | | |
| - Choice mode | | FDD | |
| - Primary CPICH info | | | |
| - Primary scrambling code | | Ref. to the Default setting in clause 6.1 (FDD) | |
| - PDSCH with SHO DCH info | | Not Present | R99 and Rel-4 only |
| - PDSCH code mapping | | Not Present | R99 and Rel-4 only |
| - Serving HS-DSCH radio link indicator | | FALSE | Rel-5 |
| - Serving E-DCH radio link indicator | | FALSE | Rel-6 |
| - Downlink DPCH info for each RL | | | |
| - Primary CPICH usage for channel estimation | | Primary CPICH may be used | |
| - DPCH frame offset | | Set to value : Default DPCH Offset Value mod 38 400 | |
| - Secondary CPICH info | | Not Present | |
| - Secondary scrambling code | | | |
| - channelisation code | | | |
| - DL channelisation code | | | |
| - Secondary scrambling code | | 3 | |
| - Spreading factor | | Reference to clause 6.10 Parameter Set | |
| - Code number | | 0 | |
| - Scrambling code change | | No change | |
| - TPC combination index | | 0 | |
| - SSDT Cell Identity | | Not Present | R99 and Rel-4 only |
| - Closed loop timing adjustment mode | | Not Present | |
| - E-AGCH Info | | Not present | Rel-6 |
| - E-HICH Information | | Not present | Rel-6 |
| - E-RGCH Information | | Not present | Rel-6 |
| - SCCPCH information for FACH | | Not Present | R99 and Rel-4 only |
| - Downlink information for each radio link | A5, A7, A8 | | |
| - Choice mode | | FDD | |
| - Primary CPICH info | | | |
| - Primary scrambling code | | Ref. to the Default setting in clause 6.1 (FDD) | |
| - PDSCH with SHO DCH info | | Not Present | R99 and Rel-4 only |
| - PDSCH code mapping | | Not Present | R99 and Rel-4 only |
| - Serving HS-DSCH radio link indicator | | FALSE | Rel-5 |
| - Serving E-DCH radio link indicator | | FALSE | Rel-6 |
| - Downlink DPCH info for each RL | | Not present | |
| - E-AGCH Info | | Not present | Rel-6 |
| - E-HICH Information | | Not present | Rel-6 |
| - E-RGCH Information | | Not present | Rel-6 |

| Information Element | Condition | Value/remark | Version |
|---|---|---|---|
| - SCCPCH information for FACH | | Not Present | R99 and Rel-4 only |
| - Downlink information for each radio link | A6, A10 | Not Present | |
| MBMS PL Service Restriction Information | A1,A2, A3,A4,A5, A6, A7, A8, A9, A10 | Not Present | Rel-6 |
| MBMS RB list released to change transfer mode | | Not Present | Rel-6 |

| Condition | Explanation | Version |
|---|---|---|
| A1 | This IE need for "Non speech in CS" | |
| A2 | This IE need for "Speech in CS" | |
| A3 | This IE need for "Packet to CELL_DCH from CELL_DCH in PS" | |
| A4 | This IE need for "Packet to CELL_DCH from CELL_FACH in PS" | |
| A5 | This IE need for "Packet to CELL_FACH from CELL_DCH in PS" | |
| A6 | This IE need for "Packet to CELL_FACH from CELL_FACH in PS" | |
| A7 | This IE need for "Non speech to CELL_FACH from CELL_DCH in CS" | |
| A8 | This IE need for "Speech to CELL_FACH from CELL_DCH in CS" | |
| A9 | This IE is needed for "Packet to CELL_DCH / HS-DSCH using three multiplexing options", or when not stated otherwise, for "Packet to CELL_DCH from CELL_DCH / HS-DSCH in PS" | Rel-5 |
| A10 | This IE is needed for "Packet to CELL_DCH / HS-DSCH using one multiplexing option", or when not stated otherwise, for "Packet to CELL_FACH from CELL_DCH / HS-DSCH in PS" | Rel-5 |

Contents of RADIO BEARER RELEASE COMPLETE message: AM

| Information Element | Value/remark |
|---|---|
| Message Type | |
| RRC transaction identifier | Checked to see the value is identical to the same IE in the downlink RADIO BEARER RELEASE message. |
| Integrity check info | |
| - Message authentication code | This IE is checked to see if it is present. The value is compared against the XMAC-I value computed by SS. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I. |
| - RRC Message sequence number | This IE is checked to see if it is present. The value is used by SS to compute the XMAC-I value. |
| Uplink integrity protection activation info | Not checked. |
| CHOICE mode | FDD |
| COUNT-C activation time | Not checked |
| Radio bearer uplink ciphering activation time info | Not checked |
| Uplink counter synchronization info | Not present |

Contents of RADIO BEARER RELEASE FAILURE message: AM

| Information Element | Value/remark |
|---|---|
| Message Type | |
| RRC transaction identitifer | Checked to see if it is set to identical value of the same IE in the downlink RADIO BEARER RELEASE message. |
| Integrity check info | |
| - Message authentication code | This IE is checked to see if it is present. The value is compared against the XMAC-I value computed by SS. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I. |
| - RRC Message sequence number | This IE is checked to see if it is present. The value is used by SS to compute the XMAC-I value. |
| Failure cause | Checked to see if it meets test requirement |
| Radio bearers for which reconfiguration would have succeeded | Not checked |

Contents of RRC CONNECTION REQUEST message: TM

| Information Element | Value/remark | Version |
|---|---|---|
| Message Type | | |

| | | |
|---|---|---|
| Predefined configuration status information | To be checked against requirement if specified | Rel-5 |
| Initial UE identity | | |
| - CHOICE UE id type | | |
| - TMSI and LAI (GSM-MAP) | Set to the UE's TMSI and LAI. | |
| Establishment cause | To be checked against requirement if specified | |
| Protocol error indicator | FALSE | |
| UE Specific Behaviour Information 1 idle | This IE will not be checked by default behaviour, but in specific test case. | |
| UE capability indication | To be checked against requirement if specified | Rel-6 |
| Measured results on RACH | To be checked against requirement if specified | Rel-4 |
| Access stratum release indicator | To be checked against requirement if specified | Rel-4 |

Contents of RRC CONNECTION REJECT message: UM

| Information Element | Value/remark |
|---|---|
| Message Type | |
| RRC transaction identifier | Arbitrarily selects an integer between 0 and 3 |
| Initial UE identity | Select the same type as in the IE "Initial UE Identity" in RRC CONNECTION REQUEST" message. |
| Rejection cause | Unspecified |
| Wait Time | 0 |
| Redirection info | Not Present |

Contents of RRC CONNECTION RELEASE message: UM

| Information Element | Value/remark | Version |
|---|---|---|
| Message Type | | |
| U-RNTI | This IE is set to the following value when the message is transmitted on the CCCH. When transmitted on DCCH, this is absent. | R99, Rel-4 |
| - SRNC identity | 0000 0000 0001B | |
| - S-RNTI | 0000 0000 0000 0000 0001B | |
| CHOICE identity type | This IE is set to the following value when the message is transmitted on the CCCH. When transmitted on DCCH, this is absent. | Rel-5 |
| - U-RNTI | | |
| - SRNC identity | 0000 0000 0001B | |
| - S-RNTI | 0000 0000 0000 0000 0001B | |
| - Group identity | [FFS] | |
| - Group release information | [FFS] | |
| RRC transaction identifier | Arbitrarily selects an integer between 0 and 3 | |
| Integrity check info | This IE is present when this message is transmitted on downlink DCCH. Else, this IE and the sub-IEs are omitted. | |
| - Message authentication code | SS calculates the value of MAC-I for this message and writes to this IE. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I. | |
| - RRC Message sequence number | SS provides the value of this IE, from its internal counter. | |
| N308 | 2 (for CELL_DCH state). Not Present (for UE in other connected mode states). | |
| Release cause | Normal event | |
| Rplmn information | Not Present | |

Contents of RRC CONNECTION RELEASE COMPLETE message: AM or UM

| Information Element | Semantics description |
|---|---|
| Message Type | |
| RRC transaction identifier | The value of this IE is checked to see that it matches the value of the same IE transmitted in the downlink RRC CONNECTION RELEASE message. |
| Integrity check info | |
| - Message authentication code | Checked to see if it's identical to the value of XMAC-I calculated by the SS. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I. |

| | |
|---|---|
| - RRC Message sequence number | Checked to see if it is present. This number is used by the SS to compute the XMAC-I |
| Error indication | Not checked |

Contents of RRC CONNECTION SETUP message: UM (Transition to CELL_DCH)

| Information Element | Value/remark | Version |
|---|---|---|
| Message Type | | |
| Initial UE identity | Select the same identity as in the IE "Initial UE Identity" in received RRC CONNECTION REQUEST" message | |
| RRC transaction identifier | Arbitrarily selects an integer between 0 and 3 | |
| Activation time | Not Present(Now) | |
| New U-RNTI | | |
| - SRNC identity | 0000 0000 0001B | |
| - S-RNTI | 0000 0000 0000 0000 0001B | |
| New C-RNTI | Not present | |
| New H-RNTI | Not present | Rel-6 |
| New Primary E-RNTI | Not present | Rel-6 |
| New Secondary E-RNTI | Not present | Rel-6 |
| RRC State Indicator | CELL_DCH | |
| UTRAN DRX cycle length coefficient | 9 | |
| Capability update requirement | | |
| - UE radio access FDD capability update requirement | TRUE | |
| - UE radio access TDD capability update requirement | FALSE | |
| - System specific capability update requirement list | GSM | |
| CHOICE *specification mode* | Complete specification | Rel-5 |
| - Complete specification | | Rel-5 |
| - Signalling RB information to setup | (UM DCCH for RRC) | |
| - RB identity | Not Present | |
| - CHOICE RLC info type | | |
| - RLC info | | |
| - CHOICE Uplink RLC mode | UM RLC | |
| - Transmission RLC discard | Not Present | |
| - CHOICE Downlink RLC mode | UM RLC | |
| - RB mapping info | | |
| - Information for each multiplexing option | 2 RBMuxOptions | |
| - RLC logical channel mapping indicator | Not Present | |
| - Number of RLC logical channels | 1 | |
| - Uplink transport channel type | DCH | |
| - UL Transport channel identity | 5 | |
| - Logical channel identity | 1 | |
| - CHOICE RLC size list | Configured | |
| - MAC logical channel priority | 1 | |
| - Downlink RLC logical channel info | | |
| - Number of RLC logical channels | 1 | |
| - Downlink transport channel type | DCH | |
| - DL DCH Transport channel identity | 10 | |
| - DL DSCH Transport channel identity | Not Present | |
| - Logical channel identity | 1 | |
| - RLC logical channel mapping indicator | Not Present | |
| - Number of RLC logical channels | 1 | |
| - Uplink transport channel type | RACH | |
| - UL Transport channel identity | Not Present | |
| - Logical channel identity | 1 | |
| - CHOICE RLC size list | Explicit List | |
| - RLC size index | According to clause 6.10.2.4.1.3 (standalone 13.6 kbps signalling radio bearer) | |
| - MAC logical channel priority | 1 | |
| - Downlink RLC logical channel info | | |

| Information Element | Value/remark | Version |
|---|---|---|
| - Number of RLC logical channels | 1 | |
| - Downlink transport channel type | FACH | |
| - DL DCH Transport channel identity | Not Present | |
| - DL DSCH Transport channel identity | Not Present | |
| - Logical channel identity | 1 | |
| - Signalling RB information to setup | (AM DCCH for RRC) | |
| - RB identity | Not Present | |
| - CHOICE RLC info type | | |
| - RLC info | | |
| - CHOICE Uplink RLC mode | AM RLC | |
| - Transmission RLC discard | | |
| - SDU discard mode | No discard | |
| - MAX_DAT | 15 | |
| - Transmission window size | 32 | |
| - Timer_RST | 500 | |
| - Max_RST | 1 | |
| - Polling info | | |
| - Timer_poll_prohibit | 200 | |
| - Timer_poll | 200 | |
| - Poll_PDU | Not Present | |
| - Poll_SDU | 1 | |
| - Last transmission PDU poll | TRUE | |
| - Last retransmission PDU poll | TRUE | |
| - Poll_Window | 99 | |
| - Timer_poll_periodic | Not Present | |
| - CHOICE Downlink RLC mode | AM RLC | |
| - In-sequence delivery | TRUE | |
| - Receiving window size | 32 | |
| - Downlink RLC status info | | |
| - Timer_status_prohibit | 200 | |
| - Timer_EPC | Not Present | |
| - Missing PDU indicator | TRUE | |
| - Timer_STATUS_periodic | Not Present | |
| - RB mapping info | | |
| - Information for each multiplexing option | 2 RBMuxOptions | |
| - RLC logical channel mapping indicator | Not Present | |
| - Number of RLC logical channels | 1 | |
| - Uplink transport channel type | DCH | |
| - UL Transport channel identity | 5 | |
| - Logical channel identity | 2 | |
| - CHOICE RLC size list | Configured | |
| - MAC logical channel priority | 2 | |
| - Downlink RLC logical channel info | | |
| - Number of RLC logical channels | 1 | |
| - Downlink transport channel type | DCH | |
| - DL DCH Transport channel identity | 10 | |
| - DL DSCH Transport channel identity | Not Present | |
| - Logical channel identity | 2 | |
| - RLC logical channel mapping indicator | Not Present | |
| - Number of RLC logical channels | 1 | |
| - Uplink transport channel type | RACH | |
| - UL Transport channel identity | Not Present | |
| - Logical channel identity | 2 | |
| - CHOICE RLC size list | Explicit List | |
| - RLC size index | According to clause 6.10.2.4.1.3 (standalone 13.6 kbps signalling radio bearer) | |
| - MAC logical channel priority | 2 | |
| - Downlink RLC logical channel info | | |
| - Number of RLC logical channels | 1 | |
| - Downlink transport channel type | FACH | |

| Information Element | Value/remark | Version |
|---|---|---|
| - DL DCH Transport channel identity | Not Present | |
| - DL DSCH Transport channel identity | Not Present | |
| - Logical channel identity | 2 | |
| - Signalling RB information to setup | (AM DCCH for NAS_DT High priority) | |
| - RB identity | Not Present | |
| - CHOICE RLC info type | | |
| - RLC info | | |
| - CHOICE Uplink RLC mode | AM RLC | |
| - Transmission RLC discard | | |
| - SDU discard mode | No discard | |
| - MAX_DAT | 15 | |
| - Transmission window size | 32 | |
| - Timer_RST | 500 | |
| - Max_RST | 1 | |
| - Polling info | | |
| - Timer_poll_prohibit | 200 | |
| - Timer_poll | 200 | |
| - Poll_PDU | Not present | |
| - Poll_SDU | 1 | |
| - Last transmission PDU poll | TRUE | |
| - Last retransmission PDU poll | TRUE | |
| - Poll_Window | 99 | |
| - Timer_poll_periodic | Not Present | |
| - CHOICE Downlink RLC mode | AM RLC | |
| - In-sequence delivery | TRUE | |
| - Receiving window size | 32 | |
| - Downlink RLC status info | | |
| - Timer_status_prohibit | 200 | |
| - Timer_EPC | Not present | |
| - Missing PDU indicator | TRUE | |
| - Timer_STATUS_periodic | Not Present | |
| - RB mapping info | | |
| - Information for each multiplexing option | 2 RBMuxOptions | |
| - RLC logical channel mapping indicator | Not Present | |
| - Number of RLC logical channels | 1 | |
| - Uplink transport channel type | DCH | |
| - UL Transport channel identity | 5 | |
| - Logical channel identity | 3 | |
| - CHOICE RLC size list | Configured | |
| - MAC logical channel priority | 3 | |
| - Downlink RLC logical channel info | | |
| - Number of RLC logical channels | 1 | |
| - Downlink transport channel type | DCH | |
| - DL DCH Transport channel identity | 10 | |
| - DL DSCH Transport channel identity | Not Present | |
| - Logical channel identity | 3 | |
| - RLC logical channel mapping indicator | Not Present | |
| - Number of RLC logical channels | 1 | |
| - Uplink transport channel type | RACH | |
| - UL Transport channel identity | Not Present | |
| - Logical channel identity | 3 | |
| - CHOICE RLC size list | Explicit List | |
| - RLC size index | According to clause 6.10.2.4.1.3 (standalone 13.6 kbps signalling radio bearer) | |
| - MAC logical channel priority | 3 | |
| - Downlink RLC logical channel info | | |
| - Number of RLC logical channels | 1 | |
| - Downlink transport channel type | FACH | |
| - DL DCH Transport channel identity | Not Present | |

| Information Element | Value/remark | Version |
|---|---|---|
| - DL DSCH Transport channel identity | Not Present | |
| - Logical channel identity | 3 | |
| - Signalling RB information to setup | (AM DCCH for NAS_DT Low priority) | |
| - RB identity | Not Present | |
| - CHOICE RLC info type | | |
| - RLC info | | |
| - CHOICE Uplink RLC mode | AM RLC | |
| - Transmission RLC discard | | |
| - SDU discard mode | No discard | |
| - MAX_DAT | 15 | |
| - Transmission window size | 32 | |
| - Timer_RST | 500 | |
| - Max_RST | 1 | |
| - Polling info | | |
| - Timer_poll_prohibit | 200 | |
| - Timer_poll | 200 | |
| - Poll_PDU | Not present | |
| - Poll_SDU | 1 | |
| - Last transmission PDU poll | TRUE | |
| - Last retransmission PDU poll | TRUE | |
| - Poll_Window | 99 | |
| - Timer_poll_periodic | Not Present | |
| - CHOICE Downlink RLC mode | AM RLC | |
| - In-sequence delivery | TRUE | |
| - Receiving window size | 32 | |
| - Downlink RLC status info | | |
| - Timer_status_prohibit | 200 | |
| - Timer_EPC | Not Present | |
| - Missing PDU indicator | TRUE | |
| - Timer_STATUS_periodic | Not Present | |
| - RB mapping info | | |
| - Information for each multiplexing option | 2 RBMuxOptions | |
| - RLC logical channel mapping indicator | Not Present | |
| - Number of RLC logical channels | 1 | |
| - Uplink transport channel type | DCH | |
| - UL Transport channel identity | 5 | |
| - Logical channel identity | 4 | |
| - CHOICE RLC size list | Configured | |
| - MAC logical channel priority | 4 | |
| - Downlink RLC logical channel info | | |
| - Number of RLC logical channels | 1 | |
| - Downlink transport channel type | DCH | |
| - DL DCH Transport channel identity | 10 | |
| - DL DSCH Transport channel identity | Not Present | |
| - Logical channel identity | 4 | |
| - RLC logical channel mapping indicator | Not Present | |
| - Number of RLC logical channels | 1 | |
| - Uplink transport channel type | RACH | |
| - UL Transport channel identity | Not Present | |
| - Logical channel identity | 4 | |
| - CHOICE RLC size list | Explicit List | |
| - RLC size index | According to clause 6.10.2.4.1.3 (standalone 13.6 kbps signalling radio bearer) | |
| - MAC logical channel priority | 4 | |
| - Downlink RLC logical channel info | | |
| - Number of RLC logical channels | 1 | |
| - Downlink transport channel type | FACH | |
| - DL DCH Transport channel identity | Not Present | |
| - DL DSCH Transport channel identity | Not Present | |

| Information Element | Value/remark | Version |
|---|---|---|
| - Logical channel identity | 4 | |
| UL Transport channel information for all transport channels | | |
| - PRACH TFCS | Not Present | |
| - CHOICE Mode | FDD | |
| - TFC subset | Nor Present | |
| - UL DCH TFCS | | |
| - CHOICE TFCI signalling | Normal | |
| - TFCI Field 1 information | | |
| - CHOICE TFCS representation | Complete | |
| - TFCS complete reconfigure | | |
| - CHOICE CTFC Size | 2bit CTFC | |
| - CTFC information | This IE is repeated for TFC numbers according to clause 6.10.2.4.1.3 (standalone 13.6 kbps signalling radio bearer) | |
| - CTFC | According to clause 6.10.2.4.1.3 (standalone 13.6 kbps signalling radio bearer) | |
| - Power offset information | | |
| - CHOICE Gain Factors | Computed Gain Factors (The last TFC is set to Signalled Gain Factors) | |
| - Gain factor ßc | 11 (below 64 kbps)<br>9 (higher than 64 kbps)<br>(Not Present if the above is set to Computed Gain Factors) | |
| - Gain factor ßd | 15<br>(Not Present if the above is set to Computed Gain Factors) | |
| - Reference TFC ID | 0 | |
| - CHOICE mode | FDD | |
| - Power offset Pp-m | Not Present | |
| Added or Reconfigured UL TrCH information | | |
| - Uplink transport channel type | DCH | |
| - UL Transport channel identity | 5 | |
| - TFS | | |
| - CHOICE Transport channel type | Dedicated transport channels | |
| - Dynamic Transport format information | | |
| - RLC size | According to clause 6.10.2.4.1.3 (standalone 13.6 kbps signalling radio bearer) | |
| - Number of TBs and TTI lists | (This IE is repeated for TFI number) | |
| - Transmission Time Interval | According to clause 6.10.2.4.1.3 (standalone 13.6 kbps signalling radio bearer) | |
| - Number of Transport blocks | According to clause 6.10.2.4.1.3 (standalone 13.6 kbps signalling radio bearer) | |
| - CHOICE Logical channel list | All | |
| - Semi-static Transport Format information | | |
| - Transmission time interval | According to clause 6.10.2.4.1.3 (standalone 13.6 kbps signalling radio bearer) | |
| - Type of channel coding | According to clause 6.10.2.4.1.3 (standalone 13.6 kbps signalling radio bearer) | |
| - Coding Rate | According to clause 6.10.2.4.1.3 (standalone 13.6 kbps signalling radio bearer) | |
| - Rate matching attribute | According to clause 6.10.2.4.1.3 (standalone 13.6 kbps signalling radio bearer) | |
| - CRC size | According to clause 6.10.2.4.1.3 (standalone 13.6 kbps signalling radio bearer) | |
| DL Transport channel information common for all transport channel | | |
| - SCCPCH TFCS | Not Present | |
| - CHOICE mode | FDD | |
| - CHOICE DL parameters | Same as UL | |
| Added or Reconfigured DL TrCH information | | |
| - Downlink transport channel type | DCH | |
| - DL Transport channel identity | 10 | |
| - CHOICE DL parameters | Same as UL | |
| - Uplink transport channel type | DCH | |
| - UL TrCH Identity | 5 | |

| Information Element | Value/remark | Version |
|---|---|---|
| - DCH quality target | | |
| - BLER Quality value | -2.0 | |
| Frequency info | Not Present | |
| Maximum allowed UL TX power | Not Present | |
| Uplink DPCH info | | |
| - Uplink DPCH power control info | | |
| - DPCCH power offset | -80dB (i.e. ASN.1 IE value of -40) | |
| - PC Preamble | 1 frame | |
| - SRB delay | 7 frames | |
| - Power Control Algorithm | Algorithm1 | |
| - TPC step size | 1dB | |
| - Scrambling code type | Long | |
| - Scrambling code number | 0 (0 to 16777215) | |
| - Number of DPDCH | Not Present(1) | |
| - Spreading factor | According to clause 6.10.2.4.1.3 (standalone 13.6 kbps signalling radio bearer) | |
| - TFCI existence | According to clause 6.10.2.4.1.3 (standalone 13.6 kbps signalling radio bearer) | |
| - Number of FBI bit | According to clause 6.10.2.4.1.3 (standalone 13.6 kbps signalling radio bearer) | |
| - Puncturing Limit | According to clause 6.10.2.4.1.3 (standalone 13.6 kbps signalling radio bearer) | |
| E-DCH Info | Not Present | Rel-6 |
| Downlink HS-PDSCH Information | Not Present | Rel-6 |
| Downlink information common for all radio links | | |
| - Downlink DPCH info common for all RL | | |
| - Timing Indication | Initialize | |
| - CFN-targetSFN frame offset | Not Present | |
| - CHOICE mode | FDD | |
| - Downlink DPCH power control information | | |
| - DPC mode | 0 (single) | |
| - Power offset $P_{Pilot-DPDCH}$ | 0 | |
| - DL rate matching restriction information | Not Present | |
| - Spreading factor | According to clause 6.10.2.4.1.3 (standalone 13.6 kbps signalling radio bearer) | |
| - Fixed or Flexible Position | According to clause 6.10.2.4.1.3 (standalone 13.6 kbps signalling radio bearer) | |
| - TFCI existence | According to clause 6.10.2.4.1.3 (standalone 13.6 kbps signalling radio bearer) | |
| - CHOICE SF | According to clause 6.10.2.4.1.3 (standalone 13.6 kbps signalling radio bearer) | |
| - DPCH compressed mode info | Not Present | |
| - TX Diversity mode | None | |
| - SSDT information | Not Present | R99 and Rel-4 only |
| - Default DPCH Offset Value | Arbitrary set to value 0..306688 by step of 512 | |
| Downlink information for each radio links list | | |
| - Downlink information for each radio links | | |
| - CHOICE mode | FDD | |
| - Primary CPICH info | | |
| - Primary scrambling code | Reference to clause 6.1 "Default settings (FDD)" | |
| - PDSCH with SHO DCH info | Not Present | R99 and Rel-4 only |
| - PDSCH code mapping | Not Present | R99 and Rel-4 only |
| - Serving E-DCH radio link indicator | FALSE | Rel-6 |
| - Downlink DPCH info for each RL | | |
| - Primary CPICH usage for channel estimation | Primary CPICH may be used | |
| - DPCH frame offset | Set to value: Default DPCH Offset Value mod 38 400 | |
| - Secondary CPICH info | Not Present | |

| Information Element | Value/remark | Version |
|---|---|---|
| - DL channelisation code | | |
| - Secondary scrambling code | 1 | |
| - Spreading factor | According to clause 6.10.2.4.1.3 (standalone 13.6 kbps signalling radio bearer) | |
| - Code number | 0 | |
| - Scrambling code change | Not Present | |
| - TPC combination index | 0 | |
| - SSDT Cell Identity | Not Present | R99 and Rel-4 only |
| - Closed loop timing adjustment mode | Not Present | |
| - E-AGCH Info | Not Present | Rel-6 |
| - E-HICH Information | Not Present | Rel-6 |
| - E-RGCH Information | Not Present | Rel-6 |
| - SCCPCH information for FACH | Not Present | R99 and Rel-4 only |

Contents of RRC CONNECTION SETUP message: UM (Transition to CELL_FACH)

| Information Element | Value/remark | Version |
|---|---|---|
| Message Type | | |
| Initial UE identity | Select the same identity as in the IE "Initial UE Identity" in received RRC CONNECTION REQUEST" message | |
| RRC transaction identifier | Arbitrarily selects an integer between 0 and 3 | |
| Activation time | Not Present (Now) | |
| New U-RNTI | | |
| - SRNC identity | 0000 0000 0001B | |
| - S-RNTI | 0000 0000 0000 0000 0001B | |
| New C-RNTI | 0000 0000 0000 0001B | |
| New H-RNTI | Not present | Rel-6 |
| New Primary E-RNTI | Not present | Rel-6 |
| New Secondary E-RNTI | Not present | Rel-6 |
| RRC state indicator | CELL_FACH | |
| UTRAN DRX cycle length coefficient | 9 | |
| Capability update requirement | | |
| - UE radio access FDD capability update requirement | TRUE | |
| - UE radio access TDD capability update requirement | FALSE | |
| - System specific capability update requirement list | GSM | |
| CHOICE *specification mode* | Complete specification | Rel-5 |
| - Complete specification | | Rel-5 |
| - Signalling RB information to setup | (UM DCCH for RRC) | |
| - RB identity | Not present | |
| - CHOICE RLC info type | RLC info | |
| - CHOICE Uplink RLC mode | UM RLC | |
| - Transmission RLC discard | timerBasedNoExplicit : dt50 | |
| - SDU discard mode | Not present | |
| - CHOICE Downlink RLC mode | UM RLC | |
| - RB mapping info | | |
| - Information for each multiplexing option | 2 RBMuxOptions | |
| - RLC logical channel mapping indicator | Not Present | |
| - Number of uplink RLC logical channels | 1 | |
| - Uplink transport channel type | DCH | |
| - UL Transport channel identity | 5 | |
| - Logical channel identity | 1 | |
| - CHOICE RLC size list | Configured | |
| - MAC logical channel priority | 1 | |
| - Downlink RLC logical channel info | | |
| - Number of downlink RLC logical channels | 1 | |
| - Downlink transport channel type | DCH | |

| Information Element | Value/remark | Version |
|---|---|---|
| - DL DCH Transport channel identity | 10 | |
| - DL DSCH Transport channel identity | Not Present | |
| - Logical channel identity | 1 | |
| - RLC logical channel mapping indicator | Not Present | |
| - Number of uplink RLC logical channels | 1 | |
| - Uplink transport channel type | RACH | |
| - UL Transport channel identity | Not Present | |
| - Logical channel identity | 1 | |
| - CHOICE RLC size list | Explicit list | |
| - RLC size index | According to clause 6.10.2.4.4.1 | |
| - MAC logical channel priority | 1 | |
| - Downlink RLC logical channel info | | |
| - Number of downlink RLC logical channels | 1 | |
| - Downlink transport channel type | FACH | |
| - DL DCH Transport channel identity | Not Present | |
| - DL DSCH Transport channel identity | Not Present | |
| - Logical channel identity | 1 | |
| - Signalling RB information to setup | (AM DCCH for RRC) | |
| - RB identity | Not Present | |
| - CHOICE RLC info type | RLC info | |
| - CHOICE Uplink RLC mode | AM RLC | |
| - Transmission RLC discard | | |
| - SDU discard mode | No Discard | |
| - MAX_DAT | 15 | |
| - Transmission window size | 32 | |
| - Timer_RST | 500 | |
| - Max_RST | 1 | |
| - Polling info | | |
| - Timer_poll_prohibit | 200 | |
| - Timer_poll | 200 | |
| - Poll_PDU | Not Present | |
| - Poll_SDU | 1 | |
| - Last transmission PDU poll | TRUE | |
| - Last retransmission PDU poll | TRUE | |
| - Poll_Windows | 99 | |
| - Timer_poll_periodic | Not Present | |
| - CHOICE Downlink RLC mode | AM RLC | |
| - In-sequence delivery | TRUE | |
| - Receiving window size | 32 | |
| - Downlink RLC status info | | |
| - Timer_status_prohibit | 200 | |
| - Timer_EPC | Not Present | |
| - Missing PDU indicator | TRUE | |
| - Timer_STATUS_periodic | Not Present | |
| - RB mapping info | | |
| - Information for each multiplexing option | 2 RBMuxOptions | |
| - RLC logical channel mapping indicator | Not Present | |
| - Number of uplink RLC logical channels | 1 | |
| - Uplink transport channel type | DCH | |
| - UL Transport channel identity | 5 | |
| - Logical channel identity | 2 | |
| - CHOICE RLC size list | Configured | |
| - MAC logical channel priority | 2 | |
| - Downlink RLC logical channel info | | |
| - Number of downlink RLC logical channels | 1 | |
| - Downlink transport channel type | DCH | |

| Information Element | Value/remark | Version |
|---|---|---|
| - DL DCH Transport channel identity | 10 | |
| - DL DSCH Transport channel identity | Not Present | |
| - Logical channel identity | 2 | |
| - RLC logical channel mapping indicator | Not Present | |
| - Number of uplink RLC logical channels | 1 | |
| - Uplink transport channel type | RACH | |
| - UL Transport channel identity | Not Present | |
| - Logical channel identity | 2 | |
| - CHOICE RLC size list | Explicit list | |
| - RLC size index | According to clause 6.10.2.4.4.1 | |
| - MAC logical channel priority | 2 | |
| - Downlink RLC logical channel info | | |
| - Number of downlink RLC logical channels | 1 | |
| - Downlink transport channel type | FACH | |
| - DL DCH Transport channel identity | Not Present | |
| - DL DSCH Transport channel identity | Not Present | |
| - Logical channel identity | 2 | |
| Signalling RB information to setup | (AM DCCH for NAS_DT High priority) | |
| - RB identity | Not present | |
| - CHOICE RLC info type | RLC info | |
| - CHOICE Uplink RLC mode | AM RLC | |
| - Transmission RLC discard | | |
| - SDU discard mode | No Discard | |
| - MAX_DAT | 15 | |
| - Transmission window size | 32 | |
| - Timer_RST | 500 | |
| - Max_RST | 1 | |
| - Polling info | | |
| - Timer_poll_prohibit | 200 | |
| - Timer_poll | 200 | |
| - Poll_PDU | Not Present | |
| - Poll_SDU | 1 | |
| - Last transmission PDU poll | TRUE | |
| - Last retransmission PDU poll | TRUE | |
| - Poll_Windows | 99 | |
| - Timer_poll_periodic | Not Present | |
| - CHOICE Downlink RLC mode | AM RLC | |
| - In-sequence delivery | TRUE | |
| - Receiving window size | 32 | |
| - Downlink RLC status info | | |
| - Timer_status_prohibit | 200 | |
| - Timer_EPC | Not Present | |
| - Missing PDU indicator | TRUE | |
| - Timer_STATUS_periodic | Not Present | |
| - RB mapping info | | |
| - Information for each multiplexing option | 2 RBMuxOptions | |
| - RLC logical channel mapping indicator | Not Present | |
| - Number of uplink RLC logical channels | 1 | |
| - Uplink transport channel type | DCH | |
| - UL Transport channel identity | 5 | |
| - Logical channel identity | 3 | |
| - CHOICE RLC size list | Configured | |
| - MAC logical channel priority | 3 | |
| - Downlink RLC logical channel info | | |
| - Number of downlink RLC logical channels | 1 | |
| - Downlink transport channel type | DCH | |

| Information Element | Value/remark | Version |
|---|---|---|
| - DL DCH Transport channel identity | 10 | |
| - DL DSCH Transport channel identity | Not Present | |
| - Logical channel identity | 3 | |
| - RLC logical channel mapping indicator | Not Present | |
| - Number of uplink RLC logical channels | 1 | |
| - Uplink transport channel type | RACH | |
| - UL DCH Transport channel identity | Not Present | |
| - Logical channel identity | 3 | |
| - CHOICE RLC size list | Explicit list | |
| - RLC size index | According to clause 6.10.2.4.4.1 | |
| - MAC logical channel priority | 3 | |
| - Downlink RLC logical channel info | | |
| - Number of downlink RLC logical channels | 1 | |
| - Downlink transport channel type | FACH | |
| - DL DCH Transport channel identity | Not Present | |
| - DL DSCH Transport channel identity | Not Present | |
| - Logical channel identity | 3 | |
| - Signalling RB information to setup | (AM DCCH for NAS_DT Low priority) | |
| - RB identity | Not Present | |
| - CHOICE RLC info type | RLC info | |
| - CHOICE Uplink RLC mode | AM RLC | |
| - Transmission RLC discard | | |
| - SDU discard mode | No Discard | |
| - MAX_DAT | 15 | |
| - Transmission window size | 32 | |
| - Timer_RST | 500 | |
| - Max_RST | 1 | |
| - Polling info | | |
| - Timer_poll_prohibit | 200 | |
| - Timer_poll | 200 | |
| - Poll_PDU | Not Present | |
| - Poll_SDU | 1 | |
| - Last transmission PDU poll | TRUE | |
| - Last retransmission PDU poll | TRUE | |
| - Poll_Windows | 99 | |
| - Timer_poll_periodic | Not Present | |
| - CHOICE Downlink RLC mode | AM RLC | |
| - In-sequence delivery | TRUE | |
| - Receiving window size | 32 | |
| - Downlink RLC status info | | |
| - Timer_status_prohibit | 200 | |
| - Timer_EPC | Not Present | |
| - Missing PDU indicator | TRUE | |
| - Timer_STATUS_periodic | Not Present | |
| - RB mapping info | | |
| - Information for each multiplexing option | 2 RBMuxOptions | |
| - RLC logical channel mapping indicator | Not Present | |
| - Number of uplink RLC logical channels | 1 | |
| - Uplink transport channel type | DCH | |
| - UL Transport channel identity | 5 | |
| - Logical channel identity | 4 | |
| - CHOICE RLC size list | Configured | |
| - MAC logical channel priority | 4 | |
| - Downlink RLC logical channel info | | |
| - Number of downlink RLC logical channels | 1 | |

| Information Element | Value/remark | Version |
|---|---|---|
| - Downlink transport channel type | DCH | |
| - DL DCH Transport channel identity | 10 | |
| - DL DSCH Transport channel identity | Not Present | |
| - Logical channel identity | 4 | |
| - RLC logical channel mapping indicator | Not Present | |
| - Number of uplink RLC logical channels | 1 | |
| - Uplink transport channel type | RACH | |
| - UL Transport channel identity | Not Present | |
| - Logical channel identity | 4 | |
| - CHOICE RLC size list | Explicit list | |
| - RLC size index | According to clause 6.10.2.4.4.1 | |
| - MAC logical channel priority | 4 | |
| - Downlink RLC logical channel info | | |
| - Number of downlink RLC logical channels | 1 | |
| - Downlink transport channel type | FACH | |
| - DL DCH Transport channel identity | Not Present | |
| - DL DSCH Transport channel identity | Not Present | |
| - Logical channel identity | 4 | |
| UL Transport channel information for all transport channels | | |
| - PRACH TFCS | Not Present | |
| - CHOICE Mode | FDD | |
| - TFC subset | Not Present | |
| - UL DCH TFCS | | |
| - CHOICE TFCI signalling | Normal | |
| - TFCI Field 1 information | | |
| - CHOICE TFCS representation | Complete | |
| - TFCS complete reconfigure | | |
| - CHOICE CTFC Size | 2bit CTFC | |
| - CTFC information | This IE is repeated for TFC numbers according to clause 6.10.2.4.1.3 (standalone 13.6 kbps signalling radio bearer) | |
| - CTFC | According to clause 6.10.2.4.1.3 (standalone 13.6 kbps signalling radio bearer) | |
| - Power offset information | | |
| - CHOICE Gain Factors | Computed Gain Factors (The last TFC is set to Signalled Gain Factors) | |
| - Gain factor ßc | 11 (below 64 kbps)<br>9 (higher than 64 kbps)<br>(Not Present if the above is set to Computed Gain Factors) | |
| - Gain factor ßd | 15<br>(Not Present if the above is set to Computed Gain Factors) | |
| - Reference TFC ID | 0 | |
| - CHOICE mode | FDD | |
| - Power offset Pp-m | Not Present | |
| Added or Reconfigured TrCH information list | TS 25.331 specifies that "Although this IE is not required when the IE "RRC state indicator" is set to "CELL_FACH", need is MP to align with ASN.1" | |
| - Added or Reconfigured UL TrCH information | | |
| - Uplink transport channel type | DCH | |
| - UL Transport channel identity | 5 | |
| - TFS | | |
| - CHOICE Transport channel type | Dedicated transport channels | |
| - Dynamic Transport format information | | |
| - RLC Size | bitMode sizeType2 {part1 2, part2 OMIT}<br>This results in an RLC size of 144 bits | |
| - Number of TBs and TTI List | List with two entry | |
| - Transmission Time Interval | Not Present | |
| - Number of Transport blocks | 0 | |

| Information Element | Value/remark | Version |
|---|---|---|
| - Transmission Time Interval | Not Present | |
| - Number of Transport blocks | 1 | |
| - CHOICE Logical channel List | ALL | |
| - Semi-static Transport Format information | | |
| - Transmission time interval | 40 ms | |
| - Type of channel coding | Convolutional | |
| - Coding Rate | 1/3 | |
| - Rate matching attribute | -170 | |
| - CRC size | 16 | |
| DL Transport channel information common for all transport channel | | |
| - SCCPCH TFCS | Not Present | |
| - CHOICE mode | FDD | |
| - CHOICE DL parameters | Same as UL | |
| Added or Reconfigured TrCH information list | TS 25.331 specifies that "Although this IE is not required when the IE "RRC state indicator" is set to "CELL_FACH", need is MP to align with ASN.1" | |
| - Added or Reconfigured DL TrCH information | | |
| - Downlink transport channel type | DCH | |
| - DL Transport channel identity | 10 | |
| - CHOICE DL parameters | Same as UL | |
| - Uplink Transport channel type | DCH | |
| - UL TrCH identity | 5 | |
| - DCH quality target | Not Present | |
| Frequency info | Not present | |
| Maximum allowed UL TX power | Not present | |
| CHOICE channel requirement | Not Present | |
| E-DCH Info | Not Present | Rel-6 |
| Downlink HS-PDSCH Information | Not Present | Rel-6 |
| Downlink information common for all radio links | Not Present | |
| Downlink information for each radio link list | Not present | |

Contents of RRC CONNECTION SETUP COMPLETE message: AM

| Information Element | Value/remark |
|---|---|
| Message Type | |
| RRC transaction identifier | The value of this IE is checked to see that it matches the value of the same IE transmitted in the downlink RRC CONNECTION SETUP message. |
| START list | This IE is checked to see if it is present. |
| UE radio access capability | Not checked |
| UE radio access capability extension | Not checked |
| UE system specific capability | Not checked |

Contents of RRC STATUS message: AM

| Information Element | Value/remark |
|---|---|
| Message Type | |
| Integrity check info | |
| - Message authentication code | This IE is checked to see if it is present. The value is compared against the XMAC-I value computed by SS. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I. |
| - RRC Message sequence number | This IE is checked to see if it is present. The value is used by SS to compute the XMAC-I value. |
| Identification of received message | Not Checked |
| Protocol error information | |
| - Protocol error cause | Refer to test requirement. |

Contents of SECURITY MODE COMMAND message: AM

| Information Element | Condition | Value/remark |
|---|---|---|

| Information Element | Condition | Value/remark |
|---|---|---|
| Message Type | A1, A2 | |
| RRC transaction identifier | | Arbitrarily selects an integer between 0 and 3 |
| Integrity check info | | |
| - Message authentication code | | Set to MAC-I value computed by the SS. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I. |
| - RRC Message Sequence Number | | Set to an arbitrarily selected integer between 0 and 15 |
| Security capability | | |
| - Ciphering algorithm capability | | |
| - UEA0 | | If the UE has indicated support for ciphering algorithm UEA0 in the IE "security capability" in the RRC CONNECTION SETUP COMPLETE message, this IE is set to TRUE. |
| - UEA1 | | If the UE has indicated support for ciphering algorithm UEA1 in the IE "security capability" in the RRC CONNECTION SETUP COMPLETE message, this IE is set to TRUE. |
| - Spare | | Spare 2-15 = FALSE |
| - Integrity protection algorithm capability | | 0000000000000010B (UIA1) |
| - UIA1 | | TRUE |
| - Spare | | Spare 0 and Spare 2-15 = FALSE |
| Ciphering mode info | | This presence of this IE is dependent on IXIT statements in TS 34.123-2. If ciphering is indicated to be active, this IE present with the values of the sub IEs as stated below. Else, this IE is omitted. |
| - Ciphering mode command | | Start/restart |
| - Ciphering algorithm | | UEA0 or UEA1. The indicated algorithm must be one of the algorithms supported by the UE as indicated in the IE "security capability" in the RRC CONNECTION SETUP COMPLETE message. |
| - Ciphering activation time for DPCH | | Not Present |
| - Radio bearer downlink ciphering activation time info | | |
| - Radio bearer activation time | | |
| - RB identity | | 1 |
| - RLC sequence number | | Current RLC SN |
| - RB identity | | 2 |
| - RLC sequence number | | Current RLC SN+2 |
| - RB identity | | 3 |
| - RLC sequence number | | Current RLC SN |
| - RB identity | | 4 |
| - RLC sequence number | | Current RLC SN |
| Integrity protection mode info | | |
| - Integrity protection mode command | | Start |
| - Downlink integrity protection activation info | | Not Present |
| - Integrity protection algorithm | | UIA1 |
| - Integrity protection initialisation number | | SS selects an arbitrary 32 bits number for FRESH |
| CN domain identity | | CS or PS |
| UE system specific security capability | A1 | Not Checked |
| UE system specific security capability | A2 | |
| - Inter-RAT UE security capability | | |
| - CHOICE *system* | | GSM |
| - GSM security capability | | The indicated algorithms must be the same as the algorithms supported by the UE as indicated in the IE " UE system specific capability " in the RRC CONNECTION SETUP COMPLETE message. |

| Condition | Explanation |
|---|---|
| A1 | UE not supporting GSM |
| A2 | UE supporting GSM |

Contents of SECURITY MODE COMPLETE message: AM

| Information Element | Value/remark |
|---|---|
| Message Type | |
| RRC transaction identifier | The value of this IE is checked to see that it matches the value of the same IE transmitted in the downlink SECURITY MODE COMMAND message. |
| Integrity check info | |
| - Message authentication code | This IE is checked to see if it is present. The value is compared against the XMAC-I value computed by SS. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I. |
| - RRC Message sequence number | This IE is checked to see if it is present. The value is used by SS to compute the XMAC-I value. |
| Uplink integrity protection activation info | Not checked. |
| Radio bearer uplink ciphering activation time info | If ciphering is not activated in SECURITY MODE COMMAND message, this IE must be absent. Else, SS checks this IE for the presence of activation times for all ciphered uplink RLC-UM and RLC-AM RBs. |

Contents of SECURITY MODE FAILURE message: AM

| Information Element | Value/remark |
|---|---|
| Message Type | |
| RRC transaction identifier | Checked to see if the value is the identical to the same IE in the downlink SECURITY MODE COMMAND message. |
| Integrity check info | |
| - Message authentication code | This IE is checked to see if it is present. The value is compared against the XMAC-I value computed by SS. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I. |
| - RRC Message sequence number | This IE is checked to see if it is present. The value is used by SS to compute the XMAC-I value. |
| Failure cause | Refer to test requirement. |

Contents of TRANSPORT CHANNEL RECONFIGURATION message: AM or UM

| Information Element | Condition | Value/remark | Version |
|---|---|---|---|
| Message Type | A1, A2, A3, A4, A5, A6 | | |
| RRC transaction identifier | | Arbitrarily selects an integer between 0 and 3 | |
| Integrity check info | | | |
| - message authentication code | | SS calculates the value of MAC-I for this message and writes to this IE. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I. | |
| - RRC message sequence number | | SS provides the value of this IE, from its internal counter. | |
| Integrity protection mode info | | Not Present | |
| Ciphering mode info | | Not Present | |
| Activation time | A1, A2, A3 | (256+CFN-(CFN MOD 8 + 8))MOD 256 | |
| Activation time | A4, A5, A6 | Not Present | |
| Delay restriction flag | A1, A2, A3, A4, A5, A6 | Not Present | Rel-6 |
| New U-RNTI | | Not Present | |
| New C-RNTI | A1, A2, A3, A4 | Not Present | |
| New C-RNTI | A5, A6 | '1010 1010 1010 1010' | |
| New DSCH-RNTI | A1, A2, A3, A4, A5, A6 | Not Present | R99 and Rel-4 only |
| New H-RNTI | A1, A2, A3, A4, A5, A6 | Not Present | Rel-5 |
| New Primary E-RNTI | | Not Present | Rel-6 |
| New Secondary E-RNTI | | Not Present | Rel-6 |
| RRC State indicator | A1, A2, A3, A4 | CELL_DCH | |
| RRC State indicator | A5, A6 | CELL_FACH | |

| Information Element | Condition | Value/remark | Version |
|---|---|---|---|
| UTRAN DRX cycle length coefficient | A1, A2, A3, A4,A5,A6 | Not Present | |
| CN information info | | Not Present | |
| URA identity | | Not Present | |
| Downlink counter synchronization info | | Not Present | |
| UL Transport channel information for all transport channels | A1, A2, A5, A6 | Not Present | |
| UL Transport channel information for all transport channels | A3, A4 | | |
| - PRACH TFCS | | Not Present | |
| - CHOICE mode | | FDD | |
| - TFC subset | | Not Present | |
| - UL DCH TFCS | | | |
| - CHOICE TFCI signalling | | Normal | |
| - TFCI Field 1 information | | | |
| - CHOICE TFCS representation | | Complete reconfiguration | |
| - TFCS complete reconfigure information | | | |
| - CHOICE CTFC Size | | Number of bits used must be enough to cover all combinations of CTFC from clause 6.10.2.4 Parameter Set. | |
| - CTFC information | | This IE is repeated for TFC numbers and reference to clause 6.10.2.4 Parameter Set | |
| - CTFC | | Reference to clause 6.10.2.4 Parameter Set | |
| - Power offset information | | | |
| - CHOICE Gain Factors | | Computed Gain Factors (The last TFC is set to Signalled Gain Factors) | |
| - Gain factor βc | | 11 (below 64 kbps)<br>9 (higher than 64 kbps)<br>(Not Present if the CHOICE Gain Factors is set to ComputedGain Factors) | |
| - Gain factor βd | | 15<br>(Not Present if the CHOICE Gain Factors is set to ComputedGain Factors) | |
| - Reference TFC ID | | 0 | |
| - CHOICE mode | | FDD | |
| - Power offset $P_{p\text{-}m}$ | | Not Present | |
| Added or Reconfigured UL TrCH information | A1, A2, A5, A6 | Not Present | |
| Added or Reconfigured UL TrCH information | A4 | 2 TrCHs(DCH for DCCH and DCH for DTCH) | |
| - Uplink transport channel type | | DCH | |
| - UL Transport channel identity | | 5 | |
| - TFS | | | |
| - CHOICE Transport channel type | | Dedicated transport channels | |
| - Dynamic Transport format information | | | |
| - RLC Size | | Reference to clause 6.10 Parameter Set | |
| - Number of TBs and TTI List | | (This IE is repeated for TFI number.) | |
| - Transmission Time Interval | | Not Present | |
| - Number of Transport blocks | | Reference to clause 6.10 Parameter Set | |
| - CHOICE Logical channel list | | All | |
| - Semi-static Transport Format information | | | |
| - Transmission time interval | | Reference to clause 6.10 Parameter Set | |
| - Type of channel coding | | Reference to clause 6.10 Parameter Set | |
| - Coding Rate | | Reference to clause 6.10 Parameter Set | |
| - Rate matching attribute | | Reference to clause 6.10 Parameter Set | |
| - CRC size | | Reference to clause 6.10 Parameter Set | |
| - Uplink transport channel type | | DCH | |
| - UL Transport channel identity | | 1 | |
| - TFS | | | |
| - CHOICE Transport channel type | | Dedicated transport channels | |
| - Dynamic Transport format information | | | |

| Information Element | Condition | Value/remark | Version |
|---|---|---|---|
| - RLC Size | | Reference to clause 6.10 Parameter Set | |
| - Number of TBs and TTI List | | (This IE is repeated for TFI number.) | |
| - Transmission Time Interval | | Not Present | |
| - Number of Transport blocks | | Reference to clause 6.10 Parameter Set | |
| - CHOICE Logical channel list | | All | |
| - Semi-static Transport Format information | | | |
| - Transmission time interval | | Reference to clause 6.10 Parameter Set | |
| - Type of channel coding | | Reference to clause 6.10 Parameter Set | |
| - Coding Rate | | Reference to clause 6.10 Parameter Set | |
| - Rate matching attribute | | Reference to clause 6.10 Parameter Set | |
| - CRC size | | Reference to clause 6.10 Parameter Set | |
| Added or Reconfigured UL TrCH information | A3 | (DCH for DTCH) | |
| - Uplink transport channel type | | DCH | |
| - UL Transport channel identity | | 1 | |
| - TFS | | | |
| - CHOICE Transport channel type | | Dedicated transport channels | |
| - Dynamic Transport format information | | | |
| - RLC Size | | Reference to clause 6.10 Parameter Set | |
| - Number of TBs and TTI List | | (This IE is repeated for TFI number.) | |
| - Transmission Time Interval | | Not Present | |
| - Number of Transport blocks | | Reference to clause 6.10 Parameter Set | |
| - CHOICE Logical channel list | | All | |
| - Semi-static Transport Format information | | | |
| - Transmission time interval | | Reference to clause 6.10 Parameter Set | |
| - Type of channel coding | | Reference to clause 6.10 Parameter Set | |
| - Coding Rate | | Reference to clause 6.10 Parameter Set | |
| - Rate matching attribute | | Reference to clause 6.10 Parameter Set | |
| - CRC size | | Reference to clause 6.10 Parameter Set | |
| CHOICE *mode* | A1,A2,A3,A4,A5,A6 | Not Present | |
| DL Transport channel information common for all transport channel | A1, A2, A5,A6 | Not Present | |
| DL Transport channel information common for all transport channel | A3,A4 | | |
| - SCCPCH TFCS | | Not Present | |
| - CHOICE mode | | FDD | |
| - CHOICE DL parameters | | Explicit | |
| - DL DCH TFCS | | | |
| - CHOICE TFCI Signalling | | Normal | |
| - TFCI Field 1 Information | | | |
| - CHOICE TFCS representation | | Complete reconfiguration | |
| - TFCS complete reconfigure | | | |
| - CHOICE CTFC Size | | Number of bits used must be enough to cover all combinations of CTFC from clause 6.10.2.4 Parameter Set. | |
| - CTFC information | | This IE is repeated for TFC numbers and reference to clause 6.10.2.4 | |
| - CTFC | | Reference to clause 6.10.2.4 Parameter Set | |
| - Power offset information | | Not Present | |
| Added or Reconfigured DL TrCH information | A1, A2, A5, A6 | Not Present | |
| Added or Reconfigured DL TrCH information | A4 | 2 TrCHs(DCH for DCCH and DCH for DTCH) | |
| - Downlink transport channel type | | DCH | |
| - DL Transport channel identity | | 10 | |
| - CHOICE DL parameters | | Same as UL | |
| - Uplink transport channel type | | DCH | |
| - UL TrCH identity | | 5 | |
| - DCH quality target | | | |
| - BLER Quality value | | Not Present | |
| - Downlink transport channel type | | DCH | |
| - DL Transport channel identity | | 6 | |

| Information Element | Condition | Value/remark | Version |
|---|---|---|---|
| - CHOICE DL parameters | | Explicit<br>Except for RAB with the symmetric DL and UL rate: Same as UL | |
| - TFS | | | |
| - CHOICE Transport channel type | | Dedicated transport channel | |
| - Dynamic transport format information | | | |
| - RLC Size | | Reference to clause 6.10 Parameter Set | |
| - Number of TBs and TTI List | | (This IE is repeated for TFI number.) | |
| - Dynamic transport format information | | | |
| - Transmission Time Interval | | Not Present | |
| - Number of Transport blocks | | Reference to clause 6.10 Parameter Set | |
| - Semi-static Transport Format information | | | |
| - Transmission time interval | | Reference to clause 6.10 Parameter Set | |
| - Type of channel coding | | Reference to clause 6.10 Parameter Set | |
| - Coding Rate | | Reference to clause 6.10 Parameter Set | |
| - Rate matching attribute | | Reference to clause 6.10 Parameter Set | |
| - CRC size | | Reference to clause 6.10 Parameter Set | |
| - DCH quality target | | | |
| - BLER Quality value | | -2.0 | |
| Added or Reconfigured DL TrCH information | A3 | | |
| - Downlink transport channel type | | DCH | |
| - DL Transport channel identity | | 6 | |
| - CHOICE DL parameters | | Explicit<br>Except for RAB with the symmetric DL and UL rate: Same as UL | |
| - TFS | | | |
| - CHOICE Transport channel type | | Dedicated transport channel | |
| - Dynamic transport format information | | | |
| - RLC Size | | Reference to clause 6.10 Parameter Set | |
| - Number of TBs and TTI List | | (This IE is repeated for TFI number.) | |
| - Dynamic transport format information | | | |
| - Transmission Time Interval | | Not Present | |
| - Number of Transport blocks | | Reference to clause 6.10 Parameter Set | |
| - Semi-static Transport Format information | | | |
| - Transmission time interval | | Reference to clause 6.10 Parameter Set | |
| - Type of channel coding | | Reference to clause 6.10 Parameter Set | |
| - Coding Rate | | Reference to clause 6.10 Parameter Set | |
| - Rate matching attribute | | Reference to clause 6.10 Parameter Set | |
| - CRC size | | Reference to clause 6.10 Parameter Set | |
| - DCH quality target | | | |
| - BLER Quality value | | -2.0 | |
| Frequency info | A1,A2,A3,A4,A5 | | |
| - UARFCN uplink (Nu) | | Not present<br>Absence of this IE is equivalent to applying the default duplex distance defined for the operating frequency according to 3GPP TS 25.101 [11] | |
| - UARFCN downlink (Nd) | | Reference to clause 5.1 Test frequencies | |
| Frequency info | A6 | Not Present | |
| Maximum allowed UL TX power | A1, A2, A3, A4, A5, A6 | 33dBm | |
| CHOICE *channel requirement* | A5, A6 | Not Present | |
| CHOICE channel requirement | A1, A2, A3, A4 | Uplink DPCH info | |
| -Uplink DPCH power control info | | | |
| - DPCCH power offset | | -80dB (i.e. ASN.1 IE value of -40) | |
| - PC Preamble | | 1 frame | |
| - SRB delay | | 7 frames | |
| - Power Control Algorithm | | Algorithm1 | |
| - TPC step size | | 1dB | |

| Information Element | Condition | Value/remark | Version |
|---|---|---|---|
| - $\Delta_{ACK}$ | | Not Present | Rel-5 |
| - $\Delta_{NACK}$ | | Not Present | Rel-5 |
| - Ack-Nack repetition factor | | Not Present | Rel-5 |
| - Scrambling code type | | Long | |
| - Scrambling code number | | 0 (0 to 16777215) | |
| - Number of DPDCH | | Not Present(1) | |
| - spreading factor | | Reference to clause 6.10 Parameter Set | |
| - TFCI existence | | Reference to clause 6.10 Parameter Set | |
| - Number of FBI bit | | Reference to clause 6.10 Parameter Set | |
| - Puncturing Limit | | Reference to clause 6.10 Parameter Set | |
| E-DCH Info | | Not Present | Rel-6 |
| CHOICE Mode | A1, A2, A3, A4, A5, A6 | FDD | R99 and Rel-4 only |
| - Downlink PDSCH information | | Not Present | R99 and Rel-4 only |
| Downlink HS-PDSCH Information | A1, A2, A3, A4, A5, A6 | Not Present | Rel-5 |
| Downlink information common for all radio links | A5, A6 | Not Present | |
| Downlink information common for all radio links | A1, A2, A3 | | |
| - Downlink DPCH info common for all RL | | | |
| - Timing indicator | | Maintain | |
| - CFN-targetSFN frame offset | | Not Present | |
| - Downlink DPCH power control information | | | |
| - DPC mode | | 0 (single) | |
| - CHOICE mode | | FDD | |
| - Power offset $P_{Pilot-DPDCH}$ | | 0 | |
| - DL rate matching restriction information | | Not Present | |
| - Spreading factor | | Reference to clause 6.10 Parameter Set | |
| - Fixed or Flexible Position | | Reference to clause 6.10 Parameter Set | |
| - TFCI existence | | Reference to clause 6.10 Parameter Set | |
| - CHOICE SF | | Reference to clause 6.10 Parameter Set | |
| - DPCH compressed mode info | | Not Present | |
| - TX Diversity mode | | None | |
| - SSDT information | | Not Present | R99 and Rel-4 only |
| - Default DPCH Offset Value | | Not Present | |
| - MAC-hs reset indicator | | Not Present | Rel-5 |
| Downlink information common for all radio links | A4 | | |
| - Downlink DPCH info common for all RL | | | |
| - Timing indicator | | Initialize | |
| - CFN-targetSFN frame offset | | Not Present | |
| - Downlink DPCH power control information | | | |
| - DPC mode | | 0 (single) | |
| - CHOICE mode | | FDD | |
| - Power offset $P_{Pilot-DPDCH}$ | | 0 | |
| - DL rate matching restriction information | | Not Present | |
| - Spreading factor | | Reference to clause 6.10 Parameter Set | |
| - Fixed or Flexible Position | | Reference to clause 6.10 Parameter Set | |
| - TFCI existence | | Reference to clause 6.10 Parameter Set | |
| - CHOICE SF | | Reference to clause 6.10 Parameter Set | |
| - DPCH compressed mode info | | Not Present | |
| - TX Diversity mode | | None | |
| - SSDT information | | Not Present | R99 and Rel-4 only |
| - Default DPCH Offset Value | | Arbitrary set to value 0..306688 by step of 512 | |
| - MAC-hs reset indicator | | Not Present | Rel-5 |

| Information Element | Condition | Value/remark | Version |
|---|---|---|---|
| Downlink information for each radio link list | A1, A2, A3 | | |
| - Downlink information for each radio links | | | |
| - CHOICE mode | | FDD | |
| - Primary CPICH info | | | |
| - Primary scrambling code | | Ref. to the Default setting in clause 6.1 (FDD) | |
| - PDSCH with SHO DCH info | | Not Present | R99 and Rel-4 only |
| - PDSCH code mapping | | Not Present | R99 and Rel-4 only |
| - Serving HS-DSCH radio link indicator | | FALSE | Rel-5 |
| - Serving E-DCH radio link indicator | | FALSE | Rel-6 |
| - Downlink DPCH info for each RL | | | |
| - Primary CPICH usage for channel estimation | | Primary CPICH may be used | |
| - DPCH frame offset | | Set to value Default DPCH Offset Value ( as currently stored in SS) mod 38 400 | |
| - Power offset $P_{Pilot-DPDCH}$ | | 0 | |
| - Secondary CPICH info | | Not Present | |
| - DL channelisation code | | | |
| - Secondary scrambling code | | 4 | |
| - Spreading factor | | Reference to clause 6.10 Parameter Set | |
| - Code number | | 0 | |
| - Scrambling code change | | No change | |
| - TPC combination index | | 0 | |
| - SSDT Cell Identity | | Not Present | R99 and Rel-4 only |
| - Closed loop timing adjustment mode | | Not Present | |
| - E-AGCH Info | | Not Present | Rel-6 |
| - E-HICH Information | | Not Present | Rel-6 |
| - E-RGCH Information | | Not Present | Rel-6 |
| - SCCPCH information for FACH | | Not Present | R99 and Rel-4 only |
| Downlink information for each radio link list | A4 | | |
| - Downlink information for each radio link | | | |
| - CHOICE mode | | FDD | |
| - Primary CPICH info | | | |
| - Primary scrambling code | | Ref. to the Default setting in clause 6.1 (FDD) | |
| - PDSCH with SHO DCH info | | Not Present | R99 and Rel-4 only |
| - PDSCH code mapping | | Not Present | R99 and Rel-4 only |
| - Serving HS-DSCH radio link indicator | | FALSE | Rel-5 |
| - Serving E-DCH radio link indicator | | FALSE | Rel-6 |
| - Downlink DPCH info for each RL | | | |
| - Primary CPICH usage for channel estimation | | Primary CPICH may be used | |
| - DPCH frame offset | | Set to value: Default DPCH Offset Value mod 38 400 | |
| - Power offset $P_{Pilot-DPDCH}$ | | 0 | |
| - Secondary CPICH info | | Not Present | |
| - DL channelisation code | | | |
| - Secondary scrambling code | | 4 | |
| - Spreading factor | | Reference to clause 6.10 Parameter Set | |
| - Code number | | 0 | |
| - Scrambling code change | | No change | |
| - TPC combination index | | 0 | |
| - SSDT Cell Identity | | Not Present | R99 and Rel-4 only |
| - Closed loop timing adjustment | | Not Present | |

| Information Element | Condition | Value/remark | Version |
|---|---|---|---|
| mode | | | |
| - E-AGCH Info | | Not Present | Rel-6 |
| - E-HICH Information | | Not Present | Rel-6 |
| - E-RGCH Information | | Not Present | Rel-6 |
| - SCCPCH information for FACH | | Not Present | R99 and Rel-4 only |
| - Downlink information for each radio link | A5 | | |
| - Choice mode | | FDD | |
| - Primary CPICH info | | | |
| - Primary scrambling code | | Ref. to the Default setting in clause 6.1 (FDD) | |
| - PDSCH with SHO DCH info | | Not Present | R99 and Rel-4 only |
| - PDSCH code mapping | | Not Present | R99 and Rel-4 only |
| - Serving HS-DSCH radio link indicator | | FALSE | Rel-5 |
| - Serving E-DCH radio link indicator | | FALSE | Rel-6 |
| - Downlink DPCH info for each RL | | Not present | |
| - E-AGCH Info | | Not Present | Rel-6 |
| - E-HICH Information | | Not Present | Rel-6 |
| - E-RGCH Information | | Not Present | Rel-6 |
| - SCCPCH information for FACH | | Not Present | R99 and Rel-4 only |
| - Downlink information for each radio link | A6 | Not Present | |
| MBMS PL Service Restriction Information | A1, A2, A3, A4, A5, A6 | Not Present | Rel-6 |

| Condition | Explanation |
|---|---|
| A1 | This IE need for "Non speech in CS" |
| A2 | This IE need for "Speech in CS" |
| A3 | This IE need for "Packet to CELL_DCH from CELL_DCH in PS" |
| A4 | This IE need for "Packet to CELL_DCH from CELL_FACH in PS" |
| A5 | This IE need for "Packet to CELL_FACH from CELL_DCH in PS" |
| A6 | This IE need for "Packet to CELL_FACH from CELL_FACH in PS" |

Contents of TRANSPORT CHANNEL RECONFIGURATION COMPLETE message: AM

| Information Element | Value/remark |
|---|---|
| Message Type | |
| RRC transaction identifier | Checked to see if the value is identical to the same IE in the downlink TRANSPORT CHANNEL RECONFIGURATION message |
| Integrity check info | |
| - Message authentication code | This IE is checked to see if it is present. The value is compared against the XMAC-I value computed by SS. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I. |
| - RRC Message sequence number | This IE is checked to see if it is present. The value is used by SS to compute the XMAC-I value. |
| Uplink integrity protection activation info | Not checked |
| CHOICE mode | FDD |
| COUNT-C activation time | Not checked |
| Radio bearer uplink ciphering activation time info | Not checked |
| Uplink counter synchronization info | Not present |

Contents of TRANSPORT CHANNEL RECONFIGURATION FAILURE message: AM

| Information Element | Value/remark |
|---|---|
| Message Type | |
| RRC transaction identitifer | Checked to see if it is set to identical value of the same IE in the downlink TRANSPORT CHANNEL RECONFIGURATION message. |
| Integrity check info | |

| | |
|---|---|
| - Message authentication code | This IE is checked to see if it is present. The value is compared against the XMAC-I value computed by SS. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I. |
| - RRC Message sequence number | This IE is checked to see if it is present. The value is used by SS to compute the XMAC-I value. |
| Failure cause | Checked to see if it meets test requirement |

Contents of TRANSPORT FORMAT COMBINATION CONTROL message: AM or UM (in CELL_DCH)

| Information Element | Value/remark |
|---|---|
| Message Type | |
| RRC transaction identifier | Arbitrarily selects an integer between 0 and 3 |
| Integrity check info | |
| - Message authentication code | SS calculates the value of MAC-I for this message and writes to this IE. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I. |
| - RRC Message sequence number | SS provides the value of this IE, from its internal counter. |
| CHOICE mode | FDD |
| DPCH/PUSCH TFCS in Uplink | |
| - CHOICE *Subset representation* | Allowed transport format combination list |
| - Allowed Transport format combination | 0 (The TFC is constructed from ALL TF0) |
| Activation time for TFC subset | Not Present |
| TFC Control duration | Not Present |

Contents of TRANSPORT FORMAT COMBINATION CONTROL FAILURE message: AM

| Information Element | Value/remark |
|---|---|
| Message Type | |
| RRC transaction identitifer | Checked to see if it is set to identical value of the same IE in the downlink TRANSPORT CHANNEL RECONFIGURATION message. |
| Integrity check info | |
| - Message authentication code | This IE is checked to see if it is present. The value is compared against the XMAC-I value computed by SS. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I. |
| - RRC Message sequence number | This IE is checked to see if it is present. The value is used by SS to compute the XMAC-I value. |
| Failure cause | Checked to see if it meets test requirement |

Contents of UE CAPABILITY ENQUIRY message: AM or UM

| Information Element | Value/remark |
|---|---|
| Message Type | |
| RRC transaction identifier | Arbitrarily selects an integer between 0 and 3 |
| Integrity check info | |
| - Message authentication code | SS calculates the value of MAC-I for this message and writes to this IE. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I. |
| - RRC Message sequence number | SS provides the value of this IE, from its internal counter. |
| Capability update requirement | |
| - UE radio access FDD capability update requirement | TRUE |
| - UE radio access TDD capability update requirement | FALSE |
| - System specific capability update requirement list | Not Present |

Contents of UE CAPABILITY INFORMATION message: AM

| Information Element | Value/remark |
|---|---|
| Message Type | |
| RRC transaction identifier | Checked to see if the value is identical to the same IE in the downlink UE CAPABILITY ENQUIRY message. |
| Integrity check info | |

| Information Element | Value/remark |
|---|---|
| - Message authentication code | This IE is checked to see if it is present. The value is compared against the XMAC-I value computed by SS. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I. |
| - RRC Message sequence number | This IE is checked to see if it is present. The value is used by SS to compute the XMAC-I value. |
| UE radio access capability | Value will be checked. Stated capability must be compatible with 3GPP TS 34.123-2 [3] (ICS statements) and the user settings |
| - Access stratum release indicator | |
| - PDCP Capability | |
| - RLC Capability | |
| - Transport channel capability | |
| - RF Capability FDD | |
| - RF Capability TDD | |
| - Physical channel capability | |
| - UE multi-mode/multi-RAT capability | |
| - Security Capability | |
| - UE positioning Capability | |
| - Measurement capability | |
| UE radio access capability extension | Value will be checked. Stated capability must be compatible with 3GPP TS 34.123-2 [3] (ICS statements) and the user settings |
| UE system specific capability | Not Checked |

Contents of UE CAPABILITY INFORMATION CONFIRM message: UM

| Information Element | Value/remark |
|---|---|
| Message Type | |
| RRC transaction identifier | Set to the same value as received in the UE CAPABILITY INFORMATON message. |
| Integrity check info | |
| - Message authentication code | SS calculates the value of MAC-I for this message and writes to this IE. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I. |
| - RRC Message sequence number | SS provides the value of this IE, from its internal counter. |

Contents of URA UPDATE message: TM

| Information Element | Value/remark |
|---|---|
| Message Type | |
| U-RNTI | |
| - SRNC identity | 0000 0000 0001B |
| - S-RNTI | 0000 0000 0000 0000 0001B |
| RRC transaction identifier | Checked to see if it is absent |
| Integrity check info | |
| - Message authentication code | This IE is checked to see if it is present. The value is compared against the XMAC-I value computed by SS. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I. |
| - RRC Message sequence number | This IE is checked to see if it is present. The value is used by SS to compute the XMAC-I value. |
| URA update cause | See the test content |
| Protocol error indicator | Checked to see if it is absent or set to 'FALSE' |
| Protocol error information | Checked to see if it is absent |

Contents of URA UPDATE CONFIRM message: UM

| Information Element | Value/remark |
|---|---|
| Message Type | |
| U-RNTI | If this message is sent on CCCH, use the following values. Else, this IE is absent. |
| - SRNC identity | 0000 0000 0001B |
| - S-RNTI | 0000 0000 0000 0000 0001B |

| | |
|---|---|
| RRC transaction identifier | Arbitrarily selects and integer between 0 and 3 |
| Integrity check info | |
| - message authentication code | SS calculates the value of MAC-I for this message and writes to this IE. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I. |
| - RRC message sequence number | SS provides the value of this IE, from its internal counter. |
| Integrity protection mode info | Not Present |
| Ciphering mode info | Not Present |
| New U-RNTI | Not Present |
| New C-RNTI | Not Present |
| RRC state indicator | URA_PCH |
| UTRAN DRX cycle length coefficient | 3 |
| CN information info | Not Present |
| URA identity | Not Present |
| Downlink counter synchronization info | Not Present |

Contents of UPLINK DIRECT TRANSFER message: AM

| Information Element | Value/remark |
|---|---|
| Message Type | |
| Integrity check info | |
| - Message authentication code | This IE is checked to see if it is present. The value is compared against the XMAC-I value computed by SS. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I. |
| - RRC Message sequence number | This IE is checked to see if it is present. The value is used by SS to compute the XMAC-I value. |
| CN domain identity | Checked to see if set to a CN domain for which a signalling connection exists |
| NAS message | Set according to that indicated in specific message content clause |
| Measured results on RACH | Not checked |

Contents of UTRAN MOBILITY INFORMATION message: AM or UM

| Information Element | Value/remark |
|---|---|
| Message Type | |
| Integrity check info | |
| - message authentication code | SS calculates the value of MAC-I for this message and writes to this IE. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I. |
| - RRC message sequence number | SS provides the value of this IE, from its internal counter. |
| RRC transaction identifier | Arbitrarily selects an integer between 0 and 3 |
| Integrity protection mode info | Not Present |
| Ciphering mode info | Not Present |
| New U-RNTI | See the test content |
| New C-RNTI | See the test content |
| UE Timers and constants in connected mode | |
| - T301 | 2 000 milliseconds |
| - N301 | 2 |
| - T302 | 4 000 milliseconds |
| - N302 | 3 |
| - T304 | 1 000 milliseconds |
| - N304 | 3 |
| - T305 | 60 minutes |
| - T307 | 50 seconds |
| - T308 | 320 milliseconds |
| - T309 | 8 seconds |
| - T310 | 320 milliseconds |
| - N310 | 5 |
| - T311 | 500 milliseconds |
| - T312 | 5 seconds |
| - N312 | 200 |
| - T313 | 10 seconds |
| - N313 | 200 |

| Information Element | Value/remark |
|---|---|
| - T314 | 20 seconds |
| - T315 | 30 seconds |
| - N315 | 200 |
| - T316 | 50 seconds |
| - T317 | 1 800 seconds |
| CN information info | Not Present |
| URA identity | Not present |
| Downlink counter synchronization info | Not Present |

Contents of UTRAN MOBILITY INFORMATION CONFIRM message: AM

| Information Element | Value/remark |
|---|---|
| Message Type | |
| RRC transaction identifier | Checked to see if it matches the value of the same IE in downlink UTRAN MOBILITY INFORMATION message |
| Integrity check info | |
| - Message authentication code | This IE is checked to see if it is present. The value is compared against the XMAC-I value computed by SS. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I. |
| - RRC Message sequence number | This IE is checked to see if it is present. The value is used by SS to compute the XMAC-I value. |
| Uplink integrity protection activation info | Not checked |
| COUNT-C activation time | Not checked |
| Radio bearer uplink ciphering activation time info | Not checked |
| Uplink counter synchronization info | Not present |

Contents of UTRAN MOBILITY INFORMATION FAILURE message: AM

| Information Element | Value/remark |
|---|---|
| Message Type | |
| RRC transaction identifier | Checked to see if it matches the value of the same IE in downlink UTRAN MOBILITY INFORMATION message |
| Integrity check info | |
| - Message authentication code | This IE is checked to see if it is present. The value is compared against the XMAC-I value computed by SS. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I. |
| - RRC Message sequence number | This IE is checked to see if it is present. The value is used by SS to compute the XMAC-I value. |
| Failure Cause | Checked to see if it meets test requirement |

## 9.1.2 Default Message Contents for Signalling (TDD)

Contents of RRC STATUS message: AM

| Information Element | Value/remark |
|---|---|
| Message Type | |
| Integrity check info | |
| - Message authentication code | This IE is checked to see if it is present. The value is compared against the XMAC-I value computed by SS. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I. |
| - RRC Message sequence number | This IE is checked to see if it is present. The value is used by SS to compute the XMAC-I value. |
| Identification of received message | Not checked |
| Protocol error information | |
| - Protocol error cause | Refer to test requirement. |

Contents of SECURITY MODE FAILURE message: AM

| Information Element | Value/remark |
|---|---|

| | |
|---|---|
| Message Type | |
| **UE information elements** | |
| RRC transaction identifier | The value of this IE is checked to see that it matches the value of the same IE transmitted in the downlink SECURITY MODE COMMAND message. |
| Integrity check info | |
| - Message authentication code | This IE is checked to see if it is present. The value is compared against the XMAC-I value computed by SS. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I. |
| - RRC Message sequence number | This IE is checked to see if it is present. The value is used by SS to compute the XMAC-I value. |
| Failure cause | Refer to test requirement. |

Contents of URA UPDATE message: TM

| Information Element | Value/remark |
|---|---|
| Message Type | |
| U-RNTI | Checked to see if it is set to the following values |
| - SRNC identity | 0000 0000 0001B |
| - S-RNTI | 0000 0000 0000 0000 0001B |
| RRC transaction identifier | Checked to see if it is absent |
| Integrity check info | |
| - Message authentication code | This IE is checked to see if it is present. The value is compared against the XMAC-I value computed by SS. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I. |
| - RRC Message sequence number | This IE is checked to see if it is present. The value is used by SS to compute the XMAC-I value. |
| URA update cause | See the test content |
| Protocol error indicator | Checked to see if it is absent or set to 'FALSE' |
| Protocol error information | Checked to see if it is absent |

Contents of URA UPDATE CONFIRM message: UM

| Information Element | Value/remark |
|---|---|
| Message Type | |
| U-RNTI | If this message is sent on CCCH, use the following values. Else, this IE is absent. |
| - SRNC identity | 0000 0000 0001B |
| - S-RNTI | 0000 0000 0000 0000 0001B |
| RRC transaction identifier | Arbitrarily selects and integer between 0 and 3 |
| Integrity check info | |
| - Message authentication code | Set to MAC-I value computed by the SS. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I. |
| - RRC Message Sequence Number | Set to an arbitrarily selected integer between 0 and 15 |
| Integrity protection mode info | Not present |
| Ciphering mode info | Not present |
| New U-RNTI | Not present |
| New C-RNTI | Not present |
| RRC State Indicator | URA_PCH |
| UTRAN DRX cycle length coefficient | 3 |
| CN Information info | Not present |
| URA identity | See the test content |
| Downlink counter synchronization info | Not present |

Contents of UPLINK DIRECT TRANSFER message: AM

| Information Element | Value/remark |
|---|---|
| Message Type | |
| Integrity check info | |
| - Message authentication code | This IE is checked to see if it is present. The value is compared against the XMAC-I value computed by SS. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I. |
| - RRC Message sequence number | This IE is checked to see if it is present. The value is used by |

| | |
|---|---|
| | SS to compute the XMAC-I value. |
| CN domain identity | Checked to see if set to a CN domain for which a signalling connection exists |
| NAS message | Set according to that indicated in specific message content for each test case |
| Measured results on RACH | Not checked |

Contents of UTRAN MOBILITY INFORMATION message: AM or UM

| Information Element | Value/remark |
|---|---|
| Message Type | |
| Integrity check info | |
| - Message authentication code | Set to MAC-I value computed by the SS. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I. |
| - RRC Message Sequence Number | Set to an arbitrarily selected integer between 0 and 15 |
| RRC transaction identifier | Arbitrarily selects and integer between 0 and 3 |
| Integrity protection mode info | Not present |
| Ciphering mode info | Not present |
| New U-RNTI | See the test content |
| New C-RNTI | See the test content |
| UE Timers and constants in connected mode | |
| - T301 | 2 000 milliseconds |
| - N301 | 2 |
| - T302 | 4 000 milliseconds |
| - N302 | 3 |
| - T304 | 1 000 milliseconds |
| - N304 | 3 |
| - T305 | 60 minutes |
| - T307 | 50 seconds |
| - T308 | 320 milliseconds |
| - T309 | 8 seconds |
| - T310 | 320 milliseconds |
| - N310 | 5 |
| - T311 | 500 milliseconds |
| - T312 | 5 seconds |
| - N312 | 200 |
| - T313 | 10 seconds |
| - N313 | 200 |
| - T314 | 20 seconds |
| - T315 | 30 seconds |
| - N315 | 200 |
| - T316 | 50 seconds |
| - T317 | 1 800 seconds |
| CN Information info | Not present |
| URA identity | Not present |
| Downlink counter synchronization info | Not present |

Contents of UTRAN MOBILITY INFORMATION CONFIRM message: AM

| Information Element | Value/remark |
|---|---|
| Message Type | |
| RRC transaction identifier | Checked to see if it matches the value of the same IE in downlink UTRAN MOBILITY INFORMATION message |
| Integrity check info | |
| - Message authentication code | This IE is checked to see if it is present. The value is compared against the XMAC-I value computed by SS. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I. |
| - RRC Message sequence number | This IE is checked to see if it is present. The value is used by SS to compute the XMAC-I value. |
| Uplink integrity protection activation info | Not checked |
| COUNT-C activation time | Not checked |
| Radio bearer uplink ciphering activation time info | Not checked |
| Uplink counter synchronization info | Not checked |

Contents of UE CAPABILITY ENQUIRY message

| Information Element | Value/remark |
|---|---|
| Message Type | UE CAPABILITY ENQUIRY |
| Integrity check info | Not Present |
| - Message authentication code | If present, SS calculates the value of MAC-I for this message and writes to this IE. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I. |
| - RRC Message sequence number | If present, SS provides the value of this IE, from its internal counter. |
| RRC transaction identifier | Arbitrarily selects an integer between 0 and 3 |
| Capability update requirement | |
| - UE radio access FDD capability update requirement | FALSE |
| - UE radio access 3.84 Mcps TDD capability update requirement | FALSE |
| - UE radio access 1.28 Mcps TDD capability update requirement | TRUE |
| - System specific capability update requirement list | Not Present |

Contents of UE CAPABILITY INFORMATION message (1.28 Mpcs TDD)

| Information Element | Value/remark |
|---|---|
| Message Type | UE CAPABILITY INFORMATION |
| Integrity check info | Not Present |
| - Message authentication code | If present, SS calculates the value of MAC-I for this message and writes to this IE. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I. |
| - RRC Message sequence number | If present, SS provides the value of this IE, from its internal counter. |
| RRC transaction identifier | Checked to see if the value is identical to the same IE in the downlink UE CAPABILITY ENQUIRY message. |
| UE radio access capability | Present |
| - Access stratum release indicator | Rel-5 |
| - DL capability with simultaneous HS-DSCH configuration | Not Present |
| - PDCP capability | |
| - Support for lossless SRNS relocation | TRUE |
| - Support for RFC2507 | TRUE |
| - Max HC context space | 512 |
| - Support for RFC3095 | FALSE |
| - RLC capability | |
| - Total RLC AM buffer size | 150 |
| - Maximum RLC AM Window Size | 2 047 |
| - Maximum number of AM entities | 30 |
| - Transport channel capability | |
| - Downlink transport channel capability information elements | |
| - Max number of bits received | 640 |
| - Max convolutionally coded bits received | 6 400 |
| - Max turbo coded bits received | 6 400 |
| - Max number of simultaneous transport channels | 8 |
| - Maximum number of simultaneous CCtrCH | 1 |
| - Max number of received transport blocks | 32 |
| - Max number of TFC | 128 |
| - Max number of TF | 64 |
| - Turbo decoding supported | TRUE |
| - Uplink transport channel capability information elements | |
| - Max number of bits transmitted | 6 400 |
| - Max convolutionally coded bits transmitted | 6 400 |
| - Max turbo coded bits transmitted | 6 400 |
| - Max number of simultaneous transport channels | 8 |
| - Max number of simultaneous CCTrCH of DCH | 1 |
| - Max number of transmitted transport blocks | 16 |
| - max number of TFC | 64 |
| - Max number of TF | 32 |

| Information Element | Value/remark |
|---|---|
| - Turbo coding supported | TRUE |
| - RF capability FDD | Not Present |
| - RF capability TDD | Present |
| - UE power class | 1 |
| - Radio frequency bands | a |
| - Chip rate capability | 1.28 Mcps |
| - Physical channel capability | |
| -Downlink physical channel capability information | |
| - FDD physical channel capability | Not Present |
| - 3.84 Mcps TDD downlink physical channel capability | Not Present |
| - 1.28 Mcps TDD downlink physical channel capability | Present |
| - maxTS per subFrame | 6 |
| - max physical channel per frame | 96 |
| - min. SF | 16 |
| - Support of PDSCH | FALSE |
| - Support of HS-PDSCH | Unsupported |
| - max. physical channel per TS | 16 |
| - Support of 8psk | FALSE |
| -Uplink physical channel capability information | |
| - FDD physical channel capability | Not Present |
| - 3.84 Mcps TDD uplink physical channel capability | Not Present |
| - 1.28 Mcps TDD uplink physical channel capability | Present |
| - maxTS per subFrame | 6 |
| - max physical channel per timeslot | 2 |
| - min. SF | 16 |
| - Support of PDSCH | FALSE |
| - max. physical channel per TS | 16 |
| - Support of 8psk | FALSE |
| - UE multi-mode/multi-RAT capability | |
| - MultiRAT capability List | |
| - Support of GSM | FALSE |
| - Support of Multicarrier | TRUE |
| - MultiMode capability | TDD |
| - Support of UTRAN to GERAN NACC | FALSE |
| - Security capability | |
| - Ciphering algorithm capability | |
| - UEA0 | FALSE |
| - UEA1 | FALSE |
| - Spare | FALSE |
| - Integrity protection algorithm | |
| - UIA1 | FALSE |
| - Spare | FALSE |
| - UE positioning capability | |
| - Standalone location method(s) supported | FALSE |
| - UE based OTDOA supported | FASLE |
| - Network Assisted GPS support | None |
| - Support for GPS timing of cell frames measurement | FALSE |
| - Support for IPDL | FALSE |
| - Support for RX-TX time difference type2 measurement | FALSE |
| - Support for Up measurement validaity in CELL-PCH and URA-PCH states | FALSE |
| - Measurement capability | Not Present |
| UE system specific capability | Not present |

Contents of UE CAPABILITY INFORMATION CONFIRM message

| Information Element | Value/remark |
|---|---|
| Message Type | UE CAPABILITY INFORMATION |
| Integrity check info | Not Present |
| - Message authentication code | If present, SS calculates the value of MAC-I for this message and writes to this IE. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I. |

| Information Element | Value/remark |
|---|---|
| - RRC Message sequence number | If present, SS provides the value of this IE, from its internal counter. |
| RRC transaction identifier | Checked to see if the value is identical to the same IE in the downlink UE CAPABILITY ENQUIRY message. |

Contents of TRANSPORT CHANNEL RECONFIGURATION message: AM or UM (3.84 Mcps TDD)

| Information Element | Condition | Value/remark | Version |
|---|---|---|---|
| Message Type | A1, A2, A3, A4, A5, A6, A7, A8, A9, A10 | | |
| RRC transaction identifier | | Arbitrarily selects an integer between 0 and 3 | |
| Integrity check info | | | |
| - message authentication code | | SS calculates the value of MAC-I for this message and writes to this IE. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I. | |
| - RRC message sequence number | | SS provides the value of this IE, from its internal counter. | |
| Integrity protection mode info | | Not Present | |
| Ciphering mode info | | Not Present | |
| Activation time | A1, A2, A3 | (256+CFN-(CFN MOD 8 + 8))MOD 256 | |
| Activation time | A4, A5, A6, A7, A8, A9, A10 | Not Present | |
| New U-RNTI | | Not Present | |
| New C-RNTI | A1, A2, A3, A4, A7, A8, A9, A10 | Not Present | |
| New C-RNTI | A5, A6 | '1010 1010 1010 1010' | |
| New DSCH-RNTI | A1, A2, A3, A4, A5, A6, A7, A8, A9, A10 | Not Present | |
| New H-RNTI | A1, A2, A3, A4, A5, A6, A7, A8, A9, A10 | Not Present | Rel-5 |
| RRC State indicator | A1, A2, A3, A4 | CELL_DCH | |
| RRC State indicator | A5, A6 | CELL_FACH | |
| RRC State indicator | A7, A8 | URA_PCH | |
| RRC State indicator | A9, A10 | CELL_PCH | |
| UTRAN DRX cycle length coefficient | A1, A2, A3, A4,A5,A6 | Not Present | |
| UTRAN DRX cycle length coefficient | A7, A8, A9, A10 | 3 | |
| CN information info | | Not Present | |
| URA identity | | Not Present | |
| Downlink counter synchronisation info | | Not Present | |
| UL Transport channel information common for all transport channels | A1, A2, A5, A6 | Not Present | |

| Information Element | Condition | Value/remark | Version |
|---|---|---|---|
| UL Transport channel information common for all transport channels | A3, A4 | | |
| - PRACH TFCS | | Not Present | |
| - CHOICE mode | | TDD | |
| - Individual UL CCTrCH information | | | |
| - UL TFCS Identity | | | |
| - TFCS ID | | 1 | |
| - Shared Channel Indicator | | FALSE | |
| - UL TFCS | | | |
| - CHOICE *TFCI signalling* | | Normal | |
| - TFCI Field 1 Information | | | |
| - CHOICE *TFCS representation* | | Complete reconfiguration | |
| - TFCS complete reconfiguration information | | | |
| - CHOICE *CTFC Size* | | Number of bits used must be enough to cover all combinations of CTFC from TS34.108 clause 6.10.3.4 Parameter Set. | |
| - CTFC information | | This IE is repeated for TFC numbers and reference to TS34.108 clause 6. 10.3.4 Parameter Set | |
| - CTFC | | Reference to TS34.108 clause 6. 10.3.4 Parameter Set | |
| - Power offset information | | | |
| - CHOICE Gain Factors | | Computed Gain Factors(The last TFC is set to Signalled Gain Factors) | |
| - Reference TFC ID | | 0 Integer(0.. 3) | |
| - CHOICE Gain Factors | | Signalled Gain Factors(Not Present if the CHOICE Gain Factors is set to ComputedGain Factors) | |
| - CHOICE mode | | TDD | |
| - Gain Factor $\beta_d$ | | 15 | |
| - Reference TFC ID | | 0 Integer(0.. 3) | |
| - CHOICE mode | | TDD | |
| - TFC subset | | | |
| - CHOICE Subset representation | | Full transport format combination set | |
| - TFC subset list | | Not Present | |
| Added or Reconfigured TrCH information list | A1, A2, A5, A6 | Not Present | |

| Information Element | Condition | Value/remark | Version |
|---|---|---|---|
| Added or Reconfigured TrCH information list | A4 | 2 TrCHs(DCH for DCCH and DCH for DTCH) | |
| - Added or Reconfigured UL TrCH information | | | |
| - Uplink transport channel type | | DCH | |
| - UL Transport channel identity | | 5 | |
| - TFS | | | |
| - CHOICE Transport channel type | | Dedicated transport channels | |
| - Dynamic Transport format information | | | |
| - RLC Size | | Reference to TS34.108 clause 6.10 Parameter Set | |
| - Number of TBs and TTI List | | This IE is repeated for maxTF number | |
| - Transmission Time Interval | | Not Present | |
| - Number of Transport blocks | | Reference to TS34.108 clause 6.10 Parameter Set | |
| - CHOICE Logical Channel list | | All | |
| - Semi-static Transport Format information | | | |
| - Transmission time interval | | Reference to TS34.108 clause 6.10 Parameter Set | |
| - Type of channel coding | | Reference to TS34.108 clause 6.10 Parameter Set | |
| - Coding Rate | | Reference to TS34.108 clause 6.10 Parameter Set | |
| - Rate matching attribute | | Reference to TS34.108 clause 6.10 Parameter Set | |
| - CRC size | | Reference to TS34.108 clause 6.10 Parameter Set | |
| - Uplink transport channel type | | DCH | |
| - UL Transport channel identity | | 1 | |
| - TFS | | | |
| - CHOICE Transport channel type | | Dedicated transport channels | |
| - Dynamic Transport format information | | | |
| - RLC Size | | Reference to TS34.108 clause 6.10 Parameter Set | |
| - Number of TBs and TTI List | | This IE is repeated for maxTF number | |
| - Transmission Time Interval | | Not Present | |
| - Number of Transport blocks | | Reference to TS34.108 clause 6.10 Parameter Set | |
| - CHOICE Logical Channel list | | All | |
| - Semi-static Transport Format information | | | |
| - Transmission time interval | | Reference to TS34.108 clause 6.10 Parameter Set | |
| - Type of channel coding | | Reference to TS34.108 clause 6.10 Parameter Set | |
| - Coding Rate | | Reference to TS34.108 clause 6.10 Parameter Set | |
| - Rate matching attribute | | Reference to TS34.108 clause 6.10 Parameter Set | |
| - CRC size | | Reference to TS34.108 clause 6.10 Parameter Set | |
| Added or Reconfigured TrCH information list | A3 | (DCH for DTCH) | |
| - Added or Reconfigured UL TrCH information | | | |
| - Uplink transport channel type | | DCH | |
| - UL Transport channel identity | | 1 | |
| - TFS | | | |
| - CHOICE Transport channel type | | Dedicated transport channels | |
| - Dynamic Transport format information | | | |
| - RLC Size | | Reference to TS34.108 clause 6.10 Parameter Set | |
| - Number of TBs and TTI List | 1 to maxTF | (This IE is repeated for TF number.) | |
| - Transmission Time Interval | | Not Present | |
| - Number of Transport blocks | | Reference to TS34.108 clause 6.10 Parameter Set | |
| - CHOICE Logical Channel list | | All | |
| - Semi-static Transport Format information | | | |

| Information Element | Condition | Value/remark | Version |
|---|---|---|---|
| - Transmission time interval | | Reference to TS34.108 clause 6.10 Parameter Set | |
| - Type of channel coding | | Reference to TS34.108 clause 6.10 Parameter Set | |
| - Coding Rate | | Reference to TS34.108 clause 6.10 Parameter Set | |
| - Rate matching attribute | | Reference to TS34.108 clause 6.10 Parameter Set | |
| - CRC size | | Reference to TS34.108 clause 6.10 Parameter Set | |
| CHOICE *mode* | A1,A2,A3, A4,A5,A6 | TDD | |
| Downlink HS-PDSCH Information | | | Rel-5 |
| DL Transport channel information common for all transport channels | A1, A2, A5,A6 | Not Present | |
| DL Transport channel information common for all transport channels | A3,A4 | | |
| - SCCPCH TFCS | | Not Present | |
| - CHOICE mode | | TDD | |
| - Individual DL CCTrCH information | | | |
| - DL TFCS Identity | | | |
| - TFCS ID | | 2 | |
| - Shared Channel Indicator | | FALSE | |
| - CHOICE DL parameters | | Independent | |
| - DL TFCS | | | |
| - CHOICE TFCI Signalling | | Normal | |
| - TFCI Field 1 Information | | | |
| - CHOICE TFCS representation | | Complete reconfiguration | |
| - TFCS complete reconfiguration information | | | |
| - CHOICE CTFC Size | | Number of bits used must be enough to cover all combinations of CTFC from clause TS34.108 clause 6. 10.3.4 Parameter Set. | |
| - CTFC information | | This IE is repeated for TFC numbers and reference to TS34.108 clause 6. 10.3.4 | |
| - CTFC | | Reference to TS34.108 clause 6. 10.3.4 Parameter Set | |
| - Power offset information | | Not Present | |
| Added or Reconfigured TrCH information list | A1, A2, A5, A6 | Not Present | |

| Information Element | Condition | Value/remark | Version |
|---|---|---|---|
| Added or Reconfigured TrCH information list | A4 | 2 TrCHs(DCH for DCCH and DCH for DTCH) | |
| - Added or Reconfigured DL TrCH information | | | |
| - Downlink transport channel type | | DCH | |
| - DL Transport channel identity | | 10 | |
| - CHOICE DL parameters | | Same as UL | |
| - Uplink transport channel type | | DCH | |
| - UL TrCH identity | | 5 | |
| - DCH quality target | | | |
| - BLER Quality value | | -2.0 Real(-6.3..0 by step of 0.1) | |
| - Transparent mode signalling info | | Not Present | |
| - Downlink transport channel type | | DCH | |
| - DL Transport channel identity | | 6 | |
| - CHOICE DL parameters | | Explicit | |
| - TFS | | | |
| - CHOICE Transport channel type | | Dedicated transport channels | |
| - Dynamic transport format information | | | |
| - RLC Size | | Reference to TS34.108 clause 6.10 Parameter Set | |
| - Number of TBs and TTI List | | (This IE is repeated for TF number.) | |
| - Transmission Time Interval | | Not Present | |
| - Number of Transport blocks | | Reference to TS34.108 clause 6.10 Parameter Set | |
| - Semi-static Transport Format information | | | |
| - Transmission time interval | | Reference to TS34.108 clause 6.10 Parameter Set | |
| - Type of channel coding | | Reference to TS34.108 clause 6.10 Parameter Set | |
| - Coding Rate | | Reference to TS34.108 clause 6.10 Parameter Set | |
| - Rate matching attribute | | Reference to TS34.108 clause 6.10 Parameter Set | |
| - CRC size | | Reference to TS34.108 clause 6.10 Parameter Set | |
| - DCH quality target | | | |
| - BLER Quality value | | -2.0 | |
| Added or Reconfigured TrCH information list | A3 | | |
| - Added or Reconfigured DL TrCH information | | | |
| - Downlink transport channel type | | DCH | |
| - DL Transport channel identity | | 6 | |
| - CHOICE DL parameters | | Explicit | |
| - TFS | | | |
| - CHOICE Transport channel type | | Dedicated transport channels | |
| - Dynamic transport format information | | | |
| - RLC Size | | Reference to TS34.108 clause 6.10 Parameter Set | |
| - Number of TBs and TTI List | | (This IE is repeated for TF number.) | |
| - Transmission Time Interval | | Not Present | |
| - Number of Transport blocks | | Reference to TS34.108 clause 6.10 Parameter Set | |
| - Semi-static Transport Format information | | | |
| - Transmission time interval | | Reference to TS34.108 clause 6.10 Parameter Set | |
| - Type of channel coding | | Reference to TS34.108 clause 6.10 Parameter Set | |
| - Coding Rate | | Reference to TS34.108 clause 6.10 Parameter Set | |
| - Rate matching attribute | | Reference to TS34.108 clause 6.10 Parameter Set | |
| - CRC size | | Reference to TS34.108 clause 6.10 Parameter Set | |
| - DCH quality target | | | |
| - BLER Quality value | | -2.0 | |
| - Transparent mode signalling info | | Not Present | |
| Frequency info | A1, A2, A3, | | |

| Information Element | Condition | Value/remark | Version |
|---|---|---|---|
| - Choice mode<br>- UARFCN (Nt) | A4, A5 | TDD<br>Reference to clause 5.1 Test frequencies | |
| Frequency info | A6, A7, A8, A9, A10 | Not Present | |
| Maximum allowed UL TX power | | 33dBm | |
| CHOICE *channel requirement* | A5, A6, A7, A8, A9, A10 | Not Present | |
| CHOICE *channel requirement*<br><br>- Uplink DPCH power control info<br>- CHOICE mode<br>- CHOICE TDD option<br>- UL target SIR<br>- CHOICE UL OL PC info<br>- CHOICE TDD option<br>- Individual timeslot interference info<br><br>- Individual timeslot interference<br>- CHOICE TDD option<br>- Timeslot number<br><br>- TDD UL interference<br><br><br>- Primary CCPCH Tx Power<br><br>- CHOICE mode<br>- Uplink Timing Advance Control<br>- CHOICE Timing Advance<br>- CHOICE TDD option | A1, A2, A3, A4 | Uplink DPCH info<br><br><br>TDD<br>3.84 Mcps TDD<br>6<br>Individually Signalled<br>3.84 Mcps TDD<br>Reference to TS34.108 clause 6.10.3 Parameter Set (for number of TS's)<br><br>3.84 Mcps TDD<br>As required by, Reference to TS34.108 clause 6.10.3 Parameter Set<br>As required by, Reference to TS34.108 clause 6.10.3 Parameter Set (if not specified -60 dBm)<br>18 Integer(6..43) (-70 dBm Received if pathloss not specified)<br>TDD<br><br>Enabled<br>3.84 Mcps TDD | Rel-4<br><br>Rel-4<br><br>Rel-4<br><br><br>

| Information Element | Condition | Value/remark | Version |
|---|---|---|---|
| - Midamble configuration | | 16 | |
| - Midamble Shift | | Not Present | |
| - CHOICE TDD option | | 3.84 Mcps TDD | Rel-4 |
| - First timeslot Code List | | Repeated (1,2) for each channelisation code assigned in the slot to meet the needs of TS34.108 clause 6 Parameter Set. | |
| - channelisation codes | | (SF/ i) where i denotes an unassigned code matching the SF specified in TS34.108 clause 6 Parameter Set. | |
| - CHOICE more timeslots | | No more timeslots | |
| - UL CCTrCH List to Remove | | Not present | |
| CHOICE Mode | A1, A2, A3, A4, A5, A6, A7, A8, A9, A10 | TDD | |
| - Downlink PDSCH information | | No data | |
| Downlink HS-PDSCH Information | A1, A2, A3, A4, A5, A6, A7, A8, A9, A10 | Not Present | |
| Downlink information common for all radio links | A1, A2, A3 | | |
| - Downlink DPCH info common for all RL | | | |
| - Timing indication | | Maintain | |
| - CFN-targetSFN frame offset | | Not Present | |
| - Downlink DPCH power control information | | | |
| - CHOICE mode | | TDD | |
| - TPC Step Size | | 1 | |
| - MAC-d HFN initial value | | Not Present | |
| - CHOICE mode | | TDD | |
| - CHOICE mode | | TDD | |
| - CHOICE TDD option | | 3.84 Mcps TDD | Rel-4 |
| - Default DPCH Offset Value | | Not Present | |
| Downlink information common for all radio links | A4 | | |
| - Downlink DPCH info common for all RL | | | |
| - Timing indication | | Initialise | |
| - CFN-targetSFN frame offset | | Not Present | |
| - Downlink DPCH power control information | | | |
| - CHOICE mode | | TDD | |
| - TPC Step Size | | 1 | |
| - MAC-d HFN initial value | | Not Present | |
| - CHOICE mode | | TDD | |
| - CHOICE mode | | TDD | |
| - CHOICE TDD option | | 3.84 Mcps TDD | Rel-4 |
| - Default DPCH Offset Value | | | |
| - CHOICE mode | | TDD | |
| - Default DPCH Offset Value | | 0 Integer(0..7) | |
| Downlink information common for all radio links | A5, A6, A7, A8, A9, A10 | Not Present | |
| Downlink information per radio link list | A1, A2, A3 | | |
| - Downlink information for each radio link | | | |
| - Choice mode | | TDD | |
| - Primary CCPCH info | | | |
| - Choice mode | | TDD | |
| - Choice TDD Option | | 3.84 Mcps TDD | Rel-4 |
| - CHOICE *SyncCase* | | Sync Case 1 | |
| - Timeslot | | Reference clause 6.1.4 Default settings for cell 1 | |
| - Cell parameters ID | | Ref. to the Default setting in TS34.108 clause 6.1 (TDD) Integer(0..127) | |
| - SCTD indicator | | FALSE | |
| - Downlink DPCH info for each RL | | | |
| - CHOICE mode | | TDD | |
| - DL CCTrCh List | | | |

| Information Element | Condition | Value/remark | Version |
|---|---|---|---|
| - TFCS ID | | 2 Integer(1.8) | |
| - Time info | | | |
| - Activation time | | Now | |
| - Duration | | Infinite | |
| - Common timeslot info | | | |
| - 2nd interleaving mode | | Default value is "Frame" | |
| - TFCI coding | | Reference to TS34.108 clause 6 Parameter set | |
| - Puncturing limit | | Reference to TS34.108 clause 6 Parameter set | |
| - Repetition period | | 1 | |
| - Repetition length | | NULL | |
| - Downlink DPCH timeslots and codes | | | |
| - First individual timeslot info | | | |
| - Timeslot number | | | |
| - CHOICE TDD option | | 3.84 Mcps TDD | Rel-4 |
| - Timeslot number | | 4 OR 5 OR 6 | |
| - TFCI existence | | TRUE | |
| - Midamble shift and burst type | | | |
| - CHOICE TDD option | | 3.84 Mcps TDD | Rel-4 |
| - CHOICE *Burst Type* | | Type 1 | |
| - Midamble allocation mode | | Default midamble | |
| - Midamble configuration | | 16 | |
| - Midamble Shift | | Not Present | |
| - CHOICE TDD option | | 3.84 Mcps TDD | Rel-4 |
| - First timeslot channelisation codes | | Repeated (1,2) for each channelisation code assigned in the slot to meet the needs of TS34.108 clause 6 Parameter Set. | |
| - CHOICE codes representation | | Bitmap | |
| - Channelisation codes bitmap | | Reference to TS34.108 clause 6.10 Parameter Set | |
| - CHOICE more timeslots | | No more timeslots | |
| - UL CCTrCH TPC List | | Default (is previous list or all defined UL CCTrCHs.) | |
| - UL TPC TFCS Identity | | | |
| - TFCS ID | | 1 | |
| - Shared Channel Indicator | | FALSE | |
| - DL CCTrCH List to Remove | | Not present | |
| - SCCPCH Information for FACH | | Not Present | R99 and Rel-4 only |
| Downlink information per radio link list | A4 | | |
| - Downlink information for each radio link | | | |
| - Choice mode | | TDD | |
| - Primary CCPCH info | | | |
| - Choice mode | | TDD | |
| - Choice TDD Option | | 3.84 Mcps TDD | Rel-4 |
| - CHOICE *SyncCase* | | Sync Case 1 | |
| - Timeslot | | Reference clause 6.1.4 Default settings for cell 1 | |
| - Cell parameters ID | | Ref. to the Default setting in TS34.108 clause 6.1 (TDD) Integer(0..127) | |
| - SCTD indicator | | FALSE | |
| - Downlink DPCH info for each RL | | | |
| - CHOICE mode | | TDD | |
| - DL CCTrCh List | | | |

| Information Element | Condition | Value/remark | Version |
|---|---|---|---|
| - TFCS ID | | 2 Integer(1.8) | |
| - Time info | | | |
| - Activation time | | Now | |
| - Duration | | Infinite | |
| - Common timeslot info | | | |
| - 2nd interleaving mode | | Default value is "Frame" | |
| - TFCI coding | | Reference to TS34.108 clause 6 Parameter set | |
| - Puncturing limit | | Reference to TS34.108 clause 6 Parameter set | |
| - Repetition period | | 1 | |
| - Repetition length | | NULL | |
| - Downlink DPCH timeslots and codes | | | |
| - First individual timeslot info | | | |
| - Timeslot number | | | |
| - CHOICE TDD option | | 3.84 Mcps TDD | Rel-4 |
| - Timeslot number | | 4 OR 5 OR 6 | |
| - TFCI existence | | TRUE | |
| - Midamble shift and burst type | | | |
| - CHOICE TDD option | | 3.84 Mcps TDD | Rel-4 |
| - CHOICE *Burst Type* | | Type 1 | |
| - Midamble allocation mode | | Default midamble | |
| - Midamble configuration | | 16 | |
| - Midamble Shift | | Not Present | |
| - CHOICE TDD option | | 3.84 Mcps TDD | Rel-4 |
| - First timeslot channelisation codes | | Repeated (1,2) for each channelisation code assigned in the slot to meet the needs of TS34.108 clause 6 Parameter Set. | |
| - CHOICE codes representation | | Bitmap | |
| - Channelisation codes bitmap | | Reference to TS34.108 clause 6.10 Parameter Set | |
| - CHOICE more timeslots | | No more timeslots | |
| - UL CCTrCH TPC List | | Default (is previous list or all defined UL CCTrCHs.) | |
| - UL TPC TFCS Identity | | | |
| - TFCS ID | | 1 | |
| - Shared Channel Indicator | | FALSE | |
| - DL CCTrCH List to Remove | | Not present | |
| - SCCPCH Information for FACH | | Not Present | R99 and Rel-4 only |
| Downlink information per radio link list | A5 | | |
| - Downlink information for each radio link | | | |
| - Choice mode | | TDD | |
| - Primary CCPCH info | | | |
| - Choice mode | | TDD | |
| - Choice TDD Option | | 3.84 Mcps TDD | Rel-4 |
| - CHOICE *SyncCase* | | Sync Case 1 | |
| - Timeslot | | Reference clause 6.1.4 Default settings for cell 1 | |
| - Cell parameters ID | | Ref. to the Default setting in TS34.108 clause 6.1 (TDD) Integer(0..127) | |
| - SCTD indicator | | FALSE | |
| - Downlink DPCH info for each RL | | Not Present | |
| - SCCPCH Information for FACH | | Not Present | R99 and Rel-4 only |
| Downlink information per radio link list | A6, A7, A8, A9, A10 | Not Present | |

| Condition | Explanation |
|---|---|
| A1 | This IE need for "Non speech in CS" |
| A2 | This IE need for "Speech in CS" |
| A3 | This IE need for "Packet to CELL_DCH from CELL_DCH in PS" |
| A4 | This IE need for "Packet to CELL_DCH from CELL_FACH in PS" |
| A5 | This IE need for "Packet to CELL_FACH from CELL_DCH in PS" |
| A6 | This IE need for "Packet to CELL_FACH from CELL_FACH in PS" |
| A7 | This IE need for "Packet to URA_PCH from CELL_FACH in PS" |
| A8 | This IE need for "Packet to URA_PCH from CELL_DCH in PS" |
| A9 | This IE need for "Packet to CELL_PCH from CELL_FACH in PS" |
| A10 | This IE need for "Packet to CELL_PCH from CELL_DCH in PS" |

Contents of TRANSPORT CHANNEL RECONFIGURATION message: AM or UM (1.28 Mcps TDD)

| Information Element | Condition | Value/remark | Version |
|---|---|---|---|
| Message Type | A1, A2, A3, A4, A5, A6, A7, A8, A9, A10 | | |
| RRC transaction identifier | | Arbitrarily selects an integer between 0 and 3 | |
| Integrity check info | | | |
| - message authentication code | | SS calculates the value of MAC-I for this message and writes to this IE. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I. | |
| - RRC message sequence number | | SS provides the value of this IE, from its internal counter. | |
| Integrity protection mode info | | Not Present | |
| Ciphering mode info | | Not Present | |
| Activation time | A1, A2, A3 | (256+CFN-(CFN MOD 8 + 8))MOD 256 | |
| Activation time | A4, A5, A6, A7, A8, A9, A10 | Not Present | |
| New U-RNTI | | Not Present | |
| New C-RNTI | A1, A2, A3, A4, A7, A8, A9, A10 | Not Present | |
| New C-RNTI | A5, A6 | '1010 1010 1010 1010' | |
| New DSCH-RNTI | A1, A2, A3, A4, A5, A6, A7, A8, A9, A10 | Not Present | |
| New H-RNTI | A1, A2, A3, A4, A5, A6, A7, A8, A9, A10 | Not Present | Rel-5 |
| RRC State indicator | A1, A2, A3, A4 | CELL_DCH | |
| RRC State indicator | A5, A6 | CELL_FACH | |
| RRC State indicator | A7, A8 | URA_PCH | |
| RRC State indicator | A9, A10 | CELL_PCH | |
| UTRAN DRX cycle length coefficient | A1, A2, A3, A4,A5,A6 | Not Present | |
| UTRAN DRX cycle length coefficient | A7, A8, A9, A10 | 3 | |
| CN information info | | Not Present | |
| URA identity | | Not Present | |
| Downlink counter synchronization info | | Not Present | |
| UL Transport channel information for all transport channels | A1, A2, A5, A6 | Not Present | |
| UL Transport channel information for all transport channels | A3, A4 | | |
| - PRACH TFCS | | Not Present | |
| - CHOICE mode | | TDD | |
| - Individual UL CCTrCH information | | | |
| - UL TFCS Identity | | | |
| - TFCS ID | | 1 | |
| - Shared Channel Indicator | | FALSE | |
| - UL TFCS | | | |
| - CHOICE *TFCI signalling* | | Normal | |
| - TFCI Field 1 Information | | | |

| Information Element | Condition | Value/remark | Version |
|---|---|---|---|
| - CHOICE *TFCS representation* | | Complete reconfiguration | |
| - TFCS complete reconfiguration information | | | |
| - CHOICE *CTFC Size* | | Number of bits used must be enough to cover all combinations of CTFC from clause 6.11.5.4 Parameter Set. | |
| - CTFC information | | This IE is repeated for TFC numbers and reference to clause 6.11.5.4 Parameter Set | |
| - CTFC | | Reference to clause 6.11.5.4 Parameter Set | |
| - Power offset information | | | |
| - CHOICE Gain Factors | | Computed Gain Factors(The last TFC is set to Signalled Gain Factors) | |
| - Reference TFC ID | | 0 Integer(0.. 3) | |
| - CHOICE Gain Factors | | Signalled Gain Factors(Not Present if the CHOICE Gain Factors is set to ComputedGain Factors) | |
| - CHOICE mode | | TDD | |
| - Gain Factor $\beta_d$ | | 15 | |
| - Reference TFC ID | | 0 Integer(0.. 3) | |
| - CHOICE mode | | TDD | |
| - TFC subset | | | |
| - CHOICE Subset representation | | Full transport format combination set | |
| - TFC subset list | | Not Present | |
| Added or Reconfigured TrCH information list | A1, A2, A5, A6 | Not Present | |
| Added or Reconfigured TrCH information list | A4 | 2 TrCHs(DCH for DCCH and DCH for DTCH) | |
| - Added or Reconfigured UL TrCH information | | | |
| - Uplink transport channel type | | DCH | |
| - UL Transport channel identity | | 5 | |
| - TFS | | | |
| - CHOICE Transport channel type | | Dedicated transport channels | |
| - Dynamic Transport format information | | | |
| - RLC Size | | Reference to clause 6.11 Parameter Set | |
| - Number of TBs and TTI List | | This IE is repeated for maxTF number | |
| - Transmission Time Interval | | Not Present | |
| - Number of Transport blocks | | Reference to clause 6.11 Parameter Set | |
| - CHOICE Logical channel list | | All | |
| - Semi-static Transport Format information | | | |
| - Transmission time interval | | Reference to clause 6.11 Parameter Set | |
| - Type of channel coding | | Reference to clause 6.11 Parameter Set | |
| - Coding Rate | | Reference to clause 6.11 Parameter Set | |
| - Rate matching attribute | | Reference to clause 6.11 Parameter Set | |
| - CRC size | | Reference to clause 6.11 Parameter Set | |
| - Uplink transport channel type | | DCH | |
| - UL Transport channel identity | | 1 | |
| - TFS | | | |
| - CHOICE Transport channel type | | Dedicated transport channels | |
| - Dynamic Transport format information | | | |
| - RLC Size | | Reference to clause 6.11 Parameter Set | |
| - Number of TBs and TTI List | | This IE is repeated for maxTF number | |
| - Transmission Time Interval | | Not Present | |
| - Number of Transport blocks | | Reference to clause 6.11 | |

| Information Element | Condition | Value/remark | Version |
|---|---|---|---|
| | | Parameter Set | |
| - CHOICE Logical channel list | | All | |
| - Semi-static Transport Format information | | | |
| - Transmission time interval | | Reference to clause 6.11 Parameter Set | |
| - Type of channel coding | | Reference to clause 6.11 Parameter Set | |
| - Coding Rate | | Reference to clause 6.11 Parameter Set | |
| - Rate matching attribute | | Reference to clause 6.11 Parameter Set | |
| - CRC size | | Reference to clause 6.11 Parameter Set | |
| Added or Reconfigured TrCH information list | A3 | (DCH for DTCH) | |
| - Added or Reconfigured UL TrCH information | | | |
| - Uplink transport channel type | | DCH | |
| - UL Transport channel identity | | 1 | |
| - TFS | | | |
| - CHOICE Transport channel type | | Dedicated transport channels | |
| - Dynamic Transport format information | | | |
| - RLC Size | | Reference to clause 6.11 Parameter Set | |
| - Number of TBs and TTI List | 1 to maxTF | (This IE is repeated for TF number.) | |
| - Transmission Time Interval | | Not Present | |
| - Number of Transport blocks | | Reference to clause 6.11 Parameter Set | |
| - CHOICE Logical channel list | | All | |
| - Semi-static Transport Format information | | | |
| - Transmission time interval | | Reference to clause 6.11 Parameter Set | |
| - Type of channel coding | | Reference to clause 6.11 Parameter Set | |
| - Coding Rate | | Reference to clause 6.11 Parameter Set | |
| - Rate matching attribute | | Reference to clause 6.11 Parameter Set | |
| - CRC size | | Reference to clause 6.11 Parameter Set | |
| CHOICE *mode* | A1,A2,A3,A4,A5,A6 | TDD | |
| Downlink HS-PDSCH Information | | | Rel-5 |
| DL Transport channel information common for all transport channels | A1, A2, A5,A6 | Not Present | |
| DL Transport channel information common for all transport channel | A3,A4 | | |
| - SCCPCH TFCS | | Not Present | |
| - CHOICE mode | | TDD | |
| - Individual DL CCTrCH information | | | |
| - DL TFCS Identity | | | |
| - TFCS ID | | 2 | |
| - Shared Channel Indicator | | FALSE | |
| - CHOICE DL parameters | | Independent | |
| - DL TFCS | | | |
| - CHOICE TFCI Signalling | | Normal | |
| - TFCI Field 1 Information | | | |
| - CHOICE TFCS representation | | Complete reconfiguration | |
| - TFCS complete reconfiguration information | | | |
| - CHOICE CTFC Size | | Number of bits used must be enough to cover all combinations of CTFC from clause 6.11.5.4 Parameter Set. | |
| - CTFC information | | This IE is repeated for TFC numbers and reference to clause 6.11.5.4 | |
| - CTFC | | Reference to clause 6.11.5.4 | |

| Information Element | Condition | Value/remark | Version |
|---|---|---|---|
| | | Parameter Set | |
| - Power offset information | | Not Present | |
| Added or Reconfigured TrCH information list | A1, A2, A5, A6 | Not Present | |
| Added or Reconfigured TrCH information list | A4 | 2 TrCHs(DCH for DCCH and DCH for DTCH) | |
| - Added or Reconfigured DL TrCH information | | | |
| - Downlink transport channel type | | DCH | |
| - DL Transport channel identity | | 10 | |
| - CHOICE DL parameters | | Same as UL | |
| - Uplink transport channel type | | DCH | |
| - UL TrCH identity | | 5 | |
| - DCH quality target | | | |
| - BLER Quality value | | -2.0 Real(-6.3..0 by step of 0.1) | |
| - Transparent mode signalling info | | Not Present | |
| - Downlink transport channel type | | DCH | |
| - DL Transport channel identity | | 6 | |
| - CHOICE DL parameters | | Explicit | |
| - TFS | | | |
| - CHOICE Transport channel type | | Dedicated transport channels | |
| - Dynamic transport format information | | | |
| - RLC Size | | Reference to clause 6.11 Parameter Set | |
| - Number of TBs and TTI List | | (This IE is repeated for TF number.) | |
| - Transmission Time Interval | | Not Present | |
| - Number of Transport blocks | | Reference to clause 6.11 Parameter Set | |
| - Semi-static Transport Format information | | | |
| - Transmission time interval | | Reference to clause 6.11 Parameter Set | |
| - Type of channel coding | | Reference to clause 6.11 Parameter Set | |
| - Coding Rate | | Reference to clause 6.11 Parameter Set | |
| - Rate matching attribute | | Reference to clause 6.11 Parameter Set | |
| - CRC size | | Reference to clause 6.11 Parameter Set | |
| - DCH quality target | | | |
| - BLER Quality value | | -2.0 | |
| Added or Reconfigured TrCH information list | A3 | | |
| - Added or Reconfigured DL TrCH information | | | |
| - Downlink transport channel type | | DCH | |
| - DL Transport channel identity | | 6 | |
| - CHOICE DL parameters | | Explicit | |
| - TFS | | | |
| - CHOICE Transport channel type | | Dedicated transport channels | |
| - Dynamic transport format information | | | |
| - RLC Size | | Reference to clause 6.11 Parameter Set | |
| - Number of TBs and TTI List | | (This IE is repeated for TF number.) | |
| - Transmission Time Interval | | Not Present | |
| - Number of Transport blocks | | Reference to clause 6.11 Parameter Set | |
| - Semi-static Transport Format information | | | |
| - Transmission time interval | | Reference to clause 6.11 Parameter Set | |
| - Type of channel coding | | Reference to clause 6.11 Parameter Set | |
| - Coding Rate | | Reference to clause 6.11 Parameter Set | |
| - Rate matching attribute | | Reference to clause 6.11 Parameter Set | |
| - CRC size | | Reference to clause 6.11 Parameter Set | |
| - DCH quality target | | | |

| Information Element | Condition | Value/remark | Version |
|---|---|---|---|
| - BLER Quality value | | -2.0 | |
| - Transparent mode signalling info | | Not Present | |
| Frequency info | A1, A2, A3, A4, A5 | | |
| - Choice mode | | TDD | |
| - UARFCN (Nt) | | Reference to clause 5.1 Test frequencies | |
| Frequency info | A6, A7, A8, A9, A10 | Not Present | |
| Maximum allowed UL TX power | | 33dBm | |
| CHOICE *channel requirement* | A5, A6, A7, A8, A9, A10 | Not Present | |
| CHOICE *channel requirement* | A1, A2, A3, A4 | Uplink DPCH info | |
| - Uplink DPCH power control info | | | |
| - CHOICE mode | | TDD | Rel-4 |
| - CHOICE TDD option | | 1.28 Mcps TDD | Rel-4 |
| - PRXPDPCHdes | | -80 Integer(-120…-58 by step of 1) | |
| - CHOICE UL OL PC info | | Individually Signalled | |
| - CHOICE TDD option | | 1.28 Mcps TDD | |
| - TPC step size | | 1 | |
| - Primary CCPCH Tx Power | | 20 Integer(6..43) | |
| - CHOICE mode | | TDD | |
| - Uplink Timing Advance Control | | | |
| - CHOICE Timing Advance | | Enabled | |
| - CHOICE TDD option | | 1.28 Mcps TDD | |
| - Uplink synchronization parameters | | | |
| - Uplink synchronization step size | | 1 | |
| - Uplink synchronization frequency | | 1 | |
| - Synchronization parameters | | | |
| - SYNC_UL codes bitmap | | 01010101 | |
| - FPACH info | | | |
| - Timeslot number | | 0 | |
| - Channelisation code | | 16/15 | |
| - Midamble Shift and burst type | | | |
| - CHOICE TDD option | | 1.28 Mcps TDD | |
| - Midamble Allocation Mode | | Default midamble | |
| - Midamble configuration | | 16 Integer(2, 4, 6, 8, 10, 12, 14, 16) | |
| - WT | | 4 Integer(1..4) | |
| - PRXUpPCHdes | | -80 dBm | |
| - SYNC_UL procedure | | | |
| - Max SYNC_UL Transmissions | | 2 | |
| - Power Ramp Step | | 2 | |
| - UL CCTrCH List | | | |
| - TFCS ID | | 1 | |
| - UL Target SIR | | Real (-11 .. 20 by step of 0.5dB) Reference to clause 6 Parameter set. | |
| - Time info | | | |
| - Activation time | | (256+CFN-(CFN MOD 8 + 8))MOD 256 | |
| - Duration | | Infinite | |
| - Common timeslot info | | | |
| - 2nd interleaving mode | | Default value is "Frame" | |
| - TFCI coding | | Reference to clause 6 Parameter set | |
| - Puncturing limit | | Reference to clause 6 Parameter set | |
| - Repetition period | | 1 | |
| - Repetition length | | | |
| - Uplink DPCH timeslots and code | | | |
| - Dynamic SF usage | | FALSE | |
| - First individual timeslot info | | | |
| - Timeslot number | | | |
| - CHOICE TDD option | | 1.28 Mcps TDD | |
| - Timeslot number | | 1 OR 2 OR 3 | |
| - TFCI existence | | TRUE | |
| - Midamble shift and burst type | | | |

| Information Element | Condition | Value/remark | Version |
|---|---|---|---|
| - CHOICE TDD option | | 1.28 Mcps TTD | |
| - Midamble allocation mode | | Default midamble | |
| - Midamble configuration | | 16 | |
| - Midamble Shift | | Not Present | |
| - CHOICE TDD option | | 1.28 Mcps TDD | |
| - Modulation | | QPSK | |
| - SS-TPC Symbols | | 1 | |
| - Additional TPC-SS Symbols | | Not present | |
| - First timeslot Code List | | Repeated (1,2) for each channelisation code assigned in the slot to meet the needs of clause 6 Parameter Set. | |
| - channelisation codes | | (SF/ i) where i denotes an unassigned code matching the SF specified in clause 6 Parameter Set. | |
| - CHOICE more timeslots | | No more timeslots | |
| - UL CCTrCH List to Remove | | Not present | |
| CHOICE Mode | A1, A2, A3, A4, A5, A6, A7, A8, A9, A10 | TDD | |
| Downlink HS-PDSCH Information | A1, A2, A3, A4, A5, A6, A7, A8, A9, A10 | Not Present | |
| Downlink information common for all radio links | A1, A2, A3 | | |
| - Downlink DPCH info common for all RL | | | |
| - Timing indication | | Maintain | |
| - CFN-targetSFN frame offset | | Not Present | |
| - Downlink DPCH power control information | | | |
| - CHOICE mode | | TDD | |
| - TPC Step Size | | 1 | |
| - MAC-d HFN initial value | | Not Present | |
| - CHOICE mode | | TDD | |
| - CHOICE mode | | TDD | |
| - CHOICE TDD option | | 1.28 Mcps TDD | |
| - TSTD indicator | | FALSE | |
| - Default DPCH Offset Value | | Not Present | |
| Downlink information common for all radio links | A4 | | |
| - Downlink DPCH info common for all RL | | | |
| - Timing indication | | Initialize | |
| - CFN-targetSFN frame offset | | Not Present | |
| - Downlink DPCH power control information | | | |
| - CHOICE mode | | TDD | |
| - TPC Step Size | | 1 | |
| - MAC-d HFN initial value | | Not Present | |
| - CHOICE mode | | TDD | |
| - CHOICE mode | | TDD | |
| - CHOICE TDD option | | 1.28 Mcps TDD | |
| - TSTD indicator | | FALSE | |
| - Default DPCH Offset Value | | | |
| - CHOICE mode | | TDD | |
| - Default DPCH Offset Value | | 0 Integer(0..7) | |
| Downlink information common for all radio links | A5, A6, A7, A8, A9, A10 | Not Present | |
| Downlink information per radio link list | A1, A2, A3 | | |
| - Downlink information for each radio link | | | |
| - Choice mode | | TDD | |
| - Primary CCPCH info | | | |
| - Choice mode | | TDD | |
| - Choice TDD Option | | 1.28 Mcps TDD | |
| - TSTD indicator | | FALSE | |
| - Cell parameters ID | | Ref. to the Default setting in clause 6.1 (TDD) Integer(0..127) | |
| - SCTD indicator | | FALSE | |
| - Downlink DPCH info for each RL | | | |
| - CHOICE mode | | TDD | |
| - DL CCTrCh List | | | |

| Information Element | Condition | Value/remark | Version |
|---|---|---|---|
| - TFCS ID | | 2 Integer(1.8) | |
| - Time info | | | |
| - Activation time | | Now | |
| - Duration | | Infinite | |
| - Common timeslot info | | | |
| - 2nd interleaving mode | | Default value is "Frame" | |
| - TFCI coding | | Reference to clause 6 Parameter set | |
| - Puncturing limit | | Reference to clause 6 Parameter set | |
| - Repetition period | | 1 | |
| - Repetition length | | NULL | |
| - Downlink DPCH timeslots and codes | | | |
| - First individual timeslot info | | | |
| - Timeslot number | | | |
| - CHOICE TDD option | | 1.28 Mcps TDD | |
| - Timeslot number | | 4 OR 5 OR 6 | |
| - TFCI existence | | TRUE | |
| - Midamble shift and burst type | | | |
| - CHOICE TDD option | | 1.28 Mcps TDD | |
| - Midamble allocation mode | | Default midamble | |
| - Midamble configuration | | 16 | |
| - Midamble Shift | | Not Present | |
| - CHOICE TDD option | | 1.28 Mcps TDD | |
| - Modulation | | QPSK | |
| - SS-TPC Symbols | | 1 | |
| - Additional TPC-SS Sysbols | | Not present | |
| - First timeslot channelisation codes | | Repeated (1,2) for each channelisation code assigned in the slot to meet the needs of clause 6 Parameter Set. | |
| - CHOICE codes representation | | | |
| - Channelisation codes bitmap | | Reference to clause 6.11 Parameter Set | |
| - CHOICE more timeslots | | No more timeslots | |
| - UL CCTrCH TPC List | | This list is not required for 1.28 Mcps TDD and is to be ignored by the UE. | |
| - UL TPC TFCS Identity | | | |
| - TFCS ID | | 1 | |
| - Shared Channel Indicator | | FALSE | |
| - DL CCTrCH List to Remove | | Not present | |
| - SCCPCH Information for FACH | | Not Present | R99 and Rel-4 only |
| Downlink information per radio link list | A4 | | |
| - Downlink information for each radio link | | | |
| - Choice mode | | TDD | |
| - Primary CCPCH info | | | |
| - Choice mode | | TDD | |
| - Choice TDD Option | | 1.28 Mcps TDD | |
| - TSTD indicator | | FALSE | |
| - Cell parameters ID | | Ref. to the Default setting in clause 6.1 (TDD) Integer(0..127) | |
| - SCTD indicator | | FALSE | |
| - Downlink DPCH info for each RL | | | |
| - CHOICE mode | | TDD | |
| - DL CCTrCh List | | | |
| - TFCS ID | | 2 Integer(1.8) | |
| - Time info | | | |
| - Activation time | | Now | |
| - Duration | | Infinite | |
| - Common timeslot info | | | |
| - 2nd interleaving mode | | Default value is "Frame" | |
| - TFCI coding | | Reference to clause 6 Parameter set | |
| - Puncturing limit | | Reference to clause 6 Parameter set | |

| Information Element | Condition | Value/remark | Version |
|---|---|---|---|
| - Repetition period | | 1 | |
| - Repetition length | | NULL | |
| - Downlink DPCH timeslots and codes | | | |
| - First individual timeslot info | | | |
| - Timeslot number | | | |
| - CHOICE TDD option | | 1.28 Mcps TDD | |
| - Timeslot number | | 4 OR 5 OR 6 | |
| - TFCI existence | | TRUE | |
| - Midamble shift and burst type | | | |
| - CHOICE TDD option | | 1.28 Mcps TDD | |
| - Midamble allocation mode | | Default midamble | |
| - Midamble configuration | | 16 | |
| - Midamble Shift | | Not Present | |
| - CHOICE TDD option | | 1.28 Mcps TDD | |
| - Modulation | | QPSK | |
| - SS-TPC Symbols | | 1 | |
| - Additional TPC-SS Sysbols | | Not present | |
| - First timeslot channelisation codes | | Repeated (1,2) for each channelisation code assigned in the slot to meet the needs of clause 6 Parameter Set. | |
| - CHOICE codes representation | | | |
| - Channelisation codes bitmap | | Reference to clause 6.11 Parameter Set | |
| - CHOICE more timeslots | | No more timeslots | |
| - UL CCTrCH TPC List | | This list is not required for 1.28 Mcps TDD and is to be ignored by the UE. | |
| - UL TPC TFCS Identity | | | |
| - TFCS ID | | 1 | |
| - Shared Channel Indicator | | FALSE | |
| - DL CCTrCH List to Remove | | Not present | |
| - SCCPCH Information for FACH | | Not Present | R99 and Rel-4 only |
| Downlink information per radio link list | A5 | | |
| - Downlink information for each radio link | | | |
| - Choice mode | | TDD | |
| - Primary CCPCH info | | | |
| - Choice mode | | TDD | |
| - Choice TDD Option | | 1.28 Mcps TDD | |
| - TSTD indicator | | FALSE | |
| - Cell parameters ID | | Ref. to the Default setting in clause 6.1 (TDD) Integer(0..127) | |
| - SCTD indicator | | FALSE | |
| - Downlink DPCH info for each RL | | Not Present | |
| - SCCPCH Information for FACH | | Not Present | R99 and Rel-4 only |
| Downlink information per radio link list | A6, A7, A8, A9, A10 | Not Present | |

| Condition | Explanation |
|---|---|
| A1 | This IE need for "Non speech in CS" |
| A2 | This IE need for "Speech in CS" |
| A3 | This IE need for "Packet to CELL_DCH from CELL_DCH in PS" |
| A4 | This IE need for "Packet to CELL_DCH from CELL_FACH in PS" |
| A5 | This IE need for "Packet to CELL_FACH from CELL_DCH in PS" |
| A6 | This IE need for "Packet to CELL_FACH from CELL_FACH in PS" |
| A7 | This IE need for "Packet to URA_PCH from CELL_FACH in PS" |
| A8 | This IE need for "Packet to URA_PCH from CELL_DCH in PS" |
| A9 | This IE need for "Packet to CELL_PCH from CELL_FACH in PS" |
| A10 | This IE need for "Packet to CELL_PCH from CELL_DCH in PS" |

Contents of TRANSPORT CHANNEL RECONFIGURATION COMPLETE message: AM (3.84 Mcps TDD)

| Information Element | Value/remark | Version |
|---|---|---|
| Message Type | | |
| RRC transaction identifier | Checked to see if it's set to identical value of the same IE in the downlink PHYSICAL CHANNEL RECONFIGURATION message | |
| Integrity check info | | |
| - Message authentication code | This IE is checked to see if it is present. The value is compared against the XMAC-I value computed by SS. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I. | |
| - RRC Message sequence number | This IE is checked to see if it is present. The value is used by SS to compute the XMAC-I value. | |
| Uplink integrity protection activation info | Not checked | |
| CHOICE mode | TDD | |
| - CHOICE TDD option | 3.84 Mcps TDD | Rel-4 |
| - Uplink Timing Advance | 0 | |
| COUNT-C activation time | Not checked | |
| Radio bearer uplink ciphering activation time info | Not checked | |
| Uplink counter synchronisation info | Not checked | |

Contents of TRANSPORT CHANNEL RECONFIGURATION COMPLETE message: AM (1.28 Mcps TDD)

| Information Element | Value/remark | Version |
|---|---|---|
| Message Type | | |
| RRC transaction identifier | Checked to see if the value is identical to the same IE in the downlink TRANSPORT CHANNEL RECONFIGURATION message | |
| Integrity check info | | |
| - Message authentication code | This IE is checked to see if it is present. The value is compared against the XMAC-I value computed by SS. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I. | |
| - RRC Message sequence number | This IE is checked to see if it is present. The value is used by SS to compute the XMAC-I value. | |
| Uplink integrity protection activation info | Not checked | |
| CHOICE mode | TDD | Rel-4 |
| CHOICE TDD option | 1.28 Mcps TDD | Rel-4 |
| COUNT-C activation time | Not checked | |
| Radio bearer uplink ciphering activation time info | Not checked | |
| Uplink counter synchronization info | Not checked | |

Contents of TRANSPORT CHANNEL RECONFIGURATION FAILURE message: AM

| Information Element | Value/remark |
|---|---|
| Message Type | |
| RRC transaction identitifer | Checked to see if it is set to identical value of the same IE in the downlink TRANSPORT CHANNEL RECONFIGURATION message. |
| Integrity check info | |
| - Message authentication code | This IE is checked to see if it is present. The value is compared against the XMAC-I value computed by SS. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I. |
| - RRC Message sequence number | This IE is checked to see if it is present. The value is used by SS to compute the XMAC-I value. |
| Failure cause | Checked to see if it meets test requirement |

Contents of TRANSPORT FORMAT COMBINATION CONTROL message: AM or UM (in CELL_DCH)

| Information Element | Value/remark |
|---|---|
| Message Type | |
| RRC transaction identifier | Arbitrarily selects an integer between 0 and 3 |
| Integrity check info | |
| - Message authentication code | SS calculates the value of MAC-I for this message and writes to this IE. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I. |
| - RRC Message sequence number | SS provides the value of this IE, from its internal counter. |
| CHOICE mode | TDD |
| - TFCS Id | |
| - TFCS ID | 1 |
| - Shared Channel Indicator | FALSE |
| DPCH/PUSCH TFCS in uplink | |
| - CHOICE *Subset representation* | Allowed transport format combination list |
| - Allowed transport format combination list | 0 (The TFC is constructed from ALL TF0) |
| Activation time for TFC subset | Now |
| TFC Control duration | Not Present |

Contents of TRANSPORT FORMAT COMBINATION CONTROL FAILURE message: AM

| Information Element | Value/remark |
|---|---|
| Message Type | |
| RRC transaction identitifer | Checked to see if it is set to identical value of the same IE in the downlink TRANSPORT CHANNEL RECONFIGURATION message. |
| Integrity check info | |
| - Message authentication code | This IE is checked to see if it is present. The value is compared against the XMAC-I value computed by SS. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I. |
| - RRC Message sequence number | This IE is checked to see if it is present. The value is used by SS to compute the XMAC-I value. |
| Failure cause | Checked to see if it meets test requirement |

Contents of RRC CONNECTION REJECT message: UM

| Information Element | Value/remark |
|---|---|
| Message Type | |
| RRC transaction identifier | Arbitrarily selects an integer between 0 and 3 |
| Initial UE identity | Select the same type as in the IE "Initial UE Identity" in RRC CONNECTION REQUEST" message. |
| Rejection cause | Unspecified |
| Wait Time | 0 |
| Redirection info | Not Present |

Contents of CELL UPDATE message: TM

| Information Element | Value/remark |
|---|---|
| Message Type | |
| U-RNTI | Checked to see if it is set to the following values |
| - SRNC identity | 0000 0000 0001B |
| - S-RNTI | 0000 0000 0000 0000 0001B |
| RRC transaction identifier | Checked to see if it is absent |
| Integrity check info | |
| - Message authentication code | This IE is checked to see if it is present. The value is compared against the XMAC-I value computed by SS. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I. |
| - RRC Message sequence number | This IE is checked to see if it is present. The value is used by SS to compute the XMAC-I value. |
| START List | Checked to see if the 'CN domain identity' and 'START' IEs are present for all CN domains supported by the UE |
| - CN domain identity | Checked to see if it is one of the supported CN domains |
| - START | Checked to see if it is present |

| Information Element | Value/remark |
|---|---|
| AM_RLC error indication (RB2, RB3 or RB4) | Checked to see if it is set to 'FALSE' |
| AM_RLC error indication (RB>4) | Checked to see if it is set to 'FALSE' |
| Cell update cause | See the test content |
| Failure cause | Checked to see if it is absent |
| RB timer indicator | |
| - T314 expired | Checked to see if it is set to 'FALSE' |
| - T315 expired | Checked to see if it is set to 'FALSE' |
| Measured results on RACH | Not checked |

Contents of CELL UPDATE CONFIRM message: UM

| Information Element | Value/remark |
|---|---|
| Message Type | |
| U-RNTI | If this message is sent on CCCH, use the following values. Else, this IE is absent. |
| - SRNC identity | 0000 0000 0001B |
| - S-RNTI | 0000 0000 0000 0000 0001B |
| RRC transaction identifier | Selects an arbitrary integer between 0 to 3 |
| Integrity check info | |
| - Message authentication code | Set to MAC-I value computed by the SS. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I. |
| - RRC Message Sequence Number | Set to an arbitrarily selected integer between 0 and 15 |
| Integrity protection mode info | Not Present |
| Ciphering mode info | Not Present |
| Activation time | Not Present - use default value |
| New U-RNTI | Not Present |
| New C-RNTI | Not Present |
| New DSCH-RNTI | Not Present |
| RRC State indicator | CELL_FACH |
| UTRAN DRX cycle length coefficient | Not Present |
| RLC re-establish indicator (RB2, RB3 and RB4) | FALSE |
| RLC re-establish indicator (RB5 and upwards) | FALSE |
| CN information info | Not Present |
| URA identity | |
| -URA identity | 0000 0000 0000 0001B |
| RB information to release list | Not Present |
| RB information to reconfigure list | Not Present |
| RB information to be affected list | Not Present |
| Downlink counter synchronization info | Not Present |
| UL Transport channel information common for all transport channels | Not Present |
| Deleted TrCH information list | Not Present |
| Added or Reconfigured TrCH information list | Not Present |
| CHOICE Mode | TDD |
| DL Transport channel information common for all transport channels | Not Present |
| Deleted TrCH information list | Not Present |
| Added or Reconfigured TrCH information list | Not Present |
| Frequency info | Not Present |
| Maximum allowed UL TX power | Not Present |
| CHOICE channel requirement | Not Present |
| CHOICE mode | TDD |
| Downlink information common for all radio links | Not Present |
| Downlink information per radio link list | Not Present |

Contents of HANDOVER FROM UTRAN COMMAND-GSM message: AM

| Information Element | Value/remark |
|---|---|
| Message Type | |
| RRC transaction identifier | Arbitrarily selects an integer between 0 and 3 |
| Integrity check info | |
| - Message authentication code | Set to MAC-I value computed by the SS. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I |
| - RRC Message sequence number | Set to an arbitrarily selected integer between 0 and 15 |

| Information Element | Value/remark |
|---|---|
| Activation time | Not Present - use default value "now" |
| RAB info | For each RAB to be handed over. In this version, the maximum size of the list of 1 shall be applied for all system types. |
| - RAB identity | 0000 0001B |
| - CN domain identity | CS domain |
| - NAS Synchronization Indicator | Not present |
| - Re-establishment time | Use T315 |
| CHOICE *System type* | GSM |
| - Frequency band | Set to "GSM/ PCS 1900" if GSM/ PCS 1900 is used in this test. Otherwise set to "GSM/DCS 1800 Band" |
| - CHOIC GSM message | Single GSM message |
| - Single GSM message | GSM HANDOVER COMMAND formatted and coded according to GSM specifications as BIT STRING (1..512). The first/ l*eftmost/ most significant* bit of the bit string contains bit 8 of the first octet of the GSM message. The contents of the HANDOVER COMMAND is to be defined in the specific test case. |

Contents of HANDOVER FROM UTRAN FAILURE message: AM

| Information Element/Group name | Value/remark |
|---|---|
| Message Type | |
| RRC transaction identifier | Checked to see if it matches the same value used in the corresponding downlink HANDOVER FROM UTRAN COMMAND - GSM message |
| Integrity check info | |
| - Message authentication code | This IE is checked to see if it is present. The value is compared against the XMAC-I value computed by SS. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I. |
| - RRC Message sequence number | This IE is checked to see if it is present. The value is used by SS to compute the XMAC-I value. |
| Inter-RAT handover failure | |
| - Inter-RAT handover failure cause | physical channel failure |
| - Protocol error information | Check to see if it is absent |
| Inter-system message | Not checked |

Contents of MEASUREMENT CONTROL Message: AM (Intra-frequence measurement) (1.28 Mcps TDD)

| Information Element | Value/remark |
|---|---|
| Message Type | |
| **UE information elements** | |
| RRC transaction identifier | Arbitrarily selects an unused integer between 0 to 3 |
| Integrity check info | |
| - Message authentication code | SS calculates the value of MAC-I for this message and writes to this IE. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I. |
| - RRC message sequence number | SS provides the value of this IE, from its internal counter. |
| **Measurement information elements** | |
| Measurement Identity | 1 |
| Measurement Command | Setup |
| Measurement Reporting Mode | |
| - Measurement Report Transfer Mode | Acknowledged mode RLC |
| - Periodical Reporting/Event Trigger Reporting Mode | Periodical reporting |
| Additional measurement list | Not Present |
| CHOICE Measurement type | Intra-frequency measurement |
| - Intra-frequency measurement | |
| **- Intra-frequency cell info list** | |
| - CHOICE intra-frequency cell removal | Not present |
| - New intra-frequency cell | |
| - Intra-frequency cell-id | 1 |
| - Cell info | |
| - Cell individual offset | 0dB |
| - Reference time difference to cell | Not Present |
| - Read SFN number | FALSE |
| - CHOICE mode | TDD |

| Information Element | Value/remark |
|---|---|
| - Primary CCPCH info | |
| - CHOICE mode | TDD |
| - CHOICE TDD option | 1.28 Mcps TDD |
| -TSTD indicator | FALSE |
| - Cell parameters ID | Reference clause 6.1.4 Default settings for cell 1(TDD) |
| - SCTD indicator | FALSE |
| - Primary CCPCH Tx power | Not present |
| - Timeslot list | Not present |
| - Cells for measurement | Not present |
| - **Intra-frequency measurement quantity** | |
| - Filter coefficient | Not present (use default 0) |
| - CHOICE mode | TDD |
| - Measurement quantity list | |
| - Measurement quantity | Primary CCPCH RSCP |
| **- Intra-frequency reporting quantity** | |
| - Reporting quantities for active set cells | |
| - Cell synchronization information reporting indicator | FALSE |
| - Cell Identity reporting indicator | TRUE |
| - CHOICE mode | TDD |
| - Timeslot ISCP reporting indicator | FALSE |
| - Proposed TGSN reporting indicator | FALSE |
| - Primary CCPCH RSCP reporting indicator | FALSE |
| - Pathloss reporting indicator | FALSE |
| - Reporting quantities for monitored set cells | |
| - Cell synchronization information reporting indicator | FALSE |
| - Cell Identity reporting indicator | TRUE |
| - CHOICE mode | TDD |
| - Timeslot ISCP reporting indicator | FALSE |
| - Proposed TGSN reporting indicator | FALSE |
| - Primary CCPCH RSCP reporting indicator | FALSE |
| - Pathloss reporting indicator | FALSE |
| - Reporting quantities for detected set cells | Not present |
| **- Reporting cell status** | Not present |
| **- Measurement validity** | Not present |
| **- CHOICE report criteria** | Intra-frequency measurement reporting criteria |
| - Parameters required for each event | |
| - Intra-frequency event identity | 1g |
| - Triggering condition 1 | Not present<br>( this IE is MP only for event "1b" or "1f", TDD should not present) |
| - Triggering condition 2 | Not present<br>(this IE is MP only for event "1c", TDD should not present) |
| - Reporting Range Constant | Not present<br>(this IE is MP only for event "1a" or "1b", TDD should not present) |
| - Cells forbidden to affect Reporting range | Not present<br>(this IE is MP only for event "1a" or "1b", TDD should not present) |
| - W | Not present<br>(this IE is MP only for event "1a" or "1b", TDD should not present) |
| - Hysteresis | 0 dBm |
| - Threshold used frequency | Not present<br>(this IE is MP only for event "1e", "1f", "1h" or "1i") |
| - Reporting deactivation threshold | Not present<br>(this IE is MP only for event "1a", TDD should not present) |
| - Replacement activation threshold | Not present<br>(this IE is MP only for event "1c" TDD should not present) |
| - Time to trigger | 0 ms |
| - Amount of reporting | Not present<br>(this IE is MP only for event "1a" or "1c" TDD should not present) |

| Information Element | Value/remark |
|---|---|
| - Reporting interval | Not present<br>(this IE is MP only for event "1a" or "1c", TDD should not present) |
| - Reporting cell status | Not present |
| **Physical channel information elements** | |
| DPCH Compressed mode status info | Not Present |

Contents of MEASUREMENT CONTROL Message: AM (Inter-frequence measurement) (1.28 Mcps TDD)

| Information Element | Value/remark |
|---|---|
| Message Type | |
| **UE information elements** | |
| RRC transaction identifier | Arbitrarily selects an unused integer between 0 to 3 |
| Integrity check info | |
| - Message authentication code | SS calculates the value of MAC-I for this message and writes to this IE. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I. |
| - RRC message sequence number | SS provides the value of this IE, from its internal counter. |
| **Measurement information elements** | |
| Measurement Identity | 2 |
| Measurement Command | Setup |
| Measurement Reporting Mode | |
| - Measurement Report Transfer Mode | Acknowledged mode RLC |
| - Periodical Reporting/Event Trigger Reporting Mode | Periodical reporting |
| Additional measurement list | Not present |
| CHOICE Measurement type | Inter-frequency measurement |
| - Inter-frequency measurement | |
| **- Inter-frequency cell info list** | |
| - CHOICE inter-frequency cell removal | Not present |
| - New inter-frequency cell | |
| - Inter-frequency cell-id | 4 |
| - Frequency info | |
| - CHOICE mode | TDD |
| - UARFCN (Nt) | Reference to table 6.1.7 for cell 4 |
| - Cell info | |
| - Cell individual offset | 0dB |
| - Reference time difference to cell | Not Present |
| - Read SFN number | FALSE |
| - CHOICE mode | TDD |
| - Primary CCPCH info | |
| - CHOICE mode | TDD |
| - CHOICE TDD option | 1.28 Mcps TDD |
| -TSTD indicator | FALSE |
| - Cell parameters ID | Reference clause 6.1.4 Default settings for cell 4(TDD) |
| - SCTD indicator | FALSE |
| - Primary CCPCH Tx power | Not present |
| - Timeslot list | Not present |
| - Cells for measurement | Not present |
| - **Inter-frequency measurement quantity** | |
| - CHOICE *reporting criteria* | Inter-frequency reporting criteria |
| - Inter-frequency reporting criteria | |
| - Filter coefficient | Not present (use default 0) |
| - CHOICE *mode* | TDD |
| - Measurement quantity for frequency quality estimate | Primary CCPCH RSCP |
| **- Inter-frequency reporting quantity** | |
| - UTRA Carrier RSSI | FALSE |
| - Frequency quality estimate | FALSE<br>This parameters is not used in this release and should be set to FALSE. It shall be ignored by the UE. |
| - Non frequency related cell reporting quantities | |
| - Cell synchronization information reporting indicator | FALSE |
| - Cell Identity reporting indicator | FALSE |
| - CHOICE mode | TDD |
| - Timeslot ISCP reporting indicator | FALSE |

| Information Element | Value/remark |
|---|---|
| - Proposed TGSN reporting indicator | FALSE |
| - Primary CCPCH RSCP reporting indicator | FALSE |
| - Pathloss reporting indicator | FASLE |
| **- Reporting cell status** | Not present |
| **- Measurement validity** | Not present |
| - **Inter-frequency set update** | Not present<br>(this IE only for FDD) |
| **- CHOICE report criteria** | Inter-frequency measurement reporting criteria |
| - Parameters required for each event | |
| - Inter-frequency event identity | 2b |
| - Threshold used frequency | -70 dBm<br>(this IE is MP for event 2b, 2d, or 2f<br>Ranges used depend on measurement quantity.<br>CPICH Ec/No -24..0dB<br>CPICH/Primary CCPCH RSCP -115..-25dBm) |
| - W used frequency | 0<br>(this IE is MP for event 2a, 2b, 2d or 2f<br>Real(0, 0.1..2.0 by step of 0.1)) |
| - Hysteresis | 1 dBm |
| - Time to trigger | 5 000 ms |
| - Reporting cell status | Within active set or within virtual active set or of the other RAT |
| - Maximum number of reporting cells | 1 |
| - Parameters required for each non-used frequency | |
| - Threshold non used frequency | -70 dBm<br>(this IE is MP for event 2a, 2b, 2c or 2e<br>Ranges used depend on measurement quantity.<br>CPICH Ec/No -24..0dB<br>CPICH/Primary CCPCH RSCP -115..-25dBm.<br>This IE is not needed if the IE "Inter-frequency event identity" is set to 2a. However, it is specified to be mandatory to align with the ASN.1) |
| - W non-used frequency | 0<br>(this IE is MP if 2a, 2b, 2c or 2e<br>Real(0, 0.1..2.0 by step of 0.1)) |
| **Physical channel information elements** | |
| DPCH Compressed mode status info | Not Present |

Contents of MEASUREMENT CONTROL FAILURE Message: AM

| Information Element | Value/remark |
|---|---|
| Message Type | |
| RRC transaction identifier | Checked to see if it's set to the identical value for the same IE in the downlink MEASUREMENT CONTROL message |
| Integrity check info | |
| - Message authentication code | This IE is checked to see if it is present. The value is compared against the XMAC-I value computed by SS. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I. |
| - RRC Message sequence number | This IE is checked to see if it is present. The value is used by SS to compute the XMAC-I value. |
| Failure cause | See the test content |

Contents of MEASUREMENT REPORT message: AM（intra-frequency measurement (3.84 Mcps TDD)

| Information Element | Value/remark | Version |
|---|---|---|
| Message Type | | |
| Integrity check info | | |
| - Message authentication code | This IE is checked to see if it is present. The value is compared against the XMAC-I value computed by SS. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I. | |
| - RRC Message sequence number | This IE is checked to see if it is present. The value is used by SS to compute the XMAC-I value. | |
| Measurement identity | 1 | |
| Measured Results | | |
| - Intra-frequency measured results | | |
| - Cell measured results | | |
| - Cell Identity | Checked that this IE is present | |
| - Cell synchronisation information | Checked that this IE is absent | |
| - CHOICE mode | TDD | |
| - Cell parameters Id | Different from the Default setting in TS34.108 clause 6.1 (TDD) | |
| - Proposed TGSN | Checked that this IE is absent | |
| - Primary CCPCH RSCP | Checked that this IE is absent | |
| - Pathloss | Checked that this IE is absent | |
| - Timeslot list | Checked that this IE is absent | |
| Measured results on RACH | Checked that this IE is absent | |
| Additional measured results | Checked that this IE is absent | |
| Event results | | |
| - CHOICE *event result* | Intra-frequency measurement event results | |
| - Intra-frequency measurement event results | | |
| - Intra-frequency event identity | Ig | |
| - Cell measurement event results | | |
| - CHOICE *mode* | TDD | |
| - Primary CCPCH info | | |
| - CHOICE mode | TDD | |
| - CHOICE TDD option | 3.84 Mcps TDD | Rel-4 |
| - CHOICE *SyncCase* | Sync Case 1 | |
| - Timeslot | Reference clause 6.1.4 Default settings for cell 1(TDD) (S/B 0) | |
| - Cell parameters ID | Reference clause 6.1.4 Default settings for cell 1(TDD) | |
| - SCTD indicator | FALSE | |

Contents of MEASUREMENT REPORT message: AM（intra-frequency measurement）(1.28 Mcps TDD)

| Information Element | Value/remark |
|---|---|
| Message Type | |
| Integrity check info | |
| - Message authentication code | This IE is checked to see if it is present. The value is compared against the XMAC-I value computed by SS. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I. |
| - RRC Message sequence number | This IE is checked to see if it is present. The value is used by SS to compute the XMAC-I value. |
| Measurement identity | 1 |
| Measured Results | |
| - Intra-frequency measured results | |
| - Cell measured results | |
| - Cell Identity | Checked that this IE is present |
| - Cell synchronization information | Checked that this IE is absent |
| - CHOICE mode | TDD |
| - Cell parameters Id | Different from the Default setting in clause 6.1 (TDD) |
| - Proposed TGSN | Checked that this IE is absent |
| - Primary CCPCH RSCP | Checked that this IE is absent |
| - Pathloss | Checked that this IE is absent |
| - Timeslot list | Checked that this IE is absent |
| Measured results on RACH | Checked that this IE is absent |

| | |
|---|---|
| Additional measured results | Checked that this IE is absent |
| Event results | |
| - CHOICE *event result* | Intra-frequency measurement event results |
| - Intra-frequency measurement event results | |
| - Intra-frequency event identity | Ig |
| - Cell measurement event results | |
| - CHOICE *mode* | TDD |
| - Primary CCPCH info | |
| - CHOICE mode | TDD |
| - CHOICE TDD option | 1.28 Mcps TDD |
| -TSTD indicator | FALSE |
| - Cell parameters ID | Reference in clause 6.1.4 Default settings for cell 1(TDD) |
| - SCTD indicator | FALSE |

Contents of MEASUREMENT REPORT message: AM (inter-frequency measurement) (3.84 Mcps TDD)

| Information Element | Value/remark | Version |
|---|---|---|
| Message Type | | |
| Integrity check info | | |
| - Message authentication code | This IE is checked to see if it is present. The value is compared against the XMAC-I value computed by SS. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I. | |
| - RRC Message sequence number | This IE is checked to see if it is present. The value is used by SS to compute the XMAC-I value. | |
| Measurement identity | 1 | |
| Measured Results | Checked that this IE is absent | |
| Measured results on RACH | Checked that this IE is absent | |
| Additional measured results | Checked that this IE is absent | |
| Event results | | |
| - CHOICE *event result* | Inter-frequency measurement event results | |
| - Inter-frequency measurement event results | | |
| - Inter-frequency event identity | 2b | |
| - Inter-frequency cells | | |
| - Frequency info | Reference to table 6.1.7 for cell 4 | |
| - Non frequency related measurement event results | | |
| - Cell measurement event results | | |
| - CHOICE *mode* | TDD | |
| - Primary CCPCH info | | |
| - CHOICE mode | TDD | |
| - CHOICE TDD option | 3.84 Mcps TDD | Rel-4 |
| - CHOICE *SyncCase* | Sync Case 1 | |
| - Timeslot | Reference clause 6.1.4 Default settings for cell 1(TDD) (S/B 0) | |
| - Cell parameters ID | Reference clause 6.1.4 Default settings for cell 1(TDD) | |
| - SCTD indicator | FALSE | |
| GSM OTD reference cell | Checked that this IE is absent | Rel-4 |

Contents of MEASUREMENT REPORT message: AM (inter-frequency measurement) (1.28 Mcps TDD)

| Information Element | Value/remark | Version |
|---|---|---|
| Message Type | | |
| Integrity check info | | |
| - Message authentication code | This IE is checked to see if it is present. The value is compared against the XMAC-I value computed by SS. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I. | |
| - RRC Message sequence number | This IE is checked to see if it is present. The value is used by SS to compute the XMAC-I value. | |
| Measurement identity | 1 | |
| Measured Results | Checked that this IE is absent | |
| Measured results on RACH | Checked that this IE is absent | |

| Information Element | Value/remark | Version |
|---|---|---|
| Additional measured results<br>Event results<br>- CHOICE *event result*<br>- Inter-frequency measurement event results<br>- Inter-frequency event identity<br>- Inter-frequency cells<br>- Frequency info<br>- Non frequency related measurement event results<br>- Cell measurement event results<br>- CHOICE *mode*<br>- Primary CCPCH info<br>- CHOICE mode<br>- CHOICE TDD option<br>-TSTD indicator<br>- Cell parameters ID<br>- SCTD indicator<br>GSM OTD reference cell | Checked that this IE is absent<br><br>Inter-frequency measurement event results<br><br><br>2b<br><br>Reference to table 6.1.7 for cell 4<br><br><br><br><br>TDD<br><br>TDD<br>1.28 Mcps TDD<br><br>FALSE<br>Reference clause 6.1.4 Default settings for cell 1(TDD)<br>FALSE<br>Checked that this IE is absent | <br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br>Rel-4 |

Contents of PHYSICAL CHANNEL RECONFIGURATION message: AM or UM (3.84 Mcps TDD)

| Information Element | Condition | Value/remark | Version |
|---|---|---|---|
| Message Type | A1, A2, A3, A4, A5, A6, A7, A8, A9, A10 | | |
| RRC transaction identifier | | Arbitrarily selects an integer between 0 and 3 | |
| Integrity check info<br>- message authentication code | | SS calculates the value of MAC-I for this message and writes to this IE. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I. | |
| - RRC message sequence number | | SS provides the value of this IE, from its internal counter. | |
| Integrity protection mode info | | Not Present | |
| Ciphering mode info | | Not Present | |
| Activation time | A1, A2, A3 | (256+CFN-(CFN MOD 8 + 8))MOD 256 | |
| Activation time | A4, A5, A6, A7, A8, A9, A10 | Not Present | |
| New U-RNTI | | Not Present | |
| New C-RNTI | A1, A2, A3, A4, A7, A8, A9, A10 | Not Present | |
| New C-RNTI | A5, A6 | '1010 1010 1010 1010' | |
| New DSCH-RNTI | A1, A2, A3, A4, A5, A6, A7, A8, A9, A10 | Not Present | |
| New H-RNTI | A1, A2, A3, A4, A5, A6, A7, A8, A9, A10 | Not Present | Rel-5 |
| RRC State indicator | A1, A2, A3, A4 | CELL_DCH | |
| RRC State indicator | A5, A6 | CELL_FACH | |
| RRC State indicator | A7, A8 | URA_PCH | |
| RRC State indicator | A9, A10 | CELL_PCH | |
| UTRAN DRX cycle length coefficient | A1, A2, A3, A4, A5, A6 | Not Present | |
| UTRAN DRX cycle length coefficient | A7, A8, A9, A10 | 3 | |

| Information Element | Condition | Value/remark | Version |
|---|---|---|---|
| CN information info | | Not Present | |
| URA identity | | Not Present | |
| Downlink counter synchronisation info | | Not Present | |
| Frequency info | A1, A2, A3, A4, A5 | | |
| - Choice mode | | TDD | |
| - UARFCN (Nt) | | Reference to clause 5.1 Test frequencies | |
| Frequency info | A6, A7, A8, A9, A10 | Not Present | |
| Maximum allowed UL TX power | | 33dBm | |
| CHOICE channel requirement | A5, A6, A7, A8, A9, A10 | Not Present | |
| CHOICE channel requirement | A1, A2, A3, A4 | Uplink DPCH info | |
| - Uplink DPCH power control info | | | |
| - CHOICE mode | | TDD | |
| - UL target SIR | | 6 | |
| - CHOICE *UL OL PC info* | | Individually Signalled | |
| - CHOICE TDD option | | 3.84 Mcps TDD | Rel-4 |
| - Individual timeslot interference info | | Reference to TS34.108 clause 6.10.3 Parameter Set | |
| - Individual timeslot interference | | | |
| - CHOICE TDD option | | 3.84 Mcps TDD | Rel-4 |
| - Timeslot number | | As required by, Reference to TS34.108 clause 6.10.3 Parameter Set | |
| - TDD UL interference | | As required by, Reference to TS34.108 clause 6.10.3 Parameter Set (if not specified -60 dBm) | |
| - Primary CCPCH Tx Power | | 18 Integer(6..43) (-70 dBm Received if pathloss not specified) | |
| - CHOICE mode | | TDD | |
| - Uplink Timing Advance Control | | | |
| - CHOICE Timing Advance | | Enabled | |
| - CHOICE TDD option | | 3.84 Mcps TDD (Default) | Rel-4 |
| - UL CCTrCH List | | | |
| - TFCS ID | | 1 | |
| - UL Target SIR | | Real (-11 .. 20 by step of 0.5dB) Reference to TS34.108 Parameter set. | |
| - Time info | | | |
| - Activation time | | (256+CFN-(CFN MOD 8 + 8))MOD 256 | |
| - Duration | | Infinite | |
| - Common timeslot info | | | |
| - 2nd interleaving mode | | Default value is "Frame" | |
| - TFCI coding | | Reference to TS34.108 clause 6 Parameter set | |
| - Puncturing limit | | Reference to TS34.108 clause 6 Parameter set | |
| - Repetition period | | 1 | |
| - Repetition length | | Null | |
| - Uplink DPCH timeslots and code | | | |
| - Dynamic SF usage | | FALSE | |
| - First individual timeslot info | | | |
| - Timeslot number | | | |
| - CHOICE TDD option | | 3.84 Mcps TDD | Rel-4 |
| - Timeslot number | | 1 OR 2 OR 3 | |
| - TFCI existence | | TRUE | |
| - Midamble shift and burst type | | | |
| - CHOICE TDD option | | 3.84 Mcps TDD | Rel-4 |
| - Midamble allocation mode | | Default midamble | |
| - Midamble configuration | | 16 | |
| - Midamble Shift | | Not Present | |
| - CHOICE TDD option | | 3.84 Mcps TDD (No Data) | Rel-4 |
| - First timeslot Code List | | Repeated (1,2) for each channelisation code assigned in the slot to meet the needs of TS34.108 clause 6 Parameter | |

| Information Element | Condition | Value/remark | Version |
|---|---|---|---|
| - channelisation codes | | Set.<br>(SF/ i) where i denotes an unassigned code matching the SF specified in TS34.108 clause 6 Parameter Set. | |
| - CHOICE more timeslots | | No more timeslots (No Data) | |
| - UL CCTrCH List to Remove | | Not present | |
| **Downlink radio resources** | | | |
| CHOICE Mode | A1, A2, A3, A4, A5, A6, A7, A8, A9, A10 | TDD | |
| - Downlink PDSCH information | | No data | |
| Downlink HS-PDSCH Information | A1, A2, A3, A4, A5, A6, A7, A8, A9, A10 | Not Present | Rel-5 |
| Downlink information common for all radio links<br>- Downlink DPCH info common for all RL<br>- Timing indication<br>- CFN-targetSFN frame offset<br>- Downlink DPCH power control information<br>- CHOICE mode<br>- TPC Step Size<br>- MAC-d HFN initial value<br>- CHOICE mode<br>- CHOICE mode<br>- CHOICE TDD option<br>- Default DPCH Offset Value | A1, A2, A3 | <br><br><br>Maintain<br>Not Present<br><br><br>TDD<br>1<br>Not Present<br>TDD<br>TDD<br>3.84 Mcps TDD (No Data)<br>Not Present | <br><br><br><br><br><br><br><br><br><br><br>Rel-4<br> |
| Downlink information common for all radio links<br>- Downlink DPCH info common for all RL<br>- Timing indication<br>- CFN-targetSFN frame offset<br>- Downlink DPCH power control information<br>- CHOICE mode<br>- TPC Step Size<br>- MAC-d HFN initial value<br>- CHOICE mode<br>- CHOICE mode<br>- CHOICE TDD option<br>- Default DPCH Offset Value<br>- CHOICE mode | A4 | <br><br><br>Initialise<br>Not Present<br><br><br>TDD<br>1<br>Not Present<br>TDD<br>TDD<br>3.84 Mcps TDD (No Data)<br><br>TDD | <br><br><br><br><br><br><br><br><br><br><br>Rel-4<br><br> |
| - Default DPCH Offset Value | | 0 Integer(0..7) | |
| Downlink information common for all radio links | A5, A6, A7, A8, A9, A10 | Not Present | |
| Downlink information per radio link list<br>- Downlink information for each radio link<br>- Choice mode<br>- Primary CCPCH info<br>- Choice mode<br>- Choice TDD Option<br>- CHOICE *SyncCase*<br>- Timeslot<br>- Cell parameters ID<br>- SCTD indicator<br>- Downlink DPCH info for each RL<br>- CHOICE mode<br>- DL CCTrCh List<br>- TFCS ID<br>- Time info<br>- Activation time | A1, A2,A3 | <br><br>TDD<br><br>TDD<br>3.84 Mcps TDD<br>Sync Case 1<br>Reference clause 6.1.4 Default settings for cell 1<br>Ref. to the Default setting in TS34.108 clause 6.1 (TDD) Integer(0..127)<br>FALSE<br><br>TDD<br><br>2 Integer(1.8)<br><br>Now | <br><br><br><br><br><br>Rel-4<br><br><br><br><br><br><br><br><br><br> |

| Information Element | Condition | Value/remark | Version |
|---|---|---|---|
| - Duration | | Infinite | |
| - Common timeslot info | | | |
| - 2nd interleaving mode | | Default value is "Frame" | |
| - TFCI coding | | Reference to TS34.108 clause 6 Parameter set | |
| - Puncturing limit | | Reference to TS34.108 clause 6 Parameter set | |
| - Repetition period | | 1 | |
| - Repetition length | | NULL | |
| - Downlink DPCH timeslots and codes | | | |
| - First individual timeslot info | | | |
| - Timeslot number | | | |
| - CHOICE TDD option | | 3.84 Mcps TDD | Rel-4 |
| - Timeslot number | | 4 OR 5 OR 6 | |
| - TFCI existence | | TRUE | |
| - Midamble shift and burst type | | | |
| - CHOICE TDD option | | 3.84 Mcps TDD | Rel-4 |
| - CHOICE *Burst Type* | | Type 1 | |
| - Midamble allocation mode | | Default midamble | |
| - Midamble configuration | | 16 | |
| - Midamble Shift | | Not Present | |
| - CHOICE TDD option | | 3.84 Mcps TDD | Rel-4 |
| - First timeslot channelisation codes | | Repeated (1,2) for each channelisation code assigned in the slot to meet the needs of TS34.108 clause 6 Parameter Set. | |
| - CHOICE codes representation | | Bitmap | |
| - Channelisation codes bitmap | | Reference to TS34.108 clause 6.10 Parameter Set | |
| - CHOICE more timeslots | | No more timeslots (No Data) | |
| - UL CCTrCH TPC List | | Default (is previous list or all defined UL CCTrCHs.) | |
| - UL TPC TFCS Identity | | | |
| - TFCS ID | | 1 | |
| - Shared Channel Indicator | | FALSE | |
| - DL CCTrCH List to Remove | | Not present | |
| - SCCPCH Information for FACH | | Not Present | R99 and Rel-4 only |
| Downlink information per radio link list | A4 | | |
| - Downlink information for each radio link | | | |
| - Choice mode | | TDD | |
| - Primary CCPCH info | | | |
| - Choice mode | | TDD | |
| - Choice TDD Option | | 3.84 Mcps TDD | Rel-4 |
| - CHOICE *SyncCase* | | Sync Case 1 | |
| - Timeslot | | Reference clause 6.1.4 Default settings for cell 1 | |
| - Cell parameters ID | | Ref. to the Default setting in TS34.108 clause 6.1 (TDD) Integer(0..127) | |
| - SCTD indicator | | FALSE | |
| - Downlink DPCH info for each RL | | | |
| - CHOICE mode | | TDD | |
| - DL CCTrCh List | | Not Present | |
| - DL CCTrCH List to Remove | | Not present | |
| - SCCPCH Information for FACH | | Not Present | R99 and Rel-4 only |
| Downlink information per radio link list | A5 | | |
| - Downlink information for each radio link | | | |
| - Choice mode | | TDD | |
| - Primary CCPCH info | | | |
| - Choice mode | | TDD | |
| - Choice TDD Option | | 3.84 Mcps TDD | Rel-4 |
| - CHOICE *SyncCase* | | Sync Case 1 | |
| - Timeslot | | Reference clause 6.1.4 Default settings for cell 1 | |
| - Cell parameters ID | | Ref. to the Default setting in TS34.108 clause 6.1 (TDD) Integer(0..127) | |

| Information Element | Condition | Value/remark | Version |
|---|---|---|---|
| - SCTD indicator | | FALSE | |
| - Downlink DPCH info for each RL | | Not Present | |
| - SCCPCH Information for FACH | | Not Present | R99 and Rel-4 only |
| Downlink information per radio link list | A6, A7, A8, A9, A10 | Not Present | |

| Condition | Explanation |
|---|---|
| A1 | This IE need for "Non speech in CS" |
| A2 | This IE need for "Speech in CS" |
| A3 | This IE need for "Packet to CELL_DCH from CELL_DCH in PS" |
| A4 | This IE need for "Packet to CELL_DCH from CELL_FACH in PS" |
| A5 | This IE need for "Packet to CELL_FACH from CELL_DCH in PS" |
| A6 | This IE need for "Packet to CELL_FACH from CELL_FACH in PS" |
| A7 | This IE need for “Packet to URA_PCH from CELL_FACH in PS” |
| A8 | This IE need for “Packet to URA_PCH from CELL_DCH in PS” |
| A9 | This IE need for “Packet to CELL_PCH from CELL_FACH in PS” |
| A10 | This IE need for “Packet to CELL_PCH from CELL_DCH in PS” |

Contents of PHYSICAL CHANNEL RECONFIGURATION message: AM or UM (1.28 Mcps TDD)

| Information Element | Condition | Value/remark | Version |
|---|---|---|---|
| Message Type | A1, A2, A3, A4, A5, A6 | | |
| RRC transaction identifier | | Arbitrarily selects an integer between 0 and 3 | |
| Integrity check info | | | |
| - message authentication code | | SS calculates the value of MAC-I for this message and writes to this IE. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I. | |
| - RRC message sequence number | | SS provides the value of this IE, from its internal counter. | |
| Integrity protection mode info | | Not Present | |
| Ciphering mode info | | Not Present | |
| Activation time | A1, A2, A3 | (256+CFN-(CFN MOD 8 + 8))MOD 256 | |
| Activation time | A4, A5, A6 | Now | |
| New U-RNTI | | Not Present | |
| New C-RNTI | A1, A2, A3, A4 | Not Present | |
| New C-RNTI | A5, A6 | '1010 1010 1010 1010' | |
| New DSCH-RNTI | A1, A2, A3, A4, A5, A6 | Not Present | |
| New H-RNTI | A1, A2, A3, A4, A5, A6 | Not Present | Rel-5 |
| RRC State indicator | A1, A2, A3, A4 | CELL_DCH | |
| RRC State indicator | A5, A6 | CELL_FACH | |
| UTRAN DRX cycle length coefficient | A1, A2, A3, A4, A5, A6 | Not Present | |
| CN information info | | Not Present | |
| URA identity | | Not Present | |
| Downlink counter synchronization info | | Not Present | |
| Frequency info | A1, A2, A3, A4, A5 | | |
| - Choice mode | | TDD | |
| - UARFCN (Nt) | | Reference to clause 5.1 Test frequencies | |
| Frequency info | A6 | Not Present | |
| Maximum allowed UL TX power | | 33dBm | |
| CHOICE channel requirement | A5, A6 | Not Present | |
| CHOICE channel requirement | A1, A2, A3, A4 | Uplink DPCH info | |
| - Uplink DPCH power control info | | | |
| - CHOICE mode | | TDD | |
| - CHOICE TDD option | | 1.28 Mcps TDD | |
| - PRXPDPCHdes | | -80 Integer(-120…-58 by step of 1) | |

| Information Element | Condition | Value/remark | Version |
|---|---|---|---|
| - CHOICE UL OL PC info | | Individually Signalled | |
| - CHOICE TDD option | | 1.28 Mcps TDD | |
| - TPC step size | | 1 | |
| - Primary CCPCH Tx Power | | 20 Integer(6..43) | |
| - CHOICE mode | | TDD | |
| - Uplink Timing Advance Control | | | |
| - CHOICE Timing Advance | | Enabled | |
| - CHOICE TDD option | | 1.28 Mcps TDD | |
| - Uplink synchronization parameters | | | |
| - Uplink synchronization step size | | 1 | |
| - Uplink synchronization frequency | | 1 | |
| - Synchronization parameters | | | |
| - SYNC_UL codes bitmap | | 01010101 | |
| - FPACH info | | | |
| - Timeslot number | | 0 | |
| - Channelisation code | | 16/15 | |
| - Midamble Shift and burst type | | | |
| - CHOICE TDD option | | 1.28 Mcps TDD | |
| - Midamble Allocation Mode | | Default midamble | |
| - Midamble configuration | | 16 Integer(2, 4, 6, 8, 10, 12, 14, 16) | |
| - WT | | 4 Integer(1..4) | |
| - PRXUpPCHdes | | -80 dBm | |
| - SYNC_UL procedure | | | |
| - Max SYNC_UL Transmissions | | 2 | |
| - Power Ramp Step | | 2 | |
| - UL CCTrCH List | | | |
| - TFCS ID | | 1 | |
| - UL Target SIR | | Real (-11 .. 20 by step of 0.5 dB)<br>Reference to clause 6 Parameter set. | |
| - Time info | | | |
| - Activation time | | (256+CFN-(CFN MOD 8 + 8))MOD 256 | |
| - Duration | | Infinite | |
| - Common timeslot info | | | |
| - 2nd interleaving mode | | Default value is "Frame" | |
| - TFCI coding | | Reference to clause 6 Parameter set | |
| - Puncturing limit | | Reference to clause 6 Parameter set | |
| - Repetition period | | 1 | |
| - Repetition length | | Null | |
| - Uplink DPCH timeslots and code | | | |
| - Dynamic SF usage | | FALSE | |
| - First individual timeslot info | | | |
| - Timeslot number | | | |
| - CHOICE TDD option | | 1.28 Mcps TDD | |
| - Timeslot number | | 1 OR 2 OR 3 | |
| - TFCI existence | | TRUE | |
| - Midamble shift and burst type | | | |
| - CHOICE TDD option | | 1.28 Mcps TDD | |
| - Midamble allocation mode | | Default midamble | |
| - Midamble configuration | | 16 | |
| - Midamble Shift | | Not Present | |
| - CHOICE TDD option | | 1.28 Mcps TDD | |
| - Modulation | | QPSK | |
| - SS-TPC Symbols | | 1 | |
| - Additional TPC-SS Symbols | | Not present | |
| - First timeslot Code List | | Repeated (1,2) for each channelisation code assigned in the slot to meet the needs of clause 6 Parameter Set. | |
| - channelisation codes | | (SF/ i) where i denotes an unassigned code matching the SF specified in clause 6 Parameter Set. | |
| - CHOICE more timeslots | | No more timeslots | |
| - UL CCTrCH List to Remove | | Not present | |
| CHOICE Mode | A1, A2, A3, A4, A5, A6 | TDD | |
| Downlink HS-PDSCH Information | A1, A2, A3, A4, A5, A6 | Not Present | Rel-5 |
| Downlink information common for all radio | A1, A2, A3 | | |

| Information Element | Condition | Value/remark | Version |
|---|---|---|---|
| links | | | |
| - Downlink DPCH info common for all RL | | | |
| - Timing indication | | Maintain | |
| - CFN-targetSFN frame offset | | Not Present | |
| - Downlink DPCH power control information | | | |
| - CHOICE mode | | TDD | |
| - TPC Step Size | | 1 | |
| - MAC-d HFN initial value | | Not Present | |
| - CHOICE mode | | TDD | |
| - CHOICE mode | | TDD | |
| - CHOICE TDD option | | 1.28 Mcps TDD | |
| - TSTD indicator | | FALSE | |
| - Default DPCH Offset Value | | Not Present | |
| Downlink information common for all radio links | A4 | | |
| - Downlink DPCH info common for all RL | | | |
| - Timing indication | | Initialize | |
| - CFN-targetSFN frame offset | | Not Present | |
| - Downlink DPCH power control information | | | |
| - CHOICE mode | | TDD | |
| - TPC Step Size | | 1 | |
| - MAC-d HFN initial value | | Not Present | |
| - CHOICE mode | | TDD | |
| - CHOICE mode | | TDD | |
| - CHOICE TDD option | | 1.28 Mcps TDD | |
| - TSTD indicator | | FALSE | |
| - Default DPCH Offset Value | | | |
| - CHOICE mode | | TDD | |
| - Default DPCH Offset Value | | 0 Integer(0..7) | |
| Downlink information common for all radio links | A5, A6 | Not Present | |
| Downlink information per radio link list | A1, A2,A3 | | |
| - Downlink information for each radio link | | | |
| - Choice mode | | TDD | |
| - Primary CCPCH info | | | |
| - Choice mode | | TDD | |
| - Choice TDD Option | | 1.28 Mcps TDD | |
| - TSTD indicator | | FALSE | |
| - Cell parameters ID | | Ref. to the Default setting in clause 6.1 (TDD) Integer(0..127) | |
| - SCTD indicator | | FALSE | |
| - Downlink DPCH info for each RL | | | |
| - CHOICE mode | | TDD | |
| - DL CCTrCh List | | | |
| - TFCS ID | | 2 Integer(1.8) | |
| - Time info | | | |
| - Activation time | | Now | |
| - Duration | | Infinite | |
| - Common timeslot info | | | |
| - 2nd interleaving mode | | Default value is "Frame" | |
| - TFCI coding | | Reference to clause 6 Parameter set | |
| - Puncturing limit | | Reference to clause 6 Parameter set | |
| - Repetition period | | 1 | |
| - Repetition length | | NULL | |
| - Downlink DPCH timeslots and codes | | | |
| - First individual timeslot info | | | |
| - Timeslot number | | | |
| - CHOICE TDD option | | 1.28 Mcps TDD | |
| - Timeslot number | | 4 OR 5 OR 6 | |
| - TFCI existence | | TRUE | |
| - Midamble shift and burst type | | | |
| - CHOICE TDD option | | 1.28 Mcps TDD | |
| - Midamble allocation mode | | Default midamble | |
| - Midamble configuration | | 16 | |
| - Midamble Shift | | Not Present | |

| Information Element | Condition | Value/remark | Version |
|---|---|---|---|
| - CHOICE TDD option | | 1.28 Mcps TDD | |
| - Modulation | | QPSK | |
| - SS-TPC Symbols | | 1 | |
| - Additional TPC-SS Sysbols | | Not present | |
| - First timeslot channelisation codes | | Repeated (1,2) for each channelisation code assigned in the slot to meet the needs of clause 6 Parameter Set. | |
| - CHOICE codes representation | | | |
| - Channelisation codes bitmap | | Reference to clause 6.11 Parameter Set | |
| - CHOICE more timeslots | | No more timeslots | |
| - UL CCTrCH TPC List | | This list is not required for 1.28 Mcps TDD and is to be ignored by the UE. | |
| - UL TPC TFCS Identity | | | |
| - TFCS ID | | 1 | |
| - Shared Channel Indicator | | FALSE | |
| - DL CCTrCH List to Remove | | Not present | |
| - SCCPCH Information for FACH | | Not Present | R99 and Rel-4 only |
| Downlink information per radio link list | A4 | | |
| - Downlink information for each radio link | | | |
| - Choice mode | | TDD | |
| - Primary CCPCH info | | | |
| - Choice mode | | TDD | |
| - Choice TDD Option | | 1.28 Mcps TDD | |
| - TSTD indicator | | FALSE | |
| - Cell parameters ID | | Ref. to the Default setting in clause 6.1 (TDD) Integer(0..127) | |
| - SCTD indicator | | FALSE | |
| - Downlink DPCH info for each RL | | | |
| - CHOICE mode | | TDD | |
| - DL CCTrCh List | | Not Present | |
| - DL CCTrCH List to Remove | | Not present | |
| - SCCPCH Information for FACH | | Not Present | R99 and Rel-4 only |
| Downlink information per radio link list | A5 | | |
| - Downlink information for each radio link | | | |
| - Choice mode | | TDD | |
| - Primary CCPCH info | | | |
| - Choice mode | | TDD | |
| - Choice TDD Option | | 1.28 Mcps TDD | |
| - TSTD indicator | | FALSE | |
| - Cell parameters ID | | Ref. to the Default setting in clause 6.1 (TDD) Integer(0..127) | |
| - SCTD indicator | | FALSE | |
| - Downlink DPCH info for each RL | | Not Present | |
| - SCCPCH Information for FACH | | Not Present | R99 and Rel-4 only |
| Downlink information per radio link list | A6 | Not Present | |

| Condition | Explanation |
|---|---|
| A1 | This IE need for "Non speech in CS" |
| A2 | This IE need for "Speech in CS" |
| A3 | This IE need for "Packet to CELL_DCH from CELL_DCH in PS" |
| A4 | This IE need for "Packet to CELL_DCH from CELL_FACH in PS" |
| A5 | This IE need for "Packet to CELL_FACH from CELL_DCH in PS" |
| A6 | This IE need for "Packet to CELL_FACH from CELL_FACH in PS" |

Contents of PHYSICAL CHANNEL RECONFIGURATION COMPLETE message: AM (3.84 Mcps TDD)

| Information Element | Value/remark | Version |
|---|---|---|
| Message Type | | |
| RRC transaction identifier | Checked to see if it's set to identical value of the same IE in the downlink PHYSICAL CHANNEL RECONFIGURATION message | |
| Integrity check info | | |
| - Message authentication code | This IE is checked to see if it is present. The value is compared against the XMAC-I value computed by SS. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I. | |
| - RRC Message sequence number | This IE is checked to see if it is present. The value is used by SS to compute the XMAC-I value. | |
| Uplink integrity protection activation info | Not checked | |
| CHOICE mode | TDD | |
| - CHOICE TDD option | 3.84 Mcps TDD | Rel-4 |
| - Uplink Timing Advance | 0 | |
| COUNT-C activation time | Not checked | |
| Radio bearer uplink ciphering activation time info | Not checked | |
| Uplink counter synchronisation info | Not checked | |

Contents of PHYSICAL CHANNEL RECONFIGURATION COMPLETE message: AM (1.28 Mcps TDD)

| Information Element | Value/remark | Version |
|---|---|---|
| Message Type | | |
| RRC transaction identifier | Checked to see if it's set to identical value of the same IE in the downlink PHYSICAL CHANNEL RECONFIGURATION message | |
| Integrity check info | | |
| - Message authentication code | This IE is checked to see if it is present. The value is compared against the XMAC-I value computed by SS. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I. | |
| - RRC Message sequence number | This IE is checked to see if it is present. The value is used by SS to compute the XMAC-I value. | |
| Uplink integrity protection activation info | Not checked | |
| CHOICE mode | TDD | |
| CHOICE TDD option | 1.28 Mcps TDD | Rel-4 |
| COUNT-C activation time | Not checked | |
| Radio bearer uplink ciphering activation time info | Not checked | |
| Uplink counter synchronization info | Not checked | |

Contents of PHYSICAL CHANNEL RECONFIGURATION FAILURE message: AM

| Information Element | Value/remark |
|---|---|
| Message Type | |
| RRC transaction identitifer | Checked to see if it is set to identical value of the same IE in the downlink PHYSICAL CHANNEL RECONFIGURATION message. |
| Integrity check info | |
| - Message authentication code | This IE is checked to see if it is present. The value is compared against the XMAC-I value computed by SS. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I. |
| - RRC Message sequence number | This IE is checked to see if it is present. The value is used by SS to compute the XMAC-I value. |
| Failure cause | Checked to see if it meets test requirement |

Contents of RADIO BEARER RECONFIGURATION message: AM or UM (3.84 Mcps TDD)

| Information Element | Condition | Value/remark | Version |
|---|---|---|---|
| Message Type | A1,A2,A3, A4,A5,A6 | | |
| **UE Information elements** | | | |

| Information Element | Condition | Value/remark | Version |
|---|---|---|---|
| RRC transaction identifier | | Arbitrarily selects an integer between 0 and 3 | |
| Integrity check info | | | |
| - message authentication code | | SS calculates the value of MAC-I for this message and writes to this IE. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I. | |
| - RRC message sequence number | | SS provides the value of this IE, from its internal counter. | |
| Integrity protection mode info | | Not Present | |
| Ciphering mode info | | Not Present | |
| Activation time | A1,A2,A3 | (256+CFN-(CFN MOD 8 + 8))MOD 256 | |
| Activation time | A4, A5,A6 | Not Present | |
| | | MD Integer(0..255) default is 'now" | |
| New U-RNTI | | Not Present | |
| New C-RNTI | A1, A2, A3, A4, | Not Present | |
| New C-RNTI | A5, A6 | '1010 1010 1010 1010' | |
| New DSCH-RNTI | A1, A2, A3, A4, A5, A6 | Not Present | |
| New H-RNTI | A1, A2, A3, A4, A5, A6 | Not Present | Rel-5 |
| RRC State indicator | A1, A2, A3, A4 | CELL_DCH<br>Indicates to a UE the RRC state to be entered. | |
| RRC State indicator | A5, A6 | CELL_FACH | |
| UTRAN DRX cycle length coefficient | A1,A2,A3, A4,A5,A6 | Not Present<br>A coefficient in the formula to count the paging occasions to be used by a specific UE | |
| **CN information elements** | | | |
| CN information info | | Not Present | |
| **UTRAN mobility information elements** | | | |
| URA identity | | Not Present | |
| CHOICE specification mode | | [FFS] | Rel-5 |
| **RB information elements** | | | |
| RAB information to reconfigure list | | Not Present | |
| RB information to reconfigure list | A1 | TS25.331 specifies that "Although this IE is not always required, need is MP to align with ASN.1". | |
| - RB information to reconfigure | | (UM DCCH for RRC) | |
| - RB identity | | 1 | |
| - PDCP info | | Not Present | |
| - PDCP SN info | | Not Present | |
| - RLC info | | Not Present | |
| - RB mapping info | | Not Present | |
| - RB stop/continue | | Not Present | |
| - RB information to reconfigure | | (AM DCCH for RRC) | |
| - RB identity | | 2 | |
| - PDCP info | | Not Present | |
| - PDCP SN info | | Not Present | |
| - RLC info | | Not Present | |
| - RB mapping info | | Not Present | |
| - RB stop/continue | | Not Present | |
| - RB information to reconfigure | | (AM DCCH for NAS_DT High priority) | |
| - RB identity | | 3 | |
| - PDCP info | | Not Present | |
| - PDCP SN info | | Not Present | |
| - RLC info | | Not Present | |
| - RB mapping info | | Not Present | |
| - RB stop/continue | | Not Present | |
| - RB information to reconfigure | | (AM DCCH for NAS_DT Low priority) | |
| - RB identity | | 4 | |
| - PDCP info | | Not Present | |
| - PDCP SN info | | Not Present | |
| - RLC info | | Not Present | |

| Information Element | Condition | Value/remark | Version |
|---|---|---|---|
| - RB mapping info | | Not Present | |
| - RB stop/continue | | Not Present | |
| - RB information to reconfigure | | (TM DTCH) | |
| - RB identity | | 10 | |
| - PDCP info | | Not Present | |
| - PDCP SN info | | Not Present | |
| - RLC info | | Not Present | |
| - RB mapping info | | Not Present | |
| - RB stop/continue | | Not Present | |
| RB information to reconfigure list | A2 | TS25.331 specifies that "Although this IE is not always required, need is MP to align with ASN.1". | |
| - RB information to reconfigure | | (UM DCCH for RRC) | |
| - RB identity | | 1 | |
| - PDCP info | | Not Present | |
| - PDCP SN info | | Not Present | |
| - RLC info | | Not Present | |
| - RB mapping info | | Not Present | |
| - RB stop/continue | | Not Present | |
| - RB information to reconfigure | | (AM DCCH for RRC) | |
| - RB identity | | 2 | |
| - PDCP info | | Not Present | |
| - PDCP SN info | | Not Present | |
| - RLC info | | Not Present | |
| - RB mapping info | | Not Present | |
| - RB stop/continue | | Not Present | |
| - RB information to reconfigure | | (AM DCCH for NAS_DT High priority) | |
| - RB identity | | 3 | |
| - PDCP info | | Not Present | |
| - PDCP SN info | | Not Present | |
| - RLC info | | Not Present | |
| - RB mapping info | | Not Present | |
| - RB stop/continue | | Not Present | |
| - RB information to reconfigure | | (AM DCCH for NAS_DT Low priority) | |
| - RB identity | | 4 | |
| - PDCP info | | Not Present | |
| - PDCP SN info | | Not Present | |
| - RLC info | | Not Present | |
| - RB mapping info | | Not Present | |
| - RB stop/continue | | Not Present | |
| - RB information to reconfigure | | (TM DTCH) | |
| - RB identity | | 10 | |
| - PDCP info | | Not Present | |
| - PDCP SN info | | Not Present | |
| - RLC info | | Not Present | |
| - RB mapping info | | Not Present | |
| - RB stop/continue | | Not Present | |
| - RB information to reconfigure | | (TM DTCH) | |
| - RB identity | | 11 | |
| - PDCP info | | Not Present | |
| - PDCP SN info | | Not Present | |
| - RLC info | | Not Present | |
| - RB mapping info | | Not Present | |
| - RB stop/continue | | Not Present | |
| - RB information to reconfigure | | (TM DTCH)<br>(This IE is needed for 12.2 kbps and 10.2 kbps) | |
| - RB identity | | 12 | |
| - PDCP info | | Not Present | |
| - PDCP SN info | | Not Present | |
| - RLC info | | Not Present | |
| - RB mapping info | | Not Present | |
| - RB stop/continue | | Not Present | |
| RB information to reconfigure list | A3,A4,A5, A6 | TS25.331 specifies that "Although this IE is not always required, need is MP to align with ASN.1". | |
| - RB information to reconfigure | | (UM DCCH for RRC) | |

| Information Element | Condition | Value/remark | Version |
|---|---|---|---|
| - RB identity<br>- PDCP info<br>- PDCP SN info<br>- RLC info<br>- RB mapping info<br>- RB stop/continue<br>- RB information to reconfigure<br>- RB identity<br>- PDCP info<br>- PDCP SN info<br>- RLC info<br>- RB mapping info<br>- RB stop/continue<br>- RB information to reconfigure<br>- RB identity<br>- PDCP info<br>- PDCP SN info<br>- RLC info<br>- RB mapping info<br>- RB stop/continue<br>- RB information to reconfigure<br>- RB identity<br>- PDCP info<br>- PDCP SN info<br>- RLC info<br>- RB mapping info<br>- RB stop/continue<br>- RB information to reconfigure<br>- RB identity<br>- PDCP info<br>- PDCP SN info<br>- RLC info<br>- RB mapping info<br>- RB stop/continue | | 1<br>Not Present<br>Not Present<br>Not Present<br>Not Present<br>Not Present<br>(AM DCCH for RRC)<br>2<br>Not Present<br>Not Present<br>Not Present<br>Not Present<br>Not Present<br>(AM DCCH for NAS_DT High priority)<br>3<br>Not Present<br>Not Present<br>Not Present<br>Not Present<br>Not Present<br>(AM DCCH for NAS_DT Low priority)<br>4<br>Not Present<br>Not Present<br>Not Present<br>Not Present<br>Not Present<br>(AM DTCH)<br>20<br>Not Present<br>Not Present<br>Not Present<br>Not Present<br>Not Present | |
| RB information to be affected | A1, A2, A3,A4,A5, A6 | Not Present | |
| **TrCH Information Elements** | | | |
| **Uplink transport channels** | | | |
| UL Transport channel information for all transport channels | A1, A2, A5,A6 | Not Present | |
| UL Transport channel information for all transport channels<br>- PRACH TFCS<br>- CHOICE mode<br>- Individual UL CCTrCH information | A3, A4 | <br><br>Not Present<br>TDD | |
| - UL TFCS Identity | | | |
| - TFCS ID | | 1 | |
| - Shared Channel Indicator | | FALSE | |
| - UL TFCS | | | |
| - CHOICE *TFCI signalling* | | Normal<br>(another option “split” only for FDD) | |
| - TFCI Field 1 Information | | | |
| - CHOICE *TFCS representation* | | Complete reconfiguration | |
| - TFCS complete reconfiguration information<br>- CHOICE *CTFC Size*<br><br><br><br>- CTFC information<br><br><br>- CTFC | | <br>Number of bits used must be enough to cover all combinations of CTFC from TS34.108 clause 6.10.3.4 Parameter Set.<br>This IE is repeated for TFC numbers and reference to TS34.108 clause 6.10.3.4 Parameter Set<br>Reference to TS34.108 clause 6.10.3.4 | |

| Information Element | Condition | Value/remark | Version |
|---|---|---|---|
| - Power offset information | | Parameter Set | |
| - CHOICE Gain Factors | | Computed Gain Factors (The last TFC is set to Signalled Gain Factors) | |
| - Reference TFC ID | | 0 Integer(0.. 3) | |
| - CHOICE Gain Factors | | Signalled Gain Factors (Not Present if the CHOICE Gain Factors is set to ComputedGain Factors) | |
| - CHOICE *mode* | | TDD | |
| - Gain Factor $\beta_d$ | | 15 | |
| - Reference TFC ID | | 0 Integer(0.. 3) | |
| - CHOICE *mode* | | TDD | |
| - TFC subset | | | |
| - CHOICE *Subset representation* | | Minimum allowed Transport format combination index | |
| - Allowed transport format combination list | | Not present | |
| - Non-allowed transport format combination list | | Not present | |
| - Non-allowed transport format combination list | | Not present | |
| - Full transport format combination set | | Not present | |
| - TFC subset list | | Not present | |
| Deleted TrCH information list | | | |
| Deleted UL TrCH information | A1, A2, A3, A4, A5,A6 | Not Present | |
| Added or Reconfigured TrCH information list | | | |
| Added or Reconfigured UL TrCH information | A1, A2, A5,A6 | Not Present | |
| Added or Reconfigured UL TrCH information | A4 | 2 TrCHs(DCH for DCCH and DCH for DTCH) | |
| - Uplink transport channel type | | DCH | |
| - UL Transport channel identity | | 5 | |
| - TFS | | | |
| - CHOICE Transport channel type | | Dedicated transport channels | |
| - Dynamic Transport format information | | | |
| - RLC Size | | Reference to TS34.108 clause 6.10.3 Parameter Set | |
| - Number of TBs and TTI List | | (This IE is repeated for TFI number.) | |
| - Transmission Time Interval | | Not Present | |
| - Number of Transport blocks | | Reference to TS34.108 clause 6.10.3 Parameter Set | |
| - CHOICE Logical Channel list | | All | |
| - Semi-static Transport Format information | | | |
| - Transmission time interval | | Reference to TS34.108 clause 6.10.3 Parameter Set | |
| - Type of channel coding | | Reference to TS34.108 clause 6.10.3 Parameter Set | |
| - Coding Rate | | Reference to TS34.108 clause 6.10.3 Parameter Set | |
| - Rate matching attribute | | Reference to TS34.108 clause 6.10.3 Parameter Set | |
| - CRC size | | Reference to TS34.108 clause 6.10.3 Parameter Set | |
| - Uplink transport channel type | | DCH | |
| - UL Transport channel identity | | 1 | |
| - TFS | | | |
| - CHOICE Transport channel type | | Dedicated transport channels | |
| - Dynamic Transport format information | | | |
| - RLC Size | | Reference to TS34.108 clause 6.10.3 Parameter Set | |
| - Number of TBs and TTI List | | (This IE is repeated for TFI number.) | |
| - Transmission Time Interval | | Not Present | |
| - Number of Transport blocks | | Reference to TS34.108 clause 6.10.3 Parameter Set | |

| Information Element | Condition | Value/remark | Version |
|---|---|---|---|
| - CHOICE Logical Channel list<br>- Semi-static Transport Format information<br>- Transmission time interval<br><br>- Type of channel coding<br><br>- Coding Rate<br><br>- Rate matching attribute<br><br>- CRC size | | All<br><br>Reference to TS34.108 clause 6.10.3 Parameter Set<br>Reference to TS34.108 clause 6.10.3 Parameter Set<br>Reference to TS34.108 clause 6.10.3 Parameter Set<br>Reference to TS34.108 clause 6.10.3 Parameter Set<br>Reference to TS34.108 clause 6.10.3 Parameter Set | |
| Added or Reconfigured UL TrCH information<br>- Uplink transport channel type<br>- UL Transport channel identity<br>- TFS<br>- CHOICE Transport channel type<br>- Dynamic Transport format information<br>- RLC Size<br><br>- Number of TBs and TTI List<br>- Transmission Time Interval<br>- Number of Transport blocks<br><br>- CHOICE Logical Channel list<br>- Semi-static Transport Format information<br>- Transmission time interval<br><br>- Type of channel coding<br><br>- Coding Rate<br><br>- Rate matching attribute<br><br>- CRC size | A3 | (DCH for DTCH)<br>DCH<br>1<br><br>Dedicated transport channels<br><br>Reference to TS34.108 clause 6.10.3 Parameter Set<br>(This IE is repeated for TFI number.)<br>Not Present<br>Reference to TS34.108 clause 6.10.3 Parameter Set<br>All<br><br>Reference to TS34.108 clause 6.10.3 Parameter Set<br>Reference to TS34.108 clause 6.10.3 Parameter Set<br>Reference to TS34.108 clause 6.10.3 Parameter Set<br>Reference to TS34.108 clause 6.10.3 Parameter Set<br>Reference to TS34.108 clause 6.10.3 Parameter Set | |
| CHOICE mode<br><br>- (no data) | A1,A2,A3, A4,A5,A6 | TDD | |
| **Downlink transport channels** | | | |
| DL Transport channel information common for all transport channel | A1, A2, A5, A6 | Not Present | |
| DL Transport channel information common for all transport channel<br>- SCCPCH TFCS<br>- CHOICE mode<br>- Individual DL CCTrCH information<br>- DL TFCS Identity<br>- TFCS ID<br>- Shared Channel Indicator<br>- CHOICE *DL parameters*<br>- DL TFCS<br>– CHOICE *TFCI signalling*<br><br>- TFCI Field 1 Information | A3,A4 | <br><br>Not Present<br>TDD<br><br><br><br><br>Independent<br><br>Normal<br>(Normal' : meaning no split in the TFCI field either 'Logical' or 'Hard') | |
| - CHOICE *TFCS representation* | | Complete reconfiguration | |
| - TFCS complete reconfiguration information<br>- CHOICE CTFC Size<br><br><br><br>- CTFC information<br><br><br>- CTFC | | <br><br>Number of bits used must be enough to cover all combinations of CTFC from clause TS34.108 clause 6.10.3.4 Parameter Set.<br>This IE is repeated for TFC numbers and reference to TS34.108 clause 6.10.3.4<br>Reference to TS34.108 clause 6.10.3.4 | |

| Information Element | Condition | Value/remark | Version |
|---|---|---|---|
| - Power offset information | | Parameter Set<br>Not Present | |
| Deleted TrCH information list | | | |
| Deleted DL TrCH information | A1, A2, A3, A4, A5,A6 | Not Present | |
| Added or Reconfigured TrCH information list | | | |
| Added or Reconfigured DL TrCH information | A1, A2, A5, A6 | Not Present | |
| Added or Reconfigured DL TrCH information<br><br>- Downlink transport channel type<br>- DL Transport channel identity<br>- CHOICE DL parameters<br>- Uplink transport channel type<br>- UL TrCH identity<br>- DCH quality target<br>- BLER Quality value<br>- Downlink transport channel type<br>- DL Transport channel identity<br>- CHOICE DL parameters<br>- TFS<br>- CHOICE Transport channel type<br>- Dynamic transport format information<br>- RLC Size<br><br>- Number of TBs and TTI List<br>- Dynamic transport format information<br>- Transmission Time Interval<br>- Number of Transport blocks<br><br>- Semi-static Transport Format information<br>- Transmission time interval<br><br>- Type of channel coding<br><br>- Coding Rate<br><br>- Rate matching attribute<br><br>- CRC size<br><br>- DCH quality target<br>- BLER Quality value | A4 | 2 TrCHs(DCH for DCCH and DCH for DTCH)<br>DCH<br>10<br>Same as UL<br>DCH<br>5<br><br>Not Present<br>DCH<br>6<br>Explicit<br><br>Dedicated transport channel<br><br>Reference to TS34.108 clause 6.10.3 Parameter Set<br>(This IE is repeated for TFI number.)<br><br>Not Present<br>Reference to TS34.108 clause 6.10.3 Parameter Set<br><br>Reference to TS34.108 clause 6.10.3 Parameter Set<br>Reference to TS34.108 clause 6.10.3 Parameter Set<br>Reference to TS34.108 clause 6.10.3 Parameter Set<br>Reference to TS34.108 clause 6.10.3 Parameter Set<br>Reference to TS34.108 clause 6.10.3 Parameter Set<br><br>-2.0 | |
| Added or Reconfigured DL TrCH information<br>- Downlink transport channel type<br>- DL Transport channel identity<br>- CHOICE DL parameters<br>- TFS<br>- CHOICE Transport channel type<br>- Dynamic transport format information<br>- RLC Size<br><br>- Number of TBs and TTI List<br>- Dynamic transport format information<br>- Transmission Time Interval<br>- Number of Transport blocks<br><br>- Semi-static Transport Format information<br>- Transmission time interval<br><br>- Type of channel coding<br><br>- Coding Rate<br><br>- Rate matching attribute | A3 | <br>DCH<br>6<br>Explicit<br><br>Dedicated transport channel<br><br>Reference to TS34.108 clause 6.10.3 Parameter Set<br>(This IE is repeated for TFI number.)<br><br>Not Present<br>Reference to TS34.108 clause 6.10.3 Parameter Set<br><br>Reference to TS34.108 clause 6.10.3 Parameter Set<br>Reference to TS34.108 clause 6.10.3 Parameter Set<br>Reference to TS34.108 clause 6.10.3 Parameter Set<br>Reference to TS34.108 clause 6.10.3 Parameter Set | |

| Information Element | Condition | Value/remark | Version |
|---|---|---|---|
| - CRC size | | Reference to TS34.108 clause 6.10.3 Parameter Set | |
| - DCH quality target | | | |
| - BLER Quality value | | -2.0 | |
| Preconfiguration | A1,A2,A3, A4,A5,A6 | [FFS] | Rel-5 |
| **PhyCH information elements** | | | |
| Frequency info | A1,A2,A3, A4,A5 | | |
| - CHOICE mode | | TDD | |
| - UARFCN (Nt) | | Reference to clause 5.1 Test frequencies | |
| Frequency info | A6 | Not Present | |
| **Uplink radio resources** | | | |
| Maximum allowed UL TX power | A1,A2,A3, A4,A5,A6 | 33dBm | |
| CHOICE channel requirement | A1, A2, A3, A4 | Uplink DPCH info | |
| -Uplink DPCH power control info | | | |
| - CHOICE mode | | TDD | |
| - UL target SIR | | 6 | |
| - CHOICE *UL OL PC info* | | Individually Signalled | |
| - CHOICE TDD option | | 3.84 Mcps TDD | Rel-4 |
| - Individual timeslot interference info | | Reference to TS34.108 clause 6.10.3 Parameter Set | |
| - Individual timeslot interference | | | |
| - CHOICE TDD option | | 3.84 Mcps TDD | Rel-4 |
| - Timeslot number | | As required by, Reference to TS34.108 clause 6.10.3 Parameter Set | |
| - TDD UL interference | | As required by, Reference to TS34.108 clause 6.10.3 Parameter Set (if not specified -60 dBm) | |
| - Primary CCPCH Tx Power | | 18 Integer(6..43) (-70 dBm Received if pathloss not specified) | |
| - CHOICE mode | | TDD | |
| - Uplink Timing Advance Control | | | |
| - CHOICE *Timing Advance* | | Enabled | |
| - CHOICE *TDD option* | | 3.84 Mcps TDD (Default) | Rel-4 |
| - UL CCTrCH List | | | |
| - TFCS ID | | 1 | |
| - UL Target SIR | | Real (-11 .. 20 by step of 0.5dB) Reference to TS34.108 Parameter set. | |
| - Time info | | | |
| - Activation time | | (256+CFN-(CFN MOD 8 + 8))MOD 256 | |
| - Duration | | infinite | |
| - Common timeslot info | | | |
| - 2nd interleaving mode | | Default value is "Frame" | |
| - TFCI coding | | Reference to TS34.108 clause 6 Parameter set | |
| - Puncturing limit | | Reference to TS34.108 clause 6 Parameter set | |
| - Repetition period | | 1 | |
| - Repetition length | | empty | |
| - Uplink DPCH timeslots and code | | | |
| - Dynamic SF usage | | FALSE | |
| - First individual timeslot info | | | |
| - Timeslot number | | | |
| - CHOICE TDD option | | 3.84 Mcps TDD | Rel-4 |
| - Timeslot number | | 1 OR 2 OR 3 | |
| - TFCI existence | | TRUE | |
| - Midamble shift and burst type | | | |
| - CHOICE TDD option | | 3.84 Mcps TDD | Rel-4 |
| - CHOICE *Burst Type* | | Type 1 | |
| - Midamble allocation mode | | Default midamble | |

| Information Element | Condition | Value/remark | Version |
|---|---|---|---|
| - Midamble configuration | | 16 | |
| - Midamble Shift | | Not Present | |
| - CHOICE TDD option | | 3.84 Mcps TDD (No Data) | Rel-4 |
| - First timeslot Code List | | Repeated (1,2) for each channelisation code assigned in the slot to meet the needs of TS34.108 clause 6 Parameter Set. | |
| - channelisation codes | | (SF/ i) where i denotes an unassigned code matching the SF specified in TS34.108 clause 6 Parameter Set. | |
| - CHOICE more timeslots | | No more timeslots (No Data) | |
| - UL CCTrCH List to Remove | | Not present | |
| CHOICE channel requirement | A5, A6 | Not Present | |
| **Downlink radio resources** | | | |
| CHOICE Mode | A1,A2,A3, A4,A5,A6 | TDD | |
| - Downlink PDSCH information | | No date | |
| Downlink HS-PDSCH Information | A1, A2, A3, A4, A5, A6 | Not Present | Rel-5 |
| Downlink information common for all radio links | A5, A6 | Not Present | |
| Downlink information common for all radio links | A1, A2, A3 | | |
| - Downlink DPCH info common for all RL | | | |
| - Timing indicaton | | Maintain | |
| - CFN-targetSFN frame offset | | Not Present | |
| - Downlink DPCH power control information | | | |
| - CHOICE *mode* | | TDD | |
| - TPC Step Size | | 1 | |
| - MAC-d HFN initial value | | Not Present | |
| - CHOICE mode | | TDD | |
| - CHOICE mode | | TDD | |
| - CHOICE *TDD option* | | 3.84 Mcps TDD (No Data) | Rel-4 |
| - Default DPCH Offset Value | | Not Present | |
| Downlink information common for all radio links | | | |
| - Downlink DPCH info common for all RL | | | |
| - Timing indication | | Initialise | |
| - CFN-targetSFN frame offset | | Not Present | |
| - Downlink DPCH power control information | | | |
| - CHOICE *mode* | | TDD | |
| - TPC Step Size | | 1 | |
| - MAC-d HFN initial value | | Not Present | |
| - CHOICE mode | | TDD | |
| - CHOICE mode | | TDD | |
| - CHOICE *TDD option* | | 3.84 Mcps TDD (no Data) | Rel-4 |
| - Default DPCH Offset Value | | | |
| - CHOICE mode | | TDD | |
| - Default DPCH Offset Value | | 0 | |
| Downlink information per radio link list | A1, A2, A3, A4 | | |
| - Downlink information for each radio link | | | |
| - Choice mode | | TDD | |
| - Primary CCPCH info | | | |
| - Choice mode | | TDD | |
| - Choice TDD Option | | 3.84 Mcps TDD | Rel-4 |
| - CHOICE *SyncCase* | | Sync Case 1 | |
| - Timeslot | | Reference clause 6.1.4 Default settings for cell 1 | |
| - SCTD indicator | | FALSE | |
| - Downlink DPCH info for each RL | | | |
| - CHOICE mode | | TDD | |
| - DL CCTrCh List | | | |
| - TFCS ID | Integer(1.8) | Identity of this CCTrCh.Default value is 1 | |
| - Time info | | | |
| - Activation time | | Now | |
| - Duration | | Infinite | |

| Information Element | Condition | Value/remark | Version |
|---|---|---|---|
| - Common timeslot info<br>- 2nd interleaving mode | | <br>Default value is "Frame" | |
| - TFCI coding<br><br>- Puncturing limit<br><br>- Repetition period<br>- Repetition length<br>- Downlink DPCH timeslots and codes<br>- First individual timeslot info<br>- Timeslot number<br>- CHOICE TDD option | | Reference to TS34.108 clause 6 Parameter set<br>Reference to TS34.108 clause 6 Parameter set<br>1<br>empty<br><br><br><br>3.84 Mcps TDD | <br><br><br><br><br><br><br><br>Rel-4 |
| - Timeslot number<br>- TFCI existence<br>- Midamble shift and burst type<br>- CHOICE TDD option<br>- CHOICE *Burst Type*<br>- Midamble allocation mode<br>- Midamble configuration<br>- Midamble Shift<br>- CHOICE TDD option<br>- First timeslot channelisation codes<br><br><br><br>- CHOICE codes representation<br>- Channelisation codes bitmap<br><br>- CHOICE more timeslots<br>- UL CCTrCH TPC List<br><br>- UL TPC TFCS Identity<br>- TFCS ID<br>- Shared Channel Indicator<br>- DL CCTrCH List to Remove<br>- SCCPCH Information for FACH | | 4 OR 5 OR 6<br>TRUE<br><br>3.84 Mcps TDD<br>Type 1<br>Default midamble<br>16<br>Not Present<br>3.84 Mcps TDD (No Data)<br>Repeated (1,2) for each channelisation code assigned in the slot to meet the needs of TS34.108 clause 6 Parameter Set.<br>Bitmap<br>Reference to TS34.108 clause 6.10 Parameter Set<br>No more timeslots (No Data)<br>Default (is previous list or all defined UL CCTrCHs.)<br><br>1<br>FALSE<br>Not present<br>Not Present | <br><br><br>Rel-4<br><br><br><br><br>Rel-4<br><br><br><br><br><br><br><br><br><br><br><br>R99 and Rel-4 only |
| Downlink information per radio link list<br>- Downlink information for each radio link<br>- Choice mode<br>- Primary CCPCH info<br>- Choice mode<br>- Choice TDD Option<br>- TSTD indicator<br>- Cell parameters ID<br><br>- SCTD indicator | A5 | <br><br>TDD<br><br>TDD<br>3.84 Mcps TDD<br>FALSE<br>Reference clause 6.1.4 Default settings for cell 1<br>FALSE | <br><br><br><br><br>Rel-4 |
| - Downlink DPCH info for each RL | | Not Present | |
| - SCCPCH Information for FACH | | Not Present | R99 and Rel-4 only |
| Downlink information per radio link list | A6 | | |
| - Downlink information for each radio link | | Not Present | |

| Condition | Explanation |
|---|---|
| A1 | This IE need for "Non speech in CS" |
| A2 | This IE need for "Speech in CS" |
| A3 | This IE need for "Packet to CELL_DCH from CELL_DCH in PS" |
| A4 | This IE need for "Packet to CELL_DCH from CELL_FACH in PS" |
| A5 | This IE need for "Packet to CELL_FACH from CELL_DCH in PS" |
| A6 | This IE need for "Packet to CELL_FACH from CELL_FACH in PS" |

Contents of RADIO BEARER RECONFIGURATION message: AM or UM (1.28 Mcps TDD)

| Information Element | Condition | Value/remark | Version |
|---|---|---|---|
| Message Type | A1,A2,A3,A4, A5,A6 | | |
| **UE Information elements** | | | |
| RRC transaction identifier | | Arbitrarily selects an integer between 0 and 3 | |
| Integrity check info | | | |
| - message authentication code | | SS calculates the value of MAC-I for this message and writes to this IE. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I. | |
| - RRC message sequence number | | SS provides the value of this IE, from its internal counter. | |
| Integrity protection mode info | | Not Present | |
| Ciphering mode info | | Not Present | |
| Activation time | A1,A2,A3 | (256+CFN-(CFN MOD 8 + 8))MOD 256 | |
| Activation time | A4, A5,A6 | Not Present<br>MD Integer(0..255) default is 'now" | |
| New U-RNTI | | Not Present | |
| New C-RNTI | A1, A2, A3, A4, | Not Present | |
| New C-RNTI | A5, A6 | '1010 1010 1010 1010' | |
| New DSCH-RNTI | A1, A2, A3, A4, A5, A6 | Not Present | |
| New H-RNTI | A1, A2, A3, A4, A5, A6 | Not Present | Rel-5 |
| RRC State indicator | A1, A2, A3, A4 | CELL_DCH<br>Indicates to a UE the RRC state to be entered. | |
| RRC State indicator | A5, A6 | CELL_FACH | |
| UTRAN DRX cycle length coefficient | A1,A2,A3,A4, A5,A6 | Not Present<br>A coefficient in the formula to count the paging occasions to be used by a specific UE | |
| **CN information elements** | | | |
| CN information info | | Not Present | |
| **UTRAN mobility information elements** | | | |
| URA identity | | Not Present | |
| CHOICE specification mode | | [FFS] | Rel-5 |
| **RB information elements** | | | |
| RAB information to reconfigure list | | Not Present | |
| RB information to reconfigure list | A1 | TS25.331 specifies that "Although this IE is not always required, need is MP to align with ASN.1". | |
| - RB information to reconfigure | | (UM DCCH for RRC) | |
| - RB identity | | 1 | |
| - PDCP info | | Not Present | |
| - PDCP SN info | | Not Present | |
| - RLC info | | Not Present | |
| - RB mapping info | | Not Present | |
| - RB stop/continue | | Not Present | |
| - RB information to reconfigure | | (AM DCCH for RRC) | |
| - RB identity | | 2 | |
| - PDCP info | | Not Present | |
| - PDCP SN info | | Not Present | |
| - RLC info | | Not Present | |
| - RB mapping info | | Not Present | |
| - RB stop/continue | | Not Present | |
| - RB information to reconfigure | | (AM DCCH for NAS_DT High priority) | |
| - RB identity | | 3 | |
| - PDCP info | | Not Present | |
| - PDCP SN info | | Not Present | |
| - RLC info | | Not Present | |
| - RB mapping info | | Not Present | |
| - RB stop/continue | | Not Present | |
| - RB information to reconfigure | | (AM DCCH for NAS_DT Low priority) | |
| - RB identity | | 4 | |

| Information Element | Condition | Value/remark | Version |
|---|---|---|---|
| - PDCP info | | Not Present | |
| - PDCP SN info | | Not Present | |
| - RLC info | | Not Present | |
| - RB mapping info | | Not Present | |
| - RB stop/continue | | Not Present | |
| - RB information to reconfigure | | (TM DTCH) | |
| - RB identity | | 10 | |
| - PDCP info | | Not Present | |
| - PDCP SN info | | Not Present | |
| - RLC info | | Not Present | |
| - RB mapping info | | Not Present | |
| - RB stop/continue | | Not Present | |
| RB information to reconfigure list | A2 | TS25.331 specifies that "Although this IE is not always required, need is MP to align with ASN.1". | |
| - RB information to reconfigure | | (UM DCCH for RRC) | |
| - RB identity | | 1 | |
| - PDCP info | | Not Present | |
| - PDCP SN info | | Not Present | |
| - RLC info | | Not Present | |
| - RB mapping info | | Not Present | |
| - RB stop/continue | | Not Present | |
| - RB information to reconfigure | | (AM DCCH for RRC) | |
| - RB identity | | 2 | |
| - PDCP info | | Not Present | |
| - PDCP SN info | | Not Present | |
| - RLC info | | Not Present | |
| - RB mapping info | | Not Present | |
| - RB stop/continue | | Not Present | |
| - RB information to reconfigure | | (AM DCCH for NAS_DT High priority) | |
| - RB identity | | 3 | |
| - PDCP info | | Not Present | |
| - PDCP SN info | | Not Present | |
| - RLC info | | Not Present | |
| - RB mapping info | | Not Present | |
| - RB stop/continue | | Not Present | |
| - RB information to reconfigure | | (AM DCCH for NAS_DT Low priority) | |
| - RB identity | | 4 | |
| - PDCP info | | Not Present | |
| - PDCP SN info | | Not Present | |
| - RLC info | | Not Present | |
| - RB mapping info | | Not Present | |
| - RB stop/continue | | Not Present | |
| - RB information to reconfigure | | (TM DTCH) | |
| - RB identity | | 10 | |
| - PDCP info | | Not Present | |
| - PDCP SN info | | Not Present | |
| - RLC info | | Not Present | |
| - RB mapping info | | Not Present | |
| - RB stop/continue | | Not Present | |
| - RB information to reconfigure | | (TM DTCH) | |
| - RB identity | | 11 | |
| - PDCP info | | Not Present | |
| - PDCP SN info | | Not Present | |
| - RLC info | | Not Present | |
| - RB mapping info | | Not Present | |
| - RB stop/continue | | Not Present | |
| - RB information to reconfigure | | (TM DTCH) (This IE is needed for 12.2 kbps and 10.2 kbps) | |
| - RB identity | | 12 | |
| - PDCP info | | Not Present | |
| - PDCP SN info | | Not Present | |
| - RLC info | | Not Present | |
| - RB mapping info | | Not Present | |
| - RB stop/continue | | Not Present | |
| RB information to reconfigure list | A3,A4,A5,A6 | TS25.331 specifies that "Although this | |

| Information Element | Condition | Value/remark | Version |
|---|---|---|---|
| | | IE is not always required, need is MP to align with ASN.1". | |
| - RB information to reconfigure | | (UM DCCH for RRC) | |
| - RB identity | | 1 | |
| - PDCP info | | Not Present | |
| - PDCP SN info | | Not Present | |
| - RLC info | | Not Present | |
| - RB mapping info | | Not Present | |
| - RB stop/continue | | Not Present | |
| - RB information to reconfigure | | (AM DCCH for RRC) | |
| - RB identity | | 2 | |
| - PDCP info | | Not Present | |
| - PDCP SN info | | Not Present | |
| - RLC info | | Not Present | |
| - RB mapping info | | Not Present | |
| - RB stop/continue | | Not Present | |
| - RB information to reconfigure | | (AM DCCH for NAS_DT High priority) | |
| - RB identity | | 3 | |
| - PDCP info | | Not Present | |
| - PDCP SN info | | Not Present | |
| - RLC info | | Not Present | |
| - RB mapping info | | Not Present | |
| - RB stop/continue | | Not Present | |
| - RB information to reconfigure | | (AM DCCH for NAS_DT Low priority) | |
| - RB identity | | 4 | |
| - PDCP info | | Not Present | |
| - PDCP SN info | | Not Present | |
| - RLC info | | Not Present | |
| - RB mapping info | | Not Present | |
| - RB stop/continue | | Not Present | |
| - RB information to reconfigure | | (AM DTCH) | |
| - RB identity | | 20 | |
| - PDCP info | | Not Present | |
| - PDCP SN info | | Not Present | |
| - RLC info | | Not Present | |
| - RB mapping info | | Not Present | |
| - RB stop/continue | | Not Present | |
| RB information to be affected | A1, A2, A3,A4,A5,A6 | Not Present | |
| **TrCH Information Elements** | | | |
| **Uplink transport channels** | | | |
| UL Transport channel information for all transport channels | A1, A2, A5,A6 | Not Present | |
| UL Transport channel information for all transport channels | A3, A4 | | |
| - PRACH TFCS | | Not Present | |
| - CHOICE mode | | TDD | |
| - Individual UL CCTrCH information | | | |
| - UL TFCS Identity | | | |
| - TFCS ID | | 1 | |
| - Shared Channel Indicator | | FALSE | |
| - UL TFCS | | | |
| - CHOICE *TFCI signalling* | | Normal<br>(another option "split" only for FDD) | |
| - TFCI Field 1 Information | | | |
| - CHOICE *TFCS representation* | | Complete reconfiguration | |
| - TFCS complete reconfiguration information | | | |
| - CHOICE *CTFC Size* | | Number of bits used must be enough to cover all combinations of CTFC from clause 6.11.5.4 Parameter Set. | |
| - CTFC information | | This IE is repeated for TFC numbers and reference to clause 6.11.5.4 Parameter Set | |

| Information Element | Condition | Value/remark | Version |
|---|---|---|---|
| - CTFC | | Reference to clause 6.11.5.4 Parameter Set | |
| - Power offset information | | | |
| - CHOICE Gain Factors | | Computed Gain Factors (The last TFC is set to Signalled Gain Factors) | |
| - Reference TFC ID | | 0 Integer(0.. 3) | |
| - CHOICE Gain Factors | | Signalled Gain Factors (Not Present if the CHOICE Gain Factors is set to ComputedGain Factors) | |
| - CHOICE *mode* | | TDD | |
| - Gain Factor $\beta_d$ | | 15 | |
| - Reference TFC ID | | 0 Integer(0.. 3) | |
| - CHOICE *mode* | | TDD | |
| - TFC subset | | | |
| - CHOICE *Subset representation* | | Minimum allowed Transport format combination index | |
| - Allowed transport format combination list | | Not present | |
| - Non-allowed transport format combination list | | Not present | |
| - Non-allowed transport format combination list | | Not present | |
| - Full transport format combination set | | Not present | |
| - TFC subset list | | Not present | |
| Deleted TrCH information list | | | |
| Deleted UL TrCH information | A1, A2, A3, A4, A5,A6 | Not Present | |
| Added or Reconfigured TrCH information list | | | |
| Added or Reconfigured UL TrCH information | A1, A2, A5,A6 | Not Present | |
| Added or Reconfigured UL TrCH information | A4 | 2 TrCHs(DCH for DCCH and DCH for DTCH) | |
| - Uplink transport channel type | | DCH | |
| - UL Transport channel identity | | 5 | |
| - TFS | | | |
| - CHOICE Transport channel type | | Dedicated transport channels | |
| - Dynamic Transport format information | | | |
| - RLC Size | | Reference to clause 6.11.5 Parameter Set | |
| - Number of TBs and TTI List | | (This IE is repeated for TFI number.) | |
| - Transmission Time Interval | | Not Present | |
| - Number of Transport blocks | | Reference to clause 6.11.5 Parameter Set | |
| - CHOICE Logical channel list | | All | |
| - Semi-static Transport Format information | | | |
| - Transmission time interval | | Reference to clause 6.11.5 Parameter Set | |
| - Type of channel coding | | Reference to clause 6.11.5 Parameter Set | |
| - Coding Rate | | Reference to clause 6.11.5 Parameter Set | |
| - Rate matching attribute | | Reference to clause 6.11.5 Parameter Set | |
| - CRC size | | Reference to clause 6.11.5 Parameter Set | |
| - Uplink transport channel type | | DCH | |
| - UL Transport channel identity | | 1 | |
| - TFS | | | |
| - CHOICE Transport channel type | | Dedicated transport channels | |
| - Dynamic Transport format information | | | |
| - RLC Size | | Reference to clause 6.11.5 Parameter Set | |
| - Number of TBs and TTI List | | (This IE is repeated for TFI number.) | |
| - Transmission Time Interval | | Not Present | |
| - Number of Transport blocks | | Reference to clause 6.11.5 Parameter | |

| Information Element | Condition | Value/remark | Version |
|---|---|---|---|
| | | Set | |
| - CHOICE Logical channel list | | All | |
| - Semi-static Transport Format information | | | |
| - Transmission time interval | | Reference to clause 6.11.5 Parameter Set | |
| - Type of channel coding | | Reference to clause 6.11.5 Parameter Set | |
| - Coding Rate | | Reference to clause 6.11.5 Parameter Set | |
| - Rate matching attribute | | Reference to clause 6.11.5 Parameter Set | |
| - CRC size | | Reference to clause 6.11.5 Parameter Set | |
| Added or Reconfigured UL TrCH information | A3 | (DCH for DTCH) | |
| - Uplink transport channel type | | DCH | |
| - UL Transport channel identity | | 1 | |
| - TFS | | | |
| - CHOICE Transport channel type | | Dedicated transport channels | |
| - Dynamic Transport format information | | | |
| - RLC Size | | Reference to clause 6.11.5 Parameter Set | |
| - Number of TBs and TTI List | | (This IE is repeated for TFI number.) | |
| - Transmission Time Interval | | Not Present | |
| - Number of Transport blocks | | Reference to clause 6.11.5 Parameter Set | |
| - CHOICE Logical channel list | | All | |
| - Semi-static Transport Format information | | | |
| - Transmission time interval | | Reference to clause 6.11.5 Parameter Set | |
| - Type of channel coding | | Reference to clause 6.11.5 Parameter Set | |
| - Coding Rate | | Reference to clause 6.11.5 Parameter Set | |
| - Rate matching attribute | | Reference to clause 6.11.5 Parameter Set | |
| - CRC size | | Reference to clause 6.11.5 Parameter Set | |
| CHOICE mode | A1,A2,A3,A4, A5,A6 | TDD | |
| - (no data) | | | |
| **Downlink transport channels** | | | |
| DL Transport channel information common for all transport channel | A1, A2, A5, A6 | Not Present | |
| DL Transport channel information common for all transport channel | A3,A4 | | |
| - SCCPCH TFCS | | Not Present | |
| - CHOICE mode | | TDD | |
| - Individual DL CCTrCH information | | | |
| - DL TFCS Identity | | | |
| - TFCS ID | | | |
| - Shared Channel Indicator | | | |
| - CHOICE *DL parameters* | | Independent | |
| - DL TFCS | | | |
| - CHOICE *TFCI signalling* | | Normal<br>(Normal' : meaning no split in the TFCI field either 'Logical' or 'Hard') | |
| - TFCI Field 1 Information | | | |
| - CHOICE *TFCS representation* | | Complete reconfiguration | |
| - TFCS complete reconfiguration information | | | |
| - CHOICE CTFC Size | | Number of bits used must be enough to cover all combinations of CTFC from clause 6.11.5.4 Parameter Set. | |
| - CTFC information | | This IE is repeated for TFC numbers and reference to clause 6.11.5.4 | |
| - CTFC | | Reference to clause 6.11.5.4 Parameter Set | |

| Information Element | Condition | Value/remark | Version |
|---|---|---|---|
| - Power offset information | | Not Present | |
| Deleted TrCH information list | | | |
| Deleted DL TrCH information | A1, A2, A3, A4, A5,A6 | Not Present | |
| Added or Reconfigured TrCH information list | | | |
| Added or Reconfigured DL TrCH information | A1, A2, A5, A6 | Not Present | |
| Added or Reconfigured DL TrCH information | A4 | 2 TrCHs(DCH for DCCH and DCH for DTCH) | |
| - Downlink transport channel type | | DCH | |
| - DL Transport channel identity | | 10 | |
| - CHOICE DL parameters | | Same as UL | |
| - Uplink transport channel type | | DCH | |
| - UL TrCH identity | | 5 | |
| - DCH quality target | | | |
| - BLER Quality value | | Not Present | |
| - Downlink transport channel type | | DCH | |
| - DL Transport channel identity | | 6 | |
| - CHOICE DL parameters | | Explicit | |
| - TFS | | | |
| - CHOICE Transport channel type | | Dedicated transport channel | |
| - Dynamic transport format information | | | |
| - RLC Size | | Reference to clause 6.11.5 Parameter Set | |
| - Number of TBs and TTI List | | (This IE is repeated for TFI number.) | |
| - Dynamic transport format information | | | |
| - Transmission Time Interval | | Not Present | |
| - Number of Transport blocks | | Reference to clause 6.11.5 Parameter Set | |
| - Semi-static Transport Format information | | | |
| - Transmission time interval | | Reference to clause 6.11.5 Parameter Set | |
| - Type of channel coding | | Reference to clause 6.11.5 Parameter Set | |
| - Coding Rate | | Reference to clause 6.11.5 Parameter Set | |
| - Rate matching attribute | | Reference to clause 6.11.5 Parameter Set | |
| - CRC size | | Reference to clause 6.11.5 Parameter Set | |
| - DCH quality target | | | |
| - BLER Quality value | | -2.0 | |
| Added or Reconfigured DL TrCH information | A3 | | |
| - Downlink transport channel type | | DCH | |
| - DL Transport channel identity | | 6 | |
| - CHOICE DL parameters | | Explicit | |
| - TFS | | | |
| - CHOICE Transport channel type | | Dedicated transport channel | |
| - Dynamic transport format information | | | |
| - RLC Size | | Reference to clause 6.11.5 Parameter Set | |
| - Number of TBs and TTI List | | (This IE is repeated for TFI number.) | |
| - Dynamic transport format information | | | |
| - Transmission Time Interval | | Not Present | |
| - Number of Transport blocks | | Reference to clause 6.11.5 Parameter Set | |
| - Semi-static Transport Format information | | | |
| - Transmission time interval | | Reference to clause 6.11.5 Parameter Set | |
| - Type of channel coding | | Reference to clause 6.11.5 Parameter Set | |
| - Coding Rate | | Reference to clause 6.11.5 Parameter Set | |
| - Rate matching attribute | | Reference to clause 6.11.5 Parameter Set | |
| - CRC size | | Reference to clause 6.11.5 Parameter | |

| Information Element | Condition | Value/remark | Version |
|---|---|---|---|
| | | Set | |
| - DCH quality target | | | |
| - BLER Quality value | | -2.0 | |
| Preconfiguration | A1,A2,A3,A4, A5,A6 | [FFS] | Rel-5 |
| **PhyCH information elements** | | | |
| Frequency info | A1,A2,A3,A4, A5 | | |
| - CHOICE mode | | TDD | |
| - UARFCN (Nt) | | Reference to clause 5.1 Test frequencies | |
| Frequency info | A6 | Not Present | |
| **Uplink radio resources** | | | |
| Maximum allowed UL TX power | A1,A2,A3,A4, A5,A6 | 33dBm | |
| CHOICE channel requirement | A1, A2, A3, A4 | Uplink DPCH info | |
| -Uplink DPCH power control info | | | |
| - CHOICE mode | | TDD | |
| - CHOICE TDD option | | 1.28 Mcps TDD | Rel-4 |
| - $PRX_{PDPCHdes}$ | | Integer(-120…-58 by step of 1) | |
| - CHOICE *UL OL PC info* | | | |
| - Broadcast UL OL PC info | | Null | |
| - CHOICE mode | | TDD | |
| - Uplink Timing Advance Control | | | |
| - CHOICE *Timing Advance* | | Enabled | |
| - CHOICE *TDD option* | | 1.28 Mcps TDD | |
| - Uplink synchronization parameters | | | |
| - Uplink synchronization step size | | 1 | |
| - Uplink synchronization frequency | | 1 | |
| - Synchronization parameters | | Not Present | |
| - UL CCTrCH List | | | |
| - TFCS ID | | 1 | |
| - UL Target SIR | | Real (-11 .. 20 by step of 0.5dB) Reference to clause 6 Parameter set. | |
| - Time info | | | |
| - Activation time | | (256+CFN-(CFN MOD 8 + 8))MOD 256 | |
| - Duration | | infinite | |
| - Common timeslot info | | | |
| - 2nd interleaving mode | | Default value is "Frame" | |
| - TFCI coding | | Reference to clause 6 Parameter set | |
| - Puncturing limit | | Reference to clause 6 Parameter set | |
| - Repetition period | | 1 | |
| - Repetition length | | empty | |
| - Uplink DPCH timeslots and code | | | |
| - Dynamic SF usage | | FALSE | |
| - First individual timeslot info | | | |
| - Timeslot number | | | |
| - CHOICE TDD option | | 1.28 Mcps TDD | |
| - Timeslot number | | 1 | |
| - TFCI existence | | TRUE | |
| - Midamble shift and burst type | | | |
| - CHOICE TDD option | | 1.28 Mcps TDD | |
| - Midamble allocation mode | | Default midamble | |
| - Midamble configuration | | 16 | |
| - Midamble Shift | | Not Present | |
| - CHOICE TDD option | | 1.28 Mcps TDD | |
| - Modulation | | QPSK | |
| - SS-TPC Symbols | | 1 | |
| - Additional TPC-SS Sysbols | | Not present | |
| - First timeslot Code List | | Repeated (1,2) for each channelisation code assigned in the slot to meet the needs of clause 6 Parameter Set. | |
| - channelisation codes | | (SF/ i) where i denotes an unassigned code matching the SF specified in clause 6 Parameter Set. | |
| - CHOICE more timeslots | | No more timeslots | |

| Information Element | Condition | Value/remark | Version |
|---|---|---|---|
| - UL CCTrCH List to Remove | | Not present | |
| CHOICE channel requirement | A5, A6 | Not Present | |
| **Downlink radio resources** | | | |
| CHOICE Mode | A1,A2,A3,A4, A5,A6 | TDD | |
| - Downlink PDSCH information | | No date | |
| Downlink HS-PDSCH Information | A1, A2, A3, A4, A5, A6 | Not Present | Rel-5 |
| Downlink information common for all radio links | A5, A6 | Not Present | |
| Downlink information common for all radio links | A1, A2, A3 | | |
| - Downlink DPCH info common for all RL | | | |
| - Timing indicaton | | Maintain | |
| - CFN-targetSFN frame offset | | Not Present | |
| - Downlink DPCH power control information | | | |
| - CHOICE *mode* | | TDD | |
| - TPC Step Size | | 1 | |
| - MAC-d HFN initial value | | Not Present | |
| - CHOICE mode | | TDD | |
| - CHOICE mode | | TDD | |
| - CHOICE *TDD option* | | 1.28 Mcps TDD | |
| - TSTD indicator | | FALSE | |
| - Default DPCH Offset Value | | Not Present | |
| Downlink information common for all radio links | A4 | | |
| - Downlink DPCH info common for all RL | | | |
| - Timing indication | | Initialize | |
| - CFN-targetSFN frame offset | | Not Present | |
| - Downlink DPCH power control information | | | |
| - CHOICE *mode* | | TDD | |
| - TPC Step Size | | 1 | |
| - MAC-d HFN initial value | | Not Present | |
| - CHOICE mode | | TDD | |
| - CHOICE mode | | TDD | |
| - CHOICE *TDD option* | | 1.28 Mcps TDD | |
| - TSTD indicator | | FALSE | |
| - Default DPCH Offset Value | | | |
| - CHOICE mode | | TDD | |
| - Default DPCH Offset Value | | 0 | |
| Downlink information per radio link list | A1, A2, A3, A4 | | |
| - Downlink information for each radio link | | | |
| - Choice mode | | TDD | |
| - Primary CCPCH info | | | |
| - Choice mode | | TDD | |
| - Choice TDD Option | | 1.28 Mcps TDD | |
| - TSTD indicator | | FALSE | |
| - Cell parameters ID | | Reference clause 6.1.4 Default settings for cell 1 | |
| - SCTD indicator | | FALSE | |
| - Downlink DPCH info for each RL | | | |
| - CHOICE mode | | TDD | |
| - DL CCTrCh List | | | |
| - TFCS ID | Integer(1.8) | Identity of this CCTrCh.Default value is 1 | |
| - Time info | | | |
| - Activation time | | Now | |
| - Duration | | Infinite | |
| - Common timeslot info | | | |
| - 2nd interleaving mode | | Default value is "Frame" | |
| - TFCI coding | | Reference to clause 6 Parameter set | |
| - Puncturing limit | | Reference to clause 6 Parameter set | |
| - Repetition period | | 1 | |
| - Repetition length | | empty | |
| - Downlink DPCH timeslots and codes | | | |
| - First individual timeslot info | | | |
| - Timeslot number | | | |
| - CHOICE TDD option | | 1.28 Mcps TDD | |
| - Timeslot number | | 4 OR 5 OR 6 | |

| Information Element | Condition | Value/remark | Version |
|---|---|---|---|
| - TFCI existence | | TRUE | |
| - Midamble shift and burst type | | | |
| - CHOICE TDD option | | 1.28 Mcps TDD | |
| - Midamble allocation mode | | Default midamble | |
| - Midamble configuration | | 16 | |
| - Midamble Shift | | Not Present | |
| - CHOICE TDD option | | 1.28 Mcps TDD | |
| - Modulation | | QPSK | |
| - SS-TPC Symbols | | 1 | |
| - Additional TPC-SS Sysbols | | Not present | |
| - First timeslot channelisation codes | | Repeated (1,2) for each channelisation code assigned in the slot to meet the needs of clause 6 Parameter Set. | |
| - CHOICE codes representation | | | |
| - Channelisation codes bitmap | | Reference to clause 6.10 Parameter Set | |
| - CHOICE more timeslots | | No more timeslots | |
| - UL CCTrCH TPC List | | This list is not required for 1.28 Mcps TDD and is to be ignored by the UE. | |
| - UL TPC TFCS Identity | | | |
| - TFCS ID | | 1 | |
| - Shared Channel Indicator | | FALSE | |
| - DL CCTrCH List to Remove | | Not present | |
| - SCCPCH Information for FACH | | Not Present | R99 and Rel-4 only |
| Downlink information per radio link list | A5 | | |
| - Downlink information for each radio link | | | |
| - Choice mode | | TDD | |
| - Primary CCPCH info | | | |
| - Choice mode | | TDD | |
| - Choice TDD Option | | 1.28 Mcps TDD | |
| - TSTD indicator | | FALSE | |
| - Cell parameters ID | | Reference clause 6.1.4 Default settings for cell 1 | |
| - SCTD indicator | | FALSE | |
| - Downlink DPCH info for each RL | | Not Present | |
| - SCCPCH Information for FACH | | Not Present | R99 and Rel-4 only |
| Downlink information per radio link list | A6 | | |
| - Downlink information for each radio link | | Not Present | |

| Condition | Explanation |
|---|---|
| A1 | This IE need for "Non speech in CS" |
| A2 | This IE need for "Speech in CS" |
| A3 | This IE need for "Packet to CELL_DCH from CELL_DCH in PS" |
| A4 | This IE need for "Packet to CELL_DCH from CELL_FACH in PS" |
| A5 | This IE need for "Packet to CELL_FACH from CELL_DCH in PS" |
| A6 | This IE need for "Packet to CELL_FACH from CELL_FACH in PS" |

Contents of RADIO BEARER RECONFIGURATION COMPLETE message: AM (3.84 Mcps TDD)

| Information Element | Value/remark | Version |
|---|---|---|
| Message Type | | |
| RRC transaction identifier | Checked to see if the value is identical to the same IE in the downlink RADIO BEARER RECONFIGURATION message | |
| Integrity check info | | |
| - Message authentication code | This IE is checked to see if it is present. The value is compared against the XMAC-I value computed by SS. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I. | |
| - RRC Message sequence number | This IE is checked to see if it is present. The value is used by SS to compute the XMAC-I value. | |

| | | |
|---|---|---|
| Uplink integrity protection activation info | Not checked | |
| CHOICE mode | TDD | |
| - CHOICE *TDD option* | 3.84 Mcps TDD | Rel-4 |
| - UL Timing Advance | 0 | |
| COUNT-C activation time | Not checked | |
| Radio bearer uplink ciphering activation time info | Not checked | |
| Uplink counter synchronisation info | Not checked | |

Contents of RADIO BEARER RECONFIGURATION COMPLETE message: AM (1.28 Mcps TDD)

| Information Element | Value/remark | Version |
|---|---|---|
| Message Type | | |
| RRC transaction identifier | Checked to see if the value is identical to the same IE in the downlink RADIO BEARER RECONFIGURATION message | |
| Integrity check info | | |
| - Message authentication code | This IE is checked to see if it is present. The value is compared against the XMAC-I value computed by SS. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I. | |
| - RRC Message sequence number | This IE is checked to see if it is present. The value is used by SS to compute the XMAC-I value. | |
| Uplink integrity protection activation info | Not checked | |
| CHOICE mode | TDD | |
| - CHOICE *TDD option* | 1.28 Mcps TDD (No data) | Rel-4 |
| COUNT-C activation time | Not checked | |
| Radio bearer uplink ciphering activation time info | Not checked | |
| Uplink counter synchronization info | Not checked | |

Contents of RADIO BEARER RECONFIGURATION FAILURE message: AM

| Information Element | Value/remark |
|---|---|
| Message Type | |
| RRC transaction identitifer | Checked to see if it is set to identical value of the same IE in the downlink RADIO BEARER RECONFIGURATION message. |
| Integrity check info | |
| - Message authentication code | This IE is checked to see if it is present. The value is compared against the XMAC-I value computed by SS. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I. |
| - RRC Message sequence number | This IE is checked to see if it is present. The value is used by SS to compute the XMAC-I value. |
| Failure cause | Checked to see if it meets test requirement |
| Radio bearers for which reconfiguration would have succeeded List | Not checked |

Contents of RADIO BEARER RELEASE message: AM or UM (3.84 Mcps TDD)

| Information Element | Condition | Value/remark | Version |
|---|---|---|---|
| Message Type | A1, A2, A3, A4, A5, A6, A7, A8 | | |
| RRC transaction identifier | | Arbitrarily selects an integer between 0 and 3 | |
| Integrity check info | | | |
| - message authentication code | | SS calculates the value of MAC-I for this message and writes to this IE. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I. | |
| - RRC message sequence number | | SS provides the value of this IE, from its internal counter. | |
| Integrity protection mode info | | Not Present | |

| Information Element | Condition | Value/remark | Version |
|---|---|---|---|
| Ciphering mode info<br>Activation time<br><br>Activation time<br>New U-RNTI | <br>A1, A2, A3, A7, A8<br>A4, A5, A6 | Not Present<br>(256+CFN-(CFN MOD 8 + 8))MOD 256<br><br>Not Present<br>Not Present | |
| New C-RNTI | A1,A2,A3, A4 | Not Present | |
| New C-RNTI | A5, A6, A7, A8 | '1010 1010 1010 1010' | |
| New DSCH-RNTI | A1, A2, A3, A4, A5, A6, A7, A8 | Not Present | |
| RRC State indicator | A1,A2, A3, A4 | CELL_DCH | |
| RRC State indicator | A5, A6, A7, A8 | CELL_FACH | |
| UTRAN DRX cycle length coefficient<br><br><br>CN information info<br>Signalling Connection release indication<br>URA identity<br>RAB information to reconfigure list | A1,A2,A3, A4,A5,A6, A7, A8 | Not Present<br><br><br>Not Present<br>Not Present<br>Not Present<br>Not Present | |
| RB information to release list | A1, A7 | | |
| RB information to release<br>- RB identity | | 10 | |
| RB information to release list | A2, A8 | | |
| RB information to release<br>- RB identity | | 10 | |
| RB information to release<br>- RB identity | | 11 | |
| RB information to release<br>- RB identity | | 12 | |
| RB information to release list | A3, A4, A5, A6 | | |
| RB information to release<br>- RB identity | | 20 | |
| RB information to be affected list | A1,A2, A3,A4,A5, A6, A7, A8 | Not Present | |
| Downlink counter synchronisation info | A1,A2,A3, A4,A5,A6, A7, A8 | Not Present | |
| UL Transport channel information common for all transport channels | A1, A2, A3, A4 | TFCS reconfigured to fit the new transport channel configuration. | |
| UL Transport channel information common for all transport channels | A5, A6, A7, A8 | Not Present | |
| Deleted TrCH information list | A1,A2, A3, A5, A7, A8 | | |
| Deleted UL TrCH Information<br><br>- Uplink transport channel type<br>- Transport channel identity | A1,A2, A3, A5, A7, A8 | <br><br>DCH<br>1 | |
| Deleted UL TrCH Information<br>- Uplink transport channel type<br>- Transport channel identity | A2, A8 | <br>DCH<br>2 | |
| Deleted UL TrCH Information<br>- Uplink transport channel type<br>- Transport channel identity | A2, A8 | <br>DCH<br>3 | |
| Deleted TrCH information list | A4, A6 | Not Present | |
| Added or Reconfigured TrCH information list | A5, A6, A7, A8 | Not Present | |
| Added or Reconfigured TrCH information list | A1, A2, A3, A4 | TrCHs (DCH for DCCH ) | |
| Added or Reconfigured UL TrCH information | | | |

| Information Element | Condition | Value/remark | Version |
|---|---|---|---|
| - Uplink transport channel type | | DCH | |
| - UL Transport channel identity | | 5 | |
| - TFS | | | |
| - CHOICE Transport channel type | | Dedicated transport channels | |
| - Dynamic Transport format information | | | |
| - RLC Size | | Reference to TS34.108 clause 6.10 Parameter Set | |
| - Number of TBs and TTI List | | (This IE is repeated for TFI number.) | |
| - Transmission Time Interval | | Not present | |
| - Number of Transport blocks | | Reference to TS34.108 clause 6.10 Parameter Set | |
| - CHOICE Logical Channel list | | All (NULL) | |
| - Semi-static Transport Format information | | | |
| - Transmission time interval | | Reference to TS34.108 clause 6.10 Parameter Set | |
| - Type of channel coding | | Reference to TS34.108 clause 6.10 Parameter Set | |
| - Coding Rate | | Reference to TS34.108 clause 6.10 Parameter Set | |
| - Rate matching attribute | | Reference to TS34.108 clause 6.10 Parameter Set | |
| - CRC size | | Reference to TS34.108 clause 6.10 Parameter Set | |
| CHOICE *mode* | | TDD (No data) | |
| DL Transport channel information common for all transport channels | A1, A2, A3, A4, | TFCS reconfigured to fit the new transport channel configuration. | |
| DL Transport channel information common for all transport channels | A5, A6, A7, A8 | Not Present | |
| Deleted TrCH information list | | | |
| - Deleted DL TrCH Information<br><br><br>- Downlink transport channel type<br>- Transport channel identity | A1, A2, A3, A5,A7, A8 | <br><br><br>DCH<br>6 | |
| - Deleted DL TrCH Information<br>- Downlink transport channel type<br>- Transport channel identity | A2, A8 | <br>DCH<br>7 | |
| - Deleted DL TrCH Information<br>- Downlink transport channel type<br>- Transport channel identity | A2, A8 | <br>DCH<br>8 | |
| Deleted TrCH information list | A4, A6 | Not Present | |
| Added or Reconfigured TrCH information list | | | |
| - Added or Reconfigured DL TrCH information | A5, A6, A7, A8 | Not Present | |
| - Added or Reconfigured DL TrCH information | A1, A2, A3, A4 | 1 TrCHs (DCH for DCCH) | |
| - Downlink transport channel type | | DCH | |
| - DL Transport channel identity | | 10 | |
| - CHOICE DL parameters | | Same as UL | |
| - Uplink transport channel type | | DCH | |
| - UL TrCH identity | | 5 | |
| - DCH quality target | | | |
| - BLER Quality value | | -2.0 Real(-6.3..0 by step of 0.1) | |
| Frequency info<br><br><br>- Choice mode<br>- UARFCN (Nt) | A1, A2, A3, A4, A5, A7, A8 | <br><br><br>TDD<br>Reference to clause 5.1 Test frequencies | |
| Frequency info | A6 | Not Present | |
| Maximum allowed UL TX power | A1, A2, A3, A4, A7, A8 | 33dBm | |
| Maximum allowed UL TX power | A5, A6 | using the default value | |
| CHOICE *channel requirement* | A5, A6 , A7, A8 | Not Present | |

| Information Element | Condition | Value/remark | Version |
|---|---|---|---|
| CHOICE *channel requirement* | A1, A2, A3, A4 | Uplink DPCH info | |
| - Uplink DPCH power control info | | Not Present | |
| - CHOICE mode | | TDD | |
| - Uplink Timing Advance Control | | Not Present | |
| - UL CCTrCH List | | | |
| - TFCS ID | | 1 | |
| - UL Target SIR | | Real (-11 .. 20 by step of 0.5dB)<br>Reference to TS34.108 Parameter set. | |
| - Time info | | | |
| - Activation time | | (256+CFN-(CFN MOD 8 + 8))MOD 256 | |
| - Duration | | Infinite | |
| - Common timeslot info | | | |
| - 2nd interleaving mode | | Default value is "Frame" | |
| - TFCI coding | | Reference to TS34.108 clause 6 Parameter set | |
| - Puncturing limit | | Reference to TS34.108 clause 6 Parameter set | |
| - Repetition period | | 1 | |
| - Repetition length | | | |
| - Uplink DPCH timeslots and code | | | |
| - Dynamic SF usage | | FALSE | |
| - First individual timeslot info | | | |
| - Timeslot number | | | |
| - CHOICE TDD option | | 3.84 Mcps TDD | |
| - Timeslot number | | 1 OR 2 OR 3 | |
| - TFCI existence | | TRUE | |
| - Midamble shift and burst type | | | |
| - CHOICE TDD option | | 3.84 Mcps TDD | |
| - CHOICE *Burst Type* | | Type 1 | |
| - Midamble allocation mode | | Default midamble | |
| - Midamble configuration | | 16 | |
| - Midamble Shift | | Not Present | |
| - CHOICE TDD option | | 3.84 Mcps TDD (no data) | |
| - First timeslot Code List | | Repeated (1,2) for each channelisation code assigned in the slot to meet the needs of TS34.108 clause 6 Parameter Set. | |
| - channelisation codes | | (SF/ i) where i denotes an unassigned code matching the SF specified in TS34.108 clause 6 Parameter Set. | |
| - CHOICE more timeslots | | No more timeslots | |
| - UL CCTrCH List to Remove | | Not present | |
| CHOICE Mode | A1, A2, A3, A4, A5, A6, A7, A8 | TDD | |
| Downlink HS-PDSCH Information | A1, A2, A3, A4, A5, A6, A7, A8 | Not Present | |
| Downlink information common for all radio links | A5, A6, A7, A8 | Not Present | |
| Downlink information common for all radio links | A1, A2, A3 | | |
| - Downlink DPCH info common for all RL | | | |
| - Timing indication | | Maintain | |
| - CFN-targetSFN frame offset | | Not Present | |
| - Downlink DPCH power control information | | | |
| - CHOICE mode | | TDD | |
| - TPC Step Size | | 1 | |
| - MAC-d HFN initial value | | Not Present | |
| - CHOICE mode | | TDD | |
| - CHOICE mode | | TDD | |

| Information Element | Condition | Value/remark | Version |
|---|---|---|---|
| - CHOICE TDD option | | 3.84 Mcps TDD | |
| - Default DPCH Offset Value | | Not Present | |
| Downlink information common for all radio links | A4 | | |
| - Downlink DPCH info common for all RL | | | |
| - Timing indication | | Initialise | |
| - CFN-targetSFN frame offset | | Not Present | |
| - Downlink DPCH power control information | | | |
| - CHOICE mode | | TDD | |
| - TPC Step Size | | 1 | |
| - MAC-d HFN initial value | | Not Present | |
| - CHOICE mode | | TDD | |
| - CHOICE mode | | TDD | |
| - CHOICE TDD option | | 3.84 Mcps TDD | |
| - Default DPCH Offset Value | | | |
| - CHOICE mode | | TDD | |
| - Default DPCH Offset Value | | 0 Integer(0..7) | |
| Downlink information per radio link list | A1, A2, A3, A4, | | |
| - Downlink information for each radio link | | | |
| - Choice mode | | TDD | |
| - Primary CCPCH info | | | |
| - Choice mode | | TDD | |
| - Choice TDD Option | | 3.84 Mcps TDD | |
| - Cell parameters ID | | Ref. to the Default setting in TS34.108 clause 6.1 (TDD) Integer(0..127) | |
| - SCTD indicator | | FALSE | |
| - Downlink DPCH info for each RL | | | |
| - CHOICE mode | | TDD | |
| - DL CCTrCh List | | | |
| - TFCS ID | | 2 Integer(1.8) | |
| - Time info | | | |
| - Activation time | | Now | |
| - Duration | | Infinite | |
| - Common timeslot info | | | |
| - 2nd interleaving mode | | Default value is "Frame" | |
| - TFCI coding | | Reference to TS34.108 clause 6 Parameter set | |
| - Puncturing limit | | Reference to TS34.108 clause 6 Parameter set | |
| - Repetition period | | 1 | |
| - Repetition length | | NULL | |
| - Downlink DPCH timeslots and codes | | | |
| - First individual timeslot info | | | |
| - Timeslot number | | | |
| - CHOICE TDD option | | 3.84 Mcps TDD | |
| - Timeslot number | | 4 OR 5 OR 6 | |
| - TFCI existence | | TRUE | |
| - Midamble shift and burst type | | | |
| - CHOICE TDD option | | 3.84 Mcps TDD | |
| - Midamble allocation mode | | Default midamble | |
| - Midamble configuration | | 16 | |
| - Midamble Shift | | Not Present | |
| - CHOICE TDD option | | 3.84 Mcps TDD (no data) | |
| - First timeslot channelisation codes | | Repeated (1,2) for each channelisation code assigned in the slot to meet the needs of TS34.108 clause 6 Parameter Set. | |
| - CHOICE codes representation | | Bitmap | |
| - Channelisation codes bitmap | | Reference to TS34.108 clause 6.10 Parameter Set | |
| - CHOICE more timeslots | | No more timeslots | |
| - UL CCTrCH TPC List | | Default is all | |
| - DL CCTrCH List to Remove | | Not present | |
| - SCCPCH Information for FACH | | Not Present7 | R99 and Rel-4 only |

| Information Element | Condition | Value/remark | Version |
|---|---|---|---|
| Downlink information per radio link list<br>- Downlink information for each radio link<br>- Choice mode<br>- Primary CCPCH info<br>- Choice mode<br>- Choice TDD Option<br>- TSTD indicator<br>- Cell parameters ID<br><br>- SCTD indicator | A5 ,A7, A8 | <br><br><br>TDD<br><br>TDD<br>3.84 Mcps TDD<br>FALSE<br>Ref. to the Default setting in TS34.108 clause 6.1 (TDD) Integer(0..127)<br>FALSE | |
| - Downlink DPCH info for each RL | | Not Present | |
| - SCCPCH Information for FACH | | Not Present | R99 and Rel-4 only |
| Downlink information per radio link list | A6 | Not Present | |

| Condition | Explanation |
|---|---|
| A1 | This IE need for "Non speech in CS" |
| A2 | This IE need for "Speech in CS" |
| A3 | This IE need for "Packet to CELL_DCH from CELL_DCH in PS" |
| A4 | This IE need for "Packet to CELL_DCH from CELL_FACH in PS" |
| A5 | This IE need for "Packet to CELL_FACH from CELL_DCH in PS" |
| A6 | This IE need for "Packet to CELL_FACH from CELL_FACH in PS" |
| A7 | This IE need for "Non speech to CELL_FACH from CELL_DCH in CS" |
| A8 | This IE need for "Speech to CELL_FACH from CELL_DCH in CS" |

Contents of RADIO BEARER RELEASE message: AM or UM (1.28 Mcps TDD)

| Information Element | Condition | Value/remark | Version |
|---|---|---|---|
| Message Type | A1, A2, A3, A4, A5, A6, A7, A8 | | |
| RRC transaction identifier | | Arbitrarily selects an integer between 0 and 3 | |
| Integrity check info<br>- message authentication code | | <br>SS calculates the value of MAC-I for this message and writes to this IE. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I. | |
| - RRC message sequence number | | SS provides the value of this IE, from its internal counter. | |
| Integrity protection mode info | | Not Present | |
| Ciphering mode info | | Not Present | |
| Activation time | A1, A2, A3, A7, A8 | (256+CFN-(CFN MOD 8 + 8))MOD 256 | |
| Activation time | A4, A5, A6 | Not Present | |
| New U-RNTI | | Not Present | |
| New C-RNTI | A1,A2,A3,A4 | Not Present | |
| New C-RNTI | A5, A6, A7, A8 | '1010 1010 1010 1010' | |
| New DSCH-RNTI | A1, A2, A3, A4, A5, A6, A7, A8 | Not Present | |
| RRC State indicator | A1,A2, A3, A4 | CELL_DCH | |
| RRC State indicator | A5, A6, A7, A8 | CELL_FACH | |
| UTRAN DRX cycle length coefficient | A1,A2,A3,A4, A5,A6, A7, A8 | Not Present | |
| CN information info | | Not Present | |
| Signalling Connection release indication | | Not Present | |
| URA identity | | Not Present | |
| RAB information to reconfigure list | | Not Present | |
| RB information to release list | A1, A7 | | |

| Information Element | Condition | Value/remark | Version |
|---|---|---|---|
| RB information to release | | | |
| - RB identity | | 10 | |
| RB information to release list | A2, A8 | | |
| RB information to release | | | |
| - RB identity | | 10 | |
| RB information to release | | | |
| - RB identity | | 11 | |
| RB information to release | | | |
| - RB identity | | 12 | |
| RB information to release list | A3, A4, A5, A6 | | |
| RB information to release | | | |
| - RB identity | | 20 | |
| RB information to be affected list | A1,A2, A3,A4,A5, A6, A7, A8 | Not Present | |
| Downlink counter synchronization info | A1,A2,A3,A4,A5,A6, A7, A8 | Not Present | |
| UL Transport channel information common for all transport channels | A1, A2, A3, A4 | TFCS reconfigured to fit the new transport channel configuration. | |
| UL Transport channel information common for all transport channels | A5, A6, A7, A8 | Not Present | |
| Deleted TrCH information list | A1,A2, A3, A5, A7, A8 | | |
| Deleted UL TrCH Information | A1,A2, A3, A5, A7, A8 | | |
| - Uplink transport channel type | | DCH | |
| - Transport channel identity | | 1 | |
| Deleted UL TrCH Information | A2, A8 | | |
| - Uplink transport channel type | | DCH | |
| - Transport channel identity | | 2 | |
| Deleted UL TrCH Information | A2, A8 | | |
| - Uplink transport channel type | | DCH | |
| - Transport channel identity | | 3 | |
| Deleted TrCH information list | A4, A6 | Not Present | |
| Added or Reconfigured TrCH information list | A5, A6, A7, A8 | Not Present | |
| Added or Reconfigured TrCH information list | A1, A2, A3, A4 | TrCHs (DCH for DCCH ) | |
| Added or Reconfigured UL TrCH information | | | |
| - Uplink transport channel type | | DCH | |
| - UL Transport channel identity | | 5 | |
| - TFS | | | |
| - CHOICE Transport channel type | | Dedicated transport channels | |
| - Dynamic Transport format information | | | |
| - RLC Size | | Reference to clause 6.11 Parameter Set | |
| - Number of TBs and TTI List | | (This IE is repeated for TFI number.) | |
| - Transmission Time Interval | | Not present | |
| - Number of Transport blocks | | Reference to clause 6.11 Parameter Set | |
| - CHOICE Logical channel list | | All (NULL) | |
| - Semi-static Transport Format information | | | |
| - Transmission time interval | | Reference to clause 6.11 Parameter Set | |
| - Type of channel coding | | Reference to clause 6.11 Parameter Set | |
| - Coding Rate | | Reference to clause 6.11 Parameter Set | |
| - Rate matching attribute | | Reference to clause 6.11 Parameter Set | |
| - CRC size | | Reference to clause 6.11 Parameter Set | |
| CHOICE *mode* | | TDD (No data) | |
| DL Transport channel information common for all transport channels | A1, A2, A3, A4, | TFCS reconfigured to fit the new transport channel configuration. | |
| DL Transport channel information common for all transport channels | A5, A6, A7, A8 | Not Present | |
| Deleted TrCH information list | | | |
| - Deleted DL TrCH Information | A1, A2, A3, A5,A7, A8 | | |
| - Downlink transport channel type | | DCH | |
| - Transport channel identity | | 6 | |

| Information Element | Condition | Value/remark | Version |
|---|---|---|---|
| - Deleted DL TrCH Information | A2, A8 | | |
| - Downlink transport channel type | | DCH | |
| - Transport channel identity | | 7 | |
| - Deleted DL TrCH Information | A2, A8 | | |
| - Downlink transport channel type | | DCH | |
| - Transport channel identity | | 8 | |
| Deleted TrCH information list | A4, A6 | Not Present | |
| Added or Reconfigured TrCH information list | | | |
| - Added or Reconfigured DL TrCH information | A5, A6, A7, A8 | Not Present | |
| - Added or Reconfigured DL TrCH information | A1, A2, A3, A4 | 1 TrCHs (DCH for DCCH) | |
| - Downlink transport channel type | | DCH | |
| - DL Transport channel identity | | 10 | |
| - CHOICE DL parameters | | Same as UL | |
| - Uplink transport channel type | | DCH | |
| - UL TrCH identity | | 5 | |
| - DCH quality target | | | |
| - BLER Quality value | | -2.0 Real(-6.3..0 by step of 0.1) | |
| Frequency info | A1, A2, A3, A4, A5, A7, A8 | | |
| - Choice mode | | TDD | |
| - UARFCN (Nt) | | Reference to clause 5.1 Test frequencies | |
| Frequency info | A6 | Not Present | |
| Maximum allowed UL TX power | A1, A2, A3, A4, A7, A8 | 33dBm | |
| Maximum allowed UL TX power | A5, A6 | using the default value | |
| CHOICE *channel requirement* | A5, A6 , A7, A8 | Not Present | |
| CHOICE *channel requirement* | A1, A2, A3, A4 | Uplink DPCH info | |
| - Uplink DPCH power control info | | Not Present | |
| - CHOICE mode | | TDD | |
| - Uplink Timing Advance Control | | Not Present | |
| - UL CCTrCH List | | | |
| - TFCS ID | | 1 | |
| - UL Target SIR | | Real (-11 .. 20 by step of 0.5dB) Reference to clause 6 Parameter set. | |
| - Time info | | | |
| - Activation time | | (256+CFN-(CFN MOD 8 + 8))MOD 256 | |
| - Duration | | Infinite | |
| - Common timeslot info | | | |
| - 2nd interleaving mode | | Default value is "Frame" | |
| - TFCI coding | | Reference to clause 6 Parameter set | |
| - Puncturing limit | | Reference to clause 6 Parameter set | |
| - Repetition period | | 1 | |
| - Repetition length | | | |
| - Uplink DPCH timeslots and code | | | |
| - Dynamic SF usage | | FALSE | |
| - First individual timeslot info | | | |
| - Timeslot number | | | |
| - CHOICE TDD option | | 1.28 Mcps TDD | |
| - Timeslot number | | 1 OR 2 OR 3 | |
| - TFCI existence | | TRUE | |
| - Midamble shift and burst type | | | |
| - CHOICE TDD option | | 1.28 Mcps TDD | |
| - Midamble allocation mode | | Default midamble | |
| - Midamble configuration | | 16 | |
| - Midamble Shift | | Not Present | |
| - CHOICE TDD option | | 1.28 Mcps TDD | |
| - Modulation | | QPSK | |
| - SS-TPC Symbols | | 1 | |
| - Additional TPC-SS Symbols | | Not present | |

| Information Element | Condition | Value/remark | Version |
|---|---|---|---|
| - First timeslot Code List | | Repeated (1,2) for each channelisation code assigned in the slot to meet the needs of clause 6 Parameter Set. | |
| - channelisation codes | | (SF/ i) where i denotes an unassigned code matching the SF specified in clause 6 Parameter Set. | |
| - CHOICE more timeslots | | No more timeslots | |
| - UL CCTrCH List to Remove | | Not present | |
| CHOICE Mode | A1, A2, A3, A4, A5, A6, A7, A8 | TDD | |
| Downlink HS-PDSCH Information | A1, A2, A3, A4, A5, A6, A7, A8 | Not Present | |
| Downlink information common for all radio links | A5, A6, A7, A8 | Not Present | |
| Downlink information common for all radio links | A1, A2, A3 | | |
| - Downlink DPCH info common for all RL | | | |
| - Timing indication | | Maintain | |
| - CFN-targetSFN frame offset | | Not Present | |
| - Downlink DPCH power control information | | | |
| - CHOICE mode | | TDD | |
| - TPC Step Size | | 1 | |
| - MAC-d HFN initial value | | Not Present | |
| - CHOICE mode | | TDD | |
| - CHOICE mode | | TDD | |
| - CHOICE TDD option | | 1.28 Mcps TDD | |
| - TSTD indicator | | FALSE | |
| - Default DPCH Offset Value | | Not Present | |
| Downlink information common for all radio links | A4 | | |
| - Downlink DPCH info common for all RL | | | |
| - Timing indication | | Initialize | |
| - CFN-targetSFN frame offset | | Not Present | |
| - Downlink DPCH power control information | | | |
| - CHOICE mode | | TDD | |
| - TPC Step Size | | 1 | |
| - MAC-d HFN initial value | | Not Present | |
| - CHOICE mode | | TDD | |
| - CHOICE mode | | TDD | |
| - CHOICE TDD option | | 1.28 Mcps TDD | |
| - TSTD indicator | | FALSE | |
| - Default DPCH Offset Value | | | |
| - CHOICE mode | | TDD | |
| - Default DPCH Offset Value | | 0 Integer(0..7) | |
| Downlink information per radio link list | A1, A2, A3, A4, | | |
| - Downlink information for each radio link | | | |
| - Choice mode | | TDD | |
| - Primary CCPCH info | | | |
| - Choice mode | | TDD | |
| - Choice TDD Option | | 1.28 Mcps TDD | |
| - TSTD indicator | | FALSE | |
| - Cell parameters ID | | Ref. to the Default setting in clause 6.1 (TDD) Integer(0..127) | |
| - SCTD indicator | | FALSE | |
| - Downlink DPCH info for each RL | | | |
| - CHOICE mode | | TDD | |
| - DL CCTrCh List | | | |
| - TFCS ID | | 2 Integer(1.8) | |
| - Time info | | | |
| - Activation time | | Now | |
| - Duration | | Infinite | |
| - Common timeslot info | | | |
| - 2nd interleaving mode | | Default value is "Frame" | |
| - TFCI coding | | Reference to clause 6 Parameter set | |
| - Puncturing limit | | Reference to clause 6 Parameter set | |
| - Repetition period | | 1 | |

| Information Element | Condition | Value/remark | Version |
|---|---|---|---|
| - Repetition length | | NULL | |
| - Downlink DPCH timeslots and codes | | | |
| - First individual timeslot info | | | |
| - Timeslot number | | | |
| - CHOICE TDD option | | 1.28 Mcps TDD | |
| - Timeslot number | | 4 OR 5 OR 6 | |
| - TFCI existence | | TRUE | |
| - Midamble shift and burst type | | | |
| - CHOICE TDD option | | 1.28 Mcps TDD | |
| - Midamble allocation mode | | Default midamble | |
| - Midamble configuration | | 16 | |
| - Midamble Shift | | Not Present | |
| - CHOICE TDD option | | 1.28 Mcps TDD | |
| - Modulation | | QPSK | |
| - SS-TPC Symbols | | 1 | |
| - Additional TPC-SS Sysbols | | Not present | |
| - First timeslot channelisation codes | | Repeated (1,2) for each channelisation code assigned in the slot to meet the needs of clause 6 Parameter Set. | |
| - CHOICE codes representation | | Bitmap | |
| - Channelisation codes bitmap | | Reference to clause 6.10 Parameter Set | |
| - CHOICE more timeslots | | No more timeslots | |
| - UL CCTrCH TPC List | | This list is not required for 1.28 Mcps TDD and is to be ignored by the UE. | |
| - DL CCTrCH List to Remove | | Not present | |
| - SCCPCH Information for FACH | | Not Present | R99 and Rel-4 only |
| Downlink information per radio link list | A5 ,A7, A8 | | |
| - Downlink information for each radio link | | | |
| - Choice mode | | TDD | |
| - Primary CCPCH info | | | |
| - Choice mode | | TDD | |
| - Choice TDD Option | | 1.28 Mcps TDD | |
| - TSTD indicator | | FALSE | |
| - Cell parameters ID | | Ref. to the Default setting in clause 6.1 (TDD) Integer(0..127) | |
| - SCTD indicator | | FALSE | |
| - Downlink DPCH info for each RL | | Not Present | |
| - SCCPCH Information for FACH | | Not Present | R99 and Rel-4 only |
| Downlink information per radio link list | A6 | Not Present | |

| Condition | Explanation |
|---|---|
| A1 | This IE need for "Non speech in CS" |
| A2 | This IE need for "Speech in CS" |
| A3 | This IE need for "Packet to CELL_DCH from CELL_DCH in PS" |
| A4 | This IE need for "Packet to CELL_DCH from CELL_FACH in PS" |
| A5 | This IE need for "Packet to CELL_FACH from CELL_DCH in PS" |
| A6 | This IE need for "Packet to CELL_FACH from CELL_FACH in PS" |
| A7 | This IE need for "Non speech to CELL_FACH from CELL_DCH in CS" |
| A8 | This IE need for "Speech to CELL_FACH from CELL_DCH in CS" |

Contents of DOWNLINK DIRECT TRANSFER message: AM

| Information Element | Value/remark |
|---|---|
| Message Type | |
| RRC transaction identifier | 0 |
| Integrity check info | |
| - Message authentication code | SS calculates the value of MAC-I for this message and writes to this IE. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I. |
| - RRC Message sequence number | SS provides the value of this IE, from its internal counter. |
| CN domain identity | CS domain or PS domain |

| | |
|---|---|
| NAS message | See Specific Message Content for each test case |

Contents of INITIAL DIRECT TRANSFER message: AM

| Information Element | Value/remark | Version |
|---|---|---|
| Message Type | | |
| Integrity check info | | |
| - Message authentication code | This IE is checked to see if it is present. The value is compared against the XMAC-I value computed by SS. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I. | |
| - RRC Message sequence number | This IE is checked to see if it is present. The value is used by SS to compute the XMAC-I value. | |
| CN domain identity | CS domain or PS domain | |
| Intra Domain NAS Node Selector | Set to the same octet string as in the IMSI stored in the USIM card | |
| NAS message | Set according to that indicated in specific message content for each test case | |
| START | This IE is checked to see if it is present. | |
| Establishment cause | See the specific test case | Rel-5 |
| Measured results on RACH | Not checked | |

Contents of PAGING TYPE 1 message: TM (Speech in CS)

| Information Element | Value/remark |
|---|---|
| Message Type | |
| Paging record list | |
| - Paging record | |
| - CHOICE Used paging identity | CN identity |
| - Paging cause | Terminating Conversational Call |
| - CN domain identity | CS domain |
| - CHOICE UE identity | |
| - IMSI (GSM-MAP) | Set to the same octet string as in the IMSI stored in the USIM card |
| BCCH modification info | Not Present |

Contents of PAGING TYPE 1 message: TM (The others of speech in CS)

| Information Element | Value/remark |
|---|---|
| Message Type | |
| Paging record list | |
| - Paging record | |
| - CHOICE Used paging identity | CN identity |
| - Paging cause | Terminating Streaming Call |
| - CN domain identity | CS domain |
| - CHOICE UE identity | |
| - IMSI (GSM-MAP) | Set to the same octet string as in the IMSI stored in the USIM card |
| BCCH modification info | Not Present |

Contents of PAGING TYPE 1 message: TM (Packet in PS)

| Information Element | Value/remark |
|---|---|
| Message Type | |
| Paging record list | |
| - Paging record | |
| - CHOICE Used paging identity | CN identity |
| - Paging cause | Terminating Interactive Call |
| - CN domain identity | PS domain |
| - CHOICE UE identity | |
| - IMSI (GSM-MAP) | Set to the same octet string as in the IMSI stored in the USIM card |
| BCCH modification info | Not Present |

Contents of RADIO BEARER SETUP message: AM or UM (Speech in CS) (3.84 Mcps TDD option)

| Information Element | Value/remark | Version |
|---|---|---|
| Message Type | | |
| RRC transaction identifier | 0 | |
| Integrity check info | | |
| - message authentication code | SS calculates the value of MAC-I for this message and writes to this IE. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I. | |
| - RRC message sequence number | SS provides the value of this IE, from its internal counter. | |
| Integrity protection mode info | Not Present | |
| Ciphering mode info | The presence of this IE is dependent on IXIT statements in 3GPP TS 34.123-2 [3]. If ciphering is indicated to be active, this IE present with the values of the sub IEs as stated below. Else, this IE is omitted. | |
| - Ciphering mode command | Start/restart | |
| - Ciphering algorithm | Use one of the supported ciphering algorithms | |
| - Ciphering activation time for DPCH | (256+CFN-(CFN MOD 8 + 8))MOD 256 | |
| - Radio bearer downlink ciphering activation time info | Not Present | |
| Activation time | (256+CFN-(CFN MOD 8 + 8))MOD 256 | |
| New U-RNTI | Not Present | |
| New C-RNTI | Not Present | |
| New DSCH-RNTI | Not Present | |
| New H-RNTI | Not Present | Rel-5 |
| RRC State indicator | CELL_DCH | |
| UTRAN DRX cycle length coefficient | Not Present | |
| CN information info | Not Present | |
| URA identity | Not Present | |
| - Signalling RB information to setup list | Not Present | |
| - RAB information for setup list | | |
| - RAB information for setup | | |
| - RAB info | | |
| - RAB identity | 0000 0001B<br>The first/ leftmost bit of the bit string contains the most significant bit of the RAB identity. | |
| - CN domain identity | CS domain | |
| - NAS Synchronization Indicator | Not Present | |
| - Re-establishment timer | UseT314 | |
| - RB information to setup | | |
| - RB identity | 10 | |
| - PDCP info | Not Present | |
| - CHOICE RLC info type | RLC info | |
| - CHOICE Uplink RLC mode | TM RLC | |
| - Transmission RLC discard | Not Present | |
| - Segmentation indication | FALSE | |
| - CHOICE Downlink RLC mode | TM RLC | |
| - Segmentation indication | FALSE | |
| - RB mapping info | | |
| - Information for each multiplexing option | | |
| - RLC logical channel mapping indicator | Not Present | |
| - Number of uplink RLC logical channels | 1 | |
| - Uplink transport channel type | DCH | |
| - UL Transport channel identity | 1 | |
| - Logical channel identity | Not Present | |
| - CHOICE RLC size list | Configured | |
| - MAC logical channel priority | 6 | |
| - Downlink RLC logical channel info | | |
| - Number of downlink RLC logical channels | 1 | |
| - Downlink transport channel type | DCH | |
| - DL DCH Transport channel identity | 6 | |
| - DL DSCH Transport channel identity | Not Present | |
| - Logical channel identity | Not Present | |
| - RB identity | 11 | |
| - PDCP info | Not Present | |

| Information Element | Value/remark | Version |
|---|---|---|
| - CHOICE RLC info type | RLC info | |
| - CHOICE Uplink RLC mode | TM RLC | |
| - Transmission RLC discard | Not Present | |
| - Segmentation indication | FALSE | |
| - CHOICE Downlink RLC mode | TM RLC | |
| - Segmentation indication | FALSE | |
| - RB mapping info | | |
| - Information for each multiplexing option | | |
| - RLC logical channel mapping indicator | Not Present | |
| - Number of uplink RLC logical channels | 1 | |
| - Uplink transport channel type | DCH | |
| - UL Transport channel identity | 2 | |
| - Logical channel identity | Not Present | |
| - CHOICE RLC size list | Configured | |
| - MAC logical channel priority | 6 | |
| - Downlink RLC logical channel info | | |
| - Number of downlink RLC logical channels | 1 | |
| - Downlink transport channel type | DCH | |
| - DL DCH Transport channel identity | 7 | |
| - DL DSCH Transport channel identity | Not Present | |
| - Logical channel identity | Not Present | |
| - RB identity | 12 | |
| - PDCP info | Not Present | |
| - CHOICE RLC info type | RLC info | |
| - CHOICE Uplink RLC mode | TM RLC | |
| - Transmission RLC discard | Not Present | |
| - Segmentation indication | FALSE | |
| - CHOICE Downlink RLC mode | TM RLC | |
| - Segmentation indication | FALSE | |
| - RB mapping info | | |
| - Information for each multiplexing option | | |
| - RLC logical channel mapping indicator | Not Present | |
| - Number of uplink RLC logical channels | 1 | |
| - Uplink transport channel type | DCH | |
| - UL Transport channel identity | 3 | |
| - Logical channel identity | Not Present | |
| - CHOICE RLC size list | Configured | |
| - MAC logical channel priority | 6 | |
| - Downlink RLC logical channel info | | |
| - Number of downlink RLC logical channels | 1 | |
| - Downlink transport channel type | DCH | |
| - DL DCH Transport channel identity | 8 | |
| - DL DSCH Transport channel identity | Not Present | |
| - Logical channel identity | Not Present | |
| RB information to be affected list | Not Present | |
| Downlink counter synchronization info | Not Present | |
| UL Transport channel information for all transport channels | | |
| - PRACH TFCS | Not Present | |
| - CHOICE mode | TDD | |
| -Individual UL CCTrCH information | | |
| - TFCS ID | (This IE is repeated for TFC number.) | |
| - Allowed Transport Format combination | 0 to MaxTFCvalue-1 (MaxTFCValue is refer to clause 6 Parameter Set.) | |
| - PRACH TFCS | (This IE is repeated for TFC number.) | |
| - CHOICE TFCI signalling | Normal | |
| - TFCI Field 1 information | | |
| - TFCS complete reconfigure information | | |
| - CHOICE TFCS Size | Number of used bits must be enough to cover all combinations of CTFC from clauses 6. Refer to clause 6 Parameter Set | |
| - CTFC information | Not Present | |
| - CHOICE mode | TDD | |
| - Individual UL CCTrCH information | Not Present | |
| Deleted TrCH information list | Not Present | |
| Added or Reconfigured TrCH information list | 3 DCHs | |

| Information Element | Value/remark | Version |
|---|---|---|
| - Added or Reconfigured UL TrCH information | | |
| - Uplink transport channel type | DCH | |
| - UL Transport channel identity | 1 | |
| - TFS | | |
| - CHOICE Transport channel type | Dedicated transport channels | |
| - Dynamic Transport format information | | |
| - RLC Size | Reference to clause 6.10 Parameter Set | |
| - Number of TBs and TTI List | (This IE is repeated for TFI number.) | |
| - Transmission Time Interval | Not Present | |
| - Number of Transport blocks | Reference to clause 6.10 Parameter Set | |
| - CHOICE Logical channel list | All | |
| - Semi-static Transport Format information | | |
| - Transmission time interval | Reference to clause 6.10 Parameter Set | |
| - Type of channel coding | Reference to clause 6.10 Parameter Set | |
| - Coding Rate | Reference to clause 6.10 Parameter Set | |
| - Rate matching attribute | Reference to clause 6.10 Parameter Set | |
| - CRC size | Reference to clause 6.10 Parameter Set | |
| - Uplink transport channel type | DCH | |
| - UL Transport channel identity | 2 | |
| - TFS | | |
| - CHOICE Transport channel type | Dedicated transport channels | |
| - Dynamic Transport format information | | |
| - RLC Size | Reference to clause 6.10 Parameter Set | |
| - Number of TBs and TTI List | (This IE is repeated for TFI number.) | |
| - Transmission Time Interval | Not Present | |
| - Number of Transport blocks | Reference to clause 6.10 Parameter Set | |
| - Transmission Time Interval | Reference to clause 6.10 Parameter Set | |
| - Number of Transport blocks | (This IE is repeated for TFI number.) | |
| - CHOICE Logical channel list | All | |
| - Semi-static Transport Format information | | |
| - Transmission time interval | Reference to clause 6.10 Parameter Set | |
| - Type of channel coding | Reference to clause 6.10 Parameter Set | |
| - Coding Rate | Reference to clause 6.10 Parameter Set | |
| - Rate matching attribute | Reference to clause 6.10 Parameter Set | |
| - CRC size | Reference to clause 6.10 Parameter Set | |
| - Uplink transport channel type | DCH | |
| - UL Transport channel identity | 3 | |
| - TFS | | |
| - CHOICE Transport channel type | Dedicated transport channels | |
| - Dynamic Transport format information | | |
| - RLC Size | Reference to clause 6.10 Parameter Set | |
| - Number of TBs and TTI List | (This IE is repeated for TFI number.) | |
| - Transmission Time Interval | Not Present | |
| - Number of Transport blocks | Reference to clause 6.10 Parameter Set | |
| - Transmission Time Interval | Reference to clause 6.10 Parameter Set | |
| - Number of Transport blocks | (This IE is repeated for TFI number.) | |
| - CHOICE Logical channel list | All | |
| - Semi-static Transport Format information | | |
| - Transmission time interval | Reference to clause 6.10 Parameter Set | |
| - Type of channel coding | Reference to clause 6.10 Parameter Set | |
| - Coding Rate | Reference to clause 6.10 Parameter Set | |
| - Rate matching attribute | Reference to clause 6.10 Parameter Set | |
| - CRC size | Reference to clause 6.10 Parameter Set | |
| CHOICE mode | TDD (no data) | |
| DL Transport channel information common for all transport channel | | |
| - SCCPCH TFCS | Not Present | |
| - CHOICE mode | TDD | |
| - CHOICE DL parameters | Same as UL | |
| Deleted TrCH information list | Not Present | |
| Added or Reconfigured TrCH information list | 3 DCHs | |
| Added or Reconfigured DL TrCH information | | |
| - Downlink transport channel type | DCH | |
| - DL Transport channel identity | 6 | |
| - CHOICE DL parameters | Same as UL | |
| - Uplink transport channel type | DCH | |
| - UL TrCH identity | 1 | |

| Information Element | Value/remark | Version |
|---|---|---|
| - DCH quality target | | |
| - BLER Quality value | -6.3 | |
| - Downlink transport channel type | DCH | |
| - DL Transport channel identity | 7 | |
| - CHOICE DL parameters | Same as UL | |
| - Uplink transport channel type | DCH | |
| - UL TrCH identity | 2 | |
| - DCH quality target | | |
| - BLER Quality value | Not Present | |
| - Downlink transport channel type | DCH | |
| - DL Transport channel identity | 8 | |
| - CHOICE DL parameters | Same as UL | |
| - Uplink transport channel type | DCH | |
| - UL TrCH identity | 3 | |
| - DCH quality target | | |
| - BLER Quality value | Not Present | |
| Frequency info | | |
| - UARFCN Nt) | Reference to clause 5.1 Test frequencies | |
| Maximum allowed UL TX power | 30dBm | |
| CHOICE channel requirement | Uplink DPCH info | |
| - Uplink DPCH power control info | | |
| - CHOICE mode | TDD | |
| - UL Target SIR | Reference to clause 6 Parameter set. | |
| - CHOICE UL OL PC info | Individually signalled | |
| - CHOICE TDD option | 3.84 Mcps | |
| - Individual timeslot interference info | | |
| - DPCH Constant Value | | |
| - CHOICE mode | TDD | |
| - Uplink Timing Advance Control | Not Present | |
| - UL CCTrCH List | | |
| - TFCS Id | 1 | |
| - Time info | | |
| - Activation time | (256+CFN-(CFN MOD 8 + 8))MOD 256 | |
| - Duration | infinite | |
| - Common timeslot info | | |
| - 2nd interleaving mode | Reference to clause 6 Parameter Set. | |
| - TFCI coding | Reference to clause 6 Parameter set. | |
| - Puncturing Limit | Reference to clause 6 Parameter set. | |
| - Repetition Period | Reference to clause 6 Parameter set. | |
| - Repetition Length | Reference to clause 6 Parameter set. | |
| - Uplink DPCH timeslots and code | | |
| - First individual timeslot info | | |
| - Timeslot number | The number of an uplink timeslot that has unassigned codes. | |
| - TFCI existence | TRUE | |
| - Midamble shift and burst type | | |
| - CHOICE TDD option | 3.84 Mcps | |
| - Midamble allocation mode | Default | |
| - Midamble configuration burst type 1 and 3 | 16 | |
| - CHOICE TDD option | 3.84 Mcps TDD (no data) | |
| - First timeslot channelisation codes | Repeated (1,2) for each channelisation code assigned in the slot to meet the needs of clause 6 Parameter Set. | |
| - Channelisation code | (i/SF) where i denotes an unassigned code matching the SF specified in clause 6 Parameter Set. | |
| - CHOICE more timeslots | The presence of this IE depends upon the number of resources specified in clause 6 and the number of slots in which they are being assigned. | |
| Downlink HS-PDSCH Information | Not Present | Rel-5 |
| Downlink information common for all radio links | | |
| - Downlink DPCH info common for all RL | | |
| - Timing indication | Maintain | |
| - CFN-targetSFN frame offset | Not Present | |
| - Downlink DPCH power control | | |

| Information Element | Value/remark | Version |
|---|---|---|
| information | | |
| - CHOICE mode | TDD | |
| - TPC step size | 1 dB | |
| - CHOICE mode | TDD | |
| - CHOICE TDD option | 3.84 Mcps (no data) | |
| - Default DPCH offset value | 0 | |
| - Downlink information for each radio link | | |
| - Choice mode | TDD | |
| - Primary CCPCH info | | |
| - CHOICE TDD option | 3.84 Mcps | |
| - CHOICE *SyncCase* | Sync Case 1 | |
| - Timeslot | PCCPCH timeslot | |
| - Cell parameters ID | 0 | |
| - SCTD indicator | | |
| - Downlink DPCH info for each RL | | |
| - CHOICE mode | TDD | |
| - DL CCTrCH List | | |
| - TFCS ID | 1 | |
| - Time info | | |
| - Activation time | (256+CFN-(CFN mod 8 + 8))mod 256 | |
| - Duration | infinite | |
| - Common timeslot info | | |
| - 2nd interleaving mode | Reference to the present document | |
| - TFCI coding | TRUE | |
| - Puncturing limit | Reference to clause 6 Parameter set | |
| - Repetition period | 1 | |
| - Repetition length | Empty | |
| - Downlink DPCH timeslots and codes | | |
| - Individual timeslot info | | |
| - Timeslot number | The number of a downlink timeslot that has unassigned codes. | |
| - TFCI existence | TRUE | |
| - Midamble shift and burst type | | |
| - CHOICE TDD option | 3.84 Mcps | |
| -CHOICE Burst Type | | |
| -Type 1 | | |
| -Midamble Allocation Mode | Default | |
| - Midamble configuration burst type 1 and 3 | As defined in 3GPP TS 25.221 [28] | |
| - First timeslot channelisation codes | | |
| - First channelisation code | (i/SF) where i is the lowest numbered code that is being assigned and SF is specified in clause 6 Parameter Set.. | |
| - Last channelisation code | (j/SF) where j is the highest numbered code that is being assigned in the slot. | |
| - Bitmap | Bitmap of the codes that are being assigned in the slot. | |
| - CHOICE more timeslots | The presence of this IE depends upon whether the requirements of clause 6 Parameter Set could be met by the codes that have been assigned in the first timeslot. | |
| - UL CCTrCH TPC List | Not Present | |
| -SCCPCH information for FACH | Not Present | R99 and Rel-4 only |

Contents of RADIO BEARER SETUP message: AM or UM (Packet to CELL_DCH from CELL_DCH in PS) (3.84 Mcps TDD option)

| Information Element | Value/remark | Version |
|---|---|---|
| Message Type | | |
| RRC transaction identifier | 0 | |
| Integrity check info | | |
| - message authentication code | SS calculates the value of MAC-I for this message and writes to this IE. The first/ leftmost bit of the bit string contains the most significant bit | |

| Information Element | Value/remark | Version |
|---|---|---|
| | of the MAC-I. | |
| - RRC message sequence number | SS provides the value of this IE, from its internal counter. | |
| Integrity protection mode info | Not Present | |
| Ciphering mode info | The presence of this IE is dependent on IXIT statements in 3GPP TS 34.123-2 [3]. If ciphering is indicated to be active, this IE present with the values of the sub IEs as stated below. Else, this IE is omitted. | |
| - Ciphering mode command | Start/restart | |
| - Ciphering algorithm | Use one of the supported ciphering algorithms | |
| - Ciphering activation time for DPCH | (256+CFN-(CFN MOD 8 + 8))MOD 256 | |
| - Radio bearer downlink ciphering activation time info | Not Present | |
| Activation time | (256+CFN-(CFN MOD 8 + 8))MOD 256 | |
| New U-RNTI | Not Present | |
| New C-RNTI | Not Present | |
| New DSCH-RNTI | Not Present | |
| New H-RNTI | Not Present | Rel-5 |
| RRC State indicator | CELL_DCH | |
| UTRAN DRX cycle length coefficient | Not Present | |
| CN information info | Not Present | |
| URA identity | Not Present | |
| - Signalling RB information to setup | Not Present | |
| - RAB information for setup | | |
| - RAB info | | |
| - RAB identity | 0000 0101B<br>The first/ leftmost bit of the bit string contains the most significant bit of the RAB identity. | |
| - CN domain identity | PS domain | |
| - NAS Synchronization Indicator | Not Present | |
| - Re-establishment timer | UseT314 | |
| - RB information to setup | | |
| - RB identity | 20 | |
| - PDCP info | Not Present | |
| - CHOICE RLC info type | RLC info | |
| - CHOICE Uplink RLC mode | AM RLC | |
| - Transmission RLC discard | | |
| - SDU discard mode | No Discard | |
| - MAX_DAT | 15 | |
| - Transmission window size | 128 | |
| - Timer_RST | 500 | |
| - Max_RST | 4 | |
| - Polling info | | |
| - Timer_poll_prohibit | 200 | |
| - Timer_poll | 200 | |
| - Poll_PDU | Not Present | |
| - Poll_SDU | 1 | |
| - Last transmission PDU poll | TRUE | |
| - Last retransmission PDU poll | TRUE | |
| - Poll_Windows | 99 | |
| - Timer_poll_periodic | Not Present | |
| - CHOICE Downlink RLC mode | AM RLC | |
| - In-sequence delivery | TRUE | |
| - Receiving window size | 128 | |
| - Downlink RLC status info | | |
| - Timer_status_prohibit | 200 | |
| - Timer_EPC | Not Present | |
| - Missing PDU indicator | TRUE | |
| - Timer_STATUS_periodic | Not Present | |
| - RB mapping info | | |
| - Information for each multiplexing option | 2 RBMuxOptions | |
| - RLC logical channel mapping indicator | Not Present | |
| - Number of uplink RLC logical channels | 1 | |
| - Uplink transport channel type | DCH | |
| - UL Transport channel identity | 1 | |
| - Logical channel identity | Not Present | |

| Information Element | Value/remark | Version |
|---|---|---|
| - CHOICE RLC size list | Configured | |
| - MAC logical channel priority | 8 | |
| - Downlink RLC logical channel info | | |
| - Number of downlink RLC logical channels | 1 | |
| - Downlink transport channel type | DCH | |
| - DL DCH Transport channel identity | 6 | |
| - DL DSCH Transport channel identity | Not Present | |
| - Logical channel identity | Not Present | |
| - RLC logical channel mapping indicator | Not Present | |
| - Number of uplink RLC logical channels | 1 | |
| - Uplink transport channel type | RACH | |
| - UL Transport channel identity | Not Present | |
| - Logical channel identity | 7 | |
| - CHOICE RLC size list | Explicit List | |
| - RLC size index | Reference to clause 6 Parameter Set | |
| - MAC logical channel priority | 8 | |
| - Downlink RLC logical channel info | | |
| - Number of downlink RLC logical channels | 1 | |
| - Downlink transport channel type | FACH | |
| - DL DCH Transport channel identity | Not Present | |
| - DL DSCH Transport channel identity | Not Present | |
| - Logical channel identity | 7 | |
| RB information to be affected list | Not Present | |
| Downlink counter synchronization info | Not Present | |
| UL Transport channel information for all transport channels | | |
| - PRACH TFCS | Not Present | |
| - CHOICE mode | TDD | |
| -Individual UL CCTrCH information | | |
| - TFCS ID | (This IE is repeated for TFC number.) | |
| - Allowed Transport Format combination | 0 to MaxTFCvalue-1 (MaxTFCValue is refer to clause 6 Parameter Set.) | |
| - PRACH TFCS | (This IE is repeated for TFC number.) | |
| - CHOICE TFCI signalling | Normal | |
| - TFCI Field 1 information | | |
| - TFCS complete reconfigure information | | |
| - CHOICE TFCS Size | Number of used bits must be enough to cover all combinations of CTFC from clauses 6. Refer to clause 6 Parameter Set | |
| - CTFC information | Not Present | |
| - CHOICE mode | TDD | |
| - Individual UL CCTrCH information | Not Present | |
| Deleted TrCH information list | Not Present | |
| Added or Reconfigured TrCH information list | | |
| - Added or Reconfigured UL TrCH information | | |
| - Uplink transport channel type | DCH | |
| - UL Transport channel identity | 1 | |
| - TFS | | |
| - CHOICE Transport channel type | Dedicated transport channels | |
| - Dynamic Transport format information | | |
| - RLC Size | Reference to clause 6.10 Parameter Set | |
| - Number of TBs and TTI List | (This IE is repeated for TFI number.) | |
| - Transmission Time Interval | Not Present | |
| - Number of Transport blocks | Reference to clause 6.10 Parameter Set | |
| - CHOICE Logical channel list | All | |
| - Semi-static Transport Format information | | |
| - Transmission time interval | Reference to clause 6.10 Parameter Set | |
| - Type of channel coding | Reference to clause 6.10 Parameter Set | |
| - Coding Rate | Reference to clause 6.10 Parameter Set | |
| - Rate matching attribute | Reference to clause 6.10 Parameter Set | |
| - CRC size | Reference to clause 6.10 Parameter Set | |
| CHOICE mode | TDD (no data) | |
| DL Transport channel information common for all transport channel | | |
| - SCCPCH TFCS | Not Present | |
| - CHOICE mode | TDD | |

| Information Element | Value/remark | Version |
|---|---|---|
| - Individual DL CCTrCH information | | |
| - DL TFCS Identity | | |
| - TFCS Id | 1 | |
| - Shared Channel Indicator | FALSE | |
| - CHOICE DL parameters | Independent | |
| - DL DCH TFCS | (This IE is repeated for TFC number.) | |
| - CHOICE TFCI signalling | Normal | |
| - TFCI Field 1 information | | |
| - CHOICE TFCS representation | Complete | |
| - TFCS complete reconfigure information | | |
| - CHOICE CTFC Size | Refer to clause 6. | |
| - CTFC information | Refer to clause 6. | |
| Added or Reconfigured TrCH information list | | |
| - Added or Reconfigured DL TrCH information | | |
| - Downlink transport channel type | DCH | |
| - DL Transport channel identity | 6 | |
| - CHOICE DL parameters | Explicit | |
| - TFS | | |
| - CHOICE Transport channel type | Dedicated transport channels | |
| - Dynamic Transport format information | (This IE is repeated for TFI number) | |
| - RLC Size | Reference to clause 6.10 Parameter Set | |
| - Number of TBs and TTI List | (This IE is repeated for TFI number.) | |
| - Transmission Time Interval | Not Present | |
| - Number of Transport blocks | Reference to clause 6.10 Parameter Set | |
| - CHOICE Logical channel list | ALL | |
| - Semi-static Transport Format information | | |
| - Transmission time interval | Reference to clause 6.10 Parameter Set | |
| - Type of channel coding | Reference to clause 6.10 Parameter Set | |
| - Coding Rate | Reference to clause 6.10 Parameter Set | |
| - Rate matching attribute | Reference to clause 6.10 Parameter Set | |
| - CRC size | Reference to clause 6.10 Parameter Set | |
| - DCH quality target | | |
| - BLER Quality value | -6.3 | |
| Frequency info | | |
| -CHOICE mode | TDD | |
| - UARFCN (Nt) | Reference to clause 5.1 Test frequencies | |
| Maximum allowed UL TX power | 30 dBm | |
| CHOICE channel requirement | Uplink DPCH info | |
| - Uplink DPCH power control info | | |
| - CHOICE mode | TDD | |
| - UL Target SIR | Reference to clause 6.10 Parameter set. | |
| - CHOICE UL OL PC info | Individually signalled | |
| - CHOICE TDD option | 3.84 Mcps | |
| - Individual timeslot interference info | | |
| - Individual timeslot interference | | |
| - DPCH Constant Value | Values are used for open loop power control, clause 8 in 3GPP TS 25.331 [34] | |
| - CHOICE mode | TDD | |
| - Uplink Timing Advance Control | Not Present | |
| - UL CCTrCH List | | |
| - TFCS Id | 1 | |
| - Time info | | |
| - Activation time | (256+CFN-(CFN MOD 8 + 8))MOD 256 | |
| - Duration | Infinite | |
| - Common timeslot info | | |
| - 2nd interleaving mode | Reference to clause 6.10 Parameter Set | |
| - TFCI coding | Reference to clause 6.10 Parameter Set | |
| - Puncturing Limit | Reference to clause 6.10 Parameter Set | |
| - Repetition Period | Reference to clause 6.10 Parameter Set | |
| - Repetition Length | Reference to clause 6.10 Parameter Set | |
| - First individual timeslot info | | |
| - Timeslot number | The number of an uplink timeslot that has unassigned codes. | |
| - TFCI existence | TRUE | |
| - Midamble shift and burst type | | |
| - CHOICE TDD option | 3.84 Mcps | |

| Information Element | Value/remark | Version |
|---|---|---|
| -CHOICE Burst Type | | |
| -Type 1 | | |
| -Midamble Allocation Mode | Default | |
| - Midamble configuration burst type 1 and 3 | As defined in 3GPP TS 25.221 [28] | |
| - First timeslot channelisation codes | Repeated (1,2) for each channelisation code assigned in the slot to meet the needs of clause 6 Parameter Set. | |
| - Channelisation code | (i/SF) where i denotes an unassigned code matching the SF specified in clause 6 Parameter Set. | |
| - CHOICE more timeslots | The presence of this IE depends upon the number of resources specified in clause 6 and the number of slots in which they are being assigned. | |
| Downlink HS-PDSCH Information | Not Present | Rel-5 |
| Downlink information common for all radio links | | |
| - Downlink DPCH info common for all RL | | |
| - Timing indication | Maintain | |
| - CFN-targetSFN frame offset | Not Present | |
| - Downlink DPCH power control information | | |
| - DPC mode | 0 (single) | |
| - CHOICE mode | TDD | |
| - CHOICE TDD option | 3.84 Mcps (no data) | |
| - Default DPCH Offset Value | Not Present | |
| Downlink information for each radio link list | | |
| - Downlink information for each radio link | | |
| - Choice mode | TDD | |
| - Primary CCPCH info | | |
| - CHOICE *SyncCase* | Sync Case 1 | |
| - Timeslot | PCCPCH timeslot | |
| - Cell parameters ID | 0 | |
| - SCTD indicator | | |
| - Downlink DPCH info for each RL | | |
| - CHOICE mode | TDD | |
| - DL CCTrCH List | | |
| - TFCS ID | 1 | |
| - Time info | | |
| - Activation time | (256+CFN-(CFN mod 8 + 8))mod 256 | |
| - Duration | infinite | |
| - Common timeslot info | | |
| - 2nd interleaving mode | Reference to the present document | |
| - TFCI coding | TRUE | |
| - Puncturing limit | Reference to clause 6 Parameter set | |
| - Repetition period | 1 | |
| - Repetition length | Empty | |
| - Downlink DPCH timeslots and codes | | |
| - Individual timeslot info | | |
| - Timeslot number | The number of a downlink timeslot that has unassigned codes. | |
| - TFCI existence | TRUE | |
| - Midamble shift and burst type | | |
| - CHOICE TDD option | 3.84 Mcps | |
| -CHOICE Burst Type | | |
| -Type 1 | | |
| -Midamble Allocation Mode | Default | |
| - Midamble configuration burst type 1 and 3 | As defined in 3GPP TS 25.221 [28] | |
| - First timeslot channelisation codes | | |
| - First channelisation code | (i/SF) where i is the lowest numbered code that is being assigned and SF is specified in clause 6 Parameter Set.. | |
| - Last channelisation code | (j/SF) where j is the highest numbered code that is being assigned in the slot. | |
| - Bitmap | Bitmap of the codes that are being assigned in the slot. | |
| - CHOICE more timeslots | The presence of this IE depends upon whether the requirements of clause 6 Parameter Set could | |

| Information Element | Value/remark | Version |
|---|---|---|
| | be met by the codes that have been assigned in the first timeslot. | |
| - UL CCTrCH TPC List | Not Present | |
| -SCCPCH information for FACH | Not Present | |

Contents of RADIO BEARER SETUP message: AM or UM (1.28 Mcps TDD)

| Information Element | Condition | Value/remark | Version |
|---|---|---|---|
| Message Type | A1, A2, A3, A4, A5, A6, A7, A8 | | |
| RRC transaction identifier | | Arbitrarily selects an integer between 0 and 3 | |
| Integrity check info | | | |
| - message authentication code | | SS calculates the value of MAC-I for this message and writes to this IE. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I. | |
| - RRC message sequence number | | SS provides the value of this IE, from its internal counter. | |
| Integrity protection mode info | | Not Present | |
| Ciphering mode info | | Not Present | |
| Activation time | A1, A2, A3, A7, A8 | (256+CFN-(CFN MOD 8 + 8))MOD 256 | |
| Activation time | A4, A5, A6 | Now | |
| New U-RNTI | A1, A2, A3, A4, A5, A6, A7, A8 | Not Present | |
| New C-RNTI | A1, A2, A3, A4, A7, A8 | Not Present | |
| New C-RNTI | A5, A6 | '1010 1010 1010 1010' | |
| New DSCH-RNTI | A1, A2, A3, A4, A5, A6, A7, A8 | Not Present | |
| New H-RNTI | | Not Present | Rel-5 |
| RRC State indicator | A1, A2, A3, A4, A7, A8 | CELL_DCH | |
| RRC State indicator | A5, A6 | CELL_FACH | |
| UTRAN DRX cycle length coefficient | A1, A2, A3, A4, A5, A6, A7, A8 | Not Present | |
| CN information info | | Not Present | |
| URA identity | | Not Present | |
| - Signalling RB information to setup list | | Not Present | |
| - RAB information for setup list | A1, A7 | | |
| - RAB info | | | |
| - RAB identity | | | |
| - CHOICE RAB identity type | | RAB identity (GSM-MAP) | |
| - RAB identity | | 0000 0001B<br>The first/ leftmost bit of the bit string contains the most significant bit of the RAB identity. | |
| - CN domain identity | | CS domain | |
| - NAS Synchronization Indicator | | Not Present | |
| - Re-establishment timer | | useT314 | |
| - RB information to setup list | | | |
| - RB information to setup | | | |
| - RB identity | | 10 | |
| - PDCP info | | Not Present | |
| - CHOICE RLC info type | | RLC info | |
| - CHOICE Uplink RLC mode | | TM RLC | |
| - Transmission RLC discard | | Not Present | |
| - Segmentation indication | | FALSE | |
| - CHOICE Downlink RLC mode | | TM RLC | |
| - Segmentation indication | | FALSE | |

| Information Element | Condition | Value/remark | Version |
|---|---|---|---|
| - RB mapping info | | | |
| - Information for each multiplexing option | | | |
| - RLC logical channel mapping indicator | | Not Present | |
| - Number of uplink RLC logical channels | | 1 | |
| - Uplink transport channel type | | DCH | |
| - UL Transport channel identity | | 1 | |
| - Logical channel identity | | Not Present | |
| - CHOICE *RLC size list* | | Configured | |
| - MAC logical channel priority | | 8 | |
| - Downlink RLC logical channel info | | | |
| - Number of downlink RLC logical channels | | 1 | |
| - Downlink transport channel type | | DCH | |
| - DL DCH Transport channel identity | | 6 | |
| - DL DSCH Transport channel identity | | Not Present | |
| - Logical channel identity | | Not Present | |
| RAB information to setup list | A2, A8 | | |
| - RAB info | | | |
| - RAB identity | | | |
| - CHOICE RAB identity type | | RAB identity (GSM-MAP) | |
| - RAB identity | | 0000 0001B<br>The first/ leftmost bit of the bit string contains the most significant bit of the RAB identity. | |
| - CN domain identity | | CS domain | |
| - NAS Synchronization Indicator | | Not Present | |
| - Re-establishment timer | | useT314 | |
| - RB information to setup list | | | |
| - RB information to setup | | | |
| - RB identity | | 10 | |
| - PDCP info | | Not Present | |
| - CHOICE RLC info type | | RLC info | |
| - CHOICE Uplink RLC mode | | TM RLC | |
| - Transmission RLC discard | | Not Present | |
| - Segmentation indication | | FALSE | |
| - CHOICE Downlink RLC mode | | TM RLC | |
| - Segmentation indication | | FALSE | |
| - RB mapping info | | | |
| - Information for each multiplexing option | | | |
| - RLC logical channel mapping indicator | | Not Present | |
| - Number of uplink RLC logical channels | | 1 | |
| - Uplink transport channel type | | DCH | |
| - UL Transport channel identity | | 1 | |
| - Logical channel identity | | Not Present | |
| - CHOICE *RLC size list* | | Configured | |
| - MAC logical channel priority | | 6 | |
| - Downlink RLC logical channel info | | | |
| - Number of downlink RLC logical channels | | 1 | |
| - Downlink transport channel type | | DCH | |
| - DL DCH Transport channel identity | | 6 | |
| - DL DSCH Transport channel identity | | Not Present | |
| - Logical channel identity | | Not Present | |
| - RB identity | | 11 | |
| - PDCP info | | Not Present | |
| - CHOICE RLC info type | | RLC info | |
| - CHOICE Uplink RLC mode | | TM RLC | |
| - Transmission RLC discard | | Not Present | |
| - Segmentation indication | | FALSE | |
| - CHOICE Downlink RLC mode | | TM RLC | |
| - Segmentation indication | | FALSE | |
| - RB mapping info | | | |
| - Information for each multiplexing option | | | |
| - RLC logical channel mapping indicator | | Not Present | |
| - Number of uplink RLC logical channels | | 1 | |
| - Uplink transport channel type | | DCH | |
| - UL Transport channel identity | | 2 | |

| Information Element | Condition | Value/remark | Version |
|---|---|---|---|
| - Logical channel identity | | Not Present | |
| - CHOICE *RLC size list* | | Configured | |
| - MAC logical channel priority | | 6 | |
| - Downlink RLC logical channel info | | | |
| - Number of downlink RLC logical channels | | 1 | |
| - Downlink transport channel type | | DCH | |
| - DL DCH Transport channel identity | | 7 | |
| - DL DSCH Transport channel identity | | Not Present | |
| - Logical channel identity | | Not Present | |
| - RB identity | | 12 | |
| - PDCP info | | Not Present | |
| - CHOICE RLC info type | | RLC info | |
| - CHOICE Uplink RLC mode | | TM RLC | |
| - Transmission RLC discard | | Not Present | |
| - Segmentation indication | | FALSE | |
| - CHOICE Downlink RLC mode | | TM RLC | |
| - Segmentation indication | | FALSE | |
| - RB mapping info | | | |
| - Information for each multiplexing option | | | |
| - RLC logical channel mapping indicator | | Not Present | |
| - Number of uplink RLC logical channels | | 1 | |
| - Uplink transport channel type | | DCH | |
| - UL Transport channel identity | | 3 | |
| - Logical channel identity | | Not Present | |
| - CHOICE *RLC size list* | | Configured | |
| - MAC logical channel priority | | 7 | |
| - Downlink RLC logical channel info | | | |
| - Number of downlink RLC logical channels | | 1 | |
| - Downlink transport channel type | | DCH | |
| - DL DCH Transport channel identity | | 8 | |
| - DL DSCH Transport channel identity | | Not Present | |
| - Logical channel identity | | Not Present | |
| RAB information for setup list | A3, A4, A5, A6 | | |
| - RAB info | | | |
| - RAB identity | | | |
| - CHOICE RAB identity type | | RAB identity (GSM-MAP) | |
| - RAB identity | | 0000 0101B<br>The first/ leftmost bit of the bit string contains the most significant bit of the RAB identity. | |
| - CN domain identity | | PS domain | |
| - NAS Synchronization Indicator | | Not Present | |
| - Re-establishment timer | | useT315 | |
| - RB information to setup list | | | |
| - RB information to setup | | | |
| - RB identity | | 20 | |
| - PDCP info | | | |
| - Support for lossless SRNS relocation | | FALSE | |
| - Max PDCP SN window size | | Not present | |
| - PDCP PDU header | | Not present | |
| - Header compression information | | Not present | |
| - CHOICE RLC info type | | RLC info | |
| - CHOICE Uplink RLC mode | | AM RLC | |
| - Transmission RLC discard | | | |
| - CHOICE *SDU Discard Mode* | | Max DAT retransmissions | |
| - MAX_DAT | | 4 | |
| - Timer_MRW | | 100 | |
| - MaxMRW | | 4 | |
| - Transmission window size | | 128 | |
| - Timer_RST | | 500 | |
| - Max_RST | | 4 | |
| - Polling info | | | |
| - Timer_poll_prohibit | | 200 | |
| - Timer_poll | | 200 | |

| Information Element | Condition | Value/remark | Version |
|---|---|---|---|
| - Poll_PDU | | Not Present | |
| - Poll_SDU | | 1 | |
| - Last transmission PDU poll | | TRUE | |
| - Last retransmission PDU poll | | TRUE | |
| - Poll_Windows | | 99 | |
| - Timer_poll_periodic | | Not Present | |
| - CHOICE Downlink RLC mode | | AM RLC | |
| - In-sequence delivery | | TRUE | |
| - Receiving window size | | 128 | |
| - Downlink RLC status info | | | |
| - Timer_status_prohibit | | 200 | |
| - Timer_EPC | | 200 | |
| - Missing PDU indicator | | TRUE | |
| - Timer_STATUS_periodic | | Not Present | |
| - RB mapping info | | | |
| - Information for each multiplexing option | | 2 RBMuxOptions | |
| - RLC logical channel mapping indicator | | Not Present | |
| - Number of uplink RLC logical channels | | 1 | |
| - Uplink transport channel type | | DCH | |
| - UL Transport channel identity | | 1 | |
| - Logical channel identity | | Not Present | |
| - CHOICE RLC size list | | Configured | |
| - MAC logical channel priority | | 8 | |
| - Downlink RLC logical channel info | | | |
| - Number of downlink RLC logical channels | | 1 | |
| - Downlink transport channel type | | DCH | |
| - DL DCH Transport channel identity | | 6 | |
| - DL DSCH Transport channel identity | | Not Present | |
| - Logical channel identity | | Not Present | |
| - RLC logical channel mapping indicator | | Not Present | |
| - Number of uplink RLC logical channels | | 1 | |
| - Uplink transport channel type | | RACH | |
| - UL Transport channel identity | | Not Present | |
| - Logical channel identity | | 7 | |
| - CHOICE RLC size list | | Explicit list | |
| - RLC size index | | Reference to clause 6 Parameter Set | |
| - MAC logical channel priority | | 8 | |
| - Downlink RLC logical channel info | | | |
| - Number of downlink RLC logical channels | | 1 | |
| - Downlink transport channel type | | FACH | |
| - DL DCH Transport channel identity | | Not Present | |
| - DL DSCH Transport channel identity | | Not Present | |
| - Logical channel identity | | 8 | |
| RB information to be affected list | A1, A2, A3, A4, A5, A6, A7, A8 | Not Present | |
| Downlink counter synchronization info | A1, A2, A3, A4, A5, A6, A7, A8 | Not Present | |
| UL Transport channel information common for all transport channels | A1, A2, A3, A4, A5, A6, A7, A8 | | |
| - PRACH TFCS | | Not Present | |
| - CHOICE mode | | TDD | |
| - Individual UL CCTrCH information | | | |
| - UL TFCS Identity | | | |
| - TFCS ID | | 1 | |
| - Shared Channel Indicator | | FALSE | |
| - UL TFCS | | | |
| - CHOICE *TFCI signalling* | | Normal | |
| - TFCI Field 1 Information | | | |
| - CHOICE *TFCS representation* | | Complete reconfiguration | |
| - TFCS complete reconfiguration information | | | |
| - CHOICE *CTFC Size* | | Number of bits used must be enough | |

| Information Element | Condition | Value/remark | Version |
|---|---|---|---|
| | | to cover all combinations of CTFC from clause 6.11.5.4 Parameter Set. | |
| - CTFC information | | This IE is repeated for TFC numbers and reference to clause 6.11.5.4 Parameter Set | |
| - CTFC | | Reference to clause 6.11.5.4 Parameter Set | |
| - Power offset information | | | |
| - CHOICE Gain Factors | | Computed Gain Factors(The last TFC is set to Signalled Gain Factors) | |
| - Reference TFC ID | | 0 Integer(0.. 3) | |
| - CHOICE Gain Factors | | Signalled Gain Factors(Not Present if the CHOICE Gain Factors is set to ComputedGain Factors) | |
| - CHOICE mode | | TDD | |
| - Gain Factor $\beta_d$ | | 15 | |
| - Reference TFC ID | | 0 Integer(0.. 3) | |
| - CHOICE mode | | TDD | |
| - TFC subset | | | |
| - CHOICE Subset representation | | Full transport format combination set | |
| - TFC subset list | | Not Present | |
| Deleted TrCH information list | A1, A2, A3, A4, A5, A6, A7, A8 | Not Present | |
| Added or Reconfigured UL TrCH information | A1, A3 A4, A5, A6, A7 | 1 DCH added, 1 DCH reconfigured | |
| - Added or Reconfigured UL TrCH information | | | |
| - Uplink transport channel type | | DCH | |
| - UL Transport channel identity | | 5 | |
| - TFS | | | |
| - CHOICE Transport channel type | | Dedicated transport channels | |
| - Dynamic Transport format information | | | |
| - RLC Size | | Reference to clause 6.11 Parameter Set | |
| - Number of TBs and TTI List | 1 to maxTF | (This IE is repeated for TF number.) | |
| - Transmission Time Interval | | Not Present | |
| - Number of Transport blocks | | Reference to clause 6.11 Parameter Set | |
| - CHOICE Logical channel list | | All | |
| - Semi-static Transport Format information | | | |
| - Transmission time interval | | Reference to clause 6.11 Parameter Set | |
| - Type of channel coding | | Reference to clause 6.11 Parameter Set | |
| - Coding Rate | | Reference to clause 6.11 Parameter Set | |
| - Rate matching attribute | | Reference to clause 6.11 Parameter Set | |
| - CRC size | | Reference to clause 6.11 Parameter Set | |
| - Uplink transport channel type | | DCH | |
| - UL Transport channel identity | | 1 | |
| - TFS | | | |
| - CHOICE Transport channel type | | Dedicated transport channels | |
| - Dynamic Transport format information | | | |
| - RLC Size | | Reference to clause 6.11 Parameter Set | |
| - Number of TBs and TTI List | 1 to maxTF | (This IE is repeated for TF number.) | |
| - Transmission Time Interval | | Not Present | |
| - Number of Transport blocks | | Reference to clause 6.11 Parameter Set | |
| - CHOICE Logical channel list | | All | |
| - Semi-static Transport Format information | | | |
| - Transmission time interval | | Reference to clause 6.11 Parameter Set | |

| Information Element | Condition | Value/remark | Version |
|---|---|---|---|
| - Type of channel coding | | Reference to clause 6.11 Parameter Set | |
| - Coding Rate | | Reference to clause 6.11 Parameter Set | |
| - Rate matching attribute | | Reference to clause 6.11 Parameter Set | |
| - CRC size | | Reference to clause 6.11 Parameter Set | |
| Added or Reconfigured TrCH information list | A2, A8 | 4 TrCHs(DCH for DCCH and 3DCHs for DTCH) | |
| - Added or Reconfigured UL TrCH information | | | |
| - Uplink transport channel type | | DCH | |
| - UL Transport channel identity | | 5 | |
| - TFS | | | |
| - CHOICE Transport channel type | | Dedicated transport channels | |
| - Dynamic Transport format information | | | |
| - RLC Size | | Reference to clause 6.11 Parameter Set | |
| - Number of TBs and TTI List | 1 to maxTF | (This IE is repeated for TF number.) | |
| - Transmission Time Interval | | Not Present | |
| - Number of Transport blocks | | Reference to clause 6.11 Parameter Set | |
| - CHOICE Logical channel list | | All | |
| - Semi-static Transport Format information | | | |
| - Transmission time interval | | Reference to clause 6.11 Parameter Set | |
| - Type of channel coding | | Reference to clause 6.11 Parameter Set | |
| - Coding Rate | | Reference to clause 6.11 Parameter Set | |
| - Rate matching attribute | | Reference to clause 6.11 Parameter Set | |
| - CRC size | | Reference to clause 6.11 Parameter Set | |
| - Uplink transport channel type | | DCH | |
| - UL Transport channel identity | | 1 | |
| - TFS | | | |
| - CHOICE Transport channel type | | Dedicated transport channels | |
| - Dynamic Transport format information | | | |
| - RLC Size | | Reference to clause 6.11 Parameter Set | |
| - Number of TBs and TTI List | 1 to maxTF | (This IE is repeated for TF number.) | |
| - Transmission Time Interval | | Not Present | |
| - Number of Transport blocks | | Reference to clause 6.11 Parameter Set | |
| - CHOICE Logical channel list | | All | |
| - Semi-static Transport Format information | | | |
| - Transmission time interval | | Reference to clause 6.11 Parameter Set | |
| - Type of channel coding | | Reference to clause 6.11 Parameter Set | |
| - Coding Rate | | Reference to clause 6.11 Parameter Set | |
| - Rate matching attribute | | Reference to clause 6.11 Parameter Set | |
| - CRC size | | Reference to clause 6.11 Parameter Set | |
| - Uplink transport channel type | | DCH | |
| - UL Transport channel identity | | 2 | |
| - TFS | | | |
| - CHOICE Transport channel type | | Dedicated transport channels | |
| - Dynamic Transport format information | | | |
| - RLC Size | | Reference to clause 6.11 Parameter Set | |
| - Number of TBs and TTI List | 1 to maxTF | (This IE is repeated for TF number.) | |

| Information Element | Condition | Value/remark | Version |
|---|---|---|---|
| - Transmission Time Interval | | Not Present | |
| - Number of Transport blocks | | Reference to clause 6.11 Parameter Set | |
| - CHOICE Logical channel list | | All | |
| - Semi-static Transport Format information | | | |
| - Transmission time interval | | Reference to clause 6.11 Parameter Set | |
| - Type of channel coding | | Reference to clause 6.11 Parameter Set | |
| - Coding Rate | | Reference to clause 6.11 Parameter Set | |
| - Rate matching attribute | | Reference to clause 6.11 Parameter Set | |
| - CRC size | | Reference to clause 6.11 Parameter Set | |
| - Uplink transport channel type | | DCH | |
| - UL Transport channel identity | | 3 | |
| - TFS | | | |
| - CHOICE Transport channel type | | Dedicated transport channels | |
| - Dynamic Transport format information | | | |
| - RLC Size | | Reference to clause 6.11 Parameter Set | |
| - Number of TBs and TTI List | 1 to maxTF | (This IE is repeated for TF number.) | |
| - Transmission Time Interval | | Not Present | |
| - Number of Transport blocks | | Reference to clause 6.11 Parameter Set | |
| - CHOICE Logical channel list | | All | |
| - Semi-static Transport Format information | | | |
| - Transmission time interval | | Reference to clause 6.11 Parameter Set | |
| - Type of channel coding | | Reference to clause 6.11 Parameter Set | |
| - Coding Rate | | Reference to clause 6.11 Parameter Set | |
| - Rate matching attribute | | Reference to clause 6.11 Parameter Set | |
| - CRC size | | Reference to clause 6.11 Parameter Set | |
| CHOICE mode | | TDD (no data) | |
| DL Transport channel information common for all transport channel | A1, A2, A7, A8 | | |
| - SCCPCH TFCS | | Not Present | |
| - CHOICE mode | | TDD | |
| - Individual DL CCTrCH information | | | |
| - DL TFCS Identity | | | |
| - TFCS ID | | 2 | |
| - Shared Channel Indicator | | FALSE | |
| - CHOICE DL parameters | | SameAsUL | |
| - UL DCH TFCS Identity | | | |
| - TFCS ID | | 1 | |
| - Shared Channel Indicator | | FALSE | |
| DL Transport channel information common for all transport channel | A3, A4, A5, A6 | | |
| - SCCPCH TFCS | | Not Present | |
| - CHOICE mode | | TDD | |
| - Individual DL CCTrCH information | | | |
| - DL TFCS Identity | | | |
| - TFCS ID | | 2 | |
| - Shared Channel Indicator | | FALSE | |
| - CHOICE DL parameters | | Independent | |
| - DL TFCS | | | |
| - CHOICE TFCI Signalling | | Normal | |
| - TFCI Field 1 Information | | | |
| - CHOICE TFCS representation | | Complete reconfiguration | |
| - TFCS complete reconfiguration | | | |

| Information Element | Condition | Value/remark | Version |
|---|---|---|---|
| information | | | |
| - CHOICE CTFC Size | | Number of bits used must be enough to cover all combinations of CTFC from clause 6.11.5.4 Parameter Set. | |
| - CTFC information | | This IE is repeated for TFC numbers and reference to clause 6.11.5.4 | |
| - CTFC | | Reference to clause 6.11.5.4 Parameter Set | |
| - Power offset information | | Not Present | |
| Deleted TrCH information list | A1, A2, A3, A4, A5, A6, A7, A8 | Not Present | |
| Added or Reconfigured TrCH information list | A1 | 1 DCH added, 1 DCH reconfigured | |
| - Added or Reconfigured DL TrCH information | | | |
| - Downlink transport channel type | | DCH | |
| - DL Transport channel identity | | 10 | |
| - CHOICE DL parameters | | Same as UL | |
| - Uplink transport channel type | | DCH | |
| - UL TrCH identity | | 5 | |
| - DCH quality target | | | |
| - BLER Quality value | | -2.0 Real(-6.3..0 by step of 0.1) | |
| - Downlink transport channel type | | DCH | |
| - DL Transport channel identity | | 6 | |
| - CHOICE DL parameters | | Same as UL | |
| - Uplink transport channel type | | DCH | |
| - UL TrCH identity | | 1 | |
| - DCH quality target | | | |
| - BLER Quality value | | -2.0 Real(-6.3..0 by step of 0.1) | |
| Added or Reconfigured TrCH information list | A3, A4, A5, A6, A7 | 2 TrCHs(DCH for DCCH and DCH for DTCH) | |
| - Added or Reconfigured DL TrCH information | | | |
| - Downlink transport channel type | | DCH | |
| - DL Transport channel identity | | 10 | |
| - CHOICE DL parameters | | Same as UL | |
| - Uplink transport channel type | | DCH | |
| - UL TrCH identity | | 5 | |
| - DCH quality target | | | |
| - BLER Quality value | | -2.0 Real(-6.3..0 by step of 0.1) | |
| - Downlink transport channel type | | DCH | |
| - DL Transport channel identity | | 6 | |
| - CHOICE DL parameters | | Explicit | |
| - TFS | | | |
| - CHOICE Transport channel type | | Dedicated transport channels | |
| - Dynamic transport format information | | | |
| - RLC Size | | Reference to clause 6.11 Parameter Set | |
| - Number of TBs and TTI List | | (This IE is repeated for TF number.) | |
| - Transmission Time Interval | | Not Present | |
| - Number of Transport blocks | | Reference to clause 6.11 Parameter Set | |
| - Semi-static Transport Format information | | | |
| - Transmission time interval | | Reference to clause 6.11 Parameter Set | |
| - Type of channel coding | | Reference to clause 6.11 Parameter Set | |
| - Coding Rate | | Reference to clause 6.11 Parameter Set | |
| - Rate matching attribute | | Reference to clause 6.11 Parameter Set | |
| - CRC size | | Reference to clause 6.11 Parameter Set | |
| - DCH quality target | | | |
| - Transparent mode signalling info | | Not Present | |
| Added or Reconfigured TrCH information list | A2, A8 | 4 TrCHs(DCH for DCCH and 3DCHs for DTCH) | |
| - Added or Reconfigured DL TrCH information | | | |

| Information Element | Condition | Value/remark | Version |
|---|---|---|---|
| - Downlink transport channel type | | DCH | |
| - DL Transport channel identity | | 10 | |
| - CHOICE DL parameters | | Same as UL | |
| - Uplink transport channel type | | DCH | |
| - UL TrCH identity | | 5 | |
| - DCH quality target | | | |
| - Transparent mode signalling info | | Not Present | |
| - Downlink transport channel type | | DCH | |
| - DL Transport channel identity | | 6 | |
| - CHOICE DL parameters | | Explicit | |
| - TFS | | | |
| - CHOICE Transport channel type | | Dedicated transport channels | |
| - Dynamic transport format information | | | |
| - RLC Size | | Reference to clause 6.11 Parameter Set | |
| - Number of TBs and TTI List | | (This IE is repeated for TF number.) | |
| - Transmission Time Interval | | Not Present | |
| - Number of Transport blocks | | Reference to clause 6.11 Parameter Set | |
| - Semi-static Transport Format information | | | |
| - Transmission time interval | | Reference to clause 6.11 Parameter Set | |
| - Type of channel coding | | Reference to clause 6.11 Parameter Set | |
| - Coding Rate | | Reference to clause 6.11 Parameter Set | |
| - Rate matching attribute | | Reference to clause 6.11 Parameter Set | |
| - CRC size | | Reference to clause 6.11 Parameter Set | |
| - DCH quality target | | | |
| - BLER Quality value | | -2.0 | |
| - Downlink transport channel type | | DCH | |
| - DL Transport channel identity | | 7 | |
| - CHOICE DL parameters | | Explicit | |
| - TFS | | | |
| - CHOICE Transport channel type | | Dedicated transport channels | |
| - Dynamic transport format information | | | |
| - RLC Size | | Reference to clause 6.11 Parameter Set | |
| - Number of TBs and TTI List | | (This IE is repeated for TF number.) | |
| - Transmission Time Interval | | Not Present | |
| - Number of Transport blocks | | Reference to clause 6.11 Parameter Set | |
| - Semi-static Transport Format information | | | |
| - Transmission time interval | | Reference to clause 6.11 Parameter Set | |
| - Type of channel coding | | Reference to clause 6.11 Parameter Set | |
| - Coding Rate | | Reference to clause 6.11 Parameter Set | |
| - Rate matching attribute | | Reference to clause 6.11 Parameter Set | |
| - CRC size | | Reference to clause 6.11 Parameter Set | |
| - DCH quality target | | | |
| - BLER Quality value | | -2.0 | |
| - Downlink transport channel type | | DCH | |
| - DL Transport channel identity | | 8 | |
| - CHOICE DL parameters | | Explicit | |
| - TFS | | | |
| - CHOICE Transport channel type | | Dedicated transport channels | |
| - Dynamic transport format information | | | |
| - RLC Size | | Reference to clause 6.11 Parameter Set | |

| Information Element | Condition | Value/remark | Version |
|---|---|---|---|
| - Number of TBs and TTI List | | (This IE is repeated for TF number.) | |
| - Transmission Time Interval | | Not Present | |
| - Number of Transport blocks | | Reference to clause 6.11 Parameter Set | |
| - Semi-static Transport Format information | | | |
| - Transmission time interval | | Reference to clause 6.11 Parameter Set | |
| - Type of channel coding | | Reference to clause 6.11 Parameter Set | |
| - Coding Rate | | Reference to clause 6.11 Parameter Set | |
| - Rate matching attribute | | Reference to clause 6.11 Parameter Set | |
| - CRC size | | Reference to clause 6.11 Parameter Set | |
| - DCH quality target | | | |
| - BLER Quality value | | -2.0 | |
| Frequency info | A1, A2, A3, A4, A5, A7, A8 | | |
| - Choice mode | | TDD | |
| - UARFCN (Nt) | | Reference to clause 5.1 Test frequencies | |
| Frequency info | A6 | Not Present | |
| Maximum allowed UL TX power | A1, A2, A3, A4, A7, A8 | 33dBm | |
| Maximum allowed UL TX power | A5, A6 | Not Present | |
| CHOICE *channel requirement* | A5, A6 | Not Present | |
| CHOICE *channel requirement* | A1, A2, A3, A4, A7, A8 | Uplink DPCH info | |
| - Uplink DPCH power control info | | | |
| - CHOICE mode | | TDD | |
| - CHOICE TDD option | | 1.28 Mcps TDD | |
| - $PRX_{PDPCHdes}$ | | Integer (-120…-58 by step of 1) | |
| - CHOICE *UL OL PC info* | | | |
| - Broadcast UL OL PC info | | Null | |
| - Uplink Timing Advance Control | | Not Present | |
| - UL CCTrCH List | | | |
| - TFCS ID | | 1 | |
| - UL Target SIR | | Real (-11 .. 20 by step of 0.5 dB) Reference to clause 6 Parameter set. | |
| - Time info | | | |
| - Activation time | | (256+CFN-(CFN MOD 8 + 8))MOD 256 | |
| - Duration | | Infinite | |
| - Common timeslot info | | | |
| - 2nd interleaving mode | | Default value is "Frame" | |
| - TFCI coding | | Reference to clause 6 Parameter set | |
| - Puncturing limit | | Reference to clause 6 Parameter set | |
| - Repetition period | | 1 | |
| - Repetition length | | | |
| - Uplink DPCH timeslots and code | | | |
| - Dynamic SF usage | | FALSE | |
| - First individual timeslot info | | | |
| - Timeslot number | | | |
| - CHOICE TDD option | | 1.28 Mcps TDD | |
| - Timeslot number | | 1 OR 2 OR 3 | |
| - TFCI existence | | TRUE | |
| - Midamble shift and burst type | | | |
| - CHOICE TDD option | | 1.28 Mcps TDD | |
| - Midamble allocation mode | | Default midamble | |
| - Midamble configuration | | 16 | |
| - Midamble Shift | | Not Present | |
| - CHOICE TDD option | | 1.28 Mcps TDD | |
| - Modulation | | QPSK | |
| - SS-TPC Symbols | | 1 | |

| Information Element | Condition | Value/remark | Version |
|---|---|---|---|
| - Additional TPC-SS Symbols | | Not present | |
| - First timeslot Code List | | Repeated (1,2) for each channelisation code assigned in the slot to meet the needs of clause 6 Parameter Set. | |
| - channelisation codes | | (SF/ i) where i denotes an unassigned code matching the SF specified in clause 6 Parameter Set. | |
| - CHOICE more timeslots | | No more timeslots | |
| - UL CCTrCH List to Remove | | Not present | |
| CHOICE Mode | A1, A2, A3, A4, A5, A6, A7, A8 | TDD | |
| Downlink HS-PDSCH Information | | Not Present | Rel-5 |
| Downlink information common for all radio links | A5, A6 | Not Present | |
| Downlink information common for all radio links | A1, A2, A3 | | |
| - Downlink DPCH info common for all RL | | | |
| - Timing indication | | Maintain | |
| - CFN-targetSFN frame offset | | Not Present | |
| - Downlink DPCH power control information | | | |
| - CHOICE mode | | TDD | |
| - TPC Step Size | | 1 | |
| - MAC-d HFN initial value | | Not Present | |
| - CHOICE mode | | TDD | |
| - CHOICE mode | | TDD | |
| - CHOICE TDD option | | 1.28 Mcps TDD | |
| - TSTD indicator | | FALSE | |
| - Default DPCH Offset Value | | Not Present | |
| Downlink information common for all radio links | A4, A7, A8 | | |
| - Downlink DPCH info common for all RL | | | |
| - Timing indication | | Initialize | |
| - CFN-targetSFN frame offset | | Not Present | |
| - Downlink DPCH power control information | | | |
| - CHOICE mode | | TDD | |
| - TPC Step Size | | 1 | |
| - MAC-d HFN initial value | | Not Present | |
| - CHOICE mode | | TDD | |
| - CHOICE mode | | TDD | |
| - CHOICE TDD option | | 1.28 Mcps TDD | |
| - TSTD indicator | | FALSE | |
| - Default DPCH Offset Value | | | |
| - CHOICE mode | | TDD | |
| - Default DPCH Offset Value | | 0 Integer(0..7) | |
| Downlink information per radio link list | A1, A2, A3, A4, A7, A8 | | |
| - Downlink information for each radio link | | | |
| - Choice mode | | TDD | |
| - Primary CCPCH info | | | |
| - Choice mode | | TDD | |
| - Choice TDD Option | | 1.28 Mcps TDD | |
| - TSTD indicator | | FALSE | |
| - Cell parameters ID | | Ref. to the Default setting in clause 6.1 (TDD) Integer(0..127) | |
| - SCTD indicator | | FALSE | |
| - Downlink DPCH info for each RL | | | |
| - CHOICE mode | | TDD | |
| - DL CCTrCh List | | | |
| - TFCS ID | | 2 Integer(1.8) | |
| - Time info | | | |
| - Activation time | | Now | |
| - Duration | | Infinite | |
| - Common timeslot info | | | |
| - 2nd interleaving mode | | Default value is "Frame" | |
| - TFCI coding | | Reference to clause 6 Parameter set | |
| - Puncturing limit | | Reference to clause 6 Parameter set | |
| - Repetition period | | 1 | |
| - Repetition length | | NULL | |
| - Downlink DPCH timeslots and codes | | | |

| Information Element | Condition | Value/remark | Version |
|---|---|---|---|
| - First individual timeslot info | | | |
| - Timeslot number | | | |
| - CHOICE TDD option | | 1.28 Mcps TDD | |
| - Timeslot number | | 4 OR 5 OR 6 | |
| - TFCI existence | | TRUE | |
| - Midamble shift and burst type | | | |
| - CHOICE TDD option | | 1.28 Mcps TDD | |
| - Midamble allocation mode | | Default midamble | |
| - Midamble configuration | | 16 | |
| - Midamble Shift | | Not Present | |
| - CHOICE TDD option | | 1.28 Mcps TDD | |
| - Modulation | | QPSK | |
| - SS-TPC Symbols | | 1 | |
| - Additional TPC-SS Sysbols | | Not present | |
| - First timeslot channelisation codes | | Repeated (1,2) for each channelisation code assigned in the slot to meet the needs of clause 6 Parameter Set. | |
| - CHOICE codes representation | | | |
| - Channelisation codes bitmap | | Reference to clause 6.11 Parameter Set | |
| - CHOICE more timeslots | | No more timeslots | |
| - UL CCTrCH TPC List | | This list is not required for 1.28 Mcps TDD and is to be ignored by the UE. | |
| - UL TPC TFCS Identity | | | |
| - TFCS ID | | 1 | |
| - Shared Channel Indicator | | FALSE | |
| - DL CCTrCH List to Remove | | Not present | |
| - SCCPCH Information for FACH | | Not Present | R99 and Rel-4 only |
| Downlink information per radio link list | A5 | | |
| - Downlink information for each radio link | | | |
| - Choice mode | | TDD | |
| - Primary CCPCH info | | | |
| - Choice mode | | TDD | |
| - Choice TDD Option | | 1.28 Mcps TDD | |
| - TSTD indicator | | FALSE | |
| - Cell parameters ID | | Ref. to the Default setting in clause 6.1 (TDD) Integer(0..127) | |
| - SCTD indicator | | FALSE | |
| - Downlink DPCH info for each RL | | Not Present | |
| - SCCPCH Information for FACH | | Not Present | R99 and Rel-4 only |
| Downlink information per radio link list | A6 | Not Present | |

| Condition | Explanation |
|---|---|
| A1 | This IE need for "Non speech to CELL_DCH from CELL_DCH in CS" |
| A2 | This IE need for "Speech to CELL_DCH from CELL_DCH in CS" |
| A3 | This IE need for "Packet to CELL_DCH from CELL_DCH in PS" |
| A4 | This IE need for "Packet to CELL_DCH from CELL_FACH in PS" |
| A5 | This IE need for "Packet to CELL_FACH from CELL_DCH in PS" |
| A6 | This IE need for "Packet to CELL_FACH from CELL_FACH in PS" |
| A7 | This IE need for "Non speech to CELL_DCH from CELL_FACH in CS" |
| A8 | This IE need for "Speech to CELL_DCH from CELL_FACH in CS" |

Contents of RADIO BEARER SETUP COMPLETE message: AM

| Information Element | Value/remark | Version |
|---|---|---|
| Message Type | | |
| RRC transaction identifier | Checked to see if the value is identical to the same IE in the downlink RADIO BEARER SETUP message. | |
| Integrity check info | | |
| - Message authentication code | This IE is checked to see if it is present. The value is compared against the XMAC-I value computed by SS. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I. | |

| | | |
|---|---|---|
| - RRC Message sequence number | This IE is checked to see if it is present. The value is used by SS to compute the XMAC-I value. | |
| Uplink integrity protection activation info | Not checked. | |
| CHOICE mode | TDD | |
| - CHOICE *TDD option* | Check that this IE is present | Rel-4 |
| START | Not checked (if ciphering is OFF), check the presence if ciphering is ON. | |
| COUNT-C activation time | The presence of this IE depends on the following 2 factors: (a) There exists RB(s) mapped to RLC-TM and (b) UE is transiting to CELL_DCH state after the RB establishment procedure. Else, this IE is absent. | |
| Radio bearer uplink ciphering activation time info | If ciphering is not activated in RADIO BEARER SETUP message, this IE must be absent. Else, SS checks this IE for the presence of activation times of all ciphered uplink RLC-UM and RLC-AM RBs. | |
| Uplink counter synchronization info | Not present | |

Contents of RADIO BEARER SETUP FAILURE message: AM

| Information Element | Value/remark |
|---|---|
| Message Type | |
| RRC transaction identitifer | Checked to see if it is set to identical value of the same IE in the downlink PHYSICAL CHANNEL RECONFIGURATION message. |
| Integrity check info | |
| - Message authentication code | This IE is checked to see if it is present. The value is compared against the XMAC-I value computed by SS. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I. |
| - RRC Message sequence number | This IE is checked to see if it is present. The value is used by SS to compute the XMAC-I value. |
| Failure cause | Checked to see if it meets test requirement |
| Radio bearers for which reconfiguration would have succeeded | Not Check |

Contents of RADIO BEARER RELEASE COMPLETE message: AM (1.28 Mcps TDD)

| Information Element | Value/remark | Version |
|---|---|---|
| Message Type | | |
| RRC transaction identifier | Checked to see the value is identical to the same IE in the downlink RADIO BEARER RELEASE message. | |
| Integrity check info | | |
| - Message authentication code | This IE is checked to see if it is present. The value is compared against the XMAC-I value computed by SS. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I. | |
| - RRC Message sequence number | This IE is checked to see if it is present. The value is used by SS to compute the XMAC-I value. | |
| Uplink integrity protection activation info | Not checked. | |
| CHOICE mode | TDD | |
| - CHOICE TDD option | 1.28 Mcps TDD (no data) | |
| COUNT-C activation time | The presence of this IE depends on the following 2 factors: (a) There exists RB(s) mapped to RLC-TM and (b) UE is transiting to CELL_DCH state after the RB release procedure. Else, this IE is absent. | |
| Radio bearer uplink ciphering activation time info | If ciphering is not activated in RADIO BEARER RELEASE message, this IE must be absent. Else, SS checks this IE for the presence of activation times of all ciphered uplink RLC-UM and RLC-AM RBs. | |
| Uplink counter synchronization info | Not checked | |

Contents of RADIO BEARER RELEASE FAILURE message: AM

| Information Element | Value/remark |
|---|---|
| Message Type | |

| | |
|---|---|
| RRC transaction identitifer | Checked to see if it is set to identical value of the same IE in the downlink RADIO BEARER RELEASE message. |
| Integrity check info | |
| - Message authentication code | This IE is checked to see if it is present. The value is compared against the XMAC-I value computed by SS. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I. |
| - RRC Message sequence number | This IE is checked to see if it is present. The value is used by SS to compute the XMAC-I value. |
| Failure cause | Checked to see if it meets test requirement |
| Radio bearers for which reconfiguration would have succeeded | Not checked |

Contents of RRC CONNECTION REQUEST message: TM

| Information Element | Value/remark | Version |
|---|---|---|
| Message Type | | |
| Predefined configuration status information | To be checked against requirement if specified | Rel-5 |
| Initial UE identity | | |
| - CHOICE UE id type | | |
| - IMSI (GSM-MAP) | Set to the UE's IMSI (GSM-MAP) or TMSI. | |
| Establishment cause | To be checked against requirement if specified | |
| Protocol error indicator | FALSE | |
| UE Specific Behaviour Information 1 idle | This IE will not be checked by default behaviour, but in specific test case. | |
| Measured results on RACH | To be checked against requirement if specified | |
| Access stratum release indicator | To be checked against requirement if specified | Rel-4 |

Contents of RRC CONNECTION REJECT message: UM

| Information Element | Value/remark |
|---|---|
| Message Type | |
| RRC transaction identifier | Arbitrarily selects an integer between 0 and 3 |
| Initial UE identity | Select the same type as in the IE "Initial UE Identity" in RRC CONNECTION REQUEST" message. |
| Rejection cause | Unspecified |
| Wait Time | 0 |
| Redirection info | Not Present |

Contents of RRC CONNECTION RELEASE message: UM

| Information Element | Value/remark | Version |
|---|---|---|
| Message Type | | |
| U-RNTI | This IE is set to the following value when the message is transmitted on the CCCH. When transmitted on DCCH, this is absent. | R99, Rel-4 |
| - SRNC identity | 0000 0000 0001B | |
| - S-RNTI | 0000 0000 0000 0000 0001B | |
| CHOICE identity type | This IE is set to the following value when the message is transmitted on the CCCH. When transmitted on DCCH, this is absent. | Rel-5 |
| - U-RNTI | | |
| - SRNC identity | 0000 0000 0001B | |
| - S-RNTI | 0000 0000 0000 0000 0001B | |
| - Group identity | [FFS] | |
| - Group release information | [FFS] | |
| RRC transaction identifier | Arbitrarily selects an integer between 0 and 3 | |
| Integrity check info | This IE is present when this message is transmitted on downlink DCCH. Else, this IE and the sub-IEs are omitted. | |
| - Message authentication code | SS calculates the value of MAC-I for this message and writes to this IE. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I. | |
| - RRC Message sequence number | SS provides the value of this IE, from its internal counter. | |

| | | |
|---|---|---|
| N308 | 2 (for CELL_DCH state). Not Present (for UE in other connected mode states). | |
| Release cause | Normal event | |
| Rplmn information | Not Present | |

Contents of RRC CONNECTION RELEASE COMPLETE message: AM or UM

| Information Element | Semantics description |
|---|---|
| Message Type | |
| RRC transaction identifier | The value of this IE is checked to see that it matches the value of the same IE transmitted in the downlink RRC CONNECTION RELEASE message. |
| Integrity check info | |
| - Message authentication code | Checked to see if it's identical to the value of XMAC-I calculated by the SS. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I. |
| - RRC Message sequence number | Checked to see if it is present. This number is used by the SS to compute the XMAC-I |
| Error indication | Not checked |

Contents of RRC CONNECTION SETUP message: UM (Transition to CELL_DCH) (3.84 Mcps TDD option)

| Information Element | Value/remark | Version |
|---|---|---|
| Message Type | | |
| Initial UE identity | Select the same identity as in the IE "Initial UE Identity" in received RRC CONNECTION REQUEST" message | |
| RRC transaction identifier | 0 | |
| Activation time | Not Present(Now) | |
| New U-RNTI | | |
| - SRNC identity | 0000 0000 0001B | |
| - S-RNTI | 0000 0000 0000 0000 0001B | |
| New C-RNTI | Not Present | |
| RRC State Indicator | CELL_DCH | |
| UTRAN DRX cycle length coefficient | 9 | |
| Capability update requirement | | |
| - UE radio access FDD capability update requirement | FALSE | |
| - UE radio access TDD capability update requirement | TRUE | |
| - System specific capability update requirement list | GSM | |
| CHOICE *specification mode* | Complete specification | Rel-5 |
| - Complete specification | | Rel-5 |
| - Signalling RB information to setup | (UM DCCH for RRC) | |
| - RB identity | Not Present | |
| - CHOICE RLC info type | | |
| - RLC info | | |
| - CHOICE Uplink RLC mode | UM RLC | |
| - Transmission RLC discard | Not Present | |
| - CHOICE Downlink RLC mode | UM RLC | |
| - RB mapping info | | |
| - Information for each multiplexing option | 2 RBMuxOptions | |
| - RLC logical channel mapping indicator | Not Present | |
| - Number of RLC logical channels | 1 | |
| - Uplink transport channel type | DCH | |
| - UL Transport channel identity | 5 | |
| - Logical channel identity | 1 | |
| - CHOICE RLC size list | Configured | |
| - MAC logical channel priority | 1 | |
| - Downlink RLC logical channel info | | |
| - Number of RLC logical channels | 1 | |
| - Downlink transport channel type | DCH | |

| Information Element | Value/remark | Version |
|---|---|---|
| - DL DCH Transport channel identity | 10 | |
| - DL DSCH Transport channel identity | Not Present | |
| - Logical channel identity | 1 | |
| - RLC logical channel mapping indicator | Not Present | |
| - Number of RLC logical channels | 1 | |
| - Uplink transport channel type | RACH | |
| - UL Transport channel identity | Not Present | |
| - Logical channel identity | 1 | |
| - CHOICE RLC size list | Explicit List | |
| - RLC size index | According to clause 6 for standalone 13.6 kbps signalling radio bearer | |
| - MAC logical channel priority | 1 | |
| - Downlink RLC logical channel info | | |
| - Number of RLC logical channels | 1 | |
| - Downlink transport channel type | FACH | |
| - DL DCH Transport channel identity | Not Present | |
| - DL DSCH Transport channel identity | Not Present | |
| - Logical channel identity | 1 | |
| - Signalling RB information to setup | (AM DCCH for RRC) | |
| - RB identity | Not Present | |
| - CHOICE RLC info type | | |
| - RLC info | | |
| - CHOICE Uplink RLC mode | AM RLC | |
| - Transmission RLC discard | | |
| - SDU discard mode | No Discard | |
| - MAX_DAT | 15 | |
| - Transmission window size | 128 | |
| - Timer_RST | 500 | |
| - Max_RST | 1 | |
| - Polling info | | |
| - Timer_poll_prohibit | 200 | |
| - Timer_poll | 200 | |
| - Poll_PDU | Not present | |
| - Poll_SDU | 1 | |
| - Last transmission PDU poll | TRUE | |
| - Last retransmission PDU poll | TRUE | |
| - Poll_Window | 99 | |
| - Timer_poll_periodic | Not Present | |
| - CHOICE Downlink RLC mode | AM RLC | |
| - In-sequence delivery | TRUE | |
| - Receiving window size | 128 | |
| - Downlink RLC status info | | |
| - Timer_status_prohibit | 200 | |
| - Timer_EPC | Not Present | |
| - Missing PDU indicator | TRUE | |
| - Timer_STATUS_periodic | Not Present | |
| - RB mapping info | | |
| - Information for each multiplexing option | 2 RBMuxOptions | |
| - RLC logical channel mapping indicator | Not Present | |
| - Number of RLC logical channels | 1 | |
| - Uplink transport channel type | DCH | |
| - UL Transport channel identity | 5 | |
| - Logical channel identity | 2 | |
| - CHOICE RLC size list | Configure | |
| - MAC logical channel priority | 2 | |
| - Downlink RLC logical channel info | | |
| - Number of RLC logical channels | 1 | |
| - Downlink transport channel type | DCH | |

| Information Element | Value/remark | Version |
|---|---|---|
| - DL DCH Transport channel identity | 10 | |
| - DL DSCH Transport channel identity | Not Present | |
| - Logical channel identity | 2 | |
| - RLC logical channel mapping indicator | Not Present | |
| - Number of RLC logical channels | 1 | |
| - Uplink transport channel type | RACH | |
| - UL Transport channel identity | Not Present | |
| - Logical channel identity | 2 | |
| - CHOICE RLC size list | Explicit List | |
| - RLC size index | According to clause 6 for standalone 13.6 kbps signalling radio bearer | |
| - MAC logical channel priority | 2 | |
| - Downlink RLC logical channel info | | |
| - Number of RLC logical channels | 1 | |
| - Downlink transport channel type | FACH | |
| - DL DCH Transport channel identity | Not Present | |
| - DL DSCH Transport channel identity | Not Present | |
| - Logical channel identity | 2 | |
| Signalling RB information to setup | (AM DCCH for NAS_DT High priority) | |
| - RB identity | Not Present | |
| - CHOICE RLC info type | | |
| - RLC info | | |
| - CHOICE Uplink RLC mode | AM RLC | |
| - Transmission RLC discard | | |
| - SDU discard mode | No Discard | |
| - MAX_DAT | 15 | |
| - Transmission window size | 128 | |
| - Timer_RST | 500 | |
| - Max_RST | 1 | |
| - Polling info | | |
| - Timer_poll_prohibit | 200 | |
| - Timer_poll | 200 | |
| - Poll_PDU | Not present | |
| - Poll_SDU | 1 | |
| - Last transmission PDU poll | TRUE | |
| - Last retransmission PDU poll | TRUE | |
| - Poll_Windows | 99 | |
| - Timer_poll_periodic | Not Present | |
| - CHOICE Downlink RLC mode | AM RLC | |
| - In-sequence delivery | TRUE | |
| - Receiving window size | 128 | |
| - Downlink RLC status info | | |
| - Timer_status_prohibit | 200 | |
| - Timer_EPC | Not Present | |
| - Missing PDU indicator | TRUE | |
| - Timer_STATUS_periodic | Not Present | |
| - RB mapping info | | |
| - Information for each multiplexing option | 2 RBMuxOptions | |
| - RLC logical channel mapping indicator | Not Present | |
| - Number of RLC logical channels | 1 | |
| - Uplink transport channel type | DCH | |
| - UL Transport channel identity | 5 | |
| - Logical channel identity | 3 | |
| - CHOICE RLC size list | Configured | |
| - MAC logical channel priority | 3 | |
| - Downlink RLC logical channel info | | |
| - Number of RLC logical channels | 1 | |
| - Downlink transport channel type | DCH | |

| Information Element | Value/remark | Version |
|---|---|---|
| - DL DCH Transport channel identity | 10 | |
| - DL DSCH Transport channel identity | Not Present | |
| - Logical channel identity | 3 | |
| - RLC logical channel mapping indicator | Not Present | |
| - Number of RLC logical channels | 1 | |
| - Uplink transport channel type | RACH | |
| - UL Transport channel identity | Not Present | |
| - Logical channel identity | 3 | |
| - CHOICE RLC size list | Explicit List | |
| - RLC size index | According to clause 6 for standalone 13.6 kbps signalling radio bearer | |
| - MAC logical channel priority | 3 | |
| - Downlink RLC logical channel info | | |
| - Number of RLC logical channels | 1 | |
| - Downlink transport channel type | FACH | |
| - DL DCH Transport channel identity | Not Present | |
| - DL DSCH Transport channel identity | Not Present | |
| - Logical channel identity | 3 | |
| - Signalling RB information to setup | (AM DCCH for NAS_DT Low priority) | |
| - RB identity | Not Present | |
| - CHOICE RLC info type | | |
| - RLC info | | |
| - CHOICE Uplink RLC mode | AM RLC | |
| - Transmission RLC discard | | |
| - SDU discard mode | No discard | |
| - MAX_DAT | 15 | |
| - Transmission window size | 128 | |
| - Timer_RST | 500 | |
| - Max_RST | 1 | |
| - Polling info | | |
| - Timer_poll_prohibit | 200 | |
| - Timer_poll | 200 | |
| - Poll_PDU | Not present | |
| - Poll_SDU | 1 | |
| - Last transmission PDU poll | TRUE | |
| - Last retransmission PDU poll | TRUE | |
| - Poll_Windows | 99 | |
| - Timer_poll_periodic | Not Present | |
| - CHOICE Downlink RLC mode | AM RLC | |
| - In-sequence delivery | TRUE | |
| - Receiving window size | 128 | |
| - Downlink RLC status info | | |
| - Timer_status_prohibit | 200 | |
| - Timer_EPC | Not Present | |
| - Missing PDU indicator | TRUE | |
| - Timer_STATUS_periodic | Not Present | |
| - RB mapping info | | |
| - Information for each multiplexing option | 2 RBMuxOptions | |
| - RLC logical channel mapping indicator | Not Present | |
| - Number of RLC logical channels | 1 | |
| - Uplink transport channel type | DCH | |
| - UL Transport channel identity | 5 | |
| - Logical channel identity | 4 | |
| - CHOICE RLC size list | Configured | |
| - MAC logical channel priority | 4 | |
| - Downlink RLC logical channel info | | |
| - Number of RLC logical channels | 1 | |
| - Downlink transport channel type | DCH | |

| Information Element | Value/remark | Version |
|---|---|---|
| - DL DCH Transport channel identity | 10 | |
| - DL DSCH Transport channel identity | Not Present | |
| - Logical channel identity | 4 | |
| - RLC logical channel mapping indicator | Not Present | |
| - Number of RLC logical channels | 1 | |
| - Uplink transport channel type | RACH | |
| - UL Transport channel identity | Not Present | |
| - Logical channel identity | 4 | |
| - CHOICE RLC size list | Explicit List | |
| - RLC size index | According to clause 6 for standalone 13.6 kbps signalling radio bearer | |
| - MAC logical channel priority | 4 | |
| - Downlink RLC logical channel info | | |
| - Number of RLC logical channels | 1 | |
| - Downlink transport channel type | FACH | |
| - DL DCH Transport channel identity | Not Present | |
| - DL DSCH Transport channel identity | Not Present | |
| - Logical channel identity | 4 | |
| UL Transport channel information for all transport channels | | |
| - PRACH TFCS | Not Present | |
| - CHOICE mode | TDD | |
| -Individual UL CCTrCH information | | |
| - UL TFCS ID | (This IE is repeated for TFC number.) | |
| - UL TFCS | | |
| - TFC subset | Default value is the complete existing set of transport format combinations | |
| - Allowed Transport Format combination | 0 to MaxTFCvalue-1 (MaxTFCValue is refer to clause 6 Parameter Set.) | |
| - PRACH TFCS | (This IE is repeated for TFC number.) | |
| - CHOICE TFCI signalling | Normal | |
| - TFCI Field 1 information | | |
| - TFCS complete reconfigure information | | |
| - CHOICE TFCS Size | Number of used bits must be enough to cover all combinations of CTFC from clauses 6. Refer to clause 6 Parameter Set | |
| - CTFC information | Not Present | |
| - CHOICE mode | TDD | |
| - Individual UL CCTrCH information | Not Present | |
| Deleted TrCH information list | Not Present | |
| Added or Reconfigured UL TrCH information | | |
| - Uplink transport channel type | DCH | |
| - UL Transport channel identity | 5 | |
| - TFS | | |
| - CHOICE Transport channel type | Dedicated transport channels | |
| - Dynamic Transport format information | | |
| - RLC size | According to clause 6 for standalone 13.6 kbps signalling radio bearer | |
| - Number of TBs and TTI lists | (This IE is repeated for TFI number) | |
| - CHOICE mode | TDD | |
| - Transmission Time Interval | According to clause 6 for standalone 13.6 kbps signalling radio bearer | |
| - CHOICE Logical channel list | All | |
| - Semi-static Transport Format information | | |
| DL Transport channel information common for all transport channel | | |

| Information Element | Value/remark | Version |
|---|---|---|
| - SCCPCH TFCS | Not Present | |
| - CHOICE mode | TDD | |
| -Individual DL CCTrCH information | | |
| - DL TFCS Identity | | |
| - TFCS ID | 1 | |
| - Shared Channel Indicator | | |
| - CHOICE DL parameters | Same as UL | |
| Added or Reconfigured TrCH information list | | |
| - Added or Reconfigured DL TrCH information | | |
| - Downlink transport channel type | DCH | |
| - DL Transport channel identity | 10 | |
| - CHOICE DL parameters | Same as UL | |
| - Uplink transport channel type | DCH | |
| - UL Transport channel identity | 5 | |
| -DCH quality target | | |
| - BLER Quality target | -6.3 | |
| Frequency info | Not Present | |
| Maximum allowed UL TX power | Not Present | |
| CHOICE channel requirement | Uplink DPCH info | |
| - Uplink DPCH power control info | | |
| - CHOICE mode | TDD | |
| - CHOICE *TDD option* | 3.84 Mcps | |
| - UL target SIR | Reference to clause 6.10 Parameter set | |
| - CHOICE mode | TDD | |
| - CHOICE *UL OL PC info* | Individually signalled | |
| - CHOICE *TDD option* | 3.84 Mcps | |
| - Individual timeslot interference info | Not Present | |
| - Individual timeslot interference | | |
| - DPCH Constant Value | | |
| - Primary CCPCH Tx Power | Not Present | |
| - Time info | | |
| - Activation time | (256+CFN-(CFN MOD 8 + 8))MOD 256 | |
| - Duration | Infinite | |
| - Common timeslot info | | |
| - 2nd interleaving mode | Reference to clause 6.10 Parameter Set | |
| - TFCI coding | Reference to clause 6.10 Parameter Set | |
| - Puncturing Limit | Reference to clause 6.10 Parameter Set | |
| - Repetition Period | Reference to clause 6.10 Parameter Set | |
| - Repetition Length | Reference to clause 6.10 Parameter Set | |
| - Uplink DPCH timeslots and codes | Default is to use the old timeslots and codes | |
| - CPCH SET Info | (no data) | R99 and Rel-4 only |
| Downlink information common for all radio links | | |
| - Downlink DPCH info common for all RL | | |
| - Timing indicator | Maintain | |
| - CFN-targetSFN frame offset | Not Present | |
| - Downlink DPCH power control information | | |
| - DPC mode | 0 (single) | |
| - CHOICE mode | TDD | |
| - CHOICE TDD option | 3.84 Mcps (no data) | |
| - Default DPCH Offset Value | Not Present | |
| Downlink information for each radio link list | | |
| - Downlink information for each radio link | | |
| - Choice mode | TDD | |
| - Primary CCPCH info | | |
| - CHOICE *SyncCase* | Sync Case 1 | |
| - Timeslot | PCCPCH timeslot | |
| - Cell parameters ID | 0 | |
| - SCTD indicator | | |
| - Downlink DPCH info for each RL | | |

| Information Element | Value/remark | Version |
|---|---|---|
| - CHOICE mode | TDD | |
| - DL CCTrCH List | | |
| - TFCS ID | 1 | |
| - Time info | | |
| - Activation time | (256+CFN-(CFN mod 8 + 8))mod 256 | |
| - Duration | infinite | |
| - Common timeslot info | | |
| - 2nd interleaving mode | Reference to the present document | |
| - TFCI coding | TRUE | |
| - Puncturing limit | Reference to clause 6 Parameter set | |
| - Repetition period | 1 | |
| - Repetition length | Empty | |
| - Downlink DPCH timeslots and codes | | |
| - CHOICE *more timeslots* | | |
| - CHOICE TDD option | 3.84 Mcps | |
| - Timeslot number | The number of a downlink timeslot that has unassigned codes in a frame. | |
| - Individual timeslot info | | |
| - TFCI existence | TRUE | |
| - Midamble shift and burst type | | |
| - CHOICE TDD option | 3.84 Mcps | |
| -CHOICE Burst Type | | |
| -Type 1 | | |
| -Midamble Allocation Mode | Default | |
| - Midamble configuration burst type 1 and 3 | As defined in 3GPP TS 25.221 [28] | |
| - First timeslot channelisation codes | | |
| - First channelisation code | (i/SF) where i is the lowest numbered code that is being assigned and SF is specified in clause 6 Parameter Set. | |
| - Last channelisation code | (j/SF) where j is the highest numbered code that is being assigned in the slot. | |
| - CHOICE more timeslots | The presence of this IE depends upon whether the requirements of clause 6 Parameter Set could be met by the codes that have been assigned in the first timeslot. | |
| - UL CCTrCH TPC List | Not Present | |
| -SCCPCH information for FACH | Not Present | R99 and Rel-4 only |

Contents of RRC CONNECTION SETUP message: UM (Transition to CELL_DCH) (1.28 Mcps TDD option)

| Information Element | Value/remark | Version |
|---|---|---|
| Message Type | | |
| Initial UE identity | Select the same identity as in the IE "Initial UE Identity" in received RRC CONNECTION REQUEST" message | |
| RRC transaction identifier | 0 | |
| Activation time | Not Present(Now) | |
| New U-RNTI | | |
| - SRNC identity | 0000 0000 0001B | |
| - S-RNTI | 0000 0000 0000 0000 0001B | |
| New C-RNTI | Not Present | |
| RRC State Indicator | CELL_DCH | |
| UTRAN DRX cycle length coefficient | 9, Integer(3...9) | |
| Capability update requirement | | |
| - UE radio access FDD capability update requirement | FALSE | |
| - UE radio access 3.84 Mcps TDD capability update requirement | FALSE | |
| - UE radio access 1.28 Mcps TDD capability update requirement | TRUE | |
| - System specific capability update requirement list | Not Present | |

| Information Element | Value/remark | Version |
|---|---|---|
| CHOICE *specification mode* | Complete specification | Rel-5 |
| - Complete specification | | Rel-5 |
| - Signalling RB information to setup list | | |
| - Signalling RB information to setup | (UM DCCH for RRC) | |
| - RB identity | 1 | |
| - CHOICE RLC info type | RLC info | |
| - CHOICE Uplink RLC mode | UM RLC | |
| - Transmission RLC discard | Not Present | |
| - CHOICE Downlink RLC mode | UM RLC | |
| - RB mapping info | | |
| - Information for each multiplexing option | 2 RBMuxOptions | |
| - RLC logical channel mapping indicator | Not Present | |
| - Number of RLC logical channels | 1 | |
| - Uplink transport channel type | DCH | |
| - UL Transport channel identity | 5 | |
| - Logical channel identity | 1 | |
| - CHOICE RLC size list | Configured | |
| - MAC logical channel priority | 1 | |
| - Downlink RLC logical channel info | | |
| - Number of RLC logical channels | 1 | |
| - Downlink transport channel type | DCH | |
| - DL DCH Transport channel identity | 10 | |
| - DL DSCH Transport channel identity | Not Present | |
| - DL HS-DSCH MAC-d flow identity | Not Present | |
| - Logical channel identity | 1 | |
| - RLC logical channel mapping indicator | Not Present | |
| - Number of RLC logical channels | 1 | |
| - Uplink transport channel type | RACH | |
| - UL Transport channel identity | Not Present | |
| - Logical channel identity | 1 | |
| - CHOICE RLC size list | Explicit List | |
| - RLC size index | Reference to clause 6 Parameter Set | |
| - MAC logical channel priority | 1 | |
| - Downlink RLC logical channel info | | |
| - Number of RLC logical channels | 1 | |
| - Downlink transport channel type | FACH | |
| - DL DCH Transport channel identity | Not Present | |
| - DL DSCH Transport channel identity | Not Present | |
| - DL HS-DSCH MAC-d flow identity | Not Present | |
| - Logical channel identity | 1 | |
| - Signalling RB information to setup | (AM DCCH for RRC) | |
| - RB identity | 2 | |
| - CHOICE RLC info type | RLC info | |
| - CHOICE Uplink RLC mode | AM RLC | |
| - Transmission RLC discard | | |
| - CHOICE SDU discard mode | No Discard | |
| - MAX_DAT | 15 | |
| - Transmission window size | 128 | |
| - Timer_RST | 500 | |
| - Max_RST | 1 | |
| - Polling info | | |
| - Timer_poll_prohibit | 200 | |
| - Timer_poll | 200 | |
| - Poll_PDU | Not present | |
| - Poll_SDU | 1 | |
| - Last transmission PDU poll | TRUE | |
| - Last retransmission PDU poll | TRUE | |
| - Poll_Window | 99 | |
| - Timer_poll_periodic | Not Present | |
| - CHOICE Downlink RLC mode | AM RLC | |
| - In-sequence delivery | TRUE | |
| - Receiving window size | 128 | |
| - Downlink RLC status info | | |
| - Timer_status_prohibit | 200 | |
| - Timer_EPC | Not Present | |
| - Missing PDU indicator | TRUE | |
| - Timer_STATUS_periodic | Not Present | |

| Information Element | Value/remark | Version |
|---|---|---|
| - RB mapping info | | |
| - Information for each multiplexing option | 2 RBMuxOptions | |
| - RLC logical channel mapping indicator | Not Present | |
| - Number of RLC logical channels | 1 | |
| - Uplink transport channel type | DCH | |
| - UL Transport channel identity | 5 | |
| - Logical channel identity | 2 | |
| - CHOICE RLC size list | Configure | |
| - MAC logical channel priority | 2 | |
| - Downlink RLC logical channel info | | |
| - Number of RLC logical channels | 1 | |
| - Downlink transport channel type | DCH | |
| - DL DCH Transport channel identity | | |
| - Transport channel identity | 10 | |
| - DL DSCH Transport channel identity | Not Present | |
| - DL HS-DSCH MAC-d flow identity | Not Present | |
| - Logical channel identity | 2 | |
| - RLC logical channel mapping indicator | Not Present | |
| - Number of RLC logical channels | 1 | |
| - Uplink transport channel type | RACH | |
| - UL Transport channel identity | Not Present | |
| - Logical channel identity | 2 | |
| - CHOICE RLC size list | Explicit List | |
| - RLC size index | Reference to clause 6 Parameter Set | |
| - MAC logical channel priority | 2 | |
| - Downlink RLC logical channel info | | |
| - Number of RLC logical channels | 1 | |
| - Downlink transport channel type | FACH | |
| - DL DCH Transport channel identity | Not Present | |
| - DL DSCH Transport channel identity | Not Present | |
| - DL HS-DSCH MAC-d flow identity | Not Present | |
| - Logical channel identity | 2 | |
| Signalling RB information to setup | (AM DCCH for NAS_DT High priority) | |
| - RB identity | 3 | |
| - CHOICE RLC info type | RLC info | |
| - CHOICE Uplink RLC mode | AM RLC | |
| - Transmission RLC discard | | |
| - CHOICE SDU discard mode | No Discard | |
| - MAX_DAT | 15 | |
| - Transmission window size | 128 | |
| - Timer_RST | 500 | |
| - Max_RST | 1 | |
| - Polling info | | |
| - Timer_poll_prohibit | 200 | |
| - Timer_poll | 200 | |
| - Poll_PDU | Not present | |
| - Poll_SDU | 1 | |
| - Last transmission PDU poll | TRUE | |
| - Last retransmission PDU poll | TRUE | |
| - Poll_Windows | 99 | |
| - Timer_poll_periodic | Not Present | |
| - CHOICE Downlink RLC mode | AM RLC | |
| - In-sequence delivery | TRUE | |
| - Receiving window size | 128 | |
| - Downlink RLC status info | | |
| - Timer_status_prohibit | 200 | |
| - Timer_EPC | Not Present | |
| - Missing PDU indicator | TRUE | |
| - Timer_STATUS_periodic | Not Present | |
| - RB mapping info | | |
| - Information for each multiplexing option | 2 RBMuxOptions | |
| - RLC logical channel mapping indicator | Not Present | |
| - Number of RLC logical channels | 1 | |
| - Uplink transport channel type | DCH | |
| - UL Transport channel identity | 5 | |
| - Logical channel identity | 3 | |
| - CHOICE RLC size list | Configured | |

| Information Element | Value/remark | Version |
|---|---|---|
| - MAC logical channel priority | 3 | |
| - Downlink RLC logical channel info | | |
| - Number of RLC logical channels | 1 | |
| - Downlink transport channel type | DCH | |
| - DL DCH Transport channel identity | | |
| - Transport channel identity | 10 | |
| - DL DSCH Transport channel identity | Not Present | |
| - DL HS-DSCH MAC-d flow identity | Not Present | |
| - Logical channel identity | 3 | |
| - RLC logical channel mapping indicator | Not Present | |
| - Number of RLC logical channels | 1 | |
| - Uplink transport channel type | RACH | |
| - UL Transport channel identity | Not Present | |
| - Logical channel identity | 3 | |
| - CHOICE RLC size list | Explicit List | |
| - RLC size index | Reference to clause 6 Parameter Set | |
| - MAC logical channel priority | 3 | |
| - Downlink RLC logical channel info | | |
| - Number of RLC logical channels | 1 | |
| - Downlink transport channel type | FACH | |
| - DL DCH Transport channel identity | Not Present | |
| - DL DSCH Transport channel identity | Not Present | |
| - DL HS-DSCH MAC-d flow identity | Not Present | |
| - Logical channel identity | 3 | |
| - Signalling RB information to setup | (AM DCCH for NAS_DT Low priority) | |
| - RB identity | 4 | |
| - CHOICE RLC info type | RLC info | |
| - CHOICE Uplink RLC mode | AM RLC | |
| - Transmission RLC discard | | |
| - CHOICE SDU discard mode | No discard | |
| - MAX_DAT | 15 | |
| - Transmission window size | 128 | |
| - Timer_RST | 500 | |
| - Max_RST | 1 | |
| - Polling info | | |
| - Timer_poll_prohibit | 200 | |
| - Timer_poll | 200 | |
| - Poll_PDU | Not present | |
| - Poll_SDU | 1 | |
| - Last transmission PDU poll | TRUE | |
| - Last retransmission PDU poll | TRUE | |
| - Poll_Windows | 99 | |
| - Timer_poll_periodic | Not Present | |
| - CHOICE Downlink RLC mode | AM RLC | |
| - In-sequence delivery | TRUE | |
| - Receiving window size | 128 | |
| - Downlink RLC status info | | |
| - Timer_status_prohibit | 200 | |
| - Timer_EPC | Not Present | |
| - Missing PDU indicator | TRUE | |
| - Timer_STATUS_periodic | Not Present | |
| - RB mapping info | | |
| - Information for each multiplexing option | 2 RBMuxOptions | |
| - RLC logical channel mapping indicator | Not Present | |
| - Number of RLC logical channels | 1 | |
| - Uplink transport channel type | DCH | |
| - UL Transport channel identity | 5 | |
| - Logical channel identity | 4 | |
| - CHOICE RLC size list | Configured | |
| - MAC logical channel priority | 4 | |
| - Downlink RLC logical channel info | | |
| - Number of RLC logical channels | 1 | |
| - Downlink transport channel type | DCH | |
| - DL DCH Transport channel identity | | |
| - Transport channel identity | 10 | |
| - DL DSCH Transport channel identity | Not Present | |
| - DL HS-DSCH MAC-d flow identity | Not Present | |

| Information Element | Value/remark | Version |
|---|---|---|
| - Logical channel identity | 4 | |
| - RLC logical channel mapping indicator | Not Present | |
| - Number of RLC logical channels | 1 | |
| - Uplink transport channel type | RACH | |
| - UL Transport channel identity | Not Present | |
| - Logical channel identity | 4 | |
| - CHOICE RLC size list | Explicit List | |
| - RLC size index | Reference to clause 6 Parameter Set | |
| - MAC logical channel priority | 4 | |
| - Downlink RLC logical channel info | | |
| - Number of RLC logical channels | 1 | |
| - Downlink transport channel type | FACH | |
| - DL DCH Transport channel identity | Not Present | |
| - DL DSCH Transport channel identity | Not Present | |
| - DL HS-DSCH MAC-d flow identity | Not Present | |
| - Logical channel identity | 4 | |
| - UL Transport channel information for all transport channels | | |
| - PRACH TFCS | Not Present | |
| - CHOICE mode | TDD | |
| -Individual UL CCTrCH information | | |
| - UL TFCS ID | (This IE is repeated for TFC number.) | |
| - TFCS ID | 1 | |
| - Shared Channel Indicator | FALSE | |
| - UL TFCS | | |
| - CHOICE TFCI signalling | Normal | |
| - TFCI Field 1 Information | | |
| - CHOICE TFCS representation | Complete reconfiguration | |
| - TFCS complete reconfiguration information | | |
| - CHOICE CTFC Size | Configured, Number of bits used must be enough to cover all combinations of CTFC from clause 6.11.5.4 Parameter Set. | |
| - CTFC information | This IE is repeated for TFC numbers and reference to clause 6.11.5.4 Parameter Set | |
| - CTFC | Reference to clause 6.11.5.4 Parameter Set | |
| - Power offset Information | | |
| - CHOICE Gain Factors | Computed Gain Factors(The last TFC is set to Signalled Gain Factors) | |
| - Reference TFC ID | 0, Integer(0.. 3) | |
| - CHOICE Gain Factors | Signalled Gain Factors(Not Present if the CHOICE Gain Factors is set to ComputedGain Factors) | |
| - CHOICE mode | TDD | |
| - Gain Factor d | 15 | |
| - Reference TFC ID | 0, Integer (0..3) | |
| - CHOICE mode | TDD | |
| - TFC subset | Default value is the complete existing set of transport format combinations | |
| - CHOICE *Subset representation* | Allowed transport format combination list | |
| - Allowed Transport Format combination | 0 to MaxTFCvalue-1 (MaxTFCValue is refer to clause 6 Parameter Set.) | |
| - Transport format combination | Integer (0.. 1023) | |
| - TFC subset list | Not present | |
| - Added or Reconfigured UL TrCH information list | | |
| - Added or Reconfigured UL TrCH information | | |
| - Uplink transport channel type | DCH | |
| - UL Transport channel identity | 5 | |
| - TFS | | |
| - CHOICE Transport channel type | Dedicated transport channels | |
| - Dynamic Transport format information | | |
| - RLC size | According to clause 6 for standalone 13.6 kbps signalling radio bearer | |
| - Number of TBs and TTI lists | (This IE is repeated for TFI number) | |
| - Transmission Time Interval | Not Present | |
| - Number of Transport blocks | Reference to clause 6.11 Parameter Set | |
| - CHOICE Logical channel list | All | |
| - Semi-static Transport Format information | | |

| Information Element | Value/remark | Version |
|---|---|---|
| - Transmission time interval | Reference to clause 6.11 Parameter Set | |
| - Type of channel coding | Reference to clause 6.11 Parameter Set | |
| - Coding Rate | Reference to clause 6.11 Parameter Set | |
| - Rate matching attribute | Reference to clause 6.11 Parameter Set | |
| DL Transport channel information common for all transport channel | | |
| - SCCPCH TFCS | Not Present | |
| - CHOICE mode | TDD | |
| -Individual DL CCTrCH information | | |
| - DL TFCS Identity | | |
| - TFCS ID | 1 | |
| - Shared Channel Indicator | FALSE | |
| - CHOICE DL parameters | Same as UL | |
| - Added or Reconfigured TrCH information list | | |
| - Added or Reconfigured DL TrCH information | | |
| - Downlink transport channel type | DCH | |
| - DL Transport channel identity | 10 | |
| - CHOICE DL parameters | Same as UL | |
| - Uplink transport channel type | DCH | |
| - UL Transport channel identity | 5 | |
| -DCH quality target | | |
| - BLER Quality target | -6.3 | |
| Frequency info | Not Present | |
| Maximum allowed UL TX power | 33dBm | |
| CHOICE channel requirement | Uplink DPCH info | |
| - Uplink DPCH power control info | | |
| - CHOICE mode | TDD | |
| - CHOICE *TDD option* | 1.28 Mcps TDD | |
| - $PRX_{PDPCHdes}$ | Reference to clause 6.11 Parameter set | |
| - CHOICE *UL OL PC info* | Individually signalled | |
| - CHOICE *TDD option* | 1.28 Mcps TDD | |
| - TPC step size | 1 dB | |
| - Primary CCPCH Tx Power | Not Present | |
| - CHOICE mode | TDD | |
| - Uplink Timing Advance Control | | |
| - CHOICE Timing Advance | Enabled | |
| - CHOICE TDD option | 1.28 Mcps TDD | |
| - Uplink synchronization parameters | | |
| - Uplink synchronization step size | 1 | |
| - Uplink synchronization frequency | 1 | |
| - Synchronization parameters | Not present | |
| - UL CCTrCH List | | |
| - TFCS ID | 1 | |
| - UL Target SIR | Real (-11 .. 20 by step of 0.5 dB)<br>Reference to clause 6.11 Parameter set. | |
| - Time info | | |
| - Activation time | (256+CFN-(CFN MOD 8 + 8))MOD 256 | |
| - Duration | Infinite | |
| - Common timeslot info | | |
| - 2nd interleaving mode | Reference to clause 6 Parameter Set | |
| - TFCI coding | Reference to clause 6 Parameter Set | |
| - Puncturing Limit | Reference to clause 6 Parameter Set | |
| - Repetition Period | | |
| - Repetition Length | null | |
| - Uplink DPCH timeslots and codes | | |
| - Dynamic SF usage | FALSE | |
| - First individual timeslot info | | |
| - Timeslot number | | |
| - CHOICE TDD option | 1.28 Mcps TDD | |
| - Timeslot number | 1 OR 2 OR 3 | |
| - TFCI existence | TRUE | |
| - Midamble shift and burst type | | |
| - CHOICE TDD option | 1.28 Mcps TDD | |
| - Midamble allocation mode | Default midamble | |
| - Midamble configuration | 16 | |
| - Midamble Shift | Not Present | |
| - CHOICE TDD option | 1.28 Mcps TDD | |

| Information Element | Value/remark | Version |
|---|---|---|
| - Modulation | QPSK | |
| - SS-TPC Symbols | 1 | |
| - Additional TPC-SS Sysbols | Not present | |
| - First timeslot Code List | Repeated (1,2) for each channelisation code assigned in the slot to meet the needs of clause 6 Parameter Set. | |
| - channelisation codes | (SF/ i) where i denotes an unassigned code matching the SF specified in clause 6 Parameter Set. | |
| - CHOICE more timeslots | No more timeslots | |
| - UL CCTrCH List to Remove | Not present | |
| Downlink information common for all radio links | | |
| - Downlink DPCH info common for all RL | | |
| - Timing indication | Initialize | |
| - CFN-targetSFN frame offset | Not Present | |
| - Downlink DPCH power control information | | |
| - CHOICE *mode* | TDD | |
| - TPC Step Size | 1 dB | |
| - MAC-d HFN initial value | Not Present | |
| - CHOICE mode | TDD (no data) | |
| - CHOICE mode | TDD | |
| - CHOICE TDD option | 1.28 Mcps TDD | |
| - TSTD indicator | FALSE | |
| - Default DPCH Offset Value | Not Present | |
| Downlink information for each radio link list | | |
| - Downlink information for each radio link | | |
| - Choice mode | TDD | |
| - Primary CCPCH info | | |
| - CHOICE *mode* | TDD | |
| - CHOICE TDD option | 1.28 Mcps TDD | |
| - TSTD indicator | FALSE | |
| - Cell parameters ID | Not present | |
| - SCTD indicator | FALSE | |
| - Downlink DPCH info for each RL | | |
| - CHOICE mode | TDD | |
| - DL CCTrCH List | | |
| - TFCS ID | 1 | |
| - Time info | | |
| - Activation time | (256+CFN-(CFN mod 8 + 8))mod 256 | |
| - Duration | infinite | |
| - Common timeslot info | | |
| - 2nd interleaving mode | Reference to clause 6.11 Parameter set | |
| - TFCI coding | Reference to clause 6.11 Parameter set | |
| - Puncturing limit | Reference to clause 6.11 Parameter set | |
| - Repetition period | 1 | |
| - Repetition length | NULL | |
| - Downlink DPCH timeslots and codes | | |
| - First Individual timeslot info | | |
| - Timeslot number | | |
| - CHOICE *more timeslots* | | |
| - CHOICE TDD option | 1.28 McpsTDD | |
| - Timeslot number | 4 OR 5 OR 6 | |
| - Individual timeslot info | | |
| - TFCI existence | TRUE | |
| - Midamble shift and burst type | | |
| - CHOICE TDD option | 1.28 Mcps TDD | |
| -Midamble Allocation Mode | Default | |
| - Midamble configuration | 16 Integer(2, 4, 6, 8, 10, 12, 14, 16) | |
| - Midamble Shift | Not present | |
| - CHOICE TDD option | 1.28 Mcps TDD | |
| - Modulation | QPSK | |
| - SS-TPC Symbols | 1 | |
| - Additional TPC-SS Symbols | Not present | |
| - First timeslot channelisation | | |

| Information Element | Value/remark | Version |
|---|---|---|
| codes | | |
| - CHOICE codes representation | Consecutive codes | |
| - First channelisation code | (i/SF) where i is the lowest numbered code that is being assigned and SF is specified in clause 6 Parameter Set.. | |
| - Last channelisation code | (j/SF) where j is the highest numbered code that is being assigned in the slot. | |
| - CHOICE more timeslots | The presence of this IE depends upon whether the requirements of clause 6 Parameter Set could be met by the codes that have been assigned in the first timeslot. | |
| - UL CCTrCH TPC List | | |
| - UL TPC TFCS Identity | 1 | |
| - DL CCTrCH List to Remove | Not present | |
| -SCCPCH information for FACH | Not Present | R99 and Rel-4 only |

Contents of RRC CONNECTION SETUP message: UM (Transition to CELL_FACH) (3.84 Mcps TDD)

| Information Element | Value/remark | Version |
|---|---|---|
| Message Type | | |
| Initial UE identity | Select the same identity as in the IE "Initial UE Identity" in received RRC CONNECTION REQUEST" message | |
| RRC transaction identifier | Arbitrarily selects an integer between 0 and 3 | |
| Activation time | Not Present(Now) | |
| New U-RNTI | | |
| - SRNC identity | 0000 0000 0001B | |
| - S-RNTI | 0000 0000 0000 0000 0001B | |
| New C-RNTI | Not Present | |
| RRC State Indicator | CELL_FACH | |
| UTRAN DRX cycle length coefficient | 9 , Integer(3...9) | |
| Capability update requirement | | |
| - UE radio access FDD capability update requirement | FALSE | |
| - UE radio access 3.84 Mcps TDD capability update requirement | FALSE | |
| - UE radio access 1.28 Mcps TDD capability update requirement | TRUE | |
| - System specific capability update requirement list | GSM | |
| CHOICE *specification mode* | Complete specification | Rel-5 |
| - Complete specification | | Rel-5 |
| - Signalling RB information to setup list | | |
| - Signalling RB information to setup | (UM DCCH for RRC) | |
| - RB identity | 1 | |
| - CHOICE RLC info type | RLC info | |
| - CHOICE Uplink RLC mode | UM RLC | |
| - Transmission RLC discard | Not Present | |
| - CHOICE Downlink RLC mode | UM RLC | |
| - RB mapping info | | |
| - Information for each multiplexing option | 2 RBMuxOptions | |
| - RLC logical channel mapping indicator | Not Present | |
| - Number of RLC logical channels | 1 | |
| - Uplink transport channel type | DCH | |
| - UL Transport channel identity | 5 | |
| - Logical channel identity | 1 | |
| - CHOICE RLC size list | Configure | |
| - MAC logical channel priority | 1 | |
| - Downlink RLC logical channel info | | |
| - Number of RLC logical channels | 1 | |
| - Downlink transport channel type | DCH | |
| - DL DCH Transport channel identity | | |
| - Transport channel identity | 10 | |
| - DL DSCH Transport channel identity | Not Present | |
| - DL HS-DSCH MAC-d flow identity | Not Present | |
| - Logical channel identity | 1 | |

| Information Element | Value/remark | Version |
|---|---|---|
| - RLC logical channel mapping indicator | Not Present | |
| - Number of RLC logical channels | 1 | |
| - Uplink transport channel type | RACH | |
| - UL Transport channel identity | | |
| - Logical channel identity | 1 | |
| - CHOICE RLC size list | Explicit List | |
| - RLC size index | Reference to clause 6 Parameter Set | |
| - MAC logical channel priority | 1 | |
| - Downlink RLC logical channel info | | |
| - Number of RLC logical channels | 1 | |
| - Downlink transport channel type | FACH | |
| - DL DCH Transport channel identity | Not Present | |
| - DL DSCH Transport channel identity | Not Present | |
| - DL HS-DSCH MAC-d flow identity | Not Present | |
| - Logical channel identity | 1 | |
| - Signalling RB information to setup | (AM DCCH for RRC) | |
| - RB identity | 2 | |
| - CHOICE RLC info type | RLC info | |
| - CHOICE Uplink RLC mode | AM RLC | |
| - Transmission RLC discard | | |
| - CHOICE SDU discard mode | No Discard | |
| - MAX_DAT | 15 | |
| - Transmission window size | 32 | |
| - Timer_RST | 500 | |
| - Max_RST | 1 | |
| - Polling info | | |
| - Timer_poll_prohibit | 200 | |
| - Timer_poll | 200 | |
| - Poll_PDU | Not present | |
| - Poll_SDU | 1 | |
| - Last transmission PDU poll | TRUE | |
| - Last retransmission PDU poll | TRUE | |
| - Poll_Window | 99 | |
| - Timer_poll_periodic | Not Present | |
| - CHOICE Downlink RLC mode | AM RLC | |
| - In-sequence delivery | TRUE | |
| - Receiving window size | 32 | |
| - Downlink RLC status info | | |
| - Timer_status_prohibit | 200 | |
| - Timer_EPC | Not Present | |
| - Missing PDU indicator | TRUE | |
| - Timer_STATUS_periodic | Not Present | |
| - RB mapping info | | |
| - Information for each multiplexing option | 2 RBMuxOptions | |
| - RLC logical channel mapping indicator | Not Present | |
| - Number of RLC logical channels | 1 | |
| - Uplink transport channel type | DCH | |
| - UL Transport channel identity | 5 | |
| - Logical channel identity | 2 | |
| - CHOICE RLC size list | Configure | |
| - MAC logical channel priority | 2 | |
| - Downlink RLC logical channel info | | |
| - Number of RLC logical channels | 1 | |
| - Downlink transport channel type | DCH | |
| - DL DCH Transport channel identity | | |
| - Transport channel identity | 10 | |
| - DL DSCH Transport channel identity | Not Present | |
| - DL HS-DSCH MAC-d flow identity | Not Present | |
| - Logical channel identity | 2 | |
| - RLC logical channel mapping indicator | Not Present | |
| - Number of RLC logical channels | 1 | |
| - Uplink transport channel type | RACH | |
| - UL Transport channel identity | Not Present | |
| - Logical channel identity | 2 | |
| - CHOICE RLC size list | Explicit List | |
| - RLC size index | Reference to clause 6 Parameter Set | |
| - MAC logical channel priority | 2 | |

| Information Element | Value/remark | Version |
|---|---|---|
| - Downlink RLC logical channel info | | |
| - Number of RLC logical channels | 1 | |
| - Downlink transport channel type | FACH | |
| - DL DCH Transport channel identity | Not Present | |
| - DL DSCH Transport channel identity | Not Present | |
| - DL HS-DSCH MAC-d flow identity | Not Present | |
| - Logical channel identity | 2 | |
| - Signalling RB information to setup | (AM DCCH for NAS_DT High priority) | |
| - RB identity | 3 | |
| - CHOICE RLC info type | RLC info | |
| - CHOICE Uplink RLC mode | AM RLC | |
| - Transmission RLC discard | | |
| - CHOICE SDU discard mode | No Discard | |
| - MAX_DAT | 15 | |
| - Transmission window size | 32 | |
| - Timer_RST | 500 | |
| - Max_RST | 1 | |
| - Polling info | | |
| - Timer_poll_prohibit | 200 | |
| - Timer_poll | 200 | |
| - Poll_PDU | Not present | |
| - Poll_SDU | 1 | |
| - Last transmission PDU poll | TRUE | |
| - Last retransmission PDU poll | TRUE | |
| - Poll_Window | 99 | |
| - Timer_poll_periodic | Not Present | |
| - CHOICE Downlink RLC mode | AM RLC | |
| - In-sequence delivery | TRUE | |
| - Receiving window size | 32 | |
| - Downlink RLC status info | | |
| - Timer_status_prohibit | 200 | |
| - Timer_EPC | Not Present | |
| - Missing PDU indicator | TRUE | |
| - Timer_STATUS_periodic | Not Present | |
| - RB mapping info | | |
| - Information for each multiplexing option | 2 RBMuxOptions | |
| - RLC logical channel mapping indicator | Not Present | |
| - Number of RLC logical channels | 1 | |
| - Uplink transport channel type | DCH | |
| - UL Transport channel identity | 5 | |
| - Logical channel identity | 3 | |
| - CHOICE RLC size list | Configure | |
| - MAC logical channel priority | 3 | |
| - Downlink RLC logical channel info | | |
| - Number of RLC logical channels | 1 | |
| - Downlink transport channel type | DCH | |
| - DL DCH Transport channel identity | | |
| - Transport channel identity | 10 | |
| - DL DSCH Transport channel identity | Not Present | |
| - DL HS-DSCH MAC-d flow identity | Not Present | |
| - Logical channel identity | 3 | |
| - RLC logical channel mapping indicator | Not Present | |
| - Number of RLC logical channels | 1 | |
| - Uplink transport channel type | RACH | |
| - UL Transport channel identity | Not Present | |
| - Logical channel identity | 3 | |
| - CHOICE RLC size list | Explicit List | |
| - RLC size index | Reference to clause 6 Parameter Set | |
| - MAC logical channel priority | 3 | |
| - Downlink RLC logical channel info | | |
| - Number of RLC logical channels | 1 | |
| - Downlink transport channel type | FACH | |
| - DL DCH Transport channel identity | Not Present | |
| - DL DSCH Transport channel identity | Not Present | |
| - DL HS-DSCH MAC-d flow identity | Not Present | |
| - Logical channel identity | 3 | |
| - Signalling RB information to setup | (AM DCCH for NAS_DT Low priority) | |

| Information Element | Value/remark | Version |
|---|---|---|
| - RB identity | 4 | |
| - CHOICE RLC info type | RLC info | |
| - CHOICE Uplink RLC mode | AM RLC | |
| - Transmission RLC discard | | |
| - CHOICE SDU discard mode | No discard | |
| - MAX_DAT | 15 | |
| - Transmission window size | 32 | |
| - Timer_RST | 500 | |
| - Max_RST | 1 | |
| - Polling info | | |
| - Timer_poll_prohibit | 200 | |
| - Timer_poll | 200 | |
| - Poll_PDU | Not present | |
| - Poll_SDU | 1 | |
| - Last transmission PDU poll | TRUE | |
| - Last retransmission PDU poll | TRUE | |
| - Poll_Window | 99 | |
| - Timer_poll_periodic | Not Present | |
| - CHOICE Downlink RLC mode | AM RLC | |
| - In-sequence delivery | TRUE | |
| - Receiving window size | 32 | |
| - Downlink RLC status info | | |
| - Timer_status_prohibit | 200 | |
| - Timer_EPC | Not Present | |
| - Missing PDU indicator | TRUE | |
| - Timer_STATUS_periodic | Not Present | |
| - RB mapping info | | |
| - Information for each multiplexing option | 2 RBMuxOptions | |
| - RLC logical channel mapping indicator | Not Present | |
| - Number of RLC logical channels | 1 | |
| - Uplink transport channel type | DCH | |
| - UL Transport channel identity | 5 | |
| - Logical channel identity | 4 | |
| - CHOICE RLC size list | Configure | |
| - MAC logical channel priority | 4 | |
| - Downlink RLC logical channel info | | |
| - Number of RLC logical channels | 1 | |
| - Downlink transport channel type | DCH | |
| - DL DCH Transport channel identity | | |
| - Transport channel identity | 10 | |
| - DL DSCH Transport channel identity | Not Present | |
| - DL HS-DSCH MAC-d flow identity | Not Present | |
| - Logical channel identity | 4 | |
| - RLC logical channel mapping indicator | Not Present | |
| - Number of RLC logical channels | 1 | |
| - Uplink transport channel type | RACH | |
| - UL Transport channel identity | Not Present | |
| - Logical channel identity | 4 | |
| - CHOICE RLC size list | Explicit List | |
| - RLC size index | Reference to clause 6 Parameter Set | |
| - MAC logical channel priority | 4 | |
| - Downlink RLC logical channel info | | |
| - Number of RLC logical channels | 1 | |
| - Downlink transport channel type | FACH | |
| - DL DCH Transport channel identity | Not Present | |
| - DL DSCH Transport channel identity | Not Present | |
| - DL HS-DSCH MAC-d flow identity | Not Present | |
| - Logical channel identity | 4 | |
| - UL Transport channel information for all transport channels | | |
| - PRACH TFCS | Not Present | |
| - CHOICE mode | TDD | |
| -Individual UL CCTrCH information | | |
| - UL TFCS Identity | | |
| - TFCS ID | 1 | |
| - Shared Channel Indicator | FALSE | |
| - UL TFCS | | |

| Information Element | Value/remark | Version |
|---|---|---|
| - CHOICE TFCI signalling | Normal | |
| - TFCI Field 1 Information | | |
| - CHOICE TFCS representation | Complete reconfiguration | |
| - TFCS complete reconfiguration information | | |
| - CHOICE CTFC Size | Configured, Number of bits used must be enough to cover all combinations of CTFC from clause 6.10.3.4 Parameter Set. | |
| - CTFC information | This IE is repeated for TFC numbers and reference to clause 6.10.3.4 Parameter Set | |
| - CTFC | Reference to clause 6.10.3.4 Parameter Set | |
| - Power offset Information | | |
| - CHOICE Gain Factors | Computed Gain Factors(The last TFC is set to Signalled Gain Factors) | |
| - Reference TFC ID | 0, Integer(0.. 3) | |
| - CHOICE mode | TDD | |
| - TFC subset | Not present Default value is the complete existing set of transport format combinations | |
| - TFC subset list | Not present | |
| - Added or Reconfigured UL TrCH information list | | |
| - Added or Reconfigured UL TrCH information | | |
| - Uplink transport channel type | DCH | |
| - UL Transport channel identity | 5 | |
| - TFS | | |
| - CHOICE Transport channel type | Dedicated transport channels | |
| - Dynamic Transport format information | | |
| - RLC size | According to clause 6 for standalone 13.6 kbps signalling radio bearer | |
| - Number of TBs and TTI lists | (This IE is repeated for TFI number) | |
| - Transmission Time Interval | According to clause 6 for standalone 13.6 kbps signalling radio bearer | |
| - Number of Transport blocks | Reference to clause 6.10 Parameter Set | |
| - CHOICE Logical channel list | All | |
| - Semi-static Transport Format information | | |
| - Transmission time interval | Reference to clause 6.10 Parameter Set | |
| - Type of channel coding | Reference to clause 6.10 Parameter Set | |
| - Coding Rate | Reference to clause 6.10 Parameter Set | |
| - Rate matching attribute | Reference to clause 6.10 Parameter Set | |
| - CRC size | Reference to clause 6.10 Parameter Set | |
| - DL Transport channel information common for all transport channel | | |
| - SCCPCH TFCS | Not Present | |
| - CHOICE mode | TDD | |
| -Individual DL CCTrCH information | | |
| - DL TFCS Identity | | |
| - TFCS ID | 1 | |
| - Shared Channel Indicator | FALSE | |
| - CHOICE DL parameters | Same as UL | |
| - UL DCH TFCS Identity | 1 | |
| - Shared Channel Indicator | FALSE | |
| - Added or Reconfigured TrCH information list | | |
| - Added or Reconfigured DL TrCH information | | |
| - Downlink transport channel type | DCH | |
| - DL Transport channel identity | 10 | |
| - CHOICE DL parameters | Same as UL | |
| - Uplink transport channel type | DCH | |
| - UL Transport channel identity | 5 | |
| -DCH quality target | | |
| - BLER Quality target | -6.3 | |
| Frequency info | Not Present | |

| Information Element | Value/remark | Version |
|---|---|---|
| Maximum allowed UL TX power | Not Present Default value is the existing maximum UL TX power | |
| CHOICE channel requirement | Not present | |
| Downlink information common for all radio links | Not present | |
| Downlink information for each radio link list | Not present | |

Contents of RRC CONNECTION SETUP message: UM (Transition to CELL_FACH) (1.28 Mcps TDD)

| Information Element | Value/remark | Version |
|---|---|---|
| Message Type | | |
| Initial UE identity | Select the same identity as in the IE "Initial UE Identity" in received RRC CONNECTION REQUEST" message | |
| RRC transaction identifier | Arbitrarily selects an integer between 0 and 3 | |
| Activation time | Not Present(Now) | |
| New U-RNTI | | |
| - SRNC identity | 0000 0000 0001B | |
| - S-RNTI | 0000 0000 0000 0000 0001B | |
| New C-RNTI | 0000 0000 0000 0001B | |
| RRC State Indicator | CELL_FACH | |
| UTRAN DRX cycle length coefficient | 9 , Integer(3...9) | |
| Capability update requirement | | |
| - UE radio access FDD capability update requirement | | |
| - UE radio access 3.84 Mcps TDD capability update requirement | | |
| - UE radio access 1.28 Mcps TDD capability update requirement | | |
| - System specific capability update requirement list | | |
| CHOICE specification mode | | |
| - Complete specification | | |
| - Signalling RB information to setup list | | |
| - Signalling RB information to setup | (UM DCCH for RRC) | |
| - RB identity | 1 | |
| - CHOICE RLC info type | RLC info | |
| - CHOICE Uplink RLC mode | UM RLC | |
| - Transmission RLC discard | Not Present | |
| - CHOICE Downlink RLC mode | UM RLC | |
| - RB mapping info | | |
| - Information for each multiplexing option | 2 RBMuxOptions | |
| - RLC logical channel mapping indicator | Not Present | |
| - Number of RLC logical channels | 1 | |
| - Uplink transport channel type | DCH | |
| - UL Transport channel identity | 5 | |
| - Logical channel identity | 1 | |
| - CHOICE RLC size list | Configure | |
| - MAC logical channel priority | 1 | |
| - Downlink RLC logical channel info | | |
| - Number of RLC logical channels | 1 | |
| - Downlink transport channel type | DCH | |
| - DL DCH Transport channel identity | | |
| - Transport channel identity | 10 | |
| - DL DSCH Transport channel identity | Not Present | |
| - DL HS-DSCH MAC-d flow identity | Not Present | |
| - Logical channel identity | 1 | |
| - RLC logical channel mapping indicator | Not Present | |
| - Number of RLC logical channels | 1 | |
| - Uplink transport channel type | RACH | |
| - UL Transport channel identity | | |
| - Logical channel identity | 1 | |
| - CHOICE RLC size list | Explicit List | |
| - RLC size index | Reference to clause 6 Parameter Set | |
| - MAC logical channel priority | 1 | |
| - Downlink RLC logical channel info | | |
| - Number of RLC logical channels | 1 | |
| - Downlink transport channel type | FACH | |

| Information Element | Value/remark | Version |
|---|---|---|
| - DL DCH Transport channel identity | Not Present | |
| - DL DSCH Transport channel identity | Not Present | |
| - DL HS-DSCH MAC-d flow identity | Not Present | |
| - Logical channel identity | 1 | |
| - Signalling RB information to setup | (AM DCCH for RRC) | |
| - RB identity | 2 | |
| - CHOICE RLC info type | RLC info | |
| - CHOICE Uplink RLC mode | AM RLC | |
| - Transmission RLC discard | | |
| - CHOICE SDU discard mode | No Discard | |
| - MAX_DAT | 15 | |
| - Transmission window size | 32 | |
| - Timer_RST | 500 | |
| - Max_RST | 1 | |
| - Polling info | | |
| - Timer_poll_prohibit | 200 | |
| - Timer_poll | 200 | |
| | Not present | |
| - Poll_SDU | 1 | |
| - Last transmission PDU poll | TRUE | |
| - Last retransmission PDU poll | TRUE | |
| - Poll_Window | 99 | |
| - Timer_poll_periodic | Not Present | |
| - CHOICE Downlink RLC mode | AM RLC | |
| - In-sequence delivery | TRUE | |
| - Receiving window size | 32 | |
| - Downlink RLC status info | | |
| - Timer_status_prohibit | 200 | |
| - Timer_EPC | Not Present | |
| - Missing PDU indicator | TRUE | |
| - Timer_STATUS_periodic | Not Present | |
| - RB mapping info | | |
| - Information for each multiplexing option | 2 RBMuxOptions | |
| - RLC logical channel mapping indicator | Not Present | |
| - Number of RLC logical channels | 1 | |
| - Uplink transport channel type | DCH | |
| - UL Transport channel identity | 5 | |
| - Logical channel identity | 2 | |
| - CHOICE RLC size list | Configure | |
| - MAC logical channel priority | 2 | |
| - Downlink RLC logical channel info | | |
| - Number of RLC logical channels | 1 | |
| - Downlink transport channel type | DCH | |
| - DL DCH Transport channel identity | | |
| - Transport channel identity | 10 | |
| - DL DSCH Transport channel identity | Not Present | |
| - DL HS-DSCH MAC-d flow identity | Not Present | |
| - Logical channel identity | 2 | |
| - RLC logical channel mapping indicator | Not Present | |
| - Number of RLC logical channels | 1 | |
| - Uplink transport channel type | RACH | |
| - UL Transport channel identity | Not Present | |
| - Logical channel identity | 2 | |
| - CHOICE RLC size list | Explicit List | |
| - RLC size index | Reference to clause 6 Parameter Set | |
| - MAC logical channel priority | 2 | |
| - Downlink RLC logical channel info | | |
| - Number of RLC logical channels | 1 | |
| - Downlink transport channel type | FACH | |
| - DL DCH Transport channel identity | Not Present | |
| - DL DSCH Transport channel identity | Not Present | |
| - DL HS-DSCH MAC-d flow identity | Not Present | |
| - Logical channel identity | 2 | |
| - Signalling RB information to setup | (AM DCCH for NAS_DT High priority) | |
| - RB identity | 3 | |
| - CHOICE RLC info type | RLC info | |
| - CHOICE Uplink RLC mode | AM RLC | |

| Information Element | Value/remark | Version |
|---|---|---|
| - Transmission RLC discard | | |
| - CHOICE SDU discard mode | No Discard | |
| - MAX_DAT | 15 | |
| - Transmission window size | 32 | |
| - Timer_RST | 500 | |
| - Max_RST | 1 | |
| - Polling info | | |
| - Timer_poll_prohibit | 200 | |
| - Timer_poll | 200 | |
| - Poll_SDU | 1 | |
| - Last transmission PDU poll | TRUE | |
| - Last retransmission PDU poll | TRUE | |
| - Poll_Window | 99 | |
| - Timer_poll_periodic | Not Present | |
| - CHOICE Downlink RLC mode | AM RLC | |
| - In-sequence delivery | TRUE | |
| - Receiving window size | 32 | |
| - Downlink RLC status info | | |
| - Timer_status_prohibit | 200 | |
| - Timer_EPC | Not Present | |
| - Missing PDU indicator | TRUE | |
| - Timer_STATUS_periodic | Not Present | |
| - RB mapping info | | |
| - Information for each multiplexing option | 2 RBMuxOptions | |
| - RLC logical channel mapping indicator | Not Present | |
| - Number of RLC logical channels | 1 | |
| - Uplink transport channel type | DCH | |
| - UL Transport channel identity | 5 | |
| - Logical channel identity | 3 | |
| - CHOICE RLC size list | Configure | |
| - MAC logical channel priority | 3 | |
| - Downlink RLC logical channel info | | |
| - Number of RLC logical channels | 1 | |
| - Downlink transport channel type | DCH | |
| - DL DCH Transport channel identity | | |
| - Transport channel identity | 10 | |
| - DL DSCH Transport channel identity | Not Present | |
| - DL HS-DSCH MAC-d flow identity | Not Present | |
| - Logical channel identity | 3 | |
| - RLC logical channel mapping indicator | Not Present | |
| - Number of RLC logical channels | 1 | |
| - Uplink transport channel type | RACH | |
| - UL Transport channel identity | Not Present | |
| - Logical channel identity | 3 | |
| - CHOICE RLC size list | Explicit List | |
| - RLC size index | Reference to clause 6 Parameter Set | |
| - MAC logical channel priority | 3 | |
| - Downlink RLC logical channel info | | |
| - Number of RLC logical channels | 1 | |
| - Downlink transport channel type | FACH | |
| - DL DCH Transport channel identity | Not Present | |
| - DL DSCH Transport channel identity | Not Present | |
| - DL HS-DSCH MAC-d flow identity | Not Present | |
| - Logical channel identity | 3 | |
| - Signalling RB information to setup | (AM DCCH for NAS_DT Low priority) | |
| - RB identity | 4 | |
| - CHOICE RLC info type | RLC info | |
| - CHOICE Uplink RLC mode | AM RLC | |
| - Transmission RLC discard | | |
| - CHOICE SDU discard mode | No discard | |
| - MAX_DAT | 15 | |
| - Transmission window size | 32 | |
| - Timer_RST | 500 | |
| - Max_RST | 1 | |
| - Polling info | | |
| - Timer_poll_prohibit | 200 | |

| Information Element | Value/remark | Version |
|---|---|---|
| - Timer_poll | 200 | |
| - Poll_SDU | 1 | |
| - Last transmission PDU poll | TRUE | |
| - Last retransmission PDU poll | TRUE | |
| - Poll_Window | 99 | |
| - Timer_poll_periodic | Not Present | |
| - CHOICE Downlink RLC mode | AM RLC | |
| - In-sequence delivery | TRUE | |
| - Receiving window size | 32 | |
| - Downlink RLC status info | | |
| - Timer_status_prohibit | 200 | |
| - Timer_EPC | Not Present | |
| - Missing PDU indicator | TRUE | |
| - Timer_STATUS_periodic | Not Present | |
| - RB mapping info | | |
| - Information for each multiplexing option | 2 RBMuxOptions | |
| - RLC logical channel mapping indicator | Not Present | |
| - Number of RLC logical channels | 1 | |
| - Uplink transport channel type | DCH | |
| - UL Transport channel identity | 5 | |
| - Logical channel identity | 4 | |
| - CHOICE RLC size list | Configure | |
| - MAC logical channel priority | 4 | |
| - Downlink RLC logical channel info | | |
| - Number of RLC logical channels | 1 | |
| - Downlink transport channel type | DCH | |
| - DL DCH Transport channel identity | | |
| - Transport channel identity | 10 | |
| - DL DSCH Transport channel identity | Not Present | |
| - DL HS-DSCH MAC-d flow identity | Not Present | |
| - Logical channel identity | 4 | |
| - RLC logical channel mapping indicator | Not Present | |
| - Number of RLC logical channels | 1 | |
| - Uplink transport channel type | RACH | |
| - UL Transport channel identity | Not Present | |
| - Logical channel identity | 4 | |
| - CHOICE RLC size list | Explicit List | |
| - RLC size index | Reference to clause 6 Parameter Set | |
| - MAC logical channel priority | 4 | |
| - Downlink RLC logical channel info | | |
| - Number of RLC logical channels | 1 | |
| - Downlink transport channel type | FACH | |
| - DL DCH Transport channel identity | Not Present | |
| - DL DSCH Transport channel identity | Not Present | |
| - DL HS-DSCH MAC-d flow identity | Not Present | |
| - Logical channel identity | 4 | |
| - UL Transport channel information for all transport channels | | |
| - PRACH TFCS | Not Present | |
| - CHOICE mode | TDD | |
| -Individual UL CCTrCH information | | |
| - UL TFCS Identity | | |
| - TFCS ID | 1 | |
| - Shared Channel Indicator | FALSE | |
| - UL TFCS | | |
| - CHOICE TFCI signalling | Normal | |
| - TFCI Field 1 Information | | |
| - CHOICE TFCS representation | Complete reconfiguration | |
| - TFCS complete reconfiguration information | | |
| - CHOICE CTFC Size | Configured, Number of bits used must be enough to cover all combinations of CTFC from clause 6.11.5.4 Parameter Set. | |
| - CTFC information | This IE is repeated for TFC numbers and reference to clause 6.11.5.4 Parameter Set | |
| - CTFC | Reference to clause 6.11.5.4 Parameter Set | |

| Information Element | Value/remark | Version |
|---|---|---|
| - Power offset Information | | |
| - CHOICE Gain Factors | Computed Gain Factors(The last TFC is set to Signalled Gain Factors) | |
| - Reference TFC ID | 0, Integer(0.. 3) | |
| - CHOICE mode | TDD | |
| - TFC subset | Not present. Default value is the complete existing set of transport format combinations | |
| - TFC subset list | Not present | |
| - Added or Reconfigured UL TrCH information list | | |
| - Added or Reconfigured UL TrCH information | | |
| - Uplink transport channel type | DCH | |
| - UL Transport channel identity | 5 | |
| - TFS | | |
| - CHOICE Transport channel type | Dedicated transport channels | |
| - Dynamic Transport format information | | |
| - RLC size | According to clause 6 for standalone 13.6 kbps signalling radio bearer | |
| - Number of TBs and TTI lists | (This IE is repeated for TFI number) | |
| - Transmission Time Interval | According to clause 6 for standalone 13.6 kbps signalling radio bearer | |
| - Number of Transport blocks | Reference to clause 6.11 Parameter Set | |
| - CHOICE Logical channel list | All | |
| - Semi-static Transport Format information | | |
| - Transmission time interval | Reference to clause 6.11 Parameter Set | |
| - Type of channel coding | Reference to clause 6.11 Parameter Set | |
| - Coding Rate | Reference to clause 6.11 Parameter Set | |
| - Rate matching attribute | Reference to clause 6.11 Parameter Set | |
| - CRC size | Reference to clause 6.11 Parameter Set | |
| - DL Transport channel information common for all transport channel | | |
| - SCCPCH TFCS | Not Present | |
| - CHOICE mode | TDD | |
| -Individual DL CCTrCH information | | |
| - DL TFCS Identity | | |
| - TFCS ID | 1 | |
| - Shared Channel Indicator | FALSE | |
| - CHOICE DL parameters | Same as UL | |
| - UL DCH TFCS Identity | 1 | |
| - Shared Channel Indicator | FALSE | |
| - Added or Reconfigured TrCH information list | | |
| - Added or Reconfigured DL TrCH information | | |
| - Downlink transport channel type | DCH | |
| - DL Transport channel identity | 10 | |
| - CHOICE DL parameters | Same as UL | |
| - Uplink transport channel type | DCH | |
| - UL Transport channel identity | 5 | |
| -DCH quality target | | |
| - BLER Quality target | -6.3 | |
| Frequency info | Not Present | |
| Maximum allowed UL TX power | Not Present. Default value is the existing maximum UL TX power | |
| CHOICE channel requirement | Not present | |
| Downlink information common for all radio links | Not present | |
| Downlink information for each radio link list | Not Present | |

Contents of RRC CONNECTION SETUP COMPLETE message: AM

| Information Element | Value/remark |
|---|---|
| Message Type | |
| RRC transaction identifier | The value of this IE is checked to see that it matches the value of the same IE transmitted in the downlink RRC CONNECTION SETUP message. |

| | |
|---|---|
| START list | This IE is checked to see if it is present. |
| UE radio access capability | Not checked |
| UE radio access capability extension | Not checked |
| UE system specific capability | Not checked |

Contents of SECURITY MODE COMMAND message: AM

| Information Element | Condition | Value/remark |
|---|---|---|
| Message Type | A1, A2 | |
| RRC transaction identifier | | Arbitrarily selects an integer between 0 and 3 |
| Integrity check info | | |
| - Message authentication code | | Set to an arbitrarily selected 32-bits integer. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I. |
| - RRC Message Sequence Number | | Set to an arbitrarily selected integer between 0 and 15 |
| Security capability | | |
| - Ciphering algorithm capability | | |
| - UEA0 | | If ciphering is not indicated to be active on IXIT statements in 3GPP TS 34.123-2 [3], set this IE to TRUE. |
| - UEA1 | | If ciphering is indicated to be active on IXIT statements in 3GPP TS 34.123-2 [3], set this IE to TRUE. |
| - Spare | | FALSE |
| - Integrity protection algorithm capability | | 0000000000000010B (UIA1) |
| - UIA1 | | TRUE |
| - Spare | | FALSE |
| Ciphering mode info | | This presence of this IE is dependent on IXIT statements in 3GPP TS 34.123-2 [3]. If ciphering is indicated to be active, this IE present with the values of the sub IEs as stated below. Else, this IE is omitted. |
| - Ciphering mode command | | Start/restart |
| - Ciphering algorithm | | Use the same ciphering algorithm specified in "ciphering algorithm capability" IE in this message. |
| - Ciphering activation time for DPCH | | Not Present |
| - Radio bearer downlink ciphering activation time info | | |
| - Radio bearer activation time | | |
| - RB identity | | 1 |
| - RLC sequence number | | Current RLC SN+2 |
| - RB identity | | 2 |
| - RLC sequence number | | Current RLC SN+2 |
| - RB identity | | 3 |
| - RLC sequence number | | Current RLC SN + 2 |
| - RB identity | | 4 |
| - RLC sequence number | | Current RLC SN + 2 |
| Integrity protection mode info | | |
| - Integrity protection mode command | | Start |
| - Downlink integrity protection activation info | | Not Present |
| - Integrity protection algorithm | | UIA1 |
| - Integrity protection initialisation number | | SS selects an arbitrary 32 bits number for FRESH |
| CN domain identity | | Supported domain |
| UE system specific security capability | A1 | Not Checked |
| UE system specific security capability | A2 | |
| - Inter-RAT UE security capability | | |
| - CHOICE *system* | | GSM |
| - GSM security capability | | The indicated algorithms must be the same as the algorithms supported by the UE as indicated in the IE " UE system specific capability " in the RRC CONNECTION SETUP COMPLETE message. |

| Condition | Explanation |
|---|---|

| | |
|---|---|
| A1 | UE not supporting GSM |
| A2 | UE supporting GSM |

Contents of SECURITY MODE COMPLETE message: AM

| Information Element | Value/remark |
|---|---|
| Message Type | |
| RRC transaction identifier | The value of this IE is checked to see that it matches the value of the same IE transmitted in the downlink SECURITY MODE COMMAND message. |
| Integrity check info | |
| - Message authentication code | This IE is checked to see if it is present. The value is compared against the XMAC-I value computed by SS. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I. |
| - RRC Message sequence number | This IE is checked to see if it is present. The value is used by SS to compute the XMAC-I value. |
| Uplink integrity protection activation info | Not checked. |
| Radio bearer uplink ciphering activation time info | If ciphering is not activated in SECURITY MODE COMMAND message, this IE must be absent. Else, SS checks this IE for the presence of activation times for all ciphered uplink RLC-UM and RLC-AM RBs. |

Contents of UPLINK DIRECT TRANSFER message: AM

| Information Element | Value/remark |
|---|---|
| Message Type | |
| Integrity check info | |
| - Message authentication code | This IE is checked to see if it is present. The value is compared against the XMAC-I value computed by SS. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I. |
| - RRC Message sequence number | This IE is checked to see if it is present. The value is used by SS to compute the XMAC-I value. |
| CN domain identity | Checked to see if set to supported CN domain as specified in the IXIT statements |
| NAS message | Set according to that indicated in specific message content clause |
| Measured results on RACH | Not checked |

# 9.2 Default Message Contents for RF

This clause contains the default values of common messages for RF test. The parameters of the UL/DL reference measurement channel 12.2 kbps, the DL reference measurement channel for BTFD, UE test loop mode 1 without Dummy DCCH transmission and UE test loop mode 2 with Dummy DCCH transmission are set to default message contents.

## 9.2.1 Default Message Contents for RF (FDD)

Contents of Activate RB Test Mode message

| Information Element | Value/remark |
|---|---|
| Protocol discriminator | F (Length 1/2) |
| Skip indicator | 0 (Length 1/2) |
| Message Type | 44h |

Contents of Close UE Test Loop message (UE test loop mode 1 without Dummy DCCH transmission)

| Information Element | Value/remark |
|---|---|
| Protocol discriminator | F (Length 1/2) |
| Skip indicator | 0 (Length 1/2) |

| | |
|---|---|
| Message Type | 40h |
| UE test loop mode | 00h |
| UE test loop mode 1 LB setup | 03h 00h F4h 0Ah |

Contents of Close UE Test Loop message (UE test loop mode 2 without Dummy DCCH transmission)

| Information Element | Value/remark |
|---|---|
| Protocol discriminator | F (Length 1/2) |
| Skip indicator | 0 (Length 1/2) |
| Message Type | 40h |
| UE test loop mode | 01h |

Contents of Open UE Test Loop message

| Information Element | Value/remark |
|---|---|
| Protocol discriminator | F (Length 1/2) |
| Skip indicator | 0 (Length 1/2) |
| Message Type | 42h |

Contents of PAGING TYPE 1 message: TM (CS)

| Information Element | Value/remark |
|---|---|
| Message Type | |
| Paging record list | |
| -Paging record | |
| - CHOICE Used paging identity | CN identity |
| - Paging cause | Terminating Streaming Call |
| - CN domain identity | CS domain |
| - CHOICE UE identity | |
| - IMSI (GSM-MAP) | Set to the same octet string as in the IMSI stored in the USIM card |
| BCCH modification info | Not Present |

Contents of PAGING TYPE 1 message: TM (PS)

| Information Element | Value/remark |
|---|---|
| Message Type | |
| Paging record list | |
| -Paging record | |
| - CHOICE Used paging identity | CN identity |
| - Paging cause | Terminating Interactive Call |
| - CN domain identity | PS domain |
| - CHOICE UE identity | |
| - IMSI (GSM-MAP) | Set to the same octet string as in the IMSI stored in the USIM card |
| BCCH modification info | Not Present |

Contents of RADIO BEARER SETUP message: AM or UM (Test Loop Mode1)

| Information Element | Condition | Value/remark | Version |
|---|---|---|---|
| Message Type | A1, A3, A4, A5, A6, A7, A8 | | |
| RRC transaction identifier | | Arbitrarily selects an integer between 0 and 3 | |
| Integrity check info | | | |
| - message authentication code | | SS calculates the value of MAC-I for this message and writes to this IE. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I. | |
| - RRC message sequence number | | SS provides the value of this IE, from its internal counter. | |
| Integrity protection mode info | | Not Present | |
| Ciphering mode info | | Not Present | |

| Information Element | Condition | Value/remark | Version |
|---|---|---|---|
| Activation time | | (256+CFN-(CFN MOD 8 + 8))MOD 256 | |
| New U-RNTI | | Not Present | |
| New C-RNTI | | Not Present | |
| New DSCH-RNTI | | Not Present | R99 and Rel-4 only |
| New H-RNTI | | Not Present | Rel-5 |
| New Primary E-RNTI | | Not Present | Rel-6 |
| New Secondary E-RNTI | | Not Present | Rel-6 |
| RRC State indicator | | CELL_DCH | |
| UTRAN DRX cycle length coefficient | | Not Present | |
| CN information info | | Not Present | |
| URA identity | | Not Present | |
| CHOICE specification mode | | Complete specification | Rel-6 |
| - Signalling RB information to setup | | Not Present | |
| - RAB information for setup list | A1, A3, A4, A5 | | |
| - RAB information for setup | | | |
| - RAB info | | | |
| - RAB identity | | 0000 0001B<br>The first/ leftmost bit of the bit string contains the most significant bit of the RAB identity. | |
| - CN domain identity | | CS domain | |
| - NAS Synchronization Indicator | | Not Present | |
| - Re-establishment timer | | UseT314 | |
| - RB information to setup list | | | |
| - RB information to setup | | | |
| - RAB information for setup list | A6, A7, A8 | | |
| - RAB information for setup | | | |
| - RAB info | | | |
| - RAB identity | | 0000 0101B<br>The first/ leftmost bit of the bit string contains the most significant bit of the RAB identity. | |
| - CN domain identity | | PS domain | |
| - NAS Synchronization Indicator | | Not Present | |
| - Re-establishment timer | | UseT315 | |
| - RB information to setup list | | | |
| - RB information to setup | | | |
| - RB identity | A1 | 10 | |
| - PDCP info | | Not Present | |
| - CHOICE RLC info type | | RLC info | |
| - CHOICE Uplink RLC mode | | TM RLC | |
| - Transmission RLC discard | | Not Present | |
| - Segmentation indication | | FALSE | |
| - CHOICE Downlink RLC mode | | TM RLC | |
| - Segmentation indication | | FALSE | |
| - One sided RLC re-establishment | | FALSE | Rel-5 |
| - RB mapping info | | | |
| - Information for each multiplexing option | | | |
| - RLC logical channel mapping indicator | | Not Present | |
| - Number of uplink RLC logical channels | | 1 | |
| - Uplink transport channel type | | DCH | |
| - UL Transport channel identity | | 1 | |
| - Logical channel identity | | Not Present | |
| - CHOICE RLC size list | | Configured | |
| - MAC logical channel priority | | 7 | |
| - Downlink RLC logical channel info | | | |
| - Number of downlink RLC logical channels | | 1 | |
| - Downlink transport channel type | | DCH | |
| - DL DCH Transport channel identity | | 6 | |

| Information Element | Condition | Value/remark | Version |
|---|---|---|---|
| - DL DSCH Transport channel identity | | Not Present | |
| - Logical channel identity | | Not Present | |
| - RB identity | A3, A4, A5 | 10 | |
| - PDCP info | | Not Present | |
| - CHOICE RLC info type | | RLC info | |
| - CHOICE Uplink RLC mode | | AM RLC | |
| - Transmission RLC discard | | | |
| - CHOICE SDU discard mode | | No Discard | |
| - MAX_DAT | | 15 | |
| - Transmission window size | | Selected with Total RLC AM Buffer Size | |
| - Timer_RST | | 500 | |
| - Max_RST | | 4 | |
| - Polling info | | | |
| - Timer_poll_prohibit | | 400 | |
| - Timer_poll | | 400 | |
| - Poll_PDU | | Not Present | |
| - Poll_SDU | | 1 | |
| - Last transmission PDU poll | | TRUE | |
| - Last retransmission PDU poll | | TRUE | |
| - Poll_Windows | | 99 | |
| - Timer_poll_periodic | | Not Present | |
| - CHOICE Downlink RLC mode | | AM RLC | |
| - In-sequence delivery | | TRUE | |
| - Receiving window size | | Selected with Total RLC AM Buffer Size | |
| - Downlink RLC status info | | | |
| - Timer_status_prohibit | | 330 | |
| - Timer_EPC | | Not Present | |
| - Missing PDU indicator | | TRUE | |
| - Timer_STATUS_periodic | | Not Present | |
| - One sided RLC re-establishment | | FALSE | Rel-5 |
| - RB mapping info | | | |
| - Information for each multiplexing option | | | |
| - RLC logical channel mapping indicator | | Not Present | |
| - Number of uplink RLC logical channels | | 1 | |
| - Uplink transport channel type | | DCH | |
| - UL Transport channel identity | | 1 | |
| - Logical channel identity | | Not Present | |
| - CHOICE RLC size list | | Configured | |
| - MAC logical channel priority | | 7 | |
| - Downlink RLC logical channel info | | | |
| - Number of downlink RLC logical channels | | 1 | |
| - Downlink transport channel type | | DCH | |
| - DL DCH Transport channel identity | | 6 | |
| - DL DSCH Transport channel identity | | Not Present | |
| - Logical channel identity | | Not Present | |
| - RB identity | A6, A7, A8 | 20 | |
| - PDCP info | | Not present | |
| - CHOICE RLC info type | | RLC info | |
| - CHOICE Uplink RLC mode | | AM RLC | |
| - Transmission RLC discard | | | |
| - CHOICE SDU discard mode | | No Discard | |
| - MAX_DAT | | 15 | |
| - Transmission window size | | Selected with Total RLC AM Buffer Size | |
| - Timer_RST | | 500 | |
| - Max_RST | | 4 | |
| - Polling info | | | |
| - Timer_poll_prohibit | | 400 | |
| - Timer_poll | | 400 | |
| - Poll_PDU | | Not Present | |
| - Poll_SDU | | 1 | |

| Information Element | Condition | Value/remark | Version |
|---|---|---|---|
| - Last transmission PDU poll | | TRUE | |
| - Last retransmission PDU poll | | TRUE | |
| - Poll_Windows | | 99 | |
| - Timer_poll_periodic | | Not Present | |
| - CHOICE Downlink RLC mode | | AM RLC | |
| - In-sequence delivery | | TRUE | |
| - Receiving window size | | Selected with Total RLC AM Buffer Size | |
| - Downlink RLC status info | | | |
| - Timer_status_prohibit | | 330 | |
| - Timer_EPC | | Not Present | |
| - Missing PDU indicator | | TRUE | |
| - Timer_STATUS_periodic | | Not Present | |
| - One sided RLC re-establishment | | FALSE | Rel-5 |
| - RB mapping info | | | |
| - Information for each multiplexing option | | 2 RBMuxOptions | |
| - RLC logical channel mapping indicator | | Not Present | |
| - Number of uplink RLC logical channels | | 1 | |
| - Uplink transport channel type | | DCH | |
| - UL Transport channel identity | | 1 | |
| - Logical channel identity | | Not Present | |
| - CHOICE RLC size list | | Configured | |
| - MAC logical channel priority | | 8 | |
| - Downlink RLC logical channel info | | | |
| - Number of downlink RLC logical channels | | 1 | |
| - Downlink transport channel type | | DCH | |
| - DL DCH Transport channel identity | | 6 | |
| - DL DSCH Transport channel identity | | Not Present | |
| - Logical channel identity | | Not Present | |
| - RLC logical channel mapping indicator | | Not Present | |
| - Number of uplink RLC logical channels | | 1 | |
| - Uplink transport channel type | | RACH | |
| - UL Transport channel identity | | Not Present | |
| - Logical channel identity | | 7 | |
| - CHOICE RLC size list | | Explicit list | |
| - RLC size index | | Reference to clause 6 Parameter Set | |
| - MAC logical channel priority | | 8 | |
| - Downlink RLC logical channel info | | | |
| - Number of downlink RLC logical channels | | 1 | |
| - Downlink transport channel type | | FACH | |
| - DL DCH Transport channel identity | | Not Present | |
| - DL DSCH Transport channel identity | | Not Present | |
| - Logical channel identity | | 7 | |
| RB information to reconfigure list | A1, A3, A4, A5, A6, A7, A8 | Not Present | Rel-6 |
| RB information to be affected list | | Not Present | |
| Downlink counter synchronization info | | Not Present | |
| PDCP ROHC target mode | | Not Present | Rel-5 |
| UL Transport channel information for all transport channels | | | |
| - PRACH TFCS | | Not Present | |
| - CHOICE mode | | FDD | |
| - TFC subset | | Not Present | |
| - UL DCH TFCS | | | |
| - CHOICE TFCI signalling | | Normal | |
| - TFCI Field 1 information | | | |

| Information Element | Condition | Value/remark | Version |
|---|---|---|---|
| - CHOICE TFCS representation | | Complete reconfiguration | |
| - TFCS complete reconfigure information | | | |
| - CHOICE CTFC Size | | 2 bit CTFC | |
| - CTFC information | | 4 TFCs | |
| - 2bit CTFC | | 0 | |
| -Power offset Information | | | |
| - CHOICE Gain Factors | | Computed Gain Factors | |
| - Reference TFC ID | | 0 | |
| - CHOICE mode | | FDD | |
| - Power offset $P_{p-m}$ | | Not Present | |
| - 2bit CTFC | | 2 | |
| - Power offset Information | | | |
| - CHOICE Gain Factors | | Computed Gain Factors | |
| - Reference TFC ID | | 0 | |
| - CHOICE mode | | FDD | |
| - Power offset $P_{p-m}$ | | Not Present | |
| - 2bit CTFC | | 1 | |
| - Power offset Information | | | |
| - CHOICE Gain Factors | | Computed Gain Factors | |
| - Reference TFC ID | | 0 | |
| - CHOICE mode | | FDD | |
| - Power offset $P_{p-m}$ | | Not Present | |
| - 2bit CTFC | | 3 | |
| - Power offset Information | | | |
| - CHOICE Gain Factors | | Signalled Gain Factors | |
| - CHOICE mode | | FDD | |
| - Gain factor ßc | | 8 | |
| - Gain factor ßd | | 15 | |
| - Reference TFC ID | | 0 | |
| - CHOICE mode | | FDD | |
| - Power offset $P_{p-m}$ | | Not Present | |
| Deleted UL TrCH information list | | Not Present | |
| Added or Reconfigured TrCH information list | A1 | 1 | |
| - Added or Reconfigured UL TrCH information | | | |
| - Uplink transport channel type | | DCH | |
| - UL Transport channel identity | | 1 | |
| - TFS | | | |
| - CHOICE Transport channel type | | Dedicated transport channels | |
| - Dynamic Transport Format Information | | | |
| - RLC size | | 244 bits | |
| - Number of TBs and TTI List | | 2 | |
| - Transmission Time Interval | | Not Present | |
| - Number of Transport blocks | | 0 | |
| - Transmission Time Interval | | Not Present | |
| - Number of Transport blocks | | 1 | |
| - CHOICE Logical channel List | | ALL | |
| - Semi-static Transport Format Information | | | |
| - Transmission time interval | | 20 | |
| - Type of channel coding | | Convolutional | |
| - Coding Rate | | 1/3 | |
| - Rate matching attribute | | 256 | |
| - CRC size | | 16 | |
| Added or Reconfigured TrCH information list | A3, A4, A5, A6, A7, A8 | 1 | |
| - Added or Reconfigured UL TrCH information | | | |
| - Uplink transport channel type | | DCH | |
| - UL Transport channel identity | | 1 | |
| - TFS | | | |
| - CHOICE Transport channel type | | Dedicated transport channels | |
| - Dynamic Transport Format | | | |

| Information Element | Condition | Value/remark | Version |
|---|---|---|---|
| Information<br>- RLC size<br>- Number of TBs and TTI List<br>- Transmission Time Interval<br>- Number of Transport blocks<br>- Transmission Time Interval<br>- Number of Transport blocks<br>- CHOICE Logical channel List<br>- Semi-static Transport Format Information<br>- Transmission time interval<br>- Type of channel coding<br>- Coding Rate<br>- Rate matching attribute<br>- CRC size | | <br>240 bits<br>2<br>Not Present<br>0<br>Not Present<br>1<br>ALL<br><br><br>20<br>Convolutional<br>1/3<br>256<br>16 | |
| CHOICE mode<br><br><br>DL Transport channel information common for all transport channel<br>- SCCPCH TFCS<br>- CHOICE mode<br>- CHOICE DL parameters<br>Deleted DL TrCH information list | A1, A3, A4, A5, A6, A7, A8 | Not Present<br><br><br><br><br>Not Present<br>FDD<br>Same as UL<br>Not Present | |
| Added or Reconfigured TrCH information list<br>- Added or Reconfigured DL TrCH information<br>- Downlink transport channel type<br>- DL Transport channel identity<br>- CHOICE DL parameters<br>- Uplink transport channel type<br>- UL TrCH identity<br>- DCH quality target<br>- BLER Quality value | A1 | 1<br><br><br><br>DCH<br>6<br>Same as UL<br>DCH<br>1<br><br>-2.0 | |
| Added or Reconfigured TrCH information list<br>- Added or Reconfigured DL TrCH information<br>- Downlink transport channel type<br>- DL Transport channel identity<br>- CHOICE DL parameters<br>- TFS<br>- CHOICE Transport channel type<br>- Dynamic transport format information<br>- RLC Size<br>- Number of TBs and TTI List<br>- Transmission Time Interval<br>- Number of Transport blocks<br>- Transmission Time Interval<br>- Number of Transport blocks<br>- CHOICE Logical channel list<br>- Semi-static Transport Format information<br>- Transmission time interval<br>- Type of channel coding<br>- Rate matching attribute<br>- CRC size<br>- DCH quality target<br>- BLER Quality value | A3, A6 | 1<br><br><br><br>DCH<br>6<br>Explicit<br><br>Dedicated transport channels<br><br><br>1280 bits<br>2<br>Not Present<br>0<br>Not Present<br>1<br>ALL<br><br><br>20<br>Turbo<br>256<br>16<br><br>-2.0 | |
| Added or Reconfigured TrCH information list<br>- Added or Reconfigured DL TrCH information<br>- Downlink transport channel type<br>- DL Transport channel identity | A4, A7 | 1<br><br><br><br>DCH<br>6 | |

| Information Element | Condition | Value/remark | Version |
|---|---|---|---|
| - CHOICE DL parameters | | Explicit | |
| - TFS | | | |
| - CHOICE Transport channel type | | Dedicated transport channels | |
| - Dynamic transport format information | | | |
| - RLC Size | | 2880 bits | |
| - Number of TBs and TTI List | | 2 | |
| - Transmission Time Interval | | Not Present | |
| - Number of Transport blocks | | 0 | |
| - Transmission Time Interval | | Not Present | |
| - Number of Transport blocks | | 1 | |
| - CHOICE Logical channel list | | ALL | |
| - Semi-static Transport Format information | | | |
| - Transmission time interval | | 20 | |
| - Type of channel coding | | Turbo | |
| - Rate matching attribute | | 256 | |
| - CRC size | | 16 | |
| - DCH quality target | | | |
| - BLER Quality value | | -2.0 | |
| Added or Reconfigured TrCH information list | A5, A8 | 1 | |
| - Added or Reconfigured DL TrCH information | | | |
| - Downlink transport channel type | | DCH | |
| - DL Transport channel identity | | 6 | |
| - CHOICE DL parameters | | Explicit | |
| - TFS | | | |
| - CHOICE Transport channel type | | Dedicated transport channels | |
| - Dynamic transport format information | | | |
| - RLC Size | | 3840 bits | |
| - Number of TBs and TTI List | | 2 | |
| - Transmission Time Interval | | Not Present | |
| - Number of Transport blocks | | 0 | |
| - Transmission Time Interval | | Not Present | |
| - Number of Transport blocks | | 1 | |
| - CHOICE Logical channel list | | ALL | |
| - Semi-static Transport Format information | | | |
| - Transmission time interval | | 10 | |
| - Type of channel coding | | Turbo | |
| - Rate matching attribute | | 256 | |
| - CRC size | | 16 | |
| - DCH quality target | | | |
| - BLER Quality value | | -2.0 | |
| Frequency info | A1, A3, A4, A5, A6, A7, A8 | Not Present | |
| Maximum allowed UL TX power | | 33dBm | |
| CHOICE channel requirement | | Uplink DPCH info | Rel-5 and earlier |
| Uplink DPCH info | | | Rel-6 |
| - Uplink DPCH power control info | | | |
| - CHOICE mode | | FDD | |
| - DPCCH power offset | | -80dB IE value will have no effect on the UE UL power when closed loop power control is active | |
| - PC Preamble | | 1 frame | |
| - SRB delay | | 7 frames | |
| - Power Control Algorithm | | Algorithm1 | |
| - TPC step size | | 1dB | |
| - $\Delta_{ACK}$ | | Not Present | Rel-5 |
| - $\Delta_{NACK}$ | | Not Present | Rel-5 |
| - Ack-Nack repetition factor | | Not Present | Rel-5 |
| - CHOICE mode | | FDD | |

| Information Element | Condition | Value/remark | Version |
|---|---|---|---|
| - Scrambling code type<br>- Scrambling code number<br>- Number of DPDCH | | Long<br>0 (0 to 16777215)<br>1 | |
| - spreading factor | A1, A3, A4, A5, A6, A7, A8 | 64 | |
| - TFCI existence<br>- Number of FBI bit<br>- Puncturing Limit | A1, A3, A4, A5, A6, A7, A8 | TRUE<br>Not Present(0)<br>1 | |
| CHOICE Mode<br>- Downlink PDSCH information | | FDD<br>Not Present | R99 and Rel-4 only<br>R99 and Rel-4 only |
| E-DCH Info | | Not Present | Rel-6 |
| Downlink HS-PDSCH Information | | Not Present | Rel-5 |
| Downlink information common for all radio links<br>- Downlink DPCH info common for all RL<br>- Timing indicator<br>- CFN-targetSFN frame offset<br>- Downlink DPCH power control information<br>- CHOICE mode<br>- DPC mode<br>- CHOICE mode<br>- Power offset $P_{Pilot-DPDCH}$<br>- DL rate matching restriction information | | <br><br>Maintain<br>Not Present<br><br>FDD<br>0 (single)<br>FDD<br>0<br>Not Present | |
| - Spreading factor<br>- Fixed or Flexible Position<br>- TFCI existence<br>- CHOICE SF<br>- Number of bits for Pilot bits | A1 | 128<br>Fixed<br>TRUE<br>128<br>8 | |
| - Spreading factor<br>- Fixed or Flexible Position<br>- TFCI existence<br>- CHOICE SF | A3, A6 | 32<br>Fixed<br>TRUE<br>32 | |
| - Spreading factor<br>- Fixed or Flexible Position<br>- TFCI existence<br>- CHOICE SF | A4, A7 | 16<br>Fixed<br>TRUE<br>16 | |
| - Spreading factor<br>- Fixed or Flexible Position<br>- TFCI existence<br>- CHOICE SF | A5, A8 | 8<br>Fixed<br>TRUE<br>8 | |
| - CHOICE mode<br>- DPCH compressed mode info<br>- TX Diversity mode<br>- SSDT information<br>- Default DPCH Offset Value<br>- MAC-hs reset indicator | A1, A3, A4, A5, A6, A7, A8 | FDD<br>Not Present<br>None<br>Not Present<br>Not Present<br>Not Present | R99 and Rel-4 only<br><br>Rel-5 |
| Downlink information per radio link list<br>- Downlink information for each radio link<br>- CHOICE mode<br>- Primary CPICH info<br>- Primary scrambling code<br>- PDSCH with SHO DCH info | | <br><br>FDD<br><br>Reference to clause 6.1 "Default settings (FDD)"<br>Not Present | <br><br><br><br><br>R99 and |

| Information Element | Condition | Value/remark | Version |
|---|---|---|---|
| - PDSCH code mapping | | Not Present | Rel-4 only<br>R99 and Rel-4 only |
| - Serving HS-DSCH radio link indicator | | FALSE | Rel-5 |
| - Downlink DPCH info for each RL | | | |
| - CHOICE mode | | FDD | |
| - Primary CPICH usage for channel estimation | | Primary CPICH may be used | |
| - DPCH frame offset | | Set to value Default DPCH Offset Value (as currently stored in SS) mod 38 400 | |
| - Secondary CPICH info | | Not Present | |
| - DL channelisation code | | | |
| - Secondary scrambling code | | Not Present | |
| - Spreading factor | A1 | 128 | |
| - Code number | | 96 | |
| - Spreading factor | A3, A6 | 32 | |
| - Code number | | 24 | |
| - Spreading factor | A4, A7 | 16 | |
| - Code number | | 12 | |
| - Spreading factor | A5, A8 | 8 | |
| - Code number | | 6 | |
| - Scrambling code change | A1, A3, A4, A5, A6, A7, A8 | No change | |
| - TPC combination index | | 0 | |
| - SSDT Cell Identity | | Not Present | R99 and Rel-4 only |
| - Closed loop timing adjustment mode | | Not Present | |
| - SCCPCH information for FACH | | Not Present | R99 and Rel-4 only |
| MBMS PL Service Restriction Information | | Not Present | Rel-6 |

| Condition | Explanation | Version |
|---|---|---|
| A1 | This IE is needed for "UE supports CS RAB for Test Loop Mode1 RMC 12.2/12.2 (TM)" | |
| A2 | Not used | |
| A3 | This IE is needed for "UE supports CS RAB for Test Loop Mode1 AMC 12.2/64 (AM)" | |
| A4 | This IE is needed for "UE supports CS RAB for Test Loop Mode1 AMC 12.2/144 (AM)" | |
| A5 | This IE is needed for "UE supports CS RAB for Test Loop Mode1 AMC 12.2/384 (AM)" | |
| A6 | This IE is needed for "UE supports PS RAB for Test Loop Mode1 AMC 12.2/64 (AM)" | |
| A7 | This IE is needed for "UE supports PS RAB for Test Loop Mode1 AMC 12.2/144 (AM)" | |
| A8 | This IE is needed for "UE supports PS RAB for Test Loop Mode1 AMC 12.2/384 (AM)" | |

Contents of RADIO BEARER SETUP message: AM or UM (UE supports PS RAB only)

| Information Element | Value/remark | Version |
|---|---|---|
| Message Type | | |
| RRC transaction identifier | Arbitrarily selects an integer between 0 and 3 | |
| Integrity check info | | |
| - message authentication code | SS calculates the value of MAC-I for this message and writes to this IE. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I. | |
| - RRC message sequence number | SS provides the value of this IE, from its internal counter. | |
| Integrity protection mode info | Not Present | |
| Ciphering mode info | Not Present | |
| Activation time | Not Present | |
| New U-RNTI | Not Present | |

| Information Element | Value/remark | Version |
|---|---|---|
| New C-RNTI | Not Present | |
| New DSCH-RNTI | Not Present | R99 and Rel-4 only |
| New H-RNTI | Not Present | Rel-5 |
| New Primary E-RNTI | Not Present | Rel-6 |
| New Secondary E-RNTI | Not Present | Rel-6 |
| RRC State indicator | CELL_DCH | |
| UTRAN DRX cycle length coefficient | Not Present | |
| CN information info | Not Present | |
| URA identity | Not Present | |
| CHOICE specification mode | Complete specification | Rel-6 |
| - Signalling RB information to setup | Not Present | |
| - RAB information for setup list | | |
| - RAB information for setup | | |
| - RAB info | (AM DTCH for PS domain) | |
| - RAB identity | 0000 0101B<br>The first/ leftmost bit of the bit string contains the most significant bit of the RAB identity. | |
| - CN domain identity | PS domain | |
| - NAS Synchronization Indicator | Not Present | |
| - Re-establishment timer | useT315 | |
| - RB information to setup | | |
| - RB identity | 20 | |
| - PDCP info | | |
| - Support for lossless SRNS relocation | FALSE | |
| - Max PDCP SN window size | Not present | |
| - PDCP PDU header | Absent | |
| - Header compression information | Not present | |
| - CHOICE RLC info type | RLC info | |
| - CHOICE Uplink RLC mode | AM RLC | |
| - Transmission RLC discard | | |
| - CHOICE SDU discard mode | No Discard | |
| - MAX_DAT | 15 | |
| - Transmission window size | 128 | |
| - Timer_RST | 500 | |
| - Max_RST | 4 | |
| - Polling info | | |
| - Timer_poll_prohibit | 200 | |
| - Timer_poll | 200 | |
| - Poll_PDU | Not Present | |
| - Poll_SDU | 1 | |
| - Last transmission PDU poll | TRUE | |
| - Last retransmission PDU poll | TRUE | |
| - Poll_Windows | 99 | |
| - Timer_poll_periodic | Not Present | |
| - CHOICE Downlink RLC mode | AM RLC | |
| - In-sequence delivery | TRUE | |
| - Receiving window size | 128 | |
| - Downlink RLC status info | | |
| - Timer_status_prohibit | 200 | |
| - Timer_EPC | Not Present | |
| - Missing PDU indicator | TRUE | |
| - Timer_STATUS_periodic | Not Present | |
| - RB mapping info | | |
| - Information for each multiplexing option | 2 RBMuxOptions | |
| - RLC logical channel mapping indicator | Not Present | |
| - Number of uplink RLC logical channels | 1 | |
| - Uplink transport channel type | DCH | |
| - UL Transport channel identity | 1 | |
| - Logical channel identity | Not Present | |
| - CHOICE RLC size list | Configured | |
| - MAC logical channel priority | 8 | |
| - Downlink RLC logical channel info | | |
| - Number of downlink RLC logical channels | 1 | |
| - Downlink transport channel type | DCH | |

| Information Element | Value/remark | Version |
|---|---|---|
| - DL DCH Transport channel identity | 6 | |
| - DL DSCH Transport channel identity | Not Present | |
| - Logical channel identity | Not Present | |
| - RLC logical channel mapping indicator | Not Present | |
| - Number of uplink RLC logical channels | 1 | |
| - Uplink transport channel type | RACH | |
| - UL Transport channel identity | Not Present | |
| - Logical channel identity | 7 | |
| - CHOICE RLC size list | Explicit list | |
| - RLC size index | Reference to clause 6 Parameter Set | |
| - MAC logical channel priority | 8 | |
| - Downlink RLC logical channel info | | |
| - Number of downlink RLC logical channels | 1 | |
| - Downlink transport channel type | FACH | |
| - DL DCH Transport channel identity | Not Present | |
| - DL DSCH Transport channel identity | Not Present | |
| - Logical channel identity | 7 | |
| RB information to reconfigure list | Not Present | Rel-6 |
| RB information to be affected list | Not Present | |
| Downlink counter synchronization info | Not Present | |
| UL Transport channel information for all transport channels | | |
| - PRACH TFCS | Not Present | |
| - CHOICE mode | FDD | |
| - TFC subset | Not Present | |
| - UL DCH TFCS | | |
| - CHOICE TFCI signalling | Normal | |
| - TFCI Field 1 information | | |
| - CHOICE TFCS representation | Complete reconfiguration | |
| - TFCS complete reconfigure information | | |
| - CHOICE CTFC Size | Number of bits used must be enough to cover all combinations of CTFC from clause 6.10.2.4 Parameter Set. | |
| - CTFC information | This IE is repeated for TFC numbers and reference to clause 6.10.2.4 Parameter Set | |
| - CTFC | Reference to clause 6.10.2.4 Parameter Set | |
| - Power offset information | | |
| - CHOICE Gain Factors | Computed Gain Factors(The last TFC is set to Signalled Gain Factors) | |
| - Gain factor βc | 11 (below 64 kbps)<br>9 (higher than 64 kbps) (Not Present if the CHOICE Gain Factors is set to Computed Gain Factors) | |
| - Gain factor βd | 15<br>(Not Present if the CHOICE Gain Factors is set to Computed Gain Factors) | |
| - Reference TFC ID | 0 | |
| - CHOICE mode | FDD | |
| - Power offset $P_{p\text{-}m}$ | Not Present | |
| Deleted UL TrCH information list | Not Present | |
| Added or Reconfigured UL TrCH information list | 1 | |
| Added or Reconfigured UL TrCH information | 1 DCH added, 1 DCH reconfigured | |
| - Uplink transport channel type | DCH | |
| - UL Transport channel identity | 1 | |
| - TFS | | |
| - CHOICE Transport channel type | Dedicated transport channels | |
| - Dynamic Transport format information | | |
| - RLC Size | Reference to clause 6.10 Parameter Set | |
| - Number of TBs and TTI List | (This IE is repeated for TFI number.) | |
| - Transmission Time Interval | Not Present | |
| - Number of Transport blocks | Reference to clause 6.10 Parameter Set | |
| - CHOICE Logical channel list | All | |
| - Semi-static Transport Format information | | |
| - Transmission time interval | Reference to clause 6.10 Parameter Set | |

| Information Element | Value/remark | Version |
|---|---|---|
| - Type of channel coding | Reference to clause 6.10 Parameter Set | |
| - Coding Rate | Reference to clause 6.10 Parameter Set | |
| - Rate matching attribute | Reference to clause 6.10 Parameter Set | |
| - CRC size | Reference to clause 6.10 Parameter Set | |
| - Uplink transport channel type | DCH | |
| - UL Transport channel identity | 5 | |
| - TFS | | |
| - CHOICE Transport channel type | Dedicated transport channels | |
| - Dynamic Transport format information | | |
| - RLC Size | Reference to clause 6.10 Parameter Set | |
| - Number of TBs and TTI List | (This IE is repeated for TFI number.) | |
| - Transmission Time Interval | Not Present | |
| - Number of Transport blocks | Reference to clause 6.10 Parameter Set | |
| - CHOICE Logical channel list | All | |
| - Semi-static Transport Format information | | |
| - Transmission time interval | Reference to clause 6.10 Parameter Set | |
| - Type of channel coding | Reference to clause 6.10 Parameter Set | |
| - Coding Rate | Reference to clause 6.10 Parameter Set | |
| - Rate matching attribute | Reference to clause 6.10 Parameter Set | |
| - CRC size | Reference to clause 6.10 Parameter Set | |
| CHOICE mode | Not Present | |
| DL Transport channel information common for all transport channel | | |
| - SCCPCH TFCS | Not Present | |
| - CHOICE mode | FDD | |
| - CHOICE DL parameters | Explicit | |
| - DL DCH TFCS | | |
| - CHOICE TFCI Signalling | Normal | |
| - TFCI Field 1 Information | | |
| - CHOICE TFCS representation | Complete reconfiguration | |
| - TFCS complete reconfigure | | |
| - CHOICE CTFC Size | Number of bits used must be enough to cover all combinations of CTFC from clause 6.10.2.4 Parameter Set. | |
| - CTFC information | This IE is repeated for TFC numbers and reference to clause 6.10.2.4 | |
| - CTFC | Reference to clause 6.10.2.4 Parameter Set | |
| - Power offset information | Not Present | |
| Added or Reconfigured DL TrCH information list | 1 | |
| Added or Reconfigured DL TrCH information | 2 TrCHs(DCH for DCCH and DCH for DTCH) | |
| - Downlink transport channel type | DCH | |
| - DL Transport channel identity | 10 | |
| - CHOICE DL parameters | Same as UL | |
| - Uplink transport channel type | DCH | |
| - UL TrCH identity | 5 | |
| - DCH quality target | | |
| - BLER Quality value | -2.0 | |
| - Downlink transport channel type | DCH | |
| - DL Transport channel identity | 6 | |
| - CHOICE DL parameters | Explicit | |
| - TFS | | |
| - CHOICE Transport channel type | Dedicated transport channel | |
| - Dynamic transport format information | | |
| - RLC Size | Reference to clause 6.10 Parameter Set | |
| - Number of TBs and TTI List | (This IE is repeated for TFI number.) | |
| - Dynamic transport format information | | |
| - Transmission Time Interval | Not Present | |
| - Number of Transport blocks | Reference to clause 6.10 Parameter Set | |
| - CHOICE Logical channel list | All | |
| - Semi-static Transport Format information | | |
| - Transmission time interval | Reference to clause 6.10 Parameter Set | |
| - Type of channel coding | Reference to clause 6.10 Parameter Set | |
| - Coding Rate | Reference to clause 6.10 Parameter Set | |
| - Rate matching attribute | Reference to clause 6.10 Parameter Set | |
| - CRC size | Reference to clause 6.10 Parameter Set | |

| Information Element | Value/remark | Version |
|---|---|---|
| - DCH quality target | | |
| - BLER Quality value | -2.0 | |
| Frequency info | Not Present | |
| Maximum allowed UL TX power | 33dBm | |
| CHOICE channel requirement | Uplink DPCH info | Rel-5 and earlier |
| Uplink DPCH info | | Rel-6 |
| - Uplink DPCH power control info | | |
| - CHOICE mode | FDD | |
| - DPCCH power offset | -80dB IE value will have no effect on the UE UL power when closed loop power control is active | |
| - PC Preamble | 1 frame | |
| - SRB delay | 7 frames | |
| - Power Control Algorithm | Algorithm1 | |
| - TPC step size | 1dB | |
| - $\Delta_{ACK}$ | Not Present | Rel-5 |
| - $\Delta_{NACK}$ | Not Present | Rel-5 |
| - Ack-Nack repetition factor | Not Present | Rel-5 |
| - CHOICE mode | FDD | |
| - Scrambling code type | Long | |
| - Scrambling code number | 0 (0 to 16777215) | |
| - Number of DPDCH | 1 | |
| - spreading factor | 64 | |
| - TFCI existence | TRUE | |
| - Number of FBI bit | Not Present(0) | |
| - Puncturing Limit | 1 | |
| CHOICE Mode | FDD | R99 and Rel-4 only |
| E-DCH Info | Not Present | Rel-6 |
| - Downlink PDSCH information | Not Present | R99 and Rel-4 only |
| Downlink HS-PDSCH Information | Not Present | Rel-5 |
| Downlink information common for all radio links | | |
| - Downlink DPCH info common for all RL | | |
| - Timing indicator | Maintain | |
| - CFN-targetSFN frame offset | Not Present | |
| - Downlink DPCH power control information | | |
| - CHOICE mode | FDD | |
| - DPC mode | 0 (single) | |
| - CHOICE mode | FDD | |
| - Power offset $P_{Pilot-DPDCH}$ | 0 | |
| - DL rate matching restriction information | Not Present | |
| - Spreading factor | Reference to clause 6.10 Parameter Set | |
| - Fixed or Flexible Position | Reference to clause 6.10 Parameter Set | |
| - TFCI existence | Reference to clause 6.10 Parameter Set | |
| - CHOICE SF | Reference to clause 6.10 Parameter Set | |
| - CHOICE mode | FDD | |
| - DPCH compressed mode info | Not Present | |
| - TX Diversity mode | None | |
| - SSDT information | Not Present | R99 and Rel-4 only |
| - Default DPCH Offset Value | Not Present | |
| Downlink information per radio link list | | |
| - Downlink information for each radio link | | |
| - CHOICE mode | FDD | |
| - Primary CPICH info | | |
| - Primary scrambling code | Reference to clause 6.1 "Default settings (FDD)" | |
| - PDSCH with SHO DCH info | Not Present | R99 and Rel-4 only |
| - PDSCH code mapping | Not Present | R99 and Rel-4 only |
| - Downlink DPCH info for each RL | | |
| - CHOICE mode | FDD | |
| - Primary CPICH usage for channel | Primary CPICH may be used | |

| Information Element | Value/remark | Version |
|---|---|---|
| estimation | | |
| - DPCH frame offset | Set to value Default DPCH Offset Value (as currently stored in SS) mod 38 400 | |
| - Secondary CPICH info | Not Present | |
| - DL channelisation code | | |
| - Secondary scrambling code | Not present | |
| - Spreading factor | Reference to clause 6.10 Parameter Set | |
| - Code number | Depends upon radio bearer used. | |
| - Scrambling code change | No change | |
| - TPC combination index | 0 | |
| - SSDT Cell Identity | Not Present | R99 and Rel-4 only |
| - Closed loop timing adjustment mode | Not Present | |
| - SCCPCH information for FACH | Not Present | R99 and Rel-4 only |
| MBMS PL Service Restriction Information | Not Present | Rel-6 |

Contents of RADIO BEARER SETUP message: AM or UM (UE supports CS RAB for Test Loop Mode 2)

| Information Element | Value/remark | Version |
|---|---|---|
| Message Type | | |
| RRC transaction identifier | Arbitrarily selects an integer between 0 and 3 | |
| Integrity check info | | |
| - message authentication code | SS calculates the value of MAC-I for this message and writes to this IE. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I. | |
| - RRC message sequence number | SS provides the value of this IE, from its internal counter. | |
| Integrity protection mode info | Not Present | |
| Ciphering mode info | Not Present | |
| Activation time | (256+CFN-(CFN MOD 8 + 8))MOD 256 | |
| New U-RNTI | Not Present | |
| New C-RNTI | Not Present | |
| New DSCH-RNTI | Not Present | R99 and Rel-4 only |
| New H-RNTI | Not Present | Rel-5 |
| New Primary E-RNTI | Not Present | Rel-6 |
| New Secondary E-RNTI | Not Present | Rel-6 |
| RRC State indicator | CELL_DCH | |
| UTRAN DRX cycle length coefficient | Not Present | |
| CN information info | Not Present | |
| URA identity | Not Present | |
| CHOICE specification mode | Complete specification | Rel-6 |
| Signalling RB information to setup | Not Present | |
| RAB information for setup list | | |
| - RAB information for setup | | |
| - RAB info | | |
| - RAB identity | 0000 0001B<br>The first/ leftmost bit of the bit string contains the most significant bit of the RAB identity. | |
| - CN domain identity | CS domain | |
| - NAS Synchronization Indicator | Not Present | |
| - Re-establishment timer | UseT314 | |
| - RB information to setup list | | |
| - RB information to setup | | |
| - RB identity | 10 | |
| - PDCP info | Not Present | |
| - CHOICE RLC info type | RLC info | |
| - CHOICE Uplink RLC mode | TM RLC | |
| - Transmission RLC discard | Not Present | |
| - Segmentation indication | FALSE | |
| - CHOICE Downlink RLC mode | TM RLC | |
| - Segmentation indication | FALSE | |

| Information Element | Value/remark | Version |
|---|---|---|
| - RB mapping info | | |
| - Information for each multiplexing option | | |
| - RLC logical channel mapping indicator | Not Present | |
| - Number of uplink RLC logical channels | 1 | |
| - Uplink transport channel type | DCH | |
| - UL Transport channel identity | 1 | |
| - Logical channel identity | Not Present | |
| - CHOICE RLC size list | Configured | |
| - MAC logical channel priority | 7 | |
| - Downlink RLC logical channel info | | |
| - Number of downlink RLC logical channels | 1 | |
| - Downlink transport channel type | DCH | |
| - DL DCH Transport channel identity | 6 | |
| - DL DSCH Transport channel identity | Not Present | |
| - Logical channel identity | Not Present | |
| RB information to reconfigure list | Not Present | Rel-6 |
| RB information to be affected list | Not Present | |
| Downlink counter synchronization info | Not Present | |
| UL Transport channel information for all transport channels | | |
| - PRACH TFCS | Not Present | |
| - CHOICE mode | FDD | |
| - TFC subset | Not Present | |
| - UL DCH TFCS | | |
| - CHOICE TFCI signalling | Normal | |
| - TFCI Field 1 information | | |
| - CHOICE TFCS representation | Complete reconfiguration | |
| - TFCS complete reconfigure information | | |
| - CHOICE CTFC Size | 2 bit CTFC | |
| - CTFC information | 4 TFCs | |
| - 2bit CTFC | 0 | |
| - Power offset Information | | |
| - CHOICE Gain Factors | Computed Gain Factors | |
| - Reference TFC ID | 0 | |
| - CHOICE mode | FDD | |
| - Power offset $P_{p-m}$ | Not Present | |
| - 2bit CTFC | 2 | |
| - Power offset Information | | |
| - CHOICE Gain Factors | Computed Gain Factors | |
| - Reference TFC ID | 0 | |
| - CHOICE mode | FDD | |
| - Power offset $P_{p-m}$ | Not Present | |
| - 2bit CTFC | 1 | |
| - Power offset Information | | |
| - CHOICE Gain Factors | Computed Gain Factors | |
| - Reference TFC ID | 0 | |
| - CHOICE mode | FDD | |
| - Power offset $P_{p-m}$ | Not Present | |
| - 2bit CTFC | 3 | |
| - Power offset Information | | |
| - CHOICE Gain Factors | Signalled Gain Factors | |
| - CHOICE mode | FDD | |
| - Gain factor ßc | 8 | |
| - Gain factor ßd | 15 | |
| - Reference TFC ID | 0 | |
| - CHOICE mode | FDD | |
| - Power offset $P_{p-m}$ | Not Present | |
| Deleted UL TrCH information list | Not Present | |
| Added or Reconfigured UL TrCH information list | 1 | |
| - Added or Reconfigured UL TrCH information | | |
| - Uplink transport channel type | DCH | |
| - UL Transport channel identity | 1 | |
| - TFS | | |
| - CHOICE Transport channel type | Dedicated transport channels | |

| Information Element | Value/remark | Version |
|---|---|---|
| - Dynamic Transport Format Information | | |
| - RLC size | 260 bits | |
| - Number of TBs and TTI List | 2 | |
| - Transmission Time Interval | Not Present | |
| - Number of Transport blocks | 0 | |
| - Transmission Time Interval | Not Present | |
| - Number of Transport blocks | 1 | |
| - CHOICE Logical channel List | ALL | |
| - Semi-static Transport Format Information | | |
| - Transmission time interval | 20 | |
| - Type of channel coding | Convolutional | |
| - Coding Rate | 1/3 | |
| - Rate matching attribute | 256 | |
| - CRC size | 0 | |
| CHOICE mode | Not Present | |
| DL Transport channel information common for all transport channel | | |
| - SCCPCH TFCS | Not Present | |
| - CHOICE mode | FDD | |
| - CHOICE DL parameters | Same as UL | |
| Deleted DL TrCH information list | Not Present | |
| Added or Reconfigured DL TrCH information list | 1 | |
| - Added or Reconfigured DL TrCH information | | |
| - Downlink transport channel type | DCH | |
| - DL Transport channel identity | 6 | |
| - CHOICE DL parameters | | |
| - CHOICE Transport channel type | Dedicated transport channels | |
| - Dynamic Transport Format Information | | |
| - RLC size | 244 bits | |
| - Number of TBs and TTI List | 2 | |
| - Transmission Time Interval | Not Present | |
| - Number of Transport blocks | 0 | |
| - Transmission Time Interval | Not Present | |
| - Number of Transport blocks | 1 | |
| - CHOICE Logical channel List | ALL | |
| - Semi-static Transport Format Information | | |
| - Transmission time interval | 20 | |
| - Type of channel coding | Convolutional | |
| - Coding Rate | 1/3 | |
| - Rate matching attribute | 256 | |
| - CRC size | 16 | |
| - Uplink transport channel type | DCH | |
| - UL TrCH identity | 1 | |
| - DCH quality target | | |
| - BLER Quality value | -2.0 | |
| Frequency info | Not Present | |
| Maximum allowed UL TX power | 33dBm | |
| CHOICE channel requirement | Uplink DPCH info | Rel-5 and earlier |
| Uplink DPCH info | | Rel-6 |
| - Uplink DPCH power control info | | |
| - CHOICE mode | FDD | |
| - DPCCH power offset | -80dB IE value will have no effect on the UE UL power when closed loop power control is active | |
| - PC Preamble | 1 frame | |
| - SRB delay | 7 frames | |
| - Power Control Algorithm | Algorithm1 | |
| - TPC step size | 1dB | |
| - $\Delta_{ACK}$ | Not Present | Rel-5 |
| - $\Delta_{NACK}$ | Not Present | Rel-5 |
| - Ack-Nack repetition factor | Not Present | Rel-5 |
| - CHOICE mode | FDD | |
| - Scrambling code type | Long | |
| - Scrambling code number | 0 (0 to 16777215) | |
| - Number of DPDCH | 1 | |
| - spreading factor | 64 | |

| Information Element | Value/remark | Version |
|---|---|---|
| - TFCI existence | TRUE | |
| - Number of FBI bit | Not Present(0) | |
| - Puncturing Limit | 1 | |
| CHOICE Mode | FDD | R99 and Rel-4 only |
| - Downlink PDSCH information | Not Present | R99 and Rel-4 only |
| E-DCH Info | Not Present | Rel-6 |
| Downlink HS-PDSCH Information | Not Present | Rel-5 |
| Downlink information common for all radio links | | |
| - Downlink DPCH info common for all RL | | |
| - Timing indicator | Maintain | |
| - CFN-targetSFN frame offset | Not Present | |
| - Downlink DPCH power control information | | |
| - CHOICE mode | FDD | |
| - DPC mode | 0 (single) | |
| - CHOICE mode | FDD | |
| - Power offset $P_{Pilot-DPDCH}$ | 0 | |
| - DL rate matching restriction information | Not Present | |
| - Spreading factor | 128 | |
| - Fixed or Flexible Position | Fixed | |
| - TFCI existence | TRUE | |
| - CHOICE SF | 128 | |
| - Number of bits for Pilot bits | 8 | |
| - CHOICE mode | FDD | |
| - DPCH compressed mode info | Not Present | |
| - TX Diversity mode | None | |
| - SSDT information | Not Present | R99 and Rel-4 only |
| - Default DPCH Offset Value | Not Present | |
| Downlink information for per radio link list | | |
| - Downlink information for each radio link | | |
| - CHOICE mode | FDD | |
| - Primary CPICH info | | |
| - Primary scrambling code | Reference to clause 6.1 "Default settings (FDD)" | |
| - PDSCH with SHO DCH info | Not Present | R99 and Rel-4 only |
| - PDSCH code mapping | Not Present | R99 and Rel-4 only |
| - Downlink DPCH info for each RL | | |
| - CHOICE mode | FDD | |
| - Primary CPICH usage for channel estimation | Primary CPICH may be used | |
| - DPCH frame offset | Set to value Default DPCH Offset Value (as currently stored in SS) mod 38 400 | |
| - Secondary CPICH info | Not Present | |
| - DL channelisation code | | |
| - Secondary scrambling code | Not Present | |
| - Spreading factor | 128 | |
| - Code number | 96 | |
| - Scrambling code change | No change | |
| - TPC combination index | 0 | |
| - SSDT Cell Identity | Not Present | R99 and Rel-4 only |
| - Closed loop timing adjustment mode | Not Present | |
| - SCCPCH information for FACH | Not Present | R99 and Rel-4 only |
| MBMS PL Service Restriction Information | Not Present | Rel-6 |

Contents of RADIO BEARER SETUP message: AM or UM (HSDPA)

| Information Element | Value/remark | Version |
|---|---|---|
| Message Type | | |
| RRC transaction identifier | Arbitrarily selects an integer between 0 and 3 | |
| Integrity check info | | |

| Information Element | Value/remark | Version |
|---|---|---|
| - message authentication code | SS calculates the value of MAC-I for this message and writes to this IE. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I. | |
| - RRC message sequence number | SS provides the value of this IE, from its internal counter. | |
| Integrity protection mode info | Not Present | |
| Ciphering mode info | Not Present | |
| Activation time | Not Present | |
| New U-RNTI | Not Present | |
| New C-RNTI | Not Present | |
| New H-RNTI | '1010 1010 1010 1010' | Rel-5 |
| New Primary E-RNTI | Not Present | Rel-6 |
| New Secondary E-RNTI | Not Present | Rel-6 |
| RRC State indicator | CELL_DCH | |
| UTRAN DRX cycle length coefficient | Not Present | |
| CN information info | Not Present | |
| URA identity | Not Present | |
| CHOICE specification mode | Complete specification | Rel-6 |
| Signalling RB information to setup | Not Present | |
| RAB information for setup list | | |
| - RAB information for setup | | |
| - RAB info | (high-speed UM DTCH for PS domain) | |
| - RAB identity | 0000 0110B<br>The first/ leftmost bit of the bit string contains the most significant bit of the RAB identity. | |
| - CN domain identity | PS domain | |
| - NAS Synchronization Indicator | Not Present | |
| - Re-establishment timer | UseT315 | |
| - RB information to setup | | |
| - RB identity | 25 | |
| - PDCP info | Not Present | |
| - CHOICE RLC info type | RLC info | |
| - CHOICE Uplink RLC mode | Not Present | |
| - CHOICE Downlink RLC mode | UM RLC | |
| - DL UM RLC LI size | 15 | Rel-5 |
| - One sided RLC re-establishment | FALSE | Rel-5 |
| - RB mapping info | | |
| - Information for each multiplexing option | 1 RBMuxOptions | |
| - RLC logical channel mapping indicator | Not Present | |
| - Downlink RLC logical channel info | | |
| - Number of downlink RLC logical channels | 1 | |
| - Downlink transport channel type | HS-DSCH | |
| - DL DCH Transport channel identity | Not Present | |
| - DL DSCH Transport channel identity | Not Present | |
| - DL HS-DSCH MAC-d flow identity | 0 | |
| - Logical channel identity | Not Present | |
| RB information to reconfigure list | Not Present | Rel-6 |
| RB information to be affected list | Not Present | |
| Downlink counter synchronization info | Not Present | |
| PDCP ROHC target mode | Not Present | Rel-5 |
| UL Transport channel information for all transport channels | | |
| - PRACH TFCS | Not Present | |
| - CHOICE mode | FDD | |
| - TFC subset | Not Present | |
| - UL DCH TFCS | | |
| - CHOICE TFCI signalling | Normal | |
| - TFCI Field 1 information | | |
| - CHOICE TFCS representation | Complete reconfiguration | |
| - TFCS complete reconfigure information | | |
| - CHOICE CTFC Size | 2 bit CTFC | |
| - CTFC information | 4 TFCs | |
| - CTFC | Reference to clause TS 34.121 clause C.2.1 Parameter Set | |
| - Power offset information | | |
| - CHOICE Gain Factors | Computed Gain Factors(The last TFC is set to Signalled Gain Factors) | |

| Information Element | Value/remark | Version |
|---|---|---|
| - Gain factor βc | 8<br>(Not Present if the CHOICE Gain Factors is set to Computed Gain Factors) | |
| - Gain factor βd | 15<br>(Not Present if the CHOICE Gain Factors is set to Computed Gain Factors) | |
| - Reference TFC ID | 0 | |
| - CHOICE mode | FDD | |
| - Power offset P p-m | Not Present | |
| Deleted UL TrCH information list | Not Present | |
| Added or Reconfigured TrCH information list | Not Present | |
| CHOICE mode | Not Present | |
| DL Transport channel information common for all transport channel | | |
| - SCCPCH TFCS | Not Present | |
| - CHOICE mode | FDD | |
| - CHOICE DL parameters | Explicit | |
| - DL DCH TFCS | | |
| - CHOICE TFCI Signalling | Normal | |
| - TFCI Field 1 Information | | |
| - CHOICE TFCS representation | Complete reconfiguration | |
| - TFCS complete reconfigure | | |
| - CHOICE CTFC Size | 2 bit CTFC | |
| - CTFC information | 4 TFCs | |
| - CTFC | Reference to clause TS 34.121 clause C.3.1 Parameter Set | |
| - Power offset information | Not Present | |
| Deleted DL TrCH information | Not Present | |
| Added or Reconfigured DL TrCH information list | 1 TrCHs added | |
| - Added or Reconfigured DL TrCH information | (HS-DSCH for DTCH) | |
| - Downlink transport channel type | HS-DSCH | Rel-5 |
| - DL Transport channel identity | Not Present | |
| - CHOICE DL parameters | HS-DSCH | |
| - HARQ Info | | Rel-5 |
| - Number of Processes | Reference to TS34.121 [2] Annex C Fixed Reference Channels | |
| - CHOICE *Memory Partitioning* | Explicit | |
| - Memory size | Reference to TS34.121 [2] Annex C Fixed Reference Channels parameter “Number of HARQ Processes”. | |
| - Process Memory Size | Reference to TS34.121 [2] Annex C Fixed Reference Channels parameter “Number of SML’s per HARQ Proc.”. | |
| - Added or reconfigured MAC-d flow | | |
| - MAC-hs queue to add or reconfigure list | (one queue) | Rel-5 |
| - MAC-hs queue Id | 0 | |
| - MAC-d Flow Identity | 0 | |
| - T1 | 50 | |
| - MAC-hs window size | 16 | |
| - MAC-d PDU size Info | | |
| - MAC-d PDU size | Reference to TS34.121 [2] Annex C Fixed Reference Channels | |
| - MAC-d PDU size index | 0 | |
| - MAC-hs queue to delete list | Not present | |
| - DCH quality target | Not present | |
| Frequency info | Not Present | |
| Maximum allowed UL TX power | 33dBm | |
| CHOICE channel requirement | Uplink DPCH info | Rel-5 and earlier |
| Uplink DPCH info | | Rel-6 |
| - Uplink DPCH power control info | | |
| - CHOICE mode | FDD | |
| - DPCCH power offset | -80dB IE value will have no effect on the UE UL power when closed loop power control is active | |
| - PC Preamble | 1 frame | |
| - SRB delay | 7 frames | |
| - Power Control Algorithm | Algorithm1 | |

| Information Element | Value/remark | Version |
|---|---|---|
| - TPC step size | 1dB | |
| - $\Delta_{ACK}$ | 3 | Rel-5 |
| - $\Delta_{NACK}$ | 3 | Rel-5 |
| - Ack-Nack repetition factor | 1 | Rel-5 |
| - CHOICE mode | FDD | |
| - Scrambling code type | Long | |
| - Scrambling code number | 0 (0 to 16777215) | |
| - Number of DPDCH | Not Present (1) | |
| - spreading factor | 64 | |
| - TFCI existence | TRUE | |
| - Number of FBI bit | Not Present(0) | |
| - Puncturing Limit | 1 | |
| CHOICE Mode | FDD | R99 and Rel-4 only |
| - Downlink PDSCH information | Not Present | R99 and Rel-4 only |
| E-DCH Info | Not Present | Rel-6 |
| Downlink HS-PDSCH Information | | |
| - HS-SCCH Info | | |
| - CHOICE mode | FDD | |
| - DL Scrambling Code | | |
| - HS-SCCH Channelisation Code Information | | |
| - HS-SCCH Channelisation Code | 2 | |
| - HS-SCCH Channelisation Code | 3 | |
| - HS-SCCH Channelisation Code | 6 | |
| - HS-SCCH Channelisation Code | 7 | |
| - Measurement Feedback Info | | |
| - CHOICE mode | FDD | |
| - POhsdsch | 6 dB | Rel-5 |
| - CQI Feedback cycle, k | 2 ms | Rel-5 |
| - CQI repetition factor | 1 | Rel-5 |
| - $\Delta_{CQI}$ | 5 (corresponds to 0dB in relative power offset) | Rel-5 |
| - CHOICE mode | FDD (no data) | |
| Downlink information common for all radio links | Not Present | |
| Downlink information per radio link list | | |
| - Downlink information for each radio link | | |
| - CHOICE mode | FDD | |
| - Primary CPICH info | | |
| - Primary scrambling code | Reference to clause 6.1 "Default settings (FDD)" | |
| - PDSCH with SHO DCH info | Not Present | R99 and Rel-4 only |
| - PDSCH code mapping | Not Present | R99 and Rel-4 only |
| - Serving HS-DSCH radio link indicator | TRUE | Rel-5 |
| - Downlink DPCH info for each RL | | |
| - CHOICE mode | FDD | |
| - Primary CPICH usage for channel estimation | Primary CPICH may be used | |
| - DPCH frame offset | Set to value Default DPCH Offset Value (as currently stored in SS) mod 38 400 | |
| - Secondary CPICH info | Not Present | |
| - DL channelisation code | | |
| - Secondary scrambling code | Not present | |
| - Spreading factor | 128 | |
| - Code number | 96 | |
| - Scrambling code change | No change | |
| - TPC combination index | 0 | |
| - SSDT Cell Identity | Not Present | R99 and Rel-4 only |
| - Closed loop timing adjustment mode | Not Present | |
| - SCCPCH information for FACH | Not Present | R99 and Rel-4 only |
| MBMS PL Service Restriction Information | Not Present | Rel-6 |

For a transition period until RAN5#31 the RADIO BEARER SETUP message: AM or UM (HSDPA) below can be used. This alternative RADIO BEARER SETUP message will be removed at RAN5#31.

Contents of RADIO BEARER SETUP message: AM or UM (HSDPA)

| Information Element | Value/remark | Version |
|---|---|---|
| Message Type | | |
| RRC transaction identifier | Arbitrarily selects an integer between 0 and 3 | |
| Integrity check info | | |
| - message authentication code | SS calculates the value of MAC-I for this message and writes to this IE. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I. | |
| - RRC message sequence number | SS provides the value of this IE, from its internal counter. | |
| Integrity protection mode info | Not Present | |
| Ciphering mode info | Not Present | |
| Activation time | Not Present | |
| New U-RNTI | Not Present | |
| New C-RNTI | Not Present | |
| New H-RNTI | '1010 1010 1010 1010' | Rel-5 |
| New Primary E-RNTI | Not Present | Rel-6 |
| New Secondary E-RNTI | Not Present | Rel-6 |
| RRC State indicator | CELL_DCH | |
| UTRAN DRX cycle length coefficient | Not Present | |
| CN information info | Not Present | |
| URA identity | Not Present | |
| CHOICE specification mode | Complete specification | Rel-6 |
| Signalling RB information to setup | Not Present | |
| RAB information for setup list | | |
| - RAB information for setup | | |
| - RAB info | (high-speed UM DTCH for PS domain) | |
| - RAB identity | 0000 0110B<br>The first/ leftmost bit of the bit string contains the most significant bit of the RAB identity. | |
| - CN domain identity | PS domain | |
| - NAS Synchronization Indicator | Not Present | |
| - Re-establishment timer | UseT315 | |
| - RB information to setup | | |
| - RB identity | 25 | |
| - PDCP info | Not Present | |
| - CHOICE RLC info type | RLC info | |
| - CHOICE Uplink RLC mode | Not Present | |
| - CHOICE Downlink RLC mode | UM RLC | |
| - DL UM RLC LI size | 15 | Rel-5 |
| - One sided RLC re-establishment | FALSE | Rel-5 |
| - RB mapping info | | |
| - Information for each multiplexing option | 1 RBMuxOptions | |
| - RLC logical channel mapping indicator | Not Present | |
| - Downlink RLC logical channel info | | |
| - Number of downlink RLC logical channels | 1 | |
| - Downlink transport channel type | HS-DSCH | |
| - DL DCH Transport channel identity | Not Present | |
| - DL DSCH Transport channel identity | Not Present | |
| - DL HS-DSCH MAC-d flow identity | 0 | |
| - Logical channel identity | Not Present | |
| - RAB information for setup | | |
| - RAB info | (TM DTCH for PS domain) | |
| - RAB identity | 0000 0001B<br>The first/ leftmost bit of the bit string contains the most significant bit of the RAB identity. | |
| - CN domain identity | PS domain | |
| - NAS Synchronization Indicator | Not Present | |
| - Re-establishment timer | UseT314 | |
| - RB information to setup | | |
| - RB identity | 10 | |
| - PDCP info | Not Present | |
| - CHOICE RLC info type | RLC info | |
| - CHOICE Uplink RLC mode | TM RLC | |
| - Transmission RLC discard | Not Present | |
| - Segmentation indication | FALSE | |
| - CHOICE Downlink RLC mode | TM RLC | |

| Information Element | Value/remark | Version |
|---|---|---|
| - Segmentation indication | FALSE | |
| - One sided RLC re-establishment | FALSE | Rel-5 |
| - RB mapping info | | |
| - Information for each multiplexing option | 1 RBMuxOptions | |
| - RLC logical channel mapping indicator | Not Present | |
| - Number of uplink RLC logical channels | 1 | |
| - Uplink transport channel type | DCH | |
| - UL Transport channel identity | 1 | |
| - Logical channel identity | Not Present | |
| - CHOICE RLC size list | Configured | |
| - MAC logical channel priority | 7 | |
| - Downlink RLC logical channel info | | |
| - Number of downlink RLC logical channels | 1 | |
| - Downlink transport channel type | DCH | |
| - DL DCH Transport channel identity | 6 | |
| - DL DSCH Transport channel identity | Not Present | |
| - DL HS-DSCH MAC-d flow identity | Not Present | Rel-5 |
| - Logical channel identity | Not Present | |
| RB information to reconfigure list | Not Present | Rel-6 |
| RB information to be affected list | Not Present | |
| Downlink counter synchronization info | Not Present | |
| PDCP ROHC target mode | Not Present | Rel-5 |
| UL Transport channel information for all transport channels | | |
| - PRACH TFCS | Not Present | |
| - CHOICE mode | FDD | |
| - TFC subset | Not Present | |
| - UL DCH TFCS | | |
| - CHOICE TFCI signalling | Normal | |
| - TFCI Field 1 information | | |
| - CHOICE TFCS representation | Complete reconfiguration | |
| - TFCS complete reconfigure information | | |
| - CHOICE CTFC Size | 2 bit CTFC | |
| - CTFC information | 4 TFCs | |
| - CTFC | Reference to clause TS 34.121 clause C.2.1 Parameter Set | |
| - Power offset information | | |
| - CHOICE Gain Factors | Computed Gain Factors(The last TFC is set to Signalled Gain Factors) | |
| - Gain factor βc | 8 (Not Present if the CHOICE Gain Factors is set to Computed Gain Factors) | |
| - Gain factor βd | 15 (Not Present if the CHOICE Gain Factors is set to Computed Gain Factors) | |
| - Reference TFC ID | 0 | |
| - CHOICE mode | FDD | |
| - Power offset $P_{p\text{-}m}$ | Not Present | |
| Deleted UL TrCH information list | Not Present | |
| Added or Reconfigured TrCH information list | 1 TrCH added | |
| - Added or Reconfigured UL TrCH information | (DCH for DTCH) | |
| - Uplink transport channel type | DCH | |
| - UL Transport channel identity | 1 | |
| - TFS | | |
| - CHOICE Transport channel type | Dedicated transport channels | |
| - Dynamic Transport format information | | |
| - RLC Size | 244 | |
| - Number of TBs and TTI List | 2 | |
| - Transmission Time Interval | Not Present | |
| - Number of Transport blocks | 0 | |
| - Transmission Time Interval | Not Present | |
| - Number of Transport blocks | 1 | |
| - CHOICE Logical channel list | All | |
| - Semi-static Transport Format information | | |
| - Transmission time interval | 20 ms | |
| - Type of channel coding | Convolutional | |
| - Coding Rate | 1/3 | |

| Information Element | Value/remark | Version |
|---|---|---|
| - Rate matching attribute | 256 | |
| - CRC size | 16 | |
| CHOICE mode | Not Present | |
| DL Transport channel information common for all transport channel | | |
| - SCCPCH TFCS | Not Present | |
| - CHOICE mode | FDD | |
| - CHOICE DL parameters | Explicit | |
| - DL DCH TFCS | | |
| - CHOICE TFCI Signalling | Normal | |
| - TFCI Field 1 Information | | |
| - CHOICE TFCS representation | Complete reconfiguration | |
| - TFCS complete reconfigure | | |
| - CHOICE CTFC Size | 2 bit CTFC | |
| - CTFC information | 4 TFCs | |
| - CTFC | Reference to clause TS 34.121 clause C.3.1 Parameter Set | |
| - Power offset information | Not Present | |
| Deleted DL TrCH information | Not Present | |
| Added or Reconfigured TrCH information list | 2 TrCHs added | |
| - Added or Reconfigured DL TrCH information | (DCH for DTCH) | |
| - Downlink transport channel type | DCH | |
| - DL Transport channel identity | 6 | |
| - CHOICE DL parameters | Same as UL | |
| - Uplink transport channel type | DCH | |
| - UL TrCH identity | 1 | |
| - DCH quality target | | |
| - BLER Quality value | -2.0 | |
| - Added or Reconfigured DL TrCH information | (HS-DSCH for DTCH) | |
| - Downlink transport channel type | HS-DSCH | Rel-5 |
| - DL Transport channel identity | Not Present | |
| - CHOICE DL parameters | HS-DSCH | |
| - HARQ Info | | Rel-5 |
| - Number of Processes | Reference to TS34.121 [2] Annex C Fixed Reference Channels | |
| - CHOICE *Memory Partitioning* | Explicit | |
| - Memory size | Reference to TS34.121 [2] Annex C Fixed Reference Channels parameter “Number of HARQ Processes”. | |
| - Process Memory Size | Reference to TS34.121 [2] Annex C Fixed Reference Channels parameter “Number of SML’s per HARQ Proc.”. | |
| - Added or reconfigured MAC-d flow | | |
| - MAC-hs queue to add or reconfigure list | (one queue) | Rel-5 |
| - MAC-hs queue Id | 0 | |
| - MAC-d Flow Identity | 0 | |
| - T1 | 50 | |
| - MAC-hs window size | 16 | |
| - MAC-d PDU size Info | | |
| - MAC-d PDU size | Reference to TS34.121 [2] Annex C Fixed Reference Channels | |
| - MAC-d PDU size index | 0 | |
| - MAC-hs queue to delete list | Not present | |
| - DCH quality target | Not present | |
| Frequency info | Not Present | |
| Maximum allowed UL TX power | 33dBm | |
| CHOICE channel requirement | Uplink DPCH info | Rel-5 and earlier |
| Uplink DPCH info | | Rel-6 |
| - Uplink DPCH power control info | | |
| - CHOICE mode | FDD | |
| - DPCCH power offset | -80dB IE value will have no effect on the UE UL power when closed loop power control is active | |
| - PC Preamble | 1 frame | |
| - SRB delay | 7 frames | |
| - Power Control Algorithm | Algorithm1 | |
| - TPC step size | 1dB | |

| Information Element | Value/remark | Version |
|---|---|---|
| - $\Delta_{ACK}$ | 3 | Rel-5 |
| - $\Delta_{NACK}$ | 3 | Rel-5 |
| - Ack-Nack repetition factor | 1 | Rel-5 |
| - CHOICE mode | FDD | |
| - Scrambling code type | Long | |
| - Scrambling code number | 0 (0 to 16777215) | |
| - Number of DPDCH | Not Present (1) | |
| - spreading factor | 64 | |
| - TFCI existence | TRUE | |
| - Number of FBI bit | Not Present(0) | |
| - Puncturing Limit | 1 | |
| CHOICE Mode | FDD | R99 and Rel-4 only |
| - Downlink PDSCH information | Not Present | R99 and Rel-4 only |
| E-DCH Info | Not Present | Rel-6 |
| Downlink HS-PDSCH Information | | |
| - HS-SCCH Info | | |
| - CHOICE mode | FDD | |
| - DL Scrambling Code | | |
| - HS-SCCH Channelisation Code Information | | |
| - HS-SCCH Channelisation Code | 2 | |
| - HS-SCCH Channelisation Code | 3 | |
| - HS-SCCH Channelisation Code | 6 | |
| - HS-SCCH Channelisation Code | 7 | |
| - Measurement Feedback Info | | |
| - CHOICE mode | FDD | |
| - POhsdsch | 6 dB | Rel-5 |
| - CQI Feedback cycle, k | 2 ms | Rel-5 |
| - CQI repetition factor | 1 | Rel-5 |
| - $\Delta_{CQI}$ | 5 (corresponds to 0dB in relative power offset) | Rel-5 |
| - CHOICE mode | FDD (no data) | |
| Downlink information common for all radio links | | |
| - Downlink DPCH info common for all RL | | |
| - Timing indicator | Maintain | |
| - CFN-targetSFN frame offset | Not Present | |
| - Downlink DPCH power control information | | |
| - CHOICE mode | FDD | |
| - DPC mode | 0 (single) | |
| - CHOICE mode | FDD | |
| - Power offset $P_{Pilot-DPDCH}$ | 0 | |
| - DL rate matching restriction information | Not Present | |
| - Spreading factor | 128 | |
| - Fixed or Flexible Position | Fixed | |
| - TFCI existence | TRUE | |
| - CHOICE SF | 128 | |
| - Number of bits for Pilot bits | 8 | |
| - CHOICE mode | FDD | |
| - DPCH compressed mode info | Not Present | |
| - TX Diversity mode | None | |
| - SSDT information | Not Present | R99 and Rel-4 only |
| - Default DPCH Offset Value | Not Present | |
| Downlink information per radio link list | | |
| - Downlink information for each radio link | | |
| - CHOICE mode | FDD | |
| - Primary CPICH info | | |
| - Primary scrambling code | Reference to clause 6.1 "Default settings (FDD)" | |
| - PDSCH with SHO DCH info | Not Present | R99 and Rel-4 only |
| - PDSCH code mapping | Not Present | R99 and Rel-4 only |
| - Serving HS-DSCH radio link indicator | TRUE | Rel-5 |
| - Downlink DPCH info for each RL | | |
| - CHOICE mode | FDD | |
| - Primary CPICH usage for channel estimation | Primary CPICH may be used | |

| Information Element | Value/remark | Version |
|---|---|---|
| - DPCH frame offset | Set to value Default DPCH Offset Value (as currently stored in SS) mod 38 400 | |
| - Secondary CPICH info | Not Present | |
| - DL channelisation code | | |
| - Secondary scrambling code | Not present | |
| - Spreading factor | 128 | |
| - Code number | 96 | |
| - Scrambling code change | No change | |
| - TPC combination index | 0 | |
| - SSDT Cell Identity | Not Present | R99 and Rel-4 only |
| - Closed loop timing adjustment mode | Not Present | |
| - SCCPCH information for FACH | Not Present | R99 and Rel-4 only |
| MBMS PL Service Restriction Information | Not Present | Rel-6 |

Contents of RADIO BEARER SETUP message: BTFD RMC for Test Loop Mode 2

| Information Element | Value/remark | Version |
|---|---|---|
| Message Type | | |
| RRC transaction identifier | Arbitrarily selects an integer between 0 and 3 | |
| Integrity check info | | |
| - message authentication code | SS calculates the value of MAC-I for this message and writes to this IE. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I. | |
| - RRC message sequence number | SS provides the value of this IE, from its internal counter. | |
| Integrity protection mode info | Not Present | |
| Ciphering mode info | Not Present.<br>For correct operation of test loop mode 2 this IE shall be omitted. | |
| Activation time | Set by operator | |
| New U-RNTI | Not Present | |
| New C-RNTI | Not Present | |
| New DSCH-RNTI | Not Present | R99 and Rel-4 only |
| New H-RNTI | Not Present | Rel-5 |
| New Primary E-RNTI | Not Present | Rel-6 |
| New Secondary E-RNTI | Not Present | Rel-6 |
| RRC State indicator | CELL_DCH | |
| UTRAN DRX cycle length coefficient | Not Present | |
| CN information info | Not Present | |
| URA identity | Not Present | |
| CHOICE *specification mode* | Complete specification | Rel-5 |
| - RAB information for setup | | |
| - RAB info | | |
| - RAB identity | 0000 0001B<br>The first/ leftmost bit of the bit string contains the most significant bit of the RAB identity. | |
| - CN domain identity | CS domain | |
| - NAS Synchronization Indicator | Not Present | |
| - Re-establishment timer | UseT314 | |
| - RB information to setup | | |
| - RB identity | 10 | |
| - PDCP info | Not Present | |
| - CHOICE RLC info type | RLC info | |
| - CHOICE Uplink RLC mode | TM RLC | |
| - Transmission RLC discard | Not Present | |
| - Segmentation indication | FALSE | |
| - CHOICE Downlink RLC mode | TM RLC | |
| - Segmentation indication | FALSE | |
| - RB mapping info | | |
| - Information for each multiplexing option | | |

| Information Element | Value/remark | Version |
|---|---|---|
| - RLC logical channel mapping indicator | Not Present | |
| - Number of uplink RLC logical channels | 1 | |
| - Uplink transport channel type | DCH | |
| - UL Transport channel identity | 1 | |
| - Logical channel identity | Not Present | |
| - CHOICE RLC size list | Configured | |
| - MAC logical channel priority | 7 | |
| - Downlink RLC logical channel info | | |
| - Number of downlink RLC logical channels | 1 | |
| - Downlink transport channel type | DCH | |
| - DL DCH Transport channel identity | 6 | |
| - DL DSCH Transport channel identity | Not Present | |
| - Logical channel identity | Not Present | |
| RB information to reconfigure list | Not Present | Rel-6 |
| RB information to be affected | Not Present | |
| Downlink counter synchronization info | Not Present | |
| | RMC for BTFD | |
| UL Transport channel information for all transport channels | | |
| - PRACH TFCS | Not Present | |
| - CHOICE mode | FDD | |
| - TFC subset | Not Present | |
| - UL DCH TFCS | | |
| - CHOICE TFCI signalling | Normal | |
| - TFCI Field 1 information | | |
| - CHOICE TFCS representation | Complete reconfiguration | |
| - TFCS complete reconfigure information | | |
| - CHOICE CTFC Size | ctfc6Bit | |
| - ctfc6Bit | 22 | |
| - ctfc6 | 0 | |
| -powerOffsetInformation(OP) | | |
| -gainFactorInformation | ComputedGainFactors | |
| - Reference TFC ID | 0 | |
| - ctfc6 | 11 | |
| -powerOffsetInformation(OP) | | |
| -gainFactorInformation | ComputedGainFactors | |
| - Reference TFC ID | 0 | |
| - ctfc6 | 1 | |
| -powerOffsetInformation(OP) | | |
| -gainFactorInformation | ComputedGainFactors | |
| - Reference TFC ID | 0 | |
| - ctfc6 | 12 | |
| -powerOffsetInformation(OP) | | |
| -gainFactorInformation | SignalledGainFactors | |
| -modeSpecificInfo | Fdd | |
| -fdd | | |
| - Gain factor ßc | 8 | |
| - Gain factor ßd | 15 | |
| - Reference TFC ID | 0 | |
| - ctfc6 | 2 | |
| -powerOffsetInformation(OP) | | |
| -gainFactorInformation | ComputedGainFactors | |
| - Reference TFC ID | 0 | |
| - ctfc6 | 13 | |
| -powerOffsetInformation(OP) | | |
| -gainFactorInformation | ComputedGainFactors | |
| - Reference TFC ID | 0 | |
| - ctfc6 | 3 | |
| -powerOffsetInformation(OP) | | |
| -gainFactorInformation | ComputedGainFactors | |
| - Reference TFC ID | 0 | |
| - ctfc6 | 14 | |
| -powerOffsetInformation(OP) | | |
| -gainFactorInformation | ComputedGainFactors | |
| - Reference TFC ID | 0 | |
| - ctfc6 | 4 | |
| -powerOffsetInformation(OP) | | |
| -gainFactorInformation | ComputedGainFactors | |

| Information Element | Value/remark | Version |
|---|---|---|
| - Reference TFC ID | 0 | |
| - ctfc6 | 15 | |
| -powerOffsetInformation(OP) | | |
| -gainFactorInformation | ComputedGainFactors | |
| - Reference TFC ID | 0 | |
| - ctfc6 | 5 | |
| -powerOffsetInformation(OP) | | |
| -gainFactorInformation | ComputedGainFactors | |
| - Reference TFC ID | 0 | |
| - ctfc6 | 16 | |
| -powerOffsetInformation(OP) | | |
| -gainFactorInformation | ComputedGainFactors | |
| - Reference TFC ID | 0 | |
| - ctfc6 | 6 | |
| -powerOffsetInformation(OP) | | |
| -gainFactorInformation | ComputedGainFactors | |
| - Reference TFC ID | 1 | |
| - ctfc6 | 17 | |
| -powerOffsetInformation(OP) | | |
| -gainFactorInformation | SignalledGainFactors | |
| -modeSpecificInfo | Fdd | |
| -fdd | | |
| - Gain factor ßc | 11 | |
| - Gain factor ßd | 15 | |
| - Reference TFC ID | 1 | |
| - ctfc6 | 7 | |
| -powerOffsetInformation(OP) | | |
| -gainFactorInformation | ComputedGainFactors | |
| - Reference TFC ID | 1 | |
| - ctfc6 | 18 | |
| -powerOffsetInformation(OP) | | |
| -gainFactorInformation | ComputedGainFactors | |
| - Reference TFC ID | 1 | |
| - ctfc6 | 8 | |
| -powerOffsetInformation(OP) | | |
| -gainFactorInformation | ComputedGainFactors | |
| - Reference TFC ID | 1 | |
| - ctfc6 | 19 | |
| -powerOffsetInformation(OP) | | |
| -gainFactorInformation | ComputedGainFactors | |
| - Reference TFC ID | 1 | |
| - ctfc6 | 9 | |
| -powerOffsetInformation(OP) | | |
| -gainFactorInformation | ComputedGainFactors | |
| - Reference TFC ID | 1 | |
| - ctfc6 | 20 | |
| -powerOffsetInformation(OP) | | |
| -gainFactorInformation | ComputedGainFactors | |
| - Reference TFC ID | 1 | |
| - ctfc6 | 10 | |
| -powerOffsetInformation(OP) | | |
| -gainFactorInformation | ComputedGainFactors | |
| - Reference TFC ID | 1 | |
| - ctfc6 | 21 | |
| -powerOffsetInformation(OP) | | |
| -gainFactorInformation | ComputedGainFactors | |
| - Reference TFC ID | 1 | |
| Added or Reconfigured UL TrCH information list | 1 | |
| - Added or Reconfigured UL TrCH information | | |
| - Uplink transport channel type | DCH | |
| - UL Transport channel identity | 1 | |
| - TFS | | |
| - CHOICE Transport channel type | Dedicated transport channels | |
| -DedicatedDynamicTF-Info | | |
| RLC size | 256 | |
| -numberOfTbSizeList | | |
| -NumberOfTransportBlocks | Zero | |

| Information Element | Value/remark | Version |
|---|---|---|
| -NumberOfTransportBlocks | One | |
| - Choice Logical channel List | ALL | |
| RLC size | 216 | |
| -numberOfTbSizeList | | |
| -NumberOfTransportBlocks | One | |
| - Choice Logical channel List | ALL | |
| RLC size | 171 | |
| -numberOfTbSizeList | | |
| -NumberOfTransportBlocks | One | |
| - Choice Logical channel List | ALL | |
| RLC size | 160 | |
| -numberOfTbSizeList | | |
| -NumberOfTransportBlocks | One | |
| - Choice Logical channel List | ALL | |
| RLC size | 146 | |
| -numberOfTbSizeList | | |
| -NumberOfTransportBlocks | One | |
| - Choice Logical channel List | ALL | |
| RLC size | 130 | |
| -numberOfTbSizeList | | |
| -NumberOfTransportBlocks | One | |
| - Choice Logical channel List | ALL | |
| RLC size | 115 | |
| -numberOfTbSizeList | | |
| -NumberOfTransportBlocks | One | |
| - Choice Logical channel List | ALL | |
| RLC size | 107 | |
| -numberOfTbSizeList | | |
| -NumberOfTransportBlocks | One | |
| - Choice Logical channel List | ALL | |
| RLC size | 51 | |
| -numberOfTbSizeList | | |
| -NumberOfTransportBlocks | One | |
| - Choice Logical channel List | ALL | |
| RLC size | 12 | |
| -numberOfTbSizeList | | |
| -NumberOfTransportBlocks | One | |
| - Choice Logical channel List | ALL | |
| -Semistatic Transport Format Information | | |
| -Transmission Time interval | 20 ms | |
| -channelCodingType | Convolutional | |
| -convolutional | 1/3 | |
| - Rate matching attribute | 256 | |
| - CRC size | 0 | |
| DL Transport channel information common for all transport channel | | |
| - SCCPCH TFCS | Not Present | |
| - CHOICE mode | FDD | |
| - CHOICE DL parameters | Explicit | |
| - DL DCH TFCS | | |
| - CHOICE TFCI signalling | Normal | |
| - TFCI Field 1 information | | |
| - CHOICE TFCS representation | Complete reconfiguration | |
| - TFCS complete reconfigure information | | |
| - CHOICE CTFC Size | Ctfc6Bit | |
| - ctfc6Bit | 18 | |
| - ctfc6 | 9 | |
| - ctfc6 | 0 | |
| - ctfc6 | 10 | |
| - ctfc6 | 1 | |
| - ctfc6 | 11 | |
| - ctfc6 | 2 | |
| - ctfc6 | 12 | |
| - ctfc6 | 3 | |
| - ctfc6 | 13 | |
| - ctfc6 | 4 | |
| - ctfc6 | 14 | |

| Information Element | Value/remark | Version |
|---|---|---|
| - ctfc6 | 5 | |
| - ctfc6 | 15 | |
| - ctfc6 | 6 | |
| - ctfc6 | 16 | |
| - ctfc6 | 7 | |
| - ctfc6 | 17 | |
| - ctfc6 | 8 | |
| Deleted DL TrCH information | Not Present | |
| Added or Reconfigured DL TrCH information list | 1 | |
| - Added or Reconfigured DL TrCH information | | |
| - Downlink transport channel type | DCH | |
| - DL Transport channel identity | 6 | |
| - CHOICE DL parameters | Explicit | |
| - TFS | | |
| - CHOICE Transport channel type | Dedicated transport channels | |
| -DedicatedDynamicTF-Info | | |
| RLC size | 244 | |
| -numberOfTbSizeList | | |
| -NumberOfTransportBlocks | One | |
| - Choice Logical channel List | ALL | |
| RLC size | 204 | |
| -numberOfTbSizeList | | |
| -NumberOfTransportBlocks | One | |
| - Choice Logical channel List | ALL | |
| RLC size | 159 | |
| -numberOfTbSizeList | | |
| -NumberOfTransportBlocks | One | |
| - Choice Logical channel List | ALL | |
| RLC size | 148 | |
| -numberOfTbSizeList | | |
| -NumberOfTransportBlocks | One | |
| - Choice Logical channel List | ALL | |
| RLC size | 134 | |
| -numberOfTbSizeList | | |
| -NumberOfTransportBlocks | One | |
| - Choice Logical channel List | ALL | |
| RLC size | 118 | |
| -numberOfTbSizeList | | |
| -NumberOfTransportBlocks | One | |
| - Choice Logical channel List | ALL | |
| RLC size | 103 | |
| -numberOfTbSizeList | | |
| -NumberOfTransportBlocks | One | |
| - Choice Logical channel List | ALL | |
| RLC size | 95 | |
| -numberOfTbSizeList | | |
| -NumberOfTransportBlocks | One | |
| - Choice Logical channel List | ALL | |
| RLC size | 39 | |
| -numberOfTbSizeList | | |
| -NumberOfTransportBlocks | One | |
| - Choice Logical channel List | ALL | |
| -Semistatic Transport Format Information | | |
| -Transmission Time interval | 20 ms | |
| -channelCodingType | Convolutional | |
| -convolutional | 1/3 | |
| - Rate matching attribute | 256 | |
| - CRC size | 12 | |
| - DCH quality target | | |
| - BLER Quality value | -2.0 | |
| - Transparent mode signalling info | Not Present | |
| Frequency info | Not Present | |
| Maximum allowed UL TX power | 33 dBm | |
| CHOICE channel requirement | Uplink DPCH info | Rel-5 and earlier |
| Uplink DPCH info | | Rel-6 |
| - Uplink DPCH power control info | | |

| Information Element | Value/remark | Version |
|---|---|---|
| - DPCCH power offset | -80dB IE value will have no effect on the UE UL power when closed loop power control is active | |
| - PC Preamble | 1 frame | |
| - SRB delay | 7 frames | |
| - Power Control Algorithm | Algorithm1 | |
| - TPC step size | 1dB | |
| - $\Delta_{ACK}$ | Not Present | Rel-5 |
| - $\Delta_{NACK}$ | Not Present | Rel-5 |
| - Ack-Nack repetition factor | Not Present | Rel-5 |
| - Scrambling code type | Long | |
| - Scrambling code number | 0 | |
| - Number of DPDCH | 1 | |
| - spreading factor | 64 | |
| - TFCI existence | TRUE | |
| - Number of FBI bit | Not Present(0) | |
| - Puncturing Limit | 1 | |
| CHOICE Mode | FDD | R99 and Rel-4 only |
| - Downlink PDSCH information | Not Present(0) | R99 and Rel-4 only |
| E-DCH Info | Not Present | Rel-6 |
| Downlink HS-PDSCH Information | Not Present | Rel-5 |
| Downlink information common for all radio links | | |
| - Downlink DPCH info common for all RL | FDD | |
| - Timing indicator | Maintain | |
| - CFN-targetSFN frame offset | Not Present | |
| - Downlink DPCH power control information | | |
| - DPC mode | 0 (single) | |
| - CHOICE mode | FDD | |
| - Power offset $P_{Pilot-DPDCH}$ | 0 | |
| - DL rate matching restriction information | Not Present | |
| - Spreading factor | 128 | |
| - Number of bits for Pilot bits(SF=128,256) | 4 | |
| - Fixed or Flexible Position | Fixed | |
| - TFCI existence | FALSE | |
| - DPCH compressed mode info | Not Present | |
| - TX Diversity mode | None | |
| - SSDT information | Not Present | R99 and Rel-4 only |
| - Default DPCH Offset Value | Not Present | |
| Downlink information for each radio link list | | |
| - Primary CPICH info | | |
| - Primary scrambling code | Reference to clause 6.1 "Default settings (FDD)" | |
| - PDSCH with SHO DCH info | Not Present | R99 and Rel-4 only |
| - PDSCH code mapping | Not Present | R99 and Rel-4 only |
| - Downlink DPCH info for each RL | | |
| - Primary CPICH usage for channel estimation | Primary CPICH may be used | |
| - DPCH frame offset | Set to value Default DPCH Offset Value (as currently stored in SS) mod 38 400 | |
| - Secondary CPICH info | Not Present | |
| - DL channelisation code | | |
| - Secondary scrambling code | Not Present | |
| - Spreading factor | 128 | |
| - Code number | 96 | |
| - Scrambling code change | No change | |
| - TPC combination index | 0 | |
| - SSDT Cell Identity | Not Present | R99 and Rel-4 only |
| - Closed loop timing adjustment mode | Not Present | |
| - SCCPCH information for FACH | Not Present | R99 and Rel-4 only |
| MBMS PL Service Restriction Information | Not Present | Rel-6 |

Contents of RADIO BEARER SETUP message: AM or UM (E-DCH and HSDPA)

| Information Element | Value/remark | Version |
|---|---|---|
| Message Type | | |
| RRC transaction identifier | Arbitrarily selects an integer between 0 and 3 | |
| Integrity check info | | |
| - message authentication code | SS calculates the value of MAC-I for this message and writes to this IE. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I. | |
| - RRC message sequence number | SS provides the value of this IE, from its internal counter. | |
| Integrity protection mode info | Not Present | |
| Ciphering mode info | Not Present | |
| Activation time | Not Present | |
| New U-RNTI | Not Present | |
| New C-RNTI | Not Present | |
| New DSCH-RNTI | Not Present | R99 and Rel-4 only |
| New H-RNTI | '1010 1010 1010 1010' | Rel-5 |
| New Primary E-RNTI | '1010 1010 1010 1010' | Rel-6 |
| New Secondary E-RNTI | Not Present | Rel-6 |
| RRC State indicator | CELL_DCH | |
| UTRAN DRX cycle length coefficient | Not Present | |
| CN information info | Not Present | |
| URA identity | Not Present | |
| CHOICE specification mode | Complete specification | Rel-6 |
| - Signalling RB information to setup | Not Present | |
| - RAB information for setup list | | |
| - RAB information for setup | | |
| - RAB info | (high-speed UM DTCH for PS domain) | |
| - RAB identity | 0000 0110B<br>The first/ leftmost bit of the bit string contains the most significant bit of the RAB identity. | |
| - CN domain identity | PS domain | |
| - NAS Synchronization Indicator | Not Present | |
| - Re-establishment timer | useT315 | |
| - RB information to setup | | |
| - RB identity | 25 | |
| - PDCP info | Not present | |
| - CHOICE RLC info type | RLC info | |
| - CHOICE Uplink RLC mode | UM RLC | |
| - Transmission RLC discard | Not present | |
| - CHOICE Downlink RLC mode | UM RLC | |
| - DL UM RLC LI size | 15 | Rel-5 |
| - DL Reception Window Size | Not present | Rel-6 |
| - One sided RLC re-establishment | FALSE | |
| - Alternative E-bit interpretation | Not present | Rel-6 |
| - RB mapping info | | |
| - Information for each multiplexing option | 1 RBMuxOptions | |
| - RLC logical channel mapping indicator | Not Present | |
| - Number of uplink RLC logical channels | 1 | |
| - Uplink transport channel type | E-DCH | |
| - Logical channel identity | 7 | |
| - E-DCH MAC-d flow identity | 2 | |
| - DDI | 5 | |
| - RLC PDU size list | 1 RLC PDU size | |
| - RLC PDU size | 336 bits | |
| - Include in scheduling info | TRUE | |
| - MAC logical channel priority | 8 | |
| - Downlink RLC logical channel info | | |
| - Number of downlink RLC logical channels | 1 | |
| - Downlink transport channel type | HS-DSCH | |
| - DL DCH Transport channel identity | Not Present | |

| Information Element | Value/remark | Version |
|---|---|---|
| - DL DSCH Transport channel identity<br>- DL HS-DSCH MAC-d flow identity<br>- Logical channel identity | Not Present<br>0<br>Not Present | |
| RB information to reconfigure list<br>RB information to be affected<br>Downlink counter synchronization info<br>PDCP ROHC target mode | Not Present<br>Not Present<br>Not Present<br>Not Present | Rel-6<br><br><br>Rel-5 |
| UL Transport channel information for all transport channels | Not Present | |
| Deleted UL TrCH information | Not Present | |
| Added or Reconfigured TrCH information list<br>- Added or Reconfigured UL TrCH information<br>- Uplink transport channel type<br>- CHOICE UL parameters<br>- E-DCH Transmission Time Interval<br>- HARQ info for E-DCH<br>- HARQ RV Configuration<br>- Added or reconfigured E-DCH MAC-d flow<br>- E-DCH MAC-d flow identity<br>- E-DCH MAC-d flow power offset<br>- E-DCH MAC-d flow maximum number of retransmissions<br>- E-DCH MAC-d flow multiplexing list<br>- CHOICE transmission grant type | 1 TrCH added<br>1 E-DCH added<br>E-DCH<br>E-DCH<br>10 ms<br><br><br>Not Present<br><br><br>2<br>0<br>7<br><br>Not Present<br>Scheduled grant info | |
| CHOICE *mode* | Not Present | R99 and Rel-4 only |
| DL Transport channel information common for all transport channels | Not Present | |
| Deleted TrCH information list | Not Present | |
| Added or Reconfigured TrCH information list<br>- Added or Reconfigured DL TrCH information<br>- Downlink transport channel type<br>- DL Transport channel identity<br>- CHOICE DL parameters<br>- HARQ Info<br>- Number of Processes<br>- CHOICE *Memory Partitioning*<br>- Memory size<br>- Process Memory Size<br>- Added or reconfigured MAC-d flow<br>- MAC-hs queue to add or reconfigure list<br>- MAC-hs queue Id<br>- MAC-d Flow Identity<br>- T1<br>- MAC-hs window size<br>- MAC-d PDU size Info<br>- MAC-d PDU size<br>- MAC-d PDU size index<br>- MAC-hs queue to delete list<br>- DCH quality target | 1 TrCH added<br>HS-DSCH for DTCH added<br><br>HS-DSCH<br>Not Present<br>HS-DSCH<br><br>Reference to TS34.121 [2] Annex C Fixed Reference Channels<br>Explicit<br>Reference to TS34.121 [2] Annex C Fixed Reference Channels parameter “Number of HARQ Processes".<br>Reference to TS34.121 [2] Annex C Fixed Reference Channels parameter “Number of SML’s per HARQ Proc.”.<br><br>(one queue)<br>0<br>0<br>50<br>16<br><br>Reference to TS34.121 [2] Annex C Fixed Reference Channels<br>0<br>Not present<br>Not present | |
| Frequency info | Not present | |
| Maximum allowed UL TX power | 33dBm | |
| CHOICE channel requirement | Uplink DPCH info | Rel-5 and earlier |
| Uplink DPCH info<br>- Uplink DPCH power control info<br>- DPCCH power offset | <br><br>-80dB | Rel-6 |

| Information Element | Value/remark | Version |
|---|---|---|
| - PC Preamble | 1 frame | |
| - SRB delay | 7 frames | |
| - Power Control Algorithm | Algorithm1 | |
| - TPC step size | 1dB | |
| - $\Delta_{ACK}$ | 3 | |
| - $\Delta_{NACK}$ | 3 | |
| - Ack-Nack repetition factor | 1 | |
| - Scrambling code type | Long | |
| - Scrambling code number | 0 (0 to 16777215) | |
| - Number of DPDCH | Not Present(1) | |
| - spreading factor | Reference to clause 6.10 Parameter Set | |
| - TFCI existence | Reference to clause 6.10 Parameter Set | |
| - Number of FBI bit | Reference to clause 6.10 Parameter Set | |
| - Puncturing Limit | Reference to clause 6.10 Parameter Set | |
| E-DCH info | NOTE: The values in this IE need to be adapted to the E-DCH RF test cases. | Rel-6 |
| - MAC-es/e reset indicator | TRUE | |
| - E-DPCCH info | | |
| - E-DPCCH/DPCCH power offset | 0 | |
| - Happy bit delay condition | 100 ms | |
| - E-DPDCH info | | |
| - E-TFCI table index | 0 | |
| - E-DCH minimum set E-TFCI | 9 | |
| - Reference E-TFCIs | 4 E-TFCIs | |
| - Reference E-TFCI | 11 | |
| - Reference E-TFCI PO | 4 | |
| - Reference E-TFCI | 83 | |
| - Reference E-TFCI PO | 16 | |
| - Reference E-TFCI | 110 | |
| - Reference E-TFCI PO | 19 | |
| - Reference E-TFCI | 115 | |
| - Reference E-TFCI PO | 21 | |
| - Maximum channelisation codes | 2sf4 | |
| - PLnon-max | 0.84 | |
| - Scheduling Information Configuration | | |
| - Periodicity for Scheduling Info – no grant | Not present | |
| - Periodicity for Scheduling Info – grant | Not present | |
| - Power Offset for Scheduling Info | 0 | |
| - Scheduled Transmission configuration | | |
| - 2ms scheduled transmission grant HARQ process allocation | Not present | |
| - Serving Grant | Not present | |
| - Primary/Secondary Grant Selector | Primary | |
| - 3-Index-Step Threshold | Not present | |
| - 2-Index-Step Threshold | Not present | |
| CHOICE Mode | FDD | R99 and Rel-4 only |
| - Downlink PDSCH information | Not Present | R99 and Rel-4 only |
| Downlink HS-PDSCH Information | | |
| - HS-SCCH Info | | |
| - CHOICE mode | FDD | |
| - DL Scrambling Code | Not present | |
| - HS-SCCH Channelisation Code Information | | |
| - HS-SCCH Channelisation Code | 2 | |
| - HS-SCCH Channelisation Code | 3 | |
| - Measurement Feedback Info | | |
| - CHOICE mode | FDD | |
| - POhsdsch | 6 dB | |
| - CQI Feedback cycle, k | 2 ms | |
| - CQI repetition factor | 1 | |
| - $\Delta_{CQI}$ | 5 (corresponds to 0dB in relative power offset) | |
| - CHOICE mode | FDD (no data) | |
| Downlink information common for all radio links | Not Present | |
| Downlink information for each radio link list | | |

| Information Element | Value/remark | Version |
|---|---|---|
| - Downlink information for each radio link | | |
| - Choice mode | FDD | |
| - Primary CPICH info | | |
| - Primary scrambling code | Ref. to clause 6.1 "Default settings (FDD)" | |
| - PDSCH with SHO DCH info | Not Present | R99 and Rel-4 only |
| - PDSCH code mapping | Not Present | R99 and Rel-4 only |
| - Serving HS-DSCH radio link indicator | TRUE | |
| - Serving E-DCH radio link indicator | TRUE | |
| - Downlink DPCH info for each RL | | |
| - CHOICE mode | FDD | |
| - Primary CPICH usage for channel estimation | Primary CPICH may be used | |
| - DPCH frame offset | Set to value Default DPCH Offset Value (as currently stored in SS) mod 38 400 | |
| - Secondary CPICH info | Not Present | |
| - DL channelisation code | | |
| - Secondary scrambling code | 1 | |
| - Spreading factor | Reference to clause 6.10 Parameter Set | |
| - Code number | 0 | |
| - Scrambling code change | No code change | |
| - TPC combination index | 0 | |
| - SSDT Cell Identity | Not Present | R99 and Rel-4 only |
| - Closed loop timing adjustment mode | Not Present | |
| - E-AGCH Info | | Rel-6 |
| - E-AGCH Channelisation Code | 14 | |
| - CHOICE E-HICH Information | | Rel-6 |
| - E-HICH Information | | |
| - DL Scrambling code | Not Present (default is primary) | |
| - Channelisation code | 6 | |
| - Signature sequence | 1 | |
| - CHOICE E-RGCH Information | | Rel-6 |
| - E-RGCH Information | | |
| - Signature Sequence | 0 | |
| - RG combination index | 0 | |
| - SCCPCH information for FACH | Not Present | R99 and Rel-4 only |
| MBMS PL Service Restriction Information | Not Present | Rel-6 |

Contents of RRC CONNECTION RELEASE message: UM

| Information Element | Value/remark | Version |
|---|---|---|
| Message Type | | |
| U-RNTI | This IE is set to the following value when the message is transmitted on the CCCH. When transmitted on DCCH, this is absent. | R99, Rel-4 |
| - SRNC identity | 0000 0000 0001B | |
| - S-RNTI | 0000 0000 0000 0000 0001B | |
| CHOICE identity type | This IE is set to the following value when the message is transmitted on the CCCH. When transmitted on DCCH, this is absent. | Rel-5 |
| - U-RNTI | | |
| - SRNC identity | 0000 0000 0001B | |
| - S-RNTI | 0000 0000 0000 0000 0001B | |
| RRC transaction identifier | Arbitrarily selects an integer between 0 and 3 | |
| Integrity check info | This IE is present when this message is transmitted on downlink DCCH. Else, this IE and the sub-IEs are omitted. | |
| - Message authentication code | SS calculates the value of MAC-I for this message and writes to this IE. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I. | |
| - RRC Message sequence number | SS provides the value of this IE, from its internal counter. | |
| N308 | 2 (for CELL_DCH state). Not Present (for UE in other connected mode states). | |
| Release cause | Normal event | |
| Rplmn information | Not Present | |

| | | |
|---|---|---|
| Redirection info | Not Present | Rel-6 |

Contents of RRC CONNECTION SETUP message: UM

| Information Element | Value/remark | Version |
|---|---|---|
| Message Type | | |
| Initial UE identity | Select the same identity as in the IE "Initial UE Identity" in received "RRC CONNECTION REQUEST" message | |
| RRC transaction identifier | Arbitrarily selects an integer between 0 and 3 | |
| Activation time | Not Present(Now) | |
| New U-RNTI | | |
| - SRNC identity | 0000 0000 0001B | |
| - S-RNTI | 0000 0000 0000 0000 0001B | |
| New C-RNTI | Not Present | |
| New H-RNTI | Not Present | Rel-6 |
| New Primary E-RNTI | Not Present | Rel-6 |
| New Secondary E-RNTI | Not Present | Rel-6 |
| RRC State Indicator | CELL_DCH | |
| UTRAN DRX cycle length coefficient | 9 | |
| Capability update requirement | | |
| - UE radio access FDD capability update requirement | TRUE | |
| - UE radio access TDD capability update requirement | FALSE | |
| - System specific capability update requirement list | GSM | |
| CHOICE *specification mode* | Complete specification | Rel-5 |
| - Complete specification | | Rel-5 |
| - Signalling RB information to setup list | 4 SRBs | |
| - Signalling RB information to setup | (UM DCCH for RRC) | |
| - RB identity | Not Present | |
| - CHOICE RLC info type | RLC info | |
| - CHOICE Uplink RLC mode | UM RLC | |
| - Transmission RLC discard | Not Present | |
| - CHOICE Downlink RLC mode | UM RLC | |
| - RB mapping info | | |
| - Information for each multiplexing option | 2 RBMuxOptions | |
| - RLC logical channel mapping indicator | Not Present | |
| - Number of RLC logical channels | 1 | |
| - Uplink transport channel type | DCH | |
| - UL Transport channel identity | 5 | |
| - Logical channel identity | 1 | |
| - CHOICE RLC size list | Configured | |
| - MAC logical channel priority | 1 | |
| - Downlink RLC logical channel info | | |
| - Number of RLC logical channels | 1 | |
| - Downlink transport channel type | DCH | |
| - DL DCH Transport channel identity | 10 | |
| - DL DSCH Transport channel identity | Not Present | |
| - Logical channel identity | 1 | |
| - RLC logical channel mapping indicator | Not Present | |
| - Number of RLC logical channels | 1 | |
| - Uplink transport channel type | RACH | |
| - UL Transport channel identity | Not Present | |
| - Logical channel identity | 1 | |
| - CHOICE RLC size list | Explicit List | |
| - RLC size index | Reference to clause 6 Parameter Set | |
| - MAC logical channel priority | 1 | |
| - Downlink RLC logical channel info | | |
| - Number of RLC logical channels | 1 | |
| - Downlink transport channel type | FACH | |
| - DL DCH Transport channel identity | Not Present | |
| - DL DSCH Transport channel identity | Not Present | |
| - Logical channel identity | 1 | |
| - Signalling RB information to setup | (AM DCCH for RRC) | |
| - RB identity | Not Present | |
| - CHOICE RLC info type | | |
| - RLC info | | |

| Information Element | Value/remark | Version |
|---|---|---|
| - CHOICE Uplink RLC mode | AM RLC | |
| - Transmission RLC discard | | |
| - SDU discard mode | No Discard | |
| - MAX_DAT | 15 | |
| - Transmission window size | 128 | |
| - Timer_RST | 500 | |
| - Max_RST | 1 | |
| - Polling info | | |
| - Timer_poll_prohibit | 200 | |
| - Timer_poll | 200 | |
| - Poll_PDU | Not Present | |
| - Poll_SDU | 1 | |
| - Last transmission PDU poll | TRUE | |
| - Last retransmission PDU poll | TRUE | |
| - Poll_Windows | 99 | |
| - Timer_poll_periodic | Not Present | |
| - CHOICE Downlink RLC mode | AM RLC | |
| - In-sequence delivery | TRUE | |
| - Receiving window size | 128 | |
| - Downlink RLC status info | | |
| - Timer_status_prohibit | 200 | |
| - Timer_EPC | Not Present | |
| - Missing PDU indicator | TRUE | |
| - Timer_STATUS_periodic | Not Present | |
| - RB mapping info | | |
| - Information for each multiplexing option | 2 RBMuxOptions | |
| - RLC logical channel mapping indicator | Not Present | |
| - Number of RLC logical channels | 1 | |
| - Uplink transport channel type | DCH | |
| - UL Transport channel identity | 5 | |
| - Logical channel identity | 2 | |
| - CHOICE RLC size list | Configured | |
| - MAC logical channel priority | 2 | |
| - Downlink RLC logical channel info | | |
| - Number of RLC logical channels | 1 | |
| - Downlink transport channel type | DCH | |
| - DL DCH Transport channel identity | 10 | |
| - DL DSCH Transport channel identity | Not Present | |
| - Logical channel identity | 2 | |
| - RLC logical channel mapping indicator | Not Present | |
| - Number of RLC logical channels | 1 | |
| - Uplink transport channel type | RACH | |
| - UL Transport channel identity | Not Present | |
| - Logical channel identity | 2 | |
| - CHOICE RLC size list | Explicit List | |
| - RLC size index | Reference to clause 6 Parameter Set | |
| - MAC logical channel priority | 2 | |
| - Downlink RLC logical channel info | | |
| - Number of RLC logical channels | 1 | |
| - Downlink transport channel type | FACH | |
| - DL DCH Transport channel identity | Not Present | |
| - DL DSCH Transport channel identity | Not Present | |
| - Logical channel identity | 2 | |
| - Signalling RB information to setup | (AM DCCH for NAS_DT High priority) | |
| - RB identity | Not Present | |
| - CHOICE RLC info type | | |
| - RLC info | | |
| - CHOICE Uplink RLC mode | AM RLC | |
| - Transmission RLC discard | | |
| - SDU discard mode | No Discard | |
| - MAX_DAT | 15 | |
| - Transmission window size | 128 | |
| - Timer_RST | 500 | |
| - Max_RST | 1 | |
| - Polling info | | |
| - Timer_poll_prohibit | 200 | |
| - Timer_poll | 200 | |

| Information Element | Value/remark | Version |
|---|---|---|
| - Poll_PDU | Not Present | |
| - Poll_SDU | 1 | |
| - Last transmission PDU poll | TRUE | |
| - Last retransmission PDU poll | TRUE | |
| - Poll_Windows | 99 | |
| - Timer_poll_periodic | Not Present | |
| - CHOICE Downlink RLC mode | AM RLC | |
| - In-sequence delivery | TRUE | |
| - Receiving window size | 128 | |
| - Downlink RLC status info | | |
| - Timer_status_prohibit | 200 | |
| - Timer_EPC | Not Present | |
| - Missing PDU indicator | TRUE | |
| - Timer_STATUS_periodic | Not Present | |
| - RB mapping info | | |
| - Information for each multiplexing option | 2 RBMuxOptions | |
| - RLC logical channel mapping indicator | Not Present | |
| - Number of RLC logical channels | 1 | |
| - Uplink transport channel type | DCH | |
| -UL Transport channel identity | 5 | |
| - Logical channel identity | 3 | |
| - CHOICE RLC size list | Configured | |
| - MAC logical channel priority | 3 | |
| - Downlink RLC logical channel info | | |
| - Number of RLC logical channels | 1 | |
| - Downlink transport channel type | DCH | |
| - DL DCH Transport channel identity | 10 | |
| - DL DSCH Transport channel identity | Not Present | |
| - Logical channel identity | 3 | |
| - RLC logical channel mapping indicator | Not Present | |
| - Number of RLC logical channels | 1 | |
| - Uplink transport channel type | RACH | |
| - UL Transport channel identity | Not Present | |
| - Logical channel identity | 3 | |
| - CHOICE RLC size list | Explicit List | |
| - RLC size index | Reference to clause 6 Parameter Set | |
| - MAC logical channel priority | 3 | |
| - Downlink RLC logical channel info | | |
| - Number of RLC logical channels | 1 | |
| - Downlink transport channel type | FACH | |
| - DL DCH Transport channel identity | Not Present | |
| - DL DSCH Transport channel identity | Not Present | |
| - Logical channel identity | 3 | |
| - Signalling RB information to setup | (AM DCCH for NAS_DT Low priority) | |
| - RB identity | Not Present | |
| - CHOICE RLC info type | | |
| - RLC info | | |
| - CHOICE Uplink RLC mode | AM RLC | |
| - Transmission RLC discard | | |
| - SDU discard mode | No Discard | |
| - MAX_DAT | 15 | |
| - Transmission window size | 128 | |
| - Timer_RST | 500 | |
| - Max_RST | 1 | |
| - Polling info | | |
| - Timer_poll_prohibit | 200 | |
| - Timer_poll | 200 | |
| - Poll_PDU | Not Present | |
| - Poll_SDU | 1 | |
| - Last transmission PDU poll | TRUE | |
| - Last retransmission PDU poll | TRUE | |
| - Poll_Windows | 99 | |
| - Timer_poll_periodic | Not Present | |
| - CHOICE Downlink RLC mode | AM RLC | |
| - In-sequence delivery | TRUE | |
| - Receiving window size | 128 | |
| - Downlink RLC status info | | |

| Information Element | Value/remark | Version |
|---|---|---|
| - Timer_status_prohibit | 200 | |
| - Timer_EPC | Not Present | |
| - Missing PDU indicator | TRUE | |
| - Timer_STATUS_periodic | Not Present | |
| - RB mapping info | | |
| - Information for each multiplexing option | 2 RBMuxOptions | |
| - RLC logical channel mapping indicator | Not Present | |
| - Number of RLC logical channels | 1 | |
| - Uplink transport channel type | DCH | |
| - UL Transport channel identity | 5 | |
| - Logical channel identity | 4 | |
| - CHOICE RLC size list | Configured | |
| - MAC logical channel priority | 4 | |
| - Downlink RLC logical channel info | | |
| - Number of RLC logical channels | 1 | |
| - Downlink transport channel type | DCH | |
| - DL DCH Transport channel identity | 10 | |
| - DL DSCH Transport channel identity | Not Present | |
| - Logical channel identity | 4 | |
| - RLC logical channel mapping indicator | Not Present | |
| - Number of RLC logical channels | 1 | |
| - Uplink transport channel type | RACH | |
| - UL Transport channel identity | Not Present | |
| - Logical channel identity | 4 | |
| - CHOICE RLC size list | Explicit List | |
| - RLC size index | Reference to clause 6 Parameter Set | |
| - MAC logical channel priority | 4 | |
| - Downlink RLC logical channel info | | |
| - Number of RLC logical channels | 1 | |
| - Downlink transport channel type | FACH | |
| - DL DCH Transport channel identity | Not Present | |
| - DL DSCH Transport channel identity | Not Present | |
| - Logical channel identity | 4 | |
| UL Transport channel information for all transport channels | | |
| - PRACH TFCS | Not Present | |
| - CHOICE Mode | FDD | |
| - TFC subset | Not Present | |
| - UL DCH TFCS | | |
| - CHOICE TFCI signalling | Normal | |
| - TFCI Field 1 information | | |
| - CHOICE TFCS representation | Complete reconfiguration | |
| - TFCS complete reconfiguration information | | |
| - CHOICE CTFC Size | 2 bit CTFC | |
| - CTFC information | 2 TFCs | |
| - 2bit CTFC | 0 | |
| - Power offset Information | | |
| - CHOICE Gain Factors | computedGainFactors | |
| - Reference TFC ID | 0 | |
| - CHOICE mode | FDD | |
| - Power offset Pp-m | Not Present | |
| - 2bit CTFC | 1 | |
| - Power offset Information | | |
| - CHOICE Gain Factors | signalledGainFactors | |
| - CHOICE mode | FDD | |
| - Gain factor ßc | 15 | |
| - Gain factor ßd | 15 | |
| - Reference TFC ID | 0 | |
| - CHOICE mode | FDD | |
| - Power offset Pp-m | Not Present | |
| Added or Reconfigured UL TrCH information list | 1 | |
| - Added or Reconfigured UL TrCH information | | |
| - Uplink transport channel type | DCH | |
| - UL Transport channel identity | 5 | |
| - TFS | | |
| - CHOICE Transport channel type | Dedicated transport channels | |
| - Dynamic Transport Format Information | | |
| - RLC size | 96 bits | |

| Information Element | Value/remark | Version |
|---|---|---|
| - Number of TBs and TTI List | 2 | |
| - Transmission Time Interval | Not Present | |
| - Number of Transport blocks | 0 | |
| - Transmission Time Interval | Not Present | |
| - Number of Transport blocks | 1 | |
| - CHOICE Logical channel List | ALL | |
| - Semi-static Transport Format Information | | |
| - Transmission time interval | 40 | |
| - Type of channel coding | Convolutional | |
| - Coding Rate | 1/3 | |
| - Rate matching attribute | 256 | |
| - CRC size | 12 | |
| DL Transport channel information common for all transport channel | | |
| - SCCPCH TFCS | Not Present | |
| - CHOICE mode | FDD | |
| - CHOICE DL parameters | Same as UL | |
| Added or Reconfigured DL TrCH information list | 1 | |
| - Added or Reconfigured DL TrCH information | | |
| - Downlink transport channel type | DCH | |
| - DL Transport channel identity | 10 | |
| - CHOICE DL parameters | SameAasUL | |
| - Uplink transport channel type | DCH | |
| - UL TrCH Identity | 5 | |
| - DCH quality target | | |
| - BLER Quality value | -2.0 | |
| Frequency info | Not Present | |
| Maximum allowed UL TX power | Not Present | |
| CHOICE channel requirement | Uplink DPCH info | Rel-5 and earlier |
| Uplink DPCH info | | Rel-6 |
| - Uplink DPCH power control info | | |
| - DPCCH power offset | -80dB | |
| - PC Preamble | 1 frame | |
| - SRB delay | 7 frames | |
| - Power Control Algorithm | Algorithm1 | |
| - TPC step size | 1dB | |
| - $\Delta_{ACK}$ | Not Present | Rel-5 |
| - $\Delta_{NACK}$ | Not Present | Rel-5 |
| - Ack-Nack repetition factor | Not Present | Rel-5 |
| - CHOICE mode | FDD | |
| - Scrambling code type | Long | |
| - Scrambling code number | 0 (0 to 16777215) | |
| - Number of DPDCH | Not Present (1) | |
| - Spreading factor | 256 | |
| - TFCI existence | TRUE | |
| - Number of FBI bit | Not Present(0) | |
| - Puncturing Limit | 1 | |
| E-DCH Info | Not Present | Rel-6 |
| Downlink HS-PDSCH Information | Not Present | Rel-6 |
| Downlink information common for all radio links | | |
| - Downlink DPCH info common for all RL | | |
| - Timing Indication | Initialize | |
| - CFN-targetSFN frame offset | Not Present | |
| - Downlink DPCH power control information | | |
| - CHOICE mode | FDD | |
| - DPC mode | 0 (single) | |
| - CHOICE mode | FDD | |
| - Power offset $P_{Pilot-DPDCH}$ | 0 | |
| - DL rate matching restriction information | Not Present | |
| - Spreading factor | 256 | |
| - Fixed or Flexible Position | Fixed | |
| - TFCI existence | FALSE | |
| - CHOICE SF | | |
| - Number of bits for Pilot bits | 8 | |
| - DPCH compressed mode info | Not Present | |
| - TX Diversity mode | None | |

| Information Element | Value/remark | Version |
|---|---|---|
| - SSDT information | Not Present | R99 and Rel-4 only |
| - Default DPCH Offset Value | Arbitrary set to value 0..306688 by step of 512 | |
| Downlink information for per radio links list | | |
| -Downlink information for each radio links | | |
| - CHOICE mode | FDD | |
| - Primary CPICH info | | |
| - Primary scrambling code | Reference to clause 6.1 "Default settings (FDD)" | |
| - PDSCH with SHO DCH info | Not Present | R99 and Rel-4 only |
| - PDSCH code mapping | Not Present | R99 and Rel-4 only |
| - Downlink DPCH info for each RL | | |
| - CHOICE mode | FDD | |
| - Primary CPICH usage for channel estimation | Primary CPICH may be used | |
| - DPCH frame offset | Set to value : Default DPCH Offset Value mod 38 400 | |
| - Secondary CPICH info | Not Present | |
| - DL channelisation code | | |
| - Secondary scrambling code | Not Present | |
| - Spreading factor | 256 | |
| - Code number | 192 | |
| - Scrambling code change | Not Present | |
| - TPC combination index | 0 | |
| - SSDT Cell Identity | Not Present | R99 and Rel-4 only |
| - Closed loop timing adjustment mode | Not Present | |
| - SCCPCH information for FACH | Not Present | R99 and Rel-4 only |

Contents of SECURITY MODE COMMAND message: AM

| Information Element | Condition | Value/remark |
|---|---|---|
| Message Type | A1, A2 | |
| RRC transaction identifier | | Arbitrarily selects an integer between 0 and 3 |
| Integrity check info | | |
| - Message authentication code | | Set to an arbitrarily selected 32-bits integer. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I. |
| - RRC Message Sequence Number | | Set to an arbitrarily selected integer between 0 and 15 |
| Security capability | | |
| - Ciphering algorithm capability | | |
| - UEA0 | | If the UE has indicated support for ciphering algorithm UEA0 in the IE "security capability" in the RRC CONNECTION SETUP COMPLETE message, this IE is set to TRUE. |
| - UEA1 | | If the UE has indicated support for ciphering algorithm UEA1 in the IE "security capability" in the RRC CONNECTION SETUP COMPLETE message, this IE is set to TRUE. |
| - Spare | | Spare 2-15 = FALSE |
| - Integrity protection algorithm capability | | 0000000000000010B (UIA1) |
| - UIA1 | | TRUE |
| - Spare | | Spare 0 and Spare 2-15 = FALSE |
| Ciphering mode info | | This presence of this IE is dependent on IXIT statements in TS 34.123-2. If ciphering is indicated to be active, this IE present with the values of the sub IEs as stated below. Else, |

| | | |
|---|---|---|
| | | this IE is omitted. |
| - Ciphering mode command | | Start/restart |
| - Ciphering algorithm | | UEA0 or UEA1. The indicated algorithm must be one of the algorithms supported by the UE as indicated in the IE "security capability" in the RRC CONNECTION SETUP COMPLETE message. |
| - Ciphering activation time for DPCH | | Not Present |
| - Radio bearer downlink ciphering activation time info | | |
| - Radio bearer activation time | | |
| - RB identity | | 1 |
| - RLC sequence number | | Current RLC SN+2 |
| - RB identity | | 2 |
| - RLC sequence number | | Current RLC SN+2 |
| - RB identity | | 3 |
| - RLC sequence number | | Current RLC SN + 2 |
| - RB identity | | 4 |
| - RLC sequence number | | Current RLC SN + 2 |
| Integrity protection mode info | | |
| - Integrity protection mode command | | Start |
| - Downlink integrity protection activation info | | Not Present |
| - Integrity protection algorithm | | UIA1 |
| - Integrity protection initialisation number | | SS selects an arbitrary 32 bits number for FRESH.<br>The first/ leftmost bit of the bit string contains the most significant bit of the FRESH.A1 |
| CN domain identity | | CS or PS |
| UE system specific security capability | A1 | Not Present |
| UE system specific security capability | A2 | |
| - Inter-RAT UE security capability | | |
| - CHOICE *system* | | GSM |
| - GSM security capability | | The indicated algorithms must be the same as the algorithms supported by the UE as indicated in the IE " UE system specific capability " in the RRC CONNECTION SETUP COMPLETE message. |

| Condition | Explanation |
|---|---|
| A1 | UE not supporting GSM |
| A2 | UE supporting GSM |

## 9.2.2 Default Message Contents for RF (TDD)

Contents of Activate RB Test Mode message

| Information Element | Value/remark |
|---|---|
| Protocol discriminator | F (Length 1/2) |
| Skip indicator | 0 (Length 1/2) |
| Message Type | 44h |

Contents of Close UE Test Loop message

| Information Element | Value/remark |
|---|---|
| Protocol discriminator | F (Length 1/2) |
| Skip indicator | 0 (Length 1/2) |
| Message Type | 40h |
| UE test loop mode | 00h |
| UE test loop mode 1 LB setup | 03h 00h F4h 0Ah |

Contents of Open UE Test Loop message

| Information Element | Value/remark |
|---|---|

| | |
|---|---|
| Protocol discriminator | F (Length 1/2) |
| Skip indicator | 0 (Length 1/2) |
| Message Type | 42h |

Contents of PAGING TYPE 1 message: TM (CS)

| Information Element | Value/remark |
|---|---|
| Message Type | |
| Paging record list | |
| -Paging record | |
| - CHOICE Used paging identity | CN identity |
| - Paging cause | Terminating Streaming Call |
| - CN domain identity | CS domain |
| - CHOICE UE identity | |
| - IMSI (GSM-MAP) | Set to the same octet string as in the IMSI stored in the USIM card |
| BCCH modification info | Not Present |

Contents of PAGING TYPE 1 message: TM (PS)

| Information Element | Value/remark |
|---|---|
| Message Type | |
| Paging record list | |
| -Paging record | |
| - CHOICE Used paging identity | CN identity |
| - Paging cause | Terminating Interactive Call |
| - CN domain identity | PS domain |
| - CHOICE UE identity | |
| - IMSI (GSM-MAP) | Set to the same octet string as in the IMSI stored in the USIM card |
| BCCH modification info | Not Present |

Contents of RADIO BEARER SETUP message: AM or UM (3.84 Mcps TDD)

| Information Element | Condition | Value/remark | Version |
|---|---|---|---|
| Message Type | A1,A3 | | |
| RRC transaction identifier | | Arbitrarily selects an integer between 0 and 3 | |
| Integrity check info | | | |
| - message authentication code | | SS calculates the value of MAC-I for this message and writes to this IE. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I. | |
| - RRC message sequence number | | SS provides the value of this IE, from its internal counter. | |
| Integrity protection mode info | | Not Present | |
| Ciphering mode info | | Not Present | |
| Activation time | | (256+CFN-(CFN MOD 8 + 8))MOD 256 | |
| New U-RNTI | | Not Present | |
| New C-RNTI | | Not Present | |
| New DSCH-RNTI | | Not Present | R99 and Rel-4 only |
| New H-RNTI | | Not Present | Rel-5 |
| RRC State indicator | | CELL_DCH | |
| UTRAN DRX cycle length coefficient | | Not Present | |
| CN information info | | Not Present | |
| URA identity | | Not Present | |
| - Signalling RB information to setup | | Not Present | |
| - RAB information for setup list | A1 | | |
| - RAB information for setup | | | |
| - RAB info | | | |
| - RAB identity | | 0000 0001B<br>The first/ leftmost bit of the bit string contains the most significant bit of the RAB identity. | |
| - CN domain identity | | CS domain | |
| - NAS Synchronization Indicator | | Not Present | |

| Information Element | Condition | Value/remark | Version |
|---|---|---|---|
| - Re-establishment timer | | UseT314 | |
| - RB information to setup list | | | |
| - RB information to setup | | | |
| - RB identity | | 10 | |
| - PDCP info | | Not Present | |
| - CHOICE RLC info type | | RLC info | |
| - CHOICE Uplink RLC mode | | TM RLC | |
| - Transmission RLC discard | | Not Present | |
| - Segmentation indication | | FALSE | |
| - CHOICE Downlink RLC mode | | TM RLC | |
| - Segmentation indication | | FALSE | |
| - RB mapping info | | | |
| - Information for each multiplexing option | | | |
| - RLC logical channel mapping indicator | | Not Present | |
| - Number of uplink RLC logical channels | | 1 | |
| - Uplink transport channel type | | DCH | |
| - UL Transport channel identity | | 1 | |
| - Logical channel identity | | Not Present | |
| - CHOICE RLC size list | | Configured | |
| - MAC logical channel priority | | 7 | |
| - Downlink RLC logical channel info | | | |
| - Number of downlink RLC logical channels | | 1 | |
| - Downlink transport channel type | | DCH | |
| - DL DCH Transport channel identity | | 6 | |
| - DL DSCH Transport channel identity | | Not Present | |
| - Logical channel identity | | Not Present | |
| RAB information for setup list | A3 | | |
| - RAB information for setup | | | |
| - RAB info | | | |
| - RAB identity | | 0000 0101B<br>The first/ leftmost bit of the bit string contains the most significant bit of the RAB identity. | |
| - CN domain identity | | PS domain | |
| - NAS Synchronization Indicator | | Not Present | |
| - Re-establishment timer | | UseT314 | |
| - RB information to setup list | | | |
| - RB information to setup | | | |
| - RB identity | | 20 | |
| - PDCP info | | Not Present | |
| - CHOICE RLC info type | | RLC info | |
| - CHOICE Uplink RLC mode | | AM RLC | |
| - Transmission RLC discard | | | |
| - CHOICE SDU discard mode | | No discard | |
| - MAX_DAT | | 15 | |
| - Transmission window size | | 128 | |
| - Timer_RST | | 500 | |
| - Max_RST | | 4 | |
| - Polling info | | | |
| - Timer_poll_prohibit | | 200 | |
| - Timer_poll | | 200 | |
| - Poll_SDU | | 1 | |
| - Last transmission PDU poll | | TRUE | |
| - Last retransmission PDU poll | | TRUE | |
| - Poll_Windows | | 99 | |
| - Timer_poll_periodic | | Not Present | |
| - CHOICE Downlink RLC mode | | AM RLC | |
| - In-sequence delivery | | TRUE | |
| - Receiving window size | | 128 | |
| - Downlink RLC status info | | | |
| - Timer_status_prohibit | | 200 | |

| Information Element | Condition | Value/remark | Version |
|---|---|---|---|
| - Timer_EPC | | 200 | |
| - Missing PDU indicator | | TRUE | |
| - Timer_STATUS_periodic | | Not Present | |
| - RB mapping info | | | |
| - Information for each multiplexing option | | 2RBMuxOptions | |
| - RLC logical channel mapping indicator | | Not Present | |
| - Number of uplink RLC logical channels | | 1 | |
| - Uplink transport channel type | | DCH | |
| - UL Transport channel identity | | 1 | |
| - Logical channel identity | | Not Present | |
| - CHOICE RLC size list | | Configured | |
| - MAC logical channel priority | | 8 | |
| - Downlink RLC logical channel info | | | |
| - Number of downlink RLC logical channels | | 1 | |
| - Downlink transport channel type | | DCH | |
| - DL DCH Transport channel identity | | 6 | |
| - DL DSCH Transport channel identity | | Not Present | |
| - Logical channel identity | | Not Present | |
| - RLC logical channel mapping indicator | | Not Present | |
| - Number of uplink RLC logical channels | | 1 | |
| - Uplink transport channel type | | RACH | |
| - UL Transport channel identity | | Not Present | |
| - Logical channel identity | | 7 | |
| - CHOICE RLC size list | | Explicit List | |
| - RLC size index | | Reference to clause 6 Parameter Set | |
| - MAC logical channel priority | | 8 | |
| - Downlink RLC logical channel info | | | |
| - Number of downlink RLC logical channels | | 1 | |
| - Downlink transport channel type | | FACH | |
| - DL DCH Transport channel identity | | Not Present | |
| - DL DSCH Transport channel identity | | Not Present | |
| - Logical channel identity | | Not Present | |
| RB information to be affected list | A1,A3 | Not Present | |
| Downlink counter synchronization info | | Not Present | |
| UL Transport channel information for all transport channels | A1,A3 | | |
| - PRACH TFCS | | Not Present | |
| - CHOICE mode | | TDD | |
| -Individual UL CCTrCH information | | | |
| - TFCS ID | | (This IE is repeated for TFC number.) | |
| - Allowed Transport Format combination | | 0 to MaxTFCvalue-1 (MaxTFCValue is refer to clause 6 Parameter Set.) | |
| - PRACH TFCS | | (This IE is repeated for TFC number.) | |
| - CHOICE TFCI signalling | | Normal | |
| - TFCI Field 1 information | | | |
| - TFCS complete reconfigure information | | | |
| - CHOICE TFCS Size | | Number of used bits must be enough to cover all combinations of CTFC from clauses 6. Refer to clause 6 Parameter Set | |
| - CTFC information | | Not Present | |
| - CHOICE mode | | TDD | |
| - Individual UL CCTrCH | | Not Present | |

| Information Element | Condition | Value/remark | Version |
|---|---|---|---|
| information<br>Deleted UL TrCH information list | | <br>Not Present | |
| Added or Reconfigured UL TrCH information list<br>- Added or Reconfigured UL TrCH information<br>- Uplink transport channel type<br>- UL Transport channel identity<br>- TFS<br>- CHOICE Transport channel type | A1 | 1<br><br><br><br>DCH<br>1<br><br>Dedicated transport channels | |
| - Dynamic Transport Format Information<br>- RLC size<br>- Number of TBs and TTI List<br>- Transmission Time Interval<br>- Number of Transport blocks<br>- Transmission Time Interval<br>- Number of Transport blocks<br>- CHOICE Logical channel List<br>- Semi-static Transport Format Information<br>- Transmission time interval<br>- Type of channel coding<br>- Coding Rate<br>- Rate matching attribute<br>- CRC size | | <br><br>Reference to clause 6.10 Parameter Set<br>(This IE is repeated for TFI number.)<br>Not Present<br>Reference to clause 6.10 Parameter Set<br>Not Present<br>1<br>ALL<br><br><br>Reference to clause 6.10 Parameter Set<br>Reference to clause 6.10 Parameter Set<br>Reference to clause 6.10 Parameter Set<br>Reference to clause 6.10 Parameter Set<br>Reference to clause 6.10 Parameter Set | |
| CHOICE mode | A1, A3 | TDD (no data) | |
| DL Transport channel information common for all transport channel<br>- SCCPCH TFCS<br>- CHOICE mode<br>- CHOICE DL parameters | A1,A3 | <br><br>Not Present<br>TDD<br>Independent (Refer to clause 6) | |
| Deleted DL TrCH information list<br>Added or Reconfigured DL TrCH information list<br>- Added or Reconfigured DL TrCH information<br>- Downlink transport channel type<br>- DL Transport channel identity<br>- CHOICE DL parameters<br>- Uplink transport channel type<br>- UL TrCH identity<br>- DCH quality target<br>- BLER Quality value | A1,A3 | Not Present<br>1<br><br><br><br>DCH<br>6<br>Same as UL<br>DCH<br>1<br><br>Reference to clause 6 | |
| Frequency info<br>Maximum allowed UL TX power<br>CHOICE channel requirement<br>- Uplink DPCH power control info<br>- CHOICE mode<br>- UL Target SIR<br>- CHOICE UL OL PC info<br>- CHOICE TDD option<br>- Individual timeslot interference info<br>- Individual timeslot interference<br>- DPCH Constant Value<br><br>- CHOICE mode<br>- Uplink Timing Advance Control<br>- UL CCTrCH List<br>- TFCS Id<br>- Time info<br>- Activation time<br>- Duration<br>- Common timeslot info<br>- 2nd interleaving mode | A1,A3 | Not Present<br>30dBm<br>Uplink DPCH info<br><br>TDD<br>Reference to clause 6 Parameter set.<br>Individually signalled<br>3.84 Mcps<br><br><br><br>Values are used for open loop power control, clause 8 in 3GPP TS 25.331 [34]<br>TDD<br>Not Present<br><br>1<br><br>(256+CFN-(CFN MOD 8 + 8))MOD 256<br>Infinite<br><br>Reference to clause 6.10 Parameter Set | |

| Information Element | Condition | Value/remark | Version |
|---|---|---|---|
| - TFCI coding | | Reference to clause 6.10 Parameter Set | |
| - Puncturing Limit | | Reference to clause 6.10 Parameter Set | |
| - Repetition Period | | Reference to clause 6.10 Parameter Set | |
| - Repetition Length | | Reference to clause 6.10 Parameter Set | |
| - First individual timeslot info | | | |
| - Timeslot number | | The number of an uplink timeslot that has unassigned codes. | |
| - TFCI existence | | TRUE | |
| - Midamble shift and burst type | | | |
| - CHOICE TDD option | | 3.84 Mcps | |
| -CHOICE Burst Type | | | |
| -Type 1 | | | |
| -Midamble Allocation Mode | | Default | |
| - Midamble configuration burst type 1 and 3 | | As defined in 3GPP TS 25.221 [28] | |
| - First timeslot channelisation codes | | Repeated (1,2) for each channelisation code assigned in the slot to meet the needs of clause 6 Parameter Set. | |
| - Channelisation code | | (i/SF) where i denotes an unassigned code matching the SF specified in clause 6 Parameter Set. | |
| - CHOICE more timeslots | | The presence of this IE depends upon the number of resources specified in clause 6 and the number of slots in which they are being assigned. | |
| CHOICE Mode | | TDD (no data) | |
| Downlink HS-PDSCH Information | A1,A3 | Not Present | Rel-5 |
| Downlink information common for all radio links | A1,A3 | | |
| - Downlink DPCH info common for all RL | | | |
| - Timing indicator | | Maintain | |
| - CFN-targetSFN frame offset | | Not Present | |
| - Downlink DPCH power control information | | | |
| - CHOICE mode | | TDD | |
| - DPC mode | | 0 (single) | |
| - CHOICE TDD mode | | 3.84 Mcps (no data) | |
| - Default DPCH Offset Value | | Not Present | |
| Downlink information for per radio link list | A1,A3 | | |
| - Downlink information for each radio link | | | |
| - CHOICE mode | | TDD | |
| - Primary CCPCH info | | | |
| - CHOICE *SyncCase* | | Sync Case 1 | |
| - Timeslot | | PCCPCH timeslot | |
| - Cell parameters ID | | 0 | |
| - SCTD indicator | | | |
| - Downlink DPCH info for each RL | | | |
| - CHOICE mode | | TDD | |
| - DL CCTrCH List | | | |
| - TFCS ID | | 1 | |
| - Time info | | | |
| - Activation time | | (256+CFN-(CFN mod 8 + 8))mod 256 | |
| - Duration | | infinite | |
| - Common timeslot info | | | |
| - 2nd interleaving mode | | Reference to the present document | |
| - TFCI coding | | TRUE | |
| - Puncturing limit | | Reference to clause 6 Parameter set | |
| - Repetition period | | 1 | |
| - Repetition length | | Empty | |
| - Downlink DPCH timeslots and codes | | | |
| - Individual timeslot info | | | |
| - Timeslot number | | The number of a downlink timeslot that has unassigned codes. | |
| - TFCI existence | | TRUE | |

| Information Element | Condition | Value/remark | Version |
|---|---|---|---|
| - Midamble shift and burst type | | | |
| - CHOICE TDD option | | 3.84 Mcps | |
| -CHOICE Burst Type | | | |
| -Type 1 | | | |
| -Midamble Allocation Mode | | Default | |
| - Midamble configuration burst type 1 and 3 | | As defined in 3GPP TS 25.221 [28] | |
| - First timeslot channelisation codes | | | |
| - First channelisation code | | (i/SF) where i is the lowest numbered code that is being assigned and SF is specified in clause 6 Parameter Set.. | |
| - Last channelisation code | | (j/SF) where j is the highest numbered code that is being assigned in the slot. | |
| - Bitmap | | Bitmap of the codes that are being assigned in the slot. | |
| - CHOICE more timeslots | | The presence of this IE depends upon whether the requirements of clause 6 Parameter Set could be met by the codes that have been assigned in the first timeslot. | |
| - UL CCTrCH TPC List | | Not Present | |
| -SCCPCH information for FACH | | Not Present | R99 and Rel-4 only |

| Condition | Explanation |
|---|---|
| A1<br>A3 | This IE is needed for transparent mode. In the case of TX and RX test cases, this IE is selected.<br>This IE is needed for acknowledged mode. |
| NOTE: In the case of Performance Requirement and RRM test cases, A1 or A3 is selected according to the combination of UL and DL channels or test requirements. | |

Contents of RADIO BEARER SETUP message: AM or UM (1.28 Mcps TDD)

| Information Element | Condition | Value/remark | Version |
|---|---|---|---|
| Message Type | A1,A3 | | |
| RRC transaction identifier | | Arbitrarily selects an integer between 0 and 3 | |
| Integrity check info | | | |
| - message authentication code | | SS calculates the value of MAC-I for this message and writes to this IE. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I. | |
| - RRC message sequence number | | SS provides the value of this IE, from its internal counter. | |
| Integrity protection mode info | | Not Present | |
| Ciphering mode info | | Not Present | |
| Activation time | | (256+CFN-(CFN MOD 8 + 8))MOD 256 | |
| New U-RNTI | | Not Present | |
| New C-RNTI | | Not Present | |
| New DSCH-RNTI | | Not Present | R99 and Rel-4 only |
| New H-RNTI | | Not Present | Rel-5 |
| RRC State indicator | | CELL_DCH | |
| UTRAN DRX cycle length coefficient | | Not Present | |
| CN information info | | Not Present | |
| URA identity | | Not Present | |
| - Signalling RB information to setup | | Not Present | |
| - RAB information for setup list | A1 | | |
| - RAB information for setup | | | |
| - RAB info | | | |
| - RAB identity | | 0000 0001B<br>The first/ leftmost bit of the bit string contains the most significant bit of the RAB identity. | |
| - CN domain identity | | CS domain | |
| - NAS Synchronization Indicator | | Not Present | |

| Information Element | Condition | Value/remark | Version |
|---|---|---|---|
| - Re-establishment timer | | UseT314 | |
| - RB information to setup list | | | |
| - RB information to setup | | | |
| - RB identity | | 10 | |
| - PDCP info | | Not Present | |
| - CHOICE RLC info type | | RLC info | |
| - CHOICE Uplink RLC mode | | TM RLC | |
| - Transmission RLC discard | | Not Present | |
| - Segmentation indication | | FALSE | |
| - CHOICE Downlink RLC mode | | TM RLC | |
| - Segmentation indication | | FALSE | |
| - RB mapping info | | | |
| - Information for each multiplexing option | | | |
| - RLC logical channel mapping indicator | | Not Present | |
| - Number of uplink RLC logical channels | | 1 | |
| - Uplink transport channel type | | DCH | |
| - UL Transport channel identity | | 1 | |
| - Logical channel identity | | Not Present | |
| - CHOICE RLC size list | | Configured | |
| - MAC logical channel priority | | 7 | |
| - Downlink RLC logical channel info | | | |
| - Number of downlink RLC logical channels | | 1 | |
| - Downlink transport channel type | | DCH | |
| - DL DCH Transport channel identity | | 6 | |
| - DL DSCH Transport channel identity | | Not Present | |
| - Logical channel identity | | Not Present | |
| RAB information for setup list | A3 | | |
| - RAB information for setup | | | |
| - RAB info | | | |
| - RAB identity | | 0000 0101B<br>The first/ leftmost bit of the bit string contains the most significant bit of the RAB identity. | |
| - CN domain identity | | PS domain | |
| - NAS Synchronization Indicator | | Not Present | |
| - Re-establishment timer | | UseT315 | |
| - RB information to setup list | | | |
| - RB information to setup | | | |
| - RB identity | | 20 | |
| - PDCP info | | Not Present | |
| - CHOICE RLC info type | | RLC info | |
| - CHOICE Uplink RLC mode | | AM RLC | |
| - Transmission RLC discard | | | |
| - CHOICE SDU discard mode | | No discard | |
| - MAX_DAT | | 15 | |
| - Transmission window size | | 128 | |
| - Timer_RST | | 500 | |
| - Max_RST | | 4 | |
| - Polling info | | | |
| - Timer_poll_prohibit | | 200 | |
| - Timer_poll | | 200 | |
| - Poll_SDU | | 1 | |
| - Last transmission PDU poll | | TRUE | |
| - Last retransmission PDU poll | | TRUE | |
| - Poll_Windows | | 99 | |
| - Timer_poll_periodic | | Not Present | |
| - CHOICE Downlink RLC mode | | AM RLC | |
| - In-sequence delivery | | TRUE | |
| - Receiving window size | | 128 | |
| - Downlink RLC status info | | | |
| - Timer_status_prohibit | | 200 | |
| - Timer_EPC | | 200 | |

| Information Element | Condition | Value/remark | Version |
|---|---|---|---|
| - Missing PDU indicator | | TRUE | |
| - Timer_STATUS_periodic | | Not Present | |
| - RB mapping info | | | |
| - Information for each multiplexing option | | 2RBMuxOptions | |
| - RLC logical channel mapping indicator | | Not Present | |
| - Number of uplink RLC logical channels | | 1 | |
| - Uplink transport channel type | | DCH | |
| - UL Transport channel identity | | 1 | |
| - Logical channel identity | | Not Present | |
| - CHOICE RLC size list | | Configured | |
| - MAC logical channel priority | | 8 | |
| - Downlink RLC logical channel info | | | |
| - Number of downlink RLC logical channels | | 1 | |
| - Downlink transport channel type | | DCH | |
| - DL DCH Transport channel identity | | 6 | |
| - DL DSCH Transport channel identity | | Not Present | |
| - Logical channel identity | | Not Present | |
| - RLC logical channel mapping indicator | | Not Present | |
| - Number of uplink RLC logical channels | | 1 | |
| - Uplink transport channel type | | RACH | |
| - UL Transport channel identity | | Not Present | |
| - Logical channel identity | | 7 | |
| - CHOICE RLC size list | | Explicit List | |
| - RLC size index | | Reference to clause 6 Parameter Set | |
| - MAC logical channel priority | | 8 | |
| - Downlink RLC logical channel info | | | |
| - Number of downlink RLC logical channels | | 1 | |
| - Downlink transport channel type | | FACH | |
| - DL DCH Transport channel identity | | Not Present | |
| - DL DSCH Transport channel identity | | Not Present | |
| - Logical channel identity | | Not Present | |
| RB information to be affected list | A1,A3 | Not Present | |
| Downlink counter synchronization info | | Not Present | |
| UL Transport channel information for all transport channels | A1,A3 | | |
| - PRACH TFCS | | Not Present | |
| - CHOICE mode | | TDD | |
| -Individual UL CCTrCH information | | | |
| - TFCS ID | | (This IE is repeated for TFC number.) | |
| - Allowed Transport Format combination | | 0 to MaxTFCvalue-1 (MaxTFCValue is refer to clause 6 Parameter Set.) | |
| - PRACH TFCS | | (This IE is repeated for TFC number.) | |
| - CHOICE TFCI signalling | | Normal | |
| - TFCI Field 1 information | | | |
| - TFCS complete reconfigure information | | | |
| - CHOICE TFCS Size | | Number of used bits must be enough to cover all combinations of CTFC from clauses 6. Refer to clause 6 Parameter Set | |
| - CTFC information | | Not Present | |
| - CHOICE mode | | TDD | |
| - Individual UL CCTrCH information | | Not Present | |
| Deleted UL TrCH information list | | Not Present | |
| Added or Reconfigured UL TrCH information list | A1,A3 | 1 | |

| Information Element | Condition | Value/remark | Version |
|---|---|---|---|
| - Added or Reconfigured UL TrCH information | | | |
| - Uplink transport channel type | | DCH | |
| - UL Transport channel identity | | 1 | |
| - TFS | | | |
| - CHOICE Transport channel type | | Dedicated transport channels | |
| - Dynamic Transport Format Information | | | |
| - RLC size | | Reference to clause 6 Parameter Set | |
| - Number of TBs and TTI List | | (This IE is repeated for TFI number.) | |
| - Transmission Time Interval | | Not Present | |
| - Number of Transport blocks | | Reference to clause 6 Parameter Set | |
| - Transmission Time Interval | | Not Present | |
| - Number of Transport blocks | | 1 | |
| - CHOICE Logical channel List | | ALL | |
| - Semi-static Transport Format Information | | | |
| - Transmission time interval | | Reference to clause 6 Parameter Set | |
| - Type of channel coding | | Reference to clause 6 Parameter Set | |
| - Coding Rate | | Reference to clause 6 Parameter Set | |
| - Rate matching attribute | | Reference to clause 6 Parameter Set | |
| - CRC size | | Reference to clause 6 Parameter Set | |
| CHOICE mode | A1, A3 | TDD (no data) | |
| DL Transport channel information common for all transport channel | A1,A3 | | |
| - SCCPCH TFCS | | Not Present | |
| - CHOICE mode | | TDD | |
| - CHOICE DL parameters | | Independent (Refer to clause 6) | |
| Deleted DL TrCH information list | A1,A3 | Not Present | |
| Added or Reconfigured DL TrCH information list | | 1 | |
| - Added or Reconfigured DL TrCH information | | | |
| - Downlink transport channel type | | DCH | |
| - DL Transport channel identity | | 6 | |
| - CHOICE DL parameters | | Same as UL | |
| - Uplink transport channel type | | DCH | |
| - UL TrCH identity | | 1 | |
| - DCH quality target | | | |
| - BLER Quality value | | Reference to clause 6 | |
| Frequency info | A1,A3 | Not Present | |
| Maximum allowed UL TX power | | 30dBm | |
| CHOICE channel requirement | | Uplink DPCH info | |
| - Uplink DPCH power control info | | | |
| - CHOICE mode | | TDD | |
| - UL Target SIR | | Reference to clause 6 Parameter set. | |
| - CHOICE UL OL PC info | | Individually signalled | |
| - CHOICE TDD option | | 1.28 Mcps | |
| - TPC step size | | 1 dB | |
| - Primary CCPCH Tx Power | | Not Present | |
| - CHOICE mode | | TDD | |
| - Uplink Timing Advance Control | | Not Present | |
| - UL CCTrCH List | | | |
| - TFCS Id | | 1 | |
| - Time info | | | |
| - Activation time | | (256+CFN-(CFN MOD 8 + 8))MOD 256 | |
| - Duration | | Infinite | |
| - Common timeslot info | | | |
| - 2nd interleaving mode | | Reference to clause 6 Parameter Set | |
| - TFCI coding | | Reference to clause 6 Parameter Set | |
| - Puncturing Limit | | Reference to clause 6 Parameter Set | |
| - Repetition Period | | Reference to clause 6 Parameter Set | |
| - Repetition Length | | Reference to clause 6 Parameter Set | |
| - First individual timeslot info | | | |
| - Timeslot number | | The number of an uplink timeslot that has unassigned codes. | |
| - TFCI existence | | TRUE | |

| Information Element | Condition | Value/remark | Version |
|---|---|---|---|
| - Midamble shift and burst type | | | |
| - CHOICE TDD option | | 1.28 Mcps | |
| - Midamble allocation mode | | Default | |
| - Midamble configuration | | 16 | |
| - CHOICE TDD option | | 1.28 Mcps TDD | |
| - Modulation | | QPSK | |
| - SS-TPC Symbols | | 1 | |
| - CHOICE Mode | | TDD | |
| - First timeslot channelisation codes | | Repeated (1,2) for each channelisation code assigned in the slot to meet the needs of clause 6 Parameter Set. | |
| - Channelisation code | | (i/SF) where i denotes an unassigned code matching the SF specified in clause 6 Parameter Set. | |
| - CHOICE more timeslots | | The presence of this IE depends upon the number of resources specified in clause 6 and the number of slots in which they are being assigned. | |
| CHOICE Mode | | TDD (no data) | |
| Downlink HS-PDSCH Information | A1,A3 | Not Present | Rel-5 |
| Downlink information common for all radio links | A1,A3 | | |
| - Downlink DPCH info common for all RL | | | |
| - Timing indicator | | Maintain | |
| - CFN-targetSFN frame offset | | Not Present | |
| - Downlink DPCH power control information | | | |
| - CHOICE mode | | TDD | |
| - TPC step size | | 1 dB | |
| - CHOICE TDD mode | | 1.28 Mcps | |
| - TSTD indicator | | TRUE | |
| - Default DPCH Offset Value | | Not Present | |
| Downlink information for per radio link list | A1,A3 | | |
| - Downlink information for each radio link | | | |
| - CHOICE mode | | TDD | |
| - Primary CCPCH info | | | |
| - CHOICE TDD option | | 1.28 Mcps | |
| - TSTD indicator | | TRUE | |
| - Cell parameters ID | | 0 | |
| - Block STTD indicator | | FALSE | |
| - Downlink DPCH info for each RL | | | |
| - CHOICE mode | | TDD | |
| - DL CCTrCH List | | | |
| - TFCS ID | | 1 | |
| - Time info | | | |
| - Activation time | | (256+CFN-(CFN mod 8 + 8))mod 256 | |
| - Duration | | Infinite | |
| - Common timeslot info | | | |
| - 2nd interleaving mode | | Reference to the present document | |
| - TFCI coding | | TRUE | |
| - Puncturing limit | | Reference to clause 6 Parameter set | |
| - Repetition period | | 1 | |
| - Repetition length | | Empty | |
| - Downlink DPCH timeslots and codes | | | |
| - Individual timeslot info | | | |
| - Timeslot number | | The number of a downlink timeslot that has unassigned codes. | |
| - TFCI existence | | TRUE | |
| - Midamble shift and burst type | | | |
| - CHOICE TDD option | | 1.28 Mcps | |
| -Midamble Allocation Mode | | Default | |

| Information Element | Condition | Value/remark | Version |
|---|---|---|---|
| - Midamble configuration | | 16 | |
| - Modulation | | QPSK | |
| - SS-TPC Symbols | | 1 | |
| - First timeslot channelisation codes | | | |
| - First channelisation code | | (i/SF) where i is the lowest numbered code that is being assigned and SF is specified in clause 6 Parameter Set. | |
| - Last channelisation code | | (j/SF) where j is the highest numbered code that is being assigned in the slot. | |
| - Bitmap | | Bitmap of the codes that are being assigned in the slot. | |
| - CHOICE more timeslots | | The presence of this IE depends upon whether the requirements of clause 6 Parameter Set could be met by the codes that have been assigned in the first timeslot. | |
| - UL CCTrCH TPC List | | Not Present | |
| -SCCPCH information for FACH | | Not Present | R99 and Rel-4 only |

| Condition | Explanation |
|---|---|
| A1 | This IE is needed for CS RAB |
| A3 | This IE is needed for PS RAB. |

Contents of RRC CONNECTION RELEASE message: UM

| Information Element | Value/remark | Version |
|---|---|---|
| Message Type | | |
| U-RNTI | This IE is set to the following value when the message is transmitted on the DCCCH. When transmitted on CDCCH, this is absent. | R99, Rel-4 |
| - SRNC identity | 0000 0000 0001B | |
| - S-RNTI | 0000 0000 0000 0000 0001B | |
| CHOICE identity type | This IE is set to the following value when the message is transmitted on the CCCH. When transmitted on DCCH, this is absent. | Rel-5 |
| - U-RNTI | | |
| - SRNC identity | 0000 0000 0001B | |
| - S-RNTI | 0000 0000 0000 0000 0001B | |
| - Group identity | [FFS] | |
| - Group release information | [FFS] | |
| RRC transaction identifier | Arbitrarily selects an integer between 0 and 3 | |
| Integrity check info | This IE is present when this message is transmitted on downlink DCCH. Else, this IE and the sub-IEs are omitted. | |
| - Message authentication code | SS calculates the value of MAC-I for this message and writes to this IE. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I. | |
| - RRC Message sequence number | SS provides the value of this IE, from its internal counter. | |
| N308 | 2 (for CELL_DCH state). Not Present (for UE in other connected mode states). | |
| Release cause | Normal event | |
| Rplmn information | Not Present | |

Contents of RRC CONNECTION SETUP message: UM (3.84 Mcps TDD)

| Information Element | Value/remark | Version |
|---|---|---|
| Message Type | | |
| Initial UE identity | Select the same identity as in the IE "Initial UE Identity" in received RRC CONNECTION REQUEST" message | |
| RRC transaction identifier | Arbitrarily selects an integer between 0 and 3 | |

| Information Element | Value/remark | Version |
|---|---|---|
| Activation time | Not Present(Now) | |
| New U-RNTI | | |
| - SRNC identity | 0000 0000 0001B | |
| - S-RNTI | 0000 0000 0000 0000 0001B | |
| New C-RNTI | Not Present | |
| RRC State Indicator | CELL_DCH | |
| UTRAN DRX cycle length coefficient | 9 | |
| Capability update requirement | | |
| - UE radio access FDD capability update requirement | FALSE | |
| - UE radio access TDD capability update requirement | TRUE | |
| - System specific capability update requirement list | GSM | |
| CHOICE *specification mode* | Complete specification | Rel-5 |
| - Complete specification | | Rel-5 |
| - Signalling RB information to setup list | 4 SRBs | |
| - Signalling RB information to setup | (UM DCCH for RRC) | |
| - RB identity | Not Present | |
| - CHOICE RLC info type | RLC info | |
| - CHOICE Uplink RLC mode | UM RLC | |
| - Transmission RLC discard | Not Present | |
| - CHOICE Downlink RLC mode | UM RLC | |
| - RB mapping info | | |
| - Information for each multiplexing option | 2 RBMuxOptions | |
| - RLC logical channel mapping indicator | Not Present | |
| - Number of RLC logical channels | 1 | |
| - Uplink transport channel type | DCH | |
| - UL Transport channel identity | 5 | |
| - Logical channel identity | 1 | |
| - CHOICE RLC size list | Configured | |
| - MAC logical channel priority | 1 | |
| - Downlink RLC logical channel info | | |
| - Number of RLC logical channels | 1 | |
| - Downlink transport channel type | DCH | |
| - DL DCH Transport channel identity | 10 | |
| - DL DSCH Transport channel identity | Not Present | |
| - Logical channel identity | 1 | |
| - RLC logical channel mapping indicator | Not Present | |
| - Number of RLC logical channels | 1 | |
| - Uplink transport channel type | RACH | |
| - UL Transport channel identity | Not Present | |
| - Logical channel identity | 1 | |
| - CHOICE RLC size list | Configured | |
| - RLC size index | Reference to clause 6 Parameter Set | |
| - MAC logical channel priority | 1 | |
| - Downlink RLC logical channel info | | |
| - Number of RLC logical channels | 1 | |
| - Downlink transport channel type | FACH | |
| - DL DCH Transport channel identity | Not Present | |
| - DL DSCH Transport channel identity | Not Present | |
| - Logical channel identity | 1 | |
| - Signalling RB information to setup | (AM DCCH for RRC) | |
| - RB identity | Not Present | |
| - CHOICE RLC info type | | |
| - RLC info | | |
| - CHOICE Uplink RLC mode | AM RLC | |
| - Transmission RLC discard | | |
| - SDU discard mode | No Discard | |
| - MAX_DAT | 415 | |
| - Transmission window size | 128 | |
| - Timer_RST | 500 | |
| - Max_RST | 4 | |
| - Polling info | | |
| - Timer_poll_prohibit | 200 | |
| - Timer_poll | 200 | |
| - Poll_PDU | Not Present | |

| Information Element | Value/remark | Version |
|---|---|---|
| - Poll_SDU | 1 | |
| - Last transmission PDU poll | TRUE | |
| - Last retransmission PDU poll | TRUE | |
| - Poll_Windows | 99 | |
| - Timer_poll_periodic | Not Present | |
| - CHOICE Downlink RLC mode | AM RLC | |
| - In-sequence delivery | TRUE | |
| - Receiving window size | 128 | |
| - Downlink RLC status info | | |
| - Timer_status_prohibit | 200 | |
| - Timer_EPC | Not Present | |
| - Missing PDU indicator | TRUE | |
| - Timer_STATUS_periodic | Not Present | |
| - RB mapping info | | |
| - Information for each multiplexing option | 2 RBMuxOptions | |
| - RLC logical channel mapping indicator | Not Present | |
| - Number of RLC logical channels | 1 | |
| - Uplink transport channel type | DCH | |
| - UL Transport channel identity | 5 | |
| - Logical channel identity | 2 | |
| - CHOICE RLC size list | Configured | |
| - MAC logical channel priority | 2 | |
| - Downlink RLC logical channel info | | |
| - Number of RLC logical channels | 1 | |
| - Downlink transport channel type | DCH | |
| - DL DCH Transport channel identity | 10 | |
| - DL DSCH Transport channel identity | Not Present | |
| - Logical channel identity | 2 | |
| - RLC logical channel mapping indicator | Not Present | |
| - Number of RLC logical channels | 1 | |
| - Uplink transport channel type | RACH | |
| - UL Transport channel identity | Not Present | |
| - Logical channel identity | 2 | |
| - CHOICE RLC size list | Explicit List | |
| - RLC size index | Reference to clause 6 Parameter Set | |
| - MAC logical channel priority | 2 | |
| - Downlink RLC logical channel info | | |
| - Number of RLC logical channels | 1 | |
| - Downlink transport channel type | FACH | |
| - DL DCH Transport channel identity | Not Present | |
| - DL DSCH Transport channel identity | Not Present | |
| - Logical channel identity | 2 | |
| - Signalling RB information to setup | (AM DCCH for NAS_DT High priority) | |
| - RB identity | Not Present | |
| - CHOICE RLC info type | | |
| - RLC info | | |
| - CHOICE Uplink RLC mode | AM RLC | |
| - Transmission RLC discard | | |
| - SDU discard mode | No Discard | |
| - MAX_DAT | 415 | |
| - Transmission window size | 128 | |
| - Timer_RST | 500 | |
| - Max_RST | 4 | |
| - Polling info | | |
| - Timer_poll_prohibit | 200 | |
| - Timer_poll | 200 | |
| - Poll_PDU | Not Present | |
| - Poll_SDU | 1 | |
| - Last transmission PDU poll | TRUE | |
| - Last retransmission PDU poll | TRUE | |
| - Poll_Windows | 99 | |
| - Timer_poll_periodic | Not Present | |
| - CHOICE Downlink RLC mode | AM RLC | |
| - In-sequence delivery | TRUE | |
| - Receiving window size | 128 | |
| - Downlink RLC status info | | |
| - Timer_status_prohibit | 200 | |

| Information Element | Value/remark | Version |
|---|---|---|
| - Timer_EPC | Not Present | |
| - Missing PDU indicator | TRUE | |
| - Timer_STATUS_periodic | Not Present | |
| - RB mapping info | | |
| - Information for each multiplexing option | 2 RBMuxOptions | |
| - RLC logical channel mapping indicator | Not Present | |
| - Number of RLC logical channels | 1 | |
| - Uplink transport channel type | DCH | |
| -UL Transport channel identity | 5 | |
| - Logical channel identity | 3 | |
| - CHOICE RLC size list | Configured | |
| - MAC logical channel priority | 3 | |
| - Downlink RLC logical channel info | | |
| - Number of RLC logical channels | 1 | |
| - Downlink transport channel type | DCH | |
| - DL DCH Transport channel identity | 10 | |
| - DL DSCH Transport channel identity | Not Present | |
| - Logical channel identity | 3 | |
| - RLC logical channel mapping indicator | Not Present | |
| - Number of RLC logical channels | 1 | |
| - Uplink transport channel type | RACH | |
| - UL Transport channel identity | Not Present | |
| - Logical channel identity | 3 | |
| - CHOICE RLC size list | Explicit List | |
| - RLC size index | Reference to clause 6 Parameter Set | |
| - MAC logical channel priority | 3 | |
| - Downlink RLC logical channel info | | |
| - Number of RLC logical channels | 1 | |
| - Downlink transport channel type | FACH | |
| - DL DCH Transport channel identity | Not Present | |
| - DL DSCH Transport channel identity | Not Present | |
| - Logical channel identity | 3 | |
| - Signalling RB information to setup | (AM DCCH for NAS_DT Low priority) | |
| - RB identity | Not Present | |
| - CHOICE RLC info type | | |
| - RLC info | | |
| - CHOICE Uplink RLC mode | AM RLC | |
| - Transmission RLC discard | | |
| - SDU discard mode | No Discard | |
| - MAX_DAT | 15 | |
| - Transmission window size | 128 | |
| - Timer_RST | 500 | |
| - Max_RST | 4 | |
| - Polling info | | |
| - Timer_poll_prohibit | 200 | |
| - Timer_poll | 200 | |
| - Poll_PDU | Not Present | |
| - Poll_SDU | 1 | |
| - Last transmission PDU poll | TRUE | |
| - Last retransmission PDU poll | TRUE | |
| - Poll_Windows | 99 | |
| - Timer_poll_periodic | Not Present | |
| - CHOICE Downlink RLC mode | AM RLC | |
| - In-sequence delivery | TRUE | |
| - Receiving window size | 128 | |
| - Downlink RLC status info | | |
| - Timer_status_prohibit | 200 | |
| - Timer_EPC | Not Present | |
| - Missing PDU indicator | TRUE | |
| - Timer_STATUS_periodic | Not Present | |
| - RB mapping info | | |
| - Information for each multiplexing option | 2 RBMuxOptions | |
| - RLC logical channel mapping indicator | Not Present | |
| - Number of RLC logical channels | 1 | |
| - Uplink transport channel type | DCH | |
| - UL Transport channel identity | 5 | |
| - Logical channel identity | 4 | |

| Information Element | Value/remark | Version |
|---|---|---|
| - CHOICE RLC size list | Configured | |
| - MAC logical channel priority | 4 | |
| - Downlink RLC logical channel info | | |
| - Number of RLC logical channels | 1 | |
| - Downlink transport channel type | DCH | |
| - DL DCH Transport channel identity | 10 | |
| - DL DSCH Transport channel identity | Not Present | |
| - Logical channel identity | 4 | |
| - RLC logical channel mapping indicator | Not Present | |
| - Number of RLC logical channels | 1 | |
| - Uplink transport channel type | RACH | |
| - UL Transport channel identity | Not Present | |
| - Logical channel identity | 4 | |
| - CHOICE RLC size list | Explicit List | |
| - RLC size index | Reference to clause 6 Parameter Set | |
| - MAC logical channel priority | 4 | |
| - Downlink RLC logical channel info | | |
| - Number of RLC logical channels | 1 | |
| - Downlink transport channel type | FACH | |
| - DL DCH Transport channel identity | Not Present | |
| - DL DSCH Transport channel identity | Not Present | |
| - Logical channel identity | 4 | |
| UL Transport channel information for all transport channels | | |
| - PRACH TFCS | Not Present | |
| - CHOICE Mode | TDD | |
| -Individual UL CCTrCH information | | |
| - UL TFCS ID | (This IE is repeated for TFC number.) | |
| - UL TFCS | | |
| - TFC subset | Default value is the complete existing set of transport format combinations | |
| - Allowed Transport Format combination | 0 to MaxTFCvalue-1 (MaxTFCValue is refer to clause 6 Parameter Set.) | |
| - PRACH TFCS | (This IE is repeated for TFC number.) | |
| - CHOICE TFCI signalling | Normal | |
| - TFCI Field 1 information | | |
| - TFCS complete reconfigure information | | |
| - CHOICE TFCS Size | Number of used bits must be enough to cover all combinations of CTFC from clauses 6. Refer to clause 6 Parameter Set | |
| - CTFC information | Not Present | |
| - CHOICE mode | TDD | |
| - Individual UL CCTrCH information | Not Present | |
| Deleted TrCH information list | Not Present | |
| Added or Reconfigured UL TrCH information list | 1 | |
| - Added or Reconfigured UL TrCH information | | |
| - Uplink transport channel type | DCH | |
| - UL Transport channel identity | 5 | |
| - TFS | | |
| - CHOICE Transport channel type | Dedicated transport channels | |
| - Dynamic Transport Format Information | | |
| - RLC size | According to clause 6 | |
| - Number of TBs and TTI List | (This IE is repeated for TFI number) | |
| - CHOICE mode | TDD | |
| - Transmission Time Interval | According to clause 6 | |
| - CHOICE Logical channel list | All | |
| - Semi-static Transport Format information | | |
| DL Transport channel information common for all transport channel | | |
| - SCCPCH TFCS | Not Present | |
| - CHOICE mode | TDD | |
| - CHOICE DL parameters | Same as UL | |
| Added or Reconfigured DL TrCH information list | 1 | |
| - Added or Reconfigured DL TrCH information | | |
| - Downlink transport channel type | DCH | |
| - DL Transport channel identity | 10 | |

| Information Element | Value/remark | Version |
|---|---|---|
| - CHOICE DL parameters | Same as UL | |
| - Uplink transport channel type | DCH | |
| - UL TrCH Identity | 5 | |
| - DCH quality target | | |
| - BLER Quality value | Reference to the present document | |
| Frequency info | Not Present | |
| Maximum allowed UL TX power | Not Present | |
| CHOICE channel requirement | Uplink DPCH info | |
| - Uplink DPCH power control info | | |
| - CHOICE mode | TDD | |
| - CHOICE *TDD option* | 3.84 Mcps | |
| - UL target SIR | Reference to clause 6 Parameter set | |
| - CHOICE mode | TDD | |
| - CHOICE *UL OL PC info* | Individually signalled | |
| - CHOICE *TDD option* | 3.84 Mcps | |
| - Individual timeslot interference info | Not Present | |
| - Individual timeslot interference | | |
| - DPCH Constant Value | | |
| - Primary CCPCH Tx Power | Not Present | |
| - Time info | | |
| - Activation time | (256+CFN-(CFN MOD 8 + 8))MOD 256 | |
| - Duration | Infinite | |
| - Common timeslot info | | |
| - 2nd interleaving mode | Reference to clause 6.10 Parameter Set | |
| - TFCI coding | Reference to clause 6.10 Parameter Set | |
| - Puncturing Limit | Reference to clause 6.10 Parameter Set | |
| - Repetition Period | Reference to clause 6.10 Parameter Set | |
| - Repetition Length | Reference to clause 6.10 Parameter Set | |
| - Uplink DPCH timeslots and codes | Default is to use the old timeslots and codes | |
| - CPCH SET Info | (no data) | R99 and Rel-4 only |
| Downlink information common for all radio links | | |
| - Downlink DPCH info common for all RL | | |
| - Timing Indication | Initialize | |
| - CFN-targetSFN frame offset | Not Present | |
| - Downlink DPCH power control information | | |
| - DPC mode | 0 (single) | |
| - CHOICE mode | TDD | |
| - CHOICE TDD option | 3.84 Mcps (no data) | |
| - Default DPCH Offset Value | Arbitrary set to value 0..306688 by step of 512 | |
| Downlink information for per radio links list | | |
| -Downlink information for each radio links | | |
| - CHOICE mode | TDD | |
| - Primary CCPCH info | | |
| - CHOICE *SyncCase* | Sync Case 1 | |
| - Timeslot | PCCPCH timeslot | |
| - Cell parameters ID | 0 | |
| - SCTD indicator | | |
| - Downlink DPCH info for each RL | | |
| - CHOICE mode | TDD | |
| - DL CCTrCH List | | |
| - TFCS ID | 1 | |
| - Time info | | |
| - Activation time | (256+CFN-(CFN mod 8 + 8))mod 256 | |
| - Duration | infinite | |
| - Common timeslot info | | |
| - 2nd interleaving mode | Reference to the present document | |
| - TFCI coding | TRUE | |
| - Puncturing limit | Reference to clause 6 Parameter set | |
| - Repetition period | 1 | |
| - Repetition length | Empty | |
| - Downlink DPCH timeslots and codes | | |
| - CHOICE *more timeslots* | | |
| - CHOICE TDD option | 3.84 Mcps | |
| - Timeslot number | The number of a downlink timeslot that has unassigned codes in a frame. | |
| - Individual timeslot info | | |

| Information Element | Value/remark | Version |
|---|---|---|
| - TFCI existence | TRUE | |
| - Midamble shift and burst type | | |
| - CHOICE TDD option | 3.84 Mcps | |
| -CHOICE Burst Type | | |
| -Type 1 | | |
| -Midamble Allocation Mode | Default | |
| - Midamble configuration burst type 1 and 3 | As defined in 3GPP TS 25.221 [28] | |
| - First timeslot channelisation codes | | |
| - First channelisation code | (i/SF) where i is the lowest numbered code that is being assigned and SF is specified in clause 6 Parameter Set.. | |
| - Last channelisation code | (j/SF) where j is the highest numbered code that is being assigned in the slot. | |
| - CHOICE more timeslots | The presence of this IE depends upon whether the requirements of clause 6 Parameter Set could be met by the codes that have been assigned in the first timeslot. | |
| - UL CCTrCH TPC List | Not Present | |
| -SCCPCH information for FACH | Not Present | R99 and Rel-4 only |

Contents of RRC CONNECTION SETUP message: UM (1.28 Mcps TDD)

| Information Element | Value/remark | Version |
|---|---|---|
| Message Type | | |
| Initial UE identity | Select the same identity as in the IE "Initial UE Identity" in received RRC CONNECTION REQUEST" message | |
| RRC transaction identifier | Arbitrarily selects an integer between 0 and 3 | |
| Activation time | Not Present(Now) | |
| New U-RNTI | | |
| - SRNC identity | 0000 0000 0001B | |
| - S-RNTI | 0000 0000 0000 0000 0001B | |
| New C-RNTI | Not Present | |
| RRC State Indicator | CELL_DCH | |
| UTRAN DRX cycle length coefficient | 9 | |
| Capability update requirement | | |
| - UE radio access FDD capability update requirement | FALSE | |
| - UE radio access TDD capability update requirement | TRUE | |
| - System specific capability update requirement list | GSM | |
| CHOICE *specification mode* | Complete specification | Rel-5 |
| - Complete specification | | Rel-5 |
| - Signalling RB information to setup list | 4 SRBs | |
| - Signalling RB information to setup | (UM DCCH for RRC) | |
| - RB identity | Not Present | |
| - CHOICE RLC info type | RLC info | |
| - CHOICE Uplink RLC mode | UM RLC | |
| - Transmission RLC discard | Not Present | |
| - CHOICE Downlink RLC mode | UM RLC | |
| - RB mapping info | | |
| - Information for each multiplexing option | 2 RBMuxOptions | |
| - RLC logical channel mapping indicator | Not Present | |
| - Number of RLC logical channels | 1 | |
| - Uplink transport channel type | DCH | |
| - UL Transport channel identity | 5 | |
| - Logical channel identity | 1 | |
| - CHOICE RLC size list | Configured | |
| - MAC logical channel priority | 1 | |
| - Downlink RLC logical channel info | | |
| - Number of RLC logical channels | 1 | |
| - Downlink transport channel type | DCH | |
| - DL DCH Transport channel identity | 10 | |

| Information Element | Value/remark | Version |
|---|---|---|
| - DL DSCH Transport channel identity | Not Present | |
| - Logical channel identity | 1 | |
| - RLC logical channel mapping indicator | Not Present | |
| - Number of RLC logical channels | 1 | |
| - Uplink transport channel type | RACH | |
| - UL Transport channel identity | Not Present | |
| - Logical channel identity | 1 | |
| - CHOICE RLC size list | Configured | |
| - RLC size index | Reference to clause 6 Parameter Set | |
| - MAC logical channel priority | 1 | |
| - Downlink RLC logical channel info | | |
| - Number of RLC logical channels | 1 | |
| - Downlink transport channel type | FACH | |
| - DL DCH Transport channel identity | Not Present | |
| - DL DSCH Transport channel identity | Not Present | |
| - Logical channel identity | 1 | |
| - Signalling RB information to setup | (AM DCCH for RRC) | |
| - RB identity | Not Present | |
| - CHOICE RLC info type | | |
| - RLC info | | |
| - CHOICE Uplink RLC mode | AM RLC | |
| - Transmission RLC discard | | |
| - SDU discard mode | No Discard | |
| - MAX_DAT | 415 | |
| - Transmission window size | 128 | |
| - Timer_RST | 500 | |
| - Max_RST | 4 | |
| - Polling info | | |
| - Timer_poll_prohibit | 200 | |
| - Timer_poll | 200 | |
| - Poll_PDU | Not Present | |
| - Poll_SDU | 1 | |
| - Last transmission PDU poll | TRUE | |
| - Last retransmission PDU poll | TRUE | |
| - Poll_Windows | 99 | |
| - Timer_poll_periodic | Not Present | |
| - CHOICE Downlink RLC mode | AM RLC | |
| - In-sequence delivery | TRUE | |
| - Receiving window size | 128 | |
| - Downlink RLC status info | | |
| - Timer_status_prohibit | 200 | |
| - Timer_EPC | Not Present | |
| - Missing PDU indicator | TRUE | |
| - Timer_STATUS_periodic | Not Present | |
| - RB mapping info | | |
| - Information for each multiplexing option | 2 RBMuxOptions | |
| - RLC logical channel mapping indicator | Not Present | |
| - Number of RLC logical channels | 1 | |
| - Uplink transport channel type | DCH | |
| - UL Transport channel identity | 5 | |
| - Logical channel identity | 2 | |
| - CHOICE RLC size list | Configured | |
| - MAC logical channel priority | 2 | |
| - Downlink RLC logical channel info | | |
| - Number of RLC logical channels | 1 | |
| - Downlink transport channel type | DCH | |
| - DL DCH Transport channel identity | 10 | |
| - DL DSCH Transport channel identity | Not Present | |
| - Logical channel identity | 2 | |
| - RLC logical channel mapping indicator | Not Present | |
| - Number of RLC logical channels | 1 | |
| - Uplink transport channel type | RACH | |
| - UL Transport channel identity | Not Present | |
| - Logical channel identity | 2 | |
| - CHOICE RLC size list | Explicit List | |
| - RLC size index | Reference to clause 6 Parameter Set | |
| - MAC logical channel priority | 2 | |

| Information Element | Value/remark | Version |
|---|---|---|
| - Downlink RLC logical channel info | | |
| - Number of RLC logical channels | 1 | |
| - Downlink transport channel type | FACH | |
| - DL DCH Transport channel identity | Not Present | |
| - DL DSCH Transport channel identity | Not Present | |
| - Logical channel identity | 2 | |
| - Signalling RB information to setup | (AM DCCH for NAS_DT High priority) | |
| - RB identity | Not Present | |
| - CHOICE RLC info type | | |
| - RLC info | | |
| - CHOICE Uplink RLC mode | AM RLC | |
| - Transmission RLC discard | | |
| - SDU discard mode | No Discard | |
| - MAX_DAT | 415 | |
| - Transmission window size | 128 | |
| - Timer_RST | 500 | |
| - Max_RST | 4 | |
| - Polling info | | |
| - Timer_poll_prohibit | 200 | |
| - Timer_poll | 200 | |
| - Poll_PDU | Not Present | |
| - Poll_SDU | 1 | |
| - Last transmission PDU poll | TRUE | |
| - Last retransmission PDU poll | TRUE | |
| - Poll_Windows | 99 | |
| - Timer_poll_periodic | Not Present | |
| - CHOICE Downlink RLC mode | AM RLC | |
| - In-sequence delivery | TRUE | |
| - Receiving window size | 128 | |
| - Downlink RLC status info | | |
| - Timer_status_prohibit | 200 | |
| - Timer_EPC | Not Present | |
| - Missing PDU indicator | TRUE | |
| - Timer_STATUS_periodic | Not Present | |
| - RB mapping info | | |
| - Information for each multiplexing option | 2 RBMuxOptions | |
| - RLC logical channel mapping indicator | Not Present | |
| - Number of RLC logical channels | 1 | |
| - Uplink transport channel type | DCH | |
| -UL Transport channel identity | 5 | |
| - Logical channel identity | 3 | |
| - CHOICE RLC size list | Configured | |
| - MAC logical channel priority | 3 | |
| - Downlink RLC logical channel info | | |
| - Number of RLC logical channels | 1 | |
| - Downlink transport channel type | DCH | |
| - DL DCH Transport channel identity | 10 | |
| - DL DSCH Transport channel identity | Not Present | |
| - Logical channel identity | 3 | |
| - RLC logical channel mapping indicator | Not Present | |
| - Number of RLC logical channels | 1 | |
| - Uplink transport channel type | RACH | |
| - UL Transport channel identity | Not Present | |
| - Logical channel identity | 3 | |
| - CHOICE RLC size list | Explicit List | |
| - RLC size index | Reference to clause 6 Parameter Set | |
| - MAC logical channel priority | 3 | |
| - Downlink RLC logical channel info | | |
| - Number of RLC logical channels | 1 | |
| - Downlink transport channel type | FACH | |
| - DL DCH Transport channel identity | Not Present | |
| - DL DSCH Transport channel identity | Not Present | |
| - Logical channel identity | 3 | |
| - Signalling RB information to setup | (AM DCCH for NAS_DT Low priority) | |
| - RB identity | Not Present | |
| - CHOICE RLC info type | | |
| - RLC info | | |

| Information Element | Value/remark | Version |
|---|---|---|
| - CHOICE Uplink RLC mode | AM RLC | |
| - Transmission RLC discard | | |
| - SDU discard mode | No Discard | |
| - MAX_DAT | 415 | |
| - Transmission window size | 128 | |
| - Timer_RST | 500 | |
| - Max_RST | 4 | |
| - Polling info | | |
| - Timer_poll_prohibit | 200 | |
| - Timer_poll | 200 | |
| - Poll_PDU | Not Present | |
| - Poll_SDU | 1 | |
| - Last transmission PDU poll | TRUE | |
| - Last retransmission PDU poll | TRUE | |
| - Poll_Windows | 99 | |
| - Timer_poll_periodic | Not Present | |
| - CHOICE Downlink RLC mode | AM RLC | |
| - In-sequence delivery | TRUE | |
| - Receiving window size | 128 | |
| - Downlink RLC status info | | |
| - Timer_status_prohibit | 200 | |
| - Timer_EPC | Not Present | |
| - Missing PDU indicator | TRUE | |
| - Timer_STATUS_periodic | Not Present | |
| - RB mapping info | | |
| - Information for each multiplexing option | 2 RBMuxOptions | |
| - RLC logical channel mapping indicator | Not Present | |
| - Number of RLC logical channels | 1 | |
| - Uplink transport channel type | DCH | |
| - UL Transport channel identity | 5 | |
| - Logical channel identity | 4 | |
| - CHOICE RLC size list | Configured | |
| - MAC logical channel priority | 4 | |
| - Downlink RLC logical channel info | | |
| - Number of RLC logical channels | 1 | |
| - Downlink transport channel type | DCH | |
| - DL DCH Transport channel identity | 10 | |
| - DL DSCH Transport channel identity | Not Present | |
| - Logical channel identity | 4 | |
| - RLC logical channel mapping indicator | Not Present | |
| - Number of RLC logical channels | 1 | |
| - Uplink transport channel type | RACH | |
| - UL Transport channel identity | Not Present | |
| - Logical channel identity | 4 | |
| - CHOICE RLC size list | Explicit List | |
| - RLC size index | Reference to clause 6 Parameter Set | |
| - MAC logical channel priority | 4 | |
| - Downlink RLC logical channel info | | |
| - Number of RLC logical channels | 1 | |
| - Downlink transport channel type | FACH | |
| - DL DCH Transport channel identity | Not Present | |
| - DL DSCH Transport channel identity | Not Present | |
| - Logical channel identity | 4 | |
| UL Transport channel information for all transport channels | | |
| - PRACH TFCS | Not Present | |
| - CHOICE Mode | TDD | |
| -Individual UL CCTrCH information | | |
| - UL TFCS ID | (This IE is repeated for TFC number.) | |
| - UL TFCS | | |
| - TFC subset | Default value is the complete existing set of transport format combinations | |
| - Allowed Transport Format combination | 0 to MaxTFCvalue-1 (MaxTFCValue is refer to clause 6 Parameter Set.) | |
| - PRACH TFCS | (This IE is repeated for TFC number.) | |
| - CHOICE TFCI signalling | Normal | |
| - TFCI Field 1 information | | |

| Information Element | Value/remark | Version |
|---|---|---|
| - TFCS complete reconfigure information | | |
| - CHOICE TFCS Size | Number of used bits must be enough to cover all combinations of CTFC from clauses 6. Refer to clause 6 Parameter Set | |
| - CTFC information | Not Present | |
| - CHOICE mode | TDD | |
| - Individual UL CCTrCH information | Not Present | |
| Deleted TrCH information list | Not Present | |
| Added or Reconfigured UL TrCH information list | 1 | |
| - Added or Reconfigured UL TrCH information | | |
| - Uplink transport channel type | DCH | |
| - UL Transport channel identity | 5 | |
| - TFS | | |
| - CHOICE Transport channel type | Dedicated transport channels | |
| - Dynamic Transport Format Information | | |
| - RLC size | According to clause 6 | |
| - Number of TBs and TTI List | (This IE is repeated for TFI number) | |
| - CHOICE mode | TDD | |
| - Transmission Time Interval | According to clause 6 | |
| - CHOICE Logical channel list | All | |
| - Semi-static Transport Format information | | |
| DL Transport channel information common for all transport channel | | |
| - SCCPCH TFCS | Not Present | |
| - CHOICE mode | TDD | |
| - CHOICE DL parameters | Same as UL | |
| Added or Reconfigured DL TrCH information list | 1 | |
| - Added or Reconfigured DL TrCH information | | |
| - Downlink transport channel type | DCH | |
| - DL Transport channel identity | 10 | |
| - CHOICE DL parameters | Same as UL | |
| - Uplink transport channel type | DCH | |
| - UL TrCH Identity | 5 | |
| - DCH quality target | | |
| - BLER Quality value | Reference to the present document | |
| Frequency info | Not Present | |
| Maximum allowed UL TX power | Not Present | |
| CHOICE channel requirement | Uplink DPCH info | |
| - Uplink DPCH power control info | | |
| - CHOICE mode | TDD | |
| - CHOICE *TDD option* | 1.28 Mcps | |
| - $PRX_{PDPCHdes}$ | Reference to clause 6 Parameter set | |
| - CHOICE mode | TDD | |
| - CHOICE *UL OL PC info* | Individually signalled | |
| - CHOICE *TDD option* | 1.28 Mcps | |
| - TPC step size | Not Present | |
| - Primary CCPCH Tx Power | Not Present | |
| - Primary CCPCH Tx Power | Not Present | |
| - Time info | | |
| - Activation time | (256+CFN-(CFN MOD 8 + 8))MOD 256 | |
| - Duration | Infinite | |
| - Common timeslot info | | |
| - 2nd interleaving mode | Reference to clause 6 Parameter Set | |
| - TFCI coding | Reference to clause 6 Parameter Set | |
| - Puncturing Limit | Reference to clause 6 Parameter Set | |
| - Repetition Period | Reference to clause 6 Parameter Set | |
| - Repetition Length | Reference to clause 6 Parameter Set | |
| - Uplink DPCH timeslots and codes | Default is to use the old timeslots and codes | |
| - CPCH SET Info | (no data) | R99 and Rel-4 only |
| Downlink information common for all radio links | | |
| - Downlink DPCH info common for all RL | | |
| - Timing Indication | Initialize | |
| - CFN-targetSFN frame offset | Not Present | |
| - Downlink DPCH power control information | | |

| Information Element | Value/remark | Version |
|---|---|---|
| - DPC mode | 0 (single) | |
| - CHOICE mode | TDD | |
| - CHOICE TDD option | 1.28 Mcps | |
| - TSTD indicator | TRUE | |
| - Default DPCH Offset Value | Arbitrary set to value 0..306688 by step of 512 | |
| Downlink information for per radio links list | | |
| -Downlink information for each radio links | | |
| - CHOICE mode | TDD | |
| - Primary CCPCH info | | |
| - CHOICE *SyncCase* | Sync Case 1 | |
| - Timeslot | PCCPCH timeslot | |
| - Cell parameters ID | 0 | |
| - SCTD indicator | | |
| - Downlink DPCH info for each RL | | |
| - CHOICE mode | TDD | |
| - DL CCTrCH List | | |
| - TFCS ID | 1 | |
| - Time info | | |
| - Activation time | (256+CFN-(CFN mod 8 + 8))mod 256 | |
| - Duration | infinite | |
| - Common timeslot info | | |
| - 2nd interleaving mode | Reference to the present document | |
| - TFCI coding | TRUE | |
| - Puncturing limit | Reference to clause 6 Parameter set | |
| - Repetition period | 1 | |
| - Repetition length | Empty | |
| - Downlink DPCH timeslots and codes | | |
| - CHOICE *more timeslots* | | |
| - CHOICE TDD option | 1.28 Mcps | |
| - Timeslot number | The number of a downlink timeslot that has unassigned codes in a subframe. | |
| - Individual timeslot info | | |
| - TFCI existence | TRUE | |
| - Midamble shift and burst type | | |
| - CHOICE TDD option | 1.28 Mcps | |
| -CHOICE Burst Type | | |
| -Midamble Allocation Mode | Default | |
| - Midamble configuration | As defined in 3GPP TS 25.221 [28] | |
| - First timeslot channelisation codes | | |
| - First channelisation code | (i/SF) where i is the lowest numbered code that is being assigned and SF is specified in clause 6 Parameter Set. | |
| - Last channelisation code | (j/SF) where j is the highest numbered code that is being assigned in the slot. | |
| - CHOICE more timeslots | The presence of this IE depends upon whether the requirements of clause 6 Parameter Set could be met by the codes that have been assigned in the first timeslot. | |
| - UL CCTrCH TPC List | Not Present | |
| -SCCPCH information for FACH | Not Present | R99 and Rel-4 only |

Contents of SECURITY MODE COMMAND message: AM

| Information Element | Condition | Value/remark |
|---|---|---|
| Message Type | A1, A2 | |
| RRC transaction identifier | | Arbitrarily selects an integer between 0 and 3 |
| Integrity check info | | |
| - Message authentication code | | Set to an arbitrarily selected 32-bits integer. The first/ leftmost bit of the bit string contains the most significant bit of the MAC-I. |
| - RRC Message Sequence Number | | Set to an arbitrarily selected integer between 0 and 15 |

| Information Element | Condition | Value/remark |
|---|---|---|
| Security capability | | |
| - Ciphering algorithm capability | | |
| - UEA0 | | If the UE has indicated support for ciphering algorithm UEA0 in the IE "security capability" in the RRC CONNECTION SETUP COMPLETE message, this IE is set to TRUE. |
| - UEA1 | | If the UE has indicated support for ciphering algorithm UEA1 in the IE "security capability" in the RRC CONNECTION SETUP COMPLETE message, this IE is set to TRUE. |
| - Spare | | Spare 2-15 = FALSE |
| - Integrity protection algorithm capability | | 0000000000000010B (UIA1) |
| - UIA1 | | TRUE |
| - Spare | | Spare 0 and Spare 2-15 = FALSE |
| Ciphering mode info | | This presence of this IE is dependent on IXIT statements in TS 34.123-2. If ciphering is indicated to be active, this IE present with the values of the sub IEs as stated below. Else, this IE is omitted. |
| - Ciphering mode command | | Start/restart |
| - Ciphering algorithm | | UEA0 or UEA1. The indicated algorithm must be one of the algorithms supported by the UE as indicated in the IE "security capability" in the RRC CONNECTION SETUP COMPLETE message.Use the same ciphering algorithm specified in "ciphering |
| - Ciphering activation time for DPCH | | Not Present |
| - Radio bearer downlink ciphering activation time info | | |
| - Radio bearer activation time | | |
| - RB identity | | 1 |
| - RLC sequence number | | Current RLC SN+2 |
| - RB identity | | 2 |
| - RLC sequence number | | Current RLC SN+2 |
| - RB identity | | 3 |
| - RLC sequence number | | Current RLC SN + 2 |
| - RB identity | | 4 |
| - RLC sequence number | | Current RLC SN + 2 |
| Integrity protection mode info | | |
| - Integrity protection mode command | | Start |
| - Downlink integrity protection activation info | | Not Present |
| - Integrity protection algorithm | | UIA1 |
| - Integrity protection initialisation number | | SS selects an arbitrary 32 bits number for FRESH |
| CN domain identity | | CS or PS |
| UE system specific security capability | A1 | Not Checked |
| UE system specific security capability | A2 | |
| - Inter-RAT UE security capability | | |
| - CHOICE *system* | | GSM |
| - GSM security capability | | The indicated algorithms must be the same as the algorithms supported by the UE as indicated in the IE " UE system specific capability " in the RRC CONNECTION SETUP COMPLETE message. |

| Condition | Explanation |
|---|---|
| A1 | UE not supporting GSM |
| A2 | UE supporting GSM |

# 10 A-GPS GPS Scenarios and Assistance Data

## 10.1 General

This clause defines the GPS scenarios and assistance data IEs which shall be available for use as specified in all A-GPS test cases in 3GPP TS 34.171 [41] and 3GPP TS 34.123-1 [1].

The information elements are given with reference to 3GPP TS 25.331 [34], where the details are defined.

Clauses 10.2 and 10.3 list the assistance data IEs required for performance testing of UE-based mode detailed in 3GPP TS 34.171 [41], and clauses 10.4 and 10.5 list the assistance data available for performance testing of UE-assisted mode detailed in 3GPP TS 34.171 [41]. Clause 10.6 lists the values of the assistance data IE fields for performance testing detailed in 3GPP TS 34.171 [41].

Clause 10.7 details the GPS scenario and the values of the assistance data IE fields for signalling testing detailed in 3GPP TS 34.123-1 [1], clause 17.2.

The A-GPS minimum performance requirements are defined by assuming that all relevant and valid assistance data is received by the UE in order to perform GPS measurements and/or position calculation. This clause does not include nor consider delays occurring in the various signalling interfaces of the network.

### 10.1.1 Satellite constellations and assistance data for performance testing

The satellite constellations for performance testing shall consist of 24 satellites. Almanac assistance data shall be available for all these 24 satellites. At least 9 of the satellites shall be visible to the UE (that is above 5 degrees elevation with respect to the UE). Other assistance data shall be available for 9 of these visible satellites. In each test, signals are generated for only a sub-set of these satellites for which other assistance data is available. The number of satellites in this sub-set is specified in the test. The satellites in this sub-set shall all be above 15 degrees elevation with respect to the UE. The HDOP for the test shall be calculated using this sub-set of satellites. The selection of satellites for this sub-set shall be random and consistent with achieving the required HDOP for the test.

### 10.1.2 GPS Scenarios for performance testing

This section defines the GPS scenarios that shall be used for all Assisted GPS performance tests defined in TS 34.171 [41].

They have been selected to be consistent with achieving the required HDOP for the Test Cases and so that for each test instance the reference location shall change sufficiently such that the UE shall have to use the new assistance data.

The satellites to be simulated in each test case are specified in clause 10.1.2.5.

The viable running time during which the scenario maintains the required HDOP or HDOPs is given. Once this time has been reached the scenario shall be restarted from its nominal start time.

#### 10.1.2.1 GPS Scenario #1

The following GPS scenario #1 shall be used during the TTFF tests defined in TS 34.171 [41]. The assistance data specified in the following sections for GPS scenario #1 is consistent with this GPS scenario.

Yuma Almanac data: see file GPS 1 Yuma.txt in the GPS data perf zip file specified in annex C.2.

UE location: the UE location is calculated as a random offset from the reference location using the method described in section 10.1.2.4. The reference location is: latitude: 33 degrees 45 minutes 0.019 seconds north, longitude: 84 degrees 23 minutes 0.011 seconds west, (Atlanta USA), height: = 300m.

Nominal start time: 22nd January 2005 (Saturday) 00:08:00.

Viable running time to maintain specified HDOP values: 19 minutes.

Visible satellites available for simulation: PRNs: 2, 6, 10, 17, 18, 21, 26, 29, 30.

Ionospheric model: see values in section 10.6.6.

Tropospheric model: STANAG with SRI equal to 324.8, as defined in [43].

### 10.1.2.2 GPS Scenario #2

The following GPS scenario #2 shall be used during the TTFF tests defined in TS 34.171 [41]. The assistance data specified in the following sections for GPS scenario #2 is consistent with this GPS scenario.

Yuma Almanac data: see file GPS 2 Yuma.txt in the GPS data perf zip file specified in annex C.2.

UE location: the UE location is calculated as a random offset from the reference location using the method described in section 10.1.2.4. The reference location is: latitude: 37 degrees 48 minutes 59.988 seconds south, longitude: 144 degrees 58 minutes 0.013 seconds east, (Melbourne Australia), height: = 100m.

Nominal start time: 22nd January 2004 (Thursday) 00:08:00.

Viable running time to maintain specified HDOP values: 19 minutes.

Visible satellites available for simulation: PRNs: 3, 11, 14, 15, 18, 22, 23, 25, 31.

Ionospheric model: see values in section 10.6.6.

Tropospheric model: STANAG with SRI equal to 324.8, as defined in [43].

### 10.1.2.3 GPS Scenario #3

The following GPS scenario #3 shall be used during the Moving Scenario and Periodic Update test case defined in TS 34.171 [41]. The assistance data specified in the following sections for GPS scenario #3 is consistent with this GPS scenario.

Yuma Almanac data: see file GPS 3 Yuma.txt in the GPS data perf zip file specified in annex C.2.

UE location: the UE location is given as a trajectory as shown in Figure 5.6.1 of TS 34.171 [41]. The reference location is at the centre of the trajectory and is at: latitude: 37 degrees 48 minutes 59.988 seconds south, longitude: 144 degrees 58 minutes 0.013 seconds east, (Melbourne Australia), height: = 100m.

Start time: 22nd January 2004 (Thursday) 00:08:00.

Start location: at the point betweeh $l_{11}$ and $l_{12}$ in Figure 5.6.1 of TS 34.171 [41], going in a clock-wise direction.

Visible satellites simulated: PRNs: 3, 14, 15, 22, 25.

Viable running time to maintain specified HDOP values: 19 minutes.

Ionospheric model: see values in section 10.6.6.

Tropospheric model: STANAG with SRI equal to 324.8, as defined in [43].

### 10.1.2.4 UE Location for TTFF test cases

This section defines the method for generating the random UE locations that are required to be used for the TTFF tests defined in TS 34.171 [41].

For every Test Instance in each TTFF test case, the UE location shall be randomly selected to be within 3 km of the Reference Location. The Altitude of the UE shall be randomly selected between 0 m to 1 000 m above WGS-84 reference ellipsoid. These values shall have uniform random distributions.

The UE location is calculated as an offset from the Reference Location.

#### 10.1.2.4.1 UE Location Offset

The UE location offset shall be calculated by selecting the next pair of random numbers, representing a pair of latitude and longitude offsets in degrees, from a standard uniform random number generator, with the following properties:

> The ranges of the latitude and longitude offsets values shall be such that when translated onto the surface of the earth they shall lie within a 3km radius circle, centred on the Reference location specified for the GPS scenario under consideration. For the purposes of this calculation make the following assumptions:

a) Over the 3km radius circle at the Reference location the earth is flat and the meridians and parallels form a rectangular grid

b) The earth is spherical with a radius of 6371141m (equal to the WGS 84 value at 35 degrees latitude)

The resolution used for the latitude and longitude offsets values shall be 90/2E23 for the latitude offset values and 360/2E24 for the longitude offset values, representing the coding resolution in degrees specified in TS23.032 [42].

#### 10.1.2.4.2 UE Altitude

The UE altitude value shall be calculated by selecting the next random number from a standard uniform random number generator, in the range 0 to 1000, representing meters. The resolution used for the random number shall be 1, representing 1 meter.

### 10.1.2.5 Satellites to be simulated in each test case

The satellites to be simulated in each test case have been selected in order to achieve the required HDOP for that test case.

**Satellites to be simulated**

| Test case | PRNs GPS #1 | PRNs GPS #2 | PRNs GPS #3 |
|---|---|---|---|
| Sensitivity Coarse Time Assistance | 2, 6, 10, 17, 18, 21, 26, 29 | 3, 11, 14, 15, 22, 23, 25, 31 | - |
| Sensitivity Fine Time Assistance | 2, 6, 10, 17, 18, 21, 26, 29 | 3, 11, 14, 15, 22, 23, 25, 31 | - |
| Nominal Accuracy | 2, 6, 10, 17, 18, 21, 26, 29 | 3, 11, 14, 15, 22, 23, 25, 31 | - |
| Dynamic Range | 2, 6, 10, 17, 26, 29 | 3, 14, 15, 22, 25, 31 | - |
| Multi-path Performance | 2, 6, 17, 21, 26 | 3, 14, 15, 22, 25 | - |
| Moving Scenario and Periodic Update Performance | - | - | 3, 14, 15, 22, 25 |

## 10.2 Information elements required for normal UE based testing

The following A-GPS assistance data IEs and fields shall be present for each test. Fields not specified shall not be present. The values of the fields are specified in clause 10.6.

**a) UE positioning GPS reference time IE.** This information element is defined in clause 10.3.7.96 of 3GPP TS 25.331 [34].

| Name of the IE | Fields of the IE |
|---|---|
| UE positioning GPS reference time | |
| | GPS Week |
| | GPS TOW msec |
| | GPS TOW Assist |
| | SatID |
| | TLM Message |
| | TLM Reserved |
| | Alert |
| | Anti-Spoof |

**b) UE positioning GPS reference UE position IE.** This information element is defined in clause 10.3.8.4c of 3GPP TS 25.331 [34].

| Name of the IE | Fields of the IE |
|---|---|
| Reference Location | Ellipsoid point with Altitude and uncertainty ellipsoid |

**c) UE positioning GPS navigation model IE**. This information element is defined in clause 10.3.7.94 of 3GPP TS 25.331 [34].

| Name of the IE | Fields of the IE |
|---|---|

| | |
|---|---|
| Navigation Model | All satellite information |

**d) UE positioning GPS ionospheric model IE.** This information element is defined in clause 10.3.7.92 of 3GPP TS 25.331 [34].

| Name of the IE | Fields of the IE |
|---|---|
| Ionospheric Model | All |

## 10.3 Information elements required for UE based Sensitivity Fine Time Assistance test case

The A-GPS assistance data IEs and fields that shall be present for the Sensitivity Fine Time Assistance test case shall be those specified in clause 10.2 with the following exception. Fields not specified shall not be present. The values of the fields are specified in clause 10.6.

**UE positioning GPS reference time IE.** This information element is defined in clause 10.3.7.96 of 3GPP TS 25.331 [34].

| Name of the IE | Fields of the IE |
|---|---|
| UE positioning GPS reference time | |
| | GPS Week |
| | GPS TOW msec |
| | UTRAN GPS reference time |
| | >UTRAN GPS timing of cell frames |
| | >CHOICE mode |
| | >>FDD |
| | >>>Primary CPICH Info |
| | >SFN |
| | SFN-TOW Uncertainty |
| | TUTRAN-GPS drift rate |
| | GPS TOW Assist |
| | SatID |
| | TLM Message |
| | TLM Reserved |
| | Alert |
| | Anti-Spoof |

## 10.4 Information elements available for normal UE assisted testing

The following A-GPS assistance data IEs and fields shall be available for use in each test. Fields not specified shall not be present. The values of the fields are specified in clause 10.6.

**a) UE positioning GPS reference time IE.** This information element is defined in clause 10.3.7.96 of 3GPP TS 25.331 [34].

| Name of the IE | Fields of the IE |
|---|---|
| UE positioning GPS reference time | |
| | GPS Week |
| | GPS TOW msec |
| | GPS TOW Assist |
| | SatID |
| | TLM Message |
| | TLM Reserved |
| | Alert |
| | Anti-Spoof |

**b) UE positioning GPS reference UE position IE.** This information element is defined in clause 10.3.8.4c of 3GPP TS 25.331 [34].

| Name of the IE | Fields of the IE |
|---|---|
| Reference Location | Ellipsoid point with Altitude and uncertainty ellipsoid |

**c) UE positioning GPS almanac** This information element is defined in clause 10.3.7.89 of 3GPP TS 25.331 [34].

| Name of the IE | Fields of the IE |
|---|---|
| UE positioning GPS almanac | |
| | Almanac Reference Week |
| | All Satellite information |

**d) UE positioning GPS navigation model IE.** This information element is defined in clause 10.3.7.94 of 3GPP TS 25.331 [34].

| Name of the IE | Fields of the IE |
|---|---|
| Navigation Model | All satellite information |

**e) UE positioning GPS acquisition assistance IE.** This information element is defined in clause 10.3.7.88 of 3GPP TS 25.331 [34].

| Name of the IE | Fields of the IE |
|---|---|
| Acquisition Assistance | |
| | GPS TOW msec |
| | Satellite information |
| | >SatID |
| | >Doppler ($0^{th}$ order term) |
| | >Extra Doppler |
| | >>Doppler ($1^{st}$ order term) |
| | >>Doppler Uncertainty |
| | >Code Phase |
| | >Integer Code Phase |
| | >GPS Bit number |
| | >Code Phase Search Window |
| | >Azimuth and Elevation |
| | >> Azimuth |
| | >> Elevation |

## 10.5 Information elements available for UE assisted Sensitivity Fine Time Assistance test case

The A-GPS assistance data IEs and fields that shall be available for use for the Sensitivity Fine Time Assistance test case shall be those specified in clause 10.4 with the following exceptions. Fields not specified shall not be present. The values of the fields are specified in clause 10.6.

**a) UE positioning GPS reference time IE.** This information element is defined in clause 10.3.7.96 of 3GPP TS 25.331 [34].

| Name of the IE | Fields of the IE |
|---|---|
| UE positioning GPS reference time | |
| | GPS Week |
| | GPS TOW msec |
| | UTRAN GPS reference time |
| | >UTRAN GPS timing of cell frames |
| | >CHOICE mode |
| | >>FDD |
| | >>>Primary CPICH Info |
| | >SFN |
| | SFN-TOW Uncertainty |
| | TUTRAN-GPS drift rate |
| | GPS TOW Assist |
| | SatID |

| | |
|---|---|
| | TLM Message |
| | TLM Reserved |
| | Alert |
| | Anti-Spoof |

**b) UE positioning GPS acquisition assistance IE.** This information element is defined in clause 10.3.7.88 of 3GPP TS 25.331 [34].

| Name of the IE | Fields of the IE |
|---|---|
| Acquisition Assistance | |
| | GPS TOW msec |
| | UTRAN GPS reference time |
| | >UTRAN GPS timing of cell frames |
| | >CHOICE mode |
| | >>FDD |
| | >>>Primary CPICH Info |
| | >SFN |
| | Satellite information |
| | >SatID |
| | >Doppler ($0^{th}$ order term) |
| | >Extra Doppler |
| | >>Doppler ($1^{st}$ order term) |
| | >>Doppler Uncertainty |
| | >Code Phase |
| | >Integer Code Phase |
| | >GPS Bit number |
| | >Code Phase Search Window |
| | >Azimuth and Elevation |
| | >> Azimuth |
| | >> Elevation |

# 10.6 Contents of Information elements for performance testing

## 10.6.1 General

This section defines the assistance data values that shall be used for all Assisted GPS performance tests defined in TS 34.171 [41]. It is given for GPS scenarios #1, #2 and #3 where it is different for each scenario; otherwise it is marked "All" where the same value is used for all scenarios.

Where assistance data is required on a per-satellite basis, or where the values of the data also varies with time it is specified in comma-separated-variable files with suffixes XX in the GPS data perf zip file specified in annex C.2, where XX is 01, 02 and 03 for GPS scenarios #1, #2 and #3 respectively. These files specify the values to be used for each satellite, indexed by satellite PRN, and, where applicable, the values to be used indexed by both time and satellite PRN.

Assistance data that is marked as "time varying" is specified and used in [80] ms increments. Interpolation between these values shall not be used.

Assistance data Information Elements and fields that are not specified shall not be used.

## 10.6.2 IE Random Offset Values

This section defines the methods for generating the random offsets that are required to be applied to one or two assistance data IEs for certain tests defined in TS 34.171 [41].

### 10.6.2.1 GPS TOW msec

For every Test Instance in each TTFF test case, the IE GPS TOW msec shall have a random offset, relative to GPS system time, within the allowed error range of Coarse Time Assistance defined in the test case. This offset value shall have a uniform random distribution.

Note: For the Moving Scenario and Periodic Update Test Case the value of the IE GPS TOW msec shall be set to the nominal value, i.e. no offset shall be used.

The offset value shall be calculated by selecting the next random number from a standard uniform random number generator, in the range specified for the GPS Coarse Time assistance error range in the Test Requirements, Test parameters table for the test under consideration. The resolution used for the random number shall be 0.01, representing 10ms.

#### 10.6.2.1 UTRAN GPS timing of cell frames

In addition, for every Fine Time Assistance Test Instance the IE UTRAN GPS timing of cell frames shall have a random offset, relative to the true value of the relationship between the two time references, within the allowed error range of Fine Time Assistance defined in the test case. This offset value shall have a uniform random distribution.

The offset value shall be calculated by selecting the next random number from a standard uniform random number generator with the following properties:

The range shall be the number of UMTS chips whose duration is less than the range specified for the GPS Fine Time assistance error range in the Test Requirements, Test parameters table for the test under consideration.

The resolution used for the random number shall be 1, representing 1 UMTS chip.

### 10.6.3 Assistance Data Reference Time

Contents of UE positioning GPS reference time IE

**Reference Time (Fields occurring once per message)**

| Parameter | Units | Value/remark GPS #1 | Value/remark GPS #2 | Value/remark GPS #3 |
|---|---|---|---|---|
| GPS Week | weeks | 282 | 230 | 230 |
| GPS TOW msec | msec | 518880000. Start time. Add number of ms as required. (Note 1) | 346080000. Start time. Add number of ms as required. (Note 1) | 346080000. Start time. Add number of ms as required. (Note 1) |
| UTRAN GPS reference time | | Present for Sensitivity Fine Time Assistance test case. Absent otherwise | Present for Sensitivity Fine Time Assistance test case. Absent otherwise | Absent |
| >UTRAN GPS timing of cell frames | | Note 2 | Note 2 | - |
| >CHOICE mode | | Present for Sensitivity Fine Time Assistance test case. Absent otherwise | Present for Sensitivity Fine Time Assistance test case. Absent otherwise | - |
| >>FDD | | - | - | - |
| >>>Primary CPICH Info | | 100 | 100 | - |
| >SFN | | Note 2 | Note 2 | - |
| SFN-TOW Uncertainty | | lessThan10 | lessThan10 | - |
| TUTRAN-GPS drift rate | | 0 | 0 | - |

**Note 1: GPS TOW msec**

This is the value in ms of GPS TOW msec when the GPS scenario is initially started in the GPS simulator. For all TTFF test cases, each time a GPS scenario is used, the GPS start time shall be advanced by 120 seconds from the value last used so that, at the time the fix is made, it is at least 2 minutes later than the previous fix made with that scenario.

The actual value of GPS TOW msec to be used in the Reference Time IE (before the addition of the random offset, if applicable) shall be calculated at the time the IE is required by adding the elapsed time since the time the scenario was started in the GPS simulator to this value. The accuracy shall be such that the Maximum Test System Uncertainty for Coarse Time Assistance, specified in Table F.1.2 of TS 34.171 [41], shall be met.

For all TTFF test cases a random offset is then added to the value of GPS TOW msec as described in clause 10.6.2

**Note 2: UTRAN GPS timing of cell frames and SFN**

The values of UTRAN GPS timing of cell frames (before the addition of the random offset) and SFN shall be calculated at the time the IE is required. The accuracy of the relationship between the two fields shall be such that the Maximum Test System Uncertainty for Fine Time Assistance, specified in Table F.1.2 of TS 34.171 [41], shall be met.

A random offset is then added to the value of UTRAN GPS timing of cell frames as described in clause 10.6.2

**Satellite Information**

| Parameter | Units | Value/remark GPS All |
|---|---|---|
| Number of satellites | --- | 9 |

**Reference Time - GPS TOW Assist (Fields occurring once per satellite)**

| Parameter | Units | Value/remark GPS #1 | Value/remark GPS #2 | Value/remark GPS #3 |
|---|---|---|---|---|
| SatID | | PRNs: 2, 6, 10, 17, 18, 21, 26, 29, 30 | PRNs: 3, 11, 14, 15, 18, 22, 23, 25, 31 | PRNs: 3, 11, 14, 15, 18, 22, 23, 25, 31 |

**Reference Time - GPS TOW Assist (Fields occurring once per satellite)**

| Parameter | Units | Value/remark GPS All |
|---|---|---|
| TLM Message | Bit string | 10922 |
| TLM Reserved | Bit string | 2 |
| Alert | Boolean | 0 |
| Anti-Spoof | Boolean | 1 |

## 10.6.4 Assistance Data Reference Position

Contents of UE positioning GPS reference UE position IE

The uncertainty of the semi-major axis is 3 km. The uncertainty of the semi-minor axis is 3 km. The orientation of the major axis is 0 degrees. The uncertainty of the altitude information is 500 m. The confidence factor is 68%.

**Reference Position**

| Parameter | Units | Value/remark GPS #1 | Value/remark GPS #2 | Value/remark GPS #3 |
|---|---|---|---|---|
| Type of Shape | Bit field | Ellipsoid point with altitude and uncertainty Ellipsoid | Ellipsoid point with altitude and uncertainty Ellipsoid | Ellipsoid point with altitude and uncertainty Ellipsoid |
| Degrees of latitude | degrees | 33.750005 | -37.816663 | -37.816663 |
| Degrees of longitude | degrees | -84.383517 | 144.966670 | 144.966670 |
| Altitude | m | +300 | +100 | +100 |
| Uncertainty semi-major | m | 3000 | 3000 | 3000 |
| Uncertainty semi-minor | m | 3000 | 3000 | 3000 |
| Orientation of major axis | degrees | 0 | 0 | 0 |
| Uncertainty altitude | m | 500 | 500 | 500 |
| Confidence | % | 68 | 68 | 68 |

## 10.6.5 Assistance Data Navigation Model

Contents of UE positioning GPS navigation model IE

**Satellite Information**

| Parameter | Units | Value/remark GPS All |
|---|---|---|
| Number of satellites | --- | 9 |

**Navigation Model (Fields occurring once per satellite)**

| Parameter | Units | Value/remark GPS #1 | Value/remark GPS #2 | Value/remark GPS #3 |
|---|---|---|---|---|
| SatID | --- | PRNs: 2, 6, 10, 17, 18, 21, 26, 29, 30 | PRNs: 3, 11, 14, 15, 18, 22, 23, 25, 31 | PRNs: 3, 11, 14, 15, 18, 22, 23, 25, 31 |
| Satellite Status | Boolean | 0 (Note) | 0 (Note) | 0 (Note) |

Note: For consistency Satellite Status is also given in file: Navigation model XX.csv

**Ephemeris and Clock Correction parameters (Fields occurring once per satellite)**

| Parameter | Units | Value/remark GPS All |
|---|---|---|
| C/A or P on L2 | Boolean | See file: Navigation model XX.csv |
| URA Index | Boolean | See file: Navigation model XX.csv |
| SV Health | Boolean | See file: Navigation model XX.csv |
| IODC | --- | See file: Navigation model XX.csv |
| L2 P Data Flag | Boolean | See file: Navigation model XX.csv |
| SF 1 Reserved | --- | See file: Navigation model XX.csv |
| $T_{GD}$ | sec | See file: Navigation model XX.csv |
| $t_{oc}$ | sec | See file: Navigation model XX.csv |
| $af_2$ | $sec/sec^2$ | See file: Navigation model XX.csv |
| $af_1$ | sec/sec | See file: Navigation model XX.csv |
| $af_0$ | sec | See file: Navigation model XX.csv |
| $C_{rs}$ | meters | See file: Navigation model XX.csv |
| $\Delta n$ | semi-circles/sec | See file: Navigation model XX.csv |
| $M_0$ | semi-circles | See file: Navigation model XX.csv |
| $C_{uc}$ | radians | See file: Navigation model XX.csv |
| e | --- | See file: Navigation model XX.csv |
| $C_{us}$ | radians | See file: Navigation model XX.csv |
| $(A)^{1/2}$ | $meters^{1/2}$ | See file: Navigation model XX.csv |
| $t_{oe}$ | sec | See file: Navigation model XX.csv |
| Fit Interval Flag | Boolean | See file: Navigation model XX.csv |
| AODO | sec | See file: Navigation model XX.csv |
| $C_{ic}$ | radians | See file: Navigation model XX.csv |
| $OMEGA_0$ | semi-circles | See file: Navigation model XX.csv |
| $C_{is}$ | radians | See file: Navigation model XX.csv |
| $i_0$ | semi-circles | See file: Navigation model XX.csv |
| $C_{rc}$ | meters | See file: Navigation model XX.csv |
| $\omega$ | semi-circles | See file: Navigation model XX.csv |
| OMEGAdot | semi-circles/sec | See file: Navigation model XX.csv |
| Idot | semi-circles/sec | See file: Navigation model XX.csv |

## 10.6.6 Assistance Data Ionospheric Model

Contents of UE positioning GPS ionospheric model IE

**Ionospheric Model**

| Parameter | Units | Value/remark GPS All |
|---|---|---|
| $\alpha_0$ | seconds | 4.6566129 10E-9 |
| $\alpha_1$ | sec/semi-circle | 1.4901161 10E-8 |
| $\alpha_2$ | $sec/(semi\text{-}circle)^2$ | -5.96046 10E-8 |
| $\alpha_3$ | $sec/(semi\text{-}circle)^3$ | -5.96046 10E-8 |
| $\beta_0$ | seconds | 79872 |
| $\beta_1$ | sec/semi-circle | 65536 |
| $\beta_2$ | $sec/(semi\text{-}circle)^2$ | -65536 |
| $\beta_3$ | $sec/(semi\text{-}circle)^3$ | -393216 |

## 10.6.7 Assistance Data Almanac

Contents of UE positioning GPS almanac

**Almanac (Field occurring once per message)**

| Parameter | Units | Value/remark GPS #1 | Value/remark GPS #2 | Value/remark GPS #3 |
|---|---|---|---|---|
| $WN_a$ | weeks | 283 | 230 | 230 |

**Satellite Information**

| Parameter | Units | Value/remark GPS All |
|---|---|---|
| Number of satellites | --- | 24 |

**Almanac (Fields occurring once per satellite)**

| Parameter | Units | Value/remark |
|---|---|---|
| DataID | --- | See file: Almanac XX.csv |

**Almanac (Fields occurring once per satellite)**

| Parameter | Units | Value/remark GPS #1 | Value/remark GPS #2 | Value/remark GPS #3 |
|---|---|---|---|---|
| SatID | --- | PRNs: 1, 2, 4, 5, 6, 7, 9, 10, 11, 14, 15, 16, 17, 18, 19, 20, 21, 22, 24, 25, 26, 27, 29, 30 | PRNs: 1, 2, 3, 4, 5, 6, 7, 8, 11, 13, 14, 15, 16, 17, 18, 20, 21, 22, 23, 25, 27, 28, 30, 31 | PRNs: 1, 2, 3, 4, 5, 6, 7, 8, 11, 13, 14, 15, 16, 17, 18, 20, 21, 22, 23, 25, 27, 28, 30, 31 |

**Almanac (Fields occurring once per satellite)**

| Parameter | Units | Value/remark |
|---|---|---|
| e | dimensionless | See file: Almanac XX.csv |
| $t_{oa}$ | sec | See file: Almanac XX.csv |
| $\delta i$ | semi-circles | See file: Almanac XX.csv |
| OMEGADOT | semi-circles/sec | See file: Almanac XX.csv |
| SV Health | Boolean | See file: Almanac XX.csv |
| $A^{1/2}$ | $meters^{1/2}$ | See file: Almanac XX.csv |
| $OMEGA_0$ | semi-circles | See file: Almanac XX.csv |
| $M_0$ | semi-circles | See file: Almanac XX.csv |
| $\omega$ | semi-circles | See file: Almanac XX.csv |
| $af_0$ | seconds | See file: Almanac XX.csv |
| $af_1$ | sec/sec | See file: Almanac XX.csv |

## 10.6.8 Assistance Data Acquisition Assistance

Contents of UE positioning GPS acquisition assistance IE

**GPS Acquisition Assistance (Fields occurring once per message)**

| Parameter | Units | Value/remark GPS #1 | Value/remark GPS #2 | Value/remark GPS #3 |
|---|---|---|---|---|
| GPS TOW msec | msec | 51888000 ms. Start time. Add number of ms as required. (Note 1) | 346080000 ms. Start time. Add number of ms as required. (Note 1) | 346080000 ms. Start time. Add number of ms as required. (Note 1) |
| UTRAN GPS reference time | | Present for Sensitivity Fine Time Assistance test case. Absent otherwise | Present for Sensitivity Fine Time Assistance test case. Absent otherwise | Absent |
| >UTRAN GPS timing of cell frames | | Note 2 | Note 2 | - |
| >CHOICE mode | | Present for Sensitivity Fine Time Assistance test case. Absent otherwise | Present for Sensitivity Fine Time Assistance test case. Absent otherwise | - |
| >>FDD | | - | - | - |
| >>>Primary CPICH Info | | 100 | 100 | - |
| >SFN | | Note 2 | Note 2 | - |

**Note 1: GPS TOW msec**
This is the value in ms of GPS TOW msec when the GPS scenario is initially started in the GPS simulator. For all TTFF test cases, each time a GPS scenario is used, the GPS start time shall be advanced by 120 seconds from the value last used so that, at the time the fix is made, it is at least 2 minutes later than the previous fix made with that scenario.

The actual value of GPS TOW msec to be used in the Acquisition Assistance IE (before the addition of the random offset, if applicable) shall be calculated at the time the IE is required by adding the elapsed time since the time the scenario was started in the GPS simulator to this value. The accuracy shall be such that the Maximum Test System Uncertainty for Coarse Time Assistance, specified in Table F.1.2 of TS 34.171 [41], shall be met.

For all TTFF test cases a random offset is then added to the value of GPS TOW msec as described in clause 10.6.2

This “final GPS TOW msec” value is then also used to determine the value of the Acquisition Assistance parameters marked as “Time varying” in clause 10.6.8

**Note 2: UTRAN GPS timing of cell frames and SFN**
The values of UTRAN GPS timing of cell frames (before the addition of the random offset) and SFN shall be calculated at the time the IE is required. The accuracy of the relationship between the two fields shall be such that the Maximum Test System Uncertainty for Fine Time Assistance, specified in Table F.1.2 of TS 34.171 [41], shall be met.

A random offset is then added to the value of UTRAN GPS timing of cell frames as described in clause 10.6.2

**Satellite Information**

| Parameter | Units | Value/remark GPS All |
|---|---|---|
| Number of satellites | --- | 9 |

**GPS Acquisition Assistance (Fields occurring once per satellite)**

| Parameter | Units | Value/remark GPS #1 | Value/remark GPS #2 | Value/remark GPS #3 |
|---|---|---|---|---|
| SatID | --- | PRNs: 2, 6, 10, 17, 18, 21, 26, 29, 30 | PRNs: 3, 11, 14, 15, 18, 22, 23, 25, 31 | PRNs: 3, 11, 14, 15, 18, 22, 23, 25, 31 |

**GPS Acquisition Assistance (Fields occurring once per satellite)**

| Parameter | Units | Value/remark GPS All |
|---|---|---|
| Doppler ($0^{th}$ order term) | Hz | Time varying. See file: Acquisition assist XX.csv (Note) |
| Doppler ($1^{st}$ order term) | Hz/sec | Time varying. See file: Acquisition assist XX.csv (Note) |
| Doppler Uncertainty | Hz | Time varying. See file: Acquisition assist XX.csv (Note) |
| Code Phase | chips | Time varying. See file: Acquisition assist XX.csv (Note) |
| Integer Code Phase | --- | Time varying. See file: Acquisition assist XX.csv (Note) |
| GPS Bit number | --- | Time varying. See file: Acquisition assist XX.csv (Note) |
| Code Phase Search Window | chips | Time varying. See file: Acquisition assist XX.csv (Note) |
| Azimuth | deg | Time varying. See file: Acquisition assist XX.csv (Note) |
| Elevation | deg | Time varying. See file: Acquisition assist XX.csv (Note) |
| **Note: Acquisition Assistance parameters**<br>This field is “Time varying” and its value depends on the “final GPS TOW msec” as described in clause 10.6.8. The value of this field to be used shall be determined by taking the “final GPS TOW msec” value and selecting the nearest field value in the Acquisition assist.csv file corresponding to the value of “final current GPS TOW msec”. | | |

# 10.7 GPS Scenario and values of Information Elements for signalling testing

## 10.7.1 General

This clause defines the GPS scenario and the associated assistance data values that shall be used for all Assisted GPS signalling tests defined in 3GPP TS 34.123-1 [1] clause 17.2.

Where assistance data is required on a per-satellite basis, or where the values of the data also varies with time it is specified in comma-separated-variable files in the GPS data sig zip file specified in annex C.1. These files specify the values to be used for each satellite, indexed by satellite PRN, and, where applicable, the values to be used indexed by both time and satellite PRN.

Assistance data that is marked as "time varying", and the GPS TOW msec field are only specified and used in 1 second increments. Interpolation between these values shall not be used.

The accuracy of the GPS TOW msec and assistance data that is marked as "time varying" in the provided assistance data shall be within ±2 s relative to the GPS time in the system simulator.

Assistance data Information Elements and fields that are not specified shall not be used.

## 10.7.2 GPS Scenario

The following GPS scenario shall be used. The assistance data specified in the following clauses is consistent with this GPS scenario:

- Yuma Almanac data: see file Tokyo Yuma.txt in the GPS data sig zip file specified in annex C.1.
- UE location and Reference location: static at latitude: 35 degrees 40 minutes north, longitude: 139 degrees 45 minutes east, (Tokyo) height: = 50m.
- Start time: 12th September 2003 21:30:00.
- Visible satellites simulated: PRNs: 4, 6, 9, 10, 13, 22.
- Ionospheric model: see values in clause 10.7.6.

## 10.7.3 Assistance Data Reference Time

Contents of UE positioning GPS reference time IE

**Reference Time (Fields occurring once per message)**

| Parameter | Units | Value/remark |
|---|---|---|
| GPS Week | weeks | 211 |
| GPS TOW msec | msec | 509 400 s. Start time. Add integer number of 1 seconds as required (see note) |
| NOTE: This is the value in seconds of GPS TOW msec when the GPS scenario is started in the GPS simulator. The value of GPS TOW msec to be used in the Reference Time IE shall be calculated at the time the IE is required by adding the elapsed time since the time the scenario was started in the GPS simulator to this value, rounded up to the next 1 second interval. | | |

## 10.7.4 Assistance Data Reference Position

Contents of UE positioning GPS reference UE position IE

**Reference Position**

| Parameter | Units | Value/remark |
|---|---|---|
| Type of Shape | Bit field | Ellipsoid point with altitude and uncertainty Ellipsoid |
| Degrees of latitude | degrees | +35.666667 |
| Degrees of longitude | degrees | +139.75 |
| Altitude | m | +50 |
| Uncertainty semi-major | m | 3 000 |
| Uncertainty semi-minor | m | 3 000 |
| Orientation of major axis | degrees | 0 |
| Uncertainty altitude | m | 500 |
| Confidence | % | 68 |

## 10.7.5 Assistance Data Navigation Model

Contents of UE positioning GPS navigation model IE

**Satellite Information**

| Parameter | Units | Value/remark |
|---|---|---|
| Number of Satellites | --- | 6 |

**Navigation Model (Fields occurring once per satellite)**

| Parameter | Units | Value/remark |
|---|---|---|
| SatID | --- | PRNs: 4, 6, 9, 10, 13, 22. |
| Satellite Status | Boolean | 0 (see note) |
| NOTE: For consistency Satellite Status is also given in file: Navigation model.csv. | | |

**Ephemeris and Clock Correction parameters (Fields occurring once per satellite)**

| Parameter | Units | Value/remark |
|---|---|---|
| C/A or P on L2 | Boolean | See file: Navigation model.csv |
| URA Index | Boolean | See file: Navigation model.csv |
| SV Health | Boolean | See file: Navigation model.csv |
| IODC | --- | See file: Navigation model.csv |
| L2 P Data Flag | Boolean | See file: Navigation model.csv |
| SF 1 Reserved | --- | See file: Navigation model.csv |
| $T_{GD}$ | sec | See file: Navigation model.csv |
| $t_{oc}$ | sec | See file: Navigation model.csv |
| $af_2$ | $sec/sec^2$ | See file: Navigation model.csv |
| $af_1$ | sec/sec | See file: Navigation model.csv |

| | | |
|---|---|---|
| $af_0$ | sec | See file: Navigation model.csv |
| $C_{rs}$ | meters | See file: Navigation model.csv |
| $\Delta n$ | semi-circles/sec | See file: Navigation model.csv |
| $M_0$ | semi-circles | See file: Navigation model.csv |
| $C_{uc}$ | radians | See file: Navigation model.csv |
| e | --- | See file: Navigation model.csv |
| $C_{us}$ | radians | See file: Navigation model.csv |
| $(A)^{1/2}$ | $meters^{1/2}$ | See file: Navigation model.csv |
| $t_{oe}$ | sec | See file: Navigation model.csv |
| Fit Interval Flag | Boolean | See file: Navigation model.csv |
| AODO | sec | See file: Navigation model.csv |
| $C_{ic}$ | radians | See file: Navigation model.csv |
| $OMEGA_0$ | semi-circles | See file: Navigation model.csv |
| $C_{is}$ | radians | See file: Navigation model.csv |
| $i_0$ | semi-circles | See file: Navigation model.csv |
| $C_{rc}$ | meters | See file: Navigation model.csv |
| $\omega$ | semi-circles | See file: Navigation model.csv |
| OMEGAdot | semi-circles/sec | See file: Navigation model.csv |
| Idot | semi-circles/sec | See file: Navigation model.csv |

## 10.7.6 Assistance Data Ionospheric Model

Contents of UE positioning GPS ionospheric model IE

**Ionospheric Model**

| Parameter | Units | Value/remark |
|---|---|---|
| $\alpha_0$ | seconds | 4.6566129 10E-9 |
| $\alpha_1$ | sec/semi-circle | 1.4901161 10E-8 |
| $\alpha_2$ | $sec/(semi\text{-}circle)^2$ | -5.96046 10E-8 |
| $\alpha_3$ | $sec/(semi\text{-}circle)^3$ | -5.96046 10E-8 |
| $\beta_0$ | seconds | 79872 |
| $\beta_1$ | sec/semi-circle | 65536 |
| $\beta_2$ | $sec/(semi\text{-}circle)^2$ | -65536 |
| $\beta_3$ | $sec/(semi\text{-}circle)^3$ | -393216 |

## 10.7.7 Assistance Data Almanac

Contents of UE positioning GPS almanac

**Almanac (Field occurring once per message)**

| Parameter | Units | Value/remark |
|---|---|---|
| $WN_a$ | weeks | 212 |

**Satellite Information**

| Parameter | Units | Value/remark |
|---|---|---|
| Number of Satellites | --- | 24 |

**Almanac (Fields occurring once per satellite)**

| Parameter | Units | Value/remark |
|---|---|---|
| DataID | --- | See file: Almanac.csv |
| SatID | --- | PRNs: 1 to 24 |
| e | dimensionless | See file: Almanac.csv |

| | | |
|---|---|---|
| $t_{oa}$ | sec | See file: Almanac.csv |
| δi | semi-circles | See file: Almanac.csv |
| OMEGADOT | semi-circles/sec | See file: Almanac.csv |
| SV Health | Boolean | See file: Almanac.csv |
| $A^{1/2}$ | $meters^{1/2}$ | See file: Almanac.csv |
| $OMEGA_0$ | semi-circles | See file: Almanac.csv |
| $M_0$ | semi-circles | See file: Almanac.csv |
| ω | semi-circles | See file: Almanac.csv |
| $af_0$ | seconds | See file: Almanac.csv |
| $af_1$ | sec/sec | See file: Almanac.csv |

## 10.7.8 Assistance Data Acquisition Assistance

Contents of UE positioning GPS acquisition assistance IE

**GPS Acquisition Assist (Field occurring once per message)**

| Parameter | Units | Value/remark |
|---|---|---|
| GPS TOW msec | msec | 509 400 s. Start time. Add integer number of 1 seconds as required (see note) |
| NOTE: This is the value in seconds of GPS TOW msec when the GPS scenario is started in the GPS simulator. The value of GPS TOW msec to be used in the Acquisition Assistance IE shall be calculated at the time the IE is required by adding the elapsed time since the time the scenario was started in the GPS simulator to this value, rounded up to the next 1 second interval. This "current GPS TOW msec" is then also used to determine the value of the Acquisition Assistance parameters marked as "Time varying" in clause 10.7.8. | | |

**Satellite Information**

| Parameter | Units | Value/remark |
|---|---|---|
| Number of Satellites | --- | 6 |

**GPS Acquisition Assist (Fields occurring once per satellite)**

| Parameter | Units | Value/remark |
|---|---|---|
| SatID | --- | PRNs: 4, 6, 9, 10, 13, 22. |
| Doppler ($0^{th}$ order term) | Hz | Time varying. See file: Acquisition assist .csv (see note) |
| Doppler ($1^{st}$ order term) | Hz/sec | Time varying. See file: Acquisition assist .csv (see note) |
| Doppler Uncertainty | Hz | Time varying. See file: Acquisition assist .csv (see note) |
| Code Phase | chips | Time varying. See file: Acquisition assist .csv (see note) |
| Integer Code Phase | --- | Time varying. See file: Acquisition assist .csv (see note) |
| GPS Bit number | --- | Time varying. See file: Acquisition assist .csv (see note) |
| Code Phase Search Window | chips | Time varying. See file: Acquisition assist .csv (see note) |
| Azimuth | deg | Time varying. See file: Acquisition assist .csv (see note) |
| Elevation | deg | Time varying. See file: Acquisition assist .csv (see note) |
| NOTE: This field is "Time varying" and its value depends on the "current GPS TOW msec" as described in clause 10.7.3. The value of this field to be used shall be determined by taking the "current GPS TOW msec" value and selecting the field value in the Acquisition assist.csv file corresponding to the value of "current GPS TOW msec". | | |

# Annex A (informative):
# Void

# Annex B (informative):
# Void

# Annex C (normative): GPS Data Files

## C.1 GPS data files for signalling tests

The GPS data files for use in signalling tests defined in 3GPP TS 34.123-1 [1] are contained in archive GPS Data Sig V2.zip which accompanies the present document.

## C.2 GPS data files for performance tests

The GPS data files for use in performance tests defined in 3GPP TS 34.171 [41]are contained in archive GPS Data Perf V3.zip which accompanies the present document.

# Annex D (informative): Change history

| Meeting-1st-Level | Doc-1st-Level | CR | Rev | Subject | Cat | Version-Current | Version-New | Doc-2nd-Level |
|---|---|---|---|---|---|---|---|---|
| TP-08 | | | | Approval of the specification | | 2.0.0 | 3.0.0 | |
| TP-09 | TP-000131 | 001 | | RRC Message Contents: RLCSize | C | 3.0.1 | 3.1.0 | T1-000190 |
| TP-09 | TP-000131 | 002 | | RRC Message Contents: RLCParam | C | 3.0.1 | 3.1.0 | T1-000191 |
| TP-09 | TP-000131 | 003 | | RRC Message Contents: PCPreamble | C | 3.0.1 | 3.1.0 | T1-000192 |
| TP-09 | TP-000131 | 004 | | RRC Message Contents: RBIdentity | C | 3.0.1 | 3.1.0 | T1-000193 |
| TP-09 | TP-000131 | 005 | | RRC Message Contents: TrCHParam | C | 3.0.1 | 3.1.0 | T1-000194 |
| TP-09 | TP-000131 | 006 | | RRC Message Contents: UECapability | C | 3.0.1 | 3.1.0 | T1-000195 |
| TP-09 | TP-000131 | 007 | | RRC Message Contents: RBMapping | C | 3.0.1 | 3.1.0 | T1-000196 |
| TP-09 | TP-000131 | 008 | | RRC Message Contents: PagingCause | C | 3.0.1 | 3.1.0 | T1-000197 |
| TP-09 | TP-000131 | 009 | | RRC Message Contents: CipheringAndIntegrity | C | 3.0.1 | 3.1.0 | T1-000198 |
| TP-09 | TP-000131 | 010 | | RRC Message Contents: RLCInfo | C | 3.0.1 | 3.1.0 | T1-000199 |
| TP-09 | TP-000131 | 011 | | RRC Message Contents: CompressedMode | C | 3.0.1 | 3.1.0 | T1-000200 |
| TP-09 | TP-000131 | 012 | | RRC Message Contents: SIB | C | 3.0.1 | 3.1.0 | T1-000201 |
| TP-09 | TP-000131 | 013 | | RRC Message Contents: PhyCH | D | 3.0.1 | 3.1.0 | T1-000202 |
| TP-09 | TP-000131 | 014 | | RRC Message Contents: Measurement | C | 3.0.1 | 3.1.0 | T1-000203 |
| TP-09 | TP-000131 | 015 | | RRC Message Contents: TFCS | C | 3.0.1 | 3.1.0 | T1-000204 |
| TP-09 | TP-000131 | 016 | | RRC Message Contents: DPCHFrameOffset | C | 3.0.1 | 3.1.0 | T1-000205 |
| TP-09 | TP-000131 | 017 | | Test USIM Parameters | F | 3.0.1 | 3.1.0 | T1-000215 |
| TP-09 | TP-000131 | 018 | | Correction to definition of the test algorithm for authentication (clause 8.1.2) | F | 3.0.1 | 3.1.0 | T1-000164 |
| TP-09 | TP-000131 | 019 | | Reference Radio Bearer Configurations | F | 3.0.1 | 3.1.0 | T1-000212 |
| TP-09 | TP-000131 | 020 | | TDD Single mode | F | 3.0.1 | 3.1.0 | T1-000220 |
| TP-10 | TP-000215 | 021 | | Common generic procedure for AS testing | B | 3.1.0 | 3.2.0 | T1-000294 |
| TP-10 | TP-000215 | 022 | | Requirements for the system simulator for support of Tcell parameter | F | 3.1.0 | 3.2.0 | T1-000303 |
| TP-10 | TP-000215 | 023 | | Minimum Performance Levels | F | 3.1.0 | 3.2.0 | T1-000306 |
| TP-10 | TP-000215 | 024 | | Downlink signal conditions and propagation conditions | D | 3.1.0 | 3.2.0 | T1-000307 |
| TP-10 | TP-000215 | 025 | | Updating 34.108 v3.1.0 to TDD single mode | F | 3.1.0 | 3.2.0 | T1-000281 |
| TP-10 | TP-000215 | 026 | | Application of integrity mode protection to signalling message by default | F | 3.1.0 | 3.2.0 | T1-000296 |
| TP-10 | TP-000215 | 027 | | Updates to the default message contents in clause 9 | C | 3.1.0 | 3.2.0 | T1-000282 |
| TP-10 | TP-000215 | 028 | | Updates to System Information Block (SIB) and Master Information Block (MIB) messages | C | 3.1.0 | 3.2.0 | T1-000283 |
| TP-10 | TP-000215 | 029 | | Application of ciphering during conformance testing | C | 3.1.0 | 3.2.0 | T1-000285 |
| TP-10 | TP-000215 | 030 | | Addition for System Information parameters (34.108 clause 6.1) | F | 3.1.0 | 3.2.0 | T1-000304 |
| TP-10 | TP-000215 | 031 | | Correction for Generic Setup Procedures (34.108 clause 7.2) | F | 3.1.0 | 3.2.0 | T1-000305 |
| TP-11 | TP-010018 | 032 | | Default radio conditions for multi-cell environment | F | 3.2.0 | 3.3.0 | T1-010078 |
| TP-11 | TP-010018 | 033 | | Correction for Generic Setup Procedures (34.108 clause 7.2) | F | 3.2.0 | 3.3.0 | T1-010079 |
| TP-11 | TP-010018 | 034 | | Corrections for Test USIM Parameters(34.108 clause 8) | F | 3.2.0 | 3.3.0 | T1-010080 |
| TP-11 | TP-010018 | 035 | | Correction of clause number in TS 34.108. | D | 3.2.0 | 3.3.0 | T1-010081 |
| TP-11 | TP-010018 | 036 | | Update of authentication test algorithm | C | 3.2.0 | 3.3.0 | T1-010082 |
| TP-11 | TP-010018 | 037 | | Updates to clause 9 of TS 34.108 v3.2.0 | F | 3.2.0 | 3.3.0 | T1-010084 |
| TP-11 | TP-010018 | 038 | | Updating to TDD single mode | F | 3.2.0 | 3.3.0 | T1-010088 |
| TP-11 | TP-010018 | 039 | | Simulated network environments for TDD mode (SIB) | F | 3.2.0 | 3.3.0 | T1-010089 |
| TP-12 | TP-010118 | 040 | | Corrections to clause 6.10 FDD parameters | F | 3.3.0 | 3.4.0 | T1-010205 |
| TP-12 | TP-010118 | 041 | | Corrections to clause 6.10 TDD parameters | F | 3.3.0 | 3.4.0 | T1-010206 |
| TP-12 | TP-010118 | 042 | | Adding section for radio bearer configurations intended for functional testing | D | 3.3.0 | 3.4.0 | T1-010210 |
| TP-12 | TP-010118 | 043 | | Update of list of abbreviations | D | 3.3.0 | 3.4.0 | T1-010211 |
| TP-12 | TP-010118 | 044 | | Updates to clause 6.1 and 9 | F | 3.3.0 | 3.4.0 | T1-010212 |
| TP-12 | TP-010118 | 045 | | Updates to clause 7.4 | F | 3.3.0 | 3.4.0 | T1-010213 |
| TP-12 | TP-010118 | 046 | | clause 6.1: System Information Blocks for TDD Mode | F | 3.3.0 | 3.4.0 | T1-010214 |
| TP-12 | TP-010118 | 047 | | Editorial corrections and removal of a reference document | F | 3.3.0 | 3.4.0 | T1-010215 |
| TP-13 | TP-010215 | 048 | | Correction to reference | F | 3.4.0 | 3.5.0 | T1-010275 |
| TP-13 | TP-010215 | 049 | | Editorial modification for References | F | 3.4.0 | 3.5.0 | T1-010276 |
| TP-13 | TP-010215 | 050 | | Some corrections in clause 5 | F | 3.4.0 | 3.5.0 | T1-010277 |
| TP-13 | TP-010215 | 051 | | Update to Scope Statement | F | 3.4.0 | 3.5.0 | T1-010278 |
| TP-13 | TP-010215 | 052 | | Clause 6.10 Definition of RB configurations, TDD parameters | F | 3.4.0 | 3.5.0 | T1-010279 |
| TP-13 | TP-010215 | 053 | | Updates to clause 6.1, clause 7.4 and clause 9 | F | 3.4.0 | 3.5.0 | T1-010280 |
| TP-13 | TP-010215 | 054 | | Clause 6.1: Default radio conditions for Signalling tests | F | 3.4.0 | 3.5.0 | T1-010281 |

| Meeting-1st-Level | Doc-1st-Level | CR | Rev | Subject | Cat | Version-Current | Version-New | Doc-2nd-Level |
|---|---|---|---|---|---|---|---|---|
| TP-13 | TP-010215 | 055 | | Correction of Radio Bearer Configurations for FDD Mode | F | 3.4.0 | 3.5.0 | T1-010282 |
| TP-13 | TP-010215 | 056 | | Correction of Radio Bearer Configurations for TDD Mode | F | 3.4.0 | 3.5.0 | T1-010283 |
| TP-13 | TP-010215 | 057 | | Changes to Signalling Radio Bearer (SRB) numbering | F | 3.4.0 | 3.5.0 | T1-010284 |
| TP-13 | TP-010215 | 058 | | Missing bearers in tables 6.10.2.1.1 and 6.10.3.1.1 | F | 3.4.0 | 3.5.0 | T1-010285 |
| TP-13 | TP-010215 | 059 | | Correction of system information block 5 | F | 3.4.0 | 3.5.0 | T1-010286 |
| TP-13 | TP-010215 | 060 | | Introducing of 1.28 Mcps TDD Mode in clauses 4, 5 and 6 | F | 3.4.0 | 4.0.0 | T1-010287 |
| TP-13 | TP-010215 | 061 | | Introduction of System Information Blocks for 1.28 Mcps TDD Mode | F | 3.4.0 | 4.0.0 | T1-010288 |
| TP-13 | TP-010215 | 062 | | Introduction of typical radio parameters for 1.28 McpsTDD | F | 3.4.0 | 4.0.0 | T1-010289 |
| TP-13 | TP-010215 | 063 | | Clause 6.11 RBs for RLC and PDCP testing | F | 3.4.0 | 3.5.0 | T1-010290 |
| TP-14 | TP-010285 | 065 | 1 | Correction to 6.1 Contents of System Information Blocks | A | 4.0.0 | 4.1.0 | T1-010475 |
| TP-14 | TP-010285 | 067 | 1 | Corrections to clause 6.1, 7.4 and 9 | A | 4.0.0 | 4.1.0 | T1-010473 |
| TP-14 | TP-010258 | 069 | | Reference Radio Conditions | A | 4.0.0 | 4.1.0 | T1-010461 |
| TP-14 | TP-010258 | 071 | | Modification of Test procedures for RF tests | A | 4.0.0 | 4.1.0 | T1-010463 |
| TP-14 | TP-010258 | 073 | | Default message contents for RF tests | A | 4.0.0 | 4.1.0 | T1-010465 |
| TP-14 | TP-010258 | 075 | | Correction to 6.10 Reference Radio Bearer configurations | A | 4.0.0 | 4.1.0 | T1-010467 |
| TP-14 | TP-010258 | 077 | | Definition of default value of rate matching attribute | A | 4.0.0 | 4.1.0 | T1-010469 |
| TP-14 | TP-010258 | 079 | | Update of clause 7.4 and 6.10 | A | 4.0.0 | 4.1.0 | T1-010471 |
| TP-14 | TP-010292 | 081 | | Correction on introduction of clause 6.10 | A | 4.0.0 | 4.1.0 | -- |
| TP-15 | TP-020038 | 083 | | Replacement of Block STTD by Space Code Transmit Diversity (SCTD) (Rel-4) | A | 4.1.0 | 4.2.0 | T1-020092 |
| TP-15 | TP-020038 | 085 | | Update of reference radio conditions (Rel-4) | A | 4.1.0 | 4.2.0 | T1-020098 |
| TP-15 | TP-020038 | 087 | | Update of system reference configurations and default messages (Rel-4) | A | 4.1.0 | 4.2.0 | T1-020100 |
| TP-15 | TP-020038 | 089 | | Corrections to 34108-410 | A | 4.1.0 | 4.2.0 | T1-020102 |
| TP-15 | TP-020038 | 091 | | Introduction of new Reference RABs (Rel-4) | A | 4.1.0 | 4.2.0 | T1-020195 |
| TP-15 | TP-020038 | 094 | | Update of SIBs for TDD (both modes) in TS 34.108 (Rel4) | F | 4.1.0 | 4.2.0 | T1-020107 |
| TP-15 | TP-020038 | 095 | | Clarification of bit rate of Interactive/Background PS RAB function (Rel-4) | A | 4.1.0 | 4.2.0 | T1-020184 |
| | | | | Correction of CR implementation errors in clauses: 6.10.2.2 and 6.10.2.4.1.58.2.1.1 | | 4.2.0 | 4.2.1 | |
| TP-16 | TP-020141 | 108 | | Clause 7(reference) Update of generic setup procedures to use 13.6 kbps SRB in RRC connection establishment TDD (3.84 Mcps and 1.28 Mcps) | F | 4.2.1 | 4.3.0 | T1-020289 |
| TP-16 | TP-020141 | 109 | | Correction to clause 7.3.3.4 RADIO BEARER SETUP message | A | 4.2.1 | 4.3.0 | T1-020291 |
| TP-16 | TP-020141 | 110 | | Change of RM attribute of DL:3.4 kbps SRBs for DCCH in for REL4 | A | 4.2.1 | 4.3.0 | T1-020292 |
| TP-16 | TP-020141 | 111 | | New additional RAB configuration ( R1-020669) for REL4 | A | 4.2.1 | 4.3.0 | T1-020293 |
| TP-16 | TP-020141 | 112 | | Correction of Puncturing Limit for RABs for REL4 | A | 4.2.1 | 4.3.0 | T1-020294 |
| TP-16 | TP-020141 | 113 | | Test USIM | A | 4.2.1 | 4.3.0 | T1-020295 |
| TP-16 | TP-020141 | 114 | | Clause 6.1 (SIBs)Rel 4 (3.84 Mcps and 1.28 Mcps TDD) | F | 4.2.1 | 4.3.0 | T1-020296 |
| TP-16 | TP-020141 | 115 | | Clause 6.10 References for TDD about Clarification of bit rate of Interactive/Background PS RAB | A | 4.2.1 | 4.3.0 | T1-020297 |
| TP-16 | TP-020141 | 116 | | Correction to default message in clause 9 for Rel4 | A | 4.2.1 | 4.3.0 | T1-020298 |
| TP-16 | TP-020141 | 117 | | Correction to clause 6.1 for Rel4 | A | 4.2.1 | 4.3.0 | T1-020299 |
| TP-16 | TP-020141 | 118 | | WCDMA1800 additions for Rel4 | A | 4.2.1 | 4.3.0 | T1-020300 |
| TP-16 | TP-020141 | 119 | | Clause 9.1 Default message contents for TDD ( 3.84 Mcps and 1.28 Mcps) R4 | F | 4.2.1 | 4.3.0 | T1-020301 |
| TP-16 | TP-020141 | 121 | | Update of generic setup procedures to use 13.6 kbps SRB in RRC connection establishment | A | 4.2.1 | 4.3.0 | T1-020434 |
| TP-17 | TP-020184 | 123 | - | Alignment of reference configurations on S-CCPCH with default system information messages | A | 4.3.0 | 4.4.0 | T1-020503 |
| TP-17 | TP-020184 | 125 | - | Addition of reference compressed mode pattern | A | 4.3.0 | 4.4.0 | T1-020505 |
| TP-17 | TP-020184 | 127 | - | Corrections to default message contents as T1S-020347rev1 | A | 4.3.0 | 4.4.0 | T1-020507 |
| TP-17 | TP-020184 | 129 | - | Additional default message contents for RF Testing | A | 4.3.0 | 4.4.0 | T1-020509 |
| TP-17 | TP-020184 | 131 | - | Corrections related to SIB11, SIB12 and to the MEASUREMENT CONTROL message | A | 4.3.0 | 4.4.0 | T1-020527 |
| TP-17 | TP-020184 | 133 | - | Corrections to clause 6.1 (T1S-020349rev1) | A | 4.3.0 | 4.4.0 | T1-020530 |
| TP-17 | TP-020184 | 135 | - | Introduction of reference configurations on S-CCPCH and PRACH with two interactive PS domain RABs | A | 4.3.0 | 4.4.0 | T1-020539 |
| TP-17 | TP-020184 | 137 | - | Removal of reference radio bearer configurations for unidirectional streaming CS RABa above 64 kbps | A | 4.3.0 | 4.4.0 | T1-020541 |

| Meeting-1st-Level | Doc-1st-Level | CR | Rev | Subject | Cat | Version-Current | Version-New | Doc-2nd-Level |
|---|---|---|---|---|---|---|---|---|
| TP-17 | TP-020184 | 140 | - | Some corrections and updates in clause 6.1 for TDD mode | F | 4.3.0 | 4.4.0 | T1-020576 |
| TP-17 | TP-020184 | 142 | - | Inclusion of default message contents for RF in clause 9.2 for TDD mode | F | 4.3.0 | 4.4.0 | T1-020578 |
| TP-18 | TP-020293 | 144 | - | Correction to default messages in 9.1 and 9.2 | A | 4.4.0 | 4.5.0 | T1-020658 |
| TP-18 | TP-020293 | 146 | - | Corrections in the TDD test frequencies according to core specs | A | 4.4.0 | 4.5.0 | T1-020674 |
| TP-18 | TP-020293 | 148 | - | Addition of alternative configuration using Turbo Coding for Interactive or background / UL:8 DL:8 kbps / PS RAB + UL:3.4 DL:3.4 kbps SRBs for DCCH | A | 4.4.0 | 4.5.0 | T1-020694 |
| TP-18 | TP-020293 | 150 | - | Correction to content of clause 6.10.2. | A | 4.4.0 | 4.5.0 | T1-020709 |
| TP-18 | TP-020293 | 152 | - | Correction to SIB 11/12 definition | A | 4.4.0 | 4.5.0 | T1-020712 |
| TP-18 | TP-020293 | 154 | - | Reference Measurement Channels | A | 4.4.0 | 4.5.0 | T1-020768 |
| TP-18 | TP-020293 | 156 | - | Transferring system information definition using ASN.1 description to PRD | A | 4.4.0 | 4.5.0 | T1-020778 |
| TP-18 | TP-020293 | 158 | - | Correction to RLC RAB TFCS | A | 4.4.0 | 4.5.0 | T1-020780 |
| TP-18 | TP-020293 | 160 | - | Default Message contents : Correction from CRs approved in RP17meeting | A | 4.4.0 | 4.5.0 | T1-020783 |
| TP-18 | TP-020293 | 162 | - | Corrections to SIB1 to SIB6 | A | 4.4.0 | 4.5.0 | T1-020799 |
| TP-18 | TP-020293 | 164 | - | Correction to RAB configurations as revision of T1S020756 | A | 4.4.0 | 4.5.0 | T1-020801 |
| TP-18 | TP-020293 | 166 | - | Parameter addition for Reference RABs based on LS from RAN2 | A | 4.4.0 | 4.5.0 | T1-020803 |
| TP-18 | TP-020293 | 168 | - | Addition to clause 7.4 for multi call as T1S-020577rev2 (revision to T1S020820) | A | 4.4.0 | 4.5.0 | T1-020818 |
| TP-18 | TP-020293 | 169 | - | RAB Combinations for IMS Services | F | 4.4.0 | 4.5.0 | T1-020819 |
| TP-18 | TP-020293 | 171 | - | Correction to Contents of the Scheduling Block Syste Information in clause 6.1.3. | F | 4.4.0 | 4.5.0 | T1-020844 |
| TP-19 | TP-030044 | 173 | - | RAB Removal from Rel 4 TS 34.108 as T1S030002rev1 | A | 4.5.0 | 4.6.0 | T1-030037 |
| TP-19 | TP-030044 | 175 | - | Combine all Radio Bearer Setup messages into one table | A | 4.5.0 | 4.6.0 | T1-030040 |
| TP-19 | TP-030044 | 177 | - | Corrections to SB and SIB configurations in clause 6.1 as T1S030046rev1 | A | 4.5.0 | 4.6.0 | T1-030042 |
| TP-19 | TP-030044 | 179 | - | Correction to TS 34.108 Rel-4 ; PAGING TYPE1 message (Packet in PS) | A | 4.5.0 | 4.6.0 | T1-030044 |
| TP-19 | TP-030044 | 181 | - | Clarification of autentication test algorithm and GSM cipher key | A | 4.5.0 | 4.6.0 | T1-030046 |
| TP-19 | TP-030044 | 183 | - | Addition of simulated network environment for inter-RAT test cases | A | 4.5.0 | 4.6.0 | T1-030048 |
| TP-19 | TP-030044 | 185 | - | Corrections to SIB1 to align with default values for LAC and RAC in 51.010-1. | A | 4.5.0 | 4.6.0 | T1-030050 |
| TP-19 | TP-030044 | 187 | - | Addition of default inter-RAT handover messages | A | 4.5.0 | 4.6.0 | T1-030052 |
| TP-19 | TP-030044 | 189 | - | Correction of activation time IEs in default messages | A | 4.5.0 | 4.6.0 | T1-030054 |
| TP-19 | TP-030044 | 191 | - | Correction to default SECURITY MODE COMMAND message | A | 4.5.0 | 4.6.0 | T1-030056 |
| TP-19 | TP-030044 | 193 | - | Addition of option for UL CM only in default reference CM patterns | A | 4.5.0 | 4.6.0 | T1-030058 |
| TP-19 | TP-030044 | 195 | - | Introduction of a reference RB configuration for RMC for BTFD tests (Rel4) | A | 4.5.0 | 4.6.0 | T1-030060 |
| TP-19 | TP-030044 | 197 | - | Update of the RRC connection request messages in 34.108 Rel4 | A | 4.5.0 | 4.6.0 | T1-030063 |
| TP-19 | TP-030043 | 198 | - | Introduction of Conversational PS RABs in Rel 4 TS 34.108 as T1S030003rev1 | F | 4.5.0 | 4.6.0 | T1-030107 |
| TP-19 | TP-030043 | 200 | - | Update of default parameters for 1 to 8 cell environments (TDD), clause 6.1.4, Rel 4 | A | 4.5.0 | 4.6.0 | T1-030208 |
| TP-19 | TP-030043 | 202 | - | Update of Multi-cell environment for default radio conditions (TDD), clause 6.1.6 (Inclusion of cell 4), Rel 4 | A | 4.5.0 | 4.6.0 | T1-030210 |
| TP-19 | TP-030043 | 204 | - | Modification to Generic Registration Procedures | A | 4.5.0 | 4.6.0 | T1-030222 |
| TP-19 | TP-030043 | 206 | - | Update of default configurations to enable testing of low end UE | A | 4.5.0 | 4.6.0 | T1-030228 |
| TP-20 | TP-030098 | 208 | - | Reinstate parameters for Interactive or background /UL:64 kbps / PS RAB | A | 4.6.0 | 4.7.0 | T1-030437 |
| TP-20 | TP-030098 | 210 | - | Correction to Figure 7.4.1.1 (Rel-4) | A | 4.6.0 | 4.7.0 | T1-030483 |
| TP-20 | TP-030098 | 212 | - | Update of SIB 11 and 12 in clause 6.1.0b in TS 34.108 (TDD) | A | 4.6.0 | 4.7.0 | T1-030507 |
| TP-20 | TP-030098 | 214 | - | Update of Default parameters for 1 to 8 cell environments in TS 34.108 (TDD) | A | 4.6.0 | 4.7.0 | T1-030509 |
| TP-20 | TP-030098 | 216 | - | Correction of default messages according to 25331 CR1823 | A | 4.6.0 | 4.7.0 | T1-030632 |
| TP-20 | TP-030098 | 218 | - | Clause 8.2: Definition of default values for authentication key K on test USIM | A | 4.6.0 | 4.7.0 | T1-030644 |
| TP-20 | TP-030098 | 219 | - | Update of Reconfiguration messages | A | 4.6.0 | 4.7.0 | T1-030692 |

| Meeting-1st-Level | Doc-1st-Level | CR | Rev | Subject | Cat | Version-Current | Version-New | Doc-2nd-Level |
|---|---|---|---|---|---|---|---|---|
| TP-20 | TP-030098 | 221 | - | Correction to RADIO BEARER RELEASE and RRC CONNECTION SETUP messages (Revision of T1-030569) | A | 4.6.0 | 4.7.0 | T1-030699 |
| TP-20 | TP-030140 | 226 | - | Correction to default SIB5 (FDD) | A | 4.6.0. | 4.7.0 | T1-030745 |
| TP-21 | TP-030191 | 228 | - | CR to 34.108, Rel-4, Clarification of seg_count in 6.1.0a.3 | A | 4.7.0 | 4.8.0 | T1-030827 |
| TP-21 | TP-030191 | 230 | - | General correction in clause 7.4 for Common generic procedures for AS testing | A | 4.7.0 | 4.8.0 | T1-030976 |
| TP-21 | TP-030191 | 233 | - | Incorrect activation time in CELL_FACH state . | A | 4.7.0 | 4.8.0 | T1-031064 |
| TP-21 | TP-030191 | 235 | - | Incorrect Transport channel Parameters | A | 4.7.0 | 4.8.0 | T1-031066 |
| TP-21 | TP-030191 | 237 | - | Corrections to TS 34.108 common procedures in clause 7.4 of Rel-4 of TS 34.108 | A | 4.7.0 | 4.8.0 | T1-031095 |
| TP-21 | TP-030191 | 239 | - | Removal of RLC AM in the Default Message Content | A | 4.7.0 | 4.8.0 | T1-031151 |
| TP-21 | TP-030191 | 242 | - | CR 34.108 Rel-4: Manual attach in State 7 Registrated Idle Mode on CS/PS | A | 4.7.0 | 4.8.0 | T1-031175 |
| TP-21 | TP-030191 | 244 | - | URA Identity in Cell Update Confirm and URA Update Confirm | A | 4.7.0 | 4.8.0 | T1-031179 |
| TP-21 | TP-030191 | 246 | - | CR to 34.108 R4; Correction to specification to reflect a change already approved in TTCN CR T1-030396 | A | 4.7.0 | 4.8.0 | T1-031241 |
| TP-21 | TP-030191 | 248 | - | CR to 34.108 REL-4; Correction to clause 7.3 Test procedures for RF test | A | 4.7.0 | 4.8.0 | T1-031251 |
| TP-21 | TP-030191 | 240 | - | RB configuration for the support of wideband AMR speech telephony services | F | 4.7.0 | 4.8.0 | T1-031154 |
| TP-22 | TP-030279 | 251 | 1 | Addition of Default message contents for TDD | F | 4.8.0 | 4.9.0 | T1-031659 |
| TP-22 | TP-030279 | 252 | 1 | Addition of Default message contents for TDD | F | 4.8.0 | 4.9.0 | T1-031660 |
| TP-22 | TP-030279 | 253 | 1 | Addition of Default message contents for TDD | F | 4.8.0 | 4.9.0 | T1-031661 |
| TP-22 | TP-030279 | 254 | 1 | Addition of Default message contents for TDD | F | 4.8.0 | 4.9.0 | T1-031662 |
| TP-22 | TP-030279 | 255 | 1 | Addition of Default message contents for TDD | F | 4.8.0 | 4.9.0 | T1-031663 |
| TP-22 | TP-030279 | 256 | 1 | Addition of Default message contents for TDD | F | 4.8.0 | 4.9.0 | T1-031664 |
| TP-22 | TP-030279 | 257 | 1 | Addition of Default message contents for TDD | F | 4.8.0 | 4.9.0 | T1-031665 |
| TP-22 | TP-030279 | 258 | 1 | Addition of Default message contents for TDD | F | 4.8.0 | 4.9.0 | T1-031666 |
| TP-22 | TP-030279 | 260 | 2 | CR on PAGING TYPE 1, RRC CONNECTION REQUEST and RRC CONNECTION SETUP messages for MT RR Connection | A | 4.8.0 | 4.9.0 | T1-031596 |
| TP-22 | TP-030279 | 262 | | CR 34.108 Rel-4: EFRPLMNACT (RPLMN Last used Access Technology) removed | A | 4.8.0 | 4.9.0 | T1-031381 |
| TP-22 | TP-030279 | 264 | 1 | Update of default messages for RRC CONNECTION SETUP and SECURITY MODE COMMAND | A | 4.8.0 | 4.9.0 | T1-031547 |
| TP-22 | TP-030279 | 266 | 1 | Description and corrections of channels for minimum performance levels, TDD mode. | F | 4.8.0 | 4.9.0 | T1-031645 |
| TP-22 | TP-030279 | 268 | 1 | Test frequencies of UMTS800MHz band VI | A | 4.8.0 | 4.9.0 | T1-031555 |
| TP-22 | TP-030279 | 269 | | CR 34.108 Rel-4: Addition of Bearer combination for Interactive/background UL 64 kbps DL 768 kbps for Rel-5 | F | 4.8.0 | 4.9.0 | T1-031441 |
| TP-22 | TP-030279 | 271 | 1 | Update of generic test procedure for TX, RX and Performance Requirement | A | 4.8.0 | 4.9.0 | T1-031610 |
| TP-22 | TP-030279 | 273 | 1 | Introduction of generic test procedure for RRM handover test cases | A | 4.8.0 | 4.9.0 | T1-031608 |
| TP-22 | TP-030279 | 275 | 1 | Correction of CM TGD parameter | A | 4.8.0 | 4.9.0 | T1-031591 |
| TP-22 | TP-030279 | 277 | 1 | Corrections to default message contents of Radio Bearer Release | F | 4.8.0 | 4.9.0 | T1-031594 |
| TP-22 | TP-030279 | 279 | 1 | Modification to default DPCCH_Power_offset value | A | 4.8.0 | 4.9.0 | T1-031598 |
| TP-22 | TP-030279 | 283 | | Correction of TFCS for radio bearer combination 6.10.2.4.1.51b | A | 4.8.0 | 4.9.0 | T1-031527 |
| TP-23 | TP-040037 | 284 | - | New Radio Bearer Setup (FDD) message for RF (Revision of T1-040258) | F | 4.9.0 | 4.10.0 | T1-040417 |
| TP-23 | TP-040037 | 287 | - | Corrections to default message contents of RRC Connection Setup message -> 2nd change not implemented (not implementable) | A | 4.9.0 | 4.10.0 | T1-040080 |
| TP-23 | TP-040037 | 289 | - | Correction to Default parameters for Cells 1 to 8 in MultiPLMN cell environments - Rel-4 | A | 4.9.0 | 4.10.0 | T1-040095 |
| TP-23 | TP-040037 | 291 | - | Corrections to TDD HCR RABs | A | 4.9.0 | 4.10.0 | T1-040103 |
| TP-23 | TP-040037 | 296 | - | LCR Corrections to TDD RABs merge of T1-040104 , T1-040201 and T1-040203 | F | 4.9.0 | 4.10.0 | T1-040299 |
| TP-23 | TP-040037 | 298 | - | Correction to handling of Entered Parameter IE in default contents for Initial Direct Transfer | A | 4.9.0 | 4.10.0 | T1-040411 |
| TP-23 | TP-040037 | 300 | - | The diverse operation in TDD mode updating according to the core specification | A | 4.9.0 | 4.10.0 | T1-040368 |
| TP-23 | TP-040037 | 302 | - | correction of measurement control default message contents for TDD -> Not implemented (not implementable) | F | 4.9.0 | 4.10.0 | T1-040370 |
| TP-23 | TP-040037 | 303 | - | correction of RADIO BEARER SETUP default message | F | 4.9.0 | 4.10.0 | T1-040371 |

| Meeting-1st-Level | Doc-1st-Level | CR | Rev | Subject | Cat | Version-Current | Version-New | Doc-2nd-Level |
|---|---|---|---|---|---|---|---|---|
| | | | | contents for 1.28 Mcps TDD | | | | |
| TP-23 | TP-040037 | 304 | - | Correction of RADIO BEARER RELEASE default message contents for TDD: AM or UM (1.28 Mcps TDD) | F | 4.9.0 | 4.10.0 | T1-040372 |
| TP-23 | TP-040037 | 305 | - | Contents of RRC CONNECTION SETUP message: UM (Transition to CELL_DCH) (1.28 Mcps TDD) -> Not implemented (not implementable) | F | 4.9.0 | 4.10.0 | T1-040373 |
| TP-23 | TP-040037 | 292 | - | New I/B UL:64 DL:768 kbps PS RAB misplaced | F | 4.10.0 | 5.0.0 | T1-040109 |
| TP-23 | TP-040037 | 294 | - | Generic setup procedure and default message contents for HSDPA (as of T1-040069rev1) | F | 4.10.0 | 5.0.0 | T1-040271 |
| TP-23 | TP-040037 | 295 | - | Baseline radio bearer combination for HSDPA support | B | 4.10.0 | 5.0.0 | T1-040273 |
| TP-24 | TP-040112 | 308 | - | Correction to IEs "START" and "ul_CounterSynchronisationInfo". | F | 5.0.0 | 5.1.0 | T1-040512 |
| TP-24 | TP-040112 | 309 | - | Correction to HSDPA reference radio bearer configurations | F | 5.0.0 | 5.1.0 | T1-040522 |
| TP-24 | TP-040112 | 310 | - | Addition of test procedure for HSDPA RF testing | F | 5.0.0 | 5.1.0 | T1-040546 |
| TP-24 | TP-040112 | 315 | - | Corrections to default RRC messages | F | 5.0.0 | 5.1.0 | T1-040593 |
| TP-24 | TP-040112 | 318 | - | Change of default LAC/RAC for inter-RAT test cases | A | 5.0.0 | 5.1.0 | T1-040656 |
| TP-24 | TP-040112 | 319 | - | Contents of Physical channel Reconfiguration message modified to incorporate transition to URA_PCH or CELL_PCH | F | 5.0.0 | 5.1.0 | T1-040673 |
| TP-24 | TP-040112 | 320 | - | Correction of reference test frequencies for UMTS800(band VI) | F | 5.0.0 | 5.1.0 | T1-040701 |
| TP-24 | TP-040112 | 325 | - | Update of generic setup procedures in clauses 7.3.4 and 7.3.5. | A | 5.0.0 | 5.1.0 | T1-040754 |
| TP-24 | TP-040112 | 326 | - | Physical channel parameters for AM RLC 7 bit Length Indicator TestCases (Rel-5) | F | 5.0.0 | 5.1.0 | T1-040902 |
| TP-24 | TP-040112 | 327 | - | Corrections to the default contents of Security Mode Command (Rel-5) | F | 5.0.0 | 5.1.0 | T1-040903 |
| TP-24 | TP-040112 | 330 | - | Corrections to Contents of Scheduling Block 1 (FDD) | F | 5.0.0 | 5.1.0 | T1-040909 |
| TP-24 | TP-040112 | 331 | - | Corrections to Contents of PHYSICAL CHANNEL RECONFIGURATION message: AM or UM | F | 5.0.0 | 5.1.0 | T1-040911 |
| TP-24 | TP-040112 | 332 | - | Corrections to Contents of RRC CONNECTION SETUP message: UM | F | 5.0.0 | 5.1.0 | T1-040913 |
| TP-24 | TP-040112 | 333 | - | RADIO BEARER SETUP message (FDD) for Test Loop Mode2. | F | 5.0.0 | 5.1.0 | T1-040917 |
| TP-24 | TP-040112 | 335 | - | Changes to establish one version of 34.108 covering all releases | A | 5.0.0 | 5.1.0 | T1-040931 |
| TP-24 | TP-040112 | 338 | - | Addition of generic test procedure for AS test cases using the test loop | A | 5.0.0 | 5.1.0 | T1-040934 |
| TP-24 | TP-040112 | 339 | - | Corrections to LCR TDD RABs | F | 5.0.0 | 5.1.0 | T1-040935 |
| TP-25 | TP-040157 | 343 | - | Correction to generic test procedure in clause 7.4.2.6a. | F | 5.1.0 | 5.2.0 | T1-041040 |
| TP-25 | TP-040157 | 344 | - | Addition of default messages for Signalling (FDD) | F | 5.1.0 | 5.2.0 | T1-041044 |
| TP-25 | TP-040157 | 345 | - | Minor change to terminology in SRB tables of clause 6.10 | F | 5.1.0 | 5.2.0 | T1-041140 |
| TP-25 | TP-040157 | 346 | - | Default Message Content for System Information Block type 5 (FDD) and type 6 (FDD) | F | 5.1.0 | 5.2.0 | T1-041154 |
| TP-25 | TP-040157 | 347 | - | Corrections to DCCH Transport channel Parameters for HSDPA RAB | D | 5.1.0 | 5.2.0 | T1-041171 |
| TP-25 | TP-040157 | 348 | - | Corrections to clause 9 | F | 5.1.0 | 5.2.0 | T1-041223 |
| TP-25 | TP-040157 | 349 | - | Corrections to HCR TDD RAB combinations | F | 5.1.0 | 5.2.0 | T1-041235 |
| TP-25 | TP-040157 | 350 | - | Adding missing clause 6.10.2.4.1.62.1 | F | 5.1.0 | 5.2.0 | T1-041252 |
| TP-25 | TP-040157 | 351 | - | Modification of AICH power offset in SysInfo 5 and 6. | F | 5.1.0 | 5.2.0 | T1-041253 |
| TP-25 | TP-040157 | 352 | - | Correction to Default Message Content for Radio Bearer Setup Message. | F | 5.1.0 | 5.2.0 | T1-041259 |
| TP-25 | TP-040157 | 353 | - | Correction to Default Message Content for Radio Bearer Reconfiguration Message for Condition A6 | F | 5.1.0 | 5.2.0 | T1-041266 |
| TP-25 | TP-040157 | 354 | - | CR to 34.108: introduction of default RB SETUP message from cell_FACH state for HSDPA | F | 5.1.0 | 5.2.0 | T1-041298 |
| TP-25 | TP-040157 | 355 | - | Corrections to Contents of RADIO BEARER SETUP message: BTFD RMC | F | 5.1.0 | 5.2.0 | T1-041317 |
| TP-25 | TP-040157 | 340 | - | Resolution of downlink code conflict between OCNS DPCH and S-CCPCH | F | 5.1.0 | 5.2.0 | T1-041327 |
| TP-25 | TP-040157 | 361 | - | Correction to test procedure for test cases using Cell_PCH or URA_PCH state | F | 5.1.0 | 5.2.0 | T1-041346 |
| TP-25 | TP-040157 | 362 | - | Removal of DCCH dummy transmission for RF testing | F | 5.1.0 | 5.2.0 | T1-041350 |
| TP-25 | TP-040157 | 341 | - | Correct title to test procedure for test cases using Cell_PCH or URA_PCH state | F | 5.1.0 | 5.2.0 | T1-041354 |
| TP-25 | TP-040157 | 363 | - | Addition of intra frequency cell to cell environments | F | 5.1.0 | 5.2.0 | T1-041356 |
| TP-25 | TP-040157 | 342 | - | Correct primary scrambling code usage in default message contents in clause 9.2.1 | F | 5.1.0 | 5.2.0 | T1-041365 |
| TP-25 | TP-040157 | 356 | - | HSDPA downlink code allocation | F | 5.1.0 | 5.2.0 | T1-041374 |
| TP-25 | TP-040157 | 357 | - | Correction to test procedure for test cases using | F | 5.1.0 | 5.2.0 | T1-041376 |

| Meeting-1st-Level | Doc-1st-Level | CR | Rev | Subject | Cat | Version-Current | Version-New | Doc-2nd-Level |
|---|---|---|---|---|---|---|---|---|
| | | | | CELL_FACH state | | | | |
| TP-25 | TP-040157 | 358 | - | Varying DPCH Power Offset according to data transmission rate | F | 5.1.0 | 5.2.0 | T1-041416 |
| TP-25 | TP-040157 | 359 | - | Corrections to default message for RADIO BEARER SETUP message in clause 9.2.1 (HSDPA RF) | F | 5.1.0 | 5.2.0 | T1-041418 |
| TP-25 | TP-040157 | 360 | - | Test SIB schedule for two S-CCPCH or two PRACH in 34.108 | F | 5.1.0 | 5.2.0 | T1-041422 |
| TP-25 | TP-040157 | 364 | - | Correction to Default Message Content for Radio Bearer Setup Message re: RM Attribute values | F | 5.1.0 | 5.2.0 | T1-041433 |
| TP-26 | TP-040233 | 365 | - | CR to 34.108 Rel-5:Correction to default value of Qrxlevmin | F | 5.2.0 | 5.3.0 | T1-041532 |
| TP-26 | TP-040233 | 366 | - | CR to 34.108 Rel-5: Corrections of the values in 6.11.5.4 for LCR TDD | F | 5.2.0 | 5.3.0 | T1-041573 |
| TP-26 | TP-040233 | 367 | - | Alignment of Prose to TTCN for SCH power level | F | 5.2.0 | 5.3.0 | T1-041584 |
| TP-26 | TP-040233 | 368 | - | Addition of new HSDPA RAB configurations with UL 64 kbps | F | 5.2.0 | 5.3.0 | T1-041651 |
| TP-26 | TP-040233 | 369 | - | Correction to initial conditions and references in clause 7.3 | F | 5.2.0 | 5.3.0 | T1-041654 |
| TP-26 | TP-040233 | 370 | - | Introduction of reference radio bearer combination for PS streaming and downlink rate up to 128 kbps | F | 5.2.0 | 5.3.0 | T1-041685 |
| TP-26 | TP-040233 | 371 | - | Correction of clause 6.1 (Simulated network environment) | F | 5.2.0 | 5.3.0 | T1-041686 |
| TP-26 | TP-040233 | 372 | - | Correction to generic Call Setup procedure for mobile terminating circuit switched calls | F | 5.2.0 | 5.3.0 | T1-041699 |
| TP-26 | TP-040233 | 373 | - | CR to 34.108 Rel-5; Corrections to the default RADIO BEARER SETUP message for HSDPA | F | 5.2.0 | 5.3.0 | T1-041754 |
| TP-26 | TP-040233 | 374 | - | Physical layer multiplexing configuration in case of AMR and two PS RABs | F | 5.2.0 | 5.3.0 | T1-041801 |
| TP-26 | TP-040233 | 375 | - | Addition of new HSDPA RAB configurations | F | 5.2.0 | 5.3.0 | T1-041802 |
| TP-26 | TP-040233 | 376 | - | Introduction of information for tests for Performance requirements for A-GPS. | B | 5.2.0 | 5.3.0 | T1-041850 |
| TP-26 | TP-040233 | 377 | - | Introduction of UMTS-850 MHz band V | F | 5.2.0 | 5.3.0 | T1-041874 |
| TP-26 | TP-040233 | 378 | - | CR to TS 34.108 Rel-5; Adding a new test condition for RADIO BEARER RELEASE Procedure (Revision of T1-041716). | F | 5.2.0 | 5.3.0 | T1-041933 |
| TP-26 | TP-040233 | 379 | - | Update of Reference Radio Bearer for Conversational / speech / UL:5.9 DL:5.9 kbps / CS RAB for DL SF=256 | F | 5.2.0 | 5.3.0 | T1-041942 |
| TP-26 | TP-040233 | 380 | - | CR to 34.108: Correction to the maximum bit rate for HS-PDSCH | F | 5.2.0 | 5.3.0 | T1-041943 |
| TP-26 | TP-040233 | 381 | - | Alignment of Prose to TTCN for RRC Connection Release (Cell DCH state) and RRC Connection Setup Message (Cell FACH State). | F | 5.2.0 | 5.3.0 | T1-041965 |
| TP-27 | TP-050032 | 382 | - | Updates from core specification changes | F | 5.3.0 | 5.4.0 | T1-050095 |
| TP-27 | TP-050032 | 383 | - | Correction to Hand over test procedure in CELL_DCH | F | 5.3.0 | 5.4.0 | T1-050350 |
| TP-27 | TP-050032 | 384 | - | CR to 34.108: Changes to test frequencies for UMTS 850 Band | B | 5.3.0 | 5.4.0 | T1-050380 |
| TP-27 | TP-050032 | 385 | - | Correction to default SIB configurations | F | 5.3.0 | 5.4.0 | T1-050019 |
| TP-27 | TP-050032 | 386 | - | Editorial corrections in HSDPA RAB configurations 6.10.2.4.5.2 and 6.10.2.4.5.4. | D | 5.3.0 | 5.4.0 | T1-050052 |
| TP-27 | TP-050032 | 387 | - | CR to 34.108 Rel-5: Update to the contents of PHYSICAL CHANNEL RECONFIGURATION message for 1.28 Mcps TDD | F | 5.3.0 | 5.4.0 | T1-050064 |
| TP-27 | TP-050032 | 388 | - | CR to 34.108 Rel-5: Update to the contents of TRANSPORT CHANNEL RECONFIGURATION message for 1.28 Mcps TDD | F | 5.3.0 | 5.4.0 | T1-050065 |
| TP-27 | TP-050032 | 389 | - | CR to 34.108 Rel-5: Update to the contents of RRC CONNECTION REQUEST message for TDD | F | 5.3.0 | 5.4.0 | T1-050066 |
| TP-27 | TP-050032 | 390 | - | Correction to the HSDPA RB Identity in Radio Bearer Setup & Radio Bearer Release message contents | F | 5.3.0 | 5.4.0 | T1-050072 |
| TP-27 | TP-050032 | 391 | - | CR to TS 34.108 v5.3.0 - Correction to Default RADIO BEARER RELEASE message (FDD) | F | 5.3.0 | 5.4.0 | T1-050202 |
| TP-27 | TP-050032 | 392 | - | Addition of reference radio bearer configuration for MAC-hs testing | F | 5.3.0 | 5.4.0 | T1-050239 |
| TP-27 | TP-050032 | 393 | - | CR to 34.108 Rel-5: Update to the contents of RRC CONNECTION REQUEST message for TDD | F | 5.3.0 | 5.4.0 | T1-050295 |
| TP-27 | TP-050032 | 394 | - | CR to 34.108 Rel-5: Update to the contents of Default System Information Block Messages for TDD | F | 5.3.0 | 5.4.0 | T1-050296 |
| TP-27 | TP-050032 | 395 | - | CR to 34.108 Rel-5: Add the contents of SIB 5 & 6 for HCR TDD | F | 5.3.0 | 5.4.0 | T1-050297 |
| TP-27 | TP-050032 | 396 | - | Correction to TFCS ordering | F | 5.3.0 | 5.4.0 | T1-050451r1 |
| TP-27 | TP-050032 | 397 | - | Addition of GPS scenario and A-GPS assistance data values for signalling tests to 34.108 | F | 5.3.0 | 5.4.0 | T1-050458 |

| Meeting-1st-Level | Doc-1st-Level | CR | Rev | Subject | Cat | Version-Current | Version-New | Doc-2nd-Level |
|---|---|---|---|---|---|---|---|---|
| TP-27 | TP-050032 | 398 | - | CR to TS 34.108 Rel-5; Correction to the physical channel parameters (Revison of T1-050176) | F | 5.3.0 | 5.4.0 | T1-050469 |
| RP-28 | RP-050267 | 399 | - | Additional call setup procedures for inter RAT RRM testing | F | 5.4.0 | 5.5.0 | R5-050618 |
| RP-28 | RP-050267 | 400 | - | CR to 34.108: Correction to RADIO BEARER SETUP message for BTFD RMC | F | 5.4.0 | 5.5.0 | R5-050704 |
| RP-28 | RP-050267 | 401 | - | CR to 34.108: Correction to reference radio conditions for GSM | F | 5.4.0 | 5.5.0 | R5-050811 |
| RP-28 | RP-050267 | 402 | - | Addition of RADIO BEARER SETUP Messages for Auxiliary Measurement | F | 5.4.0 | 5.5.0 | R5-050856 |
| RP-28 | RP-050267 | 404 | - | CR 34.108 Addition of specific message content to A-GPS performance test procedures in clause 7.5 | F | 5.4.0 | 5.5.0 | R5-050709 |
| RP-28 | RP-050267 | 405 | - | CR to 34.108 Rel-5: Clarification of generic setup procedures in section 7.3.4 | F | 5.4.0 | 5.5.0 | R5-050663 |
| RP-28 | RP-050267 | 406 | - | Removal of TGPL2 | F | 5.4.0 | 5.5.0 | R5-050513 |
| RP-28 | RP-050267 | 407 | - | Addition of compressed mode pattern for Inter Frequency FDD measurement & Inter RAT measurement GSM | F | 5.4.0 | 5.5.0 | R5-050525 |
| RP-28 | RP-050267 | 408 | - | Correction to MIB, PLMN and Cell Value Tag Value Definition to 34.108 | F | 5.4.0 | 5.5.0 | R5-050608 |
| RP-28 | RP-050267 | 409 | - | CR to 34.108 Rel-5: Corrections to the contents of System Information Block type 11 (3.84 Mcps and 1.28 Mcps TDD) in section 6.1.0b | F | 5.4.0 | 5.5.0 | R5-050613 |
| RP-28 | RP-050267 | 410 | - | CR to 34.108 Rel-5: Corrections to the usage of 'Cell info' IE in System Information Block type 11 in section 6.1.4 for TDD cell | F | 5.4.0 | 5.5.0 | R5-050619 |
| RP-28 | RP-050267 | 411 | - | CR to 34.108 Rel-5: Corrections to the contents of System Information Block type 5 (1.28 Mcps TDD) | F | 5.4.0 | 5.5.0 | R5-050620 |
| RP-28 | RP-050267 | 412 | - | Update to clause 8 Test USIM Parameters | F | 5.4.0 | 5.5.0 | R5-050638 |
| RP-28 | RP-050267 | 413 | - | CR to 34.108 Rel-5: Update of SIB3, SIB4, SIB11 and SIB12 for TDD in section 6.1.0b | F | 5.4.0 | 5.5.0 | R5-050662 |
| RP-28 | RP-050267 | 414 | - | CR to 34.108: Correction to TFCS | F | 5.4.0 | 5.5.0 | R5-050677 |
| RP-28 | RP-050267 | 415 | - | CR to TS34.108 Rel-5; Correction to the physical channel parameter | F | 5.4.0 | 5.5.0 | R5-050724 |
| RP-28 | RP-050267 | 416 | - | Correction to default SIB configurations | F | 5.4.0 | 5.5.0 | R5-050947 |
| RP-28 | RP-050267 | 417 | - | CR to 34.108: Missing Rel-5 IE's in the default Radio Bearer Setup message at section 9.1.1. | F | 5.4.0 | 5.5.0 | R5-050600 |
| RP-28 | RP-050267 | 418 | - | CR to TS34.108 Rel-5; Clarification of the reference TFCS for three RB multiplexing option (condition A9) | F | 5.4.0 | 5.5.0 | R5-050913 |
| RP-28 | RP-050268 | 419 | - | Addition of new HSDPA Streaming RAB configurations | F | 5.4.0 | 5.5.0 | R5-050880 |
| RP-28 | RP-050268 | 420 | - | CR to 34.108 Rel-5: Content Correction of RRC CONNECTION SETUP message for LCR TDD in 9.1.2 | F | 5.4.0 | 5.5.0 | R5-050585 |
| RP-28 | RP-050268 | 421 | - | Add Default RADIO BEARER RELEASE message (3.84 Mcps TDD) | F | 5.4.0 | 5.5.0 | R5-050680 |
| RP-28 | RP-050268 | 422 | - | Add Default Contents of RADIO BEARER RECONFIGURATION COMPLETE message: AM (3.84 Mcps TDD) | F | 5.4.0 | 5.5.0 | R5-050681 |
| RP-28 | RP-050268 | 423 | - | Add Default Contents of RADIO BEARER RECONFIGURATION message: AM or UM (3.84 Mcps TDD) | F | 5.4.0 | 5.5.0 | R5-050682 |
| RP-28 | RP-050268 | 424 | - | Add Default Contents of PHYSICAL CHANNEL RECONFIGURATION message: AM or UM (3.84 Mcps TDD) | F | 5.4.0 | 5.5.0 | R5-050683 |
| RP-28 | RP-050268 | 425 | - | Add Default Contents of PHYSICAL CHANNEL RECONFIGURATION COMPLETE message: AM (3.84 Mcps TDD) | F | 5.4.0 | 5.5.0 | R5-050684 |
| RP-28 | RP-050268 | 426 | - | Add Default Contents of TRANSPORT CHANNEL RECONFIGURATION message: AM or UM (3.84 Mcps TDD) | F | 5.4.0 | 5.5.0 | R5-050685 |
| RP-28 | RP-050268 | 427 | - | Add Default Contents of TRANSPORT CHANNEL RECONFIGURATION COMPLETE message: AM (3.84 Mcps TDD) | F | 5.4.0 | 5.5.0 | R5-050686 |
| RP-28 | RP-050268 | 428 | - | Add Default Contents of MEASUREMENT REPORT message: AM (intra/inter-frequency measurement (3.84 Mcps TDD) | F | 5.4.0 | 5.5.0 | R5-050956 |
| RP-28 | RP-050268 | 430 | - | Correction to RADIO BEARER SETUP message for HSDPA RF testing | F | 5.4.0 | 5.5.0 | R5-050879 |
| RP-28 | RP-050349 | 403 | - | Addition of GPS scenario and assistance data for A-GPS performance tests in 34.108 | B | 5.4.0 | 5.5.0 | R5-050836 |
| RP-28 | RP-050350 | 429 | - | Corrections to section 10.7 and GPS data file for 34.108 | F | 5.4.0 | 5.5.0 | R5-050969 |
| RP-29 | RP-050600 | 431 | - | Feature Clean Up: Removal of DRAC from section 9 of 34.108 | F | 5.5.0 | 6.0.0 | R5-051312 |

| Meeting-1st-Level | Doc-1st-Level | CR | Rev | Subject | Cat | Version-Current | Version-New | Doc-2nd-Level |
|---|---|---|---|---|---|---|---|---|
| RP-29 | RP-050600 | 432 | - | Feature Clean Up: Removal of SSDT from 34.108 | F | 5.5.0 | 6.0.0 | R5-051356 |
| RP-29 | RP-050600 | 433 | - | Feature Clean Up: Removal of 80 ms TTI for DCH for all cases except when the UE supports SF512 from 34.108 | F | 5.5.0 | 6.0.0 | R5-051379 |
| RP-29 | RP-050600 | 434 | - | Feature Clean Up: Removal of CPCH from section 4 of 34.108 | C | 5.5.0 | 6.0.0 | R5-051543 |
| RP-29 | RP-050600 | 435 | - | Feature Clean Up: Removal of CPCH from section 6 of 34.108 | C | 5.5.0 | 6.0.0 | R5-051544 |
| RP-29 | RP-050600 | 436 | - | Feature Clean Up: Removal of CPCH from section 7 & 8 of 34.108 | C | 5.5.0 | 6.0.0 | R5-051545 |
| RP-29 | RP-050600 | 437 | - | Feature Clean Up: Removal of CPCH from section 9 of 34.108 | C | 5.5.0 | 6.0.0 | R5-051546 |
| RP-29 | RP-050600 | 438 | - | Feature Clean Up: Removal of DSCH ( FDD mode) from 34.108 | F | 5.5.0 | 6.0.0 | R5-051548 |
| RP-29 | RP-050600 | 439 | - | Modification to PS setup procedure for inter RAT RRM testing | F | 5.5.0 | 6.0.0 | R5-051161 |
| RP-29 | RP-050600 | 440 | - | CR to 34.108: RRC CONNECTION SETUP exception for HSDPA testing | F | 5.5.0 | 6.0.0 | R5-051430 |
| RP-29 | RP-050600 | 441 | - | CR to 34.108: Correction to the RADIO BEARER SETUP message for HSDPA testing | F | 5.5.0 | 6.0.0 | R5-051112 |
| RP-29 | RP-050512 | 442 | - | Changes to GPS Scenarios and Assistance data in TS 34.108 | F | 5.5.0 | 6.0.0 | R5-051076 |
| RP-29 | RP-050514 | 443 | - | CR to 34.108 Rel-5: Correction of contents of RADIO BEARER SETUP message: AM or UM (1.28 Mcps TDD) in 9.2.2 | F | 5.5.0 | 6.0.0 | R5-051212 |
| RP-29 | RP-050514 | 444 | - | CR to 34.108 Rel-5: SIB default schedule in 6.1.0a - Default Master Information Block and Scheduling Block messages | F | 5.5.0 | 6.0.0 | R5-051213 |
| RP-29 | RP-050514 | 445 | - | CR to 34.108 Rel-5: Corrections to the IE"Midamble shift and burst type" of System Information Block type 5/6 (3.84Mcps TDD) in section 6.1.0b | F | 5.5.0 | 6.0.0 | R5-051222 |
| RP-29 | RP-050514 | 446 | - | CR to 34.108 Rel-5: Corrections to the contents of System Information Block type 5 (3.84 Mcps TDD) in section 6.1.1 | F | 5.5.0 | 6.0.0 | R5-051344 |
| RP-29 | RP-050514 | 447 | - | CR to 34.108 Rel-5: Corrections to the value of Sintrasearch and Sintersearch in "Cell selection and reselection quality measure" of System Information Block type 3/4 (1.28Mcps TDD and 3.84Mcps TDD) in section 6.1.0b | F | 5.5.0 | 6.0.0 | R5-051536 |
| RP-29 | RP-050600 | 448 | - | Use 'Same as UL' for the Added or Reconfigured DL TrCH information of the added or reconfigured PS RAB | F | 5.5.0 | 6.0.0 | R5-051041 |
| RP-29 | RP-050600 | 449 | - | Correction to the default contents for Radio Bearer Setup message | F | 5.5.0 | 6.0.0 | R5-051044 |
| RP-29 | RP-050600 | 450 | - | Corrections to default parameters of UL:384kbps PS Bearer | F | 5.5.0 | 6.0.0 | R5-051058 |
| RP-29 | RP-050600 | 451 | - | Correction to NB AMR Radio Bearer Configurations | F | 5.5.0 | 6.0.0 | R5-051318 |
| RP-29 | RP-050600 | 452 | - | Correction to default contents of Cell Update and Initial Direct transfer message for Rel-5 | F | 5.5.0 | 6.0.0 | R5-051325 |
| RP-29 | RP-050600 | 453 | - | Correction to DPCCH Power Offset IE in default contents for RRC Connection Setup and Radio Bearer Setup message | F | 5.5.0 | 6.0.0 | R5-051365 |
| RP-29 | RP-050515 | 454 | - | Using Test USIM for VSTK generation of VGCS/VBS ciphering | B | 5.5.0 | 6.0.0 | R5-051553 |
| RP-29 | RP-050600 | 455 | - | Correction to default contents of RADIO BEARER SETUP MESSAGE for the IE "Number of Processes" | F | 5.5.0 | 6.0.0 | R5-051324 |
| RP-29 | RP-050600 | 456 | - | Correction of DL channelisation code in RADIO BEARER SETUP for HSDPA configurations | F | 5.5.0 | 6.0.0 | R5-051345 |
| RP-29 | RP-050513 | 457 | - | Clarifiation of reference radio bearer configuration for MAC-hs test case 7.1.5.2. | F | 5.5.0 | 6.0.0 | R5-051164 |
| RP-29 | RP-050600 | 458 | - | Replacement of the technical content of 34.108 Rel-5 by a pointer to Rel-6 document | F | 5.5.0 | 6.0.0 | R5-051584 |
| RP-29 | RP-050600 | 459 | - | Introduction of HSDPA + Wideband AMR radio bearer combination | F | 5.5.0 | 6.0.0 | R5-051588 |
| RP-29 | RP-050600 | 460 | - | Feature Clean Up: Removal of DRAC from section 6 of 34.108 | F | 5.5.0 | 6.0.0 | R5-051311 |
| RP-30 | RP-050767 | 461 | - | Correction to 34.108: RF Radio Bearer setup message for HSDPA testing | F | 6.0.0 | 6.1.0 | R5-052320 |
| RP-30 | RP-050720 | 462 | - | Changes, additions and corrections to GPS scenarios and assistance data in TS 34.108 | F | 6.0.0 | 6.1.0 | R5-052017 |
| RP-30 | RP-050716 | 463 | - | Generic test procedure for EDCH RF testing | F | 6.0.0 | 6.1.0 | R5-052345 |
| RP-30 | RP-050769 | 464 | - | Correction of UE test states for RF testing. | F | 6.0.0 | 6.1.0 | R5-052302 |
| RP-30 | RP-050769 | 465 | - | Removal of temporary BLER measurement configuration | F | 6.0.0 | 6.1.0 | R5-051933 |

| Meeting-1st-Level | Doc-1st-Level | CR | Rev | Subject | Cat | Version-Current | Version-New | Doc-2nd-Level |
|---|---|---|---|---|---|---|---|---|
| RP-30 | RP-050780 | 466 | - | Introduction of UMTS1700 for TS34.108 | B | 6.0.0 | 6.1.0 | R5-052329 |
| RP-30 | RP-050767 | 467 | - | Re-definition of reference radio bearer configuration for MAC-hs test case 7.1.5.2 | F | 6.0.0 | 6.1.0 | R5-051857 |
| RP-30 | RP-050767 | 468 | - | Introduction of additional HSDPA radio bearer combination | F | 6.0.0 | 6.1.0 | R5-052159 |
| RP-30 | RP-050717 | 469 | - | CR to 34.108; Correction to default message content in INITIAL DIRECT TRANSFER | F | 6.0.0 | 6.1.0 | R5-052179 |
| RP-30 | RP-050716 | 470 | - | CR to 34.108 Rel-6; Generic Setup Procedure and Default RRC Messages for Enhanced Uplink Tests | F | 6.0.0 | 6.1.0 | R5-052192 |
| RP-30 | RP-050716 | 471 | - | Generic reference bearer radio configurations for E-DCH testing | F | 6.0.0 | 6.1.0 | R5-052167 |
| RP-30 | RP-050716 | 472 | - | Addition of basic radio bearer combinations for E-DCH testing | F | 6.0.0 | 6.1.0 | R5-052144 |
| RP-30 | RP-050718 | 473 | - | CR to TS34.108; Correction to the default system information block type3 for DSAC | F | 6.0.0 | 6.1.0 | R5-052145 |
| RP-30 | RP-050769 | 474 | - | Editorial corrections to TS 34.108 | D | 6.0.0 | 6.1.0 | R5-051840 |
| RP-30 | RP-050769 | 475 | - | Removal of deprecated values from the default contents for RRC Connection Setup (Cell FACH) and RRC Connection Release (Cell DCH) messages. | F | 6.0.0 | 6.1.0 | R5-051848 |
| RP-30 | RP-050769 | 476 | - | Correction to the default RRC message contents for the IE “UARFCN uplink (Nu)” | F | 6.0.0 | 6.1.0 | R5-052155 |
| RP-30 | RP-050769 | 477 | - | Correction of references to IB UL:8 DL:8 kbps transport channel parameters in reference radio bearer configuration 6.10.2.4.1.58a | F | 6.0.0 | 6.1.0 | R5-051858 |
| RP-30 | RP-050769 | 478 | - | Correction of UE states tables for Generic setup procedures. | F | 6.0.0 | 6.1.0 | R5-051942 |
| RP-30 | RP-050769 | 479 | - | Corrections to default message contents of ‘HANDOVER FROM UTRAN COMMAND-GSM’ message. | F | 6.0.0 | 6.1.0 | R5-051955 |
| RP-30 | RP-050776 | 480 | - | Addition of multi-rate AMR-NB configuration with SRB#5 | F | 6.0.0 | 6.1.0 | R5-052176 |
| RP-30 | RP-050769 | 481 | - | Proposed CR to 34.108 [R6 version, R99 affected] to change slot format for AMR 5.9 mono rate RAB | F | 6.0.0 | 6.1.0 | R5-052055 |
| RP-30 | RP-050769 | 482 | - | Correction to puncturing limit in radio bearer configuration 6.10.2.4.1.38a | F | 6.0.0 | 6.1.0 | R5-052096 |
| RP-30 | RP-050777 | 483 | - | Introduction of third RAB sublfow to WB-AMR test configurations | F | 6.0.0 | 6.1.0 | R5-052140 |
| RP-30 | RP-050833 | 484 | - | Correction to RADIO BEARER SETUP message for HSDPA RF testing | F | 6.0.0 | 6.1.0 | - |
| RP-30 | RP-050832 | 485 | - | Correction to test procedure for HSDPA RF testing | F | 6.0.0 | 6.1.0 | - |
| | | | | 2006-01: Editorial conversion to make file compatible with Word 2000. | | 6.1.0 | 6.1.1 | |
| RP-31 | RP-060154 | 486 | - | Corrections to GPS data files for signalling tests. | F | 6.1.1 | 6.2.0 | R5-060522 |
| RP-31 | RP-060163 | 487 | - | Corrections for reference RABs | F | 6.1.1 | 6.2.0 | R5-060273 |
| RP-31 | RP-060144 | 488 | - | Corrections to the RADIO BEARER SETUP message for Enhanced uplink | F | 6.1.1 | 6.2.0 | R5-060335 |
| RP-31 | RP-060150 | 489 | - | Correction to default message content in INITIAL DIRECT TRANSFER | F | 6.1.0 | 6.2.0 | R5-060396 |
| RP-31 | RP-060154 | 490 | - | Corrections to default message contents for signaling | F | 6.1.1 | 6.2.0 | R5-060274 |
| RP-31 | RP-060163 | 491 | - | Update of RB configuration 6.10.2.4.1.4b to increase test coverage for multi-mode AMR configurations | F | 6.1.1 | 6.2.0 | R5-060122 |
| RP-31 | RP-060144 | 492 | - | Correction of UE RRC states table for common procedures (section 7.4.1) | F | 6.1.1 | 6.2.0 | R5-060268 |
| RP-31 | RP-060154 | 493 | - | Correction to DPCCH power offset value in RADIO BEARER SETUP messages | F | 6.1.1 | 6.2.0 | R5-060459 |
| RP-31 | RP-060153 | 494 | - | Introduction of Band VII and Band VIII to TS34.108 Chapter 5 and introduction of new UARFCN scheme | F | 6.1.1 | 6.2.0 | R5-060440 |
| RP-31 | RP-060154 | 495 | - | Clarification of RB Test Mode State for RF testing | F | 6.1.1 | 6.2.0 | R5-060446 |
| RP-31 | RP-060144 | 496 | - | Common message content for E-DCH RF testing | F | 6.1.1 | 6.2.0 | R5-060439 |
| RP-31 | RP-060144 | 497 | - | Generic test procedure for E-DCH RF testing | F | 6.1.1 | 6.2.0 | R5-060438 |
| RP-31 | RP-060166 | 498 | - | Correction to GPS Assistance Data for Performance tests | F | 6.1.1 | 6.2.0 | R5-060007 |
| RP-31 | RP-060154 | 499 | - | Corrections to default message contents for RF | F | 6.1.1 | 6.2.0 | R5-060257 |
| RP-31 | RP-060163 | 500 | - | Adding of RB Configuration 6.11.4b to clause 6.11 to increase test coverage for Interactive or background PS RAB UL 0kbps/DL 0kbps. | F | 6.1.1 | 6.2.0 | R5-060571 |

# History

| Document history | | |
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